# Self-Adaptive Kernels as Evolving Measures: Transport Adaptive Control

Mathematical Foundations of Self-Adaptive Kernels, version 3: a complete mathematical treatment

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October 2026

**Version 3.** This document supersedes version 2 and, before it, version 1, *Mathematical Foundations of Self-Adaptive Kernels Achieving Lyapunov Convergence*. Section 2 states what survived from version 1 and what was withdrawn; version 3 adds Lemma E.0 (explicit constants for Theorem E), Theorems C′, D′, I and J and Proposition K (Table 1); a second increment adds the sharp slow-regime rate (Lemma E.0′), uniform-in-time propagation of chaos in the leak-dominated regime (Theorem D″) and the stationary law of the dithered finite-size loop with its non-equilibrium current (Theorem J′, Proposition J′.2); a third increment adds the explicit stationary parameter spread of the dithered linearised loop (Corollary E.0′.3) and the slow-regime Lyapunov function for non-periodic persistently exciting regressors (Lemma E.0″) and uniform-in-time propagation of chaos in the learning regime for the averaged slow loop (Theorem D‴, with the finite-$m$ weight-sensitivity law of Proposition D‴.5, the real loop on a warm start of Corollary D‴.7, the leak's prior as a design input of the window, Proposition D‴.9, and the barycentric OT leak's mass constraint, which removes the finite-size floor, Proposition D‴.12 — in the real loop too, where the collapsed subspace carries one $K_0$-atom orbit for every commensurate size, Corollary D‴.14 and Proposition D‴.15, and Theorem J holds uniformly in the number of kernels in the per-particle norm, Theorem D‴.18, while a soft (entropic) mass constraint equalises the masses only down to a critical excess, Proposition D‴.20; and the mass-constrained energy satisfies a Polyak–Łojasiewicz inequality whose constant does not depend on the number of kernels, which gives a measure-level basin with an exponential rate and, at the nearest-atom leak, explains the floor as a spurious minimum at every admissible count vector, Theorem D‴.21). This is research in progress and has not been peer reviewed. Code, tests, experiments and review records: https://github.com/Dutta-06/Lyapunov-Convergence-Kernels-Showcase.

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**Abstract**

This is a complete mathematical treatment of adaptive tracking control with Gaussian radial-basis-function (RBF) networks whose weights, centres and widths are adapted online. Part I is the projection-based adaptive RBF controller of version 1. An audit found its stability argument to be a correct but standard composition of known results, with two claims wrong as written (a non-convex "projection" for centre separation and an implicit-function-theorem argument that is not a proof); it contains no new theorem. We state it with explicit assumptions, a Hessian-free disturbance bound and an RMS bound, and list the errata to version 1. Part II develops transport adaptive control: the network is an empirical measure on a lifted parameter space, and the adaptive law is its transport. Theorems A–F, C′ and D′ are proved; A, B, C, C′, E and F are also checked by tests. Theorem E is self-contained: an explicit strict Lyapunov function (Lemma E.0) gives the radius, overshoot and rate of the label-free convergence under tangential excitation. Theorem C′ is the OT $\sigma$-modification toward a prior with a different number of atoms, through the barycentric projection of the optimal plan (Lloyd's algorithm in continuous time); Theorem D′ is propagation of chaos for the dithered closed loop, at the dimension-free rate $m^{-1/2}$. Proposition G is formal

and checked directly. Theorem H shows that the averaged mean-field loop with saturated features, $\sigma$-modification and dither is mean-field Langevin dynamics and converges globally and exponentially to the regularised optimum; this is proved for the averaged system, the reduction to it is formal, and the real loop is a conjecture. At finite gain the averaged loop is in general not a gradient flow of any free energy in the $W_{G^{-1}}$ metric: Proposition H.7 (frozen-measure part proved) exhibits a semi-gradient term, whose performative defect is non-reciprocal in the checked instances ($n = 1, 2$), and an orbit-area circulation kernel ($n \geq 2$). With a second-order averaging term they predict a closed-loop effect for which a targeted literature search found no precedent: traversing the reference orbit backwards changes the learned kernels by $O(1/k_e)$, although the occupation measure is unchanged. At $k_e = 40$ the prediction matches to 3% in size and 11% as a vector (cosine 0.994). Proposition H.8, proved assuming the chain rule for the perturbed flow, gives input-to-state stability of the averaged system. Theorem I characterises the non-reciprocity at order $k_e^{-2}$: for $n = 1$ the averaged loop is reciprocal exactly when the residual vanishes on the orbit, and no open set of measures is reciprocal for any $n$. Theorem J proves the deterministic finite-$m$ case of the averaging conjecture: every hyperbolic critical point of the regularised loss has a unique nearby periodic orbit of the real closed loop with the same stability type, whose period mean is the first-order prediction up to $O(k_e^{-2})$. Proposition K shows that a collapsed configuration is a strict local minimum iff the leak gain exceeds a critical value $\sigma_{\mathrm{crit}}$, and that in a window around $\sigma_{\mathrm{crit}}$ the direction in which the reference is traversed decides whether the kernels separate. A second increment adds the sharp slow-regime rate (Lemma E.0′), uniform-in-time propagation of chaos in the leak-dominated regime (Theorem D″) and the stationary law of the dithered finite-size loop with its non-equilibrium current (Theorem J′, Proposition J′.2). A third increment adds the explicit stationary parameter spread of the dithered linearised loop (Corollary E.0′.3) and the slow-regime Lyapunov function for non-periodic persistently exciting regressors (Lemma E.0″) and uniform-in-time propagation of chaos in the learning regime for the averaged slow loop (Theorem D‴, with the finite-$m$ weight-sensitivity law of Proposition D‴.5, the real loop on a warm start of Corollary D‴.7, the leak's prior as a design input of the window, Proposition D‴.9, and the barycentric OT leak's mass constraint, which removes the finite-size floor, Proposition D‴.12 — in the real loop too, where the collapsed subspace carries one $K_0$-atom orbit for every commensurate size, Corollary D‴.14 and Proposition D‴.15, and Theorem J holds uniformly in the number of kernels in the per-particle norm, Theorem D‴.18, while a soft (entropic) mass constraint equalises the masses only down to a critical excess, Proposition D‴.20; and the mass-constrained energy satisfies a Polyak–Łojasiewicz inequality whose constant does not depend on the number of kernels, which gives a measure-level basin with an exponential rate and, at the nearest-atom leak, explains the floor as a spurious minimum at every admissible count vector, Theorem D‴.21). A fourth increment adds adaptive priors, whose learned atoms remove a misplaced prior's bias and, at every representable sample, the finite-size floor (Theorem D‴.23, Proposition D‴.24) — a property of the amplitude–mass gauge that survives an approximation error, Proposition D‴.28, with a projection that restores the leak's a-priori bound, Proposition D‴.27 —, perfect tracking in the real loop with a Floquet spectrum that does not depend on the number of kernels (Corollary D‴.26, Lemma D‴.26a), second-order plants and a cogging-torque servo (Proposition D‴.29), measurement noise, which is coherent across the kernels and high-passed by the loop, and against which the desired regressor bounds the noise's own persistent force and keeps the replicated kernels together (Propositions D‴.30 and D‴.31, Corollary D‴.33), the cohesion leak, a leak on the unexcited subspace without a prior, which does not select the number of kernels (Proposition D‴.32), a replicator weight law, the mass transfer of the cohesion leak (Proposition D‴.34), the split of a class, an $S_n$-symmetric transcritical bifurcation that sheds the extreme particle (Proposition D‴.35), the gauge under dither, along which the replicator's masses diffuse and which an entropic leak on the masses pins (Proposition D‴.36), the dither's drift along it, the replicator flow of a log-determinant free energy that drives the masses into the saturation (Proposition D‴.37), a Fixman correction on the masses, which cancels that drift to leading order or, with a gain above one, reverses it toward the D-optimal exact fit (Proposition D‴.38), the dissipativity of the dithered gauge, lost with the prior and restored by a dead zone on a box of the unconstrained coordinates, without bias for the exact fits inside the box (Proposition D‴.39), measurement noise on the gauge, which selects nothing among the exact fits in continuous time

and, under an update that draws it afresh, selects through the increment's second moment, from the Hessian's trace to its log-determinant as a Markov noise slows (Proposition D‴.41), several particles per class, whose class masses carry the gauge's laws under an isotropic dither while the split within a class is frozen (Proposition D‴.42), the colour of the measurement noise as the sparsity prior of a sampled-data adaptation on a kernel dictionary, from a weighted $\ell^1$ norm for white noise to a log-determinant that counts the kernels for a slow one (Proposition D‴.43), the free mass–amplitude law as a mirror flow, with a conserved quantity per kernel and an implicit bias, a Bregman projection set by the starting masses, from the minimum-norm fit to weighted basis pursuit (Proposition D‴.45), a leak in the mirror coordinates, whose bias is a Bregman-regularised fit, a ridge from large starting masses and a LASSO from small ones (Proposition D‴.46), the free law's branches, on which small starts freeze the signs and the law is an exponentiated gradient whose limit is the Bregman projection onto the sign-constrained least-squares fits (Proposition D‴.47), and the rich transit, in which from small masses the kernels enter one at a time along an inverse scale space flow and, with the paradigm's moving centres, slide onto the target's features, one kernel per feature (Proposition D‴.48).

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# 1 Introduction

## 1.1 The idea

An adaptive approximator with nonlinearly parametrised features can be read as an *empirical measure* $\mu = \frac{1}{m}\sum_i \delta_{\vartheta_i}$ on a lifted parameter space. Here the features are Gaussian RBF kernels with adaptive weights, centres and widths. The adaptive law transports this measure, and the control output $\hat{f} = \int \Phi \, \mathrm{d}\mu$ is linear in $\mu$. Stability, robustness modifications, excitation, the many-kernel limit and the slow evolution of kernel geometry all acquire formulations on Wasserstein space. So does the global optimality of kernel placement. These formulations do not depend on how the kernels are labelled.

## 1.2 Why the classical theory needs it

This work identifies two structural defects of the classical labelled analysis, that of version 1 of this manuscript and of the literature it builds on.

- The classical Lyapunov analyses depend on labels. A network that is a relabelling of the ideal has zero function error but a large parameter error.
- There is no object that describes where the kernels go, or whether their placement is good.

A third apparent defect was that constants grow with the number of kernels. It turned out to be an artefact of the normalisation. The mean-field normalisation removes it for labelled and label-free analyses alike (Remark A.1).

## 1.3 What the closed loop actually is

- At infinite feedback gain, the averaged closed loop is a Wasserstein gradient flow. With dither it is a mean-field Langevin dynamics (Theorem H).
- At finite gain it is in general *not* a gradient flow of any free energy in the $W_{G^{-1}}$ metric. This already holds for $n = 1$, so it is not a two-dimensional effect (Proposition H.7). The first correction has two parts at a frozen measure, and a third along the real loop.
  - The first part is a *semi-gradient* term. The adaptive law differentiates the loss on the closed-loop state distribution while ignoring that this distribution depends on the parameters, as in performative prediction. Like semi-gradient TD, the resulting update is not the gradient of any objective. Its performative defect is present for every $n$; for $n = 1$ it is non-reciprocal exactly when the residual is non-zero on the orbit (Theorem I(iii)), and it is non-reciprocal in the checked $n = 2$ instance.
  - The second part is an antisymmetric interaction between kernels ($n \geq 2$). Its kernel is an integral of $\sum_k \det[\nabla\tilde{\Phi}_k(\vartheta), \nabla\tilde{\Phi}_k(\vartheta')]$ over the region that the reference orbit encloses (Stokes).
  - Along the real loop, a third part comes from the parameters' oscillation within each period. It is a second-order averaging term (Remark H.7a).
- Together these predict a new closed-loop effect: traversing the reference orbit backwards changes the learned kernels by $O(1/k_e)$, although the occupation measure is unchanged. At $k_e = 40$ the prediction matches to 3% in size and 11% as a vector (cosine 0.994). The vector error halves with each doubling of $k_e$ (`experiments/exp_orbit_orientation.py`).
- The gradient structure survives as an input-to-state Lyapunov function (Proposition H.8). This reduces the global conjecture for the real loop to an averaging estimate.

### 1.4 Conventions and verification

**Status labels.** Every result of Part II is labelled *proved*, *formal* or *conjectural*, and the label is given in a status line after the result.

- "Proved" means proved here. "Formal" means a derivation without error control.
- "Checked" means a test evaluates the statement, or every inequality in it, along simulated trajectories or on sampled states.
- The novelty column of Table 1 follows the verdicts in `LITERATURE_TRANSPORT.md`.

In Part I, every step of the stability argument is either proved or verified numerically by `tests/test_theorem_to_code.py`.

**Verification and review.**

- Every inequality the tests can evaluate is checked numerically along simulated trajectories, and every number in this document is produced by a script in the accompanying repository (the experiments are in `experiments/`); no table is edited by hand.
- The theory went through three rounds of automated multi-agent adversarial review, the manuscript through a fourth that compared it line by line with the theory, the results added in version 3 (Lemma E.0, Theorems C′, D′, I and J, Proposition K) through a fifth, and the second increment — the sharp slow-regime rate (Lemma E.0′), uniform-in-time propagation of chaos in the leak-dominated regime (Theorem D″) and the stationary law of the dithered finite-size loop with its non-equilibrium current (Theorem J′, Proposition J′.2) — through a sixth, and the third increment — the explicit stationary parameter spread of the dithered linearised loop (Corollary E.0′.3) and the slow-regime Lyapunov function for non-periodic persistently exciting regressors (Lemma E.0″) and uniform-in-time propagation of chaos in the learning regime for the averaged slow loop (Theorem D‴, with the finite-$m$ weight-sensitivity law of Proposition D‴.5, the real loop on a warm start of Corollary D‴.7, the leak's prior as a design input of the window, Proposition D‴.9, and the barycentric OT leak's mass constraint, which removes the finite-size floor, Proposition D‴.12 — in the real loop too, where the collapsed subspace carries one $K_0$-atom orbit for every commensurate size, Corollary D‴.14 and Proposition D‴.15, and Theorem J holds uniformly in the number of kernels in the per-particle norm, Theorem D‴.18, while a soft (entropic) mass constraint equalises the masses only down to a critical excess, Proposition D‴.20; and the mass-constrained energy satisfies a Polyak–Łojasiewicz inequality whose constant does not depend on the number of kernels, which gives a measure-level basin with an exponential rate and, at the nearest-atom leak, explains the floor as a spurious minimum at every admissible count vector, Theorem D‴.21) — through a seventh (the first two) and an eighth to a sixteenth (the third), and the fourth increment — adaptive priors and the floor removal (Theorem D‴.23, Proposition D‴.24, the gauge of Proposition D‴.28, the projection of Proposition D‴.27), perfect tracking in the real loop (Corollary D‴.26, Lemma D‴.26a), second-order plants and the cogging servo (Proposition D‴.29), measurement noise (Propositions D‴.30 and D‴.31, Corollary D‴.33), the cohesion leak, a leak on the unexcited subspace without a prior that does not select the number of kernels (Proposition D‴.32), a replicator weight law, the mass transfer of the cohesion leak (Proposition D‴.34), the split of a class, an $S_n$-symmetric transcritical bifurcation that sheds the extreme particle (Proposition D‴.35), the gauge under dither, along which the replicator's masses diffuse and which an entropic leak on the masses pins (Proposition D‴.36), the dither's drift along it, the replicator flow of a log-determinant free energy that drives the masses into the saturation (Proposition D‴.37), a Fixman correction on the masses, which cancels that drift to leading order or, with a gain above one, reverses it toward the D-optimal exact fit (Proposition D‴.38), the dissipativity of the dithered gauge, lost with the prior and

restored by a dead zone on a box of the unconstrained coordinates, without bias for the exact fits inside the box (Proposition D‴.39), measurement noise on the gauge, which selects nothing among the exact fits in continuous time and, under an update that draws it afresh, selects through the increment's second moment, from the Hessian's trace to its log-determinant as a Markov noise slows (Proposition D‴.41), several particles per class, whose class masses carry the gauge's laws under an isotropic dither while the split within a class is frozen (Proposition D‴.42), the colour of the measurement noise as the sparsity prior of a sampled-data adaptation on a kernel dictionary, from a weighted $\ell^1$ norm for white noise to a log-determinant that counts the kernels for a slow one (Proposition D‴.43), the free mass–amplitude law as a mirror flow, with a conserved quantity per kernel and an implicit bias, a Bregman projection set by the starting masses, from the minimum-norm fit to weighted basis pursuit (Proposition D‴.45), a leak in the mirror coordinates, whose bias is a Bregman-regularised fit, a ridge from large starting masses and a LASSO from small ones (Proposition D‴.46), the free law's branches, on which small starts freeze the signs and the law is an exponentiated gradient whose limit is the Bregman projection onto the sign-constrained least-squares fits (Proposition D‴.47), and the rich transit, in which from small masses the kernels enter one at a time along an inverse scale space flow and, with the paradigm's moving centres, slide onto the target's features, one kernel per feature (Proposition D‴.48) — through a seventeenth to a thirty-seventh. Every confirmed issue and its fix is logged in `review/REVIEW_LOG.md`, and the raw reports of rounds 1–3, 5–33 and 35–37 are kept beside it.

- The claim-by-claim novelty audit is in `LITERATURE_TRANSPORT.md`.

Appendix A maps each checked statement to the test or experiment that checks it.

### 1.5 Contributions and their status

Table 1 lists the results of Part II with their status, in the conventions of Section 1.4, and with the novelty verdict of the literature audit.

Table 1: Contributions of Part II and their honest status. Status conventions: Section 1.4.

| result | statement | status | novelty (audit) |
|---|---|---|---|
| A | The adaptive law is the particle form of the Wasserstein gradient flow of a $\mu$-linear potential. Projected rates coincide only for a per-kernel projection ($m \geq 2$). | proved and checked | interpretive; ingredients known [14], [20], [51], [69] |
| B | $V_{\mathrm{OT}} = \frac{1}{2}\|e\|^2 + \frac{1}{2}W^2$ satisfies a Dini inequality with a two-regime matched remainder. It gives conditional RMS bounds along (L), and along ($\mathrm{L}_\sigma$) with a leak term. | proved and checked, including a moving ideal, $\varepsilon \neq 0$, switches, an active projection, and the failure of the plain bound along ($\mathrm{L}_\sigma$) | packaging new; ingredients known [84], [1], [68], [16] |
| C | OT $\sigma$-modification: exact dissipation for selection solutions; a gap-protected strict inequality with $\sqrt{C} \leq b + 2\eta$; a local exponential estimate, $m$-uniform for a single-atom prior and $O(m^{-1/2})$ otherwise; reshuffling by the labelled leak | proved and checked; the C(iv) instance satisfies every hypothesis | novel as a control construction; C(i) and the obstruction are known [80], [9] |

*continued on the next page*

*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| C′ | The OT leak toward a prior with any number of atoms, through the barycentric projection of the optimal plan: dissipation of $W(\mu,\nu)$ up to the within-cell variance of the plan; the fixed points of the pure leak are the equal-mass centroidal configurations of the prior (Lloyd's algorithm in continuous time, the $G$-gradient flow of the quantisation energy); the induced-coupling bound $\sqrt{C_{\mathrm{bar}}} \le b + 2\eta$; a Lyapunov inequality and a local exponential estimate with the bias $b_{\mathrm{bar}}$ := distance of the ideal from the centroids of its prior cells ($\le b$) | proved for vertex-selection solutions (existence assumed) and for generalised solutions (which exist); checked (dissipation at every step, convergence to a Lloyd fixed point, the coupling bound and plan stability, the Lyapunov inequality along the closed loop) | the law is the continuous-time Lloyd descent of coverage control [19] with equal-mass (capacity-constrained) OT plans [5], used as a $\sigma$-modification leak; new here are the induced-coupling bound, the Lyapunov inequality with the bias $b_{\mathrm{bar}}$ and the transfer reading; the quantisation side is Lloyd/CVT theory [57]; it reduces to Theorem C for $m_0 = m$ |
| D | Local well-posedness of the mean-field loop; a Dobrushin estimate; a tube bound and global existence under a self-consistent growth condition; the mean-field limit on the existence interval | proved; consistency in $m$ checked numerically (the constant is not checked) | technique anticipated [29, 72, 73] |
| E | Label-free local exponential convergence under tangential PE, in a uniform tube, at fixed $m$, with explicit radius, overshoot and rate (Lemma E.0); exponential merging of solutions. In the slow-adaptation regime a second strict Lyapunov function (Lemma E.0′: second-order slaving and a second-order averaging correction, with the peak factors of the regressor as data) has the averaged rate $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ up to a factor $1-\delta$, and under parameter dither bounds the stationary parameter spread by $(1+O(\varepsilon))k\|\Sigma\|_F^2/(2(1-\delta)\lambda_{\min})$ (Corollary E.0′.3); for non-periodic PE regressors the same construction with an exponential average over a window gives the rate $(1-\delta-\delta_2)\lambda_T/k$, $\lambda_T$ the weighted PE level of the window (Lemma E.0″) | proved, self-contained; linearisation checked; the strict Lyapunov inequalities checked pointwise (E.0′ by a certificate over all states) and the explicit envelope checked on the linearised and on the implemented loop; convergence observed. The rate of E.0 is conservative in a quantified way (Remark E.0a); the rate of E.0′ is within 12% of the exact rate above its threshold | partially anticipated [16], [34]; the strict Lyapunov function is of the Mazenc–Malisoff type |
| F | Along (L), the Wasserstein length of learning is $\le \lambda_G^{1/2} L_1 \int\|e\|$; leak versions | proved and checked, including the leak clause, a tight leak instance with $e \equiv 0$ and a tight velocity instance | elementary |
| G | Averaged joint descent on the occupation loss. The slow geometry follows the gradient flow of the profile loss in the metric induced on the slow manifold, which is plain variable projection only when that manifold is flat ($S \ll 1$). | formal; checked directly, including a counterexample to plain VarPro at $S = O(1)$ | a fixed-$m$, closed-loop analogue of the two-timescale/VarPro training of [59], [6] |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| H | The averaged mean-field loop with saturated features, $\sigma$-modification and dither is mean-field Langevin dynamics. It converges globally and exponentially to the regularised optimum, with the log-Sobolev constant given in control quantities. | proved for ($\mathfrak{A}$); the reduction to ($\mathfrak{A}$) is formal (Remark H.0); real loop: Conjecture H.6 | a bridge to [15], [64]; new in adaptive control |
| H.7 | At a frozen measure, the exactly averaged loop is ($\mathfrak{A}$) plus an explicit $O(1/k_e)$ correction: a semi-gradient term, whose performative defect is present for every $n$, and an orbit-area circulation kernel ($n \geq 2$). Along the real loop, a second-order averaging term is added (Remark H.7a). | Frozen-measure part proved. Stokes identity and parity hold to machine precision. The expansion is checked through its $O(1/k_e)$ convergence. Non-reciprocity of the exact second variation is checked to 3% (asserted; 1–1.5% measured) and characterised by Theorem I. The full first-order theory predicts the orientation dependence of the learned kernels in the real closed loop to 3% in size and 11% as a vector (cosine 0.994, $k_e = 40$). | no precedent found (targeted search in the third review round); ingredients known (higher-order averaging; performative prediction [71]; semi-gradient TD [88], [85]) |
| I | Reciprocity at order $k_e^{-2}$ is a pairwise property: the antisymmetric kernel $A_{\mathrm{even}} + 2K$ is a coboundary iff it vanishes. For $n = 1$, $A_\mu \equiv 0$ iff the residual vanishes on the orbit, and the performative non-reciprocity is bounded by the residual. For $n \geq 2$ at perfect fit, $A_\mu \equiv 0$ iff the oriented orbit carries no circulation ($n = 2$: zero winding function) and the residual's Jacobian is symmetric along the orbit; $A_\mu = 2K$ when the residual vanishes near the orbit. $A_{\mathrm{even}}$ is the curl of the residual paired with the features plus the residual paired with the Lie bracket of the feature fields. No open set of measures is reciprocal, for any $n$: the averaged loop is nowhere a gradient flow at order $k_e^{-2}$. | proved; every formula checked against the generic second-variation code and, by the round-5 referee, against the exact second variation; the characterisations checked on perfect-fit, curl and zero-current instances | new; the analytic tools (injectivity of Gaussian convolution; Stokes with winding numbers) are classical |
| D′ | Finite-horizon propagation of chaos for the dithered closed loop ($\mathrm{L}_\tau$): by synchronous coupling, $\mathbb{E}\sup_{t\leq T}\big(\|x^m - \bar{x}\| + \frac{1}{m}\sum_i \|\vartheta_i - \bar{\vartheta}_i\|\big) \leq TK_\Phi e^{\Lambda T} m^{-1/2}$, with $\Lambda$ independent of $m$ and $\tau$, and a per-particle sup bound of the same order; the plant and the learned function converge at the rate $m^{-1/2}$, whose exponent is dimension-free, and only $W_1(\mu^m, \bar{\mu})$ carries $m^{-1/p}$; under a dominant leak (small-gain condition $\kappa_e \kappa_\vartheta > L_{b'} L_\vartheta$) the bound is uniform in time (Theorem D″), a leak-dominated regime in which the network does not learn (at $\tau = 0$ it collapses to one kernel; at $\tau > 0$ it stays in the dither cloud of the prior) | proved; checked: the coupled distance to a 4096-particle reference decays like $m^{-0.47\pm0.06}$ (24 seeds, $\tau = 0$; $-0.46 \pm 0.05$ with dither) and like $(1/m - 1/M)^{0.94\pm0.08}$, the law of the shared-particle coupling | the technique is Sznitman's, and the dimension-free exponent for the state is its standard output; new in that the plant is in the loop, with constants in control quantities |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴ | Uniform-in-time propagation of chaos *in the learning regime* for the averaged slow loop ($\mathfrak{A}_m$) at $\tau = 0$ (Theorem J′'s leading order): under the synchronous coupling with the mean-field characteristic flow, the interaction term of the coupling identity is $-\|\hat{f}_\Theta - \hat{f}_{\bar\Theta}\|^2_{L^2(\rho)} + O(D^2)$ — flat convexity of the loss makes it contract whatever the gains — and only the residual-weighted feature curvature $R[\mu](\vartheta) = \langle r_\mu \nabla^2_\vartheta \tilde\Phi\rangle_\rho$ (Proposition K's $R_2$) can expand; if along the mean-field flow $\sigma_0 I - G^{1/2} R G^{1/2} \succeq \kappa I$ on a neighbourhood of the support from a time $s_1$ on (the strict second-order optimality condition of the fitted configuration, compatible with $\sigma_0 < \sigma_{\rm crit}$), then $\sup_s \mathbb{E}\frac{1}{m}\sum\|\vartheta_i - \bar\vartheta_i\|^2_{G^{-1}} \le 4e^{2\Lambda_1}(L^G_\vartheta)^2 K^2_\Phi/(\kappa^2 m)$ plus a tail exponentially small in $m$ (uniformly in time under exponential convergence of the flow, over horizons $e^{cm}$ without it), no particle changes basin, and the learned function converges at $m^{-1/2}$ uniformly in time while the network learns; the window $\kappa_R[\mu_\infty] < \sigma_0 < \sigma_{\rm crit}$ is where learning and uniformity coexist; at finite $m$ the stationary coupled distance follows the weight-sensitivity law $\mathbb{E}D_\infty = (1/m - 1/M)\sum_k \bar w_k \operatorname{tr}(\tilde S_k \tilde C \tilde S_k^\top)$, $S = \partial\Theta^*/\partial w$ the sensitivity of the stationary atoms to the cluster weights (Proposition D‴.5, the sharp form of the theorem's constant); the leak's prior is a design input of the window: with a measure-valued prior (nearest-atom leak) placed on the plant's structure the stationary residual vanishes and the margin is maximal, $\kappa = \sigma_0$, while the floors at parameter and function level rise (1.6–2.2× on the instance, with the amplitudes), and the sign of the residual at a cluster sets its admissible weights — overshoot splits, undershoot stabilises, so a shrunk prior enlarges the admissible set (Proposition D‴.9, Remark D‴.11); | proved for the averaged slow loop (synchronous coupling, a bootstrap on the mean square and the per-particle distance, McDiarmid for the reference's fluctuation); the hypothesis is a computable property of the mean-field flow; it fails for a connected cold start whose mass splits between basins (the support threads the ridges), where the coupling carries a floor; the real loop at finite $k_e$: settled on a warm start by Corollary D‴.7 through Theorem J (both loops converge to their periodic orbits, whose distance is Proposition D‴.5's up to $O(1/k_e)$), open for a cold start (Remark D‴.4); dither: the entropy method of [11] (imported); the theorem's regime $m \ge m_0$ is asymptotic ($m_0 \sim 10^{23}$ on the instance) and Proposition D‴.5 is formal; numerics: on a three-bump instance at $\sigma_0 = 0.15\sigma_{\rm crit}$ the window is realised ($\kappa_R[\mu^*](N_{0.1}) = 0.0051 < \sigma_0 = 0.0071 < \sigma_{\rm crit} = 0.047$) and (H) holds along a warm-start run; the coupled distance saturates (window maxima within 15% over 900 slow units) with slope $-1.28 \pm 0.30$ in $m$, and the per-seed weight-sensitivity law predicts it within 25% in 9 of 12 cases with per-cluster cosines 0.92–1.00; the exact stationary distance of the replicated weighted configuration reproduces the measured one within 2.3% in all pairs, one $m = 16$ seed with inadmissible weights sitting on a saddle; a connected cold start grows and carries a floor of pairs in different basins whose fraction 0.20, 0.08, 0.04 ($m = 16, 64, 256$) decays like $m^{-0.64\pm0.08}$; the real loop at $k_e = 25$ follows its averaged slow dynamics on the same clouds to within 0.7%; | the two ingredients — the sign of the interaction from flat convexity, and the curvature at the actual rather than the worst-case residual — are new in propagation of chaos for adaptive control; the decomposition Gram + residual×curvature is the one of the interpolation literature ([56]); the window between Proposition K's collapsed instability and the fitted configuration's stability is new; |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴, cont. | the barycentric OT leak of Theorem C′ toward the same prior constrains the cluster masses to the prior's — the nearest-atom leak conserves them, and the floor is the memory of the initial cloud that this conservation law carries — so on the sizes commensurate with the masses the finite-$m$ stationary measure is the mean-field one exactly, both floors vanish and the fit loses the weight bias, while on other sizes at most $K_0 - 1$ split particles bridge the clusters, with floors $O(1/m^2)$ at function level and $O(1/m)$ in $W^2$ against the nearest-atom leak's $1/m$ and $m^{-1/2}$ (Proposition D‴.12); in the real loop the collapsed-by-class subspace is invariant and carries the weighted $K_0$-atom loop, so with the OT leak the lift of one $K_0$-atom periodic orbit is a periodic orbit of every commensurate size — the same orbit measure, learned function and tracking error exactly, the real loop's finite-size floor being zero — while with the nearest-atom leak it is the lift of the orbit of the realised weights, whose distance between sizes is Corollary D‴.7's floor (Corollary D‴.14); the Floquet spectrum of a lifted orbit is the $K_0$-atom orbit's together with each class's ghost spectrum of multiplicity $n_k - 1$, independent of $m$ (Proposition D‴.15); with dither the nearest-atom leak equalises the masses only by Arrhenius crossings of the Voronoi ridges (barrier $\approx 0.05$ on the instance: frozen for $350/\sigma_0$ at $\tau_0 = 0.005$), while under the OT leak a crossing particle stays assigned to its own atom and returns, so the counts are balanced 88–99.6% of the time at the same crossing barrier (Remark D‴.16); every trajectory of the OT dynamics converges to a Clarke-critical point of its energy (Kurdyka–Łojasiewicz for the definable, semiconcave-plus-smooth $J_m^{\mathrm{OT}}$), so the stationary statement is reached from every initial condition in the basin (Proposition D‴.17) | for the OT leak the frozen-plan equivalence and the stationary statement are proved (total unimodularity, $c$-cyclical monotonicity, the Hessian of D‴.5), the quantisation bounds formal, and Theorem D‴'s coupling applies only to a balanced sample (probability $\approx 0.83/m$ under an i.i.d. start, and even there formal: Step 8 is written for i.i.d. particles) — for a generic sample the shared-particle coupling keeps the transit's mismatch for ever and a coupling theorem for the OT dynamics is open; numerics on the same clouds under the two leaks: cluster weights conserved exactly against equalised by $s \approx 100$ slow units ($\ln 2/\sigma_0 = 98$), function-level and $W^2$ distances at $10^{-4}$ and $10^{-3}$ of the nearest-atom leak's after $4.2/\sigma_0$ and still falling on the commensurate sizes while the synchronous $D$ sits at 0.95 of its mismatch plateau, the quantisation laws $m \cdot W^2(\mu_m, \nu) \approx 20$ (slope $-0.98 \pm 0.03$) and $m \cdot \mathrm{fit} \approx 0.55$ (squared slope $-1.81 \pm 0.05$) on the incommensurate ones, the reference's fit error $0.012 \to 8 \cdot 10^{-5}$; the real loop at $k_e = 25$ on the same clouds: learned functions of $m = 15$, 60, 510 within squared distances $7.5 \cdot 10^{-9}$–$2.7 \cdot 10^{-8}$ of each other under the OT leak against $10^{-4}$–$10^{-3}$ under the nearest-atom leak (the lifted orbits' distance to 1.0003 at the one pair with admissible weights; three of the six nearest-atom systems sit at saddle weights and have split), rms tracking error $0.9$–$7 \cdot 10^{-6}$ against $3 \cdot 10^{-4}$–$1.9 \cdot 10^{-3}$ (on a planar plant with four kernels: $10^{-7}$ against $10^{-2}$ and $4 \cdot 10^{-6}$–$1.6 \cdot 10^{-5}$ against $2$–$7 \cdot 10^{-3}$); the lifted orbit's transverse spectrum equals the ghosts' and the collective block the $K_0$-atom loop's to $10^{-4}$ relative (invariance defects $\leq 10^{-7}$) | the finite-size floor as a conservation law of the leak, broken by the plan's mass constraint so that the particle system's stationary measure is exactly the mean-field one, is new; the lift of a weighted $K_0$-atom orbit as the exact periodic orbit of every commensurate size, with an $m$-independent Floquet spectrum (the orbit-level form of Proposition K's block structure), is new in adaptive control, where the size of the approximator is usually a nuisance parameter |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.18 | Theorem J uniformly in $m$ at a replicated configuration: for weights with margins (collective block and separation blocks $\geq \lambda_0$) the real loop's orbit near $\Theta^*_m(w)$ is the lift of the weighted $K_0$-atom orbit, unique in the per-particle ball of radius $\rho/2$ (inside the $\|\|\|\cdot\|\|\|$-ball of radius $\rho$ where the period map contracts) and attracting every solution started in it at the rate $(1-\kappa)\min\{\lambda_{\mathrm{coll}}, \lambda_{\mathrm{sep}}\}/k_e$, with $k_0$, $\rho$, $C$ independent of $m$; in the unnormalised norm of Theorem J the basin is $m$-independent too but shrinks per particle like $m^{-1/2}$ and the threshold $k_0$ grows like $\sqrt{m}$, and for the nearest-atom leak no mean-square ($W_2$-type) basin is possible — one particle in another cell changes the weights (Remark D‴.19) | proved as an adaptation of Theorem J's proof in the per-particle norm (three structural facts: the plant sees the class means up to second order in the separations; each particle's drift depends on the others only through the plant; the linearisation is block diagonal with an $m$-independent spectrum); numerics: the Floquet exponents within 0.56–3.22% of the averaged prediction and the ghosts' to $3 \cdot 10^{-4}$; on the matching prior fitted rates 0.95–1.05 of the prediction for $m \geq 15$ and constants 1.03–1.29, on a moving orbit 0.96–1.16 and 1.04–1.23, all from radii 31–261 times the proved one; across a cell boundary the nearest-atom leak ends at other weights (two of them inadmissible) while the OT leak restores the masses $q$ | uniform-in-$m$ averaging for an adaptive law with a mean-field interaction: the size of the approximator drops out of the stability constants — new in adaptive control, where Lyapunov and averaging constants usually grow with the number of parameters |
| D‴.20 | The entropic OT leak (Sinkhorn plan, regularisation $\varepsilon$) is a smooth gradient flow of $m\bar{L} + (\sigma_0 m/2)\mathrm{OT}_\varepsilon$ that interpolates between the OT leak ($\varepsilon \to 0$) and the $\sigma$-modification toward the barycentre of the prior ($\varepsilon \to \infty$); at a collapsed configuration the separation block is $\sigma_0(I - (2/\varepsilon)\,\mathrm{Cov}^G_k) - R^G_k$ with $\mathrm{Cov}^G_k$ the conditional covariance of the atoms, so where $A_k := \sigma_0 I - R^G_k \succ 0$ an over-weighted class is a saddle exactly below the critical regularisation $\varepsilon_c = 2\sigma_0 \lambda_{\max}(A_k^{-1/2}\,\mathrm{Cov}^G_k\,A_k^{-1/2})$ (implicit, along the branch $\Theta^*_\varepsilon(w)$), and a split moves its excess toward an atom its conditional plan charges, which need not reduce it; at fixed $\varepsilon$ the masses are equalised only down to a critical excess $\delta_c(\varepsilon) \approx w_k p_c(\varepsilon)$, and at large $m$ the floor returns as Proposition D‴.5's law at the entropic stationary point | (i)–(iii) proved, the separation block checked entrywise against the finite-difference Hessian; (iv) proved under $A_k \succ 0$ (which fails at large excesses and small $\varepsilon$), the conditional plan charging both under-weighted atoms ($\mathrm{Cov}^G_k$ of rank two, the rank-one formula 1.22–1.31 times $\varepsilon_c$); (v) formal; numerics: $\varepsilon_c = 5.1$, 13.5, 23.1, 65.0 for excesses 0.02–0.2; at $\varepsilon = 1$ every size equalises; at $\varepsilon = 4$ every system ends, after up to 6000 slow units, at stable weights with residual excesses up to 0.017, of the order of $\delta_c(4) = 0.012$–$0.016$; at $\varepsilon = 16$ and 64 the stable systems keep their sampled weights, with floors 0.23–0.91 of the nearest-atom leak's | the critical temperature of deterministic annealing is the precedent ([77]), with the same roles (particles as codevectors, atoms as data), and the $\varepsilon \to 0/\infty$ interpolation is that of the Sinkhorn divergences ([21]); new are its role as the threshold of mass equalisation in an adaptive law, where the conditional covariance of a collapsed class is an effect of its excess, and the consequence that on the commensurate sizes only the hard constraint is floor-free for every $m$ |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.21 | A measure-level basin for the OT leak: at commensurate sizes two exact identities along the plan segments — the energy, and its gradient paired with the displacement toward the stationary atom of one's own class — give a Polyak–Łojasiewicz inequality $\|\nabla J\|^2 \geq \lambda \cdot m(\mathcal{J} - \mathcal{J}^*)$ with $\lambda$ set by the residual-weighted curvature along the segments and independent of $m$ (sharp, $2\sigma_0$, at a matching prior); the excess energy decays exponentially from an $m$-independent sublevel set (tiny on the instance: none of its configurations has a particle in another cell below $m \approx 2 \cdot 10^5$), and a posteriori wherever the plan-segment margin is positive, which covers configurations with a fixed fraction of the particles in other cells; under a curvature hypothesis at the stationary residual the replicated configuration is the unique global minimum among the empirical measures for every $m$, while the nearest-atom energy has a spurious minimum at every admissible count vector (the energy-landscape form of the floor) | (i), (ii), (iv), (v), (vi) proved, (iii) proved for vertex-selection solutions (their existence assumed); the mean-field form and the transfer to the real loop formal (Remark D‴.22); numerics: the identities to $6 \cdot 10^{-13}$ along 64 runs, the margin positive from $s \leq 300$, the observed rate at least $\lambda$ throughout and $\lambda$ within 1.5% of the observed $2\sigma_0$ at the end (matching prior); the nearest-atom runs at admissible counts stop at the energy of the replicated configuration of their sampled weights, at inadmissible ones (5 of 8 per prior at $m \leq 60$) they split below it | the Polyak–Łojasiewicz inequality ([74]), the generalised geodesics of AGS [2, §9.2] and PL inequalities through optimal maps for functionals that are not geodesically convex ([7]) are known; new is the PL inequality along the generalised geodesic based at the prior, with the loss linearised at the stationary residual and the counts making the constant independent of $m$ |
| D‴.23 | Adaptive priors: the OT leak toward a prior whose atoms move toward the class barycentres at a rate $\eta > 0$ is the gradient flow of $\bar{L} + (\sigma_0/2)W^2(\mu_\Theta, \nu_Z)$ in a weighted metric; its replicated stationary points carry no leak bias (the critical points of the unregularised $K_0$-atom loss), its non-replicated ones the $K_0$-cluster bias, and boundedness becomes a hypothesis (the leak no longer confines the parameters); at an exact representation it linearises, for every $m$, into the separations (rate $\sigma_0$) and $2 \times 2$ collective pairs whose slower rates satisfy $\eta h_j/(h_j + \sigma_0 + \eta) < \lambda_j^- < \min\{h_j, \eta\}$ ($h_j, \eta > 0$), $h_j$ the measure-space excitation spectrum — a learned prior removes the fixed prior's bias $\sigma_0(H_L + \sigma_0 I)^{-1}x^0$ mode by mode, at the rate of the excitation; as $\eta \to \infty$ a capacity-constrained $K_0$-means regulariser; the transit drags the prior, so the adaptation should start from the fixed-prior stationary point | (i)–(iii) proved ((i) for bounded solutions), (iv) formal, (v) sketched (the radius stated, its constants not derived), (vi) the monotonicity of $\mathcal{J}$ proved and the drag numerical; numerics: the $K_0$-atom modes decay at $\lambda_j^-$; from separated starts the $m$-particle systems depart from the $K_0$-atom run at second order in the separations, consistent with the same expected size at $m = 15$ and 510 (five draws); started from the fixed-prior stationary point every size $m = 15$–510 follows the $K_0$-atom run (per-seed ratios 0.995–1.000 at $\eta = \sigma_0$, 0.968–1.001 at $\eta = \infty$), with fit errors $1.6 \cdot 10^{-4}$ and $3.4$–$3.5 \cdot 10^{-5}$ against the fixed prior's $9.78 \cdot 10^{-3}$ | proximal point / iterated Tikhonov regularisation (its per-mode filter factors), elastic averaging (its quadratic analysis) and soft weight-sharing are the precedents ([90]); new are their transfer to the mass-constrained particle system uniformly in $m$ ($H_L$ as the measure-space excitation spectrum, the separations decoupled at $\sigma_0$), and the transit's drag with its gate; the limits (the lost a-priori bound, the $K_0$-cluster bias) are the precedents' own, specific here in their particle form and in the run-off from an exact fit |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.24 | The finite-size floor is the leak's bias at the sampled weights: under the nearest-atom leak toward a fixed matching prior the function-level weight-sensitivity is $F = \hat{J}\sigma_0(H_L + \sigma_0 I)^{-1}\partial_w x^0$, the weight-derivative of the bias, and the floor law decomposes on the excitation spectrum with the band-pass weight $h_j\sigma_0^2/(h_j+\sigma_0)^2 \le \sigma_0/4$, centred at $h = \sigma_0$ — at the samples where the fixed prior's replicated point is stable (D‴.5-admissible); the nearest-atom leak toward an adaptive prior is the OT leak with the masses minimised out (the cell masses), and its stationary replicated configurations are the critical points of the unregularised weighted loss, the exact representation of the sampled weights among them — no function-level floor at any representable sample, every separation relaxing at $\sigma_0$ (no splitting), perfect tracking an invariant set of the real loop — while the coupled distance of the amplitude–mass gauge grows (the sample's weight fluctuation moved from the bias into the parameters) | (i) $F$ exact at $q$, the bias formula first order, the law (B) formal (the delta method, as D‴.9), (ii) proved inside the cells and across crossings for vertex-selection solutions, (iii) proved for the stationary points, (iv) the boundary case formal, (vi) the invariance proved; numerics: at the stable samples the exact bias at the sampled weights reproduces the observed floor (ratio 1.0000), while 5 of the 12 samples are fixed-prior saddles and split; the law is asymptotic (the exact floor's expectation 2.8, 1.5, 1.1 times it at $m = 15$, 60, 240); under the adaptive prior the function-level distance decreases without a floor (at $s = 6000$ at most $2.0 \cdot 10^{-6}$ at $\eta = \sigma_0$ over the representable samples) and the coupled distance is 1.7–4.8 times the fixed prior's, while $W^2$ is the cross-class transport of order $\|\delta w\|_1$ under both priors; the samples outside $W$ have probability 0.36, 0.074, 0.0021 at $m = 15$, 60, 240; in the real loop the tracking error follows the averaged prediction (ratios 0.987–1.001) | the $1/m$ fluctuations of mean-field networks and their vanishing at an interpolating limit ([89], [91]), the delta method and the filter-factor bound of ridge regression are classical; new are the floor as the bias's weight-slope, mode by mode, a leak toward a learned prior as the way to restore the interpolation property under regularisation, and the design table of Remark D‴.25 |
| D‴.26 | Perfect tracking with an adaptive prior in the real loop: the lift of the exact representation of representable sampled weights, with the prior at its atoms and the plant on the reference, is a $T_0$-periodic solution whose Floquet exponents are $\sigma_0/k_e$ exactly on the $(m - K_0)p$ separations — every separation stable, also where the fixed prior's replicated orbit is a saddle — and the weighted $K_0$-atom loop's on the collapsed subspace for every $m$; where $\partial_x \hat{f} = \partial_x f$ on the orbit (an exact representation) these are $(1/k_e)[\lambda_j^{\pm}(w) + c_j^{\pm}(w)/k_e^2 + O(k_e^{-4})]$ — in slow time the odd orders vanish (the variational equation's symmetry with the slow block at $-\varepsilon$, the time reversal of the reversed reference's: the slow generator's spectrum is even; the first-order correction is skew), and $c_j$, explicit, contains the error filter's attenuation ($\le 0$), the slaving's part ($\ge 0$) and averaging terms, its sign not fixed; so perfect tracking is exponentially stable (locally, for each $m$) at the rate $(1/k_e)\min\{\sigma_0, \lambda_1^-(w) + c_1^-(w)/k_e^2 + \dots\}$ | (a), (b) proved, exactly; (c) proved for $x_m \in C^4$ at simple eigenvalues, the stability where $H_L(w) \succ 0$ (Lemma D‴.26a: the slaving exact, the $O(k_e^{-4})$ remainder bounded, the parity an identity), $c_j$ explicit; the basin uniform in $m$ not proved; numerics: at $(6, 5, 4)/15$, $m = 15$ and 30, $k_e = 25$ and 50, $\eta = \sigma_0$ and $\infty$: the separations at $\sigma_0/k_e$ to $\le 1.7 \cdot 10^{-6}$, the collective exponents the $K_0$-atom loop's to $\le 6.4 \cdot 10^{-4}$; by the variational equation to $k_e = 400$ on two references (one with a non-zero first-order correction): the first-order deviations falling by $\to 4$ per doubling (every exponent below its prediction on the instance; 13.6% in the slowest mode at $k_e = 25$), the second-order residuals by $\to 16$, $c_1/\lambda_1 = -100.6$ of which the filter's $-102.1 = -0.989\omega_1^2$; with the gain scaled by 30 two $c_j > 0$, those exponents above their prediction by $c_j/(\lambda_j k_e^2)$ | Floquet theory, averaging and the Magnus expansion's commutator term, the averaging analysis of adaptive loops (the PE level), and leaks compatible with exact tracking (*e*-modification; a centre that follows the parameters) are classical, and (a)–(b) are Corollary D‴.14(ii) and Proposition D‴.15 at zero residual; new are the zero-residual lift at every representable sample with its exactly $m$-independent spectrum, and the explicit second-order coefficient of this loop (the even expansion is the classical consequence of time symmetry, [99]) |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴. 26a | The exact slow subspace of the perfect-tracking linearisation: Chang's periodic decoupling $P$ (a contraction) makes the slaving of D‴.26 an exact block-triangularisation, so the slow Floquet multipliers are exactly the slow block's; its generator $\log\Phi(\varepsilon)/(\varepsilon T_0)$ has an expansion to $O(\varepsilon^4)$ with a bounded remainder (the Magnus series convergent for $\varepsilon T_0\|A\| < \pi$); the slow block at $-\varepsilon$, defined by the same contraction, is the time reversal of the reversed reference's at $\varepsilon$, so with the variational equation's symmetry the slow generator's whole spectrum is even, $\operatorname{eig}\bar{A}(-\varepsilon) = \operatorname{eig}\bar{A}(\varepsilon)$, and the rates are exactly even; and Chang's second transformation decouples the slow coordinates' drift from the fast ones exactly, their noise $\varepsilon b\nu$, $b = -(\dot{K}+aK)^\top - \varepsilon(\ddot{K}+KK^\top K + K\Lambda^\top + a\dot{K})^\top + O(\varepsilon^2)$, shared with the fast block — at leading order the high-passed $-\varepsilon(\dot{K}+aK)^\top\nu$ —, which gives D‴.30(ii)'s floor as a period average | proved (the constants not explicit; $\varepsilon_0$ also set by the spectral gap and $\bar{A}_0$'s margin): D‴.26(c) with a bounded $O(k_e^{-4})$ remainder, D‴.30(ii) for the linearised loop under white noise as a period average with the relative remainder $O(1/k_e)$ (at $\eta = 0$ with $K = \hat{J}$, $\Lambda = \sigma_0 I$); numerics ($\eta = 0$, $\sigma_0$, $\infty$; $k_e = 12.5$–$1600$): the invariance residual $\le 3.8\cdot10^{-15}$, $P - P_2 = O(\varepsilon^3)$ and $P - P_{(3)} = O(\varepsilon^4)$, the slow block's rates the full equation's to $1.0\cdot10^{-8}$, the rates at $\pm\varepsilon$ equal to $5.9\cdot10^{-12}$ of the largest ($1.1\cdot10^{-9}$ per mode) while the slowest mode's first-order deviation reaches 29.6%, a mismatch breaking it at $O(1/k_e)$; $b - b_0 - \varepsilon b_1 = O(\varepsilon^2)$; the weakest mode's covariance deviation the slow noise's attenuation 2.4–2.5 times the rate's correction, of opposite sign; the floor's period average at the formula to $3\cdot10^{-5}$ at $k_e = 1600$, pointwise ±1% (the jitter) | Chang's decoupling — with its time-varying and stochastic versions (Kokotović–Khalil–O'Reilly Ch. 4–5 and App. A, Haddad–Kokotović, Yang–Zhu; Berglund–Gentz's slow manifolds under noise) —, the Magnus series' convergence and the reversal argument (the adjoint relation of symmetric methods) are classical ([102], [99]); new is their application here: the exact slaving, the parity as an identity at negative $\varepsilon$, the slow noise identified exactly |
| D‴.27 | A projected adaptive prior restores the leak's a-priori bound: with the prior's atoms started in and confined to a compact convex set $\mathcal{Z}$ (the projected flow; at $\eta = \infty$ the projected class barycentres), every particle's distance from $\mathcal{Z}$ is at most $B/\sigma_0 + (d(0) - B/\sigma_0)_+ e^{-\sigma_0 s}$, $B$ the worst-case learning force, for every $m$, $\eta$ and target; $\mathcal{J}$ is non-increasing (at $\eta = \infty$ in its selected form, along a Lloyd-consistent plan; the reduced energy need not decrease); inside $\mathcal{Z}$ nothing changes (Theorem D‴.23's bias removal and linearisation), and where the limit has an atom on the boundary — necessarily when no stationary point has its prior inside — a bounded bias remains; at $\eta = \infty$ on collapsed classes the leak is the gradient of the exterior penalty $(\sigma_0/2)\operatorname{dist}^2(\vartheta, \mathcal{Z})$, a switching $\sigma$-modification toward $\mathcal{Z}$ | (i), (iii) proved ((iii)'s local convergence as Theorem D‴.23(v), sketched); (ii) the monotonicity proved, the convergence argued at $\eta = \infty$ and formal at finite $\eta$; (iv) the identification at $\eta = \infty$ proved, the boundary atoms and the bias outside numerical; numerics: beyond the saturation both $\eta$ converge to the same bounded point (fit error 0.1415; the projected midpoint rule consistent, the trajectory error $6.3\cdot10^{-7}$–$3.8\cdot10^{-8}$ at $\mathrm{d}t = 0.25$–$0.0625$), particles within 0.92 of $\mathcal{Z}$ (bound $\le 3686$); the box's amplitude bound 0.75–0.99 of $a_{\max}$ trading the fit error 0.1676–0.1101 between the fixed prior's 0.1750 and the unprojected limit 0.1050, tighter boxes (which exclude the exact atoms) worse than the fixed prior; at $\eta = \infty$ the reduced energy rising by 26% along a Lloyd-consistent plan while the selected energy falls; on the four-bump target the box stops the width's run-off at its face; at the misplaced prior the box never approached | parameter projection, the exterior quadratic penalty and the switching $\sigma$-modification (which keeps the ideal properties inside its set) are classical, and at $\eta = \infty$ on collapsed classes the construction is the latter; new is its transfer to the learned centre of a mass-constrained $K_0$-atom prior at finite $\eta$, compatible with Theorem D‴.23's bias removal inside $\mathcal{Z}$ |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.28 | The adaptive prior's floor removal is a gauge property: the masses enter the network only through $w_k a(\alpha_k)$, so for any nondegenerate critical point $\vartheta^b(q)$ of the unregularised loss — exact or not — its gauge transform $G_w\vartheta^b(q)$ is a critical point at every sampled weight vector of $W_g$ with the same function: at a strict local minimum no function-level floor at the samples of $W_g$ with positive margins, whatever the approximation error; the fixed matching prior's floor is $\sigma_0\hat{J}(\tilde{A}+\sigma_0 I)^{-1}\tilde{g}$ with the full Hessian $\tilde{A}=H_L-\tilde{R}$; for a scalar amplitude per kernel the residual curvature has a zero amplitude row at a critical point and scales exactly by $q_k/w_k$ under the gauge, so the adaptive prior's splitting margins are $\sigma_0-(q_k/w_k)\lambda_k^+$ | (i), (iii) proved exactly, about the stationary points; (ii) as D‴.24(i); the dynamics' convergence, and with it the floor's vanishing, numerical; numerics: with a fourth bump that three kernels cannot represent (fit errors 0.4–2.5%), the critical point continued in $w$ from $\vartheta^b(q)$ is the gauge transform to $2\cdot10^{-11}$ with $\hat{f}$ unchanged to $6\cdot10^{-13}$, the adaptive prior's sensitivity $10^{-11}$–$3\cdot10^{-10}$ of the fixed prior's, the fixed prior's equal to the full-Hessian formula to $2\cdot10^{-9}$ (the Gauss–Newton form off by $7\cdot10^{-4}$–$5.6\cdot10^{-3}$), the margin formula exact to $10^{-14}$ and the dynamics' separation eigenvalues to $6\cdot10^{-12}$; $P(w^m\in W_g)=0.52$ at $m=15$; the dynamics: under the fixed prior the floor stays and the fit error above the approximant's; under the adaptive priors the floor falls by orders of magnitude and the fit error tends to the approximant's at the samples inside $W_g$; outside it the systems approach the box minimum on the branch (1.11–1.59 times the approximant's fit error) | the homogeneity of the conic view ([16], [23]) and the rescaling symmetries of networks ([98]) are classical; new are the gauge as the mechanism of the floor removal, robust to approximation error, and the exact margin formula for a scalar amplitude |
| D‴.29 | Second-order plants: with the sliding variable $s=\dot{e}+\lambda e$ and $u=\ddot{x}_m-\lambda\dot{e}-k_s s-\hat{f}_\Theta(z)$, the error channel is $\dot{s}=-k_s s+r_\Theta(z)$, $\dot{e}=-\lambda e+s$ — Theorem J's with $s$ in place of $e$ — so as $k_s\to\infty$ the parameters follow the averaged slow loop along the regressor's orbit $z_m(t)$, every averaged-loop statement of Section 18 transfers with the occupation measure of $z_m$, and the position error is the residual filtered by $(\mathrm{d}/\mathrm{d}t+\lambda)^{-1}(\mathrm{d}/\mathrm{d}t+k_s)^{-1}$; kernels on the coordinate $f$ depends on see the orbit as an interval, while isotropic phase-plane kernels on a circular orbit ($\omega_0=1$ in isotropic units) are over-parametrised (von Mises restrictions: in general position one zero Gauss–Newton eigenvalue per kernel; on other orbits weak excitation) | (i), (iv) exact; (ii), (iii) formal (first-order averaging with the fast pair $(s,e)$; Theorem J's steps carry over for the smooth fixed-prior leak once the error-channel steps are redone; switching targets beyond the first-order theory); numerics: a position servo with a cogging torque whose four-kernel local minimum $\vartheta^b(q)$ (fit error 1.46%; the loss's infimum lies at the saturation, $\le 0.64\%$) is Proposition D‴.28's regime: from clouds around $\vartheta^b(q)$ the fixed prior's floor $1.85\cdot10^{-3}$–$4.49\cdot10^{-5}$ ($m=16$–$128$), the adaptive prior's 38–780 times lower, the OT leak at the approximant, $\sigma$-modification toward 0 biased (0.328), and no leak as good (0.0180–0.0146); from an engineer's model (fit error 0.200) the fixed designs stay biased (0.048–0.10) and the adaptive prior removes most of the bias, slowly (0.022–0.027 at $s=2000$; the $K_0$-atom flow 0.0155 at $s=10^5$); the real servo's position error at the exactly filtered prediction to 1.0001–1.0029 (the slaved one 0.931–0.996), no design drifting; noise-free, so the leaks' robustness is not exercised | the filtered error and Gaussian networks in adaptive control are classical (Sanner–Slotine), and so is the partial excitation of Gaussian networks along an orbit; new are the transfer of the transport leaks, their floors and the adaptive and projected priors to second-order plants, and the exact fibre of isotropic phase-plane kernels on a circular orbit |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.30 | Measurement noise at perfect tracking: the noise is coherent — a replicated start stays replicated under every noise path, so the floor does not average out with $m$ (invariance, not attraction: near the lift the separations grow above a noise level, which drifts along the trajectory with the developing errors-in-variables deviation); to first order the loop high-passes the measured error's noise, $((\mathrm{d}/\mathrm{d}t - a)/(\mathrm{d}/\mathrm{d}t + k_e))\nu$, and the function-level floor is $(N_0/2k_e)[\sum_j Q_{jj}(h_j+\eta)/(h_j+\eta+\sigma_0) + \operatorname{tr} H^2]$, $Q = \langle(\dot{\hat{J}} + a\hat{J})^\top(\dot{\hat{J}} + a\hat{J})\rangle$ — the prior's mobility enters the Tikhonov filter factor as $h_j + \eta$, and the adaptive prior's floor is $(N_0/2k_e)\operatorname{tr} Q$ whatever the spectrum; to second order the noise inside the regressor exerts a mean force (the errors in variables) whose stationary response is $(H + \sigma_0 I)^{-1}$ under the fixed prior and $H^{-1}$ under the adaptive prior, for every $\eta > 0$ | (i) exact as invariance, its threshold formal at the lift, the long-time threshold open; (ii) proved for the linearised loop under white noise as a period average (Lemma D‴.26a), (iii) formal, their closed forms exact for the averaged system; numerics: the floor per unit $N_0$ 0.1597 (fixed) to 0.1757 ($\eta = \infty$), the weakest mode's variance $\times 24.1$ ($\times 28.0$ in the exact linearised loop at $k_e = 25$); the exact periodic covariance of the linearised loop at the prediction to 0.976–1.00001 ($k_e = 25$–800: the relative error $O(1/k_e)$, its first-order term from the slope $a$, the slope-free control numerically even in $1/k_e$); the errors-in-variables bias 491 (fixed) and 2397 (adaptive) per $(\mathbb{E}\nu^2)^2$ averaged, 504 and 2653 in the exact linearised loop, the parameters' fluctuation negligible; the real loop (12 common noise paths, two noise levels) within 1.75 standard errors of the prediction, the mean part 89% at std 0.01; the lift $m = 15$ identical to the $K_0$-atom run to rounding; the separations' threshold std 0.031 at the lift; on antithetic pairs the noise-induced separation rate $\lambda_{\mathrm{eff}} = 0.327$, 0.318, 0.294 at std 0.01, 0.03, 0.05 against the lift's 0.298 and the transient's 0.326–0.328, the small-noise limit $0.329 \pm 0.003$, the rate vanishing at std 0.0299 over 300 time units | measurement noise, the errors-in-variables bias, the diffusion approximation, the LMS misadjustment, leaky LMS, common noise and the lack-of-excitation drift are classical; new are the floor's interpolation in the prior's mobility ($h_j + \eta$ in the filter factor), the driving matrix $Q$ of the sensitivity-filtered measurement, and their combination with the coherence of the $K_0$-atom identification and its noise threshold at the lift |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.31 | The adaptive prior is a washout leak: the prior lags the class barycentre by its exponentially forgotten increments, so the leak acts in full on the separations and, for integral plans, on the barycentre through the high-pass $i\omega/(i\omega+\eta)$ — mode $j$'s stiffness $h_j+\sigma_0 i\omega/(i\omega+\eta)$ — which is its bias removal and its lost robustness to persistent forces on the barycentres; along every affine time-invariant prior the stationary response $R$ and the bias $B$ toward a misplaced anchor satisfy $RH+B=I$ (per mode $h_j r_{jj}+b_{jj}=1$), so no prior gives both; a force's function-level response is at most the norm of its smallest equivalent target perturbation, under every leak with $S+S^\top\succeq 0$; and the desired regressor (the adaptive law's regressor at the reference) keeps the lift, the linearisation and the first-order noise response while turning the noise's own persistent force into the least-squares force of $g=k_e\mathbb{E}[e]$, its response at most $\frac{1}{4}\|\partial_x^2 f\|_\rho^2(\mathbb{E}\nu^2)^2$ uniformly in $k_e$, $B$ and the spectrum — the parameters' fluctuation adding a force of the same order that is not the learning force of a mean tracking error, so no bound in the noise's moments applies | (i)–(iii) proved; (iv)(a)–(b) proved at the lift and for the linearised equations, (iv)(c) formal, its noise part's consequences exact for the averaged system, the parameters' part numerical; numerics: the adaptive prior's bias 2397, 0.471, 0.0526 per $(\mathbb{E}\nu^2)^2$ (noise part; MR, DR, DR*) at $k_e=25$, with the parameters' fluctuation 2398, 0.486, 0.0481 (DR +56% at $k_e=B=100$, where its noise part nearly cancels); MR over DR $3.0\cdot10^3$–$2.1\cdot10^8$ over $k_e=25$–$200$; $R$ and $B$ from the priors' dynamic blocks, $RH+B=I$ to $1.4\cdot10^{-14}$, also for non-commuting leaks, where $HR+B-I$ reaches 0.47–0.79; the real loop on common noise paths: $\mathbb{E}\|\hat{f}-f\|^2$ 8.7 times lower under DR (predicted 9.2), the MR − DR mean difference at its prediction to 0.5–5.3% (0.4–1.8 sampling errors) | washout filters, low-frequency learning, the desired regressor, the Tikhonov / Kalman split and the minimum-norm bound are classical; new are their transfer to the transport leaks (the washout reading of the adaptive prior, uniform in $m$) and the quantification; the parameters' own errors in variables is the classical second-order bias of a constant-gain stochastic approximation ([100]: Benveniste–Métivier–Priouret, Dieuleveut–Durmus–Bach, Box), new only the observation that the desired regressor does not remove it |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.32 | The cohesion leak: the particles attract each other with a finite-range or Gaussian kernel — the leak is $\sigma_0\times$ the kernel mass $\times$ the mean-shift vector of the network's own parameter distribution, no prior, no atom count, no plan — and the averaged loop is the gradient flow of $m\bar{L} + (\sigma_0 m/2)\iint W\,\mathrm{d}\mu\,\mathrm{d}\mu$ (in the mean field an aggregation equation in the fit's external potential with two interactions, the fit's and the cohesion's), its empirical measure the mean-field solution exactly; with the finite range, at a clustered configuration whose atoms are $R$ apart the leak vanishes, so every critical point of the unregularised weighted loss is stationary, for every number of classes and every set of masses, and it acts exactly on the unexcited subspace: the separations of class $k$ at $\sigma_0 w_k$, the class means the unregularised flow's; a class can split iff $\sigma_0 w_k^2 < q_k\lambda_k^+$ at a gauge transform, so $\sigma_0$ sets the resolution and an exact representation is never over-split, and a split sheds single particles (the unstable separations degenerate, the kernel concave); the Gaussian kernel pays an exponentially small bias, and twins with no third class within reach contract at $\sigma_0\psi(\Delta\alpha^2/(G_\alpha R^2))(w_k a_k' + w_l a_l')^2/(w_k a_k'^2 + w_l a_l'^2)$ — a third class within reach can win; in the real loop the lift is periodic with the separation exponents $\sigma_0 w_k/k_e$ and the $\eta=\infty$ adaptive prior's collective spectrum | (i)–(iii) proved; (iv) proved for the stationary points and their linearisation, the split's pattern explained; (v) the twins' rate to leading order, their merger only without a third class within reach (a counterexample), the dead class's motion and time scale stated, its merger not proved; (vi) the lift exact, the collapsed exponents by Corollary D‴.26 and Lemma D‴.26a; not proved: the transit, and with it any selection of the number of kernels — the count is the transit's and the schedule's; numerics: at the 15 count vectors of $m=15$ the lift of $\vartheta^e(w)$ stationary ($2.8\cdot10^{-17}$) with (iii)'s spectrum to $2.8\cdot10^{-11}$ (the finite differences' error), the margins the Jacobian's to $2\cdot10^{-11}$; the twins' rate to 0.04%, opposite-sign twins with a third class within reach not merging ($4\cdot10^5$ slow units, a biased four-class fit); the floor removed from warm starts as by the adaptive prior; from broad starts the annealed Gaussian kernel at the target's three classes in 9 of 9 runs ($\le 2.3\cdot10^{-6}$), at a fixed scale not (4–6 classes at $R=1$, three biased at $R=2$); the real loop's 45 slow Floquet exponents to $5.5\cdot10^{-6}$; the dither's within-class spread 0.063–0.068 against $\sqrt{(1-1/n_k)\tau/(\sigma_0 w_k)} = 0.0635$; $\sigma_0$ lowered: no class with a positive margin split and all 8 below $-5\cdot10^{-5}$ did, by shedding single particles (8–10 classes at $\sigma_0/64$ for four bumps); five and three heavy bumps biased (the transit's masses), two bumps closer than the broad scale merged into one | mean shift, aggregation equations, bounded-confidence and polarized consensus-based clustering, clustering in self-attention dynamics, Fukumizu–Amari's lines of critical points, splitting steepest descent, convex clustering and sum-of-norms identification, deterministic annealing and self-organising RBF networks are classical ([103]); new is their combination as a leak in mean-field adaptive control: the exact blocks at clustered configurations with the mass factor, the gauge-scaled splitting margin, the twins' rate, the real loop's exponents |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.33 | The desired regressor keeps the replicated configuration attracting under measurement noise: with the adaptive law's regressor at the reference the regressor's own term $\langle \mathbb{E}[\varepsilon_m^2] G^{1/2} \partial_x \nabla_\vartheta^2 \tilde{\Phi} G^{1/2} \rangle$ — which carries the measured regressor's drift and does not decrease with $k_e$ — leaves the separations' second-order drift; what remains, the mean error's term $\langle \mathbb{E}[e] G^{1/2} \nabla_\vartheta^2 \tilde{\Phi} G^{1/2} \rangle$ and, once the parameters fluctuate, the cross term, is $O(1/k_e)$ like the separations' stiffness, so the noise threshold is bounded below uniformly in $k_e$ and constant when the noise's bandwidth scales with the gain (at a fixed bandwidth DR's sign reverses at $k_e \approx B$); at the adaptive prior's stationary point the mean error carries the parameters' part of the force, not only the residual $(I - P)g$; under DR the compensation's rectification of the noise makes the drift at the lift negative in the centre directions of separated kernels, the whole matrix on the instance — not for same-sign, wide or edge kernels —, and along the trajectory the drift turns indefinite | (i) and (iv)'s formula formal at D‴.30(i)'s order, (ii)'s bound proved, (iii)'s balance formal with the exact periodic mean, (iv)'s sign proved in the centre directions of separated Gaussian kernels at the lift where the saturation's term is dominated, the whole matrix's numerical; numerics: at the lift DR's drift negative definite, DR*'s largest eigenvalue 60 times below MR's (the threshold std 0.238 against 0.031); $k_e \cdot \lambda_{\max}$ constant for DR and DR* at $B = 4k_e$ while MR's threshold falls like $k_e^{-1/2}$, at $B = 100$ DR positive definite from $k_e = 200$, the floor std $\geq 0.069$; at the stationary point rms $\mathbb{E}[e]$ $6.7 \cdot 10^{-4}$ (DR), its parameters' part $6.3 \cdot 10^{-4}$, the drift indefinite ($\leq +3.7 \cdot 10^{-5}$); along DR's transient the drift first positive at $t = 2714$ and at most $+1.05 \cdot 10^{-4}$ (the formal threshold std 1.64); the real loop on common noise paths: MR's separations growing, DR's decaying faster than $\sigma_0/k_e$ at the transient's prediction (0.97–0.98 of its noise part, the lift's 0.81–0.82), along DR's weakest directions by 0.14–0.36% of $\sigma_0/k_e$, DR*'s slower | the desired compensation law's insensitivity to measurement noise and the synchronisation of identical systems by a common noise are classical ([96], [104]); new is the removal of the regressor's term in the $K_0$-atom identification — a threshold bounded below uniformly in $k_e$ — and the instance's quantification (the sign at the lift, its loss along the trajectory, the stationary point) |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.34 | Mass transfer: the particles carry weights adapted by the replicator law $\mathrm{d}\beta_i/\,\mathrm{d}s = \gamma\beta_i(\langle r\tilde{\Phi}_i\rangle - \lambda)$ (in real time the exponentiated-gradient law $\dot{\beta}_i = \gamma\beta_i e(\tilde{\Phi}_i - \hat{f})$) beside the transport law with the cohesion leak; the pair is the gradient flow of the fit plus the cohesion energy in the metric Euclidean on the positions and Fisher–Rao on the weights (gradient descent on $(\sqrt{\beta}, \Theta)$), $\sum\beta$ and the saturation's bound kept without a projection; at fixed positions the relative entropy of the weights is a Lyapunov function of the real loop ($\dot{V} = -k_e|e|^2$), the negative-entropy instance of a mirror-descent adaptation law; at clustered configurations under the finite range the stationary points are Proposition D‴.32's for every vector of masses, under the Gaussian kernel the replicator selects the masses and moves the bias; the saturated transients — a class short of mass creeping to $a_{\max}$ with its residual — are left at the replicator's rate, so the transit's masses no longer decide whether a target is fitted: at fixed positions bounded trajectories reach the exact fit iff $\sum_k|A_k|/a_{\max} < 1$ | (i), (ii), (iii)'s stationarity at clustered configurations under the finite range, (v)'s zero modes proved; (iv) proved for bounded trajectories (boundedness not proved); the Gaussian kernel's mass selection and the transit numerical; numerics: at fixed positions the exact fit reached with the replicator ($3.0\cdot10^{-14}$ at $\gamma = 10^{-2}$) where the transport alone stalls at 0.388 (the least error at those masses 0.387), and not when $\sum|A_k|/a_{\max} = 1.09$ (0.071, creeping to the $\ell^1$ ball's 0.0684); from broad starts the five-bump target at five classes in 3 of 3 runs ($6.0\cdot10^{-4}$–$1.4\cdot10^{-3}$, levelling off at $2.8\cdot10^{-4}$ near the Gaussian kernel's stationary point; 6–9 biased classes without), the three heavy bumps at three ($\le 1.6\cdot10^{-7}$; 3–5 biased without), the instance unchanged; two bumps closer than the broad scale merged as without, and the shedding at $\sigma_0/64$ unchanged (the fitness vanishes at critical points); at the class level the Gaussian kernel's stationary fit error $3.85\cdot10^{-4}$ with the masses held and $2.63\cdot10^{-4}$ with them free, the compact kernel's exact; the real loop's $V$ monotone with $\dot{V} = -k_e e^2$ to $1.1\cdot10^{-16}$; at the lift 15 zero modes in $(\Theta, \log\beta)$ — 12 redistributions, 3 gauges | Wasserstein–Fisher–Rao distances and gradient flows, conic particle gradient descent and the $\sqrt{\beta}$ lift, neuron birth–death, the exponentiated-gradient law and EM's multiplicative weights, mirror-descent adaptation laws with Bregman Lyapunov functions and the replicator equation's Shahshahani gradient structure are classical ([105]); new is the replicator as the mass transfer of the cohesion leak in mean-field adaptive control: the stationary set unchanged at clustered configurations under the finite range, the saturation trap removed, the representability criterion |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.35 | The split of a class under the cohesion leak is an $S_n$-symmetric transcritical bifurcation: on the class's separations along the splitting direction $v$ the flow reduces to $\mathrm{d}\xi_i/\,\mathrm{d}s = \mu\xi_i + (c_3/2)(\xi_i^2 - \sum \xi^2/n)$, $\mu$ the negative margin, $c_3$ the residual's third derivative along $v$ (the cohesion, even, enters at third order); every small split state ($q$ particles against $n-q$) is a saddle with the eigenvalues $\mp\mu n/(n-2q)$ and $-\mu$, the $S_{n/2} \times S_{n/2}$ states cubic, so the split is a jump; the particles keep their order along $v$ (equivariance), and once its excursion exceeds $2\mu/\lvert c_3\rvert$ the extreme particle of $\mathrm{sgn}(c_3)\xi$ runs away on the side $\mathrm{sgn}(c_3)v$ — alone when its lead is large, with a near-tied neighbour as a bound pair —, at a time the reduced equation gives; after a departure the margin is $\sigma_0(w_k - k/m) - \lambda^+(r', \vartheta')$, the stiffness's drop and the curvature's gauge rise both favouring further departures, the residual's change stopping them; the mass-free leak (the normalised mean shift) has the thresholds $\lambda_k^+$, a factor $w_k$ lower, and delays the cascade without stopping it | (i)–(ii) proved to leading order in $\mu$ (the centre manifold, the $S_n$ normal form), (iii) the order by equivariance, the escape time in the quadratic truncation, (iv) a margin statement, (v) the blocks proved; the cascade's extent numerical; numerics: the split states found by Newton within 0.3%–3.0% of the prediction at $\mu = 5 \cdot 10^{-7}$, their eigenvalues converging to $\pm\mu n/(n-2q)$ and $-\mu$ (errors falling fourfold with $\mu$); the first excursion's particle the reduced quadratic equation's in 12 of 12 runs (the linear equation's in 6, the flipped one's in 2), its time to 1.1% (the linear equation's to 22.4%); bound pairs in 3 of 6 runs at $\sigma_0/16$; under Part G's schedule 8–10 classes (the cohesion leak), 6–7 with the replicator at $\gamma = 0.1$, 5–8 at $\gamma = 1$, 9, 8, 9 classes (the fit error 0.0028, the cohesion leak's end state) under the mass-free leak continued to $\sigma_0/256$ | $S_N$-equivariant bifurcation theory — the standard representation's quadratic equivariant, transcritical branches with no stable small branch, the jump of sympatric speciation, the permutation-symmetric critical points of networks — is classical, as are order preservation and the Riccati escape ([106]); new is the identification for the cohesion leak: the quadratic coefficient as the residual's third derivative, the cohesion at third order, the margin after a departure, and the instance's quantification |
| D‴.36 | The gauge under dither: at an exact representation with one particle per class the fits through the configuration form a manifold $M$ of dimension $K-1$ parametrised by the masses ($\hat{f}$ constant on it); with the replicator and the positions dithered the masses diffuse along $M$ (Katzenberger's limit) with the covariance $D = (T^\top gT)^{-1}T^\top g\Sigma gT(T^\top gT)^{-1}$ in the flow's metric, for two classes $2\tau T_\alpha/(T_\alpha + T_W)^2$, vanishing as $\gamma \to 0$ and at the saturation, where the amplitude's excursion is unbounded; the entropic leak $(\varepsilon/\gamma)\sum \beta \log \beta$ on the masses is a leak on the unexcited subspace: it keeps the exact representation at $W = w$ stationary (no bias where $w$ is representable) and makes the gauge an Ornstein–Uhlenbeck direction with stiffness $\kappa = \varepsilon(T^\top gT)^{-1}T_W$, its spread solving $\kappa C + C\kappa^\top = D$ ($\propto \varepsilon^{-1/2}$), the dither's drift term shifting it by $O(\tau/\varepsilon)$ | (i) proved, (ii) the diffusion coefficient by Katzenberger's theorem (the drift term not computed), (iii) the stationarity, the stiffness and the linearised spread proved; numerics: the masses' diffusion measured over 64 seeds at 0.95, 1.36, 0.90 times the prediction (the displacement's variance; within two standard errors); without the leak the spread 0.05 by $2 \cdot 10^5$ slow units and the first amplitude at $\lvert a\rvert/a_{\max} = 0.9994$; with it the spreads 0.0044, 0.0032, 0.0034 at $\varepsilon = 10^{-3}$ against the predicted 0.0044, 0.0031, 0.0033 (4 seeds; at $\varepsilon = \tau$ inconclusive) | Diffusions forced onto a manifold by a large drift (Katzenberger) and SGD's noise-driven motion along the manifold of minimisers (Blanc et al., Li–Wang–Arora) are classical ([107]); new: the amplitude–mass gauge of Proposition D‴.28 as such a manifold for the replicator of D‴.34, the diffusion in the flow's metric with its closed form and its excursion into the saturation, and the entropic leak on the masses as the pin |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
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| D‴.37 | The dither's drift along the gauge: the drift of Katzenberger's limit on the manifold of exact fits, from the second-order identities of the limit map in the linearisation's eigenbasis, is of the same order $\tau$ as the diffusion; in the slow-replicator limit it is the replicator (Fisher–Rao) flow of the entropic free energy $(\tau/2)\log\det\nabla^2_\Theta J$ — the positions' Gibbs law averaged and Laplace —, which within a class sees only the class's total mass and diverges to $-\infty$ at the saturation, so dither drives the masses toward the saturated, flattest fits; a boundary layer of width $\delta^* = \frac{1}{2}\sqrt{\gamma a_{\max}\lvert A_k\rvert/(mG_\alpha)} = O(\sqrt{\gamma})$, where the saturating amplitude's relaxation falls below the mass channel's, slows but does not stop it (inside, the drift is the Itô term of the amplitude's driftless diffusion, so the masses' law is not normalisable there); the entropic leak's pinned state is shifted by $\kappa^{-1}b(w) = O(\tau/\varepsilon)$ | (i) proved (given Katzenberger), (ii) the averaging and Laplace steps sketched, (iii) the divergence and the layer's scale, the inside law sketched, (iv) first order; numerics: the identities against the integrated limit map to $10^{-3}$; the drift the amplitudes' noise's; the $\gamma \to 0$ limit equal to the log-determinant law to 0.3%; the layer's zero at $0.57\delta^*$ as $\gamma \to 0$; seven zeros at $\gamma = 10^{-2}$ (three attractors of the mean drift, not settling states); the 64-seed ensemble mean on the mean path; the leak's shift and spread predicted for $\varepsilon \gtrsim 3\tau$; without a leak the excursions into the saturation grow over $10^6$ slow units | The limit map's second derivative and the noise-driven drift along manifolds of minima (Katzenberger; Li–Wang–Arora, [107]), averaging and the potential of mean force, Laplace, flat minima ([108]) are classical; new: the replicator's gauge as the slow variable with the log-determinant free energy, the dither's drive into the saturation and the $\sqrt{\gamma}$ layer that does not stop it, the finite-$\gamma$ eigenbasis formula, the leak's bias |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.38 | A Fixman correction for the replicator under dither: adding $\kappa_F(\tau/2)\partial_\beta\ell$ to the replicator's fitness — $\ell = \log\det\nabla^2_\Theta J$ along the manifold of exact fits, in closed form on the masses (the masses form), $\kappa_F(\tau/m)(1+\hat{W}_k^2/W_k^2)/(W_k-\hat{W}_k^2/W_k)$ with $\hat{W}_k = \lvert\hat{A}_k\rvert/a_{\max}$ from the classes' contributions — makes the corrected loop the gradient flow of $J - \kappa_F(\tau/2)\ell(W)$, a log-barrier on the representable masses, and adds $(T^\top gT)^{-1}\nabla_W[\kappa_F(\tau/2)\ell]$ to the dither's drift, in the slow-replicator limit the replicator flow ascending $\kappa_F(\tau/2)\ell$: at $\kappa_F = 1$ the entropic drift is cancelled to leading order (the remainder $O(\gamma^2\tau)$, of the order of the diffusion; at finite $\gamma$ over-compensated once $\gamma$ is a few $10^{-3}$), for $\kappa_F > 1$ the masses ascend the strictly concave $\ell$ to its unique maximiser $W^\sharp$, the D-optimal (largest-determinant) exact fit, their spread $O(\sqrt{\gamma/(\kappa_F-1)})$ (with an estimate $\hat{\tau}$ of the dither's level the net gain is $\kappa_F\hat{\tau}/\tau$, and $\hat{\tau}$ can be read from the rms fit error by equipartition); near a saturation the unfloored correction gives the saturating amplitude the drift $-2\kappa_F G_\alpha\tau/a_{\max}$, so the density goes like $\delta^{\kappa_F-1}$ — normalisable for $\kappa_F > 0$, the saturation repelling for $\kappa_F > 1$ —, while the implemented floor $0.01W_k$ restores $\delta^{-1}$ within $\delta_{\text{cap}} = W_{\min}(1/\sqrt{0.99}-1)$; with the entropic leak the pinned state's mean is the drift balance $\kappa^{-1}b_c$ plus the fluctuations' term through the leak's curvature, of the same order; the plug-in form, at the current amplitudes, equals it on $M$ but reads the positions' escapes off $M$ as masses at their saturation | (i) the slow-limit cancellation proved, the finite-$\gamma$ formula exact, the remainder numerical; (ii) the concavity, uniqueness and slow-limit convergence proved; (iii) the projected drift on $M$ proved, the exponents checked in Katzenberger's limit (the finite-$\tau$ statement untested at the instance's $\tau$), the floor's cap checked; (iv) second order; (v) a mechanism, numerical; numerics: the corrected drift over the uncorrected 0.31, 0.07, $2\cdot10^{-3}$, $7\cdot10^{-4}$ at $\gamma = 10^{-2}$, $5\cdot10^{-3}$, $10^{-3}$, $10^{-4}$ at $W = 1/3$, reversed from $\gamma = 2\cdot10^{-3}$ on; $\kappa_F = 2$'s zero within $2\cdot10^{-4}$ of $W^\sharp$ for $\gamma \le 10^{-2}$, the linearised spread there $\propto\sqrt{\gamma}$; the density exponents $-0.99$, $-0.00$, $+0.98$ for $\kappa_F = 0$, 1, 2, and below the floor's cap the unfloored ones minus $\kappa_F(1-(1-W_{\min}^2/W^2)/0.01)$ to $10^{-3}$; in the full loop over $10^6$ slow units the masses at $W^\sharp$ within 1.0 standard error at $\kappa_F = 2$ with the spread of the linearised limit and $\lvert\alpha\rvert \le 8.1$, at $\kappa_F = 1$ free excursions along $M$ into a layer ($\lvert a\rvert/a_{\max}$ to 0.998); $p_{\text{eff}} = 8.92$–$9.03$ coordinates holding fit energy with the correction, 8.06 without; with the leak the bias reversed as predicted (within 1.3 s.e.); the plug-in form moving the masses over 0.21–0.46 where the positions escape, the masses form not; inside the leak's window the shifts within 1.03 s.e. of the second order at $\kappa_F = 0$, 1, 2 and the paired differences within 0.93 s.e. (the first order 1.2–4.8 s.e. off) | The log-determinant correction of stiff constraints (Fixman), stiff bonds against rigid constraints (Hinch), projected diffusions and mean force (Ciccotti–Lelièvre–Vanden-Eijnden), the stiff limit's zero-point term (Helfand; Waszkiewicz–Lisicki), adaptive biasing force and well-tempered metadynamics' bias factor, the log-barrier and its D-optimal maximiser, the replicator's noise-induced drifts are classical ([109]); new: the correction for the replicator's gauge, online and in closed form on the masses, the masses-versus-plug-in distinction and the plug-in form's failure off $M$; the D-optimal selection with its gain and the confinement exponent $\delta^{\kappa_F-1}$ are its applications |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
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| D‴.39 | Dissipativity of the dithered gauge: under the cohesion leak $J$ is bounded on the unconstrained coordinates (bounded physical coordinates, a bounded aggregation energy), so the positions' Gibbs measure $e^{-J/\tau}$ has infinite mass at every $\tau$: the dithered loop has no stationary law, the time spent in any bounded set tends to zero — the prior leak's coercivity, which gave Theorem J′'s Gaussian tails, went with the prior; where a coordinate saturates its force vanishes, so approximate fits with saturated amplitudes form plateaus on which those coordinates diffuse freely, and the exact fit's well holds the positions for $e^{\Delta J_D/\tau+o(1/\tau)}$ (Freidlin–Wentzell); a dead zone $(\sigma_B/2)\sum\|\vartheta_i-\Pi_B\vartheta_i\|^2_{G^{-1}}$ on a box $B$ of the unconstrained coordinates restores a unique stationary law, leaves the deterministic objects inside $B$ unchanged (no bias for the exact fits there; a trajectory is the unboxed one until a coordinate leaves $B$ — the stationary law still weighs the plateau where it meets $B$, and there the box's force enters the masses' fitness), replaces the gauge's saturation by a non-degenerate wall $W_k=\lvert A_k\rvert/(a_{\max}\tanh(L_\alpha/a_{\max}))$ at which the entropic drift ends (at vertices of the representable polytope), and keeps Proposition D‴.38's selection of $W^\sharp$ when $W^\sharp$ lies in the box | (i) proved; (ii) the plateau, its level and the path numerical, the exit time's order Freidlin–Wentzell's; (iii) (a), (b) proved ((b) for the deterministic objects), (c) sketched (at leading order: $V$ is $C^{1,1}$, a boundary layer at $P_B$'s faces), (d) given D‴.38(ii); numerics: the plateau $7.6$–$7.8\tau$ above the exact fit at $W=1/3$ and $W^\sharp$ (fit error 0.035; at the box's edge $7.66\tau$: the box bounds the plateau, it does not raise it); the minimum-energy path at $W=1/3$ a monotone climb without a saddle; the escapes at fixed masses at $1.3\cdot10^{-6}$–$1.1\cdot10^{-5}$ per slow unit for $\tau=1.5$–$3\cdot10^{-4}$ (the Arrhenius slope $(0.81\pm0.14)\Delta J$ at $\Delta J/\tau=2.6$–$5.1$); the dead zone making the plateau's occupation stationary (0.09–0.19 against $0.18\to0.57$) with $\lvert\alpha\rvert\le5$; the full loop with the box at $\kappa_F=2$ within 0.41 s.e. of $W^\sharp$, at $\kappa_F=0$ the masses' spread halved, on their way to $P_B$'s vertices; the two channels to the saturation separated in the stored runs (on $M$ without a leak, off it with one) | Dissipativity and invariant laws of Langevin dynamics (Mattingly–Stuart–Higham; Raginsky–Rakhlin–Telgarsky), the exit problem and metastability (Freidlin–Wentzell; Bovier–Eckhoff–Gayrard–Klein), parameter projection with a Lipschitz boundary layer (Ioannou–Sun; Pomet–Praly), Langevin sampling through the Moreau–Yosida envelope of a convex set's indicator (MYULA) are classical ([110]); new: the loss of the dithered loop's stationary law with the prior, the plateau of saturated approximate fits and the monotone climb to it on the instance, the dead zone's wall on the gauge and its compatibility with the Fixman correction |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
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| D‴.41 | Measurement noise on the gauge: with the measured error $r+\xi$ driving both laws (the noise entering along the descent's own vector fields $U_x = m\mathsf{M}\nabla\hat{f}(x)$), the fields are orthogonal to the manifold of exact fits in the flow's metric — no diffusion of the masses along it —, and the limit map's invariance makes the bilinear form $\langle \nabla r \otimes \nabla(\partial\Phi \cdot U)\rangle$ antisymmetric, so for a noise white on the orbit's resolution (homoscedastic, independent over the nodes) the drift along $M$ vanishes at order $v$ in continuous time, in Stratonovich's sense and for smooth noise of any temporal spectrum: Katzenberger's Itô drift and the Wong–Zakai term cancel; with noise drawn afresh at each update of an Euler step (sampled-data adaptation, label-noise SGD) the drift is $-(v/2)\partial\Phi\langle (U\cdot\nabla)U\rangle_\rho$ — label-noise SGD's flat-minimum selection, its form depending on the update's coordinates —, on the replicator's masses the replicator flow of a strictly convex potential with an interior minimiser; for correlated draws the selection is Katzenberger's drift of the increment's second moment, each pair of normal modes weighted by a linear-response factor (for any step $h < 2/\lambda_{\max}$): $(1-\rho)/(1+\rho)$ of the fresh noise's only when $\lambda_{\max}\tau_c \ll 1$, on a flat-metric gradient flow the descent of $\operatorname{tr}\Psi(\nabla^2 J)$ with $\Psi = -(2/h)\log(1-h\lambda/2)$ for fresh draws (Damian–Ma–Lee's finite-step regulariser) and $\log(1+\lambda\tau_c)/\tau_c$ for a finely sampled slow Markov noise — from the trace to the log-determinant —, $\Psi$ concave for every colour (a smooth noise's bounded); in the slow limit a Markov measurement noise selects as a dither of the flow's own mobility ($\tau_c\Sigma_f \to m\cdot Q\mathsf{M}Q^\top$, a fluctuation–dissipation identity), toward the saturation in the slow-replicator limit; at a finite replicator rate the coloured selection is multistable (three stable targets, $W^\sharp$ unstable) | (i) proved; (ii) proved at order $v$ (given Katzenberger's and the Wong–Zakai theorems), the $O(\gamma)$ of the residual for other node covariances proved at leading order in $\gamma$, its size numerical; (iii) proved at leading order (the closed forms in the slow-replicator limit); (iv) proved at order $v$ and linear response for any step $h < 2/\lambda_{\max}$ and correlation, (c)'s concavity in the continuum limit, (e)'s finite-step limit, its identity with the dither's direction in the slow-replicator limit away from the saturations; the zero set at finite $\gamma$ numerical; (v) a corollary; numerics: Itô + Wong–Zakai $= 0$ to $10^{-8}$; the full loop against the linear-response prediction within 0.8% (fresh, $\rho = \pm 1/3, -1/2, 0.6$) and 1–2% at $h = 8$ ($h\lambda_{\max} \approx 1.6$), the per-step law rejected at $\rho \geq 0.9$; $f = uv$'s rates within 0.3% to $\rho = 0.999$; the selection multistable from $\tau_c \approx 100$ | Katzenberger's limit for noise along a manifold of minimisers (Li–Wang–Arora), label-noise SGD's selection of flat minima and its finite-step regulariser (Blanc–Gupta–Valiant–Valiant, Damian–Ma–Lee, HaoChen–Wei–Lee–Ma, Shalova–Schlichting–Peletier), the Itô–Stratonovich distinction, the Wong–Zakai theorem and the coloured-noise crossover (Wong–Zakai, van Kampen, Kupferman–Pavliotis–Stuart), the active Ornstein–Uhlenbeck particle's filter $1/(1+k\tau)$ and the coloured-noise approximation's $\det(I+\tau\nabla^2 U)$ (Jung–Hänggi, Maggi et al.), SGD's Itô SDE models (Li–Tai–E, Li–Malladi–Arora), correlated and anti-correlated noise (Orvieto et al., Lucchi et al., Kühn–Rosenow, Beneventano), the law of balance (Ziyin et al.), the diagonal network's label-noise Lasso (Pillaud-Vivien–Reygner–Flammarion) are classical ([111], [107]); new: the cancellation for noise entering through the error, for any temporal spectrum, by the antisymmetry the limit map's invariance imposes; the coloured filter of label noise's drift along $M$, its cross term cancelling by the same antisymmetry, from the trace to the log-determinant, concave for every colour, a slow Markov noise acting as a dither; the closed forms on the replicator's masses and the coloured selection's multistability there |



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*Table 1, continued*

| result | statement | status | novelty (audit) |
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| D‴.42 | Several particles per class: with each class's particles together at its exact-fit position, $\log\det\nabla^2_\Theta J$ on the manifold of exact fits is the one-particle-per-class $\ell_1$ at the class masses plus a constant (a unit-Jacobian change to the class's barycentre and internal differences makes the Hessian block-diagonal: the fit sees the barycentres, the cohesion the differences, the masses' split neither), so under the isotropic dither the class masses follow Proposition D‴.37's law with $m$ the number of particles (the one-per-class dynamics slowed by $K/m$), the fractions within a class neither drift at $O(\gamma\tau)$ nor diffuse (the diffusion vanishing at any $\gamma$ for any position noise), and Proposition D‴.38's correction at the class masses acts as with one particle per class; an anisotropic dither (Proposition D‴.40's masks) breaks this at unequal splits — the barycentre and the differences coupled, the positions' law no longer Gibbs —: the fractions drift at $O(\gamma\tau)$ and the class masses depart from the per-class law; with measurement noise the equal split is an invariant subspace on which the loop is the one-per-class loop (the laws $m$-independent, the cancellation exact), at unequal splits the cohesion breaks Proposition D‴.41's cancellation (up to 10% of the Itô drift) on the class masses and the fractions, and noise drawn afresh equalises the fractions through the full trace | (i) proved; (ii) proved for the isotropic dither in the slow limit, the diffusion's vanishing at every $\gamma$ for any position noise, the anisotropic mechanism proved and its size numerical; (iii) proved under the isotropic dither; (iv) the invariant subspace and the residual's mechanism proved, its size numerical; numerics: the determinant blind to the split to $4\cdot10^{-7}$; the ratios 0.5000 and 0.3333 at $\gamma=10^{-5}$; the per-class law at the equal split under every mask to $4\cdot10^{-4}$ (extrapolated in the finite differences' step); at unequal splits an anisotropic dither moving the class masses ($5.9\cdot10^4$ times the per-class law for the centres-and-widths mask) and the fractions at $O(\gamma\tau)$; measurement noise's residual up to 9.6% per component there | Jacobi coordinates, the block structure of a translation-invariant interaction, and the non-Gibbs stationary law of a non-isotropically driven linear system are classical; new: the gauge's laws (the dither's selection, its correction) acting on the class masses with $m$ the number of particles exactly when the dither is isotropic, the split felt at Katzenberger's order otherwise; the equal split's invariant subspace under measurement noise — the open item of Propositions D‴.36–D‴.41 |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
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| D‴.43 | The colour of the measurement noise selects the kernels: on a kernel dictionary whose coefficients are the difference of two adaptive masses per atom ($w = u \odot u - v \odot v$, the unbalanced Fisher–Rao law of a signed kernel measure's two parts), label noise under an explicit (Euler) update moves the parameters along the exact fits by the gradient flow of $R = \operatorname{tr}\Psi(K)$, $K = (4/N)\Phi\,\mathrm{diag}(u^2+v^2)\Phi^\top$, $\Psi' = f$, Proposition D‴.41's filter — the products $u_i v_i$ decaying at the rates $\omega_i = \varphi_i^\top f(K)\varphi_i$ —, the step's Itô drift at a rate $\propto h\sigma^2$ (absent in continuous time and for Stratonovich-consistent updates; Heun's scheme selects $O((h\lambda)^2)$ of it) and the descent's small-noise limit ($\sigma$ small against the truth's least coefficient times $\sigma_{\min}(\Phi_S)$; beyond it the descent prunes on its own, Lasso-like): as $h\lambda_{\max}, \sigma \to 0$ white noise selects the weighted $\ell^1$ norm (basis pursuit; at a finite step Damian–Ma–Lee's strictly convex potential, whose minimiser on $M$ is not basis pursuit's solution), a slow Markov noise $(1/\tau_c)\log\det(I+\tau_c K)$ — Fazel's log-det heuristic, sparse Bayesian learning's log-det term but not its penalty —, which by Cauchy–Binet counts the support ($k\log(\tau_c)/\tau_c$: the sparsest exact fits its global minimisers as $\tau_c \to \infty$, not at every finite $\tau_c$), every colour in the continuum a concave penalty with many local minima (at a finite step the concavity fails, for weakly or anti-correlated draws already at small $K$; a smooth noise selects slowly, $f \approx (\lambda\tau_c)^{-2}$); a $k$-sparse exact fit on fewer than $N$ atoms is held by the slow selection once $\tau_c$ passes a threshold that grows without bound with the dictionary's coherence (a weighted dual certificate, sufficient, its ratio $O(1/\tau_c)$ beyond it; for white noise Fuchs's, conservative), by the white one only where weighted basis pursuit recovers it; the slow selection recovers sparse kernel expansions where basis pursuit fails, and label-noise descent at small $\sigma$ identifies their support where its reduced flow does (the coefficients after a noiseless phase) | (i) proved at order $v$ (given D‴.41(iv)) where $\operatorname{rank} K = N$ and $\lambda_{\min}(K) \gg v_h$; the pruning phase ($\lambda_{\min}(K) \to 0$, the sparse fits singular points of $M$) and (v)'s stability statements about the reduced flow, numerical for the descent; (ii) the potential proved, the end at basis pursuit's solution sketched, in the limits $h\lambda_{\max}, \sigma \to 0$; (iii), (iv) proved ((iv) in the continuum, its failure at a finite step from the filter's expansion); (v) proved for the reduced flow (the certificate sufficient); the basins numerical; numerics: the identities to $2.6 \cdot 10^{-15}$ (Cauchy–Binet) and $4 \cdot 10^{-9}$ (the gradient), $R$ concave along every random segment (the continuum filters); on $10 \times 40$ Gaussian designs the slow flow ($\tau_c = 100$) recovers 0.75 and 0.60 of the 3- and 4-sparse truths against basis pursuit's 0.38 and 0.17 (the white flow, run to convergence, ends at basis pursuit's solution), at $20 \times 80$ 0.75 against 0.35 ($k = 6$), sparse Bayesian learning's EM 0.88 and 0.55 ($k = 3$, 4); at $\tau_c = 10$, 0.39, 0.68 and 0.85 of the slow flow's failures ($k = 4$, 5, 6) end below the truth's $R$; label-noise descent with AR(1) draws ($\sigma = 0.1$, eight instances where basis pursuit fails) recovers the truth on all 16 paths of the four whose reduced flow recovers it, on 0–2 of 16 on the other four, at $\sigma = 0.5$ on one; on Gaussian kernel dictionaries (coherence 0.96) 1.00 against 0.85 at $k = 4$; the certificate's median ratio $\propto 1/\tau_c$ (0.439, 0.0814, 0.0087 at $\tau_c = 10$, 100, 1000), on kernel dictionaries the certified fraction rising with $\tau_c$ to 0.85–1.00 at $\tau_c = 10^5$ (white: 0–0.60); from a large start ($\alpha = 2$, run to convergence) basis pursuit's rate, and atoms the white flow has dropped do not revive; at the same physical noise $\Delta\log P$ off the truth's support $-2.39$ under Euler's step 0.5, $-0.097$ under Heun's (25 times less) | the label-noise $\ell^1$/Lasso selection on quadratic parametrisations (HaoChen–Wei–Lee–Ma, Pillaud-Vivien–Reygner–Flammarion), the finite-step regulariser (Damian–Ma–Lee), the kernel and rich regimes (Woodworth et al.), sparse Bayesian learning and its log-determinant (Tipping; Wipf–Rao) and its non-separable reweighting succeeding where $\ell^1$ fails (Wipf–Nagarajan), the log-det heuristic (Fazel–Hindi–Boyd), reweighted $\ell^1$ (Candès–Wakin–Boyd), stochastic collapse (Chen et al.), sparse optimisation on measures by Fisher–Rao flows (Chizat), the dual certificate (Fuchs, Tropp) are classical ([111], [112], [113]); new: the noise's temporal correlation as the sparsity prior of the explicit update's selection along the exact fits — the trace for white noise, a support-counting log-determinant for a slow Markov one, concave for every colour in the continuum —, its stability law (every sparse fit on fewer than $N$ atoms held beyond a coherence-dependent $\tau_c$), and recovery beyond basis pursuit by the reduced flow and by label-noise descent at small $\sigma$ — the mechanism, not a better sparse solver |

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| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.45 | The free mass–amplitude law is a mirror flow: without the masses' constraint each particle's mass and amplitude are driven by one scalar signal, so $Q = \beta - (\gamma a_{\max}^2/4G)\cosh(2\alpha/a_{\max})$ is conserved for every drive (beyond the cohesion's range) — the particle moves on a fixed curve, the free replicator reparametrises the amplitude and lifts its saturation —, and on a dictionary $w = \beta \odot a/m$ follows the mirror flow $\mathrm{d}\eta = \Phi^\top(r+\xi)\,\mathrm{d}t/N$ of an explicit separable potential ($\eta = \int \mathrm{d}\alpha/(G\beta a')$ in closed form, its conjugate the mass gained): the noiseless loop converges from every start with $Q > -c$, without persistent excitation, to the start's Bregman projection on the exact fits — the mass-weighted minimum norm for large starting masses, weighted basis pursuit with weights $1/\sqrt{\beta^0}$ for small ones at zero amplitudes (when unique; approached like $(\beta^0)^{3/2}$, the off-support coefficients on the core's tangent profile with the certificate's slacks; the transit in a time $\propto (\beta^0)^{-1/2}$, D‴.48, the last digits $\propto 1/(\beta^0)^2$ where the support has fewer than $N$ atoms) —; noise with smooth paths (Stratonovich), of any colour and covariance over the nodes, never moves that projection, Itô's white noise adds the Euler step's drift; an Euler (sampled-data) update changes $Q$ by a closed-form amount per step — the conserved share of D‴.44's selection —, a mirror update on the curves keeps it; in the real loop $\frac{1}{2}e^2 + \sum$ Bregman divergences is a Lyapunov function of the joint mass–amplitude adaptation; the conserved law shifts every level by $-\gamma\beta\lambda\,\mathrm{d}t$ | (i), (ii), (iv)(a), (v) and (vi)'s identity proved; (iii) the convergence and the projection proved for $Q > -c$, the lazy limit proved, the rich limit proved when weighted basis pursuit is unique, its leading order at an $N$-atom vertex derived and checked (Part F); (iv)(b) the formula proved, the selection's end numerical; (iv)(c) in expectation; the branches: D‴.47; numerics: $\eta$'s closed form to $1.5 \cdot 10^{-14}$, $Q$ kept to $1.4 \cdot 10^{-11}$ noiseless and $3.6 \cdot 10^{-10}$ under a smooth noise mixed over the nodes, the conserved law's shift to $1.5$–$4.4 \cdot 10^{-7}$; on $5 \times 12$ Gaussian designs from masses $s \cdot m/d$ the loop's end the projection to $4.7 \cdot 10^{-11}$ (no projection failed), for 2-sparse truths 0.5% from the minimum-norm fit at $s = 10$ and with $\|w\|_1$ 1.0004 times basis pursuit's at $s = 10^{-4}$, within $10^{-8}$ at $s = 10^{-3}$ only from $t \approx 8.5 \cdot 10^6$; with unequal masses weighted basis pursuit to $1.7 \cdot 10^{-9}$ ($s = 10^{-7}$); at $N$-atom vertices the leading order to the rounding floor, the distance falling by 1068 and 984 from $\varepsilon = 10^{-7}$ to $10^{-9}$ ($\varepsilon^{3/2}$: 1000), a small slack ($\mu_{\min} = 2.7 \cdot 10^{-4}$) still 0.17 away at $10^{-7}$; Euler–Maruyama's mean change of $Q$ its Itô correction's integral (ratio 1.0011 at $h = 0.01$), Heun's $1.4 \cdot 10^{-6}$, the mirror update's $2.2 \cdot 10^{-16}$; D‴.44's reduced flow ending on basis pursuit where it is the truth (6 instances), elsewhere on its support or on another sparse fit of larger $\ell^1$ norm; at $h = 0.5$ the Euler update moving the projection by 0.030–0.043, the mirror update keeping it to $9.0 \cdot 10^{-12}$; in the real loop $\dot{V} = -k_e e^2$ to $3.0 \cdot 10^{-9}$; the conserved law's limits $1.6$–$4.1 \cdot 10^{-3}$ from the free law's projections | Classical ([113]): a gradient flow under a commuting reparametrisation is a mirror flow (Li–Wang–Lee–Arora), with one conserved function per unit (Marcotte–Gribonval–Peyré; Azulay et al.), the kernel and rich regimes and the hyperbolic entropy's shape (Woodworth et al.; Ghai–Hazan–Singer), exponentiated gradient as mirror descent and as reparametrised gradient descent (Kivinen–Warmuth; Amid–Warmuth), stochastic mirror descent's invariance in the dual span (Gunasekar et al.; Azizan–Hassibi), SGD's Itô noise shrinking the effective initialisation and isotropic noise not selecting (Pesme–Pillaud-Vivien–Flammarion; Even et al.; HaoChen–Wei–Lee–Ma), Bregman Lyapunov functions of mirror-descent adaptation laws (Boffi–Slotine); new: this saturating unit's potential in closed form (the conserved $Q$, the wall, the entropy tail, the $\ell^1$ weights $1/\sqrt{\beta^0}$ with their $-(3/2)K|w|^{2/3}$ correction, the branches), the Euler update's change of $Q$ as the conserved share of D‴.44's selection, the mirror update as a sampled-data law whose noise never selects |

*continued on the next page*

*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.46 | A leak in the mirror coordinates: adding $-\sigma(\eta_i - \eta_{p,i})$ to each particle's scalar drive in the free mass–amplitude law ($Q$ kept, $\eta$ explicit) keeps Proposition D‴.45's Lyapunov function with $\dot{V} = -k_e e^2 + ed - \sigma \sum [B(w^*, w) + B(w, w_p) - B(w^*, w_p)]$ (the three-point identity), so $V(t) \le V(0)e^{-\lambda t} + (C/\lambda)(1 - e^{-\lambda t})$ and the parameters are ultimately bounded under bounded disturbances and approximation errors, which the free masses alone are not (a decaying excitation can drive them to infinity); the averaged loop, with the error filter's Gram $H_L$, has a unique, globally attracting equilibrium — a Bregman-regularised fit, the mirror flow's minimiser to second order in $1/k_e$ —, so the leak's bias is explicit: a generalised ridge from large starting masses, the LASSO of penalty $\sigma\lambda_1(\beta^0) \propto \sigma/\sqrt{\beta^0}$ from small ones at zero amplitudes, which keeps exactly the target's kernels under the irrepresentable condition | (i) proved; (ii) proved for the averaged loop, the real loop's mean by the classical averaging theorem; (iii) the two limits proved, the support recovery by the LASSO's theory; numerics: the identity to $9.5 \cdot 10^{-11}$; on random $H$ the averaged equilibrium the ridge to $1.3 \cdot 10^{-8}$ ($\beta^0 = 10^4$), 0.15 off it at $\beta^0 = 3$ where the saturation's wall lies inside the coefficients' range, and the LASSO to $1.5 \cdot 10^{-4}$ ($\beta^0 = 10^{-8}$) with its support in every draw; on a persistently exciting orbit under an approximation error and a biased disturbance the free law creeping toward the biased fit (its largest coefficient 0.309, 0.330, 0.337, 0.341 by quarters), $\sigma = 0.01$ and 0.1 holding the coefficients at 0.081 and 0.040, $V$ below the comparison bound at every sample, the rms error rising from 0.024 to 0.031 and 0.037; without excitation the free law's coefficient 23.3 at $t = 10^6$, the leak's 0.46; the real loop's period means within $1.6 \cdot 10^{-4}$ and $1.7 \cdot 10^{-4}$ of the filtered equilibrium, 4.4 and 2.6 times closer than to the idealised one; on an orbit's Gram the equilibrium's support exactly the truth's for 415 of the 1760 3-sparse patterns, the LASSO's prediction holding for 416 (agreement on 1745), the rich start's equilibrium reached at $t \approx 3.2 \cdot 10^5$, at $2 \cdot 10^6$ where the irrepresentable condition fails | Classical ([114]): the $\sigma$-modification and its ultimate bound (Ioannou–Kokotović; Ioannou–Sun), averaging of adaptive systems with the filtered regressor (Anderson et al.; Bodson et al.; Sastry–Bodson), continuous-time discounted mirror descent — a dual-coordinate leak — with its regulariser-perturbed rest points and Lyapunov functions (Gao–Pavel; Romano–Pavel), its entropic case for the replicator (Coucheney–Gaujal–Mertikopoulos), regularisation toward a reference (Pérolat et al.; Abe et al.), the vanishing-leak selection (Alvarez–Bolte–Brahic; Attouch), the LASSO and its irrepresentable condition (Tibshirani; Zhao–Yu; Wainwright), mirror-descent adaptation laws (Boffi–Slotine); new: the dual-coordinate leak on the paradigm's $(\alpha, \beta)$ law with $Q$ kept, its bound in the adaptive loop, the starting masses' LASSO weights $\lambda_1 \propto (\beta^0)^{-1/2}$ |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.47 | The free law's branches: a start with $\beta(0) < 2c\sinh^2(\alpha(0)/a_{\max})$ — every small start at nonzero amplitude — puts the particle on a branch of its curve, where the sign of $w$ is frozen and the potential is an entropy at the branch point with the temperature $1/(\gamma a_0)$ ($\mathrm{d}\eta/\mathrm{d}\log\lvert w\rvert = a/(\gamma a^2 + G\beta a'^2)$), finite there, $B(0,w) = \beta/(\gamma m)$: the law is an exponentiated gradient. From every start the dictionary loop converges to the Bregman projection of its start onto the sign-constrained least-squares fits, the coordinates vanishing on them going extinct (exponentially when strictly complementary); the rich limit on branches is the sign-constrained basis pursuit weighted by $1/\lvert a(0)\rvert$, at a logarithmic temperature — approached like $\varepsilon^{\mu}$, $\mu$ the certificate's slack, against $\varepsilon^{3/2}$ from zero amplitudes —; a $\pm$ pair per atom is the hyperbolic entropy; on positive kernel dictionaries with a unique nonnegative fit the loop selects the target's kernels at every mass scale, algebraically fast as the certificate's margin allows | (i), (iv)'s conjugate and (vi) exact; (ii) proved for every start; (iii) the limit and the off-support rate $\varepsilon^{\mu_j}$ at an $N$-atom vertex proved, the leading order computed and checked; (iv)'s hyperbolic entropy at leading order; (v) proved, the uniqueness certified by linear programming; numerics: (i)'s slope to $2.3 \cdot 10^{-9}$, the temperature $1/(\gamma a_0)$ to ten digits, $B(\bar{w}, w) \to \beta/(\gamma m)$ ($5 \cdot 10^{-10}$ at $\bar{\delta} = 10^{-12}$); on random $6 \times 12$ designs the flow at $t = 3 \cdot 10^7$ the Bregman projection onto the sign-constrained least-squares fits to $5.0 \cdot 10^{-10}$, where the frozen signs cannot fit $y$ the wrong-signed particles extinct (masses below $10^{-103}$) at the predicted rates to 0.5–10%, with duplicated atoms $1.6 \cdot 10^{-4}$–0.20 from the equal split; the rich limit from branches $1.1 \cdot 10^{-9}$ from the sign-constrained weighted basis pursuit at $\varepsilon = 10^{-64}$ ($\varepsilon^{\mu_j}$, the leading order to $3.4 \cdot 10^{-14}$), from $\alpha(0) = 0$ $4.3 \cdot 10^{-11}$ from basis pursuit at $\varepsilon = 10^{-10}$ ($\varepsilon^{3/2}$); one projection failed (a mixed start at $\varepsilon = 10^{-12}$, residual 1.95); on positive Gaussian dictionaries the target from every start scale $\beta(0) = 10^{-6}$ to 10 at the margin 0.015 (within $6.0 \cdot 10^{-5}$ at $t = 10^9$), still 1.0–$1.3 \cdot 10^{-3}$ away at the margin 0.003; over all three-atom supports of three grids the median margins $4.0 \cdot 10^{-6}$, $3.8 \cdot 10^{-9}$, $1.2 \cdot 10^{-3}$, the loop still 0.37–0.83 away at $t = 10^9$ where the margin is below $10^{-4}$; the real loop's $V$ monotone, the coefficients still 0.080 from $w^*$ after 4000 time units | Classical ([115]): exponentiated gradient and mirror descent's Bregman-projection bias (Kivinen–Warmuth; Gunasekar et al.), Hessian–Riemannian gradient flows on constrained domains (Alvarez–Bolte–Brahic), nonnegative least squares by overparametrised gradient flow (Chou–Maly–Verdun et al.), the $u^2 - v^2$ parametrisation and the hyperbolic entropy (Woodworth et al.; Ghai–Hazan–Singer; Vaškevičius et al.), nonnegative sparse uniqueness and super-resolution without separation (Donoho–Tanner; Bruckstein–Elad–Zibulevsky; Schiebinger–Robeva–Recht; Eftekhari et al.), the exponential penalty in linear programming (Cominetti–San Martín; Weed); new: the paradigm's law as an exponentiated gradient on its branches with the temperature read off the conservation law, the limit for every start (inexact fits, extinction) completing D‴.45(iii), $B(0,w) = \beta/(\gamma m)$, the two rich limits' temperatures, an atom's particles as an infimal convolution, the scale-independent selection on positive kernel dictionaries |

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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| D‴.48 | The rich transit: from small masses the free law's kernels enter one at a time. At $\alpha = 0$ the unit near its critical curve is an effective depth-3 unit ($w \approx 2c\alpha^3/(ma_{\max}^2)$), and as the masses $\varepsilon b_i \to 0$ the loop in the time $t/\Lambda$, $\Lambda \propto \varepsilon^{-1/2}$, converges to the inverse scale space flow of $\sum\lvert w_i\rvert/\sqrt{b_i}$ — the adaptive inverse scale space path: piecewise-linear dual coordinates, kernels entering when their integrated correlation reaches its weight, ending at weighted basis pursuit; on branches (nonzero starting amplitudes) the same on the logarithmic scale $\log(1/\varepsilon)$, sign-constrained (a ± pair: Pesme–Flammarion's depth-2 limit). With the paradigm's moving centres (the mass in the position law's drift) the kernels that have not entered stay where they started and the entered ones slide — a sliding inverse scale space that forms one kernel per well-separated feature at its centre (a tie — a feature midway between two kernels — shared by both; near ties and lingering kernels creep at the core's profile $(\pi a_{\max}^2/(4c))\varepsilon b^{3/2}$ $\cos(\pi\mu/2)/\sin^3(\pi\mu/2)\cdot s\langle r\partial_c\varphi\rangle$, frozen only while $\varepsilon \ll \mu^3$); with a Wasserstein–Fisher–Rao metric the massless kernels instead climb the correlation's peaks before entering and condense there — in the limit still one kernel per feature, but several at a finite $\varepsilon$, down to much smaller masses | (o) exact expansions; (i), (ii) proved (Mosco convergence of the explicit dual potentials, the semigroups' convergence), $u$'s rates numerical; (iii)(a) proved (the creep's profile to leading order), (iii)(b) formal (two time scales; ties, near ties), the sliding path numerical; (iv) the inactive kernels' limit derived to leading order and checked along the trajectory with unequal masses, the count as $\varepsilon \to 0$ numerical; (v) numerical; numerics: $u$'s distance to the path $\propto \varepsilon^{1/2}$ on three designs (log-log slopes 0.500 between $\varepsilon = 10^{-6}$ and $10^{-8}$; $6.7\cdot10^{-6}$, $8.8\cdot10^{-6}$, $2.0\cdot10^{-5}$ at $10^{-12}$), the end weighted basis pursuit to $1.1\cdot10^{-16}$–$6.1\cdot10^{-15}$ at $10^{-12}$; on branches $\propto 1/\log(1/\varepsilon)$ exactly (15.7, 8.0, 4.0, 2.0, 1.0 at $\varepsilon = 10^{-4}\ldots 10^{-64}$); with moving centres the sliding path fitting three bumps exactly with three kernels and the loop forming exactly these three for $\varepsilon \le 10^{-5}$ (8 at $10^{-3}$, 4 at $10^{-4}$), the others displaced $\propto \varepsilon$, on sixteen random targets 11, 13 and 16 at $\varepsilon = 10^{-5}$, $10^{-6}$, $10^{-8}$; under the Wasserstein–Fisher–Rao metric the centres on (iv)'s limit to $1.7\cdot10^{-5}$ at $\varepsilon = 10^{-4}$ and $2\cdot10^{-7}$ below $10^{-5}$, 15, 12, 8, 3, 3 kernels at $\varepsilon = 10^{-4}$, $10^{-5}$, $10^{-6}$, $10^{-8}$, $10^{-10}$, on the random targets exactly three on none at $10^{-6}$ and on all sixteen at $10^{-10}$; centres held on the grid 5 kernels, 0.019 from the target; a tie shared (0.4, 0.4), a near tie formed by two kernels at $\varepsilon = 10^{-6}$; the creep at the profile's speed to 0.990–0.996 ($\varepsilon = 10^{-8}$) and 0.9999–1.0000 ($10^{-10}$), a lingering kernel entering in place of the path's at $10^{-8}$; wider bumps along a staircase (21 entries on the path); with moving widths bumps of widths 0.2 and 0.08 formed by exactly three kernels at $\varepsilon = 10^{-6}$ | Classical ([116]): the inverse scale space flow and its adaptive algorithm (Burger et al.), linearised Bregman and differential inclusions (Yin–Osher et al.; Osher–Ruan–Xiong–Yao–Yin), saddle-to-saddle dynamics and incremental learning of diagonal networks identified with the adaptive inverse scale space method (Pesme–Flammarion; Berthier; Gissin et al.), inverse scale space limits of mirror flows by Mosco convergence (Berthier–Pillaud-Vivien), depth-$D$ potentials and polynomial scales (Woodworth et al.; Jacot et al.; Chou–Maly–Rauhut), sliding Frank–Wolfe and ADCG, conic particle descent (Chizat), condensation and early alignment at small initialisation (Luo et al.; Zhou et al.; Boursier et al.; Min–Mallada–Vidal; Kumar–Haupt), alternating gradient flows and unbalanced initialisations (Kunin et al.), deep diagonal networks as mirror flows (Labarrière et al.), feature quantisation at small initialisation (Maennel–Bousquet–Gelly); (i) follows Berthier–Pillaud-Vivien's route and (ii) is their entropic case; new: the unit as an effective depth-3 unit with an explicit potential and its transit scale $\Lambda \propto \varepsilon^{-1/2}$ — the scale set continuously by $\alpha(0)/\varepsilon^{1/2}$ —, the sliding transit under the paradigm's position law (the creep's profile, ties), and what the positions' metric decides: where the massless kernels go and how small the masses must be, not the limit's count |



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*Table 1, continued*

| result | statement | status | novelty (audit) |
|---|---|---|---|
| J | At finite $m$, without dither, in the scaling $\sigma = \sigma_0/k_e$: every hyperbolic critical point of the $m$-particle regularised loss $J_m$ has a unique nearby $T_0$-periodic orbit of the real closed loop, at distance $O(1/k_e)$, with the same stability type; the parameter Floquet exponents are $\frac{1}{k_e}\lambda(G_m^{1/2}\nabla^2 J_m G_m^{1/2}) + O(k_e^{-2})$; the basin of a minimum is uniform in $k_e$; the period mean is $\Theta^* - J^{-1}(b_{\text{frozen}} + b_{\text{osc}}) + O(k_e^{-2})$ | proved (tube and slaving, periodic Riccati block-triangularisation, second-order averaging, Poincaré map); checked by Newton shooting for the exact orbit, its monodromy matrix and its period mean against the averaged Hessian and the first-order prediction, at $k_e = 10$–$80$ | the first step of Section 26, branch 1; the tools (two-time-scale averaging under attraction) are classical [48, Ch. 10–11], [78, Ch. 5], [81, Ch. 4] |
| K | At a collapsed critical point (all kernels equal) the Hessian of $J_m$ splits by permutation symmetry into a separation block $\sigma_0 G^{-1} - R_2$, with $R_2$ the Hessian of the first-variation potential at the atom (independent of $m$), and a collective block; the collapsed configuration is a strict local minimum iff the atom minimises the single-kernel objective and $\sigma_0 > \sigma_{\text{crit}} := \lambda_{\max}(G^{1/2} R_2 G^{1/2})$. The orientation-odd $k_e^{-2}$ terms of the real loop ($U_{\text{odd}}$, $b_{\text{osc}}$ and the third derivative of $J_m$ against the first-order offset) shift the separation Floquet exponents of the collapsed orbit by $O(k_e^{-2})$, identically for the $(m-1)$-fold multiplet, and in a window of width $O(1/k_e)$ around $\sigma_{\text{crit}}$ the direction in which the reference is traversed decides whether the kernels separate | proved ((iii)–(iv) given the second-order normal form, Section 23); checked by shooting on instance A: the orientation difference of the softest separation exponent scales like $k_e^{-2.18}$ and matches the prediction from the averaged data to 3% at $k_e = 80$; at a leak gain computed from that prediction, one traversal leaves the collapsed configuration stable and the other makes it unstable ($k_e = 10$ and 20) | new as a design statement; the splitting is the isotypic decomposition of the permutation action, and the stability criterion is the local Chizat–Bach condition at a Dirac mass |
| J′ | At finite $m$ with dither $\tau = \tau_0/k_e$ and leak $\sigma = \sigma_0/k_e$: the closed loop has an invariant law (unique under a smoothness hypothesis) whose parameter marginal is the Gibbs measure $\exp(-J_m/\tau_0)$ up to $O(1/k_e)$, with the explicit first-order correction $\varepsilon \int \varphi g \, \mathrm{d}\pi$ driven by the first-order drift $b$ of Theorem J as a vector field on parameter space ($g$ solves $\tau_0 \nabla \cdot (\pi G \nabla g) = \nabla \cdot (\pi b)$); the stationary tracking error is $(2/k_e^2)\pi(\bar{L}_m) + O(k_e^{-3})$; the stationary parameter process is not reversible, with a circulation $\varepsilon[\int \Theta \wedge j_1 + \pi\langle u \wedge \dot{u}\rangle] + O(\varepsilon^2)$, where $j_1 = \pi(b - \tau_0 G \nabla g)$ is the first-order stationary current, nonzero exactly when $b$ is not a $G$-gradient (Proposition J′.2: the finite-$m$ form of Theorem I's reciprocity defect plus a Hessian commutator from second-order averaging; never a gradient for $m \geq 3$) | proved (the Poisson-equation expansion of the invariant measure with the fast error and phase eliminated; three standard theorems imported and named); checked on a single-kernel instance: the first-order prediction removes the $O(1/k_e)$ discrepancy between the real loop's stationary means and the Gibbs measure, the orientation-odd part of the means (the dither version of Remark H.7a) matches its prediction, and the current machinery is checked on the first-order averaged SDE | the technique is the Pavliotis–Stuart [70] / Mattingly–Stuart–Tretyakov [60] expansion; new in its object (a hypoelliptic learning loop with a plant) and in the identification of the first-order non-equilibrium current with the reciprocity defect of Theorem I |
| H.8 | ($\mathfrak{A}$) is input-to-state stable under drift perturbations, with gain $1/(\tau\rho_\tau)$ | proved, assuming the chain rule for the perturbed flow | elementary given [15] |

### 1.6 Where the new mathematics lies

The new mathematics this viewpoint requires lies in the open problems of Section 26.

1. Infinite-horizon averaging for McKean–Vlasov closed loops. Propositions H.7 and H.8 reduce it to controlling a near-identity averaging transformation along the moving measure. Theorem J settles the deterministic finite-$m$ case.
2. Uniform-in-time propagation of chaos with a plant in the loop.
3. The theory of the non-reciprocal, semi-gradient mean-field dynamics that the closed loop actually produces. Theorem I characterises its non-reciprocity (completely for $n = 1$ and at perfect fit; by an explicit bilinear form otherwise, and nowhere reciprocal on an open set); its long-time behaviour is open. Proposition K turns one of its effects into a design rule: near the critical leak, the orientation of the reference orbit decides whether a collapsed network separates.
4. Global convergence without dither.

### 1.7 Organisation

Section 2 relates this document to version 1: what version 1 claimed, the audit's verdict, the errata, what survives and what is withdrawn, and how Part II answers the structural defects of Section 1.2. Part I treats the projection-based controller: the setting and standing assumptions (Section 3), the algorithm (Section 4), the lemmas and the uniform-ultimate-boundedness theorem (Section 5), the "no bias" claim stated precisely (Section 6), prior art and the honest novelty statement (Section 7), and the experiments (Section 8). Part II develops transport adaptive control in Sections 9–23: the setting (Section 9), Theorems A–F together with Theorem C′ and Theorem D′ (Sections 10–17), Proposition G (Section 19), Theorem H (Section 20), the finite-gain structure of Propositions H.7 and H.8 and the reciprocity criterion of Theorem I (Section 21), Theorem J (Section 22) and Proposition K (Section 23). Part III collects the numerical evidence (Section 25) and the research program (Section 26). Appendix A is the verification map.

## 2 Relation to version 1

Version 1 of this document, *Mathematical Foundations of Self-Adaptive Kernels Achieving Lyapunov Convergence* (14 pages), presented the controller of Part I. An audit (`ASSESSMENT.md`) re-examined its theory, its code and its experiments; every numerical statement of the audit was regenerated by running the code, and the scripts and raw files are named next to each claim. A revised theory (`THEORY_REVISED.md`, Part I of this document) replaced its stability argument. This section records what version 1 claimed, the audit's verdict, the errata, what survives and what is withdrawn, and how Part II answers the structural defects of Section 1.2.

### 2.1 What version 1 claimed

Version 1 was subtitled *Uniform Ultimate Boundedness via Projection-Based Adaptive RBF Control Without Persistent Bias of Leaky Modification*. It treated the plant $\dot{x} = f(x,t) + u$ on a compact domain with the certainty-equivalence control $u = \dot{x}_m - k_e e - \hat{f}(x,t)$ and the network $\hat{f} = W^\top \phi(x, c, \sigma)$, adapted weights, centres and widths by projected gradient laws, and used the Lyapunov candidate $V = \frac{1}{2} e^\top e + \frac{1}{2}\operatorname{tr}(\tilde{W}^\top \Gamma_W^{-1} \tilde{W}) + \frac{1}{2}\tilde{c}^\top \Gamma_c^{-1} \tilde{c} + \frac{1}{2}\tilde{\sigma}^\top \Gamma_\sigma^{-1} \tilde{\sigma}$. Its claims were the following.

- *Central contribution* (abstract): "a rigorous proof that a fully shape-shifting kernel architecture — with online adaptation of weights, centers, and bandwidths — achieves Uniform Ultimate Boundedness (UUB) of the tracking error without the persistent steady-state bias introduced by classical leaky modification schemes ($\sigma$-modification, e-modification)".

- *Key mechanism* (abstract, §1.3): "the strict decoupling of offline Tikhonov regularization, which guarantees the well-posedness of the ideal parameter target, from the online smooth projection operator, which provides hard parameter boundedness", with an implicit-function-theorem (IFT) argument giving finite velocity bounds $V_W, V_c, V_\sigma$ for the ideal parameters.
- *A "critical and non-trivial" result* (abstract, §7.2): the projection's hard cap on parameter energy "converts the standard boundary-only Lyapunov descent argument into a global interior argument", proving convergence to the infimal invariant set without LaSalle's invariance principle or Barbalat's lemma, with the ultimate bound

$$\limsup_{t\to\infty}\|e(t)\| \le \sqrt{R_{\mathrm{UUB}}^2 + 2V_{\max,\Theta}}.$$

- Centre separation $\|c_i - c_j\| \ge \delta > 0$, "maintained via the projection set construction" (§1.2); and a corollary that a standard persistent-excitation condition on the regressor is insufficient to guarantee parameter convergence, so that alternative mechanisms, such as integral concurrent learning (ICL), are required (§8).
- §9 summarised five architectural choices and the failure mode each was said to close: $\sigma_{\min} > 0$ (Hessian blow-up near $\sigma \to 0$); offline Tikhonov regularisation (circular self-consistency of the projection geometry); a smooth convex projection operator (a cancellation gap when the projection is active); the projection's hard energy cap (the LaSalle and Barbalat requirements); and stability from the Lyapunov function and the projection alone (the circularity of ICL, which needs bounded states).
- §10 stated that "the theoretical core of this manuscript is mathematically complete and internally consistent", and that the remaining items were "completeness additions and empirical validation requirements ..., not theoretical repairs": a constructive feasibility theorem; empirical validation (Lorenz-driven tracking against $\sigma$- and e-modification "to prove bias elimination", a Monte Carlo sweep "verifying $R_{\mathrm{UUB}}$ bound is not violated", an ablation of fixed against adaptive centres); and a novelty framing in which the architecture "eliminates this tradeoff entirely", the projection introduces "zero bias into the Lyapunov derivative", and the result is "not a marginal improvement but a structural result enabled by the strict decoupling of offline regularization from online projection".

## 2.2 The audit's verdict

The theory is a correct but standard composition of known results (Taylor-linearised inner-layer adaptation, boundary-layer projection, level-set UUB), presented with language that overstates its novelty and its conclusion. Two of its claims are wrong as written (non-convex "projection" for centre separation; an IFT argument that is not a proof) and one was not implemented (the $\Gamma$-weighted projection). The experiments contained a hand-chosen "theoretical" threshold, a confounded headline comparison against baselines that failed representationally, GP baselines handicapped by stale predictions and a mis-placed inducing grid, invalid significance tests on autocorrelated samples, and the public figure omitted the one benchmark ($n = 3$) where the method is weak and the one baseline (NTK) that beats it. After revision, the implemented algorithm is exactly the proved one and is machine-checked; the honest experiments show a real, consistent advantage of projection over $\sigma$- and e-modification on identical geometry (5/5 seeds in every cell) and a modest but consistent gain from adaptive geometry, while the a-priori bound is shown to be 915× to 10 954× loose.

**Novelty.** Table 2 compares the ingredients of version 1 with prior work. Net: no new theorem. What can honestly be claimed after revision (Section 7) is a machine-checked theorem-to-code

Table 2: Novelty of the ingredients of version 1 (audit).

| ingredient | prior source | status of the claim of version 1 |
|---|---|---|
| Adaptive RBF control with projection-type robustification | Sanner & Slotine 1992 [79]; Ioannou & Sun 1996 [47] | known |
| Adapting centres/widths by first-order Taylor expansion with bounded remainder | Lewis, Yesildirek & Liu 1996 [54]; Polycarpou 1996 [75]; Fabri & Kadirkamanathan 1996 [28]; Ge & Wang 2002 [33] | known; version 1 cites none of these |
| Smooth projection with $\varepsilon$ boundary layer and inequality (P) | Pomet & Praly 1992 [76]; Lavretsky & Gibson 2011 [52]; Lavretsky & Wise 2013 [53] | known; the showcase README called it "the core theoretical contribution" |
| Projection introduces no bias in $\dot{V}$ when $\theta^* \in \Omega$ | Ioannou & Sun 1996 [47, §4.4] | known |
| Level-set ("global interior") UUB argument | Khalil [48, Thm. 4.18]; Lewis et al. 1999 [55] | known; called "critical and non-trivial" |
| Simultaneous adaptation of $W$, $c$, $\sigma$ for Gaussian RBFs with all three projected | — | the only combinational novelty; minor |

map, an explicit Hessian-free disturbance bound, the RMS form of the bound, and a controlled mechanism-versus-geometry experiment.

**Integrity findings.** Table 3 lists the integrity findings of the audit, each verified by running code, with the evidence and the fix. File names refer to the accompanying repository.

**Recommendation.** The audit recommended that version 1 not be submitted in its form, but re-framed as an empirical-plus-verified-implementation paper: the factorial result is a clean, reproducible finding, and the theory should be the revised one (Sections 3–6) with the prior work cited.

### 2.3 Errata to version 1

Table 4 lists the errata to version 1. Section numbers in the second column refer to version 1; "README" refers to the public showcase README that accompanied it.

Table 4: Errata to version 1.

Table 3: Integrity findings of the audit of version 1 and its code (each verified by running code).

| # | finding | evidence | fix |
|---|---|---|---|
| I1 | Implemented centre update $\neq$ proved update: a pairwise repulsion term broke the projection inequality the proof needs | `tests/test_theorem_to_code.py` (oracle plant): inequality violated at 6054 of 20000 samples with the legacy term, 0 of 20000 without; the term also failed its own purpose (min separation 0.07 vs $\delta = 0.5$) | removed from the algorithm (`enforce_separation=False` default); tracking unchanged on the headline run (mean steady-state error $0.0767 \to 0.0774$, max $0.46 \to 0.40$) |
| I2 | Projection applied the isotropic formula to any $\Gamma$; the proof needs the $\Gamma$-weighted operator | `tests/test_controller_paths.py` | $\Gamma$-weighted projection implemented; fast/dense paths agree to $10^{-12}$ |
| I3 | "$R_{\mathrm{UUB}}$ threshold" in the Monte Carlo figure was the constant 0.5 | `tier1_monte_carlo.py` line 69: "A chosen arbitrary bound" | `exp_monte_carlo_honest.py`: only exactly checkable statements (200/200 hard-bound and ISS checks) |
| I4 | "Theoretical $R_{\mathrm{UUB}}$" in the $k_e$ sweep used invented constants and an underived formula | `tier2_ruub_sweep.py`, former `estimate_D_max` | `exp_oracle_ruub.py`: all constants computable; a-priori bound $915\times$–$10\,954\times$ loose; RMS bound with measured disturbance holds 48/48 |
| I5 | Headline figure: fixed-grid baselines with a grid not covering the reference; their error ($\approx 15$) exceeded open loop, so the figure showed representational failure, not "bias" | `results/tier1_lorenz_tracking.png` vs ablation figure (projection $W$-only fails identically) | `exp_bias_factorial.py`: 3 mechanisms $\times$ 3 geometries, covering grid, 5 seeds |
| I6 | Wilcoxon tests on $\sim$12 000 time samples with lag-1 autocorrelation 0.9999; the README's "$p > 0.3$ vs Exact GP" is contradicted by its own data ($p = 1.6 \times 10^{-11}$) | recomputed from `results/*.npy` | seed-level statistics only |
| I7 | $n = 3$ benchmark and NTK baseline removed from the public figure (commits "exclude n=3 and NTK") | `results/n3/*.npy`, `results/*/ntk_*.npy`; `plot_comparison_hd_complete.py` | complete table regenerated: proposed ranks 3/5, 2/5, 3/5, 2/5 |
| I8 | GP baselines held a stale prediction for 50–100 steps; sparse GP inducing points on a diagonal line over $x \in [-3, 3]$ while the reference spans $\pm 20$ | `sim1`–`sim3` | `kernel_baselines_fair.py`: per-step prediction, $9 \times 3$ covering grid, seeds |
| I9 | "$O(mn)$ per step" claim false with the $O(m^2 n)$ repulsion | `projection.py` | true after I1 |
| I10 | "Concurrent Learning" hook used the current residual only | former `controller.py` | removed |
| I11 | Lyapunov monitor requires $\theta^*$; never used by any experiment, so "Lyapunov descent verification" was never run | `integrator.py` | oracle module and tests perform it on a representable plant |

| # | section | statement in version 1 | status |
|---|---|---|---|
| E1 | 1.2, 9 | Centre separation $\|c_i - c_j\| \geq \delta$ "maintained via the projection set construction" with "convex projection boundaries" | **Wrong.** The separation set is non-convex; no projection onto it has property (P). The implemented repulsion term violated (P) in 30% of samples and did not keep the separation (min distance 0.07 vs $\delta = 0.5$). Removed from the algorithm; not needed by the proof. |
| E2 | 1.3 | Offline Tikhonov regularisation of the ideal target: "This strict separation eliminates the circularity" | **Irrelevant to the proof.** Optimality of $\theta^*$ is never used (Section 3, remarks on the assumptions). |
| E3 | 1.3 | IFT gives finite velocity bounds $V_W$, $V_c$, $V_\sigma$ | **Not a proof.** The Hessian of a Tikhonov-regularised non-convex loss at a minimiser is PSD, not necessarily PD; the IFT hypothesis is asserted, not established. Replaced by Assumption A4. |
| E4 | 2.1 | $D_{\max}$ from the Taylor remainder, "a finite geometric scalar" | **Valid but not explicit.** Replaced by the explicit, Hessian-free Lemma 2. The code's former `estimate_D_max` $= \varepsilon(1 + 2W_{\max}c_{\max})$ had no derivation and is deleted. |
| E5 | 4 | $\mathrm{tr}(\tilde{W}^\top \Gamma_W^{-1} \tilde{W})$ with $\Gamma_W$ "positive-definite learning rate matrices" | **Type error** for $\Gamma_W \in \mathbb{R}^{mn \times mn}$; the correct form is $\mathrm{vec}(\tilde{W})^\top \Gamma_W^{-1} \mathrm{vec}(\tilde{W})$. Harmless for scalar $\Gamma$. |
| E6 | 5 | Projection inequality "holds as an absolute mathematical certainty — the cancellation gap does not exist" | **True only under A3** ($\theta^*$ in the inner set) and only for the $\Gamma$-weighted operator; the previous code applied the isotropic formula to any $\Gamma$. Fixed and tested. |
| E7 | 7.2 | "A critical and non-trivial consequence of the projection operator"; its hard energy cap "converts what would be a boundary-only argument … into a global interior argument" | **Standard** level-set / comparison argument (Khalil [48, Thm. 4.18]). Correct, not new; also "strict monotone descent through every intermediate level set until the infimal set is penetrated" overstates: entry into every level above $c_{\inf}$ is finite-time, entry into $c_{\inf}$ itself only asymptotic. |
| E8 | abstract, 7.3, 10.3 | Projection gives UUB "without the persistent steady-state bias" (abstract); "no modification bias is introduced" (7.3); "zero bias into the Lyapunov derivative", "not a marginal improvement but a structural result" (10.3) | **Overclaimed.** The ultimate bound $\sqrt{R_{\mathrm{UUB}}^2 + 2V_{\max,\Theta}}$ contains the parameter-energy term; see Section 6. The tighter, bias-free statement is the RMS bound (iii). |
| E9 | 8 | "a standard Persistent Excitation (PE) condition on the regressor $\phi$ is insufficient to guarantee $\tilde{\Theta}(t) \to 0$"; convergence "requires alternative mechanisms", such as ICL | Parameter convergence is not treated; the remark is speculation and the code's former "Concurrent Learning" hook used only the current residual (not recorded data) and is removed. |
| E10 | README | "100% of the independent simulation trials converged strictly below the calculated $R_{\mathrm{UUB}}$ threshold" | **False.** The threshold 0.5 was a hand-chosen constant (`tier1_monte_carlo.py`: "A chosen arbitrary bound"). Replaced by `exp_monte_carlo_honest.py` / `exp_oracle_ruub.py`. |
| E11 | README | "$O(mn)$ time" | **Was false** with the $O(m^2 n)$ pairwise repulsion; true now that it is off by default. |

## 2.4 What survives and what is withdrawn

**What survives.**

- *The corrected theorem.* Theorem 1 of Section 5 holds under the explicit Assumptions A1–A5, for an algorithm identical to the implemented one. Its pointwise ultimate bound (ii) is the conclusion of version 1; it is the textbook level-set argument, and its radius contains the parameter-energy term $V_{\max,\Theta}$. Its RMS bound (iii) does not contain that term and is the form of the result that can be confronted with data; its reference bound (iv) holds whatever the adaptation does. The ingredients are the projection property (P) (Lemma 1), the explicit Hessian-free disturbance bound (Lemma 2) and the Lyapunov derivative (Lemma 3), verified numerically along simulated trajectories. Nothing in the

stability theorem is new (Section 7). The "no bias" claim survives only in the precise form of Section 6: projection removes the *Lyapunov-derivative* bias of leakage, not the ultimate *tracking* bound's dependence on the parameter set.

- *The projection-versus-leakage finding* (Section 8). On identical geometry, projection beats both $\sigma$- and e-modification in every cell of a mechanism-by-geometry factorial, 5/5 seeds, with paired confidence intervals excluding zero. Within projection, adaptive centres improve on the fixed grid by 0.012 (5/5) and adaptive widths by a further 0.035 (5/5). So the method's real advantage is the robustness mechanism, with geometry adaptation a smaller second-order gain, close to the reverse of the story told by the original figure.
- *The comparison with kernel regression, with its limits* (Section 8). Against kernel-regression baselines the method is never the most accurate. Its edge is cost: it is cheaper per step than every GP baseline at every $n$ and than NTK except at $n = 3$, and in the fair $n = 1$ rerun, with per-step prediction for every method, it has the cheapest step. Every kernel-regression baseline (GP and NTK, including the fair $n = 1$ rerun) is trained on noise-free samples of the true $f(x, t)$, which is equivalent to measuring the state derivative exactly, while the adaptive law uses only the tracking error.

**What is withdrawn.**

- Earlier versions of the public README stated that the method's steady-state error was "statistically indistinguishable (Wilcoxon signed-rank $p > 0.3$) from both the Exact GP and the Sparse GP", that "100% of the independent simulation trials converged strictly below the calculated $R_{UUB}$ threshold", and announced "Curse of Dimensionality Solved". Those statements are withdrawn: the significance test was invalid, the threshold was an arbitrary constant, and the omitted $n = 3$ and NTK results contradict them.
- The "zenith" language is dropped: "absolute mathematical certainty", "curse of dimensionality solved", "structural result". None of it survives the data. Claims of a "structural" advantage over leaky modifications are withdrawn.
- The claim that centre separation is "maintained via the projection set construction" is withdrawn (erratum E1); no centre-separation constraint is needed by the proof.

### 2.5 How Part II answers the structural defects

Section 1.2 named two structural defects of the classical labelled analysis, that of version 1 and of the literature it builds on: its Lyapunov analyses depend on labels, and it has no object that describes where the kernels go or whether their placement is good. Part II addresses both by working with the measure $\mu = \frac{1}{m}\sum_i \delta_{\vartheta_i}$, which does not see the labels, and with its transport.

- *Labels.* The Lyapunov function of Theorem B, $V_{\mathrm{OT}} = \frac{1}{2}\|e\|^2 + \frac{1}{2}W(\mu_t, \mu_t^*)^2$, measures the parameter error by the optimal-assignment (Wasserstein) distance $W$, a minimum over all assignments of kernels to ideal atoms (Section 9). The labelled $\sigma$-modification is not a functional of $\mu$; Theorem C replaces it by an optimal-transport $\sigma$-modification, Theorem C$'$ extends the OT leak to priors with a different number of atoms through the barycentric projection of the optimal plan, and Proposition C.5 shows that the labelled leak reshuffles kernels. Theorem E gives label-free local exponential convergence under tangential persistent excitation, with explicit radius, overshoot and rate (Lemma E.0).
- *Where the kernels go.* The measure $\mu_t$ is that object. By Theorem A the adaptive law is the particle form of the Wasserstein gradient flow of a $\mu$-linear potential; Theorems D and D$'$ treat the mean-field (many-kernel) loop, without and with dither; Theorem F bounds the Wasserstein length of learning by the integrated tracking error; Proposition G describes the slow geometry (formal). Whether the placement is good is the subject of

Theorem H: with saturated features, $\sigma$-modification and dither, the averaged mean-field loop converges globally and exponentially to the regularised optimum. This is proved for $(\mathfrak{A})$; the reduction to $(\mathfrak{A})$ is formal, and the real loop is Conjecture H.6, which Propositions H.7 and H.8 reduce to an averaging estimate. Theorem J proves its deterministic finite-$m$ case, Theorem I characterises the non-reciprocity of the averaged loop at finite gain, and Proposition K turns the orientation of the reference orbit into a design input for collapsed configurations.

- *Constants growing with $m$.* The third apparent defect is an artefact of the normalisation; the mean-field normalisation removes it for labelled and label-free analyses alike (Remark A.1).

The status of each of these results is that of Table 1.

# Part I
# The projection-based controller

## 3 Setting and standing assumptions

This part replaces the stability argument of version 1 with a statement whose assumptions are explicit, whose algorithm is exactly the one in `src/core/` of the accompanying repository, and whose every step is either proved here or verified numerically by `tests/test_theorem_to_code.py`. The errata to version 1 are listed in Section 2; Section 7 positions the result against prior work.

**Notation.** $\|\cdot\|$ is the Euclidean norm for vectors and the Frobenius norm for matrices; $\mathrm{vec}(\cdot)$ is the row-major vectorisation used throughout the code; there are $m$ kernels and $n$ states.

**Plant, reference and error.**

$$\dot{x} = f(x,t) + u, \qquad e = x - x_m(t), \qquad u = \dot{x}_m - k_e e - \hat{f}(x), \tag{3.1}$$

$$\hat{f}(x) = W^\top \varphi(x, c, \sigma), \qquad \varphi_i(x) = \exp\Big(-\frac{\|x - c_i\|^2}{2\sigma_i^2}\Big). \tag{3.2}$$

**Assumption A1** (operation on a compact set)**.** There is a compact set $X \subset \mathbb{R}^n$ with $x(t) \in X$ for all $t \geq 0$, and $F_{\max} := \sup_{x \in X,\, t \geq 0} \|f(x,t)\| < \infty$. *(Version 1 assumes "a compact spatial domain $X$" in the same way. For globally bounded $f$ the assumption is discharged by Theorem 1(iv); for the Lorenz plant it is a genuine assumption.)*

**Assumption A2** (reference)**.** $x_m$ is $C^1$ with $x_m$, $\dot{x}_m$ bounded, and $\dot{x}_m$ is available to the controller.

**Assumption A3** (admissible ideal parameters)**.** There is a curve $\theta^*(t) = (W^*(t), c^*(t), \sigma^*(t))$ lying for all $t$ in the *inner* projection sets

$$\|W^*\| \leq W_{\max} - \varepsilon, \qquad \|c^*\| \leq c_{\max} - \varepsilon, \qquad \sigma_{\min} + \varepsilon \leq \sigma_i^* \leq \sigma_{\max} - \varepsilon.$$

**Assumption A4** (ideal-parameter regularity)**.** $\theta^*(\cdot)$ is $C^1$ with $\|\dot{W}^*\| \leq V_W$, $\|\dot{c}^*\| \leq V_c$, $\|\dot{\sigma}^*\| \leq V_\sigma$. (For a time-invariant plant one may take $\theta^*$ constant, $V_W = V_c = V_\sigma = 0$.)

**Assumption A5** (gains and geometry)**.** $k_e > 0$; $\Gamma_W$, $\Gamma_c$ symmetric positive definite on $\mathbb{R}^{mn}$ (a zero $\Gamma$ freezes that group, see Remark 3); $\Gamma_\sigma$ diagonal positive; $0 < \varepsilon < \min\big(W_{\max}, c_{\max}, (\sigma_{\max} - \sigma_{\min})/2\big)$; $\sigma_{\min} > 0$.

**Remarks on the assumptions.**

- **The proof never uses optimality of $\theta^*$.** $\theta^*$ is an analytic device; any admissible $C^1$ curve works, and the approximation quality of the network enters the result only through the *actual* size of the disturbance $d(t)$ defined below, not through the bound. The Tikhonov-regularised definition of $\theta^*$ in version 1 and its implicit-function-theorem (IFT) argument for A4 are therefore unnecessary — and, as written, not a proof (erratum E3, Section 2).
- **No centre-separation constraint is needed.** The set $\{c : \|c_i - c_j\| \geq \delta \ \forall\, i \neq j\}$ is not convex, so no projection onto it has property (P) below; the claim of version 1 that separation is "maintained via the projection set construction" is withdrawn (erratum E1, Section 2). The bounded-Jacobian property that the proof needs follows from $\sigma \geq \sigma_{\min}$ alone.

# 4 The algorithm

The algorithm is identical to the one implemented in `src/core/controller.py`:

$$\begin{aligned} \dot{W} &= \mathrm{Proj}_W\big(W,\ \Gamma_W\,\varphi e^\top\big), \\ \dot{c} &= \mathrm{Proj}_c\big(c,\ \Gamma_c\,\varphi_c'^\top W e\big), \\ \dot{\sigma} &= \mathrm{Proj}_\sigma\big(\sigma,\ \Gamma_\sigma\,\varphi_\sigma'^\top W e\big), \end{aligned} \tag{4.1}$$

with $\varphi_c' = \partial\varphi/\partial\,\mathrm{vec}(c) \in \mathbb{R}^{m\times mn}$ (row $i$ has the single non-zero block $\varphi_i(x-c_i)^\top/\sigma_i^2$) and $\varphi_\sigma' = \mathrm{diag}\big(\varphi_i\|x-c_i\|^2/\sigma_i^3\big)$.

**Norm-ball projection ($W$ and $c$), boundary layer of width $\varepsilon$.**

$$\mathrm{Proj}(\theta,\Gamma y) = \begin{cases} \Gamma y - h(\theta)\,\Gamma\theta\,\dfrac{\theta^\top\Gamma y}{\theta^\top\Gamma\theta} & \text{if } h(\theta)>0 \text{ and } \theta^\top\Gamma y>0, \\ \Gamma y & \text{otherwise,} \end{cases} \tag{4.2}$$

$$h(\theta) = \begin{cases} 0 & \text{for } \|\theta\| \le \theta_{\max}-\varepsilon, \\ \big((\|\theta\|-\theta_{\max}+\varepsilon)/\varepsilon\big)^2 & \text{in between,} \\ 1 & \text{for } \|\theta\| \ge \theta_{\max}. \end{cases} \tag{4.3}$$

**Box projection ($\sigma$, component-wise).** The update component is multiplied by $\mathrm{clip}\big((\sigma_i-\sigma_{\min})/\varepsilon, 0, 1\big)$ when it points below the lower face and by $\mathrm{clip}\big((\sigma_{\max}-\sigma_i)/\varepsilon, 0, 1\big)$ when it points above the upper face.

**Discrete time.** The simulations integrate with fixed-step RK4 and re-clip the parameters to their sets after every stage (`src/core/integrator.py`). The theorem is a continuous-time statement; the clip is a safety net that is inactive if the step is small enough (property (P) already removes the outward component on the boundary).

# 5 Lemmas and the boundedness theorem

## 5.1 Lemmas

**Lemma 1** (projection property (P))**.** *Under A3 and A5, for every $\theta$ in its set and every $y$,*

$$\tilde\theta^\top\Gamma^{-1}\,\mathrm{Proj}(\theta,\Gamma y) \le \tilde\theta^\top y, \qquad \tilde\theta := \theta-\theta^*. \tag{P}$$

*Proof.* If the projection is inactive the two sides are equal. Otherwise

$$\tilde\theta^\top\Gamma^{-1}\big(\mathrm{Proj}\ -\Gamma y\big) = -h\,\frac{\theta^\top\Gamma y}{\theta^\top\Gamma\theta}\,\tilde\theta^\top\theta,$$

with $h>0$ and $\theta^\top\Gamma y>0$, and $\tilde\theta^\top\theta = \|\theta\|^2 - \theta^{*\top}\theta \ge \|\theta\|\big(\|\theta\|-\|\theta^*\|\big) \ge 0$ because $h>0$ forces $\|\theta\| \ge \theta_{\max}-\varepsilon \ge \|\theta^*\|$. For the box, component $i$, with $s_i$ the clip factor applied to that component: $\tilde\sigma_i(s_i-1)y_i \le 0$ because $s_i \le 1$, and on the lower face $y_i<0$ while $\tilde\sigma_i = \sigma_i-\sigma_i^* \le \sigma_{\min}+\varepsilon-\sigma_i^* \le 0$ (symmetrically on the upper face). □

**Status.** Proved.

Checked by: `tests/test_projection.py` (randomised; isotropic and non-isotropic $\Gamma$; vector, matrix and box), together with the counterexample showing that (P) fails when $\theta^*$ is outside the inner set.

**Lemma 2** (lumped disturbance)**.** *Define*

$$d := f - \hat{f} + \tilde{W}^\top \varphi + W^\top \varphi_c' \,\tilde{c} + W^\top \varphi_\sigma' \,\tilde{\sigma}. \tag{5.1}$$

*Then identically*

$$\dot{e} = -k_e e - \tilde{W}^\top \varphi - W^\top \varphi_c' \,\tilde{c} - W^\top \varphi_\sigma' \,\tilde{\sigma} + d, \tag{5.2}$$

*and under A1, A3,*

$$\|d\| \le D_{\max} := F_{\max} + 3W_{\max}\sqrt{m} + W_{\max}\big(2c_{\max}J_c + \sqrt{m}\,(\sigma_{\max} - \sigma_{\min})J_\sigma\big), \tag{5.3}$$
$$J_c = \mathrm{e}^{-1/2}/\sigma_{\min}, \qquad J_\sigma = 2\mathrm{e}^{-1}/\sigma_{\min}.$$

*Proof.* The identity is the definition of $d$ substituted into $\dot{e} = -k_e e + f - \hat{f}$. For the bound: $\|f\| \le F_{\max}$; $\|\hat{f}\| \le \|W\|\|\varphi\| \le W_{\max}\sqrt{m}$ $(0 < \varphi_i \le 1)$; $\|\tilde{W}\| \le 2W_{\max}$; $\|\tilde{c}\| \le 2c_{\max}$; $\|\tilde{\sigma}\| \le \sqrt{m}\,(\sigma_{\max} - \sigma_{\min})$; the rows of $\varphi_c'$ have disjoint supports, so

$$\|\varphi_c'\|_2 = \max_i \frac{\varphi_i \|x - c_i\|}{\sigma_i^2} \le \sup_{r \ge 0} \frac{r\,\mathrm{e}^{-r^2/(2\sigma^2)}}{\sigma^2} = \frac{\mathrm{e}^{-1/2}}{\sigma} \le J_c,$$

and $\|\varphi_\sigma'\|_2 = \max_i \varphi_i \|x - c_i\|^2/\sigma_i^3 \le 2\mathrm{e}^{-1}/\sigma \le J_\sigma$. □

**Status.** Proved.

Checked by: `tests/test_rbf_kernel.py` checks the two suprema by brute force.

**Remark 1** (local form)**.** Writing $\varphi^* = \varphi(x; c^*, \sigma^*)$ and expanding to first order in $(\tilde{c}, \tilde{\sigma})$ gives

$$d = \varepsilon + W^{*\top} O_{\mathrm{ho}} + \tilde{W}^\top \varphi_c' \,\tilde{c} + \tilde{W}^\top \varphi_\sigma' \,\tilde{\sigma}, \qquad \varepsilon := f - W^{*\top}\varphi^*,$$

with $O_{\mathrm{ho}}$ the Taylor remainder — the decomposition of version 1. Near convergence ($\tilde{\theta} \to 0$) this shows $\|d\| \lesssim \|\varepsilon\| + O(\|\tilde{\theta}\|^2)$, which is why the controller performs far better than $D_{\max}$ suggests; but it does not yield a smaller global constant than Lemma 2, which needs no Hessian and no Taylor expansion.

**Lemma 3** (Lyapunov derivative)**.** *Let*

$$V = \tfrac{1}{2} e^\top e + \tfrac{1}{2}\,\mathrm{vec}(\tilde{W})^\top \Gamma_W^{-1}\,\mathrm{vec}(\tilde{W}) + \tfrac{1}{2}\tilde{c}^\top \Gamma_c^{-1}\tilde{c} + \tfrac{1}{2}\tilde{\sigma}^\top \Gamma_\sigma^{-1}\tilde{\sigma}.$$

*Along trajectories of the algorithm,*

$$\begin{aligned}
\dot{V} &= -k_e\|e\|^2 + e^\top d + \mathrm{cross} - \Omega^*, \\
\mathrm{cross} &= \langle \tilde{W},\, \Gamma_W^{-1}\dot{W} - \varphi e^\top \rangle + \tilde{c}^\top\big(\Gamma_c^{-1}\dot{c} - \varphi_c'^\top W e\big) + \tilde{\sigma}^\top\big(\Gamma_\sigma^{-1}\dot{\sigma} - \varphi_\sigma'^\top W e\big) \le 0, \\
\Omega^* &= \langle \tilde{W},\, \Gamma_W^{-1}\dot{W}^* \rangle + \tilde{c}^\top \Gamma_c^{-1}\dot{c}^* + \tilde{\sigma}^\top \Gamma_\sigma^{-1}\dot{\sigma}^*, \qquad |\Omega^*| \le C_\Omega, \\
C_\Omega &:= 2W_{\max}\|\Gamma_W^{-1}\|_2 V_W + 2c_{\max}\|\Gamma_c^{-1}\|_2 V_c + \sqrt{m}\,(\sigma_{\max} - \sigma_{\min})\|\Gamma_\sigma^{-1}\|_2 V_\sigma.
\end{aligned} \tag{5.4}$$

*Proof.* Differentiate $V$, substitute the identity of Lemma 2 for $\dot{e}$, and use $e^\top \tilde{W}^\top \varphi = \langle \tilde{W},\, \varphi e^\top \rangle$, $e^\top W^\top \varphi_c' \,\tilde{c} = \tilde{c}^\top \varphi_c'^\top W e$, $e^\top W^\top \varphi_\sigma' \,\tilde{\sigma} = \tilde{\sigma}^\top \varphi_\sigma'^\top W e$. That cross $\le 0$ is Lemma 1 applied to the three groups; the bound on $\Omega^*$ is Cauchy–Schwarz with A3, A4. □

**Status.** Proved.

Checked by: `tests/test_theorem_to_code.py`: the identity (to machine precision) and cross $\le 0$ (at every sample) are verified along simulated trajectories, on a plant that is exactly representable with known constant $\theta^*$.

### 5.2 Theorem 1: uniform ultimate boundedness

**Theorem 1** (uniform ultimate boundedness)**.** *Assume A1–A5, let*

$$R_{\mathrm{UUB}} := \frac{D_{\max}}{2k_e} + \sqrt{\frac{D_{\max}^2}{4k_e^2} + \frac{C_\Omega}{k_e}},$$

*and let $V_{\max,\Theta}$ be the supremum over the projection sets of the parameter part of $V$.*

*(i)* ***Descent outside a ball.*** *$\dot V \le -k_e\|e\|^2 + D_{\max}\|e\| + C_\Omega$, hence $\dot V < 0$ whenever $\|e\| > R_{\mathrm{UUB}}$.*

*(ii)* ***Pointwise ultimate bound.*** *Every solution satisfies $V(t) \le \max\{V(0), c_{\inf}\}$ with $c_{\inf} := \frac{1}{2}R_{\mathrm{UUB}}^2 + V_{\max,\Theta}$, every level $c > c_{\inf}$ is reached in finite time, and*

$$\limsup_{t\to\infty} \|e(t)\| \le \sqrt{R_{\mathrm{UUB}}^2 + 2V_{\max,\Theta}}.$$

*(iii)* ***RMS bound (the quantitatively meaningful one).*** *For every $T_1 < T_2$,*

$$\mathrm{RMS}_{[T_1,T_2]}(\|e\|) \le \frac{D_{\max}}{2k_e} + \sqrt{\frac{D_{\max}^2}{4k_e^2} + \frac{C_\Omega}{k_e} + \frac{V(T_1)}{k_e(T_2-T_1)}},$$

*so $\limsup_{T_2\to\infty} \mathrm{RMS}_{[T_1,T_2]}(\|e\|) \le R_{\mathrm{UUB}}$ — without the parameter-energy term of (ii). The same inequality holds with $D_{\max}$ and $C_\Omega$ replaced by the trajectory-wise suprema of $\|d(t)\|$ and $|\Omega^*(t)|$ on $[T_1, T_2]$.*

*(iv)* ***Reference bound (adaptation-free).*** *Under A1 and the hard parameter bounds alone, $\dot e = -k_e e + (f - \hat f)$ with $\|f - \hat f\| \le F_{\max} + W_{\max}\sqrt{m} =: \tilde F$, so*

$$\|e(t)\| \le \mathrm{e}^{-k_e t}\|e(0)\| + \tilde F/k_e \qquad \text{for all } t,$$

*whatever the adaptation does.*

*Proof.* (i) Lemma 3 with Cauchy–Schwarz on $e^\top d$.

(ii) If $V > c_{\inf}$ then, since the parameter part of $V$ is at most $V_{\max,\Theta}$, $\frac{1}{2}\|e\|^2 > \frac{1}{2}R_{\mathrm{UUB}}^2$, so $\|e\| > R_{\mathrm{UUB}}$ and $\dot V < 0$; more precisely, with $\Delta := V - c_{\inf} > 0$ one has $\|e\|^2 \ge R_{\mathrm{UUB}}^2 + 2\Delta$ and therefore

$$\dot V \le -g\Big(\sqrt{R_{\mathrm{UUB}}^2 + 2\Delta}\Big) =: -\psi(\Delta), \qquad g(r) = k_e r^2 - D_{\max} r - C_\Omega,$$

with $g$ increasing for $r \ge R_{\mathrm{UUB}}$ and $\psi$ of class $\mathcal{K}$. The comparison lemma gives $V(t) \le \max\{V(0), c_{\inf}\}$, finite-time entry into every sublevel set $\{V \le c\}$ with $c > c_{\inf}$, and $\limsup V \le c_{\inf}$, i.e. the stated bound on $\|e\|$ ($\|e\|^2 \le 2V$).

(iii) Integrate (i):

$$\begin{aligned} k_e \int_{T_1}^{T_2} \|e\|^2 \,\mathrm{d}t &\le V(T_1) - V(T_2) + D_{\max}\int_{T_1}^{T_2} \|e\|\,\mathrm{d}t + C_\Omega(T_2 - T_1) \\ &\le V(T_1) + D_{\max}\sqrt{T_2 - T_1}\,\sqrt{\int_{T_1}^{T_2} \|e\|^2\,\mathrm{d}t} + C_\Omega(T_2 - T_1) \end{aligned}$$

by Cauchy–Schwarz; with $S := \mathrm{RMS}_{[T_1,T_2]}(\|e\|)$ this reads

$$k_e S^2 - D_{\max} S - \big(C_\Omega + V(T_1)/(T_2 - T_1)\big) \le 0,$$

and $S$ is below the positive root.

(iv) Variation of constants. □

**Status.** Proved.

**What (ii)–(iv) say together.** Part (ii) is the conclusion of version 1; it is the textbook level-set argument (Khalil [48, Thm. 4.18]; Lewis et al. [55]) and its radius contains the parameter-energy reservoir $V_{\max,\Theta}$ — a bias-like term — so it does **not** establish that projection yields a tighter tracking bound than $\sigma$-modification. Part (iii) is tighter and is the statement that can be checked without unknown constants: replace $D_{\max}$ by the measured $\sup\|d\|$, see `experiments/exp_oracle_ruub.py`, where it is satisfied in 48/48 runs (Section 8). Part (iv) shows that mere boundedness of $e$ is already a consequence of projection plus a stable error channel, independent of the adaptation law; the content of adaptive control is in making the *actual* $d(t)$ small, which no a-priori constant captures. In the oracle experiments the a-priori $R_{\mathrm{UUB}}$ exceeds the observed RMS error by factors of 915 to 10954.

**Remark 2** (no LaSalle / Barbalat)**.** Version 1's claim that the argument needs neither LaSalle's invariance principle nor Barbalat's lemma is correct, and standard: UUB statements never need them. Nothing here claims asymptotic convergence of $e$ or of $\tilde{\theta}$.

**Remark 3** (frozen groups)**.** If $\Gamma_c = 0$ (fixed centres), drop the $\tilde{c}$ term from $V$; the identity of Lemma 2 still holds with $-W^\top \varphi_c' \tilde{c}$ absorbed into $d$, with the same bound. Likewise for $\sigma$.

**Remark 4** (what is not proved)**.** Parameter convergence (it requires persistent excitation of a nonlinear-in-parameter regressor — not addressed); exact tracking; any statement about the size of the ideal approximation error $\varepsilon$.

## 6 The no-bias claim, stated precisely

With $\sigma$-modification,

$$\dot{V} \le -k_e\|e\|^2 - \tfrac{1}{2}\sigma_{\mathrm{mod}}\|\tilde{\theta}\|_\Gamma^2 + \tfrac{1}{2}\sigma_{\mathrm{mod}}\|\theta^*\|_\Gamma^2 + D\|e\| + C_\Omega:$$

the leak adds a positive constant and removes the parameter-energy reservoir, which gives exponential convergence of $V$ to a ball whose radius grows with $\sigma_{\mathrm{mod}}\|\theta^*\|^2$. With projection,

$$\dot{V} \le -k_e\|e\|^2 + D\|e\| + C_\Omega:$$

no positive constant is added, but nothing dissipates parameter energy, which is why the pointwise bound (ii) of Theorem 1 keeps $V_{\max,\Theta}$. Projection therefore removes the *Lyapunov-derivative* bias of leakage (true, and known since Pomet–Praly [76] and Ioannou–Sun [47]), not the dependence of the ultimate *tracking* bound on the parameter set (false as stated in Sections 7.3 and 10.3 of version 1). Empirically the two mechanisms are compared on identical geometry in `experiments/exp_bias_factorial.py` (Section 8).

## 7 Prior art

- Adaptive Gaussian-network control with robust modification: Sanner and Slotine [79].
- Adaptation of inner-layer parameters via first-order Taylor expansion of the feature map with bounded remainder — the exact device of Sections 4 and 5: Lewis, Yesildirek and Liu [54]; Polycarpou [75]; Ge and Wang [33]; dynamic-structure RBF networks with tuned centres and widths: Fabri and Kadirkamanathan [28].
- Smooth projection with a boundary layer and the inequality (P), including the "no bias when $\theta^* \in \Omega$" observation: Pomet and Praly [76]; Ioannou and Sun [47, §4.4]; Lavretsky and Gibson [52]; Lavretsky and Wise [53, Ch. 11].
- $\sigma$- and $e$-modification and their bias: Ioannou and Kokotovic [46]; Narendra and Annaswamy [63].

- UUB via level sets: Khalil [48, Thm. 4.18]; Lewis, Jagannathan and Yesildirek [55].
- Concurrent learning (recorded-data stacks, finite excitation): Chowdhary and Johnson [17, 18].

Table 5 sets the ingredients of version 1 against these sources.

Table 5: Novelty of the ingredients of the stability argument.

| Ingredient | Prior source | Status of the claim of version 1 |
|---|---|---|
| Adaptive RBF control with projection-type robustification | Sanner and Slotine [79]; Ioannou and Sun [47] | known |
| Adapting centres/widths by first-order Taylor expansion with bounded remainder | Lewis, Yesildirek and Liu [54]; Polycarpou [75]; Fabri and Kadirkamanathan [28]; Ge and Wang [33] | known; version 1 cites none of these |
| Smooth projection with $\varepsilon$ boundary layer and inequality (P) | Pomet and Praly [76]; Lavretsky and Gibson [52]; Lavretsky and Wise [53] | known; the public showcase README calls it "the core theoretical contribution" |
| Projection introduces no bias in $\dot{V}$ when $\theta^* \in \Omega$ | Ioannou and Sun [47, §4.4] | known |
| Level-set ("global interior") UUB argument | Khalil [48, Thm. 4.18]; Lewis et al. [55] | known; called "critical and non-trivial" |
| Simultaneous adaptation of $W$, $c$, $\sigma$ for Gaussian RBFs with all three projected | — | the only combinational novelty; minor |

Net: no new theorem.

**What is new here.** Nothing in the stability theorem: Theorem 1(i)–(ii) is the composition of known pieces (Taylor-linearised inner-layer adaptation, boundary-layer projection and level-set UUB), applied to Gaussian RBFs with simultaneously adaptive weights, centres and widths. The contribution of the accompanying repository, after the revision of version 1, is (a) a theorem-to-code map in which the implemented and the proved algorithm coincide and are machine-checked (Lemmas 1–3 verified numerically along trajectories), (b) the explicit Hessian-free disturbance bound of Lemma 2 and the RMS bound (iii) of Theorem 1, which is the form of the result that can actually be confronted with data, and (c) a controlled empirical separation of "projection vs leakage" from "adaptive vs fixed geometry" (Section 6; `results/factorial/`). Claims of a "structural" advantage over leaky modifications are withdrawn.

# 8 What the experiments show

Every numerical statement in this section was regenerated from the scripts of the accompanying repository; the result file and the generating script are named with each table. Statistics are computed over seeds, not over autocorrelated time samples.

## 8.1 Robustness mechanism versus geometry

The factorial experiment `experiments/exp_bias_factorial.py` separates the two factors that the original headline figure (`results/tier1_lorenz_tracking.png`) confounded: there, the fixed-grid baselines used a grid not covering the reference, and their error ($\approx 15$) exceeded open loop, so the figure showed representational failure, not "bias". Here three robustness mechanisms (projection, $\sigma$-modification, $e$-modification, the leaky ones implemented on the same architecture

in `src/baselines/leaky_geometry.py`) are crossed with three geometries (fixed grid with only $W$ adapted; adaptive $W, c$; adaptive $W, c, \sigma$) on Lorenz tracking with a covering grid, $m = 27$, $k_e = 10$ and 5 seeds (Tables 6 and 7, Figure 1).

Table 6: Robustness mechanism × geometry (Lorenz tracking, $m = 27$, $k_e = 10$). Seeds: 5; metric = steady-state $\|e\|$ over the last 40% of a 30 s run; mean ± 95% CI over seeds; max SS = maximum over the window, mean over seeds; "div" = diverged runs. Source: `results/factorial/table.md`.

| geometry | mechanism | mean SS $\|e\|$ | median SS | max SS | max $\|W\|_F$ | min sep | div |
|---|---|---|---|---|---|---|---|
| fixed $(W)$ | projection | $0.1584 \pm 0.0072$ | 0.1288 | 0.559 | 47.5 | 1.365 | 0 |
| fixed $(W)$ | $\sigma$-mod | $0.7327 \pm 0.0223$ | 0.5647 | 2.720 | 38.5 | 1.469 | 0 |
| fixed $(W)$ | $e$-mod | $0.5619 \pm 0.0306$ | 0.4264 | 2.192 | 26.7 | 1.469 | 0 |
| adaptive $(W, c)$ | projection | $0.1468 \pm 0.0038$ | 0.1020 | 0.751 | 40.9 | 0.039 | 0 |
| adaptive $(W, c)$ | $\sigma$-mod | $0.4004 \pm 0.0050$ | 0.3468 | 1.863 | 32.3 | 0.078 | 0 |
| adaptive $(W, c)$ | $e$-mod | $0.2327 \pm 0.0024$ | 0.1993 | 0.718 | 28.1 | 0.099 | 0 |
| adaptive $(W, c, \sigma)$ | projection | $0.1119 \pm 0.0043$ | 0.0852 | 0.512 | 38.5 | 0.172 | 0 |
| adaptive $(W, c, \sigma)$ | $\sigma$-mod | $0.3862 \pm 0.0057$ | 0.3422 | 1.655 | 30.9 | 0.215 | 0 |
| adaptive $(W, c, \sigma)$ | $e$-mod | $0.2385 \pm 0.0029$ | 0.2103 | 0.820 | 27.7 | 0.239 | 0 |

Table 7: Paired (same seed) differences of mean steady-state $\|e\|$ for the factorial experiment. Top: projection minus baseline. Bottom: geometry effect within the projection mechanism. Source: `results/factorial/table.md`.

| geometry | baseline | mean diff | 95% CI | projection better in |
|---|---|---|---|---|
| fixed $(W)$ | $\sigma$-mod | $-0.5743$ | $\pm 0.0232$ | 5/5 seeds |
| fixed $(W)$ | $e$-mod | $-0.4035$ | $\pm 0.0310$ | 5/5 seeds |
| adaptive $(W, c)$ | $\sigma$-mod | $-0.2536$ | $\pm 0.0021$ | 5/5 seeds |
| adaptive $(W, c)$ | $e$-mod | $-0.0858$ | $\pm 0.0025$ | 5/5 seeds |
| adaptive $(W, c, \sigma)$ | $\sigma$-mod | $-0.2743$ | $\pm 0.0043$ | 5/5 seeds |
| adaptive $(W, c, \sigma)$ | $e$-mod | $-0.1266$ | $\pm 0.0031$ | 5/5 seeds |
| comparison | | mean diff | 95% CI | adaptive better in |
| adaptive $(W, c)$ − fixed $(W)$ | | $-0.0116$ | $\pm 0.0056$ | 5/5 |
| adaptive $(W, c, \sigma)$ − adaptive $(W, c)$ | | $-0.0349$ | $\pm 0.0026$ | 5/5 |

Projection beats both leaky baselines in every cell, in 5/5 seeds, with paired confidence intervals excluding zero. Within projection, adaptive centres improve on the fixed grid by $0.0116 \pm 0.0056$ (5/5 seeds) and adaptive widths by a further $0.0349 \pm 0.0026$ (5/5 seeds). So the method's real advantage is the robustness mechanism, with geometry adaptation a smaller second-order gain — close to the reverse of the story told by the original headline figure.

### 8.2 Kernel-regression baselines

Table 8 is the comparison against kernel-regression baselines from saved single runs, with nothing excluded (in particular the benchmark $n = 3$ and the NTK baseline are included). The proposed method is never first: it ranks 3/5, 2/5, 3/5 and 2/5 at $n = 1, 2, 3, 6$. NTK is first at $n = 2$ and $n = 6$ (mean steady-state $\|e\|$ 0.0015 and 0.0013, against 0.0372 and 0.0044 for the proposed method), and the sparse GP is first at $n = 3$ (3.2729), where the error of

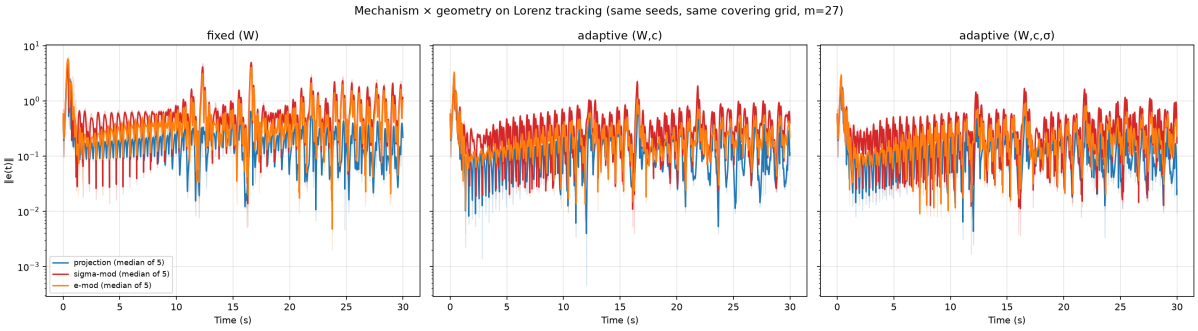



Figure 1: Mechanism × geometry on Lorenz tracking (same seeds, same covering grid, $m = 27$): $\|e(t)\|$ against time, logarithmic scale, median of the 5 seeds for projection, $\sigma$-modification and $e$-modification; shaded bands: minimum to maximum over the 5 seeds. Panels: fixed $(W)$, adaptive $(W, c)$, adaptive $(W, c, \sigma)$. Generated by `experiments/exp_bias_factorial.py`.

the proposed method is 7.7190. Its genuine edge is cost against the GP baselines: the mean update time of the proposed method is 0.10, 0.22 and 0.60 ms per step at $n = 1, 2, 6$ and 4.11 ms at $n = 3$, against 52.92–1888.80 ms per update for the GP baselines, which update every 50–100 steps. NTK takes 1.59–13.22 ms per update; it updates every 50 steps at $n = 1$ and every step at $n = 2, 3, 6$ (`experiments/sim4_ntk_control.py`, `experiments/sim4_ntk_n2.py`, `experiments/sim4_ntk_n3.py`, `experiments/sim4_ntk_n6.py`), so it is cheaper than the proposed method only at $n = 3$ (2.35 against 4.11 ms per step). The comparison is not information-matched: every baseline, here and in the fair re-run below, is trained on noise-free samples of the true $f(x, t)$, which amounts to measuring the state derivative exactly, whereas the adaptive law uses only the tracking error $e$. With one run per cell no valid significance test is available.

In the original baseline scripts (`sim1`–`sim3`) the GP baselines held a stale prediction for 50–100 steps, and the inducing points of the sparse GP lay on a diagonal line over $x \in [-3, 3]$ while the reference spans $\pm 20$. The fair re-run at $n = 1$ (`experiments/kernel_baselines_fair.py`, `experiments/run_fair_n1.py`, `experiments/plot_fair_n1.py`) uses per-step prediction for all methods, a covering inducing grid and 3 seeds (Table 9, Figure 2). The mean steady-state $\|e\|$ is 0.0591 for the exact GP, 0.1867 for the sparse GP, 0.1978 for the proposed method, 0.3087 for the deep GP and 0.4700 for NTK; the median online step is 6.309, 1.353, 0.060, 4.461 and 0.212 ms respectively. Once it is allowed to predict every step, the exact GP is 3–4× more accurate than the proposed method (mean steady-state $\|e\|$ 0.0591 against 0.1978), and more accurate in every seed (the proposed method has the lower mean in 0/3 seeds). The proposed method is on par with a fixed-grid sparse GP (the proposed method has the lower mean in 2/3 seeds) at 22.6× lower cost (median step 0.060 ms against 1.353 ms); its step is the cheapest of the five methods. This is the version of the comparison to quote.

### 8.3 The bound against data

**Oracle bound study.** The oracle experiment (`experiments/exp_oracle_ruub.py`) uses a plant that is exactly representable with constant $\theta^*$ ($m = 8$, $n = 2$), so every constant of Theorem 1 is computable; with $\theta^*$ constant, $C_\Omega = 0$ and $R_{\mathrm{UUB}} = D_{\max}/k_e$, reported as $D_{\mathrm{theory}}/k_e$ with $D_{\max}$ from Lemma 2. The "full-state bound" is $\sqrt{R_{\mathrm{UUB}}^2 + 2V_{\max,\Theta}}$ of Theorem 1(ii), and $D_{\mathrm{meas}}$ is the measured $\sup_{t\in[10,30]}\|d(t)\|$ over the evaluation window. Table 10 and Figure 3 cover $k_e \in \{2, 5, 10, 20\}$ and four learning-rate scales, 3 seeds each. The Lyapunov identity of Lemma 3 holds to $3.55 \times 10^{-15}$ (maximal $|\dot{V} - (-k_e\|e\|^2 + e^\top d + \mathrm{cross})|$), property (P) is never violated, the hard bounds held, and the RMS bound of Theorem 1(iii) with the measured disturbance holds in 48/48 runs and tracks $1/k_e$. The a-priori $R_{\mathrm{UUB}}$ is 26.1–261.1 against observed RMS

Table 8: Higher-dimensional comparison, complete (nothing excluded). Single trajectory per cell; steady state = last 40% of 30 s. Step time = mean wall-clock time of the method's update call (GP: per update, every 50–100 steps; NTK: per update, every 50 steps at $n = 1$ and every step at $n = 2, 3, 6$; proposed: per step). No valid significance test is available (one run per cell). Generated by `experiments/plot_comparison_hd_complete.py`; source: `results/comparison_complete_table.md`.

| benchmark | method | mean SS $\|e\|$ | median SS | max SS | rank (mean SS) | mean update time (ms) |
|---|---|---|---|---|---|---|
| $n = 1$ (1-D plant, Lorenz ref.) | Proposed (Proj) | 0.2342 | 0.1496 | 1.290 | 3/5 | 0.10 |
| | Exact GP | 0.2206 | 0.1415 | 1.016 | 1/5 | 273.78 |
| | Sparse GP | 0.2285 | 0.1914 | 0.874 | 2/5 | 52.92 |
| | Deep GP | 0.3963 | 0.2994 | 1.436 | 4/5 | 94.34 |
| | NTK | 0.4286 | 0.3318 | 1.324 | 5/5 | 13.22 |
| $n = 2$ (Van der Pol) | Proposed (Proj) | 0.0372 | 0.0137 | 0.310 | 2/5 | 0.22 |
| | Exact GP | 0.0555 | 0.0370 | 0.278 | 3/5 | 498.50 |
| | Sparse GP | 0.1771 | 0.0929 | 0.886 | 5/5 | 150.30 |
| | Deep GP | 0.1758 | 0.1169 | 1.097 | 4/5 | 415.37 |
| | NTK | 0.0015 | 0.0009 | 0.006 | 1/5 | 1.59 |
| $n = 3$ (Lorenz plant) | Proposed (Proj) | 7.7190 | 5.7197 | 30.476 | 3/5 | 4.11 |
| | Exact GP | 10.2774 | 8.8437 | 34.467 | 5/5 | 1888.80 |
| | Sparse GP | 3.2729 | 2.5099 | 19.176 | 1/5 | 264.64 |
| | Deep GP | 9.5344 | 8.3305 | 33.755 | 4/5 | 802.93 |
| | NTK | 5.7642 | 4.2944 | 18.240 | 2/5 | 2.35 |
| $n = 6$ (coupled Duffing) | Proposed (Proj) | 0.0044 | 0.0013 | 0.026 | 2/5 | 0.60 |
| | Exact GP | 0.0841 | 0.0695 | 0.279 | 3/5 | 668.01 |
| | Sparse GP | 0.1444 | 0.0872 | 0.670 | 5/5 | 523.83 |
| | Deep GP | 0.0953 | 0.0553 | 0.342 | 4/5 | 1607.48 |
| | NTK | 0.0013 | 0.0010 | 0.004 | 1/5 | 4.75 |

Table 9: Fair 1-D kernel comparison: per-step prediction for every method, covering inducing grid, seed-level statistics. Seeds: 3 (different reference and plant initial conditions). Steady state = last 40% of 30 s. Step time = median full online step (prediction + amortised training) after warm-up, same process type, 2 BLAS threads. Source: `results/fair_n1/table.md`.

| method | mean SS $\|e\|$ (mean $\pm$ 95% CI over seeds) | max SS $\|e\|$ (mean) | median step (ms) | mean step (ms) | seeds where proposed has lower mean SS |
|---|---|---|---|---|---|
| Proposed (Proj) | $0.1978 \pm 0.0503$ | 1.147 | 0.060 | 0.069 | — |
| Exact GP | $0.0591 \pm 0.0061$ | 0.780 | 6.309 | 13.936 | 0/3 |
| Sparse GP ($9 \times 3$ grid) | $0.1867 \pm 0.0210$ | 0.699 | 1.353 | 2.732 | 2/3 |
| Deep GP | $0.3087 \pm 0.1121$ | 1.262 | 4.461 | 5.495 | 2/3 |
| NTK | $0.4700 \pm 0.0250$ | 1.438 | 0.212 | 0.503 | 3/3 |

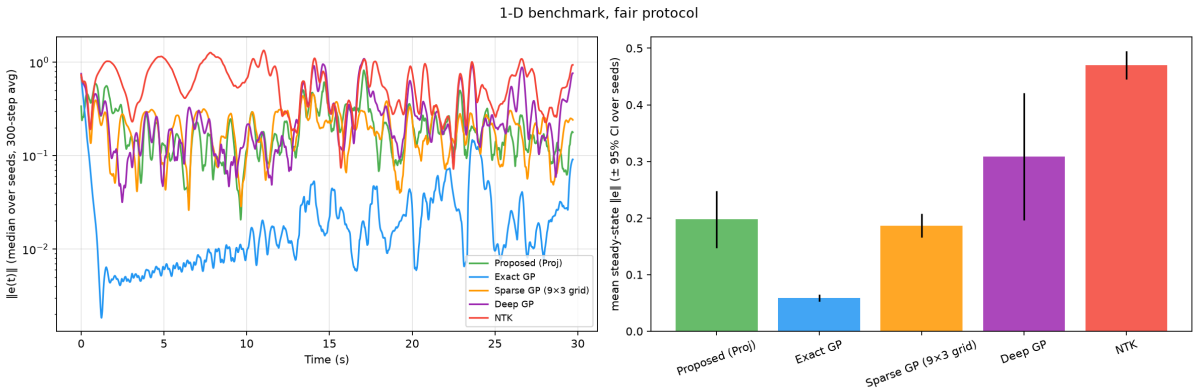



Figure 2: 1-D benchmark, fair protocol. Left: $\|e(t)\|$ (median over seeds, 300-step average), logarithmic scale, for the proposed method (Proj), the exact GP, the sparse GP ($9 \times 3$ grid), the deep GP and NTK. Right: mean steady-state $\|e\|$ ($\pm$ 95% CI over seeds). Generated by `experiments/plot_fair_n1.py`; numbers in Table 9.

errors of 0.0087–0.0908; the ratio of the a-priori bound to the observed RMS error ranges from 915× to 10954×.

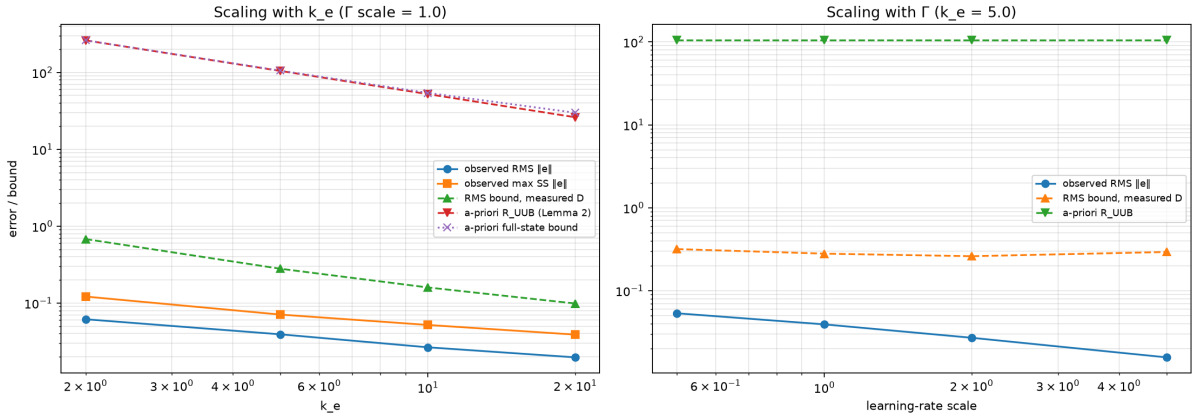



Figure 3: Oracle bound study, logarithmic axes. Left: scaling with $k_e$ at learning-rate scale 1.0 — observed RMS $\|e\|$, observed maximal steady-state $\|e\|$, the RMS bound with the measured disturbance, the a-priori $R_{\mathrm{UUB}}$ (Lemma 2) and the a-priori full-state bound. Right: scaling with the learning-rate scale at $k_e = 5$ — observed RMS $\|e\|$, the RMS bound with the measured disturbance and the a-priori $R_{\mathrm{UUB}}$. Generated by `experiments/exp_oracle_ruub.py`; numbers in Table 10.

**Monte Carlo over initial conditions.** On a plant that is not exactly representable only exactly checkable statements are tested (`experiments/exp_monte_carlo_honest.py`, Table 11): all 200 trajectories are finite, the hard parameter bounds held at every recorded step in 200/200 trials, and the a-posteriori ISS bound held in 200/200 trials (maximal ratio 0.979). No a-priori $R_{\mathrm{UUB}}$ threshold is drawn: $D_{\max}$ and $C_\Omega$ are not computable for this plant (the threshold 0.5 of the original figure was an arbitrary constant, not a calculated bound).

## 8.4 Summary

- On identical geometry, projection beats $\sigma$- and $e$-modification in every cell and in every seed (5/5), with paired confidence intervals excluding zero.

Table 10: Oracle $R_{\mathrm{UUB}}$ study (exactly representable plant, constant $\theta^*$, $m = 8$, $n = 2$). Window $[10, 30]$ s; mean over 3 seeds. All bounds in the same units as $\|e\|$. "P viol" = samples with cross $> 0$ (must be 0); "inside" = hard bounds held. RMS bound (measured disturbance) satisfied in all 48 runs: True. Maximal identity error $|\dot{V} - (-k_e\|e\|^2 + e^\top d + \mathrm{cross})|$: $3.55 \times 10^{-15}$. Source: `results/oracle_ruub/table.md`.

| $k_e$ | $\Gamma$ scale | RMS $\|e\|$ | max SS $\|e\|$ | RMS bound ($D_{\mathrm{meas}}$, finite window) | $D_{\mathrm{meas}}/k_e$ | $D_{\mathrm{theory}}/k_e$ | full-state bound | ratio theory/RMS | P viol | inside |
|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 0.5 | 0.0908 | 0.1704 | 0.6791 | 0.5435 | 261.1 | 261.9 | 2875× | 0 | True |
| 2 | 1 | 0.0616 | 0.1215 | 0.6801 | 0.6123 | 261.1 | 261.5 | 4239× | 0 | True |
| 2 | 2 | 0.0383 | 0.0796 | 0.7139 | 0.6808 | 261.1 | 261.3 | 6824× | 0 | True |
| 2 | 5 | 0.0238 | 0.0476 | 1.1766 | 1.1649 | 261.1 | 261.2 | 10954× | 0 | True |
| 5 | 0.5 | 0.0533 | 0.1054 | 0.3170 | 0.1979 | 104.4 | 106.4 | 1958× | 0 | True |
| 5 | 1 | 0.0392 | 0.0708 | 0.2793 | 0.2123 | 104.4 | 105.4 | 2664× | 0 | True |
| 5 | 2 | 0.0270 | 0.0539 | 0.2601 | 0.2240 | 104.4 | 104.9 | 3866× | 0 | True |
| 5 | 5 | 0.0157 | 0.0329 | 0.2925 | 0.2780 | 104.4 | 104.6 | 6662× | 0 | True |
| 10 | 0.5 | 0.0398 | 0.0793 | 0.1970 | 0.0981 | 52.2 | 56.1 | 1313× | 0 | True |
| 10 | 1 | 0.0266 | 0.0519 | 0.1596 | 0.1003 | 52.2 | 54.2 | 1965× | 0 | True |
| 10 | 2 | 0.0194 | 0.0369 | 0.1395 | 0.1057 | 52.2 | 53.2 | 2688× | 0 | True |
| 10 | 5 | 0.0117 | 0.0240 | 0.1335 | 0.1188 | 52.2 | 52.6 | 4466× | 0 | True |
| 20 | 0.5 | 0.0285 | 0.0522 | 0.1290 | 0.0501 | 26.1 | 33.2 | 915× | 0 | True |
| 20 | 1 | 0.0197 | 0.0389 | 0.0987 | 0.0494 | 26.1 | 29.9 | 1328× | 0 | True |
| 20 | 2 | 0.0132 | 0.0260 | 0.0799 | 0.0503 | 26.1 | 28.0 | 1971× | 0 | True |
| 20 | 5 | 0.0087 | 0.0168 | 0.0676 | 0.0534 | 26.1 | 26.9 | 3011× | 0 | True |

Table 11: Monte Carlo over random initial conditions (Van der Pol, $m = 16$, $k_e = 5.0$, 200 trials). Source: `results/monte_carlo_honest/summary.md`.

| check | result |
|---|---|
| all trajectories finite | 200/200 |
| hard parameter bounds held at every recorded step | 200/200 |
| a-posteriori ISS bound $\|e(t)\| \le \mathrm{e}^{-k_e(t-T_1)}\|e(T_1)\| + \sup\|f - \hat{f}\|/k_e$ held | 200/200 (max ratio 0.979) |
| steady-state max $\|e\|$ (last 25%): median / 95th pct / max | 0.0568 / 0.0938 / 0.1075 |
| steady-state mean $\|e\|$: median / 95th pct / max | 0.0302 / 0.0484 / 0.0593 |
| $\sup\|f - \hat{f}\|$ in steady state / $k_e$: median / max | 0.0627 / 0.1475 |
| transient max $\|e\|$: max over trials | 4.903 |

- Adaptive geometry is a smaller but consistent gain: 0.0116 for adaptive centres and a further 0.0349 for adaptive widths in mean steady-state $\|e\|$, 5/5 seeds each.
- Against kernel-regression baselines the proposed method is never the most accurate. Its edge is cost against the GP baselines (NTK is cheaper only at $n = 3$), and in the fair re-run at $n = 1$ it has the cheapest step of the five methods.
- In the fair re-run at $n = 1$ the exact GP is 3–4$\times$ more accurate than the proposed method.
- In the oracle study the a-priori bound exceeds the observed RMS error by factors of 915 to 10954, while the RMS bound with the measured disturbance holds in 48/48 runs.

# Part II
# Transport adaptive control

## 9 Setting

**Plant, reference, control, error.**

$$\dot{x} = f(x,t) + u, \qquad e = x - x_m(t), \qquad u = \dot{x}_m - k_e e - \hat{f}(x).$$

**Lifted parameter.** $\vartheta = (a, c, s) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}_{>0}$, and $p := 2n + 1$. The feature is $\Phi(x;\vartheta) := a\,\varphi(x;c,s) \in \mathbb{R}^n$, where $\varphi(x;c,s) = \exp\bigl(-\|x-c\|^2/(2s^2)\bigr)$.

**Network as a measure.**

$$\hat{f}(x) = \frac{1}{m}\sum_{i=1}^{m} \Phi(x;\vartheta_i) = \int \Phi(x;\vartheta)\,\mathrm{d}\mu(\vartheta), \qquad \mu := \frac{1}{m}\sum_{i=1}^{m} \delta_{\vartheta_i}.$$

The map $\mu \mapsto \hat{f}$ is linear.

**Metric.** $G = \mathrm{diag}(g_a I_n, g_c I_n, g_s) \succ 0$ is the gain matrix. Distances are measured in $\|v\|_{G^{-1}}^2 = v^\top G^{-1} v$, and $\lambda_G := \lambda_{\max}(G)$.

**Admissible box and projection.**

- **The box.** $\Theta_K := \{\|a\| \le a_{\max}\} \times \{\|c\| \le c_{\max}\} \times [s_{\min}, s_{\max}]$.
- **Layer and inner set.** $\varepsilon_K$ is the width of the boundary layer. Shrinking each bound of $\Theta_K$ by $\varepsilon_K$ gives the inner set $\Theta_K^\varepsilon$, with interior $\operatorname{int}\Theta_K^\varepsilon$.
- **The operator.** $\mathrm{Proj}_K(\vartheta, y)$ is the per-particle, factor-wise projection (implemented in `src/core/projection.py` in the accompanying repository). It uses Pomet–Praly norm-ball operators [76] for $a$ and $c$, with a quadratic ramp $h \in [0,1]$ (defined in Section 4), and a box operator for $s$, with a linear ramp.
- **Regularity.** The operator is locally Lipschitz in $(\vartheta, y)$. The norm-ball factors are $C^1$ at the inner edge of their layers. The box factor, and every factor at the outer edge, is only Lipschitz. The proofs use only Lipschitz continuity and ramp values in $[0,1]$.

**Property (P).** For $\vartheta \in \Theta_K$, $\vartheta' \in \Theta_K^\varepsilon$ and all $y$,

$$\langle \vartheta - \vartheta',\, G^{-1}\,\mathrm{Proj}_K(\vartheta, Gy)\rangle \le \langle \vartheta - \vartheta',\, y\rangle. \tag{P}$$

On $\partial\Theta_K$ the projected velocity has no outward component, so $\Theta_K$ is forward invariant (Nagumo; Pomet–Praly 1992 [76]).

**Lemma P.1** ($\mathrm{Proj}_K$ is non-expansive)**.** *For every $\vartheta = (a,c,s) \in \Theta_K$ and $y = (y_a, y_c, y_s)$,*

$$\|\mathrm{Proj}_K(\vartheta, Gy)\|_{G^{-1}}^2 = \|y\|_G^2 - \sum_{b\in\{a,c,s\}} h_b(2-h_b)\,\frac{\bigl[(n_b^\top G_b y_b)_+\bigr]^2}{n_b^\top G_b n_b} \;\le\; \|y\|_G^2.$$

*The terms are as follows.*

- *For the balls, $n_a = a$ and $n_c = c$, with the quadratic ramps $h_a = h(\|a\|; a_{\max}, \varepsilon_K)$ and $h_c = h(\|c\|; c_{\max}, \varepsilon_K)$.*

- *For $s < s_{\min} + \varepsilon_K$, $n_s = -1$ and $h_s = 1 - \mathrm{clip}\big((s - s_{\min})/\varepsilon_K, 0, 1\big)$.*
- *For $s > s_{\max} - \varepsilon_K$, $n_s = +1$ and $h_s = 1 - \mathrm{clip}\big((s_{\max} - s)/\varepsilon_K, 0, 1\big)$.*
- *Otherwise $h_s = 0$.*

*Proof.* $G$ is block-diagonal, so both squared norms split over the blocks. On an active ball block, $\mathrm{Proj}_b = G_b q_b$ with $q_b = y_b - h_b n_b\,(n_b^\top G_b y_b)/(n_b^\top G_b n_b)$. Expanding $q_b^\top G_b q_b$ gives the subtracted term. On an inactive block ($h_b = 0$ or $n_b^\top G_b y_b \le 0$), $\mathrm{Proj}_b = G_b y_b$. The $s$-block is the one-dimensional case. Finally $h_b \in [0,1]$, so $h_b(2 - h_b) \in [0,1]$. □

Checked by: `tests/test_transport.py`: `test_lemma_P1_projection_is_nonexpansive_and_tight_velocity_bound`.

**Potential.** $g_t(\vartheta) := e(t)^\top \Phi(x(t);\vartheta)$, and $F_t(\mu) := -\int g_t \,\mathrm{d}\mu = -e(t)^\top \hat{f}(x(t))$.

**Adaptive law (the algorithm).**

$$\dot{\vartheta}_i = \mathrm{Proj}_K\big(\vartheta_i,\, G\nabla_\vartheta g_t(\vartheta_i)\big), \qquad i = 1, \dots, m. \tag{L}$$

**Ideal measure.** For $j = 1, \dots, m$, $\vartheta_j^*(t) \in \Theta_K^\varepsilon$ is $C^1$ in $t$, and $\mu_t^* := \frac{1}{m}\sum_j \delta_{\vartheta_j^*(t)}$. Write $\varepsilon(x,t) := f(x,t) - \int \Phi(x;\cdot)\,\mathrm{d}\mu_t^*$, with $\|\varepsilon\| \le \varepsilon_{\max}$ on the operating set $X$, and set

$$V_*(t) := \Big(\frac{1}{m}\sum_j \|\dot{\vartheta}_j^*\|_{G^{-1}}^2\Big)^{1/2}, \qquad V_* := \sup_t V_*(t).$$

The proofs never use optimality of $\mu^*$. It is an analytic device, and any admissible $C^1$ curve may be used.

**Optimal-transport distance.** For uniform atomic $\mu = \frac{1}{m}\sum \delta_{\vartheta_i}$ and $\nu = \frac{1}{m}\sum \delta_{\vartheta'_j}$,

$$W(\mu,\nu)^2 := \min_{\pi \in S_m} \frac{1}{m}\sum_i \|\vartheta_i - \vartheta'_{\pi(i)}\|_{G^{-1}}^2.$$

This is the Kantorovich 2-Wasserstein distance with ground cost $\|\cdot\|_{G^{-1}}^2$. For equal masses the optimum is attained at a permutation (Birkhoff). It is computed by linear assignment in $O(m^3)$. For general measures on $\Theta_K$, $W$ denotes the same Kantorovich distance.

**Label-free Lyapunov function.**

$$V_{\mathrm{OT}}(t) := \tfrac{1}{2}\|e(t)\|^2 + \tfrac{1}{2}W(\mu_t, \mu_t^*)^2 = \min_\pi V_\pi(t),$$
$$V_\pi := \tfrac{1}{2}\|e\|^2 + \frac{1}{2m}\sum_i \|\vartheta_i - \vartheta^*_{\pi(i)}\|_{G^{-1}}^2.$$

**Standing hypotheses (S).**

(S0) $k_e > 0$; $G = \mathrm{diag}(g_a I_n, g_c I_n, g_s) \succ 0$; $s_{\min} > 0$; and $0 < \varepsilon_K < \min\{a_{\max}, c_{\max}, (s_{\max} - s_{\min})/2\}$ (`src/core/projection.py` enforces the last condition).

(S1) $f$ is continuous in $(x,t)$ and locally Lipschitz in $x$, uniformly in $t$ on bounded time intervals.

(S2) $x_m \in C^1$, and $x_m$ and $\dot{x}_m$ are bounded.

(S3) $\vartheta_i(0) \in \Theta_K$.

(S4) The $\vartheta_j^*$ are $C^1$ with values in $\Theta_K^\varepsilon$, and $V_* < \infty$.

(S5) $x(t) \in X$ on the interval considered. For bounded $f$, take $X := \bigcup_t \{x : \|x - x_m(t)\| \le r(F)\}$ with $F = \sup\|f\|$ (Theorem D(b)). Then $\varepsilon_{\max}$ is the supremum over this $X$, and it depends on $\|e(0)\|$.

Under (S0)–(S3) the right-hand side of (L) is continuous in $t$ and locally Lipschitz in $(x, \vartheta)$. Solutions are classical ($C^1$) and unique on their maximal interval. Under (S4) as well, each $V_\pi$ is $C^1$.

**Notation.**

- Times are written $t_0 \le t_1$ (or $T_1 < T_2$), and $r$ is the integration variable in time.
- $\mathrm{RMS}_{[T_1,T_2]}\|e\| := \Big((T_2 - T_1)^{-1} \int_{T_1}^{T_2} \|e\|^2 \,\mathrm{d}r\Big)^{1/2}$.

**Constants of the Gaussian.** Closed forms (`gaussian_constants` in `src/core/transport.py` of the accompanying repository). With $u := \|x - c\|^2/s^2$ and $\varphi = e^{-u/2}$, on $s \ge s_{\min}$:

$$\|\nabla_c \varphi\| \le J_c := e^{-1/2}/s_{\min}, \qquad |\partial_s \varphi| \le J_s := 2e^{-1}/s_{\min}, \qquad J := (J_c^2 + J_s^2)^{1/2},$$
$$\|\nabla^2_{(c,s)} \varphi\| \le H_\varphi := \big(\max(h_{cc}, h_{ss}) + h_{cs}\big)/s_{\min}^2,$$
$$h_{cc} = 1, \qquad h_{ss} = 2e^{-1/2}, \qquad h_{cs} = \sqrt{u}\, e^{-u/2}|u - 2| \ \text{ evaluated at } u = (5 - \sqrt{17})/2.$$

$J$ and $H_\varphi$ are upper bounds. The true supremum of $\|\nabla_{(c,s)}\varphi\|$ is about $0.92/s_{\min}$, and that of $\|\nabla^2_{(c,s)}\varphi\|$ is about $1.57/s_{\min}^2$. The Hessian of the lifted potential is the block matrix

$$\begin{pmatrix} 0 & u\,\nabla\varphi^\top \\ \nabla\varphi\, u^\top & (u^\top a)\,\nabla^2\varphi \end{pmatrix}.$$

Hence

$$\sup_{x \in \mathbb{R}^n,\ \vartheta \in \Theta_K,\ \|u\|=1} \|\nabla^2_\vartheta \big(u^\top \Phi(x;\vartheta)\big)\| \le H_K := J + a_{\max} H_\varphi, \qquad K := \tfrac{1}{2} H_K. \tag{1}$$

A brute-force numerical check over the full lifted Hessian gives sup $\approx 0.76\, H_K$.

# 10 Theorem A: identification

**Theorem A.** *Assume* (S0).

*(i) Assume scalar classical gains $\Gamma_W = \gamma_W I$, $\Gamma_c = \gamma_c I$ and $\Gamma_\sigma = \gamma_\sigma I$, and use the dictionary $W_i = a_i/m$, $\gamma_W = g_a/m$, $\gamma_c = m g_c$, $\gamma_\sigma = m g_s$.*

- *The* raw (*unprojected*) *rates of* (L) *and of the classical law of Section 4 coincide.*
- *The* projected *rates coincide when the classical joint projections are replaced by per-kernel ones. The classical law projects $W$ onto one Frobenius ball and* vec$(c)$ *onto one Euclidean ball. The per-kernel version projects:*
  - *$W_i$ onto $\|W_i\| \le a_{\max}/m$, with layer $\varepsilon_K/m$;*
  - *$c_i$ onto $\|c_i\| \le c_{\max}$, with layer $\varepsilon_K$;*
  - *$\sigma_i$ onto $[s_{\min}, s_{\max}]$, with layer $\varepsilon_K$.*
- *For $m \ge 2$, no joint ball, of any radius $W_{\max}$ and layer $\varepsilon_W$, reproduces the per-kernel field. For $m = 1$ the two coincide, with $W_{\max} = a_{\max}$ and layer $\varepsilon_K$.*

*(ii) Fix a closed-loop trajectory $(x(\cdot), e(\cdot))$. Then:*

- *The field* $v_t(\vartheta) := \mathrm{Proj}_K(\vartheta, G\nabla g_t(\vartheta))$ *is Lipschitz in* $\vartheta$ *on* $\Theta_K$*, and* $\Theta_K$ *is forward invariant.*
- *For every* $\mu_0 \in \mathcal{P}(\Theta_K)$*,* $\mu_t := (X_t)_{\#}\mu_0$ *is a distributional solution of* $\partial_t\mu + \nabla\cdot(\mu v_t) = 0$*.*
- *On* $\operatorname{int}\Theta_K^\varepsilon$*,* $v_t = -G\nabla(\delta F_t/\delta\mu)$*. So at frozen time,* $\mu_t$ *is the Wasserstein gradient flow (metric* $G^{-1}$*) of the linear functional* $F_t$*, and* (L) *is its particle form.*
- $F_t$ *is generated by the closed loop, so the coupled system is the one studied in Theorem D.*

*(iii)* ***Dictionary****, all in the* $G^{-1}$ *metric:*

| classical device | measure-level object |
|---|---|
| projection onto $\Theta_K$ | A velocity-level (Pomet–Praly) regularisation of the support constraint $\mu_t \in \mathcal{P}(\Theta_K)$. For $\varepsilon_K > 0$ it is not the gradient flow of any functional of $\mu$. Formally, as $\varepsilon_K \to 0$, the field tends to the $G^{-1}$-projection of $G\nabla g_t$ onto the tangent cone of $\Theta_K$, which is the velocity of the $W_{G^{-1}}$-gradient flow of $F_t + \iota_{\mathcal{P}(\Theta_K)}$ ([51, Prop. 3.1] for the exact $W_2$ projection). |
| $\sigma$-modification in the $G^{-1}$ metric (leak outside the gain, $\dot\vartheta = G\nabla g - \sigma(\vartheta - \vartheta^0)$, as in ($\mathrm{L}_\sigma$) and Theorem H) | the potential energy $(\sigma/2)\int\|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu$, which is geodesically $\sigma$-convex |
| classical $\sigma$-modification $\Gamma(y - \sigma\theta)$ (leak inside the gain; `src/baselines/sigma_mod.py` implements its $W$-block, with $c$ and $\sigma$ frozen) | For the full-vector leak, under (i), the object is $(\sigma/2)\int(\vartheta - \vartheta^0)^\top D(\vartheta - \vartheta^0)\,\mathrm{d}\mu$ with $D = \mathrm{diag}(I_n/m, mI_n, m)$. In the $G^{-1}$ metric it is $\lambda$-convex with $\lambda = \sigma\cdot\min(g_a/m, mg_c, mg_s)$, which depends on $m$. For the $W$-block alone it is $(\sigma/2)\int(a - a^0)^\top(I_n/m)(a - a^0)\,\mathrm{d}\mu$. |
| $\sigma$-modification toward a labelled $(\vartheta_i^0)$ | not a functional of $\mu$; it is a functional on the labelled lift $\mathbb{R}^{mp}$ |
| OT $\sigma$-modification (Theorem C) | $(\sigma/2)W(\mu, \mu^0)^2$. For atomic $\mu$ with a unique optimal assignment, its $W$-gradient at atom $i$ is $\sigma(\vartheta_i - \vartheta^0_{\tau(i)})$. At ties, ($\mathrm{L}_\sigma$) uses a measurable selection from the superdifferential $\{\sigma(\vartheta_i - \vartheta^0_{\tau(i)}) : \tau \text{ optimal}\}$. |
| $e$-modification | the same functionals with coefficient $\sigma\|e\|$ |
| parameter dither (Theorem H) | the entropy $\tau\int\mu\log\mu$ |

**Status.** Proved and checked. Novelty (literature audit): interpretive; ingredients known [14], [20], [51], [69].

Checked by: `tests/test_transport.py`: `test_identification_with_classical_law` (raw rates in (i)); `test_identification_projected_rates_per_kernel` (projected rates and the only-if construction in (i)).

*Proof.* **(i) Gain map.** We have $\nabla_a g = \varphi e$, $\nabla_c g = (a^\top e)\nabla_c\varphi$ and $\partial_s g = (a^\top e)\partial_s\varphi$. Matching these with the classical rates gives the gain map.

**Scaling identity.** $(1/m)\,\mathrm{Proj}_{r,\varepsilon}(a, y) = \mathrm{Proj}_{r/m,\varepsilon/m}(a/m, y/m)$. Three facts give it:

- $h_{r/m,\varepsilon/m}(\|a\|/m) = h_{r,\varepsilon}(\|a\|)$;
- the direction $a(a^\top y)/\|a\|^2$ is homogeneous of degree 0 in $a$ and of degree 1 in $y$;
- the activation $a^\top y > 0$ is invariant under scaling.

If the layer is not rescaled, the ramps differ.

*Only if* $(m \geq 2)$. Let $r_{\text{in}} := (a_{\max} - \varepsilon_K)/m$, and let the joint ball have radius $W_{\max}$, layer $\varepsilon_W$ and $\rho := W_{\max} - \varepsilon_W$. In both cases below the driving term points outward.

- If $\rho < \sqrt{2}\, r_{\text{in}}$, take two rows of equal norm $r$ with $\sqrt{2}\, r \in \big(\rho, \min(\sqrt{2}\, r_{\text{in}}, W_{\max})\big]$, parallel to $e$. The per-kernel field is inactive and the joint field is active.
- If $\rho \geq \sqrt{2}\, r_{\text{in}}$, take one row with norm in $\big(r_{\text{in}}, \min(a_{\max}/m, \rho)\big)$ and the others zero. The per-kernel field is active and the joint field is inactive.

The same argument applies to $\mathrm{vec}(c)$. For $m = 1$ the two operators coincide.

The test `test_identification_projected_rates_per_kernel` checks three things: the identity with active projections; that the layer must be rescaled; and the only-if construction against `ProjectionOperator.project_matrix` on five joint balls. Four are active against an inactive per-kernel field, and one is inactive against an active per-kernel field.

**(ii)** $\mathrm{Proj}_K$ is Lipschitz in $\vartheta$. The ramp vanishes for $\|\vartheta\| \leq \theta_{\max} - \varepsilon_K$, so the direction $\vartheta/\|\vartheta\|$ is evaluated only away from 0. Also, $\nabla g_t$ is $C^1$ on $s \geq s_{\min}$.

- **Invariance.** Forward invariance follows from Nagumo's theorem, since $n_b^\top \mathrm{Proj}_b \leq 0$ on the boundary, with the outward normals $n_b$ of Lemma P.1.
- **Continuity equation.** The push-forward of $\mu_0$ by a Lipschitz flow solves the continuity equation (Golse [37]; Ambrosio–Gigli–Savaré [2, §8.1]).
- **Gradient form.** On $\operatorname{int} \Theta_K^\varepsilon$ the projection is inactive, so $v_t = G\nabla g_t = -G\nabla(\delta F_t/\delta\mu)$.

**(iii) Potential-energy rows.** These follow from the computation in the metric $G^{-1}$: for any $V$, $\nabla_{G^{-1}} V = G\nabla V$. For example, $V = (\sigma/2)\|\vartheta - \vartheta^0\|_{G^{-1}}^2$ gives $G\nabla V = \sigma(\vartheta - \vartheta^0)$.

**Classical-leak row.** Applying the gain map of (i) to the classical leak $\Gamma(y - \sigma\theta)$ gives the velocity $-\sigma GD(\vartheta - \vartheta^0)$.

**Projection row.** The $G^{-1}$-rescaled correction field has an asymmetric Jacobian. Take the example $m = 1$, $n = 1$, $G = I$, $x = 0.7$, $e = 1$, $\vartheta = (0.93, 0.2, 1.5)$, $a_{\max} = 1$, $\varepsilon_K = 0.15$ (so $h(0.93) = 0.284$) and $c_{\max} = 2$, with $a$ in the layer. A numerical evaluation gives:

- the $c$-derivative of the $a$-component is $-0.060$;
- the $a$-derivative of the $c$-component is 0.

So the field is not a gradient. □

**Remark A.1** (normalisation)**.** The algorithm's raw rates are unchanged. Two things do change: the admissible set becomes a product of per-kernel balls, and $a_{\max}$, not $\|W\|_F$, is held fixed in $m$.

- ***m*-free constants.** In this normalisation the structural constants $k_e$, $K$, $L_0$, $L_1$, $\lambda_G$ and $\Delta$ are free of $m$ for fixed $G$ and $a_{\max}$. So are the constants of Theorems B, C(i)–(iii), D and F.
- ***m*-dependent data.** $\varepsilon_{\max}$ and $V_*$ are data of the chosen $m$-atom ideal. The local constants of C(iv) and of Theorem E depend on $m$.
- **Classical convention.** Fixed classical $\gamma$'s correspond to $g_a = m\gamma_W \propto m$ and $g_c, g_s \propto 1/m$. Weights speed up, centres and widths freeze, $\lambda_G \propto m$ and $\Delta \propto \sqrt{m}$.

# 11 Theorem B: the optimal-assignment Lyapunov function

**Lemma B.1** (Dini derivative of a minimum)**.** *Suppose each $V_\pi$ is $C^1$ and $V_{\mathrm{OT}} = \min_\pi V_\pi$. Then $V_{\mathrm{OT}}$ is locally Lipschitz, and $D^+ V_{\mathrm{OT}}(t) \leq \dot{V}_\pi(t)$ for every $\pi$ active at $t$.*

*Proof.* For $h > 0$, $V_{\mathrm{OT}}(t + h) \leq V_\pi(t + h)$, and $V_{\mathrm{OT}}(t) = V_\pi(t)$. □

**Lemma B.2** (matched remainder in two regimes; gradient bound)**.** *Let* $L_0 := 2a_{\max}$ *and* $L_1 := (1 + a_{\max}^2 J^2)^{1/2}$*. For* $\vartheta, \vartheta' \in \Theta_K$*:*

*(a) the remainder* $R := g_t(\vartheta') - g_t(\vartheta) - \langle \nabla g_t(\vartheta),\, \vartheta' - \vartheta \rangle$ *satisfies*

$$|R| \le \|e(t)\| \cdot \min\bigl\{ K\|\vartheta - \vartheta'\|^2,\ L_0 + L_1\|\vartheta - \vartheta'\| \bigr\};$$

*(b)* $\|\nabla_\vartheta g_t(\vartheta)\| \le L_1 \|e(t)\|$.

*Proof. (b)* We have $\nabla_a g = \varphi e$ and $\nabla_{(c,s)} g = (a^\top e)\nabla_{(c,s)}\varphi$, with $\varphi \le 1$ and $\|\nabla_{(c,s)}\varphi\| \le J$ on $s \ge s_{\min}$. So $\|\nabla g\|^2 \le \|e\|^2(1 + a_{\max}^2 J^2)$.

*(a)* If $e(t) = 0$, both sides vanish. Otherwise set $h := \vartheta' - \vartheta$.

- *Quadratic branch.* $\Theta_K$ is convex, so the Lagrange remainder of $\tau \mapsto g_t(\vartheta + \tau h)$ gives $R = \frac{1}{2} h^\top \nabla^2 g_t(\vartheta + \tau_0 h) h$ for some $\tau_0 \in (0,1)$. Since $g_t = \|e\| \cdot (u^\top \Phi)$ with $u = e/\|e\|$, the Hessian bound (1) of Section 9 gives this branch.
- *Linear branch.* It follows from $|g_t| \le \|e\| a_{\max}$ and (b). □

**Theorem B.** *Under (S), along every solution of (L), for every* $t$*, with* $W := W(\mu_t, \mu_t^*)$*:*

$$D^+ V_{\mathrm{OT}} \le -k_e\|e\|^2 + \|e\|\Bigl(\varepsilon_{\max} + \min\bigl\{K\lambda_G W^2,\ L_0 + L_1 \lambda_G^{1/2} W\bigr\}\Bigr) + V_*(t)\, W. \tag{B}$$

**Status.** Proved and checked, including a moving ideal, $\varepsilon \neq 0$, switches, an active projection, and the failure of the plain bound along ($\mathrm{L}_\sigma$). Novelty (literature audit): packaging new; ingredients known [84], [1], [68], [16].

Checked by: `tests/test_transport.py`: `test_theorem_B_inequality_along_trajectories`, `test_theorem_B_moving_ideal_nonrepresentable_switches_and_active_projection`, `test_dini_derivative_of_V_OT_bounded_by_active_branch`, `test_permuted_ideal_labelled_vs_OT`; `experiments/exp_transport_permutation.py`.

*Proof.* Fix $t$ and an optimal $\pi$, and write $\tilde\vartheta_i := \vartheta_i - \vartheta^*_{\pi(i)}$. Then

$$\dot V_\pi = e^\top \dot e + \frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, G^{-1}\dot\vartheta_i\rangle - \frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, G^{-1}\dot\vartheta^*_{\pi(i)}\rangle.$$

(a) $e^\top \dot e = -k_e\|e\|^2 + e^\top \varepsilon + \frac{1}{m}\sum_i \bigl[g_t(\vartheta^*_{\pi(i)}) - g_t(\vartheta_i)\bigr]$.

(b) By (P) with $\vartheta' = \vartheta^*_{\pi(i)}$, $\frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, G^{-1}\dot\vartheta_i\rangle \le \frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, \nabla g_t(\vartheta_i)\rangle$.

(c) Since $\pi$ is a bijection and is optimal,

$$-\frac{1}{m}\sum_i \langle \tilde\vartheta_i,\, G^{-1}\dot\vartheta^*_{\pi(i)}\rangle \le \Bigl(\frac{1}{m}\sum_i \|\tilde\vartheta_i\|^2_{G^{-1}}\Bigr)^{1/2}\Bigl(\frac{1}{m}\sum_i \|\dot\vartheta^*_{\pi(i)}\|^2_{G^{-1}}\Bigr)^{1/2} = W \cdot V_*(t).$$

Summing gives

$$\dot V_\pi \le -k_e\|e\|^2 + \|e\|\varepsilon_{\max} + \frac{1}{m}\sum_i R_i + W\, V_*(t),$$

where each $R_i$ is a matched remainder. Average Lemma B.2(a), using three facts:

- $\frac{1}{m}\sum_i \min\{A_i, B_i\} \le \min\{\mathrm{avg}\, A, \mathrm{avg}\, B\}$;
- $\frac{1}{m}\sum_i \|\tilde\vartheta_i\|^2 \le \lambda_G W^2$;
- $\frac{1}{m}\sum_i \|\tilde\vartheta_i\| \le \lambda_G^{1/2} W$.

Lemma B.1 then completes the proof. □

**Corollary B.3** (ultimate bounds)**.** *Let* $\Delta := \operatorname{diam}_{G^{-1}}(\Theta_K)$*, so* $W \le \Delta$*. Define*

$$D^{\mathrm{OT}} := \varepsilon_{\max} + \min\{K\lambda_G\Delta^2,\ L_0 + L_1\lambda_G^{1/2}\Delta\}, \qquad C^{\mathrm{OT}} := V_*\Delta,$$
$$R^{\mathrm{OT}} := \frac{D^{\mathrm{OT}}}{2k_e} + \Big(\frac{(D^{\mathrm{OT}})^2}{4k_e^2} + \frac{C^{\mathrm{OT}}}{k_e}\Big)^{1/2}.$$

*Then:*

- $D^+V_{\mathrm{OT}} < 0$ *whenever* $\|e\| > R^{\mathrm{OT}}$*.*
- *Under (S5) on* $[0,\infty)$*,* $\limsup\|e\| \le \big((R^{\mathrm{OT}})^2 + \Delta^2\big)^{1/2}$*. This uses* $\sup \frac{1}{2}W^2 \le \Delta^2/2$*.*
- $\mathrm{RMS}_{[T_1,T_2]}\|e\| \le \dfrac{D^{\mathrm{OT}}}{2k_e} + \Big(\dfrac{(D^{\mathrm{OT}})^2}{4k_e^2} + \dfrac{C^{\mathrm{OT}}}{k_e} + \dfrac{V_{\mathrm{OT}}(T_1)}{k_e(T_2-T_1)}\Big)^{1/2}$.

*Proof.* As in Theorem 1 (Section 5). The Dini comparison lemma, applied with the continuous, $V$-independent right-hand side, gives the integrated form. □

**Remark B.4** (what OT buys and what it does not)**.**

- **Globally, the OT constant is not smaller than the classical one.** Take the test box: $a_{\max} = 8$, $c_{\max} = 4$, $s_{\min} = 0.3$, $s_{\max} = 3$, $G = \mathrm{diag}(2, 1, 0.5)$ and $F_{\max} := a_{\max}$. Compare constants on the same per-kernel admissible set.

| constant | value |
|---|---|
| classical Lemma 2 (Section 5), $F_{\max} + 3a_{\max} + 2a_{\max}c_{\max}J_c + a_{\max}(s_{\max} - s_{\min})J_s$ | 214, for every $m$ |
| OT linear branch, crude form $L_0 + L_1\lambda_G^{1/2}\Delta$ | 533 |
| OT linear branch, block split through $W$ | 382 |
| OT linear branch, bounded per particle by $\|e\|\big(2a_{\max} + \|\tilde{a}_i\| + a_{\max}(J_c\|\tilde{c}_i\| + J_s\|\tilde{s}_i\|)\big)$ | 214 |

  Bounded per particle, the OT constant equals the classical one. An earlier comparison figure of "101–138" used a joint centre ball of radius $c_{\max}$. That radius is $\sqrt{m}$ times smaller than the radius $\sqrt{m}\,c_{\max}$ of the ball that circumscribes the per-kernel product of Theorem A.

- **Locally, the excess disturbance is quadratic in the matched error** (Corollary B.5).
- **Labels.** A labelled theory can make the same local statement only for a labelling fixed in advance, and only while the optimal assignment does not switch. (B) holds uniformly, with the minimum over $\pi$ taken at each $t$.
- $m$**-uniformity** is a property of the normalisation (Remark A.1), not of OT.

**Corollary B.5** (conditional bound)**.** *Along a solution of (L), suppose* $\sup_{[T_1,T_2]} W(\mu_t, \mu_t^*) \le \bar{\eta}$*. This quantity is available in oracle simulations, since it requires* $\mu_t^*$*, or it can be controlled a priori by Theorem E. With* $\bar{D} := \varepsilon_{\max} + \min\{K\lambda_G\bar{\eta}^2,\ L_0 + L_1\lambda_G^{1/2}\bar{\eta}\}$*,*

$$\mathrm{RMS}_{[T_1,T_2]}(\|e\|) \le \frac{\bar{D}}{2k_e} + \Big(\frac{\bar{D}^2}{4k_e^2} + \frac{V_*\bar{\eta}}{k_e} + \frac{V_{\mathrm{OT}}(T_1)}{k_e(T_2-T_1)}\Big)^{1/2}. \tag{11.1}$$

*As* $\bar{\eta} \to 0$ *and* $T_2 \to \infty$*, this tends to* $\varepsilon_{\max}/k_e$*, the value of the frozen-parameter error channel.*

Variant for $(\mathrm{L}_\sigma)$. *Take a solution of* $(\mathrm{L}_\sigma)$ *in the sense of Section 12 with* $\sup_{[T_1,T_2]} W(\mu_t, \mu_t^*) \le \bar{\eta}$ *and* $4\bar{\eta}(b + \bar{\eta}) < \Delta_{\mathrm{gap}}$*. For example,* $\bar{\eta}$ *can come from Theorem C(iv). Then, with* $\bar{D} := \varepsilon_{\max} + K\lambda_G\bar{\eta}^2$*,*

$$\mathrm{RMS}_{[T_1,T_2]}(\|e\|) \le \frac{\bar{D}}{2k_e} + \Big(\frac{\bar{D}^2}{4k_e^2} + \frac{V_*\bar{\eta} + \sigma_m b^2/4}{k_e} + \frac{V_{\mathrm{OT}}(T_1)}{k_e(T_2-T_1)}\Big)^{1/2}. \tag{11.2}$$

*Proof of the variant.* Apply Theorem C(iii) with $C = b(t)^2 \le b^2$, and use $-\sigma_m\eta^2 + \sigma_m\eta b \le \sigma_m b^2/4$. Without the gap condition, replace $b^2$ by $(b + 2\bar\eta)^2$ (from $\sqrt{C} \le b + 2\eta$ in C(ii)), or by $b^2 + 4\bar\eta R$, with $R := \Delta$ as in Theorem C. □

As $\bar\eta \to 0$ and $T_2 \to \infty$ the variant tends formally to $\varepsilon_{\max}/(2k_e) + \big(\varepsilon_{\max}^2/(4k_e^2) + \sigma_m b^2/(4k_e)\big)^{1/2}$. Along ($\mathrm{L}_\sigma$) with $b > 0$, the measured $\bar\eta$ stays of order $b$ (about $0.9b$ in the test instance), so the bound is first order in $b$. The plain bound of Corollary B.5 is false along ($\mathrm{L}_\sigma$). In the test instance, every hypothesis of C(iv) is asserted and $\sup W \le \bar\eta$ is measured. There, the bound of Corollary B.5 is exceeded by more than 3×, while the variant holds (`tests/test_transport.py::test_corollary_B5_fails_along_lsigma_and_its_leak_variant_holds`). Independent numerical checks found violations of 3.8–8.3×.

**Status.** The bound along (L) is not checked in the repository; its failure along ($\mathrm{L}_\sigma$) and the ($\mathrm{L}_\sigma$) variant are checked (Appendix A).

**Corollary B.6** (approximation error as quantisation error)**.** *Suppose $f = \int \Phi \,\mathrm{d}\mu_\infty^* + \varepsilon_\infty$ for a probability measure $\mu_\infty^*$ on $\Theta_K^\varepsilon$, with $\|\varepsilon_\infty\| \le \bar\varepsilon_\infty$. Then for every $m$-atom $\mu_m^*$ supported in $\Theta_K^\varepsilon$,*

$$\varepsilon_{\max}(m) \le \bar\varepsilon_\infty + L_1 \lambda_G^{1/2}\, W(\mu_m^*, \mu_\infty^*).$$

**Status.** Not checked in the repository (Appendix A).

*Proof.* For any coupling $\gamma$,

$$\begin{aligned}\Big\| \int \Phi \,\mathrm{d}(\mu_m^* - \mu_\infty^*) \Big\| &\le \int \|\Phi(\vartheta) - \Phi(\vartheta')\| \,\mathrm{d}\gamma \le L_1 \lambda_G^{1/2} \int \|\vartheta - \vartheta'\|_{G^{-1}} \,\mathrm{d}\gamma \\ &\le L_1 \lambda_G^{1/2} \Big( \int \|\vartheta - \vartheta'\|_{G^{-1}}^2 \,\mathrm{d}\gamma \Big)^{1/2}.\end{aligned}$$

Minimise over $\gamma$. The block constant $\big(g_a + a_{\max}^2 (J_c^2 g_c + J_s^2 g_s)\big)^{1/2}$ may replace $L_1 \lambda_G^{1/2}$. □

**Rates.**

(a) **First order.** Let $\mu_\infty^*$ be compactly supported and $p \ge 3$. Then there are equal-weight $\mu_m^*$, with atoms (cell barycentres) in $\Theta_K^\varepsilon$, such that $W(\mu_m^*, \mu_\infty^*) = O(m^{-1/p})$. This follows from an equal-mass dyadic decomposition (Chevallier 2018 [13]). For $p > 4$, i.i.d. atoms also achieve the rate in expectation (Fournier–Guillin 2015 [30]). If $\mu_\infty^*$ has a non-trivial absolutely continuous part, the order cannot be improved, even with free weights (Zador; Graf–Luschgy [38, Thm. 6.2 with Lemma 3.4]).

(b) **Second order.** Suppose $\mu_m^*$ is a centroid quantiser for an optimal plan; a best equal-weight approximation is one. Then the first-order term cancels, and by the Hessian bound (1) of Section 9 and convexity of $\Theta_K^\varepsilon$,

$$\varepsilon_{\max}(m) \le \bar\varepsilon_\infty + \min\big\{ K \lambda_G W^2,\ L_1 \lambda_G^{1/2} W \big\} = \bar\varepsilon_\infty + O(m^{-2/p}).$$

(c) **Free of $p$.** I.i.d. atoms drawn from $\mu_\infty^*$ give, with positive probability, an $m$-atom ideal with

$$\varepsilon_{\max}(m) \le \bar\varepsilon_\infty + O\Big( a_{\max} \big( n \log(m \cdot \operatorname{diam} X)/m \big)^{1/2} + a_{\max}/(s_{\min}\sqrt{m}) \Big).$$

This is Maurey's empirical method; the constants are not optimised.

The exponent $1/p$ is therefore a property of controlling $\varepsilon_{\max}$ through $W$, as Theorem D(e) requires. It is not a property of the approximation error itself.

**Remark B.7** (continuum form, via couplings)**.** Let $\mu_0 \in \mathcal{P}(\Theta_K)$ and $\mu_t = (X_t)_{\#}\mu_0$, and let $\mu_t^*$ be an absolutely continuous ideal curve in $(\mathcal{P}(\Theta_K^\varepsilon), W)$ with velocity field $w_t$. Set $V_* := \operatorname{ess\,sup}_t \|w_t\|_{L^2(\mu_t^*;G^{-1})}$. Its smallest admissible value is the metric derivative $|\dot\mu_t^*|_W$ (AGS [2, Thm. 8.3.1], after the isometry $\vartheta \mapsto G^{-1/2}\vartheta$). Then (B) holds for a.e. $t$.

**Status.** Not checked in the repository (Appendix A).

*Sketch.*

1. Push an optimal plan $\gamma_t$ forward by $(X_{t,t+h}, Y_{t,t+h})$. Here $Y$ is the flow of $w$, or the superposition measure of AGS [2, Thm. 8.2.1] when $w$ is not Lipschitz.
2. This couples $(\mu_{t+h}, \mu_{t+h}^*)$, so $D^+\big(\tfrac12 W^2\big) \le \int \langle \vartheta - \vartheta',\, v_t(\vartheta) - w_t(\vartheta')\rangle_{G^{-1}}\, \mathrm{d}\gamma_t$.
3. Integrate (P) and Lemma B.2 against $\gamma_t$. □

# 12 Theorem C: optimal-transport $\sigma$-modification

Let $\mu^0 = \frac1m \sum_k \delta_{\vartheta_k^0}$ be a prior configuration with $\vartheta_k^0 \in \Theta_K^\varepsilon$. The *OT $\sigma$-modified law* is

$$\begin{aligned} \dot\vartheta_i &= \mathrm{Proj}_K\big(\vartheta_i,\ G\nabla g_t(\vartheta_i) - \sigma_m(\vartheta_i - \vartheta^0_{\tau(i)})\big), \\ \tau(t) &\in \operatorname*{argmin}_{\rho}\ \frac1m \sum_i \|\vartheta_i - \vartheta^0_{\rho(i)}\|^2_{G^{-1}}. \end{aligned} \tag{$\mathrm{L}_\sigma$}$$

**Solution concept.** The right-hand side of ($\mathrm{L}_\sigma$) is discontinuous at assignment ties.

- **Selection solutions.** A solution is an absolutely continuous $(x, \vartheta)$ such that, for a.e. $t$, $\dot\vartheta_i = \mathrm{Proj}_K\big(\vartheta_i,\ G\nabla g_t(\vartheta_i) - \sigma_m(\vartheta_i - \vartheta^0_{\tau(t)(i)})\big)$ for some $\tau(t)$ optimal at $\vartheta(t)$. This is what `src/core/transport.py` in the accompanying repository integrates.
- **Filippov solutions.** These exist by Filippov's theorem, since the right-hand side is measurable and locally bounded. But they include sliding motions on repelling ties, along which (i) below fails. For example, take ideal = prior = $\{a = 0,\ a = 1\}$, both particles at $a = \frac12$ and $e \equiv 0$: this is a stationary Filippov solution.
- **What holds for which class.**
  - Parts (ii)–(iv) hold for every Filippov solution, with $C$ taken as the maximum over optimal $\tau$.
  - Part (i) and the exact decay hold for selection solutions.
  - For Filippov solutions, (i) weakens as stated below.
- **Regularity and existence.** $V_{\mathrm{OT}}$ and $Z$ are absolutely continuous along solutions, and every inequality holds for a.e. $t$. Existence of selection solutions is proved for the pure leak (Lemma C.3). In general it is assumed: two-way ties are never attracting, by monotonicity of $\mathrm{Proj}_K$, but multi-way ties, the degenerate ramp $h = 1$ and accumulating switch times are not handled.

**Notation.**

- $c_\rho(\vartheta) := \frac1m \sum_i \|\vartheta_i - \vartheta^0_{\rho(i)}\|^2_{G^{-1}}$.
- $\eta := W(\mu_t, \mu_t^*)$, and $R := \Delta = \mathrm{diam}_{G^{-1}}(\Theta_K)$.
- $b(t) := W(\mu_t^*, \mu^0)$, and $b := \sup_t b(t)$.

- $\Delta_{\mathrm{gap}}(t) := \min\{c_\rho(\vartheta^*(t)) - b(t)^2 : c_\rho(\vartheta^*(t)) > b(t)^2\}$, with $\min\emptyset := +\infty$; and $\Delta_{\mathrm{gap}} := \inf_t \Delta_{\mathrm{gap}}(t)$.
- $A_{\tau\tau'} := \frac{1}{m}\sum_i \|\vartheta^0_{\tau(i)} - \vartheta^0_{\tau'(i)}\|^2_{G^{-1}}$.

**Theorem C.** *Under* (S)*, along every solution of* $(\mathrm{L}_\sigma)$ *in the sense above:*

(i) Dissipation toward the prior (selection solutions). *Let* $Z := \frac{1}{2}W(\mu_t,\mu^0)^2$ *and* $\Gamma := \big(\frac{1}{m}\sum_i \|G\nabla g_t(\vartheta_i)\|^2_{G^{-1}}\big)^{1/2} \le \lambda_G^{1/2} L_1 \|e\|$*. Then:*

- $D^+Z \le -2\sigma_m Z + (2Z)^{1/2}\Gamma(t)$*;*
- *hence* $W(\mu_t,\mu^0) \le \max\{W(\mu_0,\mu^0),\ \sup\Gamma/\sigma_m\}$*;*
- *for the pure-leak flow* ($\Gamma \equiv 0$)*, the decay* $W(\mu_t,\mu^0) = e^{-\sigma_m t}W(\mu_0,\mu^0)$ *is exact for every* $G \succ 0$ *(Lemma C.3).*

*For a Filippov solution with weights* $\theta_k$ *on tied optimal assignments* $\tau_k$*, the bound weakens to*

$$D^+Z \le -2\sigma_m Z + \frac{\sigma_m}{2}\min_\tau \sum_k \theta_k A_{\tau\tau_k} + (2Z)^{1/2}\Gamma,$$

*with the minimum over assignments* $\tau$ *optimal at* $t$*.*

(ii) Induced coupling. *Let* $\pi^*$ *be optimal from* $\mu$ *to* $\mu^*$*, and let* $\tau$ *be the optimal assignment from* $\mu$ *to* $\mu^0$ *that the law uses at* $t$ *(for Filippov solutions, any optimal* $\tau$*). Set* $C := \frac{1}{m}\sum_i \|\vartheta^*_{\pi^*(i)} - \vartheta^0_{\tau(i)}\|^2_{G^{-1}}$*. Then*

$$C \le b(t)^2 + 4\eta R \qquad \text{and} \qquad \sqrt{C} \le b(t) + 2\eta,$$

*and* $C = b(t)^2$ *whenever* $4\eta(b(t)+\eta) < \Delta_{\mathrm{gap}}$*.*

(iii) Lyapunov inequality. *With* $C$ *as in* (ii) *(for Filippov solutions, its maximum over optimal* $\tau$*),*

$$D^+V_{\mathrm{OT}} \le -k_e\|e\|^2 + \|e\|\big(\varepsilon_{\max} + K\lambda_G\eta^2\big) + V_*\eta - \sigma_m\eta^2 + \sigma_m\eta C^{1/2}.$$

*Under* $4\eta(b+\eta) < \Delta_{\mathrm{gap}}$ *this becomes*

$$D^+V_{\mathrm{OT}} \le -k_e\|e\|^2 + \|e\|\big(\varepsilon_{\max} + K\lambda_G\eta^2\big) + V_*\eta - \frac{\sigma_m}{2}\eta^2 + \frac{\sigma_m}{2}b^2.$$

(iv) Local exponential estimate. *Suppose* $4\bar\eta(b+\bar\eta) < \Delta_{\mathrm{gap}}$ *and* $8K^2\lambda_G^2\bar\eta^2 \le k_e\sigma_m$*. Set* $\alpha := \min(k_e, \sigma_m/2)$ *and* $\beta := \varepsilon^2_{\max}/k_e + (\sigma_m/2)b^2 + 2V_*^2/\sigma_m$*, and suppose* $\max\{V_{\mathrm{OT}}(0), \beta/\alpha\} < \bar\eta^2/2$*. Then:*

- $\{V_{\mathrm{OT}} \le c\}$ *is forward invariant for every* $c \in \big[\max\{V_{\mathrm{OT}}(0), \beta/\alpha\},\ \bar\eta^2/2\big)$*;*
- $V_{\mathrm{OT}}(t) \le e^{-\alpha t}V_{\mathrm{OT}}(0) + \beta/\alpha$*;*
- *on this set* $\|e\| < \bar\eta$*, so* (S5) *holds with* $X := \bigcup_t\{x : \|x - x_m(t)\| \le \bar\eta\}$*.*

(v) The labelled leak. *Define* $b_{\mathrm{lab}}(\pi)(t) := \big(\frac{1}{m}\sum_i \|\vartheta^*_{\pi(i)}(t) - \vartheta^0_i\|^2_{G^{-1}}\big)^{1/2}$ *and* $b_{\mathrm{lab}}(\pi) := \sup_t b_{\mathrm{lab}}(\pi)(t)$*.*

- *For any fixed* $\pi$*, the labelled* $\sigma$*-modification (leak toward* $\vartheta^0_i$*) satisfies the second display of* (iii) *at every state, with no gap condition. Here* $b$ *is replaced by* $b_{\mathrm{lab}}(\pi)$*,* $V_{\mathrm{OT}}$ *by* $V_\pi$*, and* $\eta$ *by the* $\pi$*-labelled error.*
- *Choosing* $\pi$ *to minimise* $b_{\mathrm{lab}}(\pi)$ *gives the bias* $\min_\pi \sup_t b_{\mathrm{lab}}(\pi)(t) \ge b$*. Equality holds when one assignment is optimal for* $(\mu^*_t, \mu^0)$ *at every* $t$*, for example a static ideal, or whenever* $\Delta_{\mathrm{gap}} > 0$*.*

- *The labelled leak does **not** satisfy* (iii) *for* $V_{\mathrm{OT}}$ (*Proposition C.5*).

**Status.** Proved and checked; the instance used to check (iv) satisfies every hypothesis of (iv). Not checked in the repository: the Filippov form of (i) and the inequality of (v). Novelty (literature audit): novel as a control construction; C(i) and the obstruction are known [80], [9].

Checked by: `tests/test_transport.py`:
(i) and Lemma C.3 (unprojected pure leak): `test_theorem_C_i_exact_dissipation_pure_leak`;
(ii): `test_lemma_C_ii_induced_coupling_bound`;
(iv): `test_theorem_C_iv_exponential_estimate`.
(v) and Proposition C.5, in `experiments/`: `exp_ot_sigma_mod.py` and `exp_sigma_mod_transient.py`.

**Scope.**

- **Finite gap.** If $\Delta_{\mathrm{gap}} < \infty$, that is, not all ideal–prior assignments tie, then $\Delta_{\mathrm{gap}} \le 2R^2/m$ always. Indeed, if every transposition of an optimal $\rho_0$ were cost-neutral, the matrix $S(i,j) := B(i,j) - (B(i,i) + B(j,j))/2$ built from $B(i,j) := \langle \vartheta_i^*, \vartheta^0_{\rho_0(j)} \rangle_{G^{-1}}$ would be antisymmetric, and every assignment would tie.
  - Hence (iv) needs $\bar\eta < \sqrt{\Delta_{\mathrm{gap}}}/2 \le R/\sqrt{2m}$.
  - $\beta/\alpha < \bar\eta^2/2$ forces $b < \bar\eta/\sqrt{2}$.
  - So (iv) is a near-prior local result of size $O(m^{-1/2})$ in $W$, and $O(1/m)$ in $V_{\mathrm{OT}}$.
- **Infinite gap.** If $\Delta_{\mathrm{gap}} = +\infty$, for example for a single-atom prior (the classical $\sigma$-modification), the gap condition is void. Then (iv) is $m$-uniform; $\bar\eta$ is then fixed by $8K^2\lambda_G^2\bar\eta^2 \le k_e\sigma_m$ alone.
- **Switching.** $\Delta_{\mathrm{gap}} > 0$ forces the set of optimal ideal–prior assignments to be constant in $t$. If that set switches along a moving ideal, then $\Delta_{\mathrm{gap}} = 0$, and the second clause of (ii), the second display of (iii) and all of (iv) are vacuous.

*Proof. (i)* For a selection solution, at a.e. $t$ the law uses an optimal $\tau(t)$, and $Z \le Z_\tau := \frac{1}{2}c_\tau$ with equality at $t$. By Lemma B.1 it suffices to bound $\dot Z_\tau$. Apply (P) with $\vartheta' = \vartheta^0_{\tau(i)}$ and Cauchy–Schwarz in the $G^{-1}$ inner product: $\dot Z_\tau \le (2Z)^{1/2}\Gamma - 2\sigma_m Z$. Write $y := (2Z)^{1/2}$.

- Where $Z > 0$, $D^+ y \le -\sigma_m y + \Gamma$.
- At $Z = 0$, $D^+ y \le \big(\frac{1}{m}\sum \|\dot\vartheta_i\|^2_{G^{-1}}\big)^{1/2} \le \Gamma$, by Lemma P.1 (the leak term vanishes there).
- $\Gamma \le \lambda_G^{1/2} L_1 \|e\|$ by Lemma B.2(b).

The pure-leak case is Lemma C.3, and the projection is inactive there. In the layer, $a^\top(a^0 - a) \le \|a\|(\|a^0\| - \|a\|) \le 0$, and similarly for $c$. For $s$, the leak points away from the nearby face.

For Filippov velocities, the leak part averaged over the tie is

$$-\sigma_m \frac{1}{m} \sum_i \sum_k \theta_k \langle \vartheta_i - \vartheta^0_{\tau(i)},\, \vartheta_i - \vartheta^0_{\tau_k(i)} \rangle_{G^{-1}}.$$

By polarisation, and since $c_\tau = c_{\tau_k} = 2Z$ for tied optimal assignments, this equals $-2\sigma_m Z + (\sigma_m/2)\sum_k \theta_k A_{\tau\tau_k}$. Property (P) holds termwise for the convex combination. The bound is tight at the tie equilibrium of the solution-concept example ($0 = -\sigma_m/4 + \sigma_m/4$).

*(ii)* Relabel so that $\frac{1}{m}\sum \|\vartheta_i - \vartheta_i^*\|^2_{G^{-1}} = \eta^2$. For any $\rho$:

- $|c_\rho(\vartheta) - c_\rho(\vartheta^*)| \le 2\eta R$;
- $|\sqrt{c_\rho(\vartheta)} - \sqrt{c_\rho(\vartheta^*)}| \le \eta$, by the reverse triangle inequality for the $(1/m)$-weighted $G^{-1}$ norm.

Let $\rho_0$ be optimal for $(\vartheta^*(t), \vartheta^0)$. Then

$$C = c_\tau(\vartheta^*) \le c_\tau(\vartheta) + 2\eta R \le c_{\rho_0}(\vartheta) + 2\eta R \le b(t)^2 + 4\eta R,$$
$$\sqrt{C} \le \sqrt{c_\tau(\vartheta)} + \eta \le \sqrt{c_{\rho_0}(\vartheta)} + \eta \le b(t) + 2\eta.$$

So $C - b(t)^2 \le 4\eta(b(t) + \eta) < \Delta_{\mathrm{gap}} \le \Delta_{\mathrm{gap}}(t)$. No assignment cost at time $t$ lies strictly between $b(t)^2$ and $b(t)^2 + \Delta_{\mathrm{gap}}(t)$, so $C = b(t)^2$.

*(iii)* Apply Theorem B with $y = \nabla g_t - \sigma_m G^{-1}(\vartheta_i - \vartheta^0_{\tau(i)})$ in (P). With $\tilde{\vartheta}_i := \vartheta_i - \vartheta^*_{\pi^*(i)}$ as in the proof of Theorem B, the extra term is

$$-\sigma_m \frac{1}{m} \sum \langle \tilde{\vartheta}_i,\, \tilde{\vartheta}_i + \vartheta^*_{\pi^*(i)} - \vartheta^0_{\tau(i)} \rangle_{G^{-1}} \le -\sigma_m \eta^2 + \sigma_m \eta C^{1/2}.$$

Under the gap condition, (ii) gives $C^{1/2} = b(t) \le b$, and $\sigma_m \eta b \le (\sigma_m/2)(\eta^2 + b^2)$.

*(iv)* Apply Young's inequality three times:

- $\|e\|\varepsilon_{\max} \le (k_e/4)\|e\|^2 + \varepsilon^2_{\max}/k_e$;
- $\|e\| K \lambda_G \eta^2 \le (k_e/4)\|e\|^2 + (\sigma_m/8)\eta^2$ on $\{\eta \le \bar{\eta}\}$;
- $V_* \eta \le (\sigma_m/8)\eta^2 + 2V_*^2/\sigma_m$.

This gives $D^+ V_{\mathrm{OT}} \le -\alpha V_{\mathrm{OT}} + \beta$. Each set $\{V_{\mathrm{OT}} \le c\}$ with $\beta/\alpha < c < \bar{\eta}^2/2$ lies inside $\{\eta \le \bar{\eta}\}$ and has a negative boundary derivative. Intersect over $c > \beta/\alpha$ for the endpoint, and apply the comparison lemma.

*(v)* Steps (a)–(c) of Theorem B hold for any fixed $\pi$, so use the classical completion of squares for $V_\pi$.

- *Equality case.* If $\Delta_{\mathrm{gap}} > 0$, the costs minus $b(t)^2$ take values in $\{0\} \cup [\Delta_{\mathrm{gap}}, \infty)$ and are continuous in $t$, so the optimal set is constant.
- *Strict case.* If the optimal ideal–prior assignment changes along the ideal, the inequality can be strict. For example, $a^*_1(t) = t$ and $a^*_2(t) = 1 - t$ on $[0, 1]$, with prior $a \in \{0, 1\}$, give $b = 0.5$ against 1.0.
- *Not always strict.* For $a^*(t) = (u(t),\, 1 - u(t))$ with $u(t) = -0.025 + 0.575 \sin t$, the optimal assignment alternates between the identity and the swap, yet $\min_\pi \sup_t b_{\mathrm{lab}}(\pi)(t) = \sup_t b_{\mathrm{lab}}(\mathrm{id})(t) = 0.6 = b$ (computed numerically).

The negative statement is Proposition C.5. □

**Lemma C.3** (no switching along the pure leak; exact decay for any $G \succ 0$)**.** *Let $\tau$ be optimal for $(\mu_{\mathrm{init}}, \mu^0)$ and $\dot{\vartheta}_i = -\sigma_m(\vartheta_i - \vartheta^0_{\tau(i)})$. Then $\tau$ stays optimal and $W(\mu_t, \mu^0) = e^{-\sigma_m t} W(\mu_{\mathrm{init}}, \mu^0)$.*

*Proof.* Put $d_i := \vartheta_i(0) - \vartheta^0_{\tau(i)}$, so $\vartheta_i(t) - \vartheta^0_{\tau(i)} = \lambda d_i$ with $\lambda = e^{-\sigma_m t}$. For any $\rho$, with $q_i := \vartheta^0_{\tau(i)} - \vartheta^0_{\rho(i)}$,

$$c_\rho(t) - c_\tau(t) = \frac{2\lambda}{m} \sum \langle d_i,\, q_i \rangle_{G^{-1}} + \frac{1}{m} \sum \|q_i\|^2_{G^{-1}}.$$

This is affine in $\lambda$. It is non-negative at $\lambda = 1$ by optimality and at $\lambda = 0$ trivially, hence on all of $[0, 1]$.

At an exact tie at $t = 0$, the chosen assignment becomes strictly optimal immediately against every $\rho$ with $A_\rho := \frac{1}{m} \sum \|q_i\|^2 > 0$. Assignments with $A_\rho = 0$ induce the same velocity. So no selection solution of the pure leak switches. Filippov sliding on the tie is a different solution, excluded by the solution concept. □

Checked by: `test_theorem_C_i_exact_dissipation_pure_leak` in `tests/test_transport.py`: zero switches, and exact decay to integrator accuracy.

**Proposition C.5** (the labelled leak reshuffles)**.** *There are configurations with* $b = b_{\mathrm{lab}}(\mathrm{id}) = 0$ *and* $e = 0$ *at which the labelled leak gives* $D^+V_{\mathrm{OT}} > 0$*.*

*Example.* Take $n = 1$, $G = I$ and $a_{\max} - \varepsilon_K \geq 1$. Let ideal = prior = $\{a = 0,\ a = 1\}$, with the other coordinates equal and in the inner set, and let the network weights be $(1 - \delta, \delta)$ with $0 < \delta < \frac{1}{2}$.

- The optimal assignment to the ideal is the swap.
- The labelled leak moves each particle toward the *other* ideal atom.
- So $D^+V_{\mathrm{OT}} = \sigma_m(\delta - \delta^2) > 0$. □

From a relabelled start, the labelled leak moves every kernel in a non-trivial cycle of the relabelling to its prior position. In the reshuffling experiment (`results/ot_sigma_mod/transient.md` in the accompanying repository):

- the steady states coincide;
- the labelled leak's particle path is 5.8× longer (2.26 against 0.39);
- the early RMS$\|e\|$ is only about 4% larger (0.0771 against 0.0743 on $[0, 5]$).

Theorem F does not price this motion. With a leak, Theorem F bounds transport from above by tracking error plus a leak term, and the leak can pay for all of it. With $\delta = 0$ above, $e \equiv 0$ while every particle crosses the domain.

Checked by: the reshuffling transient, `experiments/exp_sigma_mod_transient.py`. The value $D^+V_{\mathrm{OT}} = \sigma_m(\delta - \delta^2)$ is not checked in the repository.

**Remark C.6** (the obstruction is known)**.**

- **Non-convexity.** For some $\nu$, for example a two-atom $\nu$ in dimension $\geq 2$, $\frac{1}{2}W^2(\cdot, \nu)$ is not $\lambda$-convex along geodesics for any $\lambda$ (Ambrosio–Gigli–Savaré [2, Example 9.1.5]; Santambrogio [80, §4.5]). It is 1-semiconcave in every dimension [2, Thm. 7.3.2]. For $\nu = \delta_{\vartheta^0}$ it is a potential energy, and 1-convex.
- **Generalised geodesics.** $\frac{1}{2}W^2(\cdot, \nu)$ is 1-convex along generalised geodesics with base $\nu$ ([2, Lemma 9.2.1]; Carlen–Craig [9]).
- **Label-free leaks.**
  - A label-free leak that is $\lambda$-convex ($\lambda > 0$) on the lift $\mathbb{R}^{mp}$ has a unique, permutation-invariant minimiser, at which all particles coincide. So it cannot vanish at a $\mu^0$ with distinct atoms.
  - 0-convex leaks that vanish at $\mu^0$ exist, for example toward $\operatorname{conv}\operatorname{supp}\mu^0$. They do not single out $\mu^0$.

So the trade-off reflects this geometry. On one side is a global inequality for a labelled $V_\pi$, which is large at relabelled starts (hence the reshuffling of Proposition C.5). On the other is a local, gap-protected inequality for the label-free $V_{\mathrm{OT}}$.

# 13 Theorem C′: the OT leak toward a prior with a different number of atoms

Theorem C needs a prior with as many atoms as the network. Here the prior is any $\nu \in \mathcal{P}_2(\Theta_K^\varepsilon)$, for example the empirical measure of a network of a different size (the transfer problem of Section 26, branch 8), and the leak points at the barycentre of the prior mass that the optimal plan assigns to each particle. The dissipation toward the prior survives up to the quantisation

error, the fixed points of the pure leak are the equal-mass centroidal configurations of $\nu$ (the leak is Lloyd's algorithm in continuous time), and the Lyapunov bias is measured by the distance of the ideal from the *centroids* of its prior cells, not by $W(\mu^*, \nu)$.

**The law.** Let $\nu \in \mathcal{P}_2(\Theta_K^\varepsilon)$, for instance $\nu = \frac{1}{m_0}\sum_{j=1}^{m_0} \delta_{\vartheta_j^0}$ with any $m_0 \geq 1$. For $\mu = \frac{1}{m}\sum \delta_{\vartheta_i}$, let $\gamma$ be a vertex optimal plan from $\mu$ to $\nu$ for the cost $\|\cdot\|_{G^{-1}}^2$ (for atomic $\nu$ a vertex of the transportation polytope; for general $\nu$ any optimal plan), and let

$$\bar{T}_i := m \int \vartheta' \, \mathrm{d}\gamma(\vartheta_i, \vartheta')$$

(the barycentric projection of $\gamma$ at $\vartheta_i$; for atomic $\nu$, $\bar{T}_i = m\sum_j \gamma_{ij}\vartheta_j^0$),

$$V_{\mathrm{bar}}^2 := \frac{1}{m}\sum_i \|\vartheta_i - \bar{T}_i\|_{G^{-1}}^2, \qquad \mathrm{Var}_\gamma := \int \|\vartheta' - \bar{T}(\vartheta)\|_{G^{-1}}^2 \, \mathrm{d}\gamma(\vartheta, \vartheta'),$$
$$\text{so that} \qquad W(\mu,\nu)^2 = V_{\mathrm{bar}}^2 + \mathrm{Var}_\gamma.$$

The **barycentric OT leak** is

$$\dot{\vartheta}_i = \mathrm{Proj}_K\big(\vartheta_i,\ G\nabla g_t(\vartheta_i) - \sigma_m(\vartheta_i - \bar{T}_i)\big). \tag{$\mathrm{L}_{\sigma,\mathrm{bar}}$}$$

For atomic $\nu$ the plan is a vertex of the transportation polytope (row sums $1/m$, column sums $1/m_0$), computed by linear programming (`transport.optimal_plan`, `transport.barycentric_projection`); for $m_0 = m$ it is a permutation and ($\mathrm{L}_{\sigma,\mathrm{bar}}$) is ($\mathrm{L}_\sigma$). The right-hand side is discontinuous where the optimal plan is not unique, and here, unlike Section 12 (Lemma C.3), the plan does switch along solutions (numerics below), so two solution concepts are used. *Vertex-selection solutions* are absolutely continuous $\vartheta(t)$ with $\dot{\vartheta}_i = \mathrm{Proj}_K\big(\vartheta_i,\ G\nabla g_t - \sigma_m(\vartheta_i - \bar{T}_i^{\gamma(t)})\big)$ for a.e. $t$ and $\gamma(t)$ a vertex optimal plan at $\vartheta(t)$; the time-stepped implementation produces these (switches at step times), and their existence for all time is assumed. *Generalised solutions* allow $\gamma(t)$ to be any measurable selection of optimal plans. The optimal face $F(\vartheta)$ of the transportation LP is a compact convex polytope, upper semicontinuous in $\vartheta$ (finitely many vertices, continuous costs), and $\gamma \mapsto \bar{T}^\gamma$ is linear, so the set of generalised velocities is compact, convex and upper semicontinuous in $\vartheta$: it contains the Filippov regularisation of the vertex law, and generalised solutions exist on $[0,\infty)$ by the standard theorem for such differential inclusions (the projection is handled as in Section 12). Every statement below holds for both concepts, with $\bar{T}$ and $V_{\mathrm{bar}}$ computed from the plan the solution uses. Since $\Theta_K^\varepsilon$ is convex, $\bar{T}_i \in \Theta_K^\varepsilon$, so property (P) applies with $\vartheta' = \bar{T}_i$.

**Notation.** $\eta := W(\mu_t, \mu_t^*)$, $b := \sup_t W(\mu_t^*, \nu)$, and

$$b_{\mathrm{bar}}(t)^2 := \frac{1}{m}\sum_j \|\vartheta_j^*(t) - \bar{T}_j^*(t)\|_{G^{-1}}^2 = W(\mu_t^*,\nu)^2 - \mathrm{Var}_{\gamma^*(t)}, \qquad b_{\mathrm{bar}} := \sup_t b_{\mathrm{bar}}(t),$$

the *centroid defect of the ideal*, computed with an optimal plan $\gamma^*(t)$ from $\mu_t^*$ to $\nu$. $\Delta_{\mathrm{gap}}(t)$ is the gap between the optimal and the next vertex cost of the transportation polytope at $\mu_t^*$ ($\min\emptyset := +\infty$), and $\Delta_{\mathrm{gap}} := \inf_t \Delta_{\mathrm{gap}}(t)$.

**Theorem C′.** *Under* (S)*, along every vertex-selection or generalised solution of* ($\mathrm{L}_{\sigma,\mathrm{bar}}$)*:*

(i) Dissipation toward the centroidal configurations. *With* $Z := \frac{1}{2}W(\mu_t,\nu)^2$ *and* $\Gamma$ *as in Theorem C(i),*

$$D^+ Z \leq -\sigma_m V_{\mathrm{bar}}^2 + V_{\mathrm{bar}}\Gamma \leq -2\sigma_m\big(Z - \tfrac{1}{2}\mathrm{Var}_\gamma\big) + (2Z)^{1/2}\Gamma \qquad \textit{for a.e. } t.$$

*For the pure leak ($\Gamma \equiv 0$), $W(\mu_t, \nu)$ is non-increasing, strictly decreasing wherever the plan in use is not centroidal ($V_{\mathrm{bar}} > 0$), and at every $\omega$-limit point $\vartheta_\infty$ of a solution on the compact $\Theta_K$ some optimal plan $\gamma_\infty$ at $\vartheta_\infty$ is centroidal: $\vartheta_{\infty,i} = \bar{T}_i^{\gamma_\infty}$ for all $i$ — a vertex plan for a vertex-selection solution. Such configurations are the fixed points of the equal-mass Lloyd iteration $\vartheta \leftarrow \bar{T}^{\gamma}(\vartheta)$ with an optimal (vertex) plan; where the optimal plan is unique they are the critical points of the quantisation energy $\vartheta \mapsto \frac{1}{2}W^2\big(\frac{1}{m}\sum \delta_{\vartheta_i}, \nu\big)$, which is differentiable there, and the pure leak is its $G$-gradient flow with the factor $m$: $\dot{\vartheta}_i = -\sigma_m m\, G\, \partial_{\vartheta_i}\big[\frac{1}{2}W^2\big]$. They include degenerate configurations with coincident particles. For $m_0 = m$, $\mathrm{Var}_\gamma = 0$ and* (i) *is Theorem C(i).*

(ii) Induced coupling. *Let $\pi^*$ be optimal from $\mu$ to $\mu^*$, and let $\gamma$ be the plan the law uses at $t$. Put $C_{\mathrm{bar}} := \frac{1}{m}\sum_i \|\vartheta^*_{\pi^*(i)} - \bar{T}_i\|^2_{G^{-1}}$. Then*

$$\sqrt{C_{\mathrm{bar}}} \le b(t) + 2\eta \quad \textit{always},$$

*and, for atomic $\nu$, $C_{\mathrm{bar}} = b_{\mathrm{bar}}(t)^2$ whenever $4\eta(b(t)+\eta) < \Delta_{\mathrm{gap}}(t)$ and $\gamma^*(t)$ is unique.*

(iii) Lyapunov inequality. *With $C_{\mathrm{bar}}$ as in* (ii),

$$D^+ V_{\mathrm{OT}} \le -k_e\|e\|^2 + \|e\|\big(\varepsilon_{\max} + K\lambda_G\eta^2\big) + V_*\eta - \sigma_m\eta^2 + \sigma_m\eta\sqrt{C_{\mathrm{bar}}},$$

*and under the gap condition of* (ii) *the leak terms are at most $-(\sigma_m/2)\eta^2 + (\sigma_m/2)b^2_{\mathrm{bar}}$.*

(iv) Local exponential estimate. *For atomic $\nu$, Theorem C(iv) holds verbatim with $b$ replaced by $b_{\mathrm{bar}}$: under $4\bar{\eta}(b+\bar{\eta}) < \Delta_{\mathrm{gap}}$, uniqueness of $\gamma^*(t)$, and $8K^2\lambda_G^2\bar{\eta}^2 \le k_e\sigma_m$, with $\alpha := \min(k_e, \sigma_m/2)$ and $\beta := \varepsilon^2_{\max}/k_e + (\sigma_m/2)b^2_{\mathrm{bar}} + 2V_*^2/\sigma_m$, if $\max\{V_{\mathrm{OT}}(0), \beta/\alpha\} < \bar{\eta}^2/2$ then $V_{\mathrm{OT}}(t) \le e^{-\alpha t}V_{\mathrm{OT}}(0) + \beta/\alpha$ and $\{V_{\mathrm{OT}} \le c\}$ is forward invariant for $c \in \big[\max\{V_{\mathrm{OT}}(0), \beta/\alpha\},\ \bar{\eta}^2/2\big)$.*

(v) What the bias measures. *$b_{\mathrm{bar}} \le b$, with $b_{\mathrm{bar}} = 0$ exactly when the ideal is itself an equal-mass centroidal configuration of $\nu$. So a prior with many atoms whose centroidal $m$-quantisation coincides with the ideal anchors the network without bias, although $W(\mu^*, \nu) > 0$; a prior with the right number of atoms but the wrong positions has $b_{\mathrm{bar}} = b$.*

**Status.** Proved and checked: dissipation at every step and convergence of the pure leak to an equal-mass Lloyd fixed point, the coupling bound and plan stability, the Lyapunov inequality along the closed loop (`tests/test_barycentric.py`, `exp_barycentric_leak.py`). Novelty (literature audit): new as a control construction; the quantisation side is Lloyd/CVT theory; it reduces to Theorem C for $m_0 = m$.

**Scope of (ii)–(iv).** The gap condition $4\eta(b+\eta) < \Delta_{\mathrm{gap}}(t)$ asks the network to be closer to the ideal than the optimal vertex of the LP is to the next one. For a prior with many atoms, $\Delta_{\mathrm{gap}}$ is at most the cost of exchanging two atoms between two cells, which is $O(1/m_0)$, so the condition holds only in a small neighbourhood of the ideal; in the closed-loop experiment below $4\eta(0)(b+\eta(0))$ exceeds that bound by factors 83–175, and (iv) applies to no tested instance (the table reports the exchange bound on $\Delta_{\mathrm{gap}}$, $\eta(0)$ and $4\eta(0)(b+\eta(0))$ for each run). What the experiment confirms is the fixed-point structure, (i) and (v): the loop settles at the centroidal configuration of its prior, whose distance from the ideal is $b_{\mathrm{bar}}$.

*Proof. (i)* At a.e. $t$ the law uses an optimal plan $\gamma$, and for $h > 0$ the same plan couples $\mu_{t+h}$ and $\nu$, so $W(\mu_{t+h}, \nu)^2 \le \sum_{ij}\gamma_{ij}\|\vartheta_i(t+h) - \vartheta^0_j\|^2_{G^{-1}}$ (for general $\nu$, $\int\|\vartheta_i(t+h) - \vartheta'\|^2\,\mathrm{d}\gamma$). Differentiating the right-hand side at $h = 0$,

$$D^+ Z \le \sum_{ij}\gamma_{ij}\langle\vartheta_i - \vartheta^0_j,\, G^{-1}\dot{\vartheta}_i\rangle = \frac{1}{m}\sum_i\langle\vartheta_i - \bar{T}_i,\, G^{-1}\dot{\vartheta}_i\rangle,$$

because $\sum_j \gamma_{ij}(\vartheta_i-\vartheta_j^0)=\frac{1}{m}(\vartheta_i-\bar T_i)$. Apply (P) with $\vartheta'=\bar T_i\in\Theta_K^\varepsilon$ and $y=\nabla g_t-\sigma_m G^{-1}(\vartheta_i-\bar T_i)$: $\langle\vartheta_i-\bar T_i,\,G^{-1}\dot\vartheta_i\rangle\le\langle\vartheta_i-\bar T_i,\,\nabla g_t(\vartheta_i)\rangle-\sigma_m\|\vartheta_i-\bar T_i\|^2_{G^{-1}}$. Average and use Cauchy–Schwarz for the first term: $\frac{1}{m}\sum\langle\vartheta_i-\bar T_i,\,\nabla g_t\rangle\le V_{\mathrm{bar}}\Gamma\le(2Z)^{1/2}\Gamma$. The Pythagorean identity $W^2=V_{\mathrm{bar}}^2+\mathrm{Var}_\gamma$ follows from $\sum_j\gamma_{ij}(\vartheta_j^0-\bar T_i)=0$: for each $i$, $\sum_j\gamma_{ij}\|\vartheta_i-\vartheta_j^0\|^2=\sum_j\gamma_{ij}\big(\|\vartheta_i-\bar T_i\|^2+\|\bar T_i-\vartheta_j^0\|^2\big)$. For the pure leak, $Z$ is non-increasing and $D^+Z\le-\sigma_m V_{\mathrm{bar}}(t)^2$, with $V_{\mathrm{bar}}$ computed from the plan in use (the coupling inequality above holds for every optimal plan, vertex or not), so $Z$ converges and $\int_0^\infty V_{\mathrm{bar}}^2\,\mathrm{d}t\le Z(0)/\sigma_m<\infty$. $V_{\mathrm{bar}}$ itself is neither monotone nor continuous — it jumps at plan switches, upward as well as downward (numerics below) — so the $\omega$-limit statement rests on the stability of the LP instead. Let $\vartheta_\infty$ be an $\omega$-limit point and $F_\infty$ the set of optimal plans at $\vartheta_\infty$ (for a vertex-selection solution, its finitely many vertices). For a fixed plan $\gamma$ the barycentres $\bar T_i^\gamma$ do not depend on $\vartheta$, so $V_{\mathrm{bar}}^\gamma(\vartheta):=\big(\frac{1}{m}\sum\|\vartheta_i-\bar T_i^\gamma\|^2_{G^{-1}}\big)^{1/2}$ is 1-Lipschitz in $\vartheta$ for the norm $\big(\frac{1}{m}\sum\|\cdot\|^2_{G^{-1}}\big)^{1/2}$; and the optimal face is upper semicontinuous — the plan costs are continuous in $\vartheta$ and the polytope has finitely many vertices — so every plan optimal at a $\vartheta$ close enough to $\vartheta_\infty$ lies in $F_\infty$. Suppose no plan of $F_\infty$ is centroidal; then $\delta:=\min_{\gamma\in F_\infty}V_{\mathrm{bar}}^\gamma(\vartheta_\infty)>0$ (a minimum over a compact set), and for $\|\vartheta-\vartheta_\infty\|<\delta/2$, $V_{\mathrm{bar}}(\vartheta)\ge\delta/2$ whatever optimal plan is in use. The speed is bounded by $\sigma_m\cdot\mathrm{diam}_{G^{-1}}(\Theta_K)$ (Lemma P.1), so the solution, which enters the ball of radius $\delta/4$ around $\vartheta_\infty$ at arbitrarily large times, either stays in the ball of radius $\delta/2$ from some time on or spends a time $\ge\delta/(4\sigma_m\,\mathrm{diam})$ in it on each of infinitely many visits; either way $\int V_{\mathrm{bar}}^2\,\mathrm{d}t=\infty$, a contradiction. Hence some plan of $F_\infty$ — a vertex for vertex-selection solutions — is centroidal at $\vartheta_\infty$. (A two-way tie between vertex plans $\gamma$ and $\rho$ is not a sliding mode of the vertex law unless $\bar T^\gamma=\bar T^\rho$: the rate of change of $c_\rho-c_\gamma$ when following $\gamma$, plus that of $c_\gamma-c_\rho$ when following $\rho$, equals $(2\sigma_m/m)\sum_i\|\bar T_i^\gamma-\bar T_i^\rho\|^2_{G^{-1}}\ge0$, so at least one of the two plans stays optimal when followed. Ties between three or more plans are not analysed.) The gradient identity is the envelope theorem for the transportation LP at a configuration with a unique optimal plan: $\partial_{\vartheta_i}\big[\frac12\sum\gamma_{ij}\|\vartheta_i-\vartheta_j^0\|^2_{G^{-1}}\big]=\sum_j\gamma_{ij}G^{-1}(\vartheta_i-\vartheta_j^0)=\frac{1}{m}G^{-1}(\vartheta_i-\bar T_i)$, so $-\sigma_m(\vartheta_i-\bar T_i)=-\sigma_m m\,G\,\partial_{\vartheta_i}\big[\frac12W^2\big]$. (Where the plan is not unique, $\frac12W^2(\cdot,\nu)$ is not differentiable; the generalised velocities are then $-\sigma_m m\,G\,\xi$ with $\xi$ ranging over the barycentric projections of the optimal plans — the elements of the (sub)differential of $\frac12W^2(\cdot,\nu)$ at an atomic $\mu$, cf. Ambrosio–Gigli–Savaré [2, §10.4].)

*(ii)* Relabel so that $\frac{1}{m}\sum\|\vartheta_i-\vartheta_i^*\|^2_{G^{-1}}=\eta^2$. For any plan $\rho$ with row sums $1/m$, the plan cost $c_\rho(\vartheta):=\sum\rho_{ij}\|\vartheta_i-\vartheta_j^0\|^2_{G^{-1}}$ is the squared norm of $(\vartheta_i-\vartheta_j^0)$ in $L^2(\rho)$, so $|\sqrt{c_\rho(\vartheta)}-\sqrt{c_\rho(\vartheta^*)}|\le\eta$ by the triangle inequality (the labels of $\vartheta$ and $\vartheta^*$ are matched). The barycentric part is dominated by the plan cost: $C_{\mathrm{bar}}=\frac{1}{m}\sum\|\vartheta_i^*-\bar T_i\|^2\le c_\gamma(\vartheta^*)$ because $\sum_j\gamma_{ij}\|\vartheta_i^*-\vartheta_j^0\|^2=\frac{1}{m}\|\vartheta_i^*-\bar T_i\|^2+\sum_j\gamma_{ij}\|\bar T_i-\vartheta_j^0\|^2$. Let $\gamma^*$ be optimal for $(\vartheta^*,\nu)$. Then

$$\sqrt{C_{\mathrm{bar}}}\le\sqrt{c_\gamma(\vartheta^*)}\le\sqrt{c_\gamma(\vartheta)}+\eta\le\sqrt{c_{\gamma^*}(\vartheta)}+\eta\le\sqrt{c_{\gamma^*}(\vartheta^*)}+2\eta=b(t)+2\eta.$$

For the equality case: $c_\gamma(\vartheta^*)-b(t)^2\le(b+2\eta)^2-b^2=4\eta(b+\eta)<\Delta_{\mathrm{gap}}(t)$; the vertex $\gamma$ (a vertex by the choice of the law) has cost at $\vartheta^*$ strictly below the next vertex cost, so it is optimal at $\vartheta^*$; if $\gamma^*(t)$ is unique, $\gamma=\gamma^*(t)$ and $C_{\mathrm{bar}}=\frac{1}{m}\sum\|\vartheta_i^*-\bar T_i^*\|^2=b_{\mathrm{bar}}(t)^2$. [For a generalised solution whose plan at $t$ is not a vertex, the same chain applies to each vertex of the optimal face at $\vartheta$, so every such vertex is optimal at $\vartheta^*$, and uniqueness of $\gamma^*(t)$ again gives $\gamma=\gamma^*(t)$.]

*(iii)* Apply Theorem B with $y=\nabla g_t-\sigma_m G^{-1}(\vartheta_i-\bar T_i)$ in (P). The extra term is

$$-\sigma_m\frac{1}{m}\sum\langle\tilde\vartheta_i,\,\tilde\vartheta_i+\vartheta^*_{\pi^*(i)}-\bar T_i\rangle_{G^{-1}}\le-\sigma_m\eta^2+\sigma_m\eta\sqrt{C_{\mathrm{bar}}}.$$

Under the gap condition $\sqrt{C_{\mathrm{bar}}}=b_{\mathrm{bar}}(t)\le b_{\mathrm{bar}}$ and $\sigma_m\eta b_{\mathrm{bar}}\le(\sigma_m/2)(\eta^2+b_{\mathrm{bar}}^2)$.

*(iv)* is the proof of Theorem C(iv) with $b_{\mathrm{bar}}$ in place of $b$.

*(v)* $b_{\mathrm{bar}}(t)^2=W(\mu_t^*,\nu)^2-\mathrm{Var}_{\gamma^*(t)}\le W(\mu_t^*,\nu)^2$, with equality iff $\mathrm{Var}_{\gamma^*}=0$, that is, iff every cell of $\gamma^*$ consists of a single atom of $\nu$ (coincident atoms counted once) — the case $m_0=m$,

but also $m_0 \mid m$ and priors with repeated atoms — and $b_{\mathrm{bar}} = 0$ iff $\vartheta_j^* = \bar{T}_j^*$ for all $j$ (at a tie, with the plan $\gamma^*(t)$ fixed by the selection). □

**Remark C′.1** (reading)**.** The barycentric leak replaces "pull each kernel to its prior kernel" by "pull each kernel to the centre of mass of the prior it is responsible for". Three consequences.

- *Transfer across sizes.* A network of $m$ kernels can be anchored to a learned measure of $m_0 \neq m$ kernels. The bias in the Lyapunov inequality is $b_{\mathrm{bar}}$, the distance of the ideal from the centroids of its cells, not the quantisation error $W(\mu^*, \nu)$ itself.
- *Quantisation on the fly.* With $e \equiv 0$ the law is Lloyd's algorithm for the equal-mass quantisation of $\nu$ in the $G^{-1}$ metric, in continuous time — the continuous-time Lloyd descent of coverage control [19], with the equal-mass constraint of capacity-constrained quantisation [5]; so the controller's leak and the quantisation theory of Corollary B.6 (Section 26, branch 9) are the same object. Like Lloyd's algorithm it converges to a critical point of the quantisation energy, not necessarily the global minimiser.
- *Theorem F.* The leak clauses (i′)–(iii′) of Theorem F hold with $W(\mu_r, \nu)$ replaced by $V_{\mathrm{bar}}(r) \leq W(\mu_r, \nu)$, since $\|\dot{\vartheta}_i\|_{G^{-1}} \leq \|\nabla g_t\|_G + \sigma_m \|\vartheta_i - \bar{T}_i\|_{G^{-1}}$ by Lemma P.1 and Minkowski.

**Checked** (`tests/test_barycentric.py`). The plan reduces to the assignment for $m_0 = m$ ($10^{-10}$) and the Pythagorean split $W^2 = V_{\mathrm{bar}}^2 + \mathrm{Var}_\gamma$ holds ($10^{-10}$); along the pure leak $W^2$ is non-increasing at every step, the Dini bound $D^+ Z \leq -\sigma_m V_{\mathrm{bar}}^2$ holds at every state (within 5% of the first-order Euler quotient), the flow reaches $V_{\mathrm{bar}}^2 < 10^{-8}$, and the end point is a fixed point of the equal-mass Lloyd iteration with the same $W^2$; $\sqrt{C_{\mathrm{bar}}} \leq b + 2\eta$ on 90 random configurations, with $C_{\mathrm{bar}} = b_{\mathrm{bar}}^2$ at small $\eta$; the Lyapunov inequality (iii) holds at every recorded state of a closed loop with the barycentric leak toward a 24-atom prior. The numbers are in `results/barycentric_leak/table.md` (`exp_barycentric_leak.py`).

**Numerics** (`results/barycentric_leak/table.md`).

- *Pure leak toward a 48-atom prior* ($m = 6$, lifted dimension 3, $G = \mathrm{diag}(2, 1, 0.5)$, $\sigma_m = 1$, three random starts). $W^2(\mu_t, \nu)$ is non-increasing at every one of the 3000 steps; the flow stops where the centroid defect is at the rounding level ($V_{\mathrm{bar}}^2 \leq 2.2 \cdot 10^{-25}$) and $W^2 = \mathrm{Var}_\gamma$ (1.164, 1.213, 1.203 from 2.55, 2.30, 2.58), after 2–6 plan switches, at which $V_{\mathrm{bar}}$ jumps (upward as well as downward). The end points are fixed points of the equal-mass Lloyd iteration; the best of 20 Lloyd runs from random starts reaches 1.171, below two of the three end points and above the third: like Lloyd's algorithm, the leak finds a local minimiser of the quantisation energy, not always the global one. With $m_0 = m$ the same code reproduces the exact decay of Theorem C(i) to $5 \cdot 10^{-16}$.
- *Closed loop* (representable plant with an $m^* = 4$ ideal, $m = 4$, $k_e = 4$, $\sigma_m = 0.5$, three seeds). Three leaks of the same gain: (A) the OT leak toward the ideal itself, (B) the barycentric leak toward a 32-atom cloud of jittered copies of the ideal atoms, (C) the labelled leak toward a random 4-subset of the cloud. The network converges, in each case, to the configuration its leak anchors: (A) to the ideal ($W(\mu_T, \mu^*)$ and RMS$\|e\|$ below the printed precision); (B) to the equal-mass centroids of the cloud, at $W(\mu_T, \mu^*) = 0.167$, 0.097, 0.157, which equal the centroid defects $b_{\mathrm{bar}}$ of the ideal (0.168, 0.097, 0.159) although $W(\mu^*, \nu) = 0.45$, 0.33, 0.40; (C) to the subset, at $W(\mu_T, \mu^*) = 0.61$, 1.02, 0.35 against $W(\mu^*, \mathrm{subset}) = 0.65$, 1.03, 0.40. The steady tracking RMS is 0.002–0.007 for (B) and 0.03–0.04 for (C). So the loop settles at the centroidal configuration of its prior, at distance $b_{\mathrm{bar}}$ from the ideal, as the fixed-point statements (i) and (v) describe; the Lyapunov bound (iv) does not apply to these runs (scope paragraph: the exchange bound on $\Delta_{\mathrm{gap}}$ and $4\eta(0)(b + \eta(0))$ are in the table). The comparison with (C) shows the cost of a prior of the

right size at the wrong positions, not an advantage of barycentric over labelled leaks as such: a labelled leak toward the cloud's own equal-mass $m$-quantiser would have the same bias $b_{\mathrm{bar}}$ by (v).

## 14 Theorem D: the mean-field closed loop

The state is $(x, \mu) \in \mathbb{R}^n \times \mathcal{P}(\Theta_K)$, with

$$\begin{gathered}\dot{x} = f(x,t) + \dot{x}_m - k_e(x - x_m) - \int \Phi(x;\vartheta)\,\mathrm{d}\mu_t, \qquad \partial_t \mu_t + \nabla \cdot (\mu_t v_t) = 0, \\ v_t = \mathrm{Proj}_K\big(\vartheta,\, G\nabla_\vartheta\big[(x - x_m)^\top \Phi(x;\vartheta)\big]\big).\end{gathered} \tag{14.1}$$

Solutions are Lagrangian: $\mu_t = (\Xi_t)_{\#}\mu_0$ for the flow $\Xi_t$ of $v_t$. This is law (L). The OT leak is multivalued at ties and is excluded here.

**Theorem D.** *Assume* (S0)–(S2).

(a) Local well-posedness. *For every $(x_0, \mu_0) \in \mathbb{R}^n \times \mathcal{P}(\Theta_K)$ there is a unique Lagrangian solution. It continues for as long as $x$ stays bounded.*

(b) Adaptation-free tube ($m$-uniform, $\mu^*$-free). *For every $\mu \in \mathcal{P}(\Theta_K)$, $\|\int \Phi(x;\cdot)\,\mathrm{d}\mu\| \le a_{\max}$. For $F \ge 0$ let*

$$r(F) := \|e(0)\| + \frac{F + a_{\max}}{k_e}, \qquad S(F) := \sup\big\{\|f(x,t)\| : t \ge 0,\ \|x - x_m(t)\| \le r(F)\big\}.$$

*Suppose some $F \ge 0$ satisfies $S(F) \le F$. Then every solution, for every $m$ and every $\mu_0$:*

- *exists for all $t \ge 0$;*
- *satisfies $\|x(t) - x_m(t)\| \le r(F)$;*
- *satisfies $\|e(t)\| \le e^{-k_e t}\|e(0)\| + (1 - e^{-k_e t})(F + a_{\max})/k_e$.*

*Examples:*

- *For bounded $f$, $F = \sup\|f\|$ works for every $e(0)$.*
- *If $\|f(x,t)\| \le \alpha + \beta\|x\|$ with $\beta < k_e$, then*

$$F = \frac{\alpha + \beta\big(\sup\|x_m\| + \|e(0)\| + a_{\max}/k_e\big)}{1 - \beta/k_e}$$

  *works.*
- *The condition can fail for superlinear $f$. For the Van der Pol plant of the numerics, $S(F) - F \ge 51.6$ for every $F$.*

(c) Dobrushin estimate. *Let two solutions of the same loop (same $f$, $x_m$, $k_e$, $G$, $\Theta_K$ and $\varepsilon_K$) remain in a compact $X$ on $[0,T]$. Then*

$$\begin{gathered}\|x_t - x'_t\| + W(\mu_t, \mu'_t) \le e^{\Lambda t}\big(\|x_0 - x'_0\| + W(\mu_0, \mu'_0)\big), \\ \Lambda := \max(L_\vartheta + L_2,\, L_x + M_x),\end{gathered} \tag{14.2}$$

*and $\Lambda$ is independent of $m$. The constants are:*

- *$L_\vartheta$: a Lipschitz constant of $\vartheta \mapsto v$ in the $G^{-1}$ norm, uniform over $x \in X$ and $t \le T$. It is proportional to $\bar{E} := \sup_{X\times[0,T]}\|x - x_m(t)\|$, and it is $O(\bar{E}/\varepsilon_K)$ because of the ramp ($h' \le 2/\varepsilon_K$).*

- *$L_x$: a Lipschitz constant, Euclidean to $G^{-1}$, of $x \mapsto v(\vartheta; x, x - x_m(t))$, uniform over $\vartheta \in \Theta_K$, $t \le T$ and $x \in X$.*
- *$M_x$: $M_x := \sup_{t \le T} \operatorname{Lip}_X f(\cdot, t) + k_e + a_{\max} J_c$. Here $a_{\max} J_c$ bounds $\|\nabla_x \int \Phi(x; \cdot)\, \mathrm{d}\mu\|$ over $x \in \mathbb{R}^n$ and $\mu \in \mathcal{P}(\Theta_K)$.*
- *$L_2$: $L_2 := \lambda_G^{1/2} L_1$, with $L_1$ from Lemma B.2, is a $W$-Lipschitz constant of $\mu \mapsto \int \Phi(x; \cdot)\, \mathrm{d}\mu$. For any coupling, $\|\int \Phi(x; \cdot)\, \mathrm{d}(\mu - \mu')\| \le L_2 \big(\int \|\vartheta - \vartheta'\|_{G^{-1}}^2 \, \mathrm{d}\gamma\big)^{1/2}$.*

(d) Mean-field limit. *Let $(x, \mu)$ be the solution from $(x_0, \mu_0)$, and let $T$ be smaller than its existence time. Put $X_1 := \{y : \operatorname{dist}(y, x([0,T])) \le 1\}$, and let $\Lambda_1$ be the $\Lambda$ of (c) computed on $X_1 \times [0,T]$. Suppose $x_0^m \to x_0$ and $\mu_0^m \to \mu_0$ in $W$, and set $\delta_m := \|x_0^m - x_0\| + W(\mu_0^m, \mu_0)$. Then, for all large $m$:*

- *the $m$-particle loop exists on $[0,T]$ and stays in $X_1$;*
- *$\sup_{t \le T} \big(\|x_t^m - x_t\| + W(\mu_t^m, \mu_t)\big) \le e^{\Lambda_1 T} \delta_m \to 0$.*

*Under the hypothesis of (b), for example for bounded $f$, the limit is global and $T$ is arbitrary.*

(e) Passage of (B) to the limit. *Work in the setting of (d), with $X := X_1$. Let $\mu_t^{*m} = \frac{1}{m} \sum_j \delta_{\vartheta_j^{*m}(t)}$ satisfy* (S4), *and let $\varepsilon_{\max}^m \ge \sup_{X \times [0,T]} \|f - \int \Phi \, \mathrm{d}\mu_t^{*m}\|$. Assume $\mu_t^{*m} \to \mu_t^*$ uniformly in $W$ on $[0,T]$, $\sup_m \varepsilon_{\max}^m \le \varepsilon_{\max}$ and $\sup_m V_*^m \le V_*$. Only tails matter, so $\sup_m$ may be replaced by $\limsup_m$. Then:*

- *$\mu_t^*$ is $V_*$-Lipschitz in $W$, with values in $\mathcal{P}(\Theta_K^\varepsilon)$;*
- *the limit satisfies* (B) *in integrated form, and in Dini form for every $t \in [0,T)$, with constants $(k_e, \varepsilon_{\max}, K, L_0, L_1, \lambda_G, V_*)$.*

*$V_*$ depends on the approximating ideals; the intrinsic version is Remark B.7.*

**Status.** Proved; consistency in $m$ checked numerically (the constant is not checked). Not checked in the repository: (b), the constant of (c), and the one-sided constant of Remark D.1. Novelty (literature audit): technique anticipated [29, 72, 73].

*Proof. (a)* Write the loop as an ODE for $(x, \Xi) \in \mathbb{R}^n \times C(\Theta_K; \mathbb{R}^p)$, with the sup norm (and $G^{-1}$ on $\mathbb{R}^p$):

$$\dot{x} = f(x,t) + \dot{x}_m - k_e(x - x_m) - \int \Phi(x; \Xi(\vartheta))\, \mathrm{d}\mu_0(\vartheta), \qquad \partial_t \Xi(\vartheta) = v(\Xi(\vartheta); x, t), \qquad \Xi_0 = \mathrm{id},$$

so that $\mu_t = \Xi_{t\#}\mu_0$. Extend $v$ Lipschitz to a neighbourhood of $\Theta_K$ inside $\{s > s_{\min}/2\}$ (McShane). On $\{\|x\| \le R\}$, with the constants of (c), the right-hand side is Lipschitz:

- $\|v(\Xi(\cdot); x, t) - v(\Xi'(\cdot); x', t)\|_\infty \le L_\vartheta \|\Xi - \Xi'\|_\infty + L_x \|x - x'\|$;
- the $x$-field changes by at most $M_x \|x - x'\| + L_2 \|\Xi - \Xi'\|_\infty$, because $W(\Xi_\# \mu_0, \Xi'_\# \mu_0) \le \|\Xi - \Xi'\|_{L^2(\mu_0; G^{-1})}$.

Picard–Lindelöf gives a unique local solution. Each characteristic stays in $\Theta_K$ (Nagumo, as in A(ii)), so the solution continues while $x$ is bounded. Golse [37, Thms. 1.3.2 and 1.4.3], and [29, 72, 73], are structural analogues (pairwise $C^1$ kernels, $W_1$), not the proof.

*(b)* While $\|e\| \le r(F)$, $\dot{e} = -k_e e + (f - \hat{f})$ with $\|f - \hat{f}\| \le F + a_{\max}$. Variation of constants gives

$$\|e(t)\| \le e^{-k_e t} \|e(0)\| + (1 - e^{-k_e t})(F + a_{\max})/k_e < r(F)$$

for every finite $t$, since $a_{\max} > 0$. So $x$ never reaches the boundary of the tube (first exit), stays bounded, and (a) continues the solution.

*(c)* Let $\gamma_0$ be an optimal plan for $(\mu_0, \mu_0')$, let $\Xi_t$ and $\Xi_t'$ be the two flows, and set $D(t)^2 := \int \|\Xi_t(\vartheta) - \Xi_t'(\vartheta')\|_{G^{-1}}^2 \,\mathrm{d}\gamma_0$. Then $(\Xi_t, \Xi_t')_\# \gamma_0$ couples $\mu_t$ and $\mu_t'$, so $W(\mu_t, \mu_t') \le D(t)$. Split

$$v_t(\Xi) - v_t'(\Xi') = \big[v_t(\Xi) - v_t(\Xi')\big] + \big[v_t(\Xi') - v_t'(\Xi')\big]$$

and apply Cauchy–Schwarz:

$$\frac{\mathrm{d}}{\mathrm{d}t} D^2 \le 2L_\vartheta D^2 + 2L_x \|x - x'\| D, \qquad D^+ \|x - x'\| \le M_x \|x - x'\| + L_2 D.$$

Apply Grönwall to $D_\delta + \|x - x'\|$, with $D_\delta := (D^2 + \delta^2)^{1/2}$, and let $\delta \downarrow 0$.

*(d)* Let $\tau_m \le T$ be the first time $x^m$ leaves $X_1$, with $\tau_m = T$ if it never leaves. On $[0, \tau_m]$ both solutions lie in $X_1$, so (c) gives $\|x_t^m - x_t\| \le e^{\Lambda_1 T} \delta_m < 1$ once $m$ is large. Then $x^m(\tau_m)$ is interior to $X_1$, so $\tau_m = T$. By (a), the $m$-solution continues while it is bounded.

*(e)* Theorem B for each $m$ gives $V^m(t_2) - V^m(t_1) \le \int R^m \le \int \bar{R}^m$. Here $\bar{R}^m$ is $R^m$ with $\varepsilon_{\max}^m$ and $V_*^m(t)$ replaced by $\varepsilon_{\max}$ and $V_*$. By (d) and the triangle inequality for $W$, $e^m \to e$ and $W(\mu_t^m, \mu_t^{*m}) \to W(\mu_t, \mu_t^*)$ uniformly. So $V^m \to V$ and $\bar{R}^m \to \bar{R}$ uniformly, and $V(t_2) - V(t_1) \le \int \bar{R}$. $\bar{R}$ is continuous, which gives the Dini form. The identity coupling gives $W(\mu_t^{*m}, \mu_s^{*m}) \le V_* |t - s|$. This passes to the uniform limit, and $\Theta_K^\varepsilon$ is closed. □

**Remark D.1** (a one-sided constant free of $\varepsilon_K$)**.** The coupling step uses $v$ only through $\langle \Xi - \Xi',\, G^{-1}(v(\Xi) - v(\Xi'))\rangle$. So the one-sided constant

$$L_\vartheta^+ := \sup \frac{\langle \vartheta - \vartheta',\, G^{-1}(v(\vartheta) - v(\vartheta'))\rangle}{\|\vartheta - \vartheta'\|_{G^{-1}}^2}$$

may replace $L_\vartheta$. Let $N(\theta, y) := h(\|\theta\|)(\hat{\theta}^\top y)_+ \hat{\theta}$ be the norm-ball correction, with $r = \|\theta\|$, $c = \hat{\theta}^\top \hat{\theta}'$ and $\alpha = (\hat{\theta}^\top y)_+$. Then

$$\langle \theta - \theta',\, N(\theta, y) - N(\theta', y)\rangle \ge -\frac{\|y\| \|\theta - \theta'\|^2}{2(\theta_{\max} - \varepsilon_K)}.$$

This follows from three facts:

- $h$ is monotone;
- $|\alpha - \alpha'| \le \|y\| \sqrt{2(1 - c)}$;
- $\|\theta - \theta'\|^2 \ge 2\sqrt{rr'}\, |r - r'| \sqrt{2(1 - c)}$.

The box correction is monotone. Hence $L_\vartheta^+$ stays bounded as $\varepsilon_K \to 0$, and so does $\Lambda := \max(L_\vartheta^+ + L_2,\, L_x + M_x)$. On the experiment's box, a numerical measurement gives $L_\vartheta^+ \approx 88$–$98$ for $\varepsilon_K$ from 0.2 to 0.005. Over the same range the two-sided constant grows like $30/\varepsilon_K$. This measurement is not repeated in the repository's tests.

**Numerics** (`results/mean_field_m`). Halving $W(\mu_0^m, \mu_0^{2m})$ halves $W(\mu_T^m, \mu_T^{2m})$, with a nearly constant factor (1.64–1.69 over the four refinements). $\sup_t \|e^m - e^{2m}\|$ falls from $4.6 \times 10^{-3}$ (16 against 32 particles) to below $10^{-4}$ (128 against 256).

This is a consistency check only.

- On the experiment's box the guaranteed factor $e^{\Lambda T}$ is at least $e^{496}$, from $L_2 = 33.08$ alone over $T = 15$.
- The condition of (b) fails for this Van der Pol plant, so the run is covered by the localised (d) through its observed boundedness, not by (b).

Checked by: `experiments/exp_mean_field_m.py` (consistency in $m$; not the constant $e^{\Lambda T}$).

# 15 Theorem E: label-free local convergence under tangential excitation

**Assumptions.**

- (S0) holds, in particular $k_e > 0$.
- **Uniform tube.** There is $\rho > 0$ such that $\varepsilon(x,t) = 0$ whenever $\|x - x_m(t)\| < \rho$, for all $t \geq 0$.
- **Interior ideal.** $\mu^*$ is constant, with $\vartheta_j^* \in \operatorname{int}\Theta_K^\varepsilon$. Put $\delta := \min_j \operatorname{dist}(\vartheta_j^*, \partial\Theta_K^\varepsilon) > 0$.
- $x_m \in C^1$, and $x_m$ and $\dot{x}_m$ are bounded.
- Define $\Psi(t) := \big[\nabla_\vartheta \Phi(x_m(t);\vartheta_1^*), \dots, \nabla_\vartheta \Phi(x_m(t);\vartheta_m^*)\big] \in \mathbb{R}^{n\times mp}$.
- **(PE)** There are $T, \alpha > 0$ with $\int_t^{t+T} \Psi^\top \Psi \,\mathrm{d}r \succeq \alpha I_{mp}$ for all $t \geq 0$.

(PE) forces $a_j^* \neq 0$ and pairwise distinct pairs $(c_j^*, s_j^*)$.

- If $a_j^* = 0$, the $(c,s)$-columns of block $j$ vanish.
- If $(c_j^*, s_j^*) = (c_k^*, s_k^*)$, the kernel of $\int \Psi^\top \Psi \,\mathrm{d}r$ contains the vector with $v$ in the $a$-slot of block $j$, $-v$ in the $a$-slot of block $k$, and zeros elsewhere.

In particular the $\vartheta_j^*$ are pairwise distinct, and

$$d^2 := \min_{\pi\neq\pi'} \frac{1}{m}\sum_i \|\vartheta_{\pi(i)}^* - \vartheta_{\pi'(i)}^*\|_{G^{-1}}^2 = \frac{2}{m}\cdot \min_{i\neq j} \|\vartheta_i^* - \vartheta_j^*\|_{G^{-1}}^2 > 0, \tag{15.1}$$

with $d^2 := +\infty$ for $m = 1$. The closed form holds because a non-identity permutation moves at least two indices, and the transposition of the $G^{-1}$-closest pair attains the bound.

**Lemma E.0** (explicit strict Lyapunov function for the skew-symmetric mixed class)**.** *Let $\Psi : [0,\infty) \to \mathbb{R}^{n\times q}$ be $C^1$ with $\|\Psi(t)\| \leq \Psi_M$ and $\|\dot\Psi(t)\| \leq \Psi_D$ for all $t$, and persistently exciting: there are $T, \alpha > 0$ with $\int_t^{t+T} \Psi(r)^\top \Psi(r)\,\mathrm{d}r \succeq \alpha I_q$ for all $t \geq 0$. Let $k > 0$ and consider*

$$\dot e = -ke - \Psi(t)\zeta, \qquad \dot\zeta = \Psi(t)^\top e, \qquad w := (e,\zeta) \in \mathbb{R}^{n+q}. \tag{LTV}$$

*Put $M := \Psi^\top\Psi$ and*

$$Q(t) := \int_t^{t+T} (t+T-r)M(r)\,\mathrm{d}r, \qquad Q_M := \frac{T^2\Psi_M^2}{2}, \qquad c_1 := T^2\Psi_M^3 + Tk\Psi_M + T\Psi_D,$$
$$V(t,e,\zeta) := \tfrac12\|e\|^2 + \tfrac12\|\zeta\|^2 - \varepsilon\,\zeta^\top Q(t)\zeta + \varepsilon T\, e^\top\Psi(t)\zeta.$$

*(i)* Every admissible $\varepsilon$. *For every $\varepsilon > 0$ with $\varepsilon(Q_M + T\Psi_M/2) < \frac12$, $V$ is $C^1$ and, along* (LTV),

$$p_{\mathrm{lo}}\|w\|^2 \leq V \leq p_{\mathrm{hi}}\|w\|^2, \qquad p_{\mathrm{lo/hi}} := \tfrac12 \mp \varepsilon(Q_M + T\Psi_M/2),$$
$$\dot V \leq -(k - \varepsilon T\Psi_M^2)\|e\|^2 - \varepsilon\alpha\|\zeta\|^2 + \varepsilon c_1\|e\|\|\zeta\|.$$

*Hence $\dot V \leq -\gamma(\varepsilon)\|w\|^2$ with $\gamma(\varepsilon) := \lambda_{\min}\left[\begin{smallmatrix} a & -c/2 \\ -c/2 & b\end{smallmatrix}\right]$, $a := k - \varepsilon T\Psi_M^2$, $b := \varepsilon\alpha$, $c := \varepsilon c_1$, whenever this is positive, and then $\|w(t)\| \leq (p_{\mathrm{hi}}/p_{\mathrm{lo}})^{1/2} e^{-(\gamma/(2p_{\mathrm{hi}}))(t-t_0)}\|w(t_0)\|$ for all $t \geq t_0 \geq 0$.*

*(ii)* Closed form. *With $\varepsilon := \min\{1/(4(Q_M + T\Psi_M/2)),\ k/(4T\Psi_M^2),\ k\alpha/c_1^2\}$, one has $p_{\mathrm{lo}} \geq \frac14$, $p_{\mathrm{hi}} \leq \frac34$ and $\dot V \leq -\gamma\|w\|^2$ with $\gamma := \min\{k/4, \varepsilon\alpha/2\}$. Hence*

$$\|w(t)\| \leq \sqrt{3}\cdot e^{-(2\gamma/3)(t-t_0)}\cdot\|w(t_0)\| \qquad \text{for all } t \geq t_0 \geq 0.$$

*(iii)* Dependence on $\Psi_D$ and on time rescaling. *The rate tends to* 0 *as* $\Psi_D \to \infty$*, and it must:* $\Psi = \sin t^2$ *is bounded and PE with* $\Psi_D = \infty$*, and* (LTV) *is then not uniformly exponentially stable (part (f) of the proof). Under the time rescaling* $\Psi_\omega(t) := \Psi(\omega t)$*, with* $\Psi_D > 0$*, the data transform as* $(\alpha, T, \Psi_M, \Psi_D) \mapsto (\alpha/\omega, T/\omega, \Psi_M, \omega\Psi_D)$*, so* $c_1 = T\Psi_D \cdot (1 + O(1/\omega))$ *and the closed-form rate is* $k\alpha^2/(3\omega^2 T^2 \Psi_D^2) \cdot (1 + O(1/\omega))$ *as* $\omega \to \infty$*. This is the* $\omega^{-2}$ *law of Remark E.1.*

*Proof. (a) The Gramian weight.* $0 \preceq Q(t) \preceq \int_t^{t+T} (t + T - r)\Psi_M^2 \,\mathrm{d}r \cdot I = Q_M I$. Since $M$ is continuous, $Q$ is $C^1$ with $\dot{Q}(t) = \int_t^{t+T} M(r)\,\mathrm{d}r - TM(t)$: the upper limit contributes $(t + T - (t + T))M(t + T) = 0$, the lower limit contributes $-TM(t)$, and differentiating the weight under the integral gives $\int_t^{t+T} M$.
*(b) Bounds on* $V$. $|\zeta^\top Q\zeta| \le Q_M\|\zeta\|^2$ and $|e^\top\Psi\zeta| \le \Psi_M\|e\|\|\zeta\| \le (\Psi_M/2)(\|e\|^2 + \|\zeta\|^2)$.
*(c) Three derivatives along* (LTV).

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\tfrac{1}{2}(\|e\|^2 + \|\zeta\|^2) &= -k\|e\|^2 - e^\top\Psi\zeta + \zeta^\top\Psi^\top e = -k\|e\|^2,\\
\frac{\mathrm{d}}{\mathrm{d}t}(-\zeta^\top Q\zeta) &= -\zeta^\top\dot{Q}\zeta - 2\zeta^\top Q\dot{\zeta} = -\zeta^\top\Big(\int_t^{t+T} M\Big)\zeta + T\|\Psi\zeta\|^2 - 2\zeta^\top Q\Psi^\top e\\
&\le -\alpha\|\zeta\|^2 + T\|\Psi\zeta\|^2 - 2\zeta^\top Q\Psi^\top e,\\
\frac{\mathrm{d}}{\mathrm{d}t}(e^\top\Psi\zeta) &= \dot{e}^\top\Psi\zeta + e^\top\dot{\Psi}\zeta + e^\top\Psi\dot{\zeta} = -k\,e^\top\Psi\zeta - \|\Psi\zeta\|^2 + e^\top\dot{\Psi}\zeta + \|\Psi^\top e\|^2.
\end{aligned}$$

Adding them with weights 1, $\varepsilon$ and $\varepsilon T$, the $\|\Psi\zeta\|^2$ terms cancel:

$$\dot{V} \le -k\|e\|^2 - \varepsilon\alpha\|\zeta\|^2 - 2\varepsilon\zeta^\top Q\Psi^\top e - \varepsilon Tk\,e^\top\Psi\zeta + \varepsilon T\,e^\top\dot{\Psi}\zeta + \varepsilon T\|\Psi^\top e\|^2.$$

The last four terms are at most $2\varepsilon Q_M\Psi_M\|e\|\|\zeta\|$, $\varepsilon Tk\Psi_M\|e\|\|\zeta\|$, $\varepsilon T\Psi_D\|e\|\|\zeta\|$ and $\varepsilon T\Psi_M^2\|e\|^2$. This is (i); the quadratic form $-a\|e\|^2 - b\|\zeta\|^2 + c\|e\|\|\zeta\|$ is at most

$$-\lambda_{\min}\begin{bmatrix} a & -c/2 \\ -c/2 & b \end{bmatrix} \cdot \|w\|^2,$$

and $V \le p_{\mathrm{hi}}\|w\|^2$, $\|w\|^2 \le V/p_{\mathrm{lo}}$ give the estimate.
*(d) The closed form.* Young's inequality gives $\varepsilon c_1\|e\|\|\zeta\| \le (\varepsilon\alpha/2)\|\zeta\|^2 + (\varepsilon c_1^2/(2\alpha))\|e\|^2$. With $\varepsilon \le k\alpha/c_1^2$ the second coefficient is at most $k/2$, and with $\varepsilon \le k/(4T\Psi_M^2)$ the coefficient of $\|e\|^2$ in (i) is at least $k - k/4$. So $\dot{V} \le -(k/4)\|e\|^2 - (\varepsilon\alpha/2)\|\zeta\|^2 \le -\gamma\|w\|^2$. Since $V \le (3/4)\|w\|^2$, $\dot{V} \le -(4\gamma/3)V$, and $\|w\|^2 \le 4V$ gives

$$\|w(t)\|^2 \le 4V(t) \le 4e^{-(4\gamma/3)(t-t_0)}V(t_0) \le 3e^{-(4\gamma/3)(t-t_0)}\|w(t_0)\|^2.$$

*(e) Rescaling.* $\Psi_\omega$ has sup norm $\Psi_M$, derivative bound $\omega\Psi_D$, and

$$\int_t^{t+T/\omega} \Psi_\omega^\top\Psi_\omega \,\mathrm{d}r = \frac{1}{\omega}\int_{\omega t}^{\omega t+T} \Psi^\top\Psi \,\mathrm{d}r \succeq \frac{\alpha}{\omega}I.$$

Substitute into $c_1$ and the closed form.
*(f)* $\Psi = \sin t^2$ *is not uniformly exponentially stabilising* ($n = q = 1$, $k > 0$). Let $F(s) := \int_s^\infty \sin r^2\,\mathrm{d}r$, so that $\sin s^2 = -F'(s)$ and $|F(s)| \le C/s$ (Fresnel). From $w(t_0) = (0, \zeta_0)$, $e(t) = -\int_{t_0}^t e^{-k(t-s)}\sin(s^2)\zeta(s)\,\mathrm{d}s$; integrating by parts and using $|\zeta| \le \|w(t_0)\|$ (the energy $\frac{1}{2}\|w\|^2$ is non-increasing) and $|\dot{\zeta}| = |\Psi e| \le |e|$,

$$|e(t)| \le \frac{C}{t_0}(2 + 1)\|w(t_0)\| + \frac{C}{t_0}\int_{t_0}^t e^{-k(t-s)}|e(s)|\,\mathrm{d}s,$$

so $\sup_{t\ge t_0}|e| \le 3C\|w(t_0)\|/\big(t_0(1-C/(kt_0))\big)$ for $t_0 > C/k$. Hence

$$\tfrac{1}{2}\|w(t_0+\Delta)\|^2 \ge \tfrac{1}{2}\|w(t_0)\|^2 - k\Delta\Big(\frac{3C}{t_0}\Big)^2\|w(t_0)\|^2(1+o(1)) :$$

the decay over any fixed $\Delta$ tends to 0 as $t_0 \to \infty$, which excludes a uniform exponential rate. The three ratios quoted in step 3 of the proof of Theorem E (0.9715, 0.9942, 0.9980 for $t_0 = 5$, 20, 50 and $\Delta = 200$) match $1 - \frac{1}{8}\big(1/t_0 - 1/(t_0+200)\big)$ to three digits. □

Checked by: `tests/test_excitation.py`: the two bounds on $V$ and $\dot V \le -\gamma\|w\|^2$ at random $(t, w)$ for four regressors of different shapes and gains, with the closed-form $\varepsilon$ and with the $\varepsilon$ maximising the rate (`excitation.optimise_eps`; every admissible $\varepsilon$ gives a valid Lyapunov function, so the maximum over the grid is a valid bound for the given PE data; the data $\Psi_M$, $\Psi_D$, $\alpha$ are themselves grid maxima and a refinement-checked quadrature—the PE level over the $20\pi$ window is computed with 800 Gauss–Legendre nodes, and `excitation.pe_level` raises if a 2× refinement disagrees—so the ordering against the exact rate is numerical); the envelope along trajectories; the ordering explicit rate ≤ exact Floquet rate; the $\omega^{-2}$ law; the slow-adaptation limit (a no-growth check of the ratio, not a saturation check); and the explicit nonlinear envelope of Theorem E on the implemented closed loop. The round-5 referee verified the pointwise form $\lambda_{\max}(N(t)) \le -\gamma$ of $\dot V \le -\gamma\|w\|^2$ over all $w$, with an independent quadrature, for ten regressor families ($q > n$, $q < n$, $k$ from $10^{-3}$ to $10^4$, $\Psi_D/\Psi_M$ up to 100, non-periodic PE, near-degenerate PE) and eight choices of $\varepsilon$, without finding a violation.

**Remark E.0a** (how conservative the explicit rate is)**.** The construction is of the Mazenc–Malisoff type, a backward–forward integral of the PE Gramian as a cross term; its content here is the explicit, self-contained statement for the mixed class (LTV), which removes the dependence of Theorem E on [62]. The price is conservativeness, which the table of `exp_explicit_rate.py` → `results/explicit_rate/table.md` quantifies against the exact Floquet rate of the linearised loop.

- *Slow adaptation* ($T\Psi_M^2 \ll k$). To leading order in the gains, $c_1 \approx T(k\Psi_M + \Psi_D)$ and $\alpha = T\lambda_{\min}\langle\Psi^\top\Psi\rangle$ for a window equal to the period, so the closed-form bound $\frac{2}{3}\gamma_{\mathrm{cf}}$ of (ii) is $k\alpha^2/(3c_1^2)$, and the rate with the optimal $\varepsilon$ of the same quadratic form is $k\alpha^2/c_1^2$ (maximise $\varepsilon\alpha - \varepsilon^2 c_1^2/(4k)$ over $\varepsilon$). The exact rate is $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ when $k$ is also large compared with the bandwidth of the reference (Remark E.1). Hence, in that doubly asymptotic regime,

$$\frac{\text{exact}}{\text{explicit}} \approx 3\kappa_\Psi\Big(1+\frac{\Psi_D}{k\Psi_M}\Big)^2 \text{ for (ii)}, \qquad \approx \kappa_\Psi\Big(1+\frac{\Psi_D}{k\Psi_M}\Big)^2 \text{ for the optimised } \varepsilon,$$

  with $\kappa_\Psi := \Psi_M^2/\lambda_{\min}\langle\Psi^\top\Psi\rangle$ the condition number of the averaged Gramian. The exact rate pays for ill-conditioning once, through $\lambda_{\min}$; the bound pays twice, because the Young step that absorbs the cross term $\varepsilon c_1\|e\|\|\zeta\|$ compares $c_1 \propto \Psi_M$ with $\alpha \propto \lambda_{\min}$ (Lemma E.0′ below removes this by slaving the error to the slow manifold before the Young step). On the ideal of the Theorem E tests ($m = 3$, $n = 2$, $k_e = 6$), driven by a two-tone reference of period $20\pi$, with $G = g\cdot\mathrm{diag}(2,2,1)$: $\kappa_\Psi = 1.27\cdot10^5$, $(1+\Psi_D/(k_e\Psi_M))^2 = 3.84$, so the two formula values are $1.47\cdot10^6$ and $4.9\cdot10^5$. The measured ratio with the optimised $\varepsilon$ falls from $6.56\cdot10^{10}$ at $g = 30$ to $4.22\cdot10^5$ at $g = 0.01$ ($5.63\cdot10^5$ with the closed-form $\varepsilon$, which the table evaluates with the exact $2\times2$ form and the exact $p_{\mathrm{hi}}$ rather than the bound (ii)). Two corrections account for the difference from $4.9\cdot10^5$: on this instance the exact rate is 0.66 of $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ at every $g$ ($k_e = 6$ is not large against the reference's bandwidth; the two last columns of the table), and the $T^2\Psi_M^3$ term is still 13% of $c_1$ at $g = 0.01$ ($(c_1/c_1^\infty)^2 = 1.31$); their product, $0.66\times1.31 = 0.86$, brings $4.9\cdot10^5$ to $4.2\cdot10^5$. The ratio is still decreasing at $g = 0.01$ and tends to about $0.66\cdot\kappa_\Psi(1+\Psi_D/(k_e\Psi_M))^2 \approx 3.2\cdot10^5$ as $g \to 0$. In the mean-field normalisation $\Psi_G \propto m^{-1/2}$, so large networks sit in this regime at fixed geometry.

- *Fast adaptation* ($T\Psi_M^2 \gtrsim k$). The ratio grows further with the excitation strength, through the $T^2\Psi_M^3$ term of $c_1$ and the $Q_M$ term of $\varepsilon$.
- *Remark E.1's instance* ($m = n = 1$, $G = I$, $k_e = 3$, $x_m = 1.5\sin\omega t$). The ratio exact/explicit decreases from 954 at $\omega = 1$ to 17.6 at $\omega = 64$. Over $\omega \geq 4$ the log-log slopes are $-1.97$ (exact) and $-1.56$ (explicit); over the last octave, $\omega = 32 \to 64$, the explicit slope is $-1.92$, approaching the asymptotic $\omega^{-2}$ of (iii).
- *The window.* Any window length $T$ is admissible with its own $\alpha(T)$. On the test ideal the best window is half a period.
- *Answered below.* Lemma E.0′ is a strict Lyapunov function whose rate is $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ up to a factor $1-\delta$, above a threshold in $k$ set by the peak factors of the regressor: in the slow regime it pays for the ill-conditioning once.

**Lemma E.0′** (the slow-adaptation regime: a strict Lyapunov function whose rate is the averaged rate)**.** *Let $\Psi$ be $T$-periodic and $C^2$, $M := \Psi^\top\Psi$, $\bar M := \frac{1}{T}\int_0^T M\,\mathrm{d}t \succ 0$, $\|v\|_{\bar M} := (v^\top \bar M v)^{1/2}$, $\lambda_{\min} := \lambda_{\min}(\bar M)$, and*

$$\Psi_M := \sup\|\Psi\|, \qquad \Psi_D := \sup\|\dot\Psi\|, \qquad \Psi_{DD} := \sup\|\ddot\Psi\|,$$
$$\rho := \sup_{t,\,v\neq 0}\frac{\|\Psi(t)v\|^2}{\|v\|_{\bar M}^2}, \qquad \rho_D := \sup\frac{\|\dot\Psi(t)v\|^2}{\|v\|_{\bar M}^2}, \qquad \rho_{DD} := \sup\frac{\|\ddot\Psi(t)v\|^2}{\|v\|_{\bar M}^2},$$

*the* peak factors*: the largest ratio, over directions, between the instantaneous excitation of a direction (by $\Psi$, $\dot\Psi$, $\ddot\Psi$) and its average excitation; $\rho \geq 1$, and all three are finite because $\bar M \succ 0$. Write $\varepsilon := 1/k$ and let*

$$P_1(t) := -\frac{2}{T}\int_t^{t+T}(t+T-r)\big(M(r)-\bar M\big)\,\mathrm{d}r, \qquad F := P_1M + MP_1,$$
$$S := \text{the zero-mean antiderivative of } F, \qquad P_2 := -(M-\bar M) + S,$$
$$e_2(\zeta,t) := -\varepsilon\Psi\zeta + \varepsilon^2\dot\Psi\zeta - \varepsilon^3\Psi M\zeta, \qquad w := e - e_2(\zeta,t),$$
$$V(e,\zeta,t) := \tfrac12\zeta^\top\big(I + \varepsilon P_1(t) + \varepsilon^2 P_2(t)\big)\zeta + \frac{\beta}{2}\|w\|^2,$$

*where $P_1$ is the zero-mean antiderivative of $2(M-\bar M)$ and $e_2$ is the slow manifold of the error, expanded through order $\varepsilon^3$. $F$ has zero mean: $\langle P_1M + MP_1\rangle = \langle P_1\rangle\bar M + \bar M\langle P_1\rangle + \frac12\big\langle\frac{\mathrm{d}}{\mathrm{d}t}(P_1^2)\big\rangle = 0$, so $S$ exists and the second-order averaged correction $N_2 := \frac12\langle F\rangle$ vanishes identically. Put*

$$\pi_1 := \tfrac{T}{2}\Psi_M(\sqrt\rho+1), \qquad \pi_2 := \Psi_M(\sqrt\rho+1) + \tfrac{T^2}{8}\Psi_M^3(2\sqrt\rho+1),$$
$$p_1 := \tfrac{T}{2}\Psi_M^2, \qquad p_2 := \Psi_M^2 + \tfrac{T^2}{4}\Psi_M^4,$$
$$c_1 := \sqrt\rho + \varepsilon\Psi_M\pi_1 + \varepsilon^2\Psi_M\pi_2,$$
$$c_2 := \varepsilon^2\sqrt{\rho_{DD}} + \varepsilon^3\big(3\Psi_D\Psi_M\sqrt\rho + 2\Psi_M^2\sqrt{\rho_D}\big) + \varepsilon^4\big(2\Psi_M^4\sqrt\rho + \Psi_D\Psi_M\sqrt{\rho_D}\big)$$
$$\quad + \varepsilon^5\big(\Psi_M^4\sqrt{\rho_D} + \Psi_D\Psi_M^3\sqrt\rho\big) + \varepsilon^6\Psi_M^6\sqrt\rho,$$
$$d_w := \varepsilon\Psi_M^2 + \varepsilon^2\Psi_D\Psi_M + \varepsilon^3\Psi_M^4,$$
$$r_\zeta := \varepsilon^3\big(\pi_2\Psi_M\sqrt\rho + \pi_1\Psi_M\sqrt{\rho_D} + \Psi_M^2\rho\big) + \varepsilon^4\big(\pi_2\Psi_M\sqrt{\rho_D} + \pi_1\Psi_M^3\sqrt\rho\big) + \varepsilon^5\pi_2\Psi_M^3\sqrt\rho,$$
$$\beta := 2c_1^2/\delta, \qquad c_\zeta := 1 - \delta - r_\zeta/\varepsilon.$$

*Fix $\delta \in (0,1)$. If*

$$\text{(A) } c_1c_2 \leq \delta/2, \qquad \text{(B) } d_w \leq k/4, \qquad \text{(C) } c_\zeta > 0, \qquad \text{(D) } \varepsilon p_1 + \varepsilon^2 p_2 < 1,$$

*then along* (LTV)*, for all $(e,\zeta,t)$,*

$$\dot V \leq -\varepsilon c_\zeta\,\zeta^\top\bar M\zeta - \frac{\beta k}{4}\|w\|^2 \leq -\gamma V, \qquad \gamma := \min\Big\{\frac{2c_\zeta\lambda_{\min}}{k(1+\varepsilon p_1+\varepsilon^2 p_2)},\ \frac{k}{2}\Big\},$$

*and* $\|(e,\zeta)(t)\| \le Ce^{-(\gamma/2)(t-t_0)}\|(e,\zeta)(t_0)\|$ *with*

$$C := \big(1+\varepsilon\Psi_M+\varepsilon^2\Psi_D+\varepsilon^3\Psi_M^3\big)^2\cdot\Big(\frac{\max\{1+\varepsilon p_1+\varepsilon^2 p_2,\ \beta\}}{\min\{1-\varepsilon p_1-\varepsilon^2 p_2,\ \beta\}}\Big)^{1/2}.$$

*In the slow-adaptation limit ($k\to\infty$ at fixed $\Psi$) the conditions hold for every $\delta\in(0,1)$ once $k\ge k_0(\delta)$, with $k_0\approx(4\rho\rho_{DD}/\delta^2)^{1/4}$ when (A) binds (small $\delta$, or $T\Psi_M^2\lesssim 1$; otherwise (C) sets $k_0$ through $r_\zeta$, as on the sinusoidal regressor below), and the rate is $\gamma/2=(1-\delta)(1-\varepsilon p_1+O(k^{-2}))\lambda_{\min}/k$: the averaged rate $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ of Remark E.1 up to the factor $1-\delta$ and the first-order deficit $\varepsilon p_1=T\Psi_M^2/(2k)$ of the quadratic form. The bound pays for the ill-conditioning of $\bar M$ once, through $\lambda_{\min}$, as the exact rate does; the peak factors enter only the threshold $k_0$ and the overshoot $C=O((\rho/\delta)^{1/2})$, not the rate.*

**Status.** Proved (below); checked pointwise by a certificate—the supremum of $\dot V/V$ over all states and over a time grid is a generalised eigenvalue problem—on the sinusoidal regressor and on the slow-adaptation ideal of the Theorem E tests, where the guaranteed rate is within 12% ($\delta=0.1$) and 26% ($\delta=0.2$) of the exact rate at $k=400$ while Lemma E.0's is five orders of magnitude below it (`tests/test_excitation.py`, `exp_explicit_rate.py`, Table 12). The round-6 referee re-derived every step and certified the inequality independently on five regressor families without a failure (`review/REVIEW_round6_E0prime.md`).

*Proof.* Write $M$, $\Psi$, … for their values at $t$. Two facts about zero-mean antiderivatives are used throughout: if $f$ is $T$-periodic with zero mean and $g$ is its zero-mean antiderivative, then $\|g\|_\infty\le(T/4)\|f\|_\infty$—for the maximiser $t^*$ of $g$, $g(t^*)=\frac1T\int_0^T(g(t^*)-g(s))\,\mathrm{d}s\le\frac1T\int_0^T\operatorname{dist}(t^*,s)\|f\|_\infty\,\mathrm{d}s=(T/4)\|f\|_\infty$ with dist the distance on the circle, and the same for $-g$—and the same bound holds for every bilinear coordinate $u^\top gv$, hence $\|g(t)v\|\le(T/4)\sup_r\|f(r)v\|$. As $\|M-\bar M\|\le\Psi_M^2$ (two positive semidefinite matrices of norm $\le\Psi_M^2$), this gives $\|P_1\|\le T\Psi_M^2/2=p_1$ and $\|P_1v\|\le(T/2)\sup\|(M-\bar M)v\|$, and $\|S\|\le(T/4)\|F\|_\infty$, $\|Sv\|\le(T/4)\sup\|Fv\|$.
*(a) The fast variable.* From $\dot e=-ke-\Psi\zeta$ and $ke_2=-\Psi\zeta+\varepsilon\dot\Psi\zeta-\varepsilon^2\Psi M\zeta$,

$$\dot e=-kw-\varepsilon\dot\Psi\zeta+\varepsilon^2\Psi M\zeta.$$

Differentiating $e_2$ along the flow, with $\dot\zeta=\Psi^\top e=\Psi^\top w-\varepsilon M\zeta+\varepsilon^2\Psi^\top\dot\Psi\zeta-\varepsilon^3M^2\zeta$ and $\dot M=\dot\Psi^\top\Psi+\Psi^\top\dot\Psi$, the terms of order $\varepsilon$ and $\varepsilon^2$ in $\zeta$ cancel and

$$\begin{aligned}\dot w=&-kw+\varepsilon\Psi\Psi^\top w-\varepsilon^2\dot\Psi\Psi^\top w+\varepsilon^3\Psi M\Psi^\top w-\varepsilon^2\ddot\Psi\zeta+\varepsilon^3\big(2\dot\Psi M+\Psi\dot\Psi^\top\Psi+2\Psi\Psi^\top\dot\Psi\big)\zeta\\&-\varepsilon^4\big(2\Psi M^2+\dot\Psi\Psi^\top\dot\Psi\big)\zeta+\varepsilon^5\big(\Psi M\Psi^\top\dot\Psi+\dot\Psi M^2\big)\zeta-\varepsilon^6\Psi M^3\zeta.\end{aligned}$$

*(b) The slow part of $\dot V$.* With $H:=I+\varepsilon P_1+\varepsilon^2P_2$ (symmetric), $\frac{\mathrm{d}}{\mathrm{d}t}[\frac12\zeta^\top H\zeta]=\zeta^\top H\dot\zeta+\frac12\zeta^\top\dot H\zeta$ and $\dot H=2\varepsilon(M-\bar M)+\varepsilon^2(F-\dot M)$. Insert $\dot\zeta$ and collect by powers of $\varepsilon$. At order $\varepsilon$: $-\zeta^\top M\zeta+\zeta^\top(M-\bar M)\zeta=-\zeta^\top\bar M\zeta$. At order $\varepsilon^2$:

$$\begin{aligned}-\zeta^\top P_1M\zeta+\zeta^\top\Psi^\top\dot\Psi\zeta+\tfrac12\zeta^\top(F-\dot M)\zeta=&-\tfrac12\zeta^\top(P_1M+MP_1)\zeta+\tfrac12\zeta^\top\dot M\zeta\\&+\tfrac12\zeta^\top(P_1M+MP_1-\dot M)\zeta=0,\end{aligned}$$

using $\zeta^\top\Psi^\top\dot\Psi\zeta=\frac12\zeta^\top\dot M\zeta$: the choice of $P_2$ removes the whole second-order term, and this is where $N_2=0$ enters (otherwise $-\zeta^\top N_2\zeta$ would remain). The remaining terms are

$$R_\zeta:=\varepsilon^3\big(-\zeta^\top P_2M\zeta+\zeta^\top P_1\Psi^\top\dot\Psi\zeta-\zeta^\top M^2\zeta\big)+\varepsilon^4\big(\zeta^\top P_2\Psi^\top\dot\Psi\zeta-\zeta^\top P_1M^2\zeta\big)-\varepsilon^5\zeta^\top P_2M^2\zeta,$$

plus the coupling $\zeta^\top H\Psi^\top w$, so $\frac{\mathrm{d}}{\mathrm{d}t}[\frac12\zeta^\top H\zeta]=-\varepsilon\zeta^\top\bar M\zeta+R_\zeta+\zeta^\top H\Psi^\top w$.

*(c) Per-direction bounds.* For every $v$ and $t$:

$$\|\Psi v\| \le \sqrt{\rho}\|v\|_{\bar M}, \qquad \|\dot\Psi v\| \le \sqrt{\rho_D}\|v\|_{\bar M}, \qquad \|\ddot\Psi v\| \le \sqrt{\rho_{DD}}\|v\|_{\bar M};$$
$$\|Mv\| \le \Psi_M\|\Psi v\| \le \Psi_M\sqrt{\rho}\|v\|_{\bar M}; \qquad \|\bar M v\| \le \lambda_{\max}(\bar M)^{1/2}\|v\|_{\bar M} \le \Psi_M\|v\|_{\bar M};$$
$$\|\Psi^\top\dot\Psi v\| \le \Psi_M\sqrt{\rho_D}\|v\|_{\bar M}; \qquad \|P_1 v\| \le \tfrac{T}{2}\sup_r\|(M(r)-\bar M)v\| \le \pi_1\|v\|_{\bar M};$$
$$\|Fv\| \le \|P_1 Mv\| + \|MP_1 v\| \le \tfrac{T}{2}\Psi_M^3\sqrt{\rho}\|v\|_{\bar M} + \Psi_M^2\pi_1\|v\|_{\bar M} = T\Psi_M^3(\sqrt{\rho}+\tfrac12)\|v\|_{\bar M},$$

so $\|Sv\| \le (T/4)\sup\|Fv\| = (T^2/8)\Psi_M^3(2\sqrt{\rho}+1)\|v\|_{\bar M}$ and $\|P_2 v\| \le \|(M-\bar M)v\| + \|Sv\| \le \pi_2\|v\|_{\bar M}$; also

$$\|P_2\| \le \Psi_M^2 + \tfrac{T}{4}\|F\|_\infty \le \Psi_M^2 + \tfrac{T}{4}\cdot 2\cdot\tfrac{T\Psi_M^2}{2}\Psi_M^2 = p_2.$$

Hence $|R_\zeta| \le r_\zeta\|\zeta\|_{\bar M}^2$, $|\zeta^\top H\Psi^\top w| = |\langle \Psi H\zeta, w\rangle| \le c_1\|\zeta\|_{\bar M}\|w\|$, and from (a), $\beta w^\top\dot w \le \beta\big[-(k-d_w)\|w\|^2 + c_2\|\zeta\|_{\bar M}\|w\|\big]$: each $\zeta$-term of $\dot w$ is bounded through $\|\Psi\zeta\|$, $\|\dot\Psi\zeta\|$, $\|\ddot\Psi\zeta\|$, $\|M\zeta\|$ or $\|M^2\zeta\|$ (the terms of $c_2$, in the order of (a); the last one is $\|\Psi M^3\zeta\| \le \Psi_M\cdot\Psi_M^4\|M\zeta\| \le \Psi_M^6\sqrt{\rho}\|\zeta\|_{\bar M}$), the $w$-terms through operator norms ($d_w$).

*(d) Assembly.* $\dot V \le -\varepsilon\|\zeta\|_{\bar M}^2 + r_\zeta\|\zeta\|_{\bar M}^2 + (c_1+\beta c_2)\|\zeta\|_{\bar M}\|w\| - \beta(k-d_w)\|w\|^2$. By Young, $(c_1+\beta c_2)\|\zeta\|_{\bar M}\|w\| \le \delta\varepsilon\|\zeta\|_{\bar M}^2 + (c_1+\beta c_2)^2\|w\|^2/(4\delta\varepsilon)$, and $(c_1+\beta c_2)^2 \le 2c_1^2 + 2\beta^2 c_2^2$. With $\beta = 2c_1^2/\delta$ the term $2c_1^2/(4\delta\varepsilon)$ equals $\beta k/4$, and (A) gives $2\beta^2c_2^2/(4\delta\varepsilon) \le \beta k/4$, so with (B) the coefficient of $\|w\|^2$ is at most $-\beta(k-d_w-k/2) \le -\beta k/4$; the coefficient of $\|\zeta\|_{\bar M}^2$ is $-\varepsilon(1-\delta-r_\zeta/\varepsilon) = -\varepsilon c_\zeta$. This is the first inequality. Since $\frac12(1-\varepsilon p_1-\varepsilon^2 p_2)\|\zeta\|^2 + \frac{\beta}{2}\|w\|^2 \le V \le \frac12(1+\varepsilon p_1+\varepsilon^2 p_2)\|\zeta\|^2 + \frac{\beta}{2}\|w\|^2$ by (D), and $\zeta^\top\bar M\zeta \ge \lambda_{\min}\|\zeta\|^2$, $\dot V \le -\gamma V$ follows, and the envelope with $C$ from $\|e\| \le \|w\| + (\varepsilon\Psi_M + \varepsilon^2\Psi_D + \varepsilon^3\Psi_M^3)\|\zeta\|$ and its converse. □

**Remark E.0′.1** (reading)**.**

- *What is sharp and what is not.* The exact rate of (LTV) is $\lambda_{\min}/k\cdot(1+O(k^{-2}))$: the first-order averaging of $\dot\zeta = -(1/k)M\zeta$ after the slaving gives $\lambda_{\min}/k$, and the second-order correction to the symmetric part of the averaged generator is $N_2 = 0$ (Table 12 measures $1-\text{exact}/(\lambda_{\min}/k) \approx 29/k^2$ on the ideal; on $\Psi = (\sin t, \cos t)$ the deviation is of order $k^{-4}$). Lemma E.0′ recovers this rate up to the factor $c_\zeta/(1+\varepsilon p_1+\varepsilon^2 p_2) \to 1-\delta$, and $\delta$ can be taken $\to 0$ as $k\to\infty$ (condition (A) allows $\delta \gtrsim 2\sqrt{\rho\rho_{DD}}/k^2$). The overshoot constant $C = O((\rho/\delta)^{1/2})$ is an artefact of the quadratic form, not of the dynamics.
- *Two regimes, two lemmas.* Lemma E.0 needs only $(\alpha, T, \Psi_M, \Psi_D)$ and gives the $\omega^{-2}$ law of the fast regime; its rate is quadratic in $\lambda_{\min}$. Lemma E.0′ needs $k$ above the threshold $k_0 \approx (4\rho\rho_{DD}/\delta^2)^{1/4}$, set by the peak factors of $\Psi$ and its two derivatives, and gives the sharp rate. On the regressors of Theorem E the derivative peak factors are large—the directions that are poorly excited on average are excited in short bursts, so $\rho_{DD}/\rho$ is of the order of the fourth power of the effective bandwidth—and $k_0$ is in the hundreds there, while on $\Psi = (\sin t, \cos t)$ ($\rho = \rho_D = \rho_{DD} = 2$) it is about 10, set by (C) rather than (A). A third-order slaving would replace $\rho_{DD}$ by the peak factor of $\dddot\Psi$ at one order higher in $\varepsilon$ and lower $k_0$ further; this is not pursued.
- *Hypotheses.* Only the periodicity of $\Psi$ is used (through the zero-mean antiderivatives; a non-periodic PE regressor would need a windowed $\bar M(t)$ and its derivative). Theorem E's $x_m$ is periodic, so $\Psi_G$ is. Theorem E's steps 4–5 go through with this $V$ in place of Lemma E.0's (Corollary E.0′.2 below): the nonlinear rate becomes $\gamma/4 \approx (1-\delta)\lambda_{\min}/(2k)$, while the explicit ball radius carries the anisotropy $p'_{\text{lo}}/p'_{\text{hi}} \approx 1/\beta$ of the quadratic form and can be smaller than Lemma E.0's.

**Numerics** (`exp_explicit_rate.py` → `results/explicit_rate/table.md`, table 4; reproduced in Table 12). The slow-adaptation ideal of Remark E.0a (table 2 of that file) at $g = 0.01$

($T = 20\pi$): $\lambda_{\min}(\bar{M}) = 2.16 \cdot 10^{-7}$, $\kappa(\bar{M}) = 7.8 \cdot 10^{3}$, $\Psi_M = 0.168$, $\Psi_D = 0.957$, $\Psi_{DD} = 8.39$, $T\Psi_M^2 = 1.78$; peak factors $\rho = 69.9$, $\rho_D = 3.7 \cdot 10^{3}$, $\rho_{DD} = 3.7 \cdot 10^{5}$ (the poorly excited directions are excited in bursts: $\rho_D/\rho \approx 53$ and $\rho_{DD}/\rho_D \approx 99$ are effective squared bandwidths); $\|N_2\| = 3 \cdot 10^{-21}$. The threshold of (A)–(D) is $k_0 = 226$ for $\delta = 0.2$ and 321 for $\delta = 0.1$, set by (A): $c_1 c_2 = 2.07$, 0.51, 0.13, 0.056, 0.032, 0.014 at $k = 50$, 100, 200, 300, 400, 600, the $k^{-2}$ law of $\varepsilon^2\sqrt{\rho\rho_{DD}}$.

Table 12: Lemma E.0′ in the slow regime on the slow-adaptation ideal of Remark E.0a at $g = 0.01$ (`results/explicit_rate/table.md`, table 4).

| $k$ | exact Floquet rate | exact / ($\lambda_{\min}/k$) | guaranteed rate $\gamma/2$, $\delta = 0.2$ ($\delta = 0.1$) | exact / guaranteed | exact / Lemma E.0 (optimised $\varepsilon$) | $\sup \dot{V}/V$ on the grid, $-\gamma$ ($\delta = 0.1$) |
|---|---|---|---|---|---|---|
| 300 | $7.21 \cdot 10^{-10}$ | 0.9997 | $5.70 \cdot 10^{-10}$ (below threshold) | 1.26 (—) | $1.33 \cdot 10^{5}$ | — |
| 400 | $5.41 \cdot 10^{-10}$ | 0.9998 | $4.30 \cdot 10^{-10}$ ($4.84 \cdot 10^{-10}$) | 1.26 (1.12) | $1.32 \cdot 10^{5}$ | $-1.079 \cdot 10^{-9}$, $-9.68 \cdot 10^{-10}$ |
| 600 | $3.61 \cdot 10^{-10}$ | 0.9999 | $2.88 \cdot 10^{-10}$ ($3.24 \cdot 10^{-10}$) | 1.25 (1.12) | $1.30 \cdot 10^{5}$ | $-7.20 \cdot 10^{-10}$, $-6.47 \cdot 10^{-10}$ |

Above the threshold the guaranteed rate is within 26% ($\delta = 0.2$) and 12% ($\delta = 0.1$) of the exact rate—the factor $1/(1-\delta)$ of the lemma, with $c_\zeta = 0.80$ and 0.90—while Lemma E.0's optimised rate is $1.3 \cdot 10^{5}$ times smaller; the certificate $\sup \dot{V}/V \le -\gamma$ holds on the 128-point grid in every row where the conditions hold, with $\sup \dot{V}/V = -1.078 \cdot 10^{-9} \approx -2\lambda_{\min}/k$ at $k = 400$ (attained in the slow direction at $w = 0$). The exact rate deviates from $\lambda_{\min}/k$ by 1.15%, 0.29%, 0.07% at $k = 50$, 100, 200: the $k^{-2}$ law that $N_2 = 0$ predicts. The overshoot constant is $C = \sqrt{\beta} \cdot (1 + O(1/k)) = 27$ ($\delta = 0.2$) and 38 ($\delta = 0.1$). Below $k_0$ the lemma makes no claim; Lemma E.0 does (Remark E.0a; table 2 of that file).

Checked by: `tests/test_excitation.py`: on $\Psi = (\sin t, \cos t)$ the computed $\lambda_{\min}$, $\rho$, $\rho_D$, $\rho_{DD}$ equal 1/2, 2, 2, 2, $N_2 = 0$ to $10^{-12}$, the threshold of (A)–(D) is below 12 for $\delta = 0.2$, the certificate $\sup \dot{V}/V \le -\gamma$ holds at $k = 20$, 50, 200, and the guaranteed rate is below the exact Floquet rate and within a factor 2 of it, the ratio decreasing towards $1/(1-\delta)$ (`test_lemma_E0prime_sinusoidal_regressor_certificate_and_sharpness`); on the slow-adaptation ideal of the Theorem E tests at $g = 0.01$ and $k = 400$, $\delta = 0.2$: $\kappa(\bar{M}) > 10^3$, the conditions hold, the certificate holds on a 128-point grid, the guaranteed rate is within a factor 1.5 of the exact rate and exceeds Lemma E.0's optimised rate by more than $10^4$ (`test_lemma_E0prime_pays_for_ill_conditioning_once_on_the_theorem_E_ideal`).

**Corollary E.0′.2** (Theorem E in the slow regime)**.** *Under the hypotheses of Theorem E, suppose the regressor $\Psi_G$ of its step 3 satisfies the conditions (A)–(D) of Lemma E.0′ at $k = k_e$ for some $\delta$, with constants $(\beta, \gamma, p_1, p_2)$ and $L_M := \varepsilon\Psi_{G,M} + \varepsilon^2\Psi_{G,D} + \varepsilon^3\Psi_{G,M}^3$ ($\varepsilon = 1/k_e$), and that $\beta L_M^2 < 1 - \varepsilon p_1 - \varepsilon^2 p_2$. Put*

$$p'_{\mathrm{lo}} := \tfrac{1}{2}\min\{\beta/2,\ 1 - \varepsilon p_1 - \varepsilon^2 p_2 - \beta L_M^2\},$$
$$p'_{\mathrm{hi}} := \tfrac{1}{2}\max\{\beta(1+L_M),\ 1 + \varepsilon p_1 + \varepsilon^2 p_2 + \beta L_M(1+L_M)\},$$
$$c'_0 := \min\Big\{\Big[\frac{\gamma p'_{\mathrm{lo}}}{4(p'_{\mathrm{hi}})^{1/2}\tilde{C}_R}\Big]^2,\ \frac{p'_{\mathrm{lo}} r^2}{\|S^{-1}\|^2}\Big\}, \qquad r'_0 := \Big(\frac{c'_0}{p'_{\mathrm{hi}}}\Big)^{1/2}, \qquad \lambda' := \frac{\gamma}{4}, \qquad C' := \Big(\frac{2p'_{\mathrm{hi}}}{p'_{\mathrm{lo}}}\Big)^{1/2},$$

*with $r$, $\tilde{C}_R$, $S$ as in Theorem E. Then the conclusions of Theorem E hold with $(c_0, \lambda, C)$ replaced by $((r'_0)^2/2, \lambda', C')$. In the slow regime $\lambda' = (1-\delta)(1 - \varepsilon p_1 + O(k_e^{-2}))\lambda_{\min}(\bar{M}_G)/(2k_e)$, $\bar{M}_G := \langle \Psi_G^\top \Psi_G \rangle$: half the exact rate of the linearised loop, up to the factor $1-\delta$ and the first-order deficit $\varepsilon p_1$.*

*Proof.* Write $V := \frac{1}{2} w^\top \mathbf{H}(t) w$ for Lemma E.0′'s $V$ in the coordinates $w = (e, \zeta)$ of step 3:

$$\mathbf{H} = \begin{bmatrix} \beta I & -\beta L \\ -\beta L^\top & H_\zeta + \beta L^\top L \end{bmatrix}$$

with $e_2 = L(t)\zeta$, $\|L\| \le L_M$ and $H_\zeta := I + \varepsilon P_1 + \varepsilon^2 P_2$. Then $w^\top \mathbf{H} w = \beta\|e - L\zeta\|^2 + \zeta^\top H_\zeta \zeta$, and $\|e - L\zeta\|^2 \le (1 + L_M)(\|e\|^2 + L_M\|\zeta\|^2)$, $\|e - L\zeta\|^2 \ge \frac{1}{2}\|e\|^2 - L_M^2\|\zeta\|^2$ give $2p'_{\mathrm{lo}}\|w\|^2 \le w^\top \mathbf{H} w \le 2p'_{\mathrm{hi}}\|w\|^2$. Along the nonlinear loop $\dot w = [A_0 + B(t)]w + \tilde R(t, w)$ with $\|\tilde R\| \le \tilde C_R\|w\|^2$ on $\{\|w\| < r/\|S^{-1}\|\}$ (step 4), so with Lemma E.0′,

$$\dot V \le -\gamma V + \langle \mathbf{H} w, \tilde R\rangle \le -\gamma V + \|\mathbf{H}^{1/2} w\|\,\|\mathbf{H}^{1/2}\|\,\tilde C_R\|w\|^2 \le -\gamma V + \frac{2(p'_{\mathrm{hi}})^{1/2}\tilde C_R}{p'_{\mathrm{lo}}} V^{3/2},$$

using $\|\mathbf{H}^{1/2} w\| = (2V)^{1/2}$, $\|\mathbf{H}^{1/2}\| = (2p'_{\mathrm{hi}})^{1/2}$ and $\|w\|^2 \le V/p'_{\mathrm{lo}}$. Hence $\dot V \le -(\gamma/2)V$ wherever $V^{1/2} \le \gamma p'_{\mathrm{lo}}/(4(p'_{\mathrm{hi}})^{1/2}\tilde C_R)$, and the sublevel set $\{V \le c'_0\}$—inside the ball where $\tilde R$ is bounded, by the second term of $c'_0$—is forward invariant with $V(t) \le e^{-(\gamma/2)(t-t_0)}V(t_0)$ on it. It contains $\{\|w\| \le r'_0\}$ because $V \le p'_{\mathrm{hi}}\|w\|^2$, and $\|w(t)\| \le (p'_{\mathrm{hi}}/p'_{\mathrm{lo}})^{1/2} e^{-(\gamma/4)(t-t_0)}\|w(t_0)\|$ follows. Step 5 of Theorem E is unchanged: $\|w\|^2 = 2V_\pi$, so $V_{\mathrm{OT}}(t_0) \le (r'_0)^2/2$ puts the state in the ball, $\|e\| + W \le \sqrt{2}\|w\|$ gives $C'$, and uniqueness and the limits follow as there. The asymptotics of $\lambda'$ are those of $\gamma$ in Lemma E.0′. □

**Numerics** (`exp_explicit_rate.py` → `results/explicit_rate/table.md`, table 5; reproduced in Table 13). Remark E.1's single-kernel instance ($m = n = 1$, $\vartheta^* = (1.5, 0.3, 0.8)$, $G = 0.3 \cdot I$, $x_m = 1.5 \sin t$) at $k_e = 40$: $\lambda_{\min}(\bar M_G) = 5.41 \cdot 10^{-2}$, $\kappa(\bar M_G) = 6.7$, peak factors $\rho = 4.8$, $\rho_D = 30.3$, $\rho_{DD} = 461$, threshold $k_0 = 24.4$ ($\delta = 0.2$) and 33.5 ($\delta = 0.1$); $L_J = 4.62$ estimated from samples; exact linear rate $1.35 \cdot 10^{-3}$.

Table 13: Theorem E in the slow regime on Remark E.1's single-kernel instance at $k_e = 40$ (`results/explicit_rate/table.md`, table 5).

| Lyapunov function | explicit ball radius $r_0$ ($w$-coordinates) | nonlinear rate $\lambda$ | overshoot $C$ | exact linear rate / $\lambda$ |
|---|---|---|---|---|
| Lemma E.0, optimised $\varepsilon$ (Theorem E) | $6.15 \cdot 10^{-6}$ | $2.63 \cdot 10^{-5}$ | 1.43 | 51 |
| Lemma E.0′, $\delta = 0.2$ (this corollary) | $1.53 \cdot 10^{-6}$ | $4.53 \cdot 10^{-4}$ | 11.8 | 3.0 |
| Lemma E.0′, $\delta = 0.1$ (this corollary) | $8.32 \cdot 10^{-7}$ | $5.16 \cdot 10^{-4}$ | 17.1 | 2.6 |

On this well-conditioned regressor ($\kappa \approx 7$) Lemma E.0 is only an order of magnitude conservative, and the corollary trades the certified ball for the rate: $\lambda'$ is within a factor 3 of the exact linear rate (the factor 2 of the Lyapunov argument times $1/(1-\delta)$), while $r'_0$ is four times smaller than Theorem E's $r_0$, carrying the anisotropy $p'_{\mathrm{lo}}/p'_{\mathrm{hi}} \approx 1/\beta$ of the quadratic form. On ill-conditioned regressors the rate improves by the factor of Remark E.0a (Table 12); the radius comparison was not computed there.

Checked by: `tests/test_excitation.py`: on the single-kernel instance at $k_e = 40$, $\delta = 0.2$, from a start with $\|w(0)\| \le r'_0$, the implemented closed loop satisfies $\|w(t)\| \le C' e^{-\lambda' t}\|w(0)\|$ at every sample over 1.5 e-folds of the explicit envelope ($\mathrm{d}t = 0.005$: the fixed-step loop has a discretisation floor $\approx 6 \cdot 10^{-6}$ at $\mathrm{d}t = 0.02$, above the explicit ball), with $\lambda'$ more than ten times Theorem E's rate from Lemma E.0 and within a factor 4 of the exact linear rate (`test_corollary_E0prime_2_envelope_on_the_implemented_loop`).

**Corollary E.0′.3** (the stationary parameter spread of the dithered linearised loop)**.** *Under the hypotheses of Lemma E.0′ (conditions (A)–(D) at $(k, \delta)$), consider* (LTV) *with parameter dither,*

$$\dot e = -ke - \Psi(t)\zeta, \qquad \mathrm{d}\zeta = \Psi(t)^\top e\,\mathrm{d}t + \Sigma\,\mathrm{d}B_t, \tag{LTV$_\tau$}$$

*with $\Sigma \in \mathbb{R}^{q\times q'}$ constant and $B$ a standard Brownian motion. Let $V$ be the strict Lyapunov function of Lemma E.0′ and $\mathbf{H}(t)$ its quadratic form ($V = \frac{1}{2}x^\top \mathbf{H}(t)x$, $x = (e,\zeta)$), $L(t) := -\varepsilon\Psi + \varepsilon^2\dot\Psi - \varepsilon^3\Psi M$ the slaving map ($e_2 = L(t)\zeta$), $L_M := \varepsilon\Psi_M + \varepsilon^2\Psi_D + \varepsilon^3\Psi_M^3 \geq \sup\|L\|$, and*

$$\nu(t) := \tfrac{1}{2}\operatorname{tr}\big(\Sigma^\top \mathbf{H}_{\zeta\zeta}(t)\Sigma\big) = \tfrac{1}{2}\operatorname{tr}\big(\Sigma^\top(I + \varepsilon P_1 + \varepsilon^2 P_2 + \beta L^\top L)\Sigma\big),$$
$$\bar\nu := \sup_t \nu(t) \leq \tfrac{1}{2}\|\Sigma\|_F^2\big(1 + \varepsilon p_1 + \varepsilon^2 p_2 + \beta L_M^2\big).$$

*Then, for every initial law with finite second moment,*

$$\limsup_{t\to\infty} \mathbb{E}V(t,x_t) \leq \frac{\bar\nu}{\gamma}, \qquad \limsup_{t\to\infty} \mathbb{E}\|\zeta_t\|^2 \leq \frac{2\bar\nu}{\gamma(1-\varepsilon p_1 - \varepsilon^2 p_2)}, \qquad \limsup_{t\to\infty} \mathbb{E}\|w_t\|^2 \leq \frac{2\bar\nu}{\gamma\beta},$$

*with $\gamma$ the rate of Lemma E.0′; in the slow regime $\gamma = 2c_\zeta\lambda_{\min}(\bar M)/(k(1+\varepsilon p_1+\varepsilon^2 p_2))$, so that*

$$\limsup \mathbb{E}\|\zeta_t\|^2 \leq \frac{k\|\Sigma\|_F^2\,(1+\varepsilon p_1+\varepsilon^2 p_2)(1+\varepsilon p_1+\varepsilon^2 p_2+\beta L_M^2)}{2c_\zeta(1-\varepsilon p_1-\varepsilon^2 p_2)\lambda_{\min}(\bar M)} = (1+O(\varepsilon))\,\frac{k\|\Sigma\|_F^2}{2(1-\delta)\lambda_{\min}(\bar M)}.$$

*For comparison, the averaged Langevin dynamics $\dot\zeta = -\frac{1}{k}\bar M\zeta + \Sigma\dot B$ has the exact stationary covariance $\bar C = kX$ with $\bar M X + X\bar M = \Sigma\Sigma^\top$, so $\mathbb{E}\|\zeta\|^2 = k\operatorname{tr}X$, equal to $\frac{k}{2}\operatorname{tr}(\bar M^{-1}\Sigma\Sigma^\top)$ when $\Sigma\Sigma^\top$ commutes with $\bar M$: the bound is within $(1+O(\varepsilon))/(1-\delta)$ of it when $\bar M$ is isotropic, and pays the condition number of $\bar M$ in the trace otherwise (it replaces every $1/\lambda_i$ by $1/\lambda_{\min}$), as a scalar Lyapunov function must.*

**Status.** Proved (Itô's formula and the pointwise inequality of Lemma E.0′). Checked (`exp_explicit_rate.py`, table 6; `tests/test_excitation.py`) against the exact periodic stationary covariance of ($\mathrm{LTV}_\tau$), computed from the periodic Lyapunov equation.

*Proof.* $V(t,x) = \frac{1}{2}x^\top\mathbf{H}(t)x$ is $C^1$ in $t$ and quadratic in $x$, so by Itô's formula along ($\mathrm{LTV}_\tau$), with $\tilde\Sigma := [0;\Sigma] \in \mathbb{R}^{(n+q)\times q'}$,

$$\begin{aligned}\mathrm{d}V &= \big[\partial_t V + \nabla_x V\cdot A(t)x\big]\,\mathrm{d}t + \nabla_x V\cdot\tilde\Sigma\,\mathrm{d}B_t + \tfrac{1}{2}\operatorname{tr}\big(\tilde\Sigma^\top\mathbf{H}(t)\tilde\Sigma\big)\,\mathrm{d}t\\ &\leq \big[-\gamma V + \nu(t)\big]\,\mathrm{d}t + \nabla_x V\cdot\tilde\Sigma\,\mathrm{d}B_t,\end{aligned}$$

the inequality being Lemma E.0′'s $\dot V \leq -\gamma V$ pointwise in $(t,x)$, and $\tilde\Sigma^\top\mathbf{H}\tilde\Sigma = \Sigma^\top\mathbf{H}_{\zeta\zeta}\Sigma$ with $\mathbf{H}_{\zeta\zeta} = I+\varepsilon P_1+\varepsilon^2 P_2+\beta L^\top L$ from the block form of $\mathbf{H}$ (Corollary E.0′.2). The stochastic integral is a martingale because its integrand is linear in $x$ and the linear SDE has bounded second moments (Gaussian with a bounded covariance), so $\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}V \leq -\gamma\mathbb{E}V + \bar\nu$ and $\mathbb{E}V(t) \leq e^{-\gamma t}\mathbb{E}V(0) + \bar\nu/\gamma$. The bounds on $\mathbb{E}\|\zeta\|^2$ and $\mathbb{E}\|w\|^2$ follow from $V \geq \frac{1}{2}(1-\varepsilon p_1-\varepsilon^2 p_2)\|\zeta\|^2$ and $V \geq \frac{\beta}{2}\|w\|^2$; the estimate of $\bar\nu$ uses $\|P_1\| \leq p_1$, $\|P_2\| \leq p_2$, $\|L\| \leq L_M$. The averaged covariance is the stationary solution of the Lyapunov equation of the time-invariant averaged system. □

**Numerics** (`exp_explicit_rate.py` → `results/explicit_rate/table.md`, table 6; reproduced in Table 14). $\Sigma = (2\cdot 10^{-3})^{1/2}I$; the exact stationary covariance from the periodic Lyapunov equation (RK4 with $k\cdot\mathrm{d}t \leq 1/4$; the period mean and the maximum of $\mathbb{E}\|\zeta\|^2$ coincide to four digits), the averaged value $k\cdot\operatorname{tr}X$, and the bound:

Table 14: Corollary E.0′.3: the explicit bound on the stationary parameter spread of the dithered (LTV) against the exact periodic stationary covariance and the averaged value (`results/explicit_rate/table.md`, table 6).

| regressor ($\kappa(\bar{M})$) | $k$ | $\delta$ | exact $\mathbb{E}\|\zeta\|^2$ | averaged $k \cdot \operatorname{tr} X$ | bound | bound / exact | $1/(1-\delta)$ |
|---|---|---|---|---|---|---|---|
| Theorem E ideal of table 4 ($7.8 \cdot 10^3$) | 300 | 0.2 | $1.509 \cdot 10^6$ | $1.509 \cdot 10^6$ | $2.64 \cdot 10^7$ | 17.5 | 1.25 |
| | 400 | 0.2 / 0.1 | $2.012 \cdot 10^6$ | $2.012 \cdot 10^6$ | $3.51 \cdot 10^7$ / $3.11 \cdot 10^7$ | 17.4 / 15.5 | 1.25 / 1.11 |
| | 600 | 0.2 / 0.1 | $3.018 \cdot 10^6$ | $3.018 \cdot 10^6$ | $5.23 \cdot 10^7$ / $4.65 \cdot 10^7$ | 17.3 / 15.4 | 1.25 / 1.11 |
| $\Psi = (\sin t, \cos t)$ (1) | 20 | 0.2 / 0.1 | $8.025 \cdot 10^{-2}$ | $8.000 \cdot 10^{-2}$ | 0.216 / 0.202 | 2.69 / 2.52 | 1.25 / 1.11 |
| | 50 | 0.2 / 0.1 | $2.001 \cdot 10^{-1}$ | $2.000 \cdot 10^{-1}$ | 0.315 / 0.282 | 1.58 / 1.41 | 1.25 / 1.11 |
| | 200 | 0.2 / 0.1 | $8.000 \cdot 10^{-1}$ | $8.000 \cdot 10^{-1}$ | 1.051 / 0.935 | 1.31 / 1.17 | 1.25 / 1.11 |

The exact stationary spread is the averaged Langevin value up to $3 \cdot 10^{-4}$ (ideal) and $1.25/k^2$ (sinusoid): on these instances, where $\Sigma$ is isotropic so that $\Sigma\Sigma^\top$ commutes with $\bar{M}$, the averaging prediction of the dithered loop's stationary covariance is exact to $O(k^{-2})$. When $\Sigma\Sigma^\top$ does not commute with $\bar{M}$ the deviation is $O(1/k)$ (referee round 7 measured $\pm 0.25/k$ on two-dimensional sinusoids with anisotropic $\Sigma$, the bound still holding with ratios 2.7–5.1). On the isotropic sinusoid the bound exceeds the exact value by a factor that falls toward $1/(1-\delta)$ (2.7, 1.6, 1.3 at $k = 20$, 50, 200 for $\delta = 0.2$; the excess is the $O(1/k)$ of the factors $(1 \pm \varepsilon p_1) \dots$ and $\beta L_M^2$). On the ideal, where one direction is excited $7.8 \cdot 10^3$ times less than the others, the bound is 15–18 times the exact value: it charges every one of the $q = 15$ directions the price of the worst, $(q/\lambda_{\min})/\sum_i (1/\lambda_i) \approx q = 15$ (up to $1/(1-\delta)$), the trace-level price of a scalar Lyapunov function. The exact $\mathbb{E}\|e\|^2$ is $\|\Sigma\|_F^2/(2k) = \operatorname{tr}(\bar{M}X)/k$ to five digits on both instances (the error rides on its slow manifold $e \approx \Psi\zeta/k$, and $\operatorname{tr}(\bar{M}X) = \frac{1}{2}\|\Sigma\|_F^2$ by the trace of the Lyapunov equation), while the bound on $\mathbb{E}\|w\|^2$ is coarse (factor $1/\beta$): on the sinusoid the exact stationary $\mathbb{E}\|w\|^2 = \mathbb{E}\|e - L\zeta\|^2$, computed from the periodic covariance and the slaving map $L = -\varepsilon\Psi + \varepsilon^2\dot{\Psi} - \varepsilon^3\Psi M$, is more than three orders of magnitude below the bound (asserted in the test below).

Checked by: `tests/test_excitation.py`: on $\Psi = (\sin t, \cos t)$ with $\Sigma = (2 \cdot 10^{-3})^{1/2} I$, the explicit bound dominates the exact period mean and the exact maximum of $\mathbb{E}\|\zeta\|^2$ at $k = 20$, 50, 200 for $\delta = 0.1$ and 0.2, its ratio to the exact value decreases toward $1/(1-\delta)$ (within 6% at $k = 200$), the grid value of $\bar{\nu}$ lies below its explicit bound, and the exact value agrees with the averaged $k \cdot \operatorname{tr} X$ within $2/k^2$ (`test_corollary_E0prime_3_dither_covariance_bound_against_the_periodic_lyapunov_solution`).

**Lemma E.0″** (the slow regime for non-periodic persistently exciting regressors)**.** *Let $\Psi\colon [0,\infty) \to \mathbb{R}^{n\times q}$ be $C^2$ with $\|\Psi\| \le \Psi_M$, $\|\dot{\Psi}\| \le \Psi_D$, $\|\ddot{\Psi}\| \le \Psi_{DD}$, $M := \Psi^\top\Psi$, and fix a window $T > 0$. For a bounded continuous $X$ let*

$$\langle X\rangle_T(t) := \frac{1}{T}\int_0^\infty e^{-s/T} X(t+s)\,\mathrm{d}s$$

*(the forward exponential average; $T\frac{\mathrm{d}}{\mathrm{d}t}\langle X\rangle_T = \langle X\rangle_T - X$),*

*and $\bar{M} := \langle M\rangle_T$. Assume the exponentially weighted PE condition*

$$\bar{M}(t) \succeq \lambda I_q \quad \text{for all } t \ge 0, \quad \text{with } \lambda > 0, \tag{PE$_T$}$$

*which is equivalent to classical PE ($\int_t^{t+T} M \succeq \alpha I$ for all $t$ gives* (PE$_T$) *with $\lambda \ge e^{-1}\alpha/T$, and* (PE$_T$) *gives PE over windows $T' > T\log(\Lambda/\lambda)$ with level $T(\lambda - e^{-T'/T}\Lambda)$; proof below). Let $\Lambda := \sup_t \lambda_{\max}(\bar{M}(t)) \le \Psi_M^2$. Fix symmetric matrices $\hat{M}$, $\hat{F}$ (any; the long-time means of $\bar{M}$*

*and $\bar F$ minimise the time-averaged squares of $\|\bar M-\hat M\|$, $\|\bar F-\hat F\|$, not necessarily the suprema below) and define*

$$P_1:=-2T(\bar M-\hat M)\ \ (so\ \dot P_1=2(M-\bar M)),\qquad F:=P_1M+MP_1,$$
$$\bar F:=\langle F\rangle_T,\qquad S:=-T(\bar F-\hat F)\ \ (\dot S=F-\bar F),$$
$$P_2:=-(M-\bar M)+S,\qquad \bar N_2:=\tfrac12(\bar F-\dot{\bar M})=\tfrac12\big(\bar F-(\bar M-M)/T\big),$$

*and $e_2$, $w$, $V$ exactly as in Lemma E.0′ ($V=\frac12\zeta^\top(I+\varepsilon P_1+\varepsilon^2P_2)\zeta+\frac{\beta}{2}\|w\|^2$, $\varepsilon=1/k$). The constants are those of Lemma E.0′ with the period mean replaced by the exponential average, all of them computable from the regressor: the time-local peak factors*

$$\rho:=\sup_{t,v}\frac{\|\Psi(t)v\|^2}{v^\top\bar M(t)v},\qquad \rho_D:=\sup\frac{\|\dot\Psi(t)v\|^2}{v^\top\bar M(t)v},\qquad \rho_{DD}:=\sup\frac{\|\ddot\Psi(t)v\|^2}{v^\top\bar M(t)v}$$
$$(all\ \le\Psi_\cdot^2/\lambda),$$

*the operator and per-direction norms*

$$p_1:=\sup_t\|P_1(t)\|=2T\sup\|\bar M-\hat M\|,\qquad p_2:=\sup_t\|P_2(t)\|\le\Psi_M^2+T\sup\|\bar F-\hat F\|,$$
$$\pi_1:=\sup_{t,v}\frac{\|P_1(t)v\|}{\|v\|_{\bar M(t)}}\le 2T\big(\sqrt{\Lambda}+\|\hat M\|/\sqrt{\lambda}\big),$$
$$\pi_2:=\sup_{t,v}\frac{\|P_2(t)v\|}{\|v\|_{\bar M(t)}}\le\Psi_M\sqrt{\rho}+\sqrt{\Lambda}+T\big(2p_1\Psi_M^2+\|\hat F\|\big)/\sqrt{\lambda},$$
$$n_2:=\sup_t\|\bar N_2(t)\|\le\tfrac12\big(\sup\|F\|+2\Psi_M^2/T\big)\le p_1\Psi_M^2+\Psi_M^2/T,$$

*and, with these $\pi_1$, $\pi_2$, $p_1$, $p_2$, the quantities $c_1$, $c_2$, $d_w$, $r_\zeta$ and $\beta:=2c_1^2/\delta$ of Lemma E.0′ (same formulas), together with*

$$\delta_2:=\varepsilon n_2/\lambda,\qquad c_\zeta:=1-\delta-\delta_2-r_\zeta/\varepsilon.$$

*Fix $\delta\in(0,1)$. If* (A) $c_1c_2\le\delta/2$, (B) $d_w\le k/4$, (C) $c_\zeta>0$ *and* (D) $\varepsilon p_1+\varepsilon^2p_2<1$, *then along* (LTV), *for all $(e,\zeta,t)$,*

$$\dot V\le-\varepsilon c_\zeta\,\zeta^\top\bar M(t)\zeta-\frac{\beta k}{4}\|w\|^2\le-\gamma V,\qquad \gamma:=\min\Big\{\frac{2c_\zeta\lambda}{k(1+\varepsilon p_1+\varepsilon^2p_2)},\ \frac{k}{2}\Big\},$$

*and $\|(e(t),\zeta(t))\|\le Ce^{-\gamma(t-t_0)/2}\|(e(t_0),\zeta(t_0))\|$ with the $C$ of Lemma E.0′. In the slow regime the guaranteed rate is*

$$\gamma/2=(1-\delta-\delta_2)\big(1-\varepsilon p_1+O(k^{-2})\big)\,\lambda/k,\qquad \lambda=\inf_t\lambda_{\min}\big(\langle M\rangle_T(t)\big):$$

*the exponentially weighted PE level of the window divided by $k$, up to the factor $1-\delta-\delta_2$, where $\delta_2=O(1/k)$ is the price of the residual second-order term (zero for a periodic $\Psi$ with $\hat M=\langle M\rangle$ in the limit $T\to\infty$, where Lemma E.0′ applies directly).*

**Status.** Proved (the proof of Lemma E.0′ with the changes listed below). Checked (`exp_pe_general.py`, `tests/test_excitation.py`) on a quasi-periodic regressor and on a regressor driven by a Lorenz orbit, where no periodic construction applies: the conditions hold from $k=20$ on for the quasi-periodic regressor and from $k=100$ on for the chaotic one (the derivative peak factors of a chaotic orbit are large), the certificate $\sup\dot V/V\le-\gamma$ holds on the time grid wherever they hold, and the guaranteed rate is within a factor 1.3–2.5 of the exact decay rate on the quasi-periodic regressor ($k=50$–$200$) and, with the window $T=20$, within a factor 3.5–5 on the chaotic one (shorter windows pay more: 7.5–11 at $T=10$ and 14–138 at $T=5$), where the worst exponential window is three times less exciting than the long-time average. Referee round 7 (`review/REVIEW_round7_E0p3_E0dp.md`) confirmed the proof and the code and found no counterexample on an attack set of barely exciting, bursty, near-resonant and chaotic regressors.

*Proof.* Follow the proof of Lemma E.0′ with $\bar M$, $P_1$, $S$, $P_2$ as defined here. (a) is unchanged (it does not involve the averages). In (b), $\dot H = 2\varepsilon(M-\bar M)+\varepsilon^2(F-\bar F-\dot M+\dot{\bar M})$ exactly, by $\dot P_1 = 2(M-\bar M)$ and $\dot P_2 = -\dot M + \dot{\bar M} + \dot S = -\dot M + \dot{\bar M} + F - \bar F$; the order-$\varepsilon$ terms give $-\zeta^\top \bar M\zeta$ as before, and the order-$\varepsilon^2$ terms give $-\zeta^\top P_1 M\zeta + \frac12\zeta^\top \dot M\zeta + \frac12\zeta^\top(F-\bar F-\dot M+\dot{\bar M})\zeta = \frac12\zeta^\top(\dot{\bar M}-\bar F)\zeta = -\zeta^\top\bar N_2\zeta$, since $\zeta^\top P_1 M\zeta = \frac12\zeta^\top F\zeta$: the periodic cancellation $N_2 = 0$ is replaced by the residual $-\varepsilon^2\zeta^\top\bar N_2\zeta$, and $R_\zeta$ has the same form, so $\frac{\mathrm d}{\mathrm dt}[\frac12\zeta^\top H\zeta] = -\varepsilon\zeta^\top(\bar M+\varepsilon\bar N_2)\zeta + R_\zeta + \zeta^\top H\Psi^\top w$. In (c), every per-direction bound is taken in the time-local norm $\|v\|_{\bar M(t)}$: $\|\Psi(t)v\| \le \sqrt{\rho}\|v\|_{\bar M(t)}$ and likewise for $\dot\Psi$, $\ddot\Psi$ by the definition of the peak factors; $\|M(t)v\| \le \Psi_M\sqrt{\rho}\|v\|_{\bar M(t)}$; $\|\bar M(t)v\| \le \sqrt{\Lambda}\|v\|_{\bar M(t)}$; $\|P_1(t)v\| = 2T\|(\bar M(t)-\hat M)v\| \le 2T(\sqrt{\Lambda}+\|\hat M\|/\sqrt{\lambda})\|v\|_{\bar M(t)}$, using $\sqrt{\lambda}\|v\| \le \|v\|_{\bar M(t)}$ (this is where the exponential average pays off: $P_1(t)$ is a function of $\bar M(t)$ alone, with no integral over other times); for $\bar F(t)v$ and $S(t)v$, which do involve other times $r \ge t$, $\|F(r)v\| \le \|P_1(r)\|\|M(r)v\| + \|M(r)\|\|P_1(r)v\| \le 2p_1\Psi_M^2\|v\| \le (2p_1\Psi_M^2/\sqrt{\lambda})\|v\|_{\bar M(t)}$, an average of such bounds bounds $\|\bar F(t)v\|$, and $\|S(t)v\| \le T(\|\bar F(t)v\| + \|\hat F v\|)$, whence the bound on $\pi_2$; the stated bounds on $p_2$ and $n_2$ are $\|P_2\| \le \|M-\bar M\| + \|S\|$ and $\|\bar N_2\| \le \frac12(\|\bar F\| + \|\dot{\bar M}\|)$ with $\|\dot{\bar M}\| = \|\bar M - M\|/T \le 2\Psi_M^2/T$ and $\|\bar F\| \le \sup\|F\| \le 2p_1\Psi_M^2$. The bounds $|R_\zeta| \le r_\zeta\|\zeta\|^2_{\bar M}$, $|\zeta^\top H\Psi^\top w| \le c_1\|\zeta\|_{\bar M}\|w\|$ and $\beta w^\top\dot w \le \beta[-(k-d_w)\|w\|^2 + c_2\|\zeta\|_{\bar M}\|w\|]$ follow with the same formulas (the suprema over $v$ at the current time $t$ of the norms that appear). In (d), the extra term is bounded by $\varepsilon^2|\zeta^\top\bar N_2\zeta| \le \varepsilon^2 n_2\|\zeta\|^2 \le (\varepsilon^2 n_2/\lambda)\zeta^\top\bar M\zeta = \varepsilon\delta_2\zeta^\top\bar M\zeta$, which turns $1-\delta-r_\zeta/\varepsilon$ into $c_\zeta = 1-\delta-\delta_2-r_\zeta/\varepsilon$; the rest of the assembly, the sandwich $\frac12(1-\varepsilon p_1-\varepsilon^2 p_2)\|\zeta\|^2 + \frac{\beta}{2}\|w\|^2 \le V \le \frac12(1+\varepsilon p_1+\varepsilon^2 p_2)\|\zeta\|^2 + \frac{\beta}{2}\|w\|^2$ and $\zeta^\top\bar M(t)\zeta \ge \lambda\|\zeta\|^2$ give $\dot V \le -\gamma V$. The equivalence of (PE$_T$) with classical PE: if $\int_t^{t+T} M \succeq \alpha I$ then $\bar M(t) \succeq \frac1T\int_0^T e^{-s/T}M(t+s)\,\mathrm ds \succeq \frac{e^{-1}}{T}\int_t^{t+T} M \succeq (e^{-1}\alpha/T)I$; conversely (PE$_T$) gives, for every $T' > 0$,

$$\int_t^{t+T'} M \succeq \int_t^{t+T'} Me^{-(r-t)/T}\,\mathrm dr \succeq T\big(\bar M(t) - e^{-T'/T}\bar M(t+T')\big) \succeq T(\lambda - e^{-T'/T}\Lambda)I,$$

positive for $T' > T\log(\Lambda/\lambda)$ (the identity $\int_t^{t+T'} e^{-(r-t)/T}M(r)\,\mathrm dr = T[\bar M(t) - e^{-T'/T}\bar M(t+T')]$ follows from the definition of the exponential average). □

**Remark E.0″.1** (reading)**.**

- *What changes and what does not.* The construction, the conditions and the rate formula are those of Lemma E.0′; the period mean becomes the exponential average over a window, the peak factors become time-local, and one new term appears: the residual $\bar N_2$, whose price $\delta_2 = \varepsilon n_2/\lambda$ vanishes like $1/k$. For a periodic regressor the exponential average with a finite window keeps residual oscillations of relative size $O(\text{period}/T)$, so Lemma E.0′ is sharper there; Lemma E.0″ is the version for references that are not periodic — quasi-periodic (several incommensurate frequencies) or chaotic (a reference along a strange attractor, as in the Lorenz-tracking experiments of Part I), where only PE is available.
- *The window is a design parameter.* The lemma holds for every $T > 0$; the guaranteed rate is $(1-\delta-\delta_2)\lambda_T/k$ with $\lambda_T = \inf\lambda_{\min}(\langle M\rangle_T)$, which increases toward the long-time excitation level as $T$ grows, against $p_1 \propto T$, $n_2 \propto T$ and the threshold in $k$, which grow with $T$. The numerics show the optimum at a window of a few characteristic times of the regressor.
- *Exact rate.* For a non-periodic regressor the exact rate is the top Lyapunov exponent of (LTV), which the certificate does not need: the lemma is a pointwise inequality in $(t, e, \zeta)$, checked on a grid in $t$ by the same generalised eigenvalue problem as Lemma E.0′.

**Numerics** (`exp_pe_general.py` → `results/explicit_rate/general_pe.md`). Two regressors that are not periodic, on $[0, 300]$ (grid step 0.005; the exponential averages formed from

an extended grid; the certificate $\sup \dot{V}/V$ over 400 times and all states; the exact rate the top Lyapunov exponent of (LTV) over 200 time units), with the measured constants (the grid suprema of $\|P_1\|$, $\|P_2\|$, the per-direction norms and $\|\bar{N}_2\|$, computable constants of the regressor like the peak factors).

*Quasi-periodic* $\Psi = (\sin t, \cos\sqrt{2}\,t)$ ($q = 2$; the long-time mean of $M$ is $\frac{1}{2}I$, and the exact decay rate is $\lambda_\infty/k = 0.5/k$ to within 1%, 0.6% at $k = 20$ and 0.8% at $k = 400$): the window average has $\lambda_T = 0.297$, 0.391, 0.444 for $T = 2\pi$, $4\pi$, $8\pi$ ($\rho = 4.0$, $\rho_D \approx 6.1$–$6.4$, $\rho_{DD} \approx 10.0$, $p_1 \approx 2.9$, $p_2 = 8.0$–$9.2$, $\pi_1 = 4.3$–$4.7$, $\pi_2 = 11.6$–$13.0$, $\|\bar{N}_2\| = 0.71$, 0.35, 0.17); the conditions hold from $k = 20$ on for both $\delta$, the certificate holds at every row, and (Table 15)

Table 15: Lemma E.0″ on the quasi-periodic regressor $\Psi = (\sin t, \cos\sqrt{2}\,t)$: the guaranteed rate against the exact decay rate for two windows (`results/explicit_rate/general_pe.md`).

| $T$ | $k$ | $\delta_2$ | guaranteed $\gamma/2$ ($\delta = 0.2$ / 0.1) | exact | exact/guaranteed | $\lambda_T/k$ |
|---|---|---|---|---|---|---|
| $4\pi$ | 20 | 0.045 | 0.0100 / 0.0117 | 0.0249 | 2.49 / 2.13 | 0.0195 |
| $4\pi$ | 50 | 0.018 | 0.00557 / 0.00631 | 0.00995 | 1.78 / 1.58 | 0.00781 |
| $4\pi$ | 200 | 0.004 | 0.00153 / 0.00172 | 0.00249 | 1.63 / 1.45 | 0.00195 |
| $8\pi$ | 20 | 0.020 | 0.0118 / 0.0137 | 0.0249 | 2.10 / 1.81 | 0.0222 |
| $8\pi$ | 200 | 0.002 | 0.00174 / 0.00196 | 0.00249 | 1.43 / 1.27 | 0.00222 |

($T = 2\pi$: 3.69 → 2.16 for $\delta = 0.2$ and 3.10 → 1.92 for $\delta = 0.1$ from $k = 20$ to 200.) The ratio exact/guaranteed is the product of the window loss $\lambda_\infty/\lambda_T$ (1.69, 1.28, 1.13 for the three windows), the factor $1/(1 - \delta - \delta_2)$ and the $O(1/k)$ factors; the explicit closed-form constants satisfy the conditions from $k = 50$ on and give rates below the measured-constant ones by 22–79% at $k = 50$ (the longer the window, the larger the deficit), 5–19% at $k = 100$, at most 5% at $k = 200$ and about 1% at $k = 400$. All constants are suprema over $[0, 300]$ (the exponential averages are formed on an extended grid): on the quasi-periodic regressor they are identical to four digits on $[0, 10^4]$ (closed form); on the Lorenz regressor below, the suprema over $[0, 1500]$ are larger by up to 14% ($p_1$ 5.31 → 5.67, $p_2$ 19.4 → 20.7, $\pi_2$ 49.4 → 51.9, $\|\bar{N}_2\|$ 0.42 → 0.48 at $T = 20$), which lowers the guaranteed rate at $k = 100$ by 3% and keeps the certificate (referee round 7): the constants of a chaotic regressor are those of the stretch of the attractor that the run visits.

*Chaotic* $\Psi = (x_1/20, (x_3 - 25)/20)$ *along a Lorenz orbit* ($\sigma = 10$, $r = 28$, $b = 8/3$, traversed five times slower than standard Lorenz time, $\dot{\Psi}$ by the chain rule through the vector field; $q = 2$): the exact decay rates are 0.00759, 0.00304, 0.00152, 0.00076, 0.00038 at $k = 20$, 50, 100, 200, 400 (an effective excitation level 0.15), while the worst exponential window has $\lambda_T = 0.015$, 0.024, 0.051 for $T = 5$, 10, 20 ($\Lambda = 0.55$, 0.36, 0.24; $\rho \approx 20$–$26$, $\rho_D \approx 110$–$160$, $\rho_{DD} \approx 2000$–$3100$; $\|\bar{N}_2\| = 1.05$, 0.34, 0.42: the residual does not follow a monotone rule in $T$). The conditions hold from $k = 100$ on (condition (A) fails below: the derivative peak factors of a chaotic orbit are large), the certificate holds at every admissible row, and at $T = 20$ the guaranteed rate is 0.00030 / 0.00035 ($k = 100$), 0.00018 / 0.00021 ($k = 200$) against the exact 0.00152 and 0.00076: exact/guaranteed = 5.0 / 4.3 and 4.2 / 3.7, of which the window loss $\lambda_\infty/\lambda_T \approx 3$ is the dominant factor. For a chaotic reference the uniform-in-time construction pays the least exciting stretch of the attractor; a longer window ($T = 20$ against the slowed Lorenz loop time $\approx 3.75$) recovers part of it ($\lambda_T = 0.015 \to 0.051$ from $T = 5$ to 20) at the price of a larger $p_2$ and $\pi_2$ (9.9 → 19.4 and 13.7 → 49.4). An exploratory regressor with components $\tanh(x_i/8)$ traversed at standard Lorenz speed (not in the table) had $\rho_{DD}$ of order $10^6$–$10^7$ and a threshold beyond $k = 400$: the slow regime requires the reference to be slow relative to the loop, and the time scale of a chaotic reference is the designer's choice.

Checked by: `tests/test_excitation.py`: on $\Psi = (\sin t, \cos\sqrt{2}\,t)$ with $T = 4\pi$, the differential identities $T\dot{\bar{M}} = \bar{M} - M$ and $\dot{P}_1 = 2(M - \bar{M})$ on the grid ($10^{-4}$), the conditions at $k = 20$, 50, 200 for $\delta = 0.1$, 0.2 with $\delta_2 \le 5\%$, the certificate $\sup \dot{V}/V \le -\gamma$ on 400 times, and the guaranteed rate below the exact

decay rate (top Lyapunov exponent over 200 time units) within a factor 2.6, the ratio improving with $k$ (`test_lemma_E0dprime_quasi_periodic_regressor_certificate_and_rate`).

**Theorem E** (with explicit constants)**.** *Let $\Psi_G := m^{-1/2}\Psi(I_m \otimes G^{1/2}) \in \mathbb{R}^{n\times mp}$, and let $(\alpha_G, T, \Psi_{G,M}, \Psi_{G,D})$ be PE data for it: $\int_t^{t+T} \Psi_G^\top \Psi_G \,\mathrm{d}r \succeq \alpha_G I$, $\|\Psi_G\| \le \Psi_{G,M}$, $\|\dot\Psi_G\| \le \Psi_{G,D}$. One may take $\alpha_G := \alpha\lambda_{\min}(G)/m$, $\Psi_{G,M} := m^{-1/2}\lambda_G^{1/2}\sup\|\Psi\|$ and $\Psi_{G,D} := m^{-1/2}\lambda_G^{1/2}\sup\|\dot\Psi\|$; $\dot\Psi$ is bounded because $x_m \in C^1$ with bounded derivative and the Gaussian's mixed derivatives are bounded for $s \ge s_{\min}$. Let $(\varepsilon, \gamma, p_{\mathrm{lo}}, p_{\mathrm{hi}}, g_V)$ be the constants of Lemma E.0 for the data $(k_e, \alpha_G, T, \Psi_{G,M}, \Psi_{G,D})$, with any admissible $\varepsilon$ (for example the closed form of E.0(ii)), where $g_V := 1 + \varepsilon(2Q_{G,M} + T\Psi_{G,M})$ bounds $\|\partial V/\partial w\|/\|w\|$ (the Hessian of $V$ is $I$ plus a matrix of norm at most $\varepsilon T\Psi_{G,M} + 2\varepsilon Q_{G,M}$) and $Q_{G,M} := T^2\Psi_{G,M}^2/2$. Let $r := \min(\rho, \delta)$, let $L_J$ be a Lipschitz constant, uniform in $t$, of the Jacobian of the closed-loop field $F$ of step 2 on the ball $D$, and let $S := \mathrm{diag}\big(I_n,\, m^{-1/2}(I_m \otimes G^{-1/2})\big)$ be the similarity to the coordinates $w := (e, \zeta)$ of step 3. Put*

$$\tilde C_R := \|S\|\,\frac{L_J}{2}\,\|S^{-1}\|^2, \qquad r_w := \min\Big\{\frac{r}{\|S^{-1}\|},\ \frac{\gamma}{2g_V\tilde C_R}\Big\}, \qquad r_0 := r_w\Big(\frac{p_{\mathrm{lo}}}{p_{\mathrm{hi}}}\Big)^{1/2},$$

$$\lambda := \frac{\gamma}{4p_{\mathrm{hi}}}, \qquad c_0 := \frac{r_0^2}{2}, \qquad C := \Big(\frac{2p_{\mathrm{hi}}}{p_{\mathrm{lo}}}\Big)^{1/2} \quad (\le \sqrt{6} \text{ for the closed-form } \varepsilon).$$

*If $V_{\mathrm{OT}}(t_0) \le c_0$, then:*

- *the optimal assignment $\pi$ at $t_0$ is unique, and it is the unique optimal assignment for all large $t$;*
- *$\|e(t)\| + W(\mu_t, \mu^*) \le Ce^{-\lambda(t-t_0)}\big(\|e(t_0)\| + W(\mu_{t_0}, \mu^*)\big)$ for $t \ge t_0$;*
- *$\vartheta_i(t) \to \vartheta^*_{\pi(i)}$. The labelled error*

$$E_{\mathrm{lab}}(t) := \Big(\frac{1}{m}\sum_i \|\vartheta_i(t) - \vartheta_i^*\|_{G^{-1}}^2\Big)^{1/2} \quad \text{tends to} \quad \Big(\frac{1}{m}\sum_i \|\vartheta^*_{\pi(i)} - \vartheta_i^*\|_{G^{-1}}^2\Big)^{1/2},$$

  *which is non-zero for $\pi \ne \mathrm{id}$.*

*Every constant is an explicit function of $(k_e, \alpha_G, T, \Psi_{G,M}, \Psi_{G,D}, m, G, r, L_J)$, hence uniform over the class of regressors with given PE data $(T, \alpha, \sup\|\Psi\|, \sup\|\dot\Psi\|)$. With the closed-form $\varepsilon$, $\lambda \ge \frac{1}{3}\cdot\min\{k_e/4,\, \varepsilon\alpha_G/2\}$. The constants are conservative, in a quantified way (Remark E.0a); their role is to make the statement self-contained. $L_J$ is defined as a supremum; a closed-form bound follows from the third-derivative suprema of the Gaussian and is not computed here (`excitation. jacobian_ lipschitz_ estimate` estimates it from samples for the tests). The previous version proved this theorem with existence constants, conditional on Morgan–Narendra [62, Thm. 1]; Lemma E.0 removes that citation.*

**Status.** Proved, self-contained (Lemma E.0 replaces the citation of Morgan–Narendra [62]); linearisation checked; the strict Lyapunov inequalities checked pointwise (E.0′ by a certificate over all states) and the explicit envelope checked on the linearised and on the implemented loop; convergence observed. The rate of E.0 is conservative in a quantified way (Remark E.0a); the rate of E.0′ is within 12% of the exact rate above its threshold. Novelty (literature audit): partially anticipated [16], [34]; the strict Lyapunov function is of the Mazenc–Malisoff type.

*Proof.*

1. *Symmetry.* Relabelling the network, $(e, \vartheta) \mapsto (e, \vartheta\circ\sigma)$, maps solutions to solutions. It fixes $e$, is an isometry of the matched metric, and conjugates the dynamics near $(0, \vartheta^*\circ\pi)$ to those near $(0, \vartheta^*)$. So it suffices to treat $\pi = \mathrm{id}$.

2. *Domain and linearisation.* Let $r := \min(\rho, \delta)$ and $D := \{z = (e, \tilde{\vartheta}) \in \mathbb{R}^{n+mp} : \|z\|_2 < r\}$.

   - For $z \in D$ we have $\|e\| < \rho$, so $f = \hat{f}^*$.
   - Also $\|\tilde{\vartheta}_i\| < \delta$ for every $i$, so $\vartheta_i \in \operatorname{int} \Theta_K^\varepsilon$ and $\mathrm{Proj}_K = \mathrm{id}$.

   On $[0, \infty) \times D$ the closed loop is

   $$F(t, z) = \Big(-k_e e + \hat{f}^*(x) - \hat{f}(x),\ \big(G \nabla_\vartheta \Phi(x; \vartheta_i)^\top e\big)_i\Big), \qquad x = x_m(t) + e.$$

   - It is $C^1$ in $(t, z)$, since $x_m \in C^1$, and $C^\infty$ in $z$.
   - Its derivatives through order two are bounded uniformly in $t$, because the Gaussian's derivatives through order three are bounded uniformly in $x$ for $s \geq s_{\min}$.
   - Its Jacobian at 0 is

     $$A(t) = \begin{pmatrix} -k_e I & -\frac{1}{m}\Psi \\ (I_m \otimes G)\Psi^\top & 0 \end{pmatrix},$$

     and the remainder is $O(\|z\|^2)$.
   - The Jacobian and the identity $QA + A^\top Q = \mathrm{diag}(-k_e I, 0)$, with $Q = \mathrm{diag}\big(I/2,\ (I \otimes G^{-1})/(2m)\big)$, are checked numerically by `test_theorem_E_linearisation_and_lyapunov_identity`.

3. *LTV stability, with explicit constants.* Put $w := (e, \zeta)$ with $\zeta := m^{-1/2}(I_m \otimes G^{-1/2})\tilde{\vartheta}$, that is $w = Sz$, and $\Psi_G := m^{-1/2}\Psi(I \otimes G^{1/2})$.

   - The system becomes $\dot{e} = -k_e e - \Psi_G \zeta$ and $\dot{\zeta} = \Psi_G^\top e$: this is (LTV) of Lemma E.0 with $k = k_e$ and $\Psi = \Psi_G$, that is, $\dot{w} = [A_0 + B(t)]w$ with $A_0 = \mathrm{diag}(-k_e I, 0)$ and $B$ skew-symmetric, bounded, with bounded derivative.
   - The excitation transfers: $\int \Psi_G^\top \Psi_G \, \mathrm{d}r \succeq (\alpha \lambda_{\min}(G)/m) I = \alpha_G I$, and $\|\Psi_G\| \leq \Psi_{G,M}$, $\|\dot{\Psi}_G\| \leq \Psi_{G,D}$.
   - Lemma E.0 gives the strict Lyapunov function $V(t, w)$ with $p_{\mathrm{lo}}\|w\|^2 \leq V \leq p_{\mathrm{hi}}\|w\|^2$ and $\dot{V} \leq -\gamma\|w\|^2$ along the linear system, hence

     $$\|w(t)\| \leq \Big(\frac{p_{\mathrm{hi}}}{p_{\mathrm{lo}}}\Big)^{1/2} e^{-(\gamma/(2p_{\mathrm{hi}}))(t - t_0)} \|w(t_0)\| :$$

     uniform exponential stability with explicit overshoot and rate.
   - The $w$-coordinates are the coordinates of the matched Lyapunov function: $\|\zeta\|^2 = \frac{1}{m}\sum_i \|\tilde{\vartheta}_i\|^2_{G^{-1}} = W_\pi^2$, so $\|w\|^2 = \|e\|^2 + W_\pi^2 = 2V_\pi$.
   - The bound on $\dot{\Psi}$ cannot be dropped. $\Psi = \sin t^2$ is bounded and PE, but for $\dot{e} = -e - \Psi\zeta$, $\dot{\zeta} = \Psi e$, from $w(t_0) = (e, \zeta) = (0, 1)$, a numerical computation gives the ratio $\|w(t_0 + 200)\|/\|w(t_0)\|$ as 0.9715, 0.9942 and 0.9980 for $t_0 = 5$, 20 and 50, and part (f) of the proof of Lemma E.0 shows that the decay over a fixed horizon tends to 0 as $t_0 \to \infty$. So that system is not UES. Consistently, the rate of Lemma E.0 tends to 0 as $\Psi_D \to \infty$.

4. *Nonlinear system, with an explicit ball.* On the closure of $D$ (to which the Lipschitz bound extends) the field is $F(t, z) = A(t)z + R(t, z)$ with $\|R(t, z)\| \leq (L_J/2)\|z\|^2$ (Taylor with the Lipschitz Jacobian). In $w$-coordinates, $\dot{w} = [A_0 + B(t)]w + \tilde{R}(t, w)$ with $\tilde{R}(t, w) := S\,R(t, S^{-1}w)$, so $\|\tilde{R}(t, w)\| \leq \tilde{C}_R\|w\|^2$ on $\{\|w\| < r/\|S^{-1}\|\} \subset S(D)$. Along the nonlinear system, since $\|\partial V/\partial w\| \leq g_V\|w\|$,

   $$\dot{V} \leq -\gamma\|w\|^2 + g_V\tilde{C}_R\|w\|^3 \leq -\frac{\gamma}{2}\|w\|^2 \qquad \text{whenever } \|w\| \leq r_w.$$

The sublevel set $\{V \le p_{\mathrm{lo}} r_w^2\}$ is contained in $\{\|w\| \le r_w\}$, so it is forward invariant, and it contains the ball $\{\|w\| \le r_0\}$, since $V \le p_{\mathrm{hi}}\|w\|^2$. On it $\dot V \le -(\gamma/(2p_{\mathrm{hi}}))V$, so for $\|w(t_0)\| \le r_0$,

$$\|w(t)\| \le \Big(\frac{p_{\mathrm{hi}}}{p_{\mathrm{lo}}}\Big)^{1/2} e^{-\lambda(t-t_0)}\|w(t_0)\|, \qquad \lambda = \frac{\gamma}{4p_{\mathrm{hi}}}.$$

These trajectories stay in $D$, where the closed loop equals $F$. (Khalil [48, Thm. 4.13] gives the same conclusion with existence constants; the explicit form is what Lemma E.0 adds.)

5. *Transfer to* $(\|e\|, W)$. Let $\pi$ be optimal at $t_0$. Then $\|w(t_0)\|^2 = 2V_\pi(t_0) = 2V_{\mathrm{OT}}(t_0) \le 2c_0 = r_0^2$, so step 4 applies. Moreover $\|e\| + W \le \|e\| + W_\pi = \|e\| + \|\zeta\| \le \sqrt{2}\|w\|$ at every $t$, and $\|w(t_0)\| \le \|e(t_0)\| + \|\zeta(t_0)\| = \|e(t_0)\| + W(t_0)$. This gives the estimate with $C = (2p_{\mathrm{hi}}/p_{\mathrm{lo}})^{1/2}$.

   *Uniqueness.* Suppose $V_\pi, V_{\pi'} \le c_0$ with $\pi \ne \pi'$. Then the state lies in the invariant balls of step 4 around both $(0, \vartheta^* \circ \pi)$ and $(0, \vartheta^* \circ \pi')$. The solution would converge to two distinct equilibria, which is impossible.

   *Limits.* Hence $z(t) \to 0$, that is, $\vartheta_i(t) \to \vartheta^*_{\pi(i)}$. For $\pi' \ne \pi$,

$$V_{\pi'}(t) \to \frac{1}{2m}\sum_i \|\vartheta^*_{\pi(i)} - \vartheta^*_{\pi'(i)}\|^2_{G^{-1}} \ge \frac{d^2}{2} > 0,$$

   while $V_\pi(t) \to 0$. So $\pi$ is the unique optimal assignment for all large $t$, and $E_{\mathrm{lab}}$ converges as stated. □

Checked by: `tests/test_transport.py`, `tests/test_excitation.py` and `experiments/exp_pe_convergence.py`.

- Linearisation: the Jacobian of the implemented loop equals $A(t)$, and $QA + A^\top Q = \mathrm{diag}(-k_e I, 0)$ (`test_theorem_E_linearisation_and_lyapunov_identity`).
- Explicit constants: on the implemented loop, from a start inside the explicit ball, $\|w(t)\| \le (p_{\mathrm{hi}}/p_{\mathrm{lo}})^{1/2} e^{-\lambda t}\|w(0)\|$ at every sample ($L_J$ estimated from samples) (`test_theorem_E_explicit_nonlinear_envelope_on_the_implemented_loop`).
- Convergence: the matched error decays and the labelled error does not (qualitative); sliding-window excitation and fitted rates (`test_theorem_E_matched_error_converges_labelled_does_not`, `exp_pe_convergence.py`).
- Not checked in the repository: the sharpness of the explicit rate beyond the tables of Remark E.0a; Corollary E.2 and Remark E.3.

**Remark E.1** (the rate)**.**

- **What converges.** "The parameters converge" is false for the labelled error and true for the matched one.
- **The rate is not set by the PE level.** $\lambda > 0$ exists for each admissible $\Psi$, but it depends on $\Psi(\cdot)$, not only on its PE level. No positive function of $\alpha\lambda_{\min}(G)/m$, $k_e$, $T$ and $\sup\|\Psi_G\|$ alone bounds it below.
  - For $m = n = 1$, $\vartheta^* = (1.5, 0.3, 0.8)$, $G = I$, $k_e = 3$ and $x_m = 1.5\sin\omega t$ ($T = 2\pi$), these four quantities do not depend on $\omega$. The exact Floquet rate, computed numerically, falls from $4.7 \times 10^{-2}$ ($\omega = 1$) to $2.9 \times 10^{-5}$ ($\omega = 64$), like $\omega^{-2}$.
  - A lower bound must also involve $\sup\|\dot\Psi_G\|$ (step 3). Lemma E.0 gives one; it reproduces the $\omega^{-2}$ law (Remark E.0a).
  - Exact Floquet rates, computed numerically, are also not monotone in $G$ or in $k_e$, and they decrease for large $G$. When $k_e$ is large relative to the reference bandwidth, they approach $\lambda_{\min}\langle\Psi_G^\top\Psi_G\rangle/k_e$.
- **The numbers** (`results/pe_convergence`; $m = 3$, $n = 2$, six runs):

| quantity | range |
|---|---|
| min over sliding 60 s windows of $\lambda_{\min}\big(\int_t^{t+60} \Psi^\top \Psi \,\mathrm{d}r\big)$ | $5.9 \times 10^{-4}$ to $2.7 \times 10^{-2}$ |
| fitted decay rate of the matched error, second half of the run | $2.3 \times 10^{-3}$ to $4.3 \times 10^{-3}\,\mathrm{s}^{-1}$ |
| rank correlation between the two | 0.60 |

The convergence test is therefore qualitative. The machine-checked part of the proof is the linearisation.

**Corollary E.2** (uniform-in-time exponential merging under PE)**.** *Take two solutions of the same closed loop with $V_{\mathrm{OT}}(0) \le c_0$ and $V'_{\mathrm{OT}}(0) \le c_0$. Then*

$$\|x_t - x'_t\| + W(\mu_t, \mu'_t) \le Ce^{-\lambda t} R_0, \qquad R_0 := \|e(0)\| + \|e'(0)\| + W(\mu_0, \mu^*) + W(\mu'_0, \mu^*). \tag{15.2}$$

*Let $\delta_0 := \|x_0 - x'_0\| + W(\mu_0, \mu'_0)$. Combine this with the coupling estimate of Theorem D(c) on $[0, T_0]$, with $T_0 := (\Lambda + \lambda)^{-1} \log(CR_0/\delta_0)$.*

- *Both solutions stay in $D$ (proof step 4), so $\|x - x_m(t)\| < r \le \rho$.*
- *The segment between $x_t$ and $x'_t$ lies in $B(x_m(t), \rho)$, where $f = \hat{f}^*$, and $\mathrm{Proj}_K = \mathrm{id}$.*
- *Hence Theorem D(c) holds with $M_x := k_e + 2a_{\max} J_c$, with $L_x$ and $L_\vartheta$ computed for $\bar{E} = r$ and an inactive projection, and with $L_2 = \lambda_G^{1/2} L_1$. This $\Lambda$ depends neither on $T_0$ nor on $f$ outside the tube.*

*This gives uniform-in-time Hölder continuity in the basin:*

$$\sup_{t \ge 0} \big( \|x_t - x'_t\| + W(\mu_t, \mu'_t) \big) \le (CR_0)^{\Lambda/(\Lambda+\lambda)} \, \delta_0^{\lambda/(\Lambda+\lambda)}. \tag{15.3}$$

**Status.** Proved, through Theorem E (self-contained since Lemma E.0). Not checked in the repository.

*Proof.* $\|x_t - x'_t\| = \|e - e'\| \le \|e\| + \|e'\|$, and $W(\mu_t, \mu'_t) \le W(\mu_t, \mu^*) + W(\mu'_t, \mu^*)$. Bound each bracket by Theorem E. On $[0, T_0]$ use $e^{\Lambda T_0} \delta_0$; on $[T_0, \infty)$ use $CR_0 e^{-\lambda T_0}$. Note that $\delta_0 \le R_0 \le CR_0$. □

**Remark E.3** (the collapse point)**.** Take the $G^{-1}$-closest pair $(j, k)$. Set $e = 0$ and $\vartheta_j = \vartheta_k = (\vartheta_j^* + \vartheta_k^*)/2$, with all other particles ideal.

- At this point $V_{\mathrm{OT}} = d^2/8$, and there are two optimal assignments.
- By exchange symmetry, $\vartheta_j \equiv \vartheta_k$ for all $t$, so $W(\mu_t, \mu^*) \ge d/2$ forever.

The uniqueness argument of step 5 shows $c_0 < d^2/8$ automatically, and $d^2/8$ is sharp along this direction.

# 16 Theorem F: the transport cost of adaptation

**Theorem F.** *Assume* (S0)–(S3)*, and take $t_0 \le t_1$ in the maximal interval of existence (all $t \ge 0$ under the hypothesis of Theorem D(b)). $\mu^*$,* (S4) *and* (S5) *are not needed. Along* (L)*:*

(i) $\big(\frac{1}{m} \sum_i \|\dot{\vartheta}_i\|^2_{G^{-1}}\big)^{1/2} \le \lambda_G^{1/2} L_1 \|e(t)\|$*;*

(ii) $W(\mu_{t_1}, \mu_{t_0}) \le \lambda_G^{1/2} L_1 \int_{t_0}^{t_1} \|e\| \,\mathrm{d}r$*. Hence the $W$-length of $r \mapsto \mu_r$ on $[t_0, t_1]$ is at most $\lambda_G^{1/2} L_1 \int_{t_0}^{t_1} \|e\| \,\mathrm{d}r$;*

(iii) $W(\mu_{T_2}, \mu_{T_1}) \le \lambda_G^{1/2} L_1 (T_2 - T_1) \cdot \mathrm{RMS}_{[T_1, T_2]} \|e\|$.

*Along* ($\mathrm{L}_\sigma$) *with the OT leak, for a.e. $t$:*

(i′) $\big(\frac{1}{m}\sum_i \|\dot\vartheta_i\|^2_{G^{-1}}\big)^{1/2} \le \lambda_G^{1/2} L_1 \|e\| + \sigma_m W(\mu_t, \mu^0)$;

(ii′) $W(\mu_{t_1}, \mu_{t_0}) \le \int_{t_0}^{t_1} \big(\lambda_G^{1/2} L_1 \|e\| + \sigma_m W(\mu_r, \mu^0)\big)\,\mathrm{d}r$;

(iii′) $W(\mu_{T_2}, \mu_{T_1}) \le \lambda_G^{1/2} L_1 (T_2 - T_1) \cdot \mathrm{RMS}_{[T_1,T_2]} \|e\| + \sigma_m \int_{T_1}^{T_2} W(\mu_r, \mu^0)\,\mathrm{d}r$.

*For the labelled leak, replace $W(\mu_r, \mu^0)$ by $d_{\mathrm{lab}}(r) := \big(\frac{1}{m}\sum_i \|\vartheta_i(r) - \vartheta_i^0\|^2_{G^{-1}}\big)^{1/2} \ge W(\mu_r, \mu^0)$. The term $W(\mu_r, \mu^0)$ does not suffice there.*

*All bounds also hold with $L_G := \big(g_a + a_{\max}^2 (g_c J_c^2 + g_s J_s^2)\big)^{1/2} \le \lambda_G^{1/2} L_1$ in place of $\lambda_G^{1/2} L_1$. Since $\max_i \|\dot\vartheta_i\|_{G^{-1}} \le L_G \|e\|$ along* (L)*, also $W_\infty(\mu_{t_1}, \mu_{t_0}) \le L_G \int_{t_0}^{t_1} \|e\|\,\mathrm{d}r$.*

**Status.** Proved and checked, including the leak clause, a tight leak instance with $e \equiv 0$ and a tight velocity instance. Novelty (literature audit): elementary.

*Proof.* (i) By Lemma P.1 and Lemma B.2(b),

$$\|\dot\vartheta_i\|_{G^{-1}} \le \|\nabla g_t(\vartheta_i)\|_G \le \lambda_G^{1/2} \|\nabla g_t(\vartheta_i)\| \le \lambda_G^{1/2} L_1 \|e(t)\|.$$

Square and average over $i$.

(ii) Use the identity coupling $(\vartheta_i(t_0), \vartheta_i(t_1))$ and Minkowski's inequality in $L^2\big(\frac{1}{m}\sum_i; G^{-1}\big)$. For general measures, use the Lagrangian coupling $(X_{t_0}, X_{t_1})_\# \mu_0$, or Ambrosio–Gigli–Savaré [2, Thm. 8.3.1] after the isometry $\vartheta \mapsto G^{-1/2}\vartheta$. Summing over a partition gives the length statement.

(iii) Cauchy–Schwarz.

(i′) Apply Lemma P.1 with $y = \nabla g_t - \sigma_m G^{-1}(\vartheta_i - \vartheta^0_{\tau(i)})$. Then Minkowski's inequality gives $\|y\|_G \le \|\nabla g_t\|_G + \sigma_m \|\vartheta_i - \vartheta^0_{\tau(i)}\|_{G^{-1}}$. Average over $i$ with $\tau$ optimal. At ties, Filippov velocities are convex combinations of fields with $\tau$ optimal at the current point. Each such field satisfies (i′), and the norm is convex.

(ii′) and (iii′) follow as (ii) and (iii).

For $L_G$: $\|\nabla g\|_G^2 = g_a \|\varphi e\|^2 + (a^\top e)^2 \big(g_c \|\nabla_c \varphi\|^2 + g_s (\partial_s \varphi)^2\big) \le \|e\|^2 L_G^2$. □

**Reading.** Along (L), moving the kernel measure by $W$ costs at least $W/(\lambda_G^{1/2} L_1)$ of integrated tracking error. This is an exchange rate between learning and tracking.

Along a leaky law the proved statement is

$$\int_{t_0}^{t_1} \big(\lambda_G^{1/2} L_1 \|e\| + \sigma_m d_r\big)\,\mathrm{d}r \ge W(\mu_{t_1}, \mu_{t_0}),$$

with $d_r = W(\mu_r, \mu^0)$ for ($\mathrm{L}_\sigma$) and $d_r = d_{\mathrm{lab}}(r)$ for the labelled leak. This gives no lower bound on $\int \|e\|$, because the leak can pay for the whole motion. Under ($\mathrm{L}_\sigma$) there are runs with $e \equiv 0$ and $W(\mu_{10}, \mu_0) \approx 1.05$ (found numerically).

**Numerical checks.** The following are checked at every sample (with the particle path length $\ge$ the Wasserstein length):

- (i) and Lemma P.1, including a tight instance with ratio $\ge 0.99$;
- (ii) along (L);
- (ii′) along ($\mathrm{L}_\sigma$), on a grid of pairs, including an instance with $e \equiv 0$ where it is tight (ratio $\ge 0.99$) and the (L) reading fails.

Checked by: `tests/test_transport.py`:
`test_lemma_P1_projection_is_nonexpansive_and_tight_velocity_bound` ((i) and Lemma P.1);
`test_theorem_F_wasserstein_length_bounded_by_integrated_error` ((ii) along (L));
`test_theorem_F_leak_clause_along_ot_sigma_mod` and
`test_theorem_F_leak_clause_is_tight_when_the_leak_pays_for_all_motion` ((ii′) along ($\mathrm{L}_\sigma$)).

# 17 Theorem D′: propagation of chaos for the dithered closed loop on finite horizons

Theorem D covers the deterministic projected law (L). The route to Conjecture H.6 (Section 21) lists the mean-field limit of the dithered law ($\mathrm{L}_\tau$) as one of its three missing pieces. This section proves it on finite horizons, with an explicit $m^{-1/2}$ rate that does not depend on the temperature, by a synchronous coupling. The uniform-in-time statement is Section 26, branch 3.

**Setting.** The design of Theorem H: saturated features $\tilde\Phi$, $\|\tilde\Phi\| \le K_\Phi$, with all $\vartheta$-derivatives bounded uniformly in $x$, and the $m$-particle closed loop

$$\begin{aligned} \dot x &= f(x) + \dot x_m - k_e e - \frac{1}{m}\sum_j \tilde\Phi(x;\vartheta_j), \qquad e = x - x_m(t), \\ \mathrm{d}\vartheta_i &= b(x,e,\vartheta_i)\,\mathrm{d}t + \sqrt{2\tau}\,G^{1/2}\,\mathrm{d}B_i, \qquad i = 1,\dots,m, \\ b(x,e,\vartheta) &:= G\nabla_\vartheta\big[e^\top\tilde\Phi(x;\vartheta)\big] - \sigma(\vartheta-\vartheta^0), \end{aligned} \tag{$\mathrm{L}_\tau$}$$

with independent Brownian motions $B_i$, i.i.d. initial particles $\vartheta_i(0) \sim \bar\mu_0 \in \mathcal{P}_2(\mathbb{R}^p)$ independent of the $B_i$, and a deterministic $x(0) = \bar x_0$. $f$ is $C^1$ and $x_m \in C^1$ with $\dot x_m$ bounded; no autonomy or periodicity is needed. The **McKean–Vlasov closed loop** is

$$\begin{aligned} \dot{\bar x} &= f(\bar x) + \dot x_m - k_e \bar e - \int \tilde\Phi(\bar x;\vartheta)\,\mathrm{d}\bar\mu_t(\vartheta), \qquad \bar e = \bar x - x_m(t), \\ \mathrm{d}\bar\vartheta &= b(\bar x_t, \bar e_t, \bar\vartheta)\,\mathrm{d}t + \sqrt{2\tau}\,G^{1/2}\,\mathrm{d}B, \qquad \bar\mu_t := \mathrm{Law}(\bar\vartheta_t), \end{aligned} \tag{MV}$$

from the same $\bar x_0$ and $\bar\vartheta(0) \sim \bar\mu_0$. Here $\bar x_t$ is deterministic, and the particle $\bar\vartheta$ interacts with its own law only through $\bar x_t$.

**Standing bound.** Fix $T > 0$ and suppose both $x^m$ (for every $m$ and every realisation) and $\bar x$ remain in a compact set $X$ on $[0,T]$. Since $\|\frac{1}{m}\sum\tilde\Phi\| \le K_\Phi$ and $\|\int\tilde\Phi\,\mathrm{d}\bar\mu\| \le K_\Phi$ whatever the parameters, the tube bound of Theorem D(b) (with $a_{\max}$ replaced by $K_\Phi$) gives such an $X$ for every $T$ when its self-consistent growth condition holds, for example for bounded $f$; otherwise the statement is local in time, as in Theorem D(d). Put $E_X := \sup_{t\le T,\,x\in X}\|x - x_m(t)\|$ and let

$$\begin{gathered} L_f := \mathrm{Lip}_X f, \qquad L_x := \sup\|\partial_x\tilde\Phi\|, \qquad L_\vartheta := \sup\|\nabla_\vartheta\tilde\Phi\|, \\ L_b := \lambda_G\big(L_\vartheta + E_X\sup\|\partial_x\nabla_\vartheta\tilde\Phi\| + E_X\sup\|\nabla_\vartheta^2\tilde\Phi\|\big) + \sigma, \\ \Lambda := \max\{L_f + L_x + 2L_b,\ L_\vartheta + L_b\}, \end{gathered}$$

where $L_b$ is a Lipschitz constant of $(x,e,\vartheta) \mapsto b$ on $X \times \{\|e\| \le E_X\} \times \mathbb{R}^p$ for the norm $\|x\| + \|e\| + \|\vartheta\|$. All of these are finite by saturation and independent of $m$ and $\tau$; the suprema of the $\tilde\Phi$-derivatives are over all $x \in \mathbb{R}^n$ (the saturated features and their derivatives are globally bounded), so the segments between $x$ and $\bar x$ used below may leave $X$. The constants grow with the output dimension $n$ ($K_\Phi = a_{\max}\sqrt{n}$), not with the lifted dimension $p$ beyond that.

**Theorem D′.** *Under the standing bound:*

(a) Well-posedness. (MV) *has a unique solution* $(\bar x, \mathrm{Law}(\bar\vartheta))$ *on* $[0,T]$*, and* $\bar x$ *is deterministic.*

(b) Synchronous coupling. *Let* $\bar\vartheta_i$*,* $i = 1,\dots,m$*, be the solutions of* (MV)*'s particle equation driven by the same* $B_i$ *and started at the same* $\vartheta_i(0)$ *as the* $m$*-particle system. Then, pathwise,*

$$\begin{aligned} \|x_t - \bar x_t\| + \frac{1}{m}\sum_i\|\vartheta_i(t) - \bar\vartheta_i(t)\| &\le e^{\Lambda t}\int_0^t \zeta_s\,\mathrm{d}s, \\ \zeta_s := \Big\|\frac{1}{m}\sum_j\tilde\Phi(\bar x_s;\bar\vartheta_j(s)) - \int\tilde\Phi(\bar x_s;\cdot)\,\mathrm{d}\bar\mu_s\Big\|, & \end{aligned}$$

*and since the $\bar\vartheta_j(s)$ are i.i.d. with law $\bar\mu_s$ and $\bar x_s$ is deterministic, $\mathbb{E}\zeta_s^2 \le K_\Phi^2/m$. Hence*

$$\mathbb{E}\sup_{t\le T}\Big(\|x_t-\bar x_t\|+\frac{1}{m}\sum_i\|\vartheta_i(t)-\bar\vartheta_i(t)\|\Big)\le TK_\Phi e^{\Lambda T}m^{-1/2},$$
$$\mathbb{E}\sup_{t\le T}\Big(\|x_t-\bar x_t\|+\frac{1}{m}\sum_i\|\vartheta_i(t)-\bar\vartheta_i(t)\|\Big)^2\le T^2K_\Phi^2e^{2\Lambda T}m^{-1},$$

*with constants independent of $\tau\ge 0$ and of $\bar\mu_0$. The exponent $-1/2$ does not depend on the dimension; the constants do, through $K_\Phi$ and the Lipschitz bounds of $\tilde\Phi$ (standing bound).*

(c) Consequences.

- *Propagation of chaos: for each fixed $k$, the law of $(\vartheta_1,\dots,\vartheta_k)$ on $[0,T]$ converges to the $k$-fold product of the path law of $\bar\vartheta$, with the per-particle bound $\mathbb{E}\sup_{t\le T}\|\vartheta_i(t)-\bar\vartheta_i(t)\|\le 2(e^{L_bT}-1)TK_\Phi e^{\Lambda T}m^{-1/2}$ for every $i$. (Exchangeability alone gives only the fixed-$t$ bound $\mathbb{E}\|\vartheta_i(t)-\bar\vartheta_i(t)\|=\mathbb{E}\bar\Delta(t)\le TK_\Phi e^{\Lambda T}m^{-1/2}$, since $\sup_t$ of a mean is not the mean of the $\sup_t$; the per-particle sup bound is the pathwise Grönwall estimate of the proof.)*
- *The learned function converges at the rate $m^{-1/2}$: at each fixed $t$,*

$$\begin{aligned}\mathbb{E}\Big\|\hat f^m(x_t)-\int\tilde\Phi(\bar x_t;\cdot)\,\mathrm{d}\bar\mu_t\Big\|&\le(L_x+L_\vartheta)TK_\Phi e^{\Lambda T}m^{-1/2}+\mathbb{E}\zeta_t\\&\le\big((L_x+L_\vartheta)Te^{\Lambda T}+1\big)K_\Phi m^{-1/2}\end{aligned}$$

  *(the supremum over $t$ would need a chaining argument for $\zeta$ and is not pursued).*
- *The parameter measure converges in $W_1$ (Euclidean cost on $\mathbb{R}^p$; the $G^{-1}$ metric used elsewhere changes it by a factor $\lambda_G^{\pm1/2}$) at the i.i.d. rate:*

$$\mathbb{E}W_1(\mu_t^m,\bar\mu_t)\le TK_\Phi e^{\Lambda T}m^{-1/2}+\mathbb{E}W_1(\bar\mu_t^{\mathrm{emp},m},\bar\mu_t),$$

  *where $\bar\mu_t^{\mathrm{emp},m}$ is the empirical measure of $m$ i.i.d. samples of the time-$t$ marginal $\bar\mu_t$; the last term is $O(m^{-1/p})$ for $p>2$ (Fournier–Guillin [30]; applicable because $\bar\mu_t$ has a bounded second moment on $[0,T]$, the drift being linearly bounded). Only this term carries the dimension.*

(d) $\tau=0$. *The statement holds for the deterministic saturated law with i.i.d. initial particles, where it gives, for this law (not the projected law of Theorem D(d)), the explicit $m^{-1/2}$ rate in place of a rate $\delta_m\to 0$ for the state and the paired particles.*

**Status.** Proved; consistency of the $m^{-1/2}$ rate checked by coupling $m$-particle systems to a 4096-particle reference (`exp_chaos.py`). Novelty (literature audit): the technique is Sznitman's; new in that the plant is in the loop and the rate for the state is dimension-free.

*Proof. (a)* Let $X'$ be a compact neighbourhood of $X$ on which $f$ is Lipschitz. For a given curve $\bar x\in C([0,T_1];X')$, the particle equation of (MV) is an SDE with drift Lipschitz in $\vartheta$ (bounded $\nabla^2_\vartheta\tilde\Phi$ and the linear leak; the derivative bounds of $\tilde\Phi$ are global in $x$) and constant diffusion; it has a unique strong solution, whose law $\bar\mu_t[\bar x]$ depends Lipschitz-continuously on $\bar x$: by synchronous coupling of two solutions driven by $\bar x$ and $\bar x'$, $\frac{\mathrm{d}}{\mathrm{d}t}\|\bar\vartheta-\bar\vartheta'\|\le L_b\big(2\|\bar x-\bar x'\|+\|\bar\vartheta-\bar\vartheta'\|\big)$, so $W_1(\bar\mu_t[\bar x],\bar\mu_t[\bar x'])\le 2L_b\int_0^t e^{L_b(t-s)}\|\bar x_s-\bar x'_s\|\,\mathrm{d}s$. The plant input $\dot x_m-k_e\bar e-\int\tilde\Phi\,\mathrm{d}\bar\mu$ is bounded by $\sup\|\dot x_m\|+k_eE_{X'}+K_\Phi$ whatever the measure, so for $T_1$ small the plant solutions from $\bar x_0$ stay in $X'$, and the map taking $\bar x$ to the solution of the plant ODE with input $\int\tilde\Phi(\cdot\,;\vartheta)\,\mathrm{d}\bar\mu_t[\bar x]$ sends $C([0,T_1];X')$ into itself and is a contraction there, since $\mu\mapsto\int\tilde\Phi(x;\cdot)\,\mathrm{d}\mu$ is $W_1$-Lipschitz with constant $L_\vartheta$ and $f$ is Lipschitz on $X'$. This gives a unique solution on $[0,T_1]$ in the class of

curves with values in $X'$; the standing bound keeps it in $X$, so the local solutions concatenate to $[0,T]$, and uniqueness holds among solutions with $\bar{x} \in C([0,T];X)$. This is the Picard argument of Theorem D(a) with the Dobrushin estimate replaced by the coupling estimate. (Theorem D(b)'s tube bound, used in the standing bound for a law it does not formally cover, rests only on $\|\hat{f}\| \le a_{\max}$ — here $K_\Phi$ — and on the pathwise error ODE, so it applies.)

*(b)* Set $\Delta_i := \vartheta_i - \bar{\vartheta}_i$ and $\delta := x - \bar{x}$. The noise cancels in $\mathrm{d}\Delta_i$, so $\Delta_i$ is absolutely continuous with $\frac{\mathrm{d}}{\mathrm{d}t}\|\Delta_i\| \le \|b(x,e,\vartheta_i) - b(\bar{x},\bar{e},\bar{\vartheta}_i)\| \le L_b\big(\|x-\bar{x}\| + \|e-\bar{e}\| + \|\Delta_i\|\big) = L_b\big(2\|\delta\| + \|\Delta_i\|\big)$, using $e - \bar{e} = \delta$ (same $x_m$). For $\delta$,

$$\dot{\delta} = f(x) - f(\bar{x}) - k_e\delta - \Big[\frac{1}{m}\sum_j \tilde{\Phi}(x;\vartheta_j) - \int \tilde{\Phi}(\bar{x};\cdot)\,\mathrm{d}\bar{\mu}_t\Big],$$

and the bracket splits as

$$\frac{1}{m}\sum_j\big[\tilde{\Phi}(x;\vartheta_j) - \tilde{\Phi}(\bar{x};\vartheta_j)\big] + \frac{1}{m}\sum_j\big[\tilde{\Phi}(\bar{x};\vartheta_j) - \tilde{\Phi}(\bar{x};\bar{\vartheta}_j)\big] + \Big[\frac{1}{m}\sum_j \tilde{\Phi}(\bar{x};\bar{\vartheta}_j) - \int \tilde{\Phi}(\bar{x};\cdot)\,\mathrm{d}\bar{\mu}_t\Big],$$

of norms at most $L_x\|\delta\|$, $L_\vartheta \frac{1}{m}\sum\|\Delta_j\|$ and $\zeta_t$. Dropping the dissipative $-k_e\delta$, $\frac{\mathrm{d}}{\mathrm{d}t}\|\delta\| \le (L_f + L_x)\|\delta\| + L_\vartheta\bar{\Delta} + \zeta_t$ with $\bar{\Delta} := \frac{1}{m}\sum\|\Delta_j\|$, and $\frac{\mathrm{d}}{\mathrm{d}t}\bar{\Delta} \le L_b\big(2\|\delta\| + \bar{\Delta}\big)$. For $D := \|\delta\| + \bar{\Delta}$ this gives $\dot{D} \le \Lambda D + \zeta_t$ with $D(0) = 0$, and Grönwall gives the pathwise bound, whose right-hand side is non-decreasing in $t$. For the expectation, $\bar{x}_s$ is deterministic and the $\tilde{\Phi}(\bar{x}_s;\bar{\vartheta}_j(s))$ are i.i.d. bounded by $K_\Phi$, so

$$\mathbb{E}\zeta_s^2 = \frac{1}{m}\mathbb{E}\big\|\tilde{\Phi}(\bar{x}_s;\bar{\vartheta}_1(s)) - \mathbb{E}\tilde{\Phi}(\bar{x}_s;\bar{\vartheta}_1(s))\big\|^2 \le K_\Phi^2/m,$$

and $\mathbb{E}\int_0^T \zeta_s\,\mathrm{d}s \le TK_\Phi m^{-1/2}$; for the second moment, $\mathbb{E}\big(\int_0^T \zeta\big)^2 \le T\int_0^T \mathbb{E}\zeta^2 \le T^2K_\Phi^2/m$.

*(c)* For the first item, $\frac{\mathrm{d}}{\mathrm{d}t}\|\Delta_i\| \le L_b\big(2\|\delta\| + \|\Delta_i\|\big)$ with $\|\delta(s)\| \le D(s) \le e^{\Lambda s}\int_0^s \zeta$ and $\Delta_i(0) = 0$ give, by Grönwall and $\Lambda \ge L_b$,

$$\|\Delta_i(t)\| \le 2L_b\int_0^t e^{L_b(t-s)}e^{\Lambda s}\Big(\int_0^s \zeta\Big)\,\mathrm{d}s \le 2(e^{L_bt} - 1)e^{\Lambda t}\int_0^t \zeta,$$

a pathwise bound whose right-hand side is non-decreasing in $t$; take $\sup_t$ and expectations. Convergence of the $k$-particle law follows from the pathwise coupling bound. For the second, $\|\hat{f}^m(x_t) - \int \tilde{\Phi}(\bar{x}_t;\cdot)\,\mathrm{d}\bar{\mu}_t\|$ is bounded by the same three terms as the bracket in (b), the first two by $(L_x + L_\vartheta)D(t)$ and the third by $\zeta_t$. For the third, couple $\mu_t^m$ with the empirical measure of the $\bar{\vartheta}_j(t)$ through the identity pairing (cost $\frac{1}{m}\sum\|\Delta_j\|$) and use the triangle inequality.

*(d)* Nothing in the argument uses $\tau > 0$: the noise cancels in the coupling and only the i.i.d. sampling of the initial particles enters through $\zeta$. □

**Remark D′.1** (what carries the dimension)**.** The plant and the learned function see the particle measure only through $\int \tilde{\Phi}\,\mathrm{d}\mu$, the mean of a bounded function. Their fluctuation is $K_\Phi/\sqrt{m}$: the exponent does not see the dimension $p$ of the lifted parameter, and the constant $K_\Phi = a_{\max}\sqrt{n}$ grows with the output dimension only. The curse of dimensionality of Corollary B.6 (rates $m^{-1/p}$) concerns the Wasserstein distance between the parameter measures, which the closed loop never evaluates. Theorem F, by contrast, bounds the transport of $\mu$ by the tracking error and is dimension-free as well.

**Remark D′.2** (what is still missing for Conjecture H.6)**.** The constant $e^{\Lambda T}$ grows with the horizon; uniformity in time would follow from a contraction of the coupled system, which the plant coupling and the non-reciprocal averaged interaction (Proposition H.7) obstruct for the general law. For the averaged system, Remark H.5 points to the uniform-in-time result of [11]; with the plant in the loop the question is Section 26, branch 3.

**Numerics** (`exp_chaos.py` → `results/chaos/table.md`). An $M = 4096$ particle system stands in for (MV); for $m = 16, 32, \dots, 1024$ the $m$-particle system is driven by the first $m$ initial particles and Brownian motions of the reference — the coupling of (b) — with $n = 1$, $k_e = 10$, $\sigma = 0.02$, $T = 20$, at $\tau = 0$ and $\tau = 2{\cdot}10^{-3}$, and $\sup_t D_m$, $D_m := \|x^m - x^M\| + \frac{1}{m}\sum_{i\le m}\|\vartheta_i^m - \vartheta_i^M\|$, is averaged over 24 seeds. Because the $m$-system shares its particles with the reference, the sampling fluctuation that drives $D_m$ has variance proportional to $1/m - 1/M$ (the shared particles contribute $(1/m - 1/M)\sum_{i\le m}$, the others $(1/M)\sum_{i>m}$), so the predicted law for the mean is a constant times $(1/m - 1/M)^{1/2}$, and $m^{-1/2}$ for $m \ll M$.

| | $\tau = 0$ | $\tau = 2 \cdot 10^{-3}$ |
|---|---|---|
| mean $\sup_t D_m$, $m = 16 \to 1024$ | $1.49 \cdot 10^{-1} \to 2.17 \cdot 10^{-2}$ | $1.52 \cdot 10^{-1} \to 2.32 \cdot 10^{-2}$ |
| log-log slope of the mean against $m$ over $m \le 256$ (bootstrap s.e. over seeds) | $-0.47 \pm 0.06$ | $-0.46 \pm 0.05$ |
| slope of the median | $-0.46$ | $-0.45$ |
| slope of the mean against $(1/m - 1/M)^{1/2}$, all $m$ (predicted 1) | $0.94 \pm 0.08$ | $0.92 \pm 0.08$ |
| mean / $(1/m - 1/M)^{1/2}$, $m = 32 \dots 1024$ | 0.70–0.80 | 0.67–0.86 |

The rate is the $m^{-1/2}$ of (b) within one standard error, the same at both temperatures (the noise cancels in the coupling, (d)), and the two-scale law in $(1/m - 1/M)^{1/2}$ holds up to $m = M/4$. Two features of the data are worth recording. First, single seeds differ by more than an order of magnitude at fixed $m$ (the table lists the minimum and maximum over seeds): $\sup_t D_m$ is governed by the projection of the initial sampling fluctuation on a few slowly amplified directions of the closed loop, so the expectation in (b) needs tens of seeds — a first version of this experiment with three seeds gave slopes $-0.25$ and $-0.21$ and was withdrawn (`review/REVIEW_LOG.md`, round 5, D2). Second, the measured ratio $\sup_t D_m/(1/m - 1/M)^{1/2} \approx 0.7$ is astronomically smaller than the constant $TK_\Phi e^{\Lambda T}$ of (b): on this horizon the theorem is a rate statement, not a quantitative bound.

Checked by: `exp_chaos.py`: the experiment only; there is no test for Theorem D′ (its hypotheses are structural and the coupling is the experiment itself).

## Theorem D″: uniform-in-time propagation of chaos in the leak-dominated regime

**Theorem D″** (uniform-in-time propagation of chaos in the leak-dominated regime)**.** *In the setting and with the constants of Theorem D′, assume the standing bound for all $t \ge 0$ and let $L_2 := \sup\|\nabla_\vartheta^2\tilde\Phi\|$, $L_{x\vartheta} := \sup\|\partial_x\nabla_\vartheta\tilde\Phi\|$ (over all $x$ and $\vartheta$; for $n > 1$ the operator norms are contracted with unit vectors as in (1) of Section 9), and*

$$\kappa_e := k_e - L_f - L_x, \qquad \kappa_\vartheta := \sigma - \lambda_G E_X L_2,$$
$$L_{b'} := \lambda_G(L_\vartheta + E_X L_{x\vartheta}), \qquad \det A := \kappa_e\kappa_\vartheta - L_{b'}L_\vartheta.$$

*If $\kappa_e > 0$ and $\det A > 0$ (then $\kappa_\vartheta > 0$ as well) — the leak dominates the loop gain of the adaptive law — then, under the synchronous coupling of Theorem D′(b), for every $t \ge 0$ and every $\tau \ge 0$,*

$$\mathbb{E}\|x_t - \bar x_t\| \le K_\Phi\kappa_\vartheta m^{-1/2}/\det A,$$
$$\mathbb{E}\frac{1}{m}\sum_i\|\vartheta_i(t) - \bar\vartheta_i(t)\| = \mathbb{E}\|\vartheta_1(t) - \bar\vartheta_1(t)\| \le K_\Phi L_{b'} m^{-1/2}/\det A,$$

*with constants independent of $t$: propagation of chaos is uniform in time, at the rate $m^{-1/2}$. Moreover $\sup_t \mathbb{E}W_1(\mu_t^m, \bar\mu_t) \le K_\Phi L_{b'} m^{-1/2}/\det A + \sup_t \mathbb{E}W_1(\bar\mu_t^{\mathrm{emp},m}, \bar\mu_t)$, and the second term is finite: $\sup_t \int\|\vartheta - \vartheta^0\|^2\,\mathrm{d}\bar\mu_t \le \max\{\int\|\vartheta - \vartheta^0\|^2\,\mathrm{d}\bar\mu_0,\ ((\lambda_G E_X L_\vartheta)^2/\sigma + 2\tau\,\mathrm{tr}\,G)/\sigma\}$.*

**Status.** Proved; checked numerically on one slow-adaptation instance (`exp_chaos_uniform.py`), where it also shows its limit. The contraction condition $\kappa_\vartheta > 0$ makes the particle map a contraction uniformly in $t$, so in the certified regime the particles of a single system stay together: at $\tau = 0$ they synchronise exponentially and the network collapses to one kernel; at $\tau > 0$ they form the dither cloud of the prior, $\mathcal{N}(\vartheta^0, (\tau/\sigma)G)$ to leading order, and the network's output stays at the level of its prior (Remark D″.1). In both cases the leak dominates the learning: the certified regime is the one in which the network does not learn. What the theorem adds to Theorem D′ is the removal of the horizon from the bound in that regime, with explicit constants (non-trivially at $\tau > 0$, where the coupled distance is of order $m^{-1/2}$ and not zero), and a precise statement of where the contraction method stops. Referee round 6 (`review/REVIEW_round6_Dpp.md`) confirmed the proof, corrected a double-counted factor 2 in the interconnection gain $L_{b'}$ (sharp constants below), and found that the run first reported as "learning regime" was a slow collapse to the dead kernel of an odd plant; the instance was changed, and the new long-horizon run in the learning regime shows the coupled distance growing in time — the uniform statement is supported only where the theorem proves it.

*Proof.* Keep the contracting terms that Theorem D′'s proof dropped. For $\delta := x - \bar{x}$, the term $-k_e\delta$ gives $D^+\|\delta\| \le -(k_e - L_f - L_x)\|\delta\| + L_\vartheta\bar{\Delta} + \zeta_t$ with the same decomposition of the bracket as in D′(b). For $\Delta_i := \vartheta_i - \bar{\vartheta}_i$, split $b(x,e,\vartheta_i) - b(\bar{x},\bar{e},\bar{\vartheta}_i) = [b(x,e,\vartheta_i) - b(\bar{x},\bar{e},\vartheta_i)] + [b(\bar{x},\bar{e},\vartheta_i) - b(\bar{x},\bar{e},\bar{\vartheta}_i)]$. Since $e - \bar{e} = \delta$, the first bracket is $G\nabla_\vartheta\tilde{\Phi}(x;\vartheta_i)^\top(e-\bar{e}) + G[\nabla_\vartheta\tilde{\Phi}(x;\vartheta_i) - \nabla_\vartheta\tilde{\Phi}(\bar{x};\vartheta_i)]^\top\bar{e}$, of norm at most $\lambda_G L_\vartheta\|\delta\| + \lambda_G L_{x\vartheta}\|\delta\|E_X = L_{b'}\|\delta\|$ on the tube. The second bracket is $G[\nabla_\vartheta\tilde{\Phi}(\bar{x};\vartheta_i) - \nabla_\vartheta\tilde{\Phi}(\bar{x};\bar{\vartheta}_i)]^\top\bar{e} - \sigma\Delta_i$, and $\langle\Delta_i, \cdot\rangle \le \lambda_G\|\bar{e}\|L_2\|\Delta_i\|^2 - \sigma\|\Delta_i\|^2 \le -\kappa_\vartheta\|\Delta_i\|^2$, so $D^+\|\Delta_i\| \le -\kappa_\vartheta\|\Delta_i\| + L_{b'}\|\delta\|$, and the same for the average $\bar{\Delta}$. The pair $u := (\|\delta\|, \bar{\Delta})$ thus satisfies $D^+u \le Au + (\zeta_t, 0)^\top$ componentwise, $u(0) = 0$, with the Metzler matrix $A := \left[\begin{smallmatrix} -\kappa_e & L_\vartheta \\ L_{b'} & -\kappa_\vartheta \end{smallmatrix}\right]$ (non-negative off-diagonal entries), whose flow $e^{As}$ is entrywise non-negative. The comparison principle for cooperative systems gives $u(t) \le \int_0^t e^{A(t-s)}(\zeta_s, 0)^\top \mathrm{d}s$ entrywise. $A$ is Hurwitz exactly when its trace is negative and its determinant positive; with $\kappa_e > 0$ and $\det A = \kappa_e\kappa_\vartheta - L_{b'}L_\vartheta > 0$ one has $\kappa_\vartheta > L_{b'}L_\vartheta/\kappa_e \ge 0$, so both hold, and then $\int_0^\infty e^{As}\,\mathrm{d}s = -A^{-1} = (\det A)^{-1}\left[\begin{smallmatrix} \kappa_\vartheta & L_\vartheta \\ L_{b'} & \kappa_e \end{smallmatrix}\right]$, entrywise non-negative. Taking expectations, with $\mathbb{E}\zeta_s \le (\mathbb{E}\zeta_s^2)^{1/2} \le K_\Phi m^{-1/2}$ for every $s$ (the $\bar{\vartheta}_j(s)$ are i.i.d. with law $\bar{\mu}_s$ and $\bar{x}_s$ is deterministic, as in D′(b)), $\mathbb{E}u(t) \le K_\Phi m^{-1/2}(-A^{-1})e_1 = K_\Phi m^{-1/2}(\kappa_\vartheta, L_{b'})^\top/\det A$ for every $t$. The per-particle statement is exchangeability at fixed $t$ (valid here, unlike for $\sup_t$: Theorem D′(c)); the $W_1$ statement is the triangle inequality as in D′(c). For the second moment of $\bar{\mu}_t$: $z(t) := \mathbb{E}\|\bar{\vartheta}_t - \vartheta^0\|^2$ satisfies, by Itô's formula with $\|G\nabla_\vartheta(\bar{e}^\top\tilde{\Phi})\| \le \lambda_G E_X L_\vartheta =: \beta$, $\dot{z} \le -2\sigma z + 2\beta\sqrt{z} + 2\tau\,\mathrm{tr}\,G \le -\sigma z + \beta^2/\sigma + 2\tau\,\mathrm{tr}\,G$, whence $z(t) \le \max\{z(0), (\beta^2/\sigma + 2\tau\,\mathrm{tr}\,G)/\sigma\}$. The noise cancels in $\Delta_i$, so the coupling estimate does not depend on $\tau$. □

**Remark D″.1** (reading)**.**

- *The certified regime is the leak-dominated one.* The condition $\kappa_\vartheta = \sigma - \lambda_G E_X L_2 > 0$ says that $\vartheta \mapsto b(\bar{x}, \bar{e}, \vartheta)$ is a contraction uniformly in $t$. At $\tau = 0$ the same estimate applied to two particles of the *same* system gives $D^+\|\vartheta_i - \vartheta_j\| \le -\kappa_\vartheta\|\vartheta_i - \vartheta_j\|$: all particles synchronise exponentially, the network collapses to a single kernel, and the $m$-system and the $M$-system — which share their first $m$ particles — become identical, so the coupled distance is not just bounded but zero (the numerics: particle spread and coupled distance at the rounding level). At $\tau > 0$ the two Brownian increments do not cancel in $\vartheta_i - \vartheta_j$; the particles form a cloud around the common mean motion whose second moment is bounded as in the proof, $\mathcal{N}(\vartheta^0, (\tau/\sigma)G)$ to leading order when the leak dominates (measured: rms spread within 3% of $(\tau\,\mathrm{tr}\,G/\sigma)^{1/2}$ in the referee's runs), the coupled distance is of order $m^{-1/2}$ and the bound is not trivial — but the network's output stays at the level of its prior: $\sup|\hat{f}|$ is a few percent of $\sup|f|$ in the certified runs. Either way, propagation of chaos is uniform in time because the leak has stopped the learning. This is the price of the contraction method, not of the closed loop: the learning regime, in which kernels separate

and the network fits the plant ($\sigma$ below the critical leak of Proposition K at the collapsed critical point, the leak below the separating force of the residual), is where Theorems H, I and K live, where the averaged interaction is non-reciprocal, and where uniformity in time remains the open part of Section 26, branch 3.

- *What the conditions say in the scalings of the program.* On the a-priori tube of Theorem D(b) with $e(0) = 0$, $k_e E_X = F + K_\Phi$, so the certified leak is $\sigma_{\mathrm{cert}} := \lambda_G E_X L_2 + L_{b'} L_\vartheta / \kappa_e = [\lambda_G (F + K_\Phi) L_2 + \lambda_G L_\vartheta^2]/k_e + O(k_e^{-2})$: every fixed leak gain $\sigma > 0$ certifies uniform-in-time propagation of chaos for $k_e$ large, provided the tracking error starts at zero (or $\|e(0)\| = O(1/k_e)$; otherwise $E_X \geq \|e(0)\|$ and the condition does not improve with $k_e$) — and stops the learning. In the scaling $\sigma = \sigma_0/k_e$ of Theorem J the condition reads $\sigma_0 > \lambda_G (F + K_\Phi) L_2 + \lambda_G L_\vartheta^2$, which implies that $J_m$ is strongly convex in the $G^{-1}$ metric (its Hessian is bounded below by $\sigma_0 - \lambda_G \sup_\Theta \|r_\Theta\| L_2 \geq \sigma_0 - \lambda_G (F + K_\Phi) L_2$ there) with a unique, collapsed minimiser.
- *Why the constants are conservative.* $L_\vartheta$, $L_2$, $L_{x\vartheta}$ are suprema over the parameter box and the tube; for sharp kernels (small $s_{\min}$) they are large, so $\sigma_{\mathrm{cert}}$ can exceed the adaptation gains by an order of magnitude; the a-priori $E_X$ is itself several times the measured $\sup\|\bar{e}\|$ (0.23 against 0.07 in the numerics below). The gap to the learning regime is not a matter of constants, however: in the numerics the coupled distance grows over the horizon once the kernels separate.
- *Technique.* The comparison system is the standard contraction argument for McKean–Vlasov systems with a confining potential, with the plant as a second contracting channel (rate $\kappa_e$) and the two interconnection gains $L_\vartheta$ (particles $\to$ plant) and $L_{b'}$ (plant $\to$ particles); $\det A > 0$ is the small-gain condition of that interconnection.

**Numerics** (`exp_chaos_uniform.py` $\to$ `results/chaos/uniform_table.md`; reproduced in Table 16). A slow-adaptation instance ($n = 1$, $k_e = 25$, $G = 0.05 \cdot \mathrm{diag}(4, 1, 1)$, saturated features with $a_{\max} = 4$, $s \geq 0.3$, prior $\vartheta^0 = 0$), a two-bump plant that is not odd, $f(x) = 1.6e^{-(x-1)^2/0.4} - 1.2e^{-(x+1)^2/0.4} + 0.1x$, on $x_m = 1.5 \sin t + 0.5 \sin 2t$ (so that the dead kernel is not a critical point; referee round 6), $m = 16$, 64, 256 coupled to an $M = 2048$ reference. The constants are exact: the Lipschitz constants of the saturated features in closed form, $L_x = L_\vartheta = a_{\max} e^{-1/2}/s_{\min} = 8.087$ and $L_2 = L_{x\vartheta} = a_{\max}/s_{\min}^2 = 44.4$ (a numerical search over $1.2 \cdot 10^5$ parameter samples with Nelder–Mead polish reproduces them to $10^{-8}$), $L_f = 2.29$, and the a-priori tube radius of Theorem D(b) with $e(0) = 0$, $E_X = (F + K_\Phi)/k_e = 0.228$ ($F = 1.70$; the measured $\sup\|\bar{e}\|$ is 0.070): $\kappa_e = 14.6$, $L_{b'} = 3.64$, $\sigma_{\mathrm{cert}} = 4.04$. Proposition K gives the critical leak of the collapsed critical point (the best single kernel, $a = 0.136$, $c = 0.068$, $s = 1.15$): $\sigma_K = \sigma_{0,\mathrm{crit}}/k_e = 0.00144$ — the certified leak is 2800 times the leak below which the kernels separate. Four runs:

Table 16: Theorem D″ on the slow-adaptation instance: four runs at different leaks (`results/chaos/uniform_table.md`).

| $\sigma$, $\tau$ | regime | $T$, seeds | reference network at $T$: rms spread in $(a, c, s)$ (from 1.53), fit error $\|f - \hat{f}\|/\|f\|$ (from 1.00) | max $D_m$, first $\to$ last window, $m = 16$ and 256 | slope against $m$ (per-seed, $\pm$ s.e.) | bound $\sup_t \mathbb{E}D_m$, $m = 16$ / 256 |
|---|---|---|---|---|---|---|
| 4.85 ($1.2\sigma_{\text{cert}}$), 0.02 | certified, with dither | 80 s, 8 | 0.033 (the dither cloud: $(\tau \operatorname{tr} G/\sigma)^{1/2} = 0.035$), 1.000 | $6.7 \cdot 10^{-4} \to 7.9 \cdot 10^{-4}$ and $1.6 \cdot 10^{-4} \to 1.9 \cdot 10^{-4}$ | $-0.51 \pm 0.02$ | 0.55 / 0.14 |
| 4.85, 0 | certified, no dither | 80 s, 8 | $1.7 \cdot 10^{-14}$, 1.000 | $2.5 \cdot 10^{-25} \to 0$ and $1.2 \cdot 10^{-26} \to 0$ | — | 0.55 / 0.14 |
| 0.505 ($\sigma_{\text{cert}}/8$), 0 | leak-dominated, below the threshold | 80 s, 8 | $3.2 \cdot 10^{-14}$, 1.000 | $7.2 \cdot 10^{-5} \to 4.3 \cdot 10^{-11}$ and $3.1 \cdot 10^{-5} \to 1.8 \cdot 10^{-11}$ | $-0.33 \pm 0.20$ | — |
| 0.00048 ($\sigma_K/3$), 0 | learning regime | 1000 s, 6 | 1.53, 0.632 | $6.1 \cdot 10^{-2} \to 2.0 \cdot 10^{-1}$ and $1.3 \cdot 10^{-2} \to 3.9 \cdot 10^{-2}$ | $-0.62 \pm 0.16$ | — |

(Windows $(10, 20]$, $(20, 40]$, $(40, 80]$ for the 80 s runs and $(100, 200]$, $(200, 500]$, $(500, 1000]$ for the long run; the table reports $\mathbb{E} \max_t D_m$ over a window, which exceeds the theorem's $\sup_t \mathbb{E}D_m$.) At the certified leak with dither the particles stay in the cloud of the prior, the network's output stays at the level of the empty network (fit error 1.000), and the coupled distance is of order $m^{-1/2}$ (slope $-0.51 \pm 0.02$), bounded in time (the last window, twice as long, has maxima 15–18% above the first) and seven hundred times below the bound: the theorem's content at $\tau > 0$. Without dither the reference collapses (spread $10^{-14}$) and the coupled distance is zero to rounding; at $\sigma_{\text{cert}}/8$ the particles still synchronise and the distance decays with them. In the learning regime ($\sigma_K/3$, the collapsed configuration unstable) the network stays spread, fits the plant (fit error $1.00 \to 0.63$ at $T = 1000$ s with these slow gains) and the coupled distance follows the $m^{-1/2}$ law at each time but *grows* over the horizon — by a factor 3 between the windows $(100, 200]$ and $(500, 1000]$ for every $m$: on this horizon propagation of chaos in the learning regime is the finite-horizon statement of Theorem D′, not a uniform one. Whether the growth saturates on longer horizons (the leak's own time scale is $1/\sigma = 2000$ s) is open; the earlier claim that the distance stays bounded in a "spread" regime (version 4, first draft) rested on a run that was in fact a slow collapse, and is withdrawn (referee round 6, DPP-3).

Checked by: `exp_chaos_uniform.py`: the experiment only.

# 18 Theorem D‴: uniform-in-time propagation of chaos in the learning regime (the averaged slow loop)

Theorem D″ ends where the leak stops dominating the loop gain: in the learning regime its small-gain argument has nothing to say, and the long-horizon run of `exp_chaos_uniform.py` showed the coupled distance growing over the horizon. This section isolates the structure that the small-gain argument throws away. Under the synchronous coupling the interaction between the particles enters through the fitted function $\hat{f}_\Theta$, which is linear in the empirical measure; the loss $\bar{L}$ is a convex quadratic of it (flat convexity, Theorem H(i)); and the interaction term of the coupling identity is therefore the negative of a squared $L^2(\rho)$-norm at linear order — it contracts, whatever the gains. What remains and can expand is the self term, whose curvature is the Hessian of the first-variation potential, $\sigma_0 G^{-1} - R[\mu](\vartheta)$ with $R$ the *residual-weighted*

curvature of the features — the same object as Proposition K's $R_2$ at a collapsed configuration. Proposition K says that the collapsed configuration is unstable when this curvature exceeds the leak *with the collapsed residual*; at a fitted configuration the residual is the approximation error, and the second-order optimality condition of the limit measure says that the curvature is *below* the leak at every atom. The theorem below turns this into a uniform-in-time statement for the averaged slow loop at $\tau = 0$, with explicit constants, in the regime in which the network learns.

**Setting.** The averaged slow dynamics of Section 24 at $\tau_0 = 0$ (Theorem J′'s leading order $\bar{L}_0$ without dither; slow time $s = t/k_e$, leak $\sigma_0 = k_e\sigma$, $f = f(x)$ continuous, $\rho$ the occupation measure of the $T_0$-periodic reference, $F_\rho := \sup_{\operatorname{supp}\rho}\|f\|$, saturated features, $K_\Phi := \sup\|\tilde{\Phi}\|$), written as the $G$-preconditioned gradient flow of $J_m(\Theta) = m\bar{L}(\mu_\Theta) + \frac{\sigma_0}{2}\sum_i\|\vartheta_i - \vartheta^0\|^2_{G^{-1}}$:

$$\frac{\mathrm{d}\vartheta_i}{\mathrm{d}s} = -G\nabla_{\vartheta_i}J_m(\Theta) = -G\nabla_\vartheta\mathcal{V}[\mu_\Theta](\vartheta_i), \qquad i = 1..m, \tag{$\mathfrak{A}_m$}$$
$$\mathcal{V}[\mu](\vartheta) := V_L[\mu](\vartheta) + \frac{\sigma_0}{2}\|\vartheta - \vartheta^0\|^2_{G^{-1}},$$

with $V_L[\mu](\vartheta) = -\langle r_\mu^\top\tilde{\Phi}(\cdot\,;\vartheta)\rangle_\rho$ and $r_\mu = f - \hat{f}[\mu]$, $\hat{f}[\mu] := \int\tilde{\Phi}(\cdot\,;\vartheta')\,\mathrm{d}\mu(\vartheta')$, as in Section 20. Its mean-field equation is the Wasserstein gradient flow ($\mathfrak{A}$) of Section 20 at $\tau = 0$ in slow time, $\partial_s\mu_s = \nabla\cdot(\mu_s G\nabla\mathcal{V}[\mu_s])$, solved by the characteristic flow

$$\frac{\mathrm{d}X_s(\vartheta)}{\mathrm{d}s} = -G\nabla\mathcal{V}[\mu_s](X_s(\vartheta)), \qquad X_0 = \mathrm{id}, \qquad \mu_s = (X_s)_\#\mu_0 \tag{$\mathfrak{A}_0$}$$

(a McKean–Vlasov equation without noise whose vector field has linear growth through the leak, is Lipschitz in $\vartheta$ and, through $r_\mu$, Lipschitz in $\mu$ for $W_1$; existence, uniqueness and the representation by characteristics follow from Dobrushin's argument [25] — the synchronous-coupling proof of Theorem D′(b) with the plant removed — and the velocity is bounded on the invariant ball of Step 0). The constants are taken in the $G^{-1}$ metric on parameters ("whitened"):

$$L^G_\vartheta := \sup_{x,\vartheta}\|\nabla_\vartheta\tilde{\Phi}(x;\vartheta)G^{1/2}\|, \qquad L^G_2 := \sup_{x,\vartheta,\|u\|=1}\|G^{1/2}\nabla^2_\vartheta\big(u^\top\tilde{\Phi}(x;\vartheta)\big)G^{1/2}\|,$$

so that $\|\tilde{\Phi}(x;\vartheta) - \tilde{\Phi}(x;\vartheta')\| \le L^G_\vartheta\|\vartheta - \vartheta'\|_{G^{-1}}$, $\|\nabla_\vartheta\tilde{\Phi}(x;\vartheta)v\| \le L^G_\vartheta\|v\|_{G^{-1}}$ and $\|(\nabla_\vartheta\tilde{\Phi}(x;\vartheta) - \nabla_\vartheta\tilde{\Phi}(x;\vartheta'))v\| \le L^G_2\|\vartheta - \vartheta'\|_{G^{-1}}\|v\|_{G^{-1}}$ ($L^G_\vartheta \le \lambda_G^{1/2}L_\vartheta$ and $L^G_2 \le \lambda_G L_2$ in the constants of Section 17; `chaos_learning.whitened_lipschitz_constants` computes the suprema by search, with closed-form bounds). The *residual-weighted curvature* of $\mu$ at $\vartheta$ is the symmetric matrix

$$R[\mu](\vartheta) := \Big\langle\sum_k r^k_\mu(x)\nabla^2_\vartheta\tilde{\Phi}^k(x;\vartheta)\Big\rangle_\rho = -\nabla^2_\vartheta V_L[\mu](\vartheta), \qquad R^G := G^{1/2}RG^{1/2},$$
$$\kappa_R[\mu](A) := \sup_{\vartheta\in A}\lambda_{\max}\big(R^G[\mu](\vartheta)\big),$$

so that $G^{1/2}\nabla^2_\vartheta\mathcal{V}[\mu](\vartheta)G^{1/2} = \sigma_0 I - R^G[\mu](\vartheta)$; for $\mu = \delta_{\bar\vartheta}$ and $\vartheta = \bar\vartheta$ it is Proposition K's $R_2$ and $\sigma_{\mathrm{crit}}(\bar\vartheta) = \kappa_R[\delta_{\bar\vartheta}](\{\bar\vartheta\})$. It satisfies $\|R^G[\mu](\vartheta)\| \le L^G_2\langle\|r_\mu\|\rangle_\rho \le L^G_2(F_\rho + K_\Phi)$ and $\|R^G[\mu](\vartheta) - R^G[\nu](\vartheta)\| \le L^G_2\|\hat{f}[\mu] - \hat{f}[\nu]\|_{L^1(\rho)}$. $N_\varrho(A) := \{\vartheta : \mathrm{dist}_{G^{-1}}(\vartheta, A) \le \varrho\}$.

*The coupling.* $\vartheta_i(0)$, $i = 1..m$, i.i.d. with law $\mu_0$ of compact support, $R_0 := \sup_{\operatorname{supp}\mu_0}\|\vartheta - \vartheta^0\|_{G^{-1}}$; $\Theta(s)$ solves ($\mathfrak{A}_m$) and the reference particles are $\bar\vartheta_i(s) := X_s(\vartheta_i(0))$, i.i.d. with law $\mu_s$ and $\bar\vartheta_i(s) \in \operatorname{supp}\mu_s$. Let $\delta_i := \vartheta_i - \bar\vartheta_i$,

$$D(s) := \frac{1}{m}\sum_i\|\delta_i(s)\|^2_{G^{-1}}, \qquad Y(s) := \|\hat{f}_{\bar\Theta(s)} - \hat{f}[\mu_s]\|_{L^2(\rho)},$$

$Y$ being the sampling fluctuation of the reference's fitted function ($\mathbb{E}Y(s)^2 = \frac{1}{m}\langle\mathrm{Var}_{\mu_s}\tilde{\Phi}(x;\cdot)\rangle_\rho \le K^2_\Phi/m$). A priori (Step 0 below) every particle of either system stays in the ball $\|\vartheta - \vartheta^0\|_{G^{-1}} \le R_\Theta := \max\{R_0,\ L^G_\vartheta(F_\rho + K_\Phi)/\sigma_0\}$, so $D \le D_{\max} := 4R^2_\Theta$, and the characteristics move at speed at most $V_{\max} := L^G_\vartheta(F_\rho + K_\Phi) + \sigma_0 R_\Theta$.

**Theorem D‴** (uniform-in-time propagation of chaos in the learning regime, $\tau = 0$)**.** *Assume*

(H) (non-degenerate stability along the mean-field flow) *there are* $s_1 \geq 0$, $\kappa > 0$ *and* $\varrho' > 0$ *such that*

$$\kappa_R[\mu_s]\big(N_{\varrho'}(\operatorname{supp}\mu_s)\big) \leq \sigma_0 - \kappa \qquad \text{for every } s \geq s_1,$$

*and put* $a(s) := \sigma_0 - \kappa_R[\mu_s](N_{\varrho'}(\operatorname{supp}\mu_s))$ *(so* $a \geq \kappa$ *on* $[s_1,\infty)$ *and* $a \geq \sigma_0 - L_2^G(F_\rho + K_\Phi)$ *everywhere) and* $\Lambda_1 := \int_0^{s_1}(\kappa - a(s))_+\,\mathrm{d}s < \infty$. *Define*

$$\varrho := \min\big\{(2\kappa)^{1/2}/L_2^G,\ \varrho'\kappa e^{-\Lambda_1}/(4(L_\vartheta^G)^2)\big\},$$
$$\theta := \min\big\{\kappa/(8L_2^G),\ \varrho\kappa e^{-\Lambda_1}/(4L_\vartheta^G),\ \varrho'\kappa e^{-\Lambda_1}/(4L_\vartheta^G)\big\},$$
$$\Delta := \theta/(4L_\vartheta^G V_{\max}), \qquad C_1 := 4e^{2\Lambda_1}(L_\vartheta^G)^2K_\Phi^2/\kappa^2, \qquad m_0 := 16K_\Phi^2/\theta^2.$$

*Then for every* $m \geq m_0$*:*

(a) (exponentially long horizons) *for every* $S > 0$ *and every* $s \in [0,S]$,

$$\mathbb{E}D(s) \leq C_1/m + 4R_\Theta^2(S/\Delta + 2)\exp\big(-m\theta^2/(32K_\Phi^2)\big);$$

*in particular* $\sup_{s\leq S}\mathbb{E}D(s) \leq (C_1 + 12R_\Theta^2)/m$ *for all* $S \leq \Delta e^{m\theta^2/(64K_\Phi^2)}$ *and* $m \geq m_0$ *with* $e^{m\theta^2/(64K_\Phi^2)} \geq m$.

(b) (uniform in time) *if moreover* (H′) *the characteristic flow converges exponentially on the support of* $\mu_0$, $\sup_{\vartheta\in\operatorname{supp}\mu_0}\|X_s(\vartheta) - X_\infty(\vartheta)\|_{G^{-1}} \leq C_2e^{-\kappa_2 s}$ *for all* $s \geq 0$, *then for every* $s \geq 0$

$$\mathbb{E}D(s) \leq C_1/m + 4R_\Theta^2(R_{\mathrm{eff}}/\Delta + 3)\exp\big(-m\theta^2/(32K_\Phi^2)\big), \qquad R_{\mathrm{eff}} := \kappa_2^{-1}\big(\log(8L_\vartheta^GC_2/\theta)\big)_+.$$

(c) (the learned function and the particles) *in either case, with the same right-hand side* $B(s)$ *for* $\mathbb{E}D(s)$,

$$\mathbb{E}\|\hat f_{\Theta(s)} - \hat f[\mu_s]\|_{L^2(\rho)}^2 \leq 2(L_\vartheta^G)^2B(s) + 2K_\Phi^2/m,$$
$$\mathbb{E}W_2(\mu_{\Theta(s)},\mu_s) \leq B(s)^{1/2} + \mathbb{E}W_2(\mu_{\bar\Theta(s)},\mu_s),$$

*and on the event* $\mathcal{G}_S := \{\sup_{s\leq S}Y(s) \leq \theta\}$ *(case* (a)*;* $\mathcal{G}_\infty$ *in case* (b)*), whose complement is the tail above, every particle stays within* $3\varrho'/4$ *of its reference up to time* $S$*:* $\max_i\sup_{s\leq S}\|\delta_i(s)\|_{G^{-1}} \leq 3\varrho'/4$ *— no particle changes basin.*

*The constants depend on* $(\kappa,\varrho',\Lambda_1,L_\vartheta^G,L_2^G,K_\Phi,F_\rho,\sigma_0,R_0)$ *and not on* $m$, $s$ *or* $S$.

**Status.** Proved (synchronous coupling with the three-term identity of Step 1, a bootstrap on the mean square and the per-particle distance, and McDiarmid's inequality [61] for the reference's sampling fluctuation; the finite-horizon part of the transient is Theorem D′'s mechanism and appears in $\Lambda_1$). Hypothesis (H) is a property of the deterministic mean-field flow and can be computed; it is the strict second-order optimality condition of the limit configuration with a margin and a neighbourhood (Remark D‴.1), and it is compatible with the learning regime $\sigma_0 < \sigma_{\mathrm{crit}}$ of Proposition K. It requires the support of $\mu_s$ to stay away from the ridges of $\mathcal{V}[\mu_s]$ for $s \geq s_1$: for an initial cloud whose support is a union of pieces each flowing into one basin (a warm start, or a cluster per bump) this holds; for a connected cold start whose mass splits between several basins it fails for every $s$ (Remark D‴.2), and the numerics below show what the coupling does there. Numerics: `exp_chaos_learning.py` → `results/chaos/learning_table.md` (below). The theorem is about the averaged slow dynamics; its transfer to the real loop at finite $k_e$ is open (Remark D‴.4). Its constants are worst-case and its regime is asymptotic: on the instance below $m_0$ is of order $10^{23}$ (Remark D‴.6), and the finite-$m$ behaviour is described sharply by Proposition D‴.5. Referee round 8 (`review/REVIEW_round8_Dppp.md`) re-derived every constant of Steps 0–9, found no counterexample to (2.1), (3.1), (4.1) on an adversarial search, and corrected the wording of Remarks

D‴.1–D‴.2, the good event per horizon and the well-posedness sentence (all applied). Referee round 9 (`review/REVIEW_round9_D3b.md`) reviewed Proposition D‴.5, Corollary D‴.7 and the numerics: all hold with fixes, the main one being that the finite-$m$ replicated configuration is a minimiser only for admissible weights (Proposition D‴.5(iv)) — one of the four $m = 16$ warm seeds sits on a saddle, which the numerics paragraph now says. Novelty (literature audit): the technique (synchronous coupling + bootstrap + bounded differences) is standard; the sign structure of the interaction and the local, residual-weighted hypothesis — and with them a uniform-in-time statement in a regime where the network learns — are new in propagation of chaos for adaptive control; the theorem's regime is asymptotic in $m$ and the finite-$m$ law is a delta-method statement, new in its object.

**Literature audit** (`LITERATURE_TRANSPORT.md`, D‴).

- *Prior.* Uniform-in-time propagation of chaos by synchronous coupling under *global* convexity of the confinement and convexity of the interaction is Malrieu's [58]; non-convex confinements with *small* Lipschitz interactions are handled by reflection coupling [26]; flat-convex energies with dither by the entropy method [11]. The Hessian decomposition "Gram + residual-weighted feature curvature" is the standard tool of the interpolation / PL-condition literature [56] and of lazy-training analyses, where the second term vanishes at interpolation.

- *Distinction.* (i) The interaction of the fitted network is neither small nor convex in the particle coordinates; its sign comes from the flat convexity of $\bar{L}$ ($\hat{f}$ linear in the measure), which makes $-\|\hat{f}_\Theta - \hat{f}_{\bar{\Theta}}\|^2$ the leading interaction term under the coupling — the finite-$m$, two-measure version of [56]'s decomposition. (ii) The convexity needed is local: $\sigma_0 I - G^{1/2} R[\mu_s] G^{1/2} \succeq \kappa I$ on a neighbourhood of the support of the mean-field flow, i.e. the strict second-order optimality condition of the fitted configuration, evaluated at the actual residual — not a global convexity (which would force the collapsed configuration to be stable and the network not to learn, Theorem D″). (iii) The window $\kappa_R[\mu_\infty] < \sigma_0 < \sigma_{\mathrm{crit}}$ between the collapsed instability (Proposition K) and the fitted stability is a design statement about the leak that has no counterpart in the cited work. (iv) The limits are stated with the result: a connected initial support that splits between basins violates the hypothesis for all times (the characteristic flow is a homeomorphism), and the coupling then has a floor of pairs in different basins; the real loop at finite $k_e$ is not covered.

- *Proposition D‴.5 (the weight-sensitivity law).* At finite $m$ the stationary coupled distance is the multinomial fluctuation of the cluster weights propagated through the implicit function $w \mapsto \Theta^*(w)$: a delta-method statement (the Gaussian limit of a smooth function of a multinomial), standard in statistics; what is specific is its object (the shared-particle coupling of a particle approximation to a stationary $K$-atom measure, with the covariance $(1/m - 1/M)(\operatorname{diag} \bar{w} - \bar{w}\bar{w}^\top)$) and its role as the sharp form of the theorem's constant (stiffness$^{-1}$ $\times$ feature correlations against $\kappa^{-1} L^G_\vartheta K_\Phi$).

- *Proposition D‴.9 (measure-valued priors).* Leaks toward a single prior ($\sigma$-modification, Ioannou–Kokotović) are the standard of robust adaptive control; regularisation toward the nearest of several centres is standard in machine learning (mixture priors, prototype regularisers) and this document's own Theorem C′ leaks toward a prior *measure* through an optimal plan. The nearest-atom leak is the simplest measure-valued prior; what is new is its role in the window of Theorem D‴ — the prior's placement sets the stationary residual, hence the lower end of the window and the relaxation rate — and the finding that removing the bias raises the coupling's floors (on the instance the weight sensitivity rises with the amplitudes): a bias–variance trade-off stated for the particle system. Verdict: standard object, new design statement; the function-level sensitivity law is the delta method once more.

- *Proposition D‴.12 (the mass constraint).* Capacity-constrained and equal-mass quantisation through optimal plans, and semi-discrete optimal transport with its Laguerre cells, are classical ([5, 3]); the integrality of the transportation polytope and the forest structure of its vertices are textbook ([42]); $c$-cyclical monotonicity is the standard optimality criterion ([86]). What is new is the observation that the finite-$m$ floor of the mean-field limit of an adaptive law is carried by a conservation law of its leak (the cluster masses under the nearest-atom leak), and that a design of the leak — the mass constraint of the OT plan — breaks it: on the commensurate sizes the stationary measure of the particle system coincides with the mean-field one, and the finite-$m$ fluctuation of the plan is counted (at most $K_0 - 1$ split particles) rather than estimated. Propagation of chaos with an exactly $m$-independent stationary law is not in the literature we know of; the mean-field form of the barycentric leak for an atomic prior (the Laguerre-cell leak) was Section 26's open item. Numerics: `exp_ot_leak.py`. Verdict: textbook ingredients, new mechanism and new object (a conservation law of the leak as the source of the finite-size floor, broken by the plan's mass constraint).

- *Proposition D‴.17 (convergence of the OT dynamics).* The Kurdyka–Łojasiewicz route to the convergence of (sub)gradient trajectories of definable functions is standard ([8, 50]); its application to an adaptive law whose energy is a semiconcave transport cost plus a smooth loss is a routine verification (the saturated Gaussian features are exponential-analytic, the plan is a minimum over finitely many vertices). Verdict: imported tool, routine verification; the content is that the mass-constrained law converges without a rate, which separates the qualitative question (settled) from Theorem D‴'s quantitative one (open for the OT dynamics).

- *Proposition D‴.20 (the entropic OT leak).* Entropic OT, Sinkhorn's algorithm and barycentric mappings of entropic plans are standard ([21]), and so is the interpolation of the entropic cost between optimal transport and a kernel (MMD-type) discrepancy as $\varepsilon$ runs from 0 to $\infty$ (Sinkhorn divergences, [21]); size-constrained soft assignment by Sinkhorn (equipartition) is related prior art for a soft mass constraint. A critical temperature equal to twice the largest eigenvalue of a conditional covariance is deterministic annealing's ([77]), with the same roles — the particles as the codevectors, the prior's atoms as the data; here the curvature of the fit enters, and the threshold is the largest generalised eigenvalue of the conditional covariance relative to $I - R^G/\sigma_0$. What is new is the soft mass constraint as an adaptive law's leak — a smooth gradient flow between the OT leak and the $\sigma$-modification toward the prior's barycentre — in which the conditional covariance of a collapsed class is an effect of its mass excess, and its consequence for propagation of chaos: the masses equalise only down to a critical excess, so the finite-size floor returns at large $m$ for every positive regularisation. Verdict: standard ingredients, new object and new consequence (on the commensurate sizes only the hard constraint is floor-free for every $m$).

- *Theorem D‴.21 (a measure-level basin for the OT leak).* The Polyak–Łojasiewicz inequality and its linear convergence are classical ([74]); in Wasserstein space Łojasiewicz inequalities are known for $\lambda$-convex and displacement-convex functionals, and PL inequalities through optimal maps for some that are not geodesically convex — barycentre functionals, the distance of uniform point clouds to a density ([7]); the leak alone, $\frac{1}{2}W^2(\cdot,\nu)$, satisfies one with constant 2 (its metric slope is $W(\cdot,\nu)$; AGS [2, §10.4]). The theorem's plan segments are the generalised geodesic from $\mu^*$ to $\mu_\Theta$ based at the prior (AGS [2, §9.2]), along which $\frac{1}{2}W^2(\cdot,\nu)$ is 1-convex, and turning convexity along such a curve into a slope bound is AGS's device. The adaptive law's energy is semiconvex along generalised geodesics only with a negative constant (the loss's curvature at the actual residual), so a global argument is unavailable. What is new is the PL inequality along that geodesic for this energy — the loss linearised exactly in the fitted function with the curvature taken at the

stationary residual, where the fixed counts of the mass constraint make the geodesic a particle interpolation and the fitted function's increment the average of the per-particle increments: its constant is set by the residual-weighted curvature along the segments and does not depend on $m$ (sharp, $2\sigma_0$, at a matching prior); an $m$-independent energy basin, and uniqueness of the global minimum among empirical measures under a curvature hypothesis at the stationary residual; and the converse for the nearest-atom leak, whose energy has a spurious minimum at every admissible count vector — the energy-landscape form of the finite-size floor. Numerics: `exp_ot_basin.py`. Verdict: classical inequality and geometry (the generalised geodesic), new mechanism for a new energy (the counts as the source of uniformity in $m$).

- *Theorem D‴.23 (adaptive priors).* A quadratic penalty toward a centre that is moved toward the iterate is the proximal point method (iterated Tikhonov regularisation) and, in continuous time with a centre drawn to the mean of several coupled variables, elastic averaging ([90]); a learned mixture prior on the weights is soft weight-sharing, whose hard limit is a $K$-means regulariser ([90]), and the capacity-constrained form of that limit is [5]'s. The per-mode formulas are the classical quadratic analyses: $\sigma_0/(h_j+\sigma_0)$, the complement of the Tikhonov filter factor $h_j/(h_j+\sigma_0)$ (the residual factor), iterated Tikhonov's contraction by it per step (Hanke–Groetsch, [90]; in continuous time the gradient flow of the Moreau envelope, which is the slow-prior limit $\eta h_j/(h_j+\sigma_0)$ of the rate), and elastic averaging's worker–centre analysis on a quadratic, whose continuous-time form is the pair matrix $\left[\begin{smallmatrix} -(h+\sigma_0) & \sigma_0 \\ \eta & -\eta \end{smallmatrix}\right]$; empirical Bayes for the mean of a Gaussian prior is the same proximal iteration. What is new is their transfer to the mass-constrained particle system of the OT leak — the prior's atoms learned, its masses kept: $H_L$ identified as the measure-space excitation spectrum, the separations decoupled at $\sigma_0$, and hence the trade-off between the fixed prior's bias $\sigma_0/(h_j+\sigma_0)$ and the adaptive prior's rate $\eta h_j/(h_j+\sigma_0+\eta) < \lambda_j^- < \min\{h_j,\eta\}$ uniformly in $m$ — the rate half of the classical trade-off between a leak's robustness and persistent excitation, whose robustness half (the leak's a-priori bound) the adaptive prior gives up (Proposition D‴.27 restores it by projecting the prior); and the transit's drag, with the gate that starts the adaptation from the fixed-prior stationary point. The limits of the bias removal are largely the precedents' own properties — the lost a-priori bound is the proximal point method's behaviour without a minimiser ([90]), the $K_0$-cluster bias the quantisation error of soft weight-sharing's hard limit — and specific here only in their particle, mass-constrained form and in the run-off from a fit that the network attains exactly. Numerics: `exp_adaptive_prior.py`. Verdict: standard ingredients and per-mode formulas (proximal point and iterated Tikhonov, elastic averaging, soft weight-sharing), a new mechanism (their transfer to the mass-constrained particle system, uniform in $m$) and a new failure mode with its remedy (the transit's drag and the gate).

- *Proposition D‴.24 (the floor is the leak's bias).* The $m^{-1}$-size fluctuations of trained mean-field networks are classical ([89, 91]) and Proposition D‴.5/D‴.9's floor law is the delta method; the amplitude–mass degeneracy is the homogeneity of the conic view ([16, 23]); semi-relaxed transport (free target masses) reducing to the nearest-centre assignment is elementary. That the fluctuations' asymptotic deviation vanishes at an interpolating limit is Chen et al.'s ([91]); the derivation of $F$ is the chain rule applied to Theorem D‴.23(iii)'s first-order bias at $x^0(q)=0$, and the weight $h_j\sigma_0^2/(h_j+\sigma_0)^2$ is the squared complement of the Tikhonov filter factor times the mode's visibility, bounded by ridge regression's AM–GM bound. What is new is the identification of the function-level floor of a regularised adaptive law with the weight-derivative of the regulariser's bias at the sampled weights — at a matching prior $F=\hat{J}\sigma_0(H_L+\sigma_0 I)^{-1}\partial_w x^0$, decomposed by modes with the band-pass weight, whose bound is $\sigma_0/4$ times the adaptive prior's coupled-distance law — the split

of the sample's weight fluctuation between the parameters ($h_j/(h_j+\sigma_0)$) and the bias ($\sigma_0/(h_j+\sigma_0)$), and a leak toward a learned prior (the OT leak with the masses minimised out) as the way to restore the interpolation property under regularisation, which also removes the fixed prior's splitting instability; the trade-off is a larger coupled distance of the gauge, the leak's a-priori bound (restored by projecting the prior, Proposition D‴.27) and the collective rate. On the instance the law (B) is asymptotic (the exact floor 2.8 times it at $m = 15$), and the samples without an exact representation are not rare at small $m$ (0.36 at $m = 15$). Numerics: `exp_adaptive_floor.py`. Verdict: classical ingredients (the delta method, the Gauss–Newton spectrum, semi-relaxed transport, the filter-factor bound, the interpolation mechanism), new mechanism (the floor as the bias's slope, mode by mode, and its removal by a learned prior under regularisation) and a resolution of Remark D‴.10's bias–variance trade-off at the function level, at the price of the collective rate.

- *Corollary D‴.26 (perfect tracking with an adaptive prior: the Floquet spectrum).* Floquet theory and the averaging of a periodic variational equation are classical (first- and second-order averaging; the Magnus expansion, whose second term is a commutator, so that for a self-adjoint generator it is skew [95]); the averaging analysis of adaptive loops, with the rate set by the averaged regressor's persistency of excitation, is Kosut–Anderson–Mareels' and Sastry–Bodson's ([95], [81]); asymptotic tracking under exact parameterisation is the textbook goal of adaptive control, which $\sigma$-modification gives up for robustness ([46], [47]), and leaks compatible with it are known — the $e$-modification scales the leak by $|e|$ and vanishes at perfect tracking ([95]), a centre that follows the parameters has unbiased fixed points ([90]). (a) and (b) are Corollary D‴.14(ii) and Proposition D‴.15 at zero residual, with the prior's coordinates added; that an expansion is even when the problem is invariant under a symmetry reversing the small parameter's sign is the classical mechanism of the even error expansions of symmetric integrators and of the time symmetry of the Magnus exponent ([99], [95]). What is specific here is the combination: with the adaptive prior the zero-residual lift exists at every representable sample (not only at $w = q$ under the OT leak), its Floquet spectrum is exactly $m$-independent with collective rates $\lambda_j^{\pm}(w)/k_e$, and its expansion in $1/k_e$ has an explicit second-order coefficient — the error filter's attenuation against the slaving's speed-up, its sign not fixed (positive coefficients appear when the adaptation gain is raised) — which a derivative mismatch $\partial_x(f-\hat{f}) \neq 0$ would spoil at first order. Numerics: `exp_adaptive_floquet.py`. Verdict: classical tools, a new structural statement (an exact, $m$-independent spectrum at every representable sample, with the explicit second-order coefficient of this loop).

- *Lemma D‴.26a (the exact slow subspace).* Chang's decoupling of linear singularly perturbed systems — the transformation pair that block-diagonalises the system into its slow and fast parts — is classical ([102], [49]), for time-varying systems by a contraction (Yang–Zhu, [102]), with white-noise inputs and the slow subsystem's transformed noise and covariance (Kokotović–Khalil–O'Reilly Ch. 4, Haddad–Kokotović, [102]); stochastic slow manifolds and their covariance are Berglund and Gentz's ([102]); the convergence of the Magnus series is Moan and Niesen's ([102]); the evenness of an expansion under a symmetry reversing the small parameter is the mechanism of symmetric integrators, and $\Phi_K(-\varepsilon) = \Phi_{\tilde{K}}(\varepsilon)^{-1}$ is their adjoint relation ([99]). What is specific is the application to the perfect-tracking linearisation of the transport-leak loop: the slaving of Corollary D‴.26 becomes an exact block-triangularisation, so the slow multipliers are exactly the slow block's; the parity becomes the exact identity $\operatorname{eig} \bar{A}(-\varepsilon) = \operatorname{eig} \bar{A}(\varepsilon)$, the slow block at negative $\varepsilon$ defined by the same (anti-causal) contraction; and the cancellation $K^\top - (1+\varepsilon a)(K^\top + \varepsilon \dot{K}^\top) = -\varepsilon(\dot{K} + aK)^\top$ makes the slow noise of Proposition D‴.30 $\varepsilon b\nu$, at leading order the high-passed $-\varepsilon(\dot{K} + aK)^\top \nu$ (the drift decoupled exactly, the noise shared with the fast part). Numerics: `exp_adaptive_floquet.py` Part D. Verdict: classical tools; a rigorous foundation for D‴.26(c) and D‴.30(ii) (the referee round 22's audit: fair once the stochastic

and time-varying precedents are named).

- *Proposition D‴.27 (a projected adaptive prior).* Projection of the parameter estimates onto a convex set (Pomet–Praly), the exterior quadratic penalty and the switching $\sigma$-modification are the classical robustness modifications of adaptive laws ([47]), and the switching $\sigma$-modification keeps the ideal (unbiased) properties of the law when the true parameter lies in its set ([47], [95]); the a-priori bound by a projection inequality and a comparison argument is standard. On collapsed classes at $\eta = \infty$ the construction is exactly that law, toward the set, with the leak off on it. What is specific here is its transfer to the learned centre of a mass-constrained $K_0$-atom prior at finite $\eta$ — a projected adaptive law for the prior's atoms (whose velocity jumps at the faces, while the particles' law keeps its smooth gradient form) — and its compatibility with Theorem D‴.23's bias removal wherever the target is representable inside the set; at $\eta = \infty$ the plan is Lloyd-consistent rather than jointly optimal (as in Lloyd's algorithm for $K$-means, whose fixed points are local), so the monotone quantity is the selected energy, and Proposition D‴.17's convergence argument does not apply as it stands. Numerics: `exp_adaptive_prior.py` Part H. Verdict: classical ingredients and, at $\eta = \infty$, a classical law; a new transfer (the learned centre of the mass-constrained prior at finite $\eta$) that restores the robustness half of the trade-off which the adaptive prior gave up.

- *Proposition D‴.28 (the floor removal is a gauge property).* The amplitude–mass degeneracy is the positive homogeneity that the conic (Wasserstein–Fisher–Rao) view of mean-field networks exploits ([16, 23]), and rescaling symmetries of network parameters — which map critical points to critical points with the same function while changing their curvature — are classical ([98]); the implicit-function derivation of the fixed prior's sensitivity is the one of Proposition D‴.24(i). What is specific here is that the gauge, not exact representability, is the mechanism of the adaptive prior's floor removal — so it survives an approximation error, the regime of real plants, at a strict local minimum in the interior — together with the exact scaling $R_k(G_w\vartheta) = (q_k/w_k)R_k(\vartheta)$ of the residual curvature at a critical point for a scalar amplitude per kernel, which turns the adaptive prior's splitting margins into a closed formula: the scalar-amplitude instance of the reparametrisation dependence of curvature [98]. Numerics: `exp_gauge_floor.py`. Verdict: classical symmetry, a new identification of the mechanism and an exact margin formula for a scalar amplitude.

- *Proposition D‴.29 (second-order plants).* The sliding-variable (filtered-error) construction is the classical route of adaptive control to higher-order plants, and Gaussian networks with it are Sanner and Slotine's ([79]); singular-perturbation averaging with a stable fast linear subsystem is standard ([49]); the exact filter $(\mathrm{d}/\mathrm{d}t + k_s)^{-1}$ in the position-error prediction is the filtered error itself, and the weak excitation of Gaussian networks along an orbit is the classical partial persistence of excitation ([92, 93, 94]). What is specific here is the transfer of the transport leaks of Section 18 — their finite-size floors, the mass constraint, the adaptive and projected priors and the gauge — to such plants through the occupation measure of the regressor's orbit, and the exact fibre of isotropic phase-plane kernels on a circular orbit (von Mises restrictions: in general position one zero Gauss–Newton eigenvalue per kernel). The numerics are noise-free: the leaks' robustness is not exercised there, and no leak at all fits as well. Numerics: `exp_servo_cogging.py`. Verdict: a classical reduction; new applications of Section 18's mechanisms to a standard servo problem, whose case rests on the noisy setting still to be run.

- *Proposition D‴.30 (measurement noise at perfect tracking).* Measurement noise in adaptive control, the robust modifications against bounded disturbances ([47]) and the errors-in-variables bias of estimates from noisy regressors (bias compensation and instrumental variables in identification, [96]) are classical, and so is the remedy of putting the reference in the regressor (the desired compensation adaptation law, [96]); stochastic averaging is

Khas'minskii's ([96]). So are the mechanisms behind the floor and the amplification ([100]): the Lyapunov covariance as the diffusion approximation of a constant-gain algorithm (Benveniste–Métivier–Priouret), a misadjustment depending on the input's spectrum only through a trace (Widrow et al.: the $\eta = \infty$ floor's $\operatorname{tr} Q$), the leaky filter's steady state (Mayyas–Aboulnasr: the fixed prior's factor $h_j/(h_j+\sigma_0)$), a common noise not averaging out in the mean-field limit (Carmona–Delarue–Lacker: the coherence) and the drift of weakly excited directions under a persistent disturbance (Anderson: the $H^{-1}$ response). What is specific here is their combination in the transport-leak setting: the floor's interpolation $(h_j+\eta)/(h_j+\eta+\sigma_0)$ in the prior's mobility (an elementary $2\times 2$ Lyapunov computation), the driving matrix $Q = \langle(\dot{\hat{J}}+a\hat{J})^\top(\dot{\hat{J}}+a\hat{J})\rangle$ of the sensitivity-filtered measurement the law sees (built on the derivative Gram of Corollary D‴.26's filter correction), and the coherence of the $K_0$-atom identification with its noise threshold at the lift (invariance, not attraction; along the noisy trajectory the separations' drift grows with the developing errors-in-variables deviation, and the long-time threshold is open). Numerics: `exp_measurement_noise.py`. Verdict: classical ingredients; a quantified structural statement for the transport leaks under noise.

- *Proposition D‴.31 (the washout prior and the desired regressor).* Washout filters — feedback through a high-pass filter, which leaves the open loop's equilibria where they are and therefore cannot oppose a persistent input — are classical in flight control and in bifurcation control ([97]); an adaptive law kept near a low-pass filtered copy of itself is Yucelen and Haddad's low-frequency learning ([101]), which has the washout structure, and a centre moving with its workers' average is elastic averaging's ([90]); the desired compensation adaptation law, whose regressor and compensation use the reference trajectory, is Sadegh and Horowitz's ([96]; its nonlinear damping, which DR* omits, is for global stability), and its regressor's freedom from measurement noise is part of its classical motivation. That a quadratic regularisation's stationary response and its bias toward the anchor sum to the identity — $RH + B = I$, per mode the Tikhonov filter factor plus the residual factor, the Kalman-gain split of a regularised estimate — is elementary, as are the minimum-norm bound (iii) and the split of a force into the regressor's errors in variables, which bias, and the output's, of which only the mean biases. What is specific here is their transfer to the transport leaks: the reading of Theorem D‴.23's adaptive prior as a washout $\sigma$-modification of the class barycentres (for integral plans; split rows leave a non-washed-out part), uniform in $m$, and the quantification on the instance. The parameters' own errors in variables is the classical second-order bias of a constant-gain stochastic approximation from its own fluctuation (Benveniste–Métivier–Priouret; Dieuleveut–Durmus–Bach for constant-step SGD; Box's curvature bias of nonlinear least squares: [100]); what is added is its quantification here — small in absolute terms ($\le 3.9\cdot 10^{-4}$ per $(\mathbb{E}\nu^2)^2$), $-8.7\%$ to $+3.3\%$ of DR's and DR*'s biases and $+56\%$ of DR's at $k_e = B = 100$, where DR's noise part nearly cancels; its own response falling like $k_e^{-2}$ on the one-dimensional reference, where the law's leading term averages out — and the observation that the desired regressor does not remove it. Numerics: `exp_eiv_remedy.py`. Verdict: classical ingredients; a structural reading of the adaptive prior and a quantified, partial remedy for the transport leaks.

- *Proposition D‴.32 (the cohesion leak).* The leak is the mean-shift step of the parameters' own kernel density (Fukunaga–Hostetler, Cheng: mode seeking with a shadow kernel), so its mean-field form is an aggregation equation in the fit's external potential with the fit's and the cohesion's interactions — a Wasserstein gradient flow of an interaction energy (Carrillo–McCann–Villani, Bertozzi–Laurent–Rosado); particles drifting toward their own kernel-weighted local means are polarized consensus-based dynamics (Bungert–Roith–Wacker), and clustering under softmax-weighted attraction is the self-attention dynamics of Geshkovski–Letrouit–Polyanskiy–Rigollet; agents averaging within a confidence radius

form clusters at least the radius apart (Hegselmann–Krause), which is what the finite range does to the classes; a duplicated hidden unit turns a critical point into a line of critical points (Fukumizu–Amari), the twins' curves of (v); a neuron split along the top eigenvector of its residual-weighted Hessian is splitting steepest descent (Wu–Wang–Liu); a prior-free pairwise attraction whose strength trades the fit against the number of distinct parameters is convex clustering and sum-of-norms identification (Hocking–Joulin–Bach–Vert, Ohlsson–Ljung–Boyd), and a cluster splitting when an annealed parameter crosses a value set by its covariance is deterministic annealing (Rose), the structure of (iv)'s resolution and of the annealed scale; RBF units added and removed by accuracy criteria are self-organising RBF networks (Lian et al.); forgetting only in the excited directions is directional forgetting (Kulhavý), and forgetting toward a prior on the unexcited ones stabilized forgetting (Kulhavý–Zarrop); the extra modes of an isotropic Gaussian mixture need overlapping components (Carreira-Perpiñán–Williams), which (v) uses ([103]). What is specific here is the combination as a leak in mean-field adaptive control: the exact blocks at clustered configurations with the mass factor $\sigma_0 w_k$, the gauge-scaled splitting margin $\sigma_0 w_k - (q_k/w_k)\lambda_k^+$, the twins' contraction rate, and the real loop's periodic lift with the separation exponents $\sigma_0 w_k/k_e$. The number of kernels is not selected by the leak (Numerics (C), (J), (K)). Numerics: `exp_cohesion_leak.py`. Verdict: classical ingredients; a new prior-free leak on the unexcited subspace.

- *Corollary D‴.33 (the desired regressor and the coherence under noise).* That a regressor evaluated on the reference trajectory removes the measurement noise from the adaptation's regressor is part of the classical motivation of the desired compensation adaptation law (Sadegh and Horowitz, [96]). The corollary's question — whether identical subsystems driven by a common noise stay together — is that of noise-induced synchronisation: identical uncoupled oscillators driven by a common weak noise synchronise when the transverse Lyapunov exponent, to second order in the noise, is negative, and a strong noise can make it positive (Teramae–Tanaka, Goldobin–Pikovsky; with independent noise, clustering: Nakao–Arai–Kawamura; [104]). Proposition D‴.30(i)'s threshold and this corollary are that exponent for the replicated kernels, the leak their transverse stiffness, the variance term the exponent's own second-order correction. That the regressor's term leaves the separations' drift is the one-line consequence of a regressor that does not see the measurement. What is specific here: that consequence for the $K_0$-atom identification — the drift $O(1/k_e)$ like the stiffness, a threshold bounded below uniformly in $k_e$ and constant when the noise's bandwidth scales with the gain — and its quantification on the instance: the sign at the lift (negative in the centre directions of separated kernels; the whole matrix only for separated alternating-sign kernels), its loss along the trajectory as the mean error relaxes with the collective modes, and the stationary point, where the parameters' part of the force dominates the mean error. Numerics: `exp_eiv_remedy.py` Part D. Verdict: a classical remedy in a classical frame; a new structural consequence for the transport leaks (and for the cohesion leak, whose separations are the softest).

- *Proposition D‴.34 (mass transfer: the replicator weight law).* Transport and creation of mass together are the Wasserstein–Fisher–Rao (Hellinger–Kantorovich) geometry of positive measures (Liero–Mielke–Savaré); particles that move and change weight are conic particle gradient descent, with global convergence for sparse optimisation on measures under non-degeneracy (Chizat), and a non-local birth–death term added to the mean-field dynamics of a one-hidden-layer network keeps the minimiser and speeds the convergence (Rotskoff–Jelassi–Bruna–Vanden-Eijnden); the multiplicative weight update is the exponentiated-gradient law (Kivinen–Warmuth) and EM's update of mixture weights (Dempster–Laird–Rubin), adaptation laws of mirror-descent type replace the quadratic parameter error of the Lyapunov function by a Bregman divergence (Boffi–Slotine) — (ii) is their negative-entropy instance —, and the replicator equation is the gradient flow

of Shahshahani's metric with the relative entropy as its Lyapunov function at a stable state (Shahshahani, Hofbauer–Sigmund); the Wasserstein–Fisher–Rao distance has two more origins (Kondratyev–Monsaingeon–Vorotnikov, Chizat–Peyré–Schmitzer–Vialard), and gradient descent on the positions and square-root weights of a two-layer network is its particle form (Chizat–Bach) ([105]). The law here is gradient descent on $(\sqrt{\beta}, \Theta)$ — Euclidean on the positions, not the WFR metric's mass-weighted one —, with the cohesion energy of the unweighted cloud. What is specific: the replicator as the mass transfer of the cohesion leak in mean-field adaptive control — the stationary set unchanged at clustered configurations under the finite range (under the Gaussian kernel the replicator selects the masses), the saturation's bound kept without a projection, the saturation trap of the transit's masses removed, the representability criterion $\sum_k |A_k|/a_{\max} < 1$ at fixed positions for bounded trajectories, the zero modes at the lift — and the numerics (the five-bump and three-heavy-bump targets found with their kernels; bumps closer than the broad scale and the shedding not repaired). Numerics: `exp_mass_transfer.py`. Verdict: classical ingredients; a new remedy for the transit's masses.

- *Proposition D‴.35 (the split as an $S_n$-symmetric transcritical bifurcation).* Bifurcation from a fully symmetric state of $N$ identical coupled units is $S_N$-equivariant; the standard representation admits a quadratic equivariant, so the generic bifurcation is transcritical, with branches of isotropy $S_q \times S_{N-q}$ and no stable small branch — the jump of the speciation models of Stewart–Elmhirst–Cohen and Golubitsky–Stewart, the normal forms and stability of Elmhirst, the generic theory with quadratic equivariants of Arjevani–Field, who motivate it by the permutation-symmetric critical points of two-layer networks; duplicated units give such critical points and saddles (Fukumizu–Amari [103], Brea et al., Şimşek et al.) ([106]); a cluster splits at a critical parameter set by its covariance in deterministic annealing (Rose, [103]). Order preservation is a consequence of the equivariance, and the escape time is the Riccati solution. What is specific here: the identification for the cohesion leak — the quadratic coefficient the residual's third derivative along the splitting direction, the cohesion and the mean term entering only at third order or in the common direction —, the margin after a departure (the stiffness's drop and the curvature's gauge rise), and the instance's quantification (the extreme particle leaving first, alone or with a near-tied neighbour; the mass-free leak delaying the cascade without stopping it). Numerics: `exp_split_bifurcation.py`. Verdict: classical bifurcation theory; a new identification for the cohesion leak's over-splitting.

- *Proposition D‴.36 (the gauge under dither).* A diffusion pushed onto a manifold of equilibria by a large drift converges, on the slow time scale, to a diffusion on the manifold whose noise is the original projected by the derivative of the flow's limit map (Katzenberger [107]); the noise-driven motion of SGD along the manifold of minimisers of an overparameterised model is the same limit (Blanc et al.; Li–Wang–Arora [107]), and the constructive projection is Parsons–Rogers'. The entropic leak on the weights is the negative-entropy mirror map's regulariser ([105]). What is specific here: the gauge of the amplitude–mass representation (Proposition D‴.28) as such a manifold for the replicator of Proposition D‴.34; the projection computed in the flow's own metric (Euclidean–$G^{-1}$ on the positions, Fisher–Rao on the masses), with the two-class closed form; the excursion into the saturation, where the limit map's derivative degenerates; and the entropic leak on the masses as a leak on the unexcited subspace, which pins the gauge without bias where the particle fractions are representable. Numerics: `exp_gauge_dither.py`. Verdict: a classical limit theorem applied; the replicator's unpinned masses under dither, and their remedy.

- *Proposition D‴.37 (the dither's drift along the gauge).* The second derivative of the limit map and the drift it gives a diffusion forced onto a manifold of equilibria are Katzenberger's; for isotropic noise the drift along a manifold of minima is $-\frac{1}{2}$ of the gradient of the

log-pseudo-determinant of the Hessian (Li–Wang–Arora's Corollary 5.1, [107]; checked by round 27's referee on its summary only); the averaging of a slow variable over a fast ergodic one is Khasminskii's, and a slow coordinate in equilibrium with fast ones feels their free energy, the potential of mean force ([108]); that noisy gradient dynamics prefers flat minima is the flat-minima picture of Hochreiter–Schmidhuber, and Entropy-SGD makes the local entropy the objective ([108]). Not seen, named by the referee: Fixman's and Bornemann–Schütte's corrections for stiff constraints, the label-noise analyses of Damian–Ma–Lee and HaoChen et al., noisy replicator dynamics (Fudenberg–Harris, Foster–Young), Rose's deterministic annealing. What is specific here: the replicator's masses as the slow variable on the amplitude–mass gauge, with noise on the positions only, and its free energy in closed form ($2\sum \log(W_k a'_k)$); the consequence that dither drives the masses into the saturation — the trap the replicator removed (Proposition D‴.34) returning through the noise —; the boundary layer of width $O(\sqrt{\gamma})$, where the replicator outruns the saturating amplitude, which slows but does not stop it; and the entropic leak's bias $\kappa^{-1}b(w)$. Numerics: `exp_gauge_drift.py`. Verdict: classical limits (Katzenberger, Li–Wang–Arora, averaging) applied; a new mechanism for the replicator's gauge under dither.

- *Proposition D‴.38 (a Fixman correction for the replicator under dither).* That a stiff system's free energy along its soft directions carries a log-determinant, and that adding the opposite term makes the constrained dynamics sample as intended, is Fixman's (the log-det term the stiff limit's zero-point term, Helfand; disputed in the overdamped setting by Waszkiewicz–Lisicki); the drift corrections between stiff bonds and rigid constraints are Hinch's, and the projected diffusions and the mean force on submanifolds are Ciccotti–Lelièvre–Vanden-Eijnden's ([109]); removing a free-energy drift online is the adaptive biasing force method, a gain on it well-tempered metadynamics' bias factor, and the max-det maximiser of a log-barrier is classical in convex optimisation ([109]); the log-determinant drift itself is Li–Wang–Arora's for isotropic noise ([107]) and Proposition D‴.37's for the replicator. What is specific here: the correction for the replicator's gauge, applied online to the masses only, in closed form on the masses — a log-barrier on the representable masses, the corrected loop a gradient flow — and the distinction between that masses form and the form at the current amplitudes, which reads the positions' escapes off the manifold as masses at their saturation; the selection of the D-optimal exact fit $W^\sharp$ with the gain and the confinement exponent $\delta^{\kappa_F-1}$ are applications; the leak's pinned state predicted to second order (the fluctuations through the leak's curvature, which the paired seeds resolve). Numerics: `exp_gauge_fixman.py`. Verdict: a classical correction carried to a new setting, with a new design knob.

- *Proposition D‴.39 (dissipativity of the dithered gauge).* That a Langevin dynamics needs a confining potential to have an invariant probability law is classical (Mattingly–Stuart–Higham; in learning, Raginsky–Rakhlin–Telgarsky's dissipativity), as are the exit times' exponential order (Freidlin–Wentzell; sharp, Bovier–Eckhoff–Gayrard–Klein) and parameter projection with its Lipschitz boundary layer (Ioannou–Sun, Pomet–Praly), and the dead zone itself is the Moreau–Yosida envelope of the box's indicator, with which MYULA samples on a convex set (Durmus–Moulines–Pereyra; Brosse et al.) — a switching $\sigma$-modification in adaptive control ([110]). What is specific here: the cohesion leak's removal of the prior (Proposition D‴.32) removes with it the dissipativity the prior leak gave the dithered loop; on the instance the dither's way out of the exact fit's well is a monotone climb, without a saddle, to a plateau of approximate fits with saturated amplitudes; the dead zone on the unconstrained coordinates replaces the gauge's saturation by a non-degenerate wall, so the entropic drift of Proposition D‴.37 ends at the vertices of the representable polytope (at leading order: the envelope is $C^{1,1}$, so the drift's description has a boundary layer at the faces), and keeps Proposition D‴.38's selection of $W^\sharp$; it leaves the deterministic objects inside the box unchanged, not the stationary averages, which weigh the plateau where it

meets the box. Numerics: `exp_gauge_plateau.py`. Verdict: classical ingredients; the loss of dissipativity under the cohesion leak and its consequences for the gauge are new.

- *Proposition D‴.41 (measurement noise on the gauge).* That isotropic or label noise near a manifold of minimisers produces a drift along it, computed by Katzenberger's theorem, is Li–Wang–Arora's ([107]), the isotropic noise's log-determinant drift among it; that label noise drawn afresh at each step selects flat minima (the trace of the Gauss–Newton matrix), with a strength set by step × variance and, at a finite step, the regulariser $-(1/2\eta)\,\mathrm{tr}\log(I-\eta\nabla^2 L/2)$, is Blanc–Gupta–Valiant–Valiant's and Damian–Ma–Lee's, the noise covariance's shape mattering HaoChen–Wei–Lee–Ma's, the singular limits of noise injection Shalova–Schlichting–Peletier's; that the white limit of physical noise is Stratonovich's, that Itô's and Stratonovich's drifts differ by the noise-induced term, and that a coloured noise's limit is Itô's, Stratonovich's or neither according to its correlation time against the relaxation time, is classical (Wong–Zakai, van Kampen, Horsthemke–Lefever, Kupferman–Pavliotis–Stuart, Pavliotis–Stuart); the continuum filter $1/(1+k\tau)$ is the active Ornstein–Uhlenbeck particle's effective temperature and $\det(I+\tau\nabla^2 U)$ the unified coloured-noise approximation's factor (Jung–Hänggi, Maggi et al.); SGD's SDE models are Itô's (Li–Tai–E, Li–Malladi–Arora); correlated and anti-correlated noise act on the Hessian's trace, a weighted trace or the weight variances (Orvieto et al., Lucchi et al., Beneventano, Kühn–Rosenow, and an abstract by Kim et al. with a correlation-time trade-off); the examples' Itô laws are Ziyin–Li–Ueda's law of balance and Ziyin–Wang–Li–Wu's noise equilibria, the diagonal network's label-noise Lasso Pillaud-Vivien–Reygner–Flammarion's ([111]). What is specific here: for noise entering through the error, the identity that the Wong–Zakai term cancels Katzenberger's Itô drift exactly on the manifold — from the antisymmetry the limit map's invariance imposes on $\langle \nabla r \otimes \nabla(\partial\Phi\cdot U)\rangle$ — and its extension to smooth noise of any temporal spectrum; the coloured filter of label noise's drift along $M$ — the increment's second moment, the cross term $\langle \nabla\psi\cdot n\xi\rangle$ cancelling by the same antisymmetry —, which contains Damian–Ma–Lee's finite-step regulariser as its white case and passes to $(1/\tau_c)\log\det(I+\tau_c\nabla^2 J)$ for a slow Markov noise, a concave potential for every colour, a slow Markov noise acting as a dither of the flow's own mobility; the closed forms on the replicator's masses, with their coordinate dependence, and the coloured selection's multistability there. We did not find the cancellation or the coloured filter of the drift along $M$ stated. Numerics: `exp_gauge_label.py`. Verdict: classical ingredients, a new exact identity with a practical consequence (continuous-time adaptation does not select through its measurement noise) and a new law for correlated label noise (its colour sets the implicit regulariser, concave and multistable when coloured).

- *Proposition D‴.42 (several particles per class).* The change to a cluster's barycentre and internal (Jacobi) coordinates, with unit Jacobian, the block structure of a translation-invariant interaction, and the non-Gibbs stationary law of a linear system driven by a noise not proportional to its mobility are classical; that the positions' Hessian at an exact fit factors into the barycentres' fit part and the internal cohesion part follows. What is specific here: the consequence for the replicator's gauge — under the isotropic dither the log-determinant selection and its Fixman correction act on the class masses exactly as with one particle per class, with $m$ the number of particles, and the within-class fractions neither drift at Katzenberger's order nor diffuse at any $\gamma$; an anisotropic dither makes them drift at $O(\gamma\tau)$ and moves the class masses off the per-class law at unequal splits; under measurement noise the equal split is an invariant subspace carrying the one-per-class loop, and the cohesion breaks the cancellation elsewhere. Numerics: `exp_gauge_classes.py`. Verdict: a robustness result closing an open item, with its limits.

- *Proposition D‴.43 (the colour of the measurement noise selects the kernels).* That label noise drawn afresh selects a weighted $\ell^1$ norm on quadratic parametrisations — and so

recovers sparse ground truths under $\ell^1$'s conditions —, and at a finite $\sigma$ a Lasso, is HaoChen–Wei–Lee–Ma's and Pillaud-Vivien–Reygner–Flammarion's; that a finite step turns the selected trace into a strictly convex regulariser, Damian–Ma–Lee's; that the hyperbolic entropy's interpolant interpolates between the kernel and the rich regimes, Woodworth et al.'s; that sparse Bayesian learning's cost $\log\det(\lambda I+\Phi\Gamma\Phi^\top)+y^\top(\lambda I+\Phi\Gamma\Phi^\top)^{-1}y$ keeps its global minimum at the sparsest solution with few local minima, Tipping's and Wipf–Rao's, and that its non-separable reweighting succeeds where $\ell^1$ fails on a set of positive measure, Wipf–Nagarajan's; that log det is a concave surrogate of rank minimised by reweighted trace steps, Fazel–Hindi–Boyd's, and reweighted $\ell^1$ Candès–Wakin–Boyd's; that gradient noise attracts sparse invariant sets, Chen et al.'s; that Fisher–Rao and Wasserstein–Fisher–Rao particle flows solve sparse problems on measures, Chizat's; the dual certificate for $\ell^1$, Fuchs' and Tropp's ([111], [112], [113]). What is specific here: that the temporal correlation of the label noise sets the sparsity prior of the explicit (Euler) update's selection along the exact fits — the trace for white noise, Fazel's log-det heuristic on $\Phi\,\mathrm{diag}|w|\Phi^\top$ for a slow Markov noise (sparse Bayesian learning's log-det term at $\gamma\propto|w|$ and $\lambda=1/\tau_c$, not its penalty: Wipf–Rao's results on its local minima do not transfer), concave for every colour in the continuum — through Proposition D‴.41's filter; the stability law of sparse fits under it (a weighted certificate, sufficient, $\propto 1/\tau_c$ beyond a correlation time that grows with the dictionary's coherence); and that this selection recovers sparse kernel expansions where basis pursuit fails — the reduced flow, and label-noise descent at small $\sigma$ —, as Wipf–Nagarajan's theorem leads one to expect, at rates sparse Bayesian learning itself matches or beats. We did not find it stated. Numerics: `exp_colour_selection.py`. Verdict: a new mechanism — the noise's spectrum as the sparsity prior of a sampled-data adaptation, at a rate $\propto h\sigma^2$ (none in continuous time; Heun's scheme removes most of it) —, not a better sparse solver; for self-adaptive kernel networks with an explicit update a slow measurement noise prunes the dictionary, slowly (millions of updates here).

- *Proposition D‴.45 (the free mass–amplitude law as a mirror flow).* In Proposition D‴.44's flat coordinates the free law is the Euclidean gradient flow of a separable $2\to1$ reparametrisation, so its being a mirror flow with a start-dependent potential is Li–Wang–Lee–Arora's theorem, with one conserved function per unit (Marcotte–Gribonval–Peyré; Azulay et al.'s $u\odot v$); the kernel and rich regimes and the $\ell^2$-to-entropy shape (Woodworth et al.; Ghai–Hazan–Singer), exponentiated gradient as mirror descent (Kivinen–Warmuth; Amid–Warmuth), the invariance of stochastic mirror descent's limit under noise in the dual span (Gunasekar et al.; Azizan–Hassibi), SGD's Itô noise shrinking the effective initialisation (Pesme–Pillaud-Vivien–Flammarion; Even et al.) and isotropic label noise not selecting (HaoChen–Wei–Lee–Ma), and Bregman Lyapunov functions for mirror-descent adaptation laws (Boffi–Slotine) are classical ([113]). What is specific here: the paradigm's saturating unit made explicit — the conserved $Q$, the mirror coordinate in every regime, the conjugate as the mass gained, the saturation's soft wall and entropy tail, the rich limit's weights $1/\sqrt{\beta^0}$ (a power law in the starting mass) with their concave correction, the branches —; the Euler update's per-step change of $Q$ identified with the conserved share of Proposition D‴.44's selection, and the mirror update that removes it. Numerics: `exp_mirror_free.py`. Verdict: a classical mechanism made explicit for the paradigm's law, with design consequences — the starting masses as a sparsity prior (paid for in time), the integrator as a switch for the noise's selection.

- *Proposition D‴.46 (a leak in the mirror coordinates).* A leak in the dual coordinates of a continuous-time mirror flow, its regulariser-perturbed rest point, its Bregman (Fenchel-coupling) Lyapunov function and the vanishing-leak selection are established (Gao–Pavel; Romano–Pavel; Coucheney–Gaujal–Mertikopoulos for the replicator; Pérolat et al.; Abe et al.; Alvarez–Bolte–Brahic), as are the $\sigma$-modification's ultimate bound (Ioannou–Kokotović; Ioannou–Sun), the averaging of adaptive systems with the filtered regressor (Anderson et

al.; Bodson et al.; Sastry–Bodson), the LASSO and its irrepresentable condition (Tibshirani; Zhao–Yu; Wainwright) and mirror-descent adaptation laws (Boffi–Slotine) ([114]). What is specific here: the leak realised on the paradigm's $(\alpha, \beta)$ law with the conservation law kept, its ultimate bound in the adaptive loop with this law, the averaged equilibrium with the error filter's Gram as a monotone equation for this law, and the starting masses setting the regime — a ridge, or a LASSO with penalty $\sigma\lambda_1 \propto \sigma/\sqrt{\beta^0}$ that keeps the target's kernels under the irrepresentable condition. Numerics: `exp_mirror_leak.py`. Verdict: classical ingredients assembled into a robust version of the paradigm's mass law, whose small-start limit selects the target's kernels under the (weighted) irrepresentable condition.

- *Proposition D‴.47 (the free law's branches).* Exponentiated gradient and its Bregman-projection bias (Kivinen–Warmuth; Gunasekar et al.), Hessian–Riemannian gradient flows on constrained domains (Alvarez–Bolte–Brahic), nonnegative least squares by over-parametrised gradient flow (Chou–Maly–Verdun et al.), the hyperbolic entropy of $u^2 - v^2$ (Woodworth et al.; Ghai–Hazan–Singer), the uniqueness of nonnegative sparse solutions and nonnegative super-resolution without separation (Donoho–Tanner; Bruckstein–Elad–Zibulevsky; Schiebinger–Robeva–Recht; Eftekhari et al.) and the exponential penalty in linear programming (Cominetti–San Martín; Weed) are classical ([115]). What is specific here: the paradigm's own law becomes an exponentiated gradient on its branches with a temperature read off the conservation law, so that small starts freeze signs; the limit for every start — mixed free and branch particles, inexact fits — as the Bregman projection onto the sign-constrained least-squares fits with the extinction of the vanishing coordinates (completing D‴.45(iii)); $B(0, w) = \beta/(\gamma m)$; the two rich limits' temperatures; and the scale-independent selection of a positive target's kernels. Numerics: `exp_mirror_branch.py`. Verdict: classical mechanisms identified in the paradigm's law, closing its implicit-bias picture for every start.

- *Proposition D‴.48 (the rich transit).* The inverse scale space flow and its adaptive algorithm (Burger–Gilboa–Osher–Xu; Burger–Möller–Benning–Osher), saddle-to-saddle dynamics of diagonal linear networks at vanishing initialisation and their identification with the adaptive inverse scale space method (Pesme–Flammarion), incremental learning (Berthier; Gissin–Shalev-Shwartz–Daniely), inverse scale space limits of mirror flows by Mosco convergence (Berthier–Pillaud-Vivien, 2026), depth-$D$ potentials and polynomial scales (Woodworth et al.; Jacot et al.; Chou–Maly–Rauhut), sliding Frank–Wolfe and ADCG, conic particle gradient descent, condensation and early alignment at small initialisation, and alternating gradient flows are known ([116]). What is specific here: the paradigm's free mass–amplitude unit as an effective depth-3 unit with an explicit potential, so that its transit has the power scale $(\text{mass})^{-1/2}$ from zero amplitudes and the logarithmic one on its branches — a continuum in between, set by $\alpha(0)/(\text{mass})^{1/2}$ —; the sliding transit with the paradigm's moving centres (the inactive kernels frozen by the mass in the position law's drift, the creep's profile and the margin condition $\varepsilon \ll \mu^3$, ties shared); and the positions' metric deciding where the massless kernels go (frozen, or condensed at the correlation's peaks) and how small the masses must be for one kernel per feature — not the limit's count, which is one per feature under both (round 37 refuted the dichotomy of counts). Unbalanced initialisations (Kunin et al. 2024), deep diagonal networks as mirror flows (Labarrière et al. 2024) and feature quantisation at small initialisation (Maennel–Bousquet–Gelly 2018) are related. Numerics: `exp_mirror_transit.py`. Verdict: the limit theorem Berthier–Pillaud-Vivien's route for an $\varepsilon$-dependent family, (ii) their entropic case; the sliding transit, as a gradient-flow limit, apparently new (formal).

*Proof. Step 0 (a priori bounds).* For either system, $\frac{\mathrm{d}}{\mathrm{d}s}\|\vartheta - \vartheta^0\|^2_{G^{-1}} = 2\langle \vartheta - \vartheta^0,\, \langle \nabla_\vartheta (r^\top \tilde{\Phi}(\cdot\,;\vartheta))\rangle_\rho \rangle - 2\sigma_0 \|\vartheta - \vartheta^0\|^2_{G^{-1}}$ with $|\langle \vartheta - \vartheta^0,\, \langle \nabla_\vartheta (r^\top \tilde{\Phi})\rangle_\rho \rangle| = |\langle r^\top \nabla_\vartheta \tilde{\Phi}(\vartheta - \vartheta^0)\rangle_\rho| \le \langle \|r\| \rangle_\rho L^G_\vartheta \|\vartheta - \vartheta^0\|_{G^{-1}} \le (F_\rho + K_\Phi) L^G_\vartheta \|\vartheta - \vartheta^0\|_{G^{-1}}$, so $D^+ \|\vartheta - \vartheta^0\|_{G^{-1}} \le L^G_\vartheta (F_\rho + K_\Phi) - \sigma_0 \|\vartheta - \vartheta^0\|_{G^{-1}}$ and $\|\vartheta(s) - \vartheta^0\|_{G^{-1}} \le R_\Theta$;

hence $\|\delta_i\|_{G^{-1}} \le 2R_\Theta$, $D \le D_{\max}$, and $\|\,\mathrm{d}\vartheta/\,\mathrm{d}s\|_{G^{-1}} = \|G^{1/2}\langle\nabla_\vartheta(r^\top\tilde\Phi)\rangle_\rho - \sigma_0 G^{-1/2}(\vartheta-\vartheta^0)\| \le V_{\max}$. Two Lipschitz facts: for $\Theta$, $\bar\Theta$ with rows $\vartheta_j$, $\bar\vartheta_j$ and $A_j(x) := \int_0^1 \nabla_\vartheta\tilde\Phi(x;\bar\vartheta_j + t\delta_j)\,\mathrm{d}t$,

$$\hat f_\Theta - \hat f_{\bar\Theta} = \frac{1}{m}\sum_j A_j\delta_j =: h, \qquad \|h\|_{L^2(\rho)} \le L_\vartheta^G \frac{1}{m}\sum_j \|\delta_j\|_{G^{-1}} \le L_\vartheta^G D^{1/2}, \tag{0.1}$$

$$\begin{aligned} &\|R^G[\mu_{\bar\Theta}](\vartheta) - R^G[\mu_s](\vartheta)\| \le L_2^G \|r_{\bar\Theta} - r_{\mu_s}\|_{L^1(\rho)} \le L_2^G Y, \\ &\|(\nabla_\vartheta\tilde\Phi(x;\vartheta_i) - A_i(x))v\| \le \frac{L_2^G}{2}\|\delta_i\|_{G^{-1}}\|v\|_{G^{-1}}. \end{aligned} \tag{0.2}$$

*Step 1 (the identity).* Since $\mathrm{d}\delta_i/\,\mathrm{d}s = -G[\nabla\mathcal{V}[\mu_\Theta](\vartheta_i) - \nabla\mathcal{V}[\mu_s](\bar\vartheta_i)]$ and $\langle\delta,\, Gv\rangle_{G^{-1}} = \langle\delta,\, v\rangle$,

$$\begin{aligned} \tfrac12\frac{\mathrm{d}D}{\mathrm{d}s} &= -\frac1m\sum_i \langle\delta_i,\, \nabla\mathcal{V}[\mu_\Theta](\vartheta_i) - \nabla\mathcal{V}[\mu_s](\bar\vartheta_i)\rangle = T_A + T_B + T_C, \\ T_A &:= -\frac1m\sum_i \langle\delta_i,\, \nabla\mathcal{V}[\mu_{\bar\Theta}](\vartheta_i) - \nabla\mathcal{V}[\mu_{\bar\Theta}](\bar\vartheta_i)\rangle, \\ T_B &:= -\frac1m\sum_i \langle\delta_i,\, \nabla V_L[\mu_\Theta](\vartheta_i) - \nabla V_L[\mu_{\bar\Theta}](\vartheta_i)\rangle, \\ T_C &:= -\frac1m\sum_i \langle\delta_i,\, \nabla V_L[\mu_{\bar\Theta}](\bar\vartheta_i) - \nabla V_L[\mu_s](\bar\vartheta_i)\rangle \end{aligned}$$

(the leak is linear and sits in $T_A$; the three brackets telescope to $\nabla\mathcal{V}[\mu_\Theta](\vartheta_i) - \nabla\mathcal{V}[\mu_s](\bar\vartheta_i)$). The self term $T_A$ compares the *reference's* potential at the two points; the interaction term $T_B$ compares the two empirical measures at the $m$-system's point; $T_C$ is the reference's sampling error. `chaos_learning.AveragedSlowLoop.decomposition` evaluates the three terms and checks the identity to rounding.

*Step 2 (the self term is local convexity).* $\nabla\mathcal{V}[\mu_{\bar\Theta}](\vartheta_i) - \nabla\mathcal{V}[\mu_{\bar\Theta}](\bar\vartheta_i) = \int_0^1 \nabla^2\mathcal{V}[\mu_{\bar\Theta}](\bar\vartheta_i + t\delta_i)\,\mathrm{d}t\cdot\delta_i = \sigma_0 G^{-1}\delta_i - \bar R_i\delta_i$ with $\bar R_i := \int_0^1 R[\mu_{\bar\Theta}](\bar\vartheta_i + t\delta_i)\,\mathrm{d}t$, so $T_A = -\sigma_0 D + \frac1m\sum_i\langle\delta_i,\, \bar R_i\delta_i\rangle$ and $\langle\delta,\, R\delta\rangle \le \lambda_{\max}(R^G)\|\delta\|_{G^{-1}}^2$. If $\|\delta_i\|_{G^{-1}} \le \varrho'$ the segment lies in $N_{\varrho'}(\operatorname{supp}\mu_s)$ ($\bar\vartheta_i \in \operatorname{supp}\mu_s$), and by (0.2) $\lambda_{\max}(R^G[\mu_{\bar\Theta}]) \le \kappa_R[\mu_s](N_{\varrho'}(\operatorname{supp}\mu_s)) + L_2^G Y$ there. Hence, as long as $\max_i\|\delta_i\|_{G^{-1}} \le \varrho'$,

$$T_A \le -\big(a(s) - L_2^G Y(s)\big)D. \tag{2.1}$$

*Step 3 (the interaction term is a negative square at linear order).*

$$\nabla V_L[\mu_\Theta](\vartheta_i) - \nabla V_L[\mu_{\bar\Theta}](\vartheta_i) = \langle\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_i)^\top(\hat f_\Theta - \hat f_{\bar\Theta})\rangle_\rho = \langle\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_i)^\top h\rangle_\rho,$$

so, with the $L^2(\rho;\mathbb{R}^n)$ inner product $\langle\langle\cdot,\cdot\rangle\rangle$,

$$\begin{aligned} T_B &= -\Big\langle\Big\langle \frac1m\sum_i \nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_i)\delta_i,\, h\Big\rangle\Big\rangle = -\|h\|^2 - \Big\langle\Big\langle \frac1m\sum_i(\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_i) - A_i)\delta_i,\, h\Big\rangle\Big\rangle \\ &\le -\|h\|^2 + \|h\|\frac{L_2^G}{2}D \le \frac{(L_2^G)^2D^2}{16} \end{aligned} \tag{3.1}$$

by (0.2) and $\max_x(-x^2+cx) = c^2/4$. The leading part $-\|h\|^2 = -\|\hat f_\Theta - \hat f_{\bar\Theta}\|_{L^2(\rho)}^2$ is the flat convexity of $\bar L$: the interaction contracts, and only the Taylor remainder of the features, quadratic in $D$, is charged. This is the step that Theorem D″'s small-gain bound replaces by $+L_\vartheta^2 D$-type terms.

*Step 4 (the sampling term).* $\nabla V_L[\mu_{\bar\Theta}](\bar\vartheta_i) - \nabla V_L[\mu_s](\bar\vartheta_i) = \langle\nabla_\vartheta\tilde\Phi(\cdot\,;\bar\vartheta_i)^\top(\hat f_{\bar\Theta} - \hat f[\mu_s])\rangle_\rho$, so $|\langle\delta_i,\,\cdot\rangle| \le L_\vartheta^G\|\delta_i\|_{G^{-1}}Y$ and

$$T_C \le L_\vartheta^G D^{1/2} Y. \tag{4.1}$$

*Step 5 (the differential inequalities).* While $\max_i\|\delta_i\|_{G^{-1}} \le \varrho'$,

$$\tfrac{1}{2}\frac{\mathrm{d}D}{\mathrm{d}s} \le -(a - L_2^G Y)D + \frac{(L_2^G)^2 D^2}{16} + L_\vartheta^G D^{1/2} Y, \tag{5.1}$$

and per particle, from the same three brackets with $|T_{B,i}| \le \|\delta_i\| L_\vartheta^G \|h\| \le \|\delta_i\|(L_\vartheta^G)^2 D^{1/2}$ by (0.1) and $|T_{C,i}| \le \|\delta_i\| L_\vartheta^G Y$,

$$D^+\|\delta_i\|_{G^{-1}} \le -(a - L_2^G Y)\|\delta_i\|_{G^{-1}} + (L_\vartheta^G)^2 D^{1/2} + L_\vartheta^G Y. \tag{5.2}$$

*Step 6 (the bootstrap on the good event).* Fix a horizon $S \le \infty$, let $\mathcal{G}_S := \{\sup_{s\le S} Y(s) \le \theta\}$ and $T^* := \inf\{s \le S : D(s) \ge \varrho^2 \text{ or } \max_i\|\delta_i(s)\|_{G^{-1}} \ge \varrho'\}$ ($T^* > 0$ since $\delta(0) = 0$; everything below is on $[0, S]$). On $\mathcal{G}_S$ and $[0, T^*)$: $L_2^G Y \le \kappa/8$, $(L_2^G)^2 D/16 \le (L_2^G)^2\varrho^2/16 \le \kappa/8$, and $L_\vartheta^G D^{1/2} Y \le (\kappa/8)D + (2(L_\vartheta^G)^2/\kappa)Y^2$ (Young), so (5.1) gives $\mathrm{d}D/\mathrm{d}s \le -2(a - 3\kappa/8)D + (4(L_\vartheta^G)^2/\kappa)Y^2$. Since $a - 3\kappa/8 \ge \kappa/2 - (\kappa - a)_+$, $\int_r^s (a - 3\kappa/8) \ge (\kappa/2)(s - r) - \Lambda_1$ for $r \le s$, and Grönwall gives

$$\begin{aligned} D(s) \le Z(s) &:= 4e^{2\Lambda_1}(L_\vartheta^G)^2\kappa^{-1}\int_0^s e^{-\kappa(s-r)}Y(r)^2\,\mathrm{d}r \\ &\le 4e^{2\Lambda_1}(L_\vartheta^G)^2\kappa^{-2}\theta^2 \le \varrho^2/4 \qquad \text{on } \mathcal{G}_S \cap [0, T^*) \end{aligned} \tag{6.1}$$

by the second constraint on $\theta$. Per particle, (5.2) with $D^{1/2} \le \varrho/2$ and $Y \le \theta$, and $\int_r^s (a - \kappa/8) \ge (\kappa/2)(s - r) - \Lambda_1$, gives $\|\delta_i(s)\|_{G^{-1}} \le 2e^{\Lambda_1}\kappa^{-1}[(L_\vartheta^G)^2\varrho/2 + L_\vartheta^G\theta] \le \varrho'/4 + \varrho'/2 = 3\varrho'/4$ by the second constraint on $\varrho$ and the third on $\theta$. Both thresholds are therefore strict on $[0, T^*)$, so $T^* = S$ on $\mathcal{G}_S$ by continuity: $D \le Z$ and $\max_i\|\delta_i\| \le 3\varrho'/4$ on $[0, S]$ on $\mathcal{G}_S$. $Z$ does not involve the $m$-system.

*Step 7 (expectation).* For $s \le S$,

$$\begin{aligned} \mathbb{E}D(s) \le \mathbb{E}[Z(s)] + D_{\max}\mathbb{P}(\mathcal{G}_S^c) &\le 4e^{2\Lambda_1}(L_\vartheta^G)^2\kappa^{-1}\int_0^s e^{-\kappa(s-r)}\mathbb{E}Y(r)^2\,\mathrm{d}r + D_{\max}\mathbb{P}(\mathcal{G}_S^c) \\ &\le C_1/m + 4R_\Theta^2\mathbb{P}(\mathcal{G}_S^c), \end{aligned}$$

using $\mathbb{E}Y(r)^2 \le K_\Phi^2/m$.

*Step 8 (the good event).* $Y(s)$ is a function of the i.i.d. sample $(\vartheta_j(0))_{j\le m}$: $Y(s) = \|\frac{1}{m}\sum_j[\tilde\Phi(\cdot\,; X_s(\vartheta_j(0))) - \int\tilde\Phi(\cdot\,; X_s(\vartheta))\,\mathrm{d}\mu_0]\|_{L^2(\rho)}$. Changing one $\vartheta_j(0)$ changes it by at most $2K_\Phi/m$ (bounded differences), and $\mathbb{E}Y(s) \le (\mathbb{E}Y^2)^{1/2} \le K_\Phi/m^{1/2}$; McDiarmid's inequality [61] gives $\mathbb{P}(Y(s) \ge K_\Phi/m^{1/2} + t) \le \exp(-mt^2/(2K_\Phi^2))$, so for $m \ge m_0 = 16K_\Phi^2/\theta^2$ ($K_\Phi/m^{1/2} \le \theta/4$, $t = \theta/4$)

$$\mathbb{P}(Y(s) \ge \theta/2) \le \exp\bigl(-m\theta^2/(32K_\Phi^2)\bigr) \qquad \text{for every fixed } s. \tag{8.1}$$

In time, $|Y(s) - Y(s')| \le 2L_\vartheta^G V_{\max}|s - s'|$ (each $X_s(\vartheta)$ moves at speed $\le V_{\max}$ and $\tilde\Phi$ is $L_\vartheta^G$-Lipschitz, for the sample and for the mean), so on a grid of step $\Delta = \theta/(4L_\vartheta^G V_{\max})$, $Y \le \theta$ on $[s_k, s_k + \Delta]$ whenever $Y(s_k) \le \theta/2$. (a) The grid $\{k\Delta : k\Delta \le S\} \cup \{S\}$ has at most $S/\Delta + 2$ points, and a union bound with (8.1) gives $\mathbb{P}(\mathcal{G}_S^c) = \mathbb{P}(\sup_{s\le S} Y > \theta) \le (S/\Delta + 2)\exp(-m\theta^2/(32K_\Phi^2))$; with Step 7 this is (a). For $S \le \Delta e^{m\theta^2/(64K_\Phi^2)}$, $(S/\Delta + 2)e^{-m\theta^2/(32K_\Phi^2)} \le 3e^{-m\theta^2/(64K_\Phi^2)} \le 3/m$ when $e^{m\theta^2/(64K_\Phi^2)} \ge m$. (b) Under (H′), for $s \ge R_{\text{eff}}$ every $X_s(\vartheta)$ is within $\theta/(8L_\vartheta^G)$ of $X_\infty(\vartheta)$ on $\operatorname{supp}\mu_0$, so $Y(s) \le Y(\infty) + \theta/4$ with $Y(\infty) := \|\frac{1}{m}\sum_j[\tilde\Phi(\cdot\,; X_\infty(\vartheta_j(0))) - \int\tilde\Phi(\cdot\,; X_\infty)\,\mathrm{d}\mu_0]\|$, another bounded-differences statistic with $\mathbb{E}Y(\infty) \le K_\Phi/m^{1/2}$; the grid on $[0, R_{\text{eff}}]$ plus the point at infinity has at most $R_{\text{eff}}/\Delta + 3$ points, and $\mathbb{P}(\mathcal{G}_\infty^c) = \mathbb{P}(\sup_{s\ge 0} Y > \theta) \le (R_{\text{eff}}/\Delta + 3)\exp(-m\theta^2/(32K_\Phi^2))$.

*Step 9 (the learned function and the measures).* $\|\hat f_\Theta - \hat f[\mu_s]\| \le \|h\| + Y \le L_\vartheta^G D^{1/2} + Y$ by (0.1), so $\mathbb{E}\|\cdot\|^2 \le 2(L_\vartheta^G)^2\mathbb{E}D + 2K_\Phi^2/m$. The coupling $(\vartheta_i, \bar\vartheta_i)$ is a coupling of $\mu_\Theta$ and $\mu_{\bar\Theta}$, so $W_2^2(\mu_\Theta, \mu_{\bar\Theta}) \le D$ ($G^{-1}$ metric), and the triangle inequality gives (c). The per-particle statement is Step 6. □

**Remark D‴.1** (what the hypothesis says, and the window)**.** (H) at $s = \infty$ reads: on a neighbourhood of every atom of the limit measure, $G^{1/2}\nabla^2\mathcal{V}[\mu_\infty]G^{1/2} = \sigma_0 I - R^G[\mu_\infty] \succeq \kappa I$ — each atom is a non-degenerate local minimiser of the first-variation potential $\mathcal{V}[\mu_\infty]$, with margin $\kappa$: the strict form of the second-order optimality condition for measures (the local Chizat–Bach condition, as in Proposition K(ii)). Its non-strict form, $\sigma_0 G^{-1} - R[\mu_\infty](\vartheta^*) \succeq 0$ at every atom $\vartheta^*$ of positive mass, is necessary for an atomic stationary measure to be a local minimiser of $\mathcal{G}$ in the Wasserstein sense: splitting an atom of mass $w$ along a direction of negative curvature of $\mathcal{V}[\mu_\infty]$ lowers $\mathcal{G}$ at second order (Proposition K(i)–(ii) with the atom's own residual). Along the flow, (H) asks for this with the *current* residual $r_{\mu_s}$ from the time $s_1$ on. It does not mention the collapsed configuration: $\sigma_{\mathrm{crit}} = \kappa_R[\delta_{\bar\vartheta}](\{\bar\vartheta\})$ is the same curvature *with the collapsed residual*, which is the residual of the best single kernel and is large when the plant needs several. The learning regime $\sigma_0 < \sigma_{\mathrm{crit}}$ (the kernels separate) and the uniform regime (H) ($\kappa_R[\mu_\infty] \le \sigma_0 - \kappa$ at the fitted configuration) therefore coexist exactly when the fitted configuration is non-degenerately stable while the collapsed one is not — the window

$$\kappa_R[\mu_\infty]\big(N_{\varrho'}(\operatorname{supp}\mu_\infty)\big) < \sigma_0 < \sigma_{\mathrm{crit}}(\bar\vartheta)$$

of leaks in which the network learns *and* propagation of chaos is uniform in time. Theorem D″'s condition, by contrast, asks the leak to dominate the worst-case residual curvature $\lambda_G L_2(F_\rho + K_\Phi)$ *and* the interaction's Lipschitz constant $L^2_\vartheta$-type terms, which forces $\sigma_0 \ge \sigma_{\mathrm{crit}}$. The two ingredients that remove the horizon here are (i) the sign of the interaction, which is the flat convexity of $\bar L$ (Step 3), and (ii) the residual-weighted curvature evaluated at the actual residual rather than at the worst case (Step 2). Both are properties of the lifted linear parametrisation $\hat f = \int \tilde\Phi \,\mathrm{d}\mu$, not of the particular features.

**Remark D‴.2** (the transient, the support, and the floor of the coupling)**.** Two things limit the theorem, and both are visible in the numerics. (i) The constant carries $e^{2\Lambda_1}$, the finite-horizon amplification of Theorem D′ over the time the flow needs to enter the non-degenerate regime: for a warm start $\Lambda_1 = 0$ and $C_1 = 4(L^G_\vartheta)^2 K^2_\Phi/\kappa^2$; for a cold start the expanding phase ($a(s) < \kappa$ while the residual is large) lasts a time of order $1/\sigma_0$ and $\Lambda_1$ is of order $\sigma_0^{-1} \cdot \sup(\kappa_R - \sigma_0)$ — the bound is uniform but quantitatively void, as D′'s $e^{\Lambda T}$ is. (ii) (H) is a statement about the $\varrho'$-neighbourhood of the *support* of $\mu_s$. The characteristic flow is a homeomorphism, so a connected support stays connected: if the mass of a connected cold start splits between several basins of $\mathcal{V}[\mu_\infty]$, the support of $\mu_s$ threads through the ridges between them for every $s$, and the filament's points converge to the saddles of $\mathcal{V}[\mu_\infty]$ on those ridges, where $\mathcal{V}$ has a direction of negative curvature and $\kappa_R > \sigma_0$ — (H) fails for every $s_1$ (asymptotically in $s$; pointwise the filament may well lie in a locally convex region for a while, referee round 8, D3-11), however little mass the filament carries. The theorem then says nothing, and the coupling can indeed carry a floor: a particle whose reference lies near a separatrix can be pushed to the other side by the $O(m^{-1/2})$ difference between the two drifts, after which the pair sits in different basins for ever, at a distance of the order of the basin separation. A fraction $p_m$ of such pairs contributes $p_m \cdot O(R^2_\Theta)$ to $D$, so $D$ follows the law of $p_m$ (the $\mu_0$-mass of a boundary layer of width $O(m^{-1/2})$ around the separatrices' preimage, hence $p_m = O(m^{-1/2})$ when $\mu_0$ has a bounded density there), not $1/m$; the learned function is less sensitive, $\|\hat f_\Theta - \hat f_{\bar\Theta}\| \le L^G_\vartheta \frac{1}{m}\sum\|\delta_i\|_{G^{-1}} \le L^G_\vartheta(p_m \cdot 2R_\Theta + D^{1/2}_{\mathrm{good}})$, which is still $O(m^{-1/2})$. The measure-weighted form of (H) — a bad set of small $\mu_s$-mass, with the per-particle bootstrap run on the good particles only and the interaction split as in Step 3 — is the natural extension and is listed in Section 26; what the present theorem shows is where the clean statement holds (supports inside basins) and why the cold start of Theorem D″'s experiment was outside it on two counts (the transient and the connected support). (iii) The floor has a second source at small $m$, on a warm start: when the realised cluster weights fall outside the admissible set of Proposition D‴.5(iv) the replicated configuration is a saddle of $J_m$, the $m$-system lingers on it and then splits a cluster, so that pairs end in different basins although

the mean-field flow satisfies (H); its probability is exponentially small in $m$ but appreciable at $m = 16$ on the instance below.

**Remark D‴.3** (dither)**.** At $\tau_0 > 0$ the synchronous coupling cancels the noise and the identity of Step 1 holds pathwise with the same three terms, but the reference law has full support and the particles visit the ridges of $\mathcal{V}$ with the Gibbs weight of the ridge, so (H) fails in the sense of (ii) above and the coupling carries a floor of that order. Uniform-in-time propagation of chaos at $\tau_0 > 0$ is the province of the entropy method: with dither, $(\mathfrak{A}_m)$ is the $m$-particle mean-field Langevin dynamics of the free energy $F_\tau$ of Theorem H in the $G^{-1}$ metric (Corollary J′.3 is its stationary statement), and [11] proves uniform-in-time propagation of chaos for mean-field Langevin dynamics under flat convexity, regularity of the first variation and a uniform log-Sobolev inequality for the proximal Gibbs measures — the three hypotheses that Theorem H(i)–(ii) verifies for $F_\tau$, with the constant $\rho_\tau$ of H(ii). That result is imported here, not re-derived; its constants are those of Theorem H and are exponentially small in $1/\tau_0$, where the constants of Theorem D‴ are those of the leak and the residual curvature. The leak-dominated regime is covered for the real loop and every $\tau \geq 0$ by Theorem D″.

**Remark D‴.4** (the real loop)**.** Theorem D‴ is a statement about the averaged slow dynamics $(\mathfrak{A}_m)$. Theorems J and J′ identify $(\mathfrak{A}_m)$ as the leading order of the real loop's slow dynamics, with the corrections $\varepsilon \cdot b$ (Theorem J′'s $b$, not a gradient: Proposition J′.2) and the fast error eliminated at $O(\varepsilon^2)$; the transfer of a uniform-in-time statement through these corrections — a near-identity change of variables that is uniform in time — is the open item of Section 26, branch 1; on a warm start it is settled by Corollary D‴.7 below through Theorem J. What the theorem settles is the mechanism: in the learning regime the uniformity rests on the fitted configuration's non-degenerate stability and on the flat convexity of the loss, not on a dominant leak; the real-loop run on a warm start (numerics, table 5) tracks the averaged slow dynamics over the horizon run, at the same $k_e$ at which Theorem D″'s cold start grew.

**Proposition D‴.5** (the stationary coupled distance at finite $m$: the weight-sensitivity law)**.** *Let $\mu^* = \sum_{k=1}^K \bar{w}_k \delta_{\vartheta_k^*}$ be a $K$-atom stationary measure of $(\mathfrak{A}_0)$ (each $\vartheta_k^*$ a critical point of $\mathcal{V}[\mu^*]$), and for weights $w \in (0,\infty)^K$ let $\mathcal{G}_w(\Theta) := \frac{1}{2}\langle \|f - \sum_k w_k \tilde{\Phi}(\cdot\,;\vartheta_k)\|^2 \rangle_\rho + \frac{\sigma_0}{2}\sum_k w_k \|\vartheta_k - \vartheta^0\|^2_{G^{-1}}$ be the mean-field energy of $\mu_w := \sum_k w_k \delta_{\vartheta_k}$, so that $\nabla_{\vartheta_k}\mathcal{G}_w(\Theta) = w_k \nabla_\vartheta \mathcal{V}[\mu_w](\vartheta_k)$. Assume the $K$-atom Hessian $\nabla^2_\Theta \mathcal{G}_{\bar{w}}(\Theta^*)$ is positive definite. Then:*

(i) (the weighted stationary point and its replication) *there is a smooth map $w \mapsto \Theta^*(w)$ on a neighbourhood of $\bar{w}$ with $\Theta^*(\bar{w}) = \Theta^*$ and $\nabla_\vartheta \mathcal{V}[\mu_{\Theta^*(w)}](\vartheta_k^*(w)) = 0$ for every $k$ (implicit function theorem); $S(w) := \partial\Theta^*/\partial w$ ($S_k := \partial\vartheta_k^*/\partial w \in \mathbb{R}^{p\times K}$) is given by $S = -[\nabla^2_\Theta \mathcal{G}_w]^{-1}\partial_w \nabla_\Theta \mathcal{G}_w$ at $(\Theta^*(w), w)$. For $m$ particles with $n_k$ of them in cluster $k$ ($w_k^m := n_k/m$), the configuration $\Theta_m^*(w^m)$ with $n_k$ copies of $\vartheta_k^*(w^m)$ is a critical point of $J_m$ ($\mu_\Theta = \mu_{w^m}$ there). Its whitened Hessian splits (Proposition K(i), applied cluster by cluster) into the separation blocks $\sigma_0 I - R^G[\mu_{w^m}](\vartheta_k^*(w^m))$, each with multiplicity $n_k - 1$, and the collective block* $\mathrm{diag}(w^m)^{-1/2} \cdot G_K^{1/2} \nabla^2_\Theta \mathcal{G}_{w^m} G_K^{1/2} \cdot \mathrm{diag}(w^m)^{-1/2}$ *(the $K$-atom Hessian, whitened and weighted). Hence $\Theta_m^*(w^m)$ is a strict local minimiser of $J_m$ exactly when $w^m$ lies in the* admissible set

$$W_{\mathrm{adm}} := \big\{ w : \sigma_0 G^{-1} - R[\mu_{\Theta^*(w)}](\vartheta_k^*(w)) \succ 0 \text{ for every } k \text{ and } \nabla^2 \mathcal{G}_w(\Theta^*(w)) \succ 0 \big\},$$

*a neighbourhood of $\bar{w}$ (both conditions are open and hold at $\bar{w}$ by (H) and the assumption), and a saddle when the splitting condition fails at some cluster: a cluster can then split — by Proposition K's local separation inside its basin (sub-clusters along the unstable direction, the cluster weights unchanged) or by particles migrating to other basins; the numerics show both (this section, referee rounds 9 and 10).*

(ii) (the stationary distance) *Under the shared-particle coupling of Theorem D‴ with i.i.d. cluster labels of law $\bar{w}$ (the $m$ particles are the first $m$ of the $M$ reference particles),*

*suppose $w^m, w^M \in W_{\mathrm{adm}}$ and that the $m$-system converges to $\Theta^*_m(w^m)$ and the reference to $\Theta^*_M(w^M)$. Then the coupling distance converges to*

$$D_\infty = \sum_k w_k^m \|\vartheta_k^*(w^m) - \vartheta_k^*(w^M)\|^2_{G^{-1}}$$
$$= \sum_k w_k^m \|S_k(w^M)(w^m - w^M)\|^2_{G^{-1}} + O(\|w^m - w^M\|^3),$$

*and, with $\tilde{S}_k := G^{-1/2} S_k(\bar{w})$ (replacing $S(w^M)$ by $S(\bar{w})$ costs a relative $O(\|w^M - \bar{w}\|)$) and $\mathbb{E}[(w^m - w^M)(w^m - w^M)^\top] = (1/m - 1/M)\tilde{C}$, $\tilde{C} := \mathrm{diag}(\bar{w}) - \bar{w}\bar{w}^\top$ (multinomial counts of the shared labels), conditionally on the admissibility of both weights,*

$$\mathbb{E}[D_\infty \mid w^m, w^M \in W_{\mathrm{adm}}] = (1/m - 1/M) \cdot \sum_k \bar{w}_k \,\mathrm{tr}\big(\tilde{S}_k \tilde{C} \tilde{S}_k^\top\big) + O(m^{-3/2}) + O(m^{-1}M^{-1/2}) :$$

*the $1/m$ law with an explicit constant, and the $(1/m - 1/M)$ law of the shared-particle coupling observed in Theorem D′'s numerics.*

(iii) (the fluctuations) *As $M \to \infty$ ($w^M \to \bar{w}$) and then $m \to \infty$, $\sqrt{m}(w^m - \bar{w}) \to \mathcal{N}(0, \tilde{C})$ and $\sqrt{m} \cdot (\vartheta_i - \bar{\vartheta}_i) \to S_{k(i)}\xi$ with $\xi \sim \mathcal{N}(0, \tilde{C})$ for every particle $i$ of cluster $k(i)$: the fluctuations of the coupling are Gaussian, carried entirely by the cluster weights, and all particles of a cluster are displaced by the same vector.*

(iv) (admissibility) *Let $\delta_w := \sup\{\delta : \|w - \bar{w}\|_\infty \le \delta \Rightarrow w \in W_{\mathrm{adm}}\} > 0$. By Hoeffding's inequality for each multinomial coordinate, $\mathbb{P}(w^m \notin W_{\mathrm{adm}}) \le 2K \exp(-2m\delta_w^2)$: the finite-$m$ configuration is a strict local minimiser with probability exponentially close to one, the exceptional event being the one on which the realised weights make the replicated configuration a saddle: a cluster of the $m$-system then splits, locally or by migration to other basins — in the latter case with Remark D‴.2's floor on a warm start.*

**Status.** Formal: (i) and the algebra of (ii)–(iv) are exact (the block splitting is Proposition K(i) cluster by cluster, confirmed on $J_{10}$ to $8 \cdot 10^{-12}$ in referee round 9; implicit function theorem; the multinomial covariance of shared labels; Hoeffding), the convergence of both systems to the stated configurations is assumed (it is what Theorem D‴ proves for $m \ge m_0$ and what the numerics show for $m = 16$–$256$ at admissible weights; at inadmissible weights the $m$-system sits on the saddle for a long time and then leaves it, numerics below). Checked: `weighted_stationary_atoms`, `weight_sensitivity`, `predicted_stationary_D` in `chaos_learning.py`; on the instance below the replicated weighted stationary point is a critical point of $J_{16}$ to $10^{-10}$, the linearisation $\vartheta^*(\bar{w} + \mathrm{d}w) - \vartheta^* = S \cdot \mathrm{d}w$ has a second-order remainder, and the per-cluster displacements of the coupled run point along $S_k(w^m - w^M)$ (test and table below). The proposition is the sharp form of Theorem D‴'s constant: $S = -H_w^{-1}\partial_w \nabla \mathcal{G}_w$ has the structure (stiffness)$^{-1}$ × (feature correlations), of which $C_1 = 4e^{2\Lambda_1}(L_\vartheta^G)^2 K_\Phi^2/\kappa^2$ is the worst-case envelope ($\kappa^{-1}$ for the stiffness, $L_\vartheta^G K_\Phi$ for the correlations).

**Remark D‴.6** (the regime of the theorem, and what the numerics test)**.** The sample size $m_0 = 16K_\Phi^2/\theta^2$ from which the theorem's tail is in force is astronomically large on realistic instances ($10^{23}$ below): $\theta$ carries $\varrho'\kappa^2 e^{-2\Lambda_1}/(L_\vartheta^G)^3$ from the per-particle bootstrap of Step 6, because the interaction's effect on one particle is bounded by $(L_\vartheta^G)^2 D^{1/2}$ and divided by the stiffness $\kappa$. This is not an artefact of the proof's constants only: Proposition D‴.5 says the particles of a cluster are displaced by $S_k(w^m - w^M)$, of order $\|S\| m^{-1/2}$ with $\|S\| \sim$ (stiffness)$^{-1}$, which exceeds any fixed $\varrho'$ for $m$ below $(\|S\|/\varrho')^2$ — on the instance below the $m = 16$ particles sit 0.4–1.4 whitened units from their reference copies, outside the neighbourhood $N_{\varrho'}$ in which $\mathcal{V}[\mu_{\bar{\Theta}}]$ is convex, at the replicated stationary configuration $\Theta^*_{16}(w^m)$ of their own $J_{16}$ — a strict local minimiser when $w^m \in W_{\mathrm{adm}}$ and a saddle otherwise (Proposition D‴.5(i), (iv); one of the four $m = 16$ seeds below). Hypothesis (H) is nevertheless the right object: it is the condition under

which the reference configuration is stable to splitting, and the $m$-system's configuration is the same configuration with perturbed weights. What the numerics test is therefore (a) hypothesis (H) at the stationary measure and along the flow, (b) the mechanism — the coupled distance saturates (no growth) and follows the $1/m$ law — and (c) the sharp constant of Proposition D‴.5; the theorem's own constant $C_1$ is reported for the record.

**Corollary D‴.7** (the real loop on a warm start: uniform in time through Theorem J)**.** *Let $\mu^* = \sum_k \bar{w}_k \delta_{\vartheta_k^*}$ be a $K$-atom stationary measure as in Proposition D‴.5, satisfying the splitting condition $\sigma_0 G^{-1} - R[\mu^*](\vartheta_k^*) \succ 0$ at every atom and the non-degeneracy of the $K$-atom Hessian, and let $w \mapsto \Theta^*(w)$ be the weighted stationary point of D‴.5(i). Fix $m \le M$ and cluster weights $w^m, w^M \in W_{\mathrm{adm}}$ (Proposition D‴.5(i)), and let $\Theta_m^* :=$ the configuration with $n_k = m w_k^m$ copies of $\vartheta_k^*(w^m)$ (likewise $\Theta_M^*$). Both are non-degenerate local minimisers of $J_m$ and $J_M$ (Proposition D‴.5(i)), hence hyperbolic critical points, and Theorem J applies to each loop: there are $k_0$, $\rho_0$, $C$ (depending on $\Theta_m^*$ and $\Theta_M^*$, hence on $m$, $M$ and the weights) such that for $k_e \ge k_0$ the real closed loop* (CL) *with $m$ kernels started at $\|\Theta(0) - \Theta_m^*\|_{G^{-1}} \le \rho_0$, $\|e(0)\| \le R$ converges to its $T_0$-periodic orbit $\Theta_p^m$, and likewise the $M$-kernel loop to $\Theta_p^M$. Under the shared-particle coupling (the $m$ kernels are the first $m$ of the $M$, the plants are driven by the same reference) the coupled distance $D(t) := \frac{1}{m}\sum_i \|\vartheta_i(t) - \bar{\vartheta}_i(t)\|_{G^{-1}}^2$ satisfies, for all $t \ge 0$,*

$$D(t)^{1/2} \le D_{\mathrm{orb}}^{1/2} + Ce^{-\mu t}\big(\|\Theta(0) - \Theta_p^m(0)\|_{G^{-1}} + \|x(0) - x_p^m(0)\|\big) + Ce^{-\mu t}\big(\|\bar{\Theta}(0) - \Theta_p^M(0)\|_{G^{-1}} + \|\bar{x}(0) - x_p^M(0)\|\big),$$

*with $\mu = (\lambda_*/k_e)(1 - \kappa_J)$ the rate of Theorem J(c) ($\kappa_J$ its small parameter; the smaller of the two loops' rates) and*

$$\begin{aligned} D_{\mathrm{orb}} &:= \sup_t \frac{1}{m}\sum_i \|\Theta_p^m(t)_i - \Theta_p^M(t)_i\|_{G^{-1}}^2 \\ &= \sum_k w_k^m \|\vartheta_k^*(w^m) - \vartheta_k^*(w^M)\|_{G^{-1}}^2 + O(1/k_e) \\ &= \sum_k w_k^m \|S_k(w^m - w^M)\|_{G^{-1}}^2 + O(\|w^m - w^M\|^3) + O(1/k_e)\,: \end{aligned}$$

*the coupled distance of the real loop is bounded uniformly in time, converges to the distance between the two periodic orbits, and that distance is Proposition D‴.5's stationary distance up to the $O(1/k_e)$ offsets of Theorem J(a),(d) (which differ between the two loops by $O(k_e^{-1}\|w^m - w^M\|) + O(k_e^{-2})$, the first-order offset of a replicated configuration being a smooth function of the $K$-atom data $(\Theta^*(w), w)$ by permutation equivariance). In particular, for cluster labels i.i.d. of law $\bar{w}$ and conditionally on both weights being admissible and both starts lying in their basins, $\mathbb{E}D_{\mathrm{orb}} = (1/m - 1/M)\sum_k \bar{w}_k \operatorname{tr}(\tilde{S}_k \tilde{C} \tilde{S}_k^\top) + O(m^{-3/2}) + O(1/k_e)$.*

**Status.** Proved as stated (Theorem J(a),(c),(d) applied to each loop, the triangle inequality in the $G^{-1}$ metric, and Proposition D‴.5(ii) for the distance between the critical points), under Theorem J's hypotheses for both configurations: the constants $k_0$, $\rho_0$, $C$, $\mu$ are those of Theorem J at $\Theta_m^*$ and $\Theta_M^*$, and the corollary does not assert their uniformity in $m$ or $M$ (Remark D‴.8). It is the statement of Remark D‴.4 for a warm start: the real loop's coupled distance saturates, at a value set by the cluster weights. Consistent with the instance of this section's numerics (table 5: the real loop at $k_e = 25$ from the jittered mixture over 80 slow units, a sixth of the relaxation time $1/\kappa$, follows its averaged slow dynamics on the same initial clouds to within a few percent at every window end; the saturation at the orbit distance lies beyond the horizon run and is not checked).

*Proof.* Theorem J(c) for the $m$-kernel loop gives $\|\Theta(t) - \Theta_p^m(t)\|_{G^{-1}} \le Ce^{-\mu t}(\|\Theta(0) - \Theta_p^m(0)\|_{G^{-1}} + \|x(0) - x_p^m(0)\|)$ and the same for the $M$-kernel loop; the first $m$ rows of the $M$-loop are the

reference particles $\bar{\vartheta}_i$. For the $(m \times p)$-row norm $\frac{1}{m}\sum_i \|\cdot\|^2_{G^{-1}} \le \|\cdot\|^2_{G^{-1}}/m \le \|\cdot\|^2_{G^{-1}}$ (the full configuration norm dominates the mean square over the first $m$ rows), so by the triangle inequality $D(t)^{1/2} \le [\frac{1}{m}\sum_i \|\Theta^m_p(t)_i - \Theta^M_p(t)_i\|^2]^{1/2} +$ (the two transients). Theorem J(a) places each orbit within $C/k_e$ of its critical point, uniformly in $t$, so the first term is $\sum_k w^m_k \|\vartheta^*_k(w^m) - \vartheta^*_k(w^M)\|^2_{G^{-1}}$ up to $O(1/k_e)$ (the two offsets are $O(1/k_e)$ each, with $O(k_e^{-2})$ remainders; at a replicated configuration the first-order offset is the same vector for all particles of a cluster and depends smoothly on the $K$-atom data $(\Theta^*(w), w)$, by the permutation equivariance of (CL) and the block splitting of Proposition D‴.5(i), so the two loops' offsets differ by $O(k_e^{-1}\|w^m - w^M\|) + O(k_e^{-2})$ — the two loops live on $\mathbb{R}^{mp}$ and $\mathbb{R}^{Mp}$, and it is this reduction to the $K$-atom data that makes them comparable), and Proposition D‴.5(ii) expands the distance between the critical points in $w^m - w^M$. The expectation statement is D‴.5(ii)'s. □

**Remark D‴.8** (what the corollary does and does not say)**.** It transfers the uniform-in-time statement to the real loop in the regime where Theorem J applies to both loops — a warm start in the basins of two hyperbolic minimisers at a gain $k_e$ above both thresholds. Theorem J's constants at a replicated configuration are controlled by the $K$-atom data (the Hessian of $J_m$ there is the collective block $\nabla^2\mathcal{G}_{w^m} \oplus$ the separation blocks $\sigma_0 G^{-1} - R_k$ of Proposition K(i), both independent of $m$ beyond the weights), so the threshold and the basin are expected to be uniform in $m$ for configurations with the same weights; the corollary does not prove this, and the cold start — where the coupled distance of the real loop grew in Theorem D″'s experiment — is outside it on the same two counts as in Remark D‴.2.

**Proposition D‴.9** (the leak toward a measure-valued prior, and the design of the window)**.** *Let $\nu = \sum_{j=1}^{K_0} q_j \delta_{\vartheta^0_j}$ be a prior measure with distinct atoms and let the leak be the $G$-gradient of the distance-to-support potential*

$$V_\nu(\vartheta) := \tfrac{1}{2}\min_j \|\vartheta - \vartheta^0_j\|^2_{G^{-1}}, \qquad -G\nabla V_\nu(\vartheta) = -(\vartheta - \vartheta^0_{j(\vartheta)}), \qquad j(\vartheta) := \operatorname*{argmin}_j \|\vartheta - \vartheta^0_j\|_{G^{-1}}$$

*(the* nearest-atom leak*: inside each Voronoi cell of $\nu$ it is the $\sigma$-modification toward that cell's atom; the barycentric OT leak of Theorem C′ is its mass-constrained relative). Write $(\mathfrak{A}^\nu_m)$ and $(\mathfrak{A}^\nu_0)$ for $(\mathfrak{A}_m)$ and $(\mathfrak{A}_0)$ with $\sigma_0(\vartheta - \vartheta^0)$ replaced by $\sigma_0(\vartheta - \vartheta^0_{j(\vartheta)})$, and $\mathcal{V}^\nu[\mu] := V_L[\mu] + \sigma_0 V_\nu$ for the first-variation potential.*

(i) (the theorems carry over) *Assume, in addition to the hypotheses of Theorem D‴ for $(\mathfrak{A}^\nu_0)$, that the assignment is locally constant along the flow: for every $s \ge 0$ the neighbourhood $N_{\varrho'}(\operatorname{supp}\mu_s)$ meets no Voronoi boundary of $\nu$ (in particular when the gap between the distances to the two nearest atoms exceeds $2\varrho'$ on $\operatorname{supp}\mu_s$, the $G^{-1}$-distance to the boundary being at least half the gap). Then Theorem D‴, Proposition D‴.5 and Corollary D‴.7 hold for $(\mathfrak{A}^\nu_m)$ with the same constants, $R_\Theta$ replaced by $\max\{R_0, L^G_\vartheta(F_\rho + K_\Phi)/\sigma_0\} + \max_j \|\vartheta^0_j - \vartheta^0_1\|_{G^{-1}}$ and $\vartheta^0$ by $\vartheta^0_{j(\cdot)}$ in the a priori bounds, and Theorem J's basin radius $\rho_0$ at the replicated configuration taken below $\min_k \operatorname{dist}_{G^{-1}}(\vartheta^*_k, \partial\,\mathrm{cell}_k)$. The masses $q_j$ enter the nearest-atom leak only through $\operatorname{supp}\nu$; they matter in (ii), and they are what the mass constraint of the OT leak (Theorem C′, Remark D‴.10) acts on.*

(ii) (the stationary residual and the window) *If $\nu$ is placed on an exact representation of the plant on the orbit — $K_0 = K$ atoms $\vartheta^0_k$ with $\sum_k q_k \tilde{\Phi}(\cdot\,; \vartheta^0_k) = f$ on $\operatorname{supp}\rho$ — then $\mu^* := \sum_k q_k \delta_{\vartheta^0_k}$ is a stationary measure of $(\mathfrak{A}^\nu_0)$ with $r_{\mu^*} = 0$, $R[\mu^*] \equiv 0$, and the window of Remark D‴.1 has its maximal margin: $\kappa_R[\mu^*](A) = 0$ for every set $A$ ($R[\mu^*] \equiv 0$) and $\kappa = \sigma_0$, the only restriction on $\varrho'$ being (i)'s Voronoi hypothesis, against $\kappa = \sigma_0 - \kappa_R[\mu^*_\sigma](N_{\varrho'})$ for the $\sigma$-modification's stationary measure $\mu^*_\sigma$, whose residual is the leak's bias. For a prior $\nu_\delta$ displaced from the exact representation the residual of the stationary measure is $O(\delta)$ and $\kappa_R[\mu^*_{\nu_\delta}] = O(\delta)$: the margin decreases with the misplacement, and the window closes when the displacement makes the residual-weighted curvature at an atom reach $\sigma_0$.*

(iii) (the floors) *Under the shared-particle coupling the stationary distances are the weight-sensitivity laws of Proposition D‴.5 at parameter level,* $\mathbb{E}[D_\infty \mid w^m, w^M \in W_{\mathrm{adm}}] = (1/m - 1/M)\sum_k \bar{w}_k \operatorname{tr}(\tilde{S}_k \tilde{C} \tilde{S}_k^\top) + O(m^{-3/2}) + O(m^{-1}M^{-1/2})$*, and at function level,*

$$\mathbb{E}\|\hat{f}_{\Theta^*(w^m)} - \hat{f}_{\Theta^*(w^M)}\|^2_{L^2(\rho)} = (1/m - 1/M) \cdot \operatorname{tr}(F\tilde{C}F^\top)/N + O(m^{-3/2}),$$
$$F_{\cdot j} := \partial \hat{f}_w / \partial w_j = \tilde{\Phi}(\cdot\,; \vartheta_j^*) + \sum_k w_k \nabla_\vartheta \tilde{\Phi}(\cdot\,; \vartheta_k^*) S_{k\cdot j},$$

*with $S$ the sensitivity of the weighted stationary point of $(\mathfrak{A}_0^\nu)$ to the weights. Neither floor vanishes for the matching prior: away from $\bar{w}$ the weighted stationary point does not stay on the zero-residual manifold (the leak pulls the amplitudes back to the prior while the fit asks for $a_k \propto 1/w_k$); on the instance both sensitivities rise with the amplitudes of the stationary atoms along the amplitude-scaled priors — a numerical trend, not a law (at a fixed amplitude the width changes the sensitivity, Table 17).*

**Status.** (i) proved (the proofs of Theorem D‴, Proposition D‴.5 and Corollary D‴.7 use the leak only through its curvature $\sigma_0 G^{-1}$ and its a priori bound — the latter goes through with $\operatorname{dist}(\cdot, \operatorname{supp}\nu)$ in place of $\|\cdot - \vartheta^0\|$, the upper Dini derivative of the minimum of quadratics along the flow being bounded by the active branch's. The local constancy of the assignment makes every segment $[\bar{\vartheta}_i, \vartheta_i]$ of Step 2 lie in one cell, and it is needed for every $s \geq 0$, not only from $s_1$ on, because (2.1) and (5.2) are used on $[0, s_1)$ as well: a pair straddling a boundary adds $(\sigma_0/m)\langle \delta_i, \vartheta^0_{j(\vartheta_i)} - \vartheta^0_{j(\bar{\vartheta}_i)} \rangle_{G^{-1}} \geq 0$ to $\frac{1}{2}\,\mathrm{d}D/\,\mathrm{d}s$ (the sum of the two cell inequalities; $V_\nu$ is semiconcave and its leak expands across the boundaries), a forcing of size up to $\sigma_0 \operatorname{diam}(\operatorname{supp}\nu)\sqrt{D}$ that none of the theorem's constants absorbs — referee round 10 found `decomposition`'s same-cell convention and the true $\frac{1}{2}\,\mathrm{d}D/\,\mathrm{d}s$ to differ by exactly this term on a straddling pair. The same hypothesis gives the well-posedness of $(\mathfrak{A}_0^\nu)$, whose vector field is discontinuous on the boundaries, and Theorem J applies to $J_m^\nu$, which is $C^2$ away from the Voronoi boundaries, on a basin that meets none.) (ii) proved ($r_{\mu^*} = 0$ and $\nabla V_\nu(\vartheta_k^0) = 0$ give stationarity; $R[\mu^*] = \langle r_{\mu^*} \nabla^2 \tilde{\Phi} \rangle_\rho = 0$; the $O(\delta)$ statements are the smooth dependence of the weighted stationary point on the prior, by the implicit function theorem at $\mu^*$, whose $K$-atom Hessian $\mathrm{Gram} + \sigma_0 \operatorname{diag}(q) G^{-1}$ is positive definite — on the instance its whitened, weighted eigenvalues are 0.0074–0.20, all at or above $\sigma_0$, referee round 10). (iii) formal as Proposition D‴.5 (the function-level law is the delta method applied to $w \mapsto \hat{f}_{\Theta^*(w)}$); the sign of the comparison with the $\sigma$-modification is a numerical finding (`exp_prior_design.py` → `results/chaos/prior_design.md`, below), not a theorem. Checked: `tests/test_prior_design.py` (the single-atom prior reproduces $(\mathfrak{A}_m)$; the gradient of $J_m^\nu$ against finite differences; the coupling identity inside the cells; the matching prior's exact representation stationary with zero residual and zero curvature, margin $\sigma_0$ against the $\sigma$-modification's; the parameter- and function-level sensitivities, the latter against finite differences of the fitted function of the weighted stationary point).

**Remark D‴.10** (design)**.** The leak's prior is a design input of the window. The $\sigma$-modification's single $\vartheta^0$ biases every kernel toward one point and leaves a residual at the fitted configuration — the residual-weighted curvature that bounds the window from below and slows the relaxation ($\kappa = 0.00196$ on the instance of this section, the theorem's relaxation time $1/\kappa = 511$ slow units). A prior measure whose atoms sit on the plant's structure removes the bias and gives the maximal margin $\kappa = \sigma_0$ ($1/\kappa = 141$ slow units, 3.6 times shorter; the observed e-folding times of the coupled distance are 3.3 times shorter in the median, numerics below), at the price of a larger floor at both levels (the stationary atoms have the full amplitudes, to which the weight sensitivity is proportional): the classical trade-off between bias and variance, here between the window's margin and the coupling's floor (Proposition D‴.24 resolves it at the function level: the floor is the weight-slope of the bias, and an adaptive prior removes both, at the price of the collective rate, of the leak's a-priori bound — restored by projecting the prior, Proposition D‴.27 — and of a larger parameter-level floor). A misplaced prior does worse than no structured prior (centres off by 0.6, amplitudes $\times 1.2$ or widths $\times 0.7$ close the window, which the $\sigma$-modification keeps

open with margin 0.0020), and a shrunk prior does better than the matching one (amplitudes $\times 0.5$–$0.7$: margin above $\sigma_0$ at the uniform weights, both floors lower, all four tested unbalanced weights admissible — with splitting margins above $\sigma_0$ at one of them and slightly below at the other three; numerics below and Remark D‴.11). The barycentric OT leak of Theorem C′ adds a mass constraint to the same geometry — each prior atom must receive its mass — and so acts on the cluster weights themselves, the source of the floor; its interplay with the weight fluctuation (an equal-mass Lloyd dynamics against the fit) is the natural next step (Section 26, branch 8).

**Numerics** (`exp_prior_design.py` → `results/chaos/prior_design.md`). The three-bump instance of Theorem D‴'s numerics at the same leak $\sigma_0 = 0.15\sigma_{\text{crit}} = 0.00707$, with the nearest-atom leak toward a prior of three atoms placed at the exact representation $(3A_k, C_k, S_k)$ and misplaced from it, against the $\sigma$-modification ($\vartheta^0 = 0$); $\varrho' = 0.1$; the floors predicted for $m = 64$ against $M = 2048$.

*The design table (table 1).* Stationary three-atom measure $\mu_\nu^*$ under each prior — fit error, $\kappa_R$ at the atoms and over $N_{0.1}$, margin $\kappa = \sigma_0 - \kappa_R(N_{0.1})$, whitened $\|S_k\|$, predicted $\mathbb{E}D_\infty$ and $\mathbb{E}\|\hat{f}_{64} - \hat{f}_M\|^2$ (Table 17):

Table 17: Proposition D‴.9 on the three-bump instance: the stationary three-atom measure under each prior — fit error, residual curvature at the atoms and over $N_{0.1}$, margin, whitened weight sensitivities, and the predicted parameter- and function-level floors for $m = 64$ against $M = 2048$ (`results/chaos/prior_design.md`, table 1).

| prior | fit error | $\kappa_R$ at the atoms (max) | $\kappa_R$ over $N_{0.1}$ | margin $\kappa$ | $\|S_k\|$ | $\mathbb{E}D_\infty$ ($m = 64$), $\times 10^{-1}$ | $\mathbb{E}\|\hat{f}_{64} - \hat{f}_M\|^2$, $\times 10^{-4}$ |
|---|---|---|---|---|---|---|---|
| $\sigma$-modification, $\vartheta^0 = 0$ | 0.2439 | +0.00411 | +0.00512 | +0.00196 | 8.38, 4.66, 8.32 | 2.010 | 4.287 |
| matching prior (exact representation) | 0.0000 | +0.00000 | +0.00000 | +0.00707 | 13.01, 6.18, 8.91 | 3.227 | 9.334 |
| centres shifted by +0.15 / +0.3 / +0.45 / +0.6 | 0.0098 / 0.0200 / 0.0311 / 0.0436 | +0.00140 / +0.00298 / +0.00480 / +0.00698 | +0.00157 / +0.00333 / +0.00534 / +0.00770 | +0.00550 / +0.00374 / +0.00173 / −0.00063 | 12.9–13.0, 6.18, 8.92 | 3.21–3.22 | 9.40–9.51 |
| amplitudes $\times 0.5$ / $\times 0.7$ / $\times 1.2$ | 0.0921 / 0.0624 / 0.0861 | −0.00039 / −0.00028 / +0.03684 | −0.00033 / −0.00023 / +0.03875 | +0.00740 / +0.00731 / −0.03167 | 9.76, 5.01, 7.47 / 10.59, 5.40, 7.92 / 17.80, 6.96, 10.11 | 1.991 / 2.303 / 5.242 | 7.106 / 7.657 / 18.02 |
| widths $\times 0.7$ / $\times 1.5$ / $\times 2$ | 0.2195 / 0.0581 / 0.1190 | +0.03086 / +0.00113 / +0.00294 | +0.03316 / +0.00126 / +0.00306 | −0.02609 / +0.00581 / +0.00401 | 13.00, 2.82, 9.77 / 11.13, 7.19, 8.50 / 10.39, 7.27, 8.71 | 3.062 / 2.932 / 3.008 | 17.16 / 7.459 / 6.148 |

The matching prior makes the exact representation stationary with zero residual (fit error $1.2 \cdot 10^{-16}$), $\kappa_R = 0$ and the maximal margin $\kappa = \sigma_0 = 0.00707$, 3.6 times the $\sigma$-modification's 0.00196 (the theorem's relaxation time $1/\kappa$: 141 against 511 slow units); the $\sigma$-modification's bias (amplitudes 2.45, −2.29, 2.05 against $3A_k = 3.0$, −2.1, 2.4; widths 0.83, 0.70, 0.88 against 0.5, 0.4, 0.5) is removed. The floors move the other way: $\|S_k\| = 13.01$, 6.18, 8.91 against 8.38, 4.66, 8.32, $\mathbb{E}D_\infty$ 1.61 times and the function-level floor 2.18 times the $\sigma$-modification's. Misplacing the prior brings the residual curvature back and closes the window at centres shifted by +0.6 (margin −0.00063), at amplitudes $\times 1.2$ (−0.03167) and at widths $\times 0.7$ (−0.02609), while shrinking the amplitudes ($\times 0.5$, $\times 0.7$) gives margins above $\sigma_0$ (+0.00740, +0.00731: $\kappa_R$ is

negative at every atom) with floors below the matching prior's ($\mathbb{E}D_\infty$ 0.199, 0.230; function level $7.1 \cdot 10^{-4}$, $7.7 \cdot 10^{-4}$) at fit errors 0.09, 0.06. The cell margins (the gap between the two nearest prior atoms along the configuration) are 1.2–6.3, above $2\varrho' = 0.2$ everywhere.

*Admissible weights (table 2).* The splitting margin of the replicated weighted stationary configuration $\Theta^*(w)$ under the three priors (Table 18):

Table 18: Proposition D‴.9 on the three-bump instance: the splitting margin $\sigma_0 - \max_k \kappa_R$ of the replicated weighted stationary configuration $\Theta^*(w)$ at four weight vectors under the $\sigma$-modification, the matching prior and the shrunk prior (`results/chaos/prior_design.md`, table 2).

| weights $w$ | $\sigma$-modification | matching prior | amplitudes $\times 0.7$ |
|---|---|---|---|
| $(7,5,4)/16$ | $+0.00257$ | $-0.00544$ | $+0.00733$ |
| $(9,4,3)/16$ | $-0.00219$ | $-0.00579$ | $+0.00570$ |
| $(10,3,3)/16$ | $-0.00377$ | $-0.00415$ | $+0.00536$ |
| $(0.5,0.3,0.2)$ | $-0.00062$ | $-0.00653$ | $+0.00663$ |

The matching prior's admissible set is the smallest: any excess weight overshoots the plant ($\kappa_R$ at the over-weighted atom $+0.0112$ to $+0.0136$), so even $(7,5,4)/16$ is a saddle, and the set is smallest in the direction of the largest bump (an excess $\varepsilon$ on the first atom alone, $w = \bar{w} + \varepsilon(1.5e_1 - 0.5)$, has margins $+0.00279$, $-0.00031$, $-0.00392$ at $\varepsilon = 0.02$, 0.04, 0.08, while $\varepsilon = 0.08$ on the second or third atom alone keeps $+0.00208$, $+0.00316$); the $\sigma$-modification undershoots ($\hat{f}/f = 0.77$–$0.78$ at the first atom) and tolerates $(7,5,4)/16$ but not $(9,4,3)/16$ (Theorem D‴'s inadmissible seed); the shrunk prior keeps all four admissible, with margins $+0.00536$ to $+0.00733$ — above $\sigma_0$ at $(7,5,4)/16$ and slightly below it at the other three, where its over-weighted first atom overshoots the plant at the centre by 1–3% ($\hat{f}/f = 1.020$, 1.028, 1.007; $\kappa_R$ there $+0.0014$, $+0.0017$, $+0.0004$; Remark D‴.11). At the realised weights of the matching-prior coupled runs below, two of the nine (seed, $m$) pairs are saddles ($m = 16$, seed 0 with $w = (6,5,5)/16$, margin $-0.00052$; $m = 64$, seed 1 with $w = (26,17,21)/64$, $-0.00296$), the other seven admissible ($+0.00056$ to $+0.00609$).

*Coupled runs with the matching prior (table 3).* $m = 16$, 64, 256 against $M = 2048$, warm start the mixture of Gaussians of width 0.02 around the exact representation, 300 slow units (2.1 relaxation times), three seeds. Along the run $\kappa_R[\mu^*]$ over the spheres is 0 ($r^* = 0$) and with the reference's own residual at most $0.00674 < \sigma_0$ (per-seed maxima 0.00634, 0.00674, 0.00155, each at the first record; the seed-mean curve peaks at 0.00488); the reference's fit error falls from 0.029 to 0.008 and its rms drift from $3.6 \cdot 10^{-3}$ to $3.0 \cdot 10^{-5}$ (seed means; per seed the fit error 0.041, 0.036, 0.011 $\to$ 0.012, 0.009, 0.003). Window maxima of $D_m$ over $(0, 37.5]$, $(37.5, 75]$, $(75, 150]$, $(150, 300]$: $0.114 \to 0.193 \to 0.239 \to 0.250 \pm 0.017$ ($m = 16$), $0.0954 \to 0.157 \to 0.190 \to 0.197 \pm 0.12$ (64), $0.00933 \to 0.0140 \to 0.0166 \to 0.0173 \pm 0.010$ (256): last-to-previous 1.05, 1.04, 1.04 — saturated within 2.1 of the theorem's relaxation times $1/\kappa$; the observed late-time e-folding times of $D_m$ (from the geometric decay of its increments over $(150, 300]$) are 48–84 slow units, median 59, against 107–306 (median 195) for the $\sigma$-modification's warm runs of Theorem D‴'s numerics — a factor 3.3 in the medians, against 3.6 in $1/\kappa$; slope $-1.23 \pm 0.40$ in $m$ (the function-level distance $-0.63 \pm 0.21$). Per seed, the linearised law from the realised weights predicts the measured $D_m$ at $s = 300$ with ratios 0.81–1.11 at parameter level (nine pairs) and 0.61–1.69 at function level, and the exact form of Proposition D‴.5(ii), $\sum_k w_k^m \|\vartheta_k^*(w^m) - \vartheta_k^*(w^M)\|_{G^{-1}}^2$, with ratios 0.987–0.998 in all nine pairs: the linearised law's deviations are its second-order remainder (its two lowest ratios, 0.81 and 0.82, sit at $|w^m - w^M|_\infty = 0.084$ and 0.033), not the saddles' doing. The two inadmissible pairs (ratios 0.91 and 0.81; exact 0.995 and 0.994) sit on their replicated saddles at $s = 300$; continued autonomously to $s = 6000$ and 9000 (referee round 10, `review/REVIEW_round10_D3c.md`), both $m$-systems leave the saddle by Proposition K's local split — the over-weighted cluster separates into two sub-clusters along the centre direction inside

its own cell, the cluster weights unchanged — and the coupled distance stays on its plateau ($-1.3\%$ at $m = 64$), unlike the $\sigma$-modification's seed of round 9, which migrated between basins.

*The function-level law for the $\sigma$-modification (table 4).* Against the warm runs of Theorem D‴'s numerics ($M = 2048$, last window): the predicted root-mean-square $(1/m - 1/M)^{1/2}$ $(\operatorname{tr} F\tilde{C}F^\top/N)^{1/2}$ is 0.0419, 0.0207, 0.00984 against the measured means $0.0307 \pm 0.0097$, $0.0154 \pm 0.0033$, $0.00467 \pm 0.0012$ (ratios 0.73, 0.74, 0.48; a mean of norms over four seeds lies below the root-mean-square).

*Reading.* The prior is a design input on three counts. Its placement sets the stationary residual, hence the window's lower end and the relaxation rate (the matching prior: margin $\sigma_0$, the theorem's relaxation time 3.6 times shorter, the observed one 3.3 times); the amplitudes it induces set the floors (which rise with them along the amplitude-scaled priors, so removing the bias raises the floors by 1.6–2.2 times); and the sign of the residual it leaves at each cluster sets the admissible set of weights (overshoot splits, undershoot stabilises). A shrunk prior — amplitudes 0.5–0.7 of the plant's — wins on all three at a fit error of 0.06–0.09; a misplaced one (centres off by more than a bump width, amplitudes too large, widths too narrow) closes the window.

**Remark D‴.11** (undershoot stabilises, overshoot splits)**.** For a Gaussian kernel, $\partial^2_{cc}\tilde{\Phi}(x;\vartheta) = -(a/s^2)(1-(x-c)^2/s^2)\varphi$ is negative at the centre, so the $(c,c)$ entry of $R[\mu](\vartheta_k) = \langle r_\mu \nabla^2 \tilde{\Phi}\rangle_\rho$ has the sign of $-a_k \cdot r_\mu$ near the centre: a cluster whose kernel overshoots the plant there ($r$ of the sign opposite to $a_k$) gets $R \succ 0$ in the centre direction — the separation block $\sigma_0 G^{-1} - R$ loses margin, and the cluster splits once $\kappa_R$ exceeds $\sigma_0$ — while an undershooting cluster gets $R \prec 0$ and a splitting margin above $\sigma_0$. For the matching prior this is exactly what Table 18 shows: at every tested weight the over-weighted atom overshoots at its centre ($\hat{f}/f = 1.07$–$1.10$), its $R_{cc}$ is positive and the top eigenvector of $R^G$ is the centre direction, $(0.01, 1.00, -0.02)$ in $(a, c, s)$, while the under-weighted atoms have $R_{cc} < 0$; the shrunk prior undershoots at the uniform weights and at $(7,5,4)/16$, and at the three more unbalanced weights its over-weighted atom overshoots slightly ($\hat{f}/f = 1.007$–$1.028$), taking the margin just below $\sigma_0$. The two other instabilities of this section and Section 23 have a different anatomy (referee round 10): the $\sigma$-modification's inadmissible seed (weight $9/16$ on the first bump) still undershoots at its centre ($R_{cc} < 0$), its unstable direction being a mixed amplitude–width direction — the broad kernel (width 0.88 against the bump's 0.5) overshoots in its tails, through the cross-curvature $\langle r\varphi(x-c)^2/s^3\rangle_\rho$; and Proposition K's collapsed kernel owes its $R_{cc} = +0.055$ to the overshoot over the negative bump and to the undershoot of the outer bumps beyond its inflection points, where $1-(x-c)^2/s^2 < 0$, not to the region between the bumps. It is an observation about the Gaussian kernel's second derivative, not a theorem: the width and amplitude directions carry other signs, and the residual's profile matters; the splitting margin itself is the quantity to compute.

**Proposition D‴.12** (the mass constraint removes the weight-fluctuation floor)**.** *Let $\nu = \sum_{j=1}^{K_0} q_j \delta_{\vartheta_j^0}$ be a prior measure with distinct atoms and masses $q_j > 0$, $\Delta_\nu := \min_{j\neq k}\|\vartheta_j^0 - \vartheta_k^0\|_{G^{-1}}$, and let the leak be the barycentric OT leak of Theorem C′ toward $\nu$ for the cost $\|\cdot\|^2_{G^{-1}}$: with $\gamma = \gamma(\Theta)$ a vertex optimal plan from $\mu_\Theta$ to $\nu$ (row sums $1/m$, column sums $q_j$) and $\bar{T}_i := m\sum_j \gamma_{ij}\vartheta_j^0$ its barycentric projection,*

$$\frac{\mathrm{d}\vartheta_i}{\mathrm{d}s} = G\langle \nabla_\vartheta (r_\Theta \tilde{\Phi}(\cdot\,;\vartheta_i))\rangle_\rho - \sigma_0(\vartheta_i - \bar{T}_i), \tag{$\mathfrak{A}_m^{\mathrm{OT}}$}$$

*the $G$-gradient flow of $J_m^{\mathrm{OT}}(\Theta) := m\bar{L}(\mu_\Theta) + (\sigma_0 m/2)W^2_{G^{-1}}(\mu_\Theta, \nu)$ wherever the optimal plan is unique (Danskin's theorem, as in Theorem C′(i)), with the vertex-selection solutions of Section 13 where it is not. Its mean-field form is the* Laguerre-cell leak*: for $\mu$ giving no mass to their boundaries, the optimal plan from $\mu$ to the atomic $\nu$ is induced by the map $T_\mu(\vartheta) = \vartheta_j^0$ on the Laguerre cell*

$$L_j[\mu] := \big\{\vartheta : \|\vartheta - \vartheta_j^0\|^2_{G^{-1}} - \psi_j \le \|\vartheta - \vartheta_k^0\|^2_{G^{-1}} - \psi_k \text{ for all } k\big\},$$

*the dual weights $\psi$ fixed by $\mu(L_j[\mu]) = q_j$ (semi-discrete optimal transport [3]; for a disconnected support the boundaries in the gaps are not unique, but $T_\mu$ is unique on* $\operatorname{supp}\mu$*), and $(\mathfrak{A}_0^{\mathrm{OT}})$ is $(\mathfrak{A}_0)$ with the leak $-\sigma_0(\vartheta - T_{\mu_s}(\vartheta))$: the nearest-atom leak of Proposition D‴.9 with the Voronoi cells replaced by cells of prescribed masses. Call $m$* commensurate *with $\nu$ if $mq_j \in \mathbb{N}$ for every $j$.*

(i) (conservation against constraint) *Along $(\mathfrak{A}_m^\nu)$ on the good event $\mathcal{G}_S$ of Theorem D‴ (Proposition D‴.9(i)) no particle changes cell, so the cluster masses $w_k(\Theta_s) := (1/m)\#\{i : j(\vartheta_i(s)) = k\}$ are conserved: $w^m(s) = w^m(0)$, the multinomial sample of the initial cloud, and the stationary floor of Proposition D‴.5 is the memory of the initial condition that this conservation law carries. Along $(\mathfrak{A}_m^{\mathrm{OT}})$ the masses assigned to the atoms are constrained instead: $\sum_i \gamma_{ij}(s) = q_j$ for every $s$.*

(ii) (the frozen plan) *If $m$ is commensurate, every vertex of the transportation polytope is integral — $\gamma_{ij} \in \{0, 1/m\}$: each particle is assigned to one atom $\pi(i)$ and atom $j$ receives exactly $mq_j$ particles — so $\bar{T}_i = \vartheta^0_{\pi(i)}$ and $(\mathfrak{A}_m^{\mathrm{OT}})$ is $(\mathfrak{A}_m^\nu)$ with the Voronoi assignment $j(\cdot)$ replaced by the plan's assignment $\pi$, locally constant in $\Theta$ wherever the optimal plan is unique; there $J_m^{\mathrm{OT}}$ is $C^2$ with the leak's Hessian $\sigma_0 G^{-1} \otimes I$. Theorem D‴'s synchronous coupling applies to $(\mathfrak{A}_m^{\mathrm{OT}})$ only under the hypothesis of Proposition D‴.9(i) read with the plan — for every $s \geq 0$ the plan assigns each particle to the atom of its reference's Laguerre cell $L_\cdot[\mu_s]$, and $N_{\varrho'}(\operatorname{supp}\mu_s)$ meets no Laguerre boundary — and at $s = 0$ this requires the sample to be* balanced, *$w^m(0) = q$ (the mean-field flow of the mixture start has masses $q$ from the start): an event of probability of order $m^{-(K_0-1)/2}$ under an i.i.d. start — 0.053, 0.014, 0.0034, 0.0016 at $m = 15, 60, 240, 510$ on the instance, $\approx 0.83/m$. On that event Theorem D‴, Proposition D‴.5 and Corollary D‴.7 go through with the same constants except for Step 8, whose McDiarmid bound for the reference's sampling term is written for i.i.d. particles and would have to be redone for the stratified sample ($mq_k$ particles i.i.d. within each class). For a generic sample the plan reassigns at least $\frac{1}{2}\|w^m(0) - q\|_1$ of the mass at $s = 0$ and the shared-particle coupling carries that mismatch for ever ((v)); what is proved about the OT dynamics in general is the local equivalence above and the stationary statement (iii).*

(iii) (the stationary measure does not depend on $m$) *Let $\Theta^*(q) = (\vartheta_1^*, \dots, \vartheta_{K_0}^*)$ be the weighted stationary point of $\mathcal{G}_q$ with the prior atoms (Proposition D‴.5 with $w = q$, each class $k$ leaking toward $\vartheta_k^0$ as in Proposition D‴.9) and $\varepsilon := \max_k \|\vartheta_k^* - \vartheta_k^0\|_{G^{-1}}$. If $\varepsilon < \Delta_\nu/2$ then, for every commensurate $m$, the replicated configuration $\Theta_m^*(q)$ ($mq_k$ particles at $\vartheta_k^*$) has the diagonal plan (class $k \to$ atom $k$) as its unique optimal plan, is a critical point of $J_m^{\mathrm{OT}}$, and is a strict local minimiser iff $q \in W_{\mathrm{adm}}$ (its Hessian is Proposition D‴.5's, with separation blocks $\sigma_0 I - R_k^G$ of multiplicities $mq_k - 1$). The stationary measure $\mu^*(q) := \sum_k q_k \delta_{\vartheta_k^*}$ and the learned function $\hat{f}^*(q) = \sum_k q_k \tilde{\Phi}(\cdot\,; \vartheta_k^*)$ are therefore the same for every commensurate $m$, and the stationary floors vanish:*

$$W_{G^{-1}}(\mu_m^*, \mu_M^*) = 0, \qquad \|\hat{f}_m^* - \hat{f}_M^*\|_{L^2(\rho)} = 0 \qquad \text{for all commensurate } m, M,$$

*against $\mathbb{E}[D_\infty \mid \mathrm{adm}] = (1/m - 1/M)\sum_k \bar{w}_k \operatorname{tr}(\tilde{S}_k \tilde{C} \tilde{S}_k^\top)$ and $(1/m - 1/M)\operatorname{tr}(F\tilde{C}F^\top)/N$ under the nearest-atom leak (Propositions D‴.5 and D‴.9(iii)), whose stationary measures are moreover at squared $W_2$-distance $\frac{1}{2}\|w^m - w^M\|_1 \cdot O(\Delta_\nu^2) = O(\Delta_\nu^2 m^{-1/2})$ from each other ($W_2$ itself $O(\Delta_\nu m^{-1/4})$) — the cross-cluster transport of the weight mismatch, which the shared-particle distance $D$ does not see. The price is the bias of the masses: the stationary residual $r^*(q) = f - \sum_k q_k \tilde{\Phi}(\cdot\,; \vartheta_k^*)$ vanishes iff $\nu$ carries an exact representation of the plant* with its masses, *$\sum_k q_k \tilde{\Phi}(\cdot\,; \vartheta_k^0) = f$ on* $\operatorname{supp}\rho$*; for masses $q_\delta$ displaced from those of the exact representation the residual is $O(\delta)$ and the window's margin decreases with $\delta$ (Proposition D‴.9(ii) with the masses as the parameter), and the designed masses must themselves be admissible, $q \in W_{\mathrm{adm}}$: past the admissible set the over-weighted equal-mass cluster splits (Remark D‴.11).*

(iv) (incommensurate $m$: the quantisation remainder) *For any $m$ and any configuration a vertex optimal plan splits at most $K_0 - 1$ particles between two or more atoms (a vertex of the transportation polytope is supported on a forest of the bipartite graph with $m + K_0$ nodes, so on at most $m + K_0 - 1$ edges, every particle carrying at least one). At a stationary configuration whose unsplit particles are collapsed by class (the minimisers of Proposition D‴.5's type) a split particle sits where the fit force balances the pull toward its barycentric target $\bar{T}_i \in \operatorname{conv}\{\vartheta_j^0 : \gamma_{ij} > 0\}$ — a* bridge particle *between the clusters — so the stationary measure differs from $\mu^*(q)$ by at most $(K_0 - 1)/m$ of mass at the bridge positions and by the $O(K_0/m)$ shift of the class weights:*

$$W_{G^{-1}}^2(\mu_m^*, \mu^*(q)) \le 4(K_0 - 1)R_\Theta^2/m + O(K_0^2\|S\|^2/m^2),$$
$$\|\hat{f}_m^* - \hat{f}^*(q)\|_{L^2(\rho)} \le 2(K_0 - 1)K_\Phi/m + O(K_0(K_\Phi + L_\vartheta^G\|S\|)/m),$$

*$R_\Theta$ the a priori radius of Step 0 (with the hull of* $\operatorname{supp}\nu$*). The orders are the content, not the constants — on the instance $R_\Theta = 2865$ makes the first bound $4 \cdot 10^6$ against a measured 1.0–1.4 at $m = 16$, and the second 1.00 against 0.018: the function-level floor is $O(1/m^2)$ in square, one order in $m$ below the $(1/m - 1/M)$ law, and the $W^2$ floor is $O(1/m)$ against the nearest-atom leak's $O(m^{-1/2})$ of (iii) — half an order gained in $W^2$, a full order at function level — with deterministic constants set by the plan's split pattern (measured $m \cdot W^2(\mu_m^*, \nu) \approx 20 = 0.5\Delta_\nu^2$ and $m \cdot \|\hat{f}_m^* - f\|/\|f\| \approx 0.55$).*

(v) (the transit) *From a cloud with cluster masses $w^m \ne q$ the plan assigns at least $\frac{1}{2}\|w^m - q\|_1$ of the mass (up to the split particles) of the over-weighted clusters to under-weighted atoms, and these particles migrate between the clusters against the fit force at the leak's rate — for the pure leak a migrating particle crosses the Voronoi midpoint at $s = \ln 2/\sigma_0$. During the transit the synchronous coupling of Theorem D‴ sees the mass migrating in one system and not in the other at a distance of order $\Delta_\nu$, $D = O(\Delta_\nu^2\|w^m - w^M\|_1)$ — it is not a coupling the OT leak respects — $W_{G^{-1}}^2(\mu_m, \mu_M)$ is of the same order $O(\Delta_\nu^2 m^{-1/2})$, the sampling term of the cluster masses, and the function-level distance is $O(K_\Phi\|w^m - w^M\|_1) = O(m^{-1/2})$. After the transit the two measure-level distances vanish on the commensurate sizes ((iii)), while the synchronous $D$ converges from below to the plateau $\sum_{k \ne k'} p_{kk'}\Delta_{kk'}^2$ of the mass reassigned in one system and not in the other (0.95–0.96 of it at $s = 600$ on the instance): the memory of the initial masses, erased from the stationary measure, is kept by the shared-particle coupling. A mean-field flow whose clusters start with masses $w \ne q$ has the same transit — the Laguerre boundaries cut the over-weighted clusters and the mass $\frac{1}{2}\|w - q\|_1$ migrates — but the mixture start of the numerics has mean-field masses $q$, so there the $M$-references transit only as finite samples and are poor stand-ins for the mean-field flow during the transit.*

**Status.** (i) proved on $\mathcal{G}_S$ (Theorem D‴'s per-particle bound keeps every particle within $3\varrho'/4$ of its reference, whose cell has margin $\varrho'$). (ii) the local equivalence proved (the transportation constraints are the incidence matrix of a bipartite graph, totally unimodular, so with integral right-hand sides every vertex is integral — Hoffman–Kruskal [42]; a unique optimal vertex of a linear program stays optimal under small perturbations of its cost; the Laguerre form of the mean-field leak is semi-discrete optimal transport [3]); the coupling statement formal — the proofs of Theorem D‴ and Proposition D‴.5 use the leak only through its Hessian and its a priori bound, as in Proposition D‴.9(i), the constancy of the plan being needed for every $s \ge 0$ (referee round 10), which at $s = 0$ is the balanced-sample event of probability $\approx 0.83/m$, and on it Step 8 would have to be redone for the stratified sample (referee round 11). (iii) proved ($c$-cyclical monotonicity [86]: for $\varepsilon < \Delta_\nu/2$ every off-diagonal pair costs at least $\frac{1}{2}(\Delta_\nu - \varepsilon)^2 > \frac{1}{2}\varepsilon^2 \ge$ every diagonal pair, so along any cycle the diagonal plan is strictly cheaper; the diagonal plan is therefore optimal, and every other plan, which differs from it by a nonnegative combination of cycle flows, costs strictly more; the Hessian is Proposition D‴.5(i)'s block splitting with the plan frozen; the bias statements are Proposition D‴.9(ii)'s implicit-function argument with the masses

as the parameter). (iv) the count proved (the vertices of the transportation polytope are the plans supported on spanning forests, Klee–Witzgall [42]); the two bounds formal (they assume the stationary configuration's structure). (v) formal; the transit and the three distances are numerical (`exp_ot_leak.py` → `results/chaos/ot_leak.md`, below). Open: a coupling theorem for the OT dynamics from an i.i.d. start — Theorem D‴'s synchronous coupling needs a balanced sample and otherwise carries the transit's mismatch for ever (the plan reassigns $O(\sqrt{m})$ particles at $s = 0$, which the shared-particle pairing never re-matches); the natural candidate follows the plan, pairing a particle with its assigned atom's mean-field mass — so what is proved about the OT dynamics is the local equivalence (ii) and the stationary statement (iii). Checked: `tests/test_ot_leak.py` (the plan is a vertex with the prescribed masses, integral when $m$ is commensurate and with at most $K_0 - 1$ split rows otherwise, and the single-atom prior gives the $\sigma$-modification; the leak is the $G$-gradient flow of $J_m^{\mathrm{OT}}$ against finite differences, for commensurate and incommensurate $m$; the replicated weighted configuration is stationary for three commensurate sizes with the diagonal plan and the same learned function — for the matching masses and for mismatched ones — with $W^2 = 0$ between the sizes and the splitting margin of Proposition D‴.5 ($\sigma_0$ at the matching masses); the conservation of the cluster masses under the nearest-atom leak against their equalisation under the OT leak on a short run, with the mass in transit equal to $\frac{1}{2}\|w(0) - q\|_1$ and the fit error lower at the end; the forest bound on random clouds; the exact $W^2$ between empirical measures by replication against the transportation LP).

**Remark D‴.13** (conservation, constraint, and what the mean-field limit forgets)**.** Theorem D‴ bounds the coupled distance uniformly in time, and Proposition D‴.5 says what the bound's constant is made of: the $m$-system's memory of its initial cluster masses, which the nearest-atom leak conserves. The OT leak replaces the conservation law by a constraint and the memory is erased from the stationary measure — the finite-$m$ stationary measure *is* the mean-field one on the commensurate sizes, and within $(K_0 - 1)/m$ of mass of it otherwise — though not from the shared-particle coupling, whose synchronous distance keeps the plateau of the reassigned particles ((v)). This is a statement about the design of the leak, not about $m$: the same clouds under the nearest-atom leak keep their floors. Three consequences. (a) The masses join the atoms as design inputs of the leak's prior — both are read from the plant's structure (Proposition D‴.9(ii)) — and the admissible set $W_{\mathrm{adm}}$ of Proposition D‴.5, a probability statement about the realised weights under the nearest-atom leak, becomes a condition on the design, $q \in W_{\mathrm{adm}}$, under the OT leak. (b) The mean-field form of Theorem C′'s leak for an atomic prior is the Laguerre-cell leak; its finite-$m$ fluctuation is not a random displacement of the cells but the split particles — at most $K_0 - 1$ — and the migrating ones during the transit. (c) A soft mass constraint (an entropic or penalised plan) would interpolate between the two leaks, with a floor set by the stiffness of the penalty; open (Section 26, branch 8), as is the transfer of a learned measure to a plant whose representation weights are unknown, where $q$ itself would have to be learned.

**Numerics** (`exp_ot_leak.py` → `results/chaos/ot_leak.md`). The instance of Theorem D‴'s numerics at $\sigma_0 = 0.15\sigma_{\mathrm{crit}} = 0.00707$ with the matching prior of Proposition D‴.9's numerics — three atoms at the exact representation, masses $q = (1/3, 1/3, 1/3)$, the plant's representation weights, $\Delta_\nu = 6.26$ — under the nearest-atom leak and under the OT leak (plan recomputed every 0.2 slow units) on the same warm clouds (the jittered mixture of width 0.02), 600 slow units $= 4.2/\sigma_0$, four seeds; commensurate sizes $m = 15$, 60, 240 against $M = 510$ and incommensurate sizes $m = 16$, 64, 256 against $M = 512$.

*Conservation against constraint (table 1).* Under the nearest-atom leak no cluster weight changes along any run ($\max_s |w(s) - w(0)|_\infty = 0$ in all 32 systems). Under the OT leak the mass in transit at $s = 0$ is $\frac{1}{2}\|w(0) - q\|_1 = 0.117$, 0.058, 0.027, 0.024 (seed means for $m = 15$, 60, 240, 510; per system 1–3, 1–5, 2–13 and 3–21 particles), it reaches zero, and the weights by nearest atom equal $q$ exactly from $s_{\mathrm{eq}} = 93$–$164$ slow units on (median 104, against the pure leak's midpoint crossing $\ln 2/\sigma_0 = 98$); no plan is split. The fit error at the end falls from 0.067, 0.029, 0.017, 0.012 under the nearest-atom leak — the bias of the realised weights — to $2.8 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$,

$0.8 \cdot 10^{-4}$ (seed means); the migrated particles are still 0.06–0.16 whitened units from their new atoms at $s = 600$ — the pure leak's remaining approach $\Delta_{kk'} e^{-\sigma_0 \cdot 600} = 0.09$–$0.10$ — and the fit force cancels their first-order effect on $\hat{f}$, so the OT finals are not yet stationary (maximal whitened drift $5$–$8 \cdot 10^{-4}$, against $\leq 10^{-3}$ and mostly $\leq 10^{-4}$ under the nearest-atom leak).

*The three distances on the commensurate sizes (table 2, last window* $(450, 600]$*).* Function level $\|\hat{f}_m - \hat{f}_M\|^2_{L^2(\rho)}$: nearest-atom $2.5 \cdot 10^{-3}$, $5.2 \cdot 10^{-4}$, $5.9 \cdot 10^{-5}$ (stationary, last/previous window 0.97–1.00; the law $(1/m - 1/M)\,\mathrm{tr}(F\tilde{C}F^\top)/N$ predicts $4.0 \cdot 10^{-3}$, $9.1 \cdot 10^{-4}$, $1.4 \cdot 10^{-4}$) against OT $4.9 \cdot 10^{-7}$, $4.9 \cdot 10^{-8}$, $7.5 \cdot 10^{-9}$ — ratios $2 \cdot 10^{-4}$, $9 \cdot 10^{-5}$, $1.3 \cdot 10^{-4}$ — still falling by a factor 0.02–0.035 per 150 slow units (a rate of $3.0$–$3.8\sigma_0$: the displaced particles' first-order effect cancels, the class means agreeing to first order) toward the zero of (iii). $W^2_{G^{-1}}(\mu_m, \mu_M)$: nearest-atom 5.1, 2.7, 0.81 — the cross-cluster transport of the weight mismatch: the exact $K \times K$ transport between the weighted stationary measures at the realised weights reproduces it to 0.997–0.999 at the seven admissible pairs (0.85–0.99 at the five pairs with saddle weights, where the $m$-system is not at the replicated configuration), the formula $\frac{1}{2}\|w^m - w^M\|_1 \Delta^2_\nu = 4.9$, 2.5, 0.70 to $-7\%$ to $+28\%$; slope $-0.67 \pm 0.13$ in $m$, the sampling term of the cluster masses — against OT $7.9 \cdot 10^{-3}$, $2.3 \cdot 10^{-3}$, $6.9 \cdot 10^{-4}$ (ratios $1.5 \cdot 10^{-3}$, $0.9 \cdot 10^{-3}$, $0.9 \cdot 10^{-3}$), falling by 0.12 per 150 slow units ($e^{-2\sigma_0 \cdot 150} = 0.12$, the square of the spread's contraction at the margin $\sigma_0$). The synchronous $D$ is the one distance the OT leak does not improve: 5.8, 3.3, 0.94 against 0.87, 0.25, 0.017, still rising (last/previous 1.08–1.10) toward the plateau $\sum_{\text{mismatched}} \Delta^2_{kk'}/m$ of the pairs reassigned in one system and not in the other (0.95–0.96 of it at $s = 600$), as (v) says: the coupling keeps the memory that the measure loses.

*The incommensurate sizes (table 3).* Every plan has one or two split particles ($\leq K_0 - 1 = 2$); the weights by nearest atom differ from $q$ by the split particle's share, 0.042, 0.010, 0.0026, $0.0013 = (2/3)/m$; the unsplit particles in transit at $s = 0$ (0–2 at $m = 16$, 1–4 at 64, 2–11 at 256, 1–19 at 512) all arrive by the end. At the end the split particles sit between the clusters, at 0.28–0.58 $\Delta_\nu$ from the nearest atom and within 0.03–0.6 ($G^{-1}$ metric) of their barycentric target $\bar{T}_i$ — the bridge particles of (iv). Against the $M = 512$ reference, which carries its own two bridge particles: function level OT $3.4 \cdot 10^{-4}$, $3.1 \cdot 10^{-5}$, $1.1 \cdot 10^{-6}$ (the $m = 16$ value still relaxing, 0.82 per window — continued to $s = 2400$ by referee round 11 it settles at $2.8 \cdot 10^{-4}$, and $m = 64$ at $2.9 \cdot 10^{-5}$, with slope $-2.00$) against nearest-atom $2.5 \cdot 10^{-3}$, $5.0 \cdot 10^{-4}$, $4.4 \cdot 10^{-5}$, ratios 0.11–0.14, 0.063, 0.025; $W^2$ OT 1.16, 0.29, 0.041 (stationary) against nearest-atom 4.8, 2.7, 0.69, ratios 0.24, 0.10, 0.059. The quantisation laws of (iv) are read against the mean-field stationary measure $\mu^*(q)$, here $\nu$ itself: $W^2(\mu_m, \nu) = 1.21$, 0.33, 0.079, 0.038 for $m = 16$, 64, 256, 512, i.e. $m \cdot W^2 = 19.4$, 21.0, 20.1, $19.5 \approx 0.5\Delta^2_\nu$ with slope $-0.98 \pm 0.03$, and the fit error 0.028, 0.0093, 0.0023, 0.0011, i.e. $m \cdot \text{fit} = 0.45$–$0.60$ with squared slope $-1.81 \pm 0.05$ — the orders $1/m$ and $1/m^2$ of (iv), with constants set by the plan's split pattern (seeds with the same pattern reach identical plateaus, referee round 11) far below the worst case $(K_0 - 1)\Delta^2_\nu = 78$; against the nearest-atom leak's fit errors 0.061, 0.035, 0.013, 0.012.

*The masses as a design input (table 4).* For $q_\delta = (1/3 + \delta, 1/3 - \delta/2, 1/3 - \delta/2)$ the weighted stationary point has fit error 0.0069, 0.013, 0.019, 0.029, 0.049 at $\delta = 0.01$, 0.02, 0.03, 0.05, 0.1 — the bias of the mass constraint, $0.69\delta$ to $0.49\delta$, decreasing with $\delta$ — $\kappa_R$ at the over-weighted atom $+0.0023$, $+0.0043$, $+0.0060$, $+0.0086$, $+0.0119$, margin $\kappa$ $+0.0047$, $+0.0026$, $+0.0008$, $-0.0019$, $-0.0054$ and splitting margin $+0.0048$, $+0.0028$, $+0.0011$, $-0.0015$, $-0.0049$: the designed masses leave the admissible set between $\delta = 0.03$ and 0.05 — the asymmetry of the matching prior's admissible set in Proposition D‴.9's numerics (an excess of 0.04 on the largest bump is inadmissible) — while the diagonal plan stays the unique optimal plan at every row ($\varepsilon \leq 1.04 < \Delta_\nu/2 = 3.13$).

*Reading.* The two leaks start from the same clouds and end in different worlds. The nearest-atom leak keeps the clusters' sampled masses for ever, and its stationary distances are what Propositions D‴.5 and D‴.9 say: a function-level floor of order $1/m$, and a $W^2$ floor of order $m^{-1/2}$ set by the cross-cluster transport of the weight mismatch. The OT leak moves one to 21

particles across in about $\ln 2/\sigma_0$ slow units and then settles on the mean-field stationary measure: on the commensurate sizes every distance but the synchronous one decays to zero (three to four orders of magnitude below the nearest-atom leak's floors after $4.2/\sigma_0$), and the fit loses the weight bias; on the incommensurate sizes the floors are the one or two bridge particles — $O(1/m^2)$ at function level, $O(1/m)$ in $W^2$ — whose position between the clusters is the price of a mass constraint the size cannot meet. The masses are a design input with a bias and an admissibility constraint of their own.

**Corollary D‴.14** (the real loop with the OT leak: one orbit for every commensurate size)**.** *Consider the real closed loop* (CL) *of Theorem J with the leak $\sigma(\vartheta_i - T_i)$, $\sigma = \sigma_0/k_e$, where $T_i$ is the nearest-atom target $\vartheta^0_{j(\vartheta_i)}$ (Proposition D″.9) or the barycentric target $\bar{T}_i$ of the vertex plan (Proposition D‴.12). For a weight vector $w$ with $mw \in \mathbb{N}^{K_0}$ and a class map $\pi_0 : \{1,\dots,m\} \to \{1,\dots,K_0\}$ with $|\pi_0^{-1}(k)| = mw_k$, let $\mathcal{C}_m(\pi_0) := \{\Theta : \vartheta_i = \vartheta_{\pi_0(i)}\}$ be the* collapsed subspace*, on which $\hat{f}_\Theta = \sum_k w_k \tilde{\Phi}(\cdot\,;\vartheta_k)$, and let the* weighted $K_0$-atom loop ($\mathrm{CL}_w$) *be* (CL) *with $\hat{f} = \sum_k w_k \tilde{\Phi}(\cdot\,;\vartheta_k)$, the $K_0$ parameters $\vartheta_k$ and the leaks $\sigma(\vartheta_k - \vartheta^0_k)$.*

(i) (invariance) *$\mathcal{C}_m(\pi_0)$ is invariant under* (CL) *with the nearest-atom leak as long as every class atom $\vartheta_k$ stays in the Voronoi cell of $\vartheta^0_k$, and under* (CL) *with the OT leak as long as $w = q$ and $\max_k \|\vartheta_k - \vartheta^0_k\|_{G^{-1}} < \Delta_\nu/2$; on it* (CL) *is* ($\mathrm{CL}_w$)*. With the OT leak and $w \neq q$ the subspace is not invariant: the plan assigns at least $\frac{1}{2}\|w - q\|_1$ of the mass of the over-weighted classes to other atoms, and those particles leave their class.*

(ii) (one orbit for every commensurate size) *Let $q \in W_{\mathrm{adm}}$ and let $\Gamma_q$ be the $T_0$-periodic orbit of* ($\mathrm{CL}_q$) *near $\Theta^*(q)$ given by Theorem J, whose proof applies to* ($\mathrm{CL}_w$) *verbatim (its averaged drift is $-(1/k_e)(\mathrm{diag}(w)^{-1} \otimes G)\nabla\mathcal{G}_w$, the gradient of $\mathcal{G}_w$ in the metric $\mathrm{diag}(w) \otimes G^{-1}$, which is all its Steps 4–6 use), and assume $\Gamma_q$ stays inside the cells, $\max_k \|\vartheta^\Gamma_k(t) - \vartheta^0_k\|_{G^{-1}} < \Delta_\nu/2$ — the cell condition of (i), which $W_{\mathrm{adm}}$ does not contain. For every commensurate $m$ the lift of $\Gamma_q$ to $\mathcal{C}_m(\pi_0)$ is a $T_0$-periodic orbit of* (CL) *with the OT leak. It is exponentially stable iff $\Gamma_q$ is and the ghost monodromies of the classes contract (Proposition D‴.15 below) — at the matching masses, where the residual vanishes, the ghosts' exponent is exactly $\sigma_0/k_e > 0$ — and it is Theorem J's orbit $\Gamma_m$ at $\Theta^*_m(q)$, with the basin (c) of that theorem, whenever it lies in Theorem J's uniqueness ball $\|\Theta - \Theta^*_m(q)\|_{G_m^{-1}} \le \rho_0(m)$: with $\|\Theta\|^2_w := \sum_k q_k \|\vartheta_k\|^2_{G^{-1}}$ the lift's $G_m^{-1}$-distance to $\Theta^*_m(q)$ is $\sqrt{m}$ times the weighted distance, so the condition is $\sqrt{m} \cdot \sup_t \|\Theta^q_p(t) - \Theta^*(q)\|_w \le \rho_0(m)$, satisfied for $k_e \ge \max\{k_0(m), \sqrt{m}\, C_q/\rho_0(m)\}$ by Theorem J(a)'s $\sup_t \|\Theta^q_p(t) - \Theta^*(q)\|_w \le C_q/k_e$. Along the lift the orbit measure $(1/m)\sum_i \delta_{\vartheta_i(t)} = \sum_k q_k \delta_{\vartheta_k(t)}$, the learned function $\sum_k q_k \tilde{\Phi}(\cdot\,;\vartheta_k(t))$, the plant's orbit $x_p(t)$ and the tracking error $e(t)$ are the same for every commensurate $m$ at every $t$ — exactly, not up to $O(1/k_e)$: the finite-size floor of the real loop (Corollary D″.7) is zero.*

(iii) (Corollary D‴.7 sharpened) *With the nearest-atom or the $\sigma$-modification leak and both realised weight vectors admissible, $w^m, w^M \in W_{\mathrm{adm}}$ (Corollary D″.7's hypothesis: the replicated configurations are hyperbolic minima), the same argument lifts $\Gamma_{w^m}$, the orbit of* ($\mathrm{CL}_{w^m}$) *with the realised weights: Theorem J's orbit near $\Theta^*_m(w^m)$ is the lift of $\Gamma_{w^m}$ under the same proviso, so the real loop's coupled distance between two sizes is exactly the distance between two weighted $K_0$-atom orbits,*

$$D_{\mathrm{orb}}(t) = \sum_k w^m_k \|\vartheta^\Gamma_k(t; w^m) - \vartheta^\Gamma_k(t; w^M)\|^2_{G^{-1}},$$

*Corollary D″.7's $O(1/k_e)$ terms being the $k_e$-dependence of $w \mapsto \Gamma_w$ and nothing else; and the tracking error along the orbit is that of* ($\mathrm{CL}_{w^m}$) *— the weight bias of the fit is a tracking error, which the mass constraint removes for every size at once. When $w^m \notin W_{\mathrm{adm}}$ the lift is a saddle orbit (a ghost exponent is negative, Proposition D‴.15) and the $m$-system*

*leaves it by splitting its over-weighted class, as three of the six nearest-atom systems of the numerics below do.*

**Status.** (i) proved (the particle law depends on $i$ only through $\vartheta_i$ and its target; on $\mathcal{C}_m(\pi_0)$ the targets agree within a class — by the cell condition for the nearest-atom leak, by the lemma of Proposition D‴.12(iii) for the OT leak — so the vector field is tangent to $\mathcal{C}_m(\pi_0)$ and coincides there with $(\mathrm{CL}_w)$'s, and solutions are unique away from the cell boundaries and the plan switches). (ii), (iii) proved from (i), Proposition D‴.15 and Theorem J under the stated proviso and hypotheses (the cell condition along $\Gamma_q$; admissible weights in (iii)); the $m$-uniformity of Theorem J's thresholds $k_0(m)$, $\rho_0(m)$ at the replicated configuration is the open point of Corollary D‴.7 — its spectral part is settled by Proposition D‴.15 (the Floquet spectrum of the lift does not depend on $m$), the basin radius in the mean norm remains formal; without the proviso the lift is still a periodic orbit of (CL) for every commensurate $m$, with the stability Proposition D‴.15 gives it, and only its identification with Theorem J's orbit is conditional. Checked: `tests/test_ot_leak.py` (the collapsed 15-system and the 3-atom loop integrate to the same configuration and plant state to $10^{-12}$ over 300 steps of the real loop with the OT leak at masses $q$, the atoms moving; the collapsed 15-system with weights $(6,5,4)/15$ stays collapsed under the nearest-atom leak, while under the OT leak one of its particles is assigned away from its class; two commensurate lifts, $m = 15$ and 30, keep $W^2 = 0$ and the same learned function along the real loop; the lifts of two weight vectors under the nearest-atom leak are different measures).

**Numerics** (`exp_ot_real_loop.py` → `results/chaos/ot_real_loop.md`). The real loop at $k_e = 25$ ($\sigma = \sigma_0/k_e$; RK4 plant with the control and the reference evaluated at the stages and the parameters frozen over the step, Euler parameters, $\mathrm{d}t = 10^{-3}$ — an earlier zero-order hold of the control had set a tracking floor of $4 \cdot 10^{-5}$, referee round 12) for 100 time units — 16 periods, 4 slow units — from the clouds at the end of Proposition D‴.12's averaged runs (contracted where the realised weights are admissible) (seeds 0 and 1; $m = 15$, 60 and $M = 510$; the OT runs' clouds carry the masses $q$, the nearest-atom runs' their conserved sampled weights) and from the collapsed lifts (the exact representation for the OT leak — the 3-atom loop — and the replicated $\Theta^*(w^m)$ for the nearest-atom leak); statistics over the last five periods on the reference orbit. Under the OT leak the learned functions of the three sizes agree to $\|\hat{f}_m - \hat{f}_M\|^2_{L^2(\mathrm{orbit})} = 7.5 \cdot 10^{-9}$–$2.7 \cdot 10^{-8}$ and with the 3-atom lift to $5.2 \cdot 10^{-10}$–$3.6 \cdot 10^{-8}$ — the clouds' within-class spread (0.013–0.033, the migrated particles still approaching their atoms) at second order — with fit errors along the orbit $3.5 \cdot 10^{-5}$–$2.9 \cdot 10^{-4}$ and rms tracking errors $0.9 \cdot 10^{-6}$–$7.3 \cdot 10^{-6}$ (the fit error over $k_e$; the collapsed lift tracks to $8 \cdot 10^{-12}$, the scheme's floor), and $W^2(\mu_m, \mu_M) = 3.3 \cdot 10^{-4}$–$1.4 \cdot 10^{-3}$ at the end. Under the nearest-atom leak they disagree by $1.1 \cdot 10^{-4}$–$2.5 \cdot 10^{-3}$, with fit errors 0.012–0.075 and rms tracking errors $3.2 \cdot 10^{-4}$–$1.9 \cdot 10^{-3}$ — the weight bias as a tracking error, 130 to 550 times the OT leak's per system — and $W^2(\mu_m, \mu_M) = 1.6$–$5.5$, the cross-cluster transport of the weight mismatch. Three of the six nearest-atom systems have inadmissible realised weights (seed 0, $m = 15$: $(6,5,4)/15$, splitting margin $-0.0035$; seed 1, $m = 15$: $(6,3,6)/15$, $-0.0011$; seed 1, $m = 60$: $(24,15,21)/60$, $-0.0022$): their lifts are saddle orbits and the $m$-systems are not on them — the over-weighted class has split (within-class spread 0.04–0.11 against 0.0006–0.0013 for the admissible systems), the distance to the lift, $4 \cdot 10^{-8}$–$3 \cdot 10^{-6}$, being the split's second-order effect on $\hat{f}$, and the lift-to-lift ratio at those pairs 0.960, 0.998, 0.961. At the one pair whose weights are both admissible (seed 0, $m = 60$ against $M = 510$) the disagreement is the distance between the lifted orbits to a ratio 1.0003, and the admissible $m$-systems sit within $2.5 \cdot 10^{-11}$ of their lifts: (iii) as stated, with its admissibility hypothesis visibly necessary.

**Numerics in two dimensions** (`exp_ot_2d.py` → `results/chaos/ot_2d.md`). The mechanism is dimension-free, and the real loop shows it on a planar plant: $\dot{x} = f(x) + u$ on $\mathbb{R}^2$ with $f$ the sum of four Gaussian bumps with vector amplitudes, the Lissajous reference $x_m(t) =$

$(2\sin t, 1.5\sin 2t)$, a saturated network with $p = 5$ parameters per kernel, $G = \mathrm{diag}(1, 1, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, the matching prior (four atoms at the exact representation, masses $\frac{1}{4}$), $k_e = 25$ and $\sigma_0 = 0.05$ (the leak is a design input; $\ln 2/\sigma_0 = 14$ slow units), 2500 time units $= 100$ slow units $= 5/\sigma_0$ from the same jittered warm clouds with random class labels, sizes $m = 8$, 32 against $M = 128$, two seeds, the plant integrated as above. Under the nearest-atom leak the sampled cluster weights — $(3, 2, 2, 1)/8$, $(7, 4, 12, 9)/32$, … — never change, the fit errors along the orbit are 0.11–0.39, the sizes disagree by $\|\hat{f}_m - \hat{f}_M\|^2_{L^2(\mathrm{orbit})} = 5.5 \cdot 10^{-3}$–$2.3 \cdot 10^{-2}$ and the rms tracking error is $1.9 \cdot 10^{-3}$–$6.7 \cdot 10^{-3}$. Under the OT leak the weights reach $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$ at 13–14 slow units in all six systems ($\ln 2/\sigma_0 = 14$), the fit errors fall to $2.1 \cdot 10^{-4}$–$8.7 \cdot 10^{-4}$ (the within-class spread 0.012–0.018 still contracting), the sizes agree to $5.0 \cdot 10^{-8}$–$1.6 \cdot 10^{-7}$ at the stopping time $5/\sigma_0$ — five orders of magnitude below the nearest-atom leak, and still falling by about 3.5 times per 10 slow units ($1.8 \cdot 10^{-12}$ at $10/\sigma_0$ in referee round 13's continued run) — and the rms tracking error is $3.9 \cdot 10^{-6}$–$1.6 \cdot 10^{-5}$, 290 to 480 times smaller per system: the weight bias is a tracking error in any dimension, and the mass constraint removes it for every size at once.

**Proposition D‴.15** (the transverse spectrum of a lifted orbit)**.** *Let $\Gamma = (x_p, \Theta_p)$ be a $T_0$-periodic orbit of* (CL) *— with any leak whose targets are locally constant near $\Gamma$ and equal within each class — lying in the collapsed subspace $\mathcal{C}_m(\pi_0)$ with classes of sizes $n_k$, and let $\Gamma_K = (x_p, (\vartheta_{p,k})_k)$ be the $K_0$-atom orbit it lifts. For each class $k$ let*

$$S_k := \Big\{\delta \in \mathbb{R}^{n+mp} : \delta x = 0,\ \delta_i = 0 \text{ for } i \notin \pi_0^{-1}(k),\ \sum_{i \in \pi_0^{-1}(k)} \delta_i = 0\Big\} \qquad \text{(dimension } (n_k - 1)p\text{)},$$

*and let $M_k$ be the monodromy over one period of the* ghost equation $\dot{\delta} = A_k(t)\delta$, $A_k(t) := \partial_\vartheta\big[G\nabla_\vartheta\big(e_p(t)^\top \tilde{\Phi}(x_p(t);\vartheta)\big)\big]\big|_{\vartheta=\vartheta_{p,k}(t)} - \sigma I$ *— the linearisation of one particle of class $k$ driven by the orbit's plant state, with weight zero in $\hat{f}$. Then the monodromy matrix $\mathbb{M}$ of $\Gamma$ leaves every $S_k$ invariant, with $\mathbb{M}|_{S_k} = M_k \otimes I_{n_k-1}$, and on the complementary invariant subspace — the plant coordinates and the class means — it is the monodromy $\mathbb{M}_K$ of $\Gamma_K$ as an orbit of the weighted $K_0$-atom loop* ($\mathrm{CL}_w$)*. Hence the Floquet spectrum of $\Gamma$ is that of $\Gamma_K$ together with the spectrum of each $M_k$ with multiplicity $n_k - 1$, independent of $m$; $\Gamma$ is exponentially stable iff $\Gamma_K$ is and every $M_k$ is contracting; and when $\Gamma_K$ is Theorem J's orbit of* ($\mathrm{CL}_w$) *at $\Theta^*(w)$, the exponents of $M_k$ are $(1/k_e)\lambda_j(\sigma_0 I - R_k^G) + O(k_e^{-2})$, $j = 1..p$, with $R_k^G = G^{1/2}R[\mu_{\Theta^*(w)}](\vartheta_k^*)G^{1/2}$ — Proposition K's separation condition and the splitting blocks of Proposition D‴.5(i), read along the orbit.*

*Proof.* Write the particle law as $\dot{\vartheta}_i = F(x, e, \vartheta_i) - \sigma(\vartheta_i - T_i)$, $F(x, e, \vartheta) := G\nabla_\vartheta(e^\top \tilde{\Phi}(x;\vartheta))$, and note that the plant's equation depends on $\Theta$ only through $\hat{f}_\Theta(x) = (1/m)\sum_i \tilde{\Phi}(x;\vartheta_i)$. Along $\Gamma$ the linearised equations are $\delta\dot{x} = (\partial_x f - k_e I - \partial_x \hat{f})\delta x - (1/m)\sum_i \partial_\vartheta \tilde{\Phi}(x_p;\vartheta_i)\delta_i$ and $\dot{\delta}_i = (\partial_x F + \partial_e F)\delta x + (\partial_\vartheta F - \sigma I)|_{\vartheta_i}\delta_i$ ($e = x - x_m$ gives $\delta e = \delta x$). For $\delta \in S_k$, $\delta x = 0$ and all $\vartheta_i$ with $i \in \pi_0^{-1}(k)$ equal $\vartheta_{p,k}(t)$, so $\sum_i \partial_\vartheta \tilde{\Phi}(x_p;\vartheta_i)\delta_i = \partial_\vartheta \tilde{\Phi}(x_p;\vartheta_{p,k})\sum_{i\in\pi_0^{-1}(k)}\delta_i = 0$: the plant is not perturbed to first order, each $\delta_i$ obeys $\dot{\delta}_i = A_k(t)\delta_i$ with the same $A_k$, and $\sum_{i\in\pi_0^{-1}(k)}\dot{\delta}_i = A_k\sum\delta_i = 0$. So $S_k$ is invariant and the linearised flow acts on it as the ghost equation on every component, whence $\mathbb{M}|_{S_k} = M_k \otimes I_{n_k-1}$. The class-constant perturbations $\{\delta_i = u_{\pi_0(i)}\}$ together with $\delta x$ form the complementary subspace; it is invariant because particles of one class have identical right-hand sides when their perturbations coincide, and there the variation of $\hat{f}$ is $\sum_k (n_k/m)\partial_\vartheta \tilde{\Phi}(x_p;\vartheta_{p,k})u_k = \sum_k w_k \partial_\vartheta \tilde{\Phi} u_k$ — the linearisation of ($\mathrm{CL}_w$) along $\Gamma_K$. The spectral statements follow. For the last one, along Theorem J's orbit of ($\mathrm{CL}_w$) the tracking error is $e_p(t) = r_{\Theta^*(w)}(x_m(t))/k_e + O(k_e^{-2})$ and $\vartheta_{p,k}(t) = \vartheta_k^*(w) + O(1/k_e)$ (Theorem J(a) with the error equation), so the ghost matrix is $A_k(t) = (1/k_e)\big[G\partial_\vartheta^2\big(r_{\Theta^*(w)}(x_m(t))\tilde{\Phi}(x_m(t);\vartheta_k^*)\big) - \sigma_0 I\big] + O(k_e^{-2})$: a $T_0$-periodic matrix of size $O(1/k_e)$ whose period mean is $(1/k_e)(GR_k - \sigma_0 I) + O(k_e^{-2})$, with $R_k = R[\mu_{\Theta^*(w)}](\vartheta_k^*)$ the residual-weighted curvature of Theorem D‴. First-order averaging of

this linear periodic equation gives $M_k = \exp(T_0(1/k_e)(GR_k - \sigma_0 I)) + O(k_e^{-2})$, and $GR_k - \sigma_0 I$ is similar to $G^{1/2}R_kG^{1/2} - \sigma_0 I = -(\sigma_0 I - R_k^G)$: the exponents are $(1/k_e)\lambda_j(\sigma_0 I - R_k^G) + O(k_e^{-2})$, with no condition on $m$. □

**Status.** Proved (linear algebra along the orbit; first-order averaging of the ghost equation for the last claim, which needs only that $\Gamma_K$ is Theorem J's orbit of $(\mathrm{CL}_w)$ — referee round 12 confirmed the finite-difference ghost monodromy against the matrix ODE to $3 \cdot 10^{-9}$ and against $\exp(T_0\bar{A}_k)$ to $7 \cdot 10^{-9}$, the prediction's remaining error being $R$ at $\Theta^*(w)$ against $R$ along $\Theta_p(t)$, falling 6–12 times from $k_e = 25$ to 50). The $m$-uniformity of the Floquet spectrum settles the spectral part of Corollary D‴.7's open point: the stability type and the rates of Theorem J(b)–(c) at a replicated configuration do not depend on $m$. Checked: `tests/test_ot_leak.py` (the lift of the 3-atom orbit to $m = 6$ at the matching masses is a fixed point of the Poincaré map to $10^{-10}$; each class's separation subspace is invariant under its monodromy to $10^{-6}$ and carries the ghost's exponents, $\sigma_0/k_e = 2.83 \cdot 10^{-4}$, to $10^{-6}$; the complement carries the 3-atom loop's exponents to $10^{-6}$).

**Numerics** (`exp_lifted_orbit.py` → `results/chaos/lifted_orbit.md`). The real loop at $k_e = 25$, $\sigma = \sigma_0/k_e$; monodromy matrices by central differences of the Poincaré map (DOP853, rtol $10^{-11}$). (A) The OT leak at the matching masses: the 3-atom orbit is the exact representation with zero tracking error, and its lifts to $m = 6$, 9, 15 are fixed points of the Poincaré map to $10^{-15}$; every class's separation subspace is invariant to $2 \cdot 10^{-8}$ and carries the exponent $\sigma_0/k_e = 2.830 \cdot 10^{-4}$ with multiplicity $n_k - 1$ — the ghosts' exponent to $4 \cdot 10^{-5}$ relative — and the collective exponents $2.935 \cdot 10^{-4}$, $3.558 \cdot 10^{-4}$, $3.999 \cdot 10^{-4}$, $8.73 \cdot 10^{-4}$, $1.109 \cdot 10^{-3}$, $1.573 \cdot 10^{-3}$, $3.704 \cdot 10^{-3}$, $5.523 \cdot 10^{-3}$, $7.934 \cdot 10^{-3}$ are the same for $m = 3$, 6, 9 and 15 to $10^{-4}$ relative, within 0.6–3.2% of Theorem J(b)'s $\lambda_j(H_q)/k_e = 2.952 \cdot 10^{-4}, \ldots, 8.091 \cdot 10^{-3}$, $H_q$ the whitened, weighted collective block of Proposition D‴.5(i). (B) The $\sigma$-modification at $w = (7, 5, 4)/16$ (admissible, splitting margin +0.00257) and $(9, 4, 3)/16$ (a saddle, −0.00219): the weighted 3-atom orbits by shooting (residuals $10^{-14}$, period-start offsets from $\Theta^*(w)$ of $1.1 \cdot 10^{-2}$ and $1.6 \cdot 10^{-2}$ — Theorem J(a)'s $O(1/k_e)$), and their lifts to $m = 16$ fixed points to $5 \cdot 10^{-14}$. The separation exponents of each class equal the ghost's to $2 \cdot 10^{-5}$ relative (invariance defects $\leq 6 \cdot 10^{-8}$) and the prediction $(\sigma_0 - \lambda_j(R_k^G))/k_e$ within 0.2–4.1% (class 0 at $(7, 5, 4)/16$: $9.989 \cdot 10^{-5}$, $4.553 \cdot 10^{-4}$, $1.267 \cdot 10^{-3}$ against $1.029 \cdot 10^{-4}$, $4.572 \cdot 10^{-4}$, $1.275 \cdot 10^{-3}$); at $(9, 4, 3)/16$ the over-weighted class carries the negative exponent $-9.13 \cdot 10^{-5}$ (predicted $-8.77 \cdot 10^{-5}$) with multiplicity 8 — the saddle orbit of Theorem D‴'s inadmissible seed, Proposition K's split at the orbit level — while the collective exponents $(2.131 \cdot 10^{-4}, \ldots, 9.985 \cdot 10^{-3})$ are the weighted 3-atom orbit's to $2 \cdot 10^{-5}$ relative.

**Remark D‴.16** (dither and the mass constraint)**.** With dither (Theorem J′: $\tau = \tau_0/k_e$, $\sigma = \sigma_0/k_e$) the leading-order stationary law of the slow variables is the Gibbs measure $\pi \propto \exp(-J_m/\tau_0)$, and the cluster masses are no longer conserved under the nearest-atom leak: a particle crosses a Voronoi ridge at the Arrhenius rate $\exp(-\Delta V/\tau_0)$, $\Delta V$ the single-particle barrier out of the configuration — at least the leak's part $\sigma_0\Delta_\nu^2/8 = (\sigma_0/2)(\Delta_\nu/2)^2$ at the lowest ridge point, plus the fit's variation — and the counts sample the Gibbs law over the count vectors $n$: in the Laplace approximation multinomial$(n) \cdot \exp(-m\mathcal{G}^*_{n/m}/\tau_0)$ times the determinants of the separation and collective blocks of Proposition D‴.5(i), for $n/m \in W_{\mathrm{adm}}$ (at a saddle weight the split configuration lies lower, and its soft and unstable separation modes carry entropy that the quadratic approximation misses). Dither thus does on the Arrhenius time scale, and only in distribution, what the OT leak does deterministically in $\ln 2/\sigma_0$ slow units at $\tau_0 = 0$. Under the OT leak the Gibbs measure is $\pi^{\mathrm{OT}} \propto \exp(-J_m^{\mathrm{OT}}/\tau_0)$ ($J_m^{\mathrm{OT}}$ is locally Lipschitz, piecewise smooth, with Gaussian tails), the plan's masses are $q$ by construction, and a particle that crosses a Voronoi ridge stays assigned to its own atom: with integral plans an exchange of the assignments of particles $a \in$ class $j$ and $b \in$ class $k$ lowers the cost only once $\langle \vartheta_a - \vartheta_b, \vartheta_k^0 - \vartheta_j^0 \rangle_{G^{-1}} > 0$, that is, once they have passed each other. So the counts by nearest atom leave $q$ at the nearest-atom leak's ridge barrier, but only for a reversible excursion — beyond the ridge the leak toward the

particle's own atom keeps pulling it back — of stationary probability of order $\exp(-\Delta V_{\mathrm{ridge}}/\tau_0)$; exchanges of assignments (whose barrier is at most the lowest tie point of two particles) leave the masses at $q$ as well. The mass constraint does not raise the barrier of a crossing; it makes the crossing reversible.

**Status.** Formal (the Gibbs law of a locally Lipschitz potential with Gaussian tails; the Arrhenius estimate; the Laplace asymptotics).

**Numerics** (`exp_dither_ot.py` → `results/chaos/dither_ot.md`): the dithered averaged slow loop — Langevin in slow time (Euler–Maruyama, $\mathrm{d}s = 0.05$), with the Gibbs law $\exp(-J_m/\tau_0)$ — for $m = 15$ particles started at the counts $(6, 5, 4)$ under the matching prior at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$, at $\tau_0 = 0.005$, 0.01, 0.02 over 50 000, 200 000 and 50 000 slow units, two seeds. On the instance the single-particle barriers out of the balanced configuration — the lowest points of the Voronoi ridges with the fourteen other particles relaxed — are 0.040–0.074 (the leak's part 0.035), and the lowest exchange barrier between two classes is 0.083 (referee round 13). Under the nearest-atom leak the record-level label changes (every 10 slow units; about half of them undone at the next change, recrossings) occur at $1.0 \cdot 10^{-5}$, $5.1 \cdot 10^{-4}$, $5.6 \cdot 10^{-3}$ per slow unit — an Arrhenius barrier of 0.048 (0.043–0.052 at 95%) from $\tau_0 = 0.01$ and 0.02, among the ridge minima, while $\tau_0 = 0.005$ has one event in 100 000 slow units — so at $\tau_0 = 0.005$ the counts $(6, 5, 4)$ stand for $350/\sigma_0$, and at $\tau_0 = 0.01$ and 0.02 the counts sample a distribution whose weight on $(5, 5, 5)$ is 0.54 and 0.25, against 0.58 and 0.33 for the energy–multinomial law $\mathrm{multinomial}(n) \cdot \exp(-m\mathcal{G}_n^*/\tau_0)$ — a comparison that cannot discriminate at $\tau_0 = 0.01$ (an integrated autocorrelation time of 7 400–23 000 slow units leaves 9–27 effective samples per seed, a standard error of about 0.1) and puts the law 2.1 standard errors above the data at $\tau_0 = 0.02$; the law under-weights the saddle count vectors $(6, 4, 5)$ and $(6, 5, 4)$ by 3.2 and 2.7 times (total-variation distance 0.20, against 0.07 between the two seeds): their soft and unstable separation modes carry the excess, the very factors the law omits. Under the OT leak the counts by nearest atom are balanced in every record after the initial transit at $\tau_0 = 0.005$ (no excursion in 50 000 slow units; 99.6% of the whole run), and in 99.4% and 88% of the records after it at $\tau_0 = 0.01$ and 0.02; the excursions are returns of the crossing particle — all 121 at $\tau_0 = 0.01$, 298 of 336 at 0.02, the other 38 exchanges — the label changes are about as frequent as the nearest-atom leak's (ratios 1.18–1.30), and the off-balance fraction ($6.3 \cdot 10^{-3}$, 0.117) has an effective barrier of 0.058, inside the ridge range.

**Proposition D‴.17** (convergence of the OT dynamics)**.** *Let $J_m^{\mathrm{OT}}$ be the energy of Proposition D‴.12 for the saturated Gaussian features, with $\rho$ the finite quadrature of the code or the period average of an analytic reference, on the a priori ball of Step 0.*

(i) (a Lyapunov function with a semiconcave part) $J_m^{\mathrm{OT}} = m\bar{L}(\mu_\Theta) + (\sigma_0 m/2)\min_\gamma \sum_{ij} \gamma_{ij}\|\vartheta_i - \vartheta_j^0\|_{G^{-1}}^2$ *is the minimum over the finitely many vertices $\gamma$ of the transportation polytope of the smooth energies $J_m^\gamma$, so it is locally Lipschitz, the sum of a smooth function and a semiconcave one (a quadratic in $\Theta$ minus a maximum of linear functions), and its Clarke subdifferential at $\Theta$ is the convex hull of the gradients $\nabla J_m^\gamma(\Theta)$ of the active vertices. The vertex-selection solutions of $(\mathfrak{A}_m^{\mathrm{OT}})$ solve the differential inclusion $\mathrm{d}\Theta/\,\mathrm{d}s \in -G_m \partial J_m^{\mathrm{OT}}(\Theta)$; along them $J_m^{\mathrm{OT}}$ is non-increasing, with $\mathrm{d}J_m^{\mathrm{OT}}/\,\mathrm{d}s = -\|\nabla J_m^\gamma(\Theta)\|_{G_m}^2$ wherever the plan $\gamma$ is unique, and at a tie the directional derivative along either active plan's drift is at most $-\|\nabla J_m^\gamma\|_{G_m}^2$.*

(ii) (convergence) $J_m^{\mathrm{OT}}$ *is definable in the o-minimal structure $\mathbb{R}_{\mathrm{an,exp}}$ on the a priori ball — finitely many compositions of polynomials, exponentials and the saturations* tanh *and* sig*, a finite quadrature sum (or an analytic function of $\Theta$ for the period average of an analytic reference), and a minimum over finitely many such functions; for the finite quadrature it is definable in $\mathbb{R}_{\exp}$ on all of $\mathbb{R}^{mp}$, and for the period average of an analytic*

*reference in* $\mathbb{R}_{\mathrm{an}}$ *on the ball — so it satisfies the Kurdyka–Łojasiewicz inequality in the nonsmooth sense [8]. The change of variables* $Y = G_m^{-1/2}\Theta$ *maps* $\mathrm{d}\Theta/\,\mathrm{d}s \in -G_m\partial J$ *into* $\mathrm{d}Y/\,\mathrm{d}s \in -\partial(J \circ G_m^{1/2})(Y)$*, the form in which the theorems of [8] are stated; definable locally Lipschitz functions obey the chain rule along absolutely continuous curves, so along a subgradient trajectory* $\mathrm{d}J/\,\mathrm{d}s = -\|\,\mathrm{d}\Theta/\,\mathrm{d}s\|^2_{G_m^{-1}}$ *[8], and with the KL inequality every bounded subgradient trajectory has finite length and converges to a Clarke-critical point. Every vertex-selection and every generalised solution of* $(\mathfrak{A}_m^{\mathrm{OT}})$ *is such a trajectory, bounded by Step 0's a priori estimate with* $\operatorname{conv}\operatorname{supp}\nu$*; hence it converges, and so do the solutions of* $(\mathfrak{A}_m^{\nu})$ *(a minimum over the Voronoi assignments) and of* $(\mathfrak{A}_m)$ *(smooth: Łojasiewicz's gradient inequality for real-analytic functions, in Kurdyka's form for definable ones [50]).*

(iii) (the limit) *At a Clarke-critical point whose optimal plan is unique, the plan is a vertex and the point is a critical point of the frozen energy* $J_m^{\gamma}$*; for commensurate* $m$ *the plan is integral (an assignment* $\pi$*) and the limit is a critical point of the frozen energy* $J_m^{\pi}$*, every particle leaking toward its assigned atom —* $\Theta_m^*(q)$*, the replication of another critical point of* $\mathcal{G}_q$*, or a split configuration. (iii) does not classify these limits: on the instance* $\mathcal{G}_q$ *has a second local minimum (fit error 0.37), whose replication to* $m = 15$ *has the diagonal plan as its unique optimal plan and is a saddle of* $J_{15}^{\mathrm{OT}}$ *— one of its separation blocks is indefinite although* $q \in W_{\mathrm{adm}}$*, the admissible set being defined at the branch* $\Theta^*(q)$ *(referee round 13). A trajectory that enters the basin of* $\Theta_m^*(q)$ *converges to it: the stationary statement of Proposition D‴.12(iii) is reached dynamically from every initial condition in that basin, with the limit's masses the plan's masses* $q$*.*

**Status.** (i) proved (Danskin's theorem and the structure of a minimum of smooth functions; the energy decrease is checked along a run with reassigned particles, and the two-sided decrease at a constructed tie, in `tests/test_ot_leak.py`). (ii) proved given the imported Kurdyka–Łojasiewicz theorem for Clarke subgradient trajectories of definable functions [8] and the definability of restricted analytic and exponential functions [50]; the KL exponent is not quantified, so the statement carries no rate — the uniform-in-$m$ rate is Theorem D‴'s question, which this does not answer. (iii) proved. Under the pure leak an integral plan (commensurate $m$) never switches once optimal: $\Theta(s) = (1 - e^{-\sigma_0 s})T + e^{-\sigma_0 s}\Theta(0)$, and the cost difference between two plans is affine in $\Theta$ (the quadratic terms cancel), so it is the convex combination of its value at $\Theta(0)$, nonnegative, and at the targets $T$, positive because an integral plan costs nothing there and every other plan costs something for distinct atoms. A split plan can switch even under the pure leak — its barycentric targets are not atoms, and it need not be optimal at them (referee round 13: switches in 216 of 400 random incommensurate instances, none in 200 commensurate ones). With the fit, cold starts on the instance switch the plan in 2 of 6 runs at $m = 15$ and in all 6 at $m = 16$ at the experiments' $\sigma_0$, and in none at ten times that leak — the test's run, which is therefore a smooth gradient flow; the nonsmooth part of (i) is tested at the static tie.

**Theorem D‴.18** (Theorem J uniformly in $m$ at a replicated configuration)**.** *Let* (CL) *be the real closed loop of Theorem J with a leak whose targets are class-constant and locally constant near the configurations considered (the* $\sigma$*-modification; the nearest-atom leak with every class atom inside its Voronoi cell; the OT leak at* $w = q$ *with* $\varepsilon < \Delta_\nu/2$*, Proposition D‴.12(iii)), and let* $\mathcal{W}$ *be a compact set of weight vectors with* $\min_k w_k \geq w_{\min} > 0$ *such that for every* $w \in \mathcal{W}$ *the weighted stationary point* $\Theta^*(w)$ *of* $\mathcal{G}_w$ *is a non-degenerate minimum with margins*

$$\lambda_{\mathrm{coll}}(w) := \lambda_{\min}(H_w) \geq \lambda_0 > 0, \qquad \lambda_{\mathrm{sep}}(w) := \min_k \lambda_{\min}(\sigma_0 I - R_k^G(w)) \geq \lambda_0 > 0,$$

*$H_w$ the whitened, weighted collective block and* $\sigma_0 I - R_k^G(w)$ *the separation blocks of Proposition D‴.5(i) (so* $\mathcal{W} \subset W_{\mathrm{adm}}$*). For a configuration* $\Theta = (\vartheta_1, \ldots, \vartheta_m)$ *with a class map* $\pi_0$ *whose class sizes are* $n_k = mw_k$*,* $w \in \mathcal{W}$*, write* $u_i := \vartheta_i - \vartheta^*_{\pi_0(i)}(w)$*,* $\bar{u}_k$ *for the class means of* $u$*,*

$u_i^\perp := u_i - \bar u_{\pi_0(i)}$, *and*

$$|||u||| := \max\Big\{\Big(\sum_k w_k\|\bar u_k\|_{G^{-1}}^2\Big)^{1/2},\ \max_i\|u_i^\perp\|_{G^{-1}}\Big\},$$

$$\textit{so that}\qquad \max_i\|u_i\|_{G^{-1}} \le (1+w_{\min}^{-1/2})|||u||| \qquad \textit{and}\qquad |||u||| \le 2\max_i\|u_i\|_{G^{-1}}$$

*(the second constant is sharp up to the factor $1-1/\max_k n_k$). There are $k_0$, $\rho$, $C$, $\kappa$ depending on the plant, the reference, the features, $\sigma_0$, $G$, $\mathcal{W}$, $\lambda_0$ and the leak's prior (through its cell and plan margins) — but* ***not on*** *$m$ — such that for every $m$, every $w\in\mathcal{W}$ with $mw\in\mathbb{N}^{K_0}$, every class map $\pi_0$ with those class sizes and every $k_e\ge k_0$:*

(a) *the lift of the weighted $K_0$-atom orbit $\Gamma_w$ (Corollary D‴.14) is the unique $T_0$-periodic orbit of* (CL) *whose parameters satisfy $\sup_t\max_i\|\vartheta_i(t)-\vartheta^*_{\pi_0(i)}(w)\|_{G^{-1}}\le\rho/2$ and whose plant state stays in Theorem J's tube;*

(b) *every solution with $\max_i\|\vartheta_i(0)-\vartheta^*_{\pi_0(i)}(w)\|_{G^{-1}}\le\rho/2$ and $\|x(0)-x_m(0)\|\le R/2$ converges to it, and with $u(t)$ the deviation from the lifted orbit at the times $t=jT_0$,*

$$|||u(jT_0)||| \le Ce^{-(1-\kappa)\min\{\lambda_{\rm coll},\lambda_{\rm sep}\}jT_0/k_e}\big(|||u(0)||| + \|x(0)-x_p(0)\|/k_e\big);$$

(c) *its Floquet spectrum is Proposition D‴.15's — the $K_0$-atom orbit's and each class's ghost exponents with multiplicity $n_k-1$ — and the exponents are $(1/k_e)\lambda_j(H_w)$ and $(1/k_e)\lambda_j(\sigma_0 I - R_k^G(w))$ up to $O(k_e^{-2})$, uniformly in $m$.*

*In particular, for $k_e\ge k_0$ Corollary D‴.14(ii)'s proviso is replaced by a statement uniform in $m$: for every commensurate $m$ the lift of $\Gamma_q$ is the unique periodic orbit in the per-particle ball of radius $\rho/2$ around $\Theta^*_m(q)$, and it attracts that ball. (Theorem J's own orbit, unique in its unnormalised ball, coincides with the lift wherever Theorem J applies, but its threshold in that norm grows like $\sqrt{m}$, Remark D‴.19.) Corollary D‴.7's open point — the $m$-uniformity of Theorem J's thresholds — is settled in the per-particle norm.*

*Proof.* Theorem J's proof with the norm $|||\cdot|||$ in place of the unnormalised $\|\cdot\|_{G_m^{-1}}$; what has to be checked is that every constant of its Steps 1–6 is independent of $m$ in the new norm. Three structural facts do it.

(1) *The plant sees the configuration through the class means, up to second order in the separations.* $\hat f_\Theta = \sum_k w_k n_k^{-1}\sum_{i\in k}\tilde\Phi(\cdot\,;\vartheta_i)$ and $\sum_{i\in k}u_i^\perp=0$, so

$$\hat f_\Theta = \sum_k w_k\tilde\Phi(\cdot\,;\bar\vartheta_k) + \tfrac12\sum_k w_k n_k^{-1}\sum_{i\in k}\nabla^2\tilde\Phi(\cdot\,;\bar\vartheta_k)[u_i^\perp,u_i^\perp] + O(\max_i\|u_i^\perp\|^3):$$

the first-order variation of the residual $r_\Theta$, hence of the slaved error $e_\Theta=\varepsilon r_\Theta(x_m)+O(\varepsilon^2)$ of Steps 1–2, is a function of the class means alone, and the separations enter at second order with a coefficient bounded by $\sup\|\nabla^2\tilde\Phi\|$, whatever $m$. Steps 1–2 (the tube, the slaved error, the fast variable $\zeta$) involve $\Theta$ through $r_\Theta$ and — in the $\zeta$-forcing $-\partial_\Theta e_\Theta\cdot\dot\Theta$ and the remainder $\partial_e F\cdot\zeta$ — through the particles' drifts, which are $O(\varepsilon)$ per particle; by (1) and (2) their constants are uniform in $m$.

(2) *Each particle's drift depends on the others only through the plant.* The particle law is $\dot\vartheta_i = F(x,e,\vartheta_i)-\sigma(\vartheta_i-T_i)$: the right-hand side of particle $i$ is a function of $(x,e,\vartheta_i)$ and of its target. So the frozen-$\Theta$ averaged drift $\langle H\rangle_i(\Theta) = -G\nabla\mathcal{V}[\mu_\Theta](\vartheta_i)$ and the antiderivative $u_i(\Theta,t)$ of Step 3 are functions of $(\vartheta_i,\hat f_\Theta)$, bounded and $C^2$ in $\vartheta_i$ with $m$-independent bounds, and their derivatives in $\vartheta_j$ $(j\ne i)$ are $O(1/m)$ and pass through $\hat f$ — that is, through the class means at first order. Writing the second-order remainder of the drift at the replicated configuration per particle, $N_i(u)=O(\|u_i\|^2)+O(\|u_i\|\cdot(1/m)\sum_j\|u_j\|)+O((1/m)\sum_j\|u_j\|^2)$ (the second derivatives $\partial^2_{\vartheta_i\vartheta_i}$ are $O(1)$, $\partial^2_{\vartheta_i\vartheta_j}$ and $\partial^2_{\vartheta_j\vartheta_j}$ for $j\ne i$ are $O(1/m)$; $\partial^2_{\vartheta_j\vartheta_l}$ for distinct $j,l\ne i$ vanishes in $\langle H\rangle$,

whose residual is linear in the $\tilde{\Phi}$'s, and is $O(1/m^2)$ in the second-order term $H_2$, which contains products of $\hat{f}$), so the remainder is bounded in $|||\cdot|||$ by an $m$-independent multiple of $|||u|||^2$; the same holds for the near-identity transformations of Step 3 and for the $O(\varepsilon^3)$ remainders of (N), which are built from the same functions.

(3) *The linearisation at the replicated configuration is block diagonal with an $m$-independent spectrum, and contracts in $|||\cdot|||$.* By Proposition D‴.5(i) and (1)–(2), $D\langle H\rangle(\Theta_m^*(w))$ acts on each separation $u_i^\perp$ of class $k$ separately, as the matrix $GR_k - \sigma_0 I$ — the same for every particle of the class, similar to $-(\sigma_0 I - R_k^G(w))$ whose eigenvalues are $\le -\lambda_{\mathrm{sep}}$ — and on the class means as minus the collective block, self-adjoint in the weighted norm $(\sum_k w_k \|\bar{u}_k\|^2_{G^{-1}})^{1/2}$ with eigenvalues $\ge \lambda_{\mathrm{coll}}$; the two families do not mix. Hence $I + \varepsilon T_0 D\langle H\rangle(\Theta_m^*(w))$ has $|||\cdot|||$-operator norm $1 - \varepsilon T_0 \min\{\lambda_{\mathrm{coll}}, \lambda_{\mathrm{sep}}\}$ once $\varepsilon T_0$ times the largest eigenvalue is below 1, the same bound for every $m$; by (2), $D\langle H\rangle$ is Lipschitz on the $|||\cdot|||$-ball around $\Theta_m^*(w)$ in that operator norm with an $m$-independent constant $C_N$, the constant used below (the remainder of (2) is then at most $(C_N/2)|||u|||^2$, by integration).

With (1)–(3), Step 4 runs as written in the norm $|||\cdot|||$ for $P_Y$ ($P_\zeta$ is unchanged), giving the Lipschitz constant $1 - \varepsilon T_0 \min\{\lambda_{\mathrm{coll}}, \lambda_{\mathrm{sep}}\}(1-\kappa)$ of $P_Y$ on the $|||\cdot|||$-ball of radius $\rho$, $\kappa = O(\rho + \varepsilon)$, with $\rho := \min\{\lambda_0/(4C_N)$, the cell and plan margins of the leak divided by $C(1 + w_{\min}^{-1/2})\}$ — the $|||\cdot|||$-ball of radius $\rho$ reaches the per-particle distance $(1 + w_{\min}^{-1/2})\rho$, the solutions of (b) reach $C$ times that, and the leak's targets must stay constant there; the coupling constants $\mathrm{Lip}_\zeta P_Y = O(\varepsilon)$ and $\mathrm{Lip}_Y P_\zeta = O(\varepsilon^3)$ are $m$-independent by (1) and (2). Step 6's $2 \times 2$ Lipschitz matrix then makes $P$ a contraction in the adapted product norm on that ball, uniformly in $m$: its fixed point is unique there, and it is the lift of $\Gamma_w$, which is a fixed point by Corollary D‴.14(i) and lies in the ball for $k_e \ge k_0$ because $\sup_t |||\vartheta^\Gamma(t) - \vartheta^*(w)||| = O(1/k_e)$ with an $m$-independent constant (it is the weighted $K_0$-atom distance, Theorem J(a) for $(\mathrm{CL}_w)$). This is (a) and (b): the per-particle ball of radius $\rho/2$ lies in the $|||\cdot|||$-ball of radius $\rho$ (since $|||u||| \le 2\max_i \|u_i\|$), which gives the uniqueness set of (a) and the start set of (b), and the $|||\cdot|||$-ball of radius $\rho$ lies in the per-particle ball of radius $(1 + w_{\min}^{-1/2})\rho$, which is what the margins have to cover. (c) is Proposition D‴.15 with its averaging argument, the ghost matrices $A_k$ being $m$-independent. □

**Remark D‴.19** (which norm)**.** In the unnormalised norm $\|\cdot\|_{G_m^{-1}}$ of Theorem J the basin radius $\rho_0$ is $m$-independent too — the Hessian of $J_m$ has the $m$-independent spectrum of (3) and its Lipschitz constant in that norm is $O(1)$ — but a ball of fixed radius in that norm contains only configurations whose particles are within $\rho_0/\sqrt{m}$ of their atoms on average: it shrinks per particle. And Theorem J's threshold is not $m$-independent in that norm: the lifted orbit sits $\sqrt{m} \cdot O(1/k_e)$ away from $\Theta_m^*(w)$ there ($\sqrt{m}$ times the weighted $K_0$-atom distance), and so does the first averaging correction, so the implicit-function step needs $k_e \ge C\sqrt{m}/\rho_0$ — Corollary D‴.14(ii)'s proviso, and the actual reason for the per-particle norm. That norm is the right one because the dynamics is a mean-field one — each particle feels its own position and the class means — and it cannot be weakened to a mean-square ($W_2$-type) ball for the nearest-atom leak: a single particle pushed into another Voronoi cell changes the cluster weights, and the solution converges to the lift of different weights however small its mean-square displacement — for $m$ large enough that the shifted weights are admissible. At small $m$ they need not be (at $m = 15$ on the matching prior, one particle moved out of a class of five gives the weights $(6, 5, 4)/15$, whose lift is a saddle, numerics below); the solution then splits instead, and either way it does not converge to $\Gamma_q$. Under the OT leak the plan restores the masses (a transit, Proposition D‴.12(v)), and the basin at the level of the empirical measure is of a different shape: for the averaged loop Theorem D‴.21 below proves an $m$-independent energy basin, and its a-posteriori form covers configurations with a fixed fraction of the particles in other cells (the two sets are not nested, and D‴.18 concerns the real loop).

**Status.** Proved as an adaptation of Theorem J's proof, at Theorem J's level of detail: Steps 1–3 and 5 carry over with $m$-independent constants by the structural facts (1)–(2), and Steps 4 and 6 are rerun in

the norm $|||\cdot|||$ with (3); the $m$-uniformity rests on the two-sided norm equivalence and on the mean-field structure of every map built in Steps 1–4. The inputs are those of Theorem J, Corollary D‴.14(i), Proposition D‴.15, the margins $\lambda_0$ of $\mathcal{W}$ and the leak's cell and plan margins. It replaces Corollary D‴.14(ii)'s proviso by an $m$-uniform statement and settles the spectral and the basin part of Corollary D‴.7's uniformity question in the per-particle norm. Checked: `tests/test_uniform_basin.py`, on the averaged loop at the prior with centres shifted by +0.15 at the matching masses, where the residual is not zero and $\lambda_{\mathrm{sep}} = 0.80\sigma_0$ (so the per-particle terms that vanish at the matching prior are exercised): the second-order remainder of the drift is bounded by $C_N|||u|||^2$ with $C_N < 0.1$ for random per-particle displacements at $m = 6$, 30, 150, 600, not growing with $m$; for a one-particle displacement the displaced particle's remainder is exactly $N_a + N_b/m$ with $N_a$ the same for every $m$ — non-zero (above $10^{-5}$) here and zero (below $10^{-11}$) at the matching prior — and every other particle's remainder is exactly $O(1/m)$; a separation perturbation (two particles of one class moved by $\pm v$) moves only those particles at first order, with the response $(GR_k - \sigma_0 I)v$, not the leak's $-\sigma_0 v$, at $m = 6$, 60, 300; and $|||u(s)||| \le Ce^{-\mu s}|||u(0)|||$ at $\mu = \min\{\lambda_{\mathrm{coll}}, \lambda_{\mathrm{sep}}\}$ itself, with one constant $C \le 1.05$ for $m = 6$, 30, 150.

**Numerics** (`exp_uniform_basin.py` → `results/chaos/uniform_basin.md`). The real loop at $k_e = 25$ ($\sigma_0 = 0.15\sigma_{\mathrm{crit}}$; RK4 plant with the control and the reference at the stages, Euler parameters, $\mathrm{d}t = 5 \cdot 10^{-3}$) for 8000 time units — 1273 periods, 320 slow units — with all systems integrated together. *The rates.* The exponents that test (b) and (c) are Floquet exponents (Part F, by shooting). At the matching prior the 3-atom orbit's parameter exponents are $2.9355 \cdot 10^{-4}$, $3.5579 \cdot 10^{-4}$, $3.9986 \cdot 10^{-4}$, … against $\lambda_j(H_q)/k_e = 2.9521 \cdot 10^{-4}$, $3.6155 \cdot 10^{-4}$, $4.0755 \cdot 10^{-4}$, … (0.56–3.22%), and every ghost exponent is $2.830 \cdot 10^{-4} = \sigma_0/k_e$. At the prior with centres shifted by +0.15 the parameter exponents are within 0.57–3.15% of $\lambda_j(H_q)/k_e$ and the ghost exponents equal $(\sigma_0 - \lambda_j(R_k^G))/k_e$ to $3 \cdot 10^{-4}$ relative, the slowest $2.2701 \cdot 10^{-4}$ (class 2) against $\lambda_{\mathrm{sep}}/k_e = 2.2696 \cdot 10^{-4}$. *The proof's radius* is small: a sampled lower bound $C_N \ge 0.092$ on the $|||\cdot|||$-Lipschitz constant of $D\langle H\rangle$ gives $\rho \le 0.019$ at the matching prior and $\rho \le 0.015$ at the shifted one, so the runs below, at per-particle radii 0.3–2.5, start 31–261 times outside the proved start radius $\rho/2$: the clean contraction there is evidence that the basin extends to the cell margins, which the theorem does not claim. (A) The matching prior at the matching masses, where the lift is the exact representation, a constant orbit: sizes $m = 3$, 15, 60, 240, every particle displaced from its class atom by $\rho$ in a random direction of the $G^{-1}$-unit sphere, the same displacements under both leaks. Inside the cells ($\rho = 0.3$, 1.0, 2.5 $< \Delta_\nu/2 = 3.13$) no particle leaves its cell and the two leaks give identical runs; $d_\infty = \max_i \min_k \|\vartheta_i - \vartheta_k^0\|_{G^{-1}}$ decays at the fitted rates $2.69 \cdot 10^{-4}$–$2.97 \cdot 10^{-4}$ for $m \ge 15$ against $\sigma_0/k_e = 2.83 \cdot 10^{-4}$ (ratios 0.95–1.05), with $C(T) := \max_{t \le T} d_\infty(t)e^{\mu t}/d_\infty(0) = 1.03$–$1.29$ at $T = 8000$ — still growing at the horizon in the $m = 15$ runs at $\rho = 0.3$ and 1 (1.16 and 1.27), constant from $T = 2000$ on in the others. The $m = 3$ runs (one particle per class, no separation modes) are mixtures of collective modes: at $\rho = 1$ the end of the run is carried by the slowest one and the fitted rate is $2.948 \cdot 10^{-4}$ ($\lambda_{\mathrm{coll}}/k_e = 2.952 \cdot 10^{-4}$), at $\rho = 2.5$ by the third (amplitudes 0.029, 0.024, 0.086; fitted $4.06 \cdot 10^{-4}$ against $4.08 \cdot 10^{-4}$), at $\rho = 0.3$ by all three ($3.48 \cdot 10^{-4}$). At $\rho = 4.5$ some particles start in another cell (1, 3, 13 and 55 for $m = 3$, 15, 60, 240). The nearest-atom leak ends at other weights — the counts $(2, 0, 1)$, $(6, 5, 4)$, $(23, 19, 18)$, $(87, 70, 83)$ — and not only through the initial misplacement: at $m = 60$ and 240 the counts change during the transient (in 3 and 12 records, the last at $t = 420$ and 285), and $m = 240$ starts at exactly $q$. Its clouds have not collapsed at the horizon (within-class rms spreads 0.24–0.66 for $m \ge 15$), but $W^2 = 12.3$, 2.81, 2.30, 1.92 is already that of the lifts of the realised weights for $m = 15$, 60, 240 (3.21, 2.35, 1.93): the cross-cluster transport of the weight mismatch. Two of these weight vectors are inadmissible — $(6, 5, 4)/15$ and $(23, 19, 18)/60$ have splitting margins $-0.00348$ and $-0.00161$, so their lifts are saddles — while $(87, 70, 83)/240$ is admissible ($+0.00167$) and $(2, 0, 1)/3$ empties a class (Newton finds no collapsed stationary point near the prior atoms). The OT leak ends at the counts $q$ at every $m$ (the last count change at $t = 2555$, 2695, 1610, 1445), with $W^2 = 0.010$–$0.23$ still

falling at the leak's rate $2\sigma_0$ per slow unit for $m \geq 15$ (log-slopes $-0.01411$, $-0.01408$, $-0.01417$ against $2\sigma_0 = 0.01415$): the transit is over, and what remains is the leak's contraction. The plan relabels — 4, 15 and 53 particles end outside the cell of their initial class — so the OT leak restores the masses $q$, not the class map $\pi_0$. (C) Remark D‴.19's mechanism: the lift of $q$ with one particle of class 0 moved 0.6 of the way to atom 1, a mean-square displacement of 0.311 ($m = 60$) and 0.078 ($m = 240$). Under the nearest-atom leak $W^2$ grows to 0.79 and 0.20, toward the lifts of the admissible weights $(19, 21, 20)/60$ and $(79, 81, 80)/240$ ($W^2$ 0.8655 and 0.2163); under the OT leak it falls to $3.0 \cdot 10^{-3}$ and $8.2 \cdot 10^{-4}$. (B) A moving orbit: the OT leak toward the prior with centres shifted by $+0.15$ (a non-zero residual, so $\Gamma_q$ moves and the tracking error along it is $2.55 \cdot 10^{-4}$), masses $q$; sizes 15, 60, 240 started at the lift of $\Theta^*(q)$ displaced by $\rho = 0.3$, 1.0 and measured against the lift of the 3-atom loop's trajectory. The second-half fits $2.18 \cdot 10^{-4}$–$2.63 \cdot 10^{-4}$ (ratios 0.96–1.16 to $2.27 \cdot 10^{-4}$) are mixtures of separation modes whose gaps exceed the fit window; they are consistent with $(1 - \kappa) \min\{\lambda_{\mathrm{coll}}, \lambda_{\mathrm{sep}}\}/k_e$, and the Floquet exponents above are the test of the rate. $C(8000) = 1.04$–$1.23$ (still growing at the horizon at $m = 15$, $\rho = 0.3$), and every $m$-system's tracking error is $2.55$–$2.56 \cdot 10^{-4}$ at the end.

**Proposition D‴.20** (the entropic OT leak: a soft mass constraint with a critical excess)**.** *Let $\nu = \sum_j q_j \delta_{\vartheta_j^0}$, $\varepsilon > 0$, let $\pi^\varepsilon(\Theta)$ be the entropic plan from $\mu_\Theta$ to $\nu$ for the cost $\|\cdot\|^2_{G^{-1}}$ — the unique minimiser of $\langle C, \pi\rangle + \varepsilon \,\mathrm{KL}(\pi \mid \mu_\Theta \otimes \nu)$ over the plans with marginals $(1/m, q)$ — and $T_i^\varepsilon := m \sum_j \pi_{ij}^\varepsilon \vartheta_j^0$ its barycentric projection, and consider*

$$\frac{\mathrm{d}\vartheta_i}{\mathrm{d}s} = G\langle \nabla_\vartheta (r_\Theta \tilde{\Phi}(\cdot\,; \vartheta_i))\rangle_\rho - \sigma_0(\vartheta_i - T_i^\varepsilon). \tag{$\mathfrak{A}_m^\varepsilon$}$$

(i) (a smooth gradient flow between two leaks) *($\mathfrak{A}_m^\varepsilon$) is the $G$-gradient flow of $J_m^\varepsilon := m\bar{L}(\mu_\Theta) + (\sigma_0 m/2)\mathrm{OT}_\varepsilon(\mu_\Theta, \nu)$ — by the envelope theorem $\partial \mathrm{OT}_\varepsilon/\partial \vartheta_i = (2/m)G^{-1}(\vartheta_i - T_i^\varepsilon)$ — and $J_m^\varepsilon$ is real-analytic in $\Theta$ (the dual potentials are analytic in the cost: the semi-dual is strictly concave once the gauge $g_1 = 0$ is fixed, and the implicit function theorem applies), so there are no plan switches and Łojasiewicz's theorem gives the convergence of every trajectory — all are bounded, $T_i^\varepsilon$ lying in $\operatorname{conv}\operatorname{supp}\nu$ and the fit's force being bounded — without the nonsmooth machinery of Proposition D‴.17. As $\varepsilon \to 0$, $T^\varepsilon \to \bar{T}$ wherever the OT plan is unique (Proposition D‴.12); as $\varepsilon \to \infty$, $\pi^\varepsilon \to \mu_\Theta \otimes \nu$ and $T_i^\varepsilon \to \sum_j q_j \vartheta_j^0$: the leak becomes the $\sigma$-modification toward the barycentre of $\nu$.*

(ii) (collapsed configurations) *At a configuration collapsed by class with weights $w$ the entropic plan is class-symmetric, $\pi_{ij}^\varepsilon = \pi^K_{\pi_0(i)j}/n_{\pi_0(i)}$ with $\pi^K$ the $K \times K_0$ entropic plan from $\sum_k w_k \delta_{\vartheta_k}$ to $\nu$, so every particle of class $k$ has the class target $T_k = \sum_j (\pi^K_{kj}/w_k)\vartheta_j^0$, and the replicated configuration is a critical point of $J_m^\varepsilon$ iff its atoms are a critical point of the weighted $K$-atom energy with the leak $(\sigma_0/2)\mathrm{OT}_\varepsilon(\sum_k w_k \delta_{\vartheta_k}, \nu)$.*

(iii) (the separation block) *The Hessian of $J_m^\varepsilon$ at such a configuration splits as in Proposition D‴.5(i), with separation blocks*

$$S_k^\varepsilon = \sigma_0\big(I - (2/\varepsilon)\,\mathrm{Cov}_k^G\big) - R_k^G, \qquad \mathrm{Cov}_k^G := G^{-1/2}\,\mathrm{Cov}_{\pi^K(\cdot|k)}(\vartheta^0)G^{-1/2},$$

*each of multiplicity $n_k - 1$: a zero-sum perturbation of a class leaves the column marginals, hence the dual potentials, stationary to first order, and the conditional plan of a particle responds to its own displacement through its softmax, which moves its barycentric target by $(2/\varepsilon)\,\mathrm{Cov}_{\pi^K(\cdot|k)}(\vartheta^0)G^{-1}$ — a positive feedback against the leak's restoring $\sigma_0 I$.*

(iv) (the critical excess) *When $w = q$ and $\varepsilon \ll \Delta_\nu^2$, the conditional plans charge the other atoms with exponentially small mass and $S_k^\varepsilon$ is the OT leak's $\sigma_0 I - R_k^G$ up to exponentially small terms. Let class $k$ carry an excess, and assume $A_k := \sigma_0 I - R_k^G \succ 0$ at $\Theta_\varepsilon^*(w)$ — the splitting condition of Proposition D‴.5, which fails at large excesses and small $\varepsilon$, where*

*the class is a saddle for the fit's own reason. Then* $S_k^\varepsilon = A_k - (2\sigma_0/\varepsilon)\,\mathrm{Cov}_k^G$ *has a negative eigenvalue iff*

$$\varepsilon < 2\sigma_0 \lambda_{\max}\big(A_k^{-1/2}\,\mathrm{Cov}_k^G\,A_k^{-1/2}\big),$$

*twice* $\sigma_0$ *times the largest generalised eigenvalue of the conditional covariance relative to* $A_k$*.* $A_k$ *and* $\mathrm{Cov}_k^G$ *depend on* $\varepsilon$ *through* $\Theta_\varepsilon^*(w)$ *and its plan, so the critical regularisation* $\varepsilon_c$ *is the root of* $\lambda_{\min}(S_k^\varepsilon) = 0$ *along the branch* $\Theta_\varepsilon^*(w)$*. When the conditional plan sends a fraction* $p$ *to one atom* $j$ *and* $1-p$ *to its own,* $\mathrm{Cov}_k^G = p(1-p)\hat\Delta\hat\Delta^\top$ *with* $\hat\Delta := G^{-1/2}(\vartheta_j^0 - \vartheta_k^0)$*, and the criterion is the rank-one* $\varepsilon < 2\sigma_0 p(1-p)\hat\Delta^\top A_k^{-1}\hat\Delta$*. Below* $\varepsilon_c$ *the over-weighted collapsed class is a saddle and splits; its split-off particles leave toward an atom its conditional plan charges, which need not be an under-weighted one, so a split moves the excess but need not reduce it. Above* $\varepsilon_c$ *the class's separation block is positive, and the replicated configuration is a local minimum when the collective block and the other classes' blocks are positive as well; the excess is then absorbed by a bias of the class atom toward* $T_k$*.*

(v) (the floor at large $m$) *For fixed* $\varepsilon$ *the critical excess* $\delta_c(\varepsilon)$ *is positive. At small* $\varepsilon$ *and* $R \approx 0$ *the critical conditional fraction is* $p_c(\varepsilon) \approx \varepsilon/(2\|\hat\Delta\|^2)$*, and the plan then sends away exactly the excess, so* $\delta_c \approx w_k p_c$*. The sampled excess of an i.i.d. warm start has the standard deviation* $(q_k(1-q_k)/m)^{1/2}$*, so the entropic leak equalises the cluster masses only down to the critical excess, and for* $m \gg ((1-q_k)/q_k)(2\|\hat\Delta\|^2/\varepsilon)^2$ *it does not equalise them at all. The masses are then conserved, as under the nearest-atom leak, and partly absorbed by the class biases: the stationary measure depends on the sample again, and the floor returns as Proposition D‴.5's law at the entropic stationary point. Only the hard constraint* $\varepsilon \to 0$ *removes the floor — on the commensurate sizes; on the others its* $K_0 - 1$ *bridge particles leave a floor of order* $m^{-2}$ *(Proposition D‴.12's numerics).*

**Status.** (i) proved (the envelope theorem; the analyticity of the entropic potentials; the limits are the convergence of the entropic plan to an optimal plan as $\varepsilon \to 0$ — the optimal plan of maximal entropy where it is not unique [21] — and the product-measure limit as $\varepsilon \to \infty$). (ii) proved (uniqueness of the entropic plan and the class symmetry). (iii) proved (the first-order stationarity of the potentials under zero-sum class perturbations and the derivative of the conditional softmax); checked against the finite-difference Hessian of $J_m^\varepsilon$ at $m = 15$ in `tests/test_entropic_leak.py`. (iv) proved under the hypothesis $A_k \succ 0$ (the generalised-eigenvalue criterion and its rank-one form); the reading of a split as a transport of the excess is not part of it. (v) formal (the scaling of the sampled excess and the identification of the dynamics' end state); numerical below. The structure is that of the critical temperature of deterministic annealing [77], with the same roles — the particles as the codevectors, the prior's atoms as the data, $\varepsilon$ as the temperature — and twice the largest eigenvalue of the conditional covariance when $R = 0$. New are the two-sided mass constraint, under which the conditional covariance of a collapsed class is an effect of its mass excess rather than of a data spread, and the fit's curvature in $A_k$. Checked: `tests/test_entropic_leak.py` (the semi-dual Newton plan's marginals and its agreement with log-domain Sinkhorn; the gradient of $J_m^\varepsilon$ against finite differences; the two limits at their rates; the replicated configuration critical with a class-symmetric plan, and the finite-difference Hessian leaving each class's separation subspace invariant and acting there as $I \otimes (\sigma_0(I - (2/\varepsilon)\,\mathrm{Cov}_k^G) - R_k^G)$, entrywise; the over-weighted class of $(6,5,4)/15$, with $A_1 \succ 0$ checked, unstable at $\varepsilon = 16$, 17 and stable at $\varepsilon = 18$, 20, 64; every weighted stationary point to a Newton residual of $10^{-9}$).

**Numerics** (`exp_entropic_leak.py` → `results/chaos/entropic_leak.md`). The instance with the matching prior (masses 1/3) at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$; the entropic plan by Newton's method on the semi-dual; every weighted stationary point continued from a converged neighbour and checked to a residual of $10^{-9}$. *The critical excess.* For $w_\delta = q + \delta(1.5e_1 - 0.5)$ (the excess of class 1 taken from both other atoms) the branch $\Theta_\varepsilon^*(w_\delta)$, continued in $\varepsilon$ from 256 down to 1 (residuals $\le 8.7 \cdot 10^{-13}$), changes the sign of the over-weighted class's margin once, at $\varepsilon_c = 5.11$,

13.5, 23.1, 65.0 for $\delta = 0.02$, 0.05, 0.1, 0.2. There $A_1 \succ 0$ ($\lambda_{\min}$ 0.0054, 0.0050, 0.0051, 0.0020), the collective block is positive (0.0073–0.0078) and so are the other classes' margins: above $\varepsilon_c$ the replicated configuration is a local minimum. At $\varepsilon_c$ the class sends the fraction $p = 0.057$, 0.15, 0.33, 0.57 of its mass away, about equally to the two under-weighted atoms (conditional plans $(0.9434, 0.0277, 0.0289)$, $(0.8485, 0.0764, 0.0751)$, $(0.6655, 0.1850, 0.1496)$, $(0.4257, 0.3074, 0.2669)$), so $\mathrm{Cov}_1^G$ has rank two; the exact threshold $2\sigma_0\lambda_{\max}(A_1^{-1/2}\,\mathrm{Cov}_1^G\,A_1^{-1/2})$ equals $\varepsilon_c$ to the printed digits, and the rank-one formula with all of $p$ sent to one atom gives 1.22–1.31 times $\varepsilon_c$ with atom 2 and 0.70–1.22 with atom 3. With the excess taken from atom 3 alone ($w'_\delta = q + \delta(e_1 - e_3)$) $\varepsilon_c = 6.99$, 14.6, 24.3 for $\delta = 0.02$, 0.05, 0.1. At fixed $\varepsilon$ the critical excess is $\delta_c = 0.0042$, 0.0083, 0.016, 0.030, 0.062, 0.14, 0.20 at $\varepsilon = 1$, 2, 4, 8, 16, 32, 64 ($w_\delta$; 0.0035–0.18 for $w'_\delta$). Up to $\varepsilon = 8$ it equals $w_1 p_c$, with $p_c$ within 12% ($w_\delta$) and 26% ($w'_\delta$) of the small-$\varepsilon$ scale $\varepsilon/(2\|\hat\Delta\|^2)$, as (v) says; from $\varepsilon = 16$ on the plan no longer sends away exactly the excess. *Dynamics.* $m = 15$, 60, 240 against $M = 510$ from the warm clouds of Proposition D‴.12's numerics, four seeds, $\varepsilon = 1$, 4, 16, 64, 600 slow units, and the $\varepsilon = 4$, 16, 64 runs continued to 6000 slow units. At $\varepsilon = 1$ every system equalises its masses (all sixteen at $q$ by $s = 600$; fit errors $1.7$–$3.0 \cdot 10^{-4}$; function-level distances to the reference $3.7 \cdot 10^{-9}$–$1.6 \cdot 10^{-6}$ per seed on $(450, 600]$, values at the stopping time, still falling): the OT leak's behaviour. At $\varepsilon = 4$ the masses move toward $q$ slowly: at $s = 600$ five of the twelve larger systems sit at unstable weights (margins $-0.00035$ to $-0.00146$ at $\Theta_4^*(w)$, e-folding times $1/|\text{margin}|$ of 700–2900 slow units, longer than the run). Continued, every system ends at stable weights (positive margins at its $s = 6000$ weights), the larger ones with residual excesses up to 1/60, 3/240 and 7/510 ($\|w - q\|_\infty = 0.0167$, 0.0125, 0.0137), of the order of the critical excess at $\varepsilon = 4$ (0.016 for an excess taken from both other atoms, 0.012 from one). At $\varepsilon = 16$ the over-weighted $(8, 4, 3)/15$ reaches $(6, 6, 3)$ within 600 slow units, and in the continuation the three systems at unstable weights move: $(6, 5, 4)/15$ to $(5, 6, 4)$ at $s \approx 3700$ — the split-off particle joins atom 2, which was at $q$, so the excess is moved, not reduced — $(6, 6, 3)/15$ to $(5, 6, 4)$ as well, and $(25, 17, 18)/60$ to $(23, 19, 18)$; every other system keeps its sampled weights, which are stable. At $\varepsilon = 64$ only $(8, 4, 3)/15$ moves (to $(7, 5, 3)$, at $s \approx 1400$). The floors return as Proposition D‴.5's law at the entropic stationary point $\Theta_\varepsilon^*(q)$, 0.55 ($\varepsilon = 16$) and 0.74 ($\varepsilon = 64$) of the nearest-atom leak's law; the observed floors on the same clouds are 0.23–0.84 ($\varepsilon = 16$) and 0.47–0.91 ($\varepsilon = 64$) of the nearest-atom leak's, per seed; at $\varepsilon = 64$ the bias toward the barycentre of $\nu$ dominates the fit (fit errors 0.12–0.14). The soft constraint thus interpolates in two ways at once: in $\varepsilon$ between the OT leak and the $\sigma$-modification toward the barycentre, and in $m$ between equalisation (small $m$, large sampled excesses) and conserved sampled masses with a floor of the nearest-atom kind, smaller by the factor of the law (large $m$).

**Theorem D‴.21** (a measure-level basin for the OT leak: a Polyak–Łojasiewicz inequality uniform in $m$)**.** *Let $\nu = \sum_j q_j \delta_{\vartheta_j^0}$ and let $\mu^* = \sum_j q_j \delta_{\vartheta_j^*}$ be a replicated stationary measure of $(\mathfrak{A}_m^{\mathrm{OT}})$ — every atom critical for its class potential, $\nabla V_L[\mu^*](\vartheta_j^*) + \sigma_0 G^{-1}(\vartheta_j^* - \vartheta_j^0) = 0$, and the diagonal plan optimal ($\varepsilon < \Delta_\nu/2$ suffices, Proposition D‴.12(iii)) — with residual $r^*$. Put $\mathcal{J}(\mu) := \bar L(\mu) + (\sigma_0/2) W_{G^{-1}}^2(\mu, \nu)$, so that $J_m^{\mathrm{OT}}(\Theta) = m\mathcal{J}(\mu_\Theta)$, and $\mathcal{J}^* := \mathcal{J}(\mu^*)$. Let $m$ be commensurate; every vertex optimal plan is then an assignment $\pi$ with $|\pi^{-1}(j)| = mq_j$ (Proposition D‴.12(ii)), $J_m^{\mathrm{OT}} = \min_\pi J_m^\pi$ over these assignments with $J_m^\pi(\Theta) := m\bar L(\mu_\Theta) + (\sigma_0/2)\sum_i \|\vartheta_i - \vartheta_{\pi(i)}^0\|_{G^{-1}}^2$, and for a configuration $\Theta$ and such a $\pi$ write*

$$u_i := \vartheta_i - \vartheta_{\pi(i)}^*, \qquad D_\pi := \frac{1}{m}\sum_i \|u_i\|_{G^{-1}}^2, \qquad \Delta\hat f := \hat f_\Theta - \hat f[\mu^*], \qquad E := \mathcal{J}(\mu_\Theta) - \mathcal{J}^*,$$

*and, along the plan segments $S_i := \{\vartheta_{\pi(i)}^* + t u_i : t \in [0, 1]\}$,*

$$\kappa^* := \max_i \sup_{S_i} \lambda_{\max} R^G[\mu^*], \quad \kappa_-^* := \max_i \sup_{S_i} \big(-\lambda_{\min} R^G[\mu^*]\big)_+, \quad \kappa^\Theta := \max_i \sup_{S_i} \lambda_{\max} R^G[\mu_\Theta].$$

(i) (two identities along the plan segments) *For every assignment $\pi$ with the counts $mq$, optimal or not,*

$$J_m^\pi(\Theta)/m - \mathcal{J}^* = \tfrac{1}{2}\|\Delta\hat f\|^2 + (\sigma_0/2)D_\pi \\ - \frac{1}{m}\sum_i \int_0^1 (1-t)u_i^\top R[\mu^*](\vartheta^*_{\pi(i)} + tu_i)u_i \,\mathrm{d}t, \tag{E}$$

$$\frac{1}{m}\sum_i \langle \nabla_{\vartheta_i} J_m^\pi(\Theta), u_i\rangle = \|\Delta\hat f\|^2 + \sigma_0 D_\pi \\ - \frac{1}{m}\sum_i \int_0^1 u_i^\top \big[(1-t)R[\mu^*] + tR[\mu_\Theta]\big](\vartheta^*_{\pi(i)} + tu_i)u_i \,\mathrm{d}t, \tag{P}$$

*the norms in $L^2(\rho)$. The counts enter through $\Delta\hat f = (1/m)\sum_i[\tilde\Phi(\cdot\,;\vartheta_i) - \tilde\Phi(\cdot\,;\vartheta^*_{\pi(i)})]$, which holds because $|\pi^{-1}(j)| = mq_j$, and through the constant term of the leak.* (E) *holds for every coupling of $\mu_\Theta$ with the column masses $q$, fractional plans included ($(1/m)\sum_i$ replaced by $\sum_{ij}\gamma_{ij}$);* (P)*, with one test direction per particle, needs an assignment. With counts $N \neq mq$ both acquire a defect — $\sum_k (N_k/m - q_k)\big(-\langle r^*, \tilde\Phi(\cdot\,;\vartheta_k^*)\rangle + (\sigma_0/2)\|\vartheta_k^* - \vartheta_k^0\|^2_{G^{-1}}\big)$ in* (E) *and $-\langle \Delta\hat f, \hat f[\mu_N] - \hat f[\mu^*]\rangle$ in* (P)*, $\mu_N := \sum_k (N_k/m)\delta_{\vartheta_k^*}$ — and at incommensurate $m$ no PL inequality relative to $\mathcal{J}^*$ can hold: the stationary configurations of Proposition D‴.12(iv), with their bridge particles, are critical at a positive excess energy.*

(ii) (the Polyak–Łojasiewicz inequality) *For every assignment $\pi$ with the counts $mq$, $\frac{1}{2}\|\Delta\hat f\|^2 + ((\sigma_0 - \kappa^*)/2)D_\pi \le J_m^\pi(\Theta)/m - \mathcal{J}^* \le \frac{1}{2}\|\Delta\hat f\|^2 + ((\sigma_0 + \kappa^*_-)/2)D_\pi$ (the lower bound is informative when $\kappa^* \le \sigma_0$). If $\pi$ is optimal at $\Theta$, so that the middle term is $E$, and the plan-segment margin $\alpha := \sigma_0 - (\kappa^* + \kappa^\Theta)/2$ is positive, then*

$$\|\nabla J_m^\pi(\Theta)\|^2_{G_m} \ge \lambda \cdot mE, \qquad \lambda := \alpha \cdot \min\{2, 2\alpha/(\sigma_0 + \kappa^*_-)\}.$$

(iii) (a-posteriori decay) *Along every vertex-selection solution of $(\mathfrak{A}_m^{\mathrm{OT}})$ (Section 13, Proposition D‴.17(i): an absolutely continuous solution that follows an optimal integral plan $\pi(s)$ for almost every $s$; their existence is assumed, as in Section 13), $E$ is non-increasing and*

$$E(s) \le E(0)\exp\Big(-\int_0^s \lambda(\sigma)_+ \,\mathrm{d}\sigma\Big),$$

*with $\lambda(\sigma)$ computed at $\Theta(\sigma)$ with its plan ($\lambda_+ := 0$ where $\alpha \le 0$); where $\kappa^* < \sigma_0$, moreover $W^2(\mu_{\Theta(s)}, \mu^*) \le D_{\pi(s)} \le 2E(s)/(\sigma_0 - \kappa^*)$ and $\|\hat f_{\Theta(s)} - \hat f[\mu^*]\|^2 \le 2E(s)$.*

(iv) (an a-priori basin, uniform in $m$) *Assume* (H$^*$)*: $\kappa_R[\mu^*](B) \le \sigma_0 - \kappa$ for some $\kappa > 0$ on a convex set $B$, forward-invariant for $(\mathfrak{A}_m^{\mathrm{OT}})$ and containing the stationary atoms — for instance the whole parameter space, or the a-priori invariant set $N_{R_\Theta}(\operatorname{conv}\operatorname{supp}\nu)$ of Proposition D‴.17 ($\bar T_i \in \operatorname{conv}\operatorname{supp}\nu$ and the fit's force is bounded). Then for every commensurate $m$, $J_m^{\mathrm{OT}} \ge m\mathcal{J}^*$ on $B^m$, with equality exactly at the replicated configurations ($\mu^*$ is the unique global minimiser among the $m$-point empirical measures in $B$, up to the labelling); and every vertex-selection solution from $B$ with $E(0) \le E_0 := \kappa^2/(2(L_2^G)^2)$ satisfies $E(s) \le E(0)e^{-\lambda_0 s}$, $\lambda_0 := (\kappa/2)\min\{2, \kappa/(\sigma_0 + \sup_B\|R^G[\mu^*]\|)\}$. The constants depend on $\kappa$, $\sigma_0$, $L_2^G$ and $r^*$ — not on $m$.*

(v) (the matching prior; sharpness) *At the matching prior ($r^* = 0$ on $\operatorname{supp}\rho$, $\mu^* = \nu$), $R[\mu^*] \equiv 0$,* (H$^*$) *holds on the whole parameter space with $\kappa = \sigma_0$, $E = \bar L(\mu_\Theta) + (\sigma_0/2)W^2(\mu_\Theta, \nu)$, $\alpha = \sigma_0 - \kappa^\Theta/2$ and $\lambda = \alpha \cdot \min\{2, 2\alpha/\sigma_0\}$, which tends to $2\sigma_0$ as $\kappa^\Theta \to 0$. The constant $2\sigma_0$ is attained by the separations (each particle's displacement within its class decays at $\sigma_0$, the separation blocks of Theorem D‴.18), so (ii) is sharp there.*

(vi) (what the mass constraint removes) *For the nearest-atom leak the assignment is the Voronoi one, $j(\cdot)$, whose counts $N$ are conserved inside the cells (Proposition D‴.12(i)) and differ from $mq$ in general. The derivation of* (E)–(P) *breaks at the counts:* $(1/m)\sum_i \tilde\Phi(\cdot\,;\vartheta^*_{j(\vartheta_i)}) = \hat f[\mu_N]$ *with* $\mu_N := \sum_k (N_k/m)\delta_{\vartheta^*_k}$, *not* $\hat f[\mu^*]$. *And the nearest-atom energy has a critical point at the replicated configuration of the weights $N/m$ — a strict local minimum when they are admissible, a saddle when a separation margin is negative — for every count vector $N$ in the domain of the weighted stationary point $w \mapsto \Theta^*(w)$ (a neighbourhood of $q$, Proposition D‴.5(i)) whose atoms lie in their own cells, at a positive excess energy when $N \neq mq$ (under* (H*), *by (iv)). These spurious minima carry the floor of Proposition D‴.5. Under the mass constraint they are not critical points; under* (H*) $\mu^*$ *is the unique global minimiser in $B$ and the only critical point where $\alpha > 0$ (in particular in $\{E \le E_0\}$). Local minima of $J_m^{\mathrm{OT}}$ at a positive excess energy are not excluded: by (ii) they need $\kappa^\Theta \ge 2\sigma_0 - \kappa^*$ on their segments.*

*Proof.* (i) The stationarity of the atoms reads $\sigma_0 G^{-1}(\vartheta_j^* - \vartheta_j^0) = \langle \nabla_\vartheta \tilde\Phi(\cdot\,;\vartheta_j^*)^\top r^*\rangle_\rho$. In $J_m^\pi/m - \mathcal{J}^*$ the loss contributes $-\langle r^*, \Delta\hat f\rangle + \frac12\|\Delta\hat f\|^2$ and the leak $(\sigma_0/2)D_\pi + (\sigma_0/m)\sum_i \langle u_i, \vartheta^*_{\pi(i)} - \vartheta^0_{\pi(i)}\rangle_{G^{-1}} = (\sigma_0/2)D_\pi + \langle r^*, (1/m)\sum_i \nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta^*_{\pi(i)})u_i\rangle$. With the counts, $\Delta\hat f = (1/m)\sum_i[\tilde\Phi(\vartheta_i) - \tilde\Phi(\vartheta^*_{\pi(i)})]$, and Taylor's formula with integral remainder along $S_i$ turns the two $r^*$-terms into $-(1/m)\sum_i \int_0^1 (1-t)u_i^\top R[\mu^*](\vartheta^*_{\pi(i)} + tu_i)u_i\,\mathrm{d}t$: this is (E). For (P), $\nabla_{\vartheta_i} J_m^\pi = -\langle \nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_i)^\top r_\Theta\rangle + \sigma_0 G^{-1}(\vartheta_i - \vartheta^0_{\pi(i)})$ with $r_\Theta = r^* - \Delta\hat f$, and the stationarity gives $\langle \nabla_{\vartheta_i} J_m^\pi, u_i\rangle = \sigma_0\|u_i\|^2_{G^{-1}} - \langle r^{*\top}(\nabla\tilde\Phi(\vartheta_i) - \nabla\tilde\Phi(\vartheta^*_{\pi(i)}))u_i\rangle + \langle \Delta\hat f^\top \nabla\tilde\Phi(\vartheta_i)u_i\rangle$. The second term is $-\int_0^1 u_i^\top R[\mu^*](\vartheta^*_{\pi(i)} + tu_i)u_i\,\mathrm{d}t$. Averaged over $i$, the third is $\|\Delta\hat f\|^2$ plus, by Taylor's formula for $\tilde\Phi(\vartheta^*_{\pi(i)})$ around $\vartheta_i$, $(1/m)\sum_i \int_0^1 t\cdot u_i^\top R_{\Delta\hat f}(\vartheta^*_{\pi(i)} + tu_i)u_i\,\mathrm{d}t$, where $R_g := \langle \sum_k g^k \nabla^2\tilde\Phi^k\rangle$ and $R_{\Delta\hat f} = R[\mu^*] - R[\mu_\Theta]$ because $R$ is linear in the residual. Collecting the $R[\mu^*]$-terms gives (P).

(ii) The $t$-weights $1-t$ and $t$ integrate to $\frac12$ each. By (E), $E = J_m^\pi/m - \mathcal{J}^*$ lies between $\frac12\|\Delta\hat f\|^2 + ((\sigma_0 - \kappa^*)/2)D_\pi$ and $\frac12\|\Delta\hat f\|^2 + ((\sigma_0 + \kappa^*_-)/2)D_\pi$, and by (P), $Q := (1/m)\sum_i \langle \nabla_{\vartheta_i} J_m^\pi, u_i\rangle \ge \|\Delta\hat f\|^2 + \alpha D_\pi$. Hence $Q \ge cE$ with $c := \min\{2, 2\alpha/(\sigma_0 + \kappa^*_-)\}$, and $Q \ge \alpha D_\pi$. Cauchy–Schwarz in the duality of $G$ and $G^{-1}$, $mQ \le \|\nabla J_m^\pi\|_{G_m}(mD_\pi)^{1/2}$, gives $\|\nabla J_m^\pi\|^2_{G_m} \ge mQ^2/D_\pi \ge m\alpha Q \ge m\alpha cE$.

(iii) $J_m^{\mathrm{OT}} = \min_{\pi'} J_m^{\pi'} \le J_m^{\pi(s)}$ with equality at $\Theta(s)$. At every $s$ where $\Theta(\cdot)$ and $s \mapsto J_m^{\mathrm{OT}}(\Theta(s))$ are differentiable — almost every $s$, $J_m^{\mathrm{OT}}$ being locally Lipschitz — $\mathrm{d}J_m^{\mathrm{OT}}(\Theta(s))/\,\mathrm{d}s \le \langle \nabla J_m^{\pi(s)}, \dot\Theta(s)\rangle = -\|\nabla J_m^{\pi(s)}\|^2_{G_m} \le -m\lambda(s)_+ E(s)$, and Grönwall's lemma applies. The distance bounds are (E)'s lower bound and $W^2 \le D_\pi$, the assignment being a coupling of $\mu_\Theta$ and $\mu^*$.

(iv) The segments lie in $B$ by convexity, so $\kappa^* \le \sigma_0 - \kappa$, and (E) gives $E \ge \frac12\|\Delta\hat f\|^2 + (\kappa/2)D_\pi \ge 0$, with equality iff $D_\pi = 0$. Since $\|R^G[\mu_\Theta](\vartheta) - R^G[\mu^*](\vartheta)\| \le L_2^G\|\Delta\hat f\|_{L^1(\rho)} \le L_2^G(2E)^{1/2}$, $\kappa^\Theta \le \kappa^* + L_2^G(2E)^{1/2}$, and $\alpha \ge \kappa - L_2^G(E/2)^{1/2} \ge \kappa/2$ when $E \le E_0$; $E$ is non-increasing and the solution stays in $B$ (forward invariance), so this persists, and (iii) gives the rate.

(v) $R[\mu^*] = \langle r^*\nabla^2\tilde\Phi\rangle = 0$. For a separation $u$ at $\mu^*$ (zero class means), $\Delta\hat f = O(\|u\|^2)$, $\nabla_{\vartheta_i} J_m^\pi = \sigma_0 G^{-1}u_i + O(\|u\|^2)$ and $E = (\sigma_0/2)D_\pi + O(\|u\|^3)$, so $\|\nabla J_m^\pi\|^2_{G_m}/(mE) \to 2\sigma_0$.

(vi) With counts $N$ the identity behind $\Delta\hat f$ becomes $(1/m)\sum_i \tilde\Phi(\vartheta^*_{j(\vartheta_i)}) = \hat f[\mu_N]$; the replicated configuration of the weights $N/m$ is critical for the nearest-atom energy by Proposition D‴.9, a strict local minimum when the weights are admissible and a saddle when a separation margin is negative (Proposition D‴.5), and its excess energy is positive unless $N = mq$. Under (H*) a critical point with $E > 0$ has $\alpha \le 0$ by (ii), that is $\kappa^* + \kappa^\Theta \ge 2\sigma_0$. □

**Remark D‴.22** (the mean-field form, propagation of chaos, and the real loop)**.** The identities and their proof hold verbatim for a probability measure $\mu$ whose optimal plan to $\nu$ is induced by a map $\pi$ (for instance an absolutely continuous $\mu$: the Laguerre map of Proposition D‴.12), with $(1/m)\sum_i$ replaced by $\int \mathrm{d}\mu$. So formally the Laguerre-cell flow ($\mathfrak{A}_0^{\mathrm{OT}}$) has the same PL inequality,

and in the basin both the particle system and the mean-field flow converge to $\mu^*$ exponentially in $W_2$, at the same $m$-independent rate, so $W_2(\mu_{\Theta(s)}, \mu_s) \le W_2(\mu_{\Theta(s)}, \mu^*) + W_2(\mu_s, \mu^*) \to 0$: for every commensurate $m$ the particle system and the mean-field flow have the same limit — no floor, where Theorem D‴'s synchronous coupling kept the transit's mismatch for ever (Proposition D‴.12(v)). This is not propagation of chaos uniform in time, which needs $\sup_s \mathbb{E} W_2(\mu_{\Theta(s)}, \mu_s) \to 0$ as $m \to \infty$: the bound is not small in $m$ at any fixed time (an i.i.d. warm start lies in the a-priori basin only at astronomically large $m$, numerics below), and smallness in $m$ during the transit is not claimed; uniform-in-time statements of that kind are known for entropy-regularised mean-field dynamics under a uniform log-Sobolev inequality [12]. The existence of the mean-field OT flow and its chain rule are not established here. For the real loop: once $E(s) \le (\sigma_0 - \kappa^*)(\rho/2)^2/(2m)$, every particle is within $\rho/2$ of its atom ($\max_i \|u_i\|^2 \le m D_\pi$), and Theorem D‴.18 takes over; on the averaged dynamics this happens after a time $O(\lambda^{-1} \log m)$. The transit of the real loop is covered only on finite horizons, by averaging with plan switches (formal).

**Status.** (i), (ii), (iv), (v) proved. (iii) proved for vertex-selection solutions, whose existence is assumed (Section 13); for this leak a two-particle tie is never attracting — with $h := \langle \vartheta_i - \vartheta_{i'}, \vartheta^0_k - \vartheta^0_j \rangle_{G^{-1}}$, $J^{\pi_1} - J^{\pi_2} = \sigma_0 h$, and on the two sides of the tie $\mathrm{d}h/\mathrm{d}s = \ell - \sigma_0 \|\vartheta^0_k - \vartheta^0_j\|^2_{G^{-1}}$ and $\ell + \sigma_0 \|\vartheta^0_k - \vartheta^0_j\|^2_{G^{-1}}$ with the same loss term $\ell$, so no solution slides along it — while multi-way ties are not analysed; a Clarke solution in the sense of Proposition D‴.17 may stop at a kink-type Clarke-critical point where the active plans' gradients cancel (two particles at one point between two atoms), which a selection solution leaves. (vi) proved for the critical point; its minimality is Proposition D‴.5's admissibility. Remark D‴.22 is formal. The classical case is the leak alone: the metric slope of $\frac{1}{2}W^2(\cdot, \nu)$ is $W(\cdot, \nu)$ at every $\mu$, so it satisfies a PL inequality with constant 2 (AGS [2, §10.4], [74]); a particle velocity field realises the slope where the optimal plan is a map. The plan segments are the generalised geodesic from $\mu^*$ to $\mu_\Theta$ based at $\nu$ (AGS [2, §9.2]): the diagonal plan $\nu \to \mu^*$ and the assignment $\nu \to \mu_\Theta$ make $(1-t)\vartheta^*_{\pi(i)} + t\vartheta_i$ that curve, along which $\frac{1}{2}W^2(\cdot, \nu)$ is 1-convex — the $(\sigma_0/2)D_\pi$ of (E) — and the passage from convexity along a curve ending at a point to a slope bound is AGS's. PL inequalities in Wasserstein space through optimal maps are known for functionals that are not geodesically convex (barycentre functionals; the distance of uniform point clouds to a density) [7]. Here $\mathcal{J}$ is semiconvex along generalised geodesics only with a negative constant (the loss contributes $-L_2^G \sup\langle |r| \rangle$), so a global Łojasiewicz argument, as for the $\lambda$-convex functionals of [7], is unavailable. What is specific is the loss linearised exactly in $\hat{f}$ with the curvature taken at the stationary residual, the explicit local constants, and the counts, which make the generalised geodesic a particle interpolation and the constant independent of $m$. (ii) is sharp at a matching prior and not in general: it bounds $E$ with $\sigma_0 + \kappa^*_-$ and $Q$ with $(\kappa^* + \kappa^\Theta)/2$ separately (numerics below). Checked: `tests/test_ot_basin.py` ((E) and (P) to $10^{-12}$ relative for random assignments with the counts $mq$ at both priors, and with other counts the two defects exactly; the PL chain $\|\nabla J\|^2/(mE) \ge Q^2/(D_\pi E) \ge \lambda$; the constant $2\sigma_0$ attained by a separation to $10^{-4}$; from the lift with one particle moved into another cell the margin positive, the particle kept in its class, $E$ below the a-posteriori bound and decaying at $2\sigma_0$ at the end, while the nearest-atom leak keeps the floor; the nearest-atom spurious minimum at $(11, 9, 10)/30$, admissible — its separation margins and its $K$-atom Hessian positive — critical for that leak, not critical for the OT leak, with the PL inequality holding there).

**Numerics** (`exp_ot_basin.py` → `results/chaos/ot_basin.md`). The averaged slow loop at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$ from the warm clouds of Proposition D‴.12's numerics: sizes $m = 15$, 60, 240, 510, four seeds, 600 slow units, under both leaks, at the matching prior and at the prior with centres shifted by $+0.15$. The identities (E) and (P) hold along every run to $6 \cdot 10^{-13}$ relative. Under the OT leak the initial optimal plan is never changed (no particle is relabelled between records), and the margin $\alpha$ is positive from $s_+ \le 300$ on — from $s \le 25$ in 19 of the 32 runs. From then on the observed ratio $\|\nabla J\|^2/(mE)$ is at least $\lambda$ at every record (the smallest ratio/$\lambda$ is 1.00 at the matching prior and 1.61–1.72 at the shifted one, $Q^2/(D_\pi E)/\lambda$ 1.58–1.69), and $E$ falls from $4.1 \cdot 10^{-3}$–$4.0 \cdot 10^{-2}$ to $8.3 \cdot 10^{-7}$–$8.1 \cdot 10^{-6}$ (means over the seeds; $W^2(\mu_\Theta, \mu^*)$ from 0.99–5.25 to $2.3 \cdot 10^{-4}$–$2.3 \cdot 10^{-3}$), below the a-posteriori bound in every run. At the end the observed ratio

is 0.01416–0.01417 at the matching prior, against $2\sigma_0 = 0.01415$ and the theorem's $\lambda = 0.0140$–$0.0141$: the inequality is sharp. At the shifted prior $\lambda = 0.0077$–$0.0078$ against an observed 0.0137–0.0139. The local PL constant there is $2\min\{\lambda_{\mathrm{sep}}, \lambda_{\mathrm{coll}}\} = 0.01135$ ($\lambda_{\mathrm{sep}} = 0.00567$), so the curvature itself costs a factor 0.802 against $2\sigma_0$ and the inequality (ii) another 0.683 (its slack: the theorem's late $\lambda$ is 0.548 of $2\sigma_0$); the window still mixes the separation modes. *The nearest-atom contrast.* Under the nearest-atom leak no particle changes cell. At the 22 admissible count vectors (every run at $m = 240$ and 510, and $(5,4,6)/15$, $(19,21,20)/60$, $(19,18,23)/60$ at both priors) the runs sit at the replicated configuration of the sampled weights — $E(600)$ is 0.02–0.24% below its energy and still rising toward it, the classes collapsed (spreads $\le 0.006$) — and that configuration is a critical point of the nearest-atom energy (drift $\le 6 \cdot 10^{-14}$) and not of the OT energy (drift $2.3$–$4.4 \cdot 10^{-2}$): the floor is a spurious minimum. At the other 10, five per prior at $m \le 60$ — $(6,5,4)$, $(6,3,6)$, $(8,4,3)/15$ and $(24,15,21)$, $(25,17,18)/60$ — the weights are inadmissible (class-0 separation margins $-0.0004$ to $-0.0061$): the replicated configuration is a saddle, class 0 splits (spreads 0.04–0.62), and $E$ ends 0.5–16.5% below it, still falling. *The a-priori basin.* At the matching prior $\kappa = \sigma_0$ on the whole space and $E_0 = 2.0 \cdot 10^{-7}$. At the shifted prior the supremum of $\lambda_{\max} R^G[\mu^*]$ over the whole parameter space is $6.60 \cdot 10^{-3} < \sigma_0$, approached at a saturation corner ($a \to a_{\max}$, $s \to s_{\min}$): (H*) holds on the whole space with $\kappa = 4.7 \cdot 10^{-4}$ — $\mu^*$ is the unique global minimiser among all $m$-point empirical measures for every commensurate $m$ — and $E_0 = 9.1 \cdot 10^{-10}$ (over the box of the runs' end configurations, which is not forward-invariant, the supremum is $4.39 \cdot 10^{-3}$ and $E_0 = 2.9 \cdot 10^{-8}$). These basins are tiny: a particle outside its cell is at $G^{-1}$-distance at least 3.13 (matching) and 2.95 (shifted) from its class atom, so $\{E \le E_0\}$ holds a fraction $p \le 5.8 \cdot 10^{-6}$ (matching) and $4.4 \cdot 10^{-7}$ (shifted) of the particles in other cells — none below $m \approx 1.7 \cdot 10^5$ and $2.3 \cdot 10^6$. The runs start at $E(0) = 4.1 \cdot 10^{-3}$–$4.0 \cdot 10^{-2}$, far outside, reach the region $\alpha > 0$ by themselves, and are covered from there a posteriori by (iii). *The measure-level basin.* With a fraction $p$ of all the particles, taken from class 0, moved 0.6 of the way into cell 1, the plan keeps them in their class, and $\kappa^\Theta = 0.0096$, 0.019, 0.029, 0.048 for $p = 1/30$, $1/15$, $1/10$, $1/6$ (segment suprema on 129 points): the margin is $+0.0023$ at $p = 1/30$ and negative from $p = 1/15$ on, the same at $m = 30$ and $m = 300$, which are the same measures. With one particle moved, $E$ decays from $3.3 \cdot 10^{-3}$ to $2.5 \cdot 10^{-5}$ ($m = 30$) and from $2.3 \cdot 10^{-4}$ to $3.1 \cdot 10^{-6}$ ($m = 300$), at the rate $2\sigma_0$ at the end and below $E(0)\exp(-\int \lambda_+)$ with $\lambda$ evaluated at every step ($9.9 \cdot 10^{-5}$ and $3.7 \cdot 10^{-6}$; $\lambda \ge 0.00148$ throughout), and the particle returns to its own cell. Under the nearest-atom leak it stays in cell 1, and $E$ rises toward the spurious minimum of the counts $(9,11,10)/30$ and $(99,101,100)/300$, both admissible: $5.72 \cdot 10^{-3}$ and $5.56 \cdot 10^{-4}$ at $s = 300$, $6.304 \cdot 10^{-3}$ and $6.128 \cdot 10^{-4}$ at $s = 1000$, and the minimum's $6.3126 \cdot 10^{-3}$ and $6.1328 \cdot 10^{-4}$ at $s = 3000$.

**Theorem D‴.23** (adaptive priors: a learned prior removes the leak's bias, at the rate of the measure-space excitation)**.** *Let $q$ be masses, $m$ commensurate, $\eta \ge 0$, and let the prior's atoms $Z = (z_1, \dots, z_{K_0})$ move toward the barycentres of the particles that the plan assigns to them while the particles follow the OT leak toward $\nu_Z := \sum_j q_j \delta_{z_j}$:*

$$\frac{\mathrm{d}\vartheta_i}{\mathrm{d}s} = G\langle \nabla_\vartheta (r_\Theta \tilde{\Phi}(\cdot\,;\vartheta_i))\rangle_\rho - \sigma_0(\vartheta_i - z_{\pi(i)}), \qquad \frac{\mathrm{d}z_j}{\mathrm{d}s} = -\eta(z_j - \hat{T}_j),$$
$$\hat{T}_j := \frac{1}{m q_j} \sum_{i \in \pi^{-1}(j)} \vartheta_i, \tag{$\mathfrak{A}_m^{\mathrm{OT},\eta}$}$$

*$\pi$ an optimal assignment from $\mu_\Theta$ to $\nu_Z$ ($\eta = 0$ is the fixed prior of Proposition D‴.12).*

(i) (a gradient flow whose replicated stationary points carry no leak bias) *For $\eta > 0$, ($\mathfrak{A}_m^{\mathrm{OT},\eta}$) is the vertex-selection gradient flow of $\mathcal{J}(\mu_\Theta, Z) := \bar{L}(\mu_\Theta) + (\sigma_0/2) W^2_{G^{-1}}(\mu_\Theta, \nu_Z)$ in the metric that weighs $\vartheta_i$ by $(1/m)G^{-1}$ and $z_j$ by $(\sigma_0 q_j/\eta)G^{-1}$ — by the envelope theorem $\nabla_{z_j} W^2 = 2q_j G^{-1}(z_j - \hat{T}_j)$ — so $\mathcal{J}$ is non-increasing (under any schedule $\eta(s) \ge 0$: $\mathcal{J}$ does not involve $\eta$), and bounded solutions converge to Clarke-critical points (Proposition D‴.17's*

*argument: $\mathcal{J}$ is a minimum over the vertices of definable smooth functions of $(\Theta, Z)$). Boundedness is now a hypothesis (Proposition D‴.27 restores it by projecting the prior onto a compact convex set). Proposition D‴.17's a-priori bound rested on the fixed prior's support, while the leak toward an adaptive prior is an internal force between a class and its atom — with the plan held, the leak cancels from $\frac{\mathrm{d}}{\mathrm{d}s}[(1/m)\sum_{i\in\pi^{-1}(j)}\vartheta_i + (\sigma_0 q_j/\eta)z_j] = (1/m)\sum_{i\in\pi^{-1}(j)} G\langle r\nabla\tilde{\Phi}(\vartheta_i)\rangle$ — and solutions can leave every compact set (numerics, Part G(a) and (b)), even from a fit that the network attains exactly. At a stationary point $z_j = \hat{T}_j$, and each particle satisfies $G\langle r\nabla\tilde{\Phi}(\vartheta_i)\rangle = \sigma_0(\vartheta_i - \hat{T}_{\pi(i)})$: the learning forces cancel over each class, but a particle may be held off its class barycentre by the leak. A replicated configuration — the particles of each class $\pi^{-1}(j)$ at one point $\vartheta_j$ — is therefore stationary with $Z$ iff $z_j = \vartheta_j$ and $\nabla V_L[\sum_l q_l\delta_{\vartheta_l}](\vartheta_j) = 0$ for every $j$: the leak force vanishes, and the stationary replicated configurations are the critical points of the unregularised weighted $K_0$-atom loss. (The classes are the plan's: labelled by the atoms of a given configuration this reads $z_{\sigma(k)} = \vartheta_k$ for the plan's class-to-atom bijection $\sigma$, and at an exact representation a prior permuted among atoms of equal mass is stationary; a permutation between atoms of different masses is not, since a replicated configuration with $W^2 = 0$ needs $q_{\sigma(k)} = q_k$.) Non-replicated stationary points — a class held apart against the loss by the leak toward its barycentre — exist (a saddle on the instance), and carry the $K_0$-cluster bias of (iv)'s regulariser (numerics, Part G(a)). The stationary equations do not involve the prior's initial position: an exact representation with the weights $q$, with the prior at its atoms, is stationary and a global minimum ($\mathcal{J} = 0$ there); which stationary point is reached does depend on it ((vi)).*

(ii) (the linearisation at an exact representation) *At an exact representation $\mu^* = \sum_k q_k\delta_{\vartheta_k^*}$ ($r^* = 0$) with $Z = (\vartheta_k^*)$, the linearisation of $(\mathfrak{A}_m^{\mathrm{OT},\eta})$ splits, for every $m$, into the separations of each class, which decay at $\sigma_0$, and the collective pairs $(\bar{u}, \zeta)$ of class means and prior offsets. In the whitened weighted coordinates $y_k := \sqrt{q_k}G^{-1/2}\bar{u}_k$, $x_k := \sqrt{q_k}G^{-1/2}\zeta_k$ it reads $\dot{y} = -H_L y - \sigma_0(y - x)$, $\dot{x} = -\eta(x - y)$, with the Gauss–Newton block $H_L := [\sqrt{q_k q_l}G^{1/2}\langle\nabla_\vartheta\tilde{\Phi}(\cdot\,;\vartheta_k^*)^\top\nabla_\vartheta\tilde{\Phi}(\cdot\,;\vartheta_l^*)\rangle_\rho G^{1/2}]_{kl} \succeq 0$ (eigenvalues $h_1 \le \cdots \le h_{pK_0}$). On its eigenvectors every pair evolves by $\left[\begin{smallmatrix} -(h_j+\sigma_0) & \sigma_0 \\ \eta & -\eta \end{smallmatrix}\right]$, whose rates $\lambda_j^\pm$ are the roots of $\lambda^2 - (h_j + \sigma_0 + \eta)\lambda + \eta h_j = 0$; for $h_j, \eta > 0$ the slower one satisfies*

$$\eta h_j/(h_j + \sigma_0 + \eta) < \lambda_j^- < \min\{h_j, \eta\},$$

*the lower bound being tight as $\eta \to 0$; $\lambda_j^- = 0$ if $h_j\eta = 0$ and $\lambda_j^- = h_j$ at $\eta = \infty$; and $\lambda^-$ increases with $h$ ($\mathrm{d}\lambda^-/\,\mathrm{d}h = (\eta - \lambda^-)/(\lambda^+ - \lambda^-) > 0$), so the slowest collective rate is $\lambda_1^-$. So the adaptive prior's convergence in mode $j$ is never faster than the loss's own curvature $h_j$, while a fixed prior's is $h_j + \sigma_0$; in an unexcited mode ($h_j = 0$) $\lambda_j^- = 0$, and the network and the prior drift together, as an adaptive law without a leak does without persistent excitation. The pair matrix is self-adjoint in the metric of (i) (in the coordinates $(y_j, (\sigma_0/\eta)^{1/2}x_j)$ it is symmetric, with off-diagonal $(\sigma_0\eta)^{1/2}$).*

(iii) (the bias it removes) *With a fixed prior near an exact representation, the stationary point is, to first order in the prior's error $x^0$, $y^* = \sigma_0(H_L + \sigma_0 I)^{-1}x^0$: mode $j$ keeps the fraction $\sigma_0/(h_j + \sigma_0)$ of the error. The leak buys the rate $h_j + \sigma_0$ at the price of that fraction, and the adaptive prior removes it at the rate $\lambda_j^-$. Weakly excited modes ($h_j \ll \sigma_0$) keep almost all of the prior's error under a fixed prior and lose it slowly under an adaptive one; the fit, $\frac{1}{2}\sum_j h_j y_j^2$ to second order, weighs them by $h_j$, so the fit is restored faster than the parameters. The spectrum $\{h_j\}$ is the measure-space excitation level of the reference at $\mu^*$ (Section 26, branch 7).*

(iv) (infinitely fast priors: a capacity-constrained $K_0$-means regulariser) *Formally, as $\eta \to \infty$ the prior follows the class barycentres, $z_j = \hat{T}_j$, and $(\mathfrak{A}_m^{\mathrm{OT},\eta})$ becomes the gradient flow of*

$\bar{L}(\mu_\Theta) + (\sigma_0/2)\mathcal{Q}_q(\mu_\Theta)$, *with* $\mathcal{Q}_q(\mu) := \min_Z W^2(\mu, \nu_Z)$ *the capacity-constrained* $K_0$*-means distortion of* $\mu$ *— along the plan the flow carries, which is optimal for its own barycentres (Lloyd-consistent) and need not be jointly optimal: the flow follows the selected piece of* $\mathcal{Q}_q$*, and* $\mathcal{Q}_q$ *itself need not decrease (Proposition D‴.27(ii)) —: a regulariser that pulls the network toward* $K_0$ *clusters of the prescribed masses without biasing where they sit. The separations keep the rate* $\sigma_0$*, and mode* $j$ *of the class means relaxes at* $h_j$*. It does bias how many there are: a target that the network fits exactly only with a class split is fitted with* $K_0$ *clusters (numerics, Part G), so* $K_0$ *remains a design input.*

(v) (uniformity in $m$) *The blocks of (ii) and their rates do not depend on* $m$*. In the norm*

$$|||(u,\zeta)|||_\eta := \max\Big\{\Big(\sum_k q_k\big(\|\bar{u}_k\|^2_{G^{-1}} + (\sigma_0/\eta)\|\zeta_k\|^2_{G^{-1}}\big)\Big)^{1/2},\ \max_i \|u_i^\perp\|_{G^{-1}}\Big\},$$

*the per-particle norm of Theorem D‴.18 with the prior's offsets added in the metric of (i), the linearisation acts on the separations as* $-\sigma_0 I$ *and is self-adjoint on the collective part with the eigenvalues* $-\lambda_j^\pm$*; the prior is driven by the class barycentres, which are linear in the class means while the plan is diagonal, so Theorem D‴.18's facts (1)–(2) hold for the joint drift with* $m$*-independent constants, and for* $0 < \eta < \infty$ *the averaged loop converges, for every commensurate* $m$*, in the* $|||\cdot|||_\eta$*-ball of radius*

$$\rho(\eta) := \min\Big\{\lambda_0(\eta)/(4C_N),\ \mathrm{margin}/\big(C[1 + q_{\min}^{-1/2} + (\eta/(\sigma_0 q_{\min}))^{1/2}]\big)\Big\},$$
$$\lambda_0(\eta) := \min\{\sigma_0, \lambda_1^-(\eta)\},$$

*at the rate* $(1-\kappa)\lambda_0(\eta)$ *—* $C_N$ *the Lipschitz constant of the joint drift's Jacobian in the* $|||\cdot|||_\eta$*-operator norm (the prior's part of the drift is linear, so it is the particles'),* margin *the plan margin of Proposition D‴.12(iii) at the exact representation,* $C$ *as in Theorem D‴.18: the ball reaches the per-particle distance* $(1 + q_{\min}^{-1/2})\rho$ *and the prior's offsets* $\|\zeta_k\|_{G^{-1}} \le \rho(\eta/(\sigma_0 q_k))^{1/2}$*, and the plan stays diagonal while both are within the margin. At* $\eta = \infty$ *the prior is the class barycentres (*$\zeta = \bar{u}$*), the norm is Theorem D‴.18's,* $\lambda_0 = \min\{\sigma_0, h_1\}$*, and the margin term is* $\mathrm{margin}/(C(1 + 2q_{\min}^{-1/2}))$*; for large finite* $\eta$ *the bound degenerates like* $\eta^{-1/2}$*, because this norm does not use the prior's relaxation onto the class means at the rate* $\eta$ *(a singular perturbation). As* $\eta \to 0$*,* $\lambda_1^- < \eta$ *and the rate and the radius vanish with it; at* $\eta = 0$ *the prior is a parameter, and the statement is Theorem D‴.18's contraction for the fixed prior.*

(vi) (the transit: when the prior may start) *From a sample whose cluster masses differ from* $q$*, the plan's classes contain particles of other clusters until the transit of Proposition D‴.12 ends, and their barycentres* $\hat{T}_j$ *are pulled toward those clusters: an adaptive prior is dragged along, and afterwards relaxes back only at the rates* $\lambda_j^-$*. At* $\eta = \infty$ *the collective modes keep no leak curvature at all, and the drag can carry the class atoms onto a plateau of the unregularised loss, which the flow then crosses at that loss's own curvature. Freezing the prior while mass is in transit —* $\eta = 0$ *as long as some particle is assigned to an atom that is not its nearest — keeps the prior out of the drag;* $\mathcal{J}$ *stays non-increasing, as under any schedule of* $\eta$*, and at* $\eta = \infty$ *the opening is a jump of* $Z$ *to the class barycentres, which lowers* $\mathcal{J}$*. But the adaptive prior forgets a perturbation of mode* $j$ *only at* $\lambda_j^- < h_j$*, so whatever the transient leaves in the weakly excited modes when the prior starts is kept for a time of order* $1/\lambda_1^-$*. When the parameters, not only the fit, are wanted at a given horizon, the adaptation should therefore start from the fixed-prior stationary point — after the transit, and after the fixed-prior dynamics has settled, which it does at the fixed prior's rates (of order* $\sigma_0$ *and faster) — and then (ii) and (iii) apply with the prior's error as the only initial displacement. The wait is a delay, which the fit error does not always repay (numerics).*

*Proof.* (i) $W^2(\mu_\Theta,\nu_Z)=\min_\gamma\sum_{ij}\gamma_{ij}\|\vartheta_i-z_j\|^2_{G^{-1}}$; where the optimal plan is unique, $\nabla_{z_j}W^2=\sum_i\gamma_{ij}2G^{-1}(z_j-\vartheta_i)=2q_jG^{-1}(z_j-\hat T_j)$, so $\mathrm{d}z_j/\,\mathrm{d}s=-(\eta/(\sigma_0q_j))G\nabla_{z_j}\mathcal{J}$ and $\mathrm{d}\vartheta_i/\,\mathrm{d}s=-mG\nabla_{\vartheta_i}\mathcal{J}$ (Proposition D‴.12); the vertex-selection argument and the convergence are Proposition D‴.17's with $(\Theta,Z)$ in place of $\Theta$, for solutions that stay bounded. With the plan held, $(1/m)\sum_{i\in\pi^{-1}(j)}\sigma_0(\vartheta_i-z_j)=\sigma_0q_j(\hat T_j-z_j)=(\sigma_0q_j/\eta)\dot z_j$, which is the conservation law. $\dot z=0$ forces $z_j=\hat T_j$ for every $j$, and then $\dot\vartheta_i=0$ reads $G\langle r\nabla\tilde\Phi(\vartheta_i)\rangle=\sigma_0(\vartheta_i-\hat T_{\pi(i)})$. At a replicated configuration $\hat T_j$ is the class point $\vartheta_j$, so the leak term vanishes and stationarity is $z_j=\vartheta_j$ and $\nabla V_L(\vartheta_j)=0$. At an exact representation $\bar L=0$ and $W^2(\mu^*,\nu_{Z^*})=0$. (ii) At $r^*=0$, $-G\nabla V_L[\mu](\vartheta_i)=-G\langle\Delta\hat f^\top\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_i)\rangle$ with $\Delta\hat f=\sum_l q_l\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_l^*)\bar u_l+O(\|u\|^2)$ (the separations enter at second order, Theorem D‴.18's fact (1)), so a separation evolves by $-\sigma_0u^\perp$ and the class means by $\dot{\bar u}_k=-G\sum_l\langle\nabla\tilde\Phi_k^\top\nabla\tilde\Phi_l\rangle q_l\bar u_l-\sigma_0(\bar u_k-\zeta_k)$, with $\dot\zeta_k=-\eta(\zeta_k-\bar u_k)$ because $\hat T_k-\vartheta_k^*=\bar u_k$; the coordinates $y$, $x$ symmetrise it. The eigenvalues of the pair matrix are $-\lambda$ with $\lambda^2-(h+\sigma_0+\eta)\lambda+\eta h=0$. Their product is $\eta h$ and their sum $h+\sigma_0+\eta$; for $h,\eta>0$ both roots are positive, so $\lambda^+<h+\sigma_0+\eta$ and $\lambda^-=\eta h/\lambda^+>\eta h/(h+\sigma_0+\eta)$; the polynomial is $-\sigma_0h<0$ at $\lambda=h$ and $-\sigma_0\eta<0$ at $\lambda=\eta$, so both lie strictly between the roots; and differentiating the polynomial in $h$ at $\lambda^-$ gives $\mathrm{d}\lambda^-/\,\mathrm{d}h=(\eta-\lambda^-)/(\lambda^+-\lambda^-)>0$. (iii) The stationary equations of the fixed prior, linearised, read $0=-H_Ly-\sigma_0(y-x^0)$; $\bar L=\frac12\|\Delta\hat f\|^2=\frac12y^\top H_Ly+O(\|y\|^3)$. (iv) For a fixed assignment the minimiser of $W^2$ over $Z$ is the class barycentres; by the envelope theorem the particles' drift of $\bar L+(\sigma_0/2)\mathcal{Q}_q$ is $-G\nabla V_L-\sigma_0(\vartheta_i-\hat T_{\pi(i)})$, the $\eta\to\infty$ limit of $(\mathfrak{A}_m^{\mathrm{OT},\eta})$ (formal: a singular perturbation in which $Z$ is the fast variable). (v) (sketch) The symmetrisation of (ii) makes the collective part self-adjoint in $|||\cdot|||_\eta$, with the eigenvalues $-\lambda_j^\pm$, and the separations decouple at $-\sigma_0$; the rest is Theorem D‴.18's contraction argument (its fact (3) with this norm) for the averaged loop, the diagonal plan being locally constant at commensurate $m$ (Proposition D‴.12(iii)) inside the ball, whose per-particle and prior extents are as stated. The constants $\kappa$, $C$, $C_N$ and the margin are not derived. (vi) $\mathcal{J}$ does not involve $\eta$, so it is non-increasing for any schedule $\eta(s)\ge0$, the gates' included; at $\eta=\infty$ the opening sets $Z$ to the class barycentres of the current plan, the minimiser of $\mathcal{J}$ over $Z$ for that plan, so $\mathcal{J}$ drops there. □

**Status.** (i)–(iii) proved, (i) for the solutions that stay bounded (a hypothesis now); (iv) formal; (v) sketched, as an adaptation of Theorem D‴.18's contraction argument for the averaged loop in the norm $|||\cdot|||_\eta$ — the structure (the decoupled separations, the self-adjoint collective pairs, the $m$-independent blocks) verified, the radius stated, its constants not derived; (vi) the monotonicity of $\mathcal{J}$ under the gates proved (it holds under any schedule of $\eta$), the drag, the memory of the weak modes and the remedy numerical (below). The prior that tracks the network is the continuous-time form of the proximal point method, or iterated Tikhonov regularisation — the leak toward a moving centre — and the coupled flow of the particles and a centre drawn to their mean is elastic averaging; the fast limit is the $K$-means form of soft weight-sharing, a regulariser of network compression [90], here with the masses fixed (capacity-constrained, [5]). The per-mode formulas are those of the quadratic analyses: $\sigma_0/(h_j+\sigma_0)$ is the complement $1-h_j/(h_j+\sigma_0)$ of the Tikhonov filter factor in the Gauss–Newton eigenbasis (the residual factor: the fraction of the prior's error that the regularised solution keeps), iterated Tikhonov contracts mode $j$ by it per step (in continuous time, the gradient flow of the Moreau envelope, whose curvature $h_j\sigma_0/(h_j+\sigma_0)$ gives the slow-prior limit $\eta h_j/(h_j+\sigma_0)$ of $\lambda_j^-$), and the pair matrix is the continuous-time form of elastic averaging's worker–centre map on a quadratic [90]. New are their transfer to the mass-constrained particle system — $H_L$ identified as the measure-space excitation spectrum, and the separations decoupled at $\sigma_0$, which make the trade-off between the fixed prior's bias and the adaptive prior's rate hold uniformly in $m$ — and the transit's drag with its gate. The limits of the bias removal are the precedents' own properties in this form: the lost a-priori bound is the proximal point method's behaviour without a minimiser (its iterates are bounded iff a zero exists [90], cited from memory), and the $K_0$-cluster bias is the quantisation error of soft weight-sharing's hard limit [90]; specific here are their particle and mass-constrained form, and the run-off from a fit that the network attains exactly (Part G(a)). Checked: `tests/test_adaptive_prior.py` (the

joint drift against finite differences of $\mathcal{J}$ in the stated metric; the exact representation stationary with the prior at its atoms and the fixed prior's stationary point not; the spectrum of the linearisation at $m = 6$ equal to the separation rate $\sigma_0$ and the pair rates $\lambda_j^{\pm}$ for two $\eta$, and the linearisation self-adjoint in the gradient-flow metric; the fixed prior's bias at a smaller shift to the first-order formula; the transit gate frozen while mass is in transit and the settled gate opening only when the fixed-prior drift has fallen below its tolerance, with $\mathcal{J}$ non-increasing in both; the limits of (i): a non-replicated stationary saddle held by the leak, a prior permuted among atoms of equal mass stationary at the exact representation, and the adaptive prior's run-off — the force on the amplitude tending to the positive Karush–Kuhn–Tucker multiplier of the loss's minimum on the face $a = a_{\max}$, where a bounded limit would need it to vanish, and the run-off law $(a_{\max} - a)s \to a_{\max}^2(\sigma_0 + \eta)/(4\eta G_\alpha F_\infty)$ of the conservation law — against the fixed prior's convergence).

**Numerics** (`exp_adaptive_prior.py` → `results/chaos/adaptive_prior.md`). The averaged slow loop at $\sigma_0 = 0.15\sigma_{\text{crit}}$ with the OT leak toward a misplaced prior — the matching prior's centres shifted by $+0.15$, masses $1/3$. *The linear theory.* At the exact representation the whitened weighted Gauss–Newton spectrum is $h_j = 3.06 \cdot 10^{-4}$ … $0.195$. A fixed prior keeps the fraction $\sigma_0/(h_j + \sigma_0)$ of its error, 0.959 in the weakest mode and 0.035 in the strongest, with collective rates $h_j + \sigma_0 \geq 7.38 \cdot 10^{-3}$; the adaptive prior's slowest rate is $\lambda_1^- = 2.68 \cdot 10^{-5}$, $1.51 \cdot 10^{-4}$, $2.78 \cdot 10^{-4}$, $3.06 \cdot 10^{-4}$ for $\eta = \sigma_0/10$, $\sigma_0$, $10\sigma_0$, $\infty$. *(v)'s radius.* A sampled lower bound on the Jacobian-Lipschitz constant of the collective block is $C_N \geq 0.341$, so the proof's radius is at most $2.0 \cdot 10^{-5}$, $1.1 \cdot 10^{-4}$, $2.0 \cdot 10^{-4}$, $2.2 \cdot 10^{-4}$ for the four $\eta$ — $\lambda_0/\sigma_0 = 0.0038$–$0.043$ times Theorem D‴.18's at the same $C_N$ — while the $K_0$-atom runs below start at $|||\cdot|||_\eta = 1.17$, 0.37, 0.12, 0.021, 94 to $6 \cdot 10^4$ times outside, the warm clouds of the $m$-particle systems $2.2 \cdot 10^4$–$3.6 \cdot 10^5$ times and the separated starts of Part F 131–3344 times: they probe the dynamics beyond the proved ball. *The fixed prior's bias.* $\Theta^*(q)$ has the fit error $9.78 \cdot 10^{-3}$, and the first-order formula $y^* = \sigma_0(H_L + \sigma_0 I)^{-1}x^0$ describes it to 4.8% in norm; per mode the weak modes 1 and 3 hold 0.85 and 0.73 of the prediction, and the error is first order in the shift (halving the shift from 0.15 to 0.01875 takes the norm error through 4.8, 2.3, 1.2, 0.58% and modes 1 and 3 to 0.98 and 0.97 of the prediction). The weakest mode keeps 0.814 of its error, against the predicted 0.959. *The $K_0$-atom loop* from $\Theta^*(q)$, with the prior at the shifted centres, for the four $\eta$ over 6000 slow units: every mode amplitude above $10^{-6}$ — thirteen of the 36 — decays at its predicted rate (log-slopes on $[3000, 6000]$ between 0.981 and 1.003 times $\lambda_j^-$), at a ratio to the exact pair solution between 0.82 and 1.12 that changes by at most 1.04% from $s = 1000$ on for $\eta \geq \sigma_0$ (6.0% at $\sigma_0/10$; the offset is set in the first, nonlinear phase); below $10^{-6}$ the faster modes stop at the level of the second-order remainder. The fit error falls from $9.78 \cdot 10^{-3}$ to $3.6 \cdot 10^{-4}$, $1.8 \cdot 10^{-5}$, $8.0 \cdot 10^{-6}$, $6.7 \cdot 10^{-6}$ at $s = 6000$, while the weakest mode still holds $1.45 \cdot 10^{-3}$, $6.7 \cdot 10^{-4}$, $3.0 \cdot 10^{-4}$, $2.5 \cdot 10^{-4}$ of its $1.58 \cdot 10^{-3}$: the fit is restored long before the parameters, as (iii) says. *$m$-particle systems:* $m = 15$, 60, 240 and $M = 510$ from the warm clouds of Proposition D‴.12's numerics, four seeds, 3000 slow units. Under the fixed prior every system ends at the lift of $\Theta^*(q)$ ($\|\bar{y}\| = 2.10 \cdot 10^{-2}$, $W^2(\mu_\Theta, \mu_{\text{exact}}) = 4.4 \cdot 10^{-4}$, fit error $9.78 \cdot 10^{-3}$). Ungated, the adaptive prior is dragged by the transit: at $s = 500$ $\|\bar{y}\|$ is 0.032–0.32 ($\eta = \sigma_0$) and 0.048–0.71 ($\eta = \infty$) in seed means, against the $K_0$-atom run's $7.4 \cdot 10^{-3}$ and $4.3 \cdot 10^{-3}$, and at $s = 3000$ still 0.0094–0.10 and 0.012–0.59, the largest at $m = 15$. At $\eta = \infty$ one $m = 15$ system is carried onto a plateau of the unregularised loss: its split class re-collapses by $s = 7000$, onto atoms where the loss's Hessian has the eigenvalues $-1.8 \cdot 10^{-5}$ and $7.2 \cdot 10^{-5}$ (fit error $3.46 \cdot 10^{-2}$ at $s = 23000$, the drift $1.4 \cdot 10^{-5}$). The collapsed $\eta = \infty$ dynamics from there, the unregularised $K_0$-atom gradient flow, falls below 0.9 of the plateau's loss $2.52 \cdot 10^{-4}$ after $4.16 \cdot 10^4$ more slow units, halves it at $4.78 \cdot 10^4$ and falls below $10^{-12}$ at $7.97 \cdot 10^4$ (the crossings located as events), at the exact representation: the plateau is crossed on the time scale $1/1.84 \cdot 10^{-5} = 5.4 \cdot 10^4$ of its negative curvature. The transit gate opens at $s = 88$–$149$ and removes the plateau, but the transient's residue in the weak modes stays: $\|\bar{y}\| = 4.9 \cdot 10^{-3}$–$3.7 \cdot 10^{-2}$ ($\eta = \sigma_0$) at $s = 3000$, against the

$K_0$-atom run's $1.3 \cdot 10^{-3}$, decreasing with $m$. The settled gate opens at $s = 1211$–$1304$, with the systems replicated (rms separations $1.2 \cdot 10^{-4}$–$7.9 \cdot 10^{-4}$) and their class means within $6.7 \cdot 10^{-5}$ of $\Theta^*(q)$; each is then the $K_0$-atom system started from its own class means, and at $s = 3000$ $\|\bar{y}\|$ and the fit error of every system equal the $K_0$-atom run's at $s - t_{\text{open}}$ — per-seed ratios 0.995–1.000 ($\eta = \sigma_0$) and 0.968–1.001 ($\eta = \infty$) for every $m$ from 15 to 510, as the invariance of the replicated configurations requires, the departures from 1 being the opening offset in the weakest mode — with fit errors $1.6 \cdot 10^{-4}$ ($\eta = \sigma_0$) and $3.4$–$3.5 \cdot 10^{-5}$ ($\eta = \infty$) against the fixed prior's $9.78 \cdot 10^{-3}$, $W^2(\mu_\Theta, \mu_{\text{exact}}) = 6.7$–$7.2 \cdot 10^{-6}$ and $9.3$–$9.9 \cdot 10^{-7}$ against $4.4 \cdot 10^{-4}$, and the counts at $q$ in every run. The wait costs 1211–1304 of the 3000 slow units: in the seed means the settled gate's $\|\bar{y}\|$ and $W^2$ are the lower everywhere, and its fit error at $\eta = \infty$ and at $m \le 240$ (at $\eta = \sigma_0$ and $m = 510$ the transit gate's fit error is the lower, $1.55 \cdot 10^{-4}$ against $1.64 \cdot 10^{-4}$); system by system the transit gate ends lower in all three in 6 of the 32 systems. *The structure behind (v), beyond its ball* (Part F): from the lift of $\Theta^*(q)$ plus zero-mean per-particle separations of rms 0.02 and 0.1 ($m = 15$ and 510, ungated, five draws of their direction), $\|\bar{y}\|$ at $s = 1750$ departs from the $K_0$-atom run's, in the draw means, by +0.13% and +3.1% at $\eta = \sigma_0$ and by +0.77% and +18.5% at $\eta = \infty$ for $m = 15$ (+0.09%, +2.2%, +0.63% and +15.5% for $m = 510$): second order in the separations draw by draw (factors 23.8–25.0 for the factor 5), with a spread across the draws 3.3–6.0 times larger at $m = 15$, and consistent with the same expected size at both $m$ (the draw means, higher at $m = 15$ in all four columns, differ by +0.9 to +1.7 combined standard errors; five draws) — the prediction of Theorem D‴.18's facts (1)–(2) at second order, the departure being linear in the separations' class covariances, whose expectation is the same at every $m$. The starts lie 131–3344 times outside (v)'s ball, so (v) itself asserts nothing there. *The limits of (i)* (Part G). On a four-bump target that six kernels fit exactly with one class split ($K_0 = 3$, $m = 6$), the joint flow has a stationary saddle with the class split — the pair at $G^{-1}$-distance 1.64, against the exact fit's 3.29 — held by a leak force $5.8 \cdot 10^{-3}$ against an unregularised gradient of the same size (fit error $4.2 \cdot 10^{-2}$); from the exact fit the flow collapses the class within 8000 slow units (at $s = 4 \cdot 10^4$, $\eta = \infty$: the merged width 1.05, fit error $2.23 \cdot 10^{-2}$), and the merged atom then widens toward $s_{\max}$ — continued, its width coordinate runs off like $\ln s$ and the width tends to $s_{\max}$ like $1/s$, as in (b): the solution is unbounded — while its physical image approaches the minimiser of the loss on the face $s = s_{\max}$ (fit error $1.943 \cdot 10^{-2}$, a Karush–Kuhn–Tucker point with $\partial L/\partial s = -1.3 \cdot 10^{-6}$, not a critical point), reached only as $s \to \infty$: the $K_0$-cluster bias drives the solution from an exact fit to infinity while $\mathcal{J}$ decreases. With a target bump of weighted amplitude $4.8 > a_{\max} = 4$, the fixed prior's $K_0$-atom solution converges (fit error 0.175), while under the adaptive prior ($\eta = \sigma_0$ and $\infty$) the amplitude coordinate grows like $2 \ln s$ (its increments per decade $1.99 \ln 10$ by $s = 10^6$) and the amplitude tends to $a_{\max}$ like $1/s$: the solution leaves every compact set. At the exact representation (equal masses) a cyclically permuted prior is stationary (drift $1.1 \cdot 10^{-17}$), with $z_{\sigma(k)} = \vartheta_k$.

**Proposition D‴.24** (the finite-size floor is the leak's bias at the sampled weights; an adaptive prior removes it without a mass constraint)**.** *Let the target admit, for the weights $w$ in a neighbourhood $W$ of $q$, an exact weighted $K_0$-atom representation $\vartheta^e(w)$ — $f = \sum_k w_k \tilde{\Phi}(\cdot\,; \vartheta^e_k(w))$, a critical point of the unregularised weighted loss whose whitened weighted Gauss–Newton block $H_L(w)$ is invertible, smooth in $w$ — and let the prior be matching, $\vartheta^0 = \vartheta^e(q)$. (On the instance the network's amplitude–mass degeneracy gives $\vartheta^e(w)$ explicitly: the amplitudes $A_k/w_k$ with the bumps' centres and widths, on the branch of $\vartheta^e(q)$ while $|A_k|/w_k < a_{\max}$; these weights are the* representable *ones, $W$. They are not Proposition D‴.5's admissible set $W_{\text{adm}}$ of the fixed prior, where its replicated stationary point is a strict local minimiser; a sample in $W_{\text{adm}}$ is called* D‴.5-admissible.*)*

(i) (the floor is the bias) *Under the nearest-atom leak toward the fixed prior, the weighted stationary point $\Theta^*(w)$ of Proposition D‴.5 satisfies, with Theorem D‴.23's whitened*

*weighted coordinates at the weights $w$,*

$$\hat{f}_{\Theta^*(w)} - f = \hat{J}(w)\sigma_0(H_L(w) + \sigma_0 I)^{-1}x^0(w) + O(\|x^0(w)\|^2),$$
$$x^0_k(w) := \sqrt{w_k}G^{-1/2}(\vartheta^0_k - \vartheta^e_k(w)),$$

*$\hat{J}(w) := [\sqrt{w_k}\nabla_\vartheta\tilde{\Phi}(\cdot\,;\vartheta^e_k(w))G^{1/2}]_k$ (so that $\hat{J}^\top\hat{J} = H_L$): the leak's bias at the sampled weights, zero at $w = q$. The function-level weight-sensitivity of Propositions D‴.5 and D‴.9 is its derivative, $F = \hat{J}\sigma_0(H_L + \sigma_0 I)^{-1}\partial_w x^0$ at $q$ — exactly, the remainder's derivative vanishing with $x^0(q)$ — and on the eigenvectors $v_j$ of $H_L(q)$ (eigenvalues $h_j$) the function-level floor law of Proposition D‴.9 reads*

$$\mathbb{E}\|\hat{f}_m - \hat{f}_M\|^2 = (1/m - 1/M)\cdot\sum_j \frac{h_j\sigma_0^2}{(h_j+\sigma_0)^2}\cdot z_j^\top\tilde{C}z_j + o(1/m), \qquad z := V^\top\partial_w x^0, \quad \text{(B)}$$

*$\partial_w$ taken along the zero-sum directions $e_j - e_{K_0}$ and $\tilde{C}$ the multinomial covariance of $(w_1,\dots,w_{K_0-1})$ — Proposition D‴.9's law written on the eigenvectors, and like it a law for samples in $W_{\mathrm{adm}}$: at a saddle the replicated stationary point is left by a split (Proposition D‴.5), and the floor is the split configuration's. Each mode is weighted by $h_j\sigma_0^2/(h_j+\sigma_0)^2 \le \sigma_0/4$, with equality at $h_j = \sigma_0$: a band-pass centred at the level of the leak, at least half its peak for $h_j/\sigma_0 \in [3-2\sqrt{2}, 3+2\sqrt{2}]$ (a factor 5.8 either side). The law is therefore at most $(1/m-1/M)(\sigma_0/4)\,\mathrm{tr}(\partial_w x^0\tilde{C}\partial_w x^{0\top})$, whatever the excitation spectrum. A weakly excited mode keeps its bias but hardly shows it in the function; a stiff one shows little bias; which modes carry the floor is set first by the gauge's displacement $\partial_w x^0$ and then re-weighted toward $h \approx \sigma_0$.*

(ii) (the nearest-atom leak toward an adaptive prior: the masses minimised out) *Let the prior's atoms follow the means of their Voronoi cells, $\mathrm{d}z_j/\,\mathrm{d}s = -\eta(z_j - \hat{T}_j)$ ($\eta > 0$) with $\hat{T}_j$ the mean of the particles nearest to $z_j$, while the particles follow the nearest-atom leak toward $Z$. Inside the cells this is the gradient flow of*

$$\mathcal{J}_\circ(\Theta, Z) := \bar{L}(\mu_\Theta) + (\sigma_0/2m)\sum_i \min_j \|\vartheta_i - z_j\|^2_{G^{-1}} = \min_{q'\in\Delta}\mathcal{J}(\mu_\Theta, \sum_j q'_j\delta_{z_j})$$

*in the metric $(1/m)G^{-1}$ per particle and $(\sigma_0 w_j/\eta)G^{-1}$ per atom, $w$ the cell masses: the OT leak of Theorem D‴.23 with the prior's masses minimised out (semi-relaxed transport — the infinitely fast limit of a dynamics of the masses, not a learning law for them), which sends every particle to its nearest atom. The counts change only when a particle crosses into another cell; across a crossing the flow is the vertex-selection flow of $\mathcal{J}_\circ$ — a minimum over the assignments of smooth functions, in a metric that depends on the active assignment — and $\mathcal{J}_\circ$ stays non-increasing, as in Theorem D‴.23(i). A cell without particles has a degenerate metric: its atom is frozen, and the cell is refilled only if a particle's own dynamics carries it in (on the instance an empty cell stays empty; tests).*

(iii) (no function-level floor at the representable samples) *At fixed counts $N$, $w = N/m$, the stationary replicated configurations are the critical points of the unregularised weighted $K_0$-atom loss at $w$ (Theorem D‴.23(i) with $w$ in place of $q$), and for $w \in W$ the exact representation $\vartheta^e(w)$ with the prior at its atoms is one, with $\hat{f} = f$. At representable sampled weights the function-level sensitivity therefore vanishes identically, and the function-level distance between two systems at their exact representations is zero. There $r = 0$, so $R = 0$ and Proposition D‴.5's splitting margins are the maximal $\sigma_0$ for every $w \in W$: the adaptive prior also removes the fixed prior's splitting instability, whose replicated point is a saddle wherever a cluster overshoots (on the instance at $w_1 \gtrsim 0.4$). Outside $W$ (on the instance some $|A_k|/w_k \ge a_{\max}$) no exact representation exists on the branch of $\vartheta^e(q)$*

— *a permutation of the classes over the bumps can be representable, but the dynamics from the warm clouds follows the branch — and the adaptive prior removes the leak's bias without reaching $f$: the amplitude coordinates run off (Theorem D‴.23(i)), and the sample keeps a function-level error. The probability of such samples decays exponentially in $m$ (on the instance at most $\sum_k \exp(-2m(q_k - |A_k|/a_{\max})^2)$, by Hoeffding's inequality, vacuous at $m = 15$), so if every representable sample converges to its exact representation the expected limiting floor is exponentially small in $m$ rather than of order $1/m$; on the instance this asymptotic statement is not yet in force at $m \le 60$ (numerics, Part E). The parameter-level stationary measure $\sum_k w_k \delta_{\vartheta^e_k(w)}$ still carries the sample's weights — the counts fix the amplitude–mass gauge — and its coupled distance is the larger one, conditionally on representability (outside $W$ the parameters run off and the coupled distance diverges): Proposition D‴.5's coupled distance has the sensitivity $S^e := \partial_w \vartheta^e$ of the exact representation, where the fixed prior's is, in the whitened weighted coordinates and to first order, $H_L(H_L + \sigma_0 I)^{-1} S^e$. Per mode, the weight fluctuation of the exact representation divides into the fraction $h_j/(h_j + \sigma_0)$ carried by the parameters and the fraction $\sigma_0/(h_j + \sigma_0)$ left as the bias, which the function sees with the weight $h_j$; the adaptive prior moves all of it into the parameters. Since $S^e = -\partial_w x^0$, the bound of (i) is $\sigma_0/4$ times the adaptive prior's coupled-distance law: the fixed prior's function-level law is at most $\sigma_0/4$ times the adaptive prior's parameter-level law. At the measure level the two priors hardly differ: between the stationary measures of two samples, $W^2$ is the transport of the mass $\frac{1}{2}\|w - w'\|_1$ across the inter-cluster distances, of order $\|w - w'\|_1 \sim m^{-1/2}$ under both priors, plus a within-class part of order $\|w - w'\|^2$, the larger under the adaptive prior.*

(iv) (the rates) *At $\vartheta^e(w)$ the linearisation is Theorem D‴.23(ii) with $w$ in place of $q$: the collective pairs relax at $\lambda_j^-(w) < \min\{h_j(w), \eta\}$, and the fit, weighted by $h_j$, recovers faster than the parameters. On the boundary of $W$ (an amplitude $A_k/w_k$ equal to the saturation $a_{\max}$) the exact representation sits at infinity in the unconstrained coordinates and the convergence is algebraic: with $a = a_{\max}\tanh(\tilde{a}/a_{\max})$, $a_{\max} - a \approx 2a_{\max}e^{-2\tilde{a}/a_{\max}}$, the loss along the saturating coordinate is of order $e^{-4\tilde{a}/a_{\max}}$, its gradient flow gives $\tilde{a} \approx (a_{\max}/4)\log s$, and the function-level error decays like $a_{\max}^2 w_k(1 + \sigma_0/\eta)/(8G_\alpha s)$, the loss's curvature cancelling (formal).*

(v) (the design table) *Fixed prior, nearest-atom leak: the function-level floor* (B) *at the D‴.5-admissible samples (elsewhere a cluster splits) and, if the prior is misplaced, its bias. Fixed prior, OT leak (Proposition D‴.12): on the commensurate sizes no function- or measure-level floor, and the prior's bias. Adaptive prior, nearest-atom leak: at the representable samples no function-level floor and no bias, and the larger coupled distance of the gauge. Adaptive prior, OT leak (Theorem D‴.23): neither function-level floor nor bias, the masses fixed by design. Under both OT leaks the coupled distance keeps the transit's mismatch (Proposition D‴.12's* Open*). The mass constraint fixes the gauge, and the adaptive prior removes the bias — with it the function-level floor, which is its slope in $w$. This resolves Remark D‴.10's trade-off between the window's margin and the coupling's floor at the function level: at an exact representation the separation margin is the maximal $\sigma_0$ and both the bias and the function-level floor vanish. The price is the collective rate, $\lambda_j^-$ in place of $h_j + \sigma_0$ (Theorem D‴.23(ii)), the leak's a-priori bound (restored by projecting the prior, Proposition D‴.27), and a larger coupled distance.*

(vi) (the real loop: perfect tracking) *With the adaptive prior, the lift of the exact representation of representable sampled weights, with the prior at its atoms and the plant on the reference ($e = 0$), is an invariant set of the real closed loop of Theorem J: $\hat{f} = f$ on the reference orbit — the support of the occupation measure $\rho$, which is all the error equation sees at $e = 0$ — so $e \equiv 0$ solves the error equation there (its linear stability also uses $\partial_x \hat{f} = \partial_x f$ on the orbit, which the exact representation, an identity of functions, supplies: Corollary*

*D‴.26(c)), and the adaptive law's gradient term (proportional to e), the leak (toward each particle's own position) and the prior's motion (toward its own atom) all vanish: perfect tracking at every representable sample. With a fixed prior the tracking error keeps Theorem J's slaved value $e = (1/k_e) r_{\Theta^*(w)}(x_m) + O(k_e^{-2})$: the leak's bias, divided by the gain.*

*Proof.* (i) The derivation of Theorem D‴.23(iii) at the weights $w$: to first order in $u_k := \vartheta_k^*(w) - \vartheta_k^e(w)$, the stationarity of $\Theta^*(w)$ reads $H_L(w)y + \sigma_0(y - x^0(w)) = 0$ in $y_k := \sqrt{w_k} G^{-1/2} u_k$, and $\hat{f}_{\Theta^*(w)} - f = \sum_k w_k \nabla_\vartheta \tilde{\Phi}(\cdot\,; \vartheta_k^e) u_k + O(\|u\|^2) = \hat{J}y + O(\|u\|^2)$. Since $x^0(q) = 0$, the derivative at $q$ keeps only $\partial_w x^0$ (the derivatives of $\hat{J}$, of $H_L$, of the factors $\sqrt{w_k}$ and of the $O(\|x^0\|^2)$ remainder multiply $x^0(q)$ or $u(q) = 0$), so $F$ is exact at $q$; the vectors $\hat{J}v_j$ are orthogonal with $\|\hat{J}v_j\|^2 = h_j$, which with Proposition D‴.9's law gives (B); and $h\sigma_0^2/(h+\sigma_0)^2 \le \sigma_0/4$ is $(h+\sigma_0)^2 \ge 4h\sigma_0$, with $h\sigma_0^2/(h+\sigma_0)^2 \ge \sigma_0/8$ iff $h^2 - 6h\sigma_0 + \sigma_0^2 \le 0$. (ii) For fixed $Z$ and $\mu$ every coupling of $\mu$ with some $\sum_j q'_j \delta_{z_j}$ costs at least $\int \min_j \|\vartheta - z_j\|^2 \,\mathrm{d}\mu$, with equality for the nearest-atom map, whose image masses are the cell masses; inside the cells $\nabla_{z_j}(\sigma_0/2m) \sum_{i:j(i)=j} \|\vartheta_i - z_j\|^2_{G^{-1}} = \sigma_0 w_j G^{-1}(z_j - \hat{T}_j)$, and $\nabla_{\vartheta_i} \mathcal{J}_\circ$ is Proposition D‴.9's. Across a crossing, $\mathcal{J}_\circ$ is the minimum over the assignments of smooth functions; a vertex-selection solution follows the gradient flow of an active assignment, whose derivative along it is minus the squared norm of the velocity in that assignment's metric and bounds the upper Dini derivative of the minimum, so $\mathcal{J}_\circ$ is non-increasing. An empty cell's atom feels no force ($w_j = 0$) and does not move. (iii) As Theorem D‴.23(i); at $\vartheta^e(w)$ the residual vanishes, so $R[\mu](\vartheta) = \langle r \nabla^2_\vartheta \tilde{\Phi} \rangle_\rho = 0$ and the separation blocks $\sigma_0 I - R^G$ of Proposition D‴.5 are $\sigma_0 I$. The parameter level: two replicated measures with the weights $w$ and $w'$ and atoms $O(\|w - w'\|)$ apart differ by the mass $\frac{1}{2}\|w - w'\|_1$ that a coupling must move across inter-cluster distances. The split: $\vartheta^*(w) = \vartheta^e(w) + u(w)$ with $\partial_w y = -\sigma_0 (H_L + \sigma_0 I)^{-1} S_y^e$ at $q$ ($S_y^e$ the whitened weighted $S^e$), so the fixed prior's $S_y = S_y^e + \partial_w y = H_L (H_L + \sigma_0 I)^{-1} S_y^e$; and $S_y^e = -\partial_w x^0$ by the definition of $x^0$. (iv) As Theorem D‴.23(ii); the boundary case: with $\varepsilon := a_{\max} - a$ and $\alpha - z_\alpha$ slaved, the conservation law $\frac{\mathrm{d}}{\mathrm{d}s}[\alpha + (\sigma_0/\eta) z_\alpha] = G_\alpha \langle r \varphi_k \rangle \partial a / \partial \alpha$ gives $\mathrm{d}\varepsilon / \mathrm{d}s = -(\eta/(\sigma_0 + \eta)) G_\alpha \langle r \varphi_k \rangle (\partial a / \partial \alpha)^2$, with $\langle r \varphi_k \rangle = w_k \varepsilon \langle \varphi_k^2 \rangle$ and $\partial a / \partial \alpha \approx 2\varepsilon / a_{\max}$, so $\varepsilon^2 \approx a_{\max}^2 (\sigma_0 + \eta)/(8 \eta G_\alpha w_k \langle \varphi_k^2 \rangle s)$ and $\langle r^2 \rangle = w_k^2 \varepsilon^2 \langle \varphi_k^2 \rangle$. (vi) Substitution into the closed loop; the slaved error is Theorem J's Step 2. □

**Status.** (i): $F = \hat{J}\sigma_0 (H_L + \sigma_0 I)^{-1} \partial_w x^0$ is exact at $q$, the bias formula at $w \ne q$ is first order in $x^0(w)$, and the law (B) is formal — the delta method, as Proposition D‴.9 — and holds for D‴.5-admissible samples; its value is the asymptotic one (on the instance the exact floor's expectation is 2.8, 1.5 and 1.1 times it at $m = 15$, 60, 240). (ii) proved inside the cells, and across crossings for vertex-selection solutions; (iii) proved for the stationary points, the probability bound by Hoeffding's inequality, the convergence to the exact representations numerical; (iv) as Theorem D‴.23, the boundary case formal; (v) a summary of (i)–(iii) with Proposition D‴.12 and Theorem D‴.23; (vi) the invariance proved, its attractivity in the real loop not proved here (it would follow from (iv) and Theorem D‴.18's argument for the joint system, as sketched in Theorem D‴.23(v)) and checked numerically. The amplitude–mass degeneracy is the homogeneity that the conic (Wasserstein–Fisher–Rao) view of mean-field networks exploits [16], [23]; the finite-size fluctuations of trained mean-field networks around their mean-field limit are known to scale like $1/m$, with central limit theorems for the training dynamics ([89], [91]), and the asymptotic deviation is known to vanish when the mean-field dynamics converges to a measure that interpolates the data (Chen–Rotskoff–Bruna–Vanden-Eijnden, [91]). That is the mechanism by which the adaptive prior removes the function-level floor: its replicated stationary points are unregularised interpolants, and an exact fit is weight-invariant, so the absence of a function-level floor in (iii) follows at once. (i) is the chain rule applied to Theorem D‴.23(iii)'s first-order bias at $x^0(q) = 0$, and its weight is the squared complement of the Tikhonov filter factor times the mode's visibility $h_j$, bounded by $\sigma_0/4$ by the AM–GM bound familiar from ridge regression's filter factors. New are the identification of the function-level floor with the weight-derivative of the regulariser's bias at the sampled weights, with its mode-by-mode law and the bound's identity with the adaptive prior's coupled-distance law; the split of

the weight fluctuation between the parameters and the bias (iii); and a leak toward a learned prior as the way to restore the interpolation property under regularisation, which also removes the fixed prior's splitting instability. Checked: `tests/test_adaptive_floor.py` (the nearest-atom adaptive flow against finite differences of $\mathcal{J}_\circ$ with the cell masses; $\mathcal{J}_\circ$ the minimum of the OT energy over the masses, attained at the cell masses and strictly below the energies with masses moved by $\pm 1/m$; the bias formula against the finite-difference sensitivity, the mode terms summing to the law, the law below the bound, and the weight moving the shares toward $h \approx \sigma_0$ against the unweighted ones at a small and a large $\sigma_0$; at the D‴.5-admissible sample $(5, 4, 6)/15$ the exact representation, with the amplitudes $A_k/w_k$, stationary for the adaptive prior while the fixed prior's stationary point is a strict local minimiser and biased; at the saddle sample $(6, 5, 4)/15$ the fixed prior's splitting margin negative and the exact representation's margins $\sigma_0$; the parameter split; $\mathcal{J}_\circ$ decreasing across a cell crossing at $\eta = \sigma_0$ and $\infty$, and an empty cell staying empty with its atom frozen).

**Numerics** (`exp_adaptive_floor.py` → `results/chaos/adaptive_floor.md`). The three-bump instance at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$ (and $0.05\sigma_{\mathrm{crit}}$, $0.5\sigma_{\mathrm{crit}}$ for the law), the matching prior unless stated, the warm clouds of Proposition D‴.12's numerics (width 0.02, four seeds), $m = 15$, 60, 240 against $M = 510$; the checks at the sampled weights from the stored final states. *The floor is the bias* (Part A). The function-level sensitivity $F$ of the fixed prior's stationary point, by central differences in $w$, equals the bias form $\hat{J}\sigma_0(H_L + \sigma_0 I)^{-1}\partial_w x^0$ to the finite-difference error ($1.1 \cdot 10^{-6}$, $1.9 \cdot 10^{-7}$, $1.2 \cdot 10^{-8}$ at $\sigma_0/\sigma_{\mathrm{crit}} = 0.05$, 0.15, 0.5, with the step $10^{-4}$); the law (B) per unit of $1/m - 1/M$ is $1.356 \cdot 10^{-2}$, $6.166 \cdot 10^{-2}$, 0.2291, equal to the sum of its mode terms, at 0.496, 0.753, 0.839 of the bound, and the bound is $\sigma_0/4$ times the adaptive prior's coupled-distance law 46.32. Modes 2–6 carry 99.4%, 99.0%, 97.9% of the law, and 97.2% of the unweighted sum $\sum_j z_j^\top \tilde{C} z_j$ — the concentration is mostly the gauge's — while the weight moves the shares toward $h \approx \sigma_0$: the modes with $h_j/\sigma_0 \in [1/5, 5]$ carry 42%, 99%, 90% of the law against 21%, 97%, 78% unweighted. The law's value at $m = 15$, 60, 240 is $3.99 \cdot 10^{-3}$, $9.07 \cdot 10^{-4}$, $1.36 \cdot 10^{-4}$. The parameter split $S = H_L(H_L + \sigma_0 I)^{-1} S^e$ holds to $1.4 \cdot 10^{-7}$–$2.9 \cdot 10^{-8}$, and the adaptive prior's coupled-distance law is 1.44, 2.17, 5.27 times the fixed prior's. Every sampled weight vector is representable (margins $\min_k(w_k - |A_k|/a_{\max}) = 0.025$–$0.10$) except seed 3's at $m = 15$, $(8, 4, 3)/15$, which lies on the boundary ($w_3 = |A_3|/a_{\max}$). But 5 of the 12 sampled $m$-systems are not D‴.5-admissible for the fixed prior: $(6, 5, 4)/15$, $(6, 3, 6)/15$, $(24, 15, 21)/60$, $(8, 4, 3)/15$ and $(25, 17, 18)/60$ have splitting margins $-3.48$, $-1.08$, $-2.16$, $-6.14$, $-3.98 \cdot 10^{-3}$, all at class 1 with $w_1 \geq 0.4$; the other seven and the four references have margins $\geq 3.78 \cdot 10^{-3}$. *Fixed against adaptive priors* (Part B, the nearest-atom leak, 6000 slow units). Under the fixed prior $\|\hat{f}_m - \hat{f}_M\|^2$ stays at its stationary value, $3.0 \cdot 10^{-4}$–$5.5 \cdot 10^{-3}$ at $m = 15$, $1.1 \cdot 10^{-4}$–$9.9 \cdot 10^{-4}$ at $m = 60$ and $4.8 \cdot 10^{-6}$–$8.2 \cdot 10^{-5}$ at $m = 240$ over the seeds. At the seven D‴.5-admissible samples it equals the exact bias at the sampled weights, $\|\hat{f}_{\Theta^*(w^m)} - \hat{f}_{\Theta^*(w^M)}\|^2$, to four digits (ratio 1.0000), while the delta method's $\|F(w^m - w^M)\|^2$ misses it by the factors 0.86–1.72; the five saddle systems have split (class diameters 0.64–1.50 at the end, class 1 broken into sub-clusters) and sit at 0.81–1.01 of the replicated point's floor. Under the adaptive prior the function-level distance decreases without a floor: at $s = 6000$ to $4.6 \cdot 10^{-9}$–$2.0 \cdot 10^{-6}$ ($\eta = \sigma_0$) and $1.9 \cdot 10^{-10}$–$5.4 \cdot 10^{-8}$ ($\eta = \infty$) at $m = 15$, to $5.4 \cdot 10^{-10}$–$6.2 \cdot 10^{-9}$ and $2.4 \cdot 10^{-10}$–$6.2 \cdot 10^{-10}$ at $m = 60$, and to $2.5 \cdot 10^{-12}$–$1.9 \cdot 10^{-8}$ and $7.1 \cdot 10^{-14}$–$2.2 \cdot 10^{-9}$ at $m = 240$, every adaptive system replicated at the end and the class means converging to the exact representation of the sampled weights. *The test of (iii)* at the parameter level: the coupled distance $D$ of the shared particles is 0.944–1.000 of the replicated $\Theta^*(w)$'s under the fixed prior and 0.951–1.000 of the exact representations' under the adaptive ones (both samples representable), and the adaptive prior's is 1.72–4.81 times the fixed prior's, sample by sample. $W^2(\mu_m, \mu_M)$ is dominated, under both priors, by the cross-class transport of the mass $\frac{1}{2}\|w^m - w^M\|_1$ — 0.82–0.99 of $W^2$ under the fixed prior, 0.67–0.99 under the adaptive ones — so the two priors' $W^2$ differ by the factors 1.00–1.32 only, while their within-class parts differ by 1.71–4.84. On the boundary sample the

function-level distance decays algebraically ($6000 \cdot \|\hat{f}_m - \hat{f}_M\|^2 = 0.88$ at $\eta = \sigma_0$ and 0.42 at $\eta = \infty$, against the predicted 0.8 and 0.4), and its parameters run off ($D = 6.4$ and 7.8 at $s = 6000$). The counts are conserved in every run. *The design table* (Part C, the misplaced prior, 3000 slow units, seed means at $m = 15$, 60, 240). Nearest-atom leak: the fixed prior's fit error $6.8 \cdot 10^{-2}$, $3.1 \cdot 10^{-2}$, $2.0 \cdot 10^{-2}$ and floor $2.5 \cdot 10^{-3}$, $5.1 \cdot 10^{-4}$, $5.9 \cdot 10^{-5}$ — the same five samples are saddles at the misplaced prior (margins $-2.87$, $-0.38$, $-1.52$, $-5.76$, $-3.41 \cdot 10^{-3}$), four split by $s = 3000$ and the fifth partly — against the adaptive prior's ($\eta = \sigma_0$) $8.9 \cdot 10^{-3}$, $5.5 \cdot 10^{-4}$, $6.0 \cdot 10^{-4}$ and $8.0 \cdot 10^{-5}$, $1.6 \cdot 10^{-7}$, $1.1 \cdot 10^{-7}$ (at $m = 15$ one sample is on the boundary). OT leak: the fixed prior's $9.78 \cdot 10^{-3}$, the bias, with no floor (below $3 \cdot 10^{-21}$), against the adaptive prior's $1.6 \cdot 10^{-4}$ with the floor below $5 \cdot 10^{-11}$ (settled gate). At the coupling level the OT designs end with the largest $D$, 6.0, 3.5, 1.0 at $m = 15$, 60, 240 — the transit moves 7, 14, 26 of the $m$-systems' particles into another cell (over the four seeds) — against the nearest-atom leak's 0.86, 0.25, 0.017 (fixed) and 2.3, 0.48, 0.046 (adaptive), while their $W^2$ is at most $2.1 \cdot 10^{-8}$. The ranking of the fit errors depends on the horizon: until the settled gate opens ($s = 1211$–$1304$) the OT design toward the adaptive prior is the fixed one, and at $s = 1250$ the ungated nearest-atom adaptive prior leads at $m = 60$ and 240 ($3.8 \cdot 10^{-3}$ and $2.8 \cdot 10^{-3}$ against $9.2$–$9.8 \cdot 10^{-3}$), not at $m = 15$ ($1.75 \cdot 10^{-2}$ in the seed mean); from $s \approx 2000$ the OT design leads ($4.0$–$4.2 \cdot 10^{-4}$ against $1.2 \cdot 10^{-2}$, $1.6 \cdot 10^{-3}$, $1.4 \cdot 10^{-3}$). *The real loop* (Part D, $k_e = 25$, 3000 time units, seed 0's sampled weights). From the fixed prior's stationary point the rms tracking error stays at $1.17$–$1.18 \cdot 10^{-3}$ ($m = 15$) and $3.43$–$3.44 \cdot 10^{-4}$ ($m = 60$), at the end 1.001 and 0.996 times the averaged loop's $(1/k_e)\|f - \hat{f}\|_\rho$ at $s = t/k_e$ — the $m = 15$ run on a saddle orbit (Proposition D‴.15; margin $-3.48 \cdot 10^{-3}$), held there by the exact replication of its start; with the adaptive prior ($\eta = \infty$) it falls to $3.27 \cdot 10^{-4}$ and $1.52 \cdot 10^{-4}$ by $t = 3000$ ($s = 120$), along the averaged loop (ratios 0.987 and 0.992); from the exact representation of the sampled weights with the adaptive prior at its atoms it stays at the integrator's level (the five-period rms $9.0 \cdot 10^{-12}$ at the first window, falling to $1.4$–$1.5 \cdot 10^{-12}$): perfect tracking is invariant in the real loop. *The instance's probabilities* (Part E, exact multinomial sums). Outside $W$: 0.361, 0.074, 0.0021 at $m = 15$, 60, 240, and on its boundary 0.118, 0.053, 0.0012, against Hoeffding's bound 1.87, 0.60, 0.036; a permutation of the classes would be representable for 31%, 87%, 100% of the mass outside $W$. The fixed prior's replicated point is a saddle with probability 0.369, 0.260, 0.109 — a failure set comparable to the adaptive prior's (outside $W$ or on its boundary: 0.48, 0.13, 0.003) at $m = 15$ and much larger at $m \geq 60$. Taking the constrained minimum of the loss on the branch of $\vartheta^e(q)$ as the limit outside $W$ (the dynamics from the warm clouds follows the branch; a local minimum, the global infimum being 0 wherever a permutation is representable), the adaptive prior's expected limiting floor is at least $7.3 \cdot 10^{-3}$, $2.2 \cdot 10^{-4}$, $6.6 \cdot 10^{-7}$: 1.8, 0.24, 0.005 times the fixed prior's law and 0.66, 0.16, 0.004 times its Monte Carlo expectation. *The law's accuracy* (Part F, 600 nested samples, the exact replicated stationary points). The fixed prior's $\mathbb{E}\|\hat{f}_m - \hat{f}_M\|^2$ is $2.78 \pm 0.20$, $1.53 \pm 0.11$, $1.11 \pm 0.07$ times the law ($2.89 \pm 0.26$, $1.56 \pm 0.15$, $1.07 \pm 0.07$ for D‴.5-admissible pairs): the $o(1/m)$ remainder dominates at $m = 15$, where the gauge's derivative varies fast near the saturation (the matching amplitude 3 against $a_{\max} = 4$). The fixed prior's coupled distance follows its law more closely ($\mathbb{E}D$/law 0.68, 0.93, 0.95); the adaptive prior's (0.53, 1.17, 1.26) has a far heavier tail — its gauge diverges at the boundary of $W$; at $m = 240$ one sample lies at 55.7 times the law, and without it the mean is 1.17 — so its means are not reliable estimates at this sample size; it is 2.9–3.7 times the fixed prior's.

**The planar check** (`exp_adaptive_floor_2d.py` → `results/chaos/adaptive_floor_2d.md`). The planar plant of `exp_ot_2d.py` — four bumps with vector amplitudes, the Lissajous reference, $p = 5$, $G = \mathrm{diag}(1, 1, 0.25, 0.25, 0.25)$, $\sigma_0 = 0.05$ — with the averaged slow loop on $\mathbb{R}^2$ of `src/core/planar_loop.py`, the matching prior, $m = 16$, 32, 64 against $M = 128$, six seeds, 6000 slow units ($\eta = \infty$: the atoms reset to the cell means at every refresh of the cells, every slow unit, and frozen in between — a lagged prior). The excitation spectrum is $h_j/\sigma_0 = 0.004$–$1.41$

(twenty modes, all below $1.5\sigma_0$); $F$ equals the bias form to the finite-difference error ($6.0 \cdot 10^{-7}$), and the law per unit of $1/m - 1/M$ is 0.310, at 0.215 of the bound, spread over the modes (the largest share 0.18). Representability needs $w > (0.200, 0.150, 0.175, 0.150)$, whose sum 0.675 leaves little room: 7 of the 18 $m$-systems are representable, with the six references. Every sample is D‴.5-admissible for the fixed prior (splitting margins +0.016 to +0.047), and the fixed prior's floor equals the exact bias at the sampled weights in all 18 systems (ratio 1.0000): $3.2 \cdot 10^{-4}$–$4.2 \cdot 10^{-3}$ at the representable systems, $3.0 \cdot 10^{-3}$–$2.1 \cdot 10^{-2}$ at the others. At the representable systems the adaptive prior's falls without a floor, to $1.0 \cdot 10^{-7}$–$4.1 \cdot 10^{-5}$ ($\eta = \sigma_0$) and $1.2 \cdot 10^{-9}$–$1.0 \cdot 10^{-5}$ ($\eta = \infty$) at $s = 6000$ — at most 0.019 and 0.006 of the fixed prior's, still falling in every system — and at the others to 0.023–0.64 of the fixed prior's, still falling, toward the constrained minimum of the loss at their sampled weights — a floor of their own (Proposition D‴.24(iii)) — not toward zero. The counts are conserved in every run. In the real loop ($k_e = 25$, $\mathrm{d}t = 2 \cdot 10^{-3}$, 2500 time units, seed 0, $m = 64$) the fixed prior's rms tracking error stays at the averaged prediction (ratios 0.993–1.009) and the adaptive prior's ($\eta = \infty$) follows the averaged decay (0.979–1.009); from the exact representation of the sampled weights with the adaptive prior at its atoms it stays at the integrator's level, $2.0$–$4.7 \cdot 10^{-10}$.

**Remark D‴.25** (the design table of the leak)**.** Propositions D‴.9, D‴.12, D‴.20, D‴.24 and Theorem D‴.23 choose, for the averaged loop near a replicated configuration, what the leak fixes — the prior's atoms and its masses — and what each choice costs. With a matching prior unless a misplaced one is named, at commensurate sizes for the OT leaks, and with the weights $w$ sampled from an i.i.d. warm start ($\delta w := w^m - w^M$; Table 19):

Table 19: Remark D‴.25: the design table of the leak for the averaged loop near a replicated configuration — for each leak and prior, the function-level floor, the coupled distance $D$ of Proposition D‴.5, $W^2$ at the measure level, the bias of a misplaced prior, the transit, the collective rates and the a-priori bound.

| leak, prior | function-level floor | coupled distance $D$ (Proposition D‴.5) | $W^2$ (measure level) | bias of a misplaced prior | transit | collective rates | a-priori bound |
|---|---|---|---|---|---|---|---|
| nearest atom, fixed (D‴.9) | $(1/m - 1/M)$ $\sum_j h_j \sigma_0^2/(h_j+\sigma_0)^2$ $z_j^\top \tilde{C} z_j$ at D‴.5-admissible $w$ (D‴.24(i)) | D‴.5's law with $H_L(H_L+\sigma_0 I)^{-1}S^e$ | the transport of $\frac{1}{2}\|\delta w\|_1$ between the clusters, plus $O(\|\delta w\|^2)$ | $\sigma_0(H_L+\sigma_0 I)^{-1}x^0$ | none while no particle crosses a cell boundary (D‴.9(i)'s hypothesis) | $h_j+\sigma_0$ | yes |
| nearest atom, adaptive (D‴.24) | 0 at representable $w$ | D‴.5's law with $S^e$ (larger), at representable $w$ | likewise, a larger within-class part | 0 | none (likewise) | $\lambda_j^-(w) <$ $\min\{h_j(w), \eta\}$ | no (yes when projected, D‴.27) |
| OT, fixed (D‴.12) | 0 | the transit's mismatch | 0 | $\sigma_0(H_L+\sigma_0 I)^{-1}x^0$ | the excess moved by the plan | $h_j+\sigma_0$ | yes |
| OT, adaptive (D‴.23) | 0 | the transit's mismatch | 0 | 0 | gated: adapt after the fixed-prior relaxation | $\lambda_j^-(q)$ | no (yes when projected, D‴.27) |
| entropic OT, fixed (D‴.20) | D‴.9's function-level law at $\Theta_\varepsilon^*(q)$ for sampled excesses below $\delta_c(\varepsilon)$ | D‴.5's law there | — | that of $\Theta_\varepsilon^*(q)$ | equalises the masses down to $\delta_c(\varepsilon)$ | — | yes |

*Reading.* The finite-size floor has three faces: the function level (the sample's weights acting through the leak's bias), the coupling level (the shared particles displaced by the amplitude–mass gauge, or relabelled by the transit), and the measure level (the mass $\frac{1}{2}\|\delta w\|_1$ that has to move between the clusters). The mass constraint removes the function-level and the measure-level floors by fixing the weights; at the coupling level it replaces the gauge's displacement by the transit's mismatch, which relabels different particles in the two systems. The adaptive prior removes the function-level floor by removing the bias at every representable weight, enlarges the coupled distance (the gauge's whole displacement goes into the parameters) and leaves the measure level's leading term unchanged; it also removes the fixed prior's splitting instability, every separation relaxing at $\sigma_0$ at an exact representation. The bias of a misplaced prior is removed only by an adaptive prior, and every adaptive prior pays with the rate — $\lambda_j^- < h_j$, so a weakly excited collective mode converges at the loss's own curvature rather than at the leak's — and with the leak's a-priori bound on the parameters (Theorem D‴.23(i); restored by projecting the prior onto a compact convex set, Proposition D‴.27, at the price of a bias where the limit reaches its boundary). The design without a function-level floor or bias is the OT leak toward an adaptive prior started from the fixed-prior stationary point; the floor-free design when the prior is trusted is the OT leak toward a fixed one; and when the masses cannot be prescribed (the sample decides them), the nearest-atom leak toward an adaptive prior keeps the fitted function exact at the representable samples and leaves the fluctuation in the parameters.

**Corollary D‴.26** (perfect tracking with an adaptive prior in the real loop: the Floquet spectrum, uniformly in $m$)**.** *Let* (CL) *be the real closed loop of Theorem J with the nearest-atom leak toward an adaptive prior — the particles leak at $\sigma = \sigma_0/k_e$ toward the atom of their class, and the atoms follow the class means at the rate $\eta/k_e$, $\dot z_j = -(\eta/k_e)(z_j - \bar\vartheta_j)$, $0 < \eta \le \infty$ ($\eta = \infty$: $z_j = \bar\vartheta_j$) — and let $w = N/m$ be representable sampled weights ($w \in W$, Proposition D‴.24) with the exact representation $\vartheta^e(w)$. Let $\Gamma_w$ be the lift of $\vartheta^e(w)$ by the classes, with $Z = \vartheta^e(w)$ and the plant on the reference: a $T_0$-periodic solution with $e \equiv 0$ (Proposition D‴.24(vi)). Its Floquet exponents (the rates $-\log|\lambda|/T_0$ of its monodromy) are:*

(a) *$\sigma_0/k_e$ exactly, with multiplicity $(m - K_0)p$, on the separation subspaces of the classes;*

(b) *on the collapsed subspace, the exponents of the weighted $K_0$-atom loop's periodic solution at $\vartheta^e(w)$ — the same for every $m$ with these weights;*

(c) *if moreover $\partial_x \hat f_{\vartheta^e(w)} = \partial_x f$ on the reference orbit ($\hat f = f$ on a neighbourhood of it — on the instances everywhere, the exact representation being an identity of functions) and $x_m \in C^4$: $n$ exponents $k_e + O(1/k_e)$ of the error channel, summing to $nk_e - \operatorname{tr} H_L(w)/k_e + O(k_e^{-3})$, and for the $K_0 p$ collective pairs*

$$(1/k_e)\big[\lambda_j^{\pm}(w) + c_j^{\pm}(w)/k_e^2 + O(k_e^{-4})\big],$$

*$\lambda_j^{\pm}(w)$ the rates of Theorem D‴.23(ii) at the weights $w$ ($\eta = \infty$: $h_j(w)$) and $c_j^{\pm}(w)$ the explicit coefficients of the proof (for simple eigenvalues; at a degenerate one only the rates are even). The odd orders vanish because the slow block at $-\varepsilon$ ($\varepsilon := 1/k_e$) is the time reversal of the reversed reference's at $\varepsilon$, so with the variational equation's symmetry the slow generator's spectrum is even in $\varepsilon$ — an identity (Lemma D‴.26a(iv)) that a derivative mismatch $a \not\equiv 0$ breaks; term by term, every term of $\bar A_k$ has the parity $(-1)^k$ under reversal, the first order in particular because the first-order averaging correction is skew. $c_j$ is the sum of the error filter's attenuation $-(\,\mathrm{d}\lambda_j^{\pm}/\,\mathrm{d}h) v_j^\top \langle \dot{\hat J}^\top \dot{\hat J}\rangle v_j \le 0$ (at $\eta = \infty$ it is $-h_j\omega_j^2$, $\omega_j^2 = \sum_n n^2|\hat J_n v_j|^2 / \sum_n |\hat J_n v_j|^2$ the mode's mean squared frequency in the regressor, by Parseval, independent of $\eta$: the averaged rate of a mode holds when $k_e \gg \omega_j$), of the slaving's part from $-(\bar J^\top \bar J)^2$, which speeds the modes up, and of the averaging terms, of either sign (the level repulsion through $\bar A_1$ is $\ge 0$ for the slowest mode): its sign is not fixed — on the instances every $c_j < 0$, the filter's part dominating, and with the adaptation gain scaled up some $c_j > 0$.*

*Hence, where $H_L(w) \succ 0$ (so $\lambda_1^-(w) > 0$) and for $k_e$ above a threshold — which depends on the spectral gaps of the collective block through Lemma D‴.26a's $\varepsilon_0$ —, perfect tracking is exponentially stable (locally, for each $m$) at every representable sample, with the rate $(1/k_e)\min\{\sigma_0, \lambda_1^-(w) + c_1^-(w)/k_e^2 + O(k_e^{-4})\}$; the same holds for the OT leak toward an adaptive prior (Theorem D‴.23) at a commensurate $m$, with $w = q$ and the diagonal plan (locally constant at the lift). Without the derivative condition the mismatch $a(t) := \partial_x(f - \hat f)(x_m(t))$ enters the error equation and moves the collective rates at first order in $1/k_e$ (the proof).*

*Proof.* Along $\Gamma_w$ the linearisation of (CL) reads $\delta\dot e = (a(t) - k_e)\delta e - D\hat f[\delta\Theta](x_m(t))$, $\delta\dot\vartheta_i = G\nabla_\vartheta \tilde\Phi(x_m(t); \vartheta^e_{\pi(i)})^\top \delta e - (\sigma_0/k_e)(\delta\vartheta_i - \delta z_{\pi(i)})$, $\delta\dot z_j = -(\eta/k_e)(\delta z_j - \delta\bar\vartheta_j)$, with $a \equiv 0$ under the derivative condition: the second-derivative term $G\nabla^2_\vartheta(e^\top\tilde\Phi)\delta\vartheta_i$ and the $x$-derivative of the learning law vanish because $e \equiv 0$. (a) A separation (zero class means, $\delta z = 0$, $\delta e = 0$) does not move $\hat f$ at first order — $D\hat f[\delta\Theta] = \sum_k (1/m)\nabla_\vartheta\tilde\Phi(\cdot\,; \vartheta^e_k)\sum_{i\in\pi^{-1}(k)}\delta\vartheta_i = 0$ — so it stays a separation with $\delta\dot\vartheta_i = -(\sigma_0/k_e)\delta\vartheta_i$: the subspace is invariant and the exponent is $\sigma_0/k_e$, whatever $a$. (b) The collapsed subspace is invariant (Corollary D‴.14(i), with the prior's equation: the class atoms and their prior atoms follow the weighted $K_0$-atom equations), and the loop on it is the weighted $K_0$-atom loop with its prior. (c) In the whitened weighted coordinates $y_k = \sqrt{w_k} G^{-1/2}\delta\vartheta_k$ (and $\zeta_k$ likewise for $\delta z_k$) the collapsed variational equation is

the adjoint pair $\delta\dot e = (a-k_e)\delta e - \hat J(t)y$, $\dot y = \hat J(t)^\top \delta e - (\sigma_0/k_e)(y-\zeta)$, $\dot\zeta = -(\eta/k_e)(\zeta - y)$, with $\hat J(t) = [\sqrt{w_k}(G^{1/2}\nabla_\vartheta \tilde\Phi(x_m(t);\vartheta_k^e))^\top]_k$ ($\eta=\infty$: $\zeta = y$, and the leak vanishes on the collapsed subspace). Let $\varepsilon = 1/k_e$, $Y = (y,\zeta)$, $\bar J = \hat J$ on the $y$ block and $C$ the leak and the prior's constant block in slow time. The error channel is slaved: the linear periodic system has the invariant subspace $\delta e = L(t)Y$, $L = \varepsilon L_1 + \varepsilon^2 L_2 + \varepsilon^3 L_3 + \dots$, with $L_1 = -\bar J$, $L_2 = \dot{\bar J} - a\bar J$ and, for $a \equiv 0$, $L_3 = -\ddot{\bar J} + \bar J C - \bar J \bar J^\top \bar J$ (the Chang transformation, [49]: Kokotović–Khalil–O'Reilly). This leaves, in slow time $s = \varepsilon t$, the slow block $\mathrm{d}Y/\mathrm{d}s = (B_0 + \varepsilon B_1 + \varepsilon^2 B_2 + \dots)Y$ with $B_0 = C - \bar J^\top \bar J$, $B_1 = \bar J^\top(\dot{\bar J} - a\bar J)$ and, for $a \equiv 0$, $B_2 = -\bar J^\top \ddot{\bar J} + \bar J^\top \bar J C - (\bar J^\top \bar J)^2$, a system of fast period $\varepsilon T_0$. Its near-identity averaging gives the averaged generator $\bar A = \bar A_0 + \varepsilon \bar A_1 + \varepsilon^2 \bar A_2 + \dots$, whose eigenvalues are the slow-time exponents: $\bar A_0 = \langle B_0\rangle = C - \langle \bar J^\top \bar J\rangle$, the averaged loop's linearisation at the replicated configuration (Theorem D‴.23(ii) at the weights $w$, Proposition D‴.24(iv)), self-adjoint in the metric $\mathrm{diag}(I,(\sigma_0/\eta)I)$ of Theorem D‴.23(i); $\bar A_1 = \langle B_1\rangle + \langle \tilde B_0 U_1\rangle$ with $U_1' = \tilde B_0$ (zero means); $\bar A_2 = \langle B_2\rangle + \langle B_1 U_1\rangle + \langle B_0 U_2\rangle$ with $U_2' = B_1 + B_0U_1 - U_1\bar A_0 - \bar A_1$. With $a \equiv 0$, $\bar A_1$ is skew: the self-adjoint parts of $\langle \bar J^\top \dot{\bar J}\rangle$ and $\langle \tilde B_0 U_1\rangle$ are $\frac12\langle(\bar J^\top \bar J)^\cdot\rangle = 0$ and $\frac12\langle (U_1^2)'\rangle = 0$, and a skew perturbation moves no simple eigenvalue of a self-adjoint matrix at first order ($v^\top K v = 0$; on a degenerate eigenspace it splits the eigenvalue into a pair of unchanged real part). All odd orders vanish: the variational matrix $M(t)$ satisfies $M(t)^\top = DM(t)D^{-1}$ with $D = \mathrm{diag}(-I_n, I, (\sigma_0/\eta)I)$ (at $\eta=\infty$ on $(\delta e, y)$, $D = \mathrm{diag}(-I_n, I)$), so the system with its coefficients reversed in time, $M(-t)$, has the same Floquet multipliers (its monodromy is conjugate to the inverse of the adjoint system's, which is the inverse transpose of the original's); reversing time changes the sign of every fast-time derivative and antiderivative, and a term of $\bar A_k$ carries a number of them of the parity of $k$ ($\varepsilon$ is a time, $\hat J$ and $\mathrm{d}/\mathrm{d}t$ are rates, $C$ a rate in slow time), so $\bar A_k \mapsto (-1)^k \bar A_k$, and the exponents, being unchanged, are even in $\varepsilon$ — for simple eigenvalues of $\bar A_0$; at a degenerate one the first-order splitting $\pm i\varepsilon\kappa$ is imaginary and only the rates are even. The symmetry $M(t)^\top = DM(t)D^{-1}$ alone does not suffice: it persists with $a \not\equiv 0$, which enters only the $(\delta e, \delta e)$ entry, while $a$ breaks the parity count. (The mechanism is that of the even error expansions of symmetric integrators and of the time symmetry of the Magnus exponent, [99], [95].) Perturbation theory gives $c_j = -\ell_j \bar A_2 r_j - \sum_{i\neq j}(\ell_j \bar A_1 r_i)(\ell_i \bar A_1 r_j)/(\lambda_i - \lambda_j)$ ($r_j$, $\ell_j$ the right and left eigenvectors of $-\bar A_0$, $\ell_j r_j = 1$), and $\langle B_2\rangle$ contains $-\langle \bar J^\top \ddot{\bar J}\rangle = \langle \dot{\bar J}^\top \dot{\bar J}\rangle$: the error filter $k_e/(k_e + i\omega)$ attenuates the regressor's harmonics, $H_L - \varepsilon^2\langle \dot{\hat J}^\top \dot{\hat J}\rangle + O(\varepsilon^4)$ in place of $H_L$; $\langle B_2\rangle$ also contains $-\langle(\bar J^\top \bar J)^2\rangle$, from the slaving's $L_3$, which enters $c_j$ with the opposite sign, and the level repulsion $-\sum_{i\neq j}(\ell_j \bar A_1 r_i)(\ell_i \bar A_1 r_j)/(\lambda_i - \lambda_j) = \sum_{i\neq j}|\ell_j \bar A_1 r_i|^2/(\lambda_i - \lambda_j)$ ($\bar A_1$ skew) is $\geq 0$ for the slowest mode, so the sign of $c_j$ is not fixed. The error channel's exponents are $k_e + O(\varepsilon)$, and by Liouville's formula all the exponents sum to $-\langle \mathrm{tr}\, M\rangle = nk_e + (\sigma_0 + \eta)K_0 p/k_e$ ($\eta=\infty$: $nk_e$), the collective ones to $(1/k_e)\,\mathrm{tr}(-\bar A_0) + O(k_e^{-3})$. With $a \not\equiv 0$ the term $-\langle \bar J^\top a \bar J\rangle$ of $\bar A_1$ is self-adjoint and moves the rates at first order, and $a$ — a rate that does not change sign under time reversal — breaks the parity. □

**Status.** (a) and (b) proved, exactly (no averaging); (c) proved by Lemma D‴.26a for $x_m \in C^4$ — the slaving is an exact block-triangularisation, the expansion has a bounded $O(k_e^{-4})$ remainder at simple eigenvalues (the threshold and the constant depending on the spectral gaps) and the parity is the exact identity $\mathrm{eig}\,\bar A(-\varepsilon) = \mathrm{eig}\,\bar A(\varepsilon)$, the stability statement under $H_L(w) \succ 0$ — with the structure (the skewness of $\bar A_1$, $c_j$ explicit) derived here; the stability consequence local, for each $m$ (the nearest-atom cells are constant near the lift, the particles lying inside their cells, so the field is smooth there and the multipliers decide). (a) and (b) are Corollary D‴.14(ii) and Proposition D‴.15 at zero residual with the prior's coordinates added; what the adaptive prior brings is the zero-residual lift at every representable $w$, not only at $w = q$ under the OT leak. The basin uniform in $m$ in the per-particle norm would follow from Theorem D‴.18's contraction argument for the joint system, as sketched in Theorem D‴.23(v), and is not proved here. With a fixed prior the lift of $\vartheta^e(w)$ is a solution only when the prior sits at $\vartheta^e(w)$ — at a matching prior, $w = q$; at the other sampled weights the leak's bias keeps $e = O(1/k_e)$ (Theorem J) — so perfect tracking at every representable sample, with this spectrum, is specific to the adaptive prior.

(a) does not depend on the fixed prior's splitting margin at $w$: at $(6,5,4)/15$, where the fixed prior's replicated orbit is a saddle orbit (Proposition D‴.24's numerics), every separation of perfect tracking decays at $\sigma_0/k_e$. Checked: `tests/test_adaptive_floquet.py` (at the matching weights, $\eta=\infty$, $m=6$: the return of one period, the nine separation exponents at $\sigma_0/k_e$, the collective exponents the $K_0$-atom loop's, within the averaging correction of the prediction; at the weights $(2,1,1)/4$ with the prior in the state, $\eta=\sigma_0$, $m=4$: the real loop's Jacobian at the lift mapping the separations to $-(\sigma_0/k_e)$ times themselves and the collapsed subspace into itself as the collapsed variational equation, the monodromy's 3 separation exponents at $\sigma_0/k_e$ and its 18 collective ones the variational ones, within the second-order prediction and below the first-order one; on an asymmetric reference: $\bar{A}_1$ non-zero and skew, the residual of the second-order prediction falling like $k_e^{-4}$, the measured second-order coefficient the explicit $c_1$, the error channel by Liouville's formula).

**Numerics** (`exp_adaptive_floquet.py` → `results/chaos/adaptive_floquet.md`). The three-bump instance at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$, $x_m = 2\sin t$, the sampled weights $w = (6,5,4)/15$ (the amplitudes $A_k/w_k = 2.5, -2.1, 3.0$; representable, and a saddle of the fixed prior, splitting margin $-3.48\cdot10^{-3}$), $\eta=\sigma_0$ and $\infty$, $k_e = 25$ and 50; the monodromy by central differences of the period map (step $10^{-5}$ relative, DOP853 at rtol $10^{-13}$). Every lift returns after one period to $\le 1.7\cdot10^{-15}$. (a) At $m=15$ and 30 the $(m-3)\cdot3 = 36$ and 81 separation exponents equal $\sigma_0/k_e$ to $\le 1.7\cdot10^{-6}$ (relative). (b) The other 18 ($\eta=\sigma_0$) or 9 ($\eta=\infty$) exponents equal the weighted $K_0$-atom loop's to $1.4\cdot10^{-5}$–$6.4\cdot10^{-4}$ (relative, the resolution of the differences for the slowest of them). (c) Part C integrates the variational equation on the collapsed subspace over a period (its generator equals the central differences of the real loop's field to $\le 5.8\cdot10^{-11}$ relative, its exponents the differences' to $\le 4.1\cdot10^{-5}$) at $k_e = 12.5, 25, \ldots, 400$. At $\eta=\sigma_0$ the slowest exponent lies 37.5%, 13.6%, 3.84%, 0.99%, 0.25% and 0.063% below $\lambda_1^-(w)/k_e$ — every exponent lies below its first-order prediction at every $k_e$, a property of the instance — and the largest deviation falls by 2.77, 3.53, 3.86, 3.96, 3.99 per doubling of $k_e$ ($\to 4$), while the residual of the second-order prediction falls by 11.1, 14.0, 15.4, 15.8, 16.0 ($\to 16$: $O(k_e^{-4})$; $4.9\cdot10^{-7}$ at $k_e=400$). Every $c_j$ is negative here ($c_j/\lambda_j$ from $-0.47$ to $-100.6$); the slowest mode's $c_1/\lambda_1 = -100.6$ ($-101.7$ at $\eta=\infty$), of which the error filter's part is $-102.1 = -0.989\omega_1^2$ ($-103.3 = -\omega_1^2$ at $\eta=\infty$), $\omega_1 = 10.16$ the slowest mode's root-mean-square frequency in the regressor — the weakest-excited mode is carried by the regressor's high harmonics, which the error filter attenuates unless $k_e \gg \omega_1$ — and the measured $(k_e\mu_1 - \lambda_1)k_e^2/\lambda_1$ approaches it ($-99.4$, $-100.3$, $-100.5$ at $k_e = 100$, 200, 400; $\mu_1$ the slowest exponent). On $x_m = 2\sin t$ the first-order correction vanishes identically (in one dimension $\langle \hat{J}^\top \dot{\hat{J}}\rangle$ is the integral of an exact differential, and the commutator term vanishes by the symmetry $t \to \pi - t$), so this reference alone could not test the mechanism; on the asymmetric $x_m = 2\sin t + 0.8\sin 2t$ it does not vanish ($\|\bar{A}_1\| = 5.6\cdot10^{-3}$, its self-adjoint part $1.4\cdot10^{-16}$), and the deviations still fall by $\to 4$ per doubling and the second-order residuals by $\to 16$ (15.5 and 15.9 at the last two doublings), with $c_1/\lambda_1 = -108.2$ (the filter's part $-109.7 = -0.984\omega_1^2$, $\omega_1 = 10.56$). With the adaptation gain scaled by 30 two coefficients are positive ($c_8/\lambda_8 = +4.17$, $c_9/\lambda_9 = +12.65$) and those exponents lie above their first-order predictions by $c_j/(\lambda_j k_e^2)$ to three digits ($+4.17\cdot10^{-4}$ and $+1.27\cdot10^{-3}$ at $k_e = 100$), the slowest still below. At $\eta = 10\sigma_0$, where the metric $\mathrm{diag}(I, (\sigma_0/\eta)I)$ is not the identity, $\bar{A}_0$ is self-adjoint in it (to $8.7\cdot10^{-18}$; in $I$ its asymmetry is $6.4\cdot10^{-2}$) and $\bar{A}_1$ skew (to $1.4\cdot10^{-16}$). The error channel's exponent by Liouville's formula is within $O(k_e^{-3})$ of $k_e - \mathrm{tr}\, H_L(w)/k_e$ (24.98084 against 24.98037 at $k_e = 25$).

**Lemma D‴.26a** (the exact slow subspace of the perfect-tracking linearisation: Corollary D‴.26(c) and Proposition D‴.30(ii) proved for the linearised loop)**.** *Let $K : \mathbb{R} \to \mathbb{R}^{n\times q}$ be $T_0$-periodic and $C^4$, $\Lambda \in \mathbb{R}^{q\times q}$, $a : \mathbb{R} \to \mathbb{R}$ $T_0$-periodic and $C^1$, $N_0 \ge 0$, and for real $\varepsilon \ne 0$*

*consider*

$$\begin{aligned}\varepsilon\,\mathrm{d}w &= -(w+K(t)Y)\,\mathrm{d}t-(1+\varepsilon a(t))\varepsilon^{-1}\sqrt{N_0}\,\mathrm{d}W,\\ \mathrm{d}Y &= \varepsilon(K(t)^\top w-\Lambda Y)\,\mathrm{d}t+K(t)^\top\sqrt{N_0}\,\mathrm{d}W\end{aligned}\tag{★}$$

*($W$ a standard $n$-dimensional Wiener process). At $\varepsilon=1/k_e>0$ and $N_0=0$ this is the collapsed variational equation of Corollary D‴.26 under its derivative condition — $\delta e=\varepsilon w$, $K=[\hat J,0]$, $Y=(y,\zeta)$, $\Lambda=\left[\begin{smallmatrix}\sigma_0 I & -\sigma_0 I\\ -\eta I & \eta I\end{smallmatrix}\right]$ for $0<\eta<\infty$ (at $\eta=\infty$: $K=\hat J$, $Y=y$, $\Lambda=0$; at $\eta=0$, the fixed prior at the atoms: $K=\hat J$, $Y=y$, $\Lambda=\sigma_0 I$ — with the pair's $\Lambda$ the prior's block would have a zero rate there) — and at $N_0>0$ Proposition D‴.30(ii)'s linearised loop under a white measurement noise of density $N_0$ (the error equation's input $-(k_e+a)\nu$, $a=\partial_x\hat f$ on the orbit, the law's $\hat J^\top\nu$). $K\in C^4$ holds when $x_m\in C^4$. There is $\varepsilon_0>0$, depending on the $C^4$ norms of $K$, on $|\Lambda|$, $\|a\|_{C^1}$ and $T_0$ — and, for (iv), on the distance of $-\lambda_j$ from the rest of the spectrum of $\bar A_0$ and on the condition of its eigenvector, for (v) on $\bar A_0$'s stability margin, as do the constants of the remainders there — such that for $0<|\varepsilon|\le\varepsilon_0$:*

(i) (the slow subspace) *The equation $\varepsilon\dot P+P=-K-\varepsilon^2P(K^\top P-\Lambda)$ has a unique $T_0$-periodic solution $P(\cdot\,;\varepsilon)$ with $\|P+K\|_\infty=O(\varepsilon)$, and $P=\sum_{k\le3}\varepsilon^kP_k+O(\varepsilon^4)$ uniformly in $t$, $P_0=-K$, $P_1=\dot K$, $P_2=-(\ddot K+KK^\top K+K\Lambda)$, $P_k=-\dot P_{k-1}-\sum_{i+j=k-2}P_iK^\top P_j+P_{k-2}\Lambda$. In the coordinates $v:=w-P(t)Y$ the deterministic part of (★) is block-triangular,*

$$\varepsilon\dot v=-(I+\varepsilon^2PK^\top)v,\qquad \dot Y=\varepsilon A(t)Y+\varepsilon K^\top v,$$
$$A:=K^\top P-\Lambda=-K^\top K-\Lambda+\varepsilon K^\top\dot K-\varepsilon^2K^\top(\ddot K+KK^\top K+K\Lambda)+O(\varepsilon^3).$$

(ii) (the slow multipliers, exactly) *For $\varepsilon>0$, $q$ Floquet multipliers of (★) are those of the slow block $\dot Y=\varepsilon A(t)Y$, of modulus $\ge e^{-\varepsilon T_0\|A\|_\infty}$, and the other $n$ those of the fast block, of modulus $\le e^{-T_0/(2\varepsilon)}$; the fast exponents sum to $n/\varepsilon+\varepsilon\langle\operatorname{tr}PK^\top\rangle=nk_e-\langle\operatorname{tr}K^\top K\rangle/k_e+O(k_e^{-3})$.*

(iii) (the slow generator) *Let $\Phi(\varepsilon)$ be the slow block's monodromy and $\bar A(\varepsilon):=\log\Phi(\varepsilon)/(\varepsilon T_0)$ (the principal logarithm). Then $\bar A(\varepsilon)=\bar A_0+\varepsilon\bar A_1+\varepsilon^2\bar A_2+\varepsilon^3\bar A_3+O(\varepsilon^4)$ uniformly in $|\varepsilon|\le\varepsilon_0$, with $\bar A_0=-\langle K^\top K\rangle-\Lambda$, $\bar A_1=\langle K^\top\dot K\rangle+(1/(2T_0))\int_0^{T_0}\int_0^{t_1}[A_0(t_1),A_0(t_2)]\,\mathrm{d}t_2\,\mathrm{d}t_1$ ($A_0:=-K^\top K-\Lambda$) and the higher coefficients those of the Magnus series of $\varepsilon A_{(3)}$ collected by powers of $\varepsilon$ ($A_{(3)}:=K^\top P_{(3)}-\Lambda$, $P_{(3)}:=\sum_{k\le3}\varepsilon^kP_k$, a polynomial in $\varepsilon$; $A$ itself has only an asymptotic expansion); the full Magnus series of $\varepsilon A$ converges where $\varepsilon T_0\|A\|_\infty<\pi$, a sufficient form of Moan–Niesen's $\int_0^{T_0}\|\varepsilon A\|_2<\pi$ ([102]). The averaged generator of Corollary D‴.26(c)'s near-identity averaging, carried there to $\bar A_2$, is conjugate to $\bar A(\varepsilon)$ up to $O(\varepsilon^3)$ — a conjugate generator, with the same eigenvalues to that order; its coefficients differ from $\bar A(\varepsilon)$'s Taylor coefficients by commutators with $\bar A_0$.*

(iv) (the rates, and their parity proved) *At a simple real eigenvalue $-\lambda_j$ of $\bar A_0$ the slow-time rate $\mu_j(\varepsilon):=-(\varepsilon T_0)^{-1}\log|\nu_j(\varepsilon)|$ of the continued multiplier $\nu_j$ is $\mu_j(\varepsilon)=\lambda_j+\varepsilon\mu_j^1+\varepsilon^2\mu_j^2+\varepsilon^3\mu_j^3+O(\varepsilon^4)$, the coefficients those of perturbation theory for $\bar A(\varepsilon)$ ($\mu_j^1=-\ell_j\bar A_1r_j$ and $\mu_j^2=c_j$ of Corollary D‴.26(c)). If the full matrix $M_\varepsilon(t)$ of (★) satisfies $M_\varepsilon(t)^\top=DM_\varepsilon(t)D^{-1}$ for an invertible $D$ — Corollary D‴.26's symmetry, $D=\operatorname{diag}(-\varepsilon^2I,I,(\sigma_0/\eta)I)$ in $(w,y,\zeta)$ ($\operatorname{diag}(-\varepsilon^2I,I)$ at $\eta=\infty$ and at $\eta=0$), which persists with a mismatch in the error channel — and a mismatch does not enter the error channel (it breaks the parity, below), then the spectrum of $\bar A(\varepsilon)$ is even: $\operatorname{eig}\bar A(-\varepsilon)=\operatorname{eig}\bar A(\varepsilon)$ as multisets of complex numbers, the slow block at $-\varepsilon$ defined by (i). In particular $\mu_j(-\varepsilon)=\mu_j(\varepsilon)$ exactly, $\mu_j^1=\mu_j^3=0$ and $\mu_j(\varepsilon)=\lambda_j+c_j\varepsilon^2+O(\varepsilon^4)$, with a bounded remainder, and at a degenerate eigenvalue the rates are even.*

(v) (the slow noise) *For $0<\varepsilon\le\varepsilon_0$ the anti-causal equation $S-\varepsilon\dot S=\varepsilon K^\top-\varepsilon^2(SPK^\top+AS)$ has a unique $T_0$-periodic solution with $S=\varepsilon K^\top+\varepsilon^2\dot K^\top+O(\varepsilon^3)$, and $Y':=Y+\varepsilon S(t)v$*

*solves exactly*

$$\begin{aligned}\mathrm{d}Y' &= \varepsilon A(t)Y'\,\mathrm{d}t + \varepsilon b(t)\sqrt{N_0}\,\mathrm{d}W,\\ b &:= \varepsilon^{-1}[K^\top - \varepsilon^{-1}(1+\varepsilon a)S - \varepsilon SPK^\top]\\ &= -(\dot K + aK)^\top - \varepsilon(\ddot K + KK^\top K + K\Lambda^\top + a\dot K)^\top + O(\varepsilon^2):\end{aligned}$$

*the slow coordinates' drift is decoupled from the fast ones, and their noise is $\varepsilon b(t)\nu$ — at leading order the high-passed $-\varepsilon(\dot K + aK)^\top\nu$; the same Wiener process drives $v$, so $\mathbb{E}[vY'^\top] \neq 0$ — with the diffusion $\varepsilon^2 N_0 bb^\top$ per unit time (in the slow time $\varepsilon N_0\langle bb^\top\rangle = (N_0/k_e)Q + O(k_e^{-2})$, $Q := \langle(\dot K + aK)^\top(\dot K + aK)\rangle$). If $\bar A_0$ is Hurwitz, the periodic stationary second moments are $\mathrm{Cov}(v) = (N_0/(2\varepsilon^3))(I + O(\varepsilon))$, $\mathbb{E}[vY'^\top] = -N_0 b^\top + O(\varepsilon N_0)$ and $\mathrm{Cov}(Y') = \bar\Sigma + O(\varepsilon^2 N_0)$, $\bar\Sigma$ the solution of $\bar A_0\bar\Sigma + \bar\Sigma\bar A_0^\top + \varepsilon N_0\bar Q = 0$ ($\bar Q = Q$ on the $y$ block, $\bar\Sigma = O(\varepsilon N_0)$); hence $\mathrm{Cov}(Y)(t) - \mathrm{Cov}(Y')(t) = (\varepsilon N_0/2)K(t)^\top K(t) - (\varepsilon^2 N_0/2)(\mathrm{d}(K^\top K)/\mathrm{d}t + 2aK^\top K)(t) + O(\varepsilon^3 N_0)$ and $\mathrm{Cov}(Y)(t) = \bar\Sigma + (\varepsilon N_0/2)K(t)^\top K(t) + O(\varepsilon^2 N_0)$ — the slow part plus the fast jitter — and Proposition D‴.30(ii)'s floor, $(N_0/2k_e)[\sum_j Q_{jj}(h_j+\eta)/(h_j+\eta+\sigma_0) + \operatorname{tr} H^2]$, holds for the linearised loop under white noise as a period average, with a relative remainder $O(1/k_e)$ (at $\eta = 0$ with $\Lambda = \sigma_0 I$, whose filter factor is $h_j/(h_j+\sigma_0)$): at time $t$ the fast part is $(N_0/2k_e)\operatorname{tr}(H\hat J(t)^\top\hat J(t))$, whose oscillation over the period is of the floor's own order.*

*Proof.* (i) On the Banach space of continuous $T_0$-periodic matrix functions (sup norm) the periodic solution of $\varepsilon\dot u + u = g$ is $\mathcal{G}g(t) = \int_0^\infty e^{-\sigma}g(t-\varepsilon\sigma)\,\mathrm{d}\sigma$ — for $\varepsilon < 0$ the anti-causal $\int_0^\infty e^{-\sigma}g(t+|\varepsilon|\sigma)\,\mathrm{d}\sigma$ —, so $\|\mathcal{G}g\| \le \|g\|$, and $\mathcal{G}g - g = -\varepsilon\mathcal{G}\dot g$. The equation is $P = \mathcal{G}F(P)$, $F(P) := -K - \varepsilon^2 P(K^\top P - \Lambda)$. With $r := 2|\varepsilon|(\|\dot K\| + \beta)$, $\beta := (\|K\|+1)(\|K\|(\|K\|+1) + |\Lambda|)$, the ball $\|P + K\| \le r \le 1$ is mapped into itself ($\|\mathcal{G}F(P) + K\| \le \|\varepsilon\mathcal{G}\dot K\| + \varepsilon^2\beta$) and $\mathcal{G}F$ is a contraction on it with constant $\varepsilon^2 L < 1$, $L := |\Lambda| + 2\|K\|(\|K\|+1)$: Banach's theorem gives $P$, unique in the ball. The truncation $P_{(3)} := \sum_{k\le 3}\varepsilon^k P_k$ satisfies $\varepsilon\dot P_{(3)} + P_{(3)} = F(P_{(3)}) + \rho$ with $\|\rho\| = O(\varepsilon^4)$ (the recursion cancels the orders $\varepsilon^0$–$\varepsilon^3$; $\rho$ involves $K$ up to its fourth derivative), so $P_{(3)} = \mathcal{G}(F(P_{(3)}) + \rho)$ and $\|P - P_{(3)}\| \le \|\rho\|/(1-\varepsilon^2 L) = O(\varepsilon^4)$. With $v = w - PY$, $\varepsilon\dot v = -(v + PY) - KY - \varepsilon\dot P Y - \varepsilon^2 P(K^\top(v+PY) - \Lambda Y) = -(I + \varepsilon^2 PK^\top)v$, the coefficient of $Y$ vanishing by $P$'s equation; and $\dot Y = \varepsilon(K^\top(v+PY) - \Lambda Y)$. (ii) The change of coordinates is periodic, so it conjugates the monodromies; in $(v, Y)$ the monodromy is block-triangular, its eigenvalues those of the two diagonal blocks. The fast block's monodromy has norm $\le \exp(-T_0/\varepsilon + \varepsilon T_0\|P\|\|K\|) \le e^{-T_0/(2\varepsilon)}$ (a logarithmic-norm estimate), the slow one's eigenvalues modulus $\ge e^{-\varepsilon T_0\|A\|}$; by Liouville's formula the fast exponents sum to $\varepsilon^{-1}\langle\operatorname{tr}(I + \varepsilon^2 PK^\top)\rangle$, and $\langle\operatorname{tr} PK^\top\rangle = -\langle\operatorname{tr} K^\top K\rangle + \varepsilon\langle\operatorname{tr}\dot K K^\top\rangle + O(\varepsilon^2)$ with $\langle\operatorname{tr}\dot K K^\top\rangle = \frac12\langle(\operatorname{tr} KK^\top)^\cdot\rangle = 0$. (iii) $A - A_{(3)} = K^\top(P - P_{(3)}) = O(\varepsilon^4)$ for $A_{(3)} := K^\top P_{(3)} - \Lambda$, a polynomial in $\varepsilon$, so by Grönwall's inequality the monodromies of $\varepsilon A$ and $\varepsilon A_{(3)}$ differ by $O(\varepsilon^5)$; the latter, $\Phi_{(3)}(\varepsilon)$, is analytic in $\varepsilon$ with $\Phi_{(3)}(0) = I$, and the principal logarithm is analytic near $I$, so $\log\Phi_{(3)}(\varepsilon)/(\varepsilon T_0)$ is analytic and $\log\Phi(\varepsilon)/(\varepsilon T_0)$ differs from it by $O(\varepsilon^4)$. Its Taylor coefficients are obtained by any formal expansion of the logarithm of the monodromy — the Magnus series of $\varepsilon A_{(3)}$ collected by powers of $\varepsilon$ ($\Omega_1 = \varepsilon T_0\langle A\rangle$, $\Omega_2 = \frac12\int\int[\varepsilon A(t_1), \varepsilon A(t_2)]$ gives $\bar A_1$'s commutator) or the near-identity averaging: a periodic transformation $I + \varepsilon U_1 + \dots$ conjugates the system to an autonomous one up to $O(\varepsilon^4)$, so the monodromy is conjugate, by $I + \varepsilon U(0) + \dots$, to $\exp(\varepsilon T_0(\bar A_0 + \dots + \varepsilon^3\bar A_3) + O(\varepsilon^5))$. The convergence of the full Magnus series for $\int_0^{T_0}\|\varepsilon A\|_2 < \pi$ is Moan and Niesen's theorem ([102]). (iv) A simple real eigenvalue of $\bar A_0$ continues to a simple real eigenvalue of the analytic family $\log\Phi_{(3)}(\varepsilon)/(\varepsilon T_0)$, analytic in $\varepsilon$ (Kato, [102]), and simple eigenvalues are Lipschitz under the $O(\varepsilon^4)$ perturbation to $\bar A(\varepsilon)$: the expansion of $\mu_j$ with a bounded remainder, the coefficients by standard perturbation theory. Parity. Let $\tilde K(t) := K(-t)$, $\tilde a(t) := a(-t)$. The full matrix of (★) for $(\tilde K, \tilde a)$ is $M_\varepsilon(-t)$, and the system with the coefficients $M_\varepsilon(-t)$ has the multipliers of $M_\varepsilon(t)$: its monodromy is conjugate to the transpose of $M_\varepsilon$'s, by the symmetry ($\dot z = -M(s)z$ is $D$-conjugate to the adjoint system, whose monodromy is the inverse transpose). The fast multipliers being exponentially small and the slow ones near 1, the slow multipliers of $\tilde K$ at $\varepsilon$ are those of $K$

at $\varepsilon$. On the other hand $P[K](-t;-\varepsilon)$ solves the equation of (i) for $\tilde K$ at $\varepsilon$ (the substitution $t\mapsto -t$, $\varepsilon\mapsto-\varepsilon$ leaves $\varepsilon\dot P+P=F(P)$ invariant when $a$ does not enter it), so by uniqueness $A[\tilde K](t;\varepsilon)=A[K](-t;-\varepsilon)$: the slow block of $\tilde K$ at $\varepsilon$ is the time reversal of the slow block of $K$ at $-\varepsilon$, and its monodromy is the inverse, $\Phi_{\tilde K}(\varepsilon)=\Phi_K(-\varepsilon)^{-1}$, whence $\bar A[\tilde K](\varepsilon)=\bar A[K](-\varepsilon)$. Together, the slow-time exponents of $K$ at $-\varepsilon$ are those at $\varepsilon$, and the continued simple eigenvalue is even: $\mu_j(-\varepsilon)=\mu_j(\varepsilon)$ — more, $\Phi_{\tilde K}(\varepsilon)=\Phi_K(-\varepsilon)^{-1}$ as matrices and $\tilde K$'s slow multipliers are $K$'s, so $\operatorname{eig}\bar A(-\varepsilon)=\operatorname{eig}\bar A(\varepsilon)$ as multisets (a shorter route needs no $P$: the full system at $-\varepsilon$ is the time reversal of $\tilde K$'s at $\varepsilon$, so with the $D$-symmetry its slow multipliers are the inverses of $K$'s at $\varepsilon$); the expansion of (iv), valid for both signs with the same coefficients (the recursion of (i) does not depend on the sign), has no odd terms. A mismatch $a$ in the error channel ($\varepsilon\dot w=-(1-\varepsilon a)w-KY$) adds $\varepsilon aP$ to $F(P)$, which the substitution changes in sign: it breaks the parity, at first order. (v) In the coordinates $(v,Y)$, $\mathrm dv=-\varepsilon^{-1}(I+\varepsilon^2PK^\top)v\,\mathrm dt-(\varepsilon^{-2}(1+\varepsilon a)+PK^\top)\sqrt{N_0}\,\mathrm dW$ and $\mathrm dY=(\varepsilon AY+\varepsilon K^\top v)\,\mathrm dt+K^\top\sqrt{N_0}\,\mathrm dW$; for $Y'=Y+\varepsilon Sv$ the drift's coefficient of $v$ is $\varepsilon K^\top+\varepsilon\dot S-S-\varepsilon^2(SPK^\top+AS)$, zero by $S$'s equation, whose periodic solution is the anti-causal contraction as in (i); the noise of $Y'$ is $K^\top-\varepsilon S(\varepsilon^{-2}(1+\varepsilon a)+PK^\top)=K^\top-(K^\top+\varepsilon\dot K^\top)(1+\varepsilon a)+O(\varepsilon^2)=-\varepsilon(\dot K+aK)^\top+O(\varepsilon^2)$ times $\sqrt{N_0}\,\mathrm dW$. The second moments solve periodic Lyapunov equations: $v$'s, $\dot V=-\varepsilon^{-1}(\Gamma V+V\Gamma^\top)+N_0gg^\top$ with $\Gamma=I+\varepsilon^2PK^\top$, $g=\varepsilon^{-2}(1+\varepsilon a)+PK^\top$, quasi-static at the fast rate ($V=N_0(1+\varepsilon a)^2/(2\varepsilon^3)+O(N_0/\varepsilon)$); the cross moment $C=\mathbb E[vY'^\top]$, $\dot C=-\varepsilon^{-1}\Gamma C+\varepsilon CA^\top-\varepsilon N_0gb^\top$, likewise $C=-N_0b^\top+O(\varepsilon N_0)$; and $Y'$'s, $\dot\Sigma'=\varepsilon(A\Sigma'+\Sigma'A^\top)+\varepsilon^2N_0bb^\top$, a linear periodic equation of the slow rate $\varepsilon$: with $\bar\Sigma$ as stated, $Z:=\Sigma'-\bar\Sigma$ is driven by $\varepsilon[(A-\bar A_0)\bar\Sigma+\bar\Sigma(A-\bar A_0)^\top]+\varepsilon^2N_0(bb^\top-\bar Q)$, of period average $O(\varepsilon^3N_0)$ and oscillating part $O(\varepsilon^2N_0)$; the oscillating part's zero-mean antiderivative is $O(\varepsilon^2N_0)$, and the slow system's periodic response to the rest is $O(\varepsilon^{-1}\cdot\varepsilon^3N_0)$ since its Lyapunov operator is invertible at rate $\varepsilon$ ($\bar A_0$ Hurwitz): $Z=O(\varepsilon^2N_0)$. Then $\operatorname{Cov}(Y)=\operatorname{Cov}(Y')-\varepsilon(SC+C^\top S^\top)+\varepsilon^2SVS^\top$, where $-\varepsilon(SC+C^\top S^\top)=-\varepsilon^2N_0(\mathrm d(K^\top K)/\mathrm dt+2aK^\top K)+O(\varepsilon^3N_0)$ and, with $S=\varepsilon K^\top+\varepsilon^2\dot K^\top+O(\varepsilon^3)$ and $V=N_0(1+\varepsilon a)^2/(2\varepsilon^3)+O(N_0/\varepsilon)$, $\varepsilon^2SVS^\top=(\varepsilon N_0/2)K^\top K+(\varepsilon^2N_0/2)(\mathrm d(K^\top K)/\mathrm dt+2aK^\top K)+O(\varepsilon^3N_0)$: $\operatorname{Cov}(Y)=\bar\Sigma+(\varepsilon N_0/2)K^\top K+O(\varepsilon^2N_0)$. The floor proved is the period average of $\operatorname{tr}(H\operatorname{Cov}(y))$: the slow part by D‴.30(ii)'s pair blocks (at $\eta=0$ by $-(H+\sigma_0I)$) and the fast part $(\varepsilon N_0/2)\operatorname{tr}(H\langle\hat J^\top\hat J\rangle)=(N_0/2k_e)\operatorname{tr}H^2$; at time $t$ the fast part is $(\varepsilon N_0/2)\operatorname{tr}(H\hat J(t)^\top\hat J(t))$. □

**Status.** Proved (the constants not made explicit). It upgrades Corollary D‴.26(c) from a formal to a proved expansion, with a bounded $O(k_e^{-4})$ remainder and the odd orders vanishing exactly (for $x_m\in C^4$ at simple eigenvalues; the stability where $H_L(w)\succ0$), and Proposition D‴.30(ii) to proved for the linearised loop under white noise, for the floor's period average, with the relative remainder $O(1/k_e)$ — the order of round 21's observation, whose first-order term comes from the slope $a$ in the noise input (with $a=0$ in the noise the numerics show an even expansion; (v) does not prove it) — at $\eta=0$ with the identification $K=\hat J$, $\Lambda=\sigma_0I$. The real loop's noise at second order (D‴.30(iii)), the OU noise of the numerics and the real nonlinear loop stay as stated there. Chang's decoupling transformation of linear singularly perturbed systems — the pair $(P,S)$, here periodic —, its time-varying and stochastic versions (Kokotović, Khalil and O'Reilly's Ch. 4–5 and App. A, with the slow subsystem's transformed noise and its covariance; Haddad and Kokotović; Yang and Zhu's contraction for slowly time-varying systems), the slow manifolds of stochastic slow–fast systems (Berglund and Gentz) and the convergence of the Magnus series are classical ([102], [49]), and so is the reversal argument for even expansions ([99]); what the lemma adds is their application to the perfect-tracking linearisation of the transport-leak loop: the slaving of Corollary D‴.26 made an exact block-triangularisation, the parity made an exact identity $\operatorname{eig}\bar A(-\varepsilon)=\operatorname{eig}\bar A(\varepsilon)$ through the slow block's definition at negative $\varepsilon$, and the slow noise identified exactly as $\varepsilon b(t)\nu$, $b=-(\dot K+aK)^\top-\varepsilon(\ddot K+KK^\top K+K\Lambda^\top+a\dot K)^\top+O(\varepsilon^2)$ — the high-passed noise at leading order, the drift decoupled, the noise shared with the fast block. Checked by `exp_adaptive_floquet.py` Part D: the asymmetric reference at $\eta=0$, $\sigma_0$ and $\infty$, $k_e=12.5$–$1600$ — $P$'s invariance residual, the truncations against $\varepsilon^3$ and $\varepsilon^4$, the slow block's rates against the full monodromy's, the rates at $\pm\varepsilon$ per mode, a mismatch breaking their equality at first order, the Magnus parameter; under white noise the

decoupling residual, $b$ against $b_0$ and $b_0 + \varepsilon b_1$, the exact periodic covariance in the coordinates $(v, Y')$ with the weakest mode's deviation split into its two sources, the jitter with its next term, and the floor over the period. Checked separately by `tests/test_adaptive_floquet.py`: the slow block's rates against the monodromy's and $P - P_2$ against $\varepsilon^3$ on the instance; on a synthetic block with an $O(1)$, non-symmetric $\Lambda$, $P - P_2$ against $\varepsilon^3$ and $P - P_{(3)}$ against $\varepsilon^4$, with a flipped $\Lambda$ and a dropped cubic term caught; the parity and its breaking by a mismatch at $\eta = \sigma_0$ and $\infty$; and $b_0$ against $\dot{K}$ by the chain rule and the plant's own slope by differences, $b_1$ against $\varepsilon^2$, and the covariance's orders.

**Numerics** (`exp_adaptive_floquet.py` Part D → `results/chaos/adaptive_floquet.md`). Corollary D‴.26's instance ($w = (6, 5, 4)/15$) on the asymmetric reference $x_m = 2\sin t + 0.8\sin 2t$, $\eta = 0$ ($K = \hat{J}$, $\Lambda = \sigma_0 I$), $\sigma_0$ and $\infty$, $k_e = 12.5$–$1600$, a 512-point Fourier grid. The contraction for $P$ converges in 3–9 iterations to an invariance residual $\leq 3.8 \cdot 10^{-15}$ (at $\eta = 0$ and $k_e = 12.5$ it stalls at rounding: 499 iterations, the residual $3.3 \cdot 10^{-15}$). $|P - P_2|/\varepsilon^3$ tends to $3.34 \cdot 10^3$ and $|P - P_{(3)}|/\varepsilon^4$ to $5.8 \cdot 10^4$ (bounded: the $O(\varepsilon^3)$ of (i) and the third term of (iii), their constants set by the regressor's harmonics). The slow block's slow-time rates equal the full variational equation's to $\leq 1.0 \cdot 10^{-8}$ (the monodromies' accuracy). The rates at $\pm\varepsilon$ agree to $\leq 5.9 \cdot 10^{-12}$ of the largest rate and $\leq 1.1 \cdot 10^{-9}$ per mode, while their deviation from the first-order prediction is $9.8 \cdot 10^{-2}$–$8.9 \cdot 10^{-6}$ of the largest rate and, per mode, up to 29.6% — the slowest mode's at $\eta = \infty$: $-29.6\%$, $-12.5\%$, $-4.0\%$, $-1.1\%$ at $k_e = 12.5$, 25, 50, 100; at $\eta = 0$ the slowest mode's is $-1.7\%$ and the largest 12.0% at $k_e = 12.5$. The parity of (iv) is an identity, not an asymptotic statement. With the plant's slope as a mismatch in the error channel the rates at $\pm\varepsilon$ differ by $1.1 \cdot 10^{-2}$–$7.8 \cdot 10^{-5}$ of the largest rate (per mode $6.1 \cdot 10^{-2}$–$3.5 \cdot 10^{-4}$), halving per doubling of $k_e$: the parity broken at first order. The Magnus parameter is 0.48 at $k_e = 12.5$, $\leq 0.25$ from $k_e = 25$ and 0.004 at 1600, so the full Magnus series converges throughout. Under white measurement noise the second transformation decouples $Y'$ to a residual $\leq 3 \cdot 10^{-16}$. The slow noise $b$ differs from $b_0 = -(\dot{K} + aK)^\top$ by 63%, 40%, 22%, 12%, 5.9%, 3.0%, 1.5% and 0.74% ($O(\varepsilon)$, with the large constant of the regressor's harmonics: $\varepsilon\omega_1 \approx 0.4$ at $k_e = 25$), and $|b - b_0 - \varepsilon b_1|/\varepsilon^2$ tends to $3.34 \cdot 10^3$ (bounded: $b_1$ of (v)). At $t = 0$ the exact periodic covariance gives $\mathrm{Cov}(v) \cdot 2\varepsilon^3 - 1$ from $-9.6 \cdot 10^{-2}$ to $-2.4 \cdot 10^{-5}$ and $\|\mathbb{E}[vY'^\top] + b^\top\|/\|b\|$ from 0.15 to $1.2 \cdot 10^{-3}$ ($O(\varepsilon)$). $\|\mathrm{Cov}(Y') - \bar{\Sigma}\|/\|\bar{\Sigma}\|$ falls from 0.35 to $9.4 \cdot 10^{-5}$ ($\eta = \infty$; to $1.7 \cdot 10^{-4}$ at $\eta = \sigma_0$ and $2.4 \cdot 10^{-4}$ at $\eta = 0$), by 3.6–4 per doubling from $k_e = 50$ up to 200–400 and by 2–2.9 beyond: the $O(\varepsilon)$ of (v), with a small constant. In the weakest mode the deviation is $O(\varepsilon^2)$ (at $\eta = \infty$: $-34\%$, $-17\%$, $-6.0\%$, $-1.7\%$ at $k_e = 12.5$–$100$, $-6.6 \cdot 10^{-5}$ at 1600). It is the sum of two parts of opposite sign: the slow noise's attenuation, $\langle bb^\top \rangle$ below $\bar{Q}$ ($\Sigma$ with $\langle bb^\top \rangle$ in place of $\bar{Q}$: $-56\%$, …, $-1.1 \cdot 10^{-4}$), and the rate's correction ($\Sigma$ with the exact slow generator in place of $\bar{A}_0$: $+49\%$, …, $+4.6 \cdot 10^{-5}$). The attenuation is 2.4–2.5 times the correction from $k_e = 100$ at $\eta = \infty$ (1.2 times at $k_e = 12.5$) and 12–23 times at $\eta = 0$. $\mathrm{Cov}(Y) - \mathrm{Cov}(Y')$ equals the jitter $(\varepsilon/2)K(0)^\top K(0)$ to a relative 0.32, 0.16, 0.081, 0.041 and 0.020 at $k_e = 100$–$1600$ ($O(\varepsilon)$), and with the next term $-(\varepsilon^2/2)(\mathrm{d}(K^\top K)/\mathrm{d}t + 2aK^\top K)(0)$ to 0.30, 0.079, 0.020, 0.005 and 0.001 ($O(\varepsilon^2)$). The floor at $t = 0$ is 0.763, 0.905, 0.971, 0.992, 0.998, 0.9995, 0.9999 and 1.0000 of the lemma's slow-plus-fast value ($\eta = \infty$). Over the period the floor's average is 0.766, 0.907, 0.972, 0.9925, 0.9981, 0.99951, 0.99987 and 0.99997 of Proposition D‴.30(ii)'s formula. Pointwise it ranges over $[0.9952, 1.0091]$ of the formula at $k_e = 1600$, which is the range of $\mathrm{tr}(H(\bar{\Sigma} + (\varepsilon/2)\hat{J}(t)^\top \hat{J}(t)))$: the fast block's jitter, $\pm 1\%$. So the formula holds as the period average (v), not pointwise.

**Proposition D‴.27** (a projected adaptive prior restores the leak's a-priori bound)**.** *Let $\mathcal{Z} \subset \mathbb{R}^p$ be a compact convex set — on the instance a bounded box in the unconstrained coordinates — and let the prior's atoms of Theorem D‴.23 (or of Proposition D‴.24(ii)) start in it, $Z(0) \in \mathcal{Z}^{K_0}$, and be confined to it: $\mathrm{d}z_j/\,\mathrm{d}s = \Pi_{T_{\mathcal{Z}}(z_j)}(-\eta(z_j - \hat{T}_j))$, the velocity projected onto the tangent cone of $\mathcal{Z}$ at $z_j$ in the metric $G^{-1}$ (for a box and diagonal $G$: the outward components set to zero on the active faces); at $\eta = \infty$, $z_j = \Pi_{\mathcal{Z}}(\hat{T}_j)$, the $G^{-1}$-projection of the class barycentre. Let*

$B := \sup_{\Theta,\vartheta} \|G\langle r_\Theta \nabla_\vartheta \tilde{\Phi}(\cdot\,;\vartheta)\rangle_\rho\|_{G^{-1}} \le (\max_\rho |f| + a_{\max}) \cdot L^G_\vartheta$, *with* $L^G_\vartheta := \sup_{x,\vartheta} \|G^{1/2} \nabla_\vartheta \tilde{\Phi}(x;\vartheta)\|$ *(*$|\hat{f}| \le a_{\max}$ *for the saturated network).*

(i) (the a-priori bound) *Every particle satisfies* $\mathrm{dist}_{G^{-1}}(\vartheta_i(s), \mathcal{Z}) \le B/\sigma_0 + (\mathrm{dist}_{G^{-1}}(\vartheta_i(0), \mathcal{Z}) - B/\sigma_0)_+ e^{-\sigma_0 s}$, *at most* $\max\{\mathrm{dist}_{G^{-1}}(\vartheta_i(0), \mathcal{Z}), B/\sigma_0\}$, *for all* $s \ge 0$, *and the prior stays in* $\mathcal{Z}$; $\mathcal{Z}$ *being bounded, every solution is bounded, for every* $m$, *every* $\eta \in (0, \infty]$ *and every target. (For an unbounded* $\mathcal{Z}$ *the distance bound still holds but bounds nothing:* $\mathcal{Z} = \mathbb{R}^p$ *is Theorem D‴.23's prior, which runs off.)*

(ii) (the energy) $\mathcal{J} = \bar{L} + (\sigma_0/2)W^2$ — *for the nearest-atom prior of Proposition D‴.24(ii) its energy* $\mathcal{J}_\circ$, *with the cell masses — is non-increasing along the projected flow, which is the vertex-selection gradient flow of* $\mathcal{J} + \iota_{\mathcal{Z}^{K_0}}(Z)$ *in the metric of Theorem D‴.23(i) (for* $\mathcal{J}_\circ$ *with the cell masses) — at* $\eta = \infty$ *in its selected form* $\mathcal{J}(\Theta, \Pi_{\mathcal{Z}} \hat{T}(\gamma))$, $\gamma$ *the plan the flow carries, optimal for its own projected barycentres (Lloyd-consistent) but not necessarily jointly optimal, so that the reduced energy* $\min_{Z \in \mathcal{Z}^{K_0}} \mathcal{J}$ *itself need not decrease. At* $\eta = \infty$ *the bounded solutions are expected to converge, by a Kurdyka–Łojasiewicz argument on the finitely many selected pieces, to a critical point of a Lloyd-consistent piece, which need not be Clarke-critical for the reduced energy; at finite* $\eta$ *the solutions are expected to converge to Clarke-critical points of* $\mathcal{J} + \iota$. *At a critical point with a unique plan the prior is a stationary point of Theorem D‴.23(i) where it is interior, and each boundary atom satisfies the Karush–Kuhn–Tucker condition* $z_j = \Pi_{\mathcal{Z}}(\hat{T}_j)$, $\hat{T}_j - z_j$ *in the normal cone of* $\mathcal{Z}$ *at* $z_j$; *at a tie of plans the condition holds for a convex combination of the active pieces, as in Proposition D‴.17(iii).*

(iii) (nothing changes inside) *As long as the prior stays in the interior of* $\mathcal{Z}$ *the projected flow is Theorem D‴.23's. In particular, at an exact representation whose atoms lie in* $\operatorname{int} \mathcal{Z}$ *the stationary point and its linearisation are Theorem D‴.23(i)–(ii)'s, and the bias removal of (iii) and the local convergence of (v) hold unchanged for starts close enough that the prior stays in* $\operatorname{int} \mathcal{Z}$.

(iv) (the price outside) *If the limit has an atom on* $\partial\mathcal{Z}$ *with its class barycentre outside* $\mathcal{Z}$ — *necessarily so when no stationary point of Theorem D‴.23(i) has its prior in* $\mathcal{Z}$ — *the particles of that class leak toward the face, and the limit keeps a bias, bounded by (i), in place of the unprojected prior's run-off (an atom on* $\partial\mathcal{Z}$ *with* $\hat{T}_j = z_j$ *is a stationary point of Theorem D‴.23(i) whose prior lies on the boundary, with no bias). At* $\eta = \infty$ *on a collapsed class the leak is* $-\sigma_0(\vartheta - \Pi_{\mathcal{Z}}\vartheta)$, *the gradient of the exterior quadratic penalty* $(\sigma_0/2)\,\mathrm{dist}^2_{G^{-1}}(\vartheta, \mathcal{Z})$ — *zero on* $\mathcal{Z}$ *and growing linearly outside: a switching* $\sigma$*-modification toward* $\mathcal{Z}$, *the leak off on* $\mathcal{Z}$ *([47], [95]).*

*Proof.* (i) $z_{\pi(i)} \in \mathcal{Z}$ at all times, and so is the leak target $m \sum_j \gamma_{ij} z_j$ of a split row (incommensurate $m$), a convex combination. With $\Pi := \Pi_{\mathcal{Z}}$ and $d := \mathrm{dist}_{G^{-1}}(\vartheta_i, \mathcal{Z})$, $\frac{1}{2}d^2$ is $C^1$ with gradient $\vartheta - \Pi\vartheta$ in the metric $G^{-1}$, so $(\mathrm{d}/\mathrm{d}s)\frac{1}{2}d^2 = \langle \vartheta_i - \Pi\vartheta_i, G\langle r\nabla\tilde{\Phi}(\vartheta_i)\rangle\rangle_{G^{-1}} - \sigma_0\langle \vartheta_i - \Pi\vartheta_i, \vartheta_i - z\rangle_{G^{-1}}$, $z$ the leak target. The projection inequality $\langle \vartheta - \Pi\vartheta, z - \Pi\vartheta\rangle_{G^{-1}} \le 0$ for $z \in \mathcal{Z}$ gives $\langle \vartheta_i - \Pi\vartheta_i, \vartheta_i - z\rangle_{G^{-1}} \ge d^2$, so $(\mathrm{d}/\mathrm{d}s)\frac{1}{2}d^2 \le Bd - \sigma_0 d^2$, i.e. $d' \le B - \sigma_0 d$ where $d > 0$, and the comparison principle gives the bound; boundedness follows from that of $\mathcal{Z}$. The bound on $B$: $\|G\langle r\nabla\tilde{\Phi}\rangle\|_{G^{-1}} = \|\langle rG^{1/2}\nabla\tilde{\Phi}\rangle\| \le \sup|r| \cdot L^G_\vartheta$ and $|r| \le |f| + |\hat{f}|$. (ii) The particles' part of $\mathrm{d}\mathcal{J}/\mathrm{d}s$ is Theorem D‴.23(i)'s, minus the squared norm of their velocity; the prior's is $\langle \nabla_Z \mathcal{J}, \Pi_T(-c\nabla_Z \mathcal{J})\rangle = -\|\Pi_T(-c\nabla_Z \mathcal{J})\|^2/c$ ($c$ the metric's factor; Moreau's decomposition of $-c\nabla_Z \mathcal{J}$ into its tangent-cone and normal-cone parts, which are orthogonal), so $\mathcal{J}$ is non-increasing. At $\eta = \infty$ the prior minimises the $W^2$ part over $\mathcal{Z}$ for the current plan: for each plan $\gamma$, $\min_Z \mathcal{J} = \bar{L} + (\sigma_0/2)\sum_j \big[\sum_i \gamma_{ij}\|\vartheta_i - \hat{T}_j\|^2_{G^{-1}} + q_j\,\mathrm{dist}^2_{G^{-1}}(\hat{T}_j, \mathcal{Z})\big]$ (minimising $\sum_i \gamma_{ij}\|\vartheta_i - z\|^2 = q_j\|z - \hat{T}_j\|^2 + \text{const}$ over $z \in \mathcal{Z}$), each piece $C^{1,1}$ and definable (the squared distance to a box is piecewise quadratic), and its gradient in $\vartheta_i$ is the learning term and the leak toward $\Pi_{\mathcal{Z}}\hat{T}_j$ (the derivative of $q_j\,\mathrm{dist}^2(\hat{T}_j, \mathcal{Z})$ adds $\gamma_{ij}(\hat{T}_j - \Pi_{\mathcal{Z}}\hat{T}_j)$ to the within-class term).

Between plan refreshes the particles follow the gradient flow of the piece $E_\gamma(\Theta) := \min_Z \mathcal{J}(\Theta, Z; \gamma)$ of the plan $\gamma$ they carry, and at a refresh the new plan is optimal for the current prior and the prior then re-minimises, so the selected energy does not increase there either. But $\gamma$ is Lloyd-consistent — optimal for $\nu_Z$ with $Z = \Pi_{\mathcal{Z}}\hat{T}(\gamma)$ — and need not attain $\min_\gamma E_\gamma = \min_Z \mathcal{J}$: the flow is then not a subgradient trajectory of the reduced energy, and Proposition D‴.17's argument, whose fixed prior makes every selected piece the active one, does not apply as it stands (Part H(e): the reduced energy rises by 26% while the selected energy falls). A Kurdyka–Łojasiewicz argument on the finitely many definable pieces that share the limiting value should give the bounded solutions finite length, with a critical point of a Lloyd-consistent piece as the limit; it is not carried out here. At finite $\eta$ the joint energy carries the indicator of $\mathcal{Z}^{K_0}$, and the Kurdyka–Łojasiewicz argument with it is not carried out here. At a critical point the projected velocity of every atom vanishes, which is the stated condition. (iii) The projection is inactive in $\operatorname{int}\mathcal{Z}$, and Theorem D‴.23(v)'s ball keeps the prior within $\rho(\eta/(\sigma_0 q_k))^{1/2}$ of the atoms. (iv) If no stationary point of Theorem D‴.23(i) has its prior in $\operatorname{int}\mathcal{Z}$, a critical point of (ii) has an atom on $\partial\mathcal{Z}$. At $\eta = \infty$ with a class collapsed at $\vartheta$, $\hat{T} = \vartheta$ and $z = \Pi_{\mathcal{Z}}\vartheta$. □

**Status.** (i) proved; (ii) the monotonicity of the selected energy and the characterisation of the critical points proved, the convergence argued at $\eta = \infty$ (a Kurdyka–Łojasiewicz argument on the selected pieces, not carried out) and formal at finite $\eta$; (iii) proved, its local convergence as Theorem D‴.23(v) (sketched); (iv) the identification at $\eta = \infty$ on collapsed classes proved, the boundary atoms and the bias outside numerical. The bound of (i) is crude (it uses the worst-case force; $L_\vartheta^G \le 4.658$ is the closed-form bound, 4.044 by search), and its content is qualitative: boundedness is no longer a hypothesis. Parameter projection onto a convex set, the exterior quadratic penalty and the switching $\sigma$-modification are classical robustness modifications of adaptive laws, and the switching $\sigma$-modification keeps the ideal (unbiased) properties of the adaptive law when the true parameter lies in its set ([47], [95]); on collapsed classes at $\eta = \infty$ the construction is that law. New is its transfer to the learned centre of a mass-constrained $K_0$-atom prior at finite $\eta$ — a projected adaptive law for the prior's atoms, whose velocity jumps at the faces while the particles' law keeps its smooth gradient form — and its compatibility with Theorem D‴.23's bias removal wherever the target is representable inside $\mathcal{Z}$. Checked: `tests/test_adaptive_prior.py` (beyond the saturation, the projected prior at $\eta = \infty$ (LSODA) keeps the particles within 2 of the box and converges, while without the box the prior's coordinate passes 11 by $s = 10^4$; at $\eta = \sigma_0$ the projected midpoint rule at $\mathrm{d}t = 0.125$ keeps the prior in the box, brings its amplitude atom onto the face, keeps the particles bounded and $\mathcal{J}$ non-increasing; at the misplaced prior near the exact representation the projected and unprojected runs are identical).

**Numerics** (`exp_adaptive_prior.py` Part H → `results/chaos/adaptive_prior.md`). The box $\mathcal{Z}$ holds the amplitudes and centres to 0.9 of $a_{\max}$ and $c_{\max}$ ($\alpha \in [-5.889, 5.889]$, $\gamma \in [-4.417, 4.417]$) and the width coordinate in $[-6, 6]$; the instance's exact atoms are inside. (a) Part G(b)'s saturated target (a weighted amplitude $4.8 > a_{\max}$), the $K_0$-atom system from the exact atoms with the prior there: under the projected prior both $\eta = \sigma_0$ (the projected midpoint rule — positions projected inside the step — at $\mathrm{d}t = 0.25$ and 0.125, to $s = 10^5$) and $\eta = \infty$ (LSODA, rtol $10^{-10}$) converge to the same bounded stationary point (largest drift $1.8 \cdot 10^{-15}$, $3.5 \cdot 10^{-15}$ and $1.2 \cdot 10^{-16}$; the two steps' end states within $3.4 \cdot 10^{-12}$), the first prior atom's amplitude coordinate on the face of $\mathcal{Z}$ ($z_a = 5.889$) and the particle's at 6.728 (amplitude 3.7325), with the fit error 0.1415 — the two steps' agreement shows their shared fixed point; the scheme's consistency: at $s = 300$ the trajectory lies within $6.3 \cdot 10^{-7}$, $1.9 \cdot 10^{-7}$ and $3.8 \cdot 10^{-8}$ of a reference at $\mathrm{d}t = 1/64$ for $\mathrm{d}t = 0.25$, 0.125, 0.0625. The particles' largest $G^{-1}$-distance from $\mathcal{Z}$ along the run is 0.92 ($\eta = \sigma_0$) and 0.89 ($\eta = \infty$), against the bound $B/\sigma_0 \le 3686$ ($B \le 26.08$ with the closed-form $L_\vartheta^G$). (b) Part C's $K_0$-atom loop at the misplaced prior ($\eta = \sigma_0$, 6000 slow units) is identical with and without $\mathcal{Z}$ (largest difference 0.0): the box is never approached — the prior and the particles keep at least 16.9% of the box's width to every face — so the two runs

coincide by construction. (c) The box trades the fit error against the parameters' bound: with the amplitude bound 0.75, 0.8, 0.9, 0.95, 0.99 of $a_{\max}$ — boxes that contain the instance's exact atoms — the limit's fit error is 0.1676, 0.1611, 0.1415, 0.1268, 0.1101, decreasing between the fixed prior's 0.1750 and the unprojected adaptive prior's 0.1050 at $s = 10^6$ (which that prior approaches only as its amplitude coordinate runs off), the prior's amplitude atom on the face in every case; the bound 0.7 (0.1736) already excludes the exact atoms (the start is projected at once), and the tighter boxes 0.65, 0.6, 0.5 are worse than the fixed prior (0.1792, 0.1845, 0.1949): a box should contain the representation. (e) At $\eta = \infty$ on six particles in a Lloyd-consistent, non-optimal pairing (a seeded search; zero amplitudes, a small three-bump target, the box) the selected energy falls from $4.52 \cdot 10^{-3}$ to $4.95 \cdot 10^{-5}$ (its largest increment $-3.4 \cdot 10^{-6}$) while the reduced energy $\min_Z \mathcal{J}$ rises from $1.251 \cdot 10^{-3}$ to $1.574 \cdot 10^{-3}$ before the plan becomes optimal at $s = 80$. (d) On Part G(a)'s four-bump target, where without the box the merged atom's width coordinate grows like $\ln s$ and the solution leaves every compact set, the box stops it: from the exact split fit at $\eta = \infty$ (the joint flow for 40000 slow units, then the collapsed dynamics) $\kappa$ passes 2.159, 3.843, 5.529 at $s = 10^6$, $10^7$, $10^8$ as before and stays at the face $\kappa = 6$ from $s = 10^9$ (the atom within $6.6 \cdot 10^{-7}$ of it, drift $5.3 \cdot 10^{-17}$), with the fit error $1.943376 \cdot 10^{-2}$, within $9.3 \cdot 10^{-7}$ of the minimiser's on the face $s = s_{\max}$ that the unprojected solution approaches only as $s \to \infty$.

**Proposition D‴.28** (the adaptive prior's floor removal is a gauge property: it survives an approximation error)**.** *Let $\vartheta^b(q)$ be a nondegenerate critical point of the unregularised weighted loss $L_q(\Theta) := \frac{1}{2}\langle |f - \sum_k q_k \tilde{\Phi}(\cdot\,;\vartheta_k)|^2 \rangle_\rho$ — not necessarily an exact representation: $r_b := f - \hat{f}_b$ may be non-zero — and let the masses enter the network only through the products $w_k a(\alpha_k)$, as they do for the saturated network. The amplitude–mass gauge $G_w : \alpha_k \mapsto a^{-1}(q_k a(\alpha_k)/w_k)$, the other coordinates unchanged, is defined on $W_g := \{w : q_k |a_k^b| < w_k a_{\max} \text{ for every } k\}$ and satisfies $\hat{f}_w \circ G_w = \hat{f}_q$, hence $L_w \circ G_w = L_q$.*

(i) (the stationary points carry no function-level floor) *For every $w \in W_g$, $\vartheta^b(w) := G_w \vartheta^b(q)$ is a nondegenerate critical point of $L_w$ with the same function $\hat{f}_b$, the same residual $r_b$ and the same Hessian signature. With the prior at its atoms it is a stationary replicated configuration of the nearest-atom leak toward an adaptive prior (Proposition D‴.24(iii)), so two systems at their samples' gauge transforms carry the same function whatever the approximation error, with the fit error $\|r_b\|_\rho$. The function-level floor — the limit of $\mathbb{E}\|\hat{f}_m - \hat{f}_M\|^2$ along the dynamics — therefore vanishes at the samples of $W_g$ with positive margins in (iii) when $\vartheta^b(q)$ is a strict local minimum and the systems converge to the gauge transforms (numerical); $W_g$ shrinks as the approximation error grows (its thresholds $q_k|a_k^b|/a_{\max}$ rise with the amplitudes). (At an exact representation $W_g$ is Proposition D‴.24's $W$.)*

(ii) (the fixed prior's floor) *At the matching prior $\vartheta^0 = \vartheta^b(q)$ the function-level weight-sensitivity of the fixed prior's stationary point is*

$$F = \sigma_0 \hat{J}(\tilde{A} + \sigma_0 I)^{-1} \tilde{g}, \qquad \tilde{A} := H_L - \tilde{R},$$

*$\tilde{A}$ the whitened weighted Hessian of $L_q$ at $\vartheta^b(q)$ ($\tilde{R}$ the whitened weighted residual curvature) and $\tilde{g}$ the whitened gauge displacement, the direct effect of the weights on $\hat{f}$ being $\hat{J}\tilde{g}$: Proposition D‴.24(i) with the full Hessian in place of the Gauss–Newton block. Along the same derivation the adaptive prior's sensitivity is $F(\sigma)$ at $\sigma = 0$, which vanishes, as (i) says.*

(iii) (the splitting margins under the adaptive prior) *For a scalar amplitude per kernel (one output), at a critical point the residual curvature $R_k = \langle r \nabla^2_\vartheta \tilde{\Phi}(\cdot\,;\vartheta_k) \rangle_\rho$ has a zero amplitude row and column, and its block in the centre and width coordinates is $a(\alpha_k)$ times a function of those coordinates alone. Hence $R_k(\vartheta^b(w)) = (q_k/w_k) R_k(\vartheta^b(q))$ exactly, and Proposition D‴.5's splitting margins at the adaptive prior's replicated stationary point are*

$$\sigma_0 - (q_k/w_k)\lambda_k^+, \qquad \lambda_k^+ := \lambda_{\max}(G^{1/2} R_k(\vartheta^b(q)) G^{1/2}) \geq 0 :$$

*a class sampled below $q_k\lambda_k^+/\sigma_0$ can split. At an exact representation $\lambda^+ = 0$ and every margin is $\sigma_0$ (Proposition D‴.24(iii)); with an approximation error, and $\vartheta^b(q)$ a strict local minimum, the adaptive prior's admissible set is $\{w \in W_g : w_k > q_k\lambda_k^+/\sigma_0\}$. For vector amplitudes ($n \geq 2$ outputs, $a(\alpha) \in \mathbb{R}^n$ saturated componentwise) only the centre–width block scales by $q_k/w_k$: stationarity in the amplitudes gives $\Psi_r = 0$ for every output $r$ but stationarity in the centres only $\sum_r a_r \partial_\gamma \Psi_r = 0$, so the amplitude–centre block $a_r'(\alpha_r)\partial_\gamma \Psi_r$ survives, is gauge-invariant in the physical amplitudes, and no closed margin formula follows.*

*Proof.* (i) $G_w$ is a diffeomorphism from $\{\alpha : |q_k a(\alpha)/w_k| < a_{\max}\}$ onto its image with $\hat{f}_w(G_w\Theta) = \hat{f}_q(\Theta)$, so $L_w \circ G_w = L_q$, and it maps critical points to critical points with the same Hessian signature; the function and the residual are unchanged by construction. The leak toward an adaptive prior at its atoms vanishes (Theorem D‴.23(i), Proposition D‴.24(ii)). (ii) Let $u := \partial_w(G_w\vartheta^b(q))$ at $w = q$, $D := \operatorname{diag}(q_k G^{-1})$, $A := \nabla^2 L_q(\vartheta^b)$, $B := \partial_w \nabla_\Theta L_w$ at fixed $\Theta$, $J := \partial_\Theta \hat{f}$, $\Phi := \partial_w \hat{f}$ at fixed $\Theta$. Differentiating $\hat{f}_w(G_w\Theta) = \hat{f}_q(\Theta)$ gives $\Phi = -Ju$, and differentiating $\nabla L_w(G_w\vartheta^b) = 0$ gives $B = -Au$. The fixed prior's stationarity $\nabla L_w(\Theta) + \sigma_0 D(\Theta - \vartheta^0) = 0$, linearised at $(q, \vartheta^b)$, reads $(A + \sigma_0 D)\delta\Theta = -B\delta w = Au\delta w$, so $\delta\hat{f} = \Phi\delta w + J\delta\Theta = -J(A + \sigma_0 D)^{-1}[(A+\sigma_0 D) - A]u\delta w = -\sigma_0 J(A+\sigma_0 D)^{-1}Du\delta w$; in the whitened coordinates ($\hat{J} = JD^{-1/2}$, $\tilde{A} = D^{-1/2}AD^{-1/2}$, $\tilde{g} := -D^{1/2}u$) this is $\sigma_0\hat{J}(\tilde{A} + \sigma_0 I)^{-1}\tilde{g}$. With $\sigma_0 = 0$, $\delta\Theta = u\delta w$ and $\delta\hat{f} = 0$. At an exact representation $\tilde{R} = 0$, $\tilde{A} = H_L$ and $\tilde{g} = \partial_w x^0$. (iii) With a scalar amplitude and $\Psi(\gamma, \kappa) := \langle r\varphi(\cdot\,; c(\gamma), s(\kappa))\rangle_\rho$, $\langle r\tilde{\Phi}(\cdot\,; \vartheta)\rangle_\rho = a(\alpha)\Psi$, whose gradient is $(a'\Psi, a\Psi_\gamma, a\Psi_\kappa)$ and whose Hessian is $\begin{bmatrix} a''\Psi & a'\Psi_\gamma & a'\Psi_\kappa \\ a'\Psi_\gamma & a\Psi_{\gamma\gamma} & a\Psi_{\gamma\kappa} \\ a'\Psi_\kappa & a\Psi_{\gamma\kappa} & a\Psi_{\kappa\kappa} \end{bmatrix}$. At a critical point the gradient vanishes, so $\Psi = \Psi_\gamma = \Psi_\kappa = 0$ ($a' > 0$, $a \neq 0$) and only the block $a\Psi_{(\gamma,\kappa)}$ remains; $r$ and $\Psi$ are unchanged by the gauge and $a$ scales by $q_k/w_k$. The $\alpha$-direction contributes the eigenvalue 0, so $\lambda_k^+ \geq 0$. □

**Status.** (i) and (iii) proved, exactly, as statements about the stationary points and their linearisation (no linearisation in $w$: the gauge maps critical points to critical points); (ii) as Proposition D‴.24(i) — exact at $q$, first order in $w$; the convergence of the dynamics to the gauge transform numerical, and with it the floor's vanishing. The amplitude–mass degeneracy is the homogeneity of the conic view [16], [23], and the rescaling symmetries of networks, which map critical points to critical points with the same function while changing their curvature, are classical ([98]); what is new is that this symmetry, not exact representability, is the mechanism of Proposition D‴.24's floor removal — so at a strict local minimum in the interior the adaptive prior keeps the fitted function independent of the sampled weights without an exact representation — and, for a scalar amplitude per kernel, the exact scaling of the residual curvature under the gauge, which gives the adaptive prior's splitting margins in closed form. Its hypotheses are a strict local minimum in the interior and samples in $W_g$ with positive margins: on the instance $P(w^m \in W_g)$ is only 0.52 at $m = 15$ (0.87 at $m = 60$), and the branch ends at the saturation near $\varepsilon = 0.68$; targets whose best approximant lies on the saturation boundary (a sharp friction step fitted by a dipole of two kernels; the cogging servo of Proposition D‴.29, globally; Proposition D‴.27's beyond-the-saturation example) are outside it. Checked: `tests/test_gauge_floor.py` (at $\varepsilon = 0.3$, a critical point with a 1.7% fit error: the gauge transform a critical point at two sampled weight vectors, the critical point continued in $w$ from $\vartheta^b(q)$ landing on it; the adaptive prior's sensitivity, from critical points reached by Newton from $\vartheta^b(q)$, below $10^{-8}$ of the fixed prior's, the fixed prior's equal to $\sigma_0\hat{J}(\tilde{A} + \sigma_0 I)^{-1}\tilde{g}$ to $10^{-6}$ and not to the Gauss–Newton form; the residual curvature's zero amplitude row, its scaling $(q_k/w_k)$, the margins $\sigma_0 - (q_k/w_k)\lambda_k^+$ and the dynamics' separation eigenvalues $-\sigma_0 + (q_k/w_k)\operatorname{eig}(G^{1/2}R_kG^{1/2})$ to $10^{-9}$; adaptive-prior runs at $\eta = \infty$ and $\sigma_0$ converging to the gauge transform, 1.2 away from $\vartheta^b(q)$, with the approximant's function).

**Numerics** (`exp_gauge_floor.py` → `results/chaos/gauge_floor.md`). The three-bump instance with a fourth bump $\varepsilon \cdot \exp(-(x - 0.8)^2/(2 \cdot 0.3^2))$ that $K_0 = 3$ kernels cannot represent, $\varepsilon = 0$–$0.5$, the masses $q = 1/3$, $\sigma_0 = 0.15\sigma_{\text{crit}}$; $\vartheta^b(q)$ the weighted approximant, a critical

point of the unregularised loss continued in $\varepsilon$ from the exact representation (relative fit error $3.6 \cdot 10^{-3}$, $6.8 \cdot 10^{-3}$, $1.2 \cdot 10^{-2}$, $1.7 \cdot 10^{-2}$, $2.5 \cdot 10^{-2}$ at $\varepsilon = 0.05$, 0.1, 0.2, 0.3, 0.5; gradient residual $\le 1.6 \cdot 10^{-15}$; nondegenerate, the smallest eigenvalue of $\tilde{A}$ $4.6 \cdot 10^{-5}$–$3.1 \cdot 10^{-4}$). (i) At six weight vectors 0.01 inside $W_g$ the unregularised gradient at $G_w \vartheta^b(q)$ is $\le 1.6 \cdot 10^{-15}$, and Newton's method continued in $w$ from $\vartheta^b(q)$ (20 steps) lands on $G_w \vartheta^b(q)$ to $\le 1.7 \cdot 10^{-11}$ with $\hat{f}$ unchanged to $\le 5.5 \cdot 10^{-13}$ — plain Newton from $\vartheta^b(q)$ at $w$ does so in 3–5 of the 6 cases, elsewhere reaching other critical points of $L_w$; the adaptive prior's function-level sensitivity, from the critical points reached from $\vartheta^b(q)$, is $1.4 \cdot 10^{-11}$–$2.7 \cdot 10^{-10}$ of the fixed prior's. The samples fall in $W_g$ with the probability 0.52, 0.87–0.88, 0.997 at $m = 15$, 60, 240 ($\varepsilon = 0.05$–0.3; 0.26, 0.71, 0.98 at $\varepsilon = 0.5$), and the branch continued to $\varepsilon = 0.68$ reaches $\max_k |a_k^b|/a_{\max} = 0.999$, the smallest eigenvalue of $\tilde{A}$ $1.2 \cdot 10^{-8}$ and $W_g$'s threshold $0.333 = q_2$ for its class: there $w = q$ leaves $W_g$. (ii) The fixed matching prior's sensitivity equals $\sigma_0 \hat{J}(\tilde{A} + \sigma_0 I)^{-1} \tilde{g}$ to $\le 2.2 \cdot 10^{-9}$ and misses Proposition D‴.24's Gauss–Newton form by $7.0 \cdot 10^{-4}$–$5.6 \cdot 10^{-3}$, growing with $\varepsilon$ as $\|\tilde{R}\|/\|H_L\|$ does ($5.3 \cdot 10^{-4}$–$2.8 \cdot 10^{-3}$); its floor law per unit of $1/m - 1/M$ rises from $6.17 \cdot 10^{-2}$ ($\varepsilon = 0$) to $8.52 \cdot 10^{-2}$ ($\varepsilon = 0.5$). (iii) The residual curvature's amplitude row is $\le 7.4 \cdot 10^{-14}$ (its largest entry $2.6 \cdot 10^{-4}$–$2.9 \cdot 10^{-3}$), $\lambda_k^+ \le 7.2 \cdot 10^{-4}$ (at most $0.10\sigma_0$), and over every count vector of $m = 15$ and 60 in $W_g$ the adaptive prior's splitting margins equal $\sigma_0 - (q_k/w_k)\lambda_k^+$ to $\le 2.1 \cdot 10^{-14}$, none negative (the smallest $+6.23 \cdot 10^{-3}$); the separation eigenvalues of the dynamics at a replicated configuration (counts $(6, 4, 5)$, the Jacobian of the averaged drift at $\eta = \sigma_0$ and $\infty$) equal $-\sigma_0 + (q_k/w_k)\,\mathrm{eig}(G^{1/2} R_k G^{1/2})$ to $\le 5.8 \cdot 10^{-12}$. *The dynamics* (Part B: warm clouds of width 0.02 around $\vartheta^b(q)$, $m = 15$, 60, 240 against $M = 510$, four seeds, 6000 slow units, $\varepsilon = 0.1$ and 0.3, the nearest-atom leak). Under the fixed prior $\|\hat{f}_m - \hat{f}_M\|^2$ stays within 8% of its value at $s = 600$ (the ratio at $s = 6000$ 0.917–1.003; at $\varepsilon = 0.1$: $3.1 \cdot 10^{-4}$–$5.9 \cdot 10^{-3}$, $1.1 \cdot 10^{-4}$–$1.0 \cdot 10^{-3}$ and $5.0 \cdot 10^{-6}$–$8.2 \cdot 10^{-5}$ at $m = 15$, 60, 240) and the fit error above the approximant's ($9.0 \cdot 10^{-3}$–$1.2 \cdot 10^{-1}$ against $6.84 \cdot 10^{-3}$), the classes splitting at the samples where the fixed prior's point is a saddle (diameters up to 1.6). Under the adaptive priors the classes stay replicated, their means approach the gauge transform $G_w \vartheta^b(q)$, $\|\hat{f}_m - \hat{f}_M\|^2$ falls (at $m = 60$, $\varepsilon = 0.1$, from $6.9 \cdot 10^{-6}$–$1.2 \cdot 10^{-4}$ at $s = 600$ to $1.0 \cdot 10^{-9}$–$1.4 \cdot 10^{-8}$ at $\eta = \sigma_0$ and to $3.2 \cdot 10^{-11}$–$3.8 \cdot 10^{-9}$ at $\eta = \infty$ by $s = 6000$), and the fit error tends to the approximant's: at the samples at least 0.02 inside $W_g$ it exceeds it at $s = 6000$ by a median $1.2 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$, $1.6 \cdot 10^{-5}$ (relative; at most $1.1 \cdot 10^{-1}$, $1.2 \cdot 10^{-2}$, $1.3 \cdot 10^{-3}$) at $m = 15$, 60, 240. At the $m = 15$ sample $(0.533, 0.267, 0.200)$, outside $W_g$ (margins $-0.0016$ and $-0.0097$ at $\varepsilon = 0.1$ and 0.3), $\vartheta^b(q)$'s branch has no gauge transform: the adaptive priors' fit errors at $s = 6000$ ($2.2 \cdot 10^{-2}$ and $1.6 \cdot 10^{-2}$ at $\varepsilon = 0.1$, $3.7 \cdot 10^{-2}$ and $3.3 \cdot 10^{-2}$ at $\varepsilon = 0.3$, for $\eta = \sigma_0$ and $\infty$) are still falling toward the minimum of $L_w$ over the closed amplitude box on the branch, 1.107 and 1.586 times the approximant's fit error (one amplitude at the bound, its coordinate running off), and a gauge transform with two classes exchanged exists there, which the warm-cloud dynamics does not reach; at $(0.400, 0.200, 0.400)$, $6 \cdot 10^{-4}$ inside $W_g$ at $\varepsilon = 0.3$, the gauge transform's amplitude lies near the saturation and the approach is slow (2.8–3.1 from it at $s = 6000$), as at the boundary of Proposition D‴.24's $W$.

**Proposition D‴.29** (second-order plants: the sliding-variable reduction)**.** *Let the plant be $\ddot{x} = f(x, \dot{x}) + u$, $x_m$ a $T_0$-periodic $C^3$ reference, $e := x - x_m$, $s := \dot{e} + \lambda e$ ($\lambda > 0$), and let the controller and the adaptive law be*

$$u = \ddot{x}_m - \lambda \dot{e} - k_s s - \hat{f}_\Theta(z), \qquad \dot{\vartheta}_i = G \nabla_\vartheta (s \tilde{\Phi}(z; \vartheta_i)) - (\sigma_0/k_s)(\vartheta_i - T_i),$$

*$z$ the coordinates of $(x, \dot{x})$ on which $f$ depends and $T_i$ any of the leak targets of this section (a fixed prior, the nearest-atom or the OT plan, an adaptive or a projected prior).*

(i) (the error channel) *$\dot{s} = -k_s s + r_\Theta(z)$ and $\dot{e} = -\lambda e + s$, with $z = z_m + (e, \dot{e})$ the regressor at the plant's state: Theorem J's error channel with $s$ in place of $e$.*

(ii) (the averaged slow loop) *As $k_s \to \infty$ at fixed $\lambda$, $s$ and $e$ are slaved — $s = (\mathrm{d}/\mathrm{d}t + k_s)^{-1}[r_\Theta(z_m)] + O(k_s^{-2}) = r_\Theta(z_m)/k_s + O(k_s^{-2})$ and $e = (\mathrm{d}/\mathrm{d}t + \lambda)^{-1} s$, the periodic*

*solutions of the two filters — and in the slow time $\tau = t/k_s$ the parameters follow Theorem J's averaged slow loop along the regressor's orbit $z_m(t)$,*

$$\frac{\mathrm{d}\vartheta_i}{\mathrm{d}\tau} = G\langle r_\Theta \nabla_\vartheta \tilde{\Phi}(\cdot\,;\vartheta_i)\rangle_\rho - \sigma_0(\vartheta_i - T_i), \quad \rho \text{ the occupation measure of } z_m \text{ over one period,}$$

*up to $O(1/k_s)$, the prior's law at finite $\eta$ being $\mathrm{d}z_j/\mathrm{d}t = -(\eta/k_s)(z_j - \hat{T}_j)$ in real time. Both forms of the slaved error — the exact filter $(\mathrm{d}/\mathrm{d}t + k_s)^{-1}$ and $r/k_s$ — are accurate to $O(k_s^{-2})$ in $s$, i.e. to $O(1/k_s)$ relatively, the parameters' within-period motion and the nonlinearity $r(z) - r(z_m)$ entering at the same order; the exact filter keeps the error filter's attenuation of the residual's harmonics, which $r/k_s$ drops at the relative order $n^2\omega_0^2/(2k_s^2)$ in the rms of the $n$-th harmonic, and is the better prediction where that term dominates (a small residual of high frequency).*

(iii) (the transfer) *Every statement of this section about the averaged slow loop holds for the second-order plant with this $\rho$ — Theorems D‴, D‴.21 and D‴.23, Propositions D‴.5, D‴.9, D‴.12, D‴.17, D‴.24(i)–(v), D‴.27 and D‴.28 — and the tracking error of the position is the filtered residual of (ii); the real-loop statements (Proposition D‴.24(vi), Corollaries D‴.14 and D‴.26) would need the variational equation of (i).*

(iv) (where the kernels live) *When $f$ depends on one coordinate (a cogging torque on the position, a friction law on the velocity), kernels on that coordinate see the orbit as an interval, and the instance is this section's one-dimensional one with another target. Isotropic Gaussian kernels on the phase plane over a circular orbit ($x_m = R\sin\omega_0 t$ with $\omega_0 = 1$ in the units where the kernel is isotropic in $(x, \dot{x})$) are over-parametrised: on the circle $z = R(\sin\theta, \cos\theta)$ a kernel restricts to $a \cdot e^{-(R^2+|c|^2)/(2s^2)} \cdot e^{(R|c|/s^2)\cos(\theta-\theta_c)}$, a von Mises function of three effective parameters for four, so in general position the Gauss–Newton block has a zero eigenvalue for every kernel (more at degenerate atoms, as for kernels centred on the orbit's centre) — a direction that the fixed prior's leak pins at the rate $\sigma_0$, that an adaptive prior leaves neutral ($\lambda^- = 0$, Theorem D‴.23(ii)), and that a projected prior keeps bounded (Proposition D‴.27). On other orbits the same directions are weakly excited rather than unexcited: the classical partial excitation of Gaussian networks along an orbit ([92, 93, 94]).*

*Proof.* (i) $\ddot{e} = f + u - \ddot{x}_m = r_\Theta(z) - \lambda\dot{e} - k_s s$, so $\dot{s} = \ddot{e} + \lambda\dot{e} = -k_s s + r_\Theta(z)$, and $\dot{e} = s - \lambda e$ by the definition of $s$. (ii) The error subsystem $(s, e)$ is linear, exponentially stable (eigenvalues $-k_s$ and $-\lambda$) and driven by $r_\Theta(z)$, while the parameters move at the rate $O(1/k_s)$ (their velocity is $G\nabla_\vartheta(s\tilde{\Phi})$ with $s = O(1/k_s)$). Theorem J's argument — for the smooth fixed-prior leak near a hyperbolic critical point, using the error channel's exponential stability, the regressor's smoothness and its Step 3's normal form — carries over with the fast pair $(s, e)$ once its error-channel steps are redone for it ($s$ slaved at $k_s$, $e$ filtered at the fixed $\lambda$: the error channel's multipliers are $e^{-k_s T_0}$ and $e^{-\lambda T_0}$), the separation being between $\lambda$ and the parameters' rate; for switching targets (plan and cell switches, adaptive and projected priors) not even the first-order loop has more than Theorem D‴.18's locally constant case. At the slaved state $z = z_m + O(1/k_s)$, and the period average of $G\nabla_\vartheta(s\tilde{\Phi}(z_m;\vartheta_i))$ with $s = r_\Theta(z_m)/k_s$ is $(1/k_s)$ times the averaged learning term. For the $n$-th harmonic of $r$, $(\mathrm{d}/\mathrm{d}t + k_s)^{-1}$ has the gain $(k_s^2 + n^2\omega_0^2)^{-1/2} = (1/k_s)(1 - n^2\omega_0^2/(2k_s^2) + \dots)$. (iii) The statements of this section use the averaged loop only through $\rho$, the target on its support and the network. (iv) $|z - c|^2 = R^2 + |c|^2 - 2R|c|\cos(\theta - \theta_c)$ on the circle; the map $(a, |c|, \theta_c, s) \mapsto (ae^{-(R^2+|c|^2)/(2s^2)}, R|c|/s^2, \theta_c)$ has a one-dimensional fibre, along which the kernel's restriction, and hence $\hat{f}$ on $\rho$'s support, is constant. □

**Status.** (i) and (iv) exact; (ii) and (iii) formal — first-order averaging with the fast pair $(s, e)$, Theorem J's steps carrying over for the smooth fixed-prior leak near a hyperbolic critical point once the error-channel steps are redone, and switching targets beyond even the first-order theory; (iii) covers the statements about the averaged loop, not the real-loop ones (Proposition D‴.24(vi) and Corollaries D‴.14 and D‴.26 would

need the variational equation of (i)). The filtered error is the classical construction of adaptive control for higher-order plants, used with Gaussian networks by Sanner and Slotine ([79]); what is added is the transfer of the transport leaks, their floors and the adaptive and projected priors to such plants, and the exact fibre of isotropic phase-plane kernels on a circular orbit (the weak excitation of the same directions on other orbits being the classical partial excitation [92, 93, 94]). Checked: `tests/test_sliding.py` (the phase-plane loop as the gradient flow of $\bar{L}$; on a circular orbit one zero Gauss–Newton eigenvalue per kernel in general position, the next above $10^{-11}$ of the largest; the servo's parameter displacement over ten periods within 3% of the averaged loop's at $k_s = 25$ and closer at $k_s = 50$, and its position error within 1% of both predictions of (ii) at $k_s = 50$).

**Numerics** (`exp_servo_cogging.py` → `results/servo/servo_cogging.md`). The servo $\ddot{x} = -\tau_{\mathrm{cog}}(x) + u$ with the cogging torque $\tau_{\mathrm{cog}}(x) = 0.4\sin(3x + 0.3) + 0.1\sin 7x$, $x_m = \sin t$, $\lambda = 5$; $K_0 = 4$ saturated Gaussian kernels on the position ($a_{\max} = 4$, $c_{\max} = 1.6$, widths in $[0.05, 1]$), masses $1/4$, $\sigma_0 = 0.01$. (A) $\vartheta^b(q)$, the nondegenerate local minimum of the unregularised weighted loss that BFGS reaches from a fixed start, has the relative fit error $1.46 \cdot 10^{-2}$, and Proposition D‴.28 is its local analysis: the whitened unregularised Hessian $\tilde{A}$ spans $2.9 \cdot 10^{-4}\sigma_0$–$30\sigma_0$ (four modes below $0.1\sigma_0$: the adaptive prior's slowest rates $1.4 \cdot 10^{-4}\sigma_0$ at $\eta = \sigma_0$ and $2.9 \cdot 10^{-4}\sigma_0$ at $\eta = \infty$, against the fixed prior's $\approx \sigma_0$), the residual curvature's $\lambda_k^+ \le 0.0098\sigma_0$ (the adaptive prior's margins positive on all of $W_g = \{w > (0.061, 0.085, 0.119, 0.008)\}$), the fixed matching prior's floor law $1.29 \cdot 10^{-2}$ per unit of $1/m - 1/M$ and its margin $+9.9 \cdot 10^{-3}$ at $w = q$. Globally the loss has its infimum at the saturation: of 120 BFGS runs from random starts one ends at $\vartheta^b(q)$ and 33 below its fit error, all with saturated amplitudes; the best reaches $6.39 \cdot 10^{-3}$, and with its two saturated amplitudes held at $|\alpha| = 6$, 8, 10, 14 ($|a| = 3.62$–$3.99$) and the rest re-optimised the fit error is $7.14 \cdot 10^{-3}$, $6.65 \cdot 10^{-3}$, $6.49 \cdot 10^{-3}$, $6.40 \cdot 10^{-3}$ — approached, not attained. Globally the servo is Proposition D‴.27's case (critical points at infinity), and the runs below are statements about the basin of $\vartheta^b(q)$ and about an engineer's model. A misplaced prior — the approximant of the cogging model with the phase 0.5 in place of 0.3, the engineer's model, its own fit error 0.200 — has a replicated stationary point (the same from its atoms and from $\vartheta^b(q)$) with the fit error $5.45 \cdot 10^{-2}$ and the splitting margin $-3.97 \cdot 10^{-3}$: a D‴.5 saddle. (B) The averaged loop ($m = 16$, 64, 128 against $M = 256$, three seeds, 2000 slow units). From warm clouds around $\vartheta^b(q)$ — computed from the true torque — with the matching prior $\vartheta^b(q)$: the fixed prior's nearest-atom leak keeps a finite-size floor ($\|\hat{f}_m - \hat{f}_M\|^2$ $1.85 \cdot 10^{-3}$, $2.24 \cdot 10^{-4}$, $4.49 \cdot 10^{-5}$; fit error 0.174, 0.066, 0.038 against the approximant's 0.0146); the adaptive prior's floor is 38–780 times lower ($\eta = \infty$: $4.85 \cdot 10^{-5}$, $2.88 \cdot 10^{-7}$, $6.36 \cdot 10^{-8}$), with the fit error 0.0298 (the classes not yet replicated at $m = 16$ by $s = 2000$), 0.0148, 0.0147; the OT leak toward the matching prior sits at the approximant (0.01462, no floor); the $\sigma$-modification toward $\vartheta^0 = 0$ at the same $\sigma_0$ keeps a large bias (0.328) — and no leak at all ($\sigma_0 = 0$) does as well as the best of them (0.0180, 0.0147, 0.0146; floors $7.7 \cdot 10^{-6}$, $2.4 \cdot 10^{-7}$, $6.6 \cdot 10^{-8}$). In this noise-free simulation the floors and biases that the adaptive and OT leaks remove are those a fixed leak creates; the leaks' purpose — robustness to noise and disturbances, Propositions D‴.27, D‴.30, D‴.31 — is not exercised here. With the misplaced prior from that start, at $\eta = \infty$ the prior never acts (its initial position enters only through the first cells: the runs are the matching ones), at $\eta = \sigma_0$ its bias decays (0.035, 0.020, 0.019 at $s = 2000$), and the fixed designs keep a bias — their $m$-systems leave the saddle by splitting their classes (diameters 0.32–1.07) and end at 0.048 (OT) and 0.10, 0.060, 0.048 (the nearest atom; above the replicated point's 0.0545 at $m = 16$ and 64). From warm clouds around the engineer's model, with the prior at it — the start an engineer has — the fixed designs stay biased (0.102, 0.060, 0.048; OT 0.048), the adaptive prior removes most of the bias but slowly ($\eta = \infty$: 0.027, 0.025, 0.022 at $s = 2000$; $\eta = \sigma_0$: 0.032, 0.028, 0.028), and no leak reaches the basin's fit (0.0152, 0.0147, 0.0147): its $m$ particles spread and explore directions that the $K_0$-atom structure the leaks impose leaves weakly excited. The collapsed $K_0$-atom flows give the time scale: from the engineer's model at $\eta = \infty$ — where the collapsed

flow is the unregularised one — the fit error is 0.0297, 0.0213, 0.0169, 0.0155 at $s = 600$, 2000, $10^4$, $10^5$, still 0.55 from $\vartheta^b(q)$ in the parameters ($10^5$ slow units are $2.5 \cdot 10^6$ time units of the real servo at $k_s = 25$), at $\eta = \sigma_0$ 0.0345, 0.0259, 0.0181, 0.0160; from $\vartheta^b(q)$ toward the misplaced prior at $\eta = \sigma_0$ 0.0240, 0.0183, 0.0154, 0.0149. (C) The real servo ($k_s = 25$, RK4 at $\mathrm{d}t = 2 \cdot 10^{-3}$, 1000 time units, $m = 16$ and 64, seed 0) from the oracle-start designs' end states, with each design's leak (its targets refreshed, and the error recorded, every 0.02 time units; the finite-$\eta$ prior relaxed in real time): the position error's rms is $1.7$–$2.1 \cdot 10^{-5}$ for the adaptive designs, the matching OT design and no leak, $7.8 \cdot 10^{-5}$ for the misplaced OT design, $9.1 \cdot 10^{-5}$–$1.8 \cdot 10^{-4}$ for the fixed nearest-atom prior and $5.4$–$5.5 \cdot 10^{-4}$ for the $\sigma$-modification; it equals the prediction of (ii) with the exact filter to 1.0001–1.0029 at $t = 500$ and 1000 in every design (0.990–1.005 over the first five periods, where the parameters still move), and with the slaved $s = r/k_s$ to 0.931–0.996, the deficit largest for the adaptive designs, whose residual is the approximation error's broad fine scale; over these 40 slow units no design drifts, no leak included. (D) Phase-plane kernels (five isotropic kernels in general position, $\sigma_0 = 0.01$): on the circle 5 exact zero Gauss–Newton eigenvalues (one per kernel) and 8 below $10^{-6}\sigma_0$, the $(K+1)$-th eigenvalue $2.4 \cdot 10^{-10}$ of the largest; on ellipses with the velocity axis scaled by 1.1 and 2, 8 and 6 below $10^{-6}\sigma_0$ and none exact; on the orbit of $1.2 \sin t + 0.3 \sin 2t$ 3; with the five centres at the origin 16 exact zeros of 20.

**Proposition D‴.30** (measurement noise at perfect tracking: a coherent, high-passed noise, and the errors in variables)**.** *Let* (CL) *be the real loop of Corollary D‴.26 ($n = 1$) at the lift $\Gamma_w$ of an exact representation of representable weights $w$, with the prior at its atoms — fixed there ($\eta = 0$) or adaptive ($0 < \eta \le \infty$) — and $\partial_x \hat{f} = \partial_x f$, $\partial_x^2 \hat{f} = \partial_x^2 f$ on the orbit (an exact representation, an identity of functions, supplies both), and let the controller and the adaptive law use the measurement $y = x + \nu$ in place of $x$: $u = \dot{x}_m - k_e(y - x_m) - \hat{f}_\Theta(y)$, $\dot{\vartheta}_i = G \nabla_\vartheta ((y - x_m)^\top \tilde{\Phi}(y; \vartheta_i)) - (\sigma_0/k_e)(\vartheta_i - z_{\pi(i)})$, $\nu$ a stationary zero-mean noise, independent of the rest, with the spectral density $N_0$ at the frequencies of the regressor and of the error channel. Let $H := H_L(w) = \langle \hat{J}^\top \hat{J} \rangle$ (the whitened weighted Gauss–Newton matrix, Corollary D‴.26's $\hat{J}$) with the eigenpairs $(h_j, v_j)$, $a(t) := \partial_x \hat{f}(x_m(t))$, $Q := \langle (\dot{\hat{J}} + a\hat{J})^\top (\dot{\hat{J}} + a\hat{J}) \rangle$ and $Q_{jj} := v_j^\top Q v_j$.*

(i) (coherence) *The particles of a class see the same $y$, the same error and the same prior atom, so a replicated configuration stays replicated under every path of $\nu$: on the replicated subspace the $m$-system is the weighted $K_0$-atom loop driven by the same noise, for every $m$ with these weights, at every order in $\nu$. Measurement noise does not average out with $m$, as the per-particle dither of Theorem J′ does. This is invariance, not attraction. A separation $u = G^{-1/2} \delta\vartheta_i$ of class $k$ obeys $\dot{u} = (G^{1/2} \varepsilon_m \nabla_\vartheta^2 \tilde{\Phi}(y; \vartheta_k) G^{1/2} - \sigma_0/k_e)u$ along the noisy class trajectory, and to second order in $\nu$ its mean drift is $\mathbb{E}[\varepsilon_m^2] G^{1/2} \partial_x \nabla_\vartheta^2 \tilde{\Phi} G^{1/2} + \mathbb{E}[e] G^{1/2} \nabla_\vartheta^2 \tilde{\Phi} G^{1/2}$ (the measured regressor; the parameters' cross term negligible), $\mathbb{E}[e]$ the mean tracking error. At the lift — the parameters at their atoms, $\mathbb{E}[e]$ the noise's own mean error at fixed parameters — its period average $S_k$ gives the decay rate $\sigma_0/k_e - \mathbb{E}\nu^2 \lambda_{\max}(S_k)$ to leading order, so near the lift the replicated configuration attracts — and an $m$-system started near it follows the $K_0$-atom loop — only below the noise level $\mathbb{E}\nu^2 = (\sigma_0/k_e)/\max_k \lambda_{\max}(S_k)$. Along the noisy trajectory $\mathbb{E}[e]$ also carries the mean tracking error $-\hat{J}\bar{y}(t)/k_e$ of the developing errors-in-variables deviation $\bar{y}(t)$ of (iii), of the same order in $\mathbb{E}\nu^2$, so the drift $S_k(t)$ changes on $\bar{y}$'s time scale and to this order the separation follows $\dot{u} = (\mathbb{E}\nu^2 S_k(t) - \sigma_0/k_e)u$ (the first-order multiplicative noise, high-passed by the loop, adds a negligible cumulant to its exponent); the long-time threshold — where $\bar{y}$ is $O(1)$ at these noise levels and the expansion no longer applies — is open.*

(ii) (the floor at first order) *To first order in $\nu$ the measured error's noise is $((\mathrm{d}/\mathrm{d}t - a)/(\mathrm{d}/\mathrm{d}t + k_e))\nu$ — the loop high-passes it — and in the slow time the collective coordinates diffuse with the matrix $(N_0/k_e)Q$; their stationary covariance solves $\bar{A}_0 \Sigma + \Sigma \bar{A}_0^\top + (N_0/k_e)\,\mathrm{diag}(Q, 0) = 0$ ($\bar{A}_0$ Corollary D‴.26's averaged collective block), and the function-level floor, averaged over*

*a period, is*

$$\mathbb{E}\|\hat{f}-f\|_\rho^2=(N_0/2k_e)\Big[\sum_j Q_{jj}(h_j+\eta)/(h_j+\eta+\sigma_0)+\operatorname{tr}H^2\Big]+O(k_e^{-2})$$

*(at time $t$ its fast part is $(N_0/2k_e)\operatorname{tr}(H\hat{J}(t)^\top\hat{J}(t))$, whose oscillation over the period is of the floor's own order): the parameters' part, each mode's share multiplied by $(h_j+\eta)/(h_j+\eta+\sigma_0)$ — the fixed prior's Tikhonov filter factor $h_j/(h_j+\sigma_0)$ at $\eta=0$, the factor 1 at $\eta=\infty$, where the part is $(N_0/2k_e)\operatorname{tr}Q$ whatever the spectrum — and the jitter of the frequencies that the loop passes, independent of the prior (the compensation error at the plant's current point has $\langle|\hat{J}|^4\rangle$ in place of $\operatorname{tr}H^2$). Mode $j$'s parameter variance is $(N_0/2k_e)Q_{jj}(h_j+\eta)/(h_j(h_j+\eta+\sigma_0))$ — from $Q_{jj}/(2(h_j+\sigma_0))$ to $Q_{jj}/(2h_j)$ in units of $N_0/k_e$ — plus the fast part $(N_0/2k_e)h_j$.*

(iii) (the errors in variables) *To second order the noise inside the regressor $\tilde{\Phi}(y)$ exerts a mean force on the collective coordinates. Its part carried by the noise's own second moments, at fixed parameters, is $k_eF$ in the slow time with $F:=\langle\hat{J}'^\top\mathbb{E}[\varepsilon_m^2]+\hat{J}^\top\mathbb{E}[e]\rangle=O(\mathbb{E}\nu^2)$ ($\hat{J}':=\partial_x\hat{J}$, $\varepsilon_m:=y-x_m$ the measured error; for an Ornstein–Uhlenbeck noise of rate $B$, $\mathbb{E}[\varepsilon_m^2]=\mathbb{E}\nu^2(a^2+k_eB)/(k_e(k_e+B))\to\mathbb{E}\nu^2$ as $B/k_e\to\infty$); the parameters' first-order fluctuation adds a force of the same order in $N_0$ — the compensation's curvature and slope–parameter terms in the error equation, the correlation $\nabla_\vartheta^2\tilde{\Phi}\mathbb{E}[\varepsilon_m\delta\vartheta]$ in the law — negligible against $F$ for the measured regressor, not for the desired one (Proposition D‴.31(iv)). The stationary response is $k_e(H+\sigma_0I)^{-1}F$ under the fixed prior and $k_eH^{-1}F$ under the adaptive prior, for every $\eta>0$: the adaptive prior gives up the leak's bound on the response to a persistent force in the weakly excited modes — the robustness half of the classical trade-off, here against the noise, and the lack-of-excitation drift of adaptive control ([100]) — and its function-level bias $k_e^2F^\top H^{-1}F$ grows like $k_e^2(\mathbb{E}\nu^2)^2$ while the noise is white well beyond the loop's bandwidth ($k_e\ll B$), and saturates for $k_e\gg B$, where $\mathbb{E}[\varepsilon_m^2]\to\mathbb{E}\nu^2B/k_e$. The part of $F$ that the adaptive prior amplifies is the regressor's own, $\langle\hat{J}'^\top\mathbb{E}[\varepsilon_m^2]\rangle$; evaluating the regressor at the reference removes it (Proposition D‴.31).*

*Hence $\eta$ buys the bias-removal rate $\lambda_j^-(\eta)$ at the price of the factor $(h_j+\eta)/(h_j+\eta+\sigma_0)$ in the floor and of the $H^{-1}$ response to the errors in variables, and $k_e$ lowers the floor like $1/k_e$ while it raises the errors-in-variables bias like $k_e^2$ ($k_e\ll B$).*

*Proof.* (i) The equations of the particles of a class are identical — the same $y$, the same measured error and the same prior atom (the class mean at $\eta=\infty$, the same $z_k$ otherwise) enter them — and so are their initial values; the field is locally Lipschitz and the cells are constant near the lift, so the trajectories coincide, and the class means follow the weighted $K_0$-atom loop with the same noise. (ii) Linearised at $\Gamma_w$ in Corollary D‴.26's whitened coordinates: $\delta\dot{e}=-k_e\delta e-\hat{J}y-(k_e+a)\nu$ (from $f(x)-\hat{f}(x+\nu)=-D\hat{f}[\delta\Theta]-a\nu$), $\dot{y}=\hat{J}^\top(\delta e+\nu)-(\sigma_0/k_e)(y-\zeta)$, $\dot{\zeta}=-(\eta/k_e)(\zeta-y)$ (the regressor's own noise is of second order). The measured error $\varepsilon_m=\delta e+\nu$ obeys $\dot{\varepsilon}_m=-k_e\varepsilon_m-\hat{J}y+\dot{\nu}-a\nu$, so $\varepsilon_m=-(\mathrm{d}/\mathrm{d}t+k_e)^{-1}[\hat{J}y]+n$, $n:=(\mathrm{d}/\mathrm{d}t+k_e)^{-1}[\dot{\nu}-a\nu]$ (written $((\mathrm{d}/\mathrm{d}t-a)/(\mathrm{d}/\mathrm{d}t+k_e))\nu$): the first part is Corollary D‴.26's slaving, the second drives $\dot{y}=\hat{J}^\top n$. With $\xi:=(\mathrm{d}/\mathrm{d}t+k_e)^{-1}\nu$, of variance $N_0/(2k_e)$ for white noise, $n=\dot{\xi}-a\xi+(\mathrm{d}/\mathrm{d}t+k_e)^{-1}[\dot{a}\xi]$ and $\hat{J}^\top n=\mathrm{d}(\hat{J}^\top\xi)/\mathrm{d}t-(\dot{\hat{J}}+a\hat{J})^\top\xi+O(\xi/k_e)$: $y=\hat{J}^\top\xi+u$, the fast jitter $\hat{J}^\top\xi$ and the slow part $u$ driven by $-(\dot{\hat{J}}+a\hat{J})^\top\xi$, white at the slow scale with the intensity $(N_0/k_e^2)Q$ per unit time, $(N_0/k_e)Q$ in the slow time; the averaged slow equation $\mathrm{d}Y=\bar{A}_0Y\,\mathrm{d}s+(N_0/k_e)^{1/2}(Q^{1/2}\,\mathrm{d}W,0)$ (stochastic averaging, [96]; exactly, for the linearised loop, Lemma D‴.26a(v)) has the stated stationary covariance. In the eigenbasis of $H$ the block $\bar{A}_0$ is a direct sum of the pair blocks $\left[\begin{smallmatrix}-(h+\sigma_0) & \sigma_0\\ \eta & -\eta\end{smallmatrix}\right]$ ($\eta=0$: $-(h+\sigma_0)$; $\eta=\infty$: $-h$), whose Lyapunov equation with the noise $q$ on $y$ gives $\Sigma_{\zeta\zeta}=\Sigma_{y\zeta}=\eta\Sigma_{yy}/(h+\eta)$ and $\Sigma_{yy}=q(h+\eta)/(2h(h+\eta+\sigma_0))$; $Q$'s off-diagonal entries in that basis enter only the covariances of different modes, which

$\operatorname{tr}(H\Sigma_{yy}) = \sum_j h_j \Sigma_{yy,jj}$ does not see. The fast jitter gives $\mathbb{E}[(\hat{J}^\top\xi)^\top H(\hat{J}^\top\xi)] = (N_0/2k_e)\operatorname{tr} H^2$ over a period, and $\mathbb{E}[(\hat{J}(t)\hat{J}(t)^\top\xi)^2] = (N_0/2k_e)\langle|\hat{J}|^4\rangle$. (iii) At fixed parameters the second-order part of $\mathbb{E}[\varepsilon_m \nabla_\vartheta \tilde{\Phi}(y;\vartheta)]$ at the lift is $\partial_x \nabla_\vartheta \tilde{\Phi} \cdot \mathbb{E}[\varepsilon_m^2] + \nabla_\vartheta \tilde{\Phi} \cdot \mathbb{E}[e]$, $\mathbb{E}[e]$ from the error equation's second-order terms $\frac{1}{2}\partial_x^2 f \cdot e^2 - \frac{1}{2}\partial_x^2 \hat{f} \cdot (e+\nu)^2$ — one second derivative for $f$ and $\hat{f}$, by the hypothesis (otherwise $\frac{1}{2}(\partial_x^2 f - \partial_x^2 \hat{f})\operatorname{Var}(e)$ remains, of the same order); the parameters' fluctuation adds $\nabla_\vartheta^2 \tilde{\Phi} \cdot \mathbb{E}[\varepsilon_m \delta\vartheta]$ in the law and $-\partial_x D\hat{f}[\mathbb{E}[(\delta e + \nu)\delta\Theta]] - \frac{1}{2}\mathbb{E}[D_\Theta^2 \hat{f}[\delta\Theta, \delta\Theta]]$ in the error equation, of the same order in $N_0$ (computed in the numerics from the periodic covariance: the full second-order mean). For the Ornstein–Uhlenbeck noise, quasi-statically along the orbit, $\mathbb{E}[\nu e] = -(k_e + a)\mathbb{E}\nu^2/(k_e + B)$ and $\operatorname{Var}(e) = (k_e + a)^2\mathbb{E}\nu^2/(k_e(k_e + B))$, which give $\mathbb{E}[\varepsilon_m^2] = \mathbb{E}\nu^2 + 2\mathbb{E}[\nu e] + \operatorname{Var}(e)$ as stated and $\mathbb{E}[e] = -(\partial_x^2 \hat{f}/(2k_e))\mathbb{E}\nu^2(B - k_e - 2a)/(k_e + B)$. A constant slow force $F_s$ on the pair block of a mode has the stationary point $\zeta = y$ and $(h+\sigma_0)y - \sigma_0\zeta = F_s$, i.e. $y = F_s/h$ for every $\eta > 0$, and $y = F_s/(h+\sigma_0)$ under the fixed prior. □

**Status.** (i) proved, exactly, as invariance; the threshold formal at the lift (the separations' averaged second-order drift there), the drift along the noisy trajectory computed from the second-order transient and checked against the real loop's separation rates on antithetic noise pairs, the long-time threshold open; (ii) proved for the linearised loop under white noise, for the floor's period average (Lemma D‴.26a(v): the slow coordinates' drift decoupled exactly, their noise $\varepsilon b\nu$, at leading order the high-passed one; at $\eta = 0$ with $K = \hat{J}$, $\Lambda = \sigma_0 I$), with the relative remainder $O(1/k_e)$, the closed form exact for the averaged system; the OU noise of the numerics and the real loop formal — and checked against the exact periodic covariance of the linearised loop: the deficit 2.4%, 0.61%, 0.14%, 0.028% at $k_e = 25$–$200$ ($\eta = 0$) is dominated there by a term in $k_e^{-2}$, but the relative error is $O(1/k_e)$ — the slope $a$ in the noise input $-(k_e + a)\nu$, a rate that does not change sign under time reversal, breaks the parity of Corollary D‴.26(c) (with the noise input $-k_e\nu$ the deviation is numerically even in $1/k_e$, to $k_e = 800$) — and the ratio crosses 1 near $k_e = 500$; (iii) formal — a quasi-static second-moment computation for the noise's own moments, the parameters' part computed numerically — its stationary response exact for the averaged system, the exact linearised loop's 10.7% (adaptive) and 2.6% (fixed) higher at $k_e = 25$, an $O(k_e^{-2})$ excess; the real nonlinear loop checked by Monte Carlo against the sum of the two. Classical: measurement noise in adaptive control and the errors-in-variables bias of estimates from noisy regressors (bias-compensated least squares and instrumental variables in identification; bounded disturbances and robust modifications, [47]); using the reference in the regressor, as the desired compensation adaptation law does ([96]); the Lyapunov covariance of a constant-gain adaptive algorithm as its diffusion approximation (Benveniste–Métivier–Priouret), the LMS misadjustment's dependence on the input's spectrum through a trace only (Widrow et al.: the $\eta = \infty$ floor's $\operatorname{tr} Q$), the leaky LMS's steady state (Mayyas–Aboulnasr: the fixed prior's factor $h_j/(h_j + \sigma_0)$), the common noise of mean-field theory (Carmona–Delarue–Lacker), of which the coherence is an instance, and the drift of the weakly excited directions under a persistent disturbance (Anderson), of which the $H^{-1}$ response is ([100]). New, in the transport-leak setting: the interpolation $(h_j + \eta)/(h_j + \eta + \sigma_0)$ of the floor in the prior's mobility, the driving matrix $Q = \langle(\dot{\hat{J}} + a\hat{J})^\top(\dot{\hat{J}} + a\hat{J})\rangle$ — the sensitivity-filtered measurement the law sees, built on Corollary D‴.26's derivative Gram — and their combination with the coherence of the $K_0$-atom identification and its noise threshold at the lift. Checked: `tests/test_measurement_noise.py` (the closed form against the Lyapunov solution, the trace identity at $\eta = \infty$ and the monotonicity in $\eta$; the exact periodic floors of the linearised loop at $k_e = 100$ for $\eta = 0$, $\sigma_0$ and $\infty$; a lift $m = 6$ under a noise path staying exactly replicated and following the $K_0$-atom loop driven by the same path; a separation of the weakest class decaying at the noise std 0.01 and growing at 0.05 — a sign test, the rates within $10^{-4}$ of the lift's prediction —; the separations' drift against an independent quadrature of its two moments, and along the second-order transient the mean parameters moved by the averaged force and the mean error carrying the tracking error of the developing deviation; the OU noise's stationary variance and correlation, and the law seeing the measured error at the measured point; the responses $(H + \sigma_0 I)^{-1}$ and $H^{-1}$ to a constant force, the quasi-static force against the full second-order mean's noise part to $2 \cdot 10^{-3}$, the parameters' part below $10^{-4}$ of the bias; the exact periodic mean against the filtered-Gram response under both priors, the fixed prior's branch and its leak in the

bias; and the mean function's curvature term against an independent contraction).

**Numerics** (`exp_measurement_noise.py` → `results/chaos/measurement_noise.md`). The three-bump instance ($\sigma_0 = 0.15\sigma_{\text{crit}}$, $x_m = 2\sin t$, $k_e = 25$) at the lift of the exact representation of the matching weights $q = 1/3$, the prior at its atoms. (A) The linear theory: $H$'s nine eigenvalues span $3.06 \cdot 10^{-4}$–$0.195$; $\operatorname{tr} Q = 8.784$, of which 8.105 is the derivative Gram $\langle \dot{\hat{J}}^\top \dot{\hat{J}} \rangle$ and 0.679 the slope term $\langle a^2 \hat{J}^\top \hat{J} \rangle$ (the cross term vanishes, as on every one-dimensional reference: an exact differential); $Q_{jj}/h_j$ ranges from 104 (mode 1) to 5.5 (mode 6). The parameters' floor per unit $N_0$ is 0.1597 under the fixed prior, 0.1627 at $\eta = \sigma_0$, 0.1690 at $10\sigma_0$ and $0.1757 = \operatorname{tr} Q/(2k_e)$ at $\eta = \infty$ — the adaptive prior costs 10% here, the strong modes carrying the floor — while the weakest mode's parameter variance grows from 0.0865 to 2.09, $\times 24.1$ in the averaged system ($\times 28.0$ in the exact linearised loop at $k_e = 25$, Part B). The fast parts are $\operatorname{tr} H^2 = 0.0657$ and $\langle |\hat{J}|^4 \rangle = 0.299$. The second-order force of the noise's own moments (OU noise, $B = 100$) has the averaged function-level stationary bias 491 (fixed prior) and 2397 (adaptive, $\times 4.88$) per unit $(\mathbb{E}\nu^2)^2$, equal to the floor of the parameters at the noise std 0.0026 and 0.0012 respectively and dominant above; the full second-order mean (the periodic covariance contracted with the true field's second derivatives) reproduces the force to $1.7 \cdot 10^{-4}$, the parameters' fluctuation changing the bias by less than $10^{-5}$ (2397.66 → 2397.68; the mean function's 2397.65), and the exact periodic second-order mean of the linearised loop gives 504 and 2653 ($\times 5.27$). The averaged bias grows in $k_e$ with the local exponent 1.93–1.99 for $B \geq 20k_e$ but 1.69 and 1.47 at $B = 8k_e$ and $4k_e$, and saturates beyond. At the lift the separations' second-order drift has $\lambda_{\max}(S_k) = 0.0148$, 0.0101 and 0.298 per unit $\mathbb{E}\nu^2$ (real time) in the three classes: there the third class's separations stop decaying at the noise std 0.031 (1.5% of the reference's amplitude), and at Part C's stds 0.003 and 0.01 the margin $\sigma_0/k_e$ is used to 0.9% and 10.5%. (B) The exact periodic stationary covariance of the linearised loop (the monodromy and a discrete Lyapunov equation) against the prediction, at $\eta = 0$, $\sigma_0$, $\infty$ and $k_e = 25$–$800$: the function-level floor's ratio 0.976–0.980 at $k_e = 25$, 0.994–0.995 at 50, 0.9986–0.9988 at 100, 0.9997–0.9998 at 200, and above 1 at 800 ($+1.2 \cdot 10^{-5}$ at $\eta = 0$, $+1.3 \cdot 10^{-5}$ at $\eta = \infty$); the deviation's successive ratios per doubling are 3.87, 4.38, 5.26, 11.4 and a change of sign ($\eta = \infty$) — a $k_e^{-2}$ term dominating to $k_e = 200$ over a first-order one of the opposite sign — while with the noise input $-k_e\nu$ (the slope removed, the control) they are 3.67, 3.90, 3.98, 3.99, 4.00: without the slope the expansion is numerically even (to $k_e = 800$; not proved). The weakest mode's variance is 0.845 (fixed prior) and 0.980 (adaptive) of its slow-plus-fast prediction at $k_e = 25$, tending to 1. The exact stationary second-order mean of the linearised loop lies 10.7%, 2.65%, 0.66%, 0.17% above the averaged bias under the adaptive prior and 2.58%, 0.64%, 0.16%, 0.04% under the fixed one at $k_e = 25$, 50, 100, 200 — the weak modes' stiffness being the filtered Gram $H - k_e^{-2}\langle \dot{\hat{J}}^\top \dot{\hat{J}} \rangle$ of Corollary D‴.26(c). (C) The real loop (RK4, $\mathrm{d}t = 10^{-3}$, an OU noise exact at the steps, 12 noise paths of 300 time units common to the four cases) against the transient covariance of the linearised loop plus the second-order mean deviation: at the noise std 0.003 within one standard error at every record ($z$ from $-0.99$ to 0.99), the mean part 42% of the prediction at $t = 300$; at std 0.01 within 1.75 standard errors ($z$ from $-0.3$ to 1.75, the records of a run correlated), the mean part 89% at $t = 300$ and the covariance part alone 8.8–8.9 times below the Monte Carlo — the second-order force is the dominant effect of the noise there. The adaptive and the fixed prior drift apart, by 6.0% and 6.4% at $t = 300$ (predicted 5.3% and 6.1%): the weak modes, where the two responses part, equilibrate over $k_e/h_1 \approx 8 \cdot 10^4$ time units, so over 300 time units the transient checks the force (the weak modes still grow linearly), and the stationary response is checked against the exact linearised loop (B). The lift $m = 15$ (classes of five) driven by the same noise path as the $K_0$-atom run gives the same $\hat{f}$ to rounding (identical in the stored float32, $5.6 \cdot 10^{-16}$ apart in float64): the noise is coherent, as (i) states. (D) But invariance is not attraction: from a separation of size $10^{-6}$ along $S_3$'s top eigenvector, on eight antithetic pairs of noise paths of 300 time units per std ($\nu$ and $-\nu$, whose pair means cancel the first-order part of the log-growth), the lift's third class

separates at $-2.50 \cdot 10^{-4} \pm 3.8 \cdot 10^{-7}$ per time unit at the std 0.01, $+2.74 \cdot 10^{-6} \pm 3.3 \cdot 10^{-6}$ at 0.03 and $+4.53 \cdot 10^{-4} \pm 8.4 \cdot 10^{-6}$ at 0.05. The noise-induced part $\lambda_{\text{eff}} := (\text{rate} + \sigma_0/k_e)/\mathbb{E}\nu^2$ is $0.3271 \pm 0.0038$, $0.3175 \pm 0.0037$ and $0.2943 \pm 0.0034$, against the lift's drift $\lambda_{\max}(S_3) = 0.2981$ and the drift along the second-order transient from the lift, 0.3264, 0.3270 and 0.3281 (the same estimator applied to $\dot{u} = (\mathbb{E}\nu^2 S_3(t) - \sigma_0/k_e)u$). Along the transient the mean error's rms per unit $\mathbb{E}\nu^2$ grows from 0.121 to 0.602 over the six 50-unit windows to $t = 300$ and $\lambda_{\max}(S_3(t))$ from 0.307 to 0.349: the tracking error of the developing errors-in-variables deviation. The weighted fit $\lambda_{\text{eff}} = c_0 + c_1 \mathbb{E}\nu^2$ has the small-noise limit $c_0 = 0.3291 \pm 0.0033$ — 10.4% above the lift's drift, within its standard error of the transient's 0.3264 — and the next order $c_1 = -13.8 \pm 2.0$; over this window the measured rate vanishes at the std 0.0299, against 0.0308 by the lift's drift and 0.0294 by the transient's. Above it the replicated configuration repels, its within-class spread staying exactly 0.0 from a replicated start only because each particle's update is the same arithmetic.

**Proposition D‴.31** (the adaptive prior is a washout leak; the desired regressor makes the noise's persistent force a least-squares force)**.** *Let the leak be Theorem D‴.23's toward an adaptive prior,* $\mathrm{d}z_j/\mathrm{d}s = -\eta(z_j - \hat{T}_j)$ *with* $0 < \eta < \infty$ *(the nearest-atom form of Proposition D‴.24(ii) with the class barycentres likewise), in the averaged loop or in the real loop* (CL) *of Corollary D‴.26* ($s = t/k_e$)*; at an exact representation of representable weights* $w$ *let* $H$, $\hat{J}$, $a$ *and the mean* $\langle\cdot\rangle$ *over a period be Proposition D‴.30's,* $\|g\|_\rho^2 := \langle g^2\rangle$ *for a function* $g$ *along the orbit, and* $P$ *the* $L^2_\rho$*-projection onto the span of* $\hat{J}$*'s components.*

(i) (the washout) *Along every solution and every path of the plan with* $\hat{T}_j$ *of bounded variation,* $\hat{T}_j(s) - z_j(s) = \int_0^s e^{-\eta(s-\tau)}\,\mathrm{d}\hat{T}_j(\tau) + e^{-\eta s}(\hat{T}_j(0) - z_j(0))$ *— and for every measurable* $\hat{T}_j$ *(a chattering plan included) in the integrated form* $z_j(s) = e^{-\eta s}z_j(0) + \eta\int_0^s e^{-\eta(s-\tau)}\hat{T}_j(\tau)\,\mathrm{d}\tau$*: the prior lags the barycentre by the barycentre's exponentially forgotten increments. For integral plans (the nearest-atom cells, the OT plan at a commensurate* $m$*) the leak on a particle,* $-\sigma_0(\vartheta_i - z)$*, is therefore the full* $\sigma$*-modification of its offset from the barycentre plus, on the barycentre, the* $\sigma$*-modification passed through the washout filter* $(\mathrm{d}/\mathrm{d}s)(\mathrm{d}/\mathrm{d}s + \eta)^{-1}$ *— zero at zero frequency, the full leak above* $\eta$*; with split rows (an incommensurate* $m$*) the barycentres also feel* $-\sigma_0(I - M)z$*,* $M_{jl} := (m/q_j)\sum_i \gamma_{ij}\gamma_{il}$*, which is not washed out. Linearised at the exact representation, mode* $j$ *of the collective block has at the slow frequency* $\omega$ *the stiffness* $h_j + \sigma_0 i\omega/(i\omega + \eta)$*: the fixed prior's* $h_j + \sigma_0$ *for* $\omega \gg \eta$*, the bare* $h_j$ *at* $\omega = 0$*. This is Theorem D‴.23(iii)'s bias removal — feedback through a washout filter leaves the equilibria where they are ([97]), here the replicated ones — and its price: no leak opposes a persistent force on the class barycentres (the separations keep the full leak).*

(ii) (the stationary identity) *Let the prior be an affine time-invariant filter of the barycentre and of a fixed anchor* $z_a$ *— linearised,* $\zeta = \mathcal{K}[y] + (I - \mathcal{K})[\delta]$ *in the whitened weighted coordinates,* $\delta$ *the anchor's misplacement from the representation; for example* $\mathrm{d}z/\mathrm{d}s = -\eta(z - \hat{T}) - \kappa(z - z_a)$*,* $\kappa \geq 0$*, with* $\mathcal{K}(0) = \eta/(\eta + \kappa)$ *(*$\kappa = 0$ *the adaptive prior,* $\kappa = \infty$ *the fixed prior at* $z_a$*) — and* $S_0 := \sigma_0(I - \mathcal{K}(0))$ *the leak's zero-frequency gain (the anchor entering through* $I - \mathcal{K}$*: the filter returns the anchor when the barycentre sits there). A persistent force* $F$ *on the collective coordinates and the misplacement* $\delta$ *give the stationary deviation* $y = RF + B\delta$ *with* $R = (H + S_0)^{-1}$ *and* $B = (H + S_0)^{-1}S_0$*, so*

$$RH + B = I, \qquad \textit{i.e.} \qquad R = (I - B)H^{-1}:$$

*the response is the unregularised one times the share not taken by the anchor (*$HR + B = I$ *as well where* $S_0$ *commutes with* $H$*, as for the family; not for a prior fixed on some coordinates or a projected prior with an active face). On every mode, whatever* $S_0$*,* $h_j r_{jj} + b_{jj} = 1$ *(*$r_{jj} := v_j^\top R v_j$*,* $b_{jj} := v_j^\top B v_j$*): a vanishing bias forces the response* $1/h_j$*. So no such prior both removes the leak's bias and keeps its bound on the response to persistent forces — the trade binding on the modes with* $h_j < \sigma_0$*, where* $1/h_j$ *exceeds the fixed prior's* $1/\sigma_0$ *— and a remedy for the robustness loss of Proposition D‴.30(iii) has to change the force.*

(iii) (least-squares forces) *For $H \succ 0$ and every $S$ with $S+S^\top \succeq 0$, $F^\top(H+S)^{-\top}H(H+S)^{-1}F \le F^\top H^{-1}F = \min\{\|g\|_\rho^2 : \langle \hat J^\top g\rangle = F\}$, with equality at $S=0$: the function-level stationary response to a persistent force is at most the squared norm of the smallest target perturbation that exerts it ($\langle \hat J^\top g\rangle$ is the learning force of the target $f+g$), and the adaptive prior attains the bound. A least-squares force $\langle \hat J^\top g\rangle$ therefore moves the learned function by at most $\|g\|_\rho$ under every prior — under the adaptive prior by exactly $Pg$, so that the stationary point's projected residual $P(g-\hat J y)$ vanishes.*

(iv) (the desired regressor) *Let the adaptive law evaluate its regressor at the reference, $\dot\vartheta_i = G\nabla_\vartheta((y-x_m)^\top\tilde\Phi(x_m;\vartheta_i)) - (\sigma_0/k_e)(\vartheta_i - z_{\pi(i)})$ (DR), with the compensation $\hat f_\Theta(y)$ as before or also at the reference, $u = \dot x_m - k_e(y-x_m) - \hat f_\Theta(x_m)$ (DR*, the desired compensation adaptation law [96]). Then (a) without noise the lift $\Gamma_w$ is a solution and the linearisation at it is Corollary D‴.26's — for DR exactly (the regressor's variation multiplies $e \equiv 0$), for DR* with $-(k_e - a(t))$ in place of $-k_e$ in the error channel — and the averaged loop is unchanged at leading order in $1/k_e$ (the regressor moves by $x - x_m = O(1/k_e)$); (b) to first order in $\nu$, DR is Proposition D‴.30(ii)'s linearised loop driven by the same noise, path by path; DR* has the noise input $-k_e\nu$ in place of $-(k_e+a)\nu$, and the same floor up to a relative $O(1/k_e)$; (c) to second order the part of the force carried by the noise's own second moments (at fixed parameters) is the least-squares force $\langle \hat J^\top g\rangle$ (slow time) of $g := k_e\mathbb{E}[e]$, the rescaled mean tracking error at the lift; for an Ornstein–Uhlenbeck noise of rate $B$, quasi-statically along the orbit, $g = -\frac12\partial_x^2\hat f\cdot\mathbb{E}\nu^2(B-k_e-2a)/(k_e+B)$ for DR and $g = \frac12\partial_x^2 f\cdot\mathbb{E}\nu^2k_e^3/((k_e-a)^2(k_e-a+B))$ for DR* ($k_e > \max a$), so $|g| \le \frac12|\partial_x^2 f|\mathbb{E}\nu^2(1+O(|a|/k_e))$ for all $k_e$ and $B$. By (iii) its function-level response is at most $\|g\|_\rho^2 \le \frac14\|\partial_x^2 f\|_\rho^2(\mathbb{E}\nu^2)^2(1+O(1/k_e))$ under every prior — uniformly in $k_e$, in $B$ and in the spectrum of $H$ — and under the adaptive prior it leaves no projected mean tracking error. The parameters' first-order fluctuation adds a second-order force of the same order in $N_0$: in the error equation the compensation's curvature $\frac12 D_\Theta^2\hat f[\Sigma_{\Theta\Theta}]$ and, for MR and DR, its slope–parameter term (DR*'s compensation, at $x_m$, has none) — a least-squares force again, of a perturbation set by the parameters' covariance; the curvature also shifts the mean function, and the loop removes its tangent part —, in the learning law the correlation $\nabla_\vartheta^2\tilde\Phi\mathbb{E}[\varepsilon_m\delta\vartheta]$ of the regressor's parameter dependence with the measured error — an errors-in-variables term in the parameters, which the reference in the regressor does not remove and which is not the learning force of a mean tracking error, so no bound in the noise's moments applies to it. Its size is set by the parameters' stationary covariance (mode $j$'s variance $(N_0/2k_e)Q_{jj}(h_j+\eta)/(h_j(h_j+\eta+\sigma_0))$, unbounded as $h_j \to 0$ at $\eta > 0$), so the bound $\|g\|_\rho^2$ is the noise part's, not the total's. With the measured regressor the noise part is $F_1 + \langle \hat J^\top g\rangle$, $F_1 := k_e\langle \hat J'^\top\mathbb{E}[\varepsilon_m^2]\rangle$ — the regressor's errors in variables proper, the learning force of no mean tracking error — and the adaptive prior's stationary point keeps the projected mean tracking error $P\mathbb{E}[e] = -\hat J H^{-1}F_1/k_e$: what grows like $k_e^2$ while $k_e \ll B$ is $k_e$ times it, of squared norm $F_1^\top H^{-1}F_1$, the function-level bias; the mean tracking error itself, of squared norm $F_1^\top H^{-1}F_1/k_e^2 \propto (B/(k_e+B))^2$, stays bounded.*

*Hence with the desired regressor the first-order floor still falls like $1/k_e$ while the noise part of the errors-in-variables bias stays bounded, and on the instance the parameters' part's own response falls like $k_e^{-2}$ — a one-dimensional reference, where the law's leading term, the period average of a function of $x_m$ times $\dot x_m$ (a line integral $\oint M\partial_x\hat J\,\mathrm{d}x$ along the orbit), vanishes; on a curve in $\mathbb{R}^n$, $n>1$, it need not: the trade-off in $k_e$ of Proposition D‴.30 disappears, and what is left is the plant's state jitter $\mathbb{E}[e^2] = \mathbb{E}\nu^2(k_e+a)^2/(k_e(k_e+B))$ (DR; DR*: $k_e^2\mathbb{E}\nu^2/((k_e-a)(k_e-a+B))$) — the classical price of a high gain under measurement noise.*

*Proof.* (i) The integrated form is the variation of constants for $\mathrm{d}z_j/\mathrm{d}s = -\eta(z_j - \hat T_j)$; for $\hat T_j$ of bounded variation (it jumps where the plan does, finitely often on bounded intervals), $u := \hat T_j - z_j$ satisfies $\mathrm{d}u = \mathrm{d}\hat T_j - \eta u\,\mathrm{d}s$, whence $u(s) = e^{-\eta s}u(0) + \int_0^s e^{-\eta(s-\tau)}\,\mathrm{d}\hat T_j(\tau)$; and $\vartheta_i - z = (\vartheta_i - \hat T) + u$

for an integral plan (for split rows the targets are $m\sum_j \gamma_{ij} z_j$, and averaging the leak over a class gives $-\sigma_0(\hat{T} - Mz)$). Linearised, $\zeta = \eta(i\omega+\eta)^{-1}y$ on each mode (the prior and the leak are scalars in the whitened weighted coordinates, so they commute with $H$), and $\sigma_0(y-\zeta) = \sigma_0 i\omega(i\omega+\eta)^{-1}y$. (ii) At zero frequency $\zeta = \mathcal{K}(0)y + (I - \mathcal{K}(0))\delta$, so the leak on $y$ is $-\sigma_0(y-\zeta) = -S_0(y-\delta)$, and the stationary equation $Hy + S_0(y-\delta) = F$ gives $y = (H+S_0)^{-1}(F + S_0\delta)$; then $RH + B = (H+S_0)^{-1}(H+S_0) = I$, and on the diagonal in $H$'s eigenbasis $h_j r_{jj} + b_{jj} = 1$. For the family, the prior's stationary equation $\eta(y-\zeta) = \kappa(\zeta - \delta)$ gives $\mathcal{K}(0) = \eta/(\eta+\kappa)$. (iii) By congruence with $H+S$, $(H+S)^{-\top}H(H+S)^{-1} \le H^{-1} \Leftrightarrow H \le (H+S)^{\top}H^{-1}(H+S) = H + S + S^{\top} + S^{\top}H^{-1}S$, true for $S + S^{\top} \succeq 0$. The map $g \mapsto \langle \hat{J}^{\top} g\rangle$ has the adjoint $F \mapsto \hat{J}F$, so the smallest $g$ with $\langle \hat{J}^{\top} g\rangle = F$ is $\hat{J}H^{-1}F$, of squared norm $F^{\top}H^{-1}F$; for $F = \langle \hat{J}^{\top} g\rangle$, $F^{\top}H^{-1}F = \|Pg\|_{\rho}^2 \le \|g\|_{\rho}^2$, and at $S = 0$, $\hat{J}y = \hat{J}H^{-1}\langle \hat{J}^{\top} g\rangle = Pg$. (iv) (a) At $e = 0$ both laws vanish, and their variations, $\delta e \nabla_{\vartheta}\tilde{\Phi}(x;\vartheta) + e(\cdots)$, coincide at $x = x_m$; for DR*, $f(x) - \hat{f}(x_m) = a\delta e - \hat{J}y$ to first order. Off the lift $e$ is slaved, $e = r(x_m)/k_e + O(k_e^{-2})$, so the two regressors differ by $O(1/k_e)$. (b) The regressor's dependence on $y$ enters the law multiplied by $\varepsilon_m$, at second order; for DR* the compensation no longer carries $\nu$. (c) At fixed parameters the mean of $\varepsilon_m \nabla_{\vartheta}\tilde{\Phi}(x_m;\vartheta)$ is $\nabla_{\vartheta}\tilde{\Phi}(x_m)\mathbb{E}[e]$, since $\mathbb{E}\nu = 0$; the parameters' fluctuation adds $\nabla_{\vartheta}^2\tilde{\Phi}(x_m)\mathbb{E}[\varepsilon_m \delta\vartheta]$ in the law and the compensation's curvature and slope–parameter terms in the error equation (Proposition D‴.30's proof). For DR, $\mathbb{E}[e]$ is Proposition D‴.30's (the same compensation); for DR*, $\dot{e} = (a - k_e)e + \frac{1}{2}\partial_x^2 f\, e^2 - k_e\nu + O(3)$, whose linear part has $\mathrm{Var}(e) = k_e^2\mathbb{E}\nu^2/((k_e - a)(k_e - a + B))$ for the OU noise, and the mean balance is $(k_e - a)\mathbb{E}[e] = \frac{1}{2}\partial_x^2 f\,\mathrm{Var}(e)$. The bounds: $|B - k_e - 2a| \le (k_e + B)(1 + 2|a|/(k_e + B))$ and $k_e^3/((k_e - a)^2(k_e - a + B)) \le (k_e/(k_e - a))^3$. With the measured regressor, at the adaptive prior's stationary point $\langle \hat{J}^{\top}(g - \hat{J}y)\rangle + F_1 = 0$ and the mean of the error equation gives $k_e\mathbb{E}[e] = g - \hat{J}y$ at second order, so $k_e P\mathbb{E}[e] = -\hat{J}H^{-1}F_1$. □

**Status.** (i) proved, exactly (every solution of the real or the averaged loop; the differential form for $\hat{T}$ of bounded variation, the integrated one always; the leak's decomposition for integral plans); (ii) proved as $RH + B = I$ and per mode — linear algebra on the averaged linearised block; $HR + B = I$ where $S_0$ commutes with $H$; (iii) proved ($S + S^{\top} \succeq 0$, $H \succ 0$); (iv)(a) proved at the lift and for its linearisation, the leading-order equality of the averaged loops by Theorem J's averaging argument (not rewritten); (iv)(b) the first-order identity proved for the linearised equations, the floors themselves with Proposition D‴.30(ii)'s status; (iv)(c) formal and split — the noise part by Proposition D‴.30(iii)'s quasi-static second-moment computation, its consequences exact for the averaged system; the parameters' part computed numerically (the full second-order mean), not in closed form. Classical: reference regressors and the desired compensation adaptation law ([96]; DR* without Sadegh and Horowitz's nonlinear damping is a local statement), washout filters ([97]), the adaptive law kept near a low-pass filtered copy of itself — Yucelen and Haddad's low-frequency learning, which has the washout structure ([101]) — and elastic averaging's moving-average centre ([90]); the identity, per mode the Tikhonov filter factor plus the residual factor (the Kalman-gain split of a regularised estimate); the bound (iii), the minimum-norm solution with the contraction of the regularised fit; and the split of the noise's force into the regressor's errors in variables proper and the output's, of which only the mean biases; and the parameters' own errors in variables, the classical second-order bias of a constant-gain stochastic approximation from its own fluctuation — the mean field's curvature in the parameters contracted with their stationary covariance, plus the noise–parameter correlation (Benveniste–Métivier–Priouret; for constant-step stochastic gradient descent Dieuleveut–Durmus–Bach, who remove it by Richardson–Romberg extrapolation; the curvature bias of nonlinear least squares, Box's, the analogue of $\frac{1}{2}D_{\Theta}^2\hat{f}[\Sigma_{\Theta\Theta}]$: [100]). New are their transfer to the transport leaks — the washout reading of the adaptive prior in the mass-constrained, mean-field setting, uniform in $m$ — and the instance's quantification, with the observation that the desired regressor does not remove the parameters' part. Checked: `tests/test_eiv_remedy.py` (the washout identity along a real-loop solution at finite $\eta$, the barycentre's velocity integrated alongside, and on the averaged loop across a plan switch, the jump's share large; the collective block's transfer function against the washout formula, $RH + B = I$ along the prior family and, $R$ and $B$ from their dynamic blocks, for priors whose

zero-frequency leak does not commute with $H$ — fixed on the amplitudes, one active face, a matrix anchor gain, a random one with $K + K^\top \succeq 0$ — where $HR + B = I$ fails; the response bound under random leaks $S \geq 0$, the smallest equivalent perturbation and the projection; the lift exact under DR and DR*, on a common noise path DR's run within the second order of MR's, and the regressor's point in the law; DR's and DR*'s noise forces least-squares, their pointwise bounds on $g$, MR's $F_1$ three orders of magnitude above; the quasi-static forces against the full second-order mean's noise part, the parameters' part above 1% for DR and DR* and below $10^{-5}$ for MR).

**Numerics** (`exp_eiv_remedy.py` → `results/chaos/eiv_remedy.md`). The instance of Proposition D‴.30 with an Ornstein–Uhlenbeck noise of rate $B$; MR the measured regressor and compensation, DR the regressor at the reference, DR* both at the reference. (A) The averaged system's biases per unit $(\mathbb{E}\nu^2)^2$. From the noise's own moments, the adaptive prior's function-level bias at $k_e = 25$, $B = 100$ is 2397 (MR), 0.471 (DR) and 0.0526 (DR*); with the parameters' fluctuation (the full second-order mean) 2398, 0.486 (+3.3%) and 0.0481 (−8.7%), the mean function's 2398, 0.476 and 0.0511. Over $k_e = 25$–200 and $B = 100$, 1000 the parameters' part changes MR's bias by at most 0.03%, DR's by −0.75% to +3.3% except at $k_e = B = 100$ — where DR's noise part nearly cancels ($4.5 \cdot 10^{-5}$) and the parameters' part raises it by 56% — and DR*'s by −8.7% to −0.15%; its own response is $3.9 \cdot 10^{-4}$, $1.1 \cdot 10^{-4}$, $2.9 \cdot 10^{-5}$ and $7.4 \cdot 10^{-6}$ at $k_e = 25$, 50, 100, 200 ($B = 100$), like $k_e^{-2}$ — the cancellation of the law's leading term on the one-dimensional reference —, while its share of DR's bias does not fall monotonically. MR over DR (full) is $4.9 \cdot 10^3$ at $k_e = 25$, $B = 100$, and $3.0 \cdot 10^3$–$2.1 \cdot 10^8$ over the grid. MR's bias grows like $(k_e B/(k_e + B))^2$ (×46.6 from $k_e = 25$ to 200 at $B = 1000$, the law ×46.7, $k_e^2$ ×64), while DR's and DR*'s stay at most 1.18, below $\frac{1}{4}\|\partial_x^2 f\|_\rho^2 = 1.31$. DR's noise part attains its bound $\|g\|_\rho^2$ to 0.01% at $k_e = 25$ — $g$ lies almost in the tangent space, the second derivative of a Gaussian bump being in the span of its amplitude and width derivatives. The rescaled projected mean tracking error $k_e^2 \|P\mathbb{E}[e]\|_\rho^2$ under the adaptive prior is 2394 for MR (0 for DR and DR* by construction), the mean tracking error itself 3.83 at $k_e = 25$ and 0.66 at 200 ($B = 100$). The noise level at which the adaptive prior's full bias reaches its floor rises from the std 0.0012 (MR) to 0.085 (DR) and 0.27 (DR*) at $B = 100$ (stds above 0.1 lie outside the small-noise regime). (B) Along the prior family at $\eta = \sigma_0$ ($\kappa/\eta = 0$, 0.1, 1, 10 and the fixed prior; $S_0$ from 0 to $\sigma_0$) the weakest mode's response falls from 3269 to 136 while its bias factor rises from 0 to 0.959, with $h_1 r_1 + b_1 = 1$ and $\max|RH + B - I| \leq 1.1 \cdot 10^{-14}$; the trade binds on the 3 of 9 modes with $h_j < \sigma_0$; MR's errors-in-variables bias falls only from 2397 to 491, and DR's from 0.471 to 0.211 — MR over DR $2.3 \cdot 10^3$–$5.1 \cdot 10^3$ along the family. For priors whose zero-frequency leak does not commute with $H$, $R$ and $B$ again from their dynamic blocks — the prior fixed on the amplitudes or on the centres and adaptive at $\eta = \sigma_0$ elsewhere, one active face of the projected adaptive prior ($\eta = \infty$), the anchor gain $K = \eta P_{\text{amp}}$, a random anchor gain with $K + K^\top \succeq 0$ — the block's $R$ and $B$ equal $(H + S_0)^{-1}$ and $(H + S_0)^{-1} S_0$, $S_0$ by elimination of the prior, to $1.9 \cdot 10^{-15}$: $RH + B = I$ holds to $1.4 \cdot 10^{-14}$ and per mode to $8.9 \cdot 10^{-15}$, while $\max|HR + B - I|$ is 0.47–0.79. (C) The real loop ($k_e = 25$, $B = 100$, std 0.01, $\eta = \infty$, RK4 $\mathrm{d}t = 10^{-3}$, 12 runs of 300 time units, the three laws on common noise paths): at $t = 300$, $\mathbb{E}\|\hat{f} - f\|_\rho^2$ is $2.43 \cdot 10^{-6}$ under MR against $2.80 \cdot 10^{-7}$ under DR and $2.75 \cdot 10^{-7}$ under DR*, a ratio of 8.7 (predicted 9.2); DR and DR* sit at their covariance part, their mean parts 0.07% and 0.008% of it. Against the prediction the $z$-scores run from −2.04 to −0.04 (MR; Proposition D‴.30's runs of the same loop, on other paths, lay 7% above it) and from −1.8 to 0.82 (DR, DR*). The run mean of $\hat{f}_{\text{MR}} - \hat{f}_{\text{DR}}$, in which the first-order responses cancel path by path, equals the predicted $\hat{J}(\bar{y}_{\text{MR}} - \bar{y}_{\text{DR}})$ to a relative $L_\rho^2$ error of 0.5–5.3% at every record, 0.43–1.83 times the run mean's relative sampling error; the averaged prediction carries its own $O(1/k_e)$ error, and the test checks $F_1$ (through MR − DR), not DR's own force, whose mean part is $7 \cdot 10^{-4}$ of its covariance part.

**Proposition D‴.32** (the cohesion leak: a leak on the unexcited subspace, without a prior)**.** *Let* $\psi : [0, \infty) \to [0, 1]$ *be* $C^2$*, non-increasing,* $\psi(0) = 1$ — *of finite range,* $\psi(z) = (1 - z)_+^3$ *(the range*

*R), or Gaussian, $\psi(z)=e^{-z/2}$ (the scale $R$) — and let the particles attract each other instead of leaking toward a prior:*

$$\frac{\mathrm{d}\vartheta_i}{\mathrm{d}s}=G\langle\nabla_\vartheta(r_\Theta\tilde\Phi(\cdot\,;\vartheta_i))\rangle_\rho-\frac{\sigma_0}{m}\sum_j\psi(\|\vartheta_i-\vartheta_j\|^2_{G^{-1}}/R^2)(\vartheta_i-\vartheta_j)\qquad(\mathfrak{A}^{\mathrm{coh}}_m)$$

*(in the real loop* (CL)*, $\sigma_0/k_e$ in place of $\sigma_0$ and the learning law in place of the averaged one).*

(i) (an energy; the network's own modes) $(\mathfrak{A}^{\mathrm{coh}}_m)$ *is the $G$-gradient flow of $J^{\mathrm{coh}}_m(\Theta):=m\bar L(\mu_\Theta)+(\sigma_0 m/2)\iint W\,\mathrm{d}\mu_\Theta\,\mathrm{d}\mu_\Theta$, $W(u):=R^2\omega(\|u\|^2_{G^{-1}}/R^2)$ with $\omega'=\psi/2$ (so $\nabla^2W(0)=G^{-1}$), which decreases along every solution; its mean-field form is the Wasserstein gradient flow (in the metric $G^{-1}$) of $\bar L(\mu)+(\sigma_0/2)\iint W\,\mathrm{d}\mu\,\mathrm{d}\mu$ — an aggregation equation in the external potential $-\langle f\tilde\Phi(\cdot\,;\vartheta)\rangle_\rho$ with two interactions, the fit's $\langle\tilde\Phi(\cdot\,;\vartheta)\tilde\Phi(\cdot\,;\vartheta')\rangle_\rho$ and the cohesion $W$. The leak force on $\vartheta_i$ is $\sigma_0\bar\psi_i(m_i-\vartheta_i)$, $\bar\psi_i:=(1/m)\sum_j\psi_{ij}$ the kernel mass of the cloud at $\vartheta_i$ and $m_i:=\sum_j\psi_{ij}\vartheta_j/\sum_j\psi_{ij}$ the kernel-weighted mean: the mean-shift step of the cloud's own kernel density, so the leak pulls the network toward the modes of its own parameter distribution — no prior, no atom count, no plan, no cells. The self-interaction vanishes ($\psi(0)\cdot 0$), so the empirical measure of $(\mathfrak{A}^{\mathrm{coh}}_m)$ solves the mean-field equation exactly, and Theorem D's finite-horizon propagation of chaos holds (the velocity field is Lipschitz in $\vartheta$ and in $\mu$ for $W_1$, Dobrushin's estimate). With the Gaussian kernel $J^{\mathrm{coh}}_m$ is real-analytic (the saturated network is), so every bounded solution converges to a single stationary point (Łojasiewicz's theorem, as for Proposition D‴.20). The forces sum to zero: the leak is internal and gives no a-priori bound.*

(ii) (no bias at any clustered configuration) *Let $\Theta$ be replicated by $K$ classes with atoms $\vartheta_1,\dots,\vartheta_K$ and masses $w_k=n_k/m$. With the finite range and the atoms pairwise at least $R$ apart, the leak vanishes on $\Theta$, and $\Theta$ is stationary iff $(\vartheta_k)$ is a critical point of the unregularised weighted loss $L_w:=\frac12\langle|f-\sum_k w_k\tilde\Phi(\cdot\,;\vartheta_k)|^2\rangle_\rho$ — for every $K$ and every $w$, with no prior and no $K_0$ fixed in advance; in particular the lift of an exact representation $\vartheta^e(w)$ is stationary at every representable sample ($w\in W$, Proposition D‴.24), as under the adaptive prior (Theorem D‴.23(iii)). With the Gaussian kernel the classes attract each other with the forces $-\sigma_0 w_l\psi(d^2_{kl}/R^2)(\vartheta_k-\vartheta_l)$ ($d_{kl}$ the atoms' distance), and at a nondegenerate critical point ($\tilde A_w\succ 0$, the whitened weighted Hessian of $L_w$) the stationary point continues to one at the whitened distance $\tilde A_w^{-1}b+O(|b|^2)$ from it, $b$ the whitened inter-class force: a bias at most of the order $\sigma_0\|\tilde A_w^{-1}\|\max_k\sum_l w_l d_{kl}e^{-d^2_{kl}/(2R^2)}$, exponentially small in $(\min d_{kl}/R)^2$. No bias is the finite range's property; the Gaussian kernel pays this small bias for the merging of (v).*

(iii) (the leak acts exactly on the unexcited subspace) *At a configuration as in (ii) (the finite range), in the whitened weighted coordinates, the linearisation of $(\mathfrak{A}^{\mathrm{coh}}_m)$ is block diagonal: on the separations of class $k$ ($\sum_{i\in k}u_i=0$) it is $-(\sigma_0 w_k I-R^G_k)$, with multiplicity $n_k-1$, $R^G_k:=G^{1/2}\langle r\nabla^2_\vartheta\tilde\Phi(\cdot\,;\vartheta_k)\rangle_\rho G^{1/2}$ the residual curvature; on the class means it is $-\tilde A_w$, the linearised unregularised weighted $K$-atom flow, the leak absent. The separations are the directions in which the plant does not see the network ($D\hat f=0$ on them), and where $H_L(w)\succ 0$ they are all of them: the leak acts exactly on the unexcited subspace. At an exact representation the separations of class $k$ decay at $\sigma_0 w_k$ and the collective modes at $h_j(w)$ (the eigenvalues of $H_L(w)$), for every $w\in W$; the class means' stationary response to a persistent force $F$ is $\tilde A_w^{-1}F$ — the adaptive prior's (Proposition D‴.31(i)–(iii)): no leak opposes it. The restoring force is linear only near the class: under the compact kernel the pair force $\psi(d^2/R^2)d$ is largest at $d=R/\sqrt7$ and vanishes at $R$, so a separation pushed beyond $R$ is not restored.*

(iv) (splitting, and no over-splitting) *The splitting margin of class $k$ is $\sigma_0 w_k-\lambda_{\max}(R^G_k)$. For a scalar amplitude per kernel at the gauge transform $\vartheta^b(w)=G_w\vartheta^b(q)$ of a nondegenerate*

*critical point $\vartheta^b(q)$ (Proposition D‴.28) it is $\sigma_0 w_k - (q_k/w_k)\lambda_k^+$, so a class can split iff $\sigma_0 w_k^2 < q_k\lambda_k^+$ — the leak's strength sets the resolution at which a class becomes unstable where the residual's curvature asks for a kernel, light classes first; at an exact representation $\lambda^+ = 0$ and no class splits, whatever $\sigma_0 > 0$: an exact representation is never over-split (as linear stability). The unstable separations of a class are $(n_k - 1)$-fold degenerate, so the linear theory does not choose which particles leave; the kernel's concavity ($\omega'' = \psi'/2 < 0$) makes shedding single particles the cheapest split (Numerics (G)).*

(v) (twins and dead classes under the Gaussian kernel) *At a degenerate exact representation — twins (two classes with the same centre and width) or a dead class ($a_k = 0$) — the fit is neutral along a curve (the transfer of amplitude between the twins at fixed $w_k a_k + w_l a_l$; the dead class's centre and width), and along it only the cohesion acts. For twins with no third class within the kernel's reach, the fit relaxed quasi-statically along the curve, the twins' amplitude difference $\Delta\alpha$ contracts at the rate*

$$\sigma_0\psi(\Delta\alpha^2/(G_\alpha R^2)) \cdot \frac{(w_k a_k' + w_l a_l')^2}{w_k a_k'^2 + w_l a_l'^2}, \qquad a' := \mathrm{d}a/\,\mathrm{d}\alpha,$$

*so they merge, for unsaturated twins on the time scale $G_\alpha R^2 e^{\Delta\alpha^2/(2G_\alpha R^2)}/(\sigma_0(w_k + w_l)\Delta\alpha^2)$ ($\Delta\alpha^2 \gg G_\alpha R^2$). A third class within reach of one twin can win: twins need not merge, and a degenerate exact representation can be a saddle of $J_m^{\mathrm{coh}}$ that the flow leaves toward a biased configuration (Numerics (I)). A dead class moves under the attraction of the live ones, a Gaussian mixture in its coordinates whose weights carry the live classes' amplitude coordinates ($w_l e^{-\alpha_l^2/(2G_\alpha R^2)}$); when the components are farther apart than a few $R$ its local minima lie near them, where it becomes a twin of a live class — on the time scale $R^2 e^{d^2/(2R^2)}/(\sigma_0 w d^2)$, $d \gg R$ its distance from the nearest live class and $w$ that class's mass. With the finite range, dead and duplicated classes farther than $R$ from the others persist.*

(vi) (the real loop) *In* (CL) *with the cohesion leak (finite range) the lift of an exact representation $\vartheta^e(w)$ by classes pairwise at least $R$ apart is a $T_0$-periodic solution with $e \equiv 0$ at every representable sample; its Floquet exponents are $\sigma_0 w_k/k_e$ exactly on the separations of class $k$ (multiplicity $(n_k - 1)p$) and, on the collapsed subspace, those of the unregularised weighted $K$-atom loop, which are Corollary D‴.26's at $\eta = \infty$ — $(1/k_e)[h_j(w) + c_j(w)/k_e^2 + O(k_e^{-4})]$ under the derivative condition (Lemma D‴.26a). So perfect tracking is exponentially stable (locally, for each $m$) at the rate $(1/k_e)\min\{\sigma_0 \min_k w_k,\ h_1(w) + c_1(w)/k_e^2 + \dots\}$. The replicated configuration is invariant under a measurement noise common to the particles, and Proposition D‴.30(ii)–(iii) hold at $\eta = \infty$ (the floor $(N_0/2k_e)[\mathrm{tr}\,Q + \mathrm{tr}\,H^2]$, the errors-in-variables response $H^{-1}$), the separations' noise threshold scaled by $w_k$ in the variance ($\sqrt{w_k}$ in the standard deviation) — the softest of the leaks; with the desired regressor the regressor's own term leaves their noise drift, whose threshold is then bounded below uniformly in $k_e$ (Corollary D‴.33). Under a dither $\tau$ (the averaged loop with $\sqrt{2\tau G}$ per particle) the separations of class $k$ are an Ornstein–Uhlenbeck process of stiffness $\sigma_0 w_k$: the particles' stationary variance about the class mean is $(1 - 1/n_k)\tau/(\sigma_0 w_k)$ per whitened coordinate, against $(1 - 1/n_k)\tau/\sigma_0$ under a leak toward a prior and the free diffusion $2\tau s$ without a leak.*

*Proof.* (i) $W$ is even, so

$$\partial_{\vartheta_i}[(\sigma_0/(2m))\sum_{j,l} W(\vartheta_j - \vartheta_l)] = (\sigma_0/m)\sum_j \nabla W(\vartheta_i - \vartheta_j),$$

and $G\nabla W(u) = R^2\omega'(z)\cdot 2u/R^2 = \psi(z)u$ ($z = \|u\|^2_{G^{-1}}/R^2$); with $\partial_{\vartheta_i}(m\bar{L}) = -\langle r\nabla_\vartheta\tilde{\Phi}(\cdot\,;\vartheta_i)\rangle_\rho$ the right-hand side of ($\mathfrak{A}_m^{\mathrm{coh}}$) is $-G\nabla_{\vartheta_i}J_m^{\mathrm{coh}}$, and $\mathrm{d}J_m^{\mathrm{coh}}/\,\mathrm{d}s = -\sum_i\|\nabla_{\vartheta_i}J_m^{\mathrm{coh}}\|_G^2 \le 0$. $\sum_j \psi_{ij}(\vartheta_i - \vartheta_j) = m\bar{\psi}_i(\vartheta_i - m_i)$, and $\sum_i\sum_j \psi_{ij}(\vartheta_i - \vartheta_j) = 0$ by antisymmetry. The mean-field form

replaces $(1/m)\sum_j$ by the integral against $\mu$: $\bar L(\mu)=\frac12\langle f^2\rangle_\rho-\int\langle f\tilde\Phi\rangle_\rho\,\mathrm d\mu+\frac12\iint\langle\tilde\Phi\tilde\Phi'\rangle_\rho\,\mathrm d\mu\,\mathrm d\mu$, the velocity is $v[\mu](\vartheta)=G\nabla_\vartheta\langle r_\mu\tilde\Phi(\cdot\,;\vartheta)\rangle_\rho-\sigma_0\int\psi(\|\vartheta-\vartheta'\|^2_{G^{-1}}/R^2)(\vartheta-\vartheta')\,\mathrm d\mu(\vartheta')=-G\nabla_\vartheta(\delta\mathcal F/\delta\mu)$, $\mathcal F:=\bar L+(\sigma_0/2)\iint W\,\mathrm d\mu\,\mathrm d\mu$, and on $\mu_\Theta$ it is the right-hand side of $(\mathfrak A_m^{\mathrm{coh}})$ at the particles, the term $j=i$ vanishing. (ii) Inside a class $\vartheta_i-\vartheta_j=0$; across classes $\psi=0$ at distances $\ge R$. The learning drift of a particle of class $k$ is $G\langle r_w\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_k)\rangle_\rho=-Gw_k^{-1}\nabla_{\vartheta_k}L_w$, which vanishes for all $k$ iff $(\vartheta_k)$ is critical for $L_w$. With the Gaussian kernel the class drift is $-Gw_k^{-1}\nabla_{\vartheta_k}L_w+b_k$, $b_k:=-\sigma_0\sum_l w_l\psi(d_{kl}^2/R^2)(\vartheta_k-\vartheta_l)$; in the whitened weighted coordinates its linearisation at the critical point is $-\tilde A_w$ (plus the derivative of $b$, of the order of $b/R$), and the implicit function theorem gives the continuation with the stated first-order displacement and the bound. (iii) A separation $u$ ($\sum_{i\in k}u_i=0$, the other classes fixed) leaves $\hat f$ unchanged at first order — $D\hat f[u]=(w_k/n_k)\sum_{i\in k}\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_k)u_i=0$ — so the residual is unchanged and the learning drift varies by $GR_ku_i$; the leak's variation is $-(\sigma_0/m)\sum_{j\in k}\psi(0)(u_i-u_j)=-\sigma_0w_ku_i$ ($\psi'$ terms vanish at coincident particles, and across classes $\psi$ and $\psi'$ vanish at distances $\ge R$). A perturbation of the class means moves the leak not at all (inside a class all particles move together; across classes $\psi=\psi'=0$), and the learning drift's variation is the unregularised $K$-atom flow's; the two subspaces do not mix (the learning drift's cross terms vanish as in Proposition D‴.5(i)). Whitening, $y=\sqrt{w_k}G^{-1/2}\cdot$, gives the blocks; $D\hat f$ restricted to the class means is the weighted Jacobian whose Gram matrix is $H_L(w)$, so where $H_L(w)\succ0$ the kernel of $D\hat f$ is exactly the separations. At an exact representation $r=0$, $R_k=0$ and $\tilde A_w=H_L(w)$. The stationary response follows from the linearisation; $\frac{\mathrm d}{\mathrm dd}[(1-d^2/R^2)^3d]=(1-d^2/R^2)^2(1-7d^2/R^2)$. (iv) Proposition D‴.28(iii): $R_k(\vartheta^b(w))=(q_k/w_k)R_k(\vartheta^b(q))$; the separations of class $k$ are $u_i=c_iv$ with $\sum c_i=0$ along each eigenvector $v$, all with the same linear rate. (v) Twins: the network depends on their amplitudes only through $p=w_ka(\alpha_k)+w_la(\alpha_l)$, and their other coordinates coincide; with $u:=\langle r\varphi\rangle_\rho$ common to both, $\dot\alpha_k=G_\alpha a_k'u-\sigma_0w_l\psi\Delta\alpha$ and $\dot\alpha_l=G_\alpha a_l'u+\sigma_0w_k\psi\Delta\alpha$ ($\psi:=\psi(\Delta\alpha^2/(G_\alpha R^2))$; no third class within reach), so $\dot p=G_\alpha u(w_ka_k'^2+w_la_l'^2)-\sigma_0\psi\Delta\alpha w_kw_l(a_k'-a_l')$. The fit relaxes $p$ at the rate $G_\alpha\langle\varphi^2\rangle_\rho(w_ka_k'^2+w_la_l'^2)$, fast against $\sigma_0\psi$, so to leading order $\dot p=0$ fixes $u$, and $\mathrm d(\Delta\alpha)/\mathrm ds=G_\alpha u(a_k'-a_l')-\sigma_0\psi(w_k+w_l)\Delta\alpha=-\sigma_0\psi\Delta\alpha(w_ka_k'+w_la_l')^2/(w_ka_k'^2+w_la_l'^2)$; for unsaturated twins $z:=\Delta\alpha^2/(2G_\alpha R^2)$ obeys $\mathrm dz/\mathrm ds=-2\sigma_0(w_k+w_l)ze^{-z}$, so the time from $z_0\gg1$ is $e^{z_0}/(2\sigma_0(w_k+w_l)z_0)(1+O(1/z_0))$. A third class at the distance $d$ from one twin adds to that twin a force of the order $\sigma_0w\psi(d^2/R^2)d$, which wins when $d$ is below the twins' own distance. A dead class: $L_w$ does not depend on its centre and width, so $J_m^{\mathrm{coh}}$ varies there only through its attraction to the live classes, a Gaussian mixture in those coordinates (its amplitude coordinate fixed, the live classes' entering the weights); a mixture's modes away from its components need overlapping components, so for components farther apart than a few $R$ its local minima lie within $O(Re^{-d^2/(2R^2)})$ of the components, and the descent toward the nearest has the rate $\sigma_0w\psi(d^2/R^2)$: $z:=d^2/(2R^2)$ obeys $\mathrm dz/\mathrm ds\approx-2\sigma_0wze^{-z}$, whence the time scale as for twins. (vi) As Corollary D‴.26: along the lift $e\equiv0$, the separations do not move $\hat f$ at first order and the second-derivative term of the learning law vanishes with $e$, so a separation stays a separation with $\delta\dot\vartheta_i=-(\sigma_0w_k/k_e)\delta\vartheta_i$; the collapsed subspace is invariant and on it the leak vanishes, which is the $\eta=\infty$ adaptive prior's collapsed variational equation; Lemma D‴.26a applies to it. The noise: Proposition D‴.30(i) needs only that every particle of a class sees the same signals, and its (ii)–(iii) only the collapsed linearisation; the separations' drift there is opposed by $\sigma_0w_k$ instead of $\sigma_0$. The dither: the offsets from the class mean are forced by $(1-1/n_k)\cdot2\tau$ per whitened coordinate. □

**Status.** (i)–(iii) proved, exactly; (iv) proved for the stationary points and their linearisation, the split's pattern explained (the degeneracy and the kernel's concavity), its dynamics numerical; (v) the twins' contraction rate proved to leading order in $\sigma_0\psi$ against the fit's relaxation (checked to 0.04%, Numerics (I)), their merger only without a third class within reach (a counterexample, Numerics (I)); the dead class's motion and time scale stated, its merger not proved; (vi) the lift and its separation exponents exact, the collapsed exponents by Corollary D‴.26 and Lemma D‴.26a, the noise statements as in Proposition

D‴.30 (formal where it is). Not proved: the transit from a spread cloud — which classes form and with which masses — and with it any selection of the number of kernels: the count is the transit's and the schedule's. Under an annealed Gaussian scale it ended at the target's count on the instance and on well-separated targets whose transit masses were representable, not for bumps closer than the broad scale or for unrepresentable transit masses, and at a fixed scale not at all (Numerics (C), (H), (J), (K)); the replicator weight law of Proposition D‴.34 removes the mass obstruction. Nor is the basin uniform in $m$ proved. Checked: `tests/test_cohesion_leak.py` (the drift equal to $-G\nabla J_m^{\mathrm{coh}}$ by central differences and $J_m^{\mathrm{coh}}$ decreasing along the flow, for both kernels; the forces summing to zero and equal to $\sigma_0\times$ the kernel mass $\times$ the mean-shift vector; at the exact representation for the counts $(3,2,2)$ and $(4,3,3)$ the drift zero and the Jacobian's spectrum the predicted one — the separations at $-\sigma_0 w_k$, the class means the unregularised $K$-atom flow's — to $10^{-9}$; at the best one-atom fit the top separation eigenvalue $\lambda^+ - \sigma_0 w$, stable at $\sigma_0$ and split at $\lambda^+/2$; twins contracting at the predicted rate to 1% under the Gaussian kernel and persisting beyond the compact range; the kernels' range and scale (a pair at $0.99R$ attracted, at $1.01R$ not; the Gaussian force at $R\,e^{-1/2}$ of its linear part); the clusters of known configurations; the real loop's lift $T_0$-periodic and a separation contracting by $e^{-\sigma_0 w_k T_0/k_e}$ per period to $10^{-6}$).

**Numerics** (`exp_cohesion_leak.py` → `results/chaos/cohesion_leak.md`). The three-bump instance ($\sigma_0 = 0.15\sigma_{\mathrm{crit}}$), the averaged slow loop, the midpoint rule ($\mathrm{d}t = 0.25$), single-linkage clusters at 0.25. (A) At each of the 15 count vectors of $m = 15$ in $W$ the lift of the exact representation $\vartheta^e(w)$ is stationary — the largest drift $2.8\cdot10^{-17}$ (compact kernel, $R = 1$) and $6.2\cdot10^{-10}$ (Gaussian, $R = 1$) — and the Jacobian's spectrum is (iii)'s to $2.8\cdot10^{-11}$ (the finite differences' error) and to $4.6\cdot10^{-9}$ (the Gaussian kernel's cross-class terms, which (iii) omits): the separations at $\sigma_0 w_k$ ($1.41\cdot10^{-3}$–$3.77\cdot10^{-3}$), the collective modes the unregularised flow's (the slowest $2.2\cdot10^{-4}$–$4.2\cdot10^{-4}$), the atoms at least 5.80 apart. The Gaussian kernel's bias (Newton on the class drift, counts $(5,5,5)$) is $2.1\cdot10^{-8}$ (whitened; the fit error $6.6\cdot10^{-10}$) at $R = 1$, $1.3\cdot10^{-3}$ ($3.9\cdot10^{-5}$) at $R = 1.5$ and 0.068 ($2.1\cdot10^{-3}$) at $R = 2$, the first order $\tilde{A}_w^{-1}b$ to $6\cdot10^{-8}$, $9\cdot10^{-4}$ and 1.0% — and 56% off at $R = 3$, where the classes overlap. With the fourth bump ($\varepsilon = 0.3$; the three-atom approximant $\vartheta^b(q)$, the fit error 0.017, $\lambda_k^+ = 1.00\cdot10^{-4}$, $2.86\cdot10^{-4}$, $1.08\cdot10^{-4}$) the predicted margins $\sigma_0 w_k - (q_k/w_k)\lambda_k^+$ are the Jacobian's to $2\cdot10^{-11}$ (the finite differences' error) at three count vectors. At $\sigma_0/8$ the counts $(6,4,5)$ have one negative margin, $-1.22\cdot10^{-4}$, and that class splits, shedding one particle: its diameter grows from $2.2\cdot10^{-3}$ to 1.0 over $10^5$ slow units, the clusters $3 \to 4$ (the masses 0.200 and 0.067), the fit error $0.017 \to 0.0148$. At $(5,5,5)$ the smallest margin, $+8.7\cdot10^{-6}$, makes that class's diameter decay at $8.8\cdot10^{-6}$ ($6.9\cdot10^{-4} \to 3.4\cdot10^{-4}$ from $s = 2\cdot10^4$ to $10^5$); at $\sigma_0$ every class collapses to rounding. Twins (bump 3 carried by classes of masses 3/15 and 2/15, the weighted amplitude shared $3:1$, 2.31 apart) merge under the Gaussian kernel ($R = 1$: their distance $6.1\cdot10^{-3}$ at $s = 4000$, 0 by $s = 18000$, the fit error $4.5\cdot10^{-10}$), on (v)'s time scale ($\sim10^3$), and persist beyond the compact range (2.31 throughout). A dead class (amplitude 0, mass 2/15, at 5.19 from the nearest live class, of mass 4/15) joins it under the Gaussian kernel at $R = 2$ (to rounding by $s = 16000$; the fit error then $1.6\cdot10^{-3}$, the inter-class bias at that scale) and at $R = 1.5$ (at $s \approx 2.8\cdot10^4$, (v)'s time scale $1.8\cdot10^4$; the fit error $1.7\cdot10^{-5}$); at $R = 1$, where the live classes do not overlap and the time scale is $1.4\cdot10^7$, it does not move in $4\cdot10^4$ slow units, nor under the compact kernel. (I) The twins' contraction: along (A)'s twins at $R = 1$ over $s \in [200, 2000]$, $\ln(\Delta\alpha(200)/\Delta\alpha(2000)) = 1.2949$ against (v)'s rate integrated along the run, 1.2943 (0.04%). Twins far apart do not merge on the runs' time scale: with bump 3's amplitude shared $0.9:0.1$ (5.28 apart) and $0.97:0.03$ (8.19) their distance does not change in $4\cdot10^4$ slow units ((v)'s rate at their distance $1.3\cdot10^{-9}$ and $3\cdot10^{-18}$; $1.45\cdot10^{-4}$ for (A)'s twins). A third class within reach wins: opposite-sign twins of bump 3 (counts $(5,3,5,2)$, amplitudes $+3.8$ and $-3.5$, 12.7 apart), the light twin at the amplitude coordinate of the class of bump 2 (3.63 from it), jittered by $10^{-3}$, do not merge in $4\cdot10^5$ slow units (their distance 11.5–12.9); the light twin is drawn toward the class of bump 2 (its centre $1.5 \to 0.88$) and the fit error grows from $3.2\cdot10^{-4}$ to

$4.5 \cdot 10^{-3}$, a biased four-class configuration; the annealed schedule does the same (11.7 apart at the end, the fit error $5.0 \cdot 10^{-3}$). (B) From Proposition D‴.24's warm clouds ($m = 15$, 60, 240 against $M = 510$, four seeds, 6000 slow units) both kernels remove the floor as the adaptive prior does, without a prior: $\|\hat{f}_m - \hat{f}_M\|^2$ at $s = 6000$ is $1.9 \cdot 10^{-10}$–$5.4 \cdot 10^{-8}$ at $m = 15$ over the three representable samples, under both kernels and the adaptive prior alike (the fixed prior's floor $3.0 \cdot 10^{-4}$–$2.4 \cdot 10^{-3}$), and $2.4 \cdot 10^{-10}$–$7.0 \cdot 10^{-10}$ and $5.1 \cdot 10^{-14}$–$2.2 \cdot 10^{-9}$ at $m = 60$ and 240 (the adaptive prior's $2.4 \cdot 10^{-10}$–$6.2 \cdot 10^{-10}$, $7.2 \cdot 10^{-14}$–$2.2 \cdot 10^{-9}$; the fixed prior's floor $1.1 \cdot 10^{-4}$–$9.9 \cdot 10^{-4}$, $4.9 \cdot 10^{-6}$–$8.2 \cdot 10^{-5}$); the class means lie within 0.012, 0.0010 and 0.0045 (whitened) of $\vartheta^e(w^m)$. The sample of seed 3 at $m = 15$, the counts $(8, 4, 3)$, lies on the boundary of $W$ and is not representable: there every leak but the fixed prior ends at $7.0 \cdot 10^{-5}$–$7.1 \cdot 10^{-5}$ (the fit error 0.013; the fixed prior $5.5 \cdot 10^{-3}$ and 0.119) — no leak repairs an unrepresentable sample —, and the compact kernel's run ends with a fourth class (a class diameter of 0.81). (C) From broad starts ($m = 60$, six seeds; $m = 120$, three) the annealed Gaussian cohesion leak ($R = 2$ for 20000 slow units, then $R = 1$ for 20000) ends at three classes with the exact representation's centres and widths in all nine runs, the fit error $1.8 \cdot 10^{-7}$–$2.3 \cdot 10^{-6}$, the masses the transit's (0.217–0.500). The nearest-atom adaptive prior told the right count ($K = 3$) does as well ($4.2 \cdot 10^{-9}$–$2.1 \cdot 10^{-6}$); told $K = 2$ it ends at three biased classes (0.131–0.191), told $K = 4$ at four, one bump split ($5.4 \cdot 10^{-4}$–$5.6 \cdot 10^{-3}$); the $\sigma$-modification ends at three biased classes (one run four; 0.24), no leak at 40–48 classes ($8.2 \cdot 10^{-4}$–$1.6 \cdot 10^{-3}$), and the compact kernel at $R = 1$ without the broad stage at 17–22 ($6.0 \cdot 10^{-4}$–$1.5 \cdot 10^{-3}$): the transit's fragments lie farther apart than $R$, where the leak no longer acts. (J) The schedule, not the kernel, gives the count: at a fixed Gaussian scale the same broad starts end at 4–6 classes ($R = 1$; the three bumps' classes and light fragments of 0.017–0.083, the fit error $4.8 \cdot 10^{-4}$–$1.6 \cdot 10^{-3}$, six seeds) or at three biased classes ($R = 2$; $7.8 \cdot 10^{-4}$–$1.4 \cdot 10^{-3}$, the order of (A)'s bias at that scale, three seeds); the annealed schedules $R = 1.5 \to 1$, $2 \to 0.5$ and $3 \to 1$ (seed 0) end at three classes, $2.4 \cdot 10^{-8}$–$5.9 \cdot 10^{-6}$. (D) The real loop ($k_e = 25$): the lift of $\vartheta^e(w)$, $w = (6, 5, 4)/15$, returns after a period to $7.8 \cdot 10^{-16}$, and its 45 slow Floquet exponents — 36 separations at $\sigma_0 w_k / k_e = 7.55 \cdot 10^{-5}$, $9.43 \cdot 10^{-5}$, $1.13 \cdot 10^{-4}$ and 9 collective — are the predicted ones to $5.5 \cdot 10^{-6}$ (relative; median $1.5 \cdot 10^{-7}$; the monodromy's finite differences' error). From the warm clouds (seed 0, 3000 time units) the rms tracking error falls to $5.8 \cdot 10^{-4}$ ($m = 15$) and $2.4 \cdot 10^{-4}$ ($m = 60$) under the cohesion leak, against $1.3 \cdot 10^{-3}$ and $3.9 \cdot 10^{-4}$ under the fixed prior. (E) Under the dither $\tau = 10^{-5}$ ($m = 60$ from the lift of $\vartheta^e(q)$, three seeds, 20000 slow units) the within-class spread per whitened coordinate, averaged over the second half, is 0.068 (compact) and 0.063 (Gaussian) against (vi)'s $\sqrt{(1 - 1/n_k)\tau/(\sigma_0 w_k)} = 0.0635$ ($n_k = 20$) — the compact kernel softening at that spread, $\psi = (1 - z)^3 \approx 1 - 3z$ —, 0.036 under the fixed prior ($\sqrt{(1 - 1/n_k)\tau/\sigma_0} = 0.0366$) and 0.030 under the $\sigma$-modification, while without a leak it diffuses (0.44–0.56 at $s = 20000$); the class means' whitened deviation is 0.020 (the fixed prior's 0.0085), the fit error $1.9 \cdot 10^{-3}$ ($1.5 \cdot 10^{-3}$). (F) On the cogging servo of Proposition D‴.29 ($\sigma_0 = 0.01$, $m = 64$, broad starts, three seeds; the Gaussian kernel at $R = 0.7$, then 0.35), whose torque no finite mixture represents exactly, the cohesion leak settles on seven or eight classes inside the saturation ($|a|/a_{\max} \le 0.84$) with the fit error 0.14%–0.35% after 80000 slow units — below the four-kernel designs of Proposition D‴.29 (1.46% inside, 0.64% at the saturation), above the best equal-weight six- and eight-kernel fits ($9.4 \cdot 10^{-6}$ and $3.3 \cdot 10^{-7}$, both saturated or nearly): the count is the transit's, not the best configuration of that count, and its smallest classes are single particles. (G) The resolution (the fourth bump, $m = 30$ from the jittered lift of $\vartheta^b(q)$ with the counts $(10, 10, 10)$, the compact kernel at $R = 1$, $\sigma$ halved every $10^5$ slow units from $\sigma_0$ to $\sigma_0/64$ with a jitter $10^{-3}$ renewed at each stage, three seeds). At $w = q$, (iv) puts the first split in the middle class at $\sigma = 3\lambda_2^+ = 0.121\sigma_0$: down to $\sigma_0/8$ the three classes stay, the fit error 0.0170, the middle class's margin there $-2.0 \cdot 10^{-6}$ to $+9.3 \cdot 10^{-7}$ ($8.7 \cdot 10^{-6}$ without the jitter, (A)); at $\sigma_0/16$ it is $-1.24 \cdot 10^{-4}$ to $-1.40 \cdot 10^{-4}$ and the class splits in every seed. Over the 21 stages no class with a positive margin split, every class of two or more particles

whose margin was below $-5\cdot10^{-5}$ split within its stage (8 of 8), and of those between $-5\cdot10^{-5}$ and 0 — where the jitter grows by $e^{|\text{margin}|\cdot10^5}=1.2$–$140$ over a stage — 3 of 7; a class counts as split when a piece passes the clustering threshold 0.25 from the jitter $10^{-3}$, a growth by 250 within $10^5$ slow units, i.e. a margin below $-\ln(250)/10^5=-5.5\cdot10^{-5}$, which is where the tallies change. The split is not a halving: the unstable class sheds one to three single particles — (iv)'s degeneracy leaves the pattern to the nonlinear terms, and the kernel's concavity makes shedding the cheapest: for a class of ten split by '$k$ particles at $+d$, $10-k$ at $-kd/(10-k)$' at a fixed $\sum\|u_i\|^2=0.5$ the cohesion energy is 0.43, 0.62, 0.70, 0.74 of its quadratic part for $k=1$, 2, 3, 5 (compact; Gaussian 0.87, 0.93, 0.94, 0.95), the same for all $k$ at $10^{-4}$. A particle of mass 1/30 carries at most $a_{\max}/30=0.13$ of amplitude, less than the fourth bump's 0.3, so the singletons do not represent it; the middle and third classes move toward it (centres 0.19–0.22 and 1.07–1.16 at $\sigma_0/64$) and the singletons fill the flanks they leave (centres $-0.29$ to 0.02 and 1.69–1.76, $|a|/a_{\max}$ up to 0.91). The fit error falls — 0.0151–0.0152 at $\sigma_0/16$, 0.0141 at $\sigma_0/32$, 0.0028–0.0031 at $\sigma_0/64$ — but the count overshoots: 8–10 classes at $\sigma_0/64$, five to seven of them single particles, for a target of four bumps. $\sigma_0$ sets the resolution as (iv) says; which classes form, and with which masses, the shedding decides — the transit, which is not proved. (H) Other targets (the annealed Gaussian schedule of (C), broad starts, $m=60$, three seeds each): two bumps ($\sum_k|A_k|/a_{\max}=0.45$) end at two classes with the fit error $8.9\cdot10^{-9}$–$1.6\cdot10^{-8}$ in all three runs; the four-bump $f_\varepsilon$ of (A) and (G) at three classes with the fit error 0.0174 in all three, near the three-atom approximant's 0.0170 — the fourth bump lies below the resolution $\sigma_0$, and its width is the bound $s_{\min}=0.3$, so $f_\varepsilon$ has no exact representation; five bumps ($s_{\min}=0.2$; $|A_k|/a_{\max}=0.225$, 0.15, 0.175, 0.125, 0.2, $\Sigma=0.875$) at 6–9 classes with the fit error 0.045–0.118. An exact representation needs $w_k>|A_k|/a_{\max}$ for every bump, which leaves 0.125 of the mass free, and the transit's masses miss it: in every run two or three of the three inner bumps get less mass than they need (0.067–0.167 against 0.125–0.175), their classes saturated ($|a|/a_{\max}$ 0.96–1.00), and the outer bumps are carried by pairs of classes (one pair in seed 0, two in seeds 1 and 2). (K) Where the count fails and where it holds (the same schedule): three equal bumps — three classes in 3 of 3, the fit error $9.4\cdot10^{-5}$–$1.8\cdot10^{-4}$, the Gaussian kernel's bias between same-sign classes 1.5 apart in the centre: (L) it stays at $9.39\cdot10^{-5}$ for $2\cdot10^5$ more slow units and falls to $2.5\cdot10^{-13}$ under the compact kernel; four alternating bumps ($\sum_k|A_k|/a_{\max}=0.46$) — four classes in 3 of 3, $9.6\cdot10^{-3}$–$1.1\cdot10^{-2}$, a slow transient, not a bias: (L) the fit error falls geometrically, to $6.1\cdot10^{-4}$ after $2\cdot10^5$ more slow units ($3.4\cdot10^{-4}$ after the compact stage), the classes' weighted amplitudes the bumps' to $10^{-4}$; three heavy bumps ($\sum_k|A_k|/a_{\max}=0.95$, 0.05 of the mass free) — 3–5 classes, 0.058–0.20, the masses short again; two bumps closer than the broad scale ($\pm0.45$, the widths 0.4) — the broad stage merges them into one class of width 0.67–0.72 that the fine stage keeps (its margin positive, $6.5\cdot10^{-3}$–$7.1\cdot10^{-3}$: a stable biased configuration), with one or two stray particles, 1–3 classes, 0.098–0.13. So the count is the target's when the bumps are farther apart than the broad scale and the transit's masses are representable, and the approximant's below the resolution; the scale and the masses, the memory of the transit, decide the rest — the masses are Proposition D‴.34's remedy, the scale is not.

**Corollary D‴.33** (the desired regressor keeps the replicated configuration attracting under measurement noise)**.** *In the setting of Proposition D‴.30 let the adaptive law's regressor be evaluated at the reference — the desired regressor (DR) or the desired compensation adaptation law (DR*) of Proposition D‴.31(iv) — and let $s_k$ be the stiffness of class $k$'s separations: $\sigma_0/k_e$ under the fixed and the adaptive prior, $\sigma_0 w_k/k_e$ under the cohesion leak (Proposition D‴.32(vi)). Write $M_k:=G^{1/2}\nabla^2_\vartheta\tilde\Phi(x_m;\vartheta_k)G^{1/2}$.*

(i) (the drift) *With the regressor at $x_m$ the law's Jacobian on a separation of class $k$ is $\varepsilon_m G^{1/2}\nabla^2_\vartheta\tilde\Phi(x_m;\vartheta_k(t))G^{1/2}$: the regressor's own term $\langle\mathbb{E}[\varepsilon_m^2]G^{1/2}\partial_x\nabla^2_\vartheta\tilde\Phi G^{1/2}\rangle$ of Proposition D‴.30(i), which does not decrease with $k_e$ and carries the measured regressor's drift, is absent. To second order in $\nu$ the separation's mean drift is $S_k+C_k$: $S_k=\langle\mathbb{E}[e]M_k\rangle$ (at the lift $\mathbb{E}[e]$ is MR's, the regressor entering only the parameters' equation) and the*

*cross term* $C_k = \langle G^{1/2} \sum_l \mathbb{E}[\varepsilon_m \delta\vartheta_{kl}] \partial_{\vartheta_l} \nabla^2_\vartheta \tilde{\Phi} G^{1/2} \rangle$*, zero at the lift, which builds up within a period as the class atom fluctuates — negligible against the measured regressor's drift, of the order of DR's and DR*'s. A separation's exponent along a unit* $\theta$ *adds the nonnegative variance term* $\mathbb{E}\|P_\perp v_1(t)\|^2/t$*,* $v_1(t) = \int_0^t \varepsilon_m M_k \theta$ *and* $P_\perp = I - \theta\theta^\top$ *(to second order* $\mathbb{E}\log\|u(t)\|/\|u(0)\| = -s_k t + \int_0^t \theta^\top (S_k + C_k)\theta + \mathbb{E}\|P_\perp v_1(t)\|^2$*,* $\varepsilon_m M_k$ *being symmetric).*

(ii) (a threshold bounded uniformly in $k_e$) *At the lift* $k_e \mathbb{E}[e] = g$*,* $|g| \le \frac{1}{2}|\partial_x^2 f| \mathbb{E}\nu^2 (1 + O(|a|/k_e))$ *(Proposition D‴.31(iv)(c)), so* $\|S_k\| \le (\mathbb{E}\nu^2/(2k_e)) \|\partial_x^2 f\|_\infty \langle \|M_k\| \rangle (1 + O(1/k_e))$*: the drift and the stiffness are both* $O(1/k_e)$*, and the noise level up to which the replicated configuration attracts at the lift is bounded below uniformly in* $k_e$ *—* $\mathbb{E}\nu^2 \ge 2\sigma_0/(\|\partial_x^2 f\|_\infty \max_k \langle \|M_k\| \rangle)$ *up to* $(1 + O(1/k_e))$ *—, and constant when the noise's bandwidth is scaled with the gain* $(B/k_e$ *fixed, where* $k_e S_k$ *depends on* $k_e$ *only through* $a/k_e)$*. For a sensor of fixed bandwidth the factor* $(B - k_e - 2a)/(k_e + B)$ *of (iv) changes sign at* $k_e \approx B$*, and DR's drift with it; the thresholds then fall toward the bound's level. Under the measured regressor the threshold falls like* $1/k_e$ *in the variance while* $k_e \ll B$*, and levels off for* $k_e \gg B$*, where* $\mathbb{E}[\varepsilon_m^2] \to \mathbb{E}\nu^2 B/k_e$*.*

(iii) (the stationary point) *At the adaptive prior's stationary point the slow force balance carries the parameters' part of the second-order force (Proposition D‴.31(iv)(c)):* $\langle \hat{J}^\top (g + g_{\mathrm{par}} - \hat{J}\bar{y}) \rangle + k_e \langle f_{y,\mathrm{par}} \rangle = 0$*,* $g_{\mathrm{par}}$ *and* $f_{y,\mathrm{par}}$ *the parameters' parts of the error equation's and of the learning law's forcing, so (averaged)*

$$k_e \mathbb{E}[e] = (I - P)(g + g_{\mathrm{par}}) - \hat{J} H^{-1} k_e \langle f_{y,\mathrm{par}} \rangle,$$

*P the projection onto the regressor's span: the residual of the noise's own target outside the tangent space, plus the parameters' part, whose learning-law term lies in the tangent space and is not removed. The drift there is* $\langle \mathbb{E}[e] M_k \rangle + C_k$ *with the stationary covariance's cross term. Both parameters' terms are of the order of the noise's residual and need not be smaller:* $\mathbb{E}[e]$ *does not vanish at the stationary point when* $g$ *lies in the tangent space.*

(iv) (the sign under DR) *With the compensation* $\hat{f}(y)$ *the mean error at the lift is the compensation's rectification of the noise,* $\mathbb{E}[e] = -(\partial_x^2 f/(2k_e)) \mathbb{E}\nu^2 (B - k_e - 2a)/(k_e + B)$*. A Gaussian kernel depends on* $x$ *and its centre* $c$ *through* $x - c$*, so* $\partial_c^2 \tilde{\Phi} = \partial_x^2 \tilde{\Phi}$*, and the drift's entry along the centre coordinate* $\gamma$*,* $c = c(\gamma)$*, with the metric weight* $G_c := G_\gamma c'(\gamma)^2$*, is* $-(\mathbb{E}\nu^2 G_c/(2k_e)) \langle \partial_x^2 f \cdot \partial_x^2 \tilde{\Phi}_k \cdot (B - k_e - 2a)/(k_e + B) \rangle - c''(\gamma) G_\gamma \langle \mathbb{E}[e] \partial_x \tilde{\Phi}_k \rangle$*; for* $B \gg k_e$ *its first term is* $-(\mathbb{E}\nu^2 G_c/(2k_e))[w_k \langle (\partial_x^2 \tilde{\Phi}_k)^2 \rangle_\rho + \sum_{l \ne k} w_l \langle \partial_x^2 \tilde{\Phi}_l \partial_x^2 \tilde{\Phi}_k \rangle_\rho]$ *(*$f = \sum_l w_l \tilde{\Phi}_l$ *at the exact representation,* $\tilde{\Phi}_l := \tilde{\Phi}(\cdot\,; \vartheta_l)$*), negative when the kernels are separated, while the saturation's term, zero at* $c = 0$*, has no sign. So at the lift a noise fast against the loop* $(B > k_e + 2\max|a|)$ *stiffens the centres of separated kernels where that term is dominated, and a slower one* $(B < k_e - 2\max|a|)$ *reverses the factor and the sign. The other directions have no sign: the whole matrix is negative definite for separated kernels of alternating sign well inside the orbit, as on the instance, and indefinite for same-sign, wide or edge kernels (their widths and amplitudes soften). Along the trajectory* $\mathbb{E}[e]$ *relaxes with the collective modes* $(k_e \mathbb{E}[e] = g - \hat{J}\bar{y}(t))$*, the fast modes first, so it passes through the part of* $g$ *carried by the slow modes, which has no sign relation to* $\partial_x^2 f$*: the sign holds near the lift, not along the trajectory. With the compensation at the reference (DR*) the mean error is the plant's rectification of its state jitter,* $+\frac{1}{2}\partial_x^2 f \cdot \mathrm{Var}(e)/(k_e - a)$*, and the sign at the lift is reversed.*

*Proof.* (i) With the regressor at $x_m$ the law is $G\nabla_\vartheta(\varepsilon_m \tilde{\Phi}(x_m; \vartheta_i))$ minus the leak, and its Jacobian on a separation is $\varepsilon_m G^{1/2} \nabla^2_\vartheta \tilde{\Phi}(x_m; \vartheta_k) G^{1/2}$: the measurement enters only through $\varepsilon_m$, whose mean is $\mathbb{E}[e]$ ($\mathbb{E}\nu = 0$), and the class atom's fluctuation through the Hessian, whose correlation with $\varepsilon_m$ is the cross term. A separation does not enter $\hat{f}$ at first order, so neither the compensation nor the leak adds a noise term. The exponent: with $A_1 = \varepsilon_m^{(1)} M_k$, $w = u e^{s_k t}$, $\log\|w\| = \langle \theta, w_1 \rangle + \langle \theta, w_2 \rangle + \frac{1}{2}\|w_1\|^2 - \langle \theta, w_1 \rangle^2 + O(\nu^3)$, and $\mathbb{E}\langle \theta, w_2 \rangle = \int \theta^\top \mathbb{E}[A_2]\theta + \frac{1}{2}\mathbb{E}\|w_1\|^2$

for a symmetric $A_1$, whence the variance term. (ii) The mean of the error equation at second order gives $k_e\mathbb{E}[e] = g - \hat{J}\bar{y}$ (Proposition D‴.31's proof), at the lift $\bar{y} = 0$, and the bound on $g$ is Proposition D‴.31(iv)(c)'s; with $B = \beta k_e$ the factor of (iv) is $(\beta - 1 - 2a/k_e)/(1 + \beta)$. For $k_e \gg B$, $\mathbb{E}[\varepsilon_m^2] = \mathbb{E}\nu^2(a^2 + k_eB)/(k_e(k_e + B)) \to \mathbb{E}\nu^2 B/k_e$. (iii) The mean of the learning law adds the parameters' part $k_e\langle f_{y,\mathrm{par}}\rangle$ (the law's $\nabla^2_{\vartheta}\tilde{\Phi}\mathbb{E}[\varepsilon_m\delta\vartheta]$) to the least-squares force, and the error equation's mean forcing is $g + g_{\mathrm{par}}$ (Proposition D‴.31(iv)(c)); stationarity gives $H\bar{y} = \langle\hat{J}^\top(g + g_{\mathrm{par}})\rangle + k_e\langle f_{y,\mathrm{par}}\rangle$, and $\hat{J}\bar{y} = P(g + g_{\mathrm{par}}) + \hat{J}H^{-1}k_e\langle f_{y,\mathrm{par}}\rangle$. (iv) $\tilde{\Phi}$ depends on $x$ and $c$ through $x - c$, so $\partial_c\tilde{\Phi} = -\partial_x\tilde{\Phi}$ and $\partial_c^2\tilde{\Phi} = \partial_x^2\tilde{\Phi}$; for $c = c(\gamma)$, $\partial_\gamma^2\tilde{\Phi} = c'^2\partial_c^2\tilde{\Phi} + c''\partial_c\tilde{\Phi}$, and the quasi-static $\mathbb{E}[e]$ of Proposition D‴.30's proof with $\partial_x^2\hat{f} = \partial_x^2 f$ gives the entry; DR*'s $\mathbb{E}[e]$ is in Proposition D‴.31's proof. □

**Status.** (i) and (iv)'s formula exact consequences of the second-order expansion — formal at the order of Proposition D‴.30(i)'s drift, with its caveats —; (ii)'s bound proved given (i) and D‴.31(iv)(c)'s bound on $g$; (iii)'s balance formal (averaged), its numbers from the exact periodic second-order mean; (iv)'s sign proved for the centre direction of separated Gaussian kernels where the saturation's term is dominated, at the lift; the whole matrix's sign numerical and specific to the target, and lost along the trajectory. Classical: the desired compensation adaptation law's insensitivity to measurement noise ([96]); the attraction of identical uncoupled systems driven by a common noise — noise-induced synchronisation, decided by the transverse Lyapunov exponent at second order in the noise, which strong noise can make positive (Teramae–Tanaka, Goldobin–Pikovsky, Nakao–Arai–Kawamura, [104]) — is the frame of Proposition D‴.30(i)'s threshold and of this corollary; the second-order drift of a regressor without the measured state is the one-line consequence of a $y$-independent regressor. New: that consequence for the $K_0$-atom identification — the separations' drift $O(1/k_e)$ like their stiffness, a threshold bounded uniformly in $k_e$, which makes the desired regressor the natural partner of the cohesion leak, whose separations are the softest ($\sigma_0 w_k$) — and the instance's quantification: the sign at the lift, its loss along the trajectory, the stationary point. Checked: `tests/test_eiv_remedy.py` (DR's drift MR's mean-error term alone at the lift, negative definite in every class, MR's 0.30 in class 3; DR*'s $O(1/k_e)$, $k_e$ times its largest eigenvalue the same at $k_e = 25$ and 50, and DR*'s drift against an independent quadrature to $10^{-4}$; at the adaptive prior's stationary point the exact periodic mean's noise and parameters' parts summing to it, the parameters' part dominant and largely in the tangent space, the drift indefinite; the period maps continuing `second_order_transient` to $10^{-7}$ and DR's drift turning indefinite).

**Numerics** (`exp_eiv_remedy.py` Part D → `results/chaos/eiv_remedy.md`). Proposition D‴.30's instance ($k_e = 25$, the OU noise $B = 100$), per unit $\mathbb{E}\nu^2$, real time. (D1) At the lift the separations' drift has the extreme eigenvalues $[-3.7\cdot10^{-1}, +1.5\cdot10^{-2}]$; $[-2.6\cdot10^{-2}, +1.0\cdot10^{-2}]$; $[-3.5\cdot10^{-2}, +3.0\cdot10^{-1}]$ in the three classes under MR, $[-1.1\cdot10^{-2}, -5.0\cdot10^{-4}]$; $[-1.5\cdot10^{-2}, -1.5\cdot10^{-4}]$; $[-7.2\cdot10^{-3}, -3.4\cdot10^{-4}]$ under DR and $[+1.6\cdot10^{-4}, +3.8\cdot10^{-3}]$; $[+5.1\cdot10^{-5}, +5.0\cdot10^{-3}]$; $[+1.1\cdot10^{-4}, +2.4\cdot10^{-3}]$ under DR*: DR's is negative definite, DR*'s largest eigenvalue, $+4.98\cdot10^{-3}$, is 60 times below MR's 0.2982 — the threshold std 0.238 under the adaptive prior against MR's 0.031, and 0.138 against 0.018 under the cohesion leak's stiffness $\sigma_0 w_k$ ($w_k = 1/3$). (D2) With $B = 4k_e$ MR's largest eigenvalue is $0.2982 \to 0.3045$ over $k_e = 25 \to 800$, its threshold std $0.031 \to 0.0054$ ($\propto k_e^{-1/2}$), while $k_e$ times DR's is $-0.0038$ and $k_e$ times DR*'s 0.1245–0.1242 at every $k_e$ (DR*'s threshold 0.238–0.239). At the fixed $B = 100$ DR's drift is indefinite at $k_e = 100$ and positive definite from $k_e = 200$, its threshold std 0.185, 0.138, 0.121 at $k_e = 200$, 400, 800; DR*'s falls $0.238 \to 0.113$ and MR's $0.031 \to 0.014$ (its variance by 4.6 over a factor 32 in $k_e$). $\|S_k\|$ is at most 0.37 of (ii)'s bound in these cases (DR 0.253, DR* 0.084 at $B = 4k_e$), which gives the threshold the floor std $\geq 0.069$ for DR and DR*. (D3) For other exactly represented targets at the lift DR's centre entries are negative for all six; its whole drift is negative definite for the instance and for two opposite bumps, and indefinite for three separated same-sign bumps (the largest eigenvalue $+2.9\cdot10^{-4}$), wide bumps ($S = 1$; $+5.6\cdot10^{-4}$), bumps at the orbit's edge ($+2.7\cdot10^{-4}$) and two close same-sign bumps ($+5.0\cdot10^{-4}$). (D4) At the adaptive prior's

stationary point (the exact periodic second-order mean) the mean error's rms is $6.70 \cdot 10^{-4}$ under DR and $6.44 \cdot 10^{-4}$ under DR* — 2.4% and 7.0% of the lift's —, its parameters' part $6.32 \cdot 10^{-4}$ and $6.27 \cdot 10^{-4}$ and its noise part $2.11 \cdot 10^{-4}$ and $1.37 \cdot 10^{-4}$ (the round-22 averaged noise-only balance $(I - P)g/k_e$ gave $2.34 \cdot 10^{-4}$ and $2.60 \cdot 10^{-4}$); its tangent part, $5.80 \cdot 10^{-4}$ and $5.81 \cdot 10^{-4}$, is the parameters' part's. The drift there is indefinite in every class — DR's $[-4.35, +3.68] \cdot 10^{-5}$; $[-0.13, +0.30] \cdot 10^{-5}$; $[-2.82, +2.44] \cdot 10^{-5}$ —, the parameters' part of $\mathbb{E}[e]$ contributing up to $+2.6 \cdot 10^{-4}$ and the cross term down to $-2.6 \cdot 10^{-4}$ in class 2, nearly cancelling. Under the fixed prior the mean error keeps 41% of the lift's (rms $1.13 \cdot 10^{-2}$, DR), DR's drift stays negative definite (the largest eigenvalue $-3.8 \cdot 10^{-5}$) and DR*'s positive definite ($+1.50 \cdot 10^{-3}$, the threshold std 0.435). (D5) Along DR's exact second-order transient from the lift (82 000 periods, $5.2 \cdot 10^5$ time units) the drift's largest eigenvalue, $-4.7 \cdot 10^{-4}$, $-1.6 \cdot 10^{-4}$, $-3.1 \cdot 10^{-4}$ in the first period, turns positive at $t = 4103$, 2947, 2714 in classes 1, 2, 3 — with the noise part of $\mathbb{E}[e]$ alone at $t = 4072$, 2721, 2689: (iv)'s mechanism, not the parameters' terms — and stays positive, reaching $+1.05 \cdot 10^{-4}$ (class 1, $t \approx 3.1 \cdot 10^4$), formally the threshold std 1.64, far outside the expansion, before settling at the stationary point's (the period maps' limit within 0.3% of the exact periodic mean's). DR*'s largest eigenvalue is positive throughout ($3.6 \cdot 10^{-3}$ in the first period, $3.4 \cdot 10^{-5}$ at the stationary point). Over the first 300 time units the cross term moves DR's class-2 window drift by $-8\%$ ($[0, 50)$: $-1.61 \cdot 10^{-4}$, without it $-1.49 \cdot 10^{-4}$) and DR*'s by $-17\%$ ($+3.69 \cdot 10^{-3}$ against $+4.45 \cdot 10^{-3}$), so DR*'s threshold std over $[0, 50)$ is 0.277 (0.238 at the lift's quasi-static moments). (D6) The real loop (the lift $m = 15$, one class separated by $10^{-6}$ along a direction $\theta$, four antithetic pairs per family on paths common to the laws, 300 time units, the rate fitted over $t \geq 50$): along MR's top eigenvector $u$ of $S_3$ DR's separations decay at $-2.9651 \cdot 10^{-4} \pm 1.3 \cdot 10^{-7}$ (std 0.05) and $-3.3650 \cdot 10^{-4} \pm 5.3 \cdot 10^{-7}$ (std 0.1), faster than $\sigma_0/k_e = 2.8298 \cdot 10^{-4}$; their noise-induced part is $0.98 \pm 0.01$ and $0.97 \pm 0.01$ of the transient's prediction — the mean error's term along the transient, the cross term and the variance term — (2.3 and 3.5 sampling errors), against 0.82 and 0.81 of the lift's. Along DR's own top eigenvectors, its weakest directions, the separations decay faster than $\sigma_0/k_e$ by only 0.36%, 0.14%, 0.26% (classes 1–3, std 0.05; 1.4%, 0.54%, 1.0% at 0.1), at the transient's prediction to 0.9, 1.9 and 0.05 sampling errors (0.8, 1.8, 0.1); along DR*'s top eigenvector of $S_2$ they decay slower than $\sigma_0/k_e$ by 3.2% and 12.6%, the noise-induced part $0.97 \pm 0.10$ and $0.96 \pm 0.10$ of the transient's prediction (0.72 of the lift's). The pair means of DR decay faster than $\sigma_0/k_e$ in every family; no DR or DR* run grows. MR's separations grow at $+4.69 \cdot 10^{-4} \pm 1.3 \cdot 10^{-5}$ and $+1.25 \cdot 10^{-3} \pm 3.1 \cdot 10^{-5}$ (8 of 8 runs each), the latter less than half the second-order prediction (the next order of Proposition D‴.30). On the same noise paths the measured regressor's replicas separate and the desired regressor's stay together.

**Proposition D‴.34** (mass transfer: the replicator weight law removes the saturation trap of the transit's masses)**.** *Let the particles of Proposition D‴.32 carry weights $\beta_i > 0$ with $\sum_i \beta_i = m$ — the network $\hat{f} = (1/m) \sum_i \beta_i \tilde{\Phi}(\cdot\,; \vartheta_i)$, its measure $\mu_{\Theta,\beta} = (1/m) \sum_i \beta_i \delta_{\vartheta_i}$, the class masses $W_k = \sum_{i \in k} \beta_i/m$ — and let the weights adapt by the replicator law*

$$\frac{\mathrm{d}\beta_i}{\mathrm{d}s} = \gamma \beta_i (u_i - \lambda), \qquad u_i := \langle r \tilde{\Phi}(\cdot\,; \vartheta_i) \rangle_\rho, \qquad \lambda := \sum_j \beta_j u_j \Big/ \sum_j \beta_j \tag{$\mathfrak{F}$}$$

*beside $\mathrm{d}\vartheta_i/\mathrm{d}s = \beta_i G \langle \nabla_\vartheta (r \tilde{\Phi}(\cdot\,; \vartheta_i)) \rangle_\rho - (\sigma_0/m) \sum_j \psi(\|\vartheta_i - \vartheta_j\|^2_{G^{-1}}/R^2)(\vartheta_i - \vartheta_j)$ ($r = f - \hat{f}$; in the real loop, on $\sum \beta = m$, $\dot{\beta}_i = \gamma \beta_i e^\top (\tilde{\Phi}(x; \vartheta_i) - \hat{f}(x))$, the replicator dynamics with the fitness $e^\top \tilde{\Phi}_i$ — an exponentiated-gradient law).*

(i) (a gradient flow; the bound kept) *The pair is the gradient flow of $J(\Theta, \beta) := m \bar{L}(\mu_{\Theta,\beta}) + (\sigma_0 m/2) \iint W \,\mathrm{d}\mu_\Theta \,\mathrm{d}\mu_\Theta$ (the cohesion energy of the unweighted cloud, so that the weights see the fit only) in the metric $\sum_i \|\delta\vartheta_i\|^2_{G^{-1}} + \sum_i \delta\beta_i^2/(\gamma\beta_i)$ on $\{\sum_i \beta_i = m\}$ — Euclidean on the positions, Fisher–Rao (Shahshahani) on the weights; with $\varrho_i := \sqrt{\beta_i}$, gradient descent on $(\varrho, \Theta)$, with the step $\gamma/4$ in $\varrho$, on the sphere $\sum_i \varrho_i^2 = m$, $\varrho_i^2/m$ the output weights of a*

*two-layer network —, so $J$ decreases; $\beta > 0$ and $\sum\beta$ are invariant (from every start), and on $\sum\beta = m$, $|\hat f| \le (1/m)\sum\beta_i|a_i| \le a_{\max}$ as for equal weights: no projection is needed. With $\beta \equiv 1$ and $\gamma = 0$ it is the loop of Proposition D‴.32. (A Wasserstein–Fisher–Rao metric would weigh the positions by their masses, $\sum_i \beta_i\|\delta\vartheta_i\|^2_{G^{-1}}$: its drift lacks the factor $\beta_i$ on the fit and divides the cohesion force by $\beta_i$; and $J$ is not a function of $\mu_{\Theta,\beta}$ alone, its cohesion being the unweighted cloud's.)*

(ii) (the real loop at fixed positions) *With the positions at atoms where $f(x) = (1/m)\sum_i \beta_i^*\tilde\Phi(x;\vartheta_i)$ at every state the loop visits (for $n = 1$ and analytic features, on the orbit's interval), $\sum\beta^* = m$, the real loop with $(\mathfrak{F})$ has $\dot V = -k_e|e|^2$ for $V := \frac12|e|^2 + (1/(\gamma m))\sum_i \beta_i^*\log(\beta_i^*/\beta_i)$: the relative entropy of the weights takes the place of the quadratic parameter error — the negative-entropy instance of the mirror-descent adaptation laws with Bregman Lyapunov functions ([105]). The identity needs only $\sum\beta^* = m$ and the pointwise representation; $V \ge 0$ needs $\sum\beta = m$, which the law keeps (it does not restore it). $\beta \to \beta^*$ is not claimed: it needs the persistent excitation of $(\tilde\Phi(x_m(t);\vartheta_i))_i$.*

(iii) (the stationary points; the saturated transients) *$(\Theta,\beta)$ is stationary iff $\beta_i G\langle r\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_i)\rangle_\rho = (\sigma_0/m)\sum_j \psi_{ij}(\vartheta_i - \vartheta_j)$ for every $i$ and $u_i = \lambda$ on the support: the fit's force carries the factor $\beta_i$, the cohesion does not. At a clustered configuration under the finite range (the classes $R$ apart) the cohesion vanishes, the condition is $\langle r\nabla_\vartheta\tilde\Phi(\cdot\,;\vartheta_k)\rangle = 0$, whose amplitude component $(1 - a_k^2/a_{\max}^2)\langle r\varphi_k\rangle = 0$ forces $\langle r\varphi_k\rangle = 0$, so $u_k = a_k\langle r\varphi_k\rangle = 0 = \lambda$: there the stationary points are Proposition D‴.32's, for every vector of masses. Under the Gaussian kernel the inter-class forces do not vanish and the fit's factor $\beta_i$ weighs against them, so $u_k = \lambda$ fixes the masses: the replicator selects the masses of each class configuration and moves the Gaussian kernel's bias (Numerics (G)); and a class whose particles carry unequal weights is not exactly replicated at a biased stationary point. The difference is the saturated transient: when a class is short of mass ($|A_k| > W_k a_{\max}$) the equal-weight loop at fixed positions stays biased (it has no critical point) while $|a_k| \to a_{\max}$ with $\langle r\varphi_k\rangle \not\to 0$, its amplitude gradient exponentially small in $\alpha_k$ (heuristically $\alpha_k$ grows like $(a_{\max}/2)\log s$); under $(\mathfrak{F})$ the starved class's mass grows at the rate $\gamma(u_k - \lambda)$, $u_k \approx a_{\max}|\langle r\varphi_k\rangle|$ — $\mathrm{d}W_k/\,\mathrm{d}s = \gamma W_k(1 - W_k)u_k > 0$ exactly when the class's particles coincide and $u_l = 0$ for the other classes — and the others take up the gauge (Proposition D‴.28: their $p_l = W_l a_l$ kept by $|a_l|$ increasing).*

(iv) (fixed positions: representability) *With one particle at each of $K$ distinct atoms whose centres and widths are fixed and whose features $\varphi_k$ are linearly independent in $L^2_\rho$, and $f = \sum_k A_k\varphi_k$ in their span (otherwise $A$ is the least-squares coefficient vector), the flow of $(W,\alpha)$ is a gradient flow of the least-squares loss in $p_k := W_k a(\alpha_k)$, convex in $p$, on the open simplex times $\mathbb{R}^K$; its critical points are the exact fits $p = A$, which exist iff $\sum_k|A_k|/a_{\max} < 1$ (the representable set of Proposition D‴.24 is not empty); every trajectory bounded in $(\log W,\alpha)$ converges to one ($\bar L$ is real-analytic there: Łojasiewicz), so bounded trajectories reach the exact fit iff the target is representable (boundedness is not proved); when $\sum_k|A_k|/a_{\max} \ge 1$ every trajectory leaves every compact set, each class saturating or losing its mass, and the fit error stays above the least-squares error over the $\ell^1$ ball $\sum_k|p_k| \le a_{\max}$. The transit's masses no longer decide whether the target is fitted.*

(v) (the linearisation at an exact representation) *At the lift of an exact representation ($\beta \equiv 1$) the weights add, on $\{\sum\beta = m\}$, $\sum_k(n_k - 1)$ neutral directions (the redistribution of mass inside a class, which $\hat f$ does not see) and $K - 1$ gauge directions (mass moved between classes against their amplitudes, $W_k a_k$ fixed), and no unstable one — in the coordinates $(\Theta,\log\beta)$, the constraint not imposed, the common rescaling of the weights against the amplitudes is a $K$-th gauge; the separations' block is Proposition D‴.32's ($-\sigma_0 w_k$, $w_k$ the particle fraction), the collective block changes by the replicator's metric on the masses. In*

*the real loop the lift is a periodic orbit in a manifold of periodic orbits (the gauge and the redistributions), neutral along it.*

*Proof.* (i) $\partial_{\vartheta_i}(m\bar{L}) = -\beta_i\langle r\nabla_\vartheta\tilde{\Phi}(\cdot\,;\vartheta_i)\rangle_\rho$ and $\partial_{\beta_i}(m\bar{L}) = -u_i$ ($\hat{f}$ is linear in $\beta$); the cohesion energy does not depend on $\beta$; the Fisher–Rao gradient on $\{\sum\beta = \text{const}\}$ is $-\gamma\beta_i(\partial_{\beta_i}J - \lambda')$, $\lambda' = \sum_j \beta_j\partial_{\beta_j}J/\sum_j\beta_j$, which is ($\mathfrak{F}$), and $\mathrm{d}J/\mathrm{d}s = -\sum_i\|\mathrm{d}\vartheta_i/\mathrm{d}s\|^2_{G^{-1}} - \sum_i(\mathrm{d}\beta_i/\mathrm{d}s)^2/(\gamma\beta_i) \le 0$; $\mathrm{d}\log\beta_i/\mathrm{d}s$ is bounded on bounded sets, so $\beta$ stays positive, and $\sum_i \mathrm{d}\beta_i/\mathrm{d}s = \gamma\sum_i\beta_i(u_i-\lambda) = 0$. With $\beta_i = \varrho_i^2$, $\delta\beta_i^2/(\gamma\beta_i) = 4\delta\varrho_i^2/\gamma$. (ii) With $\sum\beta = m$, $\lambda(t) = e^\top\hat{f}(x)$; $\frac{\mathrm{d}}{\mathrm{d}t}\sum_i\beta_i^*\log(\beta_i^*/\beta_i) = -\gamma e^\top\sum_i\beta_i^*(\tilde{\Phi}_i - \hat{f}) = -\gamma m e^\top(f-\hat{f})$ ($f = (1/m)\sum\beta_i^*\tilde{\Phi}_i$, $\sum\beta^* = m$), and $\dot{e} = f - \hat{f} - k_e e$; $\sum_i\beta_i^*\log(\beta_i^*/\beta_i) = m\cdot\mathrm{KL}(\beta^*/m\,\|\,\beta/m) \ge 0$ when $\sum\beta = m$. (iii) The stationarity of the positions and of the weights read off (i); at a clustered configuration under the finite range $\psi = 0$ across classes and $\vartheta_i - \vartheta_j = 0$ inside them; the amplitude's derivative $\mathrm{d}a/\mathrm{d}\alpha = 1 - a^2/a_{\max}^2 = 4e^{-2|\alpha|/a_{\max}}(1+o(1))$; $\mathrm{d}W_k/\mathrm{d}s = (1/m)\sum_{i\in k}\gamma\beta_i(u_k-\lambda) = \gamma W_k(u_k-\lambda)$ and $\lambda = \sum_l W_l u_l = W_k u_k$ when $u_l = 0$ for $l \ne k$. (iv) In the variables $(W,\alpha)$ the flow is the restriction of (i)'s gradient flow; $\bar{L}$ depends on them through $p$, $\partial\bar{L}/\partial p_k = -\langle r\varphi_k\rangle =: -g_k$, and at a critical point with $W$ in the open simplex the amplitude component gives $g_k = 0$ for every $k$, i.e. $p$ is the least-squares solution $A$; $|a(\alpha)| < a_{\max}$ and $\sum W_k = 1$ make $p = A$ attainable iff $\sum_k|A_k|/a_{\max} < 1$. The flow is the gradient flow of a real-analytic function for an analytic metric in $(\log W,\alpha)$, so a bounded trajectory has a single limit, a critical point (Łojasiewicz's theorem). The dissipation $\int\|\nabla J\|^2\,\mathrm{d}s \le J(0)$ makes every $\omega$-limit point critical, so without a critical point a trajectory has no $\omega$-limit point and eventually leaves every compact set; and $\sum_k|p_k| = \sum_k W_k|a_k| < a_{\max}$ always, which bounds the fit error below by the $\ell^1$ ball's. (v) At $r = 0$ the Hessian of $m\bar{L}$ is the Gram matrix of the derivatives of $\hat{f}$, and $J$'s Hessian is the linearisation's up to the metric: inside a class $\partial\hat{f}/\partial\beta_i = \tilde{\Phi}_k/m$ is the same for every particle (the redistributions), and $\partial\hat{f}/\partial\beta_i = (a_k/a_k')\partial\hat{f}/\partial\alpha_i$ at $\beta_i = 1$ (the gauge, one per class; $K-1$ combinations of them keep $\sum\delta\beta = 0$); the cohesion energy acts on the separations only and does not see the weights; $J$'s Hessian is positive semidefinite there (the Gram matrix plus the cohesion's $\nabla^2W(0) = G^{-1}$ on the separations), so no direction is unstable; along the lift $e \equiv 0$, so $\mathrm{d}\beta/\mathrm{d}t = 0$. □

**Status.** (i), (ii), (iii)'s stationarity at clustered configurations under the finite range and (v)'s zero modes proved; under the Gaussian kernel the masses' selection numerical (G); (iii)'s transient: the rate exact under its conditions, the bias's duration heuristic; (iv) proved for bounded trajectories, boundedness not proved; the transit with the cohesion leak numerical, and with it any selection of the number of kernels: the replicator does not act on the separations at a critical point ($u_k = 0 = \lambda$ there), so the shedding of Proposition D‴.32(iv) is untouched (Numerics (C)). Classical: Wasserstein–Fisher–Rao (Hellinger–Kantorovich) distances and gradient flows of measures, conic particle gradient descent and the lift to $(\sqrt{\beta},\Theta)$ with its global convergence for sparse optimisation on measures, neuron birth–death, the exponentiated-gradient law and EM's multiplicative update of mixture weights, mirror-descent adaptation laws with Bregman Lyapunov functions — of which (ii) is the negative-entropy instance —, the replicator equation as the Shahshahani gradient flow with its relative-entropy Lyapunov function ([105]). New: the replicator as the mass transfer of the cohesion leak in mean-field adaptive control — the stationary set unchanged at clustered configurations under the finite range, the masses selected under the Gaussian kernel, the saturation trap of the transit's masses removed —, the representability criterion at fixed positions for bounded trajectories, and the numerics.

**Numerics** (`exp_mass_transfer.py` → `results/chaos/mass_transfer.md`; Proposition D‴.32's instances, the midpoint rule on $(\Theta,\log\beta)$). (A) Fixed positions — the five-bump target's atoms, one particle per class, the masses (0.15, 0.30, 0.20, 0.25, 0.10), two of them short of $|A_k|/a_{\max} = (0.225, 0.150, 0.175, 0.125, 0.200)$, the amplitudes and the weights adapted: without the replicator the fit error stays at 0.388 from $s = 4000$ on — the least fit error at

those masses is 0.387 (G) —, the two short classes at $|a|/a_{\max} = 0.999$; with $\gamma = 10^{-3}$ it falls to $2.9 \cdot 10^{-4}$ at $s = 40000$, with $\gamma = 10^{-2}$ to $3.0 \cdot 10^{-14}$ (geometrically, by about 6 per 2000 slow units), every mass above its $|A_k|/a_{\max}$ at the end (0.234, 0.199, 0.191, 0.168, 0.208). With the amplitudes scaled by 1.25 ($\sum_k |A_k|/a_{\max} = 1.094$) the fit error falls to 0.071 at both rates and still creeps down toward the $\ell^1$ ball's least error 0.0684 (G), every class at $|a|/a_{\max} \geq 0.992$, the masses already the minimiser's (0.271, 0.164, 0.194, 0.132, 0.239 against 0.270, 0.164, 0.194, 0.132, 0.239): (iv)'s dichotomy. (B) The five-bump target from broad starts ($m = 60$, the annealed Gaussian schedule): without the replicator 6–9 classes, the fit error 0.045–0.118 (Proposition D‴.32, Numerics (H)); with $\gamma = 10^{-3}$ 5–7 classes, 0.0076–0.0139; with $\gamma = 10^{-2}$ five classes in 3 of 3 runs, their centres the bumps' to 0.01 ($-1.600$, $-0.798$, $+0.004$, $+0.798$, $+1.597$ for seed 1), the fit error $6.0 \cdot 10^{-4}$–$1.4 \cdot 10^{-3}$, falling at the end by 2–4% per 1000 slow units. Continued (G, seed 1) it levels off — $5.99 \cdot 10^{-4}$, $3.62 \cdot 10^{-4}$, $3.01 \cdot 10^{-4}$, $2.87 \cdot 10^{-4}$, $2.82 \cdot 10^{-4}$, $2.80 \cdot 10^{-4}$ at $s = 0, 2, \ldots, 10 \cdot 10^4$ under the Gaussian kernel, near its class-level stationary point ($2.63 \cdot 10^{-4}$), and $2.85 \cdot 10^{-4}$ after $4 \cdot 10^4$ slow units under the compact kernel, whose class-level stationary point is exact ($2.4 \cdot 10^{-16}$): the nearly saturated classes ($|a|/a_{\max}$ up to 0.96) are slow. (D) The instance (D‴.32 (C)) with $\gamma = 10^{-2}$: three classes in 3 of 3 runs, the fit error $2.4 \cdot 10^{-7}$–$9.7 \cdot 10^{-7}$ (without the replicator, the same seeds 0–2, $1.9 \cdot 10^{-7}$–$9.0 \cdot 10^{-7}$). (C) The resolution's finest stage (the fourth bump, compact $R = 1$, $\sigma_0/64$, from the jittered lift of $\vartheta^b(q)$, $10^5$ slow units): with $\gamma = 10^{-2}$ the same 8–10 classes as without, from the same splits at nearly the same times (3 of 51 records differ by one class; the fit error 0.0055–0.0075 against 0.0059–0.0078): at a critical point of $L_w$ the fitness vanishes ($u_k = a_k \langle r\varphi_k \rangle = 0 = \lambda$), so the replicator does not act on the separations' instability, and the shedding of D‴.32 (iv) is untouched. (F) The targets where D‴.32's count failed (its Numerics (K); the annealed schedule, $\gamma = 10^{-2}$): three heavy bumps ($\sum_k |A_k|/a_{\max} = 0.95$) end at three classes at the bumps in 3 of 3 runs, the fit error $7.8 \cdot 10^{-8}$–$1.6 \cdot 10^{-7}$, every mass above its $|A_k|/a_{\max}$ (0.411–0.423, 0.308–0.313, 0.265–0.280 against 0.400, 0.300, 0.250), where without the replicator 3–5 biased classes ended at 0.058–0.20; two bumps closer than the broad scale ($\pm 0.45$) end as without it, merged (1–3 classes, 0.068–0.13, one class of mass 0.88–1.00 at the centre): the scale's obstruction is not the masses', and no mass transfer separates what the broad stage merged. (G) The stationary points at the class level (a run's classes replicated with equal weights inside each, the conditions of (iii) solved to rounding, $R = 1$): for B's seed 1 (the counts $(6, 6, 20, 10, 18)$) under the Gaussian kernel the fit error is $3.85 \cdot 10^{-4}$ with the masses held and $2.63 \cdot 10^{-4}$ with $u_k = \lambda$, the masses moving (0.1573, 0.1330, 0.2618, 0.2060, 0.2419 $\to$ 0.1579, 0.1318, 0.2661, 0.2076, 0.2366) and the classes of each sign then at a common $|a|/a_{\max}$ (0.9578, 0.8464); under the compact kernel both are exact ($2.4 \cdot 10^{-16}$–$2.5 \cdot 10^{-16}$), the masses free to $5 \cdot 10^{-4}$. For D‴.32 (L)'s three equal bumps $(28, 16, 16)$ the Gaussian kernel's stationary point has $9.39 \cdot 10^{-5}$ with the masses held and $9.17 \cdot 10^{-5}$ with them free, the masses moving from (0.467, 0.267, 0.267) toward (0.567, 0.217, 0.217) (the residual of the conditions there $1.7 \cdot 10^{-10}$: the outer classes approach the saturation), the compact kernel's exact either way: (iii)'s difference between the kernels. (E) The real loop at the instance's atoms ($m = 3$, $k_e = 25$, $\gamma = 2$, $\beta(0) = (1.6, 0.7, 0.7)$, 200 time units): $V$ from 0.0418 to $2.1 \cdot 10^{-7}$, monotone; $\dot{V}$ from the vector field equal to $-k_e e^2$ to $1.1 \cdot 10^{-16}$ of $\max k_e e^2$; $\sum \beta = m$ to $2.4 \cdot 10^{-14}$; $\beta$ at the end within $1.3 \cdot 10^{-3}$ of $\beta^* = 1$. The averaged loop's linearisation in $(\Theta, \log \beta)$ at the lift of $\vartheta^e(w)$, counts $(6, 5, 4)$, compact $R = 1$: with $\gamma = 10^{-2}$ 15 eigenvalues below $10^{-9}$ — the 12 redistributions and the 3 gauges of (v) — and the 45 others in $[-0.1834, -3.081 \cdot 10^{-4}]$; with $\gamma = 0$ the 45 are Proposition D‴.32's predicted spectrum to $2.1 \cdot 10^{-11}$ (the finite differences' error).

**Proposition D‴.35** (the split of a class is an $S_n$-symmetric transcritical bifurcation: shedding, its order, its cascade)**.** *In the setting of Proposition D‴.32 (the finite range), let class $k$ of $n \geq 3$ particles sit at an atom $\vartheta_k$ of a nondegenerate critical point of $L_w$, the classes at least $R$ apart, and let $\lambda^+ = \lambda_{\max}(R_k^G)$ be simple with the whitened unit eigenvector $v$. Write the separations*

*along $v$ as $u_i = \xi_i G^{1/2} v$ ($\sum_i \xi_i = 0$), $\mu := \lambda^+ - \sigma_0 w_k$ (the negative splitting margin), and*

$$c_3 := \partial_t^3 \langle r\tilde{\Phi}(\cdot\,; \vartheta_k + tG^{1/2}v)\rangle_\rho\big|_{t=0} \qquad (r \text{ the residual at the critical point, frozen}),$$

*the residual's third derivative along the splitting direction.*

(i) (the reduced equation) *For $|\mu|$ small against the rates of the other modes, the flow has a centre manifold over the separations along $v$ — the standard representation of the symmetric group $S_n$, which permutes the particles — and on it*

$$\frac{\mathrm{d}\xi_i}{\mathrm{d}s} = \mu\xi_i + \frac{c_3}{2}\Big(\xi_i^2 - \frac{1}{n}\sum_j \xi_j^2\Big) + O(|\xi|^3 + |\mu||\xi|^2)\,:$$

*the quadratic coefficient is the residual's alone, since the cohesion energy is even in the differences ($W(-u) = W(u)$) and the mean term is common to the particles. The splitting margin of Proposition D‴.32(iv) is the linear coefficient; the quadratic one identifies the equivariant normal form's coefficient for this flow. At third order the cohesion's concavity, the residual's own cubic term and the slaving of the class mean all enter.*

(ii) (no small split is stable) *If $c_3 \neq 0$ the bifurcation at $\mu = 0$ is transcritical. Its equilibria near 0 off the $S_{n/2} \times S_{n/2}$ symmetry are the split states with isotropy $S_q \times S_{n-q}$ — $q$ particles at $(n-q)t^*$, $n-q$ at $-qt^*$, $t^* = -2\mu/(c_3(n-2q))$, $1 \le q < n/2$ — and on the centre manifold each has the eigenvalues $-\mu n/(n-2q)$ (multiplicity $q-1$, inside the $q$-group), $+\mu n/(n-2q)$ (multiplicity $n-q-1$, inside the other group) and $-\mu$ (along the branch), all to leading order in $\mu$: every one is a saddle, for either sign of $\mu$. For even $n$ the quadratic term vanishes on the $S_{n/2} \times S_{n/2}$ states, which are cubic (at $|\xi| \sim \sqrt{|\mu|}$, on one side of the threshold only). Past the threshold ($\mu > 0$) the class leaves a fixed neighbourhood of the replicated state from almost every initial condition: the split is a jump, not a gradual separation.*

(iii) (who leaves first) *The particles keep their order along $v$ (*Fix *of a transposition is invariant under any $S_n$-equivariant flow, so two particles never cross), so the particle with the largest $\mathrm{sgn}(c_3)\xi_i$ is ahead on that side at all times; in the quadratic truncation its excursion $\zeta = \mathrm{sgn}(c_3)\xi$ obeys $\mathrm{d}\zeta/\,\mathrm{d}s = \mu\zeta + (|c_3|/2)\zeta^2$ up to the common term, with the Riccati solution $s = (1/\mu)\ln[\zeta_1(\mu + |c_3|\zeta_0/2)/(\zeta_0(\mu + |c_3|\zeta_1/2))]$ — the leading behaviour of the departure, which is measured against the class's median. The quadratic term decides once $\zeta$ exceeds $2\mu/|c_3|$: the top-ranked particle then runs away along $\mathrm{sgn}(c_3)v$, alone when its lead is large, with its neighbour as a bound pair when their $\zeta_0$ are nearly tied (within about 11% on the instance). A split sheds the extreme particle, alone or with a near-tied neighbour, because the quadratic term — the residual's, not the kernel's — makes it run away first.*

(iv) (after a departure) *After $k$ particles have left the kernel's range the class has $n-k$, and its margin is $\sigma_0(w_k - k/m) - \lambda^+(r', \vartheta')$, the curvature at the new residual and the remaining particles' raised amplitude (the gauge of Proposition D‴.28: by $n/(n-k)$ when the residual is unchanged): both the stiffness's drop and the curvature's gauge rise favour further departures, and the class stops shedding only if the residual's change lowers $\lambda^+$ by more than $\mu + k\sigma_0/m$ plus that rise. On the instance the residual's change dominates and the margin is positive by the end of each stage; the cascade resumes when $\sigma$ is lowered (Numerics (C), (D)).*

(v) (the mass-free leak) *The mass-free leak $\sigma(m_i - \vartheta_i)$ — the kernel-normalised mean shift, $\sigma/(\sigma_0\bar{\psi}_i)$ times the cohesion force — vanishes at clustered configurations under the finite range as the cohesion leak does and has its blocks with $\sigma$ in place of $\sigma w_k$ on the separations (the class means free), so its thresholds are $\sigma = \lambda_k^+$, lower than the cohesion leak's by the factor $w_k$, and (i)–(iv) hold for it with $\mu = \lambda^+ - \sigma$. It is not a gradient flow, and its forces*

*do not sum to zero. It does not stop the cascade: the curvature's change after a departure is the same.*

*Proof sketch.* (i) The centre manifold theorem for the $S_n$-equivariant flow; the expansion of the learning drift along $v$ at the replicated state (the mean term $(S_2/2m)\langle\Phi''\Phi'\rangle$ is common to the particles and leaves with the projection on $\sum\xi = 0$); $W$ even, so $\nabla^3 W(0) = 0$. (ii) Substituting the isotropy ansatz gives $\mu(n-q)t + (c_3/2)(n-q)(n-2q)t^2 = 0$; the Jacobian $J_{ij} = (\mu + c_3\xi_i)\delta_{ij} - (c_3/n)\xi_j$ on $\{\sum\xi = 0\}$ has the stated eigenvalues on the zero-sum perturbations inside each group and along the branch; on $\mathrm{Fix}(S_{n/2}\times S_{n/2})$ $\sum\xi_i^2$ is constant across the groups' values and the quadratic term is the common one. (iii) Equivariance under the transposition $(i\ j)$ makes $\{\xi_i = \xi_j\}$ invariant; the Riccati solution. (iv) Proposition D‴.32(iv) at the new configuration with Proposition D‴.28's gauge scaling of $R_k$. (v) Inside a class in range $m_i$ is the class mean, so the force is $\sigma(\bar\vartheta_k - \vartheta_i)$, $-\sigma$ on the separations, and nothing on the class mean; across classes $\psi = 0$. □

**Status.** (i)–(ii) proved to leading order in $\mu$ (the centre manifold, the $S_n$-equivariant normal form; checked: the branches by Newton on the full drift, their eigenvalues converging to the predicted ones as $\mu \to 0$); at $\sigma_0/16$ three stable modes are slower than $\mu$, so there the reduced equation's agreement (C) reflects the weak coupling of those modes, not the centre manifold; (iii) the order by equivariance, the escape time in the quadratic truncation (checked); (iv) a statement of the margin, the cascade's extent numerical; (v) the blocks proved, its cascade numerical. Classical: $S_n$-equivariant bifurcation theory — the standard representation's quadratic equivariant makes generic symmetry-breaking from the fully symmetric state transcritical, with no stable small branch (Elmhirst; Golubitsky–Stewart; Arjevani–Field), the jump of sympatric speciation (Stewart–Elmhirst–Cohen, Golubitsky–Stewart), the permutation-symmetric critical points of neural networks (Fukumizu–Amari, Brea et al., Şimşek et al., Arjevani–Field), a cluster's split at a critical parameter in deterministic annealing (Rose), order preservation (equivariance) and the Riccati escape time ([106], [103]). New: the identification for the cohesion leak — the quadratic coefficient the residual's third derivative along the splitting direction, the cohesion and the mean term entering only at third order or in the common direction —, the margin after a departure, and the instance's quantification.

**Numerics** (`exp_split_bifurcation.py` → `results/chaos/split_bifurcation.md`; Proposition D‴.32's resolution instance: the lift of $\vartheta^b(q)$ with the counts $(10, 10, 10)$, $m = 30$, the compact kernel at $R = 1$). (A) At the lift $\lambda^+ = 1.00\cdot10^{-4}$, $2.86\cdot10^{-4}$, $1.08\cdot10^{-4}$ for the three classes, $v$ the centre against the width $((0, +0.75, -0.66)$, $(0, +0.72, -0.70)$, $(0, +0.78, -0.62)$, whitened), $c_3 = 1.42\cdot10^{-5}$, $-1.257\cdot10^{-3}$, $1.20\cdot10^{-4}$ (two difference steps agreeing to $2\cdot10^{-5}$ relative); the middle class splits first, at $\sigma = \lambda^+/w_k = 0.1213\sigma_0$, and leaves on the side $-v$ (its centre lower, its width larger). (B) At $\mu = 2\cdot10^{-6}$ and $5\cdot10^{-7}$ Newton on the full particle drift (90 coordinates, the drift to $10^{-17}$) finds the four split states $q = 1, \dots, 4$ from (ii)'s $t^*$: their positions within 1.3%, 1.8%, 3.5%, 15% of the prediction at $\mu = 2\cdot10^{-6}$ and within 0.3%, 0.4%, 0.8%, 3.0% at $5\cdot10^{-7}$; on the separations along $v$ the eigenvalues are $+1.2544\mu$ ($\times8$) and $-1.094\mu$ for $q = 1$ at $2\cdot10^{-6}$ and $+1.2511\mu$ ($\times8$) and $-1.023\mu$ at $5\cdot10^{-7}$, against $+1.25\mu$ and $-\mu$, and likewise for $q = 2, 3, 4$ (the unstable ones $+1.6708\mu$, $+2.5165\mu$, $+5.1444\mu$ at $5\cdot10^{-7}$ against $5\mu/3$, $5\mu/2$, $5\mu$), the errors falling about fourfold with $\mu$, as the next order says; every split state is a saddle, with $n - q - 1 = 8, 7, 6, 5$ unstable directions of the full Jacobian. (C) From the lift with a jitter $10^{-3}$ (six seeds) at $\mu = 5\cdot10^{-5}$ and $1.39\cdot10^{-4}$ ($\sigma_0/16$, Part G's first split), an excursion being a particle 0.1 from the class's median (far inside the range $R = 1$): the reduced quadratic equation integrated from the measured $\xi(0)$ names the first excursion's particle in 12 of 12 runs and gives its time to 1.1% ($\mu = 5\cdot10^{-5}$: $6.61\cdot10^4$–$8.84\cdot10^4$ slow units; $\sigma_0/16$: $2.68\cdot10^4$–$3.35\cdot10^4$); the linear equation ($c_3 = 0$) names it in 6 of 12 and misses the time by up to 22.4%, the equation with $c_3$'s sign flipped names it in 2 of 12 — the ranking, which every equivariant law preserves, does not name the particle by itself; the quadratic term does. The first excursion is on the side $\mathrm{sgn}(c_3)v$ in 10 of 12 runs; the two on the other side ($\sigma_0/16$,

seeds 2 and 4) are the linear growth of a particle whose excursion stays below $2\mu/|c_3| = 0.22$, and neither leaves the range. After $10^5$ slow units the pieces beyond the kernel's range are, at $\sigma_0/16$, one particle in 3 of 6 runs and a bound pair in 3 of 6 — the top two of $\operatorname{sgn}(c_3)\xi(0)$ within 1.2%, 4.4% and 11% of each other; where the lead was 22% or more the leader went alone —, at 1.5–1.9 (single) and 1.1–1.2 (pairs) from the main piece; at $\mu = 5 \cdot 10^{-5}$ one particle in 3 of 6 runs and none yet in the other 3. The main piece at the clustering threshold (0.25) then has 8 or 9 particles and a positive margin ($7.5 \cdot 10^{-6}$–$1.6 \cdot 10^{-4}$): the cascade stops within the stage, and resumes at the next lower $\sigma$. (D) Part G's schedule ($\sigma$ halved every $10^5$ slow units, the jitter renewed, seeds 0–2): the cohesion leak ends at 8–10 classes at $\sigma_0/64$ (five to seven single particles, the fit error 0.0028–0.0031, Proposition D‴.32 (G)); with the replicator ($\gamma = 0.1$) at 6, 6 and 7, the first escapee's mass growing to 0.105–0.106 (three particles' worth) and the later ones to 0.029–0.060, the fit error 0.0031–0.0032. (E) With the replicator at $\gamma = 1$ the same schedule ends at 5, 6 and 8 classes, the first escapee's mass 0.134–0.136 (four particles' worth), the fit error 0.0029–0.0030: with the replicator the cascade ends at 5–8 classes against 8–10 without, and the rate makes no difference in three seeds; the jitter's seeds are not paired with Part G's. (F) The mass-free leak $\sigma(m_i - \vartheta_i)$ — the separations' stiffness $\sigma$ instead of $\sigma w_k$ — splits later, its threshold $\lambda^+ = 0.040\sigma_0$ instead of $\lambda^+/w_k = 0.121\sigma_0$: at $\sigma_0/64$, where Part G's schedule stops, it has 4 classes in 3 of 3 runs (the fit error 0.0149); continued to $\sigma_0/256$ it sheds as the cohesion leak does, to 9, 10, 9 classes at $\sigma_0/128$ and 9, 8, 9 at $\sigma_0/256$ (four to six single particles, a pair in one run), the fit error 0.0028 — the cohesion leak's end state: its 4 classes were where the schedule stopped, not a count the leak selects.

**Proposition D‴.36** (the gauge under dither: the replicator's masses diffuse along the manifold of exact fits; a leak on the masses pins them)**.** *Let the loop of Proposition D‴.34 sit at a representable exact representation with one particle per class — $K$ classes, the masses $W$, the amplitude coordinates $\alpha_k$ with $W_k a(\alpha_k) = A_k$ —, and dither the positions by $\sqrt{2\tau G}$ per particle (Proposition D‴.32(vi)).*

(i) (the gauge manifold) *The exact fits through the configuration form a manifold $M = \{W_k a(\alpha_k) = A_k,$ the centres and widths fixed$\}$ of dimension $K-1$ inside the representable set, parametrised by the masses: $J$ is constant on $M$, and where $H_L \succ 0$ the flow attracts $M$ exponentially, its tangent $T$ spanned by the gauges $\delta\alpha_k = -A_k\delta W_k/(W_k^2 a_k')$, $\sum \delta W = 0$ ($a' = 1-(a/a_{\max})^2$). With $n_k > 1$ particles in a class the redistributions of its mass among them are further zero modes (Proposition D‴.34's count), $M$ has dimension $m-1$, and $T_\alpha$ below carries the factor $n_k$: the statements here are for one particle per class.*

(ii) (the masses diffuse) *As $\tau \to 0$, on the time scale $1/\tau$, the masses perform a diffusion on $M$ (a Katzenberger limit) whose covariance per unit slow time is*

$$D(W) = (T^\top g T)^{-1} T^\top g \Sigma g T (T^\top g T)^{-1}, \qquad \Sigma = 2\tau G \text{ on the positions},$$
$$g = \operatorname{diag}\big(G^{-1} \text{ on the positions}, \ (m/\gamma)/W_k \text{ on the masses}\big),$$

*the $g$-orthogonal projection of the noise onto $M$ (the linearised gradient flow's limit map), plus a drift of the same order in $\tau$ from the second derivative of that map (computed in Proposition D‴.37). For two classes $D_{11} = 2\tau T_\alpha/(T_\alpha + T_W)^2$, $T_\alpha := \sum_k (A_k/(W_k^2 a_k'))^2/G_\alpha$, $T_W := (m/\gamma)\sum_k 1/W_k$: there is no restoring force in $D$; $D \to 0$ as $\gamma \to 0$ (with equal weights the amplitudes are pinned by the fit) and $D \to 2\tau/T_\alpha$ as $\gamma \to \infty$; at the boundary of the representable set $a_k' \to 0$, so the saturating class's row and column of $D$ vanish quadratically in $W_k - |A_k|/a_{\max}$ (the other masses keep diffusing), while in the amplitude coordinate $\alpha_k \approx -(a_{\max}/2)\log(W_k - |A_k|/a_{\max})$ the diffusion tends to $2\tau G_\alpha$ exactly: heuristically the class's excursion into the saturation is unbounded in $\alpha_k$ — outside the theorem, which holds on compact parts of $M$, and subject to the drift of Proposition D‴.37. The function $\hat{f}$ does not move: the gauge's motion changes the representation's conditioning, not the fit.*

(iii) (a leak on the masses) *The entropic leak on the masses — the energy* $(\varepsilon/\gamma)\sum_i \beta_i \log \beta_i$ *added to* $J$*, so that* $\mathrm{d}\log\beta_i/\,\mathrm{d}s$ *gains* $-\varepsilon(\log\beta_i - \sum_j \beta_j \log\beta_j/\sum_j \beta_j)$ *— is a leak on the unexcited subspace: it does not change* $\hat{f}$ *along* $M$*, so at an exact representation with* $W = w$ *(equal weights) the configuration stays stationary — no bias where* $w$ *is representable — and the gauge becomes an Ornstein–Uhlenbeck direction: linearised at* $W = w$ *the masses' limit on* $M$ *is an Ornstein–Uhlenbeck process with the stiffness* $\kappa = \varepsilon(T^\top g T)^{-1} T_W$ *(*$T_W$ *the masses' share of the metric on* $M$*,* $T^\top \operatorname{diag}(0, (m/\gamma)/W)T$*), so their stationary covariance* $C$ *solves* $\kappa C + C\kappa^\top = D$ *— for two classes* $C = D/(2\kappa)$*,* $\kappa = \varepsilon T_W/(T_\alpha + T_W)$ *— and the spread falls as* $\varepsilon^{-1/2}$*, while the drift term of (ii) shifts the pinned state by* $O(\tau/\varepsilon)$*; where* $w$ *is not representable the leak pulls the masses back toward the saturation trap that the replicator removed (Proposition D‴.34(iii)), a bias of order* $\varepsilon/\gamma$ *— the leak's energy* $(\varepsilon/\gamma)\sum \beta \log \beta$ *against the fit's —, not* $\varepsilon$*.*

*Proof.* (i) On $M$ each amplitude is a function of its class's mass, $\alpha_k = a^{-1}(A_k/W_k)$, so $M$ is a graph over the representable masses; the derivatives of $\hat{f}$ along the complement of $T$ are those of the fit at fixed masses, whose Gram matrix $H_L \succ 0$ by assumption. (ii) Near an attracting manifold of equilibria of a gradient flow $\dot{x} = -g^{-1}\nabla J$, the limit map $\Phi$ (the point of $M$ the flow converges to) has derivative $\partial\Phi =$ the $g$-orthogonal projection onto $T$ (the linearisation $g^{-1}\nabla^2 J$ is $g$-self-adjoint with kernel $T$); Katzenberger's theorem gives the limit $\mathrm{d}Y = \partial\Phi(Y)\sqrt{2\tau}B\,\mathrm{d}\mathcal{W} + \tau\partial^2\Phi(Y)[BB^\top]\,\mathrm{d}s$, $BB^\top = G$ on the positions (so $2\tau BB^\top = \Sigma$); in the masses' coordinates, the coefficient of $\delta W$ along $T$ is $(T^\top g T)^{-1}T^\top g\xi$. For $K = 2$ $T$ is one vector with $\|t\|_g^2 = T_\alpha + T_W$ and $T^\top g\xi = \sum_k t_{\alpha,k}\xi_k/G_\alpha$ of variance $2\tau T_\alpha$. Near the boundary $a_k' \approx 2(1 - |a_k|/a_{\max}) \propto W_k - |A_k|/a_{\max}$. (iii) The leak's term is the Fisher–Rao gradient of $(\varepsilon/\gamma)\sum \beta\log\beta$ on $\sum\beta = m$; at $\beta \equiv 1$ it vanishes; along $M$ $\hat{f}$ is constant, so the leak's energy is the only one that varies there, its Hessian on $T$ being $\operatorname{diag}(\varepsilon m/(\gamma W))$; the reduced flow on $M$ in the metric $T^\top g T$ is then $-(T^\top g T)^{-1}H(W - w)$ to first order, and the linear SDE's stationary covariance is the Lyapunov equation's solution. □

**Status.** (i) proved for one particle per class; (ii) the diffusion coefficient by Katzenberger's theorem on compact parts of $M$ (the drift term in Proposition D‴.37), the vanishing of $D$ at the boundary proved, the excursion in $\alpha_k$ heuristic; (iii) the stationarity, the restored stiffness and the linearised spread proved, the $O(\tau/\varepsilon)$ shift and the $O(\varepsilon/\gamma)$ bias sketched. The time-scale separation is thin on the instance: the slowest normal rate ($3.06 \cdot 10^{-4}$, the second bump's width near its lower bound) is only about $3\tau$, and at $\varepsilon = 10^{-3}$ the leak's stiffness exceeds it; the reduction holds because that mode barely couples to the masses. Classical: diffusions forced onto a manifold by a large drift (Katzenberger), the noise-driven motion along manifolds of minima of overparameterised models and its implicit regularisation (Blanc et al., Li–Wang–Arora) ([107]). New: the gauge of the amplitude–mass representation (Proposition D‴.28) as such a manifold for the replicator of Proposition D‴.34, the masses' diffusion coefficient computed in the flow's metric, the excursion into the saturation, and the entropic leak on the masses as a leak on the unexcited subspace that pins the gauge without bias where the particle fractions are representable.

**Numerics** (`exp_gauge_dither.py` → `results/chaos/gauge_dither.md`; Proposition D‴.32's three-bump instance with one particle per class at the exact representation $\vartheta^e(1/3)$, $m = 3$, $A = (1, -0.7, 0.8)$, $a_{\max} = 4$ — the representable masses $W_k \geq |A_k|/a_{\max} = 0.25$, 0.175, 0.2 —, the replicator at $\gamma$, the positions dithered at $\tau = 10^{-4}$, Euler–Maruyama at $\mathrm{d}t = 0.25$). (A) At $W = 1/3$ the predicted diffusion of the masses along $M$ is $D_{kk} = 2.85 \cdot 10^{-8}$, $1.63 \cdot 10^{-8}$, $1.81 \cdot 10^{-8}$ per unit slow time at $\gamma = 10^{-2}$, rising with $\gamma$ ($4.85 \cdot 10^{-10}$ for the first class at $\gamma = 10^{-3}$, $4.01 \cdot 10^{-7}$ at $\gamma = 1$, $4.23 \cdot 10^{-7}$ at $\gamma = 10$: saturating as $\gamma \to \infty$, as for two classes). (B) Measured over 64 seeds at $\gamma = 10^{-2}$ for $5 \cdot 10^4$ slow units, the variance of the masses' displacement (the mean square less the drift's square) grows linearly ($1.5 \cdot 10^{-4}$ at $5 \cdot 10^3$, $1.4 \cdot 10^{-3}$ at $5 \cdot 10^4$ for the first class), its slope over $[10^4, 5 \cdot 10^4]$ $(2.69 \pm 0.58) \cdot 10^{-8}$,

$(2.21 \pm 0.47) \cdot 10^{-8}$, $(1.62 \pm 0.38) \cdot 10^{-8}$ (bootstrap errors) against the predicted 2.85, 1.63, $1.81 \cdot 10^{-8}$: the ratios 0.95, 1.36, 0.90, within two standard errors ($D$'s variation along the path not included). (C) Over $2 \cdot 10^5$ slow units (10 seeds) the masses spread to standard deviations 0.057, 0.053, 0.052 over the second half (the free diffusion's $\sqrt{Ds}$ at $s = 1.5 \cdot 10^5$: 0.065, 0.049, 0.052, comparable within the 10 seeds' scatter), and the amplitude coordinates run into the saturation: $\alpha_1$ up to 16.1 ($|a|/a_{\max} = 0.9994$), against $[3.58, 4.43]$ and $|a|/a_{\max} \le 0.803$ at $\gamma = 0$ (4 seeds, the masses fixed); the fit error over the second half is the dither's in both (0.0253 with the replicator, 0.0269 without): the gauge's motion costs no fit — it changes only the representation's conditioning. (D) The entropic leak at $\varepsilon = 10^{-4}$ and $10^{-3}$ (4 seeds each, $2 \cdot 10^5$ slow units) pins the masses: their second-half standard deviations 0.0152, 0.0103, 0.0117 and 0.0044, 0.0032, 0.0034 against the Ornstein–Uhlenbeck prediction of (iii) at $W = 1/3$, 0.0139, 0.0099, 0.0105 and 0.0044, 0.0031, 0.0033; four seeds scatter by 12–23%, so only the $\varepsilon = 10\tau$ agreement is informative, and at $\varepsilon = \tau$ the linearisation is not expected to hold (the shift and the spread's variation along $M$ are comparable) — the 16-seed test is Proposition D‴.37 (G). The amplitudes stay inside the range in these four seeds ($|a|/a_{\max} \le 0.89$ and 0.82, against 0.803 at $\gamma = 0$), the maximum growing with the number of seeds; the means $(0.3252, 0.3397, 0.3351)$ and $(0.3329, 0.3338, 0.3333)$ are within 1.5 standard errors of 1/3, too few seeds to resolve the $O(\tau/\varepsilon)$ shift (Proposition D‴.37 (G)). The fit error is the dither's throughout (0.0264, 0.0265). The drift itself is visible in (B): after $5 \cdot 10^4$ slow units the masses' means are $0.3212 \pm 0.0046$, $0.3445 \pm 0.0041$, $0.3342 \pm 0.0038$ (from 1/3), the first two 2.6 and 2.7 standard errors away — the drift Proposition D‴.37 computes, which the diffusion's $\sqrt{Ds}$ (0.038 for the first class) dominates on this horizon.

**Proposition D‴.37** (the dither's drift along the gauge: in the slow-replicator limit the masses descend the log-determinant of the positions' Hessian — toward the saturation —; a boundary layer of width $O(\sqrt{\gamma})$ slows them but does not stop them)**.** *In the setting of Proposition D‴.36 (the replicator at rate $\gamma$, the positions dithered by $\sqrt{2\tau G}$, the manifold $M$ of exact fits parametrised by the masses), write $x = (\Theta, \log\beta)$, $F$ the drift, $A = \partial F$ on $M$ with the eigenpairs $(\lambda_i, e_i)$, completed on the tangent $T$ by the chart's columns, and $P$ the spectral projection onto $T$ along* range $A$ *(for this gradient flow the $g$-orthogonal projection of Proposition D‴.36(ii)).*

(i) (the drift) *The limit map's second derivative at a point of $M$ is*

$$\partial^2\Phi[e_i, e_j] = -P\partial^2 F[e_i, e_j]/(\lambda_i + \lambda_j)$$

*unless both $e_i$, $e_j$ lie in $T$, and $(I-P)\partial^2\chi[c_i, c_j]$ for two tangent vectors $e_i = \partial\chi c_i$, $e_j = \partial\chi c_j$ ($\chi$ the chart); on the time scale $1/\tau$ the masses of Katzenberger's limit move by*

$$\mathrm{d}W = b(W)\,\mathrm{d}s + D(W)^{1/2}\,\mathrm{d}B,$$
$$b = \partial W \cdot \tfrac{1}{2}\partial^2\Phi[\Sigma] + \tfrac{1}{2}\partial^2 W[P\Sigma P^\top], \qquad D = \partial W\, P\Sigma P^\top\, \partial W^\top,$$

*with $\Sigma = 2\tau\,\mathrm{diag}(G, \dots, G, 0)$ the position noise: the drift is of the same order $\tau$ as the diffusion of Proposition D‴.36(ii), so the masses do not merely diffuse.*

(ii) (the slow-replicator limit: an entropic free energy) *Let $\gamma$ be small against the positions' relaxation rates at the exact representation. At fixed masses the dithered positions are a reversible diffusion with the invariant law $\propto \exp(-J(\Theta;\beta)/\tau)$; averaged over it, the masses' fitness is $u_k = \tau\partial_{\beta_k}\log Z(\beta) = -(\tau/2)\partial_{\beta_k}\ell(\beta) + O(\tau^2)$, $\ell(\beta) := \log\det\nabla^2_\Theta J$ at the exact representation with the masses $\beta/m$ (Laplace), so that*

$$\frac{\mathrm{d}W_k}{\mathrm{d}s} = -\frac{\gamma\tau}{2}W_k\Big(\partial_{\beta_k}\ell - \sum_l W_l\partial_{\beta_l}\ell\Big) + O(\gamma^2\tau, \gamma\tau^2), \qquad D = O(\gamma^2\tau)\,:$$

*the masses follow the replicator flow — the Fisher–Rao gradient flow — of the entropic free energy $(\tau/2)\ell$, deterministically on the time scale $1/(\gamma\tau)$ (their spread there $O(\sqrt{\gamma})$), down*

*the free energy: the dither favours the masses whose exact fit is flattest in the positions, not the target's particle fractions $w$. With one particle per class the amplitude's row is the only mass-dependent part of the Hessian, and $\ell$ has the closed form*

$$\ell(W) = 2\sum_k \log(W_k a_k') + \mathit{const} = 2\sum_k \log\left(W_k - (|A_k|/a_{\max})^2/W_k\right) + \mathit{const},$$

*which has no minimiser on the representable masses: it decreases to $-\infty$ at every saturation. Here the Gibbs law is the quasi-stationary law of the positions' local well around the exact representation (the global law $\exp(-J/\tau)$ also charges other representations; the escape from the well is exponentially slow in $1/\tau$). Within a class of coincident particles $\ell$ depends on the class's total mass only — the fit's block is of rank $p$ along the class's $\beta$-weighted mean and its Schur complement is the cohesion's on the separations —, so the slow limit selects the classes' masses and leaves their split among the particles free.*

(iii) (the saturation and its layer: no stop) *As a class saturates ($|a_k| \to a_{\max}$, $W_k \to |A_k|/a_{\max}$) its amplitude's curvature $\propto W_k^2 a_k'^2 \to 0$, so $\ell \to -\infty$ there: the flattest exact fits are saturated, and the free energy's flow drives the masses toward the saturation that the replicator had removed as a trap (Proposition D‴.34(iii)). The limit (ii) needs $\gamma$ small against the saturating amplitude's relaxation rate $mG_\alpha\Gamma_{kk}W_k^2 a_k'^2 \approx 4mG_\alpha\Gamma_{kk}(W_k - |A_k|/a_{\max})^2$ ($\Gamma_{kk} = \langle\varphi_k^2\rangle_\rho$), which falls below the mass channel's $\gamma a_k^2 W_k\Gamma_{kk} \approx \gamma a_{\max}|A_k|\Gamma_{kk}$ within $\delta^* := \frac{1}{2}\sqrt{\gamma a_{\max}|A_k|/(mG_\alpha)} = O(\sqrt{\gamma})$ of the saturation. Inside that layer the residual relaxes through the masses, the amplitude diffuses freely along $M$ with the mass slaved to it ($W_k = A_k/a(\alpha_k)$), and the drift tends to the Itô term of this convex map,*

$$b_k \to \tau G_\alpha \partial_\alpha^2\big(A_k/a(\alpha_k)\big) > 0 \qquad (W_k - |A_k|/a_{\max} \ll \delta^*):$$

*the drift of $W_k$ turns outward and has a zero at a fixed fraction of $\delta^*$, but it is only the Itô term of a driftless diffusion of $\alpha_k$, with the diffusion $2\tau G_\alpha$: in the amplitude coordinate nothing restores, the zero-flux density in $W_k \propto 1/(W_k - |A_k|/a_{\max})$ is not normalisable, and the excursions into the saturation grow (null recurrence in $\alpha_k$). The attracting zeros of the mean drift are therefore not states the masses settle in; without a leak the masses' law drifts into the saturations. Within about $0.1\delta^*$ of a saturation the transverse mass fluctuations are as large as the distance itself, and the description in the coordinates $W$ breaks down.*

(iv) (the leak's bias) *With the entropic leak of Proposition D‴.36(iii) the limit SDE on $M$, $\mathrm{d}W = f(W)\,\mathrm{d}s + D^{1/2}\,\mathrm{d}B$ ($f = b$ + the leak's projected force), has the stationary mean $E[W] = W_0 - J^{-1}\cdot\frac{1}{2}\partial^2 f[C] + \ldots$, $W_0$ the zero of $f$ ($= w + \kappa^{-1}b(w)$ to first order), $J = \partial f(W_0) \approx -\kappa$, $JC + CJ^\top + D = 0$: the bias is $O(\tau/\varepsilon)$ — the drift balance $\kappa^{-1}b(w)$, whose direction is the dither's drift at the target's fractions, and the fluctuations acting through the curvature of $f$, almost all of it the leak's projected force, whose stiffness varies on the scale of the distance to the saturation (of the same order on the instance, Proposition D‴.38(iv)). The expansion needs the spread $O(\sqrt{\tau/\varepsilon})$ small against the scale on which $b$ and $D$ vary, and the leak's stiffness below the slowest normal rate ($3.06\cdot10^{-4}$ on the instance): $\tau \ll \varepsilon \ll 3\cdot10^{-4}$, which the instance barely meets. In the amplitude coordinate the leak's force on a saturating class grows as $\mathrm{d}W_k/\mathrm{d}\alpha_k \to 0$, so the leak does restore what the layer does not.*

*Proof sketch.* (i) Differentiate $\partial\Phi(x)F(x) \equiv 0$ twice at a point of $M$ ($F = 0$ there): $\partial^2\Phi[u, Av] + \partial^2\Phi[v, Au] = -P\partial^2 F[u, v]$; in the eigenbasis this is the stated division by $\lambda_i + \lambda_j \neq 0$; for two tangent vectors $\Phi\circ\chi = \chi$ gives $\partial^2\Phi[\partial\chi c, \partial\chi c'] = (I - P)\partial^2\chi[c, c']$. The limit is Katzenberger's (Proposition D‴.36(ii)), its drift $\frac{1}{2}\partial^2\Phi[\Sigma]$; Itô's formula for $W(x)$. (ii) The positions' drift is $-G\nabla_\Theta J$ and their noise $\sqrt{2\tau G}$, so at fixed $\beta$ they are a reversible diffusion with the Gibbs

law; the masses' law ($\mathfrak{F}$) has $u_k = -\partial_{\beta_k} J$; for $\gamma$ small the averaging principle replaces $u_k$ by its Gibbs mean, which is $\tau\partial_{\beta_k} \log Z$; Laplace's method at the exact representation ($J = 0$ there, the cohesion inactive between the classes) gives $\log Z = -\frac{1}{2}\ell + \text{const} + O(\tau)$. The masses' diffusion is the projection of the positions' noise through the $\gamma$-small cross-mobility, $O(\gamma^2\tau)$ (Proposition D‴.36(ii): $D \to 0$ as $\gamma \to 0$, quadratically). Within a class: in the basis of the class's mean and its separations the fit's block is $(\sum\beta)^2 Q$ on the mean, $(\sum\beta)\beta_\perp^\top \otimes Q$ off it and $\beta_\perp\beta_\perp^\top \otimes Q$ + cohesion on the separations, whose Schur complement is the cohesion's alone. (iii) The amplitude's row of $\nabla^2_\Theta J$ is $\propto W_k a'_k$, so $\det \propto a'^2_k$ and $a'_k \approx 2(W_k - |A_k|/a_{\max})/W_k$; the rates: $\partial^2 J/\partial\alpha_k^2 = m\Gamma_{kk}W_k^2 a'^2_k$ and, through the masses, $\mathrm{d}e_k/\,\mathrm{d}s = -\gamma W_k a_k^2 \Gamma_{kk} e_k$ for the residual's coefficient $e_k = W_k a_k - A_k$; inside the layer the limit map moves along $\alpha_k$, and $W = A/a(\alpha)$ is convex in $\alpha$ ($a = a_{\max}\tanh(\alpha/a_{\max})$), so its Itô term is positive. (iv) The zero $W_0$ of $f = b(W) - \kappa(W)(W - w)$ expanded in $b = O(\tau)$ is the drift balance; stationarity, $E[f(W)] = 0$, expanded at $W_0$ to second order in the spread gives $J(E[W] - W_0) + \frac{1}{2}\partial^2 f[C] = O(|C|^{3/2})$, $C$ the linearisation's stationary covariance ($JC + CJ^\top + D = 0$) — the small-noise expansion of a stationary mean. □

**Status.** (i) proved (given Katzenberger's theorem; checked against the integrated limit map); (ii) the averaging and Laplace steps sketched, the law checked to three digits at $\gamma = 10^{-5}$; (iii) the divergence proved, the layer's scale by the crossing of the rates, the inside law sketched (checked to 0.6%), the non-normalisability inside the layer from the driftless $\alpha$-diffusion (the full loop's excursions growing, (H)); (iv) the drift balance and the second-order stationary mean sketched (the small-noise expansion; its window $\tau \ll \varepsilon \ll 3 \cdot 10^{-4}$ not quantified). Classical: the second derivative of the limit map and its use for the noise-driven drift along a manifold of minima (Katzenberger; Li–Wang–Arora, [107]), averaging and the free energy of fast variables (the potential of mean force), Laplace's method, the preference of noisy gradient dynamics for flat minima ([108]); the log-pseudo-determinant as the drift for isotropic noise is Li–Wang–Arora's ([107]), here met with noise on the positions only and a slow mass channel. New: the replicator's gauge as the slow variable, its free energy in closed form; the consequence that dither drives the masses into the saturation, the layer of width $O(\sqrt{\gamma})$ that slows but does not stop it; the leak's bias.

**Numerics** (`exp_gauge_drift.py` → `results/chaos/gauge_drift.md`; Proposition D‴.36's instance: three bumps, one particle per class at an exact representation, $m = 3$, $\tau = 10^{-4}$, the first class saturating at $W_1 = 0.25$; $\gamma = 10^{-2}$ unless stated). (A) The second-order identities agree with the limit map computed by integration (LSODA to $1.2 \cdot 10^5$ slow units, second differences at $\pm 0.02$ along the nine noise coordinates) to $2.2 \cdot 10^{-5}$ at $W = 1/3$ and $1.1 \cdot 10^{-4}$ at $(0.30, 0.35, 0.35)$. (B) The drift is the amplitudes' noise's: at $W = 1/3$ each class's amplitude noise moves mass out of that class at 5.34, 3.47, $4.07 \cdot 10^{-7}$ per unit time (into the others), the centres' and widths' noise contributing at most $2.2 \cdot 10^{-10}$ ($4.9 \cdot 10^{-10}$ at $(0.30, 0.35, 0.35)$); the sum $(-2.06, +1.58, +0.48) \cdot 10^{-7}$. (C) The slow-replicator limit: the drift over $\gamma\tau$ converges as $\gamma$ decreases and at $\gamma = 10^{-5}$ agrees with the log-determinant's replicator flow $-\frac{1}{2}W_k(\partial_k\ell - \sum_l W_l\partial_l\ell)$ to 0.3% at the four points tested ($(-0.362, +0.242, +0.120)$ at $W = 1/3$ both); at $\gamma = 10^{-2}$ it is 0.07–0.65 of the limit, component by component, at the interior points and of the opposite sign at $W = (0.28, 0.36, 0.36)$, inside the first class's layer. (D) The layer: on $W_2 = W_3$ the first class's drift is positive (out of the saturation) below a zero and negative above it, the zero at $W_1^\circ - 0.25 = 0.038$, 0.019, 0.011, 0.0057, 0.0033 for $\gamma = 10^{-2}$, $3 \cdot 10^{-3}$, $10^{-3}$, $3 \cdot 10^{-4}$, $10^{-4}$: 0.66, 0.61, 0.59, 0.57, 0.57 times $\delta^*$, tending to a fixed fraction ($\approx 0.56$) as $\gamma \to 0$; deep inside the layer ($W_1 - 0.25 = 0.002$–$0.0028$ at $\gamma = 10^{-2}$) the drift is $\tau G_\alpha \partial_\alpha^2 W_1$ to 0.6% (the ratios 1.006, 1.003, 0.999). (E) At $\gamma = 10^{-2}$ the drift field on the representable masses has seven zeros: a repelling node at $(0.376, 0.299, 0.325)$; three saddles, each with one class at its layer's edge ($(0.288, 0.343, 0.369)$, $(0.422, 0.209, 0.370)$, $(0.422, 0.343, 0.235)$); and three attracting nodes with two classes at their layers' edges and the third holding the rest — $(0.284, 0.205, 0.511)$, $(0.284, 0.484, 0.232)$, $(0.562, 0.206, 0.232)$, their rates 1.7–$2.4 \cdot 10^{-5}$ — the indices summing to

3−3+1 = 1, the disc's: attractors of the mean drift, two of the three amplitudes at $|a|/a_{\max} = 0.85$–0.88 there, but not states the masses settle in (iii), as (H) shows. (F) The ensemble mean of Proposition D‴.36's 64 seeds follows the mean path $\mathrm{d}W/\,\mathrm{d}s = b(W)$ within its standard errors at $10^4$, $2\cdot10^4$, $3\cdot10^4$ and $5\cdot10^4$ slow units (at $5\cdot10^4$: $0.3212\pm0.0046$, $0.3445\pm0.0041$, $0.3342\pm0.0038$ against 0.3223, 0.3419, 0.3358), while the spread is the diffusion's (0.038 for the first class). (G) The entropic leak's pinned state (16 seeds, $4\cdot10^5$ slow units, the second half): at $\varepsilon = 3\cdot10^{-4}$ and $10^{-3}$ the masses' mean shift is $(-0.0013\pm0.0004, +0.0011\pm0.0002, +0.0002\pm0.0003)$ and $(-0.0003, +0.0003, 0.0000)\pm0.0001$ against $\kappa^{-1}b(w) = (-0.0009, +0.0007, +0.0003)$ and $(-0.0003, +0.0002, +0.0001)$, within 1–2 standard errors, the spreads within 5% of Proposition D‴.36(iii)'s; at $\varepsilon = \tau = 10^{-4}$ the shift $(-0.0051\pm0.0014, +0.0040\pm0.0007, +0.0011\pm0.0011)$ is about 1.8 times $\kappa^{-1}b$ and the spreads (0.0165, 0.0112, 0.0114) 9–19% above the linear prediction — (iv)'s drift balance alone is approximate for $\varepsilon \gtrsim 3\tau$ and fails at $\varepsilon = \tau$, where the masses' excursions reach where $b$ and $D$ vary; the second component is above $\kappa^{-1}b$ at every $\varepsilon$ (by 2.8, 2.1 and 1.7 standard errors, the same seeds for the three $\varepsilon$, so not independent checks) — (iv)'s fluctuation term: with it 1.7, within 1.03 in every component (Proposition D‴.38's Part G at $\kappa_F = 0$) and 0.8 —, and at $\varepsilon = 10^{-3}$ the leak's stiffness exceeds the slowest normal rate: the window $\tau \ll \varepsilon \ll 3\cdot10^{-4}$ of (iv) is barely met. (H) Without a leak over $10^6$ slow units (12 seeds) the masses leave 1/3 and spread toward the saturations: the fraction of seeds within 0.02 of a class's saturation grows to 0.50, 0.25, 0.33 at $10^6$ (0.17, 0, 0 at $10^5$), the amplitudes reaching $|a|/a_{\max} = 0.9999$, 0.9941, 1.0000; the excursions grow through the run, the seeds' mean of the largest $|\alpha_k|$ by quarters 6.6, 10.1, 10.9, 11.3 for the first class and 4.6, 6.2, 9.4, 11.5 for the third (the fraction with $|\alpha_k| > 10$ reaching 0.26 and 0.28), and in the last quarter the first and third masses are beyond their saturation — off the manifold of exact fits — in 11% and 12% of the records: the layer does not stop them. The limit SDE on $M$ (4000 paths on E's field) gives means within 1.6 standard errors of the 12 seeds' except the third at $10^6$ (2.2), and its near-saturation fractions 0.30, 0.18, 0.26 and spreads 0.10–0.13 against 0.08–0.15; confined to the grid ($W_k \geq |A_k|/a_{\max} + 0.004$), it cannot test the saturation itself.

**Proposition D‴.38** (a Fixman correction for the replicator under dither: the entropic drift cancelled, or reversed toward the D-optimal exact fit)**.** *In the setting of Proposition D‴.37 (one particle per class, the replicator at rate $\gamma$, the positions dithered at the level $\tau$, the manifold $M$ of exact fits with $\ell = \log\det\nabla^2_\Theta J$ along it), add to the replicator's fitness the correction*

$$c_k := \kappa_F\frac{\tau}{2}\,\partial_{\beta_k}\ell(\beta/m) = \kappa_F\frac{\tau}{m}\,\frac{1+\hat W_k^2/W_k^2}{W_k-\hat W_k^2/W_k}, \qquad W_k = \beta_k/m, \quad \hat W_k = |\hat A_k|/a_{\max},$$

*with a gain $\kappa_F \geq 0$ and $\hat A_k$ the classes' contributions to the fit (on $M$, $\hat A_k = W_k a_k = A_k$ and $\hat W_k = W_{\min,k}$): Proposition D‴.37's closed form $\ell(W) = 2\sum_k \log(W_k - W_{\min,k}^2/W_k) + \text{const}$ differentiated in the masses — the masses form. It needs the dither's level, the saturation bound and the classes' contributions, vanishes without dither, acts on the masses only ($\sum\beta = m$ is kept), and with $\hat A$ fixed is the replicator's fitness of the potential $\kappa_F(\tau/2)\ell$, a function of the masses alone, as in Fixman's correction of stiff constraints (a potential on the soft coordinates computed on the constraint manifold). The corrected loop's deterministic part is then the gradient flow of $J - \kappa_F(\tau/2)\ell(W)$ in Proposition D‴.34's metric: $\ell$, which tends to $-\infty$ at every saturation, enters as a log-barrier on the representable masses (the implementation floors the denominator at $0.01W_k$, see (iii)). The plug-in form, $\hat W_k = W_k|a_k|/a_{\max}$ at the current amplitudes, i.e. $\kappa_F(\tau/m)(1+(a_k/a_{\max})^2)/(W_k a'_k)$, $a'_k = 1-(a_k/a_{\max})^2$, equals it on $M$ and depends on the amplitudes; it is $\kappa_F\tau g'_k(\alpha_k)$ with $g_k = a_k/(\beta_k a'_k)$. Since $\partial_{\alpha_k}J = -u_k/g_k$ (the gauge), one integration by parts gives $\langle u_k\rangle = -\tau\langle g'_k\rangle$ in the positions' quasi-stationary law in the exact fit's well at fixed masses (Proposition D‴.40(i); the global Gibbs law has no such identity, $\langle g'_k\rangle$ diverging on the plateau of saturated fits): under that law the plug-in form cancels the slow-limit fitness to every order in $\tau$, the masses form to Laplace's order.*

(i) (the cancellation) *On the slow time scale the correction — an $O(\tau)$ perturbation $F_c$ of the flow — adds $\partial W \cdot P \cdot F_c$ to the drift of Proposition D‴.37(i), $P$ the projection onto the tangent of $M$ along the range of the linearisation, $F_c$ evaluated on $M$ (so (i)–(iv) hold for both forms). $F_c$ is the flow's gradient of $\Phi_c := \kappa_F(\tau/2)\ell$, a function of the masses, so exactly $P \cdot F_c = (T^\top g T)^{-1}\nabla_W \Phi_c$, $T$ the tangent of $M$ in the masses' chart and $g$ the flow's metric. In the slow-replicator limit the masses' own coordinates dominate $T^\top g T$ and $P \cdot F_c$ is the replicator flow ascending $\Phi_c$, so the masses' drift is the replicator flow of the potential $(1-\kappa_F)(\tau/2)\ell$ plus $O(\gamma^2\tau)$, on compact sets of the representable masses outside the layers of Proposition D‴.37(iii): at $\kappa_F = 1$ the entropic drift is cancelled to leading order, and what remains, $O(\gamma^2\tau)$, is of the order of the diffusion $D = O(\gamma^2\tau)$, its direction not controlled by the correction. At finite $\gamma$ the amplitudes' share of $T^\top g T$ grows, and the dither's drift falls below its slow limit faster than the correction's (to 57% and 74% of them at $\gamma = 10^{-2}$, $W = 1/3$): at $\kappa_F = 1$ the correction over-compensates once $\gamma$ is a few $10^{-3}$ — the net drift reversed from $\gamma \approx 2 \cdot 10^{-3}$ on, at 31% of the uncorrected drift's size at $\gamma = 10^{-2}$.*

(ii) (the gain: the D-optimal fit) *For $\kappa_F > 1$ the masses ascend $\ell$. Each term of $\ell$ is strictly concave in $W_k$ on the representable masses and tends to $-\infty$ at the saturation, so $\ell$ has a unique maximiser $W^\sharp$ on the simplex, interior — the exact fit whose positions' Hessian has the largest determinant (D-optimal; the condition number along $M$ is nearly flat and minimised elsewhere), the repelling node of Proposition D‴.37's drift —, and in the slow-replicator limit the masses converge to it, deterministically on the time scale $1/((\kappa_F - 1)\gamma\tau)$, their spread there $O(\sqrt{\gamma/(\kappa_F - 1)})$. With an estimate $\hat\tau$ of the dither's level in the correction the net potential is $(\kappa_F\hat\tau/\tau - 1)(\tau/2)\ell$: the selection needs only $\kappa_F\hat\tau > \tau$ — and $\hat\tau$ can be read from the rms fit error by equipartition, $\hat\tau = m\langle F^2\rangle\langle e^2\rangle/p$ with $p$ the position coordinates holding fit energy (each stores $\tau/2$ of $J = (m/2)\langle r^2\rangle$; the mean of $e$ is biased low by Jensen's inequality).*

(iii) (the saturation) *For the correction as stated (unfloored): inside the layer of Proposition D‴.37(iii), where the amplitude dominates the tangent of $M$ in the flow's metric, the projected correction moves the saturating amplitude at $\mathrm{d}\alpha_k/\mathrm{d}s = G_\alpha \partial_{\alpha_k}\Phi_c = -\kappa_F G_\alpha \tau(1 + (a_k/a_{\max})^2)/a_k \to -2\kappa_F G_\alpha \tau/a_{\max}$. With the amplitude's noise $2\tau G_\alpha$ the zero-flux density of Katzenberger's limit near the saturation behaves like $e^{-2\kappa_F \alpha_k/a_{\max}}$ in the amplitude coordinate and like $\delta^{\kappa_F - 1}$ in $\delta := W_k - W_{\min,k}$: normalisable for every $\kappa_F > 0$, flat at $\kappa_F = 1$ (excursions along $M$ into the layer are then unopposed), Proposition D‴.37's $\delta^{-1}$ at $\kappa_F = 0$; the mean drift descends $(1-\kappa_F)\ell$, so only for $\kappa_F > 1$ is the saturation repelling. The implementation floors the denominator at $0.01 W_k$: within $\delta_{\mathrm{cap}} = W_{\min,k}(1/\sqrt{0.99} - 1)$ (0.00126 for the first class) of the saturation the correction stops growing, its contribution to the exponent falls from $\kappa_F$ to $\kappa_F(1 - W_{\min,k}^2/W_k^2)/0.01 \approx 200\kappa_F\delta/W_{\min,k}$, and the density goes back to $\delta^{-1}$ — the mass near the saturation log-divergent in the $\tau \to 0$ limit, as without the correction. Normalisability needs the unfloored correction, or a floor that shrinks with $\tau$ (at the finite-$\tau$ threshold below, where the description in $W$ ends anyway). At finite $\tau$ the law needs $\delta \gg (\kappa_F\tau/(m a_{\max}^2 \Gamma_{kk}))^{1/2}$ ($\Gamma_{kk} = \langle\varphi_k^2\rangle_\rho$; $0.003\sqrt{\kappa_F}$ for the first class on the instance): there the fit's overshoot, through which the masses relax, leaves $a_k'$ at its value on $M$ and the two forms agree.*

(iv) (with the leak) *With the entropic leak of Proposition D‴.36(iii) the limit SDE on $M$, $\mathrm{d}W = f(W)\,\mathrm{d}s + D^{1/2}\,\mathrm{d}B$ with $f = b_c + P \cdot F_{\mathrm{leak}}$, has the stationary mean*

$$\mathbb{E}[W] = W_0 - J_f^{-1} \cdot \tfrac{1}{2}\partial^2 f[C] + \ldots, \qquad J_f C + C J_f^\top + D = 0,$$

*$W_0$ the zero of $f$ ($= w + \kappa^{-1} b_c(w)$ to first order, $b_c$ the corrected drift, $\kappa$ the leak's stiffness along $M$), $J_f = \partial f(W_0) \approx -\kappa$, $C$ the masses' stationary covariance: the drift balance — $O(\gamma\tau/\varepsilon)$ without the correction, $O(\gamma^2\tau/\varepsilon)$ at $\kappa_F = 1$, toward $W^\sharp$ at $\kappa_F > 1$ — and the*

*fluctuations acting through the curvature of $f$, almost all of it the leak's projected force, whose stiffness varies on the scale $\delta$ of the distance to the saturation through the finite-$\gamma$ weights of (i): $O(C/\delta) = O(\gamma^2\tau/(\varepsilon\delta))$, of the order of the drift balance at the instance (30–70% of it without the correction, larger than it at $\kappa_F = 1$). The window of Proposition D‴.37(iv), $\tau \ll \varepsilon \ll$ the slowest normal rate, keeps the expansion valid.*

(v) (off $M$: the forms differ) *Off $M$ the plug-in form reads the current amplitudes: an amplitude's excursion at fixed masses — the positions' own escape from the exact fit's well toward approximate fits with saturated amplitudes, which the dither reaches at finite $\tau$ (Part F's two escaping seeds) — is read as a mass at its saturation, and the class is fed mass up to the correction's cap: the masses leave the state the leak or the correction holds them in. The masses form does not see the excursion: the masses stay, and the escape remains the positions' own. A running average of the classes' contributions in place of $\hat{A}$ inherits the plug-in form's fault on the time scale of the average.*

*Proof sketch.* The gradient structure: $c_k = \partial_{\beta_k}\Phi_c$ with $\Phi_c = \kappa_F(\tau/2)\ell(\beta/m)$, independent of $\Theta$, so the masses' fitness is $-\partial_\beta(J - \Phi_c)$ and the positions' drift $-G\nabla_\Theta(J - \Phi_c)$ (Proposition D‴.34). (i) Katzenberger's limit with an $O(\tau)$ perturbation of the drift adds $\partial\Phi \cdot F_c = P \cdot F_c$ (Proposition D‴.36(ii)); the linearisation of a gradient flow is self-adjoint in its metric $g$, so $P$ is the $g$-orthogonal projection onto $T$, and with $gF_c = \nabla\Phi_c$, $P \cdot F_c = T(T^\top gT)^{-1}T^\top\nabla\Phi_c = T(T^\top gT)^{-1}\nabla_W\Phi_c$; in the slow-replicator limit $g$'s masses' block, of order $1/\gamma$, dominates $T^\top gT$, and $(T^\top gT)^{-1}\nabla_W\Phi_c \to \gamma W_k(c_k - \sum_l W_l c_l)$, the replicator flow ascending $\Phi_c$, which adds to Proposition D‴.37(ii)'s descent of $(\tau/2)\ell$. The closed form: $W_k a_k' = W_k - W_{\min,k}^2/W_k$ on $M$, and $\partial_{\beta_k} = m^{-1}\partial_{W_k}$. (ii) With $f(W) = \log(W - c/W)$, $c = W_{\min}^2$, $f'' = -2(W^2 + c)/(W^2 - c)^2 + 1/W^2 < 0$ for $W^2 > c$ (equivalent to $W^4 + 4cW^2 - c^2 > 0$); $f \to -\infty$ as $W \downarrow \sqrt{c}$; so $\ell$ restricted to the simplex is strictly concave with an interior maximiser, characterised by equal partial derivatives; the replicator flow ascending a strictly concave function converges to its maximiser (the Shahshahani gradient flow, with $\ell$ as its Lyapunov function). (iii) Inside the layer $|\partial\alpha_k/\partial W_k| = a_k/(W_k a_k') \to \infty$, so the amplitude's part of $T$ dominates $T^\top gT$ and $P \cdot F_c$ moves the configuration along $M$ at $\mathrm{d}\alpha_k/\mathrm{d}s = G_\alpha\partial_{\alpha_k}\Phi_c$, $\Phi_c$ read as a function of $\alpha_k$ on $M$ ($W_k = A_k/a_k$); there $\partial_{\alpha_k}\log(W_k a_k') = a_k''/a_k' - a_k'/a_k = -(1 + (a_k/a_{\max})^2)/a_k$ for $a = a_{\max}\tanh(\alpha/a_{\max})$, which gives the stated drift; a one-dimensional diffusion with constant drift $-v$ and diffusion $2\tau G_\alpha$ has the density $e^{-v\alpha/(\tau G_\alpha)}$, and $\delta \propto e^{-2\alpha_k/a_{\max}}$ near the saturation; with the floor, $c_k$ is capped at $\kappa_F(\tau/m)(1 + W_{\min,k}^2/W_k^2)/(0.01W_k)$, the fraction $(1 - W_{\min,k}^2/W_k^2)/0.01$ of the unfloored value; the finite-$\tau$ condition compares $a_k'(M) \approx 2\delta/W_{\min,k}$ with the overshoot's change of $a_k'$, $\approx 2c_k/(W_k a_{\max}^2\Gamma_{kk})$, $c_k \approx \kappa_F\tau/(m\delta)$. (iv) Stationarity of $\mathbb{E}[W]$: $\mathbb{E}[f(W)] = 0$, expanded at $W_0$ to second order in $W - W_0$. (v) The plug-in form's denominator $W_k a_k'$ vanishes with $a_k'$ at fixed $W_k$; the masses form depends on the masses alone; a running average of $W_k a_k$ over an excursion with $a_k$ near $\pm a_{\max}$ tends to $W_k(\pm a_{\max})$, so its $\hat{W}_k$ tends to $W_k$. □

**Status.** (i) the slow-limit cancellation proved (given Proposition D‴.37(ii) and Katzenberger's theorem), the finite-$\gamma$ formula exact, the $O(\gamma^2\tau)$ remainder numerical; (ii) the concavity, the uniqueness and the slow-limit convergence proved; (iii) the projected drift proved on $M$ (given Katzenberger's theorem), the exponents checked in Katzenberger's limit ($\tau \to 0$) — the finite-$\tau$ statement is untested at the instance's $\tau$ — and the floor's cap checked; (iv) the second-order expansion of the stationary mean (standard), checked on paired seeds to within one standard error; (v) a mechanism, numerical. The correction is the instance's: for one particle per class, through Proposition D‴.37's closed form of $\ell$; for other architectures its general form, $\kappa_F(\tau/2)$ times the derivative of $\log\det\nabla_\Theta^2 J$ along $M$ (through the implicit function $\Theta^*(\beta)$ of the exact fit), is the statement and the closed form its evaluation. Classical ([109]): the Fixman correction — the log-determinant of the constraint metric added to the energy of a system with stiff constraints so that the flexible and the rigid ensembles agree (Fixman), the Brownian dynamics of stiff bonds against rigid constraints (Hinch), the projection of diffusions on submanifolds and the mean force (Ciccotti–Lelièvre–Vanden-Eijnden); the log-determinant of the confining

Hessian as the stiff limit's zero-point term (Helfand; in the overdamped setting Waszkiewicz–Lisicki, the "trimer paradox"); cancelling a free-energy drift along a slow coordinate online — the adaptive biasing force method (Darve–Pohorille; Lelièvre–Rousset–Stoltz) — and a gain on such a bias, well-tempered metadynamics' bias factor (Barducci–Bussi–Parrinello); the log-barrier, its maximiser and the max-det (D-optimal) selection (Sonnevend; Nesterov–Nemirovski; Vandenberghe–Boyd–Wu); noise-induced drifts of the replicator dynamics (Fudenberg–Harris; Mertikopoulos–Viossat). New: the correction for the replicator's gauge — online, in closed form on the masses —, the distinction between its masses form and its plug-in form and the plug-in form's failure off $M$; the selection of the D-optimal fit and the confinement exponent $\delta^{\kappa_F-1}$ are their applications.

**Numerics** (`exp_gauge_fixman.py` → `results/chaos/gauge_fixman.md`; Proposition D‴.36's instance: three bumps, one particle per class at an exact representation, $m = 3$, $\tau = 10^{-4}$; the drifts from `gauge_drift.limit_drift` with the correction as the perturbation $F_1$, evaluated on $M$, where the two forms agree). (A) At $\kappa_F = 1$ the corrected drift over $\gamma\tau$ at $W = 1/3$ is $(+0.063, -0.034, -0.029)$, $(+0.020, -0.010, -0.010)$, $(-0.0007, +0.0006, +0.0001)$ and $(-0.0002, +0.0002, +0.0001)$ at $\gamma = 10^{-2}$, $5 \cdot 10^{-3}$, $10^{-3}$ and $10^{-4}$, against the uncorrected $(-0.206, +0.158, +0.048)$, $(-0.288, +0.203, +0.085)$, $(-0.350, +0.236, +0.115)$ and $(-0.361, +0.241, +0.120)$: the ratio of the largest components 0.31, 0.07, $2.1 \cdot 10^{-3}$ and $6.8 \cdot 10^{-4}$, the remainder $O(\gamma^2\tau)$ at small $\gamma$ (over $\gamma\tau$, $2.4 \cdot 10^{-4}$, $4.5 \cdot 10^{-4}$, $6.1 \cdot 10^{-4}$ at $\gamma = 1$, 2, $3 \cdot 10^{-4}$), reversed from $\gamma = 2 \cdot 10^{-3}$ on (+0.0017 in the first component, against −0.337) — the correction's projected drift at $\gamma = 10^{-2}$ is 74% of its slow limit, the uncorrected one 57% of its own (from the table) —; at $\gamma = 2 \cdot 10^{-2}$ the ratio is 1.6, and at $(0.40, 0.25, 0.35)$ and $(0.36, 0.30, 0.34)$ 1.3 and 0.78, reversed as well. (B) $W^\sharp = (0.37652, 0.29876, 0.32472)$, $|a|/a_{\max} = 0.66$, 0.59, 0.62 there, the D-optimal exact fit — not the best-conditioned: the condition number of $\nabla^2_\Theta J$ is 649.9 at $W^\sharp$ and 648.6 at its minimiser along $M$ $(0.3546, 0.3277, 0.3177)$, nearly flat; at $\kappa_F = 2$ the corrected field's zero lies within $1.8 \cdot 10^{-4}$, $4.4 \cdot 10^{-5}$ and $5.4 \cdot 10^{-6}$ of $W^\sharp$ at $\gamma = 10^{-2}$, $10^{-3}$, $10^{-4}$, attracting, its eigenvalues $-(5.7\text{–}6.8)\gamma\tau$, the stationary spread of the limit SDE linearised there 0.040, 0.037, 0.038 at $\gamma = 10^{-2}$, $\propto \sqrt{\gamma}$ (0.014 at $10^{-3}$, 0.0046 at $10^{-4}$); at $\kappa_F = 1$ the remainder's zero is attracting at $\gamma = 10^{-2}$ ($(0.379, 0.297, 0.324)$, the eigenvalues $-4.5 \cdot 10^{-7}$, $-5.5 \cdot 10^{-7}$) but repelling at $10^{-3}$ and $10^{-4}$ (near $(0.39, 0.29, 0.32)$, the eigenvalues $+16\text{–}19\gamma^2\tau$ and $+29\text{–}30\gamma^2\tau$): the neutral gain leaves the remainder's direction to the higher orders. (C) In Katzenberger's limit ($\tau \to 0$), along $W_2 = W_3$, 0.002–0.004 from the first class's saturation (inside its layer at $\gamma = 10^{-2}$; at the instance's $\tau$ these $\delta$ are below the finite-$\tau$ threshold of (iii)), the exponent $\mathrm{d}\log p / \mathrm{d}\log\delta$ of the one-dimensional zero-flux density is −0.99, −0.00 and +0.98 for $\kappa_F = 0$, 1, 2, against $\kappa_F - 1$. (H) Across the floor's cap ($\delta = 0.0015\text{–}0.0004$; $\delta_{\mathrm{cap}} = 0.00126$) the exponent at $\kappa_F = 2$ falls from +0.99 to −0.22 (+0.99 to +1.15 without the floor), at $\kappa_F = 1$ from +0.00 to −0.53: the unfloored exponent minus $\kappa_F(1 - (1 - W^2_{\min}/W^2)/0.01)$ to $10^{-3}$. (D) The full loop with the plug-in form over $10^6$ slow units at $\gamma = 10^{-2}$, 12 seeds each, with the implemented floor, on the seeds of Proposition D‴.37's Part H ($\kappa_F = 0$): the seeds' mean of the largest $|\alpha_1|$ by quarters is 6.6, 10.1, 10.9, 11.3 without the correction, 5.0, 5.7, 5.1, 4.5 at $\kappa_F = 1$ and 4.5, 4.7, 4.4, 4.1 at $\kappa_F = 2$ (the third class's $4.6 \to 11.5$, $3.7 \to 5.5$ and $3.5 \to 4.3$); the records beyond a saturation in the last quarter 11% and 12% of the first and third classes' without it, none with it, though at $\kappa_F = 1$ 1% of the third class's last-quarter records have $|\alpha_3| > 10$ (the largest over all the records 12.5, against 26.9 without the correction and 7.9 at $\kappa_F = 2$) — excursions along $M$ into its layer, as (iii)'s flat law at $\kappa_F = 1$ allows —; the masses' spread at $10^6$ 0.08–0.15 without, 0.05 at $\kappa_F = 1$, 0.03–0.04 at $\kappa_F = 2$; the second half's mean $(0.3728 \pm 0.0110, 0.3139 \pm 0.0122, 0.3133 \pm 0.0112)$ at $\kappa_F = 1$, within 1.5 standard errors both of $W^\sharp$ and of the deterministic mean path from 1/3 $(0.3563, 0.3196, 0.3241)$ — the run is shorter than that zero's relaxation, so it does not tell them apart —, and $(0.3777 \pm 0.0056$, $0.3041 \pm 0.0053$, $0.3182 \pm 0.0061)$ at $\kappa_F = 2$, within 1.07 standard errors of $W^\sharp$ and of the mean path, which has reached it. The rms fit error over the second half is 0.0267 and 0.0268

with the correction and 0.0253 without, against equipartition's $\sqrt{p\tau/(m\langle F^2\rangle)} = 0.0267$, 0.0252, 0.0236 for $p = 9$, 8, 7 position coordinates holding fit energy: $p_{\text{eff}} = m\langle F^2\rangle\langle e^2\rangle/\tau = 8.96$ and 9.04 with the correction, 8.06 without — a saturated amplitude holds none —, the dither's level read back as $\hat{\tau} = 0.996 \cdot 10^{-4}$ and $1.00 \cdot 10^{-4}$ against $\tau = 10^{-4}$. (E) The masses form ($\hat{A}$ fixed at the initial exact fit, the seeds of D): the seeds' mean of the largest $|\alpha_1|$ by quarters 5.0, 5.7, 5.1, 4.5 at $\kappa_F = 1$ and 4.5, 4.7, 4.3, 4.1 at $\kappa_F = 2$, no record beyond a saturation; at $\kappa_F = 2$ the largest $|\alpha|$ over all the records is 8.1 and the second half's mean ($0.3784 \pm 0.0057$, $0.3030 \pm 0.0056$, $0.3186 \pm 0.0063$) is within 1.0 standard error of $W^\sharp$ (and of the deterministic mean path from 1/3, which has reached $W^\sharp$), the spread at $10^6$ (0.032, 0.042, 0.037) against (B)'s 0.040, 0.037, 0.038; at $\kappa_F = 1$, where (iii)'s law is flat, the third class's amplitude makes excursions along $M$ into its layer ($|\alpha_3|$ up to 13.5, $|a_3|/a_{\max} = 0.998$), and the second half's mean ($0.3746 \pm 0.0115$, $0.3136 \pm 0.0127$, $0.3118 \pm 0.0116$) is within 1.6 standard errors both of $W^\sharp$ and of the mean path $(0.3563, 0.3196, 0.3241)$ — the run is shorter than the $\kappa_F = 1$ zero's relaxation time ($2 \cdot 10^6$ slow units), so it does not tell the two apart: the drift toward $W^\sharp$ there is (i)'s finite-$\gamma$ over-compensation, not the cancellation. Equipartition with the rms fit error: $p_{\text{eff}} = m\langle F^2\rangle\langle e^2\rangle/\tau = 8.92$ and 9.03 with the correction, 8.06 without it (one saturated amplitude holds no fit energy) — the dither's level read back as $\hat{\tau} = 0.99 \cdot 10^{-4}$ and $1.00 \cdot 10^{-4}$ against $\tau = 10^{-4}$; the correction pays the fit energy of the amplitude it keeps coupled. (F) With the entropic leak ($\gamma = 3 \cdot 10^{-3}$, $\varepsilon = 10^{-4}$, 16 seeds, $4 \cdot 10^5$ slow units): the same two seeds leave the exact fit's well — some $|\alpha_k| > 10$ from $1.13 \cdot 10^5$ and $1.27 \cdot 10^5$ slow units on — without the correction and in every form, the fit error there 0.043 against 0.027. The masses stay where the leak holds them without the correction (0.314–0.348 over all the records) and with the masses form at $\kappa_F = 2$ (0.313–0.349); the plug-in form reads the escaped amplitudes as masses at their saturation and moves the masses over 0.21–0.46; the masses form with $\hat{A}$ tracked ($T_A = 10^5$) absorbs the escaped classes' contributions ($\hat{A} = 1.38$, $-0.99$, 1.10 and 1.39, $-1.10$, 1.16 at $4 \cdot 10^5$ against 1, $-0.7$, 0.8) and starts to move them ($W_1$ up to 0.366): $\hat{A}$ is to be taken from a fit, not from a running average. On the 14 seeds without an escape the leak's pinned state is shifted by ($-0.00072 \pm 0.00035$, $+0.00062 \pm 0.00024$, $+0.00010 \pm 0.00034$) without the correction and by ($+0.00145 \pm 0.00032$, $-0.00083 \pm 0.00023$, $-0.00061 \pm 0.00033$) with the masses form at $\kappa_F = 2$ (the plug-in form's alike), against the first order $(-0.00106, +0.00072, +0.00035)$ and $(+0.00111, -0.00073, -0.00037)$ and the second order $(-0.00116, +0.00077, +0.00039)$ and $(+0.00103, -0.00069, -0.00034)$: the reversal (iv) predicts, within 1.1 (first order) and 1.3 (second order) standard errors, the fluctuation term inside the errors at this $\varepsilon$. (G) With the leak inside (iv)'s window ($\gamma = 10^{-2}$, $\varepsilon = 3 \cdot 10^{-4}$; the 16 seeds of Proposition D‴.37's Part G, whose uncorrected runs this repeats bit for bit): the pinned state's shift from 1/3 is ($-0.00129 \pm 0.00039$, $+0.00112 \pm 0.00023$, $+0.00017 \pm 0.00035$) without the correction, ($+0.00001 \pm 0.00037$, $+0.00026 \pm 0.00022$, $-0.00027 \pm 0.00034$) at $\kappa_F = 1$ and ($+0.00120 \pm 0.00035$, $-0.00054 \pm 0.00021$, $-0.00066 \pm 0.00033$) at $\kappa_F = 2$, against the second order $(-0.00134, +0.00089, +0.00045)$, $(-0.00006, +0.00004, +0.00001)$ and $(+0.00112, -0.00075, -0.00038)$, within 1.03 standard errors in every component (the first order $\kappa^{-1}b_c$, $(-0.00092, +0.00066, +0.00027)$, $(+0.00028, -0.00015, -0.00014)$ and $(+0.00149, -0.00095, -0.00054)$, is 1.9–2.1 standard errors off in $W_2$ — D‴.37's excess); the fluctuation term, $(-0.00042, +0.00023, +0.00019)$ without the correction and $(-0.00036, +0.00020, +0.00016)$ at $\kappa_F = 1$, comes from the leak's projected force ($b$'s part $2 \cdot 10^{-5}$, the correction's below $10^{-5}$). So "the bias gone at $\kappa_F = 1$" is a cancellation between the finite-$\gamma$ over-compensation, $\kappa^{-1}b_c = (+0.00028, -0.00015, -0.00014)$, and the fluctuation term, not (i)'s cancellation. On the same seeds the correction's own shift, the difference from $\kappa_F = 0$, is measured to $1$–$4 \cdot 10^{-5}$: $(+0.00129, -0.00086, -0.00044)$ and $(+0.00249, -0.00166, -0.00083)$, against the second order's $(+0.00128, -0.00085, -0.00043)$ and $(+0.00246, -0.00164, -0.00082)$ — within 0.93 standard errors — where the first order $(+0.00121, -0.00081, -0.00040)$ and $(+0.00241, -0.00161, -0.00080)$ is 1.2–4.8 standard errors off and the nonlinear fixed point 2.4–3.8: the paired design resolves the fluctuation term. The

second order also accounts for D‴.37's Part G at its other leaks: at $\varepsilon = 10^{-4}$ and $10^{-3}$ the second component's excess over $\kappa^{-1}b$ falls from 2.8 and 1.7 to 1.7 and 0.8 standard errors.

**Proposition D‴.39** (dissipativity of the dithered gauge: without a prior the dithered loop has no stationary law; the dither reaches saturated approximate fits; a dead zone restores the law, without bias for the exact fits inside it)**.** *In the setting of Proposition D‴.36 — the positions in the unconstrained coordinates $\vartheta = (\alpha, \gamma, \kappa)$ of the saturating maps $a = a_{\max}\tanh(\alpha/a_{\max})$, $c = c_{\max}\tanh(\gamma/c_{\max})$, $s = s_{\min} + (s_{\max} - s_{\min})\,\mathrm{sig}(\kappa)$, dithered at the level $\tau$ ($\mathrm{d}\vartheta_i = -G\partial_{\vartheta_i} J\,\mathrm{d}s + \sqrt{2\tau G}\,\mathrm{d}\mathcal{W}_i$), the cohesion leak, the masses fixed or following the replicator —:*

(i) (no stationary law) *$J$ is bounded on the unconstrained coordinates — the physical coordinates are bounded and the cohesion energy $W \le R^2/8$ —, so at fixed masses the Gibbs measure $e^{-J/\tau}\,\mathrm{d}\Theta$, invariant for the positions' diffusion, has infinite mass for every $\tau > 0$: the positions have no invariant probability law, and the fraction of time they spend in any bounded set of the unconstrained coordinates tends to zero. With the prior leak of Theorems D′–J′, $(\sigma_0/2)\sum_i \|\vartheta_i - \vartheta^0\|^2_{G^{-1}}$, $J$ is coercive and the Gibbs law has Gaussian tails (Theorem J′); the cohesion leak, a function of the differences $\vartheta_i - \vartheta_j$, removed the prior's bias (Proposition D‴.32) and with it this confinement.*

(ii) (the plateau and the escape) *Where a coordinate saturates, $J$ tends to its value at the bound and that coordinate's force vanishes ($\partial J/\partial\alpha_k = a_k'\partial J/\partial a_k$, $a_k' \to 0$): approximate fits with saturated coordinates form plateaus on which the saturated coordinates diffuse without drift. The exact fit's well holds the positions for a time $e^{\Delta J_D/\tau + o(1/\tau)}$ (Freidlin–Wentzell), $\Delta J_D$ the lowest energy above the exact fit on the boundary of a part $D$ of its basin that excludes the plateaus; after that the positions spend a growing fraction of the time on the plateaus, their fit error at the plateau's level and the saturated amplitudes' gain $a_k' \to 0$. On the instance the way out is a monotone climb, without a saddle, of the widths and amplitudes to a plateau $\Delta J \approx 7.7\tau$ ($\tau = 10^{-4}$) above the exact fit.*

(iii) (the dead zone) *Add to $J$ the dead zone $V(\Theta) = (\sigma_B/2)\sum_i \|\vartheta_i - \Pi_B(\vartheta_i)\|^2_{G^{-1}}$, $B = \prod_j [-L_j, L_j]$ a box in the unconstrained coordinates, $\sigma_B > 0$ (the Moreau–Yosida envelope of the box's indicator; in adaptive control a switching $\sigma$-modification) — on the drift $-\sigma_B(\vartheta_i - \Pi_B(\vartheta_i))$, zero inside $B$. Then (a) $e^{-(J+V)/\tau}$ is integrable: at fixed masses the positions have a unique stationary law, and the excursions onto a plateau, still reached at the rate of (ii) where the plateau meets $B$, end; (b) inside $B$ the deterministic objects do not change — the exact fits with their positions in $B$ are stationary without bias, and the manifold $M$, Katzenberger's limit on it and Propositions D‴.36–D‴.38 are unchanged there (a dithered trajectory is the unboxed one until a coordinate leaves $B$) —, but the stationary law is not the well's: it gives the plateau, where it meets $B$, the weight of its Laplace order (on the instance 9–19% of the records at $\tau = 2\cdot 10^{-4}$, Numerics (D)), and there the box's force enters the masses' averaged fitness at the size of the noise's own selection (Proposition D‴.40(i)); (c) the representable masses become $W_k > W_{B,k} := |A_k|/(a_{\max}\tanh(L_\alpha/a_{\max})) > W_{\min,k}$: the gauge's saturation is replaced by the box's wall, where $a_k' = \mathrm{sech}^2(L_\alpha/a_{\max}) > 0$, so $\ell$ stays finite on the representable masses, and the entropic drift of Proposition D‴.37 ends at vertices of the polytope $P_B = \{W_k \ge W_{B,k},\ \sum W_k = 1\}$ — the local minima of the strictly concave $\ell$ on it — instead of running into the saturation; $V$ is $C^{1,1}$ only (its Hessian jumps by $\sigma_B$ on $B$'s faces), so Katzenberger's limit and the entropic drift, which use third derivatives, hold away from a boundary layer at $P_B$'s faces, and the vertex statement is at leading order (a $C^3$ penalty removes the layer); (d) with Proposition D‴.38's correction at $\kappa_F > 1$ the masses go to $W^\sharp$, provided $W^\sharp \in P_B$, i.e. $L_\alpha > a_{\max}\,\mathrm{artanh}(W_{\min,k}/W^\sharp_k)$ for every $k$.*

*Proof sketch.* (i) The image of the saturating maps is bounded and $\hat{f}$ is continuous in the physical coordinates, so the residual's mean square is bounded, and $\omega \le 1/8$ (compact kernel)

or $\le 1$ (Gaussian): $J$ is bounded, $e^{-J/\tau} \ge e^{-\sup J/\tau} > 0$ and $\int e^{-J/\tau}\,\mathrm{d}\Theta = \infty$. The generator $\tau\nabla\cdot(G\nabla) - G\nabla J\cdot\nabla$ is symmetric in $L^2(e^{-J/\tau}\,\mathrm{d}\Theta)$; the drift is bounded (the saturating maps have bounded derivatives, $s \ge s_{\min}$, the cohesion force $\le \sigma_0 R$), so the diffusion does not explode, and it is elliptic. A positive-recurrent elliptic diffusion has a unique invariant measure up to a factor, finite — but $e^{-J/\tau}\,\mathrm{d}\Theta$ is invariant and infinite; so the diffusion is null recurrent or transient, and the occupation fraction of a bounded set tends to zero (the ratio limit theorem against ever larger sets, or a finite occupation time). (ii) On a plateau the saturated coordinate's drift is $O(a_k') = O(e^{-2|\alpha_k|/a_{\max}})$. For the gradient diffusion the quasi-potential is $J - J(\text{exact})$, and Freidlin–Wentzell's exit theorem gives $\tau\log\mathbb{E}[\text{exit time from } D] \to \Delta J_D$ for a domain $D$ in the basin whose closure flows to the exact fit; the path's monotone climb (B) makes the plateau's entrance the boundary's lowest part. (iii) (a) $J + V \ge (\sigma_B/2)\,\mathrm{dist}_{G^{-1}}(\Theta, B^m)^2 - \sup|J|$, integrable against $e^{-\cdot/\tau}$; $J+V$ is a Lyapunov function (its generator $\to -\infty$ outside a compact); uniqueness by ellipticity. (b) $V$ and $\nabla V$ vanish on $B$. (c) An exact fit needs $|a_k| = |A_k|/W_k \le a_{\max}\tanh(L_\alpha/a_{\max})$; below $W_{B,k}$ the fit in $B$ undershoots and the class's fitness $u_k > 0$ feeds it, so $P_B$ is invariant to leading order; $\ell = 2\sum\log(W_k - W_{\min,k}^2/W_k)$ is finite and strictly concave on $P_B$, and a local minimum of a strictly concave function on a polytope is a vertex. (d) Proposition D‴.38(ii) on $P_B$. □

**Status.** (i) proved; (ii) the plateau, its level and the monotone path numerical, the exit time's exponential order Freidlin–Wentzell's (classical), its slope measured at $(0.81 \pm 0.14)\Delta J$ at the instance's $\Delta J/\tau = 2.6$–$5.1$, within 1.3 standard errors of Freidlin–Wentzell's limit; (iii)(a), (b) proved ((b) for the deterministic objects and the dynamics before the first exit from $B$, not for the stationary averages), (c) the polytope's invariance at leading order and the vertices away from the faces' boundary layer (sketched), (d) given Proposition D‴.38(ii). Classical: dissipativity and the existence of invariant laws for Langevin dynamics (Mattingly–Stuart–Higham; Raginsky–Rakhlin–Telgarsky for learning), the exit problem and metastability (Freidlin–Wentzell; Bovier–Eckhoff–Gayrard–Klein), parameter projection and its Lipschitz boundary layer in adaptive control (Ioannou–Sun; Pomet–Praly), Langevin sampling on a convex set through the Moreau–Yosida envelope of its indicator (MYULA: Durmus–Moulines–Pereyra; Brosse–Durmus–Moulines–Pereyra) ([110]). New: that the cohesion leak's removal of the prior's bias also removed the dithered loop's stationary law; the plateau of saturated approximate fits on the instance and the monotone climb to it; the dead zone's effect on the gauge — the saturation replaced by a non-degenerate wall at which the entropic drift ends — and its compatibility with the Fixman correction.

**Numerics** (`exp_gauge_plateau.py` → `results/chaos/gauge_plateau.md`; Proposition D‴.36's instance, energies over $\tau = 10^{-4}$; the box $L = (5,3,4)$ in $(\alpha,\gamma,\kappa)$, $\sigma_B = 1$, so $|a| \le 3.39$ and the representable masses $W_k \ge (0.2947, 0.2063, 0.2358)$ against the saturation's $(0.25, 0.175, 0.2)$). (A) The lowest configurations with one amplitude fixed at $\pm 20$ (200 starts each) lie $7.72\tau$ (the first or third class) and $7.76\tau$ (the second) above the exact fit at $W = 1/3$, their fit error 0.035 — approximate fits with the widths broadened (0.58–0.75 against 0.4–0.5) and two amplitudes up, or the same with the first and third particles exchanged — and $7.64\tau$, $7.71\tau$ at $W^\sharp$; on the escape set's boundary $\max|\alpha_k| = 6$, $7.69\tau$ ($W = 1/3$) and $7.67\tau$ ($W^\sharp$); at the box's edge $7.66\tau$ and $7.63\tau$: the box does not raise the plateau, it bounds it. (B) At $W = 1/3$ the minimum-energy path to the plateau (40 images) climbs monotonically — 0, 1.86, 5.33, 7.17, 7.57, 7.66, $7.70\tau$ at every fourth image —, the second width $0.40 \to 0.75$ and the fit error $0 \to 0.035$ along it, without a saddle; at $W^\sharp$ the path to the lowest plateau (the outer amplitudes saturated, the middle width 1.17) crosses a saddle at $8.21\tau$, the escape boundary's lowest point ($7.67\tau$) lying on the other family. (C) The escapes at fixed masses $W = 1/3$ ($\gamma = 0$, 24 seeds each, up to $4\cdot 10^5$ slow units; an escape when $\max|\alpha_k| > 6$): 10, 19 and 24 of the 24 at $\tau = 1.5$, 2 and $3\cdot 10^{-4}$, the rates $(1.33 \pm 0.42)\cdot 10^{-6}$, $(4.06 \pm 0.93)\cdot 10^{-6}$ and $(1.09 \pm 0.22)\cdot 10^{-5}$ per slow unit, the fit error at the escape 0.046–0.054; $\log(\text{rate})$ against $1/\tau$ has the slope $6.31\cdot 10^{-4}$ (weighted by the escape counts $(6.25 \pm 1.09)\cdot 10^{-4}$) against the

escape boundary's lowest $\Delta J = 7.69 \cdot 10^{-4}$ — $(0.81 \pm 0.14)\Delta J$, within 1.3 standard errors of Freidlin–Wentzell's limit at these barriers ($\Delta J/\tau = 2.6$–$5.1$; the rate times $e^{\Delta J/\tau}$ is 2.25, 1.90 and $1.42 \cdot 10^{-4}$). (D) The dead zone at fixed masses ($\tau = 2 \cdot 10^{-4}$, 16 seeds, $4 \cdot 10^5$ slow units; a record on the plateau when the second width exceeds 0.6): without the box every seed reaches the plateau and the occupation grows — 0.18, 0.40, 0.54, 0.57 of the records by quarters, $\max|\alpha| > 6$ in $0.09 \to 0.56$ of them, the largest $|\alpha|$ $15.0 \to 27.6$, the fit error $0.040 \to 0.045$ —; with the box every seed still reaches it, but the occupation stays at 0.09–0.19, no record has $\max|\alpha| > 6$ ($|\alpha| \le 5.0$, the box's edge), and the fit error stays at 0.039–0.040: the excursions onto the plateau end. (E) The full loop with the box ($\gamma = 10^{-2}$, $\tau = 10^{-4}$, 12 seeds, $10^6$ slow units; $P_B$'s vertices $(0.558, 0.206, 0.236)$, $(0.295, 0.470, 0.236)$, $(0.295, 0.206, 0.499)$): each run is the unboxed one record for record until a coordinate first leaves $B$ (without the correction from $1.7 \cdot 10^4$ to $4.1 \cdot 10^5$ slow units; at $\kappa_F = 2$ from $2.4 \cdot 10^4$ on, one seed never); without the correction the masses' spread at $10^6$ is 0.049, 0.075, 0.057 against 0.145, 0.154, 0.081 without the box, each seed's masses still 0.10–0.18 from the nearest vertex — on their way, at the entropic drift's slow rate; at $\kappa_F = 2$ the masses' mean at $10^6$ ($0.3803 \pm 0.0093$, $0.2961 \pm 0.0122$, $0.3236 \pm 0.0102$) is within 0.41 standard errors of $W^\sharp$, every seed within 0.010–0.080 of it; $|\alpha| \le 5.03$, the fit error 0.026. (F) In Proposition D‴.37's runs without a leak (H) every record with $|\alpha_k| > 10$ but one (3860 of 3861, on 8, 3 and 7 of the 12 seeds) has $W_k$ within 0.01 of the saturation, and the exact fit's error (0.024): the gauge's channel; with the masses held near 1/3 by the leak and no correction (Proposition D‴.38 (F)) every such record has $W_k$ more than 0.05 above it and the fit error 0.041–0.043 (516 and 13 records, on 2 and 1 of the 16 seeds): the positions' own channel, the escape of (ii).

**Proposition D‴.41** (measurement noise on the gauge: no selection in continuous time; per update, the increment's second moment selects)**.** *In the setting of Proposition D‴.36 (one particle per class, the classes beyond the cohesion's range, the replicator at rate $\gamma$, the manifold $M$ of exact fits), let the loop be driven by the measured error: the residual $r$ replaced by $r + \xi$ in both laws — with the desired regressor of Corollary D‴.33 the measurement noise enters through the error alone —, $\xi$ a noise on the orbit's quadrature nodes, of intensity $v$ ($\langle \xi h \rangle_\rho$ has the variance $v\langle h^2 \rangle_\rho$ per unit slow time), and no dither. Unless said otherwise $\xi$ is homoscedastic and independent over the nodes — the image of a measurement noise white on the orbit's resolution (correlation time $\tau_n \ll T/N$). In the coordinates $z = (\Theta, \log \beta)$ the loop is* $\mathrm{d}z = \langle (r + \xi) U \rangle_\rho \, \mathrm{d}s$ *with*

$$U_x := m \mathsf{M} \nabla_z \hat{f}(x) = \big( \beta_i G \nabla_\vartheta \tilde{\Phi}(x; \vartheta_i), \ \gamma(\tilde{\Phi}(x; \vartheta_i) - \hat{f}(x)) \big)_i,$$

$\mathsf{M}$ *the flow's mobility ($G$ on the positions, $\gamma/\beta$ on $\log \beta$, projected on $\sum \beta = m$): the noise enters along the descent's own vector fields. Write $F = \langle rU \rangle_\rho$ for the noiseless flow, $\Phi$ for its limit map, $A = \partial F$ at a point of $M$ (eigenvalues 0 on $TM$ and $-\lambda_i < 0$), $\psi_x := \partial\Phi \cdot U_x$, and $\tilde{\Phi}_k = a_k \Phi_k$ ($\Phi_k$ the $k$-th bump's shape, $a_k = a_{\max} \tanh(\alpha_k / a_{\max})$ its amplitude).*

(i) (no diffusion along $M$) *On $M$ every $U_x$ is orthogonal to $M$ in the flow's metric $g = \mathsf{M}^{-1}$ — $\langle U_x, t \rangle_g = m \partial_t \hat{f}(x) = 0$ for $t$ tangent to $M$ —, so the masses do not diffuse along $M$ at order $v$ (under dither they do, Proposition D‴.36).*

(ii) (no drift in continuous time) *For white noise in Stratonovich's sense — the limit of noise with smooth paths (Wong–Zakai) —, and for a stationary noise with smooth paths and any correlation function in time, the drift along $M$ vanishes at order $v$: Katzenberger's Itô drift $\frac{1}{2}\partial^2\Phi[\Sigma]$, $\Sigma = v\langle UU^\top \rangle_\rho$, and the Wong–Zakai term $(v/2)\partial\Phi\langle (U \cdot \nabla)U \rangle_\rho$ cancel exactly on $M$, and the correlation adds $\langle \nabla\psi \cdot K(A)U \rangle_\rho = 0$ for every function $K$. A loop adapting in continuous time does not select among its exact fits through its measurement noise. Beyond the hypothesis — a variance profile over the nodes, a noise correlated across them (a measurement offset; a real-time noise correlated on the orbit's scale, $\tau_n \gtrsim T/N$), a noise that reaches the adaptation through the loop and arrives correlated over the orbit — the node covariance $C$ weights the bilinear form below and its antisymmetry no longer removes*

*the Stratonovich drift; on this gauge that drift is $O(\gamma)$ relative to its own Itô drift: in the slow-replicator limit the masses' part of $\psi = \partial\Phi \cdot U$ vanishes to $O(\gamma^2)$ in a neighbourhood of $M$ — the masses' noise is the amplitudes' carried along the gauge —, a second mechanism, which several particles per class break (Proposition D‴.42(iv)); the $O(\gamma)$ law is asymptotic ($\gamma \lesssim 10^{-3}$ on the instance, $\gamma \lesssim 10^{-4}$ for the strongly modulated profile $1 + 0.8 \sin t$, whose $O(\gamma)$ coefficient is small: the local exponent 1.4 between $10^{-3}$ and $10^{-4}$), and its constant grows like $1/a'$ toward a saturation.*

(iii) (the selection per update) *With noise drawn afresh at each update of an Euler step $h$ in $z$ (the Itô interpretation: sampled-data adaptation, label-noise gradient descent), the drift along $M$ is $-(v/2)\partial\Phi\langle(U \cdot \nabla)U\rangle_\rho$ up to $O(h\lambda_{\max})$ ((iv) gives the whole factor). For a Euclidean metric this is $-(mv/4)\nabla_M \operatorname{tr} \nabla^2 J$ — label-noise SGD's flat-minimum selection. On the replicator's masses, in the slow-replicator limit, it is the replicator flow of the fitness*

$$\bar{u}_k = -(v/2)\beta_k G_\alpha \langle \Phi_k^2 \rangle_\rho (a_k'^2 - a_k a_k'') = -(v/2)\beta_k G_\alpha \langle \Phi_k^2 \rangle_\rho (1 - q_k^2) = -\partial_{\beta_k} \Psi_\ell,$$
$$q_k = (a_k/a_{\max})^2, \qquad \Psi_\ell(W) = (v/4) \sum_k G_\alpha \langle \Phi_k^2 \rangle_\rho m^2 (W_k - W_{\min,k}^2/W_k)^2$$

*(the amplitudes' part of $(mv/4)\operatorname{tr}(G\nabla^2 J)$; the centres' and widths' parts are constant on $M$), strictly convex in $W$, with a unique minimiser $W_\ell$ on the simplex, interior to the representable masses ($W_{\ell,k} > W_{\min,k}$ for every $k$) — the slow limit's target. The Itô selection depends on the coordinates of the update (the Stratonovich zero of (ii) does not): an Euler step in $z' = \chi(z)$ adds $-(v/2)\partial\Phi\langle\partial^2\chi[U,U]\rangle_\rho$; in the amplitude $a$ itself instead of the network's parameter $\alpha$ the factor $a'^2 - aa''$ becomes $a'^2 - 2aa'' = (1-q)(1+3q)$, the fitness of the strictly convex $(v/2)\sum_k G_\alpha\langle\Phi_k^2\rangle_\rho(\beta_k^2/2 + 2c_k \log\beta_k + 3c_k^2/(2\beta_k^2))$, $c_k = (mA_k/a_{\max})^2$, with an interior minimiser $W_\ell^a$. Either fitness vanishes linearly at the saturation ($q \to 1$), where the dither's diverges (Proposition D‴.37): fresh measurement noise draws the masses away from the saturation, but does not hold them off it against a dither of comparable level.*

(iv) (correlated draws: the increment's second moment selects) *Let the draws be stationary, with step correlations $\rho_d$ and per-step variance $\sigma_\xi^2$ (homoscedastic and independent over the nodes), and the step $h < 2/\lambda_{\max}$. At order $v$ the Euler scheme's drift along $M$ is Katzenberger's drift of the filtered covariance*

$$\Sigma_f = \sum_{il} f_{il} \Sigma_{il} e_i e_l^\top$$

*($\Sigma_{il}$ the components of $\langle UU^\top\rangle_\rho$ in the eigenbasis $(e_i, -\lambda_i)$ of $A$ on the normal space),*

$$f_{il} = 1 - h(\lambda_i R(a_i) + \lambda_l R(a_l)) + h^2 \lambda_i \lambda_l (1 + G(a_i) + G(a_l))/(1 - a_i a_l),$$
$$a = 1 - h\lambda, \qquad G(a) = \sum_{d \geq 1} \rho_d a^d, \qquad R(a) = G(a)/a,$$

*times the per-update variance rate $v_h := h\sigma_\xi^2/N$: the second moment of the whole increment $h(F + \langle U\xi\rangle_\rho)$ — the normal deviation responding to the past draws — over that of the noise. The intensity is $v = v_h \sum_d \rho_{|d|}$ ($((1+\rho)/(1-\rho) \cdot v_h$ for AR(1) noise $\rho_d = \rho^d$). So: (a) if $h\lambda_{\max} \ll 1 - \rho$ ($\lambda_{\max}\tau_c \ll 1$, $\tau_c$ the correlation time) $f \approx 1$ and the selection is $(1-\rho)/(1+\rho)$ times (iii)'s at a fixed intensity — more for anti-correlated noise, $1/k$ of it for a draw kept $k$ steps —; (b) for noise drawn afresh $f_{ii} = 2/(2 - h\lambda_i)$, and on a flat-metric gradient flow with label noise ($\Sigma \propto \nabla^2 J$ on $M$) the selection descends $\operatorname{tr}\Psi(\nabla^2 J)$, $\Psi' = f_{ii}$: here $\Psi(\lambda) = -(2/h)\log(1 - h\lambda/2)$, Damian–Ma–Lee's finite-step regulariser, convex in $\lambda$ (the stiff modes weigh more); (c) for an AR(1) noise — a Markov noise, with white innovations — of correlation time $\tau_c$ sampled finely ($h \ll \tau_c, 1/\lambda_{\max}$), $f_{il} \to (1 + 2x_i x_l/(x_i + x_l))/((1 + x_i)(1 + x_l))$, $x = \lambda\tau_c$ (the active Ornstein–Uhlenbeck particle's filter $1/(1 + \lambda\tau_c)$ on the diagonal), and $\Psi(\lambda) \to \log(1 + \lambda\tau_c)/\tau_c$: the selected*

*potential goes from the trace of the Hessian ($\tau_c \to 0$) to $(1/\tau_c)\log\det(I+\tau_c\nabla^2 J)$ — the log-determinant for a slow noise, which the stiff modes absorb (the residual follows the noise) —, so the noise's colour reweights the normal modes and turns the selection's direction, not only its size. The log-determinant is the Markov noise's: for any stationary noise sampled finely, of normalised spectral density $\hat{s}$, $f(\lambda)=\mathbb{E}[\dot{n}^2]/\sigma_\xi^2=\int \hat{s}(\omega)\omega^2/(\lambda^2+\omega^2)\,\mathrm{d}\omega/2\pi$, the response's velocity over the noise — decreasing in $\lambda$ for every coloured noise, constant only for white noise —, so $\Psi=\int f$ is concave for every colour, and $\operatorname{tr}\Psi(\nabla^2 J)$ is a concave function of the Hessian; a smooth noise (Gaussian step correlations, finite $\mathbb{E}[\dot{\xi}^2]$) has $f\approx(\lambda\tau_c)^{-2}$ for $\lambda\tau_c\gg 1$ and a bounded $\Psi$; (d) at a fixed physical noise the selection vanishes with $h$: (ii); Heun's scheme with the same draw in both stages integrates the ODE driven by the draws' piecewise-constant path, and its $O(v)$ selection is $O((h\lambda)^2)$ of the Euler scheme's, zero only as $h\to 0$; (e)* (a slow Markov measurement noise is a dither) *for label noise $\Sigma=-m\cdot A\mathsf{M}$ on $M$, and an AR(1) noise with $\lambda_{\min}\tau_c\to\infty$, the filter turns it into $\tau_c\Sigma_f\to X_h$, $X_h-BX_hB^\top=2hS$ ($B=I+hA$, $S$ the normal part of $\Sigma$: the pair $(i,l)$ weighted by $2/(\lambda_i+\lambda_l-h\lambda_i\lambda_l)$), and as $h\to 0$ into $-A_D\Sigma Q^\top=m\cdot Q\mathsf{M}Q^\top$ ($A_D$ the group inverse, $Q=I-P$) — the flow's own mobility on the normal space, a thermal noise in the flow's metric: the noise enters through the descent's fields ($\propto A$) and the slow filter divides by $A$ —, so a slow Markov measurement noise selects as a dither of the flow's mobility does: on the replicator's masses, in the slow-replicator limit, as Proposition D‴.37's dither of temperature $\tau_{\mathrm{eff}}=m\cdot v_h/(2\tau_c)$ — the entropic selection toward the saturation, the opposite of the fresh noise's interior target, which Proposition D‴.38's correction at $\tau_{\mathrm{eff}}$ cancels. Its size is small: $\tau_{\mathrm{eff}}\approx mvh/(4\tau_c^2)$ at a fixed intensity. The limit needs $\lambda_{\min}\tau_c\gg 1$, which fails near a saturation, where the softest normal rate vanishes. At a finite replicator rate the selection is multistable instead: from $\tau_c\approx 100$ on three stable zeros, each with one class's mass large, and from $\tau_c=10^3$ on an unstable node in the middle, next to $W^\sharp$ (Numerics (G)).*

(v) (with dither) *The leading-order drifts of independent noises add (Katzenberger's drift is linear in the covariance, the Wong–Zakai terms of independent noises do not mix): with dither and measurement noise the masses' selection is the dither's (Proposition D‴.37) plus (iii)'s, the latter zero in continuous time — so Proposition D‴.38's correction needs only the dither's level — and, by (iii), unable to offset the dither's drive into the saturation near it.*

(vi) (two exact examples) *For the scalar factorisation $f=uv$ fitted to $y$ with label noise, the rescaling symmetry's charge $Q=u^2-v^2$ is conserved by the gradient flow and by the noise's field $(v,u)$, hence pathwise by the Stratonovich dynamics, while an Euler step multiplies it by exactly $1-(\xi_n-r_n)^2h^2$ whatever the noise sequence: $\mathbb{E}[Q]$ decays at $h\cdot\mathbb{E}[(\xi-r)^2]$ per unit time — the per-step variance times $f(h\lambda)$, $\lambda=u^2+v^2$ ((iv)); $\sigma^2$ for noise drawn afresh as $h\to 0$, the Itô SDE's $\mathrm{d}Q=-\sigma^2 Q\,\mathrm{d}t$ exactly. For the diagonal linear network $w=u\odot u-v\odot v$ fitted to $n<d$ samples the products $P_i=u_iv_i$ — the effective initialisation scales — are conserved likewise, and an Euler step multiplies each by exactly $1-4h^2(g_i-\tilde{\xi}_i)^2$ ($g=\nabla_w L$, $\tilde{\xi}=X^\top\xi/n$): the selection — the descent of $\operatorname{tr}\nabla^2 L=(4/n)\sum_i c_i\sqrt{w_i^2+4P_i^2}$ ($c_i=\sum_j X_{ji}^2$), a weighted $\ell^1$ norm $\sum_i c_i|w_i|$ as $P\to 0$ — runs at $4h^2\mathbb{E}[(X_i^\top e)^2]/n^2$ per step ($e=r-\xi$), the per-step variance times $\sum_j f(h\kappa_j)(X_i^\top q_j)^2/c_i$ over the residual's modes $(\kappa_j,q_j)$ of $K=(4/n)X\operatorname{diag}(u^2+v^2)X^\top$: a factor that differs from coordinate to coordinate when the noise is slow. Stratonovich noise does not move the interpolant. On a generic overparametrised network the sharpness $T=\operatorname{tr}(J^\top J)/n$ descends at $-(v_h/4)\nabla T\cdot P_T\nabla\operatorname{tr}\Psi(K)$ ($K=J^\top J/n$, $P_T$ the orthogonal projection onto the manifold's tangent), $-(v/4)|P_T\nabla T|^2$ for fresh noise at small $h$, not at all for Stratonovich noise (checked for $\rho\le 0.9$; slower noise selects below the measurement's resolution there).*

*Proof.* (i) $\langle U_x,t\rangle_g=m(\mathsf{M}\nabla\hat{f}(x))^\top\mathsf{M}^{-1}t=m\nabla\hat{f}(x)\cdot t$, and $\hat{f}$ is constant along $M$. (ii) $\Phi$ is

the limit map of $F = \langle rU\rangle_\rho$ ($r$ vanishes on $M$), $\psi_x = \partial\Phi\cdot U_x$ vanishes on $M$ by (i), $\partial\Phi$ being the $g$-orthogonal projection onto $TM$ there. The invariance $\partial\Phi\cdot F\equiv 0$ reads $\langle r\psi\rangle_\rho\equiv 0$ near $M$; differentiating twice at a point of $M$, where $r$ and $\psi$ vanish, gives $\langle\partial_v r\,\partial_w\psi + \partial_w r\,\partial_v\psi\rangle_\rho = 0$: the bilinear form $B(v,w) := \langle\partial_v r\,\partial_w\psi\rangle_\rho$ is antisymmetric. In Stratonovich's form, Itô's formula gives $\mathrm{d}\Phi(z) = \partial\Phi\cdot\circ\,\mathrm{d}z$, whose drift at order $v$ is $(v/2)\langle U\cdot\nabla\psi\rangle_\rho = -(mv/2)\sum_{ab}\mathsf{M}_{ab}B(e_b,e_a) = -(mv/2)\,\mathrm{tr}(\mathsf{M}B) = 0$, $\mathsf{M}$ symmetric, $B$ antisymmetric; the same sum is Katzenberger's $\frac12\partial^2\Phi[\Sigma]$ plus the Wong–Zakai term $(v/2)\partial\Phi\langle(U\cdot\nabla)U\rangle_\rho$ (the product rule). For a smooth stationary noise of correlation $R$ the linear response $\delta z(s) = \int_0^\infty e^{uA}\langle U\xi(s-u)\rangle_\rho\,\mathrm{d}u$ gives the drift $\langle\nabla\psi\cdot K(A)U\rangle_\rho$, $K(A) = \int_0^\infty R(u)e^{uA}\,\mathrm{d}u$, and on $M$, $A = \langle U\nabla r^\top\rangle_\rho = -\mathsf{M}\nabla^2 J$, so $\langle\nabla\psi\cdot K(A)U\rangle_\rho = -m\,\mathrm{tr}(K(-\mathsf{M}\nabla^2 J)\mathsf{M}\cdot B) = 0$, $K(-\mathsf{M}\nabla^2 J)\mathsf{M} = \mathsf{M}K(-\nabla^2 J\mathsf{M})$ being symmetric. A node covariance $C$ replaces $B$ by $B_C(v,w) = N^{-2}\sum_{xy}C_{xy}\partial_v r_x\partial_w\psi_y$, which the invariance does not make antisymmetric unless $C\propto I$. The second mechanism: in the slow-replicator limit $\partial\Phi$ reads the log-masses' component minus $(\gamma/\beta_k)(a_k/(a_k'G_\alpha))$ times the amplitude's (the $g$-orthogonal projection onto the gauge's tangent, $\delta\alpha_k = -(a_k/a_k')\delta\log\beta_k$), and the fields' two parts are proportional node by node — $U_k^\alpha = \beta_k G_\alpha a_k'\Phi_k$, $U_k^{\log\beta} = \gamma(a_k\Phi_k - \hat f)$ —, and the limit map reads them in that proportion at every point of the basin, not only on $M$: $J$ depends on $\beta_k$ and $\alpha_k$ through $\beta_k a_k$ (one particle per class), so $u_k = -\partial_{\beta_k}J = -g_k\partial_{\alpha_k}J$ with $g_k = a_k/(\beta_k a_k')$, and along the noiseless flow at frozen masses $u_k = g_k\dot\alpha_k/G_\alpha$ — the masses' $O(\gamma)$ displacement until the positions reach the exact fit is exactly $\gamma\Phi_{1,k} = (\gamma/G_\alpha)(\Gamma_k(\alpha_k^*(\beta)) - \Gamma_k(\alpha_k))$, $\Gamma_k' = g_k$, $\alpha_k^*(\beta)$ the exact fit's amplitude at the current masses. Hence $\psi_{x,k}^{\log\beta} = U_{x,k}^{\log\beta} + \gamma\partial\Phi_{1,k}\cdot U_x + O(\gamma^2) = \gamma(a_k\Phi_k - \hat f)(x) - (\gamma/G_\alpha)g_k U_{x,k}^\alpha + O(\gamma^2) = -\gamma\hat f(x) + O(\gamma^2)$ near $M$ ($\partial\alpha_k^*(\beta)$ enters through $U^{\log\beta} = O(\gamma)$), the common part removed by $\sum\beta = m$; every $C$-weighted residual is carried by $\partial\psi$, so on the masses it is $O(\gamma^2)$ against the Itô drift's $O(\gamma)$. With several particles per class $\beta_k a_k$ is no longer what $J$ sees, and the residual stays (Proposition D‴.42(iv)). (iii) The Itô drift is (ii)'s sum minus the Wong–Zakai term. On the masses $\langle(U\cdot\nabla)U\rangle_\rho$ has the log-mass component $\gamma\beta_k\langle\|\nabla_\vartheta\tilde\Phi_k\|_G^2\rangle_\rho$ and the amplitude component $(\beta_k^2 G_\alpha/2)\partial_{\alpha_k}\langle\|\nabla_\vartheta\tilde\Phi_k\|_G^2\rangle_\rho + O(\gamma)$, and with $\langle\|\nabla_\vartheta\tilde\Phi_k\|_G^2\rangle_\rho = G_\alpha a_k'^2 P_k + a_k^2 Q_k$ ($P_k = \langle\Phi_k^2\rangle_\rho$, $Q_k$ the centre's and width's part, independent of $\alpha_k$) the combination read by $\partial\Phi$ is $\gamma\beta_k G_\alpha P_k(a_k'^2 - a_k a_k'') = \gamma\beta_k G_\alpha P_k(1-q_k^2)$ for $a = a_{\max}\tanh(\alpha/a_{\max})$ ($a' = 1-q$, $aa'' = -2q(1-q)$); $\bar u_k = -\partial_{\beta_k}\Psi_\ell$ since $\partial_\beta(\beta a')^2 = 2\beta(1-q^2)$ on $M$ ($\beta a' = \beta - m^2A_k^2/(a_{\max}^2\beta)$). Each term of $\Psi_\ell$, $h(W) = (W - c/W)^2$, has $h'' = 2 + 6c^2/W^4 > 0$ and $h' = 0$ at $W = \sqrt{c} = W_{\min,k}$, increasing beyond: the minimiser on the simplex has equal $P_k h_k' > 0$, so $W_{\ell,k} > W_{\min,k}$. The coordinates: an Euler step in $z' = \chi(z)$ is $z' + D\chi\,\Delta z + \frac12\partial^2\chi[\Delta z,\Delta z] + \dots$, so the scheme's Itô drift in $z'$ carries $\frac12 v\langle\partial^2\chi[U,U]\rangle_\rho$ besides the pushed-forward one; for $\chi\colon\alpha\mapsto a$ the amplitude component gains $a''\langle(U_k^\alpha)^2\rangle_\rho = \beta_k^2 G_\alpha^2 a_k'^2 a_k'' P_k$, which the projection's amplitude weight $(\gamma/\beta_k)(a_k/(a_k'^2G_\alpha))$ turns into $-\gamma\beta_k G_\alpha P_k a_k a_k''$: the factor $a'^2 - 2aa'' = (1-q)(1+3q) = 1+2q-3q^2$; with $q = c/\beta^2$ it is $-(2/(vG_\alpha P))\partial_\beta$ of $\beta^2/2 + 2c\log\beta + 3c^2/(2\beta^2)$, whose second derivative $1-2q+9q^2$ is positive. At the saturation both factors vanish like $2(1-q)$ and $4(1-q)$, the dither's fitness like $1/(1-q)$. (iv) For the Euler step $z_{n+1} = z_n + h(F(z_n) + \langle U(z_n)\xi_n\rangle)$, with $\partial\Phi\cdot F\equiv 0$,

$$\Phi(z_{n+1}) - \Phi(z_n) = h\langle\psi(z_n)\xi_n\rangle + (h^2/2)\partial^2\Phi[F + \langle U\xi_n\rangle, F + \langle U\xi_n\rangle] + O(h^3),$$

and at linear response the normal deviation is $n_k = h\sum_{j\ge1}(I+hA)^{j-1}\langle U\xi_{k-j}\rangle$ ($F = An + O(n^2)$; the quadratic parts of $F$ and $U$, the mean shift of $n$ and $\partial^2\Phi$'s dependence on the point enter the mean through third moments only). The first term's mean is $h^2\sigma_\xi^2 N^{-1}\langle\nabla\psi\cdot K_h U\rangle_\rho$, $K_h = \sum_{d\ge1}\rho_d(I+hA)^{d-1}$, zero by (ii)'s identity ($K_h\mathsf{M}$ symmetric). In the second (the draws' third moments entering at $O(v^{3/2})$, Gaussian or not), the cross term $h^2\partial^2\Phi[An,\langle U\xi\rangle]$ — $O(h^2|F||\xi|)$, whose mean vanishes only for independent draws — and the relaxation term $(h^2/2)\partial^2\Phi[An,An]$ give, in $A$'s eigenbasis, $\mathbb{E}[n^i\xi] = h\sigma_\xi^2 R(a_i)U^i/N$ and $\mathbb{E}[n^i n^l] = h^2\sigma_\xi^2 Q_{il}\langle U^i U^l\rangle/N$ with $Q_{il} = \sum_{j,j'\ge1}a_i^{j-1}a_l^{j'-1}\rho_{|j-j'|} = (1+G(a_i)+G(a_l))/(1-a_ia_l)$: the pair $(i,l)$ carries $f_{il}$, and the mean per unit time is $\frac12\partial^2\Phi[v_h\Sigma_f]$. (a) $f_{il} = 1 + O(h\lambda/(1-\rho))$. (b) With $\rho_d = 0$, $f_{ii} = 1 + h^2\lambda^2/(1-a^2) =$

$2/(2-h\lambda)$. For a flat metric and $\Sigma \propto \nabla^2 J =: H$ on $M$ (label noise: $\langle \nabla \hat{f} \nabla \hat{f}^\top \rangle_\rho$ is the Gauss–Newton matrix), $\Sigma$ is diagonal in $H$'s eigenbasis, $\Sigma_f = \varphi(H)H = HX + XH$ with $X = \frac{1}{2}\varphi(H)$ on the normal space, $\varphi(\lambda) = f_{ii}$; differentiating $\partial\Phi \cdot F \equiv 0$ twice at a point of $M$ gives $\partial^2\Phi[AX + XA^\top] = -\partial\Phi \cdot \partial^2 F[X]$, and $\partial^2 F[X] = -\nabla \operatorname{tr}(X\nabla^2 J)|_X = -\frac{1}{2}\nabla \operatorname{tr}\Psi(H)$ ($\mathrm{d}\operatorname{tr}\Psi(H) = \operatorname{tr}(\varphi(H)\,\mathrm{d}H)$; for the Gauss–Newton form $H = JJ^\top$ the kernel block does not contribute, $J^\top P_{\ker} = 0$): the drift is $-(v_h/4)P_T \nabla \operatorname{tr}\Psi(H)$. (c) $hR(a) \to \tau_c/(1+\lambda\tau_c)$ and $h^2 Q_{il} \to \tau_c^2(\lambda_i + \lambda_l + 2/\tau_c)/((\lambda_i+\lambda_l)(1+\lambda_i\tau_c)(1+\lambda_l\tau_c))$ as $h \to 0$ at fixed $\tau_c = -h/\log\rho$, and $f_{il}$ follows; $f_{ii} \to 1/(1+\lambda\tau_c)$, $\Psi \to \log(1+\lambda\tau_c)/\tau_c$. For a general finely sampled stationary noise the mode obeys $\dot{n} = -\lambda n + \xi$ at linear response, and the increment per step is $h(\xi - \lambda n) = h\dot{n}$, so $f = \mathbb{E}[\dot{n}^2]/\mathbb{E}[\xi^2] = \int S(\omega)\omega^2/(\lambda^2+\omega^2)\,\mathrm{d}\omega / \int S(\omega)\,\mathrm{d}\omega$ ($S$ the spectral density): each $\omega^2/(\lambda^2+\omega^2)$ decreases in $\lambda$, so $f$ does, and $\Psi'' = f' \le 0$; trace functions of concave functions are concave on the symmetric matrices. For $S \propto (1+\omega^2\tau_c^2)^{-1}$ this is $1/(1+\lambda\tau_c)$; when $\mathbb{E}[\dot{\xi}^2] < \infty$, $f \approx \mathbb{E}[\dot{\xi}^2]/(\lambda^2\sigma_\xi^2)$ for large $\lambda$, and $\Psi(\infty) = \int_0^\infty f = (\pi/2)\int S(\omega)|\omega|\,\mathrm{d}\omega / \int S(\omega)\,\mathrm{d}\omega = (\pi/2)\langle|\omega|\rangle$ is finite ($\sqrt{\pi/2}/\tau_c$ for Gaussian correlations). (d) $v_h = (1-\rho)/(1+\rho)\cdot v \to hv/(2\tau_c)$. Heun: in (vi)'s scalar example one step multiplies $Q$ by exactly $1 - (h^2/4)(c-\tilde{c})^2 + (h^4/4)c^2\tilde{c}^2$ ($c$, $\tilde{c}$ the error at the step's start and at the predictor, the same draw), and $c - \tilde{c} = h\lambda c + O(h^2)$, $\lambda = u^2 + v^2$: the mean per step is $(h\lambda)^2/4$ times the Euler step's $h^2\mathbb{E}[c^2]$ at leading order. (e) As $\rho \to 1$ at fixed $h$, with $\varepsilon = 1-\rho$: $h\lambda R(a) = 1 - \varepsilon/(h\lambda) + O(\varepsilon^2)$ and $G(a) = aR(a)$, so $f_{il} = 2\varepsilon/(x_i + x_l - x_i x_l) + O(\varepsilon^2)$ ($x = h\lambda$) and $\tau_c f_{il} \to 2/(\lambda_i + \lambda_l - h\lambda_i\lambda_l)$, which is $X_h - BX_hB^\top = 2hS$ in $A$'s eigenbasis; as $h \to 0$, $2/(\lambda_i+\lambda_l)$, the Lyapunov solution of $AX + XA^\top = -2S$, which for $S = -mA\mathsf{M}$ is $m \cdot Q\mathsf{M}Q^\top$. On $M$, $\nabla^2 J = m\langle \nabla\hat{f}\nabla\hat{f}^\top\rangle_\rho$ (the cohesion's Hessian vanishing there), so $\Sigma = m^2\mathsf{M}\langle\nabla\hat{f}\nabla\hat{f}^\top\rangle_\rho\mathsf{M} = m\mathsf{M}\nabla^2 J\,\mathsf{M} = -m\cdot A\mathsf{M}$, diagonal in a $g$-orthonormal eigenbasis $V$ of $A$ ($\Sigma = -mV\Lambda V^\top$, $\mathsf{M} = VV^\top$); $\Sigma_f = -mV\varphi(\Lambda)\Lambda V^\top$, and $\varphi(\lambda)\lambda \to 1/\tau_c$ on the normal modes: $\Sigma_f \to (m/\tau_c)V_N V_N^\top = (m/\tau_c)Q\mathsf{M}Q^\top$. In the slow-replicator limit $\mathsf{M}$'s log-mass block $\gamma/\beta$ vanishes, so $v_h\Sigma_f \to (mv_h/\tau_c)Q\cdot\operatorname{diag}(G,0)\cdot Q^\top$, the normal part of the dither $2\tau_{\text{eff}}G$ with $2\tau_{\text{eff}} = mv_h/\tau_c$, whose drift is the dither's there (the tangent parts' contribution vanishing at $\gamma \to 0$). (v) Linearity of $\frac{1}{2}\partial^2\Phi[\Sigma]$ in $\Sigma$; the Wong–Zakai term is a sum over the independent noises' own fields. (vi) $\nabla Q\cdot(v,u) = 2uv - 2vu = 0$; $(u+hcv)^2 - (v+hcu)^2 = (u^2-v^2)(1-h^2c^2)$ with $c = \xi - r$, and $c$'s residual part obeys $r_{n+1} = (1-h\lambda)r_n + h\lambda\xi_n$; likewise $(u_i - 2hgu_i)(v_i + 2hgv_i) = u_iv_i(1-4h^2g^2)$ with $g = g_i - \tilde{\xi}_i = X_i^\top e/n$, the residual obeying $r_{n+1} = (I-hK)r_n + hK\xi_n$; the noise fields $(2x\odot u, -2x\odot v)$ conserve every $u_iv_i$. The generic network: (iv)(b) with $T$'s gradient. □

**Status.** (i) proved; (ii) proved at order $v$ (given Katzenberger's theorem and, for white noise, the Wong–Zakai theorem; the smooth-noise case by linear response), the residual's $O(\gamma)$ relative order beyond the hypothesis proved at leading order in $\gamma$ (one particle per class; the asymptotic range numerical), its size numerical; (iii) proved at leading order (the slow-replicator limit for the closed forms); (iv) proved at order $v$ and at linear response, for any step $h < 2/\lambda_{\max}$ and any stationary correlation (the $O(h\lambda)$ terms included; Gaussian draws — the draws' third moments enter at $O(v^{3/2})$), (a)–(e) corollaries ((c)'s concavity for any colour proved in the continuum limit; (e)'s finite-step limit proved, its identity with the dither's direction proved in the slow-replicator limit away from the saturations and numerical at $\gamma = 10^{-2}$; the zero set at finite $\gamma$ and $\tau_c$ numerical); (v) a corollary; (vi) the examples' identities exact per step, their rates by (iv), the generic network's law at leading order. Classical: Katzenberger's theorem and its use for the noise-driven drift along a manifold of minimisers (Li–Wang–Arora, [107]); label-noise SGD's selection of flat minima — Blanc–Gupta–Valiant–Valiant, Damian–Ma–Lee (with its finite-step regulariser $-(1/2\eta)\operatorname{tr}\log(I - \eta\nabla^2 L/2)$), Li–Wang–Arora, HaoChen–Wei–Lee–Ma, Shalova–Schlichting–Peletier —, all with noise drawn independently at each step, the regulariser's scaling with step × variance among them; the Itô–Stratonovich distinction, the Wong–Zakai theorem and noise-induced drifts (Wong–Zakai, van Kampen, Horsthemke–Lefever), and the crossover between Itô's and Stratonovich's limits set by the ratio of the noise's correlation time to the relaxation time (Kupferman–Pavliotis–Stuart, Pavliotis–Stuart); SDE models of SGD in Itô's form (Li–Tai–E, Li–Malladi–Arora); anti-correlated parameter

noise as a regulariser of the Hessian's trace (Orvieto et al.; perturbed gradients, a different mechanism), correlated parameter noise more generally (Lucchi–Proske–Orvieto–Bach–Kersting), the anti-correlation of epoch-based minibatch noise, which lowers the weight variance along flat directions (Kühn–Rosenow), SGD without replacement's weighted-trace regulariser with a Hessian-function weight (Beneventano), and a finite-learning-rate Hessian-dependent regulariser of anti-correlated noise with a trade-off set by the correlation time (Kim et al., an abstract, flagged); the continuum filter itself is the active Ornstein–Uhlenbeck particle's effective temperature $T/(1+k\tau)$ in a trap of stiffness $k$, and the factor $\det(I+\tau\nabla^2 U)$ of the unified coloured-noise approximation's stationary law (Jung–Hänggi, Hänggi–Jung, Maggi et al.); the isotropic noise's log-determinant drift along a manifold of minima (Li–Wang–Arora) is (e)'s target; the law of balance of rescaling symmetries under SGD (Ziyin–Li–Ueda) and the noise equilibria of parameter symmetries (Ziyin–Wang–Li–Wu), of which (vi)'s Euler identities are exact discrete cases, the Stratonovich conservation following from their Itô derivation; the diagonal network's label-noise Lasso (Pillaud-Vivien–Reygner–Flammarion, in continuous time and Itô's reading) and its effective initialisation scale (Woodworth et al.) ([111]). New (we did not find it stated): the exact cancellation for noise entering through the error, for any temporal spectrum, by the antisymmetry the limit map's invariance imposes — a loop adapting in continuous time does not select through its measurement noise —; for correlated draws the coloured filter of label noise's drift along $M$ — $f_{il}$, the cross term's mean vanishing by the same antisymmetry, Damian–Ma–Lee's regulariser its white case — and its consequences: the passage from the trace to $(1/\tau_c)\log\det(I+\tau_c\nabla^2 J)$ for a slow Markov noise, a concave selected potential for every colour, a slow Markov measurement noise selecting as a dither of the flow's own mobility does; the closed forms on the replicator's masses, with their dependence on the update's coordinates, and the selection's multistability on the gauge at finite rates.

**Numerics** (`exp_gauge_label.py` → `results/chaos/gauge_label.md`). (A) At three points of $M$ and $\gamma = 10^{-2}$, $10^{-5}$ the Itô drift and the Wong–Zakai term cancel to $4.6\cdot10^{-9}$–$1.1\cdot10^{-8}$ of the Itô drift (the finite differences' floor), the diffusion along $M$ is at most $2.1\cdot10^{-20}$, and the closed form of (iii) matches the Itô drift to 2–$4\cdot10^{-4}$ at $\gamma = 10^{-5}$ (23–35% at $\gamma = 10^{-2}$: its $O(\gamma)$ correction). (A2) Beyond the hypothesis the Stratonovich drift over its own Itô drift is $O(\gamma)$ from $\gamma \approx 10^{-3}$ on (an offset: $2.4\cdot10^{-3}$, $2.4\cdot10^{-4}$, $2.4\cdot10^{-5}$ at $\gamma = 10^{-3}$, $10^{-4}$, $10^{-5}$ at $W_0$; a noise correlated along the orbit, $\tau_n/T = 1.6$: $2.4\cdot10^{-3}$ … $2.4\cdot10^{-5}$; the profile $1+0.8\sin t$: $7.2\cdot10^{-7}$ at $\gamma = 10^{-4}$, $6\cdot10^{-8}$ at $10^{-5}$) and 0.2–12% at $\gamma = 10^{-1}$, where the law is not yet asymptotic; for white homoscedastic noise it is $3\cdot10^{-9}$–$1.3\cdot10^{-8}$ at every $\gamma$; the coloured-noise identity holds to 1.3–$2.5\cdot10^{-8}$ and the one-sided kernel's Lévy-area part is below $4\cdot10^{-13}$. (B) $W_\ell = (0.3163, 0.4115, 0.2722)$, interior ($|a_k|/a_{\max}$ 0.790, 0.425, 0.735), far from $W^\sharp = (0.3765, 0.2988, 0.3247)$; at $\gamma = 10^{-2}$ the Itô drift vanishes at $(0.3221, 0.3999, 0.2780)$; the update in $a$: Katzenberger's drift within 1.9–$3.4\cdot10^{-4}$ of the closed form with $(1-q)(1+3q)$, the target $W_\ell^a = (0.2967, 0.4513, 0.2520)$. (C) The full loop at $\gamma = 10^{-2}$, $v = 5\cdot10^{-5}$ (4 seeds × $2\cdot10^5$ slow units): the gauge coordinate's slope over the linear-response prediction averaged along the seeds' mean path is 0.996–1.008 for noise drawn afresh (steps 0.25 and 0.125) and for AR(1) draws at $\rho = -1/2$, $-1/3$, $1/3$, 0.6 (standard errors 0.001–0.005), 0.996–1.014 at $\rho = 0.8$ (± 0.005–0.012) and 0.997–1.022 at $\rho = 0.9$ (± 0.012–0.030), 1.004–1.027 at $\rho = 0.95$ (± 0.02–0.07); the prediction at $W_0$ alone is off by up to 2.2% on $W_3$ where the errors are small (2.7% at $\rho = 0.95$, ± 6.6%; the path moves it); the per-step law is rejected at $\rho = 0.9$ (the ratio to the Itô drift $0.0462 \pm 0.0006$ against $(1-\rho)/(1+\rho) = 0.0526$) and 0.95 ($0.0197 \pm 0.0005$ against 0.0256); Heun's scheme moves the gauge by $(2 \pm 7)\cdot10^{-4}$ of the Itô drift, with fresh or AR(1) draws. (C2) At the step 8 ($h\lambda_{\max} \approx 1.6$; 8 seeds × $3.2\cdot10^5$): noise drawn afresh $0.988 \pm 0.007$, $0.996 \pm 0.004$, $1.020 \pm 0.032$ of the path-averaged prediction, $\rho = 1/2$ $0.988 \pm 0.009$, $0.998 \pm 0.004$, $1.020 \pm 0.027$, $\rho = -1/2$ $0.979 \pm 0.010$, $0.992 \pm 0.004$, $1.049 \pm 0.053$ — where the step-free Itô drift is off by 42% (the ratio $1.417 \pm 0.005$). (D) $f = uv$: $\mathbb{E}[Q]$'s decay rate is within 0.3% of the linear-response prediction for every scheme (1.002 against 1.002 for fresh draws, $9.586\cdot10^{-5}$ against $9.557\cdot10^{-5}$ at $\rho = 0.999$, where it is 0.19 of the per-step law), and Heun's scheme keeps every path's $Q$ to

$1.2 \cdot 10^{-5}$. (E) The diagonal network: the products' decay rates over the prediction 0.9988–0.9998 on average (0.977–1.024 per coordinate at $\rho = 0.999$), the prediction's spread across coordinates growing with the colour (1.002 for fresh draws, 1.052 at $\rho = 0.9$, 1.434 at 0.99, 2.169 at 0.999); Heun's scheme with fresh draws decays them at $2.3 \cdot 10^{-5}$ of the Euler step's rate, its $O((h\lambda)^2)$ residue. (F) The tanh network: the sharpness's slope over the linear-response prediction 1.000 (fresh), 0.990 ($\rho = -1/3$), 1.006 (1/3), 1.014 (0.6), 0.996 (kept 4 steps), 1.060 (0.9); at $\rho = 0.99$ and 0.999 the predictions lie below the measurement's floor (Heun's slopes, $1.9 \cdot 10^{-6}$). (G) The zero set of the per-update selection at the step 0.25: at $\gamma = 10^{-2}$, for fresh draws and $\tau_c = 10$ one interior stable zero ((0.3222, 0.3997, 0.2781), (0.3134, 0.4201, 0.2665)); at $\tau_c = 100$ three stable zeros, each with one class's mass large ((0.2779, 0.4949, 0.2272), (0.2821, 0.2260, 0.4919), (0.5419, 0.2266, 0.2315)) and two saddles; from $\tau_c = 10^3$ on three stable zeros, three saddles and an unstable node at (0.3773, 0.2955, 0.3271), next to $W^\sharp$; $W_0$'s flow ends on the continued branch at $\tau_c = 100$ and from $\tau_c = 10^3$ on at the zero with class 1's mass large ((0.5905, 0.1927, 0.2169) at $10^3$). The continued branch stays interior to $\tau_c = 10^5$ at $\gamma = 10^{-2}$ ((0.2647, 0.5209, 0.2144), $W/W_{\min} - 1 = (0.059, 1.98, 0.072)$, where $\lambda_{\text{soft}} \cdot \tau_c = 30$), and at $\gamma = 10^{-5}$ approaches the saturation ($W/W_{\min} - 1 = 1.3 \cdot 10^{-4}$ and $1.6 \cdot 10^{-4}$ at $\tau_c = 10^5$); at $\gamma = 10^{-5}$ the three stable zeros, from $\tau_c = 100$ on, each hold two classes near their minimal masses (within $2 \cdot 10^{-4}$, relatively, at $\tau_c = 10^5$), an unstable node joins them from $\tau_c = 10^3$ on, and $W_0$'s flow ends on the one with class 1's mass large (at $\tau_c = 10^5$ it stops on its step budget short of it, that zero's exponents $-2.2$, $-1.9$ — the round-31 rerun of the zero search); along the branch $\lambda_{\text{soft}} \cdot \tau_c$ stays below 31 (below 0.22 at $\gamma = 10^{-5}$): the targets sit where the slow limit has not set in. The slow limit: $\tau_c \Sigma_f$ at $\tau_c = 10^6$ within $6.9\text{–}7.4 \cdot 10^{-4}$ of $X_h$ ($5.3\text{–}6.3 \cdot 10^{-3}$ of the continuum's $m \cdot Q\mathsf{M}Q^\top$, the factor $2/(2 - h\lambda)$); the slow noise's direction against the dither's cos 1.00000 at $\gamma = 10^{-5}$ and 0.9988–0.9999 at $\gamma = 10^{-2}$, its magnitude there 1.8–3.3 times the dither's normal part's at the effective temperature ($-6.9$ on one component at $W = 1/3$), the dither's own drift 2.8–4.1 times its normal part's ($-14$ on that component) — the masses' block of the mobility —, both 1.00 at $\gamma = 10^{-5}$; a smooth noise at $\tau_c = 10^4$ selects 2–5% as much as the AR(1) noise, its direction against the dither's cos 0.950–1.000.

**Proposition D‴.42** (several particles per class: under isotropic dither the class masses carry the gauge's laws and the split within a class is frozen; anisotropic dither and measurement noise feel the split)**.** *In the setting of Proposition D‴.36 let each class $k$ hold $n \geq 1$ particles ($m = Kn$, $K = 3$ classes), together at the class's exact-fit position — the amplitude $A_k/W_k$ with the class mass $W_k = B_k/m$, $B_k = \sum_{i \in k} \beta_i$ —, the classes beyond the cohesion's range of each other. The manifold $M_m$ of these exact fits is parametrised by the class masses and the fractions $s_i = \beta_i/B_k$ within the classes; $\hat{W}_k = |\hat{A}_k|/a_{\max}$ is the class's saturation mass ($\hat{A}_k$ its contribution).*

(i) (the determinant) *On $M_m$, $\log\det \nabla^2_\Theta J = \ell_1(W) + c_m$, where $\ell_1$ is the one-particle-per-class log-determinant of Proposition D‴.37 at the class masses and $c_m = pK\log(m/K) + \log\det$ of the cohesion's Hessian on the within-class differences ($p = 3$ coordinates per particle): independent of the masses and of the fractions.*

(ii) (the isotropic dither) *For the dither $\Sigma = 2\tau G$ on every particle, in the slow-replicator limit the class masses follow Proposition D‴.37's law with $m$ the number of particles — the replicator flow of the entropic free energy $(\tau/2)\ell_1$, the fitness $-(\tau/(2m))\partial_{W_k}\ell_1$ for every particle of the class: the one-particle-per-class dynamics slowed by $K/m$ —, and the fractions do not move: their drift vanishes at order $\gamma\tau$ (every particle of a class has the class's averaged fitness), and their diffusion vanishes identically at every $\gamma$, for any position noise (the fractions' tangent has no position component and is orthogonal to the class masses' tangent in the flow's metric). At finite $\gamma$ an $O(\gamma^2\tau)$ drift moves the fractions toward the equal split. For an anisotropic position noise — Proposition D‴.40's masks — the frozen split and the per-class law hold at the equal split (the per-class law to the resolution of the computation, after extrapolating its second derivatives' step) but not at*

*unequal ones: there the fractions drift at $O(\gamma\tau)$ and the class masses' drift departs from the per-class law at $O(\gamma\tau)$; a noise on the centres and widths only, which freezes the gauge with one particle per class (Proposition D‴.40), moves the class masses at $O(\gamma\tau)$.*

(iii) (the corrections) *Proposition D‴.38's correction evaluated at the class masses and the classes' contributions, $\kappa_F(\tau/m)(1+\hat{W}_k^2/W_k^2)/(W_k-\hat{W}_k^2/W_k)$ given to each particle of class $k$, is $\kappa_F(\tau/2)\partial_{\beta_i}\ell_1$: under the isotropic dither it acts on the class masses exactly as with one particle per class and leaves the fractions alone. Proposition D‴.40's statements do not carry over per class at unequal splits ((ii)).*

(iv) (measurement noise) *At the equal split the $m$-particle loop with measurement noise is Proposition D‴.41's $m=K$ loop exactly: co-located particles of equal mass get identical drifts and identical noise fields, an invariant subspace — the class masses' Itô drift is the one-per-class one ($m$-independent, not $K/m$ of it: the noise is shared, where the dither's is independent per particle) and Proposition D‴.41's cancellation holds. At unequal splits the cohesion's force, which vanishes on $M_m$ but is not the error's combination $\langle rU\rangle_\rho$, breaks the cancellation: a Stratonovich drift along $M_m$ of up to 10% (9.6% on $W_1$ at the fractions $(0.2, 0.7, 0.5)$) of the Itô drift, on the class masses and on the fractions, that does not vanish as $\gamma\to 0$. With noise drawn afresh at each update the selection equalises the fractions: along them it is the replicator flow of $(v/4)\sum_k\langle\|\nabla_\vartheta\tilde{\Phi}_k\|_G^2\rangle_\rho\sum_{i\in k}\beta_i^2$, the full trace's part — the fractions' tangent moves no position, so no amplitude correction enters —, minimal at the equal split for a fixed class mass.*

*Proof.* (i) In each class change the displacements $(\delta\vartheta_i)_{i\in k}$ to the $\beta$-weighted mean $\bar{\delta}_k=\sum_{i\in k}s_i\delta\vartheta_i$ and the differences $\eta_i=\delta\vartheta_i-\delta\vartheta_{\text{last}}$: a linear map of determinant $\pm\sum_i s_i=\pm1$. At an exact fit the Hessian of $J$ is the Gauss–Newton form $m\langle\nabla\hat{f}\nabla\hat{f}^\top\rangle_\rho$, and $\delta\hat{f}=\sum_k W_k\nabla\tilde{\Phi}_k\cdot\bar{\delta}_k$, the one-per-class $\delta\hat{f}$ with $\bar{\delta}_k$ for $\delta\vartheta_k$: the fit's block depends on the $\bar{\delta}_k$ alone and is $(m/K)$ times the one-per-class Hessian at the class masses $W$; the cohesion energy, that of the unweighted cloud, depends on the differences alone and not on the masses (translation invariance), and the classes do not interact. So $\nabla^2_\Theta J$ is block-diagonal in $(\bar{\delta},\eta)$, and $\log\det\nabla^2_\Theta J=\ell_1(W)+pK\log(m/K)+\log\det\nabla^2_\eta J_{\text{coh}}$. (ii) At fixed masses the positions' law under $\Sigma=2\tau G$ is Gibbs, $\propto e^{-J/\tau}$, so in the slow-replicator limit the masses' averaged fitness is $-\partial_{\beta_i}$ of the free energy $(\tau/2)\log\det\nabla^2_\Theta J$ to Laplace's order (Proposition D‴.37(ii)), $=-(\tau/(2m))\partial_{W_k}\ell_1$ for $i\in k$ by (i); the replicator then moves each class mass as one particle of mass $B_k$ with that fitness and leaves the ratios $\beta_i/\beta_j$ within a class fixed. The diffusion: Katzenberger's $P$ is the projection onto $TM_m$ orthogonal in the flow's metric $g$ (the positions' $G^{-1}$, the log-masses' $\beta_i/\gamma$); the fractions' tangent $t_s$ changes the masses within one class at a fixed class mass and the positions not at all, so $\langle t_s,t_c\rangle_g=\sum_{i\in k}(\beta_i/\gamma)\delta\log\beta_i^{(s)}\delta\log\beta_i^{(c)}=(\delta W/(\gamma W_k))\sum_{i\in k}\delta\beta_i^{(s)}=0$ for the class masses' tangent $t_c$ (which scales the class's masses together), and a position noise has no component along $t_s$: the fractions' rows of $P\Sigma P^\top$ vanish. The anisotropic case: at unequal masses the barycentre and the differences are coupled — the fit's force on particle $i$ is $\propto\beta_i$ (it drives $\eta$ through $\beta_i-\beta_j$), the cohesion's forces $F_i$ sum to zero but $\sum_i s_iF_i$ does not (it drives $\bar{\delta}$), and per-particle noises make $\mathrm{d}\bar{\delta}$ and $\mathrm{d}\eta$ correlated ($\propto s_i-s_j$) —; for $\Sigma=2\tau G$ the joint law is nonetheless the product $e^{-(J_{\text{fit}}(\bar{\delta})+J_{\text{coh}}(\eta))/\tau}$, for another $\Sigma$ it is the Lyapunov solution, not a product, and the particles' averaged fitnesses depend on the split. At the equal split the three couplings vanish and the exchange of a class's particles fixes the split, so the fractions do not drift; the barycentre's drift is the one-per-class particle's $(-G\beta_k^{(K)}\langle r\nabla\tilde{\Phi}_k\rangle_\rho$, $\sum s_i^2=1/n$ against $B_k=n\beta_k^{(K)}$) and its noise $1/n$ of that particle's, so its fluctuations give each particle $1/n=K/m$ of the one-per-class fitness; the internal differences' block does not depend on the masses, and numerically they add nothing (a proof is open: their law is not the barycentre's). (iii) $\partial_{\beta_i}\ell_1(B/m)=m^{-1}\partial_{W_k}\ell_1$ for $i\in k$; Proposition D‴.38's closed form at the class mass. (iv) On the subspace $\{\vartheta_i=\vartheta_k,\ \beta_i=B_k/n \text{ for } i\in k\}$: $\hat{f}=(1/m)\sum_k B_k\tilde{\Phi}(\vartheta_k)$ is the $m=K$ loop's at

$\beta_k^{(K)} = KB_k/m = KW_k$; particle $i$'s position drift $-G\beta_i\langle r\nabla\tilde{\Phi}_k\rangle_\rho$ has $\beta_i = mW_k/n = KW_k$; the cohesion's force vanishes at coincidence; the log-mass drift and the noise fields $U_x$ are the class's: the subspace is invariant under every noise path, and the loop on it is the $m = K$ loop. Off it, Proposition D‴.41's proof with $\langle r\psi\rangle_\rho = -\partial\Phi\cdot F_{\mathrm{coh}}$: $F_{\mathrm{coh}}$ vanishes on $M_m$ and its derivative maps into the normal space, so $\partial\Phi\cdot F_{\mathrm{coh}}$ is second order and its Hessian adds a symmetric part to $B$, whose trace against $\mathsf{M}$ need not vanish. The Itô selection along the fractions: $(U\cdot\nabla)U$'s log-mass component for particle $i$ is $\gamma[\beta_i\langle\|\nabla_\vartheta\tilde{\Phi}_i\|_G^2\rangle_\rho - m^{-1}\sum_j\beta_j^2\langle\|\nabla_\vartheta\tilde{\Phi}_j\|_G^2\rangle_\rho] + O(\gamma^2)$, the second term common to all particles, and $\partial\Phi$ reads the fractions from the log-masses alone ($t_s$ has no position component): the fitness $-(v/2)\beta_i\langle\|\nabla_\vartheta\tilde{\Phi}_k\|_G^2\rangle_\rho$, $\mathrm{d}s_i/\mathrm{d}s = -\gamma(v/2)\langle\|\nabla_\vartheta\tilde{\Phi}_k\|_G^2\rangle_\rho B_k s_i(s_i - \sum_{j\in k}s_j^2)$ — minus the Wong–Zakai term's fractions; the Itô drift adds the cohesion's residual. □

**Status.** (i) proved; (ii) for the isotropic dither the slow-limit statements proved (given Proposition D‴.37(ii)), the diffusion's vanishing proved at every $\gamma$ for any position noise, the finite-$\gamma$ drift numerical; the anisotropic case: the mechanism (the couplings, the non-product law, the equal split's barycentre law) proved, the $O(\gamma\tau)$ sizes and the internal differences' null contribution at the equal split numerical; (iii) proved under the isotropic dither; (iv) the invariant subspace proved, the residual's mechanism proved and its size numerical, the fractions' Itô selection proved at leading order. Classical: the unit-Jacobian change to the barycentre and the internal coordinates (Jacobi coordinates) and the block structure of a translation-invariant interaction; in population genetics' terms the fractions are neutral alleles' frequencies, to which the isotropic dither gives neither selection nor drift at Katzenberger's order; that the stationary law of a non-isotropically driven linear system is not its Gibbs measure. New: that the dither's selection, its correction and the frozen split act on the class masses as with one particle per class (with $m$ the number of particles) exactly when the dither is isotropic; that an anisotropic dither, or measurement noise at unequal splits, makes the split felt at Katzenberger's order; the invariant subspace at the equal split, where measurement noise's laws are $m$-independent.

**Numerics** (`exp_gauge_classes.py` → `results/chaos/gauge_classes.md`; Proposition D‴.36's instance with $n = 2$ and 3 particles per class, together at the class's exact-fit position). (A) $\log\det\nabla_\Theta^2 J$ minus its one-per-class value at the same class masses is $-46.131219$ ($m = 6$) and $-92.262438$ ($m = 9$) at three class-mass points and three splits each (ranges $2.8\cdot10^{-7}$ and $4.2\cdot10^{-7}$). (B) In Katzenberger's limit under the isotropic dither the class masses' drift and diffusion are 0.5000 and 0.3333 of the one-per-class ones at $\gamma = 10^{-5}$ ($n = 3$'s degenerate equal split included, by the Schur method), the diffusion's ratio exactly so at $\gamma = 10^{-2}$ too; the fractions' diffusion is at most $1.8\cdot10^{-19}$ ($8.7\cdot10^{-17}$ at one point) per $\gamma^2\tau$, their drift at most $4.7\cdot10^{-5}$ per $\gamma\tau$ at $\gamma = 10^{-5}$ against class drifts of 0.12–0.46, and $1.1$–$1.7\cdot10^{-2}$ ($n = 2$) and $1.7$–$7.2\cdot10^{-3}$ ($n = 3$) per $\gamma\tau$ at $\gamma = 10^{-2}$ at unequal splits, toward the equal split. (C) The correction per class at $\kappa_F = 1$ brings the class drift from 0.18–0.46 to $1.3$–$8.8\cdot10^{-5}$ per $\gamma\tau$, as with one particle per class, the fractions' drift unchanged. (D) Measurement noise at $m = 6$: at the equal split the Itô drift and the Wong–Zakai term cancel to $4.5\cdot10^{-9}$ and $5.7\cdot10^{-9}$ of the Itô drift and the class Itô drift is $m = 3$'s (1.0000); at unequal splits the Stratonovich residual is 2.2–2.6% of the Itô drift, up to 9.6% ($W_1$ at the fractions $(0.2, 0.7, 0.5)$) per component, at $\gamma = 10^{-5}$ as at $10^{-2}$; the fractions' Itô drift is the full trace's flow less that residual ($5.390\cdot10^{-7}$ against $5.515\cdot10^{-7}$), the amplitudes' part alone an order of magnitude smaller. (E) The full loop at $m = 6$ ($\gamma = 10^{-2}$, $\tau = 10^{-4}$, 8 seeds $\times$ $4\cdot10^5$ slow units): the class masses end at $(0.330, 0.349, 0.321)$ (std 0.04–0.07) from the equal split and $(0.334, 0.315, 0.351)$ (std 0.02–0.04) from $(0.2, 0.7, 0.5)$, against $m = 3$ at half the time $(0.340, 0.346, 0.314$; std 0.06–0.07); from the equal split the fractions do not move (changes $-0.002\pm0.062$, $+0.043\pm0.058$, $-0.018\pm0.055$); from $(0.2, 0.7, 0.5)$ $s_1$ moves by $-0.035\pm0.012$, away from the equal split, against the interior drift's $+0.005$ over the run — the runs reach $|\alpha| = 26$–28, deep in the saturation, where the local drift does not hold. (F) Anisotropic dither, the second derivatives Richardson-extrapolated in their step: at the equal split the per-class law holds for every mask (ratios 1.0000 for the masks

$(1,0,0)$ and $(1,1,1)$; 1.0000–1.0004 for $(0,1,1)$ at $\gamma \geq 10^{-5}$, 1.0033 and 1.0075 at $\gamma = 10^{-6}$, where the drift reaches the computation's limit), the fractions' drift there at most $1.2 \cdot 10^{-10}$ per $\gamma\tau$ (above the step-resolution in places: the computation's floor, against $1\text{–}6 \cdot 10^{-4}$ at the unequal split); at the unequal split the mask $(1,0,0)$ moves the class masses 1.4–3.5% off the per-class law at every $\gamma$, the isotropic dither by $1.3 \cdot 10^{-2} \to 0$ as $\gamma \to 0$ (1.0134 at $10^{-3}$, 1.0000 at $10^{-6}$), and the mask $(0,1,1)$, which freezes the gauge with one particle per class, moves them at $4.1 \cdot 10^{-3}$ per $\gamma\tau$ — $5.9 \cdot 10^{4}$ times the per-class law at $\gamma = 10^{-5}$; the fractions drift there at $1\text{–}6 \cdot 10^{-4}$ per $\gamma\tau$ under either mask as $\gamma \to 0$, and at $1.8 \cdot 10^{-6}$ per $\gamma\tau$ at $\gamma = 10^{-6}$ under the isotropic dither (its $O(\gamma)$).

**Proposition D‴.43** (the colour of the measurement noise selects the kernels: from the trace — a weighted $\ell^1$ norm — to the log-determinant — a support count)**.** *Let $\varphi_1, \dots, \varphi_d$ be a kernel dictionary (fixed centres and widths) sampled at $N < d$ points, $\Phi$ the $N \times d$ design, $y$ a target with $M = \{w : \Phi w = y\}$ non-empty, and let each atom carry two nonnegative masses $u_i^2$, $v_i^2$ with coefficient $w_i = u_i^2 - v_i^2$ — the signed kernel measure $\sum_i w_i \delta_{c_i}$ split into its two parts, the gauge of the replicator's mass and amplitude in flat coordinates —, adapted by gradient descent on $L = (1/2N)|\Phi w - y|^2$ in $(u, v)$: the unbalanced Fisher–Rao (Hellinger) law of the two parts, the masses' half of the paradigm's laws on a fixed dictionary. Let the loop see the measured error $r + \xi$ (label noise), $\xi$ independent over the samples, of per-step variance $\sigma^2$, stationary in time with step correlations $\rho_d$, and the update an explicit (Euler) step of length $h < 2/\lambda_{\max}$ — sampled-data adaptation: the selection below is that step's Itô drift, at a rate $\propto h\sigma^2$, absent in continuous time and for Stratonovich-consistent updates (Proposition D‴.41(ii); Heun's scheme with the same draw in both stages selects $O((h\lambda)^2)$ of it) —, and write $v_h = h\sigma^2/N$, $s = 4v_h t/N$, $P_i = u_i v_i$ (the cancellation between the measure's two parts at atom $i$), $D_i = u_i^2 + v_i^2 = \sqrt{w_i^2 + 4P_i^2}$ (the atom's relevance, the total mass there), $K = (4/N)\Phi \operatorname{diag}(D)\Phi^\top$.*

(i) (the selection flow) *At order $v$ and linear response the parameters drift along $M$ by*

$$\frac{\mathrm{d}P}{\mathrm{d}s} = -\omega \odot P, \qquad \frac{\mathrm{d}w}{\mathrm{d}s} = -\Pi_D(\omega \odot w),$$
$$\omega_i = \varphi_i^\top f(K)\varphi_i, \qquad \Pi_D = I - D\Phi^\top(\Phi D \Phi^\top)^{-1}\Phi,$$

*the gradient flow on $M$, in the flat metric ($\operatorname{diag}(1/(4D), 1/D)$ in $(w, P)$), of $R = \operatorname{tr}\Psi(K)$, $\Psi' = f$ the filter of Proposition D‴.41(iv): the cancellations decay at the rates $\omega_i$ (the measure's decomposition becoming minimal, $D \to |w|$) while $w$ moves along $M$, reweighted by $\omega$ — the gradient of $R$ in the relevances, $\partial R/\partial D_i = (4/N)\omega_i$. The flow is the small-noise limit of the descent ($\sigma \ll \min|w_S^*|\sigma_{\min}(\Phi_S)$ in practice; at larger $\sigma$ the descent kills atoms on its own, Lasso-like), and its time unit $s = 4v_h t/N$ costs about $10^4$ updates here ($3 \cdot 10^6$ for $s = 300$).*

(ii) (white noise: the trace) *For noise drawn afresh in the limit $h\lambda_{\max} \to 0$, $f \equiv 1$: $\omega_i = |\varphi_i|^2$, $R = (4/N)\sum_i |\varphi_i|^2 D_i$, which on $M$ is minimal at $P = 0$ and there the weighted $\ell^1$ norm $(4/N)\sum_i |\varphi_i|^2 |w_i|$: $R$ is convex, and the selection ends at weighted basis pursuit's solution when it is unique and the start's support contains it. At a finite step fresh draws select Damian–Ma–Lee's strictly convex $\Psi = -(2/h)\log(1 - h\lambda/2)$, whose minimiser on $M$ is not a vertex — not basis pursuit's solution —, and at finite $\sigma$ the descent's own pruning adds to it: the white comparator of a finite-step experiment is not basis pursuit.*

(iii) (slow Markov noise: the log-determinant) *For AR(1) draws of correlation time $\tau_c$ sampled finely, $f = (I + \tau_c K)^{-1}$ and $R = (1/\tau_c)\log\det(I + \tau_c K)$, with (Cauchy–Binet)*

$$\det(I + \tau_c K) = \sum_{S \subseteq [d],\, |S| \leq N} (4\tau_c/N)^{|S|} \det(\Phi_S^\top \Phi_S) \prod_{i \in S} D_i,$$

*a polynomial in $D$ with nonnegative coefficients whose logarithm is concave. On a $k$-sparse exact fit ($P = 0$, $D = |w|$, $S = \operatorname{supp} w$ with $\Phi_S$ of full column rank, $k \le N$)*

$$\tau_c R = k \log(4\tau_c/N) + \log\Big(\det(\Phi_S^\top \Phi_S) \prod_{i\in S} |w_i|\Big) + O(1/\tau_c)\,:$$

*the support's size times $\log(\tau_c)/\tau_c$, so as $\tau_c \to \infty$ the selection's global minimisers on $M$ are the sparsest exact fits, ties broken by $\det(\Phi_S^\top \Phi_S) \prod_{i\in S} |w_i|$ — asymptotically: at a finite $\tau_c$ an exact fit on more atoms can have the lower $R$ (numerics C). $R$ is Fazel's log-det heuristic on $\Phi \operatorname{diag}|w| \Phi^\top$ — sparse Bayesian learning's $\log|\Sigma_y|$ term at $\gamma \propto |w|$ and $\lambda = 1/\tau_c$, not its penalty (which minimises over $\gamma$ and adds $w^\top \Gamma^{-1} w$), so Wipf–Rao's results on its local minima do not transfer. The weights $\omega_i = \varphi_i^\top (I + \tau_c K)^{-1} \varphi_i$ are $O(1/\tau_c)$ for an atom in the span of the relevant ones and $|\Pi^\perp \varphi_i|^2$ for an atom outside it ($\Pi^\perp$ the projection onto the complement of that span): the relevant atoms are spared and the others decay.*

(iv) (any colour in the continuum: a concave selection) *For any finely sampled stationary noise, as $h\lambda_{\max} \to 0$, $f$ is decreasing (Proposition D‴.41(iv)(c)), so $\Psi$ is concave and $R$ is a concave function of $D$ (trace functions of concave functions are concave on symmetric matrices, and $K$ is linear in $D$); on $M$ at $P = 0$, $R$ is then concave in $|w|$ on each orthant, so its minimisers on $M$ are basic exact fits (at most $N$ atoms) and from a dense start the selection ends at one of several local minima — the multistability of Proposition D‴.41(iv)(e) on the gauge —, which one depending on where it starts. A smooth noise (finite $\mathbb{E}[\dot\xi^2]$) has $f \approx (\lambda\tau_c)^{-2}$ for $\lambda\tau_c \gg 1$ and selects correspondingly slowly. At a finite step the concavity fails: the filter's slope at $\lambda = 0$ is $(1 - 3\rho)/(2(1 - \rho))$ per unit $h\lambda$ for AR(1) draws, so for $\rho < 1/3$ (anti- or weakly correlated draws) $R$ is convex near $K = 0$, and for every colour $f$ rises again beyond $h\lambda \approx 0.7$–1; white noise is the continuum's only convex selection, linear in $D$.*

(v) (which exact fits the selection holds) *At a $k$-sparse exact fit ($P = 0$, support $S$, $\Phi_S$ of full column rank, $k < N$) the selection is at rest, and its inactive atoms decay — the fit is a stable point of the selection and a strict local minimum of $R$ on $M$ — when the weighted dual certificate*

$$|c_i| < \omega_i \text{ for every } i \notin S, \qquad c = \Phi_I^\top (\Phi_S^+)^\top (\omega_S \odot \operatorname{sign} w_S)$$

*holds (sufficient, not necessary; $I$ the atoms off $S$, $\omega$ at $D = |w|$). For white noise the weights are $|\varphi_i|^2$ and this is Fuchs's (least-squares) sufficient certificate for weighted basis pursuit, conservative. For a slow Markov noise $\omega_S = O(1/\tau_c)$ while $\omega_i \to |\Pi^\perp \varphi_i|^2 > 0$ off $S$ (no inactive column in $\operatorname{span} \Phi_S$), so $|c_i|/\omega_i = O(1/\tau_c)$ once $\tau_c \gg 1/\lambda_{\min}((4/N)\Phi_S \operatorname{diag}|w_S| \Phi_S^\top)$: every exact fit on fewer than $N$ atoms is held for $\tau_c$ beyond a threshold that grows without bound with the dictionary's coherence (numerics H) — while at a fit on $N$ atoms $\operatorname{span} \Phi_S$ is everything and both sides are $O(1/\tau_c)$. When, for generic coefficients on $S$, the truth is the only exact fit on fewer than $N$ atoms, it is the only one the slow selection holds robustly; where the flow ends is decided by the basins. An atom whose masses have died ($u_i = v_i = 0$) does not revive under the flow: started from basis pursuit's solution, the slow selection stays there.*

*Proof.* (i) Proposition D‴.41(iv)(b): for label noise on a flat gradient flow the drift along $M$ at order $v$ is $-(v_h/4) P_T \nabla \operatorname{tr} \Psi(H)$, $H = J^\top J/N$ the Gauss–Newton matrix ($\nabla^2 L$ on $M$), and here $J = [2\Phi \operatorname{diag}(u), -2\Phi \operatorname{diag}(v)]$, so $JJ^\top/N = K$ and $\operatorname{tr} \Psi(H) = \operatorname{tr} \Psi(K)$ ($\Psi(0) = 0$). In the coordinates $(w_i, P_i) = (u_i^2 - v_i^2, u_i v_i)$, whose Jacobian $\left[\begin{smallmatrix} 2u_i & -2v_i \\ v_i & u_i \end{smallmatrix}\right]$ has determinant $2D_i$, the flat metric reads $|\mathrm{d}u|^2 + |\mathrm{d}v|^2 = \sum_i (\mathrm{d}w_i^2/(4D_i) + \mathrm{d}P_i^2/D_i)$; $M$ constrains $w$ only. With $\mathrm{d} \operatorname{tr} \Psi(K) = \operatorname{tr}(f(K)\,\mathrm{d}K)$, $\partial R/\partial D_i = (4/N)\varphi_i^\top f(K) \varphi_i = (4/N)\omega_i$, $\partial D_i/\partial P_i = 4P_i/D_i$ and $\partial D_i/\partial w_i = w_i/D_i$, the gradient flow $-(v_h/4) \cdot (\text{metric})^{-1} \nabla R$ projected on $\Phi\,\mathrm{d}w =$

0 is $\mathrm{d}P_i/\mathrm{d}t = -(v_h/4)D_i(4/N)\omega_i(4P_i/D_i) = -(4v_h/N)\omega_i P_i$ and $\mathrm{d}w/\mathrm{d}t = -(v_h/4)[4D - 4D\Phi^\top(\Phi 4D\Phi^\top)^{-1}\Phi 4D](4/N)(\omega \odot w/D) = -(4v_h/N)\Pi_D(\omega \odot w)$. (ii) $f \equiv 1$ gives $R = (4/N)\operatorname{tr}(\Phi \operatorname{diag}(D)\Phi^\top) = (4/N)\sum_i |\varphi_i|^2 D_i$, increasing in each $P_i^2$, and $D_i \geq |w_i|$ with equality at $P_i = 0$; $R$ is a sum of norms $|(w_i, 2P_i)|$, convex, so its gradient flow on the affine $M$ decreases it to its infimum, the weighted $\ell^1$ minimum; with a unique minimiser and the flow's multiplicative form (an atom with $w_i = P_i = 0$ never revives) the end is that minimiser when the start's support contains it. (iii) $\det(I + \Phi_*^\top \Phi_*)$ for $\Phi_* = \Phi \operatorname{diag}(\sqrt{4\tau_c D/N})$ is the sum of the principal minors of $\Phi_*^\top \Phi_*$, which vanish beyond size $N$; log det is concave on positive matrices and $I + \tau_c K$ is affine in $D$. On the $k$-sparse fit only $S \subseteq \operatorname{supp} w$ contribute, the top term being $S = \operatorname{supp} w$. The weights: with $K$'s range spanned by the relevant atoms' columns, $(I + \tau_c K)^{-1} = \Pi^\perp + O(1/\tau_c)$ on it. (iv) $f(\lambda) = \int \hat{s}(\omega)\omega^2/(\lambda^2 + \omega^2)\,\mathrm{d}\omega/2\pi$ is decreasing (Proposition D‴.41(iv)(c)); $K \mapsto \operatorname{tr}\Psi(K)$ is concave because $\Psi$ is; $D \mapsto K$ is linear; on an orthant of $w$ at $P = 0$, $D = |w|$ is linear in $w$, and a concave function on the polytope $M \cap$ orthant attains its minimum at a vertex — a basic solution, at most $N$ atoms. At a finite step, $f = 1 - 2h\lambda R(a) + (h\lambda)^2(1 + 2G(a))/(1 - a^2)$, $a = 1 - h\lambda$, with $G(a) = \rho a/(1 - \rho a)$, $R = G/a$ for AR(1) draws, expands as $1 + h\lambda(1 - 3\rho)/(2(1 - \rho)) + O((h\lambda)^2)$. (v) At $P = 0$ the feasible directions keep $\Phi\delta w = 0$; off $\operatorname{span}\Phi_S$ only the inactive atoms can act, so $\delta w_S = -\Phi_S^+ \Phi_I \delta w_I$ and, to first order, $\delta R = (4/N)[\sum_{i \notin S} \omega_i |\delta w_i| - c^\top \delta w_I]$ ($R$ linear in $D$ to first order, $D_i = |w_i|$), positive for every $\delta w_I \neq 0$ when $|c_i| < \omega_i$; the inactive coordinates' linearised flow is the gradient flow of that convex form, so they decay. For a slow Markov noise, $(I + \tau_c K)^{-1} = \Pi^\perp + O(1/\tau_c)$ with $K$'s range $\operatorname{span}\Phi_S$ ($D_I \to 0$), giving the weights; an atom with $D_i = 0$ has zero mobility $(4D_i, D_i)$. □

**Status.** (i) proved at order $v$ (given Proposition D‴.41(iv)) where $\operatorname{rank} K = N$ and $\lambda_{\min}(K) \gg v_h$ — its spectral gap —; the pruning phase, in which atoms die, $\lambda_{\min}(K) \to 0$ and the sparse fits are singular points of $M$, and so (v)'s stability, are statements about the reduced flow (numerical for the descent: numerics B, D, J); (ii) the potential proved, the end at basis pursuit's solution sketched (the flow's convergence on the degenerate metric at $P = 0$ asymptotic), in the limits $h\lambda_{\max}, \sigma \to 0$; (iii), (iv) proved (the identities exact, the asymptotics in $\tau_c$ with an $O(1/\tau_c)$ remainder; (iv) in the continuum, its failure at a finite step from the filter's expansion); (v) proved for the reduced flow (the certificate sufficient, from the expansion of $R$ along the feasible directions at the fit); which local minimum a slow flow reaches, and how often it is the truth, numerical — from a small start ($u = v = \alpha \lesssim 0.5$, the rich regime) and, run to convergence, also from $\alpha = 2$ (numerics F). The paradigm's own mass law — the replicator's masses with adaptive amplitudes — is Proposition D‴.44's. Classical: label-noise gradient descent selecting a weighted $\ell^1$ norm on the quadratic parametrisation, and at a finite $\sigma$ a Lasso (HaoChen–Wei–Lee–Ma, Pillaud-Vivien–Reygner–Flammarion), the finite-step regulariser (Damian–Ma–Lee), the hyperbolic-entropy interpolant of the noiseless flow and the kernel and rich regimes (Woodworth et al.), sparse Bayesian learning and its log-determinant (Tipping; Wipf–Rao) and the theorem that its non-separable reweighting recovers sparse solutions where $\ell^1$ fails on a set of positive measure (Wipf–Nagarajan), the log-det heuristic (Fazel–Hindi–Boyd), reweighted $\ell^1$ (Candès–Wakin–Boyd), gradient noise attracting to sparse invariant sets (Chen–Kunin–Yamamura–Ganguli), sparse optimisation on measures by Fisher–Rao and Wasserstein–Fisher–Rao flows (Chizat; Chizat–Bach), Fuchs's certificate ([111], [112], [113]). New (not found stated): that the temporal correlation of the label noise sets the sparsity prior of the explicit update's selection along the exact fits — the trace (weighted $\ell^1$) for white noise, the log-determinant of Fazel's heuristic (a support count) for a slow Markov noise, a concave penalty for every colour in the continuum —; that this selection recovers sparse expansions where weighted basis pursuit fails — at the level of the reduced flow, and by label-noise descent at small $\sigma$ —, as Wipf–Nagarajan's theorem for non-separable reweighting leads one to expect, at rates sparse Bayesian learning itself matches or beats (numerics G): the contribution is the mechanism, not a better sparse solver.

**Numerics** (`exp_colour_selection.py` → `results/chaos/colour_selection.md`; success: within $10^{-4} \cdot \max|w^*|$ of the truth; the start: the noiseless flow's interpolant from $u = v = 0.5$).

(A) $\det(I + \tau K)$ equals its subset expansion to $2.6 \cdot 10^{-15}$, $\partial R/\partial D_i$ equals $(4/N)\omega_i$ to $4 \cdot 10^{-9}$ (central differences), $R$'s second differences along random segments are negative for every $\tau$ ($-8 \cdot 10^{-5}$ to $-0.39$) and zero for white draws ($\le 3.6 \cdot 10^{-15}$); on $k$-sparse fits $\tau_c R - k \log(4\tau_c/N)$ tends to $\log(\det(\Phi_S^\top \Phi_S) \prod |w_i|)$ ($-0.065892$ against $-0.065900$ at $\tau_c = 10^6$, $k = 3$; $-1.168220$ against $-1.168243$, $k = 6$). (B) Label-noise descent ($10 \times 40$ Gaussian designs, a 4-sparse truth where basis pursuit fails, step 0.5, $\sigma = 0.1$, eight seeds $\times$ 16 paths) follows the reduced flow — the paths' mean within 0.011–0.035 of it for fresh draws, within 0.11–0.35 for AR(1) draws ($\tau_c = 100$), whose fluctuations are larger —: fresh draws end on dense fits (supports 10–13, errors 0.80–2.4, basis pursuit's 0.80–2.98); AR(1) draws recover the truth on all 16 paths of the four seeds whose reduced flow recovers it (the end interpolants' errors $10^{-13}$–$7 \cdot 10^{-11}$, supports 4) and on 0–2 of 16 on the four whose flow does not; at $\sigma = 0.3$ the four recover on 16, 0, 16 and 13 paths (and one seed whose flow fails on 7), at $\sigma = 0.5$ one seed still recovers — beyond the small-$\sigma$ limit the descent prunes on its own. (C) Recovery rates of the reduced flows (40 instances per row at $10 \times 40$, 20 at $20 \times 80$): at $10 \times 40$, $k = 3$, basis pursuit 0.38, the white flow 0.38, the slow flows 0.62, 0.75, 0.78 for $\tau_c = 10$, 100, 1000; $k = 4$: 0.17, 0.17, 0.42, 0.60, 0.62; $k = 5$: 0, 0, 0.07, 0.15, 0.17; $k = 6$: none; at $20 \times 80$, $k = 6$: 0.35, 0.35, 0.65, 0.75, 0.75; $k = 8$: 0.10, 0.10, 0.20, 0.35, 0.30; the white flow run to convergence ends at basis pursuit's solution (336 of 340 converged, to $9.4 \cdot 10^{-9}$); the slow flow's supports are smaller than basis pursuit's ($k = 3$: 4.8 against 7.3); started from the white flow's end it does no better than basis pursuit (the atoms the white flow has dropped do not revive). Where the slow flow fails at $\tau_c = 100$ its end has a larger $R$ than the truth in every instance up to $k = 4$ (a local minimum; 0.09 and 0.35 of the failures end below the truth's $R$ at $k = 5$, 6); at $\tau_c = 10$, 0.39, 0.68 and 0.85 of them ($k = 4$, 5, 6) end below it — there the truth is not the finite-$\tau_c$ minimiser. (D) Gaussian kernel dictionaries on $[0, 1]$ (40 centres, 20 samples, widths 0.03 and 0.05, largest coherence 0.90 and 0.96): the slow flow ($\tau_c = 100$) against basis pursuit 0.95 against 0.80 (width 0.03, $k = 4$), 0.68 against 0.45 ($k = 6$), 1.00 against 0.85 (width 0.05, $k = 4$), 0.78 against 0.65 ($k = 6$), the white flow at basis pursuit's rate in every row; label-noise descent with AR(1) draws ($\tau_c = 100$) recovers the truth on all 16 paths on two of the three instances where basis pursuit fails (errors $2 \cdot 10^{-13}$–$7 \cdot 10^{-8}$); on the third it is still on 12–13 atoms at the end, where its reduced flow is at $8.8 \cdot 10^{-3}$ — and recovers the truth when run on (to $3 \cdot 10^{-13}$ at $s = 1600$); fresh draws end within 0.08–0.11 of basis pursuit's solution, on 11–15 alive atoms. (E) At the step 0.5 the other colours select as well ($10 \times 40$, 20 instances): $k = 3$, fresh 0.30, AR(1) 0.65, a smooth (Gaussian-correlated) noise of $\tau_c = 10$ 0.70, a draw kept 40 steps 0.70 (basis pursuit 0.45); $k = 4$: 0.20, 0.40, 0.55, 0.40 (0.25); a smooth noise of $\tau_c = 100$ has not begun to select at the slow horizon (0.00) and recovers 0.85 and 0.65 at ten times it. (F) The start, every flow run to convergence: from $u = v = \alpha$ with $\alpha = 0.1$, 0.5 the slow flow recovers 0.70, 0.75 ($k = 3$) and 0.40, 0.50 ($k = 4$), from $\alpha = 2$ — toward the kernel regime — 0.35 and 0.30, at basis pursuit's rate (0.45, 0.25), which the white flow matches from every $\alpha$. (G) On the same instances sparse Bayesian learning (Tipping's EM) recovers 0.97, 0.88, 0.55, 0.12, 0.05 for $k = 2$–6 against the slow flow's 0.97, 0.75, 0.60, 0.15, 0.00 (mean supports at $k = 4$: 6.6, 6.4; basis pursuit 8.9); the log-determinant's majorisation–minimisation reaches 0.70, 0.45 ($k = 3$, 4) from the flows' start and 0.53, 0.33 from basis pursuit; reweighted $\ell^1$ 0.38–0.53 and 0.17–0.25 at $\varepsilon = 0.01$, 0.1, 1 from either start. (H) The dual certificate at the truth: white noise certifies 0.38 of the 2-sparse truths and 0–0.03 from $k = 3$ on; $\tau_c = 100$ certifies all for $k = 2$–5 and 0.93 at $k = 6$, $\tau_c = 1000$ all; the median ratio $\max|c_i|/\omega_i$ falls like $1/\tau_c$ ($k = 2$: 0.439, 0.0814, 0.0087 at $\tau_c = 10$, 100, 1000); basis pursuit's own solutions where it fails (on $N = 10$ atoms) are certified for 0–30% of the instances. On the kernel dictionaries (widths 0.03–0.08, $k = 4$, 6) the certified fraction rises with $\tau_c$ to 0.85–1.00 at $\tau_c = 10^5$ (white: 0–0.60), and the slow flow returns to the truth from $P = 10^{-3}$ in 0.90–1.00 ($\tau_c = 100$) and all ($\tau_c = 1000$) of the instances. (I) Under a fit floor ($y = \Phi w^* + \varepsilon e$, $|e| = 1$) the slow flow's $k$ largest atoms are the truth's support in 0.70–0.75 ($k = 3$) and 0.50 ($k = 4$) of the instances for $\varepsilon = 10^{-3}$–0.05, basis pursuit's in 0.40–0.45 and 0.25. (J) At the same physical noise (AR(1) of correlation time

50, per-step $\sigma = 0.1$) the cancellations off the truth's support decay by $\Delta \log P = -2.39$ under Euler's step 0.5, $-0.278$ under Euler's step 0.125 and $-0.097$ under Heun's step 0.5 (25 times less): the selection is the explicit scheme's.

**Proposition D‴.45** (the free mass–amplitude law is a mirror flow: a conservation law per particle, the implicit bias in closed form, the invariance under noise with smooth paths, the real loop's Lyapunov function)**.** *In the setting of Proposition D″.34 let the masses be free — the replicator without its centring ($\lambda \equiv 0$), in the real loop the unnormalised exponentiated-gradient law $\dot\beta_i = \gamma\beta_i e\tilde\Phi_i(x)$ —, one particle per class beyond the cohesion's range (the cohesion's $\alpha$-component vanishing on it), and let each particle be driven by one scalar signal $g_i$: $\langle (r+\xi)\varphi_i\rangle$ at fixed positions on a dictionary (Proposition D‴.44's setting: $\Phi$ the $N \times d$ design of full row rank, $w = \beta \odot a/m$, $J = (m/2N)|\Phi w - y|^2$), $e(t)\varphi_i(x(t))$ in the real loop — a measurement noise read in Stratonovich's sense, as the limit of noise with smooth paths:*

$$\mathrm{d}\alpha_i = G\beta_i a_i' g_i \,\mathrm{d}t, \qquad \mathrm{d}\beta_i = \gamma\beta_i a_i g_i \,\mathrm{d}t, \qquad a_i = a_{\max}\tanh(\alpha_i/a_{\max}), \qquad c := \gamma a_{\max}^2/(4G).$$

(i) (the conservation law) $Q_i := \beta_i - c\cosh(2\alpha_i/a_{\max})$ *is constant along every trajectory, for every drive — also when the centres and widths move by any law of their own, as long as the cohesion's $\alpha$-component vanishes on the particle: each particle moves on the fixed curve $\beta = Q_i + c\cosh(2\alpha/a_{\max})$ of its $(\alpha,\beta)$ plane, and the free replicator adds no degree of freedom to the amplitude, it reparametrises it. Along the curve $w_i = \beta_i a_i/m$ increases strictly with $\alpha_i$ and is unbounded — past the saturation the mass grows like $e^{2|\alpha_i|/a_{\max}}$ and carries the coefficient; the bound $|w_i| < \beta_i a_{\max}/m$ holds only at frozen masses —; $\beta$ never reaches 0 in finite time; for $Q_i > -c$ the curve is a graph over all $\alpha \in \mathbb{R}$ and $w_i$ ranges over $\mathbb{R}$; for $Q_i \le -c$ it has two branches $|\alpha_i| > \alpha_0$ ($\cosh(2\alpha_0/a_{\max}) = -Q_i/c$), $\beta \to 0$ at $\alpha_0$, and on each the sign of $w_i$ is frozen (Proposition D″.47).*

(ii) (the mirror flow) *On a level set the mirror coordinate*

$$\eta_i := \int \frac{\mathrm{d}\alpha_i}{G\beta_i a_i'} = \frac{\alpha_i - a_{\max}\kappa_i \operatorname{artanh}(\kappa_i t_i)}{2Gc}, \qquad t_i = a_i/a_{\max}, \quad \kappa_i^2 = (Q_i - c)/(Q_i + c)$$

*($\kappa \operatorname{artanh}(\kappa t)$ read as $-\kappa' \arctan(\kappa' t)$ when $\kappa^2 = -\kappa'^2 < 0$, as $a_{\max}\coth(\alpha/a_{\max})$ on the critical curve $Q = -c$, and as $\kappa\cdot\frac{1}{2}\log((\kappa t+1)/(\kappa t-1))$ on a branch) obeys $\mathrm{d}\eta_i = g_i\,\mathrm{d}t$ exactly, and $\mathrm{d}w_i/\mathrm{d}\eta_i = mD_i$, $D_i = (\beta_i/m^2)(G\beta_i a_i'^2 + \gamma a_i^2)$ — Proposition D‴.44's relevance: $w$ is the mirror flow of the separable, strictly convex potential $\Psi_Q(w) = \sum_i \psi_{Q_i}(w_i)$, $\psi_Q' = \eta$, $\psi_Q'' = 1/(mD)$, whose conjugate is the mass gained,*

$$\psi_Q^*(\eta) = (\beta - \beta^0)/(\gamma m) = 2c\sinh^2(\alpha/a_{\max})/(\gamma m), \qquad \psi_Q(w) = \eta w - \psi_Q^*,$$
$$\beta^0 := Q + c \text{ (the mass at } \alpha = 0).$$

*Its shape: near $w = 0$ a mass-weighted $\ell^2$ core, $\psi \approx mw^2/(2G(\beta^0)^2)$ ($\eta \approx mw/(G(\beta^0)^2)$); at $|w| \approx w_\sqcap := \beta^0 a_{\max}/m$ the amplitude saturates and the mass takes over, and beyond it $\eta \approx (1/(\gamma a_{\max}))\log(|w|/w_\sqcap)$ (for $\beta^0 \gg c$): the entropy, $\psi \approx (1/(\gamma a_{\max}))|w|(\log(|w|/w_\sqcap) - 1)$ — the marginal cost rising across the saturation from $\approx a_{\max}/(G\beta^0)$ to $O(1/(\gamma a_{\max}))$, by the factor $\beta^0/(4c)$, a soft wall at $|w| = w_\sqcap$; for $\beta^0 \ll c$ the core is crossed with a jump of $\eta$ of $2\lambda_1$, $\lambda_1 = \pi a_{\max}\kappa'/(4Gc)$, $\kappa' = ((2c-\beta^0)/\beta^0)^{1/2}$ ($\lambda_1 = (\pi a_{\max}/(4G))(2/(c\beta^0))^{1/2}(1+O(\beta^0/c))$), and beyond the core the curve follows the critical one, $\eta \approx \operatorname{sign}(w)\lambda_1 - K\operatorname{sign}(w)|w|^{-1/3}$:*

$$\psi = \lambda_1|w| - (3/2)K|w|^{2/3}(1 + o(1)) \qquad ((\beta^0)^{3/2} \ll |w| \ll 1),$$
$$K = (a_{\max}^2/(2Gc))(2c/(a_{\max}^2 m))^{1/3},$$

*an $\ell^1$ norm of weight $\lambda_1$ with a concave correction that under-penalises small coefficients.*

(iii) (the implicit bias) *On the dictionary, if every $Q_i > -c$, the noiseless loop converges, from every start, to the Bregman projection of its start onto the exact fits,*

$$w_\infty = \operatorname{argmin}\{\Psi_Q(w) - \langle \eta(0), w\rangle : \Phi w = y\},$$

*which depends on the start only through $(Q, \eta(0))$ — no persistent excitation is needed. From $\alpha = 0$ and masses $\beta_i^0$ ($\eta(0) = 0$) the masses set the regime: as $\beta^0 \to \infty$ (a common scale) $w_\infty$ tends to the minimum of $\sum_i w_i^2/(\beta_i^0)^2$ on the fits — the lazy regime; the minimum-norm fit for equal masses —, as $\beta^0 \to 0$ to the minimum of $\sum_i |w_i|/\sqrt{\beta_i^0}$ — the rich regime: weighted basis pursuit, the weights the starting masses' inverse square roots (when it is unique; otherwise the minimiser selected at the next order). The approach is a power, set by the core's barrier: at a vertex $w_{\mathrm{bp}}$ with $N$ atoms ($\Phi_S$ invertible) whose certificate $\xi$ ($\Phi_S^\top \xi = \rho_S \cdot \operatorname{sign} w_{\mathrm{bp},S}$, $\rho_i = (\beta_i^0)^{-1/2}$) has the relative slacks $\mu_j = 1 - |(\Phi^\top \xi)_j|/\rho_j > 0$ off the support, every off-support coefficient sits on the core's tangent profile at $\eta_j = (1-\mu_j)\lambda_{1,j}$,*

$$w_j = \operatorname{sign}((\Phi^\top \xi)_j) \cdot \frac{a_{\max}(\beta_j^0)^{3/2}}{m(2c)^{1/2}} \cdot \frac{\cos(\pi\mu_j/2)}{\sin^3(\pi\mu_j/2)} \cdot \big(1 + O((\beta^0)^{1/2}/\mu_j)\big),$$

*and the support refits ($\Phi_S w_S = y - \Phi_{S^c} w_{S^c}$): $|w_\infty - w_{\mathrm{bp}}| = O((\beta^0)^{3/2}\mu_{\min}^{-3})$, the next order computed exactly from the support's mirror coordinates at the vertex; a small slack — a near-optimal second vertex — delays this regime until $(\beta^0)^{1/2} \ll \mu_{\min}$, and the closed form's relative error is then of order $(\beta^0)^{1/2}/\mu_j$ with a large constant. A free start must sit at $\alpha(0) = 0$ to within $\theta = |\alpha(0)|/(\beta^0/(2c))^{1/2} \to 0$: otherwise the weights are $\lambda_1(1 \mp (2/\pi)\arcsin\theta)$ (Proposition D‴.47). (On a branch the barrier is an entropy and the approach $\varepsilon^{\mu_j}$, Proposition D‴.47(iii).) Its cost is time: the transit to the support takes a time $\propto (\beta^0)^{-1/2}$ (Proposition D‴.48), the last digits a time $\propto 1/(\beta^0)^2$ when the support has fewer than $N$ atoms (the off-support curvature $\mathrm{d}w/\mathrm{d}\eta \propto (\beta^0)^2$). From nonzero amplitudes at tiny masses the particles sit on branches ($Q_i < -c$, their signs frozen): their limit is the Bregman projection onto the sign-constrained least-squares fits — Proposition D‴.47; an exact fit may cease to exist, and the bias is basis pursuit only where the frozen signs allow it.*

(iv) (noise) *(a) In continuous time a noise with smooth paths — white noise read in Stratonovich's sense —, of any colour, any covariance over the nodes and any amplitude, leaves $Q$ fixed and moves $\eta$ only within $\eta(0) + \operatorname{range}(\Phi^\top)$: the limit map (the Bregman projection of the current state) is invariant pathwise — such a noise never selects among the exact fits; it moves the state only across them, on the fixed $N$-dimensional manifold $\nabla\Psi_Q^*(\eta(0) + \operatorname{range}\Phi^\top)$; under persistent noise the stationary mean is an exact fit displaced from the projection at $O(v)$. Read in Itô's sense (Pesme et al.'s stochastic gradient flow) white noise adds (b)'s drift per unit time; Proposition D‴.41(ii)'s requirement $C \propto I$ is the conserved law's — its centring (vi) —, not the free law's. (b) An Euler step of length $h$ on $(\alpha, \log\beta)$ — sampled-data adaptation, the paradigm's update — changes $Q_i$ by*

$$\Delta Q_i = (h^2 g_i^2 \gamma\beta_i/2)\big(\gamma a_i^2 - G\beta_i a_i'(1 + a_i^2/a_{\max}^2)\big) + O(h^3),$$

*negative below the saturation's layer ($G\beta a'(1 + a^2/a_{\max}^2) > \gamma a^2$), zero where $\gamma a_{\max}^3 t^3 = Gm|w|(1 - t^4)$ ($t = a/a_{\max}$): under white label noise near $M$, per unit $s = (mv/4)t$, $\mathrm{d}Q_i/\mathrm{d}s = (2|\varphi_i|^2/(mN))\gamma\beta_i(\gamma a_i^2 - G\beta_i a_i'(1 + a_i^2/a_{\max}^2))$ — the conserved quantities' share of Proposition D‴.44's selection with the paradigm's update, which is, in these coordinates, the drift of $(Q, \eta \bmod \operatorname{range}\Phi^\top)$. Along it the state first moves away from basis pursuit ($\|w\|_1$ rising), then atoms die ($Q \to -c$) and it settles on a sparse exact fit — basis pursuit's where basis pursuit is the truth; where it is not, basis pursuit's support or another sparse fit of larger $\ell^1$ norm (numerical). The mirror update — $\eta^+ = \eta + hg$, the particle placed*

*on its own curve, an exact zero-order-hold integration of each particle — keeps $Q$ and the projection exactly at every step, whatever the noise: which update the loop runs decides whether its measurement noise selects. (c) Dither on the amplitudes ($\mathrm{d}\alpha$ gains $\sqrt{2\tau G_\alpha}\,\mathrm{d}B$) drifts $Q$: $\mathrm{d}Q_i = -\gamma\tau\cosh(2\alpha_i/a_{\max})\,\mathrm{d}t + \mathrm{d}M_i$ ($M$ a martingale), so $Q$ falls without bound in expectation while the amplitudes diffuse into their saturation, where the masses carry the coefficients with frozen signs — the free law's form of Proposition D‴.37's drive into the saturation.*

(v) (the real loop) *With $f = \sum_i w_i^*\varphi_i$ on the orbit, $\dot e = -k_e e + f(x) - \hat f(x)$ and $\hat f = \sum_i w_i\varphi_i$,*

$$V := \tfrac12 e^2 + \sum_i B_{Q_i}(w_i^*, w_i), \qquad B_Q(w^*, w) = w^*(\eta^* - \eta) - (\beta^* - \beta)/(\gamma m)$$

$$(\eta^*, \beta^* \text{ at } w^* \text{ on the particle's own curve}),$$

*has $\dot V = -k_e e^2$ for every start with $w_i^*$ in the range of its particle's curve — Boffi–Slotine's Bregman Lyapunov function for this law's potential (Proposition D‴.34(ii) has the masses' alone); $e \to 0$ and the parameters stay bounded, their convergence needing excitation. Two particles on one atom without cohesion between them receive the same mirror increment — $\eta_i - \eta_i(0) = \eta_j - \eta_j(0)$ at every time, under any common drive, a measurement noise in continuous time included —, so the start fixes how they split the atom's coefficient: the split is the Bregman split of the current total. The free law loses the conserved law's a-priori bound $|\hat f| \le a_{\max}$: under an approximation error or a disturbance nothing bounds its coefficients a priori — Proposition D‴.46's leak does.*

(vi) (the conserved law) *With $\sum\beta = m$ the replicator's centring shifts every level, $\mathrm{d}Q_i = -\gamma\beta_i\lambda\,\mathrm{d}t$, $\lambda = \sum_j w_j g_j$: noiselessly $\lambda$ vanishes on the exact fits, so the shift is the transient's, and the conserved law's limit differs from the free law's projection (numerical); with noise $\lambda \neq 0$ at first order.*

*Proof.* (i) $\mathrm{d}Q_i = \gamma\beta_i a_i g_i\,\mathrm{d}t - (2c/a_{\max})\sinh(2\alpha_i/a_{\max})G\beta_i a_i' g_i\,\mathrm{d}t$ and $\sinh(2x)\operatorname{sech}^2 x = 2\tanh x$, $4Gc/a_{\max}^2 = \gamma$: zero, for any $g_i$; Stratonovich's calculus obeys the chain rule. $\mathrm{d}w/\mathrm{d}\alpha = (\beta' a + \beta a')/m > 0$ where $\beta > 0$, $\beta' = (2c/a_{\max})\sinh(2\alpha/a_{\max})$ having $\alpha$'s sign, as $a$ does; $\beta(t) = \beta(0)\exp(\gamma\int a g\,\mathrm{d}t) > 0$. (ii) $\mathrm{d}\eta = \mathrm{d}\alpha/(G\beta a') = g\,\mathrm{d}t$ by the law; with $\cosh^2 x = (1+\cosh 2x)/2$, $1/(G\beta a') = (1/(2Gc))[1 + (c-Q)/(Q + c\cosh(2\alpha/a_{\max}))]$, and $t = \tanh(\alpha/a_{\max})$ turns the second term's integral into $(c-Q)\int a_{\max}\,\mathrm{d}t/((Q+c) + (c-Q)t^2)$; at $Q = -c$, $1/(G\beta a') = \coth^2(\alpha/a_{\max})/(2Gc)$. $\mathrm{d}w/\mathrm{d}\eta = (\mathrm{d}w/\mathrm{d}t)/(\mathrm{d}\eta/\mathrm{d}t) = mD$ by the two laws; $\psi^*$: $\mathrm{d}(\beta - \beta^0)/(\gamma m\,\mathrm{d}\eta) = \gamma\beta a/(\gamma m) = w = (\psi^*)'$, and $\psi = \eta w - \psi^*$. The shape: near 0, $\eta \approx a/(G\beta^0)$ and $w \approx \beta^0 a/m$; for $\beta^0 \gg c$ the artanh term saturates at $a_{\max}\kappa\operatorname{artanh}\kappa \approx (a_{\max}/2)\log(2\beta^0/c)$ while $\alpha/(2Gc)$ keeps growing, and $w \approx (Q + (c/2)e^{2\alpha/a_{\max}})a_{\max}/m$, so $\eta \approx (a_{\max}/(4Gc))\log(mw/(\beta^0 a_{\max})) = (1/(\gamma a_{\max}))\log(|w|/w_\sqcap)$; for $\beta^0 \ll c$ the arctan jumps by $\pi\operatorname{sign}(t)$ over $|t| \lesssim 1/\kappa'$, a range in which $|w| = O((\beta^0)^{3/2})$, and beyond it $\kappa'\arctan(\kappa' t) = \pi/2 - 1/t + O(1/(\kappa'^2 t^3))$, so $\eta = \lambda_1 + (\alpha - a_{\max}/t)/(2Gc) + \dots$ — the critical curve's $\eta\cdot w^{1/3} \to -K$ at small $\alpha$ ($w \approx 2c\alpha^3/(m a_{\max}^2)$) —, and $\psi = \int\eta\,\mathrm{d}w$. (iii) $\mathrm{d}\eta = \Phi^\top r\,\mathrm{d}t/N$ keeps $\eta(t) \in \eta(0) + \operatorname{range}(\Phi^\top)$; writing $\eta = \eta(0) + \Phi^\top z$, $\dot z = -\nabla\mathcal{D}(z)/N$ for the dual function $\mathcal{D}(z) = \sum_i \psi_{Q_i}^*(\eta_i(0) + (\Phi^\top z)_i) - y^\top z$, convex with Hessian $\Phi\operatorname{diag}(mD)\Phi^\top \succ 0$ and coercive ($\psi^*$ grows exponentially in both directions when $Q > -c$, $\Phi^\top$ injective), so $z(t)$ converges to $\mathcal{D}$'s unique minimiser (the gradient flow of a strictly convex coercive function), and $w = (\psi^*)'(\eta(0) + \Phi^\top z)$ to the point where $\Phi w = y$ and $\nabla\Psi_Q(w) - \eta(0) \in \operatorname{range}(\Phi^\top)$: the KKT conditions of the strictly convex problem. The limits: on compacts $\psi_Q(w)\cdot(2G(\beta^0)^2/m) \to w^2$ as $\beta^0 \to \infty$ and $\psi_Q(w)/\lambda_1 \to |w|$ as $\beta^0 \to 0$ (the remainder $O(|w|^{2/3}/\lambda_1)$), with the minimisers bounded; the minimisers of uniformly converging strictly convex (resp. convex, with a unique limit minimiser) objectives on a fixed affine set converge. The rate: the support's coefficients sit beyond the core, $\eta_i = \lambda_{1,i} - K|w_i|^{-1/3} + \dots = \Lambda\rho_i + O(1)$ ($\Lambda := (\pi a_{\max}/(4G))(2/(c\beta^0))^{1/2}$ for equal scales), so $\Phi_S^\top z = \eta_S - \eta_S(0)$ gives

$z = \Lambda\xi + O(1)$ and $\eta_j = (\Phi^\top z)_j = \pm(1-\mu_j)\lambda_{1,j} + O(1)$ off the support, inside the core, where $\alpha = a_{\max}(\beta^0/(2c))^{1/2}\tan(\pi\eta/(2\lambda_1))(1+O(\beta^0))$ and $w = (\beta^0 + 2c\sinh^2(\alpha/a_{\max}))a/m$: at $\tan = \cot(\pi\mu_j/2)$, the formula; the $O(1)$ in $\eta$ moves the tangent's argument by $O(1/\lambda_1)$, the relative $O((\beta^0)^{1/2}/\mu_j)$. (iv) (a) The computation of (i) and (ii) holds for any $g_i$, and $r+\xi$ enters only through $g = \Phi^\top(r+\xi)/N$; the projection depends on $\eta$ modulo range$(\Phi^\top)$ ($\langle\Phi^\top z, w\rangle = \langle z, y\rangle$ on the fits). (b) $Q(\alpha + hG\beta a' g, \beta e^{h\gamma a g})$ to second order in $h$: the first order vanishes by (i), the second is $(h^2g^2/2)(\gamma^2\beta a^2 - (4c/a_{\max}^2)\cosh(2\alpha/a_{\max})G^2\beta^2 a'^2)$ and $a'^2\cosh(2\alpha/a_{\max}) = a'(1 + a^2/a_{\max}^2)$; in Itô's reading this is $\frac{1}{2}F^\top\nabla^2 Q F$ per unit of $\langle\eta\rangle$, $F = (G\beta a', \gamma a)$ on $(\alpha, \log\beta)$; near $M$, $\mathbb{E}[g_i^2] = \sigma^2|\varphi_i|^2/N^2 + O(h\lambda)$, and $s = (mv/4)t$ with $v = h\sigma^2/N$. (c) Itô's formula for $Q$ with the additive noise $\sqrt{2\tau G_\alpha}\,\mathrm{d}B$ on $\alpha$: $\frac{1}{2}\cdot(2\tau G_\alpha)\cdot\partial_\alpha^2 Q = -(4c\tau G_\alpha/a_{\max}^2)\cosh(2\alpha/a_{\max}) = -\gamma\tau\cosh(2\alpha/a_{\max})$. (v) $\frac{\mathrm{d}}{\mathrm{d}t}B_Q(w^*, w) = -\psi''(w)\dot{w}(w^* - w) = -e\varphi_i(x)(w_i^* - w_i)$ since $\dot{w}_i = mD_i e\varphi_i(x)$ (the two laws); summed, $-e(f-\hat{f})$, and $\mathrm{d}(\frac{1}{2}e^2)/\mathrm{d}t = -k_e e^2 + e(f-\hat{f})$. The pair: the same drive gives the same $\mathrm{d}\eta$. (vi) The centring adds $-\gamma\beta_i\lambda$ to $\mathrm{d}\beta_i$ and nothing to $\mathrm{d}\alpha_i$. □

**Status.** (i), (ii), (iv)(a), (v) and (vi)'s identity proved (exact identities); (iii) the convergence and the projection proved for $Q > -c$, the lazy limit proved, the rich limit proved when weighted basis pursuit is unique, its leading order at an $N$-atom vertex derived (the core's closed form; the next order from the certificate computed and checked), the time costs' scalings derived and numerical; (iv)(b) the formula proved, the selection's end numerical; (iv)(c) in expectation; the branches: Proposition D‴.47. Classical: the law is, in Proposition D‴.44's flat coordinates $(\alpha/\sqrt{G}, 2\sqrt{\beta}/\sqrt{\gamma})$, the Euclidean gradient flow of a separable $2 \to 1$ reparametrisation, so it is a mirror flow with a start-dependent potential by Li–Wang–Lee–Arora's commuting-reparametrisation theorem, with exactly one conserved function per unit, $Q$ (Marcotte–Gribonval–Peyré; the $u \odot v$ case with $\delta = u^2 - v^2$: Azulay et al.); the kernel and rich regimes of quadratic parametrisations and the hyperbolic entropy's $\ell^2$-to-entropy shape (Woodworth et al.; Ghai–Hazan–Singer); exponentiated gradient and its reparametrisation as gradient descent (Kivinen–Warmuth; Amid–Warmuth); the invariance of stochastic mirror descent's limit under noise in the dual span (Gunasekar et al.; Azizan–Hassibi); the selection by SGD's Itô noise and the shrinking effective initialisation of diagonal networks (Pesme–Pillaud-Vivien–Flammarion; Even–Pesme–Gunasekar–Flammarion), and that isotropic label noise does not select (HaoChen–Wei–Lee–Ma); Bregman Lyapunov functions for mirror-descent adaptation laws (Boffi–Slotine) ([113]). New: the paradigm's free mass–amplitude law made explicit — the conserved $Q = \beta - c\cosh(2\alpha/a_{\max})$, the mirror coordinate's closed form in every regime, the conjugate as the mass gained, the potential's $\ell^2$ core, the saturation's soft wall and its entropy tail, the $\ell^1$ weights $1/\sqrt{\beta^0}$ of the rich regime (a power law in the starting mass, to be compared with depth-$D$ diagonal networks' polynomial rich limits) with their $-(3/2)K|w|^{2/3}$ correction, the branches —; the identification of the Euler update's change of $Q$ with Proposition D‴.44's selection; and the mirror update as a sampled-data law whose noise never selects.

**Numerics** (`exp_mirror_free.py` → `results/chaos/mirror_free.md`; $5 \times 12$ Gaussian designs with unit columns and 2- or 3-sparse truths of magnitudes in $[0.15, 0.3]$, $m = 12$, $G = 1$, $\gamma = 0.1$, $a_{\max} = 1$, so $c = 0.025$, unless stated; a projection whose residual exceeds $10^{-9}(1+|y|)$ would count as a failure — there were none). (A) $\eta$'s closed form agrees with the quadrature of $\mathrm{d}\alpha/(G\beta a')$ to $1.5\cdot10^{-14}$ (relative; $q = 100c$, 3, 0.5, $c$, $c/2$, 0, $-c/2$ and the branch $q = -2c$), $(\psi^*)' = w$ and $\mathrm{d}w/\mathrm{d}\eta = mD$ to $7.2\cdot10^{-11}$ and $4.3\cdot10^{-10}$ (finite differences). Along the loop $Q$ moves by at most $1.4\cdot10^{-11}$ noiseless ($t \le 10^4$) and $3.6\cdot10^{-10}$ under a smooth noise of 30 sinusoids (frequencies $10^{-3}$–3, mixed over the nodes, amplitude 0.3; $t \le 3\cdot10^4$) — the integrator's tolerance —; the conserved law's shift, integrated along its own trajectory under that noise ($t \le 200$), equals the quadrature of $-\gamma\beta_i\lambda$ to $1.5$–$4.4\cdot10^{-7}$ (relative); along Proposition D‴.44's reduced flow (free masses, white noise, the paradigm's update) $\mathrm{d}Q/\mathrm{d}s$ equals (iv)(b)'s closed form to $9.1\cdot10^{-16}$. (B) From $\alpha = 0$ and equal masses $s\cdot m/d$ the noiseless loop's end at $t = 10^7$ is the Bregman projection to $4.7\cdot10^{-11}$ for $s = 10$, 1, 0.1, 0.01 (residuals $\le 5.7\cdot10^{-15}$; 20 instances per $k$). The projection moves from the lazy to the rich regime with $s$: for $k = 2$ its distance to

the minimum-norm fit is 0.5% of it at $s = 10$ and its $\|w\|_1$ is 1.37 times basis pursuit's on 11.8 atoms, then 1.52 ($s = 1$), 1.17 (0.1), 1.027 (0.01), 1.0022 ($10^{-3}$) and 1.0004 ($10^{-4}$), the median distance to basis pursuit 0.14, 0.16, 0.097, 0.0235, $9.0 \cdot 10^{-4}$, $1.4 \cdot 10^{-5}$ (falling from $s = 1$ on), the atoms used to 3.0, the truth recovered nearly as often as basis pursuit recovers it (0.75 at $s = 10^{-4}$ against 0.80; for $k = 3$, 0.30 and 0.30, the median distance $1.3 \cdot 10^{-4}$). At $s = 10^{-3}$ the loop is 0.277 from its projection up to $t = 10^3$, then $6.4 \cdot 10^{-3}$ ($t = 2 \cdot 10^4$), $7.1 \cdot 10^{-4}$ ($2 \cdot 10^5$), $1.5 \cdot 10^{-5}$ ($2 \cdot 10^6$), below $10^{-8}$ from $t \approx 8.5 \cdot 10^6$ — the last digits at the time scale $1/(\beta^0)^2 = 10^6$ of a support with fewer than $N$ atoms. With unequal masses $s \cdot r_i$ ($r_i$ log-uniform in $[1/4, 4]$) the projection's median distance to the weighted basis pursuit $\min \sum |w_i|/\sqrt{r_i}$ is $6.2 \cdot 10^{-3}$, $2.3 \cdot 10^{-6}$, $1.7 \cdot 10^{-9}$ at $s = 10^{-3}$, $10^{-5}$, $10^{-7}$ ($k = 2$; $1.3 \cdot 10^{-2}$, $3.0 \cdot 10^{-5}$, $2.1 \cdot 10^{-8}$ for $k = 3$), against a median 0.084 from the unweighted one for $k = 3$ (for $k = 2$ the two coincide where both recover the 2-sparse truth), and to the weighted minimum norm $\min \sum w_i^2/r_i^2$ 2.4–$2.5 \cdot 10^{-5}$ and 2.4–$2.5 \cdot 10^{-9}$ (relative) at $s = 10^2$, $10^4$. From $\alpha(0) \sim N(0, 0.05^2)$ at tiny masses the particles land on branches — 4.2 of 12 at $s = 10^{-4}$, 11.1 and 11.4 at $10^{-6}$ —; at $10^{-4}$ the frozen signs keep an exact fit in every instance and the projection's median distance to basis pursuit is $6.6 \cdot 10^{-4}$ ($k = 2$; bimodal — 10 of 20 within $10^{-3}$, the largest 0.37) and 0.15 ($k = 3$); at $10^{-6}$ a truth atom sits on a wrong-signed branch in 13 and 18 of 20 instances, the frozen signs still fit $y$ exactly in 18 and 19, and the median distance to basis pursuit is 0.25 and 0.36 (largest 16.3): the bias is the sign-constrained one of Proposition D‴.47. (F) The rate (unequal masses $\varepsilon r_i$, the instances whose weighted basis pursuit is a vertex with $N$ atoms: 10 of 20 for $k = 2$, 17 of 20 for $k = 3$): the leading order computed from the certificate agrees with the projection to the rounding floor — $|w - w_1|_\infty$ at most $1.8 \cdot 10^{-13}$ ($k = 2$) and $2.6 \cdot 10^{-11}$ ($k = 3$) at $\varepsilon = 10^{-7}$, where the distances to basis pursuit are $1.6 \cdot 10^{-8}$ and $1.4 \cdot 10^{-8}$ (medians, the instances with $\mu_{\min} \geq 0.01$; the largest $1.1 \cdot 10^{-7}$ and $1.5 \cdot 10^{-6}$) —; the closed form off the support is within 7.6% and 6.7% (medians; the largest errors 113% and 138%) at $10^{-7}$ and 0.73% and 0.69% (the largest 11% and 8%) at $10^{-9}$ — an $O(\varepsilon^{1/2}/\mu)$ correction with a large constant, the closed form's $\mu^{-3}$ making the small slacks' coefficients large —; and the distance falls between $10^{-7}$ and $10^{-9}$ by the factors 1068 (median; range 609–1252) and 984 (185–2200) ($\varepsilon^{3/2}$: 1000). The instances with a small slack approach late: $\mu_{\min} = 2.7 \cdot 10^{-4}$ ($k = 3$, instance 13) keeps a sixth atom and stays 0.17 from basis pursuit at $\varepsilon = 10^{-5}$ and $10^{-7}$ — the leading order meaningless there (114 at $10^{-7}$) —, is $4.3 \cdot 10^{-3}$ from it at $10^{-9}$ (leading order within $3.5 \cdot 10^{-4}$) and $3.2 \cdot 10^{-8}$ at $10^{-11}$ (within $4.4 \cdot 10^{-12}$); $\mu_{\min} = 3.3 \cdot 10^{-3}$ (instance 1) is $3.3 \cdot 10^{-3}$ away at $10^{-7}$ and agrees with the leading order to $6.2 \cdot 10^{-12}$ from $10^{-9}$ on — the asymptotic regime begins once $\varepsilon^{1/2} \ll \mu_{\min}$. (C) (1) White label noise at $\sigma = 0.3$ (64 paths to $t = 200$ from random states): Euler–Maruyama on $(\alpha, \log \beta)$ changes $Q$ on average by its Itô correction's time integral (ratios 1.0106, 1.0037, 1.0011 at $h = 0.1$, 0.03, 0.01; three instances alike), Heun's (Stratonovich) mean change is $1.4 \cdot 10^{-5}$, $4.1 \cdot 10^{-6}$, $1.4 \cdot 10^{-6}$ — vanishing like $h$ —, and the mirror update keeps $Q$ to $2.2 \cdot 10^{-16}$ (by construction). (2) The Euler step's exact mean change of $Q$ (Gauss–Hermite, 80 nodes) agrees with the closed form to 3.2–3.6% at $h = 0.5$, 0.29–0.35% at 0.05 and 0.029–0.036% at 0.005 — the $O(h^3)$ term. (3) Along Proposition D‴.44's reduced flow from the lazy projection ($s = 1$; ten instances) $\|w\|_1$ first rises (0.706 → 0.781 by $s = 10^3$ on instance 0), then masses fall toward extinction — $Q$'s mean falls from 0.975 to −0.021 by $s = 10^8$, its minimum $-c$, no atom yet extinct —; where basis pursuit is the truth (6 instances) the state reaches it on its two atoms, within 1.0–$1.6 \cdot 10^{-6}$ at $s = 10^6$ and 2.7–$11.6 \cdot 10^{-9}$ at $10^8$; of the four where it is not, two end on basis pursuit's support (within $2.6 \cdot 10^{-5}$ and $6.5 \cdot 10^{-4}$) and two on other sparse fits of larger $\ell^1$ norm (0.4228 against 0.4156 on 5 atoms, 0.4714 against 0.4516 on 2). (4) At $h = 0.5$ with $\sigma = 0.3$ over $2 \cdot 10^4$ steps (16 paths) the paradigm's Euler update lowers $Q$ by 0.34–0.36 on average (at most 0.41–0.48) and moves the projection by 0.030–0.043, its $\|w\|_1$ rising 0.706 → 0.778 and 0.569 → 0.714 — as Proposition D‴.44's reduced flow to the same $s = 270$ does ($\|w\|_1$ 0.779 and 0.723, $Q$'s mean change −0.338 and −0.355; the paths' mean projection within $7.4 \cdot 10^{-4}$ and $5.7 \cdot 10^{-3}$ of its state) —; the mirror update keeps $Q$ to $1.1 \cdot 10^{-16}$ (by construction) and the

projection to $9.0 \cdot 10^{-12}$. (D) The real loop ($k_e = 5$, $\gamma = 0.5$, $m = 6$, six Gaussian atoms of width 0.4 on $[-1.2, 1.2]$ and a second particle on the third, $x_m = \sin t + 0.4 \sin 2.7t$, 2000 time units): noiseless, $V$ falls monotonically from 0.175 to 0.022 with $\dot V = -k_e e^2$ to $3.0 \cdot 10^{-9}$ (relative), $Q$ is kept to $7.6 \cdot 10^{-15}$ and the pair's mirror coordinates move together to $2.4 \cdot 10^{-15}$; with a smooth noise in the adaptation only, or a sensor noise in the feedback, the regressor and the adaptation, $V$ is no longer monotone ($0.175 \to 0.136$, 0.377) while $Q$ is kept to $6.8 \cdot 10^{-12}$ and $4.9 \cdot 10^{-13}$ and the pair's coordinates move together to $6.3 \cdot 10^{-13}$ and $9.2 \cdot 10^{-14}$; at $t = 2000$ the noiseless parameters are still 0.048 (the distinct atoms) and 0.027 (the pair's sum) from $w^*$. (E) The conserved law ($\sum \beta = m$ kept) from the same starts ($\alpha = 0$, $\beta = m/d$) reaches exact fits (residuals $\le 7.5 \cdot 10^{-17}$) at distances 1.6–$4.1 \cdot 10^{-3}$ from the free law's projections, its $Q$ moving by 0.058–0.22.

**Proposition D‴.46** (a leak in the mirror coordinates: the free mass–amplitude law made robust, its bias a Bregman-regularised fit — a ridge from large starting masses, a LASSO from small ones)**.** *In the setting of Proposition D‴.45 (free masses, one particle per class beyond the cohesion's range, each particle on its curve $Q_i$ with mirror coordinate $\eta_i$ and potential $\psi_{Q_i}$), add to each particle's scalar drive the leak $-\sigma(\eta_i - \eta_{p,i})$ toward a point $\eta_{p,i} = \eta(\alpha_{p,i}; Q_i)$ of its own curve (the coefficient $w_{p,i} = w(\alpha_{p,i}; Q_i)$; the start, say):*

$$\mathrm{d}\alpha_i = G\beta_i a_i'[g_i - \sigma(\eta_i - \eta_{p,i})]\,\mathrm{d}t, \qquad \mathrm{d}\beta_i = \gamma\beta_i a_i[g_i - \sigma(\eta_i - \eta_{p,i})]\,\mathrm{d}t, \qquad \sigma > 0,$$

*so that $\mathrm{d}\eta_i = [g_i - \sigma(\eta_i - \eta_{p,i})]\,\mathrm{d}t$ and $Q_i$ stays conserved — the leak is a law on $(\alpha, \beta)$, $\eta$ explicit and $Q$ read off the state.*

(i) (the real loop: robustness) *Let $\dot e = -k_e e + f(x) - \hat f(x) + d(t)$ along the loop's own trajectory, with $f = \sum_i w_i^* \varphi_i + \varepsilon$ (any split of $f$ into a representable part with each $w_i^*$ in its particle's range and a remainder $\varepsilon$ counted in the disturbance), $|\varepsilon + d| \le \bar d$, and $V = \frac12 e^2 + \sum_i B_i(w_i^*, w_i)$ (Proposition D‴.45(v); proper on each curve: $B_i(w_i^*, \cdot) \to \infty$ at both ends when $Q_i > -c$). With a measurement noise $n$ in the adaptation (the drive $(e + n)\varphi_i$),*

$$\begin{aligned}\dot V &= -k_e e^2 + e(\varepsilon + d) - n(f - \hat f - \varepsilon) - \sigma \sum_i [B_i(w_i^*, w_i) + B_i(w_i, w_{p,i}) - B_i(w_i^*, w_{p,i})] \\ &\le -(k_e/2)e^2 - \sigma \sum_i B_i(w_i^*, w_i) + \bar d^2/(2k_e) + \sigma B(w^*, w_p) \qquad (n = 0),\end{aligned}$$

*so $V(t) \le V(0)e^{-\lambda t} + (C/\lambda)(1 - e^{-\lambda t})$, $\lambda = \min(k_e, \sigma)$, $C = \bar d^2/(2k_e) + \sigma B(w^*, w_p)$: $e$ and the parameters are ultimately bounded, whatever the disturbance's form (the bound minimised over the split). Without the leak nothing bounds the free masses' coefficients a priori (the conserved law's $|\hat f| \le a_{\max}$ is lost with the constraint): a decaying excitation and a bounded disturbance can drive them to infinity — $\varphi(t) = (1 + t)^{-1/2}$ and a $d$ built for it give $w \propto (1 + t)^{1/4}$ with $e \to 0$ —, while on a persistently exciting orbit they converge, slowly, to the biased fit $w^* + H_L^{-1}\bar d_L$ of (ii). With $d = 0$ the leak's price is the bias below.*

(ii) (the averaged loop: the leak's bias) *Let the regressor $\varphi(t) = \varphi(x_m(t))$ be periodic and the adaptation slow, the error slaved to its frozen-parameter periodic steady state $e = L[(w^* - w)^\top \varphi + \varepsilon + d]$, $L = 1/(s + k_e)$. Then*

$$\dot\eta = (H_L(w^* - w) + \bar d_L)/k_e - \sigma(\eta - \eta_p), \qquad H_L = k_e\langle \varphi L[\varphi]^\top\rangle, \qquad \bar d_L = k_e\langle \varphi L[\varepsilon + d]\rangle,$$

*where $H_L + H_L^\top = 2k_e^2\langle L[\varphi]L[\varphi]^\top\rangle \succeq 0$ (the error filter's Gram — the classical averaged adaptive system); its equilibrium $w_\sigma$ is the unique solution of the monotone equation $(H_L(w - w^*) - \bar d_L)/k_e + \sigma(\eta(w) - \eta_p) = 0$ and attracts every start ($Q_i > -c$), with the Lyapunov function $B(w_\sigma, w)$. On a scalar periodic orbit the first-order lag vanishes*

*($\langle\varphi\dot\varphi^\top\rangle = 0$, an exact differential): $H_L = H - \langle\dot\varphi\dot\varphi^\top\rangle/k_e^2 + O(k_e^{-3})$, symmetric to that order, and to that order the averaged loop is the mirror flow of the strictly convex*

$$F_\sigma(w) = (1/(2k_e))(w-w^*)^\top H_L(w-w^*) - \langle\bar d_L, w\rangle/k_e + \sigma\sum_i B_i(w_i, w_{p,i}),$$

*$H_L \to H = \langle\varphi\varphi^\top\rangle$ and $\bar d_L \to \langle\varphi(\varepsilon + d)\rangle$ as $k_e \to \infty$. As $\sigma \to 0$, $w_\sigma \to w^* + H_L^{-1}\bar d_L$ (the excitation's consistent fit nearest $w_p$ in $\Psi$'s divergence when $H_L$ is singular) — $w^*$ only without a bias. The real loop's mean follows $w_\sigma$ to $O$(the adaptation gains) (the averaging theorem for adaptive systems, an exponentially stable averaged equilibrium).*

(iii) (the regimes) *From $\alpha = 0$ and masses $\beta_i^0$ with $w_p = 0$: for $\beta^0 \gg c$ the regulariser is the ridge $\sum_i (m/(2G(\beta_i^0)^2))w_i^2$ up to the saturation's wall $w_\sqcap = \beta^0 a_{\max}/m$ — the leak a generalised Tikhonov regularisation —; for $\beta^0 \to 0$ it is the weighted $\ell^1$ norm $\sum_i \lambda_1(\beta_i^0)|w_i|$, $\lambda_1(\beta^0) = \pi a_{\max}\kappa'/(4Gc)$, $\kappa' = ((2c-\beta^0)/\beta^0)^{1/2}$ $(= (\pi a_{\max}/(4G))(2/(c\beta^0))^{1/2}(1 + O(\beta^0/c)))$, minus the remainder $(3/2)K|w|^{2/3}$, $K = (a_{\max}^2/(2Gc))(2c/(a_{\max}^2 m))^{1/3}$ (Proposition D‴.45(ii)), which vanishes against $\sigma\lambda_1$ in the limit: the averaged loop solves the LASSO $\min (1/(2k_e))(w-w^*)^\top H(w-w^*) + \sigma\sum_i \lambda_1(\beta_i^0)|w_i|$ at a fixed penalty $\sigma\lambda_1$ ($\sigma \propto \sqrt{\beta^0}$), its regularisation path parametrised by $\sigma$. Its inactive coordinates obey the KKT condition $|(H(w^*-w))_i|/k_e \le \sigma\lambda_{1,i}$; it recovers the target's support exactly — for a small penalty — when the weighted irrepresentable condition $\max_{j\notin S}|(H_{S^cS}H_{SS}^{-1}(\lambda_{1,S} \odot \operatorname{sign} w_S^*))_j|/\lambda_{1,j} < 1$ holds (and no sign flips; $\le 1$ is necessary; with equal masses the weights cancel and it is the classical $\|H_{S^cS}H_{SS}^{-1}\operatorname{sign}(w_S^*)\|_\infty < 1$): the leak then prunes to the target's kernels, otherwise it keeps spurious atoms at every penalty. The rich equilibria are reached slowly: the averaged loop's escape from a rich start takes a time growing like $k_e\lambda_1 \propto k_e(\beta^0)^{-1/2}$.*

*Proof.* (i) D‴.45(v)'s computation with the drive's leak term and the noise: $\frac{\mathrm d}{\mathrm dt}B_i(w_i^*, w_i) = -(\mathrm d\eta_i/\mathrm dt)(w_i^* - w_i)$, $\mathrm d\eta_i/\mathrm dt = (e+n)\varphi_i - \sigma(\eta_i - \eta_{p,i})$, and the three-point identity $(\eta(w) - \eta(w_p))(w^* - w) = B(w^*, w_p) - B(w^*, w) - B(w, w_p)$; $e(\varepsilon + d) \le (k_e/2)e^2 + \bar d^2/(2k_e)$; then $\dot V \le -\lambda V + C$ and the comparison lemma. The drift example: with $\eta = \eta(0) + \kappa\log(1+t)$, $\kappa = 1/(4\gamma a_{\max})$, $e = \dot\eta/\varphi = \kappa(1+t)^{-1/2}$ and $d = \dot e + k_e e + w(\eta(t))\varphi(t)$ ($w^* = 0$) is bounded, $w(\eta)$ growing like $e^{\gamma a_{\max}\eta}$ in the entropy tail. (ii) Averaging the slow law over the period with $e$ at its steady state gives $\dot\eta$; $H_L + H_L^\top$: with $\psi = L[\varphi]$, $\dot\psi + k_e\psi = \varphi$, $\langle\varphi\psi^\top + \psi\varphi^\top\rangle = \mathrm d\langle\psi\psi^\top\rangle/\mathrm dt + 2k_e\langle\psi\psi^\top\rangle$ and the first term vanishes over a period. Monotonicity: for two points, $(R(w_1) - R(w_2))\cdot(w_1 - w_2) \ge \sigma(\eta(w_1) - \eta(w_2))(w_1 - w_2) > 0$, so the root is unique (it exists: $F$'s coercivity, or a degree argument on the coercive monotone map); $\frac{\mathrm d}{\mathrm dt}B(w_\sigma, w) = -\dot\eta\cdot(w_\sigma - w) = -(w - w_\sigma)^\top H_L(w - w_\sigma)/k_e - \sigma(\eta - \eta_\sigma)(w - w_\sigma) < 0$. The expansion: $L = (1/k_e)\sum_j(-s/k_e)^j$, $\langle\varphi\dot\varphi^\top\rangle = (1/T)\oint\varphi(x)\varphi'(x)^\top\mathrm dx = 0$, $\langle\varphi\ddot\varphi^\top\rangle = -\langle\dot\varphi\dot\varphi^\top\rangle$; the odd orders' antisymmetric parts survive from $k_e^{-3}$ on $(\langle\dot\varphi_i\ddot\varphi_j - \dot\varphi_j\ddot\varphi_i\rangle = \langle\dot x^3(\varphi_i'\varphi_j'' - \varphi_j'\varphi_i'')\rangle$, not an exact differential). As $\sigma \to 0$ the monotone equation's root tends to the root of $H_L(w - w^*) = \bar d_L$. (iii) D‴.45(ii)–(iii)'s limits of $\psi_Q$ with the corrected remainder; with $w_p = 0$, $B(w, 0) = \psi(w) - \psi(0) - \eta(0)w = \psi(w)$; the LASSO's support theory (Zhao–Yu; Wainwright); the escape time: from $w \approx 0$ the coordinate's $\eta$ must reach $\sigma\lambda_1$'s level against the drive $(H(w^* - w))_i/k_e$. □

**Status.** (i) proved (the identity, the comparison bound; the drift example by construction); (ii) proved for the averaged loop (existence, uniqueness, global attraction, the expansion to $O(k_e^{-2})$), the real loop's mean by the classical averaging theorem (cited, not re-proved), numerical: the agreement; (iii) the two limits proved, the support recovery by the LASSO's theory and computed over every 3-sparse pattern of an orbit's Gram, the escape time's scaling numerical. Classical: the $\sigma$-modification and its ultimate bound (Ioannou–Kokotović; Ioannou–Sun), averaging for adaptive systems with the filtered regressor's correlation (Anderson et al.; Bodson–Sastry–Anderson–Mareels–Bitmead; Sastry–Bodson), continuous-time discounted mirror descent and its regulariser-perturbed rest points with Bregman (Fenchel-coupling) Lyapunov functions (Gao–Pavel; Romano–Pavel), its entropic case for the replicator — exponentially

discounted payoffs with a smooth best response (Coucheney–Gaujal–Mertikopoulos) —, regularisation toward a reference with a Lyapunov function (Pérolat et al.; Abe et al.), explicit regularisation through mirror descent (Azizan–Lale–Hassibi), the vanishing-leak (Tikhonov, viscosity) selection (Alvarez–Bolte–Brahic; Attouch), the LASSO and its irrepresentable condition (Tibshirani; Zhao–Yu; Wainwright), mirror-descent adaptation laws (Boffi–Slotine) ([114]). New: the dual-coordinate leak implemented on the paradigm's $(\alpha, \beta)$ with $Q$ kept; its ultimate bound in the adaptive loop with this law; the starting masses setting the regime and the LASSO weights $\lambda_1 \propto (\beta^0)^{-1/2}$; the averaged equilibrium with the error filter's Gram written as a monotone equation for this law.

**Numerics** (`exp_mirror_leak.py` → `results/chaos/mirror_leak.md`). (A) The identity of (i), $\dot V = -k_e e^2 + ed - n(f - \hat f) - \sigma \sum [B(w^*, w) + B(w, w_p) - B(w^*, w_p)]$, holds to $9.5 \cdot 10^{-11}$ (relative) at twelve random states of the real loop with a disturbance and a measurement noise in the adaptation ($\sigma = 0$ and 0.3). On random positive definite $H$ ($10 \times 10$, $k_e = 2$, four draws) the averaged equilibrium is the ridge $(H + \rho I)^{-1}(Hw^* + \bar d)$, $\rho = k_e \sigma m/(G(\beta^0)^2) = 0.5$, to $1.3 \cdot 10^{-8}$ ($\beta^0 = 10^4$) and $1.3 \cdot 10^{-4}$ ($10^2$) — a ridge that moves $w$ by 31–55% —, while at $\beta^0 = 3$, where the saturation's wall $w_\sqcap = 0.25$ lies inside the coefficients' range ($\max|w^*| = 0.295$), it departs from the ridge by 0.15 (relative): the regulariser is a ridge only below the wall; and it is the LASSO of penalty $\sigma\lambda_1 = 0.02$ to 0.015, $1.5 \cdot 10^{-3}$, $1.5 \cdot 10^{-4}$ (relative; $\beta^0 = 10^{-4}$, $10^{-6}$, $10^{-8}$ — the remainder falls like $(\beta^0)^{1/2}$), with the LASSO's support in every draw ($\lambda_1$'s leading term exact to $1.0 \cdot 10^{-7}$ at $10^{-8}$). (B) (1) On a persistently exciting orbit ($k_e = 5$, $m = 6$, six atoms of width 0.4, $x_m$ of period $20\pi$; the filtered Gram's eigenvalues $2.7 \cdot 10^{-3}$ to 0.68) under an approximation error ($0.1 \sin 4x$) and a disturbance with a bias ($0.2 \sin 3.1t + 0.1$), the free law without the leak creeps toward the biased fit $w^* + H_L^{-1}\bar d_L$ ($\max|w|$ 0.337, $\sum B(w^*, w) = 0.465$, the averaged loop's slowest time constant $1.5 \cdot 10^4$): its largest $|w|$ by quarters of $2 \cdot 10^4$ time units 0.309, 0.330, 0.337, 0.341 (each quarter's largest value includes the parameters' oscillation along the orbit about the averaged state) and $V$ rising from 0.40 to 0.45; $\sigma = 0.01$ and 0.1 hold the coefficients at 0.081 and 0.040 with $V$ at 0.085 and 0.13, below the ultimate bounds $C/\lambda = 1.74$ and 0.30 and below the comparison bound $V(0)e^{-\lambda t} + (C/\lambda)(1 - e^{-\lambda t})$ at every sample, the late means within $2.5 \cdot 10^{-3}$ and $6.5 \cdot 10^{-4}$ of the averaged equilibrium $w_\sigma^L$; the rms error rises from 0.024 to 0.031 and 0.037 (the leak's price); $Q$ is kept to $6 \cdot 10^{-11}$. (2) Without persistent excitation ($\varphi(t) = (1 + t)^{-1/2}$, one particle, a bounded disturbance $|d| \le 0.77$ built for it) the free law's coefficient grows like $(1 + t)^{1/4}$ as constructed — 0.70, 2.31, 7.37, 23.3 at $t = 1$, $10^2$, $10^4$, $10^6$ —, while the leak bounds it: $\sigma = 0.01$ gives 0.70, 1.52, 0.52, 0.46 and $\sigma = 0.1$ gives 0.69, 0.63, 0.47, 0.46, below the comparison bound throughout ($C/\lambda = 29.5$ and 3.07). (C) In slow adaptation ($k_e = 20$, $G = 0.5$, $\gamma = 0.05$, the leak toward the start) the error filter's Gram differs from the idealised $\langle \varphi\varphi^\top \rangle$ by 0.89% (max, relative) and is symmetric to $3 \cdot 10^{-14}$ on this scalar orbit; started at the averaged equilibrium $w_\sigma^L$, the real loop's period means (period $20\pi$) stay near it — the last within $1.6 \cdot 10^{-4}$ (lazy start, $\beta^0 = 3$, $\sigma = 5 \cdot 10^{-4}$, 318 periods) and $1.7 \cdot 10^{-4}$ (intermediate, $\beta^0 = 0.3$, $\sigma = 10^{-4}$, 954 periods), 4.4 and 2.6 times closer than to the idealised equilibrium (lag $|w_\sigma^L - w_\sigma| = 8.2 \cdot 10^{-4}$ and $5.1 \cdot 10^{-4}$), against the leak's bias $|w_\sigma^L - w^*| = 0.038$ and 0.12 —, the last periods' means changing by $1.6 \cdot 10^{-10}$ and $1.1 \cdot 10^{-9}$. (D) On the Gram of an orbit (twelve atoms of width 0.25 on $[-1.4, 1.4]$, eigenvalues $1.4 \cdot 10^{-4}$ to 0.55): (1) for a 3-sparse truth (irrepresentable quantity 0.771) the equilibrium from a rich start ($\beta^0 = 10^{-6}$) is the LASSO of penalty $\mu = \sigma\lambda_1$ to $3.7 \cdot 10^{-6}$–$6.5 \cdot 10^{-4}$ (absolute) and keeps exactly the truth's three atoms for $\mu \le 10^{-2}$ (at $3 \cdot 10^{-2}$ it drops one, the LASSO's own shrinkage), while from a lazy start ($\beta^0 = 30$) it is the ridge to $3.9 \cdot 10^{-6}$–$2.5 \cdot 10^{-4}$, dense (no coefficient vanishes; 6–12 above the reporting threshold); (2) over every 3-sparse support and sign pattern (1760) at $\mu = 10^{-3}$ the irrepresentable condition holds for 416, the LASSO's prediction (the condition and no sign flip) for 416, and the equilibrium's support is exactly the truth's for 415 — prediction and equilibrium agree on 1745 of 1760; elsewhere it carries 1.76 extra atoms on average; (3) with a disturbance bias correlated with the regressors ($\bar d = Hv$, $|v| = 0.05$) the equilibrium tends to $w^* + H^{-1}\bar d$ as

$\sigma \to 0$ (distance $2.5 \cdot 10^{-2}$, $3.7 \cdot 10^{-4}$, $3.7 \cdot 10^{-6}$, $3.7 \cdot 10^{-8}$ at $\sigma = 10^{-2}$, $10^{-4}$, $10^{-6}$, $10^{-8}$, linear in $\sigma$), 0.043 from $w^*$; (4) the averaged loop from the rich start reaches its equilibrium (to $10^{-3}$ relative) at $t \approx 3.2 \cdot 10^5$ for the truth above and $2 \cdot 10^6$ for a support violating the irrepresentable condition (quantity 1.22), whose equilibrium keeps five atoms.

**Proposition D‴.47** (the free law's branches: an exponentiated gradient with frozen signs — small starts land on them; the limit is the Bregman projection onto the sign-constrained least-squares fits, the wrong-signed particles' masses going extinct; on positive kernel dictionaries the target's kernels at every scale)**.** *In the setting of Proposition D″.45 (free masses, one particle per class beyond the cohesion's range, each driven by its scalar $g_i$; $c = \gamma a_{\max}^2/(4G)$), call particle $i$* on a branch *when $Q_i < -c$, i.e. when it starts with*

$$\beta_i(0) < 2c \sinh^2(\alpha_i(0)/a_{\max})$$

*— a mass small against its amplitude's square: a start at small masses and nonzero amplitudes puts every particle on a branch, and only $|\alpha_i(0)| < a_{\max} \operatorname{asinh}((\beta_i(0)/(2c))^{1/2})$ keeps a free sign. A free sign is not enough for Proposition D‴.45(iii)'s symmetric rich limit: with $\theta_i = |\alpha_i(0)|/(\beta_i(0)/(2c))^{1/2} < 1$ the curve's mass is $\beta_i^0 = (1-\theta_i^2)\beta_i(0)$ and the start sits at $\eta_i(0) = s_i(2/\pi)\arcsin\theta_i \cdot \lambda_{1,i}$, so its weights become $\lambda_{1,i}(1 \mp (2/\pi)\arcsin\theta_i)$ — the limit needs $\theta_i \to 0$. On a branch let $s_i = \operatorname{sign}\alpha_i$, $\alpha_{0,i}$ its branch point ($\cosh(2\alpha_{0,i}/a_{\max}) = -Q_i/c$, $\beta = 0$ there) and $a_{0,i} = a_{\max}\tanh(\alpha_{0,i}/a_{\max})$.*

(i) (the potential on a branch: an entropy with a temperature) *Along every curve the mirror coordinate's slope in $\log|w|$ is*

$$\frac{\mathrm{d}\eta_i}{\mathrm{d}\log|w_i|} = \frac{a_i}{\gamma a_i^2 + G\beta_i a_i'^2},$$

*so on a branch, where $\beta \to 0$ at $\alpha_0$ with $a \to a_0$, the potential is an entropy near $w = 0$ with the temperature $1/(\gamma a_{0,i})$ — $\eta_i = s_i(1/(\gamma a_{0,i}))\log|w_i| + C_i + O(|w_i|)$, $\psi_i(w) = (1/(\gamma a_{0,i}))(|w|\log|w| - |w|) + C_i w + \psi_i(0) + O(w^2)$ — and at $|w| \to \infty$ an entropy with the temperature $1/(\gamma a_{\max})$ (D″.45(ii)'s tail). $\psi_i$ is finite at 0, $\psi_i(0) = (Q_i + c)/(\gamma m)$, with an infinite slope there (a barrier: the sign is frozen), and the Bregman divergence from 0 is the particle's mass,*

$$B_i(0, w_i) = \beta_i/(\gamma m).$$

*Near its branch point the particle runs the unnormalised exponentiated gradient: its amplitude frozen near $a_0$, $\beta_i \approx \beta_i(0)\exp(\gamma a_{0,i}\int g_i\,\mathrm{d}t)$. (On the critical curve $Q_i = -c$ the two branches meet at $\alpha = 0$, $a_0 = 0$, and the barrier is a power, $\eta \propto -|w|^{-1/3}$.)*

(ii) (the limit: the Bregman projection onto the sign-constrained least-squares fits) *On a dictionary ($\Phi$ the $N \times d$ design, labels $y$, $J(w) = (m/2N)|\Phi w - y|^2$, the noiseless loop $\mathrm{d}\eta = -\nabla J(w)\,\mathrm{d}t/m$) let $K = \{w : s_i w_i \geq 0$ for every particle on a branch$\}$ and $S = \operatorname{argmin}_K J$, the sign-constrained least-squares fits — nonempty (a convex quadratic on a polyhedron), $S = K \cap \{\Phi w = \hat{y}\}$ with $\hat{y}$ the common value of $\Phi w$ on it. From every start (any mix of free particles and branches):*

(a) *$B(\bar{w}, w(t))$ is nonincreasing for every $\bar{w} \in S$, and*

$$J(w(t)) - \min_K J \leq m \cdot \min_{\bar{w}\in S} B(\bar{w}, w(0))/t;$$

(b) *$w(t)$ converges, and its limit is the Bregman projection of the start onto $S$,*

$$w_\infty = \operatorname{argmin}\{\Psi_Q(w) - \langle \eta(0), w\rangle : w \in S\};$$

(c) *the particles on branches whose coefficient vanishes on all of $S$ go extinct — $\beta_i \to 0$, $s_i\eta_i \to -\infty$ —, exponentially at the rate $\gamma a_{0,i}|g_i(\infty)|$ when $g_i(\infty) = -(\nabla J(w_\infty))_i/m \neq 0$ (strict complementarity; on the critical curve $Q_i = -c$, $a_0 = 0$, the decay is algebraic, about $t^{-3}$); every other particle's $\eta$ converges, and no other branch coefficient vanishes in the limit (a free particle's may, e.g. on a column uncorrelated with everything, started at $\alpha = 0$).*

*When $y$ has a fit in $K$ with every branch coefficient strictly of its particle's sign, $S$ is the set of exact fits in $K$, no coefficient vanishes on all of it and nothing goes extinct — Proposition D″.45(iii) with branches —; otherwise the wrong-signed particles die and the loop fits what the frozen signs allow, in the least-squares sense.*

(iii) (the rich limit on branches: sign-constrained basis pursuit at a logarithmic temperature) *Let the masses shrink at fixed amplitudes, $\beta_i(0) = \varepsilon b_i$, $\alpha_i(0) \neq 0$. As $\varepsilon \to 0$ every particle is on a branch, $a_{0,i} \to |a_i(0)|$, the objective of (ii)(b) is $\log(1/\varepsilon)\sum_i |w_i|/(\gamma|a_i(0)|) + O(1)$ on $S$, and*

$$w_\infty(\varepsilon) \to \operatorname{argmin}\Big\{\sum_i |w_i|/|a_i(0)| : w \in S\Big\} \qquad \textit{(when unique)},$$

*the sign-constrained basis pursuit weighted by the inverse starting amplitudes. The linear program is perturbed by an entropy at the temperature $1/\log(1/\varepsilon)$, and an entropic penalty's minimiser approaches a nondegenerate vertex exponentially in the inverse temperature (Cominetti–San Martín; Weed): at a vertex with $N$ atoms whose certificate $\xi$ ($s_j(\Phi^\top\xi)_j \le \rho_j := 1/(\gamma|a_j(0)|)$, with equality on the support) has the relative slacks $\mu_j = 1 - s_j(\Phi^\top\xi)_j/\rho_j > 0$, the off-support coefficients are*

$$|w_j| = e^{O(1)}\varepsilon^{\mu_j},$$

*a power of $\varepsilon$ with the slack for its exponent (the $O(1)$ computed from the support's mirror coordinates at the vertex), and the support refits — against $\varepsilon^{3/2}$ from D″.45(iii)'s start $\alpha(0) = 0$, whose barrier is the core's tangent, not an entropy: with $\mu_{\min}$ well below 3/2 the branches' rich limit is reached far more slowly. At a vertex with fewer than $N$ atoms the approach is still a power, $\varepsilon^{\mu^*}$, $\mu^*$ the slack of the dual direction $z/\log(1/\varepsilon)$ the entropy selects (not computed here). In a mixed start the free particles' weight $\varepsilon^{-1/2}$ dominates the branches' $\log(1/\varepsilon)$: when the branch atoms' frozen signs admit an exact fit, the branches carry it and the free particles keep $O(\varepsilon^2\log(1/\varepsilon))$; when they do not, the free particles reach their walls on the power scale and the limit is basis pursuit over the free atoms (weights $\rho_i$) with the branch atoms sign-constrained and, to leading order, free of charge — their weights $1/(\gamma a_0)$ breaking ties —, the free coefficients $O(1)$.*

(iv) (a pair per atom: the infimal convolution, the hyperbolic entropy) *Particles on one atom without cohesion between them receive the same mirror increment $\Delta$ (D″.45(v)), so the atom's coefficient $W(\Delta) = \sum_j w_j(\eta_j(0) + \Delta)$ has the conjugate $\sum_j \psi_j^*(\eta_j(0) + \Delta)$: the atom's potential is the infimal convolution of its particles' tilted potentials. For a pair started at $\pm\alpha(0)$ with equal masses ($Q_+ = Q_-$), near the branch points $W = 2u\sinh(\gamma a_0\Delta)$ ($u = w_+(0) = -w_-(0)$) and the atom's potential is the hyperbolic entropy*

$$\psi_{\text{pair}}(W) = \frac{1}{\gamma a_0}\big[W\,\operatorname{asinh}(W/(2u)) - (W^2 + 4u^2)^{1/2}\big],$$

*the $u^2 - v^2$ parametrisation's: $\ell^2$ for $|W| \ll u$ (curvature $1/(2\gamma a_0 u)$), an entropy for $|W| \gg u$. A pair per atom gives the branches a free sign; its rich limit ($u \to 0$) is basis pursuit weighted by $1/a_0$, at the temperature $1/\log(1/u)$ of (iii).*

(v) (positive kernel dictionaries: the target at every scale) *Let $\Phi \geq 0$ have a strictly positive vector in its row space (positive kernels: every row of a Gaussian design is one) and every particle start on a positive branch. Then $K \cap \{\Phi w = y\}$ is a bounded polytope; if the target $w^* \geq 0$ is its only point — certified by a $v$ with $h = \Phi^\top v$ vanishing on the target's atoms and $h_i \geq \mu > 0$ on the others (for Gaussian atoms of one width and a $k$-sparse nonnegative target such a certificate exists once $N \geq 2k+1$ exact samples, whatever the atoms' separation: Schiebinger–Robeva–Recht; Eftekhari–Tanner–Thompson–Toader–Tyagi) —, then $S = \{w^*\}$ and the loop converges to $w^*$ from every positive-branch start, at every mass scale, where the free law's own limits from $\alpha = 0$ depend on the scale (the weighted minimum-norm fit of a lazy start, the equal-weight basis pursuit of a rich one, which need not be the target). The convergence is algebraic, not exponential — the residual vanishes in the limit, so $g(\infty) = 0$ —: the other atoms' total mass obeys*

$$\sum_{i \notin \operatorname{supp} w^*} w_i(t) \leq (|v|/\mu)\big(2N \cdot B(w^*, w(0))/t\big)^{1/2},$$

*so the selection's speed is the certificate's margin $\mu$, and on Gaussian grids sampled at few points $\mu$ is typically tiny: over all the three-atom supports of the instances the median margin is far below the showcase targets', a large fraction has no certificate in double precision (exact margins of order $10^{-32}$), and there nonnegative fits carrying $O(1)$ mass off the support exist to rounding — the selection then is invisible on any reachable horizon, the masses lingering on the target's neighbouring atoms (numerics E). Uniqueness alone does not make the target reachable; stability constants (Eftekhari et al.) govern it.*

(vi) (the real loop) *Proposition D″.45(v)'s $V = \frac{1}{2}e^2 + \sum_i B_i(w_i^*, w_i)$ has $\dot V = -k_e e^2$ also with particles on branches when each $w_i^*$ has its particle's sign or is 0, the atoms the target does not use contributing their masses $\beta_i/(\gamma m)$.*

*Proof.* (i) By the two laws $\mathrm{d}w/\,\mathrm{d}\eta = mD = (\beta/m)(G\beta a'^2 + \gamma a^2)$ and $w = \beta a/m$, so $\mathrm{d}\eta/\,\mathrm{d}\log|w| = w/(mD) = a/(G\beta a'^2 + \gamma a^2)$. On a branch $\beta \to 0$ at $\alpha_0$ with $a \to a_0 \neq 0$: the slope tends to $1/(\gamma a_0)$ with an $O(\beta) = O(|w|)$ correction; integrate twice. $\psi = \eta w - (\beta - \beta^0)/(\gamma m)$ ($\beta^0 := Q + c$) with $\eta w \to 0$ and $\beta \to 0$ gives $\psi(0) = (Q+c)/(\gamma m)$, and $B(0, w) = \psi(0) - \psi(w) + \eta(w)w = \beta/(\gamma m)$. At $Q = -c$: $\beta = 2c \sinh^2(\alpha/a_{\max})$, $1/(G\beta a') \approx a_{\max}^2/(2Gc\alpha^2)$, $w \approx 2c\alpha^3/(m a_{\max}^2)$. (ii)(a) $\frac{\mathrm{d}}{\mathrm{d}t} B(\bar w, w) = -\dot\eta \cdot (\bar w - w) = \nabla J(w) \cdot (\bar w - w)/m \leq (J(\bar w) - J(w))/m \leq 0$ by convexity, also for $\bar w$ on $K$'s boundary (the term $B_i(0, \cdot) = \beta_i/(\gamma m)$ has $\mathrm{d}(\beta_i/(\gamma m))/\,\mathrm{d}t = w_i g_i$, the same formula); $J$ decreases ($\mathrm{d}J/\,\mathrm{d}t = -\nabla J^\top \operatorname{diag}(mD) \nabla J/m$), and integrating gives the rate. (b) The sublevel sets of $B(\bar w, \cdot)$ are bounded ($\psi$ superlinear), the cluster points lie in $S$ by (a); at a cluster point $w^1$, $B(w^1, w(t_n)) \to 0$ ($B_i(w_i^1, \cdot)$ is continuous at $w_i^1$, also at $w_i^1 = 0$ on a branch by (i)), and $B(w^1, w(t))$ is nonincreasing, so $w(t) \to w^1 =: w_\infty$. Let $\bar w$ be the projection of the start onto $S$, $Z'$ the branch particles with $w_{\infty,i} = 0$; the others' $\eta$ converge. Since $\eta(t) - \eta(0) = \Phi^\top z(t)$ and $\Phi\bar w = \Phi w_\infty = \hat y$, $\sum_i (\eta_i(t) - \eta_i(0))(\bar w_i - w_{\infty,i}) = 0$ at every $t$; a term with $i \in Z'$ and $\bar w_i \neq 0$ (so $s_i \bar w_i > 0$) would tend to $-\infty$ while the others converge: so $\bar w$ vanishes on $Z'$ too. On the remaining coordinates $\eta_\infty - \eta(0)$ lies in the closed range of their transposed design and $\Phi w_\infty = \hat y$: the KKT conditions of the strictly convex projection onto the face $S \cap \{w_{Z'} = 0\}$, which contains $\bar w$; so $w_\infty = \bar w$. (c) The projection puts no branch coefficient at 0 unless all of $S$ does ($\psi_i' \to -s_i\infty$ at 0: moving toward a point of $S$ with $s_i w_i > 0$ lowers the objective at an infinite rate), so $Z'$ is the set $Z$ of coordinates vanishing on $S$. For $i \in Z$, $s_i g_i(\infty) \leq 0$ (the KKT conditions of $\min_K J$); if it is $< 0$, $\eta_i = g_i(\infty)t + o(t)$ and by (i) $|w_i| \propto \beta_i$ decays at the rate $\gamma a_{0,i} |g_i(\infty)|$. (iii) With $\beta_i(0) = \varepsilon b_i$ the start approaches the branch point and $Q_i$ converges, so $\psi_i$ converges uniformly on compacts of $K$ and $\eta_i(0) = -s_i \log(1/\varepsilon)/(\gamma a_{0,i}) + O(1)$: the term $-\eta_i(0) w_i$ is $\log(1/\varepsilon)|w_i|/(\gamma a_{0,i}) + O(|w_i|)$; the leading term is coercive, so the minimisers stay bounded and converge to its minimiser on $S$ ($\Gamma$-convergence); divided by $\log(1/\varepsilon)$ the objective is the linear program plus $1/\log(1/\varepsilon)$ times a term that is an entropy at 0 (by (i)). The rate: $\eta_j(w) = s_j(1/(\gamma a_{0,j})) \log|w_j| + C_j + O(|w_j|)$

near the branch point and $\eta_j(0) = -s_j \log(1/\varepsilon)/(\gamma a_{0,j}) + O(1)$; the support's equations $\Phi_S^\top z = \eta_S(w_\infty) - \eta_S(0)$, with $\eta_S(w_\infty) = O(1)$ at the vertex, give $z = \log(1/\varepsilon)\xi + O(1)$, so off the support $\log|w_j| = \log|w_j(0)| + \gamma a_{0,j} s_j (\Phi^\top z)_j + O(1) = -\mu_j \log(1/\varepsilon) + O(1)$ ($\gamma a_{0,j}\rho_j = 1 + O(\varepsilon)$). The free particles: D‴.45(ii), their $\eta = O(\log(1/\varepsilon))$ deep in the core where $w \approx G(\beta^0)^2\eta/m$. (iv) The common increment is D‴.45(v); the conjugate of an atom's $W$ as a function of $\Delta$ is the sum of its particles' ($\mathrm{d}W/\mathrm{d}\Delta = \sum_j m D_j$). With $w_+ = ue^{\gamma a_0 \Delta}$, $w_- = -ue^{-\gamma a_0 \Delta}$ near the branch points (by (i)), $W = 2u\sinh(\gamma a_0 \Delta)$, $\psi'_{\mathrm{pair}}(W) = \Delta = \operatorname{asinh}(W/(2u))/(\gamma a_0)$, and one integration gives $\psi_{\mathrm{pair}}$. (v) If $h = \Phi^\top v > 0$, every point of $K \cap \{\Phi w = y\}$ has $h^\top w = v^\top y$, so $|w|_1 \le v^\top y / \min_i h_i$; when $w^*$ is its only point, $\min_K J = 0$ is attained only there, $S = \{w^*\}$, and (ii) applies ($B(w^*, w(0)) < \infty$ by (i)). With the certificate, $v^\top y = h^\top w^* = 0$, so $\sum_{i \notin \mathrm{supp}} h_i w_i = v^\top(\Phi w - y) \le |v||\Phi w - y|$, and (ii)(a) gives $|\Phi w - y|^2 = 2NJ/m \le 2NB(w^*, w(0))/t$. The certificate for Gaussian translates is the Chebyshev-system argument (after Karlin): a nonnegative $\sum_j v_j \varphi(x_j - \cdot)$ vanishing exactly at the target's $k$ atoms exists when $N \ge 2k + 1$. (vi) D‴.45(v)'s computation with $B_i(0, \cdot) = \beta_i/(\gamma m)$: $\mathrm{d}(\beta_i/(\gamma m))/\mathrm{d}t = w_i e \varphi_i(x) = -\dot\eta_i(0 - w_i)$. □

**Status.** (i), (iv)'s conjugate and (vi) exact identities; (ii) proved for every start; (iii) the limit and the off-support rate $\varepsilon^{\mu_j}$ at an $N$-atom vertex proved, the $O(1)$ computed and checked; (iv)'s hyperbolic entropy at leading order near the branch points; (v) proved, the uniqueness for Gaussian translates cited and its certificate computed by linear programming on the instances, the observed rate ($t^{-0.9}$–$t^{-1}$) and the margins' distribution numerical. Classical: exponentiated gradient and the implicit bias of mirror descent (Kivinen–Warmuth; Gunasekar–Lee–Soudry–Srebro), Hessian–Riemannian gradient flows on constrained domains and their limits (Alvarez–Bolte–Brahic), nonnegative least squares by overparametrised gradient flow (Chou–Maly–Verdun et al.), the $u^2 - v^2$ parametrisation and the hyperbolic entropy (Woodworth et al.; Ghai–Hazan–Singer; Vaškevičius–Kanade–Rebeschini), the uniqueness of sparse nonnegative solutions (Donoho–Tanner; Bruckstein–Elad–Zibulevsky), nonnegative super-resolution without separation (Schiebinger–Robeva–Recht; Eftekhari et al.), the exponential penalty in linear programming (Cominetti–San Martín; Weed), the implicit bias of reparametrised linear models as a Bregman divergence (Chou–Maly–Stöger). (ii) is Alvarez–Bolte–Brahic's convergence with the range argument; the inconsistent case with extinction a modest corollary of it; (iv) is the $u^2 - v^2$ parametrisation's; (v)'s identifiability is Eftekhari et al.'s ([115]). New: the paradigm's own law is an exponentiated gradient on its branches, with the temperature $1/(\gamma a_0)$ read off the conservation law, so that a small start freezes the signs — completing Proposition D‴.45(iii), whose branches were numerical —; $B(0, w) = \beta/(\gamma m)$; the two rich limits' temperatures and rates ($\varepsilon^{\mu_j}$ on branches against $\varepsilon^{3/2}$ from $\alpha = 0$) and the mixed start's two scales; and, on positive kernel dictionaries, a scale-independent selection whose speed is the certificate's margin.

**Numerics** (`exp_mirror_branch.py` → `results/chaos/mirror_branch.md`; particles on branches represented without cancellation near their branch points, `mirror_free.ParticleSet`). (A) $\mathrm{d}\eta/\mathrm{d}\log|w| = a/(\gamma a^2 + G\beta a'^2)$ holds to $2.3 \cdot 10^{-9}$ (finite differences, 40 random states free and on branches); at the branch point the temperature equals $1/(\gamma a_0)$ to ten digits ($\alpha_0 = 0.05$, 0.3, 1, 2); $B(\bar w, w) \to \beta/(\gamma m)$ as $\bar w \to 0$ ($2 \cdot 10^{-4}$ relative at $\bar\delta = 10^{-6}$, $5 \cdot 10^{-10}$ at $10^{-12}$, all digits shown at $10^{-30}$ and $10^{-100}$); with $\alpha(0)$ uniform on $[-0.3, 0.3]$ the fraction of starts on branches is 0.52, 0.95, 0.995 and 0.9996 at $\beta(0) = 10^{-3}$, $10^{-5}$, $10^{-7}$, $10^{-9}$; a $\pm$ pair's coefficient follows $2u\sinh(\gamma a_0 \Delta)$ to $2 \cdot 10^{-5}$ for $\Delta \le 10$ ($6 \cdot 10^{-4}$ at $\Delta = 120$, where the members leave their branch points); and on the critical curve $\eta \cdot w^{1/3} \to -K = -3.218$ ($-3.2182$ at $\alpha = 10^{-4}$). (B) On random $6 \times 12$ Gaussian designs (the target on four atoms with signs $+, -, +, -$; masses $10^{-3}$, amplitudes 0.2–0.4; 8 seeds for each of three sign patterns) the flow at $t = 3 \cdot 10^7$ is the Bregman projection onto $S$ computed independently, to $5.0 \cdot 10^{-10}$ at most; where the frozen signs cannot fit $y$ (5 of 8 positive starts, 2 of 8 random ones; $|y - \hat y|$ up to 2.25, 7–8 coordinates vanishing on $S$) $J$ reaches its sign-constrained minimum (to rounding) and the wrong-signed particles go extinct — their masses below $10^{-103}$ —, $B(\bar w, w(t))$ never increases (by more than $2.2 \cdot 10^{-16}$), and the

extinction rates fitted where $10^{-250} < \beta < 10^{-12}$ match $\gamma a_0 |g(\infty)|$ to 0.5–10% (a fit over a finite window of an exponential with a slowly converging prefactor); the mixed starts (half the particles free) fit exactly. Where the least-squares set is not a point — $5 \times 4$ designs with every atom carried by two particles, $y$ random, one atom's frozen sign against its unconstrained least-squares coefficient (8 seeds; 2–6 coordinates vanishing on $S$ on 7 of them, negative coefficients alive on 6) — the flow at $t = 3 \cdot 10^7$ is the Bregman projection to $6.4 \cdot 10^{-10}$ at most, while the projection lies $1.6 \cdot 10^{-4}$–0.20 from the equal split of the duplicates (another point of $S$): the selection with extinction is the Bregman one, not the symmetric one. (C) The rich limits on one $8 \times 20$ design: from branches (amplitudes 0.2–0.6, masses $\varepsilon$) the projection tends to the sign-constrained basis pursuit weighted by $1/|a(0)|$ — distances 0.23, 0.030, $1.3 \cdot 10^{-3}$, $5.1 \cdot 10^{-6}$, $1.1 \cdot 10^{-9}$ at $\varepsilon = 10^{-4}$, $10^{-8}$, $10^{-16}$, $10^{-32}$, $10^{-64}$ ($6 \cdot 10^{-13}$ at $10^{-128}$, the solver's floor) —, the leading order from the certificate (slacks $\mu_{\min} = 0.104$, then 0.130, 0.187) agreeing with it to $4.1 \cdot 10^{-4}$, $2.4 \cdot 10^{-9}$, $3.4 \cdot 10^{-14}$ from $\varepsilon = 10^{-16}$ on: the approach $\varepsilon^{\mu_j}$; from $\alpha(0) = 0$ to (unweighted) basis pursuit — 0.29, 0.14, $4.0 \cdot 10^{-4}$, $5.2 \cdot 10^{-8}$, $4.3 \cdot 10^{-11}$ at $\varepsilon = 10^{-2}, \ldots, 10^{-10}$ —, the leading order agreeing to $3.5 \cdot 10^{-6}$ at $10^{-6}$ and $8.4 \cdot 10^{-13}$ at $10^{-8}$, the closed form off the support to 0.19 and 0.017 at $10^{-8}$ and $10^{-10}$ (its certificate's slack 0.016 is small): the approach $\varepsilon^{3/2}$. With the even atoms free and the odd ones on branches the free particles carry $2.5 \cdot 10^{-6}$, $7.8 \cdot 10^{-14}$, $7.8 \cdot 10^{-22}$ ($\varepsilon = 10^{-4}$, $10^{-8}$, $10^{-12}$: $\approx \varepsilon^2$) and the projection is the branch atoms' own sign-constrained basis pursuit to $2 \cdot 10^{-6}$, $1 \cdot 10^{-12}$, $2 \cdot 10^{-14}$; with the odd atoms' frozen signs against the truth's (no sign-constrained fit of their own) the free particles enter on the power scale and carry 2.27 at $\varepsilon = 10^{-4}$ and $10^{-8}$, the limit's own (basis pursuit over the free atoms, the branch atoms sign-constrained and free of charge), at the distances $4.1 \cdot 10^{-4}$ and $2.5 \cdot 10^{-11}$ — at $10^{-12}$ the projection fails (residual 1.95; reported as failed, C's one projection with a residual above the tolerance). Below $\varepsilon \approx 10^{-32}$ the float Newton step stalls — the curvatures of particles near extinction fall below the float Hessian's rounding — and the decimal-arithmetic fallback converges in 37–55 steps. (D) A $\pm$ pair per atom (amplitudes $\pm 0.4$, equal masses, $u = w(0)$): the atom's coefficient tends to basis pursuit weighted by $1/a_0$ — distances 0.29, 0.21, 0.12, 0.020, 0.011, $2.6 \cdot 10^{-4}$, $1.9 \cdot 10^{-6}$ at $u = 10^{-4}$, $10^{-8}$, $10^{-16}$, $10^{-32}$, $10^{-64}$, $10^{-128}$, $10^{-256}$ —, slowly, its certificate's slacks being small (0.016, 0.020), the leading order agreeing to $1.4 \cdot 10^{-6}$ at $10^{-128}$ and $5.7 \cdot 10^{-11}$ at $10^{-256}$; never are both members of an atom active. (E) Positive kernel dictionaries (Gaussian atoms of one width on a grid of $[-1, 1]$, samples on $[-1.1, 1.1]$, the target 1.0, 0.7, 0.5 on two adjacent atoms and one more, positive branches with $\alpha(0) = 1$ or 2; the showcase targets' margins above 89–94% of all three-atom supports'): with 21 atoms of width 0.15 and 7 samples (margin $\mu = 0.015$) the loop reaches the target from every start scale, $\beta(0) = 10^{-6}$ to 10 — within $6.0 \cdot 10^{-5}$ and on its three atoms at $t = 10^9$, the other atoms' mass at most $5.6 \cdot 10^{-5}$ (the certificate's bound 0.03–0.09) —, while from $\alpha(0) = 0$ the rich start ($\beta(0) = 10^{-6}$) ends on a seven-atom exact fit 1 away from the target (as is equal-weight basis pursuit) and the lazy one ($\beta(0) = 100$) spreads over all 21 atoms; with 41 atoms and 7 samples ($\mu = 0.003$) the selection is slow — $1.0$–$1.3 \cdot 10^{-3}$ away at $t = 10^9$ on $[10, 11, 29, 30, 31]$, the target's third atom shared with its neighbours —; with 41 atoms of width 0.1 and 11 samples ($\mu = 0.027$) every positive start ends on the target's atoms ($\leq 2.3 \cdot 10^{-5}$ at $10^9$), as does the rich free start here, where basis pursuit recovers the target. The decay is algebraic ($t^{-0.9}$–$t^{-1}$ between $10^7$ and $10^9$). Typical targets are another matter: over all three-atom supports the margins' 10, 25, 50, 75, 90% quantiles are 0, $8.4 \cdot 10^{-10}$, $4.0 \cdot 10^{-6}$, $2.0 \cdot 10^{-3}$, 0.015 (21 atoms, width 0.15), 0, 0, $3.8 \cdot 10^{-9}$, $7.0 \cdot 10^{-6}$, $8.1 \cdot 10^{-4}$ (41 atoms, width 0.15) and $1.9 \cdot 10^{-7}$, $2.8 \cdot 10^{-5}$, $1.2 \cdot 10^{-3}$, 0.014, 0.026 (41 atoms, width 0.1) — 25%, 45% and 4% at most $10^{-9}$, numerically without a certificate —, and from a positive start ($\alpha(0) = 1$, $\beta(0) = 10^{-2}$) the loop reaches the supports at the quantiles whose margins exceed $10^{-3}$ by $t = 10^9$ (to $2.3 \cdot 10^{-3}$ and $5.3 \cdot 10^{-5}$; $6.0 \cdot 10^{-3}$, $1.2 \cdot 10^{-4}$ and $2.1 \cdot 10^{-5}$; at the margin $8.1 \cdot 10^{-4}$ it is $3.6 \cdot 10^{-2}$ away, on its way), while on those below $10^{-4}$ it is still 0.37–0.83 away at $10^9$, hardly moving from $10^5$ on, with 0.45–1.33 of mass on other atoms: the selection's speed is the margin. (F) The real loop on positive branches ($k_e = 5$, $G = \gamma = 1$,

twelve atoms of width 0.25, the target on atoms $[2, 6, 9]$, $\alpha(0) = 0.5$, $\beta(0) = 10^{-3}$): $\dot{V} = -k_e e^2$ holds to $1.7 \cdot 10^{-8}$ at random states, $V$ falls monotonically from 3.64 (largest by quarters 3.64, 2.67, 1.72, 0.87), the rms error from 0.014 to 0.009 — while the masses on the unused atoms rise (largest 0.0055, 0.031, 0.20, 0.78 by quarters): over 4000 time units the coefficients are still 0.080 from $w^*$ on this weakly exciting orbit, the unused atoms helping the fit along it, their terms $\beta/(\gamma m)$ in $V$ paid for by the others.

**Proposition D‴.48** (the rich transit: from small masses the kernels enter one at a time — the inverse scale space flow, on the power scale $(\beta^0)^{-1/2}$ from zero amplitudes and on the logarithmic scale on branches; with the paradigm's moving centres the kernels that have not entered stay where they started and the entered ones slide; under a Wasserstein–Fisher–Rao metric the massless kernels condense at the correlation's peaks instead, and the count reaches one kernel per feature only at much smaller masses)**.** *In the setting of Proposition D‴.45 on a dictionary ($\Phi$ the $N \times d$ design of full row rank, $J = (m/2N)|\Phi w - y|^2$, the noiseless loop $\mathrm{d}\eta = \Phi^\top(y - \Phi w)\,\mathrm{d}t/N$):*

(o) (the unit at small mass) *On the critical curve $Q = -c$ the unit is $w = (2c/(ma_{\max}^2))\alpha^3(1 + O(\alpha^2))$ with $\dot{\alpha} = (2Gc/a_{\max}^2)\alpha^2 g(1+O(\alpha^2))$ — to leading order the gradient flow of a depth-3 diagonal unit $u^3$ ($u \propto \alpha$). A start at $\alpha = 0$ with the mass $\beta^0$ lies on $Q = \beta^0 - c$, whose core $|\alpha| \lesssim a_{\max}(\beta^0/(2c))^{1/2}$ the loop crosses with the jump $2\lambda_1$ of $\eta$ (Proposition D‴.45(ii)): $\lambda_1 \propto (\beta^0)^{-1/2}$ plays the depth-3 unit's $\alpha_0^{-(D-2)}$, and the remainder $-(3/2)K|w|^{2/3}$ is the depth-3 potential's. On a branch ($Q < -c$) the amplitude is frozen near $a_0$ and the unit is the exponentiated gradient at the temperature $1/(\gamma a_0)$ (Proposition D‴.47(i)) — the depth-2 behaviour. The scale is set by the start's amplitude against its mass, $\theta = |\alpha(0)|/(\beta^0/(2c))^{1/2}$: $\theta \to 0$ (in particular $\alpha(0) = 0$) gives the power scale $(\beta^0)^{-1/2}$, a fixed amplitude the logarithmic scale $\log(1/\beta^0)/(\gamma a(0))$, and in between there is a continuum — $\alpha(0) \propto \varepsilon^p$ with $0 < p < \frac{1}{2}$ and $\beta^0 \propto \varepsilon$ give $\varepsilon^{-p}\log(1/\varepsilon)$, and a branch start with $\alpha(0) \lesssim \varepsilon^{1/2}$ is still on the power scale.*

(i) (from $\alpha = 0$: the inverse scale space flow on the power scale) *Let $\alpha_i(0) = 0$, $\beta_i^0 = \varepsilon b_i$, $\Lambda(\varepsilon) = (\pi a_{\max}/(4G))(2/(c\varepsilon))^{1/2}$, $\rho_i = b_i^{-1/2}$ (so $\lambda_{1,i} = \Lambda\rho_i(1 + O(\varepsilon))$), $\tau = t/\Lambda$ and $u_i(\tau) = \eta_i(\Lambda\tau)/\Lambda$. As $\varepsilon \to 0$, $u^\varepsilon \to u$ uniformly on compacts of $[0, \infty)$ and $\Phi w^\varepsilon \to \Phi w$ in $L^2_{\mathrm{loc}}$ ($w^\varepsilon \to w$ in $L^2_{\mathrm{loc}}$ on the intervals where the active columns are independent), where $(u, w)$ is the inverse scale space flow of $R(w) = \sum_i \rho_i|w_i|$,*

$$\frac{\mathrm{d}u}{\mathrm{d}\tau} = \Phi^\top(y - \Phi w)/N, \qquad u(0) = 0, \qquad w(\tau) = \operatorname{argmin}\{|\Phi v - y| : v \in \mathcal{N}(u(\tau))\},$$

*$\mathcal{N}(u) = \{v : v_i = 0$ where $|u_i| < \rho_i$, $u_i v_i \geq 0$ where $|u_i| = \rho_i\}$ the box's normal cone — the adaptive inverse scale space path (Burger–Möller–Benning–Osher): $u$ piecewise linear, $w$ piecewise constant; an atom enters when $|u_i|$ reaches $\rho_i$ (the first at $\tau_1 = \min_i \rho_i N/|(\Phi^\top y)_i|$) and leaves when its least-squares coefficient vanishes with an inward velocity; the path reaches weighted basis pursuit at a finite $\tau^*$. The loop reaches the rich limit's support at $t \approx \Lambda\tau^* \propto (\beta^0)^{-1/2}$ (its last digits later, D‴.45(iii)).*

(ii) (on branches: the logarithmic scale) *Let $\alpha_i(0) \neq 0$ with the signs $s_i$ and the masses $\varepsilon b_i$ (on branches for small $\varepsilon$, D‴.47), $\rho_i = 1/(\gamma|a_i(0)|)$, $\ell = \log(1/\varepsilon)$, $\tau = t/\ell$ and $u_i = (\eta_i - \eta_i(0))/\ell$. As $\varepsilon \to 0$ the same convergence holds with the one-sided boxes $s_i u_i \leq \rho_i$: the inverse scale space flow of $\sum_i \rho_i|w_i| + \chi\{s_i w_i \geq 0\ \forall i\}$, ending at the sign-constrained weighted basis pursuit of D‴.47(iii) when $y$ has a fit with the frozen signs; $u$ converges like $1/\log(1/\varepsilon)$. A $\pm$ pair per atom (D‴.47(iv)) gives the two-sided box with the weights $1/(\gamma a_0)$ — Pesme–Flammarion's depth-2 limit. The rate $1/\log(1/\varepsilon)$ keeps this limit asymptotic: at $\varepsilon = 10^{-4}$ the dual is still 40–80% of the weights away and the entry times are off by more than the first entry.*

(iii) (moving centres, the paradigm's metric: the sliding transit) *Let the kernels' centres move by the paradigm's law (Proposition D‴.34's positions, Euclidean, the mass in*

*the drift),* $\dot{c}_i = G\beta_i a_i\langle r\partial_c\varphi_i\rangle = Gmw_i\langle r\partial_c\varphi_i\rangle$*, the free law on* $(\alpha,\beta)$ *unchanged (*$Q$ *kept for any drive). (a) A kernel whose dual coordinate stays inside its box by the margin* $\mu$*,* $|u_i| \le (1-\mu)\rho_i$ *on* $[0,\tau]$*, has the coefficient of the core's profile,* $w_i = s_i(a_{\max}(\varepsilon b_i)^{3/2}/(m(2c)^{1/2}))\cos(\pi\mu/2)/\sin^3(\pi\mu/2)(1+O(\varepsilon^{1/2}/\mu))$ *(*$s_i$ *the sign of* $u_i$*; at most the same with* $\mu^{-3}$*,* $\cos(\pi\mu/2)/\sin^3(\pi\mu/2) \le (8/\pi^3)\mu^{-3}(1+O(\mu^2))$*), and its centre creeps at*

$$\frac{\mathrm{d}c_i}{\mathrm{d}\tau} = \frac{\pi a_{\max}^2}{4c}\varepsilon b_i^{3/2}\cdot\frac{\cos(\pi\mu/2)}{\sin^3(\pi\mu/2)}\cdot s_i\langle r\partial_c\varphi_i\rangle\cdot\big(1+O(\varepsilon^{1/2}/\mu)\big),$$

*toward a larger correlation: in the limit the kernels that have not entered stay where they are — at their starts, or where they left the active block —, their dual coordinates integrating the residual's correlation there. The limit is not uniform in the margins: a kernel that loses a near-tie lingers inside its wall at a small margin* $\mu$*, so its position stays frozen only while* $\varepsilon \ll \mu^3$ *— at a finite* $\varepsilon$ *it may slide toward the residual's peak and enter in place of the predicted kernel. (b) Between entries the active block relaxes on the fast time* $t = O(1)$ *— on the critical curve, by an* $\varepsilon$*-independent flow — to a stationary point of* $J$ *over its coefficients and centres. The limit is the sliding inverse scale space:* $u_j(\tau) = u_j(\tau_k) + (\tau-\tau_k)\langle r_k\varphi(\cdot;c_j)\rangle$ *for the inactive kernels (*$c_j$ *where it is), entry at* $|u_j| = \rho_j$ *— kernels reaching their walls together (a tie, e.g. by symmetry) entering together —, the block's relaxation, an exit where a coefficient vanishes with an inward correlation; no kernel enters once the block fits the target exactly. A tie shares a feature: a bump midway between two grid kernels is formed by both, at its centre, with half its amplitude each, at every* $\varepsilon$*. Generically (no ties), a target of* $k$ *separated bumps of the kernels' width is formed by* $k$ *kernels at the bumps' centres, the others keeping their starting positions and negligible masses — in the limit* $\varepsilon \to 0$ *at fixed margins. At a finite* $\varepsilon$ *the margins the entries leave decide (Proposition D‴.51): a runner-up whose margin at an entry is* $O(\varepsilon^{1/2})$ *is pulled across its wall by the entrant's overshoot and shares the feature, and a kernel lingering near its wall in an unfitted residual creeps into it unless* $\varepsilon \ll \mu^4$*; so near a tie (a bump 0.001 off the midpoint) the loop forms an extra kernel down to* $\varepsilon \approx 10^{-6}$*, and close bumps' next kernel, creeping toward the residual's peak while it waits, enters earlier than on the path. A target that is not a few kernels of the dictionary's width (bumps wider than the kernels) is fitted along a staircase: the kernels enter one at a time at roughly geometrically growing times (the next entry waits* $\propto 1/|r|$*), the residual falling at each entry.*

(iv) (moving centres, a Wasserstein–Fisher–Rao metric: condensation) *With the positions weighted by the masses (*$\dot{c}_i = Ga_i\langle r\partial_c\varphi_i\rangle$*, D‴.34(i)'s alternative) an inactive kernel's amplitude is* $a \approx a_{\max}(\varepsilon b_i/(2c))^{1/2}\tan(\pi u_i/(2\rho_i))$ *and its centre moves at* $O(1)$ *in* $\tau$*: before the first entry the limit is*

$$\frac{\mathrm{d}u_i}{\mathrm{d}\tau} = \langle f^*\varphi(\cdot;c_i)\rangle, \qquad \frac{\mathrm{d}c_i}{\mathrm{d}\tau} = \frac{\pi a_{\max}^2}{4c\rho_i}\tan\Big(\frac{\pi u_i}{2\rho_i}\Big)\langle f^*\partial_c\varphi(\cdot;c_i)\rangle,$$

*the kernels climbing the correlation's peaks, faster as they near the wall: the massless kernels condense at the peaks before entering. In the limit the count is still one kernel per feature — the condensed kernels reach their walls at different times (their dual coordinates integrate the correlation along different paths), so generically only one enters per feature —, but the condensed kernels are near-ties at small margins, so at a finite* $\varepsilon$ *several enter with each feature, and the count falls to one per feature only at much smaller masses than under the paradigm's metric.*

(v) (moving widths) *With the widths moving by the same position law (*$\dot{\sigma}_i = G\beta_i a_i\langle r\partial_\sigma\varphi_i\rangle$*, the paradigm's full kernel law) the sliding transit fits the features' widths as well: bumps wider or narrower than the kernels' starting width are formed one per bump, at their centres, widths and amplitudes, where fixed widths spread a wider bump over several kernels and fit a narrower one with the wrong amplitude.*

*Proof.* (o) With $x = \alpha/a_{\max}$, $2c\sinh^2 x \cdot a_{\max}\tanh x/m = (2ca_{\max}/m)x^3(1+O(x^2))$ and $\beta = 2c\sinh^2 x$, $a' = 1+O(x^2)$; with $u = \kappa\alpha$, $\kappa^3 = 2c/(ma_{\max}^2)$: $w = u^3$ and $\dot u = (2Gc/(\kappa a_{\max}^2))u^2 g$, the depth-3 unit's gradient flow $3u^2\cdot hg$ with the step $h = 2Gc/(3\kappa a_{\max}^2)$. (i) In the dual variable $\zeta$ ($u = \Phi^\top\zeta$) the rescaled loop is $\mathrm{d}\zeta/\mathrm{d}\tau = -\nabla\mathcal{D}_\varepsilon(\zeta)/N$, $\mathcal{D}_\varepsilon(\zeta) = \sum_i g_{\varepsilon,i}((\Phi^\top\zeta)_i) - y^\top\zeta$, $g_{\varepsilon,i}(v) = \psi_i^*(\Lambda v)/\Lambda$ (D‴.45(iii)'s dual flow in $\tau$). In the core $\alpha = a_{\max}(\beta^0/(2c))^{1/2}\tan(\pi\eta/(2\lambda_1))(1+O(\beta^0))$ and $\psi^* = 2c\sinh^2(\alpha/a_{\max})/(\gamma m)$, so $g_{\varepsilon,i}(v) = O(\varepsilon^{3/2}\delta^{-2})$ uniformly on $|v| \le (1-\delta)\rho_i$; beyond the wall $\alpha \ge 2Gc(\eta-\lambda_1)$ exactly (arctan $\le \pi/2$), so $g_{\varepsilon,i}(v) \ge C\Lambda^{-1}e^{(4Gc/a_{\max})\Lambda\delta\rho_i}$ on $|v| \ge (1+\delta)\rho_i$. So each $g_{\varepsilon,i}$ converges to the indicator of $[-\rho_i,\rho_i]$ uniformly on the compacts off $\pm\rho_i$; for the composition, a compact set of $\mathbb{R}^N\setminus P$ lies at a positive distance from $P$, so $\max_i|(\Phi^\top\zeta)_i|/\rho_i \ge 1+\delta$ on it and, every $g_{\varepsilon,j}$ being nonnegative, $\mathcal{D}_\varepsilon \to +\infty$ uniformly there, while on the compacts of $\operatorname{int} P$ $\mathcal{D}_\varepsilon \to -y^\top\zeta$ uniformly: $\mathcal{D}_\varepsilon \to \mathcal{D}_\infty(\zeta) = \chi_P(\zeta) - y^\top\zeta$, $P = \{|\Phi^\top\zeta| \le \rho\}$, epigraphically (convex functions on $\mathbb{R}^N$, $0 \in \operatorname{int}\operatorname{dom}\mathcal{D}_\infty$: Rockafellar–Wets, Thm. 7.17) — in finite dimension, Mosco convergence. Attouch's theorem turns it into the graph convergence of the subdifferentials and Brezis–Pazy's into the convergence of the semigroups, uniformly on compacts, from $\zeta(0) = 0 \in \operatorname{int} P$ — the route Berthier–Pillaud-Vivien take for mirror flows started near the boundary of a fixed potential's domain; here the potential itself changes with the start. The energy identity $\mathcal{D}_\varepsilon(\zeta(T)) + N\int_0^T|\dot\zeta|^2 = \mathcal{D}_\varepsilon(0) = 0$ ($\mathrm{d}\zeta/\mathrm{d}\tau = -\nabla\mathcal{D}_\varepsilon/N$), with the liminf inequality at $\zeta(T)$, makes $\dot\zeta^\varepsilon \to \dot\zeta$ in $L^2(0,T)$, i.e. $\Phi w^\varepsilon \to \Phi w$; the limit's derivative is the minimal-norm section, $-N^{-1}(\Phi v - y)$ with $v$ the least-squares element of $\mathcal{N}(u)$; and $w^\varepsilon \to w$ on the compact subintervals between events where the active columns are independent and the inactive atoms' margins are bounded below. Its structure — piecewise linear, finite termination at weighted basis pursuit — is the adaptive inverse scale space method's (Burger et al.). (ii) The same with D‴.47(i)'s entropy at the branch point: $\psi_i^*(\eta_i(0)+\ell v) - \psi_i^*(\eta_i(0)) = (|w_i(0)|/(\gamma a_{0,i}))(e^{\gamma a_{0,i}\ell s_i v} - 1)(1+o(1))$ near the branch point, so $g_{\varepsilon,i}(v) \to 0$ for $s_i v < \rho_i$ (uniformly on compacts, like $e^{-\ell(1-\gamma a_{0,i}s_i v)}/\ell$) and $\to +\infty$ for $s_i v > \rho_i$; the $O(1)$ in $\eta_i(0)$ gives $u$'s $1/\ell$; the pair: D‴.47(iv)'s infimal convolution. (iii)(a) At $\eta = (1-\mu)\lambda_1$ the core's tangent profile gives D‴.45(iii)'s off-support formula, with $\cos(\pi\mu/2)/\sin^3(\pi\mu/2) = (2/(\pi\mu))^3(1+O(\mu^2))$; then $\dot c_i = Gmw_i\langle r\partial_c\varphi_i\rangle$ and $\mathrm{d}c_i/\mathrm{d}\tau = \Lambda\dot c_i$, with $\Lambda Gm\cdot a_{\max}(\varepsilon b_i)^{3/2}/(m(2c)^{1/2}) = (\pi a_{\max}^2/(4c))\varepsilon b_i^{3/2}$: the creep's speed. (b) Formal: the active block's mobility on the critical curve is $mD = 3|w|^{4/3}/K$ (from $\eta = \lambda_1 - K|w|^{-1/3}$), independent of $\varepsilon$, while the inactive kernels' and the entrants' times are $O(\Lambda)$. (iv) In the core $a \approx \alpha$, and $\Lambda G\cdot a_{\max}(\varepsilon b/(2c))^{1/2} = \pi a_{\max}^2\sqrt{b}/(4c)$, $\sqrt{b} = 1/\rho$. □

**Status.** (o) exact expansions; (i) and (ii) proved (Mosco convergence of the explicit dual potentials, the convergence of the semigroups; the path's structure classical), $u$'s rates numerical; (iii)(a) proved (the profile to leading order in $\varepsilon^{1/2}/\mu$), (iii)(b) formal (two time scales; the block's relaxation to a stationary point assumed; ties and near ties as stated), the sliding path and the recovery of the bumps numerical; (iv) the inactive kernels' limit derived to leading order and checked along the trajectory, with unequal masses; the count as $\varepsilon \to 0$ numerical; (v) numerical. Classical: the inverse scale space flow and its adaptive algorithm (Burger–Gilboa–Osher–Xu; Burger–Möller–Benning–Osher), linearised Bregman iterations and differential inclusions (Yin–Osher–Goldfarb–Darbon; Osher–Ruan–Xiong–Yao–Yin), saddle-to-saddle dynamics and incremental learning of diagonal linear networks at vanishing initialisation and their identification with the adaptive inverse scale space method (Pesme–Flammarion; Berthier; Gissin–Shalev-Shwartz–Daniely), inverse scale space limits of mirror flows by Mosco convergence (Berthier–Pillaud-Vivien), depth-$D$ potentials and the polynomial scales of deep units (Woodworth et al.; Jacot et al.; Chou–Maly–Rauhut's initialisation-weighted $\ell^1$), Mosco convergence and the convergence of gradient flows (Mosco; Attouch; Brezis–Pazy), sliding Frank–Wolfe and ADCG (Denoyelle–Duval–Peyré–Soubies; Boyd–Schiebinger–Recht), conic particle gradient descent (Chizat), condensation and early alignment at small initialisation (Luo et al.; Zhou et al.; Boursier–Pillaud-Vivien–Flammarion; Min–Mallada–Vidal; Kumar–Haupt), alternating gradient flows and unbalanced initialisations' rapid feature learning (Kunin et al.), deep diagonal networks as mirror flows (Labarrière et al.), the quantisation of features at small initialisation (Maennel–Bousquet–Gelly). (i) follows Berthier–Pillaud-Vivien's route for a family of potentials that changes with the start,

and (ii) is their entropic case ([116]). New: the paradigm's free mass–amplitude unit at small mass as an effective depth-3 unit with an explicit potential, its transit scale $\Lambda(\varepsilon) \propto (\beta^0)^{-1/2}$ and the continuum of scales set by $\alpha(0)/(\beta^0)^{1/2}$; with the paradigm's moving centres (and widths) the sliding transit — the inactive kernels frozen by the mass in the position law's drift, with the creep's exact profile and its non-uniformity $\varepsilon \ll \mu^3$, ties sharing a feature —, answering which kernels form from a spread cloud (paper 13's open question for the cohesion leak) at small masses: one per feature, at its centre and width; and what the metric on the positions decides: not the limit's count (one per feature under both) but where the massless kernels go — they stay put under the paradigm's metric, condense at the correlation's peaks under a Wasserstein–Fisher–Rao one — and how small the masses must be.

**Numerics** (`exp_mirror_transit.py` → `results/chaos/mirror_transit.md`; $m = 12$, $G = 1$, $\gamma = 0.1$, $a_{\max} = 1$, $c = 0.025$; each kernel integrated on its own curve — $\alpha$ alone, $\beta = \varepsilon b_i + 2c\sinh^2(\alpha/a_{\max})$ —, so that $Q$ is kept exactly). (A) From $\alpha = 0$ on three $6 \times 14$ Gaussian designs (3-sparse truths, masses $\varepsilon b_i$, $b_i$ log-uniform in $[1/4, 4]$) the adaptive inverse scale space path of $\sum|w_i|/\sqrt{b_i}$ ends at weighted basis pursuit (to $5 \cdot 10^{-16}$) after 3, 8 and 10 events (entries and exits); the loop in $\tau = t/\Lambda(\varepsilon)$ follows it: $\sup_\tau|u - u_{\mathrm{ISS}}|$ falls from 0.54, 0.61, 1.15 at $\varepsilon = 10^{-2}$ to $6.9 \cdot 10^{-4}$, $8.9 \cdot 10^{-4}$, $1.7 \cdot 10^{-3}$ at $10^{-8}$ and $6.7 \cdot 10^{-6}$, $8.8 \cdot 10^{-6}$, $2.0 \cdot 10^{-5}$ at $10^{-12}$ — log-log slopes 0.500 on all three between $10^{-6}$ and $10^{-8}$, 0.503, 0.501, 0.484 between $10^{-8}$ and $10^{-12}$: the rate $1/\Lambda \propto \varepsilon^{1/2}$ —; the entry times converge at the same rate down to the grid's step ($1.5 \cdot 10^{-3}$ on the first design by $\varepsilon = 10^{-8}$; 0.35 and 0.26 at $10^{-8}$ on the others, whose late entries approach their walls slowly, their steps 0.059 and 0.044 by $10^{-10}$); the residual's difference $|\Phi(w - w_{\mathrm{ISS}})|$ is $O(1)$ at an event over a $\tau$-width $\propto \varepsilon^{1/2}$ and $O(\varepsilon^{3/2})$ away from the events — its median over the samples $9.7 \cdot 10^{-8}$, $8.8 \cdot 10^{-11}$, $8.7 \cdot 10^{-14}$ at $\varepsilon = 10^{-6}$, $10^{-8}$, $10^{-10}$ on the first design, the slopes 1.51, 1.69, 1.46 between $10^{-6}$ and $10^{-10}$, the inactive atoms' coefficients $\propto \varepsilon^{3/2}$ —, so its rms over $\tau$ falls like $\varepsilon^{1/4}$ (slope 0.26 between $10^{-6}$ and $10^{-8}$ on the first design) until the width drops below the grid's step, below which the rms counts the few samples that fall within an entry; the end at $1.5\tau^*$ is weighted basis pursuit to $8.5 \cdot 10^{-12}$–$6.2 \cdot 10^{-9}$ at $10^{-8}$ and $1.1 \cdot 10^{-16}$–$6.1 \cdot 10^{-15}$ at $10^{-12}$. (B) On branches (amplitudes 0.2–0.6, the first design) the loop follows the one-sided path in $\tau = t/\log(1/\varepsilon)$ — $\sup|u - u_{\mathrm{ISS}}| = 15.7$, 8.0, 4.0, 2.0, 1.0 and the entry times' largest difference 55, 28, 14, 7.0, 3.5 at $\varepsilon = 10^{-4}$, $10^{-8}$, $10^{-16}$, $10^{-32}$, $10^{-64}$, exactly $\propto 1/\log(1/\varepsilon)$ (the path's weights $1/(\gamma|a(0)|) \approx 17$–$50$) —, ending at the sign-constrained weighted basis pursuit (to $1.3 \cdot 10^{-5}$ at $\varepsilon = 10^{-4}$, below $10^{-14}$ from $10^{-16}$ on); $\pm$ pairs follow the two-sided path the same way (21, 7.9, 4.0, 2.0, 0.99 at $u = 10^{-4} \ldots 10^{-64}$). (C) Moving centres (Gaussian kernels of width 0.12 on 61 samples of $[-1, 1]$, 21 kernels started on a grid of spacing 0.1, the target three bumps of the same width at $-0.52$, 0.07, 0.61 with amplitudes 0.9, $-0.6$, 0.75): (1) the sliding inverse scale space enters the kernels started at $-0.5$, 0.6 and 0.1 ($\tau = 10.72$, 12.85, 16.39), which slide onto the bumps and fit them exactly (rms residual $1.4 \cdot 10^{-16}$); the loop with the paradigm's metric forms 8 kernels at $\varepsilon = 10^{-3}$ (three clusters on the bumps), 4 at $10^{-4}$ (a fourth of coefficient $-0.013$), and exactly these three for $\varepsilon \le 10^{-5}$ — entering at $\tau = 10.75$, 12.88, 16.41 (the grid's step 0.048), at the bumps' centres with their amplitudes (rms residual $3.8 \cdot 10^{-7}$, $1.6 \cdot 10^{-8}$, $4.4 \cdot 10^{-10}$, $1.3 \cdot 10^{-11}$ at $\varepsilon = 10^{-5} \ldots 10^{-8}$) —, the other kernels displaced by at most 0.033, $2.6 \cdot 10^{-3}$, $2.3 \cdot 10^{-4}$, $2.3 \cdot 10^{-5}$ ($\propto \varepsilon$, the creep) with coefficients below $2.9 \cdot 10^{-5}$, $4.3 \cdot 10^{-7}$, $1.1 \cdot 10^{-8}$, $3.4 \cdot 10^{-10}$ ($\to \varepsilon^{3/2}$, the core's profile); (2) with the Wasserstein–Fisher–Rao metric the kernels' centres before the first entry follow (iv)'s limit to $1.7 \cdot 10^{-5}$ at $\varepsilon = 10^{-4}$, $3.4 \cdot 10^{-7}$ at $10^{-5}$ and $2 \cdot 10^{-7}$ below (the integrators' floor at their tolerance $10^{-9}$), all 21 gathered at three positions ($-0.521$, 0.068, 0.613) by $\tau = 10$ — with unequal masses ($b_i$ log-uniform in $[0.26, 3.8]$, $\varepsilon = 10^{-6}$) the limit with $\rho_i = b_i^{-1/2}$ to $1.1 \cdot 10^{-7}$, the one with the masses ignored 0.45 away —, and the loop forms 15, 12, 8, 3, 3 kernels at $\varepsilon = 10^{-4}$, $10^{-5}$, $10^{-6}$, $10^{-8}$, $10^{-10}$, always in three clusters at the bumps (rms residual $\le 6 \cdot 10^{-10}$): one per feature from $\varepsilon \approx 10^{-8}$; (3) with the centres held on the grid ($\varepsilon = 10^{-6}$, to $\tau = 64$) it forms 5 kernels and is still 0.019 from the target (the grid's least-squares fit $8.4 \cdot 10^{-5}$); (4) on sixteen random three-bump targets

(centres in $[-0.8, 0.8]$ at least 0.3 apart, amplitudes $\pm U(0.4, 1)$) the sliding path recovers all sixteen exactly with three kernels and no exits; the loop with the paradigm's metric 11, 13 and 16 of them at $\varepsilon = 10^{-5}$, $10^{-6}$, $10^{-8}$ (otherwise one or two extra kernels; the inactive kernels' largest displacement 0.12, 0.070, 0.0055), its formed kernels those of the path, in its order, on 15 of 16 at $10^{-8}$ (the sixteenth: (6)); with the Wasserstein–Fisher–Rao metric 4–7 kernels at $\varepsilon = 10^{-6}$ (exactly three at the bumps on none), 3–5 at $10^{-8}$ (on 10 of 16) and exactly three at the bumps on all sixteen at $10^{-10}$; with the centres held on the grid 4–6 kernels, rms residuals 0.011–0.026; (5) ties and near ties: a bump midway between two kernels enters both together, on the path and in the loop at every $\varepsilon$, each with half its amplitude at its centre (0.4, 0.4); a bump 0.001 off the midpoint (the runner-up's margin $\mu^* = 3.5 \cdot 10^{-3}$) is formed by one kernel on the path, by two at $\varepsilon = 10^{-6}$ (the runner-up with 0.062 of 0.8; $\mu^*/\varepsilon^{1/2} = 3.5$) and by one from $10^{-8}$ on, and three bumps with one at a midpoint (the tie broken by the other bumps' tails) by four kernels at $10^{-6}$ ($-0.446$ and $-0.154$ on the midpoint's bump) and by the path's three from $10^{-8}$ on — the near-tie law of Proposition D‴.51; two bumps 0.15 apart (1.25 widths) and 0.25 apart, off the grid, are resolved by two kernels at every $\varepsilon$, the second entering earlier than on the path ($\tau = 16.2$, 29.6, 33.1 at $\varepsilon = 10^{-6}$, $10^{-8}$, $10^{-10}$ against 33.5; 12.1, 16.1, 17.0 against 17.1) — it creeps toward the residual's peak while it waits; and a target of three bumps wider than the kernels (width 0.2), not three kernels of the dictionary, is fitted along a staircase: the path enters the kernels one at a time at roughly geometrically growing times (21 entries from $\tau = 9.1$ to $1.2 \cdot 10^{10}$, its rms residual falling from 0.51 to $3.3 \cdot 10^{-5}$ — 11 of its 21 relaxations stationary within the budget), and the loop at $\varepsilon = 10^{-5}$, followed on a geometric $\tau$ grid, enters the same ten kernels first (in another order, earlier: the creep at this mass) and ends with 16 kernels at the rms residual $2.4 \cdot 10^{-7}$; (6) the creep (target 10 of (4)): after the second entry an inactive kernel lingers inside its wall at a small margin $\mu$; its instantaneous speed $\mathrm{d}c/\,\mathrm{d}\tau$ (the loop's own right-hand side) is (iii)(a)'s profile to 0.990–0.996 at $\varepsilon = 10^{-8}$ ($\mu = 0.0039$–$0.0065$) and to 0.9999–1.0000 at $10^{-10}$ — the correction $O(\varepsilon^{1/2}/\mu)$; 0.55 at $\mu = 4.5 \cdot 10^{-4}$, $\varepsilon = 10^{-8}$, where $\varepsilon^{1/2}/\mu = 0.22$ and the kernel is entering — and a quarter of the bound's $\mu^{-3}$ form (0.255–0.258: $8/\pi^3 = 0.258$); at $\varepsilon = 10^{-8}$ kernel 15, 0.0065 inside its wall where $\langle r\partial_c\varphi\rangle = -0.033$, creeps by 0.16 before the third entry and enters in place of the path's kernel 12 (the formed kernels' centres unchanged), at $10^{-10}$ it moves by $8 \cdot 10^{-4}$ and the loop enters the path's kernel; a kernel left at the margin 0.0043 beside a fitted bump, where $\langle r\partial_c\varphi\rangle \approx -10^{-5}$, moves by $3.8 \cdot 10^{-3}$ and $3.2 \cdot 10^{-5}$ in all at $10^{-8}$ and $10^{-10}$. (D) With moving widths too ($\dot\sigma = G\beta a\langle r\partial_\sigma\varphi\rangle$, kernels started at width 0.12) the transit forms the three bumps at widths 0.2 and 0.08 with exactly three kernels at their centres, widths and amplitudes at $\varepsilon = 10^{-6}$ (rms residual $2.3 \cdot 10^{-8}$ and $1.2 \cdot 10^{-8}$; at $10^{-5}$ the wide bumps with a fourth kernel of coefficient 0.059), where fixed widths spread the wide bumps over six kernels (rms residual 0.024) and fit the narrow ones by three kernels with the amplitudes 0.71, $-0.48$, 0.59 (rms residual 0.094).

**Numerics** (`exp_chaos_learning.py` → `results/chaos/learning_table.md`). The instance: $n = 1$, saturated features with $a_{\max} = 4$, $c_{\max} = 3$, $s \in [0.3, 2]$, prior $\vartheta^0 = 0$, $G = \mathrm{diag}(1, 0.25, 0.25)$; a plant of three Gaussian bumps, $(A, C, S) = (1, -1.5, 0.5)$, $(-0.7, 0, 0.4)$, $(0.8, 1.5, 0.5)$, which three kernels of weights $3A_k$ represent exactly; reference $x_m = 2\sin t$, $\rho$ on a grid of 256 times; slow time, midpoint rule with step 0.02; $m \in \{16, 64, 256\}$ sharing their particles with an $M = 2048$ reference (the stand-in for the mean-field flow, as in Theorems D′ and D″); horizon 900 slow units; four seeds per start. Proposition K gives $\sigma_{\mathrm{crit}} = 0.04716$ (collapsed atom $a = 0.331$, $c = -0.057$, $s = 1.218$); the runs use $\sigma_0 = 0.15\sigma_{\mathrm{crit}} = 0.00707$ (leak time 141 slow units). Whitened constants by search: $L^G_\vartheta = 4.044$ (closed form 4.658), $L^G_2 = 11.11$ (closed form $\lambda_G a_{\max}/s^2_{\min} = 44.44$); $K_\Phi = 4$, $F_\rho = 0.999$.

*The hypothesis at the stationary measure, and the window (table 1).* The critical point of $J_3$ reached from the exact representation, atoms

$$(a, c, s) = (2.4504, -1.4161, 0.8287), \quad (-2.2879, 0.0120, 0.7009), \quad (2.0478, 1.3427, 0.8810),$$

has $\|\nabla J_3\|_\infty = 4 \cdot 10^{-17}$ (a stationary measure $\mu^*$ of the mean-field flow), fit error $\|f - \hat{f}\|/\|f\| = 0.244$ on the orbit (the leak's bias: amplitudes 2.0–2.5 against $3A_k = 2.1$–3, widths 0.70–0.88 against 0.4–0.5), and a positive whitened Hessian (eigenvalues 0.0057–0.2003). $\kappa_R$ at the atoms is 0.00004, −0.00227, 0.00411; over the spheres of radius $\varrho'$ around them (400 directions, radii $\varrho'$ and $\varrho'/2$) it is 0.00461 ($\varrho' = 0.05$), 0.00512 (0.1), 0.00619 (0.2) and 0.00730 (0.3), so (H) holds at $\mu^*$ with $\varrho' = 0.1$ and margin $\kappa = \sigma_0 - 0.00512 = 0.00195$ (relaxation time $1/\kappa = 512$) and fails from $\varrho' \approx 0.3$ on. The window of Remark D‴.1 is realised: $\kappa_R[\mu^*](N_{0.1}) = 0.00512 < \sigma_0 = 0.00707 < \sigma_{\text{crit}} = 0.04716$. At $\sigma_0 = 0.3\sigma_{\text{crit}}$, by contrast, the three-atom critical point (fit error 0.406) fails the splitting condition at its third atom ($\kappa_R = 0.01696 > \sigma_0 = 0.01415$): there the mean-field limit has more atoms, and the hypothesis has to be checked at that configuration. The theorem's constants for the warm start ($\Lambda_1 = 0$, $\kappa := \frac{1}{2}\min\{\kappa(\varrho'), \min_s a(s)\} = 0.00070$, the factor $\frac{1}{2}$ leaving room for the sampling of the supremum by finitely many directions): $R_\Theta = 2857$, $\varrho = 1.1 \cdot 10^{-6}$, $\theta = 4.6 \cdot 10^{-11}$, $C_1 = 2.2 \cdot 10^9$, $m_0 = 1.2 \cdot 10^{23}$ — the worst-case envelope of Remark D‴.6.

*The hypothesis along the flow (table 2).* On the warm start ($\mu_0$ the mixture of Gaussians of width 0.02 around the atoms) the margin $a(s) = \sigma_0 - \kappa_R[\mu^*]$(spheres of radius 0.1 around 256 reference particles, 24 directions) stays between +0.00139 and +0.00167 over the whole run, so (H) holds with $s_1 = 0$; the same statistic with the reference's own residual is larger by at most 0.00013 (its sampling fluctuation; the proof's bound (0.2) allows $L_2^G \|r_{\text{ref}} - r^*\|_{L^1(\rho)} \le 0.214$), the reference's rms drift decays from $6 \cdot 10^{-3}$ to $3 \cdot 10^{-6}$ and its fit error stays at 0.245–0.248: the reference is stationary. On the cold start ($a \sim \mathcal{N}(0, 0.8^2)$, $c \sim \mathcal{U}(-2, 2)$, $s \sim \mathcal{U}(0.4, 1)$, a connected cloud) the margin is negative over the whole run, −0.196 at the first record (−0.179 at $s = 1$) and −0.0025 at $s = 900$, while the fit error falls from 0.948 to 0.244 — the same fit as the warm limit — and the rms drift from 0.15 to $4.7 \cdot 10^{-4}$: the cold reference converges to a configuration of the same quality whose support reaches the ridges (Remark D‴.2).

*Warm start (table 3).* Window maxima of $D_m$ over $(0, 112.5]$, $(112.5, 225]$, $(225, 450]$, $(450, 900]$ (Table 20):

Table 20: Theorem D‴ on the three-bump instance, warm start: window maxima of the coupled distance $D_m$ over four windows of slow time, the mean on the last window, the theorem's constant $C_1/m$, the learned-function distance and the per-particle maximum (`results/chaos/learning_table.md`, table 3).

| $m$ | $(0, 112.5]$ | $(112.5, 225]$ | $(225, 450]$ | $(450, 900]$ | mean on $(450, 900]$ | $C_1/m$ | mean $\|\hat{f}_\Theta - \hat{f}_{\text{ref}}\|$, last window | $\max_i \|\delta_i\|_{G^{-1}}$ at $s = 900$ |
|---|---|---|---|---|---|---|---|---|
| 16 | 0.396 ± 0.24 | 0.423 ± 0.26 | 0.441 ± 0.28 | 0.448 ± 0.28 | 0.446 ± 0.28 | $1.4 \cdot 10^8$ | 0.0307 ± 0.0097 | 1.36 |
| 64 | 0.122 ± 0.046 | 0.131 ± 0.049 | 0.136 ± 0.050 | 0.138 ± 0.050 | 0.138 ± 0.051 | $3.4 \cdot 10^7$ | 0.0154 ± 0.0033 | 0.62 |
| 256 | 0.0119 ± 0.0050 | 0.0127 ± 0.0052 | 0.0134 ± 0.0055 | 0.0137 ± 0.0056 | 0.0136 ± 0.0055 | $8.4 \cdot 10^6$ | 0.00467 ± 0.0012 | 0.21 |

The distance saturates: last-to-first window ratios 1.13, 1.13, 1.15 and last-to-previous 1.02, 1.01, 1.02. Its $m$-law has slope $-1.28 \pm 0.30$ (per-seed fits; the theorem's mean square: $-1$), the learned function's $-0.69 \pm 0.16$ ($m^{-1/2}$: $-0.5$). The per-particle maxima 1.36, 0.62, 0.21 are far above $3\varrho'/4 = 0.075$: the localisation of part (c) is not in force at these $m$ (Remark D‴.6), and the values lie below the vacuous $C_1/m$ by 8.4–8.8 orders of magnitude.

*Proposition D‴.5 (the same table).* The sensitivities of the three atoms to the weights have whitened norms $\|S_k\| = 8.38$, 4.66, 8.32. The multinomial expectation $\mathbb{E}D_\infty = (1/m - 1/M)\sum_k \bar{w}_k \operatorname{tr}(\tilde{S}_k \tilde{C} \tilde{S}_k^\top)$ is 0.823, 0.201, 0.0454 against the four-seed means $0.446 \pm 0.28$, $0.138 \pm 0.051$, $0.0136 \pm 0.0055$ (ratios 0.54, 0.69, 0.30: a mean over four seeds of a heavy-tailed quantity,

$\mathbb{E}|w^m - w^M|^2$ being carried by the rare large fluctuations). The per-seed law is the sharp test: $\sum_k w_k^m \|\tilde{S}_k(w^m - w^M)\|^2$ computed from the realised cluster weights predicts the measured $D_m$ at $s = 900$ within 25% in 9 of the 12 (seed, $m$) pairs (for instance $0.123 \to 0.112$, $0.187 \to 0.186$, $0.206 \to 0.205$ at $m = 16$; $0.0303 \to 0.0261$, $0.00087 \to 0.00085$, $0.0081 \to 0.0087$, $0.0187 \to 0.0190$ at $m = 256$); the largest deviation, a ratio 0.57 at $m = 16$, is the seed with the largest weight fluctuation $|w^m - w^M| = 0.275$, where the linearisation's remainder and the saturation $a \le a_{\max}$ enter. The exact first equality of (ii) — $\sum_k w_k^m \|\vartheta_k^*(w^m) - \vartheta_k^*(w^M)\|^2$ with the weighted stationary points themselves — reproduces the measured $D_m$ within 2.3% in all 12 pairs ($1.292 \to 1.292$ for the seed just mentioned): every $m$-system sits at the replicated weighted stationary configuration. The splitting margin of $\Theta^*(w^m)$ is positive in 11 of the 12 pairs ($+0.0006$ to $+0.0050$); it is negative, $-0.0022$, for $m = 16$, seed 3, whose weights $(9, 4, 3)/16$ lie outside the admissible set: there the replicated configuration is a saddle of $J_{16}$ and the plateau at $s = 900$ is the saddle's — referee round 9 continued that run to $s = 3600$ and saw $D$ grow from 1.29 to 5.13 as two particles changed basins and the weights re-sorted to $(7, 5, 4)/16$ (Proposition D‴.5(iv); the root-mean-square weight fluctuation is 0.204, 0.102, 0.051 at $m = 16$, 64, 256 against admissible margins of 0.06–0.36 in $\|\cdot\|_\infty$). Per cluster (seed 0), the measured mean displacements of the $m$-system's particles point along $S_k(w^m - w^M)$ with cosines 0.92–1.000 and norms within 0.8–20% ($m = 16$: 0.434 against 0.455, 0.130 against 0.129, 0.338 against 0.358): the particles of a cluster move together, by the predicted vector. The ratios 0.54, 0.69, 0.30 of the four-seed means to the multinomial expectation are the sampling of a heavy-tailed quantity at $m = 16$ and 64 and, at $m = 256$, four seeds whose realised fluctuations happen to be small.

*Cold start (table 4).* Window maxima of $D_m$: $1.81 \pm 0.28 \to 3.13 \to 4.50 \to 5.33 \pm 0.16$ ($m = 16$), $0.505 \to 0.900 \to 1.42 \to 1.79 \pm 0.93$ (64), $0.0744 \to 0.162 \to 0.347 \to 0.583 \pm 0.19$ (256): the distance grows over the whole horizon (last-to-first 2.95, 3.54, 7.84; last-to-previous 1.19, 1.26, 1.68), with slope $-0.87 \pm 0.14$ in $m$ (the learned function $-0.66 \pm 0.06$, steeper than the $-0.5$ of the warm start). The per-particle deviations are bimodal: medians 0.40, 0.14, 0.067 (the $m^{-1/2}$ law of the particles that stayed with their references) against maxima 6.7, 6.3, 6.2 for every $m$ (the separation between basins), and the fraction of pairs more than one whitened unit apart is $0.203 \pm 0.047$, $0.082 \pm 0.021$, $0.037 \pm 0.010$ — a factor 2.5 and 2.2 per factor 4 in $m$, a decay like $m^{-0.64 \pm 0.08}$ (per-seed fits; the medians decay like $m^{-0.62 \pm 0.07}$) — steeper than the $m^{-1/2}$ of Remark D‴.2's boundary-layer count, which is a lower-order prediction (the flips contribute $p_m \cdot O(R^2) \approx 0.2 \cdot 38 = 7.8$, 3.1, 1.4 against the measured 5.3, 1.8, 0.58). This is Theorem D″'s negative finding reproduced and diagnosed on the averaged system: a connected cold start that splits between basins is outside (H), and its coupling distance is dominated by the pairs that chose different basins.

*The real loop (table 5).* The closed loop at $k_e = 25$ ($\sigma = \sigma_0/k_e = 2.8 \cdot 10^{-4}$, $\tau = 0$, $\mathrm{d}t = 10^{-3}$, three seeds) from the same warm start: window maxima over $(0, 500]$, $(500, 1000]$, $(1000, 2000]$ real-time units are $0.328 \pm 0.11 \to 0.467 \to 0.590 \pm 0.17$ ($m = 16$), $0.0605 \to 0.0801 \to 0.0909 \pm 0.045$ (64), $0.0188 \to 0.0253 \to 0.0294 \pm 0.011$ (256), slope $-1.13 \pm 0.15$ in $m$; the reference's fit error stays at 0.245–0.246. The horizon is 80 slow units, a sixth of the relaxation time $1/\kappa = 512$, and the growth (last-to-first 1.80, 1.50, 1.56) is the early transient of the averaged dynamics: the averaged flow started from the real loop's own initial clouds (reproduced from their seeds) gives $D_{\mathrm{real}}(t)/D_{\mathrm{averaged}}(t/k_e)$ between 0.993 and 1.004 at every window end, for every seed and every $m$ — the real loop at $k_e = 25$ follows its averaged slow dynamics to within 0.7% (across different clouds the last-window maxima are 1.57, 0.78 and 2.58 times the averaged runs' 0.377, 0.117, 0.0114, within the seed-to-seed spread). Two of the three $m = 16$ clouds have inadmissible or marginal weights (splitting margins $-0.0021$, $-0.00001$; the third $+0.0026$), so their eventual configuration is a saddle's; at $m = 64$ and 256 all margins are positive ($+0.0029$ to $+0.0049$). The saturation at the orbit distance of Corollary D‴.7 lies beyond the horizon run.

*Reading.* In the window $\kappa_R[\mu^*] < \sigma_0 < \sigma_{\mathrm{crit}}$ the network learns (the collapsed configuration is unstable, the three kernels fit the plant to 0.244) and propagation of chaos is uniform in time on

a warm start with admissible weights: the coupled distance saturates with the $1/m$ law at the stationary distance of the replicated weighted configuration, which the exact form of Proposition D‴.5(ii) gives within 2.3% — the level is the weight-sensitivity law, not the theorem's constant. At $m = 16$ the multinomial weights leave the admissible set with appreciable probability; the one such seed sits on a saddle and later splits a cluster. The cold start of a connected cloud shows what the hypothesis excludes: growth over the horizon and a floor of pairs in different basins whose fraction decays like $m^{-0.64}$, while the learned function still converges at a rate between $m^{-1/2}$ and $m^{-2/3}$ in both cases. The real loop at $k_e = 25$ follows its averaged slow dynamics to within 0.7% over the horizon run.

Checked by: `tests/test_chaos_learning.py`: $(\mathfrak{A}_m)$'s drift equals $-G\nabla J_m$ of `periodic.ClosedLoop` to machine precision (`test_averaged_slow_loop_is_the_G_gradient_flow_of_J_m`); the three-term identity of Step 1 holds to rounding on random configurations with a mean-field stand-in, the interaction term equals $-\|\hat{f}_\Theta - \hat{f}_{\bar{\Theta}}\|^2$ up to a relative $O(D^{1/2})$ correction (ratio $0.98 \to 0.9999$ as $\delta$ is scaled by $1 \to 0.01$) and obeys (3.1) at every scale (`test_coupling_identity_three_terms_and_the_sign_of_the_interaction`); $\kappa_R$ at the atoms of a collapsed configuration with the collapsed residual equals Proposition K's $\lambda_{\max}(G^{1/2}R_2G^{1/2})$, which equals $\sigma_0$ at $\sigma_0 = \sigma_{\mathrm{crit}}$, the collapsed critical point has zero drift, and the deterministic neighbourhood search over the sphere of radius $\varrho$ reproduces the first-order expansion $\lambda_{\max} + \varrho\|G^{1/2}\nabla\lambda_{\max}\|$ with a relative error $O(\varrho)$ (`test_residual_curvature_...`); the searched whitened constants reproduce the document's values and lie below their closed forms, `theorem_constants` satisfies the four constraints the bootstrap needs, and $\Lambda_1$ integrates exactly to $s_1$ (`test_whitened_constants_...`); at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$ the three-atom critical point of $J_3$ has zero drift and (H) holds at it with $\varrho' = 0.1$ inside the learning regime, the replicated weighted stationary point is a critical point of $J_{16}$ to $10^{-10}$, the linearisation in the weights has a second-order remainder, and on a coupled run from the jittered mixture ($m = 16$, 64 against $M = 512$, 60 slow units) $\kappa_R[\mu^*]$ over the neighbourhood of the reference's support stays below $\sigma_0$, the two curvature statistics differ by at most the bound (0.2), the per-cluster displacements point along $S_k(w^m - w^M)$ (cosine $\geq 0.8$) with $D$ within a factor 2 of $\sum_k w_k^m \|S_k(w^m - w^M)\|^2$, $\|\hat{f}_\Theta - \hat{f}_{\bar{\Theta}}\| \leq L_\vartheta^G \cdot \mathrm{mean}\|\delta_i\|$, $D_{64} < D_{16}$ and the (vacuous) bound $C_1/m$ holds (`test_hypothesis_H_at_the_three_atom_stationary_measure_and_the_weight_sensitivity_law`).

# 19 Proposition G: the slow geometry

This section concerns joint descent, and variable projection (VarPro) in the induced metric.

**Setting.**

- $f = f(x)$ is time-invariant and $C^2$.
- $x_m$ is $T_0$-periodic, with $\omega_{\mathrm{ref}} := 2\pi/T_0$ and occupation measure $\rho := \frac{1}{T_0}\int_0^{T_0} \delta_{x_m(t)}\,\mathrm{d}t$.
- Law (L) acts in the interior, with gains $(g_a, \delta\gamma_c, \delta\gamma_s)$. Put $\Gamma := \mathrm{diag}(\gamma_c I_n, \gamma_s)$, acting per kernel, and $G_\xi := \delta\Gamma$. Write the geometry as $\xi := (c, s)$.
- $\bar{L}(a, c, s) := \frac{1}{2}\int \|f - \frac{1}{m}\sum_i a_i\varphi(\cdot\,; c_i, s_i)\|^2\,\mathrm{d}\rho$, and $M(c, s) := \frac{1}{m^2}\int \varphi\varphi^\top\,\mathrm{d}\rho \otimes I_n$ is the Hessian of $\bar{L}$ in $a$.
- $\mathcal{K}$ is the set of geometries $\xi$ on which (H1) holds, and $\mathcal{K}' \Subset \mathcal{K}$ is compact. (The set $\mathcal{K}$ is not the functional $\mathcal{G}(\mu)$ of Section 20.)
- $a^*(\xi) := \mathrm{argmin}_a\,\bar{L}$, and $L^*(\xi) := \bar{L}(a^*(\xi), \xi)$ is the profile (variable-projection) loss.
- Three dimensionless parameters:
  - the rate ratio $R := g_a m_0 / \big(\delta \cdot \sup_{\mathcal{K}'}\|\Gamma^{1/2}\nabla^2 L^*\Gamma^{1/2}\|\big)$;
  - the slope $S := (\delta/g_a) \cdot \sup_{\mathcal{K}'}\|D_\xi a^* \cdot \Gamma^{1/2}\|^2$;

– the averaging parameter $\varepsilon_{\text{avg}} := (m/k_e)\,\lambda_{\max}(G^{1/2}\nabla^2\bar{L}\,G^{1/2})/\omega_{\text{eff}}$, the second ratio in (H2a′). The result tables in `results/variable_projection` report its inverse, with $\omega_{\text{ref}}$ in place of $\omega_{\text{eff}}$.

**Hypotheses.**

(H0) On the interval considered, the closed-loop and reduced geometries stay in $\mathcal{K}'$. The parameters stay in $\operatorname{int}\Theta_K^{\varepsilon}$, and $\sup_{\mathcal{K}'}\max_i\|a_i^*\| < a_{\max} - \varepsilon_K$.

(H1) $M \succeq m_0 I$ on $\mathcal{K}$.

(H2a′) *Two scales.* $k_e \gg \omega_{\text{eff}}$ and $\varepsilon_{\text{avg}} \ll 1$, where

$$\omega_{\text{eff}} := \max\Big\{\omega_{\text{ref}},\ \frac{\sup_t\|\dot{x}_m\|}{s_{\min}},\ \sup_X\|\partial_x(f-\hat{f})\|,\ \frac{\sup_t\|\frac{\mathrm{d}}{\mathrm{d}t}r(x_m(t))\|}{\sup\|r\|}\Big\}.$$

The last entry is the bandwidth of the residual along the reference. Also $\sup\|r\|/k_e \ll s_{\min}$.

(H2b) *Rate separation.* $R \gg 1$: the slowest weight rate $g_a m_0 m/k_e$ is much larger than the geometry rate $(m\delta/k_e)\|\Gamma^{1/2}\nabla^2 L^*\Gamma^{1/2}\|$.

(H2c) *Flat slow manifold.* $S \ll 1$. Since $Da^* = -M^{-1}\partial^2_{a,\xi}\bar{L}$, a checkable sufficient condition is $\delta\cdot\max(\gamma_c,\gamma_s)\cdot\|\partial^2_{a,\xi}\bar{L}\|^2 \ll g_a m_0^2$.

Three remarks on the hypotheses.

- **(H2b) is about $\lambda_{\min}$.** It is a condition on $\lambda_{\min}$ of the feature Gram matrix. Since $0<\varphi\le 1$, $m_0 \le 1/m^2$. So the slowest weight rate is at most $g_a/(k_e m)$, and overlapping kernels make (H2b) fail.
- **The ratios are gain choices.** $R$ and $S$ contain neither $k_e$ nor $\omega_{\text{ref}}$. The separation between weights and geometry is therefore set by the gains.
- **(H2b) does not imply (H2c).** The product

$$R\cdot S = m_0\cdot\sup\|Da^*\Gamma^{1/2}\|^2 \big/ \sup\|\Gamma^{1/2}\nabla^2 L^*\Gamma^{1/2}\|$$

does not depend on $g_a$, $\delta$ or $k_e$. It is fixed by the problem and by the shape of $\Gamma$: in the slope configuration below, a numerical check gives values from 166 to 1149 over $\gamma_s/\gamma_c \in [10^{-4}, 10^4]$. It is large whenever refitting the weights largely compensates a change of geometry.

**Proposition G** (formal reduction)**.**

(a) Two scales, under (H0) and (H2a′). *To leading order in $\omega_{\text{eff}}/k_e$ and $\varepsilon_{\text{avg}}$, the period-averaged closed loop is the joint gradient flow*

$$\dot{\vartheta}_i = -\frac{m}{k_e}\,G\,\nabla_{\vartheta_i}\bar{L}(\vartheta_1,\dots,\vartheta_m). \tag{19.1}$$

*Its measure form is $-\frac{1}{k_e}G\nabla(\delta\bar{L}/\delta\mu)$.*

- *This is the drift of $(\mathfrak{A})$ at $\sigma=\tau=0$, with $\Phi$ in place of $\tilde{\Phi}$ and $(a,c,s)$ in place of $(\alpha,\gamma,\kappa)$, by the same averaging computation as Theorem H(i).*
- *It is not literally a special case of H(i): the saturated coordinates there induce a different, state-dependent metric.*
- *$\bar{L}$ is non-increasing along the averaged flow. Along the closed loop, $\bar{L}$ at period-averaged parameters decreases while the averaged dissipation $\frac{m}{k_e}\|\nabla\bar{L}\|_G^2$ dominates the $O(\omega_{\text{eff}}/k_e)$ corrections of Proposition H.7 and Remark H.7a. Near equilibrium it can increase by $O((\omega_{\text{eff}}/k_e)^2)$, as the weights settle at their shifted equilibrium (in a numerical check: $+8\cdot10^{-7}$ over 35 periods at $k_e=20$, and $3.7\cdot10^{-8}$, about 22 times less, at $k_e=80$).*

(b′) Three scales, for the averaged system (a), under (H0), (H1), (H2a′) and (H2b). *After a boundary layer, the weights follow the slow manifold:*

$$\begin{aligned} a - a^*(\xi) &= -\frac{k_e}{m g_a} M^{-1} Da^* \cdot \dot{\xi} + \text{h.o.t.} \\ &= \frac{\delta}{g_a} M^{-1} Da^* \Big(\Gamma^{-1} + \frac{\delta}{g_a} (Da^*)^\top Da^*\Big)^{-1} \nabla L^*(\xi) + \text{h.o.t.} \end{aligned} \tag{19.2}$$

*This reduces to* $\frac{\delta}{g_a} M^{-1} Da^* \, \Gamma \, \nabla L^*(\xi)$ *when* $S \ll 1$. *The geometry follows, to leading order in* $1/R$ *and for every* $S$,

$$\dot{\xi} = -\frac{m\delta}{k_e} \Big(\Gamma^{-1} + \frac{\delta}{g_a} (Da^*)^\top Da^*\Big)^{-1} \nabla L^*(\xi). \tag{19.3}$$

- *This is the gradient flow of* $(m/k_e)L^*$ *in the metric that the* $G^{-1}$ *metric induces on the graph* $\{(a^*(\xi), \xi)\}$.
- *Its metric tensor is* $G_\xi^{-1} + g_a^{-1}(Da^*)^\top Da^*$, *and it couples all kernels through* $Da^*$.
- *In the closed loop, the weight equilibrium is further shifted by* $O(\omega_{\text{eff}}/k_e)$ *(Proposition H.7). This shift can exceed the offset displayed above: in a numerical check on the* `satisfying` *run it is* $3\text{–}6 \cdot 10^{-3}$, *against an offset of* $4 \cdot 10^{-5}\text{–}3 \cdot 10^{-4}$.

(b) Variable projection, under (H2c) in addition. *Let* $(\bar{c}, \bar{s})$ *solve* $\bar{\xi}' = -\frac{m\delta}{k_e}\Gamma \nabla L^*(\bar{\xi})$, *that is,* $\bar{c}_i' = -\frac{m\delta\gamma_c}{k_e}\nabla_{c_i} L^*$ *and* $\bar{s}_i' = -\frac{m\delta\gamma_s}{k_e}\partial_{s_i} L^*$, *with* $(\bar{c}, \bar{s})(0) = (c, s)(0)$. *Then*

$$\sup_{t \le \tau_1 k_e/(m\delta)} \|(c,s)(t) - (\bar{c},\bar{s})(t)\| \to 0 \quad \text{as } (\omega_{\text{eff}}/k_e,\ \varepsilon_{\text{avg}},\ S,\ 1/R) \to 0, \text{ with } \tau_1 \text{ fixed.} \tag{19.4}$$

- *Formally the error is* $O(\omega_{\text{eff}}/k_e + \varepsilon_{\text{avg}} + S + 1/R)$.
- $L^*$ *is non-increasing along* $\bar{\xi}$, *and the weights track* $a^*$.
- *This flow is the* $S \to 0$ *limit of (b′).*

**Status.** Formal (a derivation without error control); checked directly, including a counterexample to plain VarPro at $S = O(1)$ (see the numerics below). Novelty (literature audit): a fixed-$m$, closed-loop analogue of the two-timescale/VarPro training of [59], [6].

**No mean-field form.** For fixed $m$, on configurations of $m$ distinct geometries in $\mathcal{K}$, the flow (b) is the gradient flow of $L^*/k_e$ in the metric $(1/m)\sum_i \|\dot{\xi}_i\|^2_{G_\xi^{-1}}$. Locally, that metric is the Wasserstein metric restricted to uniform $m$-atom measures.

- $L^*$ is the unregularised profile loss. It is the $\lambda \to 0$ limit of $\lambda\mathcal{L}^\lambda$ of [6] (time rescaled by $\lambda$). It is not the $\lambda \to 0$ functional $\mathcal{L}^0$ of [6], which is a min-norm interpolation functional whose flow is an ultra-fast diffusion.
- As a functional of the geometry marginal, $L^*$ depends only on the support, because the free weights absorb the masses. So it is not $W$-continuous, and the statement has no mean-field form. This is consistent with $m_0 \le 1/m^2$.

*Derivation (formal).* Each step is a standard theorem on its own time scale. Their composition on the $O(k_e/(m\delta))$ horizon is formal.

1. *Error channel (exact identity).* For $w := e - r(x, \vartheta)/k_e$, $\dot{w} = -k_e w - \dot{r}/k_e$. So $\|w(t)\| \le e^{-k_e t}\|w(0)\| + \sup\|\dot{r}\|/k_e^2$, and

$$e = r(x_m)/k_e + O\big(\sup\|\dot{r}\|/k_e^2 + \sup\|\partial_x r\| \cdot \sup\|r\|/k_e^2\big).$$

This is $O(\omega_{\text{eff}} \cdot \sup\|r\|/k_e^2)$, with the bandwidth term in $\omega_{\text{eff}}$.

2. *Period averaging of the joint system* (Khalil [48, Thm. 10.4]; Sastry–Bodson [81, Ch. 4]). This gives (a), on $[0, b/\varepsilon_{\mathrm{avg}}]$.

3. *Weights against geometry.* Use Tikhonov in slow time (Kokotović–Khalil–O'Reilly [49]) with the uniformly exponentially stable averaged weights on $\mathcal{K}'$ (H1).

   - On the slow manifold $a = h(\xi)$, $\dot{a} = Dh\,\dot{\xi} = -\frac{mg_a}{k_e} M(h - a^*)$, so $h - a^* = -\frac{k_e}{mg_a} M^{-1} Dh\,\dot{\xi}$. This must be evaluated with the (b′) velocity, not the VarPro velocity.
   - Since $\bar{L}$ is quadratic in $a$ and $\partial^2_{\xi a}\bar{L} = -(Da^*)^\top M$, we get $\nabla_\xi \bar{L}(h, \xi) = \nabla L^* - (Da^*)^\top M(h - a^*) + O((\delta/g_a)^2)$.
   - Substituting into $\dot{\xi} = -\frac{m\delta}{k_e}\Gamma \nabla_\xi \bar{L}(h, \xi)$, with $Dh = Da^* + O(1/R)$, gives (b′). The resummed form solves the invariance equation exactly once $Dh - Da^*$ and the quadratic term are dropped, so the remaining corrections are in $1/R$ (they involve $D^2 a^*$ and $\partial_\xi M$), not in $S$.

4. *Envelope theorem.* $\nabla_\xi \bar{L}(a^*(\xi), \xi) = \nabla L^*(\xi)$. When $S \ll 1$, the metric correction in (b′) drops out, which gives (b).

The composition requires the averaged system to remain valid up to $t \sim \tau_1 k_e/(m\delta)$, which is much longer than $b/\varepsilon_{\mathrm{avg}}$. Step 2 does not certify this. A proof would combine:

- a near-identity averaging transform;
- uniform exponential stability of the averaged weights on $\mathcal{K}'$;
- Tikhonov in slow time with an $O(\varepsilon_{\mathrm{avg}})$ perturbation;
- Grönwall in $\xi$ (Sastry–Bodson [81, §4.4]; Sanders–Verhulst–Murdock [78, Ch. 5]).

□

**Numerics.** The results are in `results/variable_projection` of the accompanying repository; Table 21 summarises the comparison.

- The first two rows use centred differences of period means.
- The slope rows use the displacement between the first and last period means. They start on the slow manifold of (b′), so no weight boundary layer enters the comparison.
- (b′) was evaluated only in the slope rows.
- $k_e/\omega_{\mathrm{eff}}$ is 2.0 in the first two rows with the box's $s_{\min}$, and 2.6–4.8 with the actual widths, so (H2a′) holds only marginally there. In the `satisfying` run the residual bandwidth, about 7.7, then dominates.
- $k_e/\omega_{\mathrm{eff}}$ is about 16 in the slope rows.
- **Joint descent.** In all runs, $\bar{L}$ at the period-mean parameters is non-increasing, sampled once per period.
- **(H2b) is not enough.** The `slope` rows satisfy (H2b) by a wide margin, yet (b) fails until $S$ is small. Meanwhile (b′) tracks the closed loop at every $S$. So (H2c) is the hypothesis missing from an earlier formulation of Proposition G(b), and (b′) is the correct leading-order slow flow. The configuration is a counterexample to that earlier formulation, made reproducible in `experiments/exp_slow_manifold_slope.py`.
- **Error decomposition** (`slope`, $\delta = 0.47$, started on the slow manifold):
  - $\|(\mathrm{a})\|/\|(\mathrm{b}')\| = 1.015$. This is the intrinsic $O(1/R)$ error of (b′).

| configuration | (H2b) ratio | $S$ | G(a): closed loop against the joint flow (a) | G(b) and the induced-metric flow (b′) |
|---|---|---|---|---|
| `violating` ($m = 4$, overlapping) | 0.10–0.23 | 5.4–25 | joint cosine 0.9994 / 0.9985, size ratio 1.00 / 0.94 | (b) fails: cosine $-0.12$ / 0.39, and $L^*$ *increases* from $3.8 \cdot 10^{-3}$ to $5.9 \cdot 10^{-3}$ |
| `satisfying` ($m = 2$) | 79–87 | 0.006 | joint cosine 0.985 / 0.962, size 0.96 / 0.94; geometry block 0.999 / 0.998; the weight block fails (see below) | (b) holds: cosine 0.999, ratio 0.97; $L^*$ decreases from $1.20 \cdot 10^{-2}$ to $4.11 \cdot 10^{-3}$ |
| `slope`, $\delta = 0.47$ ($m = 1$) | 81–178 | 2.0–2.8 | cosine 0.9992, size 1.10 | (b) fails: ratio 0.42, cosine 0.93, and the centre moves the opposite way; (b′): cosine 0.9986, ratio 1.12 |
| `slope`, $\delta = 0.47$, all gains $\div 4$ | 81–178 | 2.0–2.8 | cosine 1.0000, size 1.05 | (b) fails: ratio 0.40, cosine 0.92; (b′): cosine 0.9999, ratio 1.065 |
| `slope`, $\delta = 0.047$ | 654–1779 | 0.20–0.32 | cosine 1.0000, size 1.04 | (b): ratio 0.85, cosine 0.9986; (b′): ratio 1.05, cosine 1.0000 |
| `slope`, $\delta = 0.0047$ | 13162–17790 | 0.020–0.023 | cosine 1.0000, size 1.04 | (b): ratio 1.02; (b′): ratio 1.04; both cosines 1.0000 |

Table 21: Proposition G: the closed loop against the averaged predictions. In the first two rows, an entry '$p$ / $q$' gives the medians over the first quarter ($p$) and over the last quarter ($q$) of the run.

  - $\|\text{closed}\|/\|(\text{a})\| = 1.10$. It falls to 1.05 when all gains are divided by 4, which leaves $R$ and $S$ unchanged and divides $\varepsilon_{\text{avg}} \approx 0.08$ by 4. So this part is the averaging error.
  - The remaining $\approx 4.4\%$ is the $O(1/k_e)$ error-channel factor at $k_e = 20$. It is the same at $\delta = 0.047$ and 0.0047.

- **The size ratio in the `satisfying` run.** The centred ratio of 0.97 is the error-channel factor times a $+1.4\%$ averaging correction from the weights' oscillation within each period, the analogue of Remark H.7a.
  - The frozen-parameter ratio is 0.951–0.956 along the run, and 0.996 at $k_e = 80$.
  - The fastest weight rate over $\omega_{\text{ref}}$ is about 0.14.
  - In a further numerical check, with the weights running at frozen geometry the ratio is 0.969–0.970; with $g_a/4$ it is 0.956–0.960.
  - The slope correction is below 0.01% here, because $\nabla L^*$ is nearly orthogonal to the range of $(Da^*)^\top$. An earlier estimate of 0.94–0.95 contained a 2.5% forward-difference bias.

- **The weight block.** In the `satisfying` run the weights sit at $a^* + O(1/k_e)$, within 0.1% of $a^*$.
  - Their actual drift is tiny: 0.001–0.008 of the geometry speed in the $G^{-1}$ norm.
  - The prediction $-(m/k_e) g_a \nabla_a \bar{L}$ at the period-mean weights is 0.17–0.30 of the geometry speed, 40–200 times the actual drift. It is the restoring velocity of the $O(1/k_e)$ offset, which (a) does not contain.
  - So the weight block fails (cosine $-0.21$ / 0.50, carrying 92–95% of the residual), while the geometry block matches.

Checked by: `experiments/exp_variable_projection.py`, which writes `table_violating.md` and `table_satisfying.md`; and `experiments/exp_slow_manifold_slope.py`, which writes `table_slope.md` (all in `results/variable_projection`).

**Status.**

- **What the loop supplies.** The error-channel and averaging scales, and the data distribution (the occupation measure of the reference), come from the closed loop.
- **What the designer supplies.** The separation between weights and geometry is a gain choice, as in [59], [6]: $\delta \ll g_a \cdot \min\big(m_0/\|\Gamma^{1/2}\nabla^2 L^*\Gamma^{1/2}\|,\ 1/\|Da^*\Gamma^{1/2}\|^2\big)$. Equivalently, $\|G_\xi^{1/2}\nabla^2 L^* G_\xi^{1/2}\| \ll g_a m_0$ and $\|Da^* G_\xi^{1/2}\|^2 \ll g_a$.
- **What is new here.**
  - The learner's data distribution is its own excitation, and at finite $k_e$ it moves with the parameters (Proposition H.7(c)).
  - The slow flow is variable projection in the *induced* metric, valid for every $S$. The slope $S$ measures the departure from plain VarPro.

# 20 Theorem H: Langevin adaptive control and global convergence to the regularised optimum

**Why something new is needed.** Theorems B–G are local, or describe the dynamics. A global statement about kernel placement needs either the Chizat–Bach theory or entropy.

- **Boundary conditions.** The Chizat–Bach theory for noiseless Wasserstein flows (partially 1-homogeneous case, [14]) requires boundary conditions: either Neumann conditions on a bounded convex domain, or $C^1$-uniform behaviour at infinity. Gaussian kernels with free centres and widths satisfy neither.
- **Separation.** Its Theorem 3.5 also needs the support of $\mu_0$ to separate $\{-r_0\}\times\Theta$ from $\{r_0\}\times\Theta$. A Dirac start violates this, and that is an obstruction distinct from the boundary conditions.
- **The collapsed configuration is invariant.** It remains a moving Dirac mass and never separates without noise, by permutation symmetry and uniqueness of solutions. The test `test_noiseless_symmetric_initialisation_never_separates_but_dither_does` in the accompanying repository checks this on the real closed loop with $m = 6$. The rows stay bit-identical at every snapshot, the common particle moves, and dither separates the rows within 0.1 s. (Whether the collapsed configuration is a strict local minimum of the regularised loss, so that nearby non-collapsed starts return to it, is decided by the leak gain against the critical leak $\sigma_{\mathrm{crit}}$ of Proposition K, Section 23.)

Entropy removes the obstruction: Theorem H(ii) covers Dirac starts after any positive time.

**Design.** The particles are unconstrained, $\vartheta = (\alpha, \gamma, \kappa) \in \mathbb{R}^p$, and the features are saturated coordinate by coordinate:

$$a = a_{\max}\tanh(\alpha/a_{\max}), \qquad c = c_{\max}\tanh(\gamma/c_{\max}), \qquad s = s_{\min} + (s_{\max} - s_{\min})\cdot \mathrm{sig}(\kappa).$$

Here $a_{\max}$ bounds each coordinate of $a$, unlike Section 9, where it bounds $\|a\|$. So $\tilde\Phi(x;\vartheta) = a\varphi(x;c,s)$ satisfies $\|\tilde\Phi\| \le K_\Phi := a_{\max}\sqrt{n}$, and every $\vartheta$-derivative of $\tilde\Phi$ is bounded uniformly in $x$. The law is

$$\mathrm{d}\vartheta_i = \big[\, G\nabla_\vartheta g_t(\vartheta_i) - \sigma(\vartheta_i - \vartheta^0)\,\big]\,\mathrm{d}t + \sqrt{2\tau}\, G^{1/2}\,\mathrm{d}B_i, \qquad g_t(\vartheta) := e(t)^\top \tilde\Phi(x(t);\vartheta). \tag{$\mathrm{L}_\tau$}$$

It has three parts: the classical gradient law, $\sigma$-modification written in the $G^{-1}$ metric, and parameter dither of temperature $\tau$.

**Averaged mean-field system.** Assume $f = f(x)$ is continuous and time-invariant, as in Proposition G (the functions `occupation_loss` and `first_variation` of the accompanying code assume this). Let $\rho$ be the occupation measure of a $T_0$-periodic reference, and $F_\rho := \sup_{\operatorname{supp}\rho}\|f\|$. Define

$$r_\mu := f - \int \tilde{\Phi}(\cdot\,;\vartheta')\,\mathrm{d}\mu(\vartheta'), \qquad \bar{L}(\mu) := \frac{1}{2}\int \|r_\mu\|^2\,\mathrm{d}\rho,$$
$$V_L[\mu](\vartheta) := -\int r_\mu(x)^\top \tilde{\Phi}(x;\vartheta)\,\mathrm{d}\rho(x), \qquad V[\mu] := \frac{1}{k_e}V_L[\mu] + \frac{\sigma}{2}\|\vartheta - \vartheta^0\|^2_{G^{-1}},$$

and the averaged system

$$\partial_t\mu = \nabla\cdot\big(\mu G\nabla V[\mu]\big) + \tau\nabla\cdot(G\nabla\mu). \tag{$\mathfrak{A}$}$$

**Remark H.0** (from the closed loop to ($\mathfrak{A}$): a formal reduction with a measured error)**.**

- **An algebraic identity.** Substitute $e := r_\mu(x_m)/k_e$ and $x := x_m$ in the drift of ($\mathrm{L}_\tau$), and average over one reference period at frozen $\mu$. The result is exactly $-G\nabla V[\mu]$. The test `test_averaged_drift_is_the_mean_field_langevin_drift` checks this identity to $10^{-7}$. It does not test averaging.
- **The formal reduction.** ($\mathfrak{A}$) is the leading-order description of the mean-field closed loop under two conditions:
  - *Tikhonov:* $k_e \gg \max\big(\omega_{\mathrm{ref}},\ \mathrm{Lip}(f) + K_\Phi/s_{\min},\ (F_\rho + K_\Phi)/s_{\min}\big)$, so that the shift $x - x_m = O(1/k_e)$ is small on the kernel scale.
  - *Period averaging:* $T_0\cdot\sup_\vartheta\|(1/k_e)G\nabla V_L[\mu](\vartheta)\| \ll \ell$, $\sigma T_0 \ll 1$ and $\tau\lambda_G T_0 \ll \ell^2$, where $\ell$ is the $\vartheta$-scale of $\tilde{\Phi}$. The left-hand side is at most $(T_0/k_e)\lambda_G(F_\rho + K_\Phi)\cdot\sup\|\nabla_\vartheta\tilde{\Phi}\|$. The confinement drift does not depend on time and needs no averaging.
- **The first corrections.**
  - At frozen $\mu$ (and frozen $\vartheta$) they are $O(1/k_e)$ relative to the drift, do not depend on the gains, and are given exactly by Proposition H.7.
  - Along the loop, the within-period oscillation of the parameters adds the second-order averaging term $b_{\mathrm{osc}}$ of Remark H.7a. Its size relative to the frozen terms is $O(|G|T_0)$, so it does depend on the gains.
  - In the formal limit below, both are $O(k_e^{-2})$.

  Measured relative errors of the frozen-$\mu$ averaged drift against ($\mathfrak{A}$), with $b_{\mathrm{osc}}$ excluded:

| instance | $k_e = 10$ | 20 | 40 | 80 | $k_e \times$ error |
|---|---|---|---|---|---|
| $n = 1$ | 4.2% | 2.0% | 0.91% | 0.48% | $\approx 0.4$ |
| $n = 2$, ellipse | 43% | 21% | 10% | 5.1% | $\approx 4.2$ |

  Tests: `test_averaging_error_of_the_formal_reduction_decreases_like_one_over_ke` (instance $n = 1$) and `test_averaged_drift_error_decays_like_one_over_ke_for_an_orbit_enclosing_area` (instance $n = 2$; $k_e = 10, 20, 40, 80$).
  - $k_e = 10$ is outside the averaging regime for the second instance.

– Its larger error does not come from the enclosed area. Proposition H.7 attributes it to the orientation-even state-shift term: it predicts $k_e \times \text{error} \approx 4.05$, of which $U_{\text{odd}}$ alone would give 1.5, and a numerical check on the reversed orbit gives 4.0.

– A numerical check shows that at the gains of these instances $b_{\text{osc}}$ is not small. $k_e$ times its relative size is 0.92 ($n = 1$, $G = \text{diag}(4, 1, 1)$) and 4.4 ($n = 2$, $G = I$), against 0.42 and 4.05 for the frozen terms.

- **The exact formal limit** is $k_e \to \infty$ with $G$, $k_e\sigma$ and $k_e\tau$ fixed, in slow time $t/k_e$ (Conjecture H.6).
- **Not proved here.** The mean-field limit of $(\mathrm{L}_\tau)$ itself is not proved; Theorem D covers the deterministic projected law (L).

**Status.** Formal reduction, with a measured error.

**Theorem H.** *Assume the setting above, and $\sigma > 0$, $\tau > 0$ where stated.*

(i) Identification. *$V[\mu]$ is the first variation of $\mathcal{G}(\mu) := \frac{1}{k_e}\bar{L}(\mu) + \frac{\sigma}{2}\int \|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu$. Hence $(\mathfrak{A})$ is the mean-field Langevin dynamics, in the $G^{-1}$ metric, of the free energy $F_\tau := \mathcal{G} + \tau H$, with $H(\mu) = \int \mu \log \mu$. Equivalently, with $\gamma_\tau := \mathcal{N}(\vartheta^0, (\tau/\sigma)G)$,*

$$F_\tau = \frac{1}{k_e}\bar{L} + \tau H(\cdot \,|\, \gamma_\tau) + \text{const.}$$

*$F_\tau$ is well defined on all of $\mathcal{P}(\mathbb{R}^p)$, and finite exactly when $H(\mu \,|\, \gamma_\tau) < \infty$.*

(ii) Global exponential convergence ($\sigma, \tau > 0$). *$F_\tau$ has a unique minimiser, $\mu^*_\tau \propto e^{-V[\mu^*_\tau]/\tau}$. For every $\mu_0 \in \mathcal{P}_2(\mathbb{R}^p)$ with $F_\tau(\mu_0) < \infty$,*

$$F_\tau(\mu_t) - F_\tau(\mu^*_\tau) \le e^{-2\tau\rho_\tau t}\big(F_\tau(\mu_0) - F_\tau(\mu^*_\tau)\big),$$
$$W_{G^{-1}}(\mu_t, \mu^*_\tau)^2 \le \frac{2}{\tau\rho_\tau}\, e^{-2\tau\rho_\tau t}\big(F_\tau(\mu_0) - F_\tau(\mu^*_\tau)\big).$$

*Here $\rho_\tau$ is a log-Sobolev constant, for the whitened measures, of every proximal Gibbs measure $\hat{\mu} \propto e^{-V[\mu]/\tau}$. In $\vartheta$-coordinates this means Fisher information in the $G$ metric. It satisfies*

$$\rho_\tau \ge \frac{\sigma}{\tau} \cdot \exp\Big(-\frac{2K_\Phi(F_\rho + K_\Phi)}{k_e\tau}\Big).$$

*Suppose $\mu_0$ is only in $\mathcal{P}_2(\mathbb{R}^p)$, for example a Dirac mass or an empirical measure, so that $H(\mu_0) = +\infty$. Then $F_\tau(\mu_{t_0}) < \infty$ for every $t_0 > 0$, and the estimates hold with $(\mu_0, t)$ replaced by $(\mu_{t_0}, t - t_0)$.*

(iii) Loss and tracking. *Let $\bar{e}_t := r_{\mu_t}/k_e$. Then $\langle \|\bar{e}_t\|^2 \rangle_\rho = 2\bar{L}(\mu_t)/k_e^2$, and*

$$\begin{aligned}|\bar{L}(\mu_t) - \bar{L}(\mu^*_\tau)| &\le (F_\rho + K_\Phi)L_\Phi \cdot W_{G^{-1}}(\mu_t, \mu^*_\tau) \\ &\le (F_\rho + K_\Phi)L_\Phi\Big(\frac{2}{\tau\rho_\tau}\Big)^{1/2} e^{-\tau\rho_\tau t}\big(F_\tau(\mu_0) - F_\tau(\mu^*_\tau)\big)^{1/2},\end{aligned}$$

*where $L_\Phi := \sup_{x,\vartheta,\|u\|=1} \|G^{1/2}\nabla_\vartheta(u^\top \tilde{\Phi})\|$. The regularisation bias satisfies*

$$\bar{L}(\mu^*_\tau) \le \inf_\nu \big\{\bar{L}(\nu) + k_e\tau H(\nu \,|\, \gamma_\tau)\big\}.$$

*The link $e \approx \bar{e}$ between $(\mathfrak{A})$ and the closed loop is formal (Remark H.0).*

(iv) Annealing. *Let $\mu_0 \in \mathcal{P}_2$ with $F_{\tau_{\text{init}}}(\mu_0) < \infty$. Let $(\tau_t)$ be smooth and decreasing from $\tau_{\text{init}}$, with $\tau_t = \alpha_a/\log t$ for $t \ge t_1 > 1$, where $\alpha_a > \alpha^* := 2K_\Phi(F_\rho + K_\Phi)/k_e$. Then $\mathcal{G}(\mu_t) - \inf \mathcal{G} \le C' \cdot \log\log t/\log t$, where $\inf \mathcal{G}$ is the $\sigma$-regularised optimum. This is Theorem 4.1 of [15]. Its hypotheses hold:*

- $\mathcal{G} \geq 0$;
- *Assumptions 1–3 hold for every* $\tau > 0$;
- *for* $\tau \in (0, \tau_{\text{init}}]$, *(ii) gives* $\rho_\tau \geq C_0 e^{-\alpha^*/\tau}$ *with* $C_0 = \sigma/\tau_{\text{init}}$, *which is all its proof uses.*

**Status.** Proved for ($\mathfrak{A}$); the reduction to ($\mathfrak{A}$) is formal (Remark H.0); real loop: Conjecture H.6. Novelty (literature audit): a bridge to [15], [64]; new in adaptive control.

Checked by: `tests/test_langevin.py`:
`test_first_variation_bound_used_in_holley_stroock` (the oscillation bound in (ii));
`test_lsi_lower_bound_formula` (formula transcription only);
`experiments/exp_langevin_global.py`, `exp_langevin_tradeoff.py` and `exp_gibbs_reference.py` (global behaviour, speed–bias trade-off, reference values).

*Proof. (i)* We have

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} \bar{L}\big((1-\varepsilon)\mu + \varepsilon\nu\big)\Big|_{\varepsilon=0} = -\int r_\mu^\top \Big(\int \tilde{\Phi}\, \mathrm{d}(\nu - \mu)\Big)\, \mathrm{d}\rho = \int V_L\, \mathrm{d}(\nu - \mu).$$

This is exact, because $\bar{L}$ is quadratic in $\mu$. The confinement is linear in $\mu$. In the $G^{-1}$ metric, $\nabla \cdot \big(\mu G \nabla(V + \tau \log \mu)\big) = \nabla \cdot (\mu G \nabla V) + \tau \nabla \cdot (G \nabla \mu)$. Since $\log \gamma_\tau = -(\sigma/(2\tau))\|\vartheta - \vartheta^0\|^2_{G^{-1}} + \text{const}$, we get $\tau H(\mu \,|\, \gamma_\tau) = \tau H(\mu) + (\sigma/2) \int \|\vartheta - \vartheta^0\|^2_{G^{-1}}\, \mathrm{d}\mu + \text{const}$.

*(ii)* Whiten with $\zeta := G^{-1/2}\vartheta$. The metric becomes Euclidean, the confinement becomes $(\sigma/2)\|\zeta - \zeta^0\|^2$, $W_{G^{-1}}$ becomes $W_2$, and $H$ shifts by $\frac{1}{2}\log\det G$. ($\mathfrak{A}$) becomes equation (12) of [15]. We verify Assumptions 1–3 of [15].

- *Assumption 1.* $\nabla V[\mu](\zeta)$ is Lipschitz in $(\zeta, \mu)$. The constants do not depend on $\mu$, because $\|r_\mu\| \leq F_\rho + K_\Phi$ on $\operatorname{supp}\rho$ for every $\mu \in \mathcal{P}(\mathbb{R}^p)$.
  - In $\zeta$, this holds because $\nabla^2\tilde{\Phi}$ is bounded (saturation, and $s \geq s_{\min}$).
  - In $\mu$, it holds because $\int \tilde{\Phi}\, \mathrm{d}\mu$ is $W_1$-Lipschitz, $W_1 \leq W_2$, and $\nabla\tilde{\Phi}$ is bounded.
- *Assumption 2.* $\mathcal{G}$ is convex: a linear map followed by a convex quadratic, plus a linear term. $F_\tau$ has a minimiser by confinement and lower semicontinuity (proof of [15, Prop. 5.1]).
- *Assumption 3.* $e^{-V[\mu]/\tau} = e^{-(\sigma/(2\tau))\|\zeta - \zeta^0\|^2} \cdot e^{-V_L[\mu]/(k_e\tau)}$.
  - The Gaussian factor satisfies LSI($\sigma/\tau$) (Bakry–Émery [4]).
  - $|V_L[\mu]| \leq \int \|r_\mu\| \|\tilde{\Phi}\|\, \mathrm{d}\rho \leq (F_\rho + K_\Phi) K_\Phi$, so the perturbation has oscillation at most $2K_\Phi(F_\rho + K_\Phi)/(k_e\tau)$. Here $F_\rho$ may be replaced by $\int \|f\|\, \mathrm{d}\rho$.
  - Holley–Stroock [43] gives $\rho_\tau$.

Theorem 3.2 and Corollary 3.3 of [15] (the latter with Otto–Villani [65]) give the two inequalities. Uniqueness follows from strict convexity of $H$.

$\mathcal{P}_2$ *starts.* Work in $\zeta$-coordinates and fix the solution $\mu_t$. Compare the particle SDE with the Ornstein–Uhlenbeck process $\mathrm{d}Y = -\sigma(Y - \zeta^0)\, \mathrm{d}t + \sqrt{2\tau}\, \mathrm{d}B$ started from the same law.

- The two drifts differ by $b_t = -(1/k_e) G^{1/2} \nabla V_L[\mu_t]$, with $\|b_t\| \leq B := (1/k_e)\lambda_G^{1/2}(F_\rho + K_\Phi) \cdot \sup\|\nabla_\vartheta \tilde{\Phi}\|$.
- Girsanov and data processing give $H(\mu_{t_0} \,|\, \mathrm{OU}_{t_0}) \leq B^2 t_0/(4\tau)$.
- $\mathrm{OU}_{t_0}$ has a density at most $\big(2\pi(\tau/\sigma)(1 - e^{-2\sigma t_0})\big)^{-p/2}$, so $H(\mu_{t_0}) < \infty$.
- $\mu_{t_0} \in \mathcal{P}_2$, because the drift grows linearly.

Hence $F_\tau(\mu_{t_0}) < \infty$.

*(iii)* $\bar{L}(\mu) - \bar{L}(\nu) = \frac{1}{2}\int (r_\mu - r_\nu)^\top (r_\mu + r_\nu)\,\mathrm{d}\rho$. Here $\|r_\mu + r_\nu\| \le 2(F_\rho + K_\Phi)$, and $\|r_\mu - r_\nu\| = \|\int \tilde{\Phi}\,\mathrm{d}(\nu - \mu)\| \le L_\Phi \cdot W_{1,G^{-1}} \le L_\Phi \cdot W_{G^{-1}}$. Conclude with (ii). For the bias: $(1/k_e)\bar{L}(\mu_\tau^*) + \tau H(\mu_\tau^* \,|\, \gamma_\tau) \le (1/k_e)\bar{L}(\nu) + \tau H(\nu \,|\, \gamma_\tau)$ for every $\nu$, and $H(\mu_\tau^* \,|\, \gamma_\tau) \ge 0$.

*(iv)* This is Theorem 4.1 of [15] with the stated hypotheses. □

**Remark H.1** (dither adds no output noise in the mean-field limit; its cost is the entropic bias)**.** In the limit, $\hat{f} = \int \tilde{\Phi}\,\mathrm{d}\mu$ is deterministic by the law of large numbers. The only effect of dither is the entropic bias, which Theorem H(iii) quantifies.

In the experiment ($\sigma = 0.02$, $\tau = 0.004$, $G = \mathrm{diag}(4, 1, 1)$, $k_e = 10$), a grid computation gives $\bar{L}(\mu_\tau^*) = 0.11118$. It is grid-converged (`exp_gibbs_reference.py` → `results/langevin_global/gibbs_reference.md`). At $(\sigma, \tau) = (0.002, 5 \cdot 10^{-4})$ the same computation gives 0.0085. The unregularised offline optimum over 32 kernels is about $7.6 \cdot 10^{-6}$ (best of 5 L-BFGS restarts, an upper bound), and $\bar{L}(0) = 0.311$.

At finite $m$, the fluctuation of $\hat{f}$ is $O(K_\Phi/\sqrt{m})$ at each fixed $t$ (finite-horizon propagation of chaos; Theorem D′ of Section 17 gives the rate $m^{-1/2}$ for the plant and the learned function of the dithered loop). It is uniform in time only under Remark H.5.

**Remark H.2** (the control knobs, and what the numbers say)**.**

- **The knobs.** $\sigma$ is the strong-convexity constant of the confinement, $\tau$ is the temperature, and $a_{\max}$ sets $K_\Phi$.
- **The role of $k_e$.** Since $k_e F_\tau = \bar{L} + (k_e\sigma/2)\int \|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu + k_e\tau H$, $\mu_\tau^*$ and $\rho_\tau$ depend on $(\sigma, \tau, k_e)$ only through $(k_e\sigma, k_e\tau)$. Raising $k_e$ at fixed $(\sigma, \tau)$ has three effects:
  - it improves $\rho_\tau$ and the guaranteed rate $2\sigma e^{-\alpha^*/\tau}$, since $\alpha^* \propto 1/k_e$;
  - it moves $\mu_\tau^*$ toward the prior $\mathcal{N}(\vartheta^0, (\tau/\sigma)G)$, which increases the regularisation bias;
  - it shrinks the loss-driven drift like $1/k_e$.
- **The guaranteed rate** is exponentially small in $1/\tau$, a known feature of Holley–Stroock bounds [43]:
  - $2\tau\rho_\tau \ge e^{-8401}\,\mathrm{s}^{-1}$ at $(\sigma, \tau) = (0.002, 5 \cdot 10^{-4})$;
  - $2\tau\rho_\tau \ge e^{-1053}\,\mathrm{s}^{-1}$ at $(0.02, 0.004)$.

  The experiments do not test this rate.
- **The temperature sweep** (`results/langevin_tradeoff`; $m = 32$, $\sigma = 0.002$, $k_e = 10$, $T = 1500\,\mathrm{s}$, 2 seeds):

| $\tau$ | escape time of the collapsed start (s), per seed | loss floor of the spread start |
|---|---|---|
| 0 | never, never | $7.0 \cdot 10^{-3}$ |
| $5 \cdot 10^{-4}$ | never, 1050 | $1.1 \cdot 10^{-2}$ |
| $1 \cdot 10^{-3}$ | 1480, 1005 | $1.5 \cdot 10^{-2}$ |
| $2 \cdot 10^{-3}$ | 1115, 800 | $2.3 \cdot 10^{-2}$ |
| $4 \cdot 10^{-3}$ | 970, 430 | $3.6 \cdot 10^{-2}$ |
| $8 \cdot 10^{-3}$ | 390, 295 | $6.5 \cdot 10^{-2}$ |
| annealed, hot start | 990, 530 | $3.2 \cdot 10^{-2}$ |

- **How escape is measured.** Escape is the first sample at which the collapsed run's loss falls to at most twice the spread run's final loss, at the same $\tau$ and seed. Two caveats follow:

  - the threshold grows with $\tau$, from 0.022 to 0.13, and the escape time includes convergence after escape;
  - the $\tau = 5 \cdot 10^{-4}$ point is censored and is excluded from the fit.

- **The fit.** The seed-averaged escape time fits $t_{\mathrm{esc}} \approx -1693 + 427 \cdot \log(1/\tau)$ (4.5% relative residual) better than an Arrhenius law (25%). On the eight per-seed values the two laws fit equally poorly (27% and 33%; `langevin_tradeoff/table.md`).
- **No saddle explanation.** The noiseless collapsed end point is a saddle of the averaged landscape, but, as a numerical check shows, a weak one. Its unstable rate is about $5 \cdot 10^{-5}\,\mathrm{s}^{-1}$ for $(\mathfrak{A})$, and about $2 \cdot 10^{-4}\,\mathrm{s}^{-1}$ for the exactly averaged loop at $k_e = 10$. A saddle-escape law would therefore need $\tau \lesssim 10^{-5}$, and would have a $\log(1/\tau)$ slope of $10^3$–$10^4$ s.
- **What the data show.** At the tested temperatures escape is governed by diffusion, by the approach transient and by the $\tau$-dependent threshold. The data do not identify the escape mechanism. An earlier "saddle, not barrier" reading of these data is withdrawn.
- **The annealed runs** use $\alpha_a$ between about $2 \cdot 10^{-3}$ and $2 \cdot 10^{-2}$, far below $\alpha^* = 4.2$. They are not instances of H(iv).

**Remark H.3** (why saturation and not projection)**.** The mean-field Langevin results are on $\mathbb{R}^d$. Projecting the drift with the Pomet–Praly operator [76], while keeping additive noise, leaves the noise unconfined. Reflected (Skorokhod) dynamics on the convex body $\Theta_K$ would confine it.

- Such dynamics satisfy a log-Sobolev inequality with constant of order $1/\operatorname{diam}^2$, even for $\sigma = 0$.
- The argument is a diameter-based log-Sobolev inequality for the uniform measure on a convex body, followed by Holley–Stroock [43]. The diameter-based inequality comes from $\mathrm{CD}(0,\infty)$ with a convex Neumann boundary plus a diameter bound (F.-Y. Wang 1997 [87]), or from Frieze–Kannan (1999) [31] for log-concave measures on convex bodies. Together these give $\mathrm{LSI}\big(c \cdot \operatorname{diam}^{-2} \cdot e^{-2K_\Phi(F_\rho + K_\Phi)/(k_e\tau)}\big)$.
- Bakry–Émery [4] alone gives nothing at $\sigma = 0$.
- That route connects to [69] and is left open.

**Remark H.4** (the role of $\sigma$-modification)**.** In the classical theory, $\sigma$-modification is a robustness device with a bias. In the saturated design on $\mathbb{R}^p$ it supplies the confinement that makes $e^{-V/\tau}$ integrable and log-Sobolev, so the global result rests on it. Its bias is the regularisation term of the variational problem that $\mu_\tau^*$ solves. Under reflection on $\Theta_K$, compactness would play that role instead.

**Remark H.5** (finite $m$; conditional)**.** The $m$-particle averaged system is

$$\mathrm{d}\vartheta_i = -G\nabla V[\mu_t^m](\vartheta_i)\,\mathrm{d}t + \sqrt{2\tau}\,G^{1/2}\,\mathrm{d}B_i, \qquad \mu_t^m := \frac{1}{m}\sum_i \delta_{\vartheta_i}.$$

Uniform-in-time propagation of chaos would follow from [11], under its hypotheses.

- *Available:* convexity of $\mathcal{G}$; a uniform LSI, from (ii); and a bounded, smooth second variation $(1/k_e)\int \tilde{\Phi}(x;\vartheta)^\top \tilde{\Phi}(x;\vartheta')\,\mathrm{d}\rho$.
- *Still to check:* the remaining regularity conditions of [11], verbatim (open question 4 of `LITERATURE_TRANSPORT.md` in the accompanying repository).

**Status.** Conditional on the hypotheses of [11]; their remaining regularity conditions are still to be checked.

**Conjecture H.6** (the real closed loop, in the scaling where $(\mathfrak{A})$ is the limit)**.** *Let $f = f(x)$ satisfy the tube condition of Theorem D(b), with $a_{\max}$ replaced by $K_\Phi$. Fix $G \succ 0$ and $\sigma_0, \tau_0 > 0$, and set $\sigma := \sigma_0/k_e$ and $\tau := \tau_0/k_e$.*

- *Then $\mu_\tau^*$, the minimiser of $\bar{L} + (\sigma_0/2)\int \|\vartheta - \vartheta^0\|_{G^{-1}}^2 \,\mathrm{d}\mu + \tau_0 H$, does not depend on $k_e$.*
- *In slow time $s = t/k_e$, $(\mathfrak{A})$ is the formal limit of the closed loop.*

Conjecture: *for the mean-field limit of the closed loop under $(\mathrm{L}_\tau)$, for every $\mu_0 \in \mathcal{P}_2$ and every $e(0)$, there are $S(\mu_0)$ and $r(k_e) \to 0$ such that $W_{G^{-1}}(\mu_t, \mu_\tau^*) \le r(k_e)$ for all $t \ge k_e \cdot S$.*

**Status.** Conjectural.

At fixed $k_e$ the analogous statement is false. The limit of small gains is the exactly averaged system, with the true periodic error $e_\mu$ and $x = x_m + e_\mu$.

- Its stationary state is offset from $\mu_\tau^*$: $\bar{L} = 0.11415$ against $\bar{L}(\mu_\tau^*) = 0.11118$ at $k_e = 10$ (`results/langevin_global/gibbs_reference.md`).
- In the scaling of Conjecture H.6, the $L^2(\rho)$ distance between the two learned functions is $6.8 \cdot 10^{-3}$, $1.4 \cdot 10^{-3}$ and $9.7 \cdot 10^{-4}$ at $k_e = 10$, 20 and 40. It is small and decreasing, but a clean $1/k_e$ law is not resolved in this range.
- By Proposition H.7(e) it is not the $W_{G^{-1}}$-gradient flow of any free energy. The cyclic sums of its antisymmetric second variation are non-zero in the checked instances, including the network of the experiments; by Theorem I(iii) (Section 21) they are non-zero at every $n = 1$ measure whose residual is non-zero on the orbit, and by Theorem I(vi) on no open set of measures do they all vanish.

# 21 Finite gain: semi-gradient and non-reciprocal structure

This section continues Section 20 at finite feedback gain $k_e$. The notation is that of Section 9 and Section 20: the saturated features $\tilde{\Phi}$, the residual $r_\mu$, the potential $V_L[\mu]$, the averaged system $(\mathfrak{A})$ and the law $(\mathrm{L}_\tau)$. We write $\tilde{\Phi}_\vartheta := \tilde{\Phi}(\cdot; \vartheta)$.

## 21.1 Frozen-measure structure of the exactly averaged loop

**Proposition H.7** (frozen-measure structure of the exactly averaged closed loop)**.** *Assume the following:*

- *$f \in C^2(\mathbb{R}^n)$ is autonomous;*
- *$x_m \in C^2$ is $T_0$-periodic, with oriented orbit $\mathcal{O}$;*
- *the features are the saturated features of the design (Section 20);*
- *$\mu \in \mathcal{P}(\mathbb{R}^p)$ is frozen.*

*Then for every $R > 0$ there is $k_0(R)$, independent of $\mu$, such that for $k_e \ge k_0(R)$ the following hold.*

*(a)* Periodic error. *The frozen-$\mu$ error equation $\dot{e} = r_\mu(x_m + e) - k_e e$ has exactly one $T_0$-periodic solution $e_\mu$ with $\sup_t \|e_\mu\| \le R$. It attracts exponentially every solution with $\|e(0)\| \le R$, and*

$$e_\mu = \frac{r_\mu(x_m)}{k_e} + \frac{\nabla r_\mu(x_m)\, r_\mu(x_m) - \frac{\mathrm{d}}{\mathrm{d}t} r_\mu(x_m(t))}{k_e^2} + O(k_e^{-3}), \tag{21.1}$$

*uniformly in $t$ and $\mu$.*

- *Here $\nabla r_\mu$ is the Jacobian $(\partial r_k/\partial x_l)$, and $\partial_x \tilde{\Phi} = (\partial \tilde{\Phi}_k/\partial x_l)$.*

- *If $f$ is one-sided Lipschitz, $\langle f(x)-f(y),x-y\rangle \le L_f\|x-y\|^2$, then for $k_e > L_f + a_{\max}\sqrt{n}\,e^{-1/2}/s_{\min}$ the periodic solution is unique and attracts every solution.*
- *Without such a condition uniqueness is only local. For $f(x)=x^2$ there is a second, repelling periodic solution near $e \approx k_e$, and solutions above it blow up.*

*(b)* Expansion. *The period average of the drift of* $(\mathrm{L}_\tau)$ *along* $e_\mu$ *is* $G\nabla_\vartheta P_\mu(\vartheta) - \sigma(\vartheta-\vartheta^0)$, *where* $P_\mu(\vartheta) := \langle e_\mu^\top \tilde\Phi(x_m+e_\mu;\vartheta)\rangle_t$ *satisfies*

$$P_\mu(\vartheta) = -\frac{1}{k_e}V_L[\mu](\vartheta) + \frac{1}{k_e^2}\big(U_{\mathrm{even}}[\mu](\vartheta) + U_{\mathrm{odd}}[\mu](\vartheta)\big) + O(k_e^{-3}), \tag{21.2}$$

$$U_{\mathrm{even}}[\mu](\vartheta) := \int (r_\mu\cdot\nabla_x)\big(r_\mu^\top\tilde\Phi(\cdot;\vartheta)\big)\,\mathrm{d}\rho, \qquad U_{\mathrm{odd}}[\mu](\vartheta) := -\frac{1}{T_0}\oint_{\mathcal{O}} \tilde\Phi(\cdot;\vartheta)^\top\,\mathrm{d}r_\mu, \tag{21.3}$$

*uniformly in* $\vartheta$ *and* $\mu$, *together with* $\vartheta$*-gradients.*

*(c)* Semi-gradient form.

$$-\frac{1}{k_e}V_L[\mu] + \frac{1}{k_e^2}U_{\mathrm{even}}[\mu] = \frac{1}{k_e}\int r_\mu^\top\tilde\Phi(\cdot;\vartheta)\,\mathrm{d}\rho_\mu^{\mathrm{cl}} + O(k_e^{-3}). \tag{21.4}$$

*Here* $\rho_\mu^{\mathrm{cl}}$ *is the occupation measure of the closed-loop state* $x_m + e_\mu$.

- *So, apart from the lag term* $U_{\mathrm{odd}}$, *the averaged law follows the gradient of the closed-loop loss* $\frac12\int\|r_\nu\|^2\,\mathrm{d}\rho_\mu^{\mathrm{cl}}$ *at* $\nu=\mu$, *with the closed-loop state distribution* $\rho_\mu^{\mathrm{cl}}$ *held fixed.*
- *This is a semi-gradient: the update freezes part of its own parameter dependence. In TD learning the frozen part is the bootstrapped target, and the data distribution does not depend on the parameters [85, 88].*
- *Here the frozen part is the data distribution* $\rho_\mu^{\mathrm{cl}}$. *This is the structure of performative prediction [71]: steps taken with the distribution held at its current value converge to performatively stable points, not to optima.*
- ***The performative defect carries the even non-reciprocity.*** *With* $J^{\mathrm{cl}}(\mu) := \frac12\int\|r_\mu\|^2\,\mathrm{d}\rho_\mu^{\mathrm{cl}}$,

$$\begin{aligned} &-\frac{1}{k_e}V_L[\mu] + \frac{1}{k_e^2}U_{\mathrm{even}}[\mu] \\ &\qquad = -\frac{1}{k_e}\frac{\delta J^{\mathrm{cl}}}{\delta\mu} - \frac{1}{k_e^2}\int(\tilde\Phi_\vartheta\cdot\nabla)\big(\tfrac12\|r_\mu\|^2\big)\,\mathrm{d}\rho + O(k_e^{-3}). \end{aligned} \tag{21.5}$$

*The first term is a true first variation, hence reciprocal. The second is the defect: it has the same order as* $U_{\mathrm{even}}$, *and it carries all of* $A_{\mathrm{even}}$ *in (e).*

*(d)* Circulation kernel. $U_{\mathrm{odd}}[\mu](\vartheta) = -\frac{1}{T_0}\oint_{\mathcal{O}}\tilde\Phi(\cdot;\vartheta)^\top\,\mathrm{d}f + \int K(\vartheta,\vartheta')\,\mathrm{d}\mu(\vartheta')$, *with*

$$K(\vartheta,\vartheta') := \frac{1}{T_0}\oint_{\mathcal{O}}\tilde\Phi(\cdot;\vartheta)^\top\,\mathrm{d}\tilde\Phi(\cdot;\vartheta') = \Big\langle\tilde\Phi(x_m;\vartheta)^\top\frac{\mathrm{d}}{\mathrm{d}t}\tilde\Phi(x_m;\vartheta')\Big\rangle_t. \tag{21.6}$$

- $K$ *is antisymmetric.*
- *By Stokes,* $K(\vartheta,\vartheta') = \frac{1}{T_0}\int_\Sigma\sum_k \mathrm{d}\tilde\Phi_k(\cdot;\vartheta)\wedge\mathrm{d}\tilde\Phi_k(\cdot;\vartheta')$ *for any 2-chain* $\Sigma$ *with* $\partial\Sigma=\mathcal{O}$.
- *For* $n=2$ *and a simple, positively oriented orbit bounding* $D$,

$$K(\vartheta,\vartheta') = \frac{1}{T_0}\sum_k\int_D \det\big[\nabla_x\tilde\Phi_k(\cdot;\vartheta),\nabla_x\tilde\Phi_k(\cdot;\vartheta')\big]\,\mathrm{d}A.$$

- $K$ *is an integral over the enclosed region, not the area itself. For Gaussian features,* $\sum_k\det[\nabla\tilde\Phi_k(\vartheta),\nabla\tilde\Phi_k(\vartheta')] = (a^\top a')\cdot\det[\nabla\varphi,\nabla\varphi']$, *so* $K=0$ *whenever* $a^\top a'=0$ *or* $c=c'$, *whatever the area.*

- *Hence* $U_{\mathrm{odd}} \equiv 0$ *for* $n = 1$.
- *Reversing the orbit preserves* $\rho$*, flips* $U_{\mathrm{odd}}$*, and leaves* $U_{\mathrm{even}}$ *unchanged.*

*(e)* Non-reciprocity. *Write* $S(\vartheta, \vartheta') := \delta P_\mu(\vartheta)/\delta\mu(\vartheta')$ *for the derivative in the direction* $\delta_{\vartheta'}$*, and call* $S(\vartheta, \vartheta') - S(\vartheta', \vartheta)$ *its antisymmetric part (without the factor* $\frac{1}{2}$*). Adding a function of* $\vartheta$ *alone, or a constant* $c(\mu)$*, to* $P_\mu$ *changes the antisymmetric part by a coboundary, which the criterion below allows.*

- *At order* $1/k_e$*,* $S$ *is symmetric: it equals* $-\frac{1}{k_e}\int \tilde{\Phi}(\cdot;\vartheta)^\top \tilde{\Phi}(\cdot;\vartheta')\,\mathrm{d}\rho$.
- *At order* $1/k_e^2$ *it is not. The second variation of the* $k_e^{-2}$ *coefficient* $U_{\mathrm{even}} + U_{\mathrm{odd}}$ *has antisymmetric part* $A_{\mathrm{even}} + 2K$*, with*

$$A_{\mathrm{even}}(\vartheta, \vartheta') := -\int \Big[(\tilde{\Phi}_{\vartheta'} \cdot \nabla)(r_\mu^\top \tilde{\Phi}_\vartheta) - (\tilde{\Phi}_\vartheta \cdot \nabla)(r_\mu^\top \tilde{\Phi}_{\vartheta'})\Big]\,\mathrm{d}\rho. \tag{21.7}$$

- $P_\mu$*, up to* $O(k_e^{-3})$*, can be a first variation of a functional of* $\mu$ *only if* $A_{\mathrm{even}} + 2K$ *is a coboundary* $a(\vartheta) - a(\vartheta')$*, that is, only if all its cyclic sums vanish.*
- *They do not vanish in the checked instances, for* $n = 1$ *(from* $A_{\mathrm{even}}$ *alone) and for* $n = 2$*. Theorem I shows that the cyclic condition is equivalent to* $A_{\mathrm{even}} + 2K \equiv 0$*, and characterises when that happens: for* $n = 1$ *exactly when the residual vanishes on the orbit; for* $n \geq 2$ *at perfect fit exactly when the orbit carries no circulation and the residual's Jacobian is symmetric along the orbit. It also shows (Theorem I(vi)) that no open set of measures satisfies the condition, for any* $n$*.*
- *So at finite* $k_e$*, the exactly averaged loop is a non-reciprocal McKean–Vlasov dynamics: it is not the* $W_{G^{-1}}$*-gradient flow of any free energy on any open set of measures. A different,* $\mu$*-dependent metric is not excluded by this argument.*

**Status.** The frozen-measure part is proved. Non-reciprocity of the exact second variation is checked to 3% (asserted; 1–1.5% measured) and characterised by Theorem I. The Stokes identity and the parity under orbit reversal hold to machine precision. The expansion is checked through its $O(1/k_e)$ convergence. Novelty (literature audit): no precedent found (targeted search); ingredients known (higher-order averaging; performative prediction [71]; semi-gradient TD [85, 88]).

*Proof. (a)* Write $e = r_\mu(x_m)/k_e + w$. Then $w$ solves $\dot{w} = -k_e w + g(t, w)$ with $g = \nabla r_\mu(x_m)e - \dot{r}_\mu/k_e + O(|e|^2)$.

- The tube $\{\|e\| \leq R\}$ is forward invariant once $k_e R > \sup_{\|e\|\leq R}\|r_\mu(x_m + e)\|$.
- Inside the tube, solutions contract at rate $k_e - \mathrm{Lip}_R(r_\mu)$, which gives existence, uniqueness and attraction there. A contraction on $T_0$-periodic functions gives the expansion.
- The bounds on $r_\mu$, $\nabla r_\mu$ and $\nabla^2 r_\mu$ on the $R$-neighbourhood of the orbit are uniform in $\mu$, by the saturation and $f \in C^2$.
- The global statement uses $L_f$ in place of the local Lipschitz constant of $f$.

*(b)* Taylor-expand $e^\top \tilde{\Phi}(x_m + e;\vartheta) = e^\top \tilde{\Phi}(x_m;\vartheta) + e^\top \partial_x \tilde{\Phi}(x_m;\vartheta)e + O(|e|^3)$, and insert (a).

- The period averages of the time-symmetric terms are integrals against $\rho$.
- $(\nabla r\, r)^\top \tilde{\Phi} + r^\top(\partial_x \tilde{\Phi})r = (r \cdot \nabla)(r^\top \tilde{\Phi})$.
- $\vartheta$-derivatives commute with the average.
- The remainders are uniform, because the $x$- and $\vartheta$-derivatives of $\tilde{\Phi}$ of the orders needed are bounded.

*(c)* We have

$$\int r_\mu^\top \tilde\Phi_\vartheta \,\mathrm{d}\rho_\mu^{\mathrm{cl}} = \big\langle r_\mu(x_m+e_\mu)^\top \tilde\Phi(x_m+e_\mu;\vartheta)\big\rangle = \int r_\mu^\top \tilde\Phi_\vartheta \,\mathrm{d}\rho + \big\langle \nabla_x (r_\mu^\top \tilde\Phi_\vartheta)\cdot e_\mu\big\rangle + O(k_e^{-2}).$$

By (a) this equals $\int r_\mu^\top \tilde\Phi_\vartheta \,\mathrm{d}\rho + (1/k_e)U_{\mathrm{even}} + O(k_e^{-2})$.
*(d)* $\frac{\mathrm{d}}{\mathrm{d}t} r_\mu(x_m) = \frac{\mathrm{d}}{\mathrm{d}t} f(x_m) - \int \frac{\mathrm{d}}{\mathrm{d}t}\tilde\Phi(x_m;\vartheta')\,\mathrm{d}\mu(\vartheta')$.

- Antisymmetry: $K(\vartheta,\vartheta') + K(\vartheta',\vartheta) = \frac{1}{T_0}\oint \mathrm{d}\big(\tilde\Phi_\vartheta^\top \tilde\Phi_{\vartheta'}\big) = 0$.
- The area form is Stokes, or Green's theorem for $n = 2$.
- For $n = 1$, every 1-form on $\mathbb{R}$ is exact, so its loop integrals vanish.
- Reversal changes the sign of oriented loop integrals and preserves $\rho$.

*(e)* The second variation of $U_{\mathrm{even}}$ is

$$S_{\mathrm{even}}(\vartheta,\vartheta') = -\int \Big[(\tilde\Phi_{\vartheta'}\cdot\nabla)(r_\mu^\top\tilde\Phi_\vartheta) + (r_\mu\cdot\nabla)(\tilde\Phi_{\vartheta'}^\top\tilde\Phi_\vartheta)\Big]\,\mathrm{d}\rho,$$

whose second term is symmetric. The second variation of $U_{\mathrm{odd}}$ is $K$. A kernel's antisymmetric part $A$ is a coboundary if and only if $A(\vartheta_0,\vartheta_1) + A(\vartheta_1,\vartheta_2) + A(\vartheta_2,\vartheta_0) = 0$ for all triples.

The remainder of (b) is not differentiated in $\mu$. Instead the argument uses a circulation around a small triangle of measures.

1. Fix $\vartheta_0,\vartheta_1,\vartheta_2$ and $\varepsilon \in (0,1)$. For $(a,b)$ in $T_\varepsilon := \{a,b \ge 0,\ a+b \le \varepsilon\}$, let $\mu_{a,b} := (1-\varepsilon)\mu + (\varepsilon - a - b)\delta_{\vartheta_0} + a\delta_{\vartheta_1} + b\delta_{\vartheta_2}$.
2. The circulation $\oint_{\partial T_\varepsilon} \langle Q_{\mu_{a,b}}, \mathrm{d}\mu_{a,b}\rangle$ vanishes when $Q_\mu$ is a first variation plus a constant $c(\mu)$, because $\mathrm{d}\mu_{a,b}$ has zero mass. In particular it vanishes for $-(1/k_e)V_L$.
3. $U_{\mathrm{even}} + U_{\mathrm{odd}}$ is quadratic in $\mu$, so its circulation equals $-(\varepsilon^2/2)\cdot\sum_{\mathrm{cyc}}(A_{\mathrm{even}} + 2K)(\vartheta_i,\vartheta_{i+1})$ at the centroid of $T_\varepsilon$.
4. The $O(k_e^{-3})$ remainder of (b) contributes at most $6\varepsilon\cdot\sup|R| = O(\varepsilon k_e^{-3})$.
5. Suppose that, for all large $k_e$, $P_\mu$ were a first variation up to $O(k_e^{-3})$, with constants uniform in $k_e$. Then $(\varepsilon^2/(2k_e^2))|\mathrm{cyc}| \le C\varepsilon k_e^{-3}$, that is, $|\mathrm{cyc}| \le 2C/(\varepsilon k_e)$. The cyclic sum does not depend on $k_e$, so letting $k_e \to \infty$ at fixed $\varepsilon$ gives $\mathrm{cyc} = 0$ at the centroid measure of the triangle; letting $\varepsilon \to 0$, since $A_{\mathrm{even}} + 2K$ is affine in the measure, gives $\mathrm{cyc} = 0$ at $\mu$.

This is checked numerically: for $\varepsilon = 0.05$ and $n = 2$ the circulation equals the centroid formula to all printed digits. □

Checked by: `tests/test_averaging.py` in the accompanying repository, using `src/core/averaging.py`; seven tests: four for Proposition H.7, one for Remark H.0, two for Remark H.7a. The checks of Proposition H.7 are listed in Table 22.

## 21.2 The oscillation correction and the orientation of the reference orbit

**Remark H.7a** (the full first-order correction, and a test in the real closed loop)**.** Proposition H.7 freezes the measure over a period. In the closed loop the parameters also oscillate within each period, by $O(1/k_e)$.

*The parameter equation.* In the scaling $\sigma = \sigma_0/k_e$, write

$$\dot\Theta = \frac{1}{k_e}H(\Theta,t) + \frac{1}{k_e^2}H_2(\Theta,t) + O(k_e^{-3}), \tag{21.8}$$
$$H_i(\Theta,t) := G\nabla_\vartheta\big[r_\Theta(x_m(t))^\top\tilde\Phi(x_m(t);\vartheta_i)\big] - \sigma_0(\vartheta_i - \vartheta^0).$$

Table 22: Checks of Proposition H.7 (`tests/test_averaging.py`).

| check | result |
|---|---|
| expansion: relative error of $k_e^2(P_\mu - P_0)$ against $U_{\text{even}} + U_{\text{odd}}$ | halves when $k_e$ doubles. At $k_e = 40 \to 80$ it goes $0.54 \to 0.25$ for $n = 1$, $0.038 \to 0.019$ for $n = 2$ counter-clockwise, and $0.022 \to 0.011$ clockwise. A Richardson step at $k_e = 80$ and 160 brings $n = 1$ within 3% (asserted) |
| Stokes: loop-integral $K$ against the area integral over the enclosed ellipse | agreement to $7 \cdot 10^{-15}$ relative (asserted at $10^{-6}$); $K$ antisymmetric; $K \equiv 0$ for $n = 1$ |
| parity under orbit reversal (same $\rho$) | $U_{\text{even}}$ unchanged and $U_{\text{odd}}$ flipped, to $2 \cdot 10^{-15}$ relative (asserted at $10^{-10}$) |
| non-reciprocity: the *exact* second variation $S$ ($k_e = 80$) against the prediction | the full matrix $k_e^2 S + k_e S_0$ is within 5% of $S_{\text{even}} + K$. The cyclic sums of the antisymmetric part agree to 3%: $n = 1$ gives $-0.0711$ against $-0.0704$, all from the state shift; $n = 2$ gives $-0.0471$ against $-0.0464$ (state shift $-0.089$, circulation $+0.043$) |

*Second-order averaging* (Sanders–Verhulst–Murdock [78, §2.9]) gives the dynamics of the period mean $Y$:

$$\dot{Y} = \frac{1}{k_e}\langle H\rangle(Y) + \frac{1}{k_e^2}\Big[\langle H_2\rangle(Y) + \big\langle D_\Theta H(Y,t)\cdot u(Y,t)\big\rangle\Big] + O(k_e^{-3}). \tag{21.9}$$

- $u$ is the zero-mean antiderivative in $t$ of $H - \langle H\rangle$.
- $\langle H_2\rangle$ is the frozen-measure correction of Proposition H.7.
- This is formal. The averaging theorem gives it on finite slow-time horizons.
- `averaging.oscillation_correction` computes the oscillation term $b_{\text{osc}} := \langle D_\Theta H \cdot u\rangle$.

Three properties follow.

- **Parity and value.** The oscillation term $b_{\text{osc}}$ is odd under reversal of the orbit (`test_oscillation_correction_is_odd_under_orbit_reversal`). Its value matches an independent one-period near-identity flow at $k_e = 80$: the test asserts 3%, and 0.7% is measured (`test_oscillation_correction_matches_a_one_period_flow`).
- **Size.** Relative to the frozen-measure terms it scales like $|G|T_0$, the parameter displacement per period times $k_e$. It is negligible when adaptation is slow within a period, which is the small-gain limit in the fixed-$k_e$ discussion of Conjecture H.6. It is not negligible as $k_e \to \infty$ at fixed $G$. Both terms are $O(k_e^{-2})$ and uniformly bounded, so the order of the route to Conjecture H.6 is unchanged.
- **Non-reciprocity survives.** The only orientation-even part of the full correction is $U_{\text{even}}$, and its antisymmetric second variation is not a coboundary (Proposition H.7(e)). Suppose the full first-order dynamics were reciprocal for both orientations. Then the average of the two antisymmetric parts, $A_{\text{even}}$, would be a coboundary, which it is not. So the full first-order averaged dynamics is non-reciprocal for at least one orientation.

*A prediction tested in the real closed loop* (`exp_orbit_orientation.py` → `results/orbit_orientation/`).

- **The prediction.** Traverse the same elliptic reference both ways. The occupation measure is the same, so ($\mathfrak{A}$), and any functional of $\rho$, cannot distinguish the two runs. The first-order theory predicts that the learned kernels differ by $\Theta_+ - \Theta_- \approx -2J^{-1}(b_{\text{odd}} + b_{\text{osc}}) = O(1/k_e)$. Here $J$ is the Jacobian of the leading averaged drift at the equilibrium.

- **The setting.** $n = 2$, $m = 6$, $G = \mathrm{diag}(4, 4, 1, 1, 1)$, $\sigma = 2/k_e$, no dither, and the same initial parameters for both runs. The runs converged to machine precision. The six kernels start spread out (kernel spread 1.05) and converge to a common configuration (spread at most $5.6 \cdot 10^{-17}$ at the end of every run), so the learned network is effectively a single kernel: the test probes the orientation effect at a collapsed equilibrium of the $m = 6$ system.

Table 23: Orientation dependence of the learned kernels in the real closed loop (`results/orbit_orientation/table_sigma2_slow16.md`). The size ratio is measured over predicted.

| | | frozen measure only (H.7) | | full (H.7 + oscillation) | | full |
|---|---|---|---|---|---|---|
| $k_e$ | measured $\|\Theta_+ - \Theta_-\|$ | size ratio | cosine | size ratio | cosine | vector error |
| 10 | $5.7 \cdot 10^{-3}$ | 2.67 | 0.76 | 0.94 | 0.93 | 0.36 |
| 20 | $2.8 \cdot 10^{-3}$ | 2.41 | 0.83 | 0.97 | 0.98 | 0.20 |
| 40 | $1.4 \cdot 10^{-3}$ | 2.29 | 0.86 | 0.97 | 0.994 | 0.11 |

- The measured difference scales like $k_e^{-1.01}$.
- At $k_e = 40$ the full first-order theory predicts it to within 3% in size and 11% as a vector (cosine 0.994).
- The vector error halves with each doubling of $k_e$: 36%, 20% and 11% at $k_e = 10$, 20 and 40.
- The frozen-measure terms alone account for less than half of it.
- A first attempt with a weaker leak ($\sigma = 0.5/k_e$) and a shorter horizon had not converged: its residual drift was a third of the difference. It is not used.

The direction in which the reference orbit is traversed is therefore a property of the excitation that the learned kernels retain, at order $1/k_e$. The first-order theory predicts its effect.

**Status.** The second-order averaged equation (21.9) is formal; the averaging theorem gives it on finite slow-time horizons. The oscillation term is checked against an independent one-period near-identity flow (within 3% at $k_e = 80$) and is odd under orbit reversal (to $10^{-6}$). The full first-order theory predicts the orientation dependence of the learned kernels in the real closed loop to 3% in size and 11% as a vector (cosine 0.994, $k_e = 40$).

Checked by: `tests/test_averaging.py` (the two tests named above); `exp_orbit_orientation.py` → `results/orbit_orientation/table_sigma2_slow16.md`.

## 21.3 Reciprocity at order $k_e^{-2}$: a pairwise criterion and its characterisation

**Theorem I** (reciprocity at order $k_e^{-2}$: a pairwise criterion and its characterisation)**.** *Assume the setting of Proposition H.7: saturated Gaussian features $\tilde{\Phi}(x; \vartheta) = a\varphi_\xi(x)$ with $\xi := (c, s)$, $f \in C^2(\mathbb{R}^n)$ autonomous, $x_m \in C^2$ $T_0$-periodic with oriented orbit $\mathcal{O}$ and occupation measure $\rho$, and a frozen $\mu \in \mathcal{P}(\mathbb{R}^p)$. Let $S$ be the $k_e^{-2}$ coefficient of the second variation of $P_\mu$, and $A_\mu := S - S^\top = A_{\mathrm{even}} + 2K$ its antisymmetric part, as in H.7(e). Call $\mu$* reciprocal at order *$k_e^{-2}$ if $A_\mu \equiv 0$ (the 1-form that the $k_e^{-2}$ drift defines is closed at $\mu$). Note that $A_\mu$ is affine in $\mu$, because $r_\mu$ is and $K$ does not depend on $\mu$.*

*(i)* Pairwise criterion. *$A_\mu$ is a coboundary $a(\vartheta) - a(\vartheta')$ if and only if $A_\mu \equiv 0$. Consequently, by Proposition H.7(e), whose hypothesis concerns a neighbourhood of $\mu$ (its triangle of measures), $P_\nu$ can be a first variation of a functional of $\nu$ up to $O(k_e^{-3})$, with constants uniform in $k_e$ and in $\nu$ near $\mu$, only if $A_\nu \equiv 0$ for every $\nu$ in a neighbourhood of $\mu$.*

*(ii)* Structure. *With $Dr$ the Jacobian of $r_\mu$, $(Dr)_{kl} = \partial_l r_k$, and $[X, Y] := (X \cdot \nabla)Y - (Y \cdot \nabla)X$ the Lie bracket of vector fields on $\mathbb{R}^n$,*

$$A_{\mathrm{even}}(\vartheta, \vartheta') = -\int \Big\{ \tilde{\Phi}_\vartheta^\top (Dr_\mu - Dr_\mu^\top) \tilde{\Phi}_{\vartheta'} + r_\mu^\top [\tilde{\Phi}_{\vartheta'}, \tilde{\Phi}_\vartheta] \Big\} \,\mathrm{d}\rho. \tag{21.10}$$

*For the Gaussian features $A_\mu$ is a bilinear form in the weights: $A_\mu(\vartheta, \vartheta') = a^\top B_\mu(\xi, \xi') a'$ with*

$$\begin{aligned} B_\mu(\xi, \xi') := & -\int (Dr_\mu - Dr_\mu^\top) \varphi_\xi \varphi_{\xi'} \,\mathrm{d}\rho - \int r_\mu (\nabla \varphi_\xi)^\top \varphi_{\xi'} \,\mathrm{d}\rho \\ & + \int (\nabla \varphi_{\xi'}) r_\mu^\top \varphi_\xi \,\mathrm{d}\rho + 2\kappa(\xi, \xi') I_n, \end{aligned} \tag{21.11}$$

$$\kappa(\xi, \xi') := \Big\langle \varphi_\xi(x_m(t)) \cdot \frac{\mathrm{d}}{\mathrm{d}t} \varphi_{\xi'}(x_m(t)) \Big\rangle_t, \tag{21.12}$$

*so $A_\mu \equiv 0$ if and only if $B_\mu(\xi, \xi') = 0$ for all admissible $\xi, \xi'$. The first term is the curl of the residual paired with the features, the next two are the residual paired with the Lie bracket of the feature fields, and the last is the circulation kernel $K(\vartheta, \vartheta') = (a^\top a')\kappa(\xi, \xi')$. The performative part is bounded by the residual and by the antisymmetric part of its Jacobian along the orbit:*

$$|A_{\mathrm{even}}(\vartheta, \vartheta')| \le |a||a'| \Big[ \int \|Dr_\mu - Dr_\mu^\top\| \,\mathrm{d}\rho + \frac{2e^{-1/2}}{s_{\min}} \int |r_\mu| \,\mathrm{d}\rho \Big].$$

*(iii)* $n = 1$. *$A_\mu \equiv 0$ if and only if $r_\mu = 0$ $\rho$-a.e. on the orbit. So the one-dimensional loop is reciprocal at order $k_e^{-2}$, at a frozen $\mu$, exactly when $\mu$ interpolates $f$ along the orbit, and its non-reciprocity is bounded by the residual: $|A_\mu(\vartheta, \vartheta')| \le (2e^{-1/2}/s_{\min})|a||a'| \int |r_\mu| \,\mathrm{d}\rho$.*

*(iv)* $n \ge 2$ at perfect fit. *Suppose $r_\mu = 0$ $\rho$-a.e. Then the Lie-bracket terms vanish and*

$$B_\mu(\xi, \xi') = -\int (Dr_\mu - Dr_\mu^\top) \varphi_\xi \varphi_{\xi'} \,\mathrm{d}\rho + 2\kappa(\xi, \xi') I_n,$$

*an antisymmetric matrix plus a multiple of the identity. Hence $A_\mu \equiv 0$ if and only if both*

- *(a) $K \equiv 0$, which holds if and only if the 1-current of the oriented orbit vanishes, $T_\mathcal{O} := \int_0^{T_0} \delta_{x_m(t)} \dot{x}_m(t) \,\mathrm{d}t = 0$ (an $\mathbb{R}^n$-valued measure; $\int v \cdot \mathrm{d}T_\mathcal{O} = \oint_\mathcal{O} v \cdot \mathrm{d}x$); for $n = 2$, iff the winding-number function $w_\mathcal{O}$ of the orbit vanishes a.e.; and*
- *(b) $Dr_\mu = Dr_\mu^\top$ $\rho$-a.e.: the Jacobian of the residual has no antisymmetric part along the orbit.*

*Vanishing of $r_\mu$ on the orbit constrains only its tangential derivative, so (b) is a genuine extra condition: for $f := \hat{f}_\mu + h e_1$ with $h = 0$ on the orbit and $\partial_\perp h \ne 0$, $A_\mu - 2K = -\int \varphi_\xi \varphi_{\xi'} \partial_\perp h \,\mathrm{d}\rho \cdot (e_1 e_\perp^\top - e_\perp e_1^\top) \ne 0$. When $r_\mu$ vanishes on a neighbourhood of the orbit, $Dr_\mu = 0$ there and $A_\mu = 2K$: the semi-gradient defect has vanished and the loop is non-reciprocal exactly when the orbit carries circulation, through the orbit-area kernel alone.*

*(v)* General $n \ge 2$. *In any connected finite-dimensional real-analytic family of measures, for example atomic measures with $J$ atoms parametrised by their weights and lifted positions, the set of reciprocal measures is either the whole family or a closed real-analytic subvariety of measure zero. The instances of `tests/test_averaging.py` ($n = 1$ and the $n = 2$ ellipse) have $A_\mu \not\equiv 0$, with cyclic sums $-0.0704$ and $-0.0464$ of the predicted kernel.*

*(vi)* Nowhere a gradient flow. *For every $n$, every $(f, x_m)$ and every $\mu$, every neighbourhood of $\mu$ contains measures $\nu$ with $A_\nu \not\equiv 0$: along the segments $\nu_\delta := (1 - \delta)\mu + \delta\delta_{\vartheta'}$ either the*

*zero feature $\vartheta' = \vartheta_z$ or a feature with $a' \neq 0$ gives $A_{\nu_\delta} \not\equiv 0$ for all small $\delta > 0$. Hence, by (i), there is no open set of measures on which $P_\nu$ is a first variation up to $O(k_e^{-3})$ with uniform constants: the exactly averaged closed loop is nowhere the $W_{G^{-1}}$-gradient flow of a free energy at order $k_e^{-2}$. The reciprocal measures of (iii) and (iv) are coincidences without interior.*

**Status.** Proved; every formula checked against the generic second-variation code and, by the round-5 referee, against the exact second variation; the characterisations checked on perfect-fit, curl and zero-current instances. Novelty (literature audit): new; the analytic tools (injectivity of Gaussian convolution; Stokes with winding numbers) are classical.

*Proof. (i)* Let $\vartheta_z := (\alpha, \gamma, \kappa)$ with $\alpha = 0$, an admissible point of $\mathbb{R}^p$, so $a = 0$ and $\tilde\Phi(\cdot; \vartheta_z) \equiv 0$ as a function of $x$; its $x$-derivatives vanish too. Every term of $A_{\mathrm{even}}(\vartheta, \vartheta_z)$ contains $\tilde\Phi_{\vartheta_z}$ or $(\tilde\Phi_{\vartheta_z} \cdot \nabla)$, and $K(\vartheta, \vartheta_z) = \langle \tilde\Phi_\vartheta^\top \frac{\mathrm{d}}{\mathrm{d}t} \tilde\Phi_{\vartheta_z} \rangle = 0$; so $A_\mu(\vartheta, \vartheta_z) = 0$ for every $\vartheta$. If $A_\mu(\vartheta, \vartheta') = a(\vartheta) - a(\vartheta')$, then $a(\vartheta) = a(\vartheta_z)$ for every $\vartheta$, so $A_\mu \equiv 0$. The converse is trivial. For the consequence, apply H.7(e) at any $\nu$ near $\mu$ with $\vartheta_2 := \vartheta_z$ in its triangle of measures: the cyclic sum over $(\vartheta_0, \vartheta_1, \vartheta_z)$ is $A(\vartheta_0, \vartheta_1)$, evaluated at the centroid measure of the triangle, which tends to $\nu$ as the triangle's size $\varepsilon \to 0$; $A$ is affine in the measure, so $A_\nu(\vartheta_0, \vartheta_1) = 0$.
*(ii)* Expand $(\tilde\Phi' \cdot \nabla)(r^\top \tilde\Phi) = \sum_{k,l} \tilde\Phi'_l (\partial_l r_k) \tilde\Phi_k + \sum_{k,l} \tilde\Phi'_l r_k \partial_l \tilde\Phi_k = \tilde\Phi^\top (Dr) \tilde\Phi' + r^\top (\tilde\Phi' \cdot \nabla) \tilde\Phi$, and subtract the expression with $\vartheta$ and $\vartheta'$ exchanged. For $\tilde\Phi = a\varphi$, $(\tilde\Phi' \cdot \nabla)\tilde\Phi = a(a'^\top \nabla\varphi)\varphi'$, so $r^\top[\tilde\Phi', \tilde\Phi] = (r^\top a)(a'^\top \nabla\varphi)\varphi' - (r^\top a')(a^\top \nabla\varphi')\varphi$, and $K(\vartheta, \vartheta') = (a^\top a')\kappa(\xi, \xi')$. Collect the bilinear form in $(a, a')$. Since $a$ ranges over the open cube $(-a_{\max}, a_{\max})^n$, $A_\mu \equiv 0$ iff $B_\mu \equiv 0$. The bound uses $\varphi \le 1$ and $|\nabla\varphi| \le e^{-1/2}/s_{\min}$.
*(iii)* For $n = 1$, $Dr - Dr^\top = 0$, and for equal widths $s$, $\varphi_{c'} D\varphi_c - \varphi_c D\varphi_{c'} = \varphi_c \varphi_{c'} (c - c')/s^2$, so $B_\mu(\xi, \xi') = -((c - c')/s^2) \int r_\mu \varphi_c \varphi_{c'} \, \mathrm{d}\rho$. Now $\varphi_c \varphi_{c'} = e^{-(c-c')^2/(4s^2)} \cdot e^{-(x-m)^2/s^2}$ with $m := (c + c')/2$. Put $\nu := r_\mu \rho$, a finite signed measure with compact support. The function $m \mapsto \int e^{-(x-m)^2/s^2} \, \mathrm{d}\nu(x)$ extends to an entire function. If $B_\mu \equiv 0$ on admissible pairs, it vanishes for all $m \in (-c_{\max}, c_{\max})$ (take $c = m - \delta$, $c' = m + \delta$ with $0 < \delta$ small), hence everywhere; its Fourier transform is the product of the Gaussian's, which has no zeros, with $\hat\nu$, so $\nu = 0$, that is $r_\mu = 0$ $\rho$-a.e. Conversely, if $\nu = 0$ then $A_{\mathrm{even}}(\vartheta, \vartheta') = -\int (\varphi' D\varphi - \varphi D\varphi') aa' \, \mathrm{d}\nu = 0$ for all widths.
*(iv)* If $r_\mu = 0$ $\rho$-a.e. the two Lie-bracket integrals vanish, which gives the displayed $B_\mu$; the first term is an antisymmetric matrix and the second a multiple of $I_n$, so $B_\mu \equiv 0$ iff both vanish (take the trace, which is $2n\kappa$, and the antisymmetric part). The first vanishes identically iff the matrix-valued measure $(Dr_\mu - Dr_\mu^\top)\rho$ is zero, by the argument of (iii) applied entrywise with equal widths (the entries of $\int (Dr - Dr^\top)\varphi_c \varphi_{c'} \, \mathrm{d}\rho$ are Gaussian convolutions of the entries of that measure). For (a): with equal widths, $\varphi_\xi \varphi_{\xi'} = e^{-|c-c'|^2/(4s^2)} e^{-|x-m|^2/s^2}$ and $\nabla\varphi_{\xi'} = -((x - c')/s^2)\varphi_{\xi'}$, so

$$\kappa(\xi, \xi') = -\frac{1}{s^2} e^{-|c-c'|^2/(4s^2)} \big\langle e^{-|x_m - m|^2/s^2} (x_m - c') \cdot \dot x_m \big\rangle_t .$$

Write $x_m - c' = (x_m - m) + (c - c')/2$. The first part integrates to zero, being the time derivative of $-(s^2/2) e^{-|x_m - m|^2/s^2}$ along a period. Hence

$$\kappa(\xi, \xi') = -\frac{1}{2s^2} e^{-|c-c'|^2/(4s^2)} (c - c') \cdot F_s(m),$$
$$F_s(m) := \big\langle e^{-|x_m(t) - m|^2/s^2} \dot x_m(t) \big\rangle_t = \frac{1}{T_0} \int e^{-|x-m|^2/s^2} \, \mathrm{d}T_{\mathcal{O}}(x),$$

the Gaussian convolution of the vector measure $T_{\mathcal{O}}$. If $\kappa \equiv 0$ on admissible pairs, then $F_s(m) \cdot v = 0$ for every $m$ with $\|m\| < c_{\max}$ and every direction $v$ (take $c = m - \delta v$, $c' = m + \delta v$), so $F_s \equiv 0$ on that ball; $F_s$ is entire, so $F_s \equiv 0$ on $\mathbb{R}^n$, and the Fourier argument gives $T_{\mathcal{O}} = 0$. Conversely, $\kappa(\xi, \xi') = \frac{1}{T_0} \int \varphi_\xi \nabla\varphi_{\xi'} \cdot \mathrm{d}T_{\mathcal{O}} = 0$ for all widths when $T_{\mathcal{O}} = 0$. For $n = 2$, $\oint_{\mathcal{O}} v \cdot \mathrm{d}x = \int w_{\mathcal{O}} \operatorname{curl} v \, \mathrm{d}A$

for every smooth $v$ (Stokes with winding numbers), and every smooth compactly supported $g$ is curl $v$ for some smooth $v$, so $T_{\mathcal{O}} = 0$ iff $w_{\mathcal{O}} = 0$ a.e. ($T_{\mathcal{O}}$ is compactly supported, so $v$ need not be). For the example, $r = he_1$ with $h = 0$ on $\mathcal{O}$ gives $(Dr - Dr^\top)_{1l} = \partial_l h$ and the only non-zero entries are the normal derivative; the Lie-bracket terms vanish because $r = 0$ on $\mathcal{O}$.
*(v)* The saturated features are real-analytic in the lifted coordinates and extend holomorphically to a complex neighbourhood of $\mathbb{R}^p$ (tanh and the logistic function are holomorphic on the strips $|\mathrm{Im}| < \pi a_{\max}/2$, $\pi c_{\max}/2$, $\pi$, where $s$ stays away from 0), uniformly for $x$ on the compact orbit; integrals over the orbit of such integrands are real-analytic in the parameters, and $r_\mu$ is affine in the weights. The zero set of a real-analytic function on a connected real-analytic manifold is either the whole manifold or a closed set of Lebesgue measure zero. The instances are those of `test_exactly_averaged_dynamics_is_non_reciprocal_for_n1_and_n2`.
*(vi)* Since $r_\nu$ is affine in $\nu$, $B_{\nu_\delta} = B_\mu + \delta B^{\mathrm{even}}[g]$ with $g := \hat{f}_\mu - \tilde{\Phi}_{\vartheta'}$, where $B^{\mathrm{even}}[g]$ denotes the three residual terms of $B$ with $g$ in place of $r_\mu$. The trace of $B^{\mathrm{even}}[g]$ at equal widths is $-(1/s^2)(c - c') \cdot \int g\varphi_c\varphi_{c'}\,\mathrm{d}\rho$ (the curl term is traceless). If $A_{\nu_\delta} \equiv 0$ for all small $\delta$, then $B^{\mathrm{even}}[g] \equiv 0$, so $(c - c') \cdot F[g\rho]((c + c')/2) = 0$ for all admissible $c, c'$, where $F[g\rho]$ is the Gaussian convolution of the vector measure $g\rho$; as in (iv), $g\rho = 0$, that is $g = 0$ $\rho$-a.e. For $\vartheta' = \vartheta_z$ this says $\hat{f}_\mu = 0$ on the orbit; for $\vartheta'$ with $a' \neq 0$ it then says $\tilde{\Phi}_{\vartheta'} = 0$ on the orbit, which is false since a Gaussian has no zeros. So one of the two segments has $A_{\nu_\delta} \not\equiv 0$ for all small $\delta > 0$. □

Checked by: `tests/test_nonreciprocity.py`, using `src/core/nonreciprocity.py` against `src/core/averaging.py`: the bilinear form $a^\top B_\mu a'$ reproduces the antisymmetric part of the generic second-variation coefficient ($10^{-10}$, $n = 1$ and 2); $A_\mu(\vartheta, \vartheta_z) = 0$ ($10^{-14}$) and the cyclic sum over $(\vartheta_0, \vartheta_1, \vartheta_z)$ equals $A_\mu(\vartheta_0, \vartheta_1)$; for $n = 1$ with $f := \hat{f}_\mu$ (perfect fit) $A_\mu \equiv 0$ (below $10^{-13}$) while a non-zero residual gives $A_\mu \neq 0$, equal to the closed form; the closed form of $\kappa$ ($10^{-10}$), the second component of $F_s(m)$ at $m = (0.5, 0)$, $s = 0.7$ is positive on the ellipse, and $\kappa \equiv 0$ on a segment traversed back and forth (below $10^{-12}$); with $f \equiv \hat{f}_\mu$ on $\mathbb{R}^2$ (so $Dr_\mu \equiv 0$), $B_\mu = 2\kappa I$ ($10^{-13}$), $A_\mu \neq 0$ on the ellipse and $A_\mu \equiv 0$ on the segment; with $f := \hat{f}_\mu + he_1$, $h = (x_1/1.2)^2 + (x_2/0.8)^2 - 1$ (zero on the ellipse, non-zero normal derivative), $B_\mu - 2\kappa I$ equals $-\int \varphi\varphi'\partial_2 h\,\mathrm{d}\rho \cdot (e_1 e_2^\top - e_2 e_1^\top)$ ($10^{-12}$) and is not small; the positivity witness $f \equiv 0$, $\mu = \delta_{\bar{\vartheta}}$, for which $A_\mu(\vartheta_0, \vartheta_1) = a_0 a_1 \bar{a}\,((c_0 - c_1)/s^2) \int \varphi_{\bar{c}}\varphi_{c_0}\varphi_{c_1}\,\mathrm{d}\rho$ has a strictly positive integral. The round-5 referee re-derived every formula and checked $a^\top B_\mu a'$ against the exact second variation of the averaged potential (Richardson in $k_e$) to $2 \cdot 10^{-4}$–$9 \cdot 10^{-4}$ relative, including parameters at the saturation limits.

**Remark I.1** (what Theorem I changes in the reading of Proposition H.7)**.**

- The non-reciprocity of H.7(e) is no longer an instance-by-instance check. Pointwise, for $n = 1$ it holds at every frozen $\mu$ that does not interpolate $f$ along the orbit; for $n \geq 2$ at perfect fit it holds unless the orbit carries no circulation and the residual's Jacobian is symmetric along the orbit; in general it fails only on a null set of any connected analytic family that contains one non-reciprocal member. Globally, (vi) shows that no open set of measures is reciprocal, for any $n$: the averaged loop is nowhere a gradient flow at order $k_e^{-2}$.
- The two defects scale differently. The performative (semi-gradient) defect is bounded by the residual and the antisymmetric part of its Jacobian along the orbit, so it is small at a well-fitting equilibrium whose residual is smooth across the orbit; the circulation defect does not depend on the fit at all.
- For the $n = 1$ Langevin experiments and Conjecture H.6: the non-gradient correction that falsifies H.6 at fixed $k_e$ is, by (iii), bounded by the residual, which at the regularised optimum $\mu_\tau^*$ is the regularisation bias. Both the leading drift and the correction are proportional to the residual there, so the correction is $O(1/k_e)$ relative to the leading drift and $O(k_e^{-2})$ absolute, as in Remark H.0.

- The pairwise criterion also simplifies the test of reciprocity: a single pair $(\vartheta, \vartheta')$ with $B_\mu(\xi, \xi') \neq 0$ suffices.

## 21.4 Input-to-state stability of $(\mathfrak{A})$ and the route to Conjecture H.6

**Proposition H.8** (input-to-state stability of $(\mathfrak{A})$)**.** *Assume $\sigma, \tau > 0$ and the setting of Theorem H. Let $\mu_t$ solve the perturbed equation*

$$\partial_t \mu = \nabla \cdot \big(\mu(G\nabla V[\mu] - b_t)\big) + \tau \nabla \cdot (G \nabla \mu) \tag{21.13}$$

*from $\mu_0 \in \mathcal{P}_2$ with $F_\tau(\mu_0) < \infty$. Here $b_t$ is a velocity perturbation with $\beta(t)^2 := \int \|b_t\|_{G^{-1}}^2 \, \mathrm{d}\mu_t \leq \bar{\beta}^2$. Assume the chain rule of [15, Lemma 2.2] holds along this solution:*

$$\frac{\mathrm{d}}{\mathrm{d}t} F_\tau(\mu_t) = \int \langle \nabla(\delta F_\tau / \delta \mu)(\mu_t), \, v_t \rangle \, \mathrm{d}\mu_t,$$

*in whitened coordinates, with $v_t$ the velocity. It is expected for smooth bounded perturbations (parabolic regularity and the entropy-dissipation estimate), but it is not verified here. Then*

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(F_\tau(\mu_t) - F^*\big) \leq -\tau\rho_\tau \big(F_\tau(\mu_t) - F^*\big) + \tfrac{1}{2}\beta(t)^2, \tag{21.14}$$

$$F_\tau(\mu_t) - F^* \leq e^{-\tau\rho_\tau t}\big(F_\tau(\mu_0) - F^*\big) + \frac{\bar{\beta}^2}{2\tau\rho_\tau}, \tag{21.15}$$

$$\limsup_{t\to\infty} W_{G^{-1}}(\mu_t, \mu_\tau^*) \leq \frac{\bar{\beta}}{\tau\rho_\tau}. \tag{21.16}$$

**Status.** Proved, assuming the chain rule for the perturbed flow. A proof only; not checked in the repository. Novelty (literature audit): elementary given [15].

*Proof.* Work in $\zeta = G^{-1/2}\vartheta$, and let $\xi_t := \nabla(\delta F_\tau/\delta\mu)(\mu_t) = \nabla \tilde{V}[\mu_t] + \tau \nabla \log \mu_t$. The velocity is $-\xi_t + \tilde{b}_t$, with $\|\tilde{b}_t\| = \|b_t\|_{G^{-1}}$.

1. By the chain rule and Young's inequality,

$$\frac{\mathrm{d}}{\mathrm{d}t} F_\tau(\mu_t) = -\int \|\xi_t\|^2 \, \mathrm{d}\mu_t + \int \langle \xi_t, \, \tilde{b}_t \rangle \, \mathrm{d}\mu_t \leq -\frac{1}{2}\int \|\xi_t\|^2 \, \mathrm{d}\mu_t + \frac{1}{2}\beta^2.$$

2. With $\hat{\mu}_t \propto e^{-\tilde{V}[\mu_t]/\tau}$,

$$\int \|\xi_t\|^2 \, \mathrm{d}\mu_t = \tau^2 I(\mu_t | \hat{\mu}_t) \geq 2\rho_\tau \tau^2 H(\mu_t | \hat{\mu}_t) \geq 2\rho_\tau \tau \big(F_\tau(\mu_t) - F^*\big),$$

   by the LSI of Theorem H(ii) and the entropy sandwich [15, Lemma 3.4].

3. Grönwall gives the first two lines.

4. For the third, $F_\tau(\mu) - F^* \geq \tau H(\mu|\mu_\tau^*)$ [15, Lemma 3.4]. $\mu_\tau^*$ satisfies $\mathrm{LSI}(\rho_\tau)$, hence Talagrand's inequality $W^2 \leq (2/\rho_\tau) H(\cdot|\mu_\tau^*)$ (Otto–Villani [65]). So $W^2 \leq (2/(\tau\rho_\tau))(F_\tau - F^*)$, and the limsup follows from the second line. □

The gain $1/(\tau\rho_\tau) \leq e^{\alpha^*/\tau}/\sigma$ is exponentially large in $1/\tau$, like the inverse of the rate.

**Route to Conjecture H.6.** In the scaling of Conjecture H.6:

- $\tau\rho_\tau \geq (\sigma_0/k_e) \cdot e^{-2K_\Phi(F_\rho+K_\Phi)/\tau_0}$, so $1/(\tau\rho_\tau) = O(k_e)$.
  - In this scaling the whitened proximal Gibbs measures do not depend on $k_e$, so $\tau\rho_\tau = \Theta(1/k_e)$.
  - At the experiment's $\tau_0 = 0.04$ the constant is $e^{1049}$.
- Define $b := (1/k_e^2)\big[G\nabla(U_{\text{even}} + U_{\text{odd}}) + b_{\text{osc}}\big] + O(k_e^{-3})$. The first part is the frozen-measure correction of Proposition H.7; the second is the oscillation term of Remark H.7a. Both are bounded uniformly in $\mu$ and $\vartheta$, so $\bar{\beta} = O(k_e^{-2})$.
- Proposition H.8 then confines every solution of $(\mathfrak{A}) + b$ to an asymptotic $W$-ball around $\mu_\tau^*$ of radius $O(1/k_e) \cdot e^{2K_\Phi(F_\rho+K_\Phi)/\tau_0}/\sigma_0$. This needs $F_\tau(\mu_0) < \infty$, or a $\mathcal{P}_2$ start, after the Girsanov regularisation of Theorem H(ii), which applies because $b$ is bounded.

Three things are still missing for Conjecture H.6:

1. control of the frozen-measure approximation along the moving $\mu_t$, that is, a near-identity averaging transformation for the stochastic McKean–Vlasov loop, uniformly in time;
2. the mean-field limit of $(\mathrm{L}_\tau)$, since Theorem D covers only (L);
3. the chain rule that Proposition H.8 assumes for the perturbed flow.

## 21.5 Evidence from the real closed loop

The evidence comes from the real closed loop with $m = 32$ and $n = 1$, not from the averaged system.

- **Dither is necessary.** Without dither the collapsed start is trapped forever. Its loss stays at 0.166, against 0.007 from the spread start (`results/langevin_tradeoff`, 1500 s).
- **Equalisation at moderate temperature.** At $\sigma = 0.02$ and $\tau = 0.004$ (3 seeds, `results/langevin_global/table_sigma0.02_tau0.004.md`):
  - collapsed and spread starts reach the same loss (ratio 1.03) and tracking RMS (0.0465 against 0.0452);
  - their final 32-particle configurations, driven by the same noise realisation, are 4.2× closer in $W_2$ over the physical coordinates $(a, c, s)$ than without dither (0.40 against 1.69);
  - a numerical check shows that two independent 32-point samples of $\mu_\tau^*$ are typically about 0.6 apart, so 0.40 is at the sampling floor;
  - the common loss of about 0.11 matches the grid value $\bar{L}(\mu_\tau^*) = 0.11118$. It is dominated by the regularisation bias.
- **Metastability at low temperature.** At $\sigma = 0.002$ and $\tau = 5 \cdot 10^{-4}$ (table for these values in `results/langevin_global/`), dither frees the collapsed start ($0.166 \to 0.048$). But the starts have not equalised within 900 s (loss ratio 5.2). The spread start's $9.2 \cdot 10^{-3}$ is close to the regularised optimum $\bar{L}(\mu_\tau^*) = 8.5 \cdot 10^{-3}$.
- **Limits of the evidence.**
  - The Langevin runs have $n = 1$. By Proposition H.7, the averaged dynamics is already non-reciprocal for $n = 1$, through the state shift.
  - The only $n \geq 2$ closed-loop runs are the deterministic orientation runs of Remark H.7a. They confirm the first-order theory quantitatively. No $n \geq 2$ run with dither has been made.

– The runs use fixed $k_e$, not the scaling of Conjecture H.6.

– All Langevin runs use $k_e = 10$, below the Tikhonov scale of Remark H.0 for this network ($(F_\rho + K_\Phi)/s_{\min} \approx 35$). The measured frozen-$\mu$ drift error at $k_e = 10$ is 4.2% ($n = 1$), so the condition is conservative, but the agreement with ($\mathfrak{A}$) is empirical.

– So the evidence supports Conjecture H.6 qualitatively, but the equalisation time can exceed practical horizons at low $\tau$.

# 22 Theorem J: the finite-$m$ closed loop as a perturbed gradient flow

This section proves the first step of Section 26, branch 1: at finite $m$, without dither, in the scaling $\sigma = \sigma_0/k_e$, the real closed loop has, near every hyperbolic critical point of the $m$-particle regularised occupation loss, a unique periodic orbit with the same stability type, at distance $O(1/k_e)$, whose first-order location is the one Proposition H.7 and Remark H.7a predict. It turns Remark H.7a's "formal" second-order averaging into a theorem for this system.

**Setting.** The closed loop of Theorem H's design with $\tau = 0$, at finite $m$, in the scaling $\sigma = \sigma_0/k_e$:

$$\begin{aligned} \dot{x} &= f(x) + \dot{x}_m - k_e e - \hat{f}_\Theta(x), \qquad e := x - x_m(t), \qquad \hat{f}_\Theta := \frac{1}{m}\sum_j \tilde{\Phi}(\cdot\,;\vartheta_j), \\ \dot{\vartheta}_i &= G\nabla_\vartheta\big[e^\top \tilde{\Phi}(x;\vartheta_i)\big] - \frac{\sigma_0}{k_e}(\vartheta_i - \vartheta^0), \qquad i = 1,\dots,m, \end{aligned} \tag{CL}$$

with $\Theta := (\vartheta_1,\dots,\vartheta_m) \in \mathbb{R}^{mp}$, $G_m := I_m \otimes G$, saturated features, $x_m \in C^2$ $T_0$-periodic with occupation measure $\rho$, and $f \in C^2$ on a neighbourhood of the tube $\mathcal{T}_R := \{x : \operatorname{dist}(x, x_m([0,T_0])) \le R\}$ for some $R > 0$. Put

$$r_\Theta(x) := f(x) - \hat{f}_\Theta(x), \qquad J_m(\Theta) := m\bar{L}_m(\Theta) + \frac{\sigma_0}{2}\sum_i \|\vartheta_i - \vartheta^0\|^2_{G^{-1}},$$

$$\bar{L}_m(\Theta) := \frac{1}{2}\int \|r_\Theta\|^2 \,\mathrm{d}\rho,$$

$$H_i(\Theta,t) := G\nabla_\vartheta\big[r_\Theta(x_m(t))^\top \tilde{\Phi}(x_m(t);\vartheta)\big]\Big|_{\vartheta=\vartheta_i} - \sigma_0(\vartheta_i - \vartheta^0),$$

$$\langle H\rangle(\Theta) = -G_m \nabla J_m(\Theta).$$

The averaged system of (CL) is $\dot{\Theta} = \frac{1}{k_e}\langle H\rangle(\Theta) = -\frac{1}{k_e}G_m\nabla J_m(\Theta)$: the gradient flow of $J_m$ in the $G^{-1}$ metric, which is ($\mathfrak{A}$) at $\tau = 0$ for the $m$-atom measure (Remark H.0); in slow time $s := t/k_e$ it does not depend on $k_e$. Its equilibria are the critical points of $J_m$. A critical point $\Theta^*$ is *hyperbolic* if $\nabla^2 J_m(\Theta^*)$ is non-singular; its index is the number of negative eigenvalues; it is a *non-degenerate minimum* if $\nabla^2 J_m(\Theta^*) \succ 0$. Write $\nu_1 \le \cdots \le \nu_{mp}$ for the eigenvalues of $G_m^{1/2}\nabla^2 J_m(\Theta^*)G_m^{1/2}$ (those of $G_m\nabla^2 J_m(\Theta^*)$), and $\lambda_* := \nu_1$ when $\Theta^*$ is a minimum.

Let $u(\Theta,t)$ be the zero-mean antiderivative in $t$ of $H(\Theta,t) - \langle H\rangle(\Theta)$ ($T_0$-periodic), and define, as in Proposition H.7 and Remark H.7a, at the atomic measure $\mu^* := \frac{1}{m}\sum_j \delta_{\vartheta_j^*}$:

$$J := D_\Theta\big[\tfrac{1}{k_e}\langle H\rangle\big](\Theta^*) = -\frac{1}{k_e}G_m\nabla^2 J_m(\Theta^*),$$

$$b_{\mathrm{frozen},i} := \frac{1}{k_e^2}G\nabla_\vartheta\big(U_{\mathrm{even}} + U_{\mathrm{odd}}\big)[\mu^*](\vartheta)\Big|_{\vartheta=\vartheta_i^*}, \qquad b_{\mathrm{osc}} := \frac{1}{k_e^2}\big\langle D_\Theta H(\Theta^*,t)\cdot u(\Theta^*,t)\big\rangle_t.$$

**Theorem J.** *Let $\Theta^*$ be a hyperbolic critical point of $J_m$. There are $k_0, \rho_0, C > 0$, depending on ($f$, $x_m$, $G$, $\sigma_0$, $R$, the feature bounds, $\Theta^*$) but not on $k_e$, such that for every $k_e \ge k_0$:*

(a) Existence, uniqueness, location. (CL) *has a $T_0$-periodic solution $(x_p, \Theta_p)$, unique among the $T_0$-periodic solutions with $\sup_t \|\Theta_p(t) - \Theta^*\|_{G^{-1}} \le \rho_0$ and $\sup_t \|x_p(t) - x_m(t)\| \le R$. It satisfies*

$$\sup_t \|x_p(t) - x_m(t)\| \le C/k_e, \qquad \sup_t \|\Theta_p(t) - \Theta^*\|_{G^{-1}} \le C/k_e,$$
$$\Theta_p(t) - \bar{\Theta}_p = \frac{1}{k_e} u(\Theta^*, t) + O(k_e^{-2}),$$

*with $\bar{\Theta}_p$ the period mean of $\Theta_p$.*

(b) Stability type. *The monodromy matrix of $(x_p, \Theta_p)$ has $n$ eigenvalues of modulus at most $e^{-(k_e - L)T_0}$, with $L$ independent of $k_e$ (the error channel), and $mp$ eigenvalues*

$$\lambda_j = \exp\bigl(-(T_0/k_e)\nu_j + O(k_e^{-2})\bigr), \qquad j = 1, \dots, mp.$$

*Hence the orbit is hyperbolic with the index of $\Theta^*$: exponentially stable if $\nabla^2 J_m(\Theta^*) \succ 0$, with parameter Floquet exponents $(1/k_e)\nu_j + O(k_e^{-2})$, and unstable otherwise.*

(c) Basin (non-degenerate minimum). *If $\nabla^2 J_m(\Theta^*) \succ 0$, every solution of* (CL) *with $\|\Theta(0) - \Theta^*\|_{G^{-1}} \le \rho_0$ and $\|x(0) - x_m(0)\| \le R$ converges to the periodic orbit, and*

$$\|\Theta(t) - \Theta_p(t)\|_{G^{-1}} + \|x(t) - x_p(t)\| \le C e^{-\mu t}\bigl(\|\Theta(0) - \Theta_p(0)\|_{G^{-1}} + \|x(0) - x_p(0)\|\bigr),$$
$$\mu := \frac{\lambda_*}{k_e}(1 - \kappa),$$

*with $\kappa = \kappa(\rho_0, 1/k_e) \to 0$ as $(\rho_0, 1/k_e) \to 0$. The basin radius $\rho_0$ does not depend on $k_e$; the rate is $\lambda_*/k_e$ to leading order, that is, $\lambda_*$ in slow time.*

(d) First-order location. *The period mean satisfies*

$$\begin{aligned}\bar{\Theta}_p &= \Theta^* - J^{-1}(b_{\text{frozen}} + b_{\text{osc}}) + O(k_e^{-2}) \\ &= \Theta^* + \frac{1}{k_e}\bigl(G_m \nabla^2 J_m(\Theta^*)\bigr)^{-1} \cdot k_e^2 (b_{\text{frozen}} + b_{\text{osc}}) + O(k_e^{-2}).\end{aligned}$$

*The offset is $O(1/k_e)$. Its orientation-odd part is the quantity measured in Remark H.7a, so the prediction $\Theta_+ - \Theta_- = -2J^{-1}(b_{\text{odd}} + b_{\text{osc}}) + O(k_e^{-2})$ of that remark is now a statement with error control.*

**Status.** Proved (tube and slaving, periodic Riccati block-triangularisation, second-order averaging, Poincaré map); checked by Newton shooting for the exact orbit, its monodromy matrix and its period mean against the averaged Hessian and the first-order prediction, at $k_e = 10$–$160$ (`tests/test_periodic.py`, `exp_periodic_orbit.py`). Novelty (literature audit): the first step of §8 branch 1; the tools (two-time-scale averaging under attraction) are classical.

*Proof.* Write $\varepsilon := 1/k_e$. All constants below are uniform in $\varepsilon \le 1/k_0$ and in $\Theta$ in the ball $\mathcal{K} := \{\|\Theta - \Theta^*\| \le 2\rho_0\}$, because the saturated features and their derivatives are bounded uniformly in $\Theta$ and $f \in C^2$ on the compact tube $\mathcal{T}_R$.

*Step 1 (the tube and the slaved error).* Let $M_r := \sup\{\|r_\Theta(x)\| : x \in \mathcal{T}_R,\ \Theta \in \mathcal{K}\}$ and $L_r := \sup\|\partial_x r_\Theta\|$ on the same set. For $k_e > M_r/R$ the tube $\{\|e\| \le R\}$ is forward invariant: on $\|e\| = R$, $\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\|e\|^2 = e^\top r_\Theta(x) - k_e\|e\|^2 < 0$. Put $w := e - \varepsilon r_\Theta(x)$ with $x$ the actual state. Since $r_\Theta(x) - k_e e = -k_e w$ and $\dot{x} = \dot{x}_m - k_e w$,

$$\dot{w} = -\bigl(k_e I - \partial_x r_\Theta(x)\bigr)w - \varepsilon\, \partial_x r_\Theta(x)\dot{x}_m - \varepsilon\, \partial_\Theta r_\Theta(x) \cdot \dot{\Theta},$$

and $\dot{\Theta} = F(e, \Theta, t) := \bigl(G\nabla_\vartheta[e^\top \tilde{\Phi}(x; \vartheta_i)] - \varepsilon\sigma_0(\vartheta_i - \vartheta^0)\bigr)_i$ is $O(\|e\| + \varepsilon) = O(\|w\| + \varepsilon)$. So for $k_e \ge 2L_r + C$, $\frac{\mathrm{d}}{\mathrm{d}t}\|w\| \le -(k_e/2)\|w\| + C_1\varepsilon$ on $\{\|w\| \le 1\} \cap \{\|e\| \le R\}$, whence $\|w(t)\| \le$

$e^{-k_e t/2}\|w(0)\| + 2C_1\varepsilon^2$. The set $W_\varepsilon := \{\|w\| \le 3C_1\varepsilon^2\}$ is forward invariant and absorbs every solution of the tube within $t_1 = O(\varepsilon \log(1/\varepsilon))$, during which $\Theta$ moves by $O(\varepsilon\|e(0)\| + \varepsilon^2 \log(1/\varepsilon))$ ($\dot\Theta = O(\|e\| + \varepsilon)$ and $\|e\|$ decays at rate $k_e/2$). On $W_\varepsilon$, $e = \varepsilon r_\Theta(x_m(t)) + \varepsilon[r_\Theta(x_m + e) - r_\Theta(x_m)] + w = \varepsilon r_\Theta(x_m(t)) + O(\varepsilon^2)$.

*Step 2 (the slow equation and the fast variable).* Let $e_\Theta(t)$ be the frozen-$\Theta$ periodic error of Proposition H.7(a): the unique $T_0$-periodic solution of $\dot e = r_\Theta(x_m(t) + e) - k_e e$, exponentially stable with rate at least $k_e/2$, with $e_\Theta = \varepsilon r_\Theta(x_m) + \varepsilon^2(\partial_x r_\Theta\, r_\Theta - \dot r_\Theta)(x_m) + O(\varepsilon^3)$ in sup norm (Taylor expansion of $r_\Theta$ in $x$ to second order; $f \in C^2$). Because $\Theta$ enters $r_\Theta = f - \hat f_\Theta$ only through the smooth $\hat f_\Theta$, $e_\Theta(t)$ is $C^2$ in $\Theta$ as soon as $f \in C^1$, and differentiating the frozen equation (gain $1/k_e$) gives $\partial_\Theta e_\Theta = \varepsilon\, \partial_\Theta r_\Theta(x_m) + O(\varepsilon^2)$ and $\partial_\Theta^2 e_\Theta = O(\varepsilon)$, uniformly on $\mathcal{K}$. Let $\zeta := e - e_{\Theta(t)}(t)$ be the fast variable. Then

$$\dot\zeta = \big[r_\Theta(x_m + e) - r_\Theta(x_m + e_\Theta)\big] - k_e\zeta - \partial_\Theta e_\Theta(t)\cdot\dot\Theta,$$

where the bracket is Lipschitz in $\zeta$ with constant $L_r$ and the last term is $O(\varepsilon)\cdot O(\varepsilon) = O(\varepsilon^2)$ ($\dot\Theta = F = O(\|e\| + \varepsilon)$). So for $k_e \ge 2L_r + C$ the set $Z_\varepsilon := \{\|\zeta\| \le C_2\varepsilon^3\}$ is forward invariant and absorbs every solution of the tube within $O(\varepsilon\log(1/\varepsilon))$; on $W_\varepsilon$ of Step 1 one has $\|\zeta\| = O(\varepsilon^2)$ at once and $O(\varepsilon^3)$ after this further transient. On $Z_\varepsilon$, since $F$ is $C^1$ in $e$ with $\partial_e F = O(1)$,

$$\dot\Theta = F(e_\Theta(t), \Theta, t) + R(\Theta, \zeta, t) = \varepsilon H(\Theta, t) + \varepsilon^2 H_2(\Theta, t) + O(\varepsilon^3), \qquad \|R\| \le C\|\zeta\| = O(\varepsilon^3), \tag{S}$$

where $\varepsilon H + \varepsilon^2 H_2$ is the expansion of the frozen-$\Theta$ drift along the frozen periodic error, and by Proposition H.7(b), $\langle H_2\rangle(\Theta)_i = G\nabla_\vartheta(U_{\text{even}} + U_{\text{odd}})[\mu_\Theta](\vartheta_i)$, that is, $\varepsilon^2\langle H_2\rangle(\Theta^*) = b_{\text{frozen}}$. $H$ and $H_2$ are $T_0$-periodic and $C^2$ in $\Theta$ on $\mathcal{K}$, with bounds uniform in $\varepsilon$.

*Regularity of the remainders (used in Steps 4 and 6).* The second-order terms of (S), and of (N) below, are used in sup norm only, for the location (d). The implicit-function and contraction arguments of Steps 4 and 6 need first-order $C^1$ closeness, which holds: $F(e_\Theta(t), \Theta, t)$ is $C^2$ in $\Theta$ with $\partial_\Theta[F(e_\Theta(t), \Theta, t)] = \varepsilon\, \partial_\Theta H(\Theta, t) + O(\varepsilon^2)$ (from $\partial_\Theta e_\Theta = \varepsilon\, \partial_\Theta r_\Theta(x_m) + O(\varepsilon^2)$ and the smooth dependence of $F$ on $(e, \Theta)$ through $\tilde\Phi$); the remainder $R$ is Lipschitz in $\zeta$ with constant $\sup\|\partial_e F\| = O(1)$ and Lipschitz in $\Theta$ with constant $O(\|\zeta\|) = O(\varepsilon^3)$; and the forcing $-\partial_\Theta e_\Theta\cdot\dot\Theta$ of the $\zeta$-equation is Lipschitz in $\Theta$ with constant $O(\varepsilon^2)$ (from $\partial_\Theta^2 e_\Theta = O(\varepsilon)$ and $\partial_\Theta F = O(\varepsilon)$). None of this uses more than $f \in C^2$.

*Step 3 (second-order averaging normal form).* Let $u$ be the zero-mean antiderivative of $H - \langle H\rangle$ and $\Theta = Y + \varepsilon u(Y, t)$. Then $\dot Y = \varepsilon\langle H\rangle(Y) + \varepsilon^2\big[H_2(Y, t) + D_\Theta H(Y, t)u(Y, t) - D_Y u(Y, t)\langle H\rangle(Y)\big] + O(\varepsilon^3)$. Let $u_2$ be the zero-mean antiderivative of the fluctuating part of the bracket and $Y = Y_2 + \varepsilon^2 u_2(Y_2, t)$. Since $\langle D_Y u\rangle = D_Y\langle u\rangle = 0$,

$$\dot Y_2 = \varepsilon\langle H\rangle(Y_2) + \varepsilon^2\Phi_2(Y_2) + O(\varepsilon^3), \qquad \Phi_2 := \langle H_2\rangle + \langle D_\Theta H\cdot u\rangle, \tag{N}$$

the second-order averaged equation of Remark H.7a (Sanders–Verhulst–Murdock [78, §2.9]), now with a remainder that is uniformly $O(\varepsilon^3)$ on $Z_\varepsilon \times \mathcal{K}$ because (S) holds there. The transformation $\Theta \mapsto Y_2$ is a $T_0$-periodic near-identity diffeomorphism from $\mathcal{K}$ onto its image, $O(\varepsilon)$-close to the identity with $O(\varepsilon)$ derivative, and the period mean of $\Theta$ along any curve $Y_2(t) = Y_2^0 + O(\varepsilon^2)$ equals the period mean of $Y_2$ up to $O(\varepsilon^3)$, because $u$ and $u_2$ have zero mean.

*Step 4 (the Poincaré map in the slow coordinates).* Let $P = (P_Y, P_\zeta)$ be the time-$T_0$ map of (CL) from $t = 0$, written in the coordinates $(Y_2, \zeta)$ on $\mathcal{K}\times Z_\varepsilon$. By (N) and Step 2,

$$P_Y(Y_2, \zeta) = Y_2 + \varepsilon T_0\tilde A(Y_2) + O(\varepsilon^3), \qquad \tilde A := \langle H\rangle + \varepsilon\big[\Phi_2 + (T_0/2)D\langle H\rangle\cdot\langle H\rangle\big],$$
$$\|P_\zeta(Y_2, \zeta)\| \le e^{-k_e T_0/2}\|\zeta\| + O(\varepsilon^3),$$

the $\varepsilon^2(T_0^2/2)D\langle H\rangle\langle H\rangle$ term being the second-order term of the time-$T_0$ flow of (N). The Lipschitz constants of the coupling are $\mathrm{Lip}_\zeta P_Y = O(\varepsilon)$ (a perturbation of $\zeta$ acts on $\dot\Theta$ through $\partial_e F = O(1)$

while it decays at rate $k_e/2$, so its integrated effect over a period is $O(\varepsilon)\|\zeta\|$) and $\mathrm{Lip}_Y P_\zeta = O(\varepsilon^3)$ (the forcing of the $\zeta$-equation is $O(\varepsilon^2)$-Lipschitz in $\Theta$ and is integrated against the fast decay). Hence $\zeta \mapsto P_\zeta(Y_2, \zeta)$ has a unique fixed point $\zeta^*(Y_2) = O(\varepsilon^3)$, Lipschitz in $Y_2$ with constant $O(\varepsilon^3)$, and the fixed points of $P$ are the zeros of

$$\tilde{A}_\varepsilon(Y) := \frac{P_Y(Y, \zeta^*(Y)) - Y}{\varepsilon T_0} = \tilde{A}(Y) + O(\varepsilon^2).$$

By the regularity paragraph, $D_Y P_Y = I + \varepsilon T_0 D\langle H\rangle(Y) + O(\varepsilon^2)$ (the variational equation of the slow flow over one period, whose coefficient is $\varepsilon\,\partial_\Theta H + O(\varepsilon^2)$), and the $\zeta^*$-dependence contributes $\mathrm{Lip}_\zeta P_Y \cdot \mathrm{Lip}_Y \zeta^* = O(\varepsilon^4)$, so $\tilde{A}_\varepsilon$ is Lipschitz with $\mathrm{Lip}(\tilde{A}_\varepsilon - \langle H\rangle) = O(\varepsilon)$ on $\mathcal{K}$. Since $D\langle H\rangle(\Theta^*) = -G_m \nabla^2 J_m(\Theta^*)$ is invertible, $Y \mapsto Y - D\langle H\rangle(\Theta^*)^{-1}\tilde{A}_\varepsilon(Y)$ maps a ball of $\varepsilon$-independent radius $\rho_0$ around $\Theta^*$ into itself and is a contraction there for $\varepsilon$ small (the Lipschitz form of the implicit function theorem), which gives a unique zero $Y_2^* = \Theta^* + O(\varepsilon)$, hence a unique fixed point of $P$ with $\Theta$-component within $O(\varepsilon)$ of $\Theta^*$, that is, a unique $T_0$-periodic solution in the stated set; the bound on $x_p - x_m$ is Step 1. Expanding $\tilde{A}_\varepsilon(Y_2^*) = 0$ to first order, $D\langle H\rangle(\Theta^*)(Y_2^* - \Theta^*) + \varepsilon\Phi_2(\Theta^*) = O(\varepsilon^2)$ (the term $(T_0/2)D\langle H\rangle\langle H\rangle(Y_2^*)$ is $O(\varepsilon)$ because $\langle H\rangle(Y_2^*) = O(\varepsilon)$), so

$$Y_2^* = \Theta^* - \varepsilon D\langle H\rangle(\Theta^*)^{-1}\Phi_2(\Theta^*) + O(\varepsilon^2) = \Theta^* - J^{-1}(b_{\mathrm{frozen}} + b_{\mathrm{osc}}) + O(\varepsilon^2),$$

using $J = \varepsilon D\langle H\rangle(\Theta^*)$ and $\varepsilon^2\Phi_2(\Theta^*) = b_{\mathrm{frozen}} + b_{\mathrm{osc}}$. Along the periodic orbit $Y_2(t) = Y_2^* + O(\varepsilon^2)$ (since $\langle H\rangle(Y_2^*) = O(\varepsilon)$), so by Step 3 the period mean of $\Theta_p$ is $Y_2^* + O(\varepsilon^2)$: this is (d), and $\Theta_p(t) - \bar{\Theta}_p = \varepsilon u(\Theta^*, t) + O(\varepsilon^2)$ is (a).

*Step 5 (the monodromy matrix).* Linearise (CL) along the periodic orbit: with $\delta e$ and $\delta\Theta$,

$$\delta\dot{e} = (\partial_x r - k_e I)\delta e + \partial_\Theta r \cdot \delta\Theta, \qquad \delta\dot{\Theta} = \partial_e F \cdot \delta e + \partial_\Theta F \cdot \delta\Theta,$$

all coefficients $T_0$-periodic and evaluated along the orbit; $\partial_x r$, $\partial_\Theta r$, $\partial_e F$ are $O(1)$ and $\partial_\Theta F = O(\varepsilon)$ ($F$ is linear in $e$ with coefficients smooth in $\Theta$, plus the leak $-\varepsilon\sigma_0(\vartheta_i - \vartheta^0)$, so $\partial_\Theta F = O(\|e\| + \varepsilon)$, and $e = O(\varepsilon)$ on the orbit). Let $S(t)$ be a $T_0$-periodic solution of the Riccati equation

$$\dot{S} = (\partial_x r - k_e I)S + \partial_\Theta r - S\,\partial_e F\,S - S\,\partial_\Theta F.$$

For $k_e$ large it exists, is unique with $\|S\| \le C/k_e$, and depends smoothly on the data (contraction of the periodic solution operator of the dominant part $\dot{S} = -k_e S + g$; cf. Proposition H.7(a)). In the coordinates $\eta := \delta e - S\delta\Theta$ the linearisation is block triangular:

$$\dot{\eta} = \big[(\partial_x r - k_e I) - S\,\partial_e F\big]\eta, \qquad \delta\dot{\Theta} = \partial_e F \cdot \eta + \big[\partial_e F\,S + \partial_\Theta F\big]\delta\Theta,$$

so the monodromy matrix is similar to a block-triangular matrix whose diagonal blocks are the monodromies of the two diagonal equations. The $\eta$-block has norm at most $e^{-(k_e - L)T_0}$ with $L := L_r + \sup\|S\,\partial_e F\| = O(1)$ (Grönwall): these are the $n$ error-channel multipliers. The $\delta\Theta$-block is the linearisation along the orbit of the slow equation (S) in $\Theta$-coordinates, $\varepsilon D_\Theta H(\Theta_p(t), t) + O(\varepsilon^2) = \varepsilon D_\Theta H(\Theta^*, t) + O(\varepsilon^2)$, whose monodromy is $I + \varepsilon T_0 D\langle H\rangle(\Theta^*) + O(\varepsilon^2) = I - \varepsilon T_0 G_m \nabla^2 J_m(\Theta^*) + O(\varepsilon^2)$. Its eigenvalues are $1 - \varepsilon T_0\nu_j + O(\varepsilon^2) = \exp(-\varepsilon T_0\nu_j + O(\varepsilon^2))$, those of $G_m\nabla^2 J_m(\Theta^*)$ being the $\nu_j$ by similarity with $G_m^{1/2}\nabla^2 J_m(\Theta^*)G_m^{1/2}$; the $O(\varepsilon^2)$ perturbation moves them by $O(\varepsilon^2)$ because $I - \varepsilon T_0 G_m \nabla^2 J_m(\Theta^*)$ is diagonalisable (similar to a symmetric matrix through $G_m^{1/2}$), so the Bauer–Fike theorem applies with a condition number independent of $\varepsilon$ — this covers the repeated eigenvalues of a collapsed configuration. This is (b).

*Step 6 (the basin).* Let $\nabla^2 J_m(\Theta^*) \succ 0$. In the norm $\|v\|_{G^{-1}} := \|G_m^{-1/2} v\|$, the matrix $I - \varepsilon T_0 G_m \nabla^2 J_m(\Theta^*)$ is similar to a symmetric contraction with norm exactly $1 - \varepsilon T_0\lambda_*$ once $\varepsilon T_0\nu_{mp} < 1$. By Step 4, $D_Y P_Y = I - \varepsilon T_0 G_m \nabla^2 J_m(Y) + O(\varepsilon^2)$ on $\mathcal{K}$, and $\nabla^2 J_m$ is Lipschitz, so

on the ball $\{\|Y_2 - Y_2^*\|_{G^{-1}} \le \rho_0\}$ the Lipschitz constant of $P_Y$ in $Y_2$ is at most $1 - \varepsilon T_0 \lambda_*(1-\kappa)$ with $\kappa = O(\rho_0 + \varepsilon)$; the coupling constants are $\mathrm{Lip}_\zeta P_Y = O(\varepsilon)$ and $\mathrm{Lip}_Y P_\zeta = O(\varepsilon^3)$, and $P_\zeta$ contracts $\zeta$ by $e^{-k_e T_0/2}$ (Step 4). The $2 \times 2$ matrix of Lipschitz constants

$$\begin{pmatrix} 1 - \varepsilon T_0 \lambda_*(1-\kappa) & C\varepsilon \\ C\varepsilon^3 & e^{-k_e T_0/2} \end{pmatrix}$$

has spectral radius $1 - \varepsilon T_0 \lambda_*(1-\kappa')$ with $\kappa' = O(\rho_0 + \varepsilon)$, so on $\{\|Y_2 - Y_2^*\|_{G^{-1}} \le \rho_0\} \times Z_\varepsilon$, with $\rho_0$ small but independent of $\varepsilon$, $P$ is a contraction in an adapted product norm with that factor: its iterates converge to the fixed point geometrically, uniformly in $\varepsilon$. Solutions started in the tube with $\|e(0) - e_p(0)\| \le 1$ enter $Z_\varepsilon$ within $O(\varepsilon \log(1/\varepsilon))$, during which $\Theta$ moves by $O(\varepsilon)\|e(0) - e_p(0)\|$ (Step 1), which is absorbed by halving $\rho_0$. Between the sampling times the solutions stay within a factor $C$ of each other (Grönwall on one period, with the fast variable absorbing $\|e(0) - e_p(0)\|$ into an $O(\varepsilon)\|e(0) - e_p(0)\|$ displacement of $\Theta$, Step 1). Transporting back through the near-identity transformations gives (c). □

**Numerics** (`exp_periodic_orbit.py` → `results/periodic_orbit/table.md`; `tests/test_periodic.py`). Two instances, $k_e = 10$, 20, 40, 80, 160. For each $k_e$ the periodic orbit of (CL) is found by Newton shooting on the Poincaré map (DOP853 at relative tolerance $10^{-11}$; residual below $10^{-11}$ in every case), its monodromy matrix is the finite-difference Jacobian of the map, and its period mean is compared with $\Theta^*$ and with the first-order prediction $\Theta^* - J^{-1}(b_{\mathrm{frozen}} + b_{\mathrm{osc}})$ of (d). Distances and cosines are in the labelled $G^{-1}$ metric.

- **Instance A** ($n = 2$, $m = 3$, the plant and elliptic reference of Remark H.7a, $\sigma_0 = 0.5$, $G = \mathrm{diag}(4, 4, 1, 1, 1)$). The minimiser of $J_m$ is a collapsed configuration (kernel spread $4 \cdot 10^{-14}$) with $\lambda_{\min}(\nabla^2 J_m) = 0.0148$: a non-degenerate minimum with permutation symmetry.

| $k_e$ | $\|\bar\Theta_p - \Theta^*\|$ | full prediction: size ratio, cosine, vector error | frozen-measure part only: vector error | parameter Floquet exponents against $(1/k_e)\nu_j$: max, median relative error |
|---|---|---|---|---|
| 10 | $4.09 \cdot 10^{-2}$ | 1.149, 0.9972, 0.169 | 0.398 | $8.3 \cdot 10^{-2}$, $1.2 \cdot 10^{-2}$ |
| 20 | $1.91 \cdot 10^{-2}$ | 1.074, 0.9993, 0.083 | 0.360 | $7.3 \cdot 10^{-2}$, $7.3 \cdot 10^{-3}$ |
| 40 | $9.22 \cdot 10^{-3}$ | 1.036, 0.9998, 0.041 | 0.348 | $4.5 \cdot 10^{-2}$, $4.0 \cdot 10^{-3}$ |
| 80 | $4.53 \cdot 10^{-3}$ | 1.018, 1.0000, 0.020 | 0.343 | $2.5 \cdot 10^{-2}$, $2.1 \cdot 10^{-3}$ |
| 160 | $2.24 \cdot 10^{-3}$ | 1.009, 1.0000, 0.010 | 0.342 | $1.3 \cdot 10^{-2}$, $1.1 \cdot 10^{-3}$ |

  The offset scales like $k_e^{-1.05}$ (fitted exponent; (a) predicts $-1$), the vector error of the full prediction halves with every doubling of $k_e$ (the $O(k_e^{-2})$ remainder of (d) against an $O(1/k_e)$ offset), and the frozen-measure part alone has a 35% vector error at every $k_e$: the oscillation term $b_{\mathrm{osc}}$ of Remark H.7a is an $O(1)$ fraction of the first-order location. The median relative error of the Floquet exponents is $O(1/k_e)$, as (b) states; the largest relative error sits on the $(m-1)$-fold degenerate separation pair with the smallest exponent (0.0018 at $k_e = 20$), where an $O(k_e^{-2})$ absolute error is a large fraction of a small exponent. The two error-channel multipliers are below the resolution of the finite-difference monodromy (exponents of order $k_e$); their product is recovered from Liouville's identity $\sum \log|\lambda| = \int_0^{T_0} \mathrm{tr}\, \partial F/\partial z \, \mathrm{d}t$ along the orbit, which gives a mean error-channel exponent of $k_e - 0.16$, $k_e - 0.11$, $k_e - 0.086$, $k_e - 0.074$, $k_e - 0.068$ at $k_e = 10 \ldots 160$ (instance B: $k_e + 0.05 \ldots k_e + 0.09$): the multipliers are $e^{-(k_e - L)T_0}$ with $|L| < 0.2$, as (b) states.

- **Instance B** ($n = 1$, $m = 2$, a two-bump plant $f(x) = 1.6e^{-(x-1)^2/0.4} - 1.2e^{-(x+1)^2/0.4} + 0.1x$ on $x_m = 2\sin t$, $\sigma_0 = 0.05$, $G = \mathrm{diag}(2, 1, 1)$). The minimiser has two distinct kernels

of opposite sign (spread 2.1, $\lambda_{\min}(\nabla^2 J_m) = 0.066$). Here $b_{\mathrm{osc}} = 0$ exactly: for a one-dimensional sinusoidal sweep the reflection $t \mapsto \pi - t$ maps $x_m$ to itself and $u$ to $-u$, so $\langle D_\Theta H \cdot u \rangle = 0$, and the full and frozen predictions coincide. The reference is fast on the kernel scale ($\sup\|\dot{x}_m\|/s$ is $2/s$: between 1.3 and 10 over the admissible widths, about 2.6 at the minimiser's width; the residual bandwidth is of the same order), so $k_e = 10$–$20$ is outside the averaging regime:

| $k_e$ | $\|\bar{\Theta}_p - \Theta^*\|$ | prediction: size ratio, cosine, vector error | Floquet: max, median relative error |
|---|---|---|---|
| 10 | $6.44 \cdot 10^{-2}$ | 2.818, 0.880, 1.995 | $9.5 \cdot 10^{-2}$, $4.0 \cdot 10^{-2}$ |
| 20 | $2.22 \cdot 10^{-2}$ | 1.942, 0.929, 1.078 | $3.2 \cdot 10^{-2}$, $1.4 \cdot 10^{-2}$ |
| 40 | $8.37 \cdot 10^{-3}$ | 1.464, 0.970, 0.551 | $1.2 \cdot 10^{-2}$, $5.4 \cdot 10^{-3}$ |
| 80 | $3.51 \cdot 10^{-3}$ | 1.227, 0.990, 0.277 | $4.6 \cdot 10^{-3}$, $2.2 \cdot 10^{-3}$ |
| 160 | $1.59 \cdot 10^{-3}$ | 1.111, 0.9969, 0.139 | $2.0 \cdot 10^{-3}$, $9.5 \cdot 10^{-4}$ |

The vector error and the Floquet error halve with each doubling of $k_e$, as the theorem's $O(k_e^{-2})$ remainders predict, but the constants are large: the asymptotic regime starts around $k_e \approx 80$ here. This instance shows what the theorem does not claim: the $O(1/k_e)$ location is sharp only once $k_e$ exceeds the effective bandwidth $\omega_{\mathrm{eff}}$ of Proposition G's (H2a′) (Section 19).

Checked by: `tests/test_periodic.py`: $\nabla J_m$ against finite differences ($10^{-7}$) and the non-degenerate two-kernel minimum of instance B; at $k_e = 20$ and 40 the shooting residual is below $10^{-10}$, $\|\bar{\Theta}_p - \Theta^*\|$ and $\max\|e\|$ fall by more than 35% from 20 to 40, the vector error of the first-order location falls by more than 40% and is below 0.6 at $k_e = 40$, and the maximal relative Floquet error falls by more than 40% and is below 2% at $k_e = 40$.

**Remark J.1** (what Theorem J says about the program of Section 26)**.**

- It is the deterministic, finite-$m$ first step of branch 1: near every hyperbolic critical point of the $m$-particle regularised loss $k_e \cdot \mathcal{G}$ at $\tau = 0$ the real closed loop has exactly one stationary state (a periodic orbit), an $O(1/k_e)$-perturbation of it with the same stability type, and the convergence rate is the one of the averaged gradient flow in slow time. (The theorem does not say that every periodic orbit of (CL) arises this way.) The two missing pieces of Conjecture H.6 remain: dither ($\tau > 0$, where the stationary state is a measure) and uniformity in $m$.
- It makes the stationary (period-mean) prediction of Remark H.7a a theorem for this system: the first-order location of the orbit, hence the orientation difference of the period-mean kernels, is $\Theta^* - J^{-1}(b_{\mathrm{frozen}} + b_{\mathrm{osc}})$ with an $O(k_e^{-2})$ error. When the leading offset is itself small, the $O(k_e^{-2})$ remainder can dominate its *direction* at moderate $k_e$: a collapsed saddle instance examined in the round-5 review (`review/REVIEW_round5_J.md`) has a first-order location nearly orthogonal to the true offset at $k_e = 20$ and aligned with it at $k_e = 80$, while the sizes obey the theorem's orders throughout.
- The leak is not needed for the structure, only for the hyperbolicity of $\Theta^*$. At a collapsed critical point the separation Hessian is $\sigma_0 G^{-1} - R_2$ (Proposition K, Section 23); for $n = 1$ and $\sigma_0 = 0$ the amplitude row of $R_2$ vanishes at a leak-free critical point ($\langle r\varphi \rangle = \partial_c \langle r\varphi \rangle = \partial_s \langle r\varphi \rangle = 0$ there), so the $m - 1$ amplitude-redistribution directions are exactly flat and the theorem does not apply, while for $n \geq 2$ only the amplitude block of $R_2$ vanishes and a collapsed critical point is generically non-degenerate even without a leak. Where $\nabla^2 J_m$ is singular the theorem does not apply.

- At a collapsed critical point (all $\vartheta_i^*$ equal) the permutation symmetry makes the separation modes $(m-1)$-fold degenerate; the Floquet exponents come in such multiplets (`results/periodic_orbit/table.md`); Proposition K (Section 23) works out this case.

## 23 Proposition K: collapsed configurations, the critical leak, and the orientation of the orbit as a design input

Branch 2 of Section 26 asks whether the circulation of the reference orbit can be designed. Theorem J (Section 22) makes the question precise for the collapsed configuration, the trap that Theorem H's dither was introduced to escape: its stability is decided by the eigenvalues of a Hessian that the orientation shifts at order $k_e^{-2}$.

**Setting.** As in Theorem J. A configuration is *collapsed* if $\vartheta_i = \bar\vartheta$ for all $i$. Then $\hat f_\Theta = \tilde\Phi(\cdot\,;\bar\vartheta)$, so $\Theta$ is a critical point of $J_m$ if and only if $\bar\vartheta$ is a critical point of the single-kernel objective $J_1(\vartheta) := \frac12 \int \|f - \tilde\Phi(\cdot\,;\vartheta)\|^2 \,\mathrm{d}\rho + \frac{\sigma_0}{2}\|\vartheta - \vartheta^0\|^2_{G^{-1}}$. Write $\bar r := f - \tilde\Phi(\cdot\,;\bar\vartheta)$ and

$$M_2 := \int \nabla_\vartheta \tilde\Phi(\cdot\,;\bar\vartheta)^\top \nabla_\vartheta \tilde\Phi(\cdot\,;\bar\vartheta)\,\mathrm{d}\rho \succeq 0,$$

$$R_2 := \int \sum_k \bar r_k \nabla^2_\vartheta \tilde\Phi_k(\cdot\,;\bar\vartheta)\,\mathrm{d}\rho = \nabla^2_\vartheta \big\langle \bar r^\top \tilde\Phi(\cdot\,;\vartheta)\big\rangle_\rho\Big|_{\bar\vartheta} \quad \text{(residual frozen)}.$$

$R_2$ is the Hessian of $-V_L[\bar\mu](\vartheta)$ at the atom: the second-order cost of moving a small amount of mass off the atom at frozen residual.

**Proposition K** (collapsed configurations, the critical leak, and the orientation as a design input)**.** *Let $\Theta^*$ be a collapsed critical point of $J_m$ with atom $\bar\vartheta$.*

(i) Symmetry splitting. $\nabla^2 J_m(\Theta^*) = I_m \otimes (\sigma_0 G^{-1} - R_2) + \frac1m(\mathbf{1}\mathbf{1}^\top)\otimes M_2$. *On the separation subspace $\{(u_1,\dots,u_m) : \sum_i u_i = 0\}$, of dimension $(m-1)p$, it acts as $\sigma_0 G^{-1} - R_2$; on the collective subspace $\{u_i$ all equal$\}$ as $\nabla^2 J_1(\bar\vartheta) = \sigma_0 G^{-1} - R_2 + M_2$. The same splitting holds for the Jacobian of any permutation-equivariant particle drift at a collapsed configuration, in particular for the second-order averaged drift of Remark H.7a (Section 21): it has the form $I_m \otimes D + \frac1m(\mathbf{1}\mathbf{1}^\top)\otimes E$.*

(ii) Separation stability at leading order. *$\Theta^*$ is a non-degenerate local minimum of $J_m$ iff $\nabla^2 J_1(\bar\vartheta) \succ 0$ and $\sigma_0 G^{-1} \succ R_2$, that is, iff $\bar\vartheta$ is a non-degenerate minimiser of $J_1$ and $\sigma_0 > \sigma_{\mathrm{crit}}(\bar\vartheta) := \lambda_{\max}(G^{1/2} R_2 G^{1/2})$. The separation Floquet exponents of the collapsed periodic orbit of* (CL) *are then, by Theorem J(b), $(1/k_e)\lambda_j(\sigma_0 I - G^{1/2} R_2 G^{1/2}) + O(k_e^{-2})$, each with multiplicity $m-1$, and they do not depend on $m$. The condition $\sigma_0 G^{-1} \succ R_2$ is the statement that the atom is a strict local minimiser of the first-variation potential $V_L[\bar\mu] + \frac{\sigma_0}{2}\|\cdot\|^2_{G^{-1}}$ on its support, the local form of the Chizat–Bach optimality condition for measures.*

(iii) The orientation-odd shift. *Let $\mathcal{O}$ be traversed in either direction (the same $\rho$, hence the same $\Theta^*$ and the same leading exponents). Write the second-order normal form of Theorem J's proof as*

$$\dot\Theta = -\frac{1}{k_e} G_m \nabla J_m(\Theta) + b(\Theta) + O(k_e^{-3}), \qquad b := b_{\mathrm{frozen}} + b_{\mathrm{osc}} = O(k_e^{-2}), \tag{N}$$

*with $b_{\mathrm{frozen}} = \frac{1}{k_e^2} G_m \nabla(U_{\mathrm{even}} + U_{\mathrm{odd}})$ (Proposition H.7) and $b_{\mathrm{osc}}$ of Remark H.7a.* (N) *has a unique equilibrium $\bar\Theta_2$ near $\Theta^*$, which is collapsed, with $\bar\Theta_2 - \Theta^* = -J_1^{-1} b(\Theta^*) + O(k_e^{-2})$ for $J_1 := -\frac{1}{k_e} G_m \nabla^2 J_m(\Theta^*)$ (the first-order location of Theorem J(d)); let $J_2$ be the Jacobian*

*of* (N) *at* $\bar\Theta_2$*. The parameter Floquet exponents of the collapsed orbit are the eigenvalues of* $-J_2$ *up to* $O(k_e^{-3})$*, and*

$$J_2 = J_1 + \partial b(\Theta^*) + \partial J_1(\Theta^*)[\bar\Theta_2 - \Theta^*] + O(k_e^{-3}),$$

*where the third term — the third derivative of* $J_m$ *contracted with the first-order offset — is* $O(k_e^{-2})$*, of the same order as* $\partial b$*, and must be kept: the Jacobian of* (N) *at* $\Theta^*$ *is not the right linearisation. Under orbit reversal* $\rho$*,* $\Theta^*$*,* $J_1$ *and* $U_{\mathrm{even}}$ *are unchanged while* $U_{\mathrm{odd}}$ *and* $b_{\mathrm{osc}}$ *change sign (Proposition H.7(d), Remark H.7a), so with* $b_{\mathrm{even}} := \frac{1}{k_e^2} G_m \nabla U_{\mathrm{even}}$ *and* $b_{\mathrm{odd}} := \frac{1}{k_e^2} G_m \nabla U_{\mathrm{odd}} + b_{\mathrm{osc}}$*,*

$$J_2 = J_1 + E \pm O,$$
$$E := \partial b_{\mathrm{even}}(\Theta^*) - \partial J_1(\Theta^*)\big[J_1^{-1} b_{\mathrm{even}}(\Theta^*)\big], \qquad O := \partial b_{\mathrm{odd}}(\Theta^*) - \partial J_1(\Theta^*)\big[J_1^{-1} b_{\mathrm{odd}}(\Theta^*)\big],$$

*for the two orientations.* $J_1$*,* $E$ *and* $O$ *are permutation-equivariant and preserve the separation subspace; by* (i) *their restrictions to one separation block are* $p \times p$ *matrices, written* $J_1|_{\mathrm{sep}} = -\frac{1}{k_e}(\sigma_0 I - G R_2)$*,* $D_{\mathrm{even}}$*,* $D_{\mathrm{odd}}$*. The separation Floquet exponents of the two orientations differ by*

$$\lambda_j^{(+)} - \lambda_j^{(-)} = -2\,\frac{w_j^\top D_{\mathrm{odd}} v_j}{w_j^\top v_j} + O(k_e^{-3})$$

(*simple eigenvalue* $\lambda_j$ *of* $-J_1|_{\mathrm{sep}}$*, right and left eigenvectors* $v_j$*,* $w_j$*), an* $O(k_e^{-2})$ *quantity that scales like* $k_e^{-2}$ *at fixed* $(G, \sigma_0)$*, while the exponents themselves are* $O(1/k_e)$*. The difference is a property of the excitation's orientation alone: the occupation measure, the averaged system* ($\mathfrak{A}$) *and all of Theorems B–H are blind to it.*

(iv) The critical leak as a design point. *At* $\sigma_0 = \sigma_{\mathrm{crit}}(\bar\vartheta)$ *the softest separation exponent vanishes at leading order and equals the* $k_e^{-2}$ *value* $-(w^\top D_{\mathrm{even}} v)/(w^\top v) \mp (w^\top D_{\mathrm{odd}} v)/(w^\top v) + O(k_e^{-3})$*. If* $|w^\top D_{\mathrm{odd}} v| > |w^\top D_{\mathrm{even}} v|$*, the collapsed configuration is stable for one orientation and unstable for the other: the direction in which the reference is traversed decides whether the kernels separate. More generally, for* $\sigma_0$ *in a window of width* $O(1/k_e)$ *around* $\sigma_{\mathrm{crit}}$ *— the window where the leading exponent* $(\sigma_0 - \sigma_{\mathrm{crit}})(1 + o(1))/k_e$ *lies between the two* $k_e^{-2}$ *values — the orientation decides the sign, and the centre of the window,* $\sigma_0 = \sigma_{\mathrm{crit}} + k_e (w^\top D_{\mathrm{even}} v)/(w^\top v) + O(k_e^{-1})$*, can be computed from the averaged data alone.*

**Status.** (i) and (ii) are proved (linear algebra at the collapsed point and Theorem J(b)). (iii) and (iv) are proved given the second-order normal form (N) with an $O(k_e^{-3})$ remainder, which is Theorem J's proof carried one order further (periodic averaging to second order, Sanders–Verhulst–Murdock [78, Ch. 2–3]; the first-order step is written out in the proof of Theorem J, the second-order step is Remark H.7a's derivation and is not written out here). The orientation-odd shift, its $k_e^{-2}$ scaling, the prediction and the design window are checked numerically in one instance (`exp_orientation_stability.py`).

*Proof. (i)* $\partial^2 J_m / \partial\vartheta_i \partial\vartheta_j = \frac{1}{m}\int \nabla\tilde\Phi_i^\top \nabla\tilde\Phi_j \,\mathrm{d}\rho - \delta_{ij}\int \sum_k r_k \nabla^2 \tilde\Phi_{i,k}\,\mathrm{d}\rho + \delta_{ij}\sigma_0 G^{-1}$ (the first term from the quadratic dependence of $\frac{1}{2}\int \|r\|^2\,\mathrm{d}\rho$ on the $m$ features with weights $1/m$, the second from the $\vartheta_i$-curvature of the residual term: with $r = f - \frac{1}{m}\sum \tilde\Phi_j$, $\partial_{\vartheta_i}(m\bar L_m) = -\int r^\top \nabla\tilde\Phi_i\,\mathrm{d}\rho$ and $\partial_{\vartheta_j}\partial_{\vartheta_i}(m\bar L_m) = \frac{1}{m}\int \nabla\tilde\Phi_j^\top\nabla\tilde\Phi_i\,\mathrm{d}\rho - \delta_{ij}\int\sum_k r_k\nabla^2\tilde\Phi_{i,k}\,\mathrm{d}\rho$). At a collapsed point all blocks coincide, giving the Kronecker form; the two subspaces are the isotypic components of the permutation action, on which $\frac{1}{m}(\mathbf{1}\mathbf{1}^\top)$ acts as 0 and 1 respectively. The second-order averaged drift is permutation-equivariant (each particle moves by a function of its own position and of the measure), so its Jacobian at a collapsed point has the same form with $D$ the self-derivative and $E$ the derivative through the measure.

*(ii)* Positive definiteness of a block-diagonal form; the exponents are Theorem J(b) applied to the separation block, whose eigenvalues of $G_m^{1/2}\nabla^2 J_m G_m^{1/2}$ restricted to the separation subspace are those of $\sigma_0 I - G^{1/2}R_2G^{1/2}$, each repeated $m-1$ times. For the last sentence, the first variation of $\mathcal{G} := \bar{L} + \frac{\sigma_0}{2}\int\|\cdot-\vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu$ at $\bar{\mu}=\delta_{\bar{\vartheta}}$ is $V_L[\bar{\mu}](\vartheta) + \frac{\sigma_0}{2}\|\vartheta-\vartheta^0\|^2_{G^{-1}} = -\langle \bar{r}^\top\tilde{\Phi}(\cdot\,;\vartheta)\rangle_\rho + \frac{\sigma_0}{2}\|\vartheta-\vartheta^0\|^2_{G^{-1}}$, whose Hessian at $\bar{\vartheta}$ is $-R_2+\sigma_0G^{-1}$.

*(iii)* The proof of Theorem J carries one order further: the near-identity change of variables of its Step 3 removes the oscillating terms through order $k_e^{-2}$, which turns (CL) into (N) with an $O(k_e^{-3})$ remainder (this is the normal form of Remark H.7a's derivation, with $b_{\mathrm{osc}}$ the term that the frozen-measure expansion misses). (N) is a regular perturbation of the averaged system near the hyperbolic point $\Theta^*$, so it has a unique equilibrium $\bar{\Theta}_2$ within $O(1/k_e)$ of $\Theta^*$, obtained by the implicit function theorem; by the first-order expansion, $\bar{\Theta}_2-\Theta^* = -J_1^{-1}b(\Theta^*)+O(k_e^{-2})$ ($J_1^{-1}=O(k_e)$, $b=O(k_e^{-2})$). (N) is permutation-equivariant and $\Theta^*$ is collapsed, so the equivariant Newton iterates stay collapsed and $\bar{\Theta}_2$ is collapsed. Theorem J(b) applied to (N) (same proof, remainder $O(k_e^{-3})$) gives the Floquet exponents as the eigenvalues of $-J_2+O(k_e^{-3})$. Expanding the Jacobian of (N) about $\Theta^*$: $J_2 = J_1 + \partial b(\Theta^*) + \partial J_1(\Theta^*)[\bar{\Theta}_2-\Theta^*] + O(\|\bar{\Theta}_2-\Theta^*\|^2/k_e) + O(k_e^{-2}\|\bar{\Theta}_2-\Theta^*\|)$, and both error terms are $O(k_e^{-3})$; substituting the offset gives $E$ and $O$, which are the even and odd parts under reversal because $\partial J_1$ is even and $b_{\mathrm{even}}$, $b_{\mathrm{odd}}$ have the parities named. All three operators are permutation-equivariant, hence preserve the separation subspace, and by the Kronecker structure of (i) each acts on it as $I_{m-1}\otimes$ (a $p\times p$ block). First-order eigenvalue perturbation of the simple eigenvalue $\lambda_j$ of $-J_1|_{\mathrm{sep}}$ gives

$$\lambda_j^{(\pm)} = \lambda_j - \frac{w_j^\top D_{\mathrm{even}} v_j}{w_j^\top v_j} \mp \frac{w_j^\top D_{\mathrm{odd}} v_j}{w_j^\top v_j} + O(k_e^{-3}),$$

and the difference as stated. (For the $(m-1)$-fold degenerate separation modes the degeneracy is the trivial one of the Kronecker structure, which the restriction to one $p$-dimensional block removes; a further degeneracy within the block would call for degenerate perturbation theory.)

*(iv)* is (iii) at a vanishing leading eigenvalue; the window follows because the leading exponent is $(\sigma_0-\sigma_{\mathrm{crit}})/k_e\cdot(1+o(1))$ near $\sigma_{\mathrm{crit}}$ ($\bar{\vartheta}$, $R_2$ and the eigenvectors depend smoothly on $\sigma_0$) while the shifts are $O(k_e^{-2})$; its centre is where the leading exponent cancels the even shift. □

**Remark K.1** (reading)**.** Three things.

- The collapsed configuration's stability is a single-kernel question at leading order (the Hessian $\sigma_0G^{-1}-R_2$ does not depend on $m$), which explains why the $m=6$ and $m=32$ experiments of Section 20 collapsed alike: a leak above $\sigma_{\mathrm{crit}}$ traps every network size.
- Dither (Theorem H) and the orientation (this proposition) are two different ways to leave the trap: the first adds entropy, the second tilts the Hessian by $O(k_e^{-2})$. The second is free but small; it matters only near $\sigma_{\mathrm{crit}}$, that is, when the leak is tuned to the edge of trapping, or for nearly flat separation modes such as the weak saddle of Remark H.2.
- The sign of the shift depends on the data $(f,\mathcal{O},\bar{\vartheta})$ through $D_{\mathrm{odd}}$ and has no general sign: the design rule is to compute $D_{\mathrm{odd}}$ (from the averaged data: $U_{\mathrm{odd}}$, $b_{\mathrm{osc}}$ and the third derivative of $J_m$) and choose the orientation with the smaller softest separation exponent (or the negative one); the window of (iv) is where the choice matters.

**Numerics** (`exp_orientation_stability.py` → `results/orientation_stability/table.md`). Instance A of Theorem J's numerics (Section 22; $n=2$, $m=3$, the elliptic reference traversed in both directions, $G=\mathrm{diag}(4,4,1,1,1)$). For each orientation the collapsed periodic orbit is found by Newton shooting (residual $\le 4\cdot10^{-13}$), the parameter block of its monodromy matrix is restricted to the separation and collective subspaces, and the exponents are compared

with the leading order $(\sigma_0 I - G^{1/2} R_2 G^{1/2})/k_e$ and with the second-order prediction, the eigenvalues of $-J_2$ at the equilibrium $\bar{\Theta}_2$ of (N) (Newton from $\Theta^*$; every $\bar{\Theta}_2$ found is collapsed to $10^{-8}$). The second-order equilibrium is $\|\bar{\Theta}_2 - \Theta^*\|_{G^{-1}} = 3.7 \cdot 10^{-2}$, $1.8 \cdot 10^{-2}$, $9.0 \cdot 10^{-3}$, $4.5 \cdot 10^{-3}$ away from $\Theta^*$ at $k_e = 10$, 20, 40, 80: the $O(1/k_e)$ offset of Theorem J(d).

- *The shift and its prediction* ($\sigma_0 = 0.5$, a stable collapse). The softest separation exponents of the two orientations differ by $9.0 \cdot 10^{-4}$, $1.9 \cdot 10^{-4}$, $4.1 \cdot 10^{-5}$, $9.7 \cdot 10^{-6}$ at $k_e = 10$, 20, 40, 80: fitted exponent $-2.18$ against $k_e$ ((iii): $-2$). Scaled by $k_e^2$, the measured corrections to the leading order are $-0.032$, $-0.057$, $-0.070$, $-0.077$ for $(+)$ and $-0.123$, $-0.132$, $-0.136$, $-0.139$ for $(-)$, approaching the predicted $-0.083$ and $-0.143$ with a gap that halves at every doubling of $k_e$ (the $O(k_e^{-3})$ remainder); the predicted difference at $\bar{\Theta}_2$ — $7.8 \cdot 10^{-4}$, $1.7 \cdot 10^{-4}$, $3.9 \cdot 10^{-5}$, $9.4 \cdot 10^{-6}$ — is within 14%, 9%, 5%, 3% of the measured one. The prediction with the Jacobian taken at $\Theta^*$ instead of $\bar{\Theta}_2$ (the third-derivative term omitted) reads $9.0 \cdot 10^{-4}$, $2.1 \cdot 10^{-4}$, $5.0 \cdot 10^{-5}$, $1.2 \cdot 10^{-5}$: it agrees at $k_e = 10$ by accident and its $k_e^2$-scaled value converges to 0.078 while the measured one converges to 0.060, the 30% error that the numerics exposed (`review/REVIEW_LOG.md`, round 5, K-1). The separation exponents come in exact pairs (multiplicity $m - 1 = 2$) and the collective exponents differ between the orientations by 2% at $k_e = 10$ and 0.2% at $k_e = 80$, as (i)–(iii) describe.
- *At the critical leak* ($\sigma_0 = \sigma_{\mathrm{crit}} = 0.4606$). The leading exponent vanishes and the $k_e^{-2}$ terms decide: the softest separation exponent is $-4.1 \cdot 10^{-4}$ for $(+)$ and $-1.27 \cdot 10^{-3}$ for $(-)$ at $k_e = 10$, $-4.8 \cdot 10^{-5}$ and $-8.7 \cdot 10^{-5}$ at $k_e = 40$ — unstable for both orientations in this instance, the even part dominating the odd one by a factor 2–3; the predictions at $\bar{\Theta}_2$ are within 14% and 6% of the measured values at $k_e = 40$.
- *The orientation decides* ($\sigma_0$ designed from the averaged data). Solving $\mathrm{pred}_+(\sigma_0) + \mathrm{pred}_-(\sigma_0) = 0$ for the predicted softest exponents gives $\sigma_0 - \sigma_{\mathrm{crit}} = 0.0125$ at $k_e = 10$ and 0.0061 at $k_e = 20$, the centre of the window of (iv); there the measured softest separation exponents are $+8.5 \cdot 10^{-4}$ and $-2.5 \cdot 10^{-5}$ ($k_e = 10$), $+1.4 \cdot 10^{-4}$ and $-3.7 \cdot 10^{-5}$ ($k_e = 20$): the same reference orbit, traversed one way, keeps the collapsed kernels together, and traversed the other way makes them separate. The window is narrow — the leading exponent at these $\sigma_0$ is $1.2 \cdot 10^{-3}$ and $3.0 \cdot 10^{-4}$ — and the design lands inside it although the $k_e^{-3}$ terms move both measured exponents by about $+4.7 \cdot 10^{-4}$ and $+6 \cdot 10^{-5}$ against the prediction.

Checked by: `tests/test_periodic.py`: the Hessian splitting (i) against finite differences ($10^{-5}$ relative) at the collapsed minimiser of instance A, the isotypic eigenvalue structure, the gap equation of $\sigma_{\mathrm{crit}}$ ($10^{-8}$), and the two-fold zero of the separation block at $\sigma_{\mathrm{crit}}$ with a positive definite collective block (`test_proposition_K_hessian_splitting_and_critical_leak`); the experiment above for (iii)–(iv) (`exp_orientation_stability.py` → `results/orientation_stability/table.md`).

# 24 Theorem J′: the stationary law of the dithered finite-$m$ closed loop

This section proves the second step of Section 26, branch 1 — the finite-$m$ closed loop *with dither* — and answers the first "next step" of branch 2 (the orientation effect when the stationary state is a measure). It also opens a new angle: the stationary state of the dithered loop is a non-equilibrium steady state, with a stationary probability current of order $1/k_e$ that vanishes exactly when the first-order averaged drift is reciprocal (Theorem I), and the stationary parameter process is not reversible in time.

**Setting.** The closed loop of Theorem H's design at finite $m$, in the scaling $\sigma = \sigma_0/k_e$, $\tau = \tau_0/k_e$ (the formal limit of Remark H.0: $k_e \to \infty$ with $G$, $k_e\sigma$, $k_e\tau$ fixed), $\varepsilon := 1/k_e$:

$$\begin{aligned} \dot{x} &= f(x) + \dot{x}_m - k_e e - \hat{f}_\Theta(x), \qquad e := x - x_m(t), \\ \mathrm{d}\vartheta_i &= \big[G\nabla_\vartheta\big(e^\top \tilde{\Phi}(x;\vartheta_i)\big) - (\sigma_0/k_e)(\vartheta_i - \vartheta^0)\big]\,\mathrm{d}t + \sqrt{2\tau_0/k_e}\,G^{1/2}\,\mathrm{d}B_i, \qquad i = 1..m, \end{aligned} \tag{$\mathrm{CL}_\tau$}$$

with the notation of Section 22: saturated features, $\Theta \in \mathbb{R}^{mp}$, $G_m := I_m \otimes G$, $x_m \in C^\infty$ $T_0$-periodic with occupation measure $\rho$, $f \in C^\infty$ on a neighbourhood of the tube $\mathcal{T}_R$, $F_R := \sup_{\mathcal{T}_R}\|f\|$, and $r_\Theta$, $J_m$, $H$, $u$, $\langle H\rangle = -G_m\nabla J_m$ as there. Assume $k_e \geq k_T := (F_R + K_\Phi)/R$, so that the ball $\{\|e\| \leq R\}$ is forward invariant for every parameter path ($\|\dot{e}\| \leq F_R + K_\Phi - k_e\|e\|$ on it): the plant never sees $f$ outside the tube, and $f$ may be modified outside a neighbourhood of $\mathcal{T}_R$ to be smooth with bounded derivatives on $\mathbb{R}^n$ without changing the law of the loop.

*Slow time and the three scales.* Put $s := \varepsilon t$, $\theta := t \bmod T_0 \in \mathbb{T} := \mathbb{R}/T_0\mathbb{Z}$, and $\eta := e/\varepsilon = k_e e$. Then $\mathrm{d}W_s := \sqrt{\varepsilon}\,\mathrm{d}B_t$ is a Brownian motion in slow time and ($\mathrm{CL}_\tau$) reads

$$\begin{aligned} \frac{\mathrm{d}\eta}{\mathrm{d}s} &= \varepsilon^{-2}\big[r_\Theta(x_m(\theta) + \varepsilon\eta) - \eta\big], \\ \mathrm{d}\vartheta_i &= \big[G\nabla_\vartheta\big(\eta^\top\tilde{\Phi}(x_m(\theta) + \varepsilon\eta;\vartheta_i)\big) - \sigma_0(\vartheta_i - \vartheta^0)\big]\,\mathrm{d}s + \sqrt{2\tau_0}\,G^{1/2}\,\mathrm{d}W_i, \qquad \frac{\mathrm{d}\theta}{\mathrm{d}s} = \varepsilon^{-1}: \end{aligned}$$

the error relaxes at rate $\varepsilon^{-2}$ to the slaved value $\eta^*(\Theta,\theta) := r_\Theta(x_m(\theta))$, the phase turns at rate $\varepsilon^{-1}$, and the parameters move at rate one. The generator of the Markov process $Z := (\Theta, \eta, \theta)$ on $\mathcal{S} := \mathbb{R}^{mp} \times \{\|\eta\| \leq F_R + K_\Phi\} \times \mathbb{T}$ is

$$\begin{aligned} L_\varepsilon &= \varepsilon^{-2}A_\varepsilon + \varepsilon^{-1}\partial_\theta + B_\varepsilon, \qquad A_\varepsilon := \big(r_\Theta(x_m(\theta) + \varepsilon\eta) - \eta\big)\cdot\nabla_\eta, \\ B_\varepsilon &:= \sum_i \big[G\nabla_\vartheta\big(\eta^\top\tilde{\Phi}(x_m(\theta) + \varepsilon\eta;\vartheta_i)\big) - \sigma_0(\vartheta_i - \vartheta^0)\big]\cdot\nabla_{\vartheta_i} + \tau_0\sum_i \mathrm{tr}\big(G\nabla^2_{\vartheta_i}\big), \end{aligned}$$

with $A_\varepsilon = A_0 + \varepsilon A_1 + \ldots$, $A_0 = (\eta^* - \eta)\cdot\nabla_\eta$, $A_1 = (D_x r_\Theta(x_m)\eta)\cdot\nabla_\eta$, and $B_\varepsilon = B_0 + \varepsilon B_1 + \ldots$, where $B_0$ has the drift $\sum_i[G\nabla_\vartheta(\eta^\top\tilde{\Phi}(x_m;\vartheta_i)) - \sigma_0(\vartheta_i - \vartheta^0)]$ (equal to $H(\Theta,\theta)$ at $\eta = \eta^*$) and $B_1 = \sum_i G\nabla_\vartheta\big(\eta^\top D_x\tilde{\Phi}(x_m;\vartheta_i)\eta\big)\cdot\nabla_{\vartheta_i}$.

*The averaged objects.* $\bar{L}_0 := -G_m\nabla J_m\cdot\nabla + \tau_0\,\mathrm{tr}(G_m\nabla^2)$ is the Langevin generator in the $G^{-1}$ metric with invariant measure the Gibbs measure $\pi := Z^{-1}\exp(-J_m/\tau_0)$ (Theorem H(ii) at finite $m$; $\pi$ has Gaussian tails because $J_m \geq (\sigma_0/2)\sum\|\vartheta_i - \vartheta^0\|^2_{G^{-1}} - C$). The first-order drift is the vector field on $\mathbb{R}^{mp}$

$$\begin{aligned} b(\Theta) &:= b_{\mathrm{fr}}(\Theta) + b_{\mathrm{osc}}(\Theta), \qquad b_{\mathrm{fr},i} := G\nabla_\vartheta(U_{\mathrm{even}} + U_{\mathrm{odd}})[\mu_\Theta](\vartheta)\big|_{\vartheta=\vartheta_i}, \\ b_{\mathrm{osc}} &:= \big\langle (D_\Theta H)(\Theta,\theta)\cdot u(\Theta,\theta)\big\rangle_\theta, \end{aligned}$$

with $\mu_\Theta := \frac{1}{m}\sum_j \delta_{\vartheta_j}$ and $U_{\mathrm{even}}$, $U_{\mathrm{odd}}$ the functionals of Proposition H.7 (so $b = k_e^2(b_{\mathrm{frozen}} + b_{\mathrm{osc}})$ of Theorem J, now at every $\Theta$ and not only at critical points; `periodic.second_order_drift` and `dither.SingleKernelInstance.averaged` compute it). The *first-order correction* $g\colon \mathbb{R}^{mp} \to \mathbb{R}$ is the solution with $\pi(g) = 0$ of the weighted Poisson equation

$$\tau_0\nabla\cdot(\pi G_m\nabla g) = \nabla\cdot(\pi b)$$

(equivalently $\bar{L}_0 g = -\tau_0\pi^{-1}\nabla\cdot(\pi b) = \nabla J_m\cdot b - \tau_0\nabla\cdot b$, which has zero $\pi$-mean),

the *first-order current* is $j_1 := \pi(b - \tau_0 G_m\nabla g) = \pi b_\perp$ with $b_\perp := b - \tau_0 G_m\nabla g$, so that $\nabla\cdot j_1 = 0$ and $b = G_m\nabla(\tau_0 g) + b_\perp$ is the Hodge decomposition of $b$ in $L^2(\pi; G_m^{-1})$ into a $G_m$-gradient and a $\pi$-divergence-free field. For an observable $\varphi \in C^{3,\alpha}(\mathbb{R}^{mp})$ with polynomially bounded derivatives, $\psi_\varphi$ denotes the solution of $\bar{L}_0\psi_\varphi = \varphi - \pi(\varphi)$, $\pi(\psi_\varphi) = 0$.

**Theorem J′** (the stationary law of the dithered finite-$m$ closed loop)**.** *Assume the setting above, with $G$ diagonal (as in every instance of this document; for a general symmetric $G$ replace $G_l\partial_{\vartheta_l}$ by the contraction with $G$ below). There are $\varepsilon_0 > 0$ and, for every observable $\varphi \in C^{3,\alpha}(\mathbb{R}^{mp})$ (three derivatives, the third Hölder continuous) with polynomially bounded derivatives up to order three, a constant $C_\varphi$ (depending on $\varphi$, $f$, $x_m$, $G$, $\sigma_0$, $\tau_0$, $R$, $m$, the feature bounds, and not on $\varepsilon$) such that for every $\varepsilon = 1/k_e \in (0, \varepsilon_0]$:*

(a) Invariant law. *The process $Z$ has an invariant probability measure $m_\varepsilon$ on $\mathcal{S}$, every invariant measure is supported in $\{\|\eta\| \le F_R + K_\Phi\}$ and has moments of all orders in $\Theta$ bounded uniformly in $\varepsilon$, and its $\theta$-marginal is uniform. If moreover $a_{\max} > F_R$ componentwise and the tube lies in the box of the saturated centres, the invariant measure is unique, so the law of $Z_t$ converges to $m_\varepsilon$ and time averages along a single trajectory converge to $m_\varepsilon$-expectations.*

(b) The Gibbs measure at leading order. *For every invariant measure $m_\varepsilon$, $|m_\varepsilon(\varphi) - \pi(\varphi)| \le C_\varphi \varepsilon$.*

(c) First-order correction. $m_\varepsilon(\varphi) = \pi(\varphi) - \varepsilon\,\pi(b\cdot\nabla\psi_\varphi) + O(\varepsilon^2) = \pi(\varphi) + \varepsilon\int \varphi g\,\mathrm{d}\pi + O(\varepsilon^2)$, *with $|O(\varepsilon^2)| \le C_\varphi\varepsilon^2$. In particular the stationary parameter means are $\pi(\Theta) + \varepsilon\int \Theta g\,\mathrm{d}\pi + O(\varepsilon^2)$.*

(d) Observables of the full state. *For $\varphi(\Theta, \eta, \theta)$ smooth, $T_0$-periodic in $\theta$, with polynomially bounded derivatives in $\Theta$ ($\bar\varphi$ below in the class of* (b)*), $m_\varepsilon(\varphi) = \pi(\bar\varphi) + O(\varepsilon)$, where $\bar\varphi(\Theta) := \langle \varphi(\Theta, r_\Theta(x_m(\theta)), \theta)\rangle_\theta$. In particular the stationary mean-square tracking error is*

$$\mathbb{E}_{m_\varepsilon}\|e\|^2 = \frac{2}{k_e^2}\,\pi(\bar L_m) + O(k_e^{-3}), \qquad \bar L_m(\Theta) = \tfrac{1}{2}\int \|r_\Theta\|^2\,\mathrm{d}\rho :$$

*the Gibbs average of the occupation loss, divided by $k_e^2/2$.*

(e) Circulation and irreversibility. *Let*

$$V(\Theta, \eta, \theta) := k_e \cdot (\textit{drift of } \Theta \textit{ in physical time}) = \sum_i \big[G\nabla_\vartheta\big(\eta^\top \tilde\Phi(x_m(\theta) + \varepsilon\eta; \vartheta_i)\big) - \sigma_0(\vartheta_i - \vartheta^0)\big]$$

*(so $\dot\Theta = \varepsilon V + \text{noise}$). For every pair of coordinates $(k, l)$ of $\mathbb{R}^{mp}$,*

$$\Lambda_\varepsilon^{kl} := \mathbb{E}_{m_\varepsilon}[\Theta_k V_l - \Theta_l V_k] = \varepsilon\Big[\int (\Theta_k j_{1,l} - \Theta_l j_{1,k})\,\mathrm{d}\Theta + \pi\big(\langle u_k\dot u_l - u_l\dot u_k\rangle_\theta\big)\Big] + O(\varepsilon^2),$$

*where $\dot u := \partial_\theta u = H - \langle H\rangle$. If the bracket is non-zero for some $(k, l)$, then for $\varepsilon$ small the stationary parameter process $(\Theta_t)_{t\ge 0}$ under $m_\varepsilon$ is not reversible in time: $\mathbb{E}[\Theta_{k,t}\Theta_{l,t+h}] \ne \mathbb{E}[\Theta_{l,t}\Theta_{k,t+h}]$ for small $h > 0$.*

**Status.** Proved. The proof imports three standard theorems, named where they are used: the existence, growth and regularity theory of the Poisson equation for the Langevin generator with a confining potential (Pardoux–Veretennikov [66], with interior Schauder estimates [35] for the higher derivatives), and, for the uniqueness in (a) only, Hörmander's theorem for the smoothness of the transition densities of a hypoelliptic time-periodic diffusion [44], [10], the Stroock–Varadhan support theorem [83] and the Doob–Khasminskii and Harris theorems [22], [40], [41]. Parts (b)–(e) hold for every invariant measure and do not use the uniqueness. The first-order drift $b$ is the one of Theorem J, so (c) is the thermal version of Theorem J(d): at low temperature it reduces to the $O(1/k_e)$ offset of the periodic orbit (Remark J′.1). The orientation effect of Remark H.7a and Proposition K appears in (c) as the orientation-odd part of $\int \Theta g\,\mathrm{d}\pi$, and in the numerics it is measured to within 7–8% at $k_e = 10$ and within the Monte Carlo resolution (10–30%) at $k_e = 20$–$80$. Checked numerically on the single-kernel instance below (`exp_dither_stationary.py`, `tests/test_dither.py`), where the first-order prediction removes the $O(1/k_e)$ discrepancy between the real loop and the Gibbs measure; the circulation of (e) is below the Monte Carlo resolution of the real loop at the affordable cost, and the current machinery behind it is checked on the first-order averaged SDE instead (Proposition J′.2).

*Proof.* Throughout, $C$ denotes a constant independent of $\varepsilon$, and "polynomially bounded" means bounded by $C(1+\|\Theta\|^q)$ on $\mathcal{S}$ for some $q$.

*Step 0 (invariance, moments, existence).* By the choice $k_e \geq k_T$, $\{\|e\| \leq R\}$ is forward invariant and $\|e(t)\| \leq e^{-k_e t}\|e(0)\| + (F_R + K_\Phi)/k_e$, so the set $\{\|\eta\| > F_R + K_\Phi + \delta\}$ is left in finite time and never re-entered: every invariant measure is supported in $\{\|\eta\| \leq F_R + K_\Phi\}$, and on this set the $\Theta$-drift of $(\mathrm{CL}_\tau)$ is bounded by $\lambda_G(F_R+K_\Phi)L_\vartheta + \sigma\|\Theta - \Theta^0\|$. For $W := \|\Theta - \Theta^0\|^2$ (slow time) $L_\varepsilon W \leq -2\sigma_0 W + 2\lambda_G(F_R + K_\Phi)L_\vartheta\sqrt{mW} + 2\tau_0 \operatorname{tr} G_m \leq -\sigma_0 W + C$, and likewise $L_\varepsilon W^q \leq -c_q W^q + C_q$ for every $q \geq 1$ (Itô's formula and Young's inequality). Hence $\mathbb{E}_z W^q(Z_s) \leq e^{-c_q s} W^q(z) + C_q/c_q$: the Cesàro averages of the laws of $Z_s$ are tight ($W^q$ is coercive in $\Theta$, $\eta$ and $\theta$ live in compact sets), the semigroup is Feller (the coefficients are smooth with bounded derivatives in $\eta$ and $\theta$ and of linear growth in $\Theta$), and Krylov–Bogoliubov gives an invariant probability measure $m_\varepsilon$. For any invariant measure, $0 = m_\varepsilon(L_\varepsilon W^q) \leq -c_q\, m_\varepsilon(W^q) + C_q$ gives the uniform moment bounds; the $\theta$-marginal is uniform because $\theta$ moves at constant speed. Dynkin's formula with these moment bounds shows $m_\varepsilon(L_\varepsilon \psi) = 0$ for every $\psi \in C^2(\mathcal{S})$, $T_0$-periodic in $\theta$, with polynomially bounded derivatives up to order two: the local martingale in Itô's formula is a martingale, and the stationary expectation of $\psi(Z_s) - \psi(Z_0)$ vanishes.

*Step 1 (uniqueness).* The noise acts on all of $\Theta$; the $x$-equation carries no noise. Write the SDE for $(x, \Theta)$ in physical time with the vector fields $X_0$ (drift, time-periodic) and $X_{i,k} := \sqrt{2\tau_0/k_e}\,(G^{1/2})_{\cdot k} \cdot \partial_{\vartheta_i}$. The bracket $[X_{i,\alpha_k}, X_0]$ contains, in the $x$-direction, $-\frac{1}{m}(\partial_{\alpha_k}\tilde{\Phi})(x;\vartheta_i)\cdot \partial_{x_k} = -\frac{1}{m}(\,\mathrm{d}a_k/\,\mathrm{d}\alpha_k)\varphi(x;c_i,s_i)\partial_{x_k}$ with $\mathrm{d}a_k/\,\mathrm{d}\alpha_k > 0$ and $\varphi > 0$: together with the $X_{i,k}$ these brackets span $\mathbb{R}^n \times \mathbb{R}^{mp}$ at every point and every time, so the parabolic Hörmander condition holds everywhere. By Hörmander's theorem for time-dependent hypoelliptic diffusions with smooth coefficients of bounded derivatives ([44]; the non-homogeneous case is [10]), the transition probabilities $P_{s,t}(z,\cdot)$ have densities that are smooth in $z$, so the period map $P := P_{0,T_0}$ is strong Feller. Accessibility: fix $\Theta_1$ and let $z^* := (e^*, \Theta_1)$ be the state at time $T_0$ of the deterministic skeleton that follows, on $[0, T_0/2]$, a $C^1$ parameter path along which $\hat{f}_{\Theta(t)}(x(t)) = f(x(t)) - w(t)$ with $w(t) := \lambda \cdot \beta(t)$ for a fixed bump $\beta$ supported in $(0, T_0/2)$ and $\lambda \in \mathbb{R}$ chosen so that $e(T_0/2) = 0$ (possible since $\dot{e} = -k_e e + w$ is controllable and the required $\lambda$ is $O(e^{-k_e T_0/2})$; such parameter paths exist because $a_{\max} > F_R$: $m$ kernels centred at $x(t)$ with amplitude $f(x(t)) - w(t)$ realise $\hat{f} = f - w$, and the saturated centre box contains the tube), and on $[T_0/2, T_0]$ any $C^1$ path from that configuration to $\Theta_1$. From every initial $(e_0, \Theta_0)$ in the invariant set the same construction (with $\lambda$ depending on $e_0$) reaches $e(T_0/2) = 0$ at the same parameter configuration, hence $z^*$ at time $T_0$. By the Stroock–Varadhan support theorem [83] (the skeleton paths are realised by controls $h \in L^2$: the $\Theta$-noise is non-degenerate and the skeleton's $\Theta$-path is $C^1$), $z^* \in \operatorname{supp} P(z,\cdot)$ for every $z$. The supports of two distinct ergodic invariant measures of a strong Feller kernel are disjoint, and both contain $z^*$ ([22, Prop. 4.1.1 and Thm 4.2.1]; [40, Cor. 1.14–Thm 1.16]), so $P$ has at most one invariant measure. The periodic family $P^*_{0,t} m_\varepsilon$ is then the unique invariant law of the augmented process; the strong Feller property with the accessible point gives a minorisation of $P^2$ on the sublevel sets of the Lyapunov function $W$ ([40, Lemma 1.13]), and Harris' theorem ([41]) gives the convergence of the law of $Z_{nT_0}$ to $m_\varepsilon$ in a $W$-weighted total variation norm, hence the convergence of time averages along a single trajectory.

*Step 2 (the fast operator).* At frozen $(\Theta, \theta)$, $A_0$ is the generator of the flow $\dot{\eta} = \eta^* - \eta$, which contracts to $\eta^*$ at rate one. For $g$ smooth with $g(\Theta, \eta^*, \theta) = 0$ the function

$$(A_0^{-1} g)(\Theta, \eta, \theta) := -\int_0^\infty g\big(\Theta, \eta^* + e^{-s}(\eta - \eta^*), \theta\big)\,\mathrm{d}s$$

is smooth, satisfies $A_0(A_0^{-1} g) = g$, and its derivatives in $(\Theta, \theta)$ of any order are bounded by those of $g$; if $g$ is linear in $\eta - \eta^*$, $g = (\eta - \eta^*)^\top w(\Theta, \theta)$, then $A_0^{-1} g = -(\eta - \eta^*)^\top w$. Conversely $A_0 \psi = 0$ with $\psi$ bounded in $\eta$ forces $\psi = \psi(\Theta, \theta)$.

*Step 3 (the expansion).* Let $\varphi \in C^5(\mathbb{R}^{mp})$ with polynomially bounded derivatives up to order five. Define

$$\psi_1 := -u \cdot \nabla\varphi, \qquad w_\varphi := \sum_{i,l} G_l \, \partial_{\vartheta_l} \tilde{\Phi}(x_m(\theta); \vartheta_i) \, \partial_{(i,l)}\varphi \in \mathbb{R}^n$$

$$\text{(for a symmetric } G\text{: } (w_\varphi)_k = \sum_i \big(\nabla_\vartheta \tilde{\Phi}_k(x_m; \vartheta_i)\big)^\top G \nabla_{\vartheta_i}\varphi),$$

$$\psi_2 := (\eta - \eta^*)^\top w_\varphi + \psi_2'(\Theta, \theta),$$

where $\psi_2'$ is the zero-mean antiderivative in $\theta$ of $b \cdot \nabla\varphi - [-\dot{r}^\top w_\varphi + (D_x r_\Theta \, r)^\top w_\varphi + (B_0\psi_1)|_{\eta^*} + (B_1\varphi)|_{\eta^*}]$ with $\dot{r} := \partial_\theta[r_\Theta(x_m(\theta))]$ and $r := r_\Theta(x_m(\theta))$, and $\psi_3 := A_0^{-1}$ of the $\eta$-dependent remainder defined below. Then $\psi_\varepsilon := \varphi + \varepsilon\psi_1 + \varepsilon^2\psi_2 + \varepsilon^3\psi_3$ satisfies, order by order:

- $\varepsilon^{-2}$: $A_0\varphi = 0$.
- $\varepsilon^{-1}$: $A_0\psi_1 + \partial_\theta\varphi + A_1\varphi = 0$, since $\psi_1$ does not depend on $\eta$.
- $\varepsilon^0$: $A_0\psi_2 + \partial_\theta\psi_1 + A_1\psi_1 + (B_0 + A_2)\varphi$. Here $A_1\psi_1 = A_2\varphi = 0$ and $A_0\psi_2 = -(\eta - \eta^*)^\top w_\varphi = (B_0\varphi)|_{\eta^*} - B_0\varphi$, so the sum equals $\partial_\theta\psi_1 + (B_0\varphi)|_{\eta^*} = -\dot{u} \cdot \nabla\varphi + H \cdot \nabla\varphi + \tau_0 \operatorname{tr}(G_m\nabla^2\varphi) = \langle H\rangle \cdot \nabla\varphi + \tau_0 \operatorname{tr}(G_m\nabla^2\varphi) = \bar{L}_0\varphi$, because $\dot{u} = H - \langle H\rangle$. This is Remark H.0's algebraic identity: the slaved drift averages to the Langevin drift.
- $\varepsilon^1$: $A_0\psi_3 + \partial_\theta\psi_2 + A_1\psi_2 + (B_0 + A_2)\psi_1 + (B_1 + A_3)\varphi$. At $\eta = \eta^*$: $\partial_\theta\psi_2 = -\dot{r}^\top w_\varphi + \partial_\theta\psi_2'$ (the lag of the error behind its slaved value), $A_1\psi_2 = (D_x r_\Theta(x_m)r)^\top w_\varphi$ (the state shift in the error equation), $B_0\psi_1 = -H \cdot \nabla(u \cdot \nabla\varphi) - \tau_0 \operatorname{tr}(G_m\nabla^2(u \cdot \nabla\varphi))$, $A_2\psi_1 = A_3\varphi = 0$, and $B_1\varphi = \sum_i G\nabla_\vartheta\big(r^\top D_x\tilde{\Phi}(x_m; \vartheta_i)r\big) \cdot \nabla_{\vartheta_i}\varphi$ (the state shift in the feature). By the choice of $\psi_2'$ the sum at $\eta = \eta^*$ equals $b \cdot \nabla\varphi$ pointwise in $\theta$, once the $\theta$-averages are identified: $\langle -\dot{r}^\top w_\varphi\rangle = G\nabla_\vartheta\langle -\dot{r}^\top\tilde{\Phi}(x_m; \vartheta)\rangle \cdot \nabla\varphi = (G_m\nabla U_{\text{odd}}) \cdot \nabla\varphi$ at the particles, $\langle (D_x r \, r)^\top w_\varphi + (B_1\varphi)|_{\eta^*}\rangle = \langle (r \cdot \nabla_x)(r^\top\tilde{\Phi})\rangle$-gradient $\cdot \nabla\varphi = (G_m\nabla U_{\text{even}}) \cdot \nabla\varphi$ (the two state shifts combine into $U_{\text{even}}$), so these three give $b_{\text{fr}} \cdot \nabla\varphi$; and $\langle B_0\psi_1|_{\eta^*}\rangle = -\langle (D_\Theta u)H\rangle \cdot \nabla\varphi - \langle H \otimes u\rangle : \nabla^2\varphi - \tau_0 \operatorname{tr}(G_m\nabla^2(\langle u\rangle \cdot \nabla\varphi)) = \langle (D_\Theta H)u\rangle \cdot \nabla\varphi = b_{\text{osc}} \cdot \nabla\varphi$, because $\langle u\rangle = 0$, $\langle (D_\Theta u)\dot{u}\rangle = -\langle (D_\Theta\dot{u})u\rangle = -\langle (D_\Theta H)u\rangle$ (integration by parts in $\theta$, $\langle D_\Theta u\rangle\langle H\rangle = 0$), and $\langle H_j u_k\rangle + \langle H_k u_j\rangle = \langle \partial_\theta(u_j u_k)\rangle = 0$ so the symmetric contraction with $\nabla^2\varphi$ vanishes (the same cancellation as $N_2 = 0$ in Lemma E.0′). Writing $T_1(\Theta, \eta, \theta)$ for the sum of the order-$\varepsilon$ terms other than $A_0\psi_3$, one has $T_1(\Theta, \eta^*, \theta) = b \cdot \nabla\varphi$, so $\psi_3 := A_0^{-1}(b \cdot \nabla\varphi - T_1)$ is well defined (Step 2) and $A_0\psi_3 + T_1 = b \cdot \nabla\varphi$: the order-$\varepsilon$ term is exactly $\bar{L}_1\varphi := b \cdot \nabla\varphi$.
- $\varepsilon^2$: $R_\varepsilon := \varepsilon^{-2}[L_\varepsilon\psi_\varepsilon - \bar{L}_0\varphi - \varepsilon\bar{L}_1\varphi]$ is a finite sum of products of derivatives of $f$ on the tube (orders $\le 3$), of $\tilde{\Phi}$ in $(x, \vartheta)$ (bounded), of $x_m$, of $u$, $w_\varphi$, $\psi_2'$, $\psi_3$ and of derivatives of $\varphi$: $\psi_2'$ contains $\nabla^3\varphi$ (through $\tau_0 \operatorname{tr}(G_m\nabla^2(u \cdot \nabla\varphi))$ in $(B_0\psi_1)|_{\eta^*}$) and $\psi_3$ contains $\nabla^2\varphi$, so $B_0\psi_2 \ni \tau_0 \operatorname{tr}(G_m\nabla^2\psi_2')$ carries $\nabla^5\varphi$ and $B_0\psi_3$ carries $\nabla^4\varphi$; using the exact operators $A_\varepsilon$, $B_\varepsilon$ and Taylor's theorem in $\varepsilon$ rather than their formal series, $|R_\varepsilon| \le C(1 + \|\Theta\|^q)$ on $\mathcal{S}$ uniformly in $\varepsilon \in (0, \varepsilon_0]$ for $\varphi \in C^5$ with polynomially bounded derivatives up to order five. (For the zeroth-order statement (b) alone $\psi_2'$ can be dropped — it only serves to make the order-$\varepsilon$ term exactly $b \cdot \nabla\varphi$ — and $C^3$ suffices.)

*Step 4 (the Poisson equation).* $\bar{L}_0$ is uniformly elliptic with smooth coefficients, $\nabla J_m$ has linear growth with bounded derivatives of all orders, and $\langle -G_m\nabla J_m(\Theta), \Theta - \Theta^0\rangle \le -\sigma_0\|\Theta - \Theta^0\|^2 + C\|\Theta - \Theta^0\|$: the recurrence condition of Pardoux–Veretennikov [66, Thm 1] holds with every polynomial Lyapunov function. For $\varphi \in C^{3,\alpha}(\mathbb{R}^{mp})$ with polynomially bounded derivatives up to order three there is a unique $\psi_\varphi \in C^2(\mathbb{R}^{mp})$ with $\bar{L}_0\psi_\varphi = \varphi - \pi(\varphi)$, $\pi(\psi_\varphi) = 0$, and $\psi_\varphi$, $\nabla\psi_\varphi$ are polynomially bounded ([66, Thm 1 and 2]); interior Schauder estimates on unit balls [35, Thm 6.2 and 6.17], which gain two derivatives and whose constants depend polynomially on $\|\Theta\|$ through the local Hölder norms of the coefficients, give $\psi_\varphi \in C^{5,\alpha}$ with polynomially bounded derivatives up to order five.

*Step 5 ((b) and (c)).* Apply Step 3 to $\psi := \psi_\varphi$ and Step 0's identity to $\psi_\varepsilon$: $0 = m_\varepsilon(L_\varepsilon\psi_\varepsilon) = m_\varepsilon(\bar{L}_0\psi) + \varepsilon m_\varepsilon(b\cdot\nabla\psi) + \varepsilon^2 m_\varepsilon(R_\varepsilon) = m_\varepsilon(\varphi) - \pi(\varphi) + \varepsilon m_\varepsilon(b\cdot\nabla\psi) + O(\varepsilon^2)$, the $O(\varepsilon^2)$ by the moment bounds of Step 0. Since $b\cdot\nabla\psi$ is polynomially bounded, $|m_\varepsilon(\varphi) - \pi(\varphi)| \le C\varepsilon$, which is (b). Now $b\cdot\nabla\psi \in C^4(\mathbb{R}^{mp})$ with polynomially bounded derivatives ($b$ is smooth with bounded derivatives in $\Theta$, $\psi \in C^{5,\alpha}$), so (b) applies to it (its Poisson solution is in $C^3$, which the zeroth-order expansion needs): $m_\varepsilon(b\cdot\nabla\psi) = \pi(b\cdot\nabla\psi) + O(\varepsilon)$, and $m_\varepsilon(\varphi) = \pi(\varphi) - \varepsilon\pi(b\cdot\nabla\psi_\varphi) + O(\varepsilon^2)$. The second form of (c) is Green's identity for the weighted Poisson equation:

$$\pi(b\cdot\nabla\psi) = \int b\cdot\nabla\psi\,\pi = -\int \psi\nabla\cdot(\pi b) = -\tau_0\int \psi\nabla\cdot(\pi G_m\nabla g) = -\tau_0\int g\nabla\cdot(\pi G_m\nabla\psi)$$
$$= -\int g\pi\bar{L}_0\psi = -\int g(\varphi - \pi(\varphi))\pi = -\int \varphi g\,\mathrm{d}\pi,$$

using $\tau_0\pi^{-1}\nabla\cdot(\pi G_m\nabla\psi) = \bar{L}_0\psi$ and $\pi(g) = 0$; the boundary terms vanish by the Gaussian tails of $\pi$ and the polynomial growth of $g$, $\psi$ and their gradients (the same regularity for $g$ as for $\psi_\varphi$, with $\varphi$ replaced by the polynomially bounded $\nabla J_m\cdot b - \tau_0\nabla\cdot b$).

*Step 6 ((d)).* Write $\varphi = \bar{\varphi} + \varphi_\theta + \varphi_\eta$ with $\varphi_\theta(\Theta,\theta) := \varphi(\Theta,\eta^*,\theta) - \bar{\varphi}(\Theta)$ (zero $\theta$-mean) and $\varphi_\eta := \varphi(\Theta,\eta,\theta) - \varphi(\Theta,\eta^*,\theta)$ (zero at $\eta^*$). With $\chi_1 := -$(zero-mean antiderivative in $\theta$ of $\varphi_\theta$) and $\chi_2 := -A_0^{-1}\varphi_\eta$ (Step 2), $L_\varepsilon(\varepsilon\chi_1 + \varepsilon^2\chi_2) = \partial_\theta\chi_1 + A_0\chi_2 + \varepsilon[B_0\chi_1 + (\partial_\theta + A_1)\chi_2] + O(\varepsilon^2) = -(\varphi_\theta + \varphi_\eta) + \varepsilon[\dots] + O(\varepsilon^2)$, all terms polynomially bounded. Step 0's identity gives $m_\varepsilon(\varphi) = m_\varepsilon(\bar{\varphi}) + \varepsilon m_\varepsilon(B_0\chi_1 + (\partial_\theta + A_1)\chi_2) + O(\varepsilon^2) = \pi(\bar{\varphi}) + O(\varepsilon)$ by (b). For $\varphi = \|\eta\|^2$, $\bar{\varphi} = \langle\|r_\Theta(x_m(\theta))\|^2\rangle_\theta = 2\bar{L}_m(\Theta)$, and $\mathbb{E}\|e\|^2 = \varepsilon^2\mathbb{E}\|\eta\|^2$.

*Step 7 ((e)).* Expand $V = V^0 + \varepsilon V^1 + O(\varepsilon^2)$ with $V^0$ the $B_0$-drift and $V^1$ the $B_1$-drift, and apply Step 6 to $\varphi := \Theta_k V_l - \Theta_l V_k$ keeping the order-$\varepsilon$ terms. For $\varphi^0 := \Theta\wedge V^0$ ($\wedge$ on the pair $(k,l)$): $\bar{\varphi}^0 = \Theta\wedge\langle H\rangle = \Theta\wedge(-G_m\nabla J_m)$, and $\pi(\bar{\varphi}^0) = 0$ because $-\pi G_m\nabla J_m = \tau_0 G_m\nabla\pi$ and $\int\Theta\wedge(G_m\nabla\pi) = 0$ for a symmetric $G_m$ (integration by parts: the integrand is a divergence plus $\pi\cdot(G_{lk} - G_{kl})$). Next, $\varphi^0_\theta = \Theta\wedge(H - \langle H\rangle) = \Theta\wedge\dot{u}$, so $\chi_1 = -\Theta\wedge u$ and $\langle(B_0\chi_1)|_{\eta^*}\rangle = -\langle H\cdot\nabla_\Theta(\Theta\wedge u)\rangle - \tau_0\langle\mathrm{tr}\,G_m\nabla^2(\Theta\wedge u)\rangle = -\langle H\wedge u\rangle - \langle\Theta\wedge(D_\Theta u\cdot H)\rangle = -\langle\dot{u}\wedge u\rangle + \Theta\wedge\langle(D_\Theta H)u\rangle = \langle u\wedge\dot{u}\rangle + \Theta\wedge b_{\mathrm{osc}}$ (the $\tau_0$ term and $\langle H\rangle\wedge\langle u\rangle$ vanish since $\langle u\rangle = 0$, and $\langle(D_\Theta u)H\rangle = -\langle(D_\Theta H)u\rangle$ as in Step 3). Further, $\varphi^0_\eta = \Theta\wedge(V^0(\eta) - V^0(\eta^*))$ is linear in $\eta - \eta^*$, so $\chi_2 = \varphi^0_\eta$ and $\langle((\partial_\theta + A_1)\chi_2)|_{\eta^*}\rangle = \Theta\wedge\big\langle\sum_i G\nabla_\vartheta\big((-\dot{r} + D_x r\,r)^\top\tilde{\Phi}(x_m;\vartheta_i)\big)\big\rangle = \Theta\wedge$(lag and error-shift parts of $b_{\mathrm{fr}}$); with $\bar{\varphi}^1 = \Theta\wedge\langle V^1|_{\eta^*}\rangle = \Theta\wedge$(feature-shift part of $b_{\mathrm{fr}}$) these complete $\Theta\wedge b_{\mathrm{fr}}$. Collecting, and using (c) for $\bar{\varphi}^0$ ($m_\varepsilon(\bar{\varphi}^0) = \varepsilon\int\Theta\wedge(-G_m\nabla J_m)g\,\mathrm{d}\pi + O(\varepsilon^2)$) and (b) for the order-$\varepsilon$ observables,

$$\Lambda_\varepsilon = \varepsilon\Big[\int\Theta\wedge(b - gG_m\nabla J_m)\,\mathrm{d}\pi + \pi\big(\langle u\wedge\dot{u}\rangle\big)\Big] + O(\varepsilon^2),$$

and $b - gG_m\nabla J_m = (b - \tau_0 G_m\nabla g) + (\tau_0 G_m\nabla g - gG_m\nabla J_m) = b_\perp + \tau_0\pi^{-1}G_m\nabla(g\pi)$, whose second term integrates to zero against $\Theta\wedge$ as above. This is the formula of (e). For the irreversibility: under $m_\varepsilon$ the $\Theta$-process is stationary; reversibility would give $\mathbb{E}[\Theta_{k,t}\Theta_{l,t+h}] = \mathbb{E}[\Theta_{l,t}\Theta_{k,t+h}]$ for all $h$, i.e. $\mathbb{E}[\Theta_t\wedge\Theta_{t+h}] = 0$, whereas $\mathbb{E}[\Theta_t\wedge(\Theta_{t+h} - \Theta_t)] = \mathbb{E}[\Theta_t\wedge\int_t^{t+h}\varepsilon V\,\mathrm{d}s] + 0$ (the martingale term has mean zero) $= h\varepsilon\Lambda_\varepsilon + o(h)$ by stationarity and the continuity of the integrand; so $\Lambda_\varepsilon \neq 0$ excludes reversibility. □

**Remark J′.1** (reading)**.**

- *From the periodic orbit to the thermal state.* As $\tau_0 \to 0$ with $\Theta^*$ the non-degenerate global minimiser of $J_m$, $\pi \to \delta_{\Theta^*}$, $\psi_{\Theta_k} \to$ the linear function $v_k^\top(\Theta - \Theta^*)$ with $-(G_m\nabla^2 J_m)^\top v_k = e_k$, and (c) gives $m_\varepsilon(\Theta) \to \Theta^* + \varepsilon(G_m\nabla^2 J_m(\Theta^*))^{-1}b(\Theta^*)$: Theorem J(d)'s period mean of the periodic orbit. Theorem J′(c) is the finite-temperature version: the correction is a Gibbs average of the same first-order drift, weighted by the Poisson solution instead of the inverse Hessian.

- *The orientation effect with dither (branch 2).* Reversing the direction of traversal of the reference flips $U_{\mathrm{odd}}$ and $b_{\mathrm{osc}}$ and leaves $U_{\mathrm{even}}$, $J_m$ and $\pi$ unchanged, so the stationary means of the two traversals differ at order $\varepsilon$ by $2\varepsilon\int \Theta g_{\mathrm{odd}}\,\mathrm{d}\pi$ with $g_{\mathrm{odd}}$ the correction driven by the odd part of $b$: the quantity measured in Remark H.7a for the periodic orbit is now a statement about stationary measures, and it is checked below (to within 7–8% at $k_e = 10$, and within the resolution of the runs at larger $k_e$) in the single-kernel instance (where, in one state dimension, $U_{\mathrm{odd}} \equiv 0$ — a 1-form integrated over a path that retraces itself — and the whole odd effect comes from $b_{\mathrm{osc}}$; a reference with the time-reversal symmetry $x_m(\theta_0+\theta) = x_m(\theta_0-\theta)$, such as a sinusoid, has no odd effect at all, since its reversal is a time shift). As a design rule: a time-symmetric reference (one whose Fourier series carries only cosines after a time shift, such as $x_m = X[\cos\theta + r_2\cos 2\theta]$) removes $b_{\mathrm{osc}}$ and $U_{\mathrm{odd}}$ identically, so its first-order correction is the performative term $G_m\nabla U_{\mathrm{even}}$ alone, which is reciprocal exactly when Theorem I's $A_{\mathrm{even}} \equiv 0$; a reference traversed one way carries both, and the odd part can be used (Proposition K) or cancelled by reversing the traversal. Checked: for the cosine reference the code's $b_{\mathrm{osc}}$ and $\langle u \wedge \dot{u}\rangle$ are $10^{-17}$ at every point tested.

- *Non-equilibrium.* (e) says that the dithered learning loop, at stationarity, is not in detailed balance even to first order: the periodic forcing alone (the term $\pi\langle u \wedge \dot{u}\rangle$, the signed area swept by the fast parameter loop) breaks the time-reversal symmetry, and the slow dynamics carries in addition the current $j_1$ of Proposition J′.2, nonzero exactly when the first-order drift is not reciprocal. The first is a property of any periodically forced system; the second is the signature of closed-loop learning (Theorem I).

- *Hypotheses.* Smoothness of $f$ on the tube and of the features is used through the expansion (three $x$-derivatives of $f$, five $\Theta$-derivatives of the Poisson solution, i.e. three Hölder derivatives of the observable) and, for the uniqueness only, through Hörmander's theorem; $k_e \geq k_T$ keeps the plant in the tube for every parameter path. The constants are not explicit: the result is an asymptotic expansion of the invariant measure, not a quantitative bound at a given $k_e$, and the numerics show where it applies.

**Proposition J′.2** (the first-order current, entropy production, and reciprocity)**.** *Consider the first-order averaged dynamics* $\mathrm{d}\Theta = (-G_m\nabla J_m + \varepsilon b)\,\mathrm{d}s + \sqrt{2\tau_0}\,G_m^{1/2}\,\mathrm{d}W$ *on* $\mathbb{R}^{mp}$*, with* $b$ *as above.*

(i) *Its stationary law* $m_\varepsilon^{(1)}$ *satisfies* $m_\varepsilon^{(1)}(\varphi) = \pi(\varphi) + \varepsilon\int \varphi g\,\mathrm{d}\pi + O(\varepsilon^2)$ *for every* $\varphi$ *as in Theorem J′ (the expansion* $m_\varepsilon^{(1)} = \pi(1+\varepsilon g) + O(\varepsilon^2)$ *in the weak sense; on the grid it holds in* $L^1$ *with an* $\varepsilon^2$ *remainder, Numerics below), and its stationary current* $j := m_\varepsilon^{(1)}(-G_m\nabla J_m + \varepsilon b) - \tau_0 G_m \nabla m_\varepsilon^{(1)}$ *equals* $\varepsilon j_1 + O(\varepsilon^2)$ *in the same sense, with* $j_1 = \pi b_\perp$*; its entropy production rate (in slow time) is*

$$\mathrm{EPR}_\varepsilon := \int \frac{\|j\|^2_{G_m^{-1}}}{\tau_0 m_\varepsilon^{(1)}}\,\mathrm{d}\Theta = \varepsilon^2\tau_0^{-1}\pi\big(\|b_\perp\|^2_{G_m^{-1}}\big) + O(\varepsilon^3)$$

*(formal in the remainder: the* $\varepsilon^2$ *term uses only the first-order current, the* $O(\varepsilon^3)$ *presumes the expansion of* $j$ *in* $L^2(1/\pi)$*).*

(ii) *The following are equivalent:* $j_1 \equiv 0$*;* $b_\perp \equiv 0$*;* $b$ *is a* $G_m$*-gradient field,* $b = -G_m\nabla J_1$*;* $D_\Theta(G_m^{-1}b)$ *is a symmetric matrix at every* $\Theta$*. In that case the first-order averaged dynamics* is *the Langevin dynamics of* $J_m + \varepsilon J_1$*, with stationary density exactly* $\propto \exp(-(J_m+\varepsilon J_1)/\tau_0)$*, in detailed balance.*

(iii) *The antisymmetric part of* $D_\Theta(G_m^{-1}b)$ *is, block by block (*$(i,j)$ *blocks of size* $p\times p$*),*

$$\tfrac{1}{2}\big[D(G_m^{-1}b) - D(G_m^{-1}b)^\top\big]_{ij} = \frac{1}{2m}\nabla_\vartheta\nabla_{\vartheta'}A_{\mu_\Theta}(\vartheta_i,\vartheta_j) + \tfrac{1}{2}\big\langle[\nabla^2_\Theta\mathcal{J},\nabla^2_\Theta\mathcal{U}]_{G_m}\big\rangle_{\theta,ij},$$

*where $A_\mu = S - S^\top$ is the antisymmetric kernel of Theorem I at the atomic measure $\mu_\Theta$ (derivatives in the kernel arguments, $\mu_\Theta$ frozen), $\mathcal{J}(\Theta,\theta) := \frac{m}{2}\|r_\Theta(x_m(\theta))\|^2 + \frac{\sigma_0}{2}\sum_i \|\vartheta_i - \vartheta^0\|^2_{G^{-1}}$ is the instantaneous loss (so $H = -G_m\nabla\mathcal{J}$, $\langle\mathcal{J}\rangle = J_m$), $\mathcal{U}$ its zero-mean antiderivative in $\theta$, and $[X,Y]_G := XGY - YGX$. The first term is the finite-$m$ form of the reciprocity defect of Theorem I; the second is the failure of the instantaneous Hessians at different phases to commute, the non-gradient part of second-order averaging of a time-periodic gradient flow.*

(iv) *In the setting of Theorem I (saturated Gaussian features, $f \in C^2$, $x_m \in C^2$ periodic), for $m \geq 3$, $b_{\rm fr}$ is never a $G_m$-gradient on $\mathbb{R}^{mp}$ ($b_{\rm fr}$ does not depend on $\sigma_0$ and depends on $G$ through the factor $G$ only); consequently, along the gain scale $t \mapsto tG$, the full drift $b = b_{\rm fr} + b_{\rm osc}$ is a $(tG)_m$-gradient for at most one value of $t > 0$. For $m = 1, 2$ the condition of* (ii) *is the explicit vanishing of* (iii)*, checked to fail in the single-kernel instance below (non-gradient fraction $\pi(\|b_\perp\|^2)/\pi(\|b\|^2) \approx 0.3$–$0.5$).*

*Proof. (i)* Stationary Fokker–Planck: $\nabla\cdot(mG_m\nabla J_m) - \varepsilon\nabla\cdot(mb) + \tau_0\nabla\cdot(G_m\nabla m) = 0$; at order $\varepsilon^0$ the Gibbs measure, at order $\varepsilon^1$ with $\pi_1 = g\pi$: $\nabla\cdot(g\pi G_m\nabla J_m) + \tau_0\nabla\cdot(G_m\nabla(g\pi)) = \tau_0\nabla\cdot(\pi G_m\nabla g)$ (using $\tau_0\nabla\pi = -\pi\nabla J_m$), so the equation is $\tau_0\nabla\cdot(\pi G_m\nabla g) = \nabla\cdot(\pi b)$, the defining equation of $g$; the expansion is justified by the same Poisson-equation argument as in Theorem J′ (the generator of this SDE is $\bar{L}_0 + \varepsilon b\cdot\nabla$, and Steps 4–5 apply verbatim without the fast variables). The current: $j = \pi(-G_m\nabla J_m) - \tau_0 G_m\nabla\pi + \varepsilon[\pi b - g\pi G_m\nabla J_m - \tau_0 G_m\nabla(g\pi)] + O(\varepsilon^2) = \varepsilon\,\pi(b - \tau_0 G_m\nabla g) + O(\varepsilon^2)$. The entropy production rate of a diffusion with mobility $G_m$ and temperature $\tau_0$ in a stationary state $m$ with current $j$ is $\int j^\top(\tau_0 G_m)^{-1} j/m$ [82], which gives the formula.

*(ii)* $j_1 \equiv 0$ iff $b = \tau_0 G_m\nabla g$ iff $b$ is a $G_m$-gradient (then $\tau_0 g = -J_1 + \text{const}$ and the equation for $g$ is satisfied); on the simply connected $\mathbb{R}^{mp}$ a smooth field $G_m^{-1}b$ is a gradient iff its Jacobian is symmetric (Poincaré lemma). When $b = -G_m\nabla J_1$ the drift $-G_m\nabla(J_m + \varepsilon J_1)$ is a gradient drift, so the SDE is the Langevin dynamics of $J_m + \varepsilon J_1$ with the exact Gibbs law $\propto \exp(-(J_m + \varepsilon J_1)/\tau_0)$ and $j \equiv 0$; consistently, $g = -J_1/\tau_0 + \text{const}$ solves the defining equation.

*(iii)* $G^{-1}b_{{\rm fr},i} = \nabla_\vartheta P_2[\mu_\Theta](\vartheta)|_{\vartheta_i}$ with $P_2 := U_{\rm even} + U_{\rm odd}$, a functional of $\mu$ (quadratic in $\mu$ through $r_\mu$) and of the evaluation point. Differentiating with respect to $\vartheta_j$ moves the atom $\vartheta_j$ of $\mu_\Theta$, which gives $\frac{1}{m}\nabla_\vartheta\nabla_{\vartheta'}S(\vartheta_i,\vartheta_j)$ with $S(\vartheta,\vartheta') := \delta P_2[\mu](\vartheta)/\delta\mu(\vartheta')$ (Theorem I's $k_e^{-2}$ second-variation kernel), and, for $j = i$, additionally the symmetric $\nabla^2_\vartheta P_2(\vartheta_i)$. The antisymmetric part of the $(i,j)$ block is therefore $\frac{1}{2m}\nabla\nabla'[S(\vartheta_i,\vartheta_j) - S(\vartheta_j,\vartheta_i)] = \frac{1}{2m}\nabla\nabla' A_\mu(\vartheta_i,\vartheta_j)$. For $b_{\rm osc}$: $H = -G_m\nabla\mathcal{J}$ and $u = -G_m\nabla\mathcal{U}$, so $G_m^{-1}b_{\rm osc} = \langle\nabla^2\mathcal{J}\,G_m\nabla\mathcal{U}\rangle$ and $\partial_j(G_m^{-1}b_{\rm osc})_k = \langle\partial_j\partial_k\partial_l\mathcal{J}\,G_l\partial_l\mathcal{U}\rangle + \langle(\nabla^2\mathcal{J}G_m\nabla^2\mathcal{U})_{kj}\rangle$; the first is symmetric in $(j,k)$, the antisymmetric part of the second is $\frac{1}{2}\langle\nabla^2\mathcal{J}G_m\nabla^2\mathcal{U} - \nabla^2\mathcal{U}G_m\nabla^2\mathcal{J}\rangle$.

*(iv)* Suppose $b_{\rm fr}$ is a $G_m$-gradient for some $m \geq 3$. By (iii), $\nabla_\vartheta\nabla_{\vartheta'}A_{\mu_\Theta}(\vartheta_i,\vartheta_j) = 0$ for all $\Theta$ and all $i$, $j$. $A_\mu$ is affine in $\mu$ (Theorem I): $A_\mu = A_f + B(\int\tilde{\Phi}\,\mathrm{d}\mu)$ with $B$ linear in the function $\int\tilde{\Phi}(\cdot\,;\vartheta)\,\mathrm{d}\mu(\vartheta)$, so for $\Theta = (\vartheta,\vartheta',\vartheta_3,\dots)$, $\nabla_\vartheta\nabla_{\vartheta'}[A_f + \frac{1}{m}\sum_j B(\tilde{\Phi}(\cdot\,;\vartheta_j))](\vartheta,\vartheta') = 0$ with the atom positions inside $B$ frozen. Varying $\vartheta_3$ alone shows that $\nabla_\vartheta\nabla_{\vartheta'}B(\tilde{\Phi}(\cdot\,;\vartheta_3))(\vartheta,\vartheta')$ does not depend on $\vartheta_3$; at the zero feature $\vartheta_3 = \vartheta_z$ it vanishes ($B(0) = 0$), so it vanishes for all $\vartheta_3$, and then $\nabla\nabla' A_f(\vartheta,\vartheta') = 0$ as well: $\nabla_\vartheta\nabla_{\vartheta'}A_\mu(\vartheta,\vartheta') = 0$ for every $\mu$ and all $(\vartheta,\vartheta')$. A smooth antisymmetric kernel with vanishing mixed derivative on the connected $\mathbb{R}^p\times\mathbb{R}^p$ has the form $a(\vartheta) - a(\vartheta')$, a coboundary, hence $A_\mu \equiv 0$ by Theorem I(i) — for every $\mu$, contradicting Theorem I(vi). For the gain scale: under $G \to tG$, $b_{\rm fr} \to t\,b_{\rm fr}$ and $b_{\rm osc} \to t^2 b_{\rm osc}$, so the antisymmetric part of $D((tG)^{-1}b)$ is $a_{\rm fr} + t\,a_{\rm osc}$ with $a_{\rm fr} \not\equiv 0$; it vanishes identically in $\Theta$ for at most one $t$. □

**Corollary J′.3** (the stationary law in the iterated limit $k_e \to \infty$, $m \to \infty$)**.** *Let $\pi_m$ be the Gibbs measure of Theorem J′ on $\mathbb{R}^{mp}$, and $\mu_\tau^*$ the unique minimiser of Theorem H's free energy $F_\tau$ at $\sigma = \sigma_0/k_e$, $\tau = \tau_0/k_e$; since $k_e F_\tau = \bar{L} + \frac{\sigma_0}{2}\int\|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu + \tau_0 H(\mu) = \bar{L} + \tau_0 H(\cdot\,|\,\gamma_\tau) + \text{const}$ with $\gamma_\tau = \mathcal{N}(\vartheta^0, (\tau_0/\sigma_0)G)$, $\mu_\tau^*$ does not depend on $k_e$. Then:*

(i) Chaoticity of the Gibbs measures. *Under $\pi_m$ the empirical measure $\mu_\Theta = \frac{1}{m}\sum_i \delta_{\vartheta_i}$ converges in probability to $\mu_\tau^*$ as $m \to \infty$, and for every $k$ the $k$-particle marginal of $\pi_m$ converges weakly to $(\mu_\tau^*)^{\otimes k}$.*

(ii) The iterated limit of the real loop. *For every $\varphi \in C^2(\mathbb{R}^p)$ bounded with bounded derivatives up to order two,*

$$\lim_{m\to\infty}\lim_{k_e\to\infty} \mathbb{E}_{m_\varepsilon}\Big[\frac{1}{m}\sum_i \varphi(\vartheta_i)\Big] = \int \varphi \,\mathrm{d}\mu_\tau^*,$$

*and the inner limit is attained at the rate $O(1/k_e)$ of Theorem J′(b), with a constant that depends on $m$.*

**Status.** Proved (Sanov's theorem and Varadhan's lemma, [24, Thms 6.2.10 and 4.3.1]; Theorem H(i)–(ii); Theorem J′(b)). The joint limit, with $k_e$ and $m$ related, is Conjecture H.6 and remains open; the corollary places the stationary law of the finite loop between Theorem J′ and Theorem H. Numerics: `exp_gibbs_m.py` → `results/dither/gibbs_m.md` (below).

*Proof.* $J_m(\Theta) = m[\bar L(\mu_\Theta) + \frac{\sigma_0}{2}\int\|\vartheta - \vartheta^0\|^2_{G^{-1}}\,\mathrm{d}\mu_\Theta]$, so $\pi_m(\,\mathrm{d}\Theta) \propto \exp(-m\bar L(\mu_\Theta)/\tau_0)\,\gamma_\tau^{\otimes m}(\,\mathrm{d}\Theta)$. The map $\mu \mapsto \bar L(\mu) = \frac12\int\|f - \int\tilde\Phi(\cdot\,;\vartheta)\,\mathrm{d}\mu(\vartheta)\|^2\,\mathrm{d}\rho$ is bounded and continuous for the weak topology of $\mathcal{P}(\mathbb{R}^p)$ (the features are bounded and continuous in $\vartheta$, and $\rho$ is a probability measure). Under $\gamma_\tau^{\otimes m}$ the empirical measure satisfies Sanov's large-deviation principle with rate function $H(\cdot\,|\,\gamma_\tau)$; by Varadhan's lemma (tilting by the bounded continuous $-m\bar L/\tau_0$), under $\pi_m$ it satisfies a large-deviation principle with rate function $I(\mu) = \bar L(\mu)/\tau_0 + H(\mu\,|\,\gamma_\tau) - \inf = (k_e/\tau_0)(F_\tau(\mu) - F_\tau(\mu_\tau^*))$, by Theorem H(i). $F_\tau$ is strictly convex ($\bar L$ is convex, the entropy strictly convex) with the unique minimiser $\mu_\tau^*$ (Theorem H(ii)), so $I$ has the unique zero $\mu_\tau^*$ and $\mu_\Theta \to \mu_\tau^*$ in probability. Exchangeability of $\pi_m$ then gives the convergence of the $k$-marginals to $(\mu_\tau^*)^{\otimes k}$ (Sznitman's equivalence of chaoticity and the convergence of the empirical measure). For (ii), Theorem J′(b) applied to the observable $\frac1m\sum_i\varphi(\vartheta_i) \in C^2(\mathbb{R}^{mp})$ gives $\lim_{k_e\to\infty}\mathbb{E}_{m_\varepsilon}[\frac1m\sum\varphi(\vartheta_i)] = \pi_m(\frac1m\sum\varphi(\vartheta_i)) = \pi_m^{(1)}(\varphi)$, the one-particle marginal, which converges to $\int\varphi\,\mathrm{d}\mu_\tau^*$ by (i). □

**Remark J′.4** (what the corollary says about Conjecture H.6)**.** The conjecture asks for the limit of the real loop's law when $k_e \to \infty$ with $m$ fixed or growing, uniformly in time. Theorem J′ settles the stationary part at fixed $m$, and the corollary shows that the resulting Gibbs measures are chaotic with limit $\mu_\tau^*$, the stationary state of $(\mathfrak{A})$: the stationary law of the real dithered loop is the minimiser of Theorem H's free energy in the iterated limit. What is missing for the conjecture is the uniformity in $m$ of Theorem J′'s $O(1/k_e)$ (its constant grows with $m$ through the Poisson equation on $\mathbb{R}^{mp}$) and the time-dependent statement.

**Numerics** (`exp_dither_stationary.py` → `results/dither/table.md`). The single-kernel instance: $n = 1$, $m = 1$, plant $f(x) = e^{-(x-0.5)^2/0.5} + 0.4x$, reference $x_m(t) = 1.5[\sin t' + \frac13\sin 2t']$ with $t' = \pm t$ (the two orientations), saturated amplitude and centre $\vartheta = (\alpha, \gamma)$ ($a_{\max} = 3$, $c_{\max} = 2.5$), frozen width $s = 0.6$, $G = \mathrm{diag}(2, 2)$, $\sigma_0 = 0.4$, $\vartheta^0 = 0$, $\tau_0 = 0.1$, $k_e \in \{10, 20, 40, 80\}$. The averaged data on a $201^2$ grid: $\pi$, $g$, $j_1$ (rms $b_{\mathrm{fr}} = 0.143$, rms $b_{\mathrm{osc}} = 0.225$; non-gradient fraction $\pi(\|b_\perp\|^2)/\pi(\|b\|^2) = 0.32$ and 0.46 for the two orientations; EPR coefficient $\tau_0^{-1}\pi(\|b_\perp\|^2_{G^{-1}}) = 0.144$ and 0.118; circulation coefficient $\int\Theta\wedge j_1 + \pi\langle u\wedge\dot u\rangle = +0.0117 - 0.0167 = -0.0051$ and $-0.0103 + 0.0167 = +0.0065$). The real loop: $2\cdot10^4$ independent replicas per ($k_e$, orientation), started from the first-order law $\pi(1 + \varepsilon g)$, Strang splitting with $k_e h = 1/2$, burn-in $\geq 10$ and averaging $\geq 40$ slow-time units (whole numbers of reference periods), standard errors from the spread of the per-replica time averages (the slowest mode of $\bar L_0$ relaxes in 3.5 slow-time units on this instance, so batch means in time would be correlated; referee round 6).

*The first-order law (b), (c).* The stationary means against the Gibbs value and the first-order prediction, orientation +1 ($\varepsilon$-coefficients $\int\varphi g\,\mathrm{d}\pi$: $-0.255$ for $\alpha$, $-0.426$ for $\gamma$); Table 24.

Table 24: Theorem J′(b), (c) on the single-kernel instance: the stationary means of the real loop against the Gibbs value and the first-order prediction (`results/dither/table.md`).

| $k_e$ | $m_\varepsilon(\alpha)-\pi(\alpha)$ (s.e. 0.0013) | $\times k_e$ | $m_\varepsilon(\alpha)-$ first order | $m_\varepsilon(\gamma)-\pi(\gamma)$ (s.e. 0.0018) | $\times k_e$ | $m_\varepsilon(\gamma)-$ first order |
|---|---|---|---|---|---|---|
| 10 | $-0.0276$ | $-0.28$ | $-0.0021$ | $-0.0441$ | $-0.44$ | $-0.0015$ |
| 20 | $-0.0179$ | $-0.36$ | $-0.0051$ | $-0.0269$ | $-0.54$ | $-0.0056$ |
| 40 | $-0.0089$ | $-0.36$ | $-0.0025$ | $-0.0127$ | $-0.51$ | $-0.0020$ |
| 80 | $-0.0046$ | $-0.37$ | $-0.0015$ | $-0.0083$ | $-0.67$ | $-0.0030$ |

The discrepancy between the real loop and the Gibbs measure falls like $k_e^{-0.87}$ ($\alpha$), $k_e^{-0.83}$ ($\gamma$), $k_e^{-0.72}$ ($\alpha^2$) and $k_e^{-1.09}$ ($\alpha\gamma$) ($\gamma^2$ has a first-order coefficient 0.03, below resolution), and the first-order prediction removes most of it: 93% and 97% of the $\alpha$ and $\gamma$ discrepancies at $k_e = 10$, 70–85% at $k_e = 20$–40; at $k_e = 80$ the residuals ($-0.0015 \pm 0.0013$ and $-0.0030 \pm 0.0018$) are at the Monte Carlo floor. The second-order decay of the residual is *not* resolved: the residuals at $k_e = 20$–80 are 1–3 standard errors with a common sign, compatible with an $O(\varepsilon^2)$ coefficient of order $-1$ to $-2$ and with noise; at $k_e = 10$ the even part of the $\alpha^2$ and $\gamma^2$ shifts exceeds its prediction by a factor 1.5–1.8 (an $O(\varepsilon^2)$ term of the size seen in Theorem J's instance A, where $c_2/c_1 \approx 1.4$).

*The orientation-odd part (Remark J′.1).* Measured $(m_+ - m_-)/2$ over predicted $(p_{1+} - p_{1-})/(2k_e)$; Table 25.

Table 25: Theorem J′ on the single-kernel instance: the orientation-odd part of the stationary means, measured over predicted (`results/dither/table.md`).

| $k_e$ | $\alpha$ | $\gamma$ | $\alpha\gamma$ |
|---|---|---|---|
| 10 | $0.92 \pm 0.04$ | $0.93 \pm 0.04$ | $0.89 \pm 0.06$ |
| 20 | $1.23 \pm 0.09$ | $1.10 \pm 0.07$ | $1.28 \pm 0.13$ |
| 40 | $1.28 \pm 0.17$ | $1.15 \pm 0.15$ | $1.03 \pm 0.26$ |
| 80 | $1.25 \pm 0.34$ | $1.38 \pm 0.29$ | $1.11 \pm 0.51$ |

The dither version of Remark H.7a's test: the direction in which the reference is traversed shifts the stationary kernel by $2\varepsilon \int \Theta g_{\mathrm{odd}}\, \mathrm{d}\pi$ (here entirely from $b_{\mathrm{osc}}$), measured to within 7–8% at $k_e = 10$ (where the odd shift is 0.02–0.03 against standard errors of 0.001) and consistent with the prediction within the 10–30% resolution of the runs at $k_e = 20$–80 (the ratios sit 10–30% above one with error bars of the same size: a second-order odd term of relative size $O(\varepsilon)$ is not excluded).

*The tracking error (d).* $k_e^2 \mathbb{E}\|e\|^2/(2\pi(\bar{L})) = 1.054$, 1.033, 1.017, 1.009 (orientation $+1$) and 1.023, 1.014, 1.008, 1.004 ($-1$) at $k_e = 10$, 20, 40, 80, with standard errors 0.001: the deviation from one is 0.54–0.74/$k_e$ and 0.23–0.35/$k_e$, the $O(1/k_e)$ of (d).

*The circulation (e) and Proposition J′.2.* The predicted first-order circulation $\varepsilon[\int \Theta \wedge j_1 + \pi\langle u \wedge \dot{u}\rangle]$ is $-5 \cdot 10^{-4}$ ($+1$) and $+6.5 \cdot 10^{-4}$ ($-1$) at $k_e = 10$ and scales like $1/k_e$, against a Monte Carlo standard error of $5 \cdot 10^{-4}$ at every $k_e$: the real loop's circulation is not resolved at first order for $k_e \geq 20$ (measured $-4 \cdot 10^{-4} \cdots + 1.2 \cdot 10^{-3}$, all within 1.7 standard errors of the prediction), and at $k_e = 10$ the measurement ($+3.7 \cdot 10^{-3} \pm 0.5 \cdot 10^{-3}$ for $+1$, $-0.6 \cdot 10^{-3} \pm 0.5 \cdot 10^{-3}$ for $-1$) is dominated by higher orders (an even $O(\varepsilon^2)$ term of coefficient $\approx +0.2$ and an odd one of the opposite sign to the first-order odd part). The current machinery is therefore checked where the signal is clean, on the first-order averaged SDE itself: its stationary density $m_\varepsilon$ and current $j_\varepsilon$, computed on the grid (the discrete current at $\varepsilon = 0$, $-6.6 \cdot 10^{-5}$, a residual of the central scheme, subtracted), give $\int \Theta \wedge (j_\varepsilon - j_0)/(\varepsilon \int \Theta \wedge j_1) = 1.47$, 1.21, 1.09, 1.03 at $\varepsilon = 0.2$, 0.1, 0.05, 0.025 — one plus a term linear in $\varepsilon$ — and $\|m_\varepsilon - m_0(1 + \varepsilon g)\|_1/\varepsilon^2 = 0.307$, 0.308, 0.307, 0.305: the first-order expansion of Proposition J′.2(i) with a clean $\varepsilon^2$ remainder.

*Corollary J′.3* (`exp_gibbs_m.py` → `results/dither/gibbs_m.md`). On the same instance, $\mu_\tau^*$ is computed on the $201^2$ grid as the fixed point of the first-variation condition (59 damped iterations, residual $10^{-11}$): $\int \alpha \, \mathrm{d}\mu^* = 0.4963$, $\int \gamma \, \mathrm{d}\mu^* = 0.1873$, $\int \alpha^2 \, \mathrm{d}\mu^* = 0.9353$, $\bar{L}(\mu_\tau^*) = 0.1394$. The Gibbs measures $\pi_m$ for $m = 1$, 2, 4, 8, 16, 32 are sampled by MALA (4000 replicas, acceptance 0.98–0.996, standard errors from the independent replicas; the $m = 1$ row agrees with the quadrature of $\pi_1$ within one standard error); Table 26.

Table 26: Corollary J′.3 on the single-kernel instance: the one-particle marginals of the Gibbs measures $\pi_m$ against the mean-field minimiser $\mu_\tau^*$, with the rescaled deviations (`results/dither/gibbs_m.md`).

| $m$ | $\mathbb{E}_{\pi_m}[\alpha_1^2]$ | $\mathbb{E}_{\pi_m}[\gamma_1]$ | $\mathbb{E}_{\pi_m}[\bar{L}(\mu_\Theta)]$ | $(\mathbb{E}[\alpha_1^2] - \int \alpha^2 \, \mathrm{d}\mu^*) \cdot m$ | $(\mathbb{E}[\gamma_1] - \int \gamma \, \mathrm{d}\mu^*) \cdot m$ | $(\mathbb{E}[\bar{L}] - \bar{L}(\mu^*)) \cdot m$ |
|---|---|---|---|---|---|---|
| 1 | 0.536 | 0.251 | 0.1960 | −0.40 | +0.064 | +0.057 |
| 2 | 0.729 | 0.225 | 0.1681 | −0.41 | +0.075 | +0.057 |
| 4 | 0.832 | 0.204 | 0.1540 | −0.41 | +0.067 | +0.058 |
| 8 | 0.881 | 0.194 | 0.1470 | −0.43 | +0.050 | +0.061 |
| 16 | 0.909 | 0.191 | 0.1432 | −0.42 | +0.051 | +0.060 |
| 32 | 0.922 | 0.190 | 0.1413 | −0.42 | +0.084 | +0.059 |

The one-particle marginals converge to $\mu_\tau^*$ with a clean $1/m$ law (slopes $-1.01$ for $\gamma$ and $-0.99$ for the Gibbs-averaged occupation loss, whose rescaled deviation is $0.058 \pm 0.002$ at every $m$; the second moment of $\alpha$ moves from 0.54 at $m = 1$ to 0.92 at $m = 32$ against 0.935, with rescaled deviation $-0.41 \pm 0.01$ throughout: the single kernel sits in a narrow well, while in the mean field each particle sees the wide one-body potential $V[\mu_\tau^*]$); the mean of $\alpha$ is within 0.01 of its limit already at $m = 1$ (a small $1/m$ coefficient, its slope not resolved). The stationary law of the finite dithered loop thus approaches the minimiser of Theorem H's free energy in the iterated limit at the rate $1/m$, as the corollary states.

Checked by: `tests/test_dither.py`: the single-kernel averaged data ($J$, $\nabla J$, $b_{\mathrm{fr}}$, $b_{\mathrm{osc}}$) against the general implementation of Proposition H.7 and Remark H.7a (`averaging.expansion_terms`, `averaging.oscillation_correction`, `periodic.ClosedLoop`) at three parameter points ($2 \cdot 10^{-5}$ relative); the weighted Poisson solver against the Ornstein–Uhlenbeck closed forms ($g = G^{-1} b \cdot \Theta/\tau_0$, mean shift $(G\lambda)^{-1} b$, the Green identity $\int \varphi g \, \mathrm{d}\pi = -\pi(b \cdot \nabla \psi_\varphi)$, zero current and entropy production for a gradient drift); the stationary Fokker–Planck solver of the first-order SDE (OU exact; on the instance the current ratio $\to 1$ and $\|m - \pi(1 + \varepsilon g)\|_1 \propto \varepsilon^2$); the real loop at $k_e = 20$ (8000 replicas: the stationary means of $\alpha$ and $\gamma$ closer to the first-order prediction than to $\pi$, the $\gamma$ shift resolved at $\geq 3$ standard errors, the tracking error within 10% of $2\pi(\bar{L})$); and the Corollary J′.3 numerics (the mean-field fixed point to $10^{-9}$, the Gibbs sampler against the quadrature of $\pi_1$)
(`test_single_kernel_averaged_data_matches_the_general_H7_implementation`,
`test_poisson_solver_reproduces_the_ornstein_uhlenbeck_closed_forms`,
`test_first_order_stationary_density_and_current_of_the_averaged_sde`,
`test_real_loop_means_are_closer_to_the_first_order_prediction_than_to_gibbs`,
`test_corollary_J3_mean_field_minimiser_and_gibbs_sampler_consistency`).

# Part III
# Evidence and outlook

## 25 Numerical evidence

Every number in this section is produced by the named script from raw files, and no table is edited by hand. The scripts are in the directory `experiments/` of the accompanying repository and write their output to subdirectories of `results/`; "`script.py` → `dir/`" below names a script and its output directory or file.

- The test files check the inequalities at every sample of simulated trajectories, or on sampled states: `tests/test_transport.py`, `tests/test_langevin.py`, `tests/test_averaging.py`, `tests/test_excitation.py`, `tests/test_nonreciprocity.py`, `tests/test_periodic.py` and `tests/test_barycentric.py`.
- They are run with `venv\Scripts\python -m pytest tests`. The current number of tests is given in the repository's `README.md`.

### 25.1 Theorem B: the label-free Lyapunov function

Script: `exp_transport_permutation.py` → `transport_permutation/`.

Each network starts as a random relabelling of the ideal configuration plus noise 0.05, with mean-field gains and 3 seeds per $m$. "Excess" is $|(1/m)\sum \text{Taylor remainders}|$, the part that actually enters $\dot{V}$. Table 27 and Figure 4 show the results.

Table 27: Theorem B, permuted-ideal start, mean-field gains; means over 3 seeds. The two bounds are the same bound, evaluated with the matched (OT) error and with the labelled error, along the same trajectory.

| $m$ | labelled error ($t=0$) | OT error $W$ ($t=0$) | RMS$\|e\|$ | actual excess | bound, matched error | same bound, labelled error |
|---|---|---|---|---|---|---|
| 8 | 2.675 | 0.1070 | 0.0113 | $1.96 \cdot 10^{-5}$ | $8.27 \cdot 10^{-2}$ | 2.93 |
| 16 | 2.532 | 0.1044 | 0.0065 | $9.58 \cdot 10^{-6}$ | $4.37 \cdot 10^{-2}$ | 1.86 |
| 32 | 2.684 | 0.1122 | 0.0079 | $2.69 \cdot 10^{-5}$ | $7.07 \cdot 10^{-2}$ | 2.11 |
| 64 | 2.764 | 0.1136 | 0.0094 | $1.23 \cdot 10^{-5}$ | $6.80 \cdot 10^{-2}$ | 2.43 |

- The bound is at least the actual excess at every sample of every run.
- The matched error is about 1/25 of the labelled one.
- The bound is 30–45× smaller when evaluated with the matched error.
- Nothing grows with $m$.
- The bounds remain far above the actual excess, because the Hessian supremum is global.

**Corollary B.5 along** ($L_\sigma$)**.** This is checked by the test `test_corollary_B5_fails_along_lsigma_and_its_leak_variant_holds`. The test asserts every hypothesis of C(iv) and measures $\sup W \le \bar{\eta}$. There, the plain B.5 bound is exceeded by more than 3×, while the leak variant with $\sigma_m b^2/4$ holds.

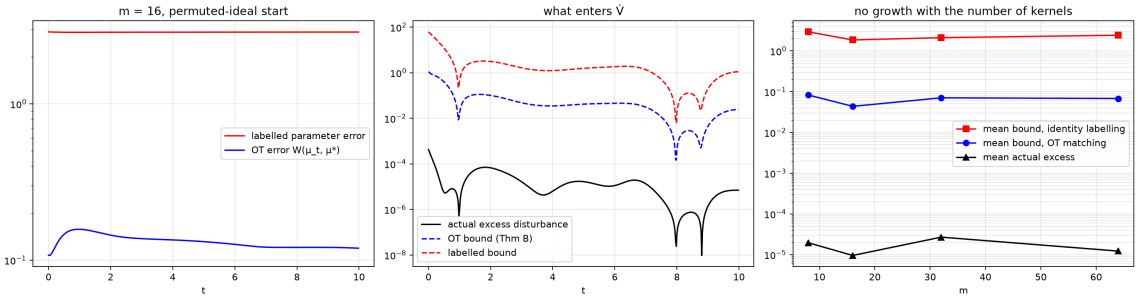



Figure 4: Theorem B, permuted-ideal start. Left: labelled parameter error and OT error $W(\mu_t, \mu^*)$ for $m = 16$. Centre: the actual excess disturbance that enters $\dot{V}$, the OT bound of Theorem B and the labelled bound, for $m = 16$. Right: the mean bound with identity labelling, the mean bound with OT matching and the mean actual excess, against $m$.

## 25.2 Theorem C: OT $\sigma$-modification

Scripts: `exp_ot_sigma_mod.py` and `exp_sigma_mod_transient.py` → `ot_sigma_mod/`.

The two starts share plant, prior, geometry and $x_0$. They differ only in particle labels (Table 28, Figure 5). Table 29 shows the reshuffling transient from the relabelled start (5 seeds).

Table 28: Theorem C: OT $\sigma$-modification against labelled $\sigma$-modification (oracle plant, $m = 6$, 5 seeds, $\sigma_m = 0.3$). RMS$\|e\|$ is taken over 20–40 s and reported as mean ± 95% confidence interval over seeds; $W_{\text{end}}$ is the OT error at the end.

| start | mechanism | RMS$\|e\|$ (20–40 s) | $W_{\text{end}}$ | labelled error at end |
|---|---|---|---|---|
| aligned | projection | $0.0051 \pm 0.0010$ | 0.2953 | 0.2953 |
| aligned | labelled $\sigma$-mod. | $0.0091 \pm 0.0018$ | 0.3040 | 0.3040 |
| aligned | OT $\sigma$-mod. | $0.0091 \pm 0.0018$ | 0.3040 | 0.3040 |
| relabelled | projection | $0.0051 \pm 0.0010$ | 0.2953 | 0.2953 |
| relabelled | labelled $\sigma$-mod. | $0.0091 \pm 0.0018$ | 0.3040 | **2.0590** |
| relabelled | OT $\sigma$-mod. | $0.0091 \pm 0.0018$ | 0.3040 | 0.3040 |

Table 29: Reshuffling transient from the relabelled start ($m = 6$, 5 seeds, $\sigma_m = 0.3$): RMS$\|e\|$ over time windows, and the particle path length.

| leak | $[0, 5]$ s | $[5, 10]$ s | $[10, 20]$ s | $[20, 40]$ s | particle path length |
|---|---|---|---|---|---|
| labelled | 0.0771 | 0.0086 | 0.0093 | 0.0091 | 2.2585 |
| OT | 0.0743 | 0.0072 | 0.0093 | 0.0091 | 0.3917 |

- The labelled leak's longer path costs only about 4% of early RMS$\|e\|$. Theorem F does not price leak-driven transport (Proposition C.5).
- The C(iv) instance in the tests satisfies every hypothesis, asserted. $V_{\text{OT}}$ stays below the envelope $e^{-\alpha t}V(0) + \beta/\alpha$ at every sample and contracts by more than 5×.
- The C(ii) test asserts $C \le b^2 + 4\eta R$, $\sqrt{C} \le b + 2\eta$, and $C = b^2$ under both gap conditions, on 300 configurations.

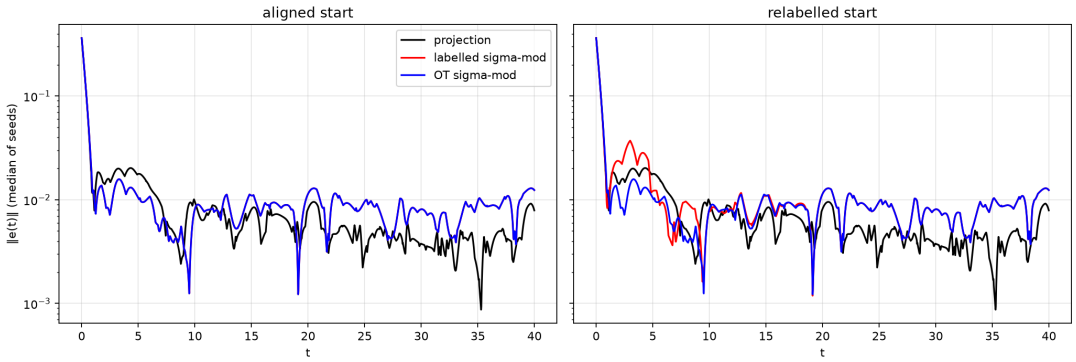



Figure 5: Theorem C: $\|e(t)\|$ (median of seeds) for projection, labelled $\sigma$-modification and OT $\sigma$-modification, from the aligned start (left) and from the relabelled start (right).

### 25.3 Theorem C′: the barycentric OT leak

Script: `exp_barycentric_leak.py` → `barycentric_leak/`. Table 30 shows the pure leak toward a prior with more atoms than the network, and Table 31 the closed loop with three leaks of equal gain toward three priors.

Table 30: Theorem C′: pure barycentric leak toward a 48-atom prior ($m = 6$, $n = 1$, lifted dimension 3, $G = \mathrm{diag}(2, 1, 0.5)$, $\sigma_m = 1$, $\mathrm{d}t = 0.01$, 3000 steps). The last column is the best of 20 equal-mass Lloyd runs.

| seed | $W^2(\mu_0, \nu)$ | $W^2(\mu_T, \nu)$ | $V^2_{\mathrm{bar}}$ at $T$ | $\mathrm{Var}_\gamma$ at $T$ | $W^2$ non-increasing | plan switches | best Lloyd |
|---|---|---|---|---|---|---|---|
| 0 | 2.5526 | 1.1642 | $1.1 \cdot 10^{-26}$ | 1.1642 | yes | 3 | 1.1708 |
| 1 | 2.3032 | 1.2128 | $4.8 \cdot 10^{-26}$ | 1.2128 | yes | 2 | 1.1708 |
| 2 | 2.5807 | 1.2030 | $2.2 \cdot 10^{-25}$ | 1.2030 | yes | 6 | 1.1708 |

Table 31: Theorem C′: closed loop with three leaks of equal gain ($\sigma_m = 0.5$) toward three priors (representable plant, $m = 4$, $k_e = 4$): (A) the OT leak to the ideal, (B) the barycentric leak to a 32-atom cloud, (C) the labelled leak to a random 4-subset. Each entry of (A)–(C) is RMS$\|e\|$ on $[20, 40]$ s, $W(\mu_T, \mu^*)$. Here $b = W(\mu^*, \mathrm{cloud})$, $b_{\mathrm{bar}}$ is the centroid defect of the ideal with respect to the cloud, and "labelled" is the labelled bias to the subset.

| seed | $b$ | $b_{\mathrm{bar}}$ | $W(\mu^*, \mathrm{subset})$ | labelled | (A) | (B) | (C) |
|---|---|---|---|---|---|---|---|
| 0 | 0.446 | 0.168 | 0.647 | 1.987 | 0.0000, 0.000 | 0.0035, 0.167 | 0.0431, 0.614 |
| 1 | 0.328 | 0.097 | 1.027 | 1.479 | 0.0000, 0.000 | 0.0024, 0.097 | 0.0430, 1.016 |
| 2 | 0.401 | 0.159 | 0.395 | 1.765 | 0.0000, 0.000 | 0.0069, 0.157 | 0.0293, 0.352 |

- The pure leak stops exactly where $W^2 = \mathrm{Var}_\gamma$ (every particle at the barycentre of its cell; $V^2_{\mathrm{bar}} \le 2.2 \cdot 10^{-25}$): an equal-mass centroidal configuration of the prior. Like Lloyd's algorithm it finds a local minimiser of the quantisation energy, not always the global one (compare the last two columns of Table 30).
- With $m_0 = m$ the same leak gives the exact decay of Theorem C(i): the maximal relative deviation of $W(\mu_t, \mu^0)$ from $(1 - \sigma_m \, \mathrm{d}t)^k W_0$ over 500 Euler steps is $5.1 \cdot 10^{-16}$.
- In the closed loop, the bias of the barycentric leak is governed by $b_{\mathrm{bar}}$, the distance of the ideal from the centroids of the prior's cells: $W(\mu_T, \mu^*) = 0.167, 0.097, 0.157$ against $b_{\mathrm{bar}} = 0.168, 0.097, 0.159$, not by $b = W(\mu^*, \nu) = 0.45, 0.33, 0.40$ (Theorem C′(ii)–(iii)).

The labelled leak to the subset ends at $W(\mu_T, \mu^*) \approx W(\mu^*, \text{subset})$. A rich prior whose centroidal quantisation is close to the ideal is a better anchor than a crude prior with the right number of atoms.

- Tests (`tests/test_barycentric.py`): the plan reduces to the assignment for $m_0 = m$ and $W^2 = V_{\text{bar}}^2 + \text{Var}_\gamma$; the pure leak dissipates at every step with $D^+ Z \le -\sigma_m V_{\text{bar}}^2$ and ends at a Lloyd fixed point; $\sqrt{C_{\text{bar}}} \le b + 2\eta$ on 90 configurations, with $C_{\text{bar}} = b_{\text{bar}}^2$ at small $\eta$; the Lyapunov inequality (iii) along a closed loop with a 24-atom prior.

## 25.4 Theorem D: convergence in $m$

Script: `exp_mean_field_m.py` → `mean_field_m/`. Table 32 and Figure 6 show the results.

Table 32: Theorem D: mean-field consistency in $m$ (Van der Pol plant, mean-field gains, smooth initial measure quantised at $m$ points). The ratio is $W(\mu_T^m, \mu_T^{2m}) / W(\mu_0^m, \mu_0^{2m})$.

| $m$ | $\sup_t \|e^m - e^{2m}\|$ | $W(\mu_0^m, \mu_0^{2m})$ | $W(\mu_T^m, \mu_T^{2m})$ | ratio |
|---|---|---|---|---|
| 16 vs 32 | 0.0046 | 0.2179 | 0.3658 | 1.68 |
| 32 vs 64 | 0.0009 | 0.1091 | 0.1788 | 1.64 |
| 64 vs 128 | 0.0001 | 0.0545 | 0.0920 | 1.69 |
| 128 vs 256 | 0.0000 | 0.0273 | 0.0459 | 1.68 |

This is a consistency check in $m$, not a check of the constant.

- The guaranteed factor $e^{\Lambda T}$ is at least $e^{496}$ on this box.
- The Van der Pol plant fails the tube condition of Theorem D(b), so the run is covered by the localised D(d).

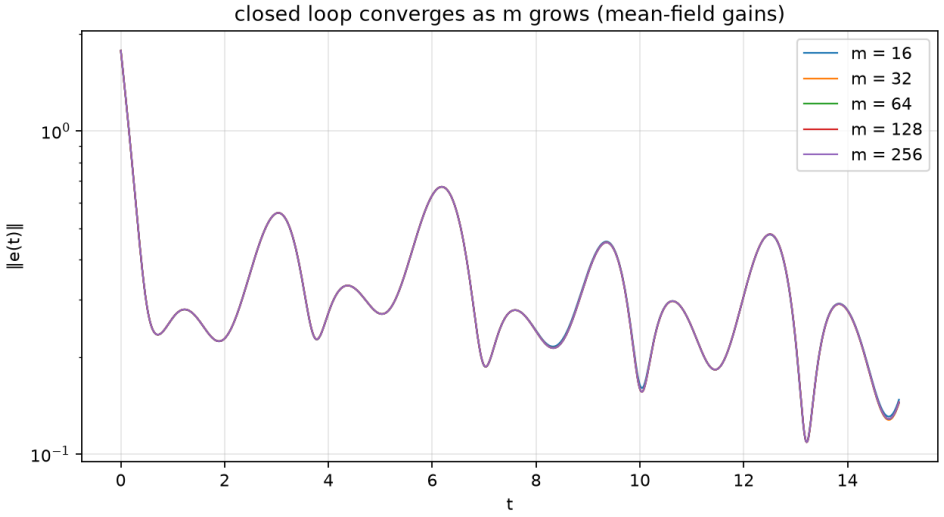



Figure 6: Theorem D: tracking error $\|e(t)\|$ of the closed loop (Van der Pol plant, mean-field gains) for $m = 16$, 32, 64, 128 and 256.

## 25.5 Theorem D′: propagation of chaos for the dithered loop

Script: `exp_chaos.py` → `chaos/` (raw per-seed values in `raw_sup_D.npz`). The $m$-particle loop ($n = 1$, $k_e = 10$, $\sigma = 0.02$, $G = \text{diag}(4, 1, 1)$, $T = 20\,\text{s}$, $dt = 0.001$; Euler–Maruyama for the particles and RK4 for the plant) is coupled synchronously to an $M = 4096$ reference that shares its

first $m$ particles and Brownian motions, and $\sup_t D_m$ with $D_m := \|x^m - x^M\| + \frac{1}{m}\sum_i \|\vartheta_i^m - \vartheta_i^M\|$ is averaged over 24 seeds (Table 33).

Table 33: Theorem D′: synchronous coupling of the $m$-particle dithered loop to an $M = 4096$ reference, 24 seeds. Slopes are log-log fits, with bootstrap standard errors over seeds.

| | $\tau = 0$ | $\tau = 2 \cdot 10^{-3}$ |
|---|---|---|
| mean $\sup_t D_m$, $m = 16 \to 1024$ | $1.49 \cdot 10^{-1} \to 2.17 \cdot 10^{-2}$ | $1.52 \cdot 10^{-1} \to 2.32 \cdot 10^{-2}$ |
| slope of the mean against $m$ ($m \le 256$) | $-0.47 \pm 0.06$ | $-0.46 \pm 0.05$ |
| slope of the median | $-0.46$ | $-0.45$ |
| slope against $(1/m - 1/M)^{1/2}$ (predicted 1) | $0.94 \pm 0.08$ | $0.92 \pm 0.08$ |

- The $m^{-1/2}$ rate of Theorem D′(b) holds within one standard error at both temperatures; the shared-particle coupling predicts the mean $\propto (1/m - 1/M)^{1/2}$, which the data follow up to $m = M/4$.
- Single seeds differ by more than an order of magnitude at fixed $m$; a three-seed version of the experiment gave slopes $-0.25$ and $-0.21$ and was withdrawn (review round 5). The constant of the theorem ($TK_\Phi e^{\Lambda T}$) is not checked.
- There is no test: the experiment is the check.
- Theorem D″ (`exp_chaos_uniform.py` → `chaos/uniform_table.md`; $n = 1$, $k_e = 25$, $G = 0.05 \cdot \mathrm{diag}(4, 1, 1)$, a two-bump plant that is not odd, $M = 2048$): exact Lipschitz constants $L_x = L_\vartheta = 8.087$, $L_2 = L_{x\vartheta} = 44.4$ and the a-priori tube $E_X = 0.228$ give $\sigma_{\mathrm{cert}} = 4.04$, against Proposition K's critical leak $\sigma_K = 0.00144$ of the collapsed critical point (ratio 2800). At $1.2\sigma_{\mathrm{cert}}$ with dither $\tau = 0.02$ (80 s, 8 seeds) the coupled distance is $6.7 \cdot 10^{-4} \to 7.9 \cdot 10^{-4}$ ($m = 16$) and $1.6 \cdot 10^{-4} \to 1.9 \cdot 10^{-4}$ ($m = 256$) between the first and the last window, slope $-0.51 \pm 0.02$, bound 0.55 and 0.14; the network stays at its prior (fit error 1.000, rms spread 0.033 = the dither cloud). Without dither the reference collapses (spread $10^{-14}$) and the distance is zero; at $\sigma_{\mathrm{cert}}/8$ it decays to $10^{-11}$. In the learning regime ($\sigma_K/3$, 1000 s, 6 seeds) the network stays spread (rms 1.53) and fits the plant (fit error 0.632), and the distance follows $m^{-1/2}$ (slope $-0.62 \pm 0.16$) but grows threefold between the windows $(100, 200]$ and $(500, 1000]$: no empirical uniformity there (the first draft's "spread regime" run was a slow collapse; review round 6).

## 25.6 Theorem D‴: uniform-in-time propagation of chaos in the learning regime

Script: `exp_chaos_learning.py` → `chaos/learning_table.md`.

- Instance: $n = 1$, $G = \mathrm{diag}(1, 0.25, 0.25)$, three-bump plant $(1, -1.5, 0.5)$, $(-0.7, 0, 0.4)$, $(0.8, 1.5, 0.5)$, $x_m = 2\sin t$, $\sigma_{\mathrm{crit}} = 0.04716$ (Proposition K), $\sigma_0 = 0.15\sigma_{\mathrm{crit}} = 0.00707$; $L_\vartheta^G = 4.044$, $L_2^G = 11.11$, $K_\Phi = 4$, $F_\rho = 0.999$. Averaged slow loop ($\mathfrak{A}_m$), $m = 16$, 64, 256 against $M = 2048$, 900 slow units, 4 seeds per start; real loop at $k_e = 25$ over 2000 time units, 3 seeds.
- The hypothesis at the three-atom stationary measure (fit error 0.244): $\kappa_R$ at the atoms 0.00004, $-0.00227$, 0.00411; over spheres of radius 0.05, 0.1, 0.2, 0.3: 0.00461, 0.00512, 0.00619, 0.00730 — (H) holds with $\varrho' = 0.1$ and margin 0.00195 ($1/\kappa = 512$), the window $\kappa_R[\mu^*](N_{0.1}) = 0.00512 < \sigma_0 = 0.00707 < \sigma_{\mathrm{crit}} = 0.04716$ is realised; at $0.3\sigma_{\mathrm{crit}}$ the three-atom measure fails splitting ($\kappa_R = 0.01696 > 0.01415$). Theorem constants: $C_1 = 2.2 \cdot 10^9$, $m_0 = 1.2 \cdot 10^{23}$ (vacuous envelopes).

- Along the warm run (jittered mixture around the atoms) the margin $a(s)$ stays in $[+0.00139, +0.00167]$; the own-residual statistic exceeds the $r^*$-one by $\le 0.00013$ ((0.2) allows 0.214); rms drift $6 \cdot 10^{-3} \to 3 \cdot 10^{-6}$. Along the cold run (connected cloud) the margin is negative throughout ($-0.196 \to -0.0025$) while the fit converges to 0.244 as well.
- Warm start: window maxima of $D_m$ 0.396 $\to$ 0.448 ($m = 16$), 0.122 $\to$ 0.138 (64), 0.0119 $\to$ 0.0137 (256) over $(0, 112.5] \to (450, 900]$ (last/first 1.13–1.15, last/previous 1.01–1.02: saturated); slope $-1.28 \pm 0.30$ (mean square; theorem $-1$), learned function $-0.69 \pm 0.16$; per-particle maxima 1.36, 0.62, 0.21 $\gg 3\varrho'/4 = 0.075$ (Remark D‴.6).
- Proposition D‴.5: $\|S_k\| = 8.38$, 4.66, 8.32; $\mathbb{E}D_\infty$ predicted 0.823, 0.201, 0.0454 against means $0.446 \pm 0.28$, $0.138 \pm 0.051$, $0.0136 \pm 0.0055$ (four seeds, heavy-tailed); per seed the linearised prediction is within 25% of the measured $D$ in 9 of 12 pairs (worst ratio 0.57 at $|w^m - w^M| = 0.275$) and the exact stationary distance of the replicated weighted configuration within 2.3% in all 12; splitting margins positive in 11 of 12 (+0.0006 to +0.0050), negative ($-0.0022$) for $m = 16$, seed 3 (weights $(9, 4, 3)/16$: a saddle, left later by particle migration — review round 9); per-cluster displacements along $S_k(w^m - w^M)$ with cosines 0.92–1.000 and norms within 0.8–20%.
- Cold start: $D$ grows 1.81 $\to$ 5.33 (16), 0.505 $\to$ 1.79 (64), 0.0744 $\to$ 0.583 (256) (last/first 2.95, 3.54, 7.84), slope $-0.87 \pm 0.14$; bimodal deviations (medians 0.40, 0.14, 0.067; maxima 6.2–6.7); fraction of pairs $> 1$ whitened unit apart $0.203 \pm 0.047$, $0.082 \pm 0.021$, $0.037 \pm 0.010$, decaying like $m^{-0.64 \pm 0.08}$ (medians $m^{-0.62 \pm 0.07}$): the floor of Remark D‴.2.
- Real loop ($k_e = 25$, warm start): window maxima 0.328 $\to$ 0.590 (16), 0.0605 $\to$ 0.0909 (64), 0.0188 $\to$ 0.0294 (256) over $(0, 500] \to (1000, 2000]$ (slow time $\le 80 = 0.16/\kappa$), slope $-1.13 \pm 0.15$; on the same initial clouds the averaged flow gives $D_{\text{real}}/D_{\text{averaged}}$ between 0.993 and 1.004 at every window end (across different clouds the last-window ratios are 1.57, 0.78, 2.58): the real loop follows its averaged slow dynamics to within 0.7% (Corollary D‴.7); two of the three $m = 16$ clouds have inadmissible or marginal weights (margins $-0.0021$, $-0.00001$).

### 25.7 Proposition D‴.9: measure-valued priors and the design of the window

Script: `exp_prior_design.py` → `chaos/prior_design.md`.

- Same instance and leak as Theorem D‴'s numerics ($\sigma_0 = 0.15\sigma_{\text{crit}} = 0.00707$), nearest-atom leak toward three prior atoms. Matching prior (exact representation): fit error 0, $\kappa_R = 0$, margin $\sigma_0 = 0.00707$ against the $\sigma$-modification's 0.00196 (the theorem's relaxation time $1/\kappa$ 141 against 511 slow units; observed late-time e-folding of the coupled distance 48–84 against 107–306); $\|S_k\| = 13.01$, 6.18, 8.91 against 8.38, 4.66, 8.32; $\mathbb{E}D_\infty(m = 64)$ 0.323 against 0.201 (1.61×); function-level floor $9.33 \cdot 10^{-4}$ against $4.29 \cdot 10^{-4}$ (2.18×).
- Misplaced priors: centres +0.15 / +0.3 / +0.45 / +0.6 → margins +0.0055 / +0.0037 / +0.0017 / −0.0006; amplitudes ×0.5 / ×0.7 / ×1.2 → +0.0074 / +0.0073 / −0.0317; widths ×0.7 / ×1.5 / ×2 → −0.0261 / +0.0058 / +0.0040; the shrunk priors (×0.5, ×0.7) have the largest margins and lower floors ($\mathbb{E}D_\infty$ 0.199, 0.230; function level $7.1 \cdot 10^{-4}$, $7.7 \cdot 10^{-4}$) at fit errors 0.09, 0.06.
- Admissible weights (splitting margin of $\Theta^*(w)$): $(7, 5, 4)/16$: +0.0026 / −0.0054 / +0.0073; $(9, 4, 3)/16$: −0.0022 / −0.0058 / +0.0057; $(10, 3, 3)/16$: −0.0038 / −0.0042 / +0.0054; $(0.5, 0.3, 0.2)$: −0.0006 / −0.0065 / +0.0066 ($\sigma$-modification / matching / amplitudes ×0.7): the shrunk prior keeps all four admissible (above $\sigma_0$ only at $(7, 5, 4)/16$; its over-weighted atom overshoots by 1–3% at the other three), and the matching prior's set is smallest toward the largest bump (an excess 0.04 on atom 1 alone is inadmissible, 0.08 on atom 2 or 3 alone is not).

- Matching-prior coupled runs ($m = 16$, 64, 256 vs $M = 2048$, 300 slow units, 3 seeds): $D$ saturates (last/previous 1.04–1.05), slope $-1.23 \pm 0.40$; per-seed linearised-law ratios 0.81–1.11 (parameter level) and 0.61–1.69 (function level), exact law 0.987–0.998 in all nine pairs; two of nine realised weight vectors are saddles (margins $-0.0005$, $-0.0030$), left by a local split inside the cell with the coupled distance on its plateau (round 10); observed e-folding times 48–84 slow units (median 59) against 107–306 (median 195) for the $\sigma$-modification.
- Function-level law for the $\sigma$-modification against Theorem D‴'s warm runs: predicted RMS 0.0419, 0.0207, 0.0098 vs measured means $0.0307 \pm 0.0097$, $0.0154 \pm 0.0033$, $0.0047 \pm 0.0012$ (ratios 0.73, 0.74, 0.48; RMS over seeds/predicted 0.84, 0.79, 0.52; per-seed law from the realised weights 0.60–1.57).

## 25.8 Proposition D‴.12: the OT leak's mass constraint against the weight fluctuation

Script: `exp_ot_leak.py` → `chaos/ot_leak.md`.

- Same instance and matching prior as Proposition D‴.9's numerics ($\sigma_0 = 0.15\sigma_{\text{crit}}$, $q = (1/3, 1/3, 1/3)$, $\Delta_\nu = 6.26$); the nearest-atom and the barycentric OT leak on the same warm clouds, 600 slow units ($4.2/\sigma_0$), 4 seeds; $m = 15$, 60, 240 vs $M = 510$ (commensurate) and $m = 16$, 64, 256 vs $M = 512$ (incommensurate).
- Cluster weights: conserved exactly under the nearest-atom leak (max deviation 0 in 32 systems); under the OT leak the mass in transit $\frac{1}{2}|w(0) - q|_1 = 0.117$, 0.058, 0.027, 0.024 (seed means; 1–21 particles per system) reaches zero and the weights equal $q$ from $s_{\text{eq}} = 93$–164 slow units ($\ln 2/\sigma_0 = 98$). Fit error at the end (seed means): 0.067, 0.029, 0.017, 0.012 (nearest) $\to 2.8 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$, $0.8 \cdot 10^{-4}$ (OT; the migrated particles still 0.06–0.16 from their atoms, the OT finals not yet stationary).
- Commensurate sizes, last window: function-level distance$^2$ $2.5 \cdot 10^{-3}$, $5.2 \cdot 10^{-4}$, $5.9 \cdot 10^{-5}$ (nearest; law predicts $4.0 \cdot 10^{-3}$, $9.1 \cdot 10^{-4}$, $1.4 \cdot 10^{-4}$) vs $4.9 \cdot 10^{-7}$, $4.9 \cdot 10^{-8}$, $7.5 \cdot 10^{-9}$ (OT, still falling $\times 0.02$–$0.035$ per 150 slow units); $W^2$ 5.1, 2.7, 0.81 (nearest; cross-cluster transport $\frac{1}{2}|w^m - w^M|_1 \Delta_\nu^2 = 4.9$, 2.5, 0.70; slope $-0.67 \pm 0.13$) vs $7.9 \cdot 10^{-3}$, $2.3 \cdot 10^{-3}$, $6.9 \cdot 10^{-4}$ (OT, $\times 0.12$ per 150 slow units); synchronous $D$ 0.87, 0.25, 0.017 (nearest) vs 5.8, 3.3, 0.94 (OT: the reassigned particles; at 0.95–0.96 of the mismatch plateau and still rising). Exact $K \times K$ transport between the weighted stationary measures at the realised weights: 0.997–0.999 of the measured $W^2$ at the seven admissible pairs.
- Incommensurate sizes: one or two split (bridge) particles per system at 0.28–0.58 $\Delta_\nu$ from the nearest atom; against $\nu$ (the mean-field stationary measure): $W^2(\mu_m, \nu) = 1.21$, 0.33, 0.079, 0.038, $m \cdot W^2 = 19.4$–21.0 (slope $-0.98 \pm 0.03$), fit error 0.028, 0.0093, 0.0023, 0.0011, $m \cdot$ fit = 0.45–0.60 (squared slope $-1.81 \pm 0.05$); against the $M = 512$ reference (two bridge particles of its own) OT $3.4 \cdot 10^{-4}$, $3.1 \cdot 10^{-5}$, $1.1 \cdot 10^{-6}$ (function level; $m = 16$ still relaxing, plateau $2.8 \cdot 10^{-4}$ in round 11's continued run) and 1.16, 0.29, 0.041 ($W^2$) vs nearest $2.5 \cdot 10^{-3}$, $5.0 \cdot 10^{-4}$, $4.4 \cdot 10^{-5}$ and 4.8, 2.7, 0.69; nearest fit errors 0.061, 0.035, 0.013, 0.012.
- Masses as a design input: $q_\delta = (1/3 + \delta, 1/3 - \delta/2, 1/3 - \delta/2)$ gives fit error 0.0069–0.049 for $\delta = 0.01$–0.1, splitting margin $+0.0048$, $+0.0028$, $+0.0011$, $-0.0015$, $-0.0049$ (inadmissible between $\delta = 0.03$ and 0.05), diagonal plan unique at every row ($\varepsilon \leq 1.04 < 3.13$).
- Tests (`tests/test_ot_leak.py`): the plan is a vertex with the prescribed masses (integral when commensurate, $\leq K_0 - 1$ split rows otherwise) and the single-atom prior is the $\sigma$-modification; the leak is the $G$-gradient flow of $J_m^{\text{OT}}$ against finite differences; the replicated weighted configuration is stationary for three commensurate sizes with the diagonal plan

and the same learned function ($W^2 = 0$ between sizes), for matching and mismatched masses; conservation against constraint on a short run; the forest bound; the exact $W^2$ by replication against the transportation LP.

### 25.9 Corollary D‴.14: the real loop with the OT leak

Script: `exp_ot_real_loop.py` → `chaos/ot_real_loop.md`.

- Real loop at $k_e = 25$ ($\sigma = \sigma_0/k_e$, RK4 plant with the control and the reference evaluated at the stages, Euler parameters, $\mathrm{d}t = 10^{-3}$, 100 time units $= 16$ periods) from the clouds at the end of Proposition D‴.12's averaged runs (contracted where the weights are admissible) (seeds 0, 1; $m = 15$, 60, $M = 510$) and from the collapsed lifts; last five periods on the reference orbit.
- OT leak: $\|\hat{f}_m - \hat{f}_M\|^2_{L^2(\mathrm{orbit})} = 7.5 \cdot 10^{-9}$–$2.7 \cdot 10^{-8}$, against the 3-atom lift $5.2 \cdot 10^{-10}$–$3.6 \cdot 10^{-8}$; fit error along the orbit $3.5 \cdot 10^{-5}$–$2.9 \cdot 10^{-4}$; rms tracking error $0.9 \cdot 10^{-6}$–$7.3 \cdot 10^{-6}$ (the lift $8 \cdot 10^{-12}$); $W^2(\mu_m, \mu_M)$ $3.3 \cdot 10^{-4}$–$1.4 \cdot 10^{-3}$; within-class spreads 0.013–0.033.
- Nearest-atom leak: $\|\hat{f}_m - \hat{f}_M\|^2 = 1.1 \cdot 10^{-4}$–$2.5 \cdot 10^{-3}$; fit errors 0.012–0.075; rms tracking errors $3.2 \cdot 10^{-4}$–$1.9 \cdot 10^{-3}$; $W^2(\mu_m, \mu_M)$ 1.6–5.5. Three of six systems have inadmissible realised weights (margins $-0.0035$, $-0.0011$, $-0.0022$): their lifts are saddle orbits and the systems have split (spread 0.04–0.11; lift-to-lift ratios 0.960, 0.998, 0.961); at the one pair with both weights admissible the disagreement is the lifted orbits' distance to 1.0003, the admissible systems within $2.5 \cdot 10^{-11}$ of their lifts (spread 0.0006–0.0013).
- Tests (`tests/test_ot_leak.py`): the collapsed 15-system and the 3-atom loop integrate identically under the OT leak at masses $q$ ($10^{-12}$ over 300 steps); the collapsed system with weights $(6, 5, 4)/15$ stays collapsed under the nearest-atom leak and loses a particle under the OT leak; two commensurate lifts keep $W^2 = 0$ and the same learned function; the lifted orbit's class separation subspaces are invariant and carry the ghost spectrum (Proposition D‴.15, $m = 6$).

### 25.10 Corollary D‴.14 in two dimensions

Script: `exp_ot_2d.py` → `chaos/ot_2d.md`.

- Planar plant (four Gaussian bumps with vector amplitudes), Lissajous reference, $p = 5$ parameters per kernel, matching prior (masses $\frac{1}{4}$), $k_e = 25$, $\sigma_0 = 0.05$, 2500 time units $= 100$ slow units, sizes 8, 32 vs 128 from the same jittered clouds, two seeds.
- OT leak: weights equalised at 13–14 slow units ($\ln 2/\sigma_0 = 14$); $\|\hat{f}_m - \hat{f}_M\|^2 = 5.0 \cdot 10^{-8}$–$1.6 \cdot 10^{-7}$; fit error along the orbit $2.1 \cdot 10^{-4}$–$8.7 \cdot 10^{-4}$; rms tracking error $3.9 \cdot 10^{-6}$–$1.6 \cdot 10^{-5}$; $W^2(\mu_m, \mu_M)$ $2.0 \cdot 10^{-4}$–$4.4 \cdot 10^{-4}$.
- Nearest-atom leak: weights conserved; $\|\hat{f}_m - \hat{f}_M\|^2 = 5.5 \cdot 10^{-3}$–$2.3 \cdot 10^{-2}$; fit errors 0.11–0.39; rms tracking errors $1.9 \cdot 10^{-3}$–$6.7 \cdot 10^{-3}$; $W^2(\mu_m, \mu_M)$ 4.7–11.8.

### 25.11 Proposition D‴.15: the transverse spectrum of a lifted orbit

Script: `exp_lifted_orbit.py` → `chaos/lifted_orbit.md`.

- Real loop at $k_e = 25$; monodromies by central differences (DOP853, rtol $10^{-11}$). OT leak at the matching masses: the 3-atom orbit (the exact representation, zero tracking error) lifted to $m = 6$, 9, 15 — return residuals $\leq 1.7 \cdot 10^{-15}$; separation exponents $\sigma_0/k_e = 2.830 \cdot 10^{-4}$ in every class (= ghosts; invariance defects $\leq 1.8 \cdot 10^{-8}$); collective exponents $2.935 \cdot 10^{-4} \ldots 7.934 \cdot 10^{-3}$ identical across $m$ to $10^{-4}$ relative, Theorem J(b)'s prediction $2.952 \cdot 10^{-4} \ldots 8.091 \cdot 10^{-3}$ (0.6–3.2%).

- $\sigma$-modification at $(7,5,4)/16$ (margin $+0.00257$) and $(9,4,3)/16$ ($-0.00219$), lifts to $m = 16$ (residuals $\le 4.9 \cdot 10^{-14}$): separation exponents = ghosts to $2 \cdot 10^{-5}$ relative, $= (\sigma_0 - \lambda_j(R_k^G))/k_e$ within 0.2–4.1% (defects $\le 5.7 \cdot 10^{-8}$); the saddle's class 0 exponent $-9.13 \cdot 10^{-5}$ (predicted $-8.77 \cdot 10^{-5}$, multiplicity 8); period-start offsets from $\Theta^*(w)$ $1.1 \cdot 10^{-2}$, $1.6 \cdot 10^{-2}$ (Theorem J(a)).

## 25.12 Remark D‴.16: dither and the mass constraint

Script: `exp_dither_ot.py` → `chaos/dither_ot.md`.

- Dithered averaged slow loop (Langevin in slow time, Gibbs law $\exp(-J_m/\tau_0)$), $m = 15$ from the counts $(6,5,4)$, matching prior, $\sigma_0 = 0.15\sigma_{\text{crit}}$; $\tau_0 = 0.005$, 0.01, 0.02 over 50 000, 200 000, 50 000 slow units; two seeds.
- Nearest-atom leak: record-level label changes $1.0 \cdot 10^{-5}$, $5.1 \cdot 10^{-4}$, $5.6 \cdot 10^{-3}$ per slow unit (about half undone at the next change; Arrhenius barrier 0.048, 0.043–0.052 at 95%, among the ridge minima 0.040–0.074); time at the balanced counts 0.00, 0.54, 0.25 against the energy–multinomial law 0.86, 0.58, 0.33 (standard errors $\approx 0.1$ at $\tau_0 = 0.01$; the law under-weights the saddle count vectors 2.7–3.2 times).
- OT leak: balanced counts in 100%, 99.4%, 88% of the records after the initial transit (99.6%, 99.2%, 88% of the whole runs); the excursions are returns of the crossing particle (121 of 121 at $\tau_0 = 0.01$, 298 of 336 at 0.02, the rest exchanges); off-balance fraction $6.3 \cdot 10^{-3}$, 0.117, effective barrier 0.058: the mass constraint makes a crossing reversible without raising its barrier.

## 25.13 Theorem D‴.18: Theorem J uniformly in $m$

Script: `exp_uniform_basin.py` → `chaos/uniform_basin.md`.

- Real loop at $k_e = 25$ ($\sigma_0 = 0.15\sigma_{\text{crit}}$, $\mathrm{d}t = 5 \cdot 10^{-3}$), 8000 time units, many systems integrated together.
- The test of the rates (Part F, by shooting): the 3-atom orbit's parameter Floquet exponents within 0.56–3.22% (matching prior) and 0.57–3.15% (prior centres +0.15) of $\lambda_j(H_q)/k_e$; the ghosts' exponents $\sigma_0/k_e = 2.830 \cdot 10^{-4}$ (matching) and $(\sigma_0 - \lambda_j(R_k^G))/k_e$ to $3 \cdot 10^{-4}$ relative (shifted; slowest $2.2701 \cdot 10^{-4}$ against $2.2696 \cdot 10^{-4}$). The proof's radius is at most 0.019 (sampled $C_N \ge 0.092$): the runs start 31–261 times outside the proved start radius.
- Matching prior at the matching masses (constant lift), sizes 3, 15, 60, 240, per-particle random displacements $\rho$: inside the cells ($\rho = 0.3$, 1.0, 2.5) no particle leaves its cell and both leaks are identical; fitted decay rates of the per-particle distance $2.69$–$2.97 \cdot 10^{-4}$ for $m \ge 15$ ($\sigma_0/k_e = 2.83 \cdot 10^{-4}$; ratios 0.95–1.05), constants $C(8000) = 1.03$–$1.29$ (still growing at the horizon for $m = 15$ at $\rho = 0.3$, 1); at $m = 3$ (no separation modes) the runs are mixtures of collective modes ($2.948 \cdot 10^{-4}$ when the slowest carries the end, $\lambda_{\text{coll}}/k_e = 2.952 \cdot 10^{-4}$; $4.06 \cdot 10^{-4}$ when the third does).
- Across a cell boundary ($\rho = 4.5$; 1, 3, 13, 55 particles start in another cell): the nearest-atom leak ends at counts $(2,0,1)$, $(6,5,4)$, $(23,19,18)$, $(87,70,83)$ — at $m = 60$, 240 through crossings during the transient — with $W^2$ 12.3, 2.81, 2.30, 1.92, already that of the lifts of the realised weights (3.21, 2.35, 1.93 for $m = 15$, 60, 240) though the clouds have not collapsed; $(6,5,4)/15$ and $(23,19,18)/60$ are inadmissible (margins $-0.00348$, $-0.00161$). The OT leak ends at $q$ for every $m$ (last count change at $t \le 2695$), $W^2$ 0.010–0.23 falling at the leak's rate $2\sigma_0$ per slow unit (the transit is over); the plan relabels 4, 15, 53 particles out of their initial class's cell.

- One particle across a boundary (Remark D‴.19; mean-square displacement 0.311, 0.078 at $m = 60, 240$): $W^2$ grows to 0.79, 0.20 under the nearest-atom leak (the lifts of $(19, 21, 20)/60$ and $(79, 81, 80)/240$: 0.8655, 0.2163) and falls to $3.0 \cdot 10^{-3}$, $8.2 \cdot 10^{-4}$ under the OT leak.
- Moving orbit (OT leak, prior centres +0.15): second-half fits $2.18$–$2.63 \cdot 10^{-4}$ against $2.27 \cdot 10^{-4}$ (ratios 0.96–1.16; mixtures of separation modes, consistent with the bound $(1 - \kappa)\mu$), constants 1.04–1.23.
- Tests (`tests/test_uniform_basin.py`, at the prior with centres shifted by +0.15, where the residual is not zero and $\lambda_{\text{sep}} = 0.80\sigma_0$): the drift's second-order remainder $\le C_N |||u|||^2$ with $C_N < 0.1$ for random per-particle displacements at $m = 6$, 30, 150, 600, not growing with $m$; the displaced particle's remainder exactly $N_a + N_b/m$ with $N_a$ the same for every $m$ (non-zero here, zero at the matching prior), the others' exactly $O(1/m)$; a separation perturbation moves only its own particles at first order, with the response $(GR_k - \sigma_0 I)v$ at $m = 6$, 60, 300; the averaged loop contracts $|||\cdot|||$ at $\mu = \min\{\lambda_{\text{coll}}, \lambda_{\text{sep}}\}$ with one constant $C \le 1.05$ for $m = 6$, 30, 150.

## 25.14 Proposition D‴.20: the entropic OT leak

Script: `exp_entropic_leak.py` → `chaos/entropic_leak.md`.

- Matching prior, $\sigma_0 = 0.15\sigma_{\text{crit}}$; the entropic plan by semi-dual Newton; every weighted stationary point continued from a converged neighbour and checked to $10^{-9}$.
- Critical excess at $w_\delta = q + \delta(1.5e_1 - 0.5)$, by continuation in $\varepsilon$ (residuals $\le 8.7 \cdot 10^{-13}$): $\varepsilon_c = 5.11$, 13.5, 23.1, 65.0 for $\delta = 0.02$, 0.05, 0.1, 0.2; at $\varepsilon_c$ the outflow $p = 0.057$, 0.15, 0.33, 0.57 splits about equally between the two under-weighted atoms ($\text{Cov}_1^G$ of rank two); the exact threshold $2\sigma_0\lambda_{\max}(A^{-1/2}\,\text{Cov}_1^G\,A^{-1/2})$ equals $\varepsilon_c$; the rank-one formula 1.22–1.31 (atom 2) and 0.70–1.22 (atom 3) times $\varepsilon_c$. Excess from atom 3 alone: $\varepsilon_c = 6.99$, 14.6, 24.3 for $\delta = 0.02$, 0.05, 0.1. At fixed $\varepsilon$: $\delta_c = 0.0042$, 0.0083, 0.016, 0.030, 0.062, 0.14, 0.20 at $\varepsilon = 1$, 2, 4, 8, 16, 32, 64, equal to $w_1 p_c$ up to $\varepsilon = 8$.
- Dynamics ($m = 15$, 60, 240 vs $M = 510$, 600 slow units, 4 seeds; $\varepsilon = 4$, 16, 64 continued to 6000): $\varepsilon = 1$ — all equalised, fit errors $1.7$–$3.0 \cdot 10^{-4}$; $\varepsilon = 4$ — five of the twelve larger systems at unstable weights at $s = 600$, every system at stable weights at $s = 6000$ with residual excesses up to 1/60, 3/240, 7/510 ($\delta_c(4) = 0.012$–$0.016$); $\varepsilon = 16$ — the stable systems keep their sampled weights, three unstable ones move ($(6, 5, 4)/15 \to (5, 6, 4)$: the excess moved to an atom at $q$, not reduced); $\varepsilon = 64$ — only $(8, 4, 3)/15$ moves, fit errors 0.12–0.14 (the barycentric bias).
- Floors: Proposition D‴.5's law at $\Theta_\varepsilon^*(q)$ is 0.55 ($\varepsilon = 16$) and 0.74 ($\varepsilon = 64$) of the nearest-atom law; observed 0.23–0.84 and 0.47–0.91 of the nearest-atom leak's floors on the same clouds, per seed.
- Tests (`tests/test_entropic_leak.py`): marginals and the agreement of the two solvers; the gradient of $J_m^\varepsilon$ against finite differences; the limits $\varepsilon \to 0$ and $\varepsilon \to \infty$ at their rates (the OT plan's uniqueness checked); the separation blocks of the finite-difference Hessian at a collapsed configuration equal to $I \otimes (\sigma_0(I - (2/\varepsilon)\,\text{Cov}^G) - R^G)$ entrywise, the separation subspace invariant; the over-weighted class of $(6, 5, 4)/15$, continued in $\varepsilon$ with $A \succ 0$ checked and every residual below $10^{-9}$, unstable at $\varepsilon = 16$, 17 and stable at 18, 20, 64.

## 25.15 Theorem D‴.21: a measure-level basin for the OT leak

Script: `exp_ot_basin.py` → `chaos/ot_basin.md`.

- Averaged slow loop at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$, the warm clouds of Proposition D‴.12, $m = 15$, 60, 240, 510, four seeds, 600 slow units, the matching prior and the prior with centres $+0.15$, both leaks. The identities (E), (P) hold along every run to $6 \cdot 10^{-13}$ relative.
- OT leak: no particle relabelled; the margin $\alpha$ positive from $s \le 300$ on (from $s \le 25$ in 19 of 32 runs); the observed ratio at least $\lambda$ at every record with $\alpha > 0$ (smallest ratio/$\lambda$ 1.00 at the matching prior, 1.61–1.72 at the shifted one); $E$ from $4.1 \cdot 10^{-3}$–$4.0 \cdot 10^{-2}$ to $8.3 \cdot 10^{-7}$–$8.1 \cdot 10^{-6}$ ($W^2(\mu_\Theta, \mu^*)$ from 0.99–5.25 to $2.3 \cdot 10^{-4}$–$2.3 \cdot 10^{-3}$), below the a-posteriori bound; at the end the observed rate is 0.01416–0.01417 against $2\sigma_0 = 0.01415$ and $\lambda = 0.0140$–$0.0141$ (matching prior: sharp); at the shifted prior $\lambda = 0.0077$–$0.0078$ against the observed 0.0137–0.0139, the local PL constant $2\lambda_{\mathrm{sep}} = 0.01135$ (curvature cost 0.802, the slack of (ii) 0.683).
- Nearest-atom leak: no particle changes cell. At the 22 admissible count vectors $E(600)$ is 0.02–0.24% below the energy of the replicated configuration of the sampled weights and rising toward it — a critical point of the nearest-atom energy (drift $\le 6 \cdot 10^{-14}$) but not of the OT energy (drift $2.3$–$4.4 \cdot 10^{-2}$); at the 10 inadmissible ones ($m \le 60$; class-0 margins $-0.0004$ to $-0.0061$) class 0 splits and $E$ ends 0.5–16.5% below it, still falling.
- The a-priori basin: $E_0 = 2.0 \cdot 10^{-7}$ at the matching prior ($\kappa = \sigma_0$); at the shifted prior (H*) holds on the whole parameter space ($\sup \lambda_{\max} R^G[\mu^*] = 6.60 \cdot 10^{-3} < \sigma_0$, at a saturation corner; $\kappa = 4.7 \cdot 10^{-4}$, $E_0 = 9.1 \cdot 10^{-10}$). $\{E \le E_0\}$ holds a fraction $\le 5.8 \cdot 10^{-6}$ (matching) and $4.4 \cdot 10^{-7}$ (shifted) of the particles in other cells.
- The measure-level basin (matching prior): a fraction $p$ of the particles moved 0.6 of the way into another cell stays in its class; $\kappa^\Theta = 0.0096$, 0.019, 0.029, 0.048 and $\alpha = +0.0023$ at $p = 1/30$, negative from $p = 1/15$ (the same at $m = 30$ and $m = 300$: the same measures). One particle moved: $E$ $3.3 \cdot 10^{-3} \to 2.5 \cdot 10^{-5}$ ($m = 30$) and $2.3 \cdot 10^{-4} \to 3.1 \cdot 10^{-6}$ ($m = 300$), at $2\sigma_0$ at the end, below $E(0) \exp(-\int \lambda_+)$, the particle back in its cell; under the nearest-atom leak $E$ rises to the spurious minima's $6.3126 \cdot 10^{-3}$ and $6.1328 \cdot 10^{-4}$ by $s = 3000$.
- Tests (`tests/test_ot_basin.py`): (E) and (P) to $10^{-12}$ for random assignments with the counts $mq$ at both priors, and their exact defects at other counts; the PL chain; the constant $2\sigma_0$ attained by a separation to $10^{-4}$; the decay from a particle in another cell, with a positive margin, below the a-posteriori bound, at $2\sigma_0$ at the end, against the nearest-atom floor; the spurious minimum $(11, 9, 10)/30$ of the nearest-atom energy, admissible (separation margins and $K$-atom Hessian positive), not critical for the OT leak.

### 25.16 Theorem D‴.23: adaptive priors

Script: `exp_adaptive_prior.py` → `chaos/adaptive_prior.md`.

- Averaged slow loop at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$, the OT leak toward the prior with centres $+0.15$ (masses 1/3); the prior's atoms move toward the class barycentres at the rate $\eta$.
- Linear theory at the exact representation: $h_j = 3.06 \cdot 10^{-4} \ldots 0.195$; a fixed prior keeps $\sigma_0/(h_j + \sigma_0) = 0.959 \ldots 0.035$ of its error and relaxes at $h_j + \sigma_0 \ge 7.38 \cdot 10^{-3}$; the adaptive prior's slowest rate $\lambda_1^- = 2.68 \cdot 10^{-5}$, $1.51 \cdot 10^{-4}$, $2.78 \cdot 10^{-4}$, $3.06 \cdot 10^{-4}$ for $\eta = \sigma_0/10$, $\sigma_0$, $10\sigma_0$, $\infty$. (v)'s radius: $C_N \ge 0.341$ (sampled), so $\rho \le 2.0 \cdot 10^{-5}$, $1.1 \cdot 10^{-4}$, $2.0 \cdot 10^{-4}$, $2.2 \cdot 10^{-4}$, while the $K_0$-atom runs start 94–$6 \cdot 10^4$ times outside, the warm clouds $2.2 \cdot 10^4$–$3.6 \cdot 10^5$ times and the separated starts 131–3344 times. The fixed prior's bias: fit error $9.78 \cdot 10^{-3}$, within 4.8% of the first-order formula in norm, 0.85 and 0.73 of it in the weak modes 1 and 3; the error is first order in the shift (0.58% in norm, 0.98 and 0.97 in modes 1 and 3, at 1/8 of the shift).

- $K_0$-atom loop from $\Theta^*(q)$, 6000 slow units: every mode above $10^{-6}$ (13 of 36) decays at $\lambda_j^-$ (log-slopes 0.981–1.003 of it; ratios to the pair solutions 0.82–1.12, constant to 1.04% from $s = 1000$ on for $\eta \geq \sigma_0$); fit error $9.78 \cdot 10^{-3} \to 3.6 \cdot 10^{-4}$, $1.8 \cdot 10^{-5}$, $8.0 \cdot 10^{-6}$, $6.7 \cdot 10^{-6}$.
- $m = 15$, 60, 240, 510 from the warm clouds, four seeds, 3000 slow units. Fixed prior: the lift of $\Theta^*(q)$ ($\|\bar{y}\| = 2.10 \cdot 10^{-2}$, $W^2 = 4.4 \cdot 10^{-4}$). Ungated: the prior dragged by the transit ($\|\bar{y}\|$ 0.032–0.32 at $\eta = \sigma_0$, 0.048–0.71 at $\eta = \infty$, at $s = 500$); one $m = 15$ system at $\eta = \infty$ on a plateau of the unregularised loss (Hessian eigenvalues $-1.8 \cdot 10^{-5}$, $7.2 \cdot 10^{-5}$; fit error $3.46 \cdot 10^{-2}$ at $s = 23000$), which the unregularised gradient flow crosses to the exact representation (below 0.9 of the plateau's loss after $4.16 \cdot 10^4$, half at $4.78 \cdot 10^4$, below $10^{-12}$ at $7.97 \cdot 10^4$ more slow units, the crossings located as events). Transit gate (opens at $s = 88$–$149$): $\|\bar{y}\|$ $4.9 \cdot 10^{-3}$–$3.7 \cdot 10^{-2}$ at $\eta = \sigma_0$ against the $K_0$-atom run's $1.3 \cdot 10^{-3}$. Settled gate (opens at $s = 1211$–$1304$, the systems replicated: rms separations $\leq 7.9 \cdot 10^{-4}$): $\|\bar{y}\|$ and the fit error equal to the $K_0$-atom run's at $s - t_{\text{open}}$ for every $m$ (per-seed ratios 0.995–1.000 at $\eta = \sigma_0$, 0.968–1.001 at $\eta = \infty$: the invariance of the replicated configurations); fit errors $1.6 \cdot 10^{-4}$ and $3.4$–$3.5 \cdot 10^{-5}$, $W^2$ 6.7–$7.2 \cdot 10^{-6}$ and 9.3–$9.9 \cdot 10^{-7}$; in the seed means the settled gate is the lower except the fit error at $\eta = \sigma_0$ and $m = 510$ ($1.64 \cdot 10^{-4}$ against the transit gate's $1.55 \cdot 10^{-4}$), and system by system the transit gate ends lower in all three of $\|\bar{y}\|$, $W^2$ and the fit error in 6 of 32.
- Separated starts (the structure behind (v), beyond its ball; five draws): $\|\bar{y}\|$ at $s = 1750$ departs from the $K_0$-atom run's by +0.13% and +3.1% (rms separation 0.02 and 0.1) at $\eta = \sigma_0$ and +0.77% and +18.5% at $\eta = \infty$ for $m = 15$ in the draw means ($m = 510$: +0.09%, +2.2%, +0.63%, +15.5%): second order in the separations draw by draw (factors 23.8–25.0), consistent with the same expected size at both $m$ (the means, higher at $m = 15$, within 0.9–1.7 combined standard errors; five draws), the spread 3.3–6.0 times larger at $m = 15$.
- The limits of (i): a non-replicated stationary saddle on a four-bump target (one class split, held by a leak force $5.8 \cdot 10^{-3}$); from the exact split fit the flow collapses the class (width 1.05, fit error $2.23 \cdot 10^{-2}$ at $s = 4 \cdot 10^4$), and the merged width coordinate then runs off like $\ln s$, the width tending to $s_{\max}$ like $1/s$: the solution is unbounded, its physical image approaching the minimiser of the loss on the face $s = s_{\max}$ (fit error $1.943 \cdot 10^{-2}$, a Karush–Kuhn–Tucker point with $\partial L/\partial s = -1.3 \cdot 10^{-6}$); beyond the saturation (weighted amplitude $4.8 > a_{\max}$) the fixed prior's solution converges while the adaptive prior's amplitude coordinate grows like $2 \ln s$; a prior permuted among atoms of equal mass is stationary at the exact representation.
- Tests (`tests/test_adaptive_prior.py`): the joint drift is the gradient flow of $\mathcal{J}$ in the stated metric (finite differences); the exact representation stationary, the fixed prior's stationary point not; the linearisation's spectrum ($\sigma_0$ and the pair rates $\lambda_j^{\pm}$) and its self-adjointness in the gradient-flow metric; the first-order bias; the transit gate and the settled gate, with $\mathcal{J}$ non-increasing; the non-replicated saddle, the permuted prior and the adaptive prior's run-off (the force on the amplitude tending to the positive multiplier of the loss's face minimum, and the run-off law $(a_{\max} - a)s \to a_{\max}^2(\sigma_0 + \eta)/(4\eta G_\alpha F_\infty)$).

## 25.17 Proposition D‴.24: the floor is the leak's bias; an adaptive prior without a mass constraint

Script: `exp_adaptive_floor.py` → `chaos/adaptive_floor.md`.

- Averaged slow loop on the three-bump instance at $\sigma_0 = 0.15\sigma_{\text{crit}}$ ($0.05\sigma_{\text{crit}}$ and $0.5\sigma_{\text{crit}}$ for the law), the nearest-atom leak toward the matching prior unless stated; the warm clouds (four seeds), $m = 15$, 60, 240 against $M = 510$; the checks at the sampled weights from the stored final states.

- The floor as the bias (no dynamics): $F$ equals $\hat{J}\sigma_0(H_L+\sigma_0 I)^{-1}\partial_w x^0$ to the finite-difference error ($1.1\cdot10^{-6}$, $1.9\cdot10^{-7}$, $1.2\cdot10^{-8}$; exact at $q$); the law (B) per unit of $1/m-1/M$ is $1.356\cdot10^{-2}$, $6.166\cdot10^{-2}$, 0.2291, the sum of its mode terms, at 0.496, 0.753, 0.839 of the bound $(\sigma_0/4)\,\mathrm{tr}(\partial_w x^0\tilde{C}\partial_w x^{0\top})$, itself $\sigma_0/4$ times the adaptive prior's coupled-distance law 46.32. Modes 2–6 carry 99.4%, 99.0%, 97.9% of the law and 97.2% unweighted; the weight (a band-pass centred at $h=\sigma_0$) moves the shares of the modes with $h_j/\sigma_0\in[1/5,5]$ from 21%, 97%, 78% to 42%, 99%, 90%. The law's value at $m=15$, 60, 240: $3.99\cdot10^{-3}$, $9.07\cdot10^{-4}$, $1.36\cdot10^{-4}$. The parameter split $S=H_L(H_L+\sigma_0 I)^{-1}S^e$ holds to $\le1.4\cdot10^{-7}$; the adaptive prior's coupled-distance law is 1.44, 2.17, 5.27 times the fixed prior's.

- Two notions of admissibility. Representable ($w>(0.25,0.175,0.2)$): every sample but $(8,4,3)/15$, on the boundary. D‴.5-admissible for the fixed prior: 7 of the 12 $m$-systems; $(6,5,4)/15$, $(6,3,6)/15$, $(24,15,21)/60$, $(8,4,3)/15$, $(25,17,18)/60$ are saddles (margins $-3.48$, $-1.08$, $-2.16$, $-6.14$, $-3.98\cdot10^{-3}$, class 1, $w_1\ge0.4$); the references and the others $\ge3.78\cdot10^{-3}$.

- Fixed against adaptive priors, 6000 slow units: the fixed prior's $\|\hat{f}_m-\hat{f}_M\|^2$ stays ($3.0\cdot10^{-4}$–$5.5\cdot10^{-3}$, $1.1\cdot10^{-4}$–$9.9\cdot10^{-4}$, $4.8\cdot10^{-6}$–$8.2\cdot10^{-5}$ at $m=15$, 60, 240); at the seven D‴.5-admissible samples it is the exact bias at the sampled weights (ratio 1.0000), the delta method missing it by 0.86–1.72; the five saddles have split (class diameters 0.64–1.50) at 0.81–1.01 of the replicated floor. The adaptive prior's decreases without a floor (at $s=6000$ at most $2.0\cdot10^{-6}$ at $\eta=\sigma_0$ and $5.4\cdot10^{-8}$ at $\eta=\infty$ over the representable samples); the boundary sample decays like $1/s$ ($6000\cdot\|\hat{f}_m-\hat{f}_M\|^2=0.88$ and 0.42 against the predicted 0.8 and 0.4); the counts are conserved.

- The parameter level: the coupled distance $D$ is 0.944–1.000 of the replicated stationary points' (fixed: $\Theta^*(w)$; adaptive: $\vartheta^e(w)$), and 1.72–4.81 times larger under the adaptive prior; $W^2$ is dominated by the cross-class transport of $\frac{1}{2}\|w^m-w^M\|_1$ (0.82–0.99 of it fixed, 0.67–0.99 adaptive), so the two priors' $W^2$ differ by 1.00–1.32 and their within-class parts by 1.71–4.84.

- The design table (the misplaced prior, 3000 slow units, $m=15$, 60, 240): nearest atom and fixed, fit error $6.8\cdot10^{-2}$–$2.0\cdot10^{-2}$ and floor $2.5\cdot10^{-3}$–$5.9\cdot10^{-5}$ (the same five samples saddles, margins $-0.38$ to $-5.76\cdot10^{-3}$); nearest atom and adaptive, $8.9\cdot10^{-3}$–$5.5\cdot10^{-4}$ and $8.0\cdot10^{-5}$–$1.1\cdot10^{-7}$; OT and fixed, $9.78\cdot10^{-3}$ and no floor; OT and adaptive (settled gate), $1.6\cdot10^{-4}$ and below $5\cdot10^{-11}$. The OT designs' coupled distance is the largest (6.0, 3.5, 1.0: the transit's mismatch) with $W^2\le2.1\cdot10^{-8}$. At $s=1250$, before the settled gate's effect, the nearest-atom adaptive prior leads at $m=60$ and 240 (fit error $3.8\cdot10^{-3}$ and $2.8\cdot10^{-3}$ against 9.2–$9.8\cdot10^{-3}$), not at $m=15$ ($1.75\cdot10^{-2}$); from $s\approx2000$ the OT design leads.

- The real loop ($k_e=25$, 3000 time units, seed 0's weights): the fixed prior's rms tracking error 1.17–$1.18\cdot10^{-3}$ ($m=15$, a saddle orbit) and 3.43–$3.44\cdot10^{-4}$ ($m=60$), at the averaged prediction (ratios 1.001, 0.996 at the end); the adaptive prior's ($\eta=\infty$) $1.16\cdot10^{-3}\to3.27\cdot10^{-4}$ and $3.40\cdot10^{-4}\to1.52\cdot10^{-4}$ along the averaged loop (0.987, 0.992); from the exact representation the five-period rms is at most $9.0\cdot10^{-12}$.

- The instance's probabilities (exact): outside $W$ 0.361, 0.074, 0.0021 and on its boundary 0.118, 0.053, 0.0012 at $m=15$, 60, 240, against Hoeffding's 1.87, 0.60, 0.036; a permutation of the classes representable for 31%, 87%, 100% of the mass outside $W$; the fixed prior's saddles 0.369, 0.260, 0.109 (comparable to the adaptive prior's failure set at $m=15$, much larger at $m\ge60$). The adaptive prior's expected limiting floor (the constrained minima on the branch of $\vartheta^e(q)$ outside $W$; the global infimum is 0 where a permutation is representable) $\ge7.3\cdot10^{-3}$, $2.2\cdot10^{-4}$, $6.6\cdot10^{-7}$: 1.8, 0.24, 0.005 times the fixed prior's law.

- Monte Carlo (600 nested samples, the exact replicated stationary points): the fixed prior's floor is $2.78 \pm 0.20$, $1.53 \pm 0.11$, $1.11 \pm 0.07$ times the law (2.89, 1.56, 1.07 for D‴.5-admissible pairs); $\mathbb{E}D$/law 0.68, 0.93, 0.95 (fixed) and 0.53, 1.17, 1.26 (adaptive: a far heavier tail, one sample at 55.7 times the law at $m = 240$, 1.17 without it — the means not reliable), the adaptive prior's 2.9–3.7 times the fixed prior's.

- In two dimensions (`exp_adaptive_floor_2d.py` → `chaos/adaptive_floor_2d.md`; four bumps with vector amplitudes, $p = 5$, $\sigma_0 = 0.05$, $m = 16$, 32, 64 against $M = 128$, six seeds): $F$ equals the bias form to $6.0 \cdot 10^{-7}$, the law 0.310 per unit of $1/m - 1/M$ at 0.215 of the bound; every sample D‴.5-admissible (margins +0.016 to +0.047) and the fixed prior's floor equal to the exact bias at the sampled weights in all 18 systems (ratio 1.0000); 7 of the 18 systems representable ($w > (0.200, 0.150, 0.175, 0.150)$), where the adaptive prior's function-level distance falls to $1.0 \cdot 10^{-7}$–$4.1 \cdot 10^{-5}$ ($\eta = \sigma_0$) and $1.2 \cdot 10^{-9}$–$1.0 \cdot 10^{-5}$ ($\eta = \infty$: the atoms reset to the cell means every slow unit), at most 0.019 and 0.006 of the fixed prior's, elsewhere to 0.023–0.64 of it (still falling, toward the constrained minimum at the sampled weights); the real loop ($k_e = 25$, $m = 64$; the adaptive prior at $\eta = \infty$) at the averaged prediction (ratios 0.979–1.009) and invariant from the exact representation ($2.0$–$4.7 \cdot 10^{-10}$).

- Tests (`tests/test_adaptive_floor.py`): the nearest-atom adaptive flow is the gradient flow of $\mathcal{J}_\circ$ with the cell masses (finite differences); $\mathcal{J}_\circ$ is the minimum of the OT energy over the masses, strictly below the energies with masses moved by $\pm 1/m$; the floor law as the bias derivative, below the bound, the weight moving the shares toward $h \approx \sigma_0$ against the unweighted ones at two $\sigma_0$; at the D‴.5-admissible sample $(5, 4, 6)/15$ the exact representation stationary under the adaptive prior and the fixed prior's stationary point a strict local minimiser, biased; at the saddle $(6, 5, 4)/15$ the exact representation's margins $\sigma_0$; the parameter split; $\mathcal{J}_\circ$ decreasing across a cell crossing; an empty cell staying empty with its atom frozen. `tests/test_planar_loop.py`: the planar averaged loop (the scalar one at $n = 1$; the learning term against finite differences of $\bar{L}$ and the orbit average of the real law), the exact representation of representable weights stationary for the adaptive prior, the floor law as the bias derivative in two dimensions, and the finite-$\eta$ branch of its integrator second order against DOP853.

## 25.18 Corollary D‴.26: perfect tracking with an adaptive prior in the real loop

Script: `exp_adaptive_floquet.py` → `chaos/adaptive_floquet.md`.

- The three-bump instance at $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$, $x_m = 2\sin t$, the sampled weights $w = (6, 5, 4)/15$ (amplitudes $A_k/w_k = 2.5$, $-2.1$, 3.0; representable, and a saddle of the fixed prior), the nearest-atom leak toward an adaptive prior in the real loop with the prior in the state; $\eta = \sigma_0$ and $\infty$, $k_e = 25$ and 50; the monodromy by central differences of the period map (step $10^{-5}$, DOP853 at rtol $10^{-13}$).

- Every lift returns after one period to $\le 1.7 \cdot 10^{-15}$. Separations: at $m = 15$ and 30 the 36 and 81 exponents equal $\sigma_0/k_e$ to $\le 1.7 \cdot 10^{-6}$ (relative). Collective exponents: the other 18 ($\eta = \sigma_0$) or 9 ($\eta = \infty$) equal the weighted $K_0$-atom loop's to $1.4 \cdot 10^{-5}$–$6.4 \cdot 10^{-4}$.

- Part C, the variational equation on the collapsed subspace (its generator the real loop's linearisation to $5.8 \cdot 10^{-11}$, its exponents the differences' to $4.1 \cdot 10^{-5}$), $k_e = 12.5$–400, on $x_m = 2\sin t$ (where the first-order averaging correction vanishes identically) and on $2\sin t + 0.8\sin 2t$ (where it does not: $\|\bar{A}_1\| = 5.6 \cdot 10^{-3}$, skew): every exponent below its first-order prediction $\lambda_j^{\pm}(w)/k_e$ here — the slowest mode by 13.6% and 3.84% at $k_e = 25$ and 50 ($\eta = \sigma_0$) — the deviations falling by $\to 4$ per doubling of $k_e$ ($O(k_e^{-2})$ relative), the residuals of the second-order prediction by $\to 16$ ($O(k_e^{-4})$: the odd orders vanish; $4.9 \cdot 10^{-7}$

at $k_e = 400$). The second-order coefficients are all negative here, $c_1/\lambda_1 = -100.6$ in the slowest mode, of which the error filter's part $-102.1 = -0.989\omega_1^2$, $\omega_1 = 10.16$ (measured $-100.5$ at $k_e = 400$); with the adaptation gain scaled by 30 two are positive ($+4.17$, $+12.65$) and those exponents lie above their first-order prediction by $c_j/(\lambda_j k_e^2)$; at $\eta = 10\sigma_0$ $\bar{A}_0$ is self-adjoint in $\mathrm{diag}(I, (\sigma_0/\eta)I)$ only. The error channel's exponent by Liouville's formula within $O(k_e^{-3})$ of $k_e - \operatorname{tr} H_L(w)/k_e$.

- Tests (`tests/test_adaptive_floquet.py`): at the matching weights, $\eta = \infty$, $m = 6$: the return of one period; the nine separation exponents at $\sigma_0/k_e$; the collective exponents equal to the weighted $K_0$-atom loop's, within the averaging correction of the prediction. At the weights $(2, 1, 1)/4$ with the prior in the state, $\eta = \sigma_0$, $m = 4$: the real loop's Jacobian at the lift maps the separations to $-\sigma_0/k_e$ times themselves and the collapsed subspace into itself as the collapsed variational equation; the monodromy's 3 separation exponents at $\sigma_0/k_e$ and its 18 collective ones the variational ones, within the second-order prediction and below the first-order one (an instance property). On an asymmetric reference: $\bar{A}_1$ non-zero and skew, $\ell_j \bar{A}_1 r_j = 0$, the second-order residual falling like $k_e^{-4}$, the measured second-order coefficient the explicit $c_1$, the error channel by Liouville's formula; at $\eta = 10\sigma_0$ $\bar{A}_0$ self-adjoint in the metric and not in $I$, $\bar{A}_1$ skew in it.

## 25.19 Lemma D‴.26a: the exact slow subspace: Corollary D‴.26(c) and Proposition D‴.30(ii) proved for the linearised loop

Script: `exp_adaptive_floquet.py` Part D → `chaos/adaptive_floquet.md`.

- Corollary D‴.26's instance on the asymmetric reference $x_m = 2\sin t + 0.8\sin 2t$, $\eta = 0$ ($K = \hat{J}$, $\Lambda = \sigma_0 I$), $\sigma_0$ and $\infty$, $k_e = 12.5$–$1600$.

- Chang's slow subspace: the contraction in 3–9 iterations to an invariance residual $\leq 3.8 \cdot 10^{-15}$ (at $\eta = 0$, $k_e = 12.5$ it stalls at rounding, 499 iterations); $|P - P_2|/\varepsilon^3 \to 3.34 \cdot 10^3$ and $|P - P_{(3)}|/\varepsilon^4 \to 5.8 \cdot 10^4$; the slow block's rates the full variational equation's to $\leq 1.0 \cdot 10^{-8}$.

- The parity as an identity: the rates at $\pm\varepsilon$ equal to $\leq 5.9 \cdot 10^{-12}$ of the largest rate and $\leq 1.1 \cdot 10^{-9}$ per mode, while their first-order deviation is $9.8 \cdot 10^{-2}$–$8.9 \cdot 10^{-6}$ of the largest rate and per mode up to 29.6% (the slowest mode at $\eta = \infty$: $-29.6\%$, $-12.5\%$, $-4.0\%$, $-1.1\%$ at $k_e = 12.5$–$100$); a mismatch in the error channel breaks the equality at $O(1/k_e)$ ($1.1 \cdot 10^{-2} \to 7.8 \cdot 10^{-5}$); the Magnus parameter $\leq 0.48$ (0.004 at $k_e = 1600$).

- The slow noise: the decoupling residual $\leq 3 \cdot 10^{-16}$; $b$ against $b_0 = -(\dot{K} + aK)^\top$ at $O(1/k_e)$ ($63\% \to 0.74\%$) and against $b_0 + \varepsilon b_1$ at $O(1/k_e^2)$ ($|b - b_0 - \varepsilon b_1|/\varepsilon^2 \to 3.34 \cdot 10^3$); at $t = 0$ $\mathrm{Cov}(v) \cdot 2\varepsilon^3 \to 1$, $\mathbb{E}[vY'^\top] \to -b^\top$ at $O(1/k_e)$, $\mathrm{Cov}(Y') \to \bar{\Sigma}$ ($0.35 \to 9.4 \cdot 10^{-5}$ at $\eta = \infty$; $O(\varepsilon^2)$ to $k_e = 200$–$400$, $O(\varepsilon)$ beyond). The weakest mode's deviation ($-34\% \to -6.6 \cdot 10^{-5}$, $O(\varepsilon^2)$) is the slow noise's attenuation, $\langle bb^\top \rangle$ below $\bar{Q}$ ($-56\% \to -1.1 \cdot 10^{-4}$), plus the rate's correction ($+49\% \to +4.6 \cdot 10^{-5}$): 2.4–2.5 to 1 and of opposite sign at $\eta = \infty$ (12–23 to 1 at $\eta = 0$). $\mathrm{Cov}(Y) - \mathrm{Cov}(Y')$ is the jitter $(\varepsilon/2)K(0)^\top K(0)$ at $O(\varepsilon)$ and with its next term at $O(\varepsilon^2)$ ($0.30 \to 0.001$ at $k_e = 100$–$1600$).

- The floor: at $t = 0$ $0.763 \to 1.0000$ of the slow-plus-fast value; over the period its average $0.766 \to 0.99997$ of Proposition D‴.30(ii)'s formula, pointwise in $[0.9952, 1.0091]$ of it at $k_e = 1600$ — the predicted range of the fast jitter $(\varepsilon/2)\operatorname{tr}(H\hat{J}(t)^\top \hat{J}(t))$: the formula holds as the period average.

- Tests (`tests/test_adaptive_floquet.py`): Chang's $P$ with the exact slow multipliers and the $\varepsilon^3$ truncation on the instance; on a synthetic block with an $O(1)$, non-symmetric $\Lambda$ the truncations at $\varepsilon^3$ and $\varepsilon^4$, a flipped $\Lambda$ and a dropped cubic term caught; the rates exactly even in $\varepsilon$ at $\eta = \sigma_0$ and $\infty$, a mismatch breaking it; the slow noise decoupled, $b_0$ against $\dot{K}$

by the chain rule and the plant's own slope by differences, $b_1$ at $\varepsilon^2$, the covariances at their orders.

## 25.20 Proposition D‴.27: a projected adaptive prior

Script: `exp_adaptive_prior.py` Part H → `chaos/adaptive_prior.md`.

- The box $\mathcal{Z}$ (unconstrained coordinates): $\alpha \in [-5.889, 5.889]$, $\gamma \in [-4.417, 4.417]$ (amplitudes and centres to 0.9 of $a_{\max}$ and $c_{\max}$), $\kappa \in [-6, 6]$; the instance's exact atoms inside.
- Beyond the saturation (Part G(b)'s target, weighted amplitude $4.8 > a_{\max}$; the $K_0$-atom system from the exact atoms with the prior there): the projected prior converges at $\eta = \sigma_0$ (the projected midpoint rule, positions projected inside the step, $\mathrm{d}t = 0.25$ and 0.125, to $s = 10^5$; the two end states within $3.4 \cdot 10^{-12}$, their shared fixed point; the trajectory at $s = 300$ within $6.3 \cdot 10^{-7}$, $1.9 \cdot 10^{-7}$, $3.8 \cdot 10^{-8}$ of a reference for $\mathrm{d}t = 0.25$, 0.125, 0.0625) and at $\eta = \infty$ (LSODA) to the same bounded stationary point (drift $\leq 3.5 \cdot 10^{-15}$): the prior's amplitude atom on the face (5.889), the particle's at 6.728, the fit error 0.1415.
- The particles' largest $G^{-1}$-distance from $\mathcal{Z}$ along the run: 0.92 ($\eta = \sigma_0$), 0.89 ($\eta = \infty$), against the a-priori bound $B/\sigma_0 \leq 3686$ ($B \leq 26.08$ with the closed-form bound 4.658 of $L^G_\vartheta$; 4.044 by search).
- The box's size: the amplitude bound 0.75, 0.8, 0.9, 0.95, 0.99 of $a_{\max}$ (boxes containing the exact atoms) gives the fit errors 0.1676, 0.1611, 0.1415, 0.1268, 0.1101 — between the fixed prior's 0.1750 and the unprojected adaptive prior's 0.1050, reached only as its coordinate runs off; 0.7 (0.1736) excludes the exact atoms, and the tighter 0.65, 0.6, 0.5 are worse than the fixed prior (0.1792, 0.1845, 0.1949): a box should contain the representation.
- At $\eta = \infty$ the plan is Lloyd-consistent, not jointly optimal: on six particles in a consistent, non-optimal pairing the selected energy falls (largest increment $-3.4 \cdot 10^{-6}$) while the reduced energy $\min_Z \mathcal{J}$ rises from $1.251 \cdot 10^{-3}$ to $1.574 \cdot 10^{-3}$ before the plan becomes optimal.
- Part G(a)'s four-bump target: the box stops the merged atom's width coordinate at the face $\kappa = 6$ from $s = 10^9$ (unprojected: $\kappa = 9.75$ at $s = 10^{10}$ and growing like $\ln s$), the fit error within $9.3 \cdot 10^{-7}$ of the face minimiser's.
- Part C's $K_0$-atom loop at the misplaced prior ($\eta = \sigma_0$, 6000 slow units) is identical with and without $\mathcal{Z}$: the box is never approached (margins at least 16.9% of its width), so the runs coincide by construction.
- Tests (`tests/test_adaptive_prior.py`): beyond the saturation the projected prior at $\eta = \infty$ keeps the particles within 2 of the box and converges, while without the box the prior's coordinate passes 11 by $s = 10^4$; at $\eta = \sigma_0$ and $\mathrm{d}t = 0.125$ the prior stays in the box with its amplitude atom on the face (the step that projects the velocity at the midpoint stalled it inside), the particles bounded and $\mathcal{J}$ non-increasing; at the misplaced prior near the exact representation the projected and unprojected runs are identical.

## 25.21 Proposition D‴.28: the floor removal is a gauge property

Script: `exp_gauge_floor.py` → `chaos/gauge_floor.md`.

- The three-bump instance plus a fourth bump $\varepsilon \cdot \exp(-(x - 0.8)^2/(2 \cdot 0.3^2))$ that three kernels cannot represent, $\varepsilon = 0$–0.5, masses 1/3, $\sigma_0 = 0.15\sigma_{\mathrm{crit}}$: the weighted approximant $\vartheta^b(q)$ continued in $\varepsilon$, a strict local minimum of the unregularised loss with the relative fit error $3.6 \cdot 10^{-3}$–$2.5 \cdot 10^{-2}$ ($\varepsilon = 0.05$–0.5).

- The gauge: at six weight vectors of $W_g$ the critical point continued in $w$ from $\vartheta^b(q)$ is the gauge transform (to $\le 1.7 \cdot 10^{-11}$, $\hat{f}$ unchanged to $\le 5.5 \cdot 10^{-13}$); the adaptive prior's sensitivity $1.4 \cdot 10^{-11}$–$2.7 \cdot 10^{-10}$ of the fixed prior's; $P(w^m \in W_g) = 0.52$, 0.87, 0.997 at $m = 15$, 60, 240 ($\varepsilon \le 0.3$); the branch reaches the saturation near $\varepsilon = 0.68$.
- The fixed prior's sensitivity equals $\sigma_0 \hat{J}(\tilde{A} + \sigma_0 I)^{-1} \tilde{g}$ (the full Hessian) to $\le 2.2 \cdot 10^{-9}$, against $7.0 \cdot 10^{-4}$–$5.6 \cdot 10^{-3}$ for the Gauss–Newton form; its floor law $6.17 \cdot 10^{-2}$ ($\varepsilon = 0$) to $8.52 \cdot 10^{-2}$ ($\varepsilon = 0.5$) per unit of $1/m - 1/M$.
- The residual curvature: the amplitude row $\le 7.4 \cdot 10^{-14}$; the margins $\sigma_0 - (q_k/w_k)\lambda_k^+$ exact to $\le 2.1 \cdot 10^{-14}$ over every count vector of $m = 15$ and 60 in $W_g$, none negative; the dynamics' separation eigenvalues at a replicated configuration equal to them to $\le 5.8 \cdot 10^{-12}$.
- The dynamics ($m = 15$, 60, 240 against $M = 510$, four seeds, 6000 slow units, $\varepsilon = 0.1$ and 0.3): the fixed prior's floor within 8% of its value at $s = 600$, its fit error above the approximant's (its bias; split classes at its saddles); the adaptive priors' floor falling ($m = 60$, $\varepsilon = 0.1$: to $1.0 \cdot 10^{-9}$–$1.4 \cdot 10^{-8}$ at $\eta = \sigma_0$, $3.2 \cdot 10^{-11}$–$3.8 \cdot 10^{-9}$ at $\eta = \infty$) and the fit error tending to the approximant's (median relative excess $1.2 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$, $1.6 \cdot 10^{-5}$ at $m = 15$, 60, 240 for the samples 0.02 inside $W_g$); at the sample outside $W_g$ the systems approach the box minimum on the branch (1.107 and 1.586 times the approximant's fit error), at the sample $6 \cdot 10^{-4}$ inside it the approach is slow.
- Tests (`tests/test_gauge_floor.py`): at $\varepsilon = 0.3$, the gauge transform a critical point at two sampled weight vectors with the same function, reached by continuation from $\vartheta^b(q)$; the adaptive prior's sensitivity negligible, the fixed prior's the full-Hessian formula and not the Gauss–Newton form; the residual curvature's zero amplitude row, its scaling by $q_k/w_k$, the margins and the dynamics' separation eigenvalues; adaptive-prior runs ($\eta = \infty$ and $\sigma_0$) converging to the gauge transform with the approximant's function.

## 25.22 Proposition D‴.29: second-order plants: the sliding-variable reduction; a cogging-torque servo

Script: `exp_servo_cogging.py` → `servo/servo_cogging.md`.

- The servo $\ddot{x} = -\tau_{\mathrm{cog}}(x) + u$, $\tau_{\mathrm{cog}}(x) = 0.4 \sin(3x + 0.3) + 0.1 \sin 7x$, $x_m = \sin t$, $s = \dot{e} + 5e$; $K_0 = 4$ saturated Gaussian kernels on the position, masses 1/4, $\sigma_0 = 0.01$.
- $\vartheta^b(q)$, the nondegenerate local minimum BFGS reaches from a fixed start (no exact representation): fit error $1.46 \cdot 10^{-2}$, $\tilde{A}$ spanning $2.9 \cdot 10^{-4}\sigma_0$–$30\sigma_0$ (four modes below $0.1\sigma_0$; the adaptive prior's slowest rates $1.4 \cdot 10^{-4}\sigma_0$ and $2.9 \cdot 10^{-4}\sigma_0$ against the fixed prior's $\approx \sigma_0$), $\lambda_k^+ \le 0.0098\sigma_0$, the floor law $1.29 \cdot 10^{-2}$. Globally the loss's infimum lies at the saturation: of 120 random starts 1 ends at $\vartheta^b(q)$ and 33 below its fit error, all saturated; the best $6.39 \cdot 10^{-3}$, approached with the saturated amplitudes held at $|\alpha| = 6$–14 ($7.14 \cdot 10^{-3}$–$6.40 \cdot 10^{-3}$), not attained. A misplaced prior (phase 0.5, the engineer's model, fit error 0.200) whose replicated stationary point is a D‴.5 saddle (margin $-3.97 \cdot 10^{-3}$, fit error 0.0545).
- The averaged loop ($m = 16$, 64, 128 against $M = 256$, three seeds, 2000 slow units) from clouds around $\vartheta^b(q)$: the fixed prior's floor $1.85 \cdot 10^{-3}$, $2.24 \cdot 10^{-4}$, $4.49 \cdot 10^{-5}$ (fit error 0.174, 0.066, 0.038); the adaptive prior's ($\eta = \infty$) 38–780 times lower, the fit error 0.0298, 0.0148, 0.0147 against the approximant's 0.0146; the OT leak at the approximant with no floor; the $\sigma$-modification toward 0 biased (0.328); no leak ($\sigma_0 = 0$) as good: 0.0180, 0.0147, 0.0146 (floors $7.7 \cdot 10^{-6}$, $2.4 \cdot 10^{-7}$, $6.6 \cdot 10^{-8}$) — noise-free, the floors and biases the leaks remove are those a fixed leak creates. Misplaced: the fixed designs split their classes and end at 0.048 (OT) and 0.10–0.048 (nearest atom); the adaptive prior is unaffected at $\eta = \infty$ (its prior never acts from this start), slower at $\eta = \sigma_0$ (0.035–0.019).

- From the engineer's model (clouds around it, the prior at it): the fixed designs 0.102, 0.060, 0.048 (OT 0.048); the adaptive prior 0.027, 0.025, 0.022 ($\eta = \infty$) and 0.032, 0.028, 0.028 ($\eta = \sigma_0$) at $s = 2000$; no leak 0.0152, 0.0147, 0.0147. The collapsed $K_0$-atom flow from the engineer's model at $\eta = \infty$: 0.0297, 0.0213, 0.0169, 0.0155 at $s = 600$, 2000, $10^4$, $10^5$.
- Phase-plane kernels (five isotropic kernels): on the circle 5 exact zero Gauss–Newton eigenvalues and 8 below $10^{-6}\sigma_0$; ellipses $\times 1.1$ and $\times 2$: 8 and 6 below $10^{-6}\sigma_0$, none exact; a two-harmonic orbit 3; centred kernels 16 exact of 20.
- The real servo ($k_s = 25$, 1000 time units; records and leak targets every 0.02 time units): position-error rms 1.7–2.1 $\cdot 10^{-5}$ (adaptive designs, matching OT, no leak), $7.8 \cdot 10^{-5}$ (misplaced OT), $9.1 \cdot 10^{-5}$–$1.8 \cdot 10^{-4}$ (fixed nearest atom), 5.4–5.5 $\cdot 10^{-4}$ ($\sigma$-modification), equal to the prediction $(\mathrm{d}/\mathrm{d}t + \lambda)^{-1}(\mathrm{d}/\mathrm{d}t + k_s)^{-1} r(x_m)$ to 1.0001–1.0029 at $t = 500$ and 1000 (with the slaved $r/k_s$: 0.931–0.996, the filter's attenuation of the residual's broad fine scale).
- Tests (`tests/test_sliding.py`): the phase-plane averaged loop is the gradient flow of $\bar{L}$; on a circular orbit isotropic phase-plane kernels in general position have one zero Gauss–Newton eigenvalue each; the servo's parameters follow the averaged loop over ten periods (within 3% at $k_s = 25$, closer at 50), its position error within 1% of both predictions.

## 25.23 Proposition D‴.30: measurement noise at perfect tracking: coherent, high-passed, and the errors in variables

Script: `exp_measurement_noise.py` → `chaos/measurement_noise.md`.

- The three-bump instance at the lift of the exact representation of $q = 1/3$ ($\sigma_0 = 0.15\sigma_{\mathrm{crit}}$, $x_m = 2\sin t$, $k_e = 25$), the measurement $y = x + \nu$ in the controller and the adaptive law.
- The linear theory: $\operatorname{tr} Q = 8.784$ (the derivative Gram 8.105, the slope term 0.679); the parameters' floor per unit $N_0$ 0.1597 (fixed prior), 0.1627 ($\eta = \sigma_0$), 0.1690 ($10\sigma_0$), $0.1757 = \operatorname{tr} Q/(2k_e)$ ($\eta = \infty$); the weakest mode's parameter variance 0.0865 → 2.09 ($\times 24.1$; $\times 28.0$ in the exact linearised loop at $k_e = 25$); the fast parts $\operatorname{tr} H^2 = 0.0657$, $\langle |\hat{J}|^4 \rangle = 0.299$.
- The errors in variables (OU noise, $B = 100$): the averaged stationary function-level bias 491 (fixed) and 2397 (adaptive, every $\eta > 0$) per unit $(\mathbb{E}\nu^2)^2$, above the floor of the parameters beyond the noise std 0.0026 and 0.0012; the parameters' fluctuation changes it by less than $10^{-5}$ (2397.68); the exact periodic second-order mean of the linearised loop 504 and 2653, the excess $O(k_e^{-2})$ (10.7%, 2.65%, 0.66%, 0.17% at $k_e = 25$–200, adaptive); the bias's growth in $k_e$ at the local exponent 1.93–1.99 for $B \geq 20k_e$, 1.69 and 1.47 at $B = 8k_e$ and $4k_e$.
- The exactly linearised loop: its periodic stationary covariance at the prediction to 0.976–0.980 ($k_e = 25$), 0.994–0.995 (50), 0.9986–0.9988 (100), 0.9997–0.9998 (200), above 1 at 800 ($+1.3 \cdot 10^{-5}$): the relative error is $O(1/k_e)$, its first-order term from the slope $a$ in the noise input — with the noise input $-k_e\nu$ the deviation is numerically even in $1/k_e$ (successive ratios 3.67 → 4.00 to $k_e = 800$; not proved).
- Invariance, not attraction: at the lift the separations' second-order drift $\lambda_{\max}(S_k) = 0.0148$, 0.0101, 0.298 per $\mathbb{E}\nu^2$ in the three classes, the third's separations growing there above the noise std 0.031; along the noisy trajectory the mean error also carries the tracking error of the developing errors-in-variables deviation (rms 0.121 → 0.602 per $\mathbb{E}\nu^2$, $\lambda_{\max}(S_3)$ 0.307 → 0.349 over 300 time units). In the real loop (the lift $m = 15$, eight antithetic pairs of noise paths of 300 time units per std) the separations' rates are $-2.50 \cdot 10^{-4}$, $+2.74 \cdot 10^{-6}$ and $+4.53 \cdot 10^{-4}$ per time unit at std 0.01, 0.03, 0.05; $\lambda_{\mathrm{eff}} = (\mathrm{rate} + \sigma_0/k_e)/\mathbb{E}\nu^2$ is $0.3271 \pm 0.0038$, $0.3175 \pm 0.0037$, $0.2943 \pm 0.0034$ against the lift's 0.2981 and the transient's

0.3264, 0.3270, 0.3281; the fit's small-noise limit $0.3291 \pm 0.0033$, the next order $-13.8 \pm 2.0$; the rate vanishes at std 0.0299 over the window (0.0308 by the lift's drift, 0.0294 by the transient's); the long-time threshold is open.

- The real loop (12 noise paths of 300 time units, common to the four cases): within 1.75 standard errors of the linear prediction plus the mean part at both noise levels, the mean part 42% (std 0.003) and 89% (std 0.01) at $t = 300$, the priors 6.0–6.4% apart (predicted 5.3–6.1%); the lift $m = 15$ on a common noise path identical to the $K_0$-atom run to rounding ($5.6 \cdot 10^{-16}$ in float64).

- Tests (`tests/test_measurement_noise.py`): the closed form against the Lyapunov solution, the trace identity at $\eta = \infty$ and the floor increasing in $\eta$; the exact periodic floors of the linearised loop at $k_e = 100$ for $\eta = 0$, $\sigma_0$, $\infty$; a lift $m = 6$ under a noise path staying replicated and following the $K_0$-atom loop; a separation of the weakest class decaying at std 0.01 and growing at 0.05 (a sign test); the separations' drift against an independent quadrature of its two moments; along the second-order transient the mean parameters moved by the averaged force and the mean error carrying the tracking error of the developing deviation; the OU noise's stationary variance and correlation, and the law seeing the measurement; the responses $(H + \sigma_0 I)^{-1}$ and $H^{-1}$ to a constant force, and the quasi-static force against the full second-order mean's noise part; the exact periodic mean against the filtered-Gram response under both priors, the fixed prior's branch and its leak in the bias; the mean function's curvature term against an independent contraction.

## 25.24 Proposition D‴.31: the adaptive prior is a washout leak; the desired regressor makes the noise's own persistent force a least-squares force

Script: `exp_eiv_remedy.py` → `chaos/eiv_remedy.md`.

- The instance of Proposition D‴.30 with an Ornstein–Uhlenbeck noise of rate $B$; MR the measured regressor (Proposition D‴.30), DR the regressor at the reference, DR* the regressor and the compensation at the reference (the desired compensation adaptation law).

- The adaptive prior's function-level errors-in-variables bias per unit $(\mathbb{E}\nu^2)^2$ at $k_e = 25$, $B = 100$: from the noise's own moments 2397 (MR), 0.471 (DR), 0.0526 (DR*); with the parameters' fluctuation (the full second-order mean) 2398, 0.486, 0.0481 — the parameters' part $\leq 0.03\%$ for MR, $-8.7\%$ to $+3.3\%$ for DR and DR* over the grid except $+56\%$ for DR at $k_e = B = 100$, where its noise part nearly cancels; its own response $3.9 \cdot 10^{-4}$, $1.1 \cdot 10^{-4}$, $2.9 \cdot 10^{-5}$, $7.4 \cdot 10^{-6}$ at $k_e = 25$–$200$ ($\propto k_e^{-2}$: on the one-dimensional reference the law's leading term, a line integral along the orbit, vanishes), its share of DR's bias not falling monotonically. This part is the classical second-order bias of a constant-gain stochastic approximation (Benveniste–Métivier–Priouret, Dieuleveut–Durmus–Bach, Box's curvature bias); the desired regressor does not remove it. MR over DR $3.0 \cdot 10^3$–$2.1 \cdot 10^8$ over $k_e = 25$–$200$ and $B = 100$, 1000; MR's growing like $(k_e B/(k_e + B))^2$, DR's and DR*'s at most 1.18, below $\frac{1}{4}\|\partial_x^2 f\|_\rho^2 = 1.31$; DR's noise part at its least-squares bound to 0.01%; $k_e^2 \|P\mathbb{E}[e]\|_\rho^2$ 2394 for MR (zero by construction for DR, DR*), the mean tracking error itself $3.83 \to 0.66$ ($k_e = 25 \to 200$). The noise std at which the bias reaches the floor: 0.0012 (MR), 0.085 (DR), 0.27 (DR*).

- The prior family $\mathrm{d}z/\,\mathrm{d}s = -\eta(z - \hat{T}) - \kappa(z - z_a)$: $RH + B = I$ to $1.1 \cdot 10^{-14}$; the weakest mode's response $3269 \to 136$ as its bias factor goes $0 \to 0.959$, the trade binding on the 3 of 9 modes with $h_j < \sigma_0$; MR's bias $2397 \to 491$, DR's $0.471 \to 0.211$. Priors whose zero-frequency leak does not commute with $H$ (fixed amplitudes or centres with $\eta = \sigma_0$ elsewhere, an active face at $\eta = \infty$, the anchor gain $\eta P_{\mathrm{amp}}$, a random anchor gain with

$K + K^\top \succeq 0$), $R$ and $B$ from their dynamic blocks: equal to $(H + S_0)^{-1}$ and $(H + S_0)^{-1} S_0$, $S_0$ by elimination, to $1.9 \cdot 10^{-15}$; $RH + B = I$ to $1.4 \cdot 10^{-14}$, $HR + B - I$ 0.47–0.79.

- The real loop on common noise paths ($k_e = 25$, std 0.01, $\eta = \infty$, 12 runs of 300 time units): $\mathbb{E}\|\hat{f} - f\|^2$ at $t = 300$ $2.43 \cdot 10^{-6}$ (MR) against $2.80 \cdot 10^{-7}$ (DR) and $2.75 \cdot 10^{-7}$ (DR*), DR and DR* at their covariance part; the MR − DR run mean at its prediction to 0.5–5.3%, 0.43–1.83 times the run mean's sampling error.

- Tests (`tests/test_eiv_remedy.py`): the washout identity along a real-loop solution at finite $\eta$ and across a plan switch of the averaged loop; the collective block's transfer function against the washout formula, $RH + B = I$ along the prior family and, $R$ and $B$ from their dynamic blocks, for non-commuting leaks ($HR + B = I$ failing there); the response bound under random leaks $S \geq 0$, the smallest equivalent target perturbation and the projection; the lift exact under DR and DR*, on a common noise path DR's run within the second order of MR's (the difference $\propto$ the noise's variance), the regressor's point in the law; DR's and DR*'s noise forces least-squares with their pointwise bounds on $g$, MR's $F_1$ three orders of magnitude above; the quasi-static forces against the full second-order mean, the parameters' part above 1% for DR and DR*, below $10^{-5}$ for MR.

## 25.25 Proposition D‴.32: the cohesion leak: a leak on the unexcited subspace, without a prior

Script: `exp_cohesion_leak.py` → `chaos/cohesion_leak.md`.

- The three-bump instance ($\sigma_0 = 0.15\sigma_{\text{crit}}$), the averaged slow loop. (A) At each of the 15 count vectors of $m = 15$ the lift of $\vartheta^e(w)$ is stationary (the largest drift $2.8 \cdot 10^{-17}$ under the compact kernel, $6.2 \cdot 10^{-10}$ under the Gaussian) with (iii)'s spectrum to $2.8 \cdot 10^{-11}$ (the finite differences' error) and $4.6 \cdot 10^{-9}$ (the Gaussian kernel's cross-class terms): the separations at $\sigma_0 w_k$, the collective modes the unregularised flow's. The Gaussian kernel's bias is $2.1 \cdot 10^{-8}$, $1.3 \cdot 10^{-3}$ and 0.068 (whitened) at $R = 1$, 1.5, 2, at its first order to $6 \cdot 10^{-8}$–1.0% (56% off at $R = 3$, where the classes overlap). The splitting margins $\sigma_0 w_k - (q_k / w_k)\lambda_k^+$ are the Jacobian's to $2 \cdot 10^{-11}$; at $\sigma_0/8$ the class with the margin $-1.22 \cdot 10^{-4}$ splits ($3 \to 4$ classes, the fit error $0.017 \to 0.0148$). Twins merge under the Gaussian kernel and persist under the compact one; a dead class joins a live one at $R = 2$ and 1.5 and does not move at $R = 1$, as (v)'s time scale says.

- (I) The twins' contraction on their gauge curve is (v)'s rate to 0.04%; twins 5.3 and 8.2 apart do not move in $4 \cdot 10^4$ slow units; opposite-sign twins with a third class within reach do not merge in $4 \cdot 10^5$ slow units and end at a biased four-class fit (the fit error $3.2 \cdot 10^{-4} \to 4.5 \cdot 10^{-3}$).

- (B) From warm clouds ($m = 15$, 60, 240) both kernels remove the floor as the adaptive prior does ($\|\hat{f}_m - \hat{f}_M\|^2$ at $s = 6000$ $1.9 \cdot 10^{-10}$–$5.4 \cdot 10^{-8}$ over the representable samples at $m = 15$, against the fixed prior's floor $3.0 \cdot 10^{-4}$–$2.4 \cdot 10^{-3}$); the one sample on the boundary of $W$ (seed 3) ends at $7 \cdot 10^{-5}$ under every leak but the fixed prior.

- (C) From broad starts the annealed Gaussian kernel ($R = 2$, then $R = 1$) ends at the target's three classes in 9 of 9 runs, the fit error $1.8 \cdot 10^{-7}$–$2.3 \cdot 10^{-6}$ — as the adaptive prior told $K_0 = 3$ ($4.2 \cdot 10^{-9}$–$2.1 \cdot 10^{-6}$); told $K_0 = 2$ it ends biased (0.13–0.19), told $K_0 = 4$ with a bump split ($5.4 \cdot 10^{-4}$–$5.6 \cdot 10^{-3}$); the $\sigma$-modification ends at 0.24, no leak at 40–48 classes, the compact kernel without the broad stage at 17–22. (J) At a fixed Gaussian scale it does not: 4–6 classes at $R = 1$, three biased classes at $R = 2$; three other annealed schedules end at three classes.

- (D) The real loop ($k_e = 25$): the lift returns after a period to $7.8 \cdot 10^{-16}$, and its 45 slow Floquet exponents (36 separations at $\sigma_0 w_k / k_e$) are the predicted ones to $5.5 \cdot 10^{-6}$; from

warm clouds the rms tracking error at $t \approx 2900$ is $5.8 \cdot 10^{-4}$ ($m = 15$) and $2.4 \cdot 10^{-4}$ ($m = 60$), against the fixed prior's $1.3 \cdot 10^{-3}$ and $3.9 \cdot 10^{-4}$.

- (E) Under the dither $\tau = 10^{-5}$ the within-class spread over the second half is 0.068 (compact) and 0.063 (Gaussian) against $\sqrt{(1 - 1/n_k)\tau/(\sigma_0 w_k)} = 0.0635$ (the fixed prior's 0.036 against 0.0366; without a leak it diffuses, 0.44–0.56).

- (F) On the cogging servo the leak settles on 7–8 classes with the fit error 0.14%–0.35%, below the four-kernel designs of Proposition D‴.29 (1.46%, 0.64%) and above the best equal-weight six- and eight-kernel fits.

- (G) $\sigma$ halved stage by stage from $\sigma_0$ to $\sigma_0/64$ (the fourth bump, $m = 30$): no split down to $\sigma_0/8$ (the predicted threshold $0.121\sigma_0$), the middle class splitting at $\sigma_0/16$ in every seed; no class with a positive margin split, all 8 with a margin below $-5 \cdot 10^{-5}$ (the detection threshold $-\ln 250/10^5$) did. A split sheds single particles (the kernel's concavity: shedding one particle costs 0.43 of the quadratic cohesion energy, halving 0.74): 8–10 classes at $\sigma_0/64$, five to seven of them of one particle, the fit error 0.0028–0.0031.

- (H) Two bumps: two classes, the fit error $\leq 1.6 \cdot 10^{-8}$; the four-bump $f_\varepsilon$: three classes (0.0174), the fourth bump below the resolution; five bumps ($\sum_k |A_k|/a_{\max} = 0.875$): 6–9 classes, biased (0.045–0.118) — the transit's masses are not representable. (K) Three equal bumps: three classes ($9.4 \cdot 10^{-5}$–$1.8 \cdot 10^{-4}$, the Gaussian kernel's bias, removed by the compact kernel: (L)); four alternating bumps: four ($9.6 \cdot 10^{-3}$–$1.1 \cdot 10^{-2}$, a slow transient: (L)); three heavy bumps (0.95): 3–5, biased (0.058–0.20); two bumps closer than the broad scale: merged into one stable class (0.098–0.13). The count is not selected by the leak: it is the transit's and the schedule's.

- Tests (`tests/test_cohesion_leak.py`): the drift equals $-G\nabla J_m^{\mathrm{coh}}$ (central differences) and $J_m^{\mathrm{coh}}$ decreases along the flow, for both kernels; the forces sum to zero and equal $\sigma_0 \times$ the kernel mass $\times$ the mean-shift vector; at the exact representation for the counts $(3, 2, 2)$ and $(4, 3, 3)$ the drift vanishes and the Jacobian's spectrum is the predicted one to $10^{-9}$; at the best one-atom fit the top separation eigenvalue is $\lambda^+ - \sigma_0 w$, stable at $\sigma_0$ and split at $\lambda^+/2$; twins contract at (v)'s rate to 1% under the Gaussian kernel and persist beyond the compact range; under the compact kernel a pair at $0.99R$ attracts and at $1.01R$ does not, and the Gaussian pair force at the distance $R$ is $e^{-1/2}$ of its linear part; single linkage recovers known clusters; the real loop's lift is $T_0$-periodic and a separation contracts by $e^{-\sigma_0 w_k T_0/k_e}$ per period to $10^{-6}$.

## 25.26 Corollary D‴.33: the desired regressor keeps the replicated configuration attracting under measurement noise

Script: `exp_eiv_remedy.py` Part D → `chaos/eiv_remedy.md`.

- Proposition D‴.30's instance ($k_e = 25$, OU noise $B = 100$), per unit $\mathbb{E}\nu^2$. At the lift the separations' second-order drift has the extreme eigenvalues $[-3.7 \cdot 10^{-1}, +1.5 \cdot 10^{-2}]$; $[-2.6 \cdot 10^{-2}, +1.0 \cdot 10^{-2}]$; $[-3.5 \cdot 10^{-2}, +3.0 \cdot 10^{-1}]$ (MR), $[-1.1 \cdot 10^{-2}, -5.0 \cdot 10^{-4}]$; $[-1.5 \cdot 10^{-2}, -1.5 \cdot 10^{-4}]$; $[-7.2 \cdot 10^{-3}, -3.4 \cdot 10^{-4}]$ (DR) and $[+1.6 \cdot 10^{-4}, +3.8 \cdot 10^{-3}]$; $[+5.1 \cdot 10^{-5}, +5.0 \cdot 10^{-3}]$; $[+1.1 \cdot 10^{-4}, +2.4 \cdot 10^{-3}]$ (DR*): DR's negative definite on the instance, DR*'s largest 60 times below MR's (the threshold std 0.238 against 0.031 under the adaptive prior, 0.138 against 0.018 under the cohesion leak's $\sigma_0 w_k$).

- Against $k_e$: at $B = 4k_e$ MR's threshold std $0.031 \to 0.0054$ ($\propto k_e^{-1/2}$, $k_e = 25 \to 800$), $k_e$ times DR's and DR*'s largest eigenvalue constant ($-0.0038$, 0.1245–0.1242); at the fixed $B = 100$ DR's drift positive definite from $k_e = 200$ (std $0.185 \to 0.121$), DR*'s threshold $0.238 \to 0.113$, MR's $0.031 \to 0.014$. $\|S_k\| \leq 0.37$ of (ii)'s bound: the threshold's floor std 0.069, uniform in $k_e$.

- Other targets at the lift: DR's centre entries negative for all six, the whole drift negative definite only for alternating-sign separated kernels (three same-sign bumps $+2.9 \cdot 10^{-4}$; wide, edge and close bumps up to $+5.6 \cdot 10^{-4}$).
- At the stationary point (the exact periodic second-order mean, adaptive prior): rms $\mathbb{E}[e]$ $6.70 \cdot 10^{-4}$ (DR) and $6.44 \cdot 10^{-4}$ (DR*), 2.4% and 7.0% of the lift's, its parameters' part $6.32 \cdot 10^{-4}$ and $6.27 \cdot 10^{-4}$ (the averaged noise-only residual $2.34 \cdot 10^{-4}$, $2.60 \cdot 10^{-4}$); the drift indefinite, DR's largest eigenvalue $+3.68 \cdot 10^{-5}$ (class 1); under the fixed prior DR's drift negative definite and DR*'s positive definite (std 0.435).
- Along DR's exact second-order transient ($5.2 \cdot 10^5$ time units) the largest eigenvalue turns positive at $t = 4103$, 2947, 2714 (classes 1–3) and reaches $+1.05 \cdot 10^{-4}$ ($t \approx 3.1 \cdot 10^4$; formally std 1.64); the cross term $-8\%$ (DR) and $-17\%$ (DR*) of the class-2 drift over $[0, 50)$, DR*'s threshold std there 0.277.
- The real loop on common noise paths (four antithetic pairs per family): MR's separations grow ($+4.69 \cdot 10^{-4}$, $+1.25 \cdot 10^{-3}$ at std 0.05, 0.1; every run), DR's decay faster than $\sigma_0/k_e = 2.83 \cdot 10^{-4}$ (along $u$ $-2.965 \cdot 10^{-4}$ and $-3.365 \cdot 10^{-4}$, the noise part 0.98 and 0.97 of the transient's prediction, 0.82 and 0.81 of the lift's; along DR's own top eigenvectors faster by 0.14–0.36% at std 0.05), DR*'s slower (by 3.2% and 12.6% along its top eigenvector, at the transient's prediction); no DR or DR* run grows.
- Tests (`tests/test_eiv_remedy.py`): DR's separation drift equals MR's mean-error term at the lift and is negative definite in every class, MR's 0.30 in class 3; DR*'s drift is $O(1/k_e)$ ($k_e$ times its largest eigenvalue the same at $k_e = 25$ and 50, $B = 4k_e$) and its independent quadrature's to $10^{-4}$; at the adaptive prior's stationary point the exact periodic mean is the sum of its noise and parameters' parts, the parameters' part dominant and largely in the tangent space, the drift indefinite; the period maps of `long_transient` continue `second_order_transient` to $10^{-7}$, and DR's drift turns indefinite within 800 periods.

## 25.27 Proposition D‴.34: mass transfer: the replicator weight law with the cohesion leak

Script: `exp_mass_transfer.py` → `chaos/mass_transfer.md`.

- Proposition D‴.32's instances, the weighted network $\hat{f} = (1/m) \sum \beta_i \tilde{\Phi}_i$, the replicator law $\mathrm{d}\beta_i/\,\mathrm{d}s = \gamma \beta_i (u_i - \lambda)$ beside the transport law. (A) At fixed positions (the five bumps' atoms, two classes short of mass) the transport alone stalls at the fit error 0.388; with the replicator it reaches $2.9 \cdot 10^{-4}$ ($\gamma = 10^{-3}$) and $3.0 \cdot 10^{-14}$ ($\gamma = 10^{-2}$) at $s = 4 \cdot 10^4$; with the amplitudes scaled by 1.25 ($\sum_k |A_k|/a_{\max} = 1.094$) it stalls at 0.071 with every class saturating, creeping toward the $\ell^1$ ball's least error 0.0684 — the representability criterion (iv) for bounded trajectories.
- (B) The five-bump target from broad starts ($m = 60$, the annealed Gaussian schedule): 6–9 biased classes without the replicator (0.045–0.118), 5–7 at $\gamma = 10^{-3}$ (0.0076–0.0139), five at the bumps in 3 of 3 runs at $\gamma = 10^{-2}$ ($6.0 \cdot 10^{-4}$–$1.4 \cdot 10^{-3}$; continued, seed 1 levels off at $2.8 \cdot 10^{-4}$). (F) The three heavy bumps of D‴.32 (K) ($\sum_k |A_k|/a_{\max} = 0.95$): three classes in 3 of 3, $7.8 \cdot 10^{-8}$–$1.6 \cdot 10^{-7}$ (3–5 biased without); two bumps closer than the broad scale stay merged (0.068–0.13): the scale's obstruction is not the masses'. (D) The instance: three classes, $2.4 \cdot 10^{-7}$–$9.7 \cdot 10^{-7}$, as without.
- (G) At the class level under the Gaussian kernel the replicator selects the masses and moves the bias (B's seed 1: $3.85 \cdot 10^{-4}$ with the masses held, $2.63 \cdot 10^{-4}$ with them free; three equal bumps: $9.39 \cdot 10^{-5}$ and $9.17 \cdot 10^{-5}$); under the compact kernel the stationary points are exact either way.

- (C) The finest resolution stage ($\sigma_0/64$): the same 8–10 classes with and without the replicator — the fitness vanishes at a critical point, so the shedding of D‴.32 (G) is untouched.
- (E) The real loop at the instance's atoms ($k_e = 25$, $\gamma = 2$): $V = \frac{1}{2}e^2 + (1/(\gamma m))\sum \beta^* \log(\beta^*/\beta)$ from 0.0418 to $2.1 \cdot 10^{-7}$, monotone, $\dot{V} = -k_e e^2$ to $1.1 \cdot 10^{-16}$ (relative), $\sum \beta = m$ to $2.4 \cdot 10^{-14}$. The averaged linearisation at the lift of $\vartheta^e(w)$, counts $(6, 5, 4)$: 15 zero modes in $(\Theta, \log \beta)$ — the 12 within-class redistributions and 3 gauges — the rest stable, at $\gamma = 0$ Proposition D‴.32's spectrum to $2.1 \cdot 10^{-11}$.
- Tests (`tests/test_mass_transfer.py`): the weighted loop's drift equals the gradient of its energy in the Euclidean–Fisher–Rao metric (central differences) and is Proposition D‴.32's at $\gamma = 0$; the weights' drift matches an independent quadrature of the fitness; in the real loop at fixed positions $\dot{V} = -k_e e^2$ pointwise from the vector field and $\sum \beta$ is conserved; at fixed positions $\sum \beta = m$ is kept and the exact fit is reached with the replicator and not without it, and not below the $\ell^1$ ball's least error when $\sum |A_k|/a_{\max} > 1$; at the lift of an exact representation the zero modes in $(\Theta, \log \beta)$ are the predicted $\sum(n_k - 1)$ redistributions and $K$ gauges, a separation keeps $-\sigma_0 w_k$ with $\gamma > 0$, and the rest of the spectrum is D‴.32's ($\gamma = 0$) and stable ($\gamma > 0$).

## 25.28 Proposition D‴.35: the split of a class is an $S_n$-symmetric transcritical bifurcation

Script: `exp_split_bifurcation.py` → `chaos/split_bifurcation.md`.

- Proposition D‴.32's resolution instance (the lift of $\vartheta^b(q)$, $m = 30$, compact $R = 1$). (A) The middle class splits first ($\lambda^+ = 2.86 \cdot 10^{-4}$, the threshold $0.1213\sigma_0$); along its splitting direction the residual's third derivative is $c_3 = -1.257 \cdot 10^{-3}$.
- (B) Near the threshold Newton on the full drift finds the $S_q \times S_{n-q}$ split states ($q = 1$–$4$) where the reduced equation puts them (0.3%–3.0% at $\mu = 5 \cdot 10^{-7}$), each a saddle, its eigenvalues converging to $\mp\mu n/(n - 2q)$ and $-\mu$ as $\mu \to 0$.
- (C) The reduced quadratic equation names the first excursion's particle in 12 of 12 runs and its time to 1.1%; the linear equation names it in 6 and misses the time by up to 22.4%, the flipped one in 2. The first excursion is on the side $\operatorname{sgn}(c_3)v$ in 10 of 12. At $\sigma_0/16$ a bound pair leaves in 3 of 6 runs (the top two within 1.2%–11%), a single particle in the others; the cascade stops within a stage once the main piece's margin is positive.
- (D) Under Part G's schedule: 8–10 classes with the cohesion leak, 6–7 with the replicator ($\gamma = 0.1$). (E) With the replicator at $\gamma = 1$: 5, 6 and 8 classes (the first escapee's mass 0.134–0.136): the rate makes no difference in three seeds. (F) The mass-free leak continued to $\sigma_0/256$: 4 classes at $\sigma_0/64$ in 3 of 3 runs (the fit error 0.0149), then 9, 10, 9 at $\sigma_0/128$ and 9, 8, 9 at $\sigma_0/256$, the fit error 0.0028 — the cohesion leak's end state: the 4 classes were where the schedule stopped.
- Tests (`tests/test_split_bifurcation.py`): the full drift along $v$ against the reduced quadratic equation (the remainder cubic); the single-particle split state a saddle with the predicted eigenvalues; the predicted particle leaving first, on the predicted side; the reduced crossing against closed forms; the mass-free leak's blocks.

## 25.29 Proposition D‴.36: the gauge under dither

Script: `exp_gauge_dither.py` → `chaos/gauge_dither.md`.

- Proposition D‴.32's three-bump instance, one particle per class at the exact representation $\vartheta^e(1/3)$, the replicator at $\gamma = 10^{-2}$, the positions dithered at $\tau = 10^{-4}$. (A–B) The masses

diffuse along the manifold of exact fits at the predicted rate (the displacement's variance at 0.95, 1.36, 0.90 times the prediction over 64 seeds, within two standard errors); their means drift by about 2.6 standard errors in $5 \cdot 10^4$ slow units (the limit's drift term, computed in Proposition D‴.37).

- (C) Without a leak the masses spread to 0.05 in $2 \cdot 10^5$ slow units and the first amplitude runs into the saturation ($|a|/a_{\max} = 0.9994$, against 0.80 with fixed weights), the fit error unchanged (the dither's).
- (D) The entropic leak on the masses pins them: at $\varepsilon = 10^{-3}$ the spreads 0.0044, 0.0032, 0.0034 against the Ornstein–Uhlenbeck prediction 0.0044, 0.0031, 0.0033; at $\varepsilon = 10^{-4}$ four seeds scatter too much to test it (0.0152, 0.0103, 0.0117 against 0.0139, 0.0099, 0.0105), the 16-seed test being Proposition D‴.37 (G).
- Tests (`tests/test_gauge_dither.py`): the leak a Fisher–Rao gradient; the two-class diffusion's closed form; the leak keeping the exact representation stationary and stabilising the gauge; the leak's spread against the two-class closed form; no weight motion at $\gamma = 0$; $D$ against the projector of the code's drift Jacobian; `run_dithered`'s noise $2\tau G\,\mathrm{d}t$.

## 25.30 Proposition D‴.37: the dither's drift along the gauge

Script: `exp_gauge_drift.py` → `chaos/gauge_drift.md`.

- Proposition D‴.36's instance at $\tau = 10^{-4}$. (A) The drift from the limit map's second-order identities agrees with the integrated limit map to $10^{-3}$; (B) it is the amplitudes' noise's (each class's drains its own mass), the centres' and widths' $10^3$ times smaller.
- (C) As $\gamma \to 0$ the drift over $\gamma\tau$ becomes the replicator flow of the entropic free energy $(\tau/2)\log\det\nabla^2_\Theta J$ (0.3% at $\gamma = 10^{-5}$), $\ell = 2\sum \log(W_k a'_k) + \text{const}$; (D) in a saturation layer of width $\propto \sqrt{\gamma}$ the drift of $W$ turns outward (its zero at $\approx 0.56\delta^*$ as $\gamma \to 0$), but it is the Itô term of the amplitude's driftless diffusion: the layer slows the descent, it does not stop it.
- (E) At $\gamma = 10^{-2}$ the mean drift has three attracting states, each with two classes near their saturation — not where the masses settle; (F) the 64-seed ensemble mean follows the mean path; (G) the leak's shift is approximately $\kappa^{-1}b$ for $\varepsilon \gtrsim 3\tau$, 1.8 times it at $\varepsilon = \tau$; (H) without a leak the excursions into the saturation grow over $10^6$ slow units (the largest $|\alpha_1|$ by quarters $6.6 \to 11.3$, 11–12% of the last quarter beyond the saturation).
- Tests (`tests/test_gauge_drift.py`): the identities on flows with known limit maps and on an integrated one; the diffusion against D‴.36's projection; the slow limit against the log-determinant law; the split-invariance of the log-determinant within a class.

## 25.31 Proposition D‴.38: a Fixman correction for the replicator under dither

Script: `exp_gauge_fixman.py` → `chaos/gauge_fixman.md`.

- Proposition D‴.36's instance at $\tau = 10^{-4}$; the correction $\kappa_F(\tau/m)(1 + \hat{W}_k^2/W_k^2)/(W_k - \hat{W}_k^2/W_k)$ on the replicator's fitness, $\hat{W}_k$ from the classes' contributions (the masses form) or from the current amplitudes (the plug-in form), equal on $M$. (A) At $\kappa_F = 1$ the corrected drift is 0.31, 0.07, $2 \cdot 10^{-3}$ and $7 \cdot 10^{-4}$ of the uncorrected at $W = 1/3$ for $\gamma = 10^{-2}$, $5 \cdot 10^{-3}$, $10^{-3}$, $10^{-4}$ ($O(\gamma^2\tau)$ against $O(\gamma\tau)$); from $\gamma = 2 \cdot 10^{-3}$ on it is reversed (the finite-$\gamma$ projection weighs the correction's drift more than the dither's: 74% and 57% of their slow limits at $\gamma = 10^{-2}$), at $\gamma = 2 \cdot 10^{-2}$ by the factor 1.6.
- (B) At $\kappa_F = 2$ the corrected field's zero is the D-optimal fit $W^\sharp = (0.3765, 0.2988, 0.3247)$ (the largest $\det\nabla^2_\Theta J$; not the best-conditioned — the condition number is nearly flat along

$M$) to $2 \cdot 10^{-4}$ or better, attracting, the linearised spread there 0.040, 0.014, 0.0046 at $\gamma = 10^{-2}$, $10^{-3}$, $10^{-4}$ ($\propto \sqrt{\gamma}$); at $\kappa_F = 1$ the remainder's zero is attracting at $\gamma = 10^{-2}$ but repelling at $10^{-3}$ and $10^{-4}$. (C) In Katzenberger's limit near the saturation the density's exponent is $-0.99$, $-0.00$, $+0.98$ for $\kappa_F = 0$, 1, 2 (predicted $\kappa_F - 1$); (H) across the implemented floor's cap ($\delta_{\text{cap}} = 0.00126$) it falls back toward $-1$, as predicted to $10^{-3}$.

- (D, E) In the full loop ($\gamma = 10^{-2}$, 12 seeds, $10^6$ slow units; the plug-in form re-run with the implemented floor, the masses form with $\hat{A}$ fixed) the correction keeps the amplitudes out of the saturation (the seeds' mean of the largest $|\alpha_1|$ 4–6 against $6.6 \to 11.3$ without, no record beyond a saturation; at $\kappa_F = 1$ excursions along $M$ into a layer, 1% of the third class's last-quarter records with $|\alpha_3| > 10$); at $\kappa_F = 2$ the second half's mean is within 1.07 s.e. of $W^\sharp$ and of the mean path (plug-in) and within 1.0 (masses form); at $\kappa_F = 1$ within 1.5 s.e. of both $W^\sharp$ and the mean path, which the run cannot tell apart; equipartition with the rms fit error: $p_{\text{eff}} = 8.96$–$9.04$ with the correction, 8.06 without (a saturated amplitude holds no fit energy), the dither's level read back to 0.4%.

- (F) With the leak ($\gamma = 3 \cdot 10^{-3}$, $\varepsilon = 10^{-4}$, 16 seeds): two seeds' positions escape the exact fit's well in every form; the masses form keeps the masses where the leak holds them (0.31–0.35), the plug-in form moves them over 0.21–0.46, a running average of $\hat{A}$ drifts with the escape; on the other 14 seeds the bias reverses as predicted (within 1.1 s.e. of the first order, 1.3 of the second).

- (G) Inside the leak's window ($\gamma = 10^{-2}$, $\varepsilon = 3 \cdot 10^{-4}$, 16 seeds): the shifts ($-0.0013$, $+0.0011$, $+0.0002$), ($+0.0000$, $+0.0003$, $-0.0003$), ($+0.0012$, $-0.0005$, $-0.0007$) at $\kappa_F = 0$, 1, 2 are within 1.03 s.e. of the second order — the drift balance plus the fluctuations' term through the leak's curvature, ($-0.0004$, $+0.0002$, $+0.0002$) at $\kappa_F = 0$ —, the paired differences within 0.93 s.e. of it (the first order 1.2–4.8 s.e. off): "the bias gone at $\kappa_F = 1$" is the finite-$\gamma$ over-compensation cancelled by the fluctuation term. The same term accounts for Proposition D‴.37's Part G at $\varepsilon = 10^{-4}$ and $10^{-3}$ (within 1.7 s.e.).

- Tests (`tests/test_gauge_fixman.py`): the correction as $(\tau/2)\partial_\beta \ell$ along $M$; no correction without dither; the slow-limit cancellation; the gain's zero at $W^\sharp$; the density exponents and the floor's cap; the general form reducing to the closed form; the masses form equal to the plug-in form on $M$ and blind to an amplitude's excursion at fixed masses; the tracked contributions fixed on $M$ and relaxing at their time constant; `kappa()` as the leak's projected stiffness; the analysis' statistics on synthetic records (`tests/test_runcache.py`: the cache key carries the code's version).

## 25.32 Proposition D‴.39: dissipativity of the dithered gauge

Script: `exp_gauge_plateau.py` → `chaos/gauge_plateau.md`.

- Proposition D‴.36's instance at $\tau = 10^{-4}$; the box $L = (5, 3, 4)$ in the unconstrained coordinates $(\alpha, \gamma, \kappa)$, $\sigma_B = 1$. (A) The plateau of approximate fits with one amplitude saturated lies 7.6–7.8$\tau$ above the exact fit at $W = 1/3$ and at $W^\sharp$ (fit error 0.035), 7.66$\tau$ at the box's edge: the box bounds the plateau, it does not raise it. (B) At $W = 1/3$ the minimum-energy path climbs to it monotonically.

- (C) At fixed masses the escapes occur at $(1.33, 4.06, 10.9) \cdot 10^{-6}$ per slow unit for $\tau = 1.5$, 2, $3 \cdot 10^{-4}$ (10, 19, 24 of 24 seeds within $4 \cdot 10^5$), the Arrhenius slope ($0.81 \pm 0.14$) of $\Delta J = 7.69 \cdot 10^{-4}$ at $\Delta J/\tau = 2.6$–$5.1$ (weighted by the escape counts), within 1.3 standard errors of Freidlin–Wentzell's limit. (D) Without the box the plateau's occupation grows ($0.18 \to 0.57$ of the records by quarters, the largest $|\alpha|$ to 27.6); with it it stays at 0.09–0.19 and $|\alpha| \leq 5$. (E) In the full loop with the box ($\gamma = 10^{-2}$, 12 seeds, $10^6$ slow units) each run is the unboxed one until a coordinate leaves $B$; at $\kappa_F = 0$ the masses' spread at $10^6$

is 0.05–0.075 against 0.08–0.15 without the box, on their way to $P_B$'s vertices; at $\kappa_F = 2$ within 0.41 s.e. of $W^\sharp$. (F) Without a leak all but one of the records with $|\alpha_k| > 10$ have their mass within 0.01 of the saturation and the exact fit's error; with the leak every such record is off $M$ ($W_k$ more than 0.05 above the saturation, the fit error 0.041–0.043).

- Tests (`tests/test_gauge_plateau.py`): the dead zone inactive inside the box, minus its $G$-gradient outside; $J$ bounded on the unconstrained coordinates and $J + V$ coercive; the box's wall and vertices; a run inside the box bit for bit the unboxed one.

### 25.33 Proposition D‴.41: measurement noise on the gauge

Script: `exp_gauge_label.py` → `chaos/gauge_label.md`.

- Proposition D‴.36's instance without dither; the measured error $r + \xi$ in both laws, $\xi$ independent over the quadrature nodes, of intensity $v$. (A) At three points of $M$ and $\gamma = 10^{-2}$, $10^{-5}$, Katzenberger's Itô drift ($\Sigma = v\langle UU^\top\rangle_\rho$) and the Wong–Zakai term $(v/2)\partial\Phi\langle(U\cdot\nabla)U\rangle_\rho$ cancel to $4.6\cdot10^{-9}$–$1.1\cdot10^{-8}$ relative; the diffusion along $M$ vanishes ($\le 2.1\cdot10^{-20}$); the closed form on the masses matches the Itô drift to 2–$4\cdot10^{-4}$ at $\gamma = 10^{-5}$. (A2) Other node covariances (an offset, an orbit-correlated noise, a variance profile): the Stratonovich drift over its Itô drift $O(\gamma)$ from $\gamma \approx 10^{-3}$ on (from $10^{-4}$ for the $1 + 0.8\sin t$ profile) ($2.4\cdot10^{-5}$ at $\gamma = 10^{-5}$ for an offset), 0.2–12% at $\gamma = 10^{-1}$; the coloured-noise identity to $2.5\cdot10^{-8}$.

- (B) The slow limit's target of the per-update selection, $W_\ell = (0.3163, 0.4115, 0.2722)$, interior ($|a|/a_{\max}$ 0.79, 0.43, 0.73), far from the dither's D-optimal $W^\sharp = (0.3765, 0.2988, 0.3247)$; at $\gamma = 10^{-2}$ the Itô drift vanishes at $(0.3221, 0.3999, 0.2780)$; an Euler step in the amplitude $a$ gives the factor $(1-q)(1+3q)$ (Katzenberger's drift in those coordinates within $3.4\cdot10^{-4}$ of it) and the target $(0.2967, 0.4513, 0.2520)$.

- (C) The full loop ($\gamma = 10^{-2}$, 4 seeds × $2\cdot10^5$ slow units): the gauge's slope over the linear-response prediction averaged along the path 0.996–1.008 for fresh draws and AR(1) draws at $\rho = -1/2 \ldots 0.6$, 0.996–1.022 at $\rho = 0.8$, 0.9 (± up to 0.03), 1.004–1.027 at $\rho = 0.95$; the per-step law rejected at $\rho = 0.9$ ($0.0462 \pm 0.0006$ of the Itô drift against 0.0526); Heun's scheme $(2 \pm 7)\cdot10^{-4}$ of the Itô drift. (C2) At the step 8 ($h\lambda_{\max} \approx 1.6$): 0.979–1.049 of the prediction (± 0.004–0.053), where the step-free Itô drift is off by 42%.

- (D) $f = uv$: $\mathbb{E}[Q]$'s decay rates within 0.3% of the linear-response prediction for every scheme ($\rho$ up to 0.999, 0.19 of the per-step law there); Heun keeps $Q$ to $1.2\cdot10^{-5}$. (E) The diagonal network: the products' decay rates over the prediction 0.9988–0.9998 on average, the prediction differing across coordinates by a factor up to 2.17 at $\rho = 0.999$. (F) A tanh network: the sharpness's slope over the prediction 0.990–1.060 to $\rho = 0.9$ (slower noise below the measurement's floor).

- (G) The per-update selection's zero set at $\gamma = 10^{-2}$: one interior target for fresh draws and $\tau_c = 10$; three stable zeros, each with one class's mass large, from $\tau_c = 100$; from $\tau_c = 10^3$ three saddles and an unstable node next to $W^\sharp$ besides; $W_0$'s flow changing target between $\tau_c = 100$ and $10^3$; at $\gamma = 10^{-5}$ the targets approach the saturation ($W/W_{\min} - 1 \approx 10^{-4}$ at $\tau_c = 10^5$). The slow limit $\tau_c\Sigma_f$ within $7\cdot10^{-4}$ of its finite-step value; the slow noise's direction the dither's (cos 1.00000) at $\gamma = 10^{-5}$; a smooth noise selects 2–5% as much as a Markov one of the same correlation time.

- Tests (`tests/test_gauge_label.py`): the noise along the fields; the fields orthogonal to the gauge in the flow's metric; the Wong–Zakai cancellation at a point; $W_\ell$ interior and minimising $\Psi_\ell$, and the amplitude coordinates' target; the schemes' reductions; the AR(1) draws and Heun steps (correlated noise off $M$; dither) against hand-coded steps; the weighted and the covariance Wong–Zakai terms; the linear-response factor against the

exact stationary moments and its continuum limits; the filtered covariance against the pairwise factors; Gaussian correlation sums; the finite-step slow limit.

### 25.34 Proposition D‴.42: several particles per class

Script: `exp_gauge_classes.py` → `chaos/gauge_classes.md`.

- Proposition D‴.36's instance with 2 and 3 particles per class, together at the class's exact-fit position. (A) $\log\det\nabla^2_\Theta J$ minus its one-per-class value at the same class masses is $-46.131219$ ($m = 6$) and $-92.262438$ ($m = 9$) at every class mass and split (to $3 \cdot 10^{-7}$, $4 \cdot 10^{-7}$). (B) In Katzenberger's limit under the isotropic dither the class masses' drift and diffusion are $3/m$ times the one-per-class ones at $\gamma = 10^{-5}$ (0.5000, 0.3333, $n = 3$'s degenerate equal split included; the diffusion at $\gamma = 10^{-2}$ too); the fractions' diffusion is at most $10^{-16}$ per $\gamma^2\tau$, their drift $10^{-4}$ of the class drift at $\gamma = 10^{-5}$ and at $\gamma = 10^{-2}$ $O(\gamma)$ relative, toward the equal split.

- (C) The correction per class at $\kappa_F = 1$ cancels the class drift as for one particle per class, the fractions' drift unchanged. (D) Measurement noise at $m = 6$: at the equal split the Itô drift and the Wong–Zakai term cancel to $6 \cdot 10^{-9}$ and the class Itô drift is $m = 3$'s (1.0000); at unequal splits the residual is 2.2–2.6% of the Itô drift (up to 9.6% per component), at $\gamma = 10^{-5}$ as at $10^{-2}$; the fractions' Itô drift is the full trace's flow less it.

- (E) The full loop at $m = 6$ (8 seeds $\times$ $4 \cdot 10^5$ slow units): the class masses at (0.330, 0.349, 0.321) and (0.334, 0.315, 0.351) against $m = 3$ at half the time (0.340, 0.346, 0.314); the fractions still from the equal split, from (0.2, 0.7, 0.5) $s_1$ moving by $-0.035 \pm 0.012$ — away from the equal split — as $|\alpha|$ reaches 26–28. (F) Anisotropic dither (Richardson-extrapolated): the per-class law at the equal split for every mask (1.0000–1.0004 at $\gamma \geq 10^{-5}$, 1.0075 at $10^{-6}$), the fractions still there; at the unequal split 1.4–3.5% off it for the amplitudes-only mask, $5.9 \cdot 10^4$ times it for the centres-and-widths mask (which freezes one particle per class), the fractions drifting at 1–$6 \cdot 10^{-4}$ per $\gamma\tau$.

- Tests (`tests/test_gauge_classes.py`): the determinant blind to the split; the class masses at the law of $m$ and the fractions without diffusion; the correction per class, off $M$ and at the floor; the degenerate equal split.

### 25.35 Proposition D‴.43: the colour of the measurement noise selects the kernels

Script: `exp_colour_selection.py` → `chaos/colour_selection.md`.

- (A) The identities: Cauchy–Binet to $2.6 \cdot 10^{-15}$, the gradient $(4/N)\omega$ to $4 \cdot 10^{-9}$, $R$'s second differences negative along every random segment ($-8 \cdot 10^{-5}$ to $-0.39$, the continuum filters; zero for white draws), $\tau_c R - k\log(4\tau_c/N) \to \log(\det(\Phi_S^\top\Phi_S)\prod|w_i|)$ ($-0.065892$ against $-0.065900$ at $\tau_c = 10^6$).

- (B) Label-noise descent ($10 \times 40$, $k = 4$ where basis pursuit fails, step 0.5, $\sigma = 0.1$, eight seeds $\times$ 16 paths) follows the reduced flow (the paths' mean within 0.011–0.035 of it for fresh draws, 0.11–0.35 for AR(1) draws at $\tau_c = 100$); fresh draws end on dense fits (supports 10–13, errors 0.80–2.4, basis pursuit's 0.80–2.98); AR(1) draws recover the truth on all 16 paths of the four seeds whose reduced flow recovers it (the end interpolants' errors $10^{-13}$–$7 \cdot 10^{-11}$) and on 0–2 of 16 on the other four; at $\sigma = 0.3$ the four recover on 16, 0, 16 and 13 paths, at $\sigma = 0.5$ one seed still recovers — beyond the small-$\sigma$ limit the descent prunes on its own.

- (C) Recovery ($10 \times 40$, 40 instances per row): $k = 3$ basis pursuit 0.38, white 0.38, slow 0.62/0.75/0.78 ($\tau_c = 10/100/1000$); $k = 4$ 0.17, 0.17, 0.42/0.60/0.62; $k = 5$ 0, 0,

0.07/0.15/0.17; $k = 6$ none; at $20 \times 80$, $k = 6$ 0.35 against 0.75; the white flow run to convergence ends at basis pursuit's solution (336 of 340 converged, to $9.4 \cdot 10^{-9}$); from the white flow's end the slow flow does no better than basis pursuit at $10 \times 40$ (dropped atoms do not revive). Where the slow flow fails at $\tau_c = 100$ its end has a larger $R$ than the truth up to $k = 4$ (a local minimum; 0.09 and 0.35 of the failures end below the truth's $R$ at $k = 5$, 6); at $\tau_c = 10$, 0.39, 0.68 and 0.85 of them ($k = 4$, 5, 6) do — there the truth is not the finite-$\tau_c$ minimiser.

- (D) Gaussian kernel dictionaries (coherence 0.90, 0.96): the slow flow 0.95 against 0.80 (width 0.03, $k = 4$), 1.00 against 0.85 (width 0.05, $k = 4$), 0.78 against 0.65 ($k = 6$), the white flow at basis pursuit's rate; label-noise descent with AR(1) draws recovers the truth on two of three instances where basis pursuit fails (all 16 paths, errors $2 \cdot 10^{-13}$–$7 \cdot 10^{-8}$); on the third it is still on 12–13 atoms at the end, its reduced flow then at $8.8 \cdot 10^{-3}$ and reaching the truth when run on ($3 \cdot 10^{-13}$ at $s = 1600$); fresh draws end within 0.08–0.11 of basis pursuit's solution.

- (E) At the step 0.5 a smooth noise of $\tau_c = 10$ and a draw kept 40 steps select as well (0.70, 0.70 against fresh 0.30 and basis pursuit's 0.45 at $k = 3$: the finite-step white comparator is not basis pursuit); a smooth noise of $\tau_c = 100$ has not begun to select at the slow horizon (0.00) and recovers 0.85 and 0.65 at ten times it. (F) The flows run to convergence: from $\alpha = 0.1$, 0.5 the slow flow recovers 0.70, 0.75 ($k = 3$) and 0.40, 0.50 ($k = 4$), from $\alpha = 2$ — toward the kernel regime — 0.35 and 0.30, basis pursuit's rate (0.45, 0.25), which the white flow matches from every $\alpha$.

- (G) Sparse Bayesian learning (Tipping's EM) 0.97, 0.88, 0.55, 0.12, 0.05 for $k = 2$–6 against the slow flow's 0.97, 0.75, 0.60, 0.15, 0.00 (mean supports at $k = 4$: 6.6, 6.4; basis pursuit 8.9); the log-det's MM 0.70, 0.45 ($k = 3$, 4) from the flows' start, 0.53, 0.33 from basis pursuit; reweighted $\ell^1$ 0.38–0.53 and 0.17–0.25 at $\varepsilon = 0.01$, 0.1, 1 from either start.

- (H) The certificate (sufficient; for white draws Fuchs's, certifying 0.38 of the 2-sparse truths and 0–0.03 from $k = 3$ on) holds at every truth for $\tau_c = 100$ at $k = 2$–5 (0.93 at $k = 6$) and for $\tau_c = 1000$, the median ratio $\propto 1/\tau_c$ ($k = 2$: 0.439, 0.0814, 0.0087 at $\tau_c = 10$, 100, 1000); on kernel dictionaries (widths 0.03–0.08, $k = 4$, 6) the certified fraction rises with $\tau_c$ to 0.85–1.00 at $\tau_c = 10^5$ (white: 0–0.60), and the slow flow returns to the truth from $P = 10^{-3}$ in 0.90–1.00 ($\tau_c = 100$) and all ($\tau_c = 1000$) of the instances.

- (I) Under a fit floor ($y = \Phi w^* + \varepsilon e$, $|e| = 1$, $\varepsilon = 10^{-3}$–0.05) the slow flow's $k$ largest atoms are the truth's support in 0.70–0.75 ($k = 3$) and 0.50 ($k = 4$) of the instances, basis pursuit's in 0.40–0.45 and 0.25. (J) At the same physical noise (AR(1) of correlation time 50, per-step $\sigma = 0.1$) $\Delta \log P$ off the truth's support is $-2.39$ under Euler's step 0.5, $-0.278$ under Euler's step 0.125 and $-0.097$ under Heun's step 0.5 (25 times less): the selection is the explicit scheme's.

- Tests (`tests/test_colour_selection.py`): the subset expansion and the gradient, also with a finite step's filter; the white and slow potentials (finely sampled AR(1) draws tending to the slow one), $R$ linear (white) and concave (slow) along segments; the support count; the projection, singular cases included; the reduced flow on $M$, $P$'s rate, finite weights at a sparse fit, an overshooting trial stage rejected, white $\to$ basis pursuit, slow $\to$ the sparse truth; the flow as $R$'s gradient flow in its metric (a doubled or halved $w$-drift fails); the exact interpolant against small-step descent, the end interpolants (at infinity flagged, converging within the alive atoms' span); label-noise descent against hand-coded AR(1) steps; the dictionary; the $(u, v)$ round trip; the experiment's success threshold and dual certificate against hand computations; sparse Bayesian learning's EM on a 2-sparse truth; the white flow run to convergence; Heun's noiseless step on $M$.

### 25.36 Proposition D‴.45: the free mass–amplitude law as a mirror flow

Script: `exp_mirror_free.py` → `chaos/mirror_free.md`.

- (A) Unless stated: $5 \times 12$ Gaussian designs with unit columns, 2- or 3-sparse truths, $m = 12$, $G = 1$, $\gamma = 0.1$, $a_{\max} = 1$ ($c = 0.025$); no projection failed. $\eta$'s closed form agrees with the quadrature of $\mathrm{d}\alpha/(G\beta a')$ to $1.5 \cdot 10^{-14}$ (relative, free curves and a branch), $(\psi^*)' = w$ and $\mathrm{d}w/\,\mathrm{d}\eta = mD$ to $7.2 \cdot 10^{-11}$ and $4.3 \cdot 10^{-10}$. Along the loop $Q$ moves by at most $1.4 \cdot 10^{-11}$ noiseless and $3.6 \cdot 10^{-10}$ under a smooth noise of 30 sinusoids mixed over the nodes (the integrator's tolerance); the conserved law's shift equals the quadrature of $-\gamma\beta_i\lambda$ to 1.5–$4.4 \cdot 10^{-7}$; along D‴.44's reduced flow $\mathrm{d}Q/\,\mathrm{d}s$ is (iv)(b)'s closed form to $9.1 \cdot 10^{-16}$.

- (B) From $\alpha = 0$ and equal masses $s \cdot m/d$ the noiseless loop's end at $t = 10^7$ is the Bregman projection to $4.7 \cdot 10^{-11}$ (20 instances per $k$). For $k = 2$ the projection is 0.5% from the minimum-norm fit at $s = 10$ and its $\|w\|_1$ 1.0004 times basis pursuit's at $s = 10^{-4}$, the truth recovered nearly as often as by basis pursuit (0.75 against 0.80); at $s = 10^{-3}$ the loop is within $10^{-8}$ of its projection only from $t \approx 8.5 \cdot 10^6$. Unequal masses $s \cdot r_i$: weighted basis pursuit to $1.7 \cdot 10^{-9}$ and $2.1 \cdot 10^{-8}$ ($k = 2$, 3) at $s = 10^{-7}$, the weighted minimum norm to 2.4–$2.5 \cdot 10^{-9}$ at $s = 10^4$; from small random amplitudes at $s = 10^{-6}$ the particles sit on branches, the bias the sign-constrained one (D‴.47). (F) At $N$-atom vertices the leading order from the certificate agrees with the projection to the rounding floor ($1.8 \cdot 10^{-13}$, $2.6 \cdot 10^{-11}$ at $\varepsilon = 10^{-7}$) and the distance falls by 1068 and 984 from $10^{-7}$ to $10^{-9}$ ($\varepsilon^{3/2}$: 1000); a small slack ($\mu_{\min} = 2.7 \cdot 10^{-4}$) keeps a sixth atom, 0.17 away at $10^{-7}$.

- (C) White label noise ($\sigma = 0.3$, 64 paths): Euler–Maruyama changes $Q$ on average by its Itô correction's integral (ratios 1.0106, 1.0037, 1.0011 at $h = 0.1$, 0.03, 0.01), Heun's (Stratonovich) mean change is $1.4 \cdot 10^{-5}$, $4.1 \cdot 10^{-6}$, $1.4 \cdot 10^{-6}$, the mirror update keeps $Q$ to $2.2 \cdot 10^{-16}$; the Euler step's exact mean change is the closed form's to 0.029–0.036% at $h = 0.005$. Along D‴.44's reduced flow from the lazy projection $\|w\|_1$ first rises, then $Q$'s mean falls to $-0.021$ by its time $s = 10^8$ (no atom yet extinct): the state ends on basis pursuit where it is the truth (6 instances), elsewhere on its support (two) or on another sparse fit of larger $\ell^1$ norm (two). At $h = 0.5$ the paradigm's Euler update lowers $Q$ by 0.34–0.36 and moves the projection by 0.030–0.043, as the reduced flow does; the mirror update keeps it to $9.0 \cdot 10^{-12}$.

- (D) The real loop ($k_e = 5$, $m = 6$, six Gaussian atoms, a second particle on the third, 2000 time units): noiseless, $V$ falls monotonically from 0.175 to 0.022 with $\dot{V} = -k_e e^2$ to $3.0 \cdot 10^{-9}$, $Q$ kept to $7.6 \cdot 10^{-15}$ and the pair's mirror coordinates moving together to $2.4 \cdot 10^{-15}$; under a smooth noise in the adaptation, or a sensor noise, $V$ is no longer monotone ($0.175 \to 0.136$, 0.377) while $Q$ is kept to $6.8 \cdot 10^{-12}$ and $4.9 \cdot 10^{-13}$; at $t = 2000$ the noiseless coefficients are still 0.048 (the distinct atoms) and 0.027 (the pair's sum) from $w^*$. (E) The conserved law ($\sum \beta = m$) from the same starts reaches exact fits 1.6–$4.1 \cdot 10^{-3}$ from the free law's projections, its $Q$ moving by 0.058–0.22.

- Tests (`tests/test_mirror_free.py`): $\eta$ against quadrature in the three cases, on both sides of a branch and on the critical curve, and against a 60-digit reference near $\kappa = 1$ and $|t| = 1$; the inverse's round trips and its range errors; the conjugate $(\psi^*)' = w$ and the metric $\mathrm{d}w/\,\mathrm{d}\eta = mD$; $Q$ under a smooth node-correlated noise and the conserved law's shift $\mathrm{d}Q = -\gamma\beta\lambda\,\mathrm{d}t$; the noiseless limit equal to the Bregman projection; the rich projection converging to weighted basis pursuit; the Euler step's second-order change of $Q$; $\dot{V} = -k_e e^2$ in the real loop (and a wrong mass law seen by $V$); the mirror update keeping $Q$ and moving $\eta$ by $hg$, on branches too.

## 25.37 Proposition D‴.46: a leak in the mirror coordinates

Script: `exp_mirror_leak.py` → `chaos/mirror_leak.md`.

- (A) The identity of (i) — the leak with a disturbance and a measurement noise — holds to $9.5 \cdot 10^{-11}$ (relative) at twelve random states of the real loop ($\sigma = 0$ and 0.3). On random positive definite $H$ ($10 \times 10$, $k_e = 2$, four draws) the averaged equilibrium is the ridge with $\rho = k_e \sigma m/(G(\beta^0)^2) = 0.5$, which moves $w$ by 31–55%, to $1.3 \cdot 10^{-8}$ ($\beta^0 = 10^4$) and $1.3 \cdot 10^{-4}$ ($10^2$), but departs from it by 0.15 at $\beta^0 = 3$, where the saturation's wall $w_\sqcap = 0.25$ lies inside the coefficients' range: a ridge only below the wall; at small masses it is the LASSO of penalty $\sigma\lambda_1 = 0.02$ to 0.015, $1.5 \cdot 10^{-3}$, $1.5 \cdot 10^{-4}$ ($\beta^0 = 10^{-4}$, $10^{-6}$, $10^{-8}$; the remainder $\propto (\beta^0)^{1/2}$), with the LASSO's support in every draw.

- (B) On a persistently exciting orbit (the filtered Gram's eigenvalues $2.7 \cdot 10^{-3}$ to 0.68) under an approximation error and a biased disturbance the free law without the leak creeps toward the biased fit $w^* + H_L^{-1}\bar{d}_L$ (its largest coefficient 0.309, 0.330, 0.337, 0.341 by quarters, $V$ rising from 0.40 to 0.45); $\sigma = 0.01$ and 0.1 hold the coefficients at 0.081 and 0.040 and $V$ at 0.085 and 0.13, below $C/\lambda = 1.74$ and 0.30 and the comparison bound at every sample, the late means within $2.5 \cdot 10^{-3}$ and $6.5 \cdot 10^{-4}$ of $w_\sigma^L$, the rms error rising from 0.024 to 0.031 and 0.037. Without persistent excitation the free law's coefficient grows as constructed, 0.70, 2.31, 7.37, 23.3 at $t = 1$, $10^2$, $10^4$, $10^6$, while $\sigma = 0.01$ and 0.1 bound it (0.46 at $t = 10^6$).

- (C) Slow adaptation ($k_e = 20$): the error filter's Gram is 0.89% from the idealised $\langle\varphi\varphi^\top\rangle$, symmetric to $3 \cdot 10^{-14}$; started at the averaged equilibrium $w_\sigma^L$, the real loop's last period means lie within $1.6 \cdot 10^{-4}$ (lazy start) and $1.7 \cdot 10^{-4}$ (intermediate) of it, 4.4 and 2.6 times closer than to the idealised equilibrium (the leak's bias 0.038 and 0.12). (D) On an orbit's Gram the rich start's equilibrium is the LASSO to $3.7 \cdot 10^{-6}$–$6.5 \cdot 10^{-4}$, keeping exactly the truth's three atoms for $\mu \le 10^{-2}$ (one dropped at $3 \cdot 10^{-2}$), the lazy start's the ridge to $3.9 \cdot 10^{-6}$–$2.5 \cdot 10^{-4}$, dense; over all 1760 3-sparse patterns the support is exactly the truth's for 415, the LASSO's prediction holds for 416, the two agree on 1745; with a correlated disturbance bias the equilibrium tends to $w^* + H^{-1}\bar{d}$ as $\sigma \to 0$, linearly; from the rich start it is reached at $t \approx 3.2 \cdot 10^5$, at $2 \cdot 10^6$ where the irrepresentable condition fails (five atoms kept).

- Tests (`tests/test_mirror_free.py`): the Lyapunov identity with the leak, a disturbance and a measurement noise; $Q$ kept by the leak; the equilibrium's ridge limit, its biased limit $w^* + H^{-1}\bar{d}$ and its LASSO limit (same support) at $k_e \ne 1$, the solver's residual asserted; the comparison bound under a worst-case disturbance, coming within a factor 2 of its ultimate value; the filtered Gram's expansion $H - \langle\dot\varphi\dot\varphi^\top\rangle/k_e^2 + O(k_e^{-3})$ and its $O(k_e^{-3})$ asymmetry.

## 25.38 Proposition D‴.47: the free law's branches

Script: `exp_mirror_branch.py` → `chaos/mirror_branch.md`.

- (A) The slope $\mathrm{d}\eta/\,\mathrm{d}\log|w| = a/(\gamma a^2 + G\beta a'^2)$ holds to $2.3 \cdot 10^{-9}$ (40 random states, free and on branches); at the branch point the temperature is $1/(\gamma a_0)$ to ten digits ($\alpha_0 = 0.05$, 0.3, 1, 2); $B(\bar{w}, w) \to \beta/(\gamma m)$ as $\bar{w} \to 0$ ($2 \cdot 10^{-4}$ relative at $\bar\delta = 10^{-6}$, $5 \cdot 10^{-10}$ at $10^{-12}$); with $\alpha(0)$ uniform on $[-0.3, 0.3]$ the fraction of starts on branches is 0.52, 0.95, 0.995, 0.9996 at $\beta(0) = 10^{-3}$, $10^{-5}$, $10^{-7}$, $10^{-9}$; a $\pm$ pair's coefficient follows $2u\sinh(\gamma a_0 \Delta)$ to $2 \cdot 10^{-5}$ for $\Delta \le 10$ ($6 \cdot 10^{-4}$ at $\Delta = 120$, where the members leave their branch points); on the critical curve $\eta \cdot w^{1/3} \to -K = -3.218$ ($-3.2182$ at $\alpha = 10^{-4}$).

- (B) Random $6 \times 12$ Gaussian designs (the target on four atoms, masses $10^{-3}$, amplitudes 0.2–0.4, 8 seeds for each of three sign patterns): the flow at $t = 3 \cdot 10^7$ is the Bregman projection onto $S$, computed independently, to $5.0 \cdot 10^{-10}$; where the frozen signs cannot

fit $y$ (5 of 8 positive starts, 2 of 8 random ones) $J$ reaches its sign-constrained minimum and the wrong-signed particles go extinct (masses below $10^{-103}$) at the rates $\gamma a_0|g(\infty)|$ to 0.5–10%, $B(\bar{w}, w(t))$ never increasing by more than $2.2 \cdot 10^{-16}$; the mixed starts fit exactly. With duplicated atoms ($S$ not a point; 8 seeds) the flow is the Bregman projection to $6.4 \cdot 10^{-10}$, $1.6 \cdot 10^{-4}$–0.20 from the equal split of the duplicates: the selection with extinction is the Bregman one, not the symmetric one.

- (C) On one $8 \times 20$ design: from branches (amplitudes 0.2–0.6, masses $\varepsilon$) the projection's distance to the sign-constrained basis pursuit weighted by $1/|a(0)|$ is 0.23, 0.030, $1.3 \cdot 10^{-3}$, $5.1 \cdot 10^{-6}$, $1.1 \cdot 10^{-9}$ at $\varepsilon = 10^{-4}$, $10^{-8}$, $10^{-16}$, $10^{-32}$, $10^{-64}$, the leading order agreeing to $4.1 \cdot 10^{-4}$, $2.4 \cdot 10^{-9}$, $3.4 \cdot 10^{-14}$ from $10^{-16}$ on ($\mu_{\min} = 0.104$): the approach $\varepsilon^{\mu_j}$; from $\alpha(0) = 0$ the distance to basis pursuit is 0.29, 0.14, $4.0 \cdot 10^{-4}$, $5.2 \cdot 10^{-8}$, $4.3 \cdot 10^{-11}$ at $\varepsilon = 10^{-2}, \ldots, 10^{-10}$: $\varepsilon^{3/2}$. In a mixed start the free particles carry $\approx \varepsilon^2$ — or, where the branch atoms' frozen signs cannot fit, 2.27 on the power scale, the projection failing at $\varepsilon = 10^{-12}$ (residual 1.95). (D) A $\pm$ pair per atom tends to basis pursuit weighted by $1/a_0$, slowly (slacks 0.016, 0.020): 0.29 at $u = 10^{-4}$, $2.6 \cdot 10^{-4}$ at $10^{-128}$, $1.9 \cdot 10^{-6}$ at $10^{-256}$; never are both members of an atom active.

- (E) Positive Gaussian dictionaries, positive branches: with 21 atoms of width 0.15 and 7 samples (margin $\mu = 0.015$) the loop reaches the target from every start scale, $\beta(0) = 10^{-6}$ to 10, within $6.0 \cdot 10^{-5}$ at $t = 10^9$, where from $\alpha(0) = 0$ the rich start ends on a seven-atom fit 1 away and the lazy one spreads over all 21 atoms; at $\mu = 0.003$ it is still $1.0$–$1.3 \cdot 10^{-3}$ away at $10^9$, at $\mu = 0.027$ on the target's atoms; the decay is algebraic ($t^{-0.9}$–$t^{-1}$). Over all three-atom supports the median margins are $4.0 \cdot 10^{-6}$, $3.8 \cdot 10^{-9}$, $1.2 \cdot 10^{-3}$ (25%, 45%, 4% at most $10^{-9}$); the loop reaches the supports at the quantiles whose margins exceed $10^{-3}$ by $t = 10^9$, and on those below $10^{-4}$ is still 0.37–0.83 away: the selection's speed is the margin. (F) The real loop on positive branches: $\dot{V} = -k_e e^2$ to $1.7 \cdot 10^{-8}$, $V$ falls monotonically from 3.64 while the unused atoms' masses rise to 0.78; after 4000 time units the coefficients are still 0.080 from $w^*$.

- Tests (`tests/test_mirror_free.py`): the branch criterion and $\delta(0)$ solved from $\beta(0)$ (masses far below $c$'s precision kept), $\eta$'s inverse to any depth, the temperature $1/(\gamma a_0)$ at a branch point, $B(\bar{w}, w) \to \beta/(\gamma m)$ as $\bar{w} \to 0$, a $\pm$ pair's $W(\Delta) = 2u \sinh(\gamma a_0 \Delta)$ at leading order; the limit theorem: from positive branches that cannot fit the labels the loop converges to the projection onto the sign-constrained fits (computed independently), the vanishing coordinates' masses below $10^{-30}$.

### 25.39 Proposition D‴.48: the rich transit

Script: `exp_mirror_transit.py` → `chaos/mirror_transit.md`.

- (A) From $\alpha = 0$ on three $6 \times 14$ Gaussian designs (3-sparse truths, masses $\varepsilon b_i$, $b_i$ log-uniform in $[1/4, 4]$) the adaptive inverse scale space path of $\sum |w_i|/\sqrt{b_i}$ ends at weighted basis pursuit after 3, 8 and 10 events, and the loop in $\tau = t/\Lambda(\varepsilon)$ follows it: $\sup_\tau |u - u_{\mathrm{ISS}}|$ falls from 0.54, 0.61, 1.15 at $\varepsilon = 10^{-2}$ to $6.7 \cdot 10^{-6}$, $8.8 \cdot 10^{-6}$, $2.0 \cdot 10^{-5}$ at $10^{-12}$ (log-log slopes 0.500 between $10^{-6}$ and $10^{-8}$: $1/\Lambda \propto \varepsilon^{1/2}$); the entry times converge at that rate down to the grid's step; the residual's difference is $O(1)$ at an event over a $\tau$-width $\propto \varepsilon^{1/2}$ and $O(\varepsilon^{3/2})$ away from the events (median slopes 1.51, 1.69, 1.46); the end at $1.5\tau^*$ is weighted basis pursuit to $1.1 \cdot 10^{-16}$–$6.1 \cdot 10^{-15}$ at $10^{-12}$.

- (B) On branches (amplitudes 0.2–0.6) the loop follows the one-sided path in $\tau = t/\log(1/\varepsilon)$: $\sup|u - u_{\mathrm{ISS}}| = 15.7$, 8.0, 4.0, 2.0, 1.0 and the entry times' largest difference 55, 28, 14, 7.0, 3.5 at $\varepsilon = 10^{-4}$, $10^{-8}$, $10^{-16}$, $10^{-32}$, $10^{-64}$, exactly $\propto 1/\log(1/\varepsilon)$ (the path's weights $1/(\gamma|a(0)|) \approx 17$–50), ending at the sign-constrained weighted basis pursuit (to $1.3 \cdot 10^{-5}$

at $\varepsilon = 10^{-4}$, below $10^{-14}$ from $10^{-16}$ on); $\pm$ pairs follow the two-sided path the same way (21, 7.9, 4.0, 2.0, 0.99 at $u = 10^{-4} \ldots 10^{-64}$).

- (C) Moving centres: the sliding path enters three kernels that slide onto the three bumps and fit them exactly; the loop with the paradigm's metric forms 8 kernels at $\varepsilon = 10^{-3}$, 4 at $10^{-4}$ and exactly these three for $\varepsilon \le 10^{-5}$, the others displaced $\propto \varepsilon$; with the Wasserstein–Fisher–Rao metric the centres follow (iv)'s limit and the loop forms 15, 12, 8, 3, 3 kernels at $\varepsilon = 10^{-4}$, $10^{-5}$, $10^{-6}$, $10^{-8}$, $10^{-10}$; with the centres held on the grid it forms 5 kernels, 0.019 from the target. On sixteen random targets the loop forms exactly three kernels at the bumps on 11, 13, 16 at $\varepsilon = 10^{-5}$, $10^{-6}$, $10^{-8}$ (Wasserstein–Fisher–Rao: none at $10^{-6}$, all at $10^{-10}$); a tie is shared (0.4, 0.4), a near tie takes two kernels at $\varepsilon = 10^{-6}$; the creep's speed is the profile's to 0.990–0.996 ($\varepsilon = 10^{-8}$) and 0.9999–1.0000 ($10^{-10}$); wider bumps are fitted along a staircase. (D) Moving widths fit bumps of widths 0.2 and 0.08 with exactly three kernels at $\varepsilon = 10^{-6}$, where fixed widths need six kernels or miss the amplitudes (0.71, −0.48, 0.59).
- Tests (`tests/test_rich_transit.py`, `tests/test_mirror_free.py`): the critical curve's coefficient and its depth-3 form; the sliding path recovers three bumps exactly; the loop at $\varepsilon = 10^{-5}$ forms the same three kernels at the path's entry times; the inactive kernels' coefficients below the core's bound before the first entry; the Wasserstein–Fisher–Rao metric's centres on their limit; the inverse scale space path ends at weighted basis pursuit (one-sided: the sign-constrained one) and the loop at $\varepsilon = 10^{-6}$ follows it.

### 25.40 Theorem E: label-free convergence under excitation

Script: `exp_pe_convergence.py` → `pe_convergence/`. Table 34 and Figure 7 show the results.

Table 34: Theorem E: tangential excitation and label-free parameter convergence ($m = 3$, $n = 2$, permuted start, $T = 200\,\mathrm{s}$). $\lambda_{\min}$ is that of the excitation Gramian.

| reference | seed | $\lambda_{\min}$, first 60 s window | min over sliding 60 s windows | $W(T)/W(0)$ | fitted rate of $W$, $t \ge 100\,\mathrm{s}$ (1/s) | labelled($T$) / labelled(0) |
|---|---|---|---|---|---|---|
| rich (6 tones) | 0 | $8.60 \cdot 10^{-3}$ | $4.12 \cdot 10^{-3}$ | 0.544 | $2.31 \cdot 10^{-3}$ | 0.992 |
| rich (6 tones) | 1 | $5.52 \cdot 10^{-3}$ | $1.02 \cdot 10^{-3}$ | 0.409 | $2.32 \cdot 10^{-3}$ | 0.962 |
| rich (6 tones) | 2 | $1.08 \cdot 10^{-3}$ | $5.91 \cdot 10^{-4}$ | 0.369 | $3.07 \cdot 10^{-3}$ | 1.007 |
| single sine | 0 | $2.54 \cdot 10^{-2}$ | $2.41 \cdot 10^{-2}$ | 0.274 | $4.05 \cdot 10^{-3}$ | 0.988 |
| single sine | 1 | $2.96 \cdot 10^{-2}$ | $2.71 \cdot 10^{-2}$ | 0.314 | $4.26 \cdot 10^{-3}$ | 0.967 |
| single sine | 2 | $6.74 \cdot 10^{-3}$ | $6.61 \cdot 10^{-3}$ | 0.226 | $2.91 \cdot 10^{-3}$ | 1.008 |

- The matched error decays in all six runs. The labelled error stays within 4% of its initial value.
- The rank correlation between the uniform excitation level and the fitted rate is 0.60 over the six runs. The PE level is a hypothesis of Theorem E, not a rate predictor (Remark E.1).
- The proof's linearisation is checked exactly: the Jacobian equals $A(t)$, and $QA + A^\top Q = \mathrm{diag}(-k_e I, 0)$.

### 25.41 Lemma E.0 and Theorem E with explicit constants

Script: `exp_explicit_rate.py` → `explicit_rate/`. The explicit rate of Lemma E.0 is compared with the exact Floquet rate of the linearised loop (Tables 35 and 36). It lies below the exact rate

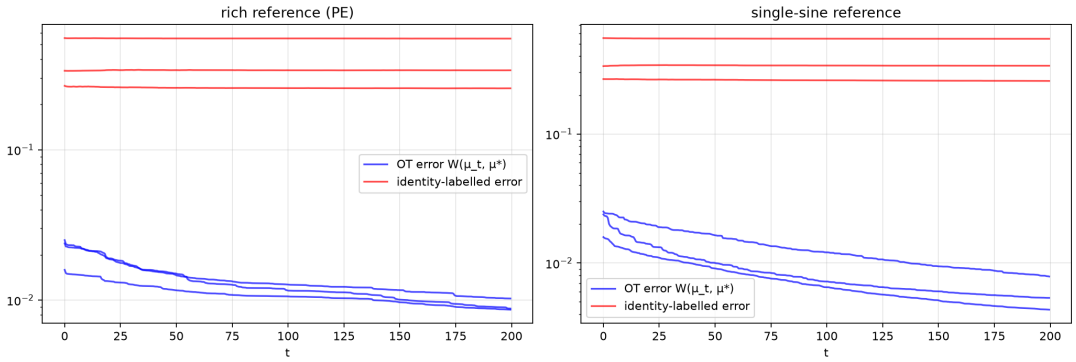



Figure 7: Theorem E: OT error $W(\mu_t, \mu^*)$ and identity-labelled error for the three seeds, with the rich reference (left) and the single-sine reference (right).

in every row; the ordering is numerical ($\Psi_M$, $\Psi_D$ are grid maxima, $\alpha$ a converged quadrature and $\varepsilon$ a grid optimum), with margins of at least $17\times$.

Table 35: Lemma E.0: the explicit rate against the exact Floquet rate of the linearised loop (`explicit_rate/table.md`).

| instance | regime | exact / explicit |
|---|---|---|
| Remark E.1 ($m = n = 1$, $G = I$, $k_e = 3$), $\omega = 1, \dots, 64$ | $\Psi_M^2/k_e \approx 1$ | $954 \to 17.6$ (decreasing in $\omega$) |
| the ideal of the Theorem E tests ($m = 3$, $n = 2$, $k_e = 6$), two-tone reference of period $20\pi$, $G = g \cdot \mathrm{diag}(2, 2, 1)$, $g = 30, \dots, 0.01$ | fast $\to$ slow | $6.56 \cdot 10^{10} \to 4.22 \cdot 10^5$ (optimised $\varepsilon$; $8.74 \cdot 10^{10} \to 5.63 \cdot 10^5$ with the closed-form $\varepsilon$) |

Table 36: Lemma E.0 in Remark E.1's instance, $x_m = 1.5 \sin \omega t$ ($m = n = 1$, $G = I$, $k_e = 3$): PE window $T$, level $\alpha(T)$, $\Psi_D$ ($\Psi_M = 1.78$ throughout), the exact Floquet rate and the explicit rate with the closed-form and with the optimised $\varepsilon$.

| $\omega$ | $T$ | $\alpha(T)$ | $\Psi_D$ | exact rate | explicit, closed-form $\varepsilon$ | explicit, optimised $\varepsilon$ |
|---|---|---|---|---|---|---|
| 1 | 6.28 | 1.13 | 3.69 | $4.660 \cdot 10^{-2}$ | $3.688 \cdot 10^{-5}$ | $4.886 \cdot 10^{-5}$ |
| 2 | 3.14 | 0.566 | 7.38 | $2.158 \cdot 10^{-2}$ | $7.858 \cdot 10^{-5}$ | $1.040 \cdot 10^{-4}$ |
| 4 | 1.57 | 0.283 | 14.8 | $6.834 \cdot 10^{-3}$ | $8.696 \cdot 10^{-5}$ | $1.152 \cdot 10^{-4}$ |
| 8 | 0.785 | 0.142 | 29.5 | $1.831 \cdot 10^{-3}$ | $4.736 \cdot 10^{-5}$ | $6.301 \cdot 10^{-5}$ |
| 16 | 0.393 | 0.0708 | 59 | $4.662 \cdot 10^{-4}$ | $1.650 \cdot 10^{-5}$ | $2.199 \cdot 10^{-5}$ |
| 32 | 0.196 | 0.0354 | 118 | $1.171 \cdot 10^{-4}$ | $4.720 \cdot 10^{-6}$ | $6.286 \cdot 10^{-6}$ |
| 64 | 0.0982 | 0.0177 | 236 | $2.931 \cdot 10^{-5}$ | $1.249 \cdot 10^{-6}$ | $1.665 \cdot 10^{-6}$ |

- In Remark E.1's instance the exact rate falls like $\omega^{-2}$ (log-log slope $-1.97$ over $\omega \geq 4$); the explicit rate approaches that law (slope $-1.56$ over $\omega \geq 4$, $-1.92$ over the last octave $\omega = 32 \to 64$). Lemma E.0(iii) derives the $\omega^{-2}$ law from $\Psi_D \propto \omega$.
- In the slow-adaptation limit ($T\Psi_M^2 \ll k_e$) the ratio decreases towards a constant of order $\kappa_\Psi(1 + \Psi_D/(k_e\Psi_M))^2$ for the optimised $\varepsilon$ ($3\kappa_\Psi(\cdots)^2$ for the closed-form bound (ii)), with $\kappa_\Psi$ the condition number of the averaged Gramian ($1.27 \cdot 10^5$ for this ideal; formula values $4.9 \cdot 10^5$ and $1.47 \cdot 10^6$; measured $4.22 \cdot 10^5$ with the optimised $\varepsilon$ and $5.63 \cdot 10^5$ with the closed-form $\varepsilon$ at $g = 0.01$). On this instance the exact rate is 0.66 of $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k_e$, because $k_e = 6$ is not large against the reference's bandwidth and the $T^2\Psi_M^3$ term is still

13% of $c_1$; the PE level is evaluated with 800 Gauss–Legendre nodes, converged over the $20\pi$ window. The exact rate pays for ill-conditioning once, the bound twice (Remark E.0a).

- Lemma E.0′ (the slow regime; table 4 of the same file): on the $g = 0.01$ ideal ($\kappa(\bar{M}) = 7.8{\cdot}10^3$; peak factors $\rho = 69.9$, $\rho_D = 3.7 \cdot 10^3$, $\rho_{DD} = 3.7 \cdot 10^5$; threshold $k_0 = 226$ for $\delta = 0.2$, 321 for $\delta = 0.1$) the guaranteed rate is within a factor 1.26 ($\delta = 0.2$) and 1.12 ($\delta = 0.1$) of the exact Floquet rate at $k = 400$ and 600, where Lemma E.0's optimised rate is $1.3 \cdot 10^5$ times smaller; the certificate $\sup \dot{V}/V \le -\gamma$ holds on the grid wherever the conditions hold; exact/$(\lambda_{\min}/k) = 0.9885$, 0.9971, 0.9993, 0.9997, 0.9998, 0.9999 at $k = 50, \dots, 600$ (the $k^{-2}$ law of $N_2 = 0$).
- Corollary E.0′.2 (Theorem E in the slow regime; table 5): on the single-kernel instance at $k_e = 40$ the nonlinear rate from Lemma E.0′'s $V$ is $4.53 \cdot 10^{-4}$ ($\delta = 0.2$) against $2.63 \cdot 10^{-5}$ from Lemma E.0 and an exact linear rate of $1.35 \cdot 10^{-3}$; the explicit ball radius is $1.53 \cdot 10^{-6}$ against $6.15 \cdot 10^{-6}$ (the anisotropy of the quadratic form), the overshoot 11.8 against 1.43.
- Tests (`tests/test_excitation.py`): the pointwise inequalities of Lemma E.0 on random states for four regressors, the envelope along trajectories, the ordering explicit $\le$ exact, the $\omega^{-2}$ law, the no-growth of the ratio as the gains shrink, and the explicit nonlinear envelope of Theorem E on the implemented closed loop (single-kernel instance with $G = 0.3\,I$, where $r_0 \approx 10^{-5}$ in the $w$-coordinates and the explicit nonlinear rate is $3.7 \cdot 10^{-5}$: the inequality holds at every sample over 8000 s, about 0.3 e-folds of the explicit envelope, while the state itself decays by far more). Lemma E.0′: the certificate and the sharpness on $\Psi = (\sin t, \cos t)$ ($k = 20$, 50, 200; ratio $\to 1/(1-\delta)$) and on the slow-adaptation ideal at $k = 400$ (ratio $\le 1.5$; more than $10^4$ above Lemma E.0's rate). Corollary E.0′.2: the explicit envelope on the implemented single-kernel loop at $k_e = 40$ over 1.5 e-folds ($\mathrm{d}t = 0.005$, below the fixed-step discretisation floor).
- Corollary E.0′.3 (table 6): the explicit bound on the stationary parameter spread of the dithered (LTV), with $\Sigma = (2 \cdot 10^{-3})^{1/2} I$, against the exact periodic stationary covariance: on $\Psi = (\sin t, \cos t)$ the ratio bound/exact is 2.69, 1.58, 1.31 ($\delta = 0.2$) and 2.52, 1.41, 1.17 ($\delta = 0.1$) at $k = 20$, 50, 200, falling toward $1/(1-\delta)$; on the Theorem E ideal ($\kappa = 7.8 \cdot 10^3$) it is 15.4–17.5, the price of $q = 15$ directions charged at the worst excitation; the exact value equals the averaged Langevin covariance $k \cdot \operatorname{tr} X$ within $3 \cdot 10^{-4}$ (ideal) and $1.25/k^2$ (sinusoid), and $\mathbb{E}\|e\|^2 = \|\Sigma\|_F^2/(2k) = \operatorname{tr}(\bar{M}X)/k$ to five digits.
- Lemma E.0″ (`exp_pe_general.py` → `explicit_rate/general_pe.md`): the exponential-average construction on two non-periodic regressors. Quasi-periodic $\Psi = (\sin t, \cos \sqrt{2}\, t)$: $\lambda_T = 0.297$, 0.391, 0.444 for windows $2\pi$, $4\pi$, $8\pi$; conditions from $k = 20$; exact/guaranteed ($\delta = 0.2$ / 0.1) 2.49/2.13 $\to$ 1.63/1.45 at $T = 4\pi$ and 2.10/1.81 $\to$ 1.43/1.27 at $T = 8\pi$ from $k = 20$ to 200; the exact rate is $0.5/k$ to within 1%. Chaotic $\Psi = (x_1/20, (x_3 - 25)/20)$ along a Lorenz orbit traversed five times slower than standard time: $\lambda_T = 0.015$, 0.024, 0.051 for windows 5, 10, 20 against an effective excitation 0.15 (exact rates $0.00759/k \cdot 20$ …); conditions from $k = 100$; exact/guaranteed 5.0/4.3 ($k = 100$) $\to$ 4.2/3.7 ($k = 200$) at $T = 20$, dominated by the window loss $\lambda_\infty/\lambda_T \approx 3$; the certificate holds at every admissible row.

### 25.42 Theorem F: transport cost

Theorem F is checked by tests.

- Along (L), the particle path length (which is at least the Wasserstein length) stays below $\lambda_G^{1/2} L_1 \int \|e\|$ at every sample.
- Along (L$_\sigma$) with the OT leak, $W(\mu_{t_1}, \mu_{t_0}) \le \int (\lambda_G^{1/2} L_1 \|e\| + \sigma_m W(\mu_r, \mu^0))$ on a grid of pairs.

- Lemma P.1 holds on 2000 adversarial states, and a tight instance of F(i) reaches ratio $\geq 0.99$.

### 25.43 Proposition G: slow geometry

Scripts: `exp_variable_projection.py` and `exp_slow_manifold_slope.py`, both → `variable_projection/`.

In Table 37, ranges are along the run (5 samples), and "$a$ / $b$" means first-quarter / last-quarter medians. Velocities are centred differences of period means.

Table 37: Proposition G: the two configurations, `violating` ($m = 4$, $k_e = 20$, $g_a = 2$, $\delta = 0.1$, $T = 800$ s) and `satisfying` ($m = 2$, $k_e = 20$, $g_a = 13$, $\delta = 0.004$, $T = 3000$ s). In the rows for $L^*$ and $\bar{L}$, "$x \to y$" is first sample → last sample.

| quantity | violating ($m = 4$) | satisfying ($m = 2$) |
|---|---|---|
| (H2a) $\omega_{\text{ref}}$ / joint fastest | 15.3–25.6 | 6.53–7.27 |
| (H2b) slowest weight / geometry | 0.0993–0.231 | 78.8–87.2 |
| (H2c) $S$ | 5.44–25 | 0.00601–0.00602 |
| G(a) direct check (cosine, $G^{-1}$ metric) | 0.9994 / 0.9985 | 0.9848 / 0.9624 (geometry block 0.9992 / 0.9982) |
| G(b) cosine | −0.118 / 0.385 | 0.999 / 0.999 |
| G(b) size ratio | 4.207 / 0.472 | 0.969 / 0.970 |
| $L^*$ | **increases** $(3.8136 \cdot 10^{-3} \to 5.8866 \cdot 10^{-3})$ | decreases $(1.1987 \cdot 10^{-2} \to 4.1073 \cdot 10^{-3})$ |
| $\bar{L}$ at period-mean weights | non-increasing $(0.21892 \to 0.0080127)$ | non-increasing |

**Slope counterexample.** Table 38 (`table_slope.md`). Here $m = 1$, $k_e = 20$, gains $(g_a, \delta, \delta)$, and (H2b) holds in every row. The closed loop and the joint flow start on the slow manifold of (b′).

Table 38: Proposition G(b): closed loop against the averaged predictions. "closed / (b)" compares the closed loop with plain variable projection (VarPro) G(b), and "closed / (b′)" with the induced-metric flow G(b′); size is the ratio of displacement norms.

| case | (H2b) ratio | $S$ | closed / (b) | | closed / (b′) | | (a) / (b′) | closed / (a) |
|---|---|---|---|---|---|---|---|---|
| | | | size | cosine | size | cosine | | |
| $\delta = 0.47$, $g_a = 20$ | 81–178 | 2–2.81 | 0.422 | 0.9303† | 1.121 | 0.9986 | 1.015 | 1.104 |
| $\delta = 0.47/4$, $g_a = 5$ (all gains ÷4) | 81–178 | 2–2.81 | 0.401 | 0.9156 | 1.065 | 0.9999 | 1.015 | 1.049 |
| $\delta = 0.047$, $g_a = 20$ | 654–1779 | 0.2–0.319 | 0.850 | 0.9986 | 1.047 | 1.0000 | 1.003 | 1.044 |
| $\delta = 0.0047$, $g_a = 20$ | 13162–17790 | 0.02–0.0229 | 1.023 | 1.0000 | 1.044 | 1.0000 | 1.000 | 1.044 |

†The centre moves the opposite way.

- (a) / (b′) isolates the $O(1/R)$ error of the induced-metric flow.
- closed / (a) isolates the averaging error, which shrinks with the gains, plus a 4.4% $O(1/k_e)$ error-channel factor.

What the tables show:

- G(b) holds in the satisfying configuration, where (H2b) and $S \ll 1$ both hold. The violating configuration fails both.
- The slope rows show that (H2b) alone is not sufficient.
- The induced-metric flow G(b′) tracks the closed loop at every $S$.
- In the satisfying configuration, the G(a) weight block is dominated by the $O(1/k_e)$ shift of the weight equilibrium: cosine $-0.21$ / $0.50$, carrying 92–95% of the residual. The geometry block matches.

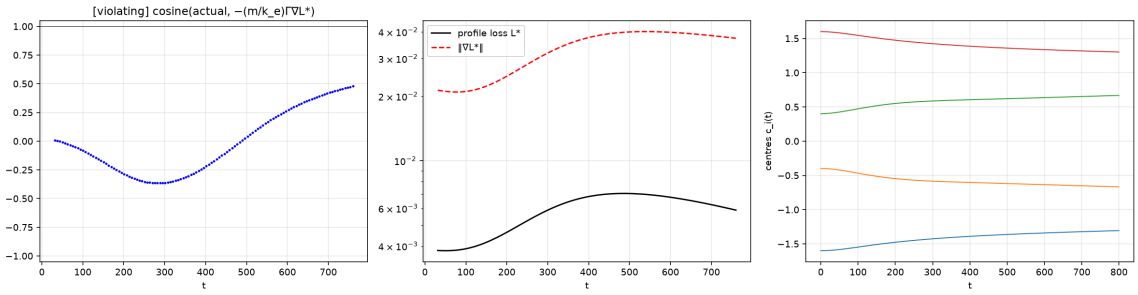



Figure 8: Proposition G, `violating` configuration. Left: the G(b) cosine between the actual geometry velocity and the VarPro direction $-(m\delta/k_e)\Gamma\nabla L^*$, along the run. Centre: the profile loss $L^*$ and $\|\nabla L^*\|$. Right: the centres $c_i(t)$.

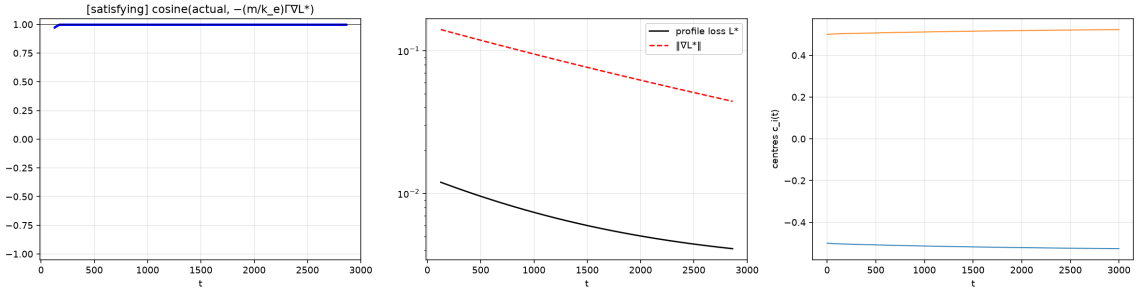



Figure 9: Proposition G, `satisfying` configuration; panels as in Figure 8.

## 25.44 Theorem H: Langevin adaptive control

Scripts: `exp_langevin_global.py` → `langevin_global/`, `exp_langevin_tradeoff.py` → `langevin_tradeoff/`, and `exp_gibbs_reference.py` → `langevin_global/gibbs_reference.md`.

The setting is the real closed loop with $m = 32$, a scalar plant and a periodic reference.

- The unregularised offline optimum of the occupation loss is about $7.6 \cdot 10^{-6}$ (best of 5 L-BFGS restarts, an upper bound). One run gave $7.0 \cdot 10^{-6}$, because of floating-point summation order.
- The regularised optimum $\bar{L}(\mu_\tau^*)$ is 0.11118 at $(\sigma, \tau) = (0.02, 0.004)$ and 0.00850 at $(0.002, 5 \cdot 10^{-4})$, grid-converged (`exp_gibbs_reference.py` → `langevin_global/gibbs_reference.md`).
- In the scaling $\sigma = 0.2/k_e$, $\tau = 0.04/k_e$ of Conjecture H.6, $\bar{L}(\mu_\tau^*) = 0.11118$ for every $k_e$. The exactly averaged stationary state gives $\bar{L} = 0.11415$, 0.11148 and 0.11101 at $k_e = 10$, 20 and 40. The $L^2(\rho)$ distance between the two learned functions is $6.81 \cdot 10^{-3}$, $1.43 \cdot 10^{-3}$ and $9.7 \cdot 10^{-4}$.

**Speed–bias trade-off.** Table 39 and Figure 10 ($\sigma = 0.002$, 2 seeds, `langevin_tradeoff/table.md`).

Table 39: Theorem H: speed–bias trade-off of parameter dither ($m = 32$, $\sigma = 0.002$, $k_e = 10$, $T = 1500\,\mathrm{s}$, 2 seeds). The escape time is the first sample at which the collapsed run's loss is at most twice the spread run's final loss (same seed); per-seed values are listed, and $\infty$ means not within the horizon. The losses are the final values of $\bar{L}$.

| $\tau$ | escape of the collapsed start (s) | loss, collapsed | loss, spread |
|---|---|---|---|
| 0 | $\infty$, $\infty$ | 0.166 | 0.00696 |
| 0.0005 | $\infty$, 1050 | 0.0249 | 0.0109 |
| 0.001 | 1480, 1005 | 0.0216 | 0.0148 |
| 0.002 | 1115, 800 | 0.0288 | 0.0233 |
| 0.004 | 970, 430 | 0.0392 | 0.0362 |
| 0.008 | 390, 295 | 0.0603 | 0.0653 |
| annealed (hot start) | 990, 530 | 0.0317 | 0.0320 |

- **Fit.** The seed-averaged escape time fits $A + B\log(1/\tau)$ with an RMS relative residual of 0.045, against 0.253 for an Arrhenius fit. On the eight per-seed values both laws fit poorly (0.274 and 0.330). The data do not identify the escape mechanism, and the "saddle" reading of an earlier version of this analysis is withdrawn (Remark H.2).
- **Caveats.** The threshold grows with $\tau$, and the censored $\tau = 5 \cdot 10^{-4}$ point is excluded from the fit.
- **Guaranteed rate.** The Holley–Stroock [43] rate $2\tau\rho_\tau$ is at least $e^{-8401}\,\mathrm{s}^{-1}$ at $(0.002, 5 \cdot 10^{-4})$ and at least $e^{-1053}\,\mathrm{s}^{-1}$ at $(0.02, 0.004)$.

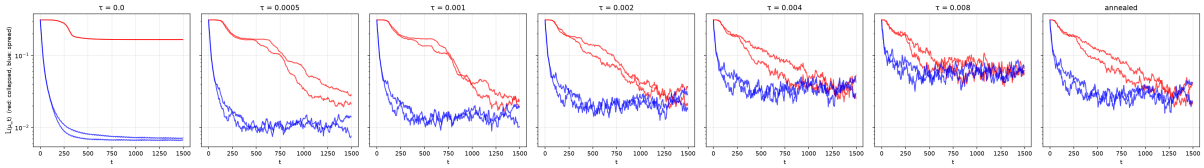


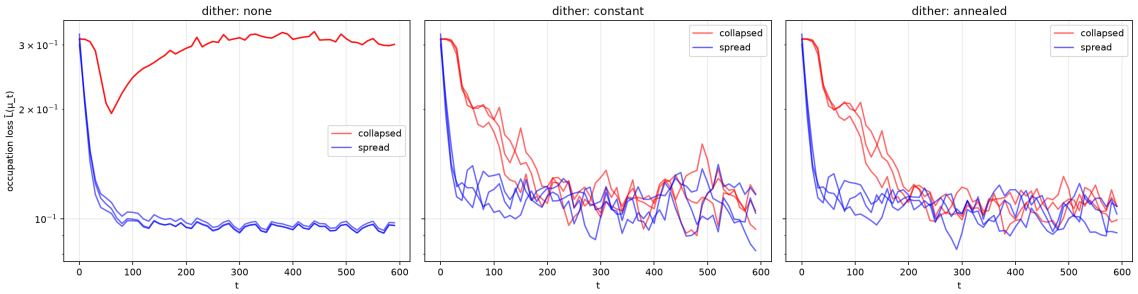

Figure 10: Theorem H, speed–bias trade-off: $\bar{L}(\mu_t)$ of the collapsed start (red) and of the spread start (blue), 2 seeds, for $\tau = 0$, 0.0005, 0.001, 0.002, 0.004, 0.008 and the annealed (hot-start) schedule.

**Two fixed settings.** Three seeds (`langevin_global/table_*.md`; Figure 11).

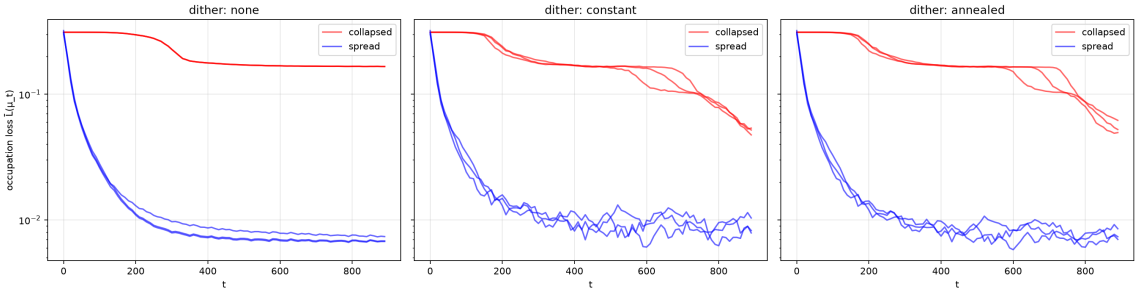
- $\sigma = 0.02$, $\tau = 0.004$**.** Both starts equalise: the loss ratio is 1.03, and the final measures are 4.2× closer in $W$ than without dither. The common loss is about 0.11, which matches the grid value $\bar{L}(\mu^*_\tau) = 0.111$. It is dominated by the regularisation bias.
- $\sigma = 0.002$, $\tau = 5 \cdot 10^{-4}$**.** Dither frees the collapsed start ($0.166 \to 0.0480$), but the starts do not equalise within 900 s (loss ratio 5.23).

## 25.45 Remark H.0 and Proposition H.7: the exactly averaged closed loop

These are checked by tests. Table 40 gives the relative error of the frozen-$\mu$ averaged drift against $(\mathfrak{A})$, and Table 41 the structure of the first correction (`test_averaging.py`).




(a) $\sigma = 0.02$, $\tau = 0.004$




(b) $\sigma = 0.002$, $\tau = 5 \cdot 10^{-4}$

Figure 11: Theorem H, global kernel placement ($m = 32$, scalar plant, $k_e = 10$, 3 seeds): the occupation loss $\bar{L}(\mu_t)$ of the collapsed and spread starts without dither, with constant dither and with annealed dither.

Table 40: Remark H.0: relative error of the frozen-$\mu$ averaged drift against ($\mathfrak{A}$).

| instance | $k_e = 10$ | 20 | 40 | 80 | $k_e \times$ error |
|---|---|---|---|---|---|
| $n = 1$ (`test_langevin.py`) | 4.2% | 2.0% | 0.91% | 0.48% | $\approx 0.4$ |
| $n = 2$, ellipse (`test_averaging.py`) | 43% | 21% | 10% | 5.1% | $\approx 4.2$ |

Table 41: Proposition H.7: structure of the first correction (`test_averaging.py`).

| check | $n = 1$ | $n = 2$ (ellipse) |
|---|---|---|
| expansion error of $k_e^2(P_\mu - P_0)$ against $U_{\text{even}} + U_{\text{odd}}$, $k_e = 40 \to 80$ | $0.54 \to 0.25$ | $0.038 \to 0.019$ |
| circulation kernel $K$ | $\equiv 0$ (asserted at $10^{-12}$) | antisymmetric; equals the enclosed-area integral to $7 \cdot 10^{-15}$ (asserted at $10^{-6}$) |
| orbit reversal (same $\rho$) | $U_{\text{odd}} \equiv 0$ | $U_{\text{even}}$ unchanged, $U_{\text{odd}}$ flipped, to $2 \cdot 10^{-15}$ (asserted at $10^{-10}$) |
| $k_e^2 \cdot$ cyclic sum of the antisymmetric second variation, exact ($k_e = 80$) against predicted (asserted at 3%; the full matrix at 5%) | $-0.0711$ against $-0.0704$ | $-0.0471$ against $-0.0464$ |

## 25.46 Remark H.7a: the orientation of the reference orbit

Script: `exp_orbit_orientation.py` → `orbit_orientation/`.

The setup is deterministic: $n = 2$, $m = 6$, $G = \text{diag}(4, 4, 1, 1, 1)$, $\sigma = 2/k_e$, no dither. The same elliptic reference is traversed both ways, so the occupation measure is the same, and so is

the initial configuration. The runs converged to machine precision: the residual drift over the last 10 periods is about $10^{-16}$ (between $1.0 \cdot 10^{-16}$ and $5.0 \cdot 10^{-16}$). The six kernels converge to a common configuration (kernel spread 1.05 at the start, at most $5.6 \cdot 10^{-17}$ at the end of every run), so the learned network is effectively a single kernel and the test probes the effect at a collapsed equilibrium. Predictions are linear responses at the (+) equilibrium (Table 42).

Table 42: Remark H.7a: orientation dependence of the learned kernels in the real closed loop. Size ratios are measured / predicted.

| | | frozen-measure prediction (H.7) | | full prediction (H.7 + oscillation term) | |
|---|---|---|---|---|---|
| $k_e$ | measured $\|\Theta_+ - \Theta_-\|$ | size ratio | cosine | size ratio | cosine |
| 10 | $5.729 \cdot 10^{-3}$ | 2.670 | 0.7577 | 0.937 | 0.9314 |
| 20 | $2.843 \cdot 10^{-3}$ | 2.414 | 0.8302 | 0.965 | 0.9809 |
| 40 | $1.403 \cdot 10^{-3}$ | 2.294 | 0.8631 | 0.969 | 0.9944 |

- The vector error $\|\text{measured} - \text{predicted}\|/\|\text{predicted}\|$ of the full prediction is 0.364, 0.195 and 0.109 at $k_e = 10$, 20 and 40.
- The measured difference scales like $k_e^{-1.01}$, against a predicted $-1$.
- The full first-order theory accounts for it. The frozen-measure terms alone account for less than half.
- A first attempt with $\sigma = 0.5/k_e$ and half the horizon had not converged (residual drift about a third of the difference). It was discarded and rerun.

### 25.47 Theorem I: reciprocity at order $k_e^{-2}$

There is no new experiment. Every statement of Theorem I is checked against the generic second-variation code of `src/core/averaging.py` by the tests of `tests/test_nonreciprocity.py` (Table 43).

### 25.48 Theorem J: periodic orbits of the finite-$m$ closed loop

Script: `exp_periodic_orbit.py` → `periodic_orbit/`. Newton shooting on the Poincaré map finds the exact periodic orbit of the real closed loop near the critical point $\Theta^*$ of $J_m$ (residual $< 10^{-11}$); its period mean $\bar{\Theta}_p$ and its monodromy matrix are compared with the averaged system (Table 44). Instance A has $n = 2$, $m = 3$, $\sigma_0 = 0.5$, $G = \mathrm{diag}(4, 4, 1, 1, 1)$ and a collapsed minimiser ($J_m(\Theta^*) = 0.762775$, $\lambda_{\min}(\nabla^2 J_m) = 0.0148$); instance B has $n = 1$, $m = 2$, $\sigma_0 = 0.05$, $G = \mathrm{diag}(2, 1, 1)$, two distinct kernels ($J_m(\Theta^*) = 0.258249$, $\lambda_{\min}(\nabla^2 J_m) = 0.0661$) and a fast sweep. Distances and cosines are in the labelled $G^{-1}$ metric.

- Fitted exponent of the offset $\|\bar{\Theta}_p - \Theta^*\|$ against $k_e$: $-1.05$ (A) and $-1.33$ (B; not yet asymptotic below $k_e \approx 80$); Theorem J gives $-1$.
- The vector error of the first-order location halves with every doubling of $k_e$ in both instances. In A the frozen-measure terms alone leave a 35% error at every $k_e$, so the oscillation term of Remark H.7a is an $O(1)$ fraction of the offset.
- The median relative error of the parameter Floquet exponents falls from $1.2 \cdot 10^{-2}$ to $1.1 \cdot 10^{-3}$ (A) and from $4.0 \cdot 10^{-2}$ to $9.5 \cdot 10^{-4}$ (B) between $k_e = 10$ and 160.
- Error channel: the error-channel multipliers are below the resolution of the finite-difference monodromy (exponents of order $k_e$), so their product is recovered from Liouville's identity $\sum \log|\lambda| = \int_0^{T_0} \mathrm{tr}\, \partial F/\partial z \, \mathrm{d}t$ along the orbit. The mean error-channel exponent is $k_e - L$

Table 43: Theorem I: the checks of `tests/test_nonreciprocity.py`.

| check | result |
|---|---|
| $A(\vartheta,\vartheta') = a^\top B(\xi,\xi')a'$ against the antisymmetric part of $S_{\text{even}} + K$, $n = 1$ and 2 | agreement to $10^{-10}$ |
| $A(\vartheta,\vartheta_z) = 0$ for the zero feature; the cyclic sum over $(\vartheta_0,\vartheta_1,\vartheta_z)$ equals $A(\vartheta_0,\vartheta_1)$ | $10^{-14}$ |
| $n = 1$, $f := \hat{f}_\mu$ (perfect fit on the orbit) | $\max\|A\| < 10^{-13}$ over 20 random pairs |
| $n = 1$, non-zero residual | $\|A\| > 10^{-3}$ found; equals the closed form $-aa' \frac{c-c'}{s^2} \int r\varphi_c\varphi_{c'}\,\mathrm{d}\rho$ to $10^{-12}$ |
| $\kappa(\xi,\xi')$ against its closed form for $n = 2$: the Gaussian factor $-\frac{1}{2s^2}e^{-\lvert c-c'\rvert^2/(4s^2)}$ times $(c-c')\cdot F_s$ at the midpoint of $c$ and $c'$ | $10^{-10}$ |
| $F_2((0.5,0)) > 0$ on the ellipse; $\kappa \equiv 0$ on a segment traversed back and forth | $F_2 > 10^{-3}$; $\|\kappa\| < 10^{-12}$ |
| $n = 2$ with $f \equiv \hat{f}_\mu$ on $\mathbb{R}^2$ (residual and its Jacobian vanish on the orbit): $B = 2\kappa I$ | $10^{-13}$; $A \neq 0$ on the ellipse, $A \equiv 0$ ($< 10^{-13}$) on the segment |
| $n = 2$ with $f = \hat{f}_\mu + he_1$, $h$ zero on the ellipse (residual zero on the orbit, Jacobian not symmetric): $B - 2\kappa I = -\int \varphi\varphi'\partial_2 h\,\mathrm{d}\rho \cdot (e_1 e_2^\top - e_2 e_1^\top)$ | $10^{-12}$, and not small (the round-5 correction of Theorem I(iv)) |
| positivity witness $f \equiv 0$, $\mu = \delta_{\bar{\vartheta}}$ | integral $> 0$; the generic code equals the closed form to $10^{-12}$ |

Table 44: Theorem J: the exact periodic orbit against the averaged system. "vec. error" is the vector error of the first-order location $\Theta^* - J^{-1}(b_{\text{frozen}} + b_{\text{osc}})$ of Proposition H.7 and Remark H.7a; "frozen only" omits $b_{\text{osc}}$ (in instance B, $b_{\text{osc}} = 0$ by symmetry). "Floquet" is the maximal relative error of the parameter Floquet exponents against $\frac{1}{k_e}\lambda(G_m^{1/2}\nabla^2 J_m G_m^{1/2})$.

| | instance A (collapsed minimiser) | | | | instance B (two distinct kernels) | | |
|---|---|---|---|---|---|---|---|
| $k_e$ | $\\|\bar{\Theta}_p - \Theta^*\\|$ | vec. error | frozen only | Floquet | $\\|\bar{\Theta}_p - \Theta^*\\|$ | vec. error | Floquet |
| 10 | $4.09 \cdot 10^{-2}$ | 0.169 | 0.398 | $8.31 \cdot 10^{-2}$ | $6.44 \cdot 10^{-2}$ | 1.995 | $9.48 \cdot 10^{-2}$ |
| 20 | $1.91 \cdot 10^{-2}$ | 0.083 | 0.360 | $7.34 \cdot 10^{-2}$ | $2.22 \cdot 10^{-2}$ | 1.078 | $3.21 \cdot 10^{-2}$ |
| 40 | $9.22 \cdot 10^{-3}$ | 0.041 | 0.348 | $4.50 \cdot 10^{-2}$ | $8.37 \cdot 10^{-3}$ | 0.551 | $1.16 \cdot 10^{-2}$ |
| 80 | $4.53 \cdot 10^{-3}$ | 0.020 | 0.343 | $2.46 \cdot 10^{-2}$ | $3.51 \cdot 10^{-3}$ | 0.277 | $4.63 \cdot 10^{-3}$ |
| 160 | $2.24 \cdot 10^{-3}$ | 0.010 | 0.342 | $1.28 \cdot 10^{-2}$ | $1.59 \cdot 10^{-3}$ | 0.139 | $2.03 \cdot 10^{-3}$ |

with $L = 0.160, 0.111, 0.086, 0.074, 0.068$ (A) and $L = -0.054, -0.074, -0.084, -0.089, -0.092$ (B) at $k_e = 10, 20, 40, 80, 160$, as Theorem J(b) states ($L = O(1)$).

- Tests (`tests/test_periodic.py`): $\nabla J_m$ against finite differences; shooting residual $< 10^{-10}$; the offset, the vector error and the Floquet error all fall by the required factors from $k_e = 20$ to 40 on instance B.

### 25.49 Proposition K: collapsed configurations and the orientation of the orbit

Script: `exp_orientation_stability.py` → `orientation_stability/`. The setting is instance A of Theorem J ($n = 2$, $m = 3$, $G = \mathrm{diag}(4, 4, 1, 1, 1)$) with the ellipse $(1.2\cos t, \pm 0.8 \sin t)$ traversed both ways. The Floquet exponents of the collapsed periodic orbit of the real closed loop are computed by Newton shooting and restricted to the separation subspace $\sum_i u_i = 0$; an exponent is $-\log|\lambda|/T_0$, so positive means stable. The leading-order separation exponents are the eigenvalues of $(\sigma_0 I - G^{1/2} R_2 G^{1/2})/k_e$, identical for both orientations. The second-order prediction is the linearisation of $\frac{1}{k_e}\langle H \rangle + b_{\mathrm{frozen}} + b_{\mathrm{osc}}$ at its own equilibrium $\bar{\Theta}_2$ (every $\bar{\Theta}_2$ found is collapsed, kernel spread below $1.2 \cdot 10^{-8}$); the prediction "at $\Theta^*$" uses the Jacobian at the collapsed critical point instead, which omits the third-derivative term (Table 45).

Table 45: Proposition K: the softest separation Floquet exponent of the collapsed orbit for the two orientations at $\sigma_0 = 0.5$, and its orientation difference, measured against the second-order prediction (`orientation_stability/table.md`).

| | softest separation exponent | | difference (+) − (−) | | | $k_e^2 \times$ difference | |
|---|---|---|---|---|---|---|---|
| $k_e$ | (+) | (−) | measured | pred. at $\bar{\Theta}_2$ | pred. at $\Theta^*$ | measured | pred. at $\bar{\Theta}_2$ |
| 10 | $3.563 \cdot 10^{-3}$ | $2.660 \cdot 10^{-3}$ | $9.03 \cdot 10^{-4}$ | $7.77 \cdot 10^{-4}$ | $9.01 \cdot 10^{-4}$ | 0.090 | 0.078 |
| 20 | $1.800 \cdot 10^{-3}$ | $1.614 \cdot 10^{-3}$ | $1.86 \cdot 10^{-4}$ | $1.69 \cdot 10^{-4}$ | $2.08 \cdot 10^{-4}$ | 0.075 | 0.068 |
| 40 | $9.277 \cdot 10^{-4}$ | $8.863 \cdot 10^{-4}$ | $4.14 \cdot 10^{-5}$ | $3.93 \cdot 10^{-5}$ | $4.98 \cdot 10^{-5}$ | 0.066 | 0.063 |
| 80 | $4.737 \cdot 10^{-4}$ | $4.640 \cdot 10^{-4}$ | $9.71 \cdot 10^{-6}$ | $9.44 \cdot 10^{-6}$ | $1.22 \cdot 10^{-5}$ | 0.062 | 0.060 |

- Fitted exponent of the difference against $k_e$: $-2.18$ (Proposition K(iii): $-2$). The prediction at $\bar{\Theta}_2$ converges to the measured $k_e^2$-scaled limit; the prediction at $\Theta^*$, which omits the third-derivative term, converges to a value 30% too large.
- At $\sigma_{\mathrm{crit}} = 0.4606$ both orientations are unstable ($-4.1 \cdot 10^{-4}$ and $-1.27 \cdot 10^{-3}$ at $k_e = 10$); the even part dominates. The two separation exponents coincide (multiplicity $m - 1 = 2$, Proposition K(ii)).
- Designed $\sigma_0$, chosen so that the predicted softest exponents of the two orientations are opposite ($\mathrm{pred}_+ + \mathrm{pred}_- = 0$): $\sigma_0 - \sigma_{\mathrm{crit}} = 0.0125$ ($k_e = 10$) and $0.0061$ ($k_e = 20$). The measured softest exponents are $+8.5 \cdot 10^{-4} / -2.5 \cdot 10^{-5}$ and $+1.4 \cdot 10^{-4} / -3.7 \cdot 10^{-5}$: one orientation of the same reference orbit (same occupation measure, same averaged system) leaves the collapsed configuration stable and the other makes it unstable.
- Test `test_proposition_K_hessian_splitting_and_critical_leak` in `tests/test_periodic.py`: the Hessian splitting at the collapsed point ($10^{-5}$), the gap equation of $\sigma_{\mathrm{crit}}$ ($10^{-8}$), the two-fold zero separation mode at $\sigma_{\mathrm{crit}}$.

### 25.50 Theorem J′: the stationary law of the dithered single-kernel loop

Script: `exp_dither_stationary.py` → `dither/`. Single-kernel instance ($n = 1$, $m = 1$, $\vartheta = (\alpha, \gamma)$, frozen width, $G = \mathrm{diag}(2, 2)$, $\sigma_0 = 0.4$, $\tau_0 = 0.1$), reference traversed both ways; $2 \cdot 10^4$ independent replicas per ($k_e$, orientation) started from the first-order law, $k_e = 10, 20, 40, 80$;

the Gibbs measure, the first-order correction $g$, the current $j_1$ and the circulation coefficient on a $201^2$ grid. Full tables in `results/dither/table.md` (raw in `raw.npz`); Table 46.

Table 46: Theorem J′: the stationary law of the dithered single-kernel loop against the Gibbs measure and the first-order prediction (`dither/table.md`).

| | $k_e = 10$ | 20 | 40 | 80 |
|---|---|---|---|---|
| $(m_\varepsilon(\alpha) - \pi(\alpha)) \cdot k_e$, prediction $-0.255$ | $-0.28$ | $-0.36$ | $-0.36$ | $-0.37$ |
| $(m_\varepsilon(\gamma) - \pi(\gamma)) \cdot k_e$, prediction $-0.426$ | $-0.44$ | $-0.54$ | $-0.51$ | $-0.67$ |
| residual $m_\varepsilon(\alpha)$ − first order (s.e. 0.0013) | $-0.0021$ | $-0.0051$ | $-0.0025$ | $-0.0015$ |
| residual $m_\varepsilon(\gamma)$ − first order (s.e. 0.0018) | $-0.0015$ | $-0.0056$ | $-0.0020$ | $-0.0030$ |
| orientation-odd shift of $\gamma$, measured/predicted | $0.93 \pm 0.04$ | $1.10 \pm 0.07$ | $1.15 \pm 0.15$ | $1.38 \pm 0.29$ |
| $k_e^2 \mathbb{E}e^2/(2\pi(\bar{L}))$, orientation +1 | 1.054 | 1.033 | 1.017 | 1.009 |
| circulation $\Lambda_\varepsilon$ (+1), measured $\pm 0.0005$ / predicted | $+0.0037$ / $-0.0005$ | $+0.0006$ / $-0.0003$ | $-0.0004$ / $-0.0001$ | $+0.0003$ / $-0.0001$ |

- The Gibbs discrepancy decays like $k_e^{-0.87}$ ($\alpha$), $k_e^{-0.83}$ ($\gamma$), $k_e^{-1.09}$ ($\alpha\gamma$), and the first-order prediction removes 93–97% of it at $k_e = 10$ and 70–85% at $k_e = 20$–40; the residual is at the Monte Carlo floor at $k_e = 80$ and its second-order decay is not resolved.
- The orientation-odd shift (the dither version of Remark H.7a) is predicted to 7–8% at $k_e = 10$ and within the 10–30% resolution at larger $k_e$.
- The tracking error is the Gibbs-averaged occupation loss over $k_e^2/2$ with an $O(1/k_e)$ correction of $0.5$–$0.7/k_e$.
- The first-order circulation is below the Monte Carlo resolution for $k_e \geq 20$ and dominated by higher orders at $k_e = 10$; the current machinery of Proposition J′.2 is checked on the first-order averaged SDE on the grid: $\int \Theta \wedge (j_\varepsilon - j_0)/(\varepsilon \int \Theta \wedge j_1) = 1.47$, 1.21, 1.09, 1.03 at $\varepsilon = 0.2$, 0.1, 0.05, 0.025 and $\|m_\varepsilon - m_0(1 + \varepsilon g)\|_1/\varepsilon^2 = 0.31$ constant.
- Referee round 6 (`review/REVIEW_round6_Jprime.md`): the first run's batch standard errors were biased low by 1.4–1.9 and its burn-in too short (JP-4–JP-6); the run reported here is the corrected one.
- Corollary J′.3 (`exp_gibbs_m.py` → `dither/gibbs_m.md`): $\mu_\tau^*$ on the grid ($\int \alpha \, \mathrm{d}\mu^* = 0.4963$, $\int \gamma \, \mathrm{d}\mu^* = 0.1873$, $\int \alpha^2 \, \mathrm{d}\mu^* = 0.9353$, $\bar{L}(\mu^*) = 0.1394$); the Gibbs measures $\pi_m$ for $m = 1$–32 by MALA (4000 replicas): $\mathbb{E}[\bar{L}(\mu_\Theta)] = 0.1960$, 0.1681, 0.1540, 0.1470, 0.1432, 0.1413 with $(\mathbb{E}[\bar{L}] - \bar{L}(\mu^*)) \cdot m = 0.057$–$0.061$ (slope $-0.99$), $\mathbb{E}[\gamma_1]$ slope $-1.01$, $\mathbb{E}[\alpha_1^2]$ from 0.536 to 0.922 with rescaled deviation $-0.41 \pm 0.01$; the $m = 1$ row matches the quadrature of $\pi_1$ within one standard error.

# 26 Research program

Each branch states what is proved, what is open, and the first concrete step.

1. **Infinite-horizon averaging for McKean–Vlasov closed loops (Conjecture H.6).**
   - *Proved:* the averaged system converges globally and exponentially (H(ii)); the exact first correction at frozen measure (H.7); input-to-state stability of the averaged system (H.8).

- *Open:* a near-identity averaging transformation for the stochastic McKean–Vlasov loop that keeps the frozen-measure correction valid along the moving measure, uniformly in time.
- *Done (Theorem J, Section 22):* the deterministic case ($\tau = 0$, $\sigma = \sigma_0/k_e$) at fixed $m$: near every hyperbolic critical point of $J_m$ the real loop has a unique periodic orbit with the same stability type, a $k_e$-uniform basin, and the first-order location of Proposition H.7 and Remark H.7a.
- *Done (Theorem J′, Section 24):* the dithered case at fixed $m$. The invariant law of the loop has the Gibbs measure $\exp(-J_m/\tau_0)$ as its parameter marginal up to $O(1/k_e)$, with the first-order correction driven by the same drift $b$ as Theorem J; the fast error and the phase are eliminated by a Poisson-equation expansion of the generator rather than a transformation of the Fokker–Planck equation.
- *Next steps:* uniformity in $m$ — the mean-field limit of the stationary law (the Gibbs measures of $J_m$ converge to the minimiser of the mean-field free energy $F_\tau$ of Theorem H, a propagation-of-chaos question for Gibbs measures which is where Conjecture H.6 now lives), the second-order correction, and quantitative constants in place of the $O(\varepsilon^2)$.

2. **Non-reciprocal mean-field dynamics from closed-loop learning (new; Proposition H.7).**

   - *Proved at a frozen measure:* the averaged adaptive dynamics is a semi-gradient flow plus a lag term (Proposition H.7). Along the real loop a second-order averaging term is added (Remark H.7a).
   - Its interaction has two antisymmetric parts:
     - the performative defect $A_{\text{even}}$ of the state shift, which is present for every $n$ and is the only part for $n = 1$;
     - the circulation kernel $K$, an integral over the region that the reference orbit encloses ($n \geq 2$).
   - *Proved (Theorem I, Section 21):* reciprocity at order $k_e^{-2}$ is equivalent to $A_\mu \equiv 0$; for $n = 1$ it holds exactly at perfect fit on the orbit, and for $n \geq 2$ at perfect fit exactly when the orbit carries no circulation and the residual's Jacobian is symmetric along the orbit; no open set of measures is reciprocal, for any $n$. The performative defect is bounded by the residual and the antisymmetric part of its Jacobian; the circulation defect depends on neither.
   - *Open questions:*
     - Are stationary states unique beyond the perturbative regime?
     - Can circulation be *designed*, choosing the orbit's orientation and area, to break symmetric traps or to speed escape from the collapsed configuration? (Partly answered by Proposition K, below: the orientation decides the stability of the collapsed configuration in a window around the critical leak.)
     - Is there a Lyapunov functional other than $F_\tau$ for the non-reciprocal flow?
   - *Context:* the closest analogues are performative prediction [71], semi-gradient TD learning [88], [85] and non-reciprocal phase transitions [32].
   - *Done (Remark H.7a):* deterministic $n = 2$ closed-loop runs with the reference traversed both ways (same $\rho$). The full first-order theory predicts the $O(1/k_e)$ difference of the learned kernels to 3% in size and 11% as a vector.
   - *Done (Proposition K, Section 23):* at a collapsed configuration the orientation-odd $k_e^{-2}$ terms shift the separation Floquet exponents of the real loop by $O(k_e^{-2})$, with a

closed-form prediction from the averaged data; at the critical leak $\sigma_{\text{crit}}$ the direction of traversal decides whether the kernels separate, and the $\sigma_0$ at which it does can be computed in advance.

- *Done (Theorem J′):* the test with dither. The orientation-odd part of the stationary parameter means is $2\varepsilon \int \Theta g_{\text{odd}} \, \mathrm{d}\pi$, with $g_{\text{odd}}$ driven by the odd part of $b$; measured on the single-kernel instance of Section 24 (where, in one state dimension, the whole odd effect is $b_{\text{osc}}$).
- *New (Proposition J′.2): the stationary state is a non-equilibrium steady state.* The first-order averaged dynamics carries a stationary current $j_1 = \pi(b - \tau_0 G \nabla g)$ exactly when $b$ is not a $G$-gradient, which for $m \geq 3$ is always the case; its entropy production rate is $\varepsilon^2 \tau_0^{-1} \pi(\|b_\perp\|^2_{G^{-1}})$; and the real loop's stationary parameter process is not reversible. The Lyapunov-functional question for the non-reciprocal flow is therefore a question about a non-equilibrium steady state (its free energy is not a Lyapunov functional of the real loop at order $\varepsilon$).
- *Next steps:* the design of the orbit's area (not only its orientation) as a control input; the orientation effect at non-collapsed configurations with several kernels; measuring the current of the real loop (below the Monte Carlo resolution in Section 24's instance; a variance-reduced estimator or a stronger instance is needed); the mean-field limit of the current.

3. **Uniform-in-time propagation of chaos with a plant in the loop.**

- Theorems D and D′ are finite-horizon; D′ gives the $m^{-1/2}$ rate for the dithered loop by synchronous coupling, with constants independent of the temperature and an exponent independent of the dimension for the state and the learned function. Uniform-in-time results exist for mean-field Langevin dynamics [11] and for shallow networks [36].
- *Done (Theorem D″, Section 17):* uniform in time when the leak dominates the loop gain of the adaptive law (a small-gain condition between the plant channel and the particle channel, explicit in Lipschitz constants). The certified regime is leak-dominated: the network does not learn (without dither it collapses to one kernel; with dither it stays in the cloud of the prior, where the bound has content: the coupled distance is of order $m^{-1/2}$ and bounded in time). The result marks where the contraction method stops rather than settling the question.
- *Open, and now with a negative indication:* the learning regime (kernels separated, leak below the critical value of Proposition K), where the plant couples all particles through $e$ and (by H.7) the averaged interaction is not reciprocal. The long-horizon run of `exp_chaos_uniform.py` ($\sigma = \sigma_K/3$, 1000 s) shows the coupled distance growing threefold between the windows $(100, 200]$ and $(500, 1000]$ while keeping the $m^{-1/2}$ law at each time: on that horizon only the finite-horizon Theorem D′ is supported. Whether the growth saturates (the leak's time scale is $1/\sigma = 2000$ s), and whether a uniform statement holds at all in the learning regime, is the open question; the first draft's claim of empirical uniformity was based on a run that was in fact a slow collapse and is withdrawn (review round 6).
- *Done (Theorem D‴, Section 18):* for the averaged slow loop at $\tau = 0$ the question is settled where the fitted configuration is non-degenerately stable: the interaction term of the synchronous coupling is a negative square at linear order (flat convexity), the self term carries the residual-weighted curvature at the actual residual, and uniform-in-time propagation of chaos at the rate $1/m$ (mean square) holds under the strict second-order optimality condition along the mean-field flow, with explicit constants carrying the transient's $e^{2\Lambda_1}$; the window $\kappa_R[\mu_\infty] < \sigma_0 < \sigma_{\text{crit}}$ is where learning and uniformity coexist. The cold start of Theorem D″'s experiment was outside it on two counts (the

expanding transient, and a connected support threading the ridges, where the coupling has a floor of pairs in different basins). At finite $m$ the stationary coupled distance is the weight-sensitivity law of Proposition D‴.5, and Corollary D‴.7 carries the uniform statement to the real loop on a warm start through Theorem J.

- *Done (Corollary D‴.14, Proposition D‴.15):* in the real loop the collapsed-by-class subspace is invariant and carries the weighted $K_0$-atom loop, so Theorem J's orbit at a replicated configuration is, under its uniqueness-ball proviso, the lift of one $K_0$-atom orbit — with the OT leak the same orbit for every commensurate size (the real loop's finite-size floor is zero), with the nearest-atom leak the orbit of the realised weights (Corollary D‴.7's floor is exactly the distance between two weighted $K_0$-atom orbits); the Floquet spectrum of a lifted orbit is the $K_0$-atom orbit's together with each class's single-particle (ghost) spectrum of multiplicity $n_k - 1$, independent of $m$ — the spectral part of Corollary D‴.7's uniformity question.
- *Done (Theorem D‴.18):* Theorem J uniformly in $m$ at a replicated configuration with admissible weights (margins $\lambda_0$ on a compact set $\mathcal{W}$), in the per-particle norm — one threshold $k_0$, one basin radius $\rho$ and one rate for every size; for $k_e \geq k_0$ the proviso of Corollary D‴.14(ii) is replaced by this $m$-uniform statement. In the mean-square norm no $m$-independent basin exists for the nearest-atom leak (Remark D‴.19).
- *Done (Theorem D‴.21):* the measure-level basin of the OT leak, for the averaged loop. The OT energy satisfies a Polyak–Łojasiewicz inequality whose constant is set by the residual-weighted curvature along the plan segments and does not depend on $m$ (sharp, $2\sigma_0$, at a matching prior); the excess energy decays exponentially from an $m$-independent sublevel set (tiny on the instance) and a posteriori wherever the plan-segment margin is positive, which covers configurations with a fixed fraction of the particles in other cells; under a curvature hypothesis at the stationary residual the replicated configuration is the unique global minimum among the empirical measures, while the nearest-atom energy has a spurious minimum at every admissible count vector. *Open:* the transit of the real loop (averaging with plan switches) and the existence of the mean-field OT flow (Remark D‴.22); the dithered real loop with the OT leak (Remark D‴.16, checked on the averaged loop: under the OT leak a ridge crossing is reversible — the particle stays assigned to its own atom — while under the nearest-atom leak dither equalises the masses only on the Arrhenius time scale) — Theorem J′'s numerics with the two leaks.
- *Open:* the measure-weighted form of hypothesis (H) (a bad set of small $\mu_s$-mass, the bootstrap on the good particles, the flip fraction $p_m$ and its $m$-law); the transfer to the real loop at finite $k_e$ through Theorem J′'s $O(\varepsilon)$ corrections, uniformly in time; dither (the entropy method of [11], whose hypotheses Theorem H verifies, against the coupling's floor).

4. **Geometry of two-time-scale adaptation (Proposition G(b′)).**

- *Proved formally, and checked:* the slow geometry follows the gradient flow of the profile loss in the metric induced on the slow manifold, and the slope $S$ measures the departure from plain variable projection.
- *Open:* a rigorous long-horizon statement; the $O(1/R)$ corrections, which involve $D^2a^*$ and $\partial_\xi M$; gain designs that shape the induced metric.

5. **Noiseless global convergence.**

- *Done (Proposition K):* the local criterion for the collapsed trap: a collapsed configuration is a strict local minimum of the regularised loss iff $\sigma_0 > \sigma_{\mathrm{crit}}(\bar{\vartheta}) = \lambda_{\max}(G^{1/2}R_2G^{1/2})$, where $R_2$ is the Hessian of the first-variation potential at the atom; below $\sigma_{\mathrm{crit}}$ the

noiseless flow leaves the trap along the softest separation mode, without dither, and the residual at the atom is what drives the separation ($R_2$ is linear in it).

- *Open:* a closed-loop separation condition under which the noiseless flow avoids spurious stationary points, despite the failure of the Chizat–Bach [14] boundary conditions. Candidates are a compactified width coordinate and reflecting boundaries (Remark H.3).

6. **Unbalanced transport, meaning online structure adaptation.**

   - Treat the weights as masses (Hellinger–Kantorovich / Wasserstein–Fisher–Rao) to obtain birth and death of kernels with Lyapunov guarantees. The conic particle descent of [16] and [23] is the static analogue.

7. **Excitation theory in measure space.**

   - *Done (Lemma E.0):* a quantitative, uniform UES estimate for the skew-symmetric class, explicit in $(k, \alpha, T, \Psi_M, \Psi_D)$, which reproduces the $\omega^{-2}$ law of Remark E.1.
   - *Done (Lemma E.0′):* a strict Lyapunov function whose rate is the averaged rate $\lambda_{\min}\langle\Psi^\top\Psi\rangle/k$ up to a factor $1-\delta$, for $k$ above a threshold set by the peak factors of $\Psi$, $\dot\Psi$, $\ddot\Psi$ (second-order slaving of the error to its slow manifold and a second-order averaging correction, whose mean part vanishes identically); the ill-conditioning of the averaged Gramian is paid once, as in the exact rate.
   - *Done (Corollary E.0′.3):* the same Lyapunov function bounds the stationary parameter spread of the dithered linearised loop, $\limsup \mathbb{E}\|\zeta\|^2 \le (1+O(\varepsilon))k\|\Sigma\|_F^2/(2(1-\delta)\lambda_{\min})$, within $1/(1-\delta)$ of the exact (averaged Langevin) value on isotropic regressors and paying the condition number in the trace otherwise: an explicit design formula for how much dither a given tracking budget tolerates.
   - *Done (Lemma E.0″):* the non-periodic case. The period mean is replaced by a forward exponential average over a window $T$ (whose derivative identity $T\dot{\bar M} = \bar M - M$ gives the antiderivatives in closed form), the peak factors become time-local, and one residual second-order term $\bar N_2$ appears, costing $\delta_2 = O(1/k)$ in the rate; the guaranteed rate is $(1-\delta-\delta_2)\lambda_T/k$ with $\lambda_T$ the exponentially weighted PE level of the window. Checked on a quasi-periodic regressor and on a regressor driven by a Lorenz orbit, where no periodic construction applies.
   - *Done (Theorem D″.23(ii)):* at an exact representation the measure-space excitation spectrum $h_j$ — the whitened weighted Gauss–Newton spectrum of the $K_0$-atom loss — sets both a fixed prior's bias, the fraction $\sigma_0/(h_j+\sigma_0)$ of its error kept in mode $j$, and an adaptive prior's rates $\lambda_j^- < \min\{h_j, \eta\}$ $(h_j, \eta > 0)$: removing the leak's bias costs the excitation of every collective mode, and the leak's a-priori bound on the parameters (the leak toward an adaptive prior is an internal force, and the solutions can run off where no exact representation exists; projecting the prior onto a compact convex set restores the bound, Proposition D‴.27) — the measure-space form of the classical trade-off between a leak's robustness and persistent excitation (an unexcited mode, $h_j = 0$, drifts with the prior).
   - *Done (Proposition D″.24(i)):* the same spectrum sets the finite-size floor of a fixed prior, each mode weighted by the band-pass $h_j\sigma_0^2/(h_j+\sigma_0)^2 \le \sigma_0/4$, centred at $h = \sigma_0$ — the floor sits in the modes that carry the gauge's displacement, re-weighted toward the level of the leak.
   - *Open:* the continuum form of (PE); the trajectory kernel

$$K_T(\vartheta, \vartheta') = \int \Phi(x(s);\vartheta)\Phi(x(s);\vartheta')\,\mathrm{d}s;$$

identifiability of measures up to symmetry; the relation to [39]. Remark E.1 shows that the PE level alone does not predict the rate. For Lemmas E.0′ and E.0″: lowering the threshold $k_0$ (a third-order slaving trades $\rho_{DD}$ for the peak factor of $\dddot{\Psi}$ one order higher); the choice of the window $T$ as an optimisation; the nonlinear Theorem E with the constants of E.0″ (Corollary E.0′.2's steps carry over verbatim, with $\lambda_T$ in place of $\lambda_{\min}(\bar{M})$).

8. **Transport-regularised adaptive laws.**

- Wasserstein proximal (JKO) discrete-time adaptive laws.
- *Done (Theorem C′, Section 13):* the OT leak toward priors with a different number of atoms, using optimal plans with barycentric projection; it is Lloyd's algorithm in continuous time, and its bias is the centroid defect of the ideal.
- *Open (in part):* its mean-field form — done for an atomic prior (the Laguerre-cell leak, Proposition D‴.12), open for a general prior $\nu$ and a general $\mu$ (where the barycentric projection is the superdifferential of $\frac{1}{2}W^2(\cdot,\nu)$; Ambrosio–Gigli–Savaré [2, Thm. 10.4.12]) — and adaptive priors ($\nu$ itself learned: its atoms, Theorem D‴.23; its masses, Proposition D‴.24).
- *Done (Proposition D‴.9, Section 18):* the nearest-atom leak toward a prior measure inherits Theorem D‴, Proposition D‴.5 and Corollary D‴.7 inside the Voronoi cells; the prior is a design input of the window (maximal margin $\kappa = \sigma_0$ at a matching prior, the window closing as the prior is misplaced) and of the floors (which grow with the stationary amplitudes).
- *Done (Proposition D‴.12):* the barycentric OT leak's mass constraint against the weight fluctuation. The nearest-atom leak conserves the cluster masses, and the finite-$m$ floor of Proposition D‴.5 is the memory of the initial cloud that this conservation carries; the OT leak constrains the masses to the prior's, so on the sizes commensurate with the masses the finite-$m$ stationary measure is the mean-field one exactly (the replicated weighted configuration is a critical point for every such $m$, by total unimodularity and $c$-cyclical monotonicity), and otherwise differs from it by at most $K_0 - 1$ split particles; for an atomic prior the mean-field form of the barycentric leak is the Laguerre-cell leak. *Open:* a coupling theorem for the OT dynamics from an i.i.d. start (Theorem D‴'s synchronous coupling needs a balanced sample, of probability $\approx 0.83/m$, and otherwise keeps the transit's mismatch for ever; the convergence of every trajectory to a Clarke-critical point is Proposition D‴.17, without a rate; Theorem D‴.21 gives the rate in an $m$-independent energy basin, with the plan as the test direction, and formally (Remark D‴.22) propagation of chaos with no floor there; smallness in $m$ during the transit from an i.i.d. start remains open); a penalised (unbalanced) mass constraint — the entropic one is Proposition D‴.20, which equalises the masses only down to a critical excess and conserves smaller ones, so that its floor returns at large $m$ (Proposition D‴.5's law at the entropic stationary point); the mean-field form for a general prior $\nu$ (the barycentric projection as the superdifferential of $\frac{1}{2}W^2(\cdot,\nu)$, Ambrosio–Gigli–Savaré [2, Thm. 10.4.12]) and adaptive priors with learned masses beyond the cell masses (Proposition D‴.24).
- *Done (Theorem D‴.23):* adaptive priors with the masses fixed. The prior's atoms move toward the class barycentres of the plan; the joint flow is the gradient flow of $\bar{L} + (\sigma_0/2)W^2(\mu_\Theta, \nu_Z)$, its replicated stationary points are the unregularised $K_0$-atom critical points (no leak bias; the non-replicated ones carry the $K_0$-cluster bias), and at an exact representation every collective mode relaxes at $\lambda_j^- < \min\{h_j, \eta\}$, for every commensurate $m$; the fast limit is a capacity-constrained $K_0$-means regulariser. The transit drags the prior and its weak modes remember the drag, so the adaptation starts from the fixed-prior stationary point (a gate). *Open:* the real loop with an adaptive

prior over the adaptation's own time scale beyond the linearisation (Corollary D‴.26); the plateaus of the unregularised loss that the fast limit exposes; the constants of (v).

- *Done (Corollary D‴.26):* perfect tracking with an adaptive prior is an exponentially stable periodic solution of the real loop (locally, for each $m$) at every representable sample: its Floquet exponents are $\sigma_0/k_e$ exactly on the separations and the weighted $K_0$-atom loop's on the collapsed subspace, for every $m$, and the latter are $(1/k_e)[\lambda_j^{\pm}(w) + c_j^{\pm}(w)/k_e^2 + O(k_e^{-4})]$ — the odd orders vanish (the time-reversal parity of the averaged terms), $c_j$ explicit — the error filter slowing, the slaving speeding up, its sign not fixed. *Open:* its basin uniformly in $m$ (Theorem D‴.18's contraction for the joint system).
- *Done (Lemma D‴.26a):* the slow Floquet expansion proved for the linearised loop. Chang's periodic decoupling makes the slaving exact, so the slow multipliers are exactly the slow block's, with an expansion to $O(k_e^{-4})$ whose remainder is bounded; defining the slow block at negative $\varepsilon$ by the same contraction, the time-reversal symmetry becomes the identity $\operatorname{eig} \bar{A}(-\varepsilon) = \operatorname{eig} \bar{A}(\varepsilon)$ — the odd orders vanish exactly; and Chang's second transformation decouples the slow coordinates' drift exactly, their noise $\varepsilon b \nu$ with $b$ the high-passed $-(\dot{K} + aK)^{\top}$ at leading order and its next term explicit, which proves Proposition D‴.30(ii)'s floor for the linearised loop under white noise as a period average (pointwise the fast jitter oscillates by $\pm 1\%$ on the instance). *Open:* the constants explicit; the evenness of the noise floor without the slope (observed to $k_e = 800$, not proved); the real loop's nonlinear noise response.
- *Done (Proposition D‴.28):* the adaptive prior's floor removal is a property of the amplitude–mass gauge, not of exact representability: at any nondegenerate critical point of the unregularised loss the gauge transform is a critical point at the sampled weights, with the same function, so at a strict local minimum the function-level floor vanishes at the samples of the gauge set $W_g$ with positive margins, whatever the approximation error ($W_g$ shrinking as the error grows); the fixed prior's floor is the bias formula with the full Hessian; for a scalar amplitude the residual curvature scales exactly by $q_k/w_k$ under the gauge, so the adaptive prior's splitting margins are $\sigma_0 - (q_k/w_k)\lambda_k^{+}$. *Open:* critical points at infinity (the best approximant on the saturation boundary, as for a sharp friction step, a cogging function that two kernels fit as a dipole, or Proposition D‴.29's servo globally), where only the projected prior keeps the parameters bounded; vector amplitudes, where the margins have no closed form.
- *Done (Proposition D‴.48):* the rich transit. From small masses the kernels enter one at a time: from zero amplitudes the loop, in the time $t/\Lambda$ with $\Lambda \propto (\text{starting mass})^{-1/2}$, follows the inverse scale space path of the weighted $\ell^1$ norm — the unit near its critical curve is an effective depth-3 unit —, and on branches the same on the logarithmic scale, sign-constrained. With the paradigm's moving centres the kernels that have not entered stay where they started and the entered ones slide onto the target's features: one kernel per well-separated feature at its centre (a tie shared by both kernels); a Wasserstein–Fisher–Rao metric on the positions condenses the massless kernels at the correlation's peaks instead, reaching one kernel per feature only at much smaller masses (on three bumps the paradigm's metric forms exactly three kernels for $\varepsilon \leq 10^{-5}$, the Wasserstein–Fisher–Rao metric 15, 12 and 8 kernels at $\varepsilon = 10^{-4}$, $10^{-5}$, $10^{-6}$ and three from $\varepsilon \approx 10^{-8}$; on sixteen random targets exactly three kernels at the bumps on all sixteen at $\varepsilon = 10^{-8}$ under the paradigm's metric, on 10 of 16 under the Wasserstein–Fisher–Rao one). *Open:* the sliding limit proved (the fast block's relaxation to a stationary point); at a finite mass the near ties and the creep are Proposition D‴.51's.
- *Done (Proposition D‴.47):* the free law's branches. A small start at nonzero amplitude puts a particle on a branch of its curve, where its sign is frozen and the law is an exponentiated gradient — an entropy at the branch point with the temperature $1/(\gamma a_0)$,

read off the conservation law. For every start the loop converges to the Bregman projection of its start onto the sign-constrained least-squares fits; particles whose sign cannot help die out (exponentially under strict complementarity off the critical curve); the rich limit is a sign-constrained basis pursuit, and a pair of opposite particles per atom restores the free sign as a hyperbolic entropy. On positive kernel dictionaries the frozen signs make the selection of the target's kernels independent of the starting scale, at a speed set by the certificate's margin — typically tiny on Gaussian grids sampled at few points, where the selection is not reached (with 21 atoms, 7 samples and the margin 0.015 the loop reaches the target from every start scale, $\beta(0) = 10^{-6}$ to 10, within $6.0 \cdot 10^{-5}$ at $t = 10^9$; over all three-atom supports of three Gaussian grids the median margins are $4.0 \cdot 10^{-6}$, $3.8 \cdot 10^{-9}$ and $1.2 \cdot 10^{-3}$, and below $10^{-4}$ the loop is still 0.37–0.83 away at $t = 10^9$). *Open:* the extinction's rate without strict complementarity; branches in the real loop's slow modes; moving centres on branches.

- *Done (Proposition D‴.46):* a leak for the free mass–amplitude law. Written in the mirror coordinates — each particle's drive minus $\sigma$ times its mirror coordinate's distance to a prior point of its own curve —, the leak keeps the conservation law and the Lyapunov function, now with the three-point identity's $-\sigma B$ terms, so the parameters stay ultimately bounded under bounded disturbances and approximation errors, which the free masses alone are not (a decaying excitation can drive them to infinity). Its bias is explicit: the averaged loop — with the error filter's Gram — settles on the fit plus $\sigma$ times the Bregman divergence to the prior, a ridge from large starting masses, a LASSO from small ones, which keeps exactly the target's kernels under the irrepresentable condition (on random $H$ the equilibrium matches the ridge at $\beta^0 = 10^4$ and the LASSO at $\beta^0 = 10^{-8}$ to the relative distances $1.3 \cdot 10^{-8}$ and $1.5 \cdot 10^{-4}$; over all 1760 3-sparse patterns of an orbit's Gram its support is exactly the truth's on 415, the LASSO's prediction holds on 416, and the two agree on 1745). *Open:* the real loop's bias beyond averaging; a leak rate adapted to the excitation; the rich equilibria's slow approach; moving centres and widths.
- *Done (Proposition D‴.45):* the free mass–amplitude law is a mirror flow. Without the masses' constraint each particle's mass and amplitude are driven by one scalar signal, so a closed-form quantity $Q = \beta - (\gamma a_{\max}^2/4G)\cosh(2\alpha/a_{\max})$ is conserved whatever the drive: the free replicator only reparametrises the amplitude, and lifts its saturation. On a dictionary the coefficients follow the mirror flow of an explicit potential — a mass-weighted $\ell^2$ core, a soft wall at the saturation, an entropy tail —, so the noiseless loop converges from every start, without persistent excitation, to the start's Bregman projection on the exact fits: the minimum-norm fit for large starting masses, weighted basis pursuit with weights $1/\sqrt{\beta^0}$ for small ones, paid for in time ($\propto (\beta^0)^{-1/2}$ for the transit, $\propto 1/(\beta^0)^2$ for the last digits where the support has fewer than $N$ atoms) (from masses $s \cdot m/d$ on $5 \times 12$ Gaussian designs with 2-sparse truths the projection lies within 0.5% of the minimum-norm fit at $s = 10$ and has 1.0004 times basis pursuit's $\ell^1$ norm at $s = 10^{-4}$; at $s = 10^{-3}$ the loop is within $10^{-8}$ of it only from $t \approx 8.5 \cdot 10^6$). Noise with smooth paths (Stratonovich) never moves that projection; a sampled-data update — like Itô's white noise — does, through a closed-form change of $Q$ per step, the conserved share of D‴.44's selection, which a mirror update on the curves removes; in the real loop $\frac{1}{2}e^2$ plus the Bregman divergences is a Lyapunov function of the joint adaptation. *Open:* the conserved law's implicit bias (its level shift couples the particles); moving centres and widths ($Q$ stays conserved while the cohesion's $\alpha$-component vanishes, the mirror structure does not); the sampled-data drift of the projection in closed form.
- *Done (Proposition D‴.43):* which kernels the colour of the measurement noise keeps. On a kernel dictionary with coefficients that are differences of adaptive masses, label noise under an explicit (Euler) update moves the fit along the exact fits by the gradient

flow of $\operatorname{tr}\Psi(K)$ — the step's Itô drift, at a rate $\propto h\sigma^2$, absent in continuous time and 25 times less under Heun's scheme at the step 0.5 —: the trace — a weighted $\ell^1$ norm, basis pursuit as the step and the noise vanish — for white noise, the log-determinant of Fazel's heuristic (sparse Bayesian learning's log-det term) — a support count — for a slow Markov noise, a concave penalty for every colour in the continuum. A sparse exact fit on fewer than $N$ atoms is held by the slow selection once the correlation time passes a threshold that grows with the dictionary's coherence, by the white one only where weighted basis pursuit recovers it. The slow selection recovered the sparse truth where basis pursuit fails (its reduced flow on $10 \times 40$ Gaussian designs 0.75 and 0.60 of the 3- and 4-sparse truths against basis pursuit's 0.38 and 0.17, on Gaussian kernel dictionaries 1.00 against 0.85), and label-noise descent at small $\sigma$ identified the support where its reduced flow does (four of eight instances; the coefficients after a noiseless phase); sparse Bayesian learning matches or beats it (0.88 and 0.55): the contribution is the mechanism. The replicator's own masses: Proposition D‴.44. *Open:* the descent's pruning phase (fewer than $N$ atoms alive, $\lambda_{\min}(K) \to 0$); the basins; noise that reaches the adaptation through a closed loop.

- *Done (Proposition D‴.42):* several particles per class. With a class's particles together, the positions' Hessian on the manifold of exact fits factors into the one-particle-per-class Hessian at the class masses and the cohesion's within-class block, whatever the split of the class's mass: under the isotropic dither the selection and its correction act on the class masses as with one particle per class (with $m$ the number of particles) and the fractions within a class are frozen at Katzenberger's order — but an anisotropic dither makes the split felt at unequal splits, and measurement noise's cancellation holds on the equal split's invariant subspace only, the cohesion breaking it by up to 10% elsewhere. In the full loop the class masses follow the per-class law at half the speed; from an unequal split the fractions moved away from the equal split while the amplitudes visited the saturation, where the local laws do not hold. *Open:* separated particles of one class (the split's onset, Proposition D‴.35, under dither); the order-$\gamma^2\tau$ drift of the fractions; the real loop.
- *Done (Proposition D‴.41):* what measurement noise does to the gauge. Noise in the error enters along the descent's own vector fields, which are orthogonal to the manifold of exact fits in the flow's metric; the limit map's invariance then makes the Wong–Zakai term cancel Katzenberger's Itô drift exactly: in continuous time, for a noise white on the orbit's resolution and of any temporal spectrum, the masses neither diffuse nor drift along the manifold at leading order (other noise structures over the nodes leave an $O(\gamma)$ residual). Only an update that draws the noise afresh selects — label-noise SGD's flat-minimum bias, on the replicator's masses a strictly convex potential whose minimiser is interior, its form depending on the coordinates of the update —, and for correlated draws the increment's second moment selects, not the noise's variance: each pair of normal modes is weighted by a linear-response factor, the potential passing from the trace (Damian–Ma–Lee's finite-step form for fresh draws) to $(1/\tau_c)\log\det(I+\tau_c\nabla^2 J)$ as a Markov noise slows — concave for every colour —; a slow Markov measurement noise selects as a dither of the flow's own mobility does, and at a finite replicator rate the coloured selection is multistable: three stable targets, each with one class's mass large. *Open:* the order-$v^2$ motion; the real loop's sampled-data implementation; which target a given start reaches, beyond the instance.
- *Done (Proposition D‴.39):* the dithered loop's stationary law. Without a prior the energy is bounded on the unconstrained coordinates, so the dithered positions have no stationary law at all: the prior leak's confinement went with its bias. The way out of the exact fit's well is a climb, without a saddle on the instance, to a plateau of approximate fits with saturated amplitudes about $7.7\tau$ higher, on which those amplitudes diffuse

freely. The escape runs at a rate close to Arrhenius's at the instance's moderate barriers (the weighted slope $0.81 \pm 0.14$ of the barrier), and in the stored runs the two channels to the saturation separate cleanly: along the manifold without a leak, off it with the masses held by the leak. A dead zone on a box of the unconstrained coordinates restores a unique stationary law and leaves the deterministic objects inside the box — the exact fits, the noiseless flow — unchanged (not the stationary law's occupation, which the box sets); the gauge's saturation becomes a non-degenerate wall at which the entropic drift ends, and the Fixman correction still selects the D-optimal fit. *Open:* the plateau's occupation in the box's stationary law as a function of the box; a box chosen from data; several particles per class (the within-class masses are neutral at leading order and have their own unbounded direction, the log-ratios).

- *Done (Proposition D‴.38):* a remedy for the dither's drive into the saturation. Adding to the replicator's fitness the derivative of the log-determinant free energy with a gain $\kappa_F$ — a Fixman correction, in closed form on the masses, which makes the corrected loop a gradient flow with a log-barrier on the representable masses — cancels the entropic drift to leading order at $\kappa_F = 1$ and, for $\kappa_F > 1$, reverses it: the masses climb the strictly concave log-determinant to its unique maximiser, the D-optimal exact fit (in the full loop within 1.1 standard errors, with the spread of the linearised limit). Near the saturation the density goes like $\delta^{\kappa_F - 1}$ instead of $\delta^{-1}$ for the unfloored correction (the implemented floor restores $\delta^{-1}$ within 0.00126 of it); with a leak the pinned state is predicted to second order — the fluctuations acting through the leak's curvature, which also explains D‴.37's excess. Evaluated at the current amplitudes (the plug-in form) the correction reads the positions' escapes off the manifold as masses at their saturation and moves them; on the masses alone it does not. *Open:* several particles per class; the real loop; $\hat{A}$ from data on a changing target; a floor that shrinks with $\tau$.
- *Done (Proposition D‴.37):* where dither drives the replicator's masses. The drift of the masses along the manifold of exact fits is of the order of their diffusion; in the slow-replicator limit it is the replicator flow of the entropic free energy $(\tau/2) \log\det$ of the positions' Hessian (checked to 0.3%), which diverges at the saturation: dither prefers the flattest exact fits, the saturated ones — the trap the replicator removed returns through the noise. A boundary layer of width $O(\sqrt{\gamma})$, where the replicator outruns the saturating amplitude, slows it but does not stop it — inside, the amplitude diffuses without drift, and over $10^6$ slow units the excursions into the saturation grow; the mean drift's three attracting states are not where the masses settle. Only the entropic leak holds them; its bias is the drift over its stiffness, $O(\tau/\varepsilon)$, approximately, in the narrow window $\tau \ll \varepsilon \ll 3 \cdot 10^{-4}$. *Open:* the stationary law of the masses; several particles per class at finite $\gamma$; whether a leak in the amplitudes (not the masses) removes the selection; the real loop.
- *Done (Proposition D‴.36):* the replicator's gauge under dither. At an exact representation the fits through the configuration form a manifold parametrised by the masses, on which $\hat{f}$ does not move; dithering the positions makes the masses diffuse along it (Katzenberger's limit, the noise projected in the flow's own metric), measured at the predicted rate, and their amplitudes run into the saturation ($|a|/a_{\max} = 0.9994$) at no cost in fit — the replicator removed the masses' obstruction (Proposition D‴.34) and with it their stiffness. An entropic leak on the masses restores it without biasing the representable target: the spread is the Ornstein–Uhlenbeck one at $\varepsilon = 10\tau$ (to 3%; at $\varepsilon = \tau$ four seeds do not resolve it). *Open:* the limit's drift term (visible in the means, a selection of the masses by the dither); the gauge with several particles per class; the real loop's dither.
- *Done (Proposition D‴.35):* how a class splits. The class's separations along the splitting

direction carry the standard representation of $S_n$, and the reduced flow has the quadratic equivariant with the residual's third derivative as its coefficient (the cohesion, even, enters at third order): the bifurcation is transcritical, every small split state a saddle, the split a jump. The particles keep their order, and once its excursion passes $2\mu/|c_3|$ the extreme particle runs away on the side the quadratic term picks, alone or with a near-tied neighbour — the quadratic equation names the first excursion's particle in 12 of 12 runs and its time to 1.1%, where the linear law names it in 6 and misses the time by 22% —; after a departure the stiffness's drop and the curvature's gauge rise favour more, the residual's change stops them. A mass-free leak has thresholds a factor $w_k$ lower and only delays the cascade: continued to $\sigma_0/256$ it ends at 8–9 classes and the fit error 0.0028, as the cohesion leak does; its 4 classes at $\sigma_0/64$ were the schedule's. *Open:* steering the escape (the sign of $c_3$ against the residual's location); a schedule-free count; the transit beyond one split.

- *Done (Proposition D‴.34):* mass transfer for the cohesion leak. The particles carry weights adapted by the replicator law (in real time an exponentiated-gradient law with the relative entropy of the weights as its Lyapunov function, the negative-entropy instance of a mirror-descent adaptation law); with the transport law the pair is a gradient flow (Euclidean on the positions, Fisher–Rao on the weights: gradient descent on $(\sqrt{\beta}, \Theta)$), $\sum \beta$ and the saturation's bound kept. At clustered configurations under the finite range its stationary points are D‴.32's; under the Gaussian kernel it selects the masses and moves the bias. The saturated transients that froze the transit's masses are left: at fixed positions bounded trajectories reach the exact fit iff the target is representable ($\sum_k |A_k|/a_{\max} < 1$), and from broad starts the five-bump and the three-heavy-bump targets of D‴.32 (H), (K) are found with their kernels. It does not separate bumps closer than the broad scale, nor act on the shedding (the fitness vanishes at a critical point). *Open:* boundedness at fixed positions; the transit with the replicator (which classes form; the shedding of D‴.32 (G)); the neutral directions under noise (a common noise random-walks the gauge; the redistributions move only when the positions are dithered); the real loop with the transport.
- *Done (Corollary D‴.33):* the desired regressor keeps the replicated configuration attracting under measurement noise. With the regressor at the reference the regressor's own term leaves the separations' second-order drift; what remains — the mean error's term and the cross term — is $O(1/k_e)$ like the separations' stiffness, so the noise threshold is bounded below uniformly in $k_e$ (constant when the noise's bandwidth scales with the gain; at a fixed bandwidth DR's sign reverses at $k_e \approx B$). Under DR near the lift the compensation's rectification of the noise stiffens the separations of separated alternating-sign kernels (the instance; same-sign, wide or edge kernels soften); along the trajectory the drift turns indefinite as the mean error relaxes with the collective modes, and at the stationary point the parameters' part of the force dominates the mean error. It makes the desired regressor the partner of the cohesion leak, whose separations are the softest ($\sigma_0 w_k$). *Open:* the long-time threshold (the drift indefinite along the trajectory and at the stationary point, formally at noise levels outside the expansion); the higher orders; the real loop beyond 300 time units; references in $\mathbb{R}^n$.
- *Done (Proposition D‴.32):* the cohesion leak, a prior-free leak on the unexcited subspace. The particles attract each other toward the modes of their own distribution (the mean shift); the averaged loop is the gradient flow of the fit plus an aggregation energy; with a finite-range kernel, at a clustered configuration whose atoms are $R$ apart the leak vanishes, so every critical point of the unregularised loss is stationary, for every count and every set of masses, and it acts exactly on the unexcited subspace (the separations at $\sigma_0 w_k$, the class means free); a class can split iff $\sigma_0 w_k^2 < q_k \lambda_k^+$, so $\sigma_0$ sets the resolution and an exact representation is never over-split, and a split sheds single particles. A

Gaussian kernel pays an exponentially small bias, and twins with no third class within reach contract at an explicit rate (checked to 0.04%) — a third class within reach can win. The number of kernels is not selected by the leak: an annealed Gaussian scale found the instance's three kernels from every broad start, a fixed scale did not, and on other targets the count failed where the transit's masses were not representable or the bumps were closer than the broad scale. *Open:* the transit (which classes form, with which masses), and with it any selection; the shedding below the resolution's threshold (Part G); the scale $R$ chosen from the data (Parts J, K); the a-priori bound (the leak is internal: a projection). The masses' obstruction is removed by Proposition D‴.34.

- *Done (Proposition D‴.31):* the adaptive prior is a washout leak, and the remedy for the errors in variables. The prior lags the class barycentre by its exponentially forgotten increments, so the leak acts in full on the separations and on the barycentre through the high-pass $i\omega/(i\omega+\eta)$ — the bias removal and the lost robustness to persistent forces in one transfer function; along every affine time-invariant prior the stationary response $R$ and the bias $B$ satisfy $RH+B=I$, so no prior gives both; the response to a force is at most the norm of its smallest equivalent target perturbation, under every leak; and the desired regressor (the adaptive law's regressor at the reference) keeps the lift, the linearisation and the first-order floor while turning the noise's own persistent force into a least-squares one, bounded uniformly in $k_e$ and in the spectrum — the parameters' fluctuation adding a force that is not the learning force of a mean tracking error (the classical second-order bias of a constant-gain stochastic approximation): small in absolute terms on the instance ($\le 3.9\cdot10^{-4}$ per $(\mathbb{E}\nu^2)^2$), $-8.7\%$ to $+3.3\%$ of DR's and DR*'s biases and $+56\%$ of DR's at $k_e=B=100$, its own response falling like $k_e^{-2}$ on a one-dimensional reference, where the law's leading term averages out — which on a curve in $\mathbb{R}^n$ it need not. *Open:* the second-order computation made rigorous, the parameters' part in closed form, and in $\mathbb{R}^n$; the stationary biases in the real loop (the weak modes need $\sim10^5$ time units); a sensor model with quantisation and delay.
- *Done (Proposition D‴.30):* measurement noise at perfect tracking. The noise is coherent — a replicated start stays replicated, so its floor does not average out with $m$ (invariance, not attraction: near the lift the separations grow above a noise level, which drifts along the trajectory with the developing errors-in-variables deviation); at first order the loop high-passes the measured error's noise and the function-level floor is $(N_0/2k_e)[\sum_j Q_{jj}(h_j+\eta)/(h_j+\eta+\sigma_0)+\operatorname{tr}H^2]$, the adaptive prior costing at most the filter factor; at second order the noise inside the regressor exerts a mean force (the errors in variables) whose stationary response is $(H+\sigma_0 I)^{-1}$ under the fixed prior and $H^{-1}$ under the adaptive prior for every $\eta>0$ — the robustness half of the leak, lost in the weakly excited modes. *Open:* the real loop's noise response beyond the linearisation (the linearised loop under white noise: Lemma D‴.26a); a projected or partially fixed prior on the weak modes; the optimal $k_e$ and $\eta$ for a given noise level.
- *Done (Proposition D‴.29):* second-order plants. With the sliding variable $s=\dot e+\lambda e$ the error channel is Theorem J's with $s$ in place of $e$, so the averaged slow loop — and with it every averaged-loop statement of Section 18 — holds (formally) along the regressor's orbit, the position error being the residual filtered by $(\mathrm{d}/\mathrm{d}t+\lambda)^{-1}(\mathrm{d}/\mathrm{d}t+k_s)^{-1}$. On a cogging-torque servo whose torque four kernels cannot represent — the loss's infimum lies at the saturation, and its local minimum $\vartheta^b(q)$ is Proposition D‴.28's regime — the adaptive prior removes the fixed prior's floor and, from an engineer's misplaced model, most of its bias, at the rate of the weakest excitation; the real servo follows the exactly filtered prediction to 0.3%. Noise-free, no leak at all does as well: the floors and biases the leaks remove there are those a fixed leak creates. *Open:* the real-loop statements (Corollaries D‴.14, D‴.26) for second-order plants; the servo with measurement noise and disturbances, where the leaks' robustness is the point, against the classical baselines

and a Fourier regressor of the known harmonics; velocity-dependent friction with the projected prior in the real loop.

- *Done (Proposition D‴.27):* the a-priori bound restored. Confining the adaptive prior's atoms to a compact convex set keeps every particle within $B/\sigma_0$ of it (eventually) whatever the target and the energy non-increasing (at $\eta = \infty$ in its selected form, the plan being Lloyd-consistent, not jointly optimal); inside the set the bias removal of Theorem D‴.23 is unchanged, and where the limit has an atom on the boundary a bounded bias remains — the box trading the fit error against the parameters' bound, as long as it contains the representation; at $\eta = \infty$ on collapsed classes it is a switching $\sigma$-modification toward the set. *Open:* the convergence (at $\eta = \infty$ the Kurdyka–Łojasiewicz argument on the selected pieces, at finite $\eta$ with the indicator), the choice of the set as a design input (how large a box keeps every representable target inside), and a sharper bound than the worst-case force $B$.
- *Done (Proposition D‴.24):* the masses minimised out of the leak's own energy are the cell masses ($\min_q W^2(\mu, \nu_{Z,q})$ is the nearest-atom distortion), so the nearest-atom leak toward an adaptive prior is the OT leak with free masses — and the function-level floor does not return at the representable samples (those outside $W$, of probability exponentially small in $m$ but 0.36 at $m = 15$ on the instance, keep an error): it is the weight-derivative of the leak's bias at the sampled weights, which the adaptive atoms remove. What remains is the coupled distance of the amplitude–mass gauge, conditional on representability, which the mass constraint replaces by the transit's mismatch, and at the measure level the cross-class transport common to both priors; the adaptive prior also removes the fixed prior's splitting instability. With an adaptive prior the real loop has perfect tracking as an invariant set at every representable sample. *Open:* the gauge as a design input (the masses set the excitation spectrum $H_L(q)$); the floor law and its removal for a misplaced prior beyond first order.
- Transfer: transporting a learned measure to a new plant.

9. **Quantisation and the curse of dimensionality (Corollary B.6).**
   - *Done (Theorem C′):* an adaptive law whose leak tracks the equal-mass centroidal quantisers of a prior measure (centroidal Voronoi tessellations [57]; coverage control [19], [45]), and Theorem D′'s observation that the state and the learned function converge at the rate $m^{-1/2}$, whose exponent is dimension-free: the curse of dimensionality of Corollary B.6 lives in the $W$-distance of the parameter measures, which the loop never evaluates.
   - *Open:* second-order (centroid) quantisation rates for the ideal measure on the lifted space; approximation rates that do not go through $W$ (Maurey-type).
10. **Symmetric systems beyond kernels.** The OT quotient applies verbatim to swarms of identical agents, mixtures and multi-head models ([27] and [45] are the control-side precedents).
11. **Deep approximators.** Hierarchical (multi-layer) mean-field measures, with a layer-wise OT Lyapunov function.

# A Verification map

Table 47 lists the tests and experiments of the accompanying repository that check each statement; statements not checked are listed after it. Test files are abbreviated: `T` is `tests/test_transport.py`, `Lg` is `tests/test_langevin.py`, `Av` is `tests/test_averaging.py`, `Ex` is `tests/test_excitation.py`, `NR` is `tests/test_nonreciprocity.py`, `Pe` is `tests/test_periodic.py` and `Ba` is `tests/test_barycentric.py`.

Table 47: Verification map.

| statement | what is checked | where |
|---|---|---|
| constants (1) of Section 9, closed forms | $\nabla^2_{(c,s)}\varphi \le H_\varphi$ by brute force | `T::test_gaussian_constants_are_suprema` |
| Lemma P.1 | $\|\mathrm{Proj}_K(\vartheta, Gy)\|_{G^{-1}} \le \|y\|_G$ on 2000 adversarial states, with equality when inactive; a tight instance of F(i) with ratio $\ge 0.99$ | `T::test_lemma_P1_projection_is_nonexpansive_and_tight_velocity_bound` |
| A(i) raw rates | lifted raw rates equal classical raw rates under the gain map ($10^{-10}$) | `T::test_identification_with_classical_law` |
| A(i) projected rates, only-if | per-kernel projection with rescaled layer matches to $10^{-10}$ with active projections; an unrescaled layer does not; five joint balls are each distinguished (two-case construction: four active against an inactive per-kernel field, one inactive against an active one) | `T::test_identification_projected_rates_per_kernel` |
| B with constant ideal and $\varepsilon = 0$ | the identity to floating-point precision; cross term $\le 0$; remainder $\le$ Lemma B.2; $\dot V_\pi \le$ (B) at every sample | `T::test_theorem_B_inequality_along_trajectories` |
| B with a moving ideal, $\varepsilon \ne 0$, switches and active projection | the same plus $V_* W$; at least one switch; projection active at $\ge 10$ samples; Dini quotient $\le$ (B) | `T::test_theorem_B_moving_ideal_nonrepresentable_switches_and_active_projection` |
| B, Dini form | forward-difference quotient of $V_{\mathrm{OT}} \le$ (B) | `T::test_dini_derivative_of_V_OT_bounded_by_active_branch` |
| B against labelling | matched error $\ll$ labelled error; the bound is 30–45$\times$ smaller | `T::test_permuted_ideal_labelled_vs_OT`, `exp_transport_permutation.py` |
| B.5: failure of the plain bound along ($\mathrm{L}_\sigma$); the ($\mathrm{L}_\sigma$) variant holds | with every C(iv) hypothesis asserted, the plain bound fails by $> 3\times$ and the leak variant holds | `T::test_corollary_B5_fails_along_lsigma_and_its_leak_variant_holds` |
| C(i), Lemma C.3 (unprojected pure leak) | zero switches and exact $e^{-\sigma t}$ decay (explicit Euler, pure leak) | `T::test_theorem_C_i_exact_dissipation_pure_leak` |
| C(ii) | $C \le b^2 + 4\eta R$; $\sqrt{C} \le b + 2\eta$; $C = b^2$ under both gap conditions (300 configurations) | `T::test_lemma_C_ii_induced_coupling_bound` |
| C(iv) | an instance satisfying **every** hypothesis, asserted, including the gap condition $4\bar\eta(b + \bar\eta) < \Delta_{\mathrm{gap}}$; the envelope holds at every sample, with actual contraction | `T::test_theorem_C_iv_exponential_estimate` |
| C(v), Proposition C.5 | label invariance of the OT leak; the reshuffling transient of the labelled leak | `exp_ot_sigma_mod.py`, `exp_sigma_mod_transient.py` |
| C′(i) | the plan reduces to the assignment for $m_0 = m$ ($10^{-10}$) and $W^2 = V^2_{\mathrm{bar}} + \mathrm{Var}_\gamma$ ($10^{-10}$); along the pure leak $W^2$ is non-increasing at every step and $D^+ Z \le -\sigma_m V^2_{\mathrm{bar}}$ holds at every state; the flow reaches $V^2_{\mathrm{bar}} < 10^{-8}$ and ends at a fixed point of the equal-mass Lloyd iteration | `Ba::test_plan_reduces_to_assignment_and_pythagorean_split`, `Ba::test_pure_barycentric_leak_dissipates_to_a_centroidal_configuration`, `exp_barycentric_leak.py` → `results/barycentric_leak/table.md` |
| C′(ii) | $\sqrt{C_{\mathrm{bar}}} \le b + 2\eta$ on 90 random configurations; $C_{\mathrm{bar}} = b^2_{\mathrm{bar}}$ at small $\eta$ (plan stability) | `Ba::test_induced_coupling_bound_and_plan_stability` |
| C′(iii) | the Lyapunov inequality at every recorded state of a closed loop with the barycentric leak toward a 24-atom prior | `Ba::test_lyapunov_inequality_along_the_barycentric_leak_closed_loop` |
| D | consistency in $m$ (nearly constant amplification 1.64–1.69; not the constant $e^{\Lambda T}$) | `exp_mean_field_m.py` |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D′(b), (d) | the $m^{-1/2}$ rate of the synchronous coupling to a 4096-particle reference: slopes $-0.47 \pm 0.06$ ($\tau = 0$) and $-0.46 \pm 0.05$ ($\tau = 2 \cdot 10^{-3}$) over 24 seeds, $0.94 \pm 0.08$ and $0.92 \pm 0.08$ against $(1/m - 1/M)^{1/2}$; not the constant | `exp_chaos.py` → `results/chaos/table.md` |
| D″ | with exact Lipschitz constants and the a-priori tube, $\sigma_{\text{cert}} = 4.04$ against Proposition K's $\sigma_K = 0.00144$; at $1.2\sigma_{\text{cert}}$ with dither the coupled distance is of order $m^{-1/2}$ (slope $-0.51 \pm 0.02$), bounded in time and 700× below the bound, with the network at its prior (fit error 1.000); without dither the reference collapses (spread $10^{-14}$) and the distance is zero; at $\sigma_{\text{cert}}/8$ it decays to $10^{-11}$; in the learning regime ($\sigma_K/3$, 1000 s) the network stays spread and fits the plant (fit error 0.63) and the distance follows $m^{-1/2}$ ($-0.62 \pm 0.16$) but grows threefold between the windows $(100, 200]$ and $(500, 1000]$ | `exp_chaos_uniform.py` → `results/chaos/uniform_table.md` |
| D‴, D‴.5, D‴.7 | the window realised at $\sigma_0 = 0.15\sigma_{\text{crit}}$ ($\kappa_R[\mu^*](N_{0.1}) = 0.0051 < 0.0071 < 0.047$), (H) along the warm run (margin $\geq 0.0014$), saturation of the coupled distance (last/first window ratios 1.13–1.15 over 900 slow units), slope $-1.28 \pm 0.30$ in $m$; the weight-sensitivity law per seed within 25% in 9 of 12 pairs, per-cluster cosines 0.92–1.00; the exact stationary distance of Proposition D‴.5(ii) within 2.3% in all 12 pairs, splitting margins positive in 11 (one $m = 16$ seed a saddle); cold start: growth (2.9–7.8×) and the floor's fraction 0.20, 0.08, 0.04 decaying like $m^{-0.64 \pm 0.08}$; real loop at $k_e = 25$ within 0.7% of its averaged slow dynamics on the same clouds; not the constants $C_1$, $m_0$ (vacuous: $2 \cdot 10^9$, $10^{23}$) nor the saturation of Corollary D‴.7 | `exp_chaos_learning.py` → `results/chaos/learning_table.md` |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.9, D‴.11 | the matching prior: zero residual, $\kappa_R = 0$, margin $\sigma_0 = 0.00707$ against the $\sigma$-modification's 0.00196, floors 1.61× and 2.18× higher; misplacement closes the window (centres +0.6, amplitudes ×1.2, widths ×0.7); shrunk amplitudes ×0.5–0.7 give margins above $\sigma_0$ at the uniform weights and lower floors; admissible-weight table under three priors (overshoot splits, undershoot stabilises; the shrunk prior keeps all four tested weights admissible, +0.0054 to +0.0073); matching-prior coupled runs saturate within 2.1 of the theorem's relaxation times (observed e-folding 48–84 slow units against 107–306 for the $\sigma$-modification) with per-seed linearised-law ratios 0.81–1.11 (parameter) and 0.61–1.69 (function) and the exact law within 0.987–0.998 in all nine pairs; the two saddles among them are left by a local split with the coupled distance on its plateau; the function-level law for the $\sigma$-modification within ratios 0.48–0.74 of the measured means (RMS over seeds 0.52–0.84; per-seed law from the realised weights 0.60–1.57) | `exp_prior_design.py` → `results/chaos/prior_design.md` |
| D‴.12 | the same clouds under the nearest-atom and the OT leak: cluster weights conserved (max deviation 0 in 32 systems) against equalised ($s_{\mathrm{eq}} = 93$–$164$ slow units; $\ln 2/\sigma_0 = 98$; 1–21 particles moved); commensurate sizes: function-level distance$^2$ $4.9 \cdot 10^{-7}$–$7.5 \cdot 10^{-9}$ against $2.5 \cdot 10^{-3}$–$5.9 \cdot 10^{-5}$ and $W^2$ $7.9 \cdot 10^{-3}$–$6.9 \cdot 10^{-4}$ against 5.1–0.81 (the exact $K \times K$ transport between the weighted stationary measures within 0.997–0.999 at admissible weights; the formula $\frac{1}{2}\|w^m - w^M\|_1 \Delta_\nu^2$ within −7% to +28%), both OT distances still falling (0.02–0.035 and 0.12 per 150 slow units) while the synchronous $D$ rises to 0.95–0.96 of its mismatch plateau; incommensurate sizes: one or two bridge particles at 0.28–0.58 $\Delta_\nu$ from the atoms; against $\nu$: $m \cdot W^2 = 19.4$–$21.0$ (slope $-0.98 \pm 0.03$) and $m \cdot \mathrm{fit} = 0.45$–$0.60$ (squared slope $-1.81 \pm 0.05$); against the $M = 512$ reference slopes $-2.07 \pm 0.05$ and $-1.21 \pm 0.04$ (the reference carries its own bridge particles; the $m = 16$ value still relaxing); fit error of the reference $0.012 \to 8 \cdot 10^{-5}$; the designed masses inadmissible between $\delta = 0.03$ and 0.05 | `exp_ot_leak.py` → `results/chaos/ot_leak.md`; `tests/test_ot_leak.py` (the vertex plan, the gradient flow, the $m$-independent stationary configuration, conservation against constraint, the forest bound, the exact $W^2$) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.14 | the real loop at $k_e = 25$ from the clouds at the end of `exp_ot_leak.py` (contracted where the weights are admissible) and from the collapsed lifts: under the OT leak $\|\hat{f}_m - \hat{f}_M\|^2_{L^2(\text{orbit})} = 7.5 \cdot 10^{-9}$–$2.7 \cdot 10^{-8}$ ($m = 15$, 60 against 510) and $5.2 \cdot 10^{-10}$–$3.6 \cdot 10^{-8}$ against the 3-atom lift, rms tracking error $0.9 \cdot 10^{-6}$–$7.3 \cdot 10^{-6}$ (the lift $8 \cdot 10^{-12}$); under the nearest-atom leak $1.1 \cdot 10^{-4}$–$2.5 \cdot 10^{-3}$ with tracking errors $3.2 \cdot 10^{-4}$–$1.9 \cdot 10^{-3}$; at the one pair with both weights admissible the disagreement equals the lifted orbits' distance to 1.0003 and the admissible systems sit within $2.5 \cdot 10^{-11}$ of their lifts; three systems at saddle weights (margins $-0.0035$, $-0.0011$, $-0.0022$) have split (spread 0.04–0.11) and sit off their lifts | `exp_ot_real_loop.py` → `results/chaos/ot_real_loop.md`; `tests/test_ot_leak.py` (the collapsed subspace's invariance, the lifts' equality) |
| D‴.15 | lifts of the 3-atom orbit to $m = 6$, 9, 15 (OT leak, matching masses) and to $m = 16$ ($\sigma$-modification at $(7, 5, 4)/16$ and $(9, 4, 3)/16$): fixed points of the Poincaré map to $10^{-14}$–$10^{-15}$; class separation subspaces invariant to $\leq 6 \cdot 10^{-8}$; separation exponents equal to the ghosts' to $2 \cdot 10^{-5}$ relative and to $(\sigma_0 - \lambda_j(R_k^G))/k_e$ within 0.2–4.1%; collective exponents equal to the $K_0$-atom orbit's to $10^{-4}$ relative for every $m$ and to Theorem J(b)'s prediction within 0.6–3.2%; the saddle's exponent $-9.13 \cdot 10^{-5}$ (multiplicity 8) at $(9, 4, 3)/16$ | `exp_lifted_orbit.py` → `results/chaos/lifted_orbit.md`; `tests/test_ot_leak.py` (the $m = 6$ lift against the ghosts and the 3-atom loop) |
| D‴.16 | the dithered averaged slow loop, $m = 15$ from the counts $(6, 5, 4)$: nearest-atom record-level label changes $1.0 \cdot 10^{-5}$, $5.1 \cdot 10^{-4}$, $5.6 \cdot 10^{-3}$ per slow unit at $\tau_0 = 0.005$, 0.01, 0.02 (Arrhenius barrier 0.048, among the ridge minima 0.040–0.074); time at the balanced counts 0.00, 0.54, 0.25 against the energy–multinomial law 0.86, 0.58, 0.33 (standard errors $\approx 0.1$ at $\tau_0 = 0.01$) and against the OT leak 0.996, 0.992, 0.88, whose excursions are returns of the crossing particle (121 of 121, 298 of 336) with the effective barrier 0.058 | `exp_dither_ot.py` → `results/chaos/dither_ot.md` |
| D‴.14 ($n = 2$) | a planar plant (four vector-valued bumps, a Lissajous reference, $p = 5$), $k_e = 25$, $\sigma_0 = 0.05$, sizes 8, 32 against 128 from the same clouds: the OT leak equalises the weights at 13–14 slow units ($\ln 2/\sigma_0 = 14$), the sizes agree to $5.0 \cdot 10^{-8}$–$1.6 \cdot 10^{-7}$ with rms tracking errors $3.9 \cdot 10^{-6}$–$1.6 \cdot 10^{-5}$, against $5.5 \cdot 10^{-3}$–$2.3 \cdot 10^{-2}$ and $1.9 \cdot 10^{-3}$–$6.7 \cdot 10^{-3}$ under the nearest-atom leak (fit errors along the orbit 2–$9 \cdot 10^{-4}$ against 0.11–0.39) | `exp_ot_2d.py` → `results/chaos/ot_2d.md` |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.17 | $J_m^{\mathrm{OT}}$ non-increasing along 300 steps of a run with reassigned particles (strong leak; the plan does not switch there, so this is a smooth gradient flow), $\mathrm{d}J/\,\mathrm{d}s = -\|\mathrm{drift}\|^2_{G^{-1}}$ where the plan is unique, and at a constructed tie between two optimal plans (mirror-image particles) $J^{\mathrm{OT}}$ equals both frozen energies and decreases along either plan's drift | `tests/test_ot_leak.py` |
| D‴.18 | the real loop at $k_e = 25$, 8000 time units, sizes 3–240: the 3-atom orbit's Floquet exponents within 0.56–3.22% of $\lambda_j(H_q)/k_e$ and the ghosts' equal to $(\sigma_0 - \lambda_j(R_k^G))/k_e$ to $3 \cdot 10^{-4}$ (matching and shifted priors); inside the cells the per-particle distance to the lift decays at $2.69$–$2.97 \cdot 10^{-4}$ for $m \geq 15$ (prediction $2.83 \cdot 10^{-4}$) with constants 1.03–1.29 under both leaks, the $m = 3$ runs mixtures of collective modes; across a cell boundary ($\rho = 4.5$) the nearest-atom leak ends at other counts ($W^2$ 1.9–12.3, that of the lifts of the realised weights; two of them inadmissible) and the OT leak at $q$ ($W^2$ 0.010–0.23, falling at $2\sigma_0$); one particle across a boundary: $W^2$ grows to 0.79, 0.20 (nearest) and falls to $3.0 \cdot 10^{-3}$, $8.2 \cdot 10^{-4}$ (OT); on a moving orbit (shifted prior) $2.18$–$2.63 \cdot 10^{-4}$ against $2.27 \cdot 10^{-4}$, constants 1.04–1.23; the proof's radius at most 0.019 | `exp_uniform_basin.py` → `results/chaos/uniform_basin.md`; `tests/test_uniform_basin.py` (at the shifted prior: the remainder constant and the exact $N_a + N_b/m$ structure of the per-particle term, the separation response $(GR_k - \sigma_0 I)v$, the contraction at $\mu = \min\{\lambda_{\mathrm{coll}}, \lambda_{\mathrm{sep}}\}$ with one constant, the same for every $m$) |
| D‴.20 | the critical excess by continuation in $\varepsilon$ (Newton residuals $\leq 8.7 \cdot 10^{-13}$): $\varepsilon_c = 5.11$, 13.5, 23.1, 65.0 for $\delta = 0.02$–$0.2$ (the exact threshold equal to $\varepsilon_c$; the rank-one formula 1.22–1.31 and 0.70–1.22 times $\varepsilon_c$ with the two atoms), 6.99, 14.6, 24.3 for an excess taken from one atom; $\delta_c(\varepsilon) = 0.0042$–$0.20$ for $\varepsilon = 1$–$64$; dynamics for $m = 15$–$510$ at $\varepsilon = 1$, 4, 16, 64 over 600 slow units, continued to 6000: equalised at $\varepsilon = 1$; at $\varepsilon = 4$ stable end weights with residual excesses up to 1/60, 3/240, 7/510; at $\varepsilon = 16$ and 64 the stable systems keep their sampled weights and the four unstable ones move (one moving its excess to an atom that was at $q$); floors 0.55 and 0.74 of the nearest-atom law, 0.23–0.84 and 0.47–0.91 of the observed nearest-atom floors; fit errors 0.12–0.14 at $\varepsilon = 64$ | `exp_entropic_leak.py` → `results/chaos/entropic_leak.md`; `tests/test_entropic_leak.py` (the solvers, the gradient flow, the limits at their rates, the separation block entrywise with the invariance of the subspace, the critical regularisation by continuation in $\varepsilon$ with $A \succ 0$ checked and every residual below $10^{-9}$) |

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*Table 47, continued*

| statement | what is checked | where |
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| D‴.21 | the averaged loop from the warm clouds ($m = 15$–$510$, four seeds, matching and shifted priors, both leaks, 600 slow units): (E), (P) to $6 \cdot 10^{-13}$; under the OT leak no relabelling, $\alpha > 0$ from $s \leq 300$, ratio/$\lambda \geq 1.00$ (matching) and $\geq 1.61$ (shifted), $E$ from $4.1 \cdot 10^{-3}$–$4.0 \cdot 10^{-2}$ to $8.3 \cdot 10^{-7}$–$8.1 \cdot 10^{-6}$ below the a-posteriori bound, the end rate 0.01416–0.01417 against $2\sigma_0 = 0.01415$ and $\lambda = 0.0140$–$0.0141$, at the shifted prior the curvature's cost 0.802 and (ii)'s slack 0.683; the nearest-atom $E$ at the spurious minima's energy at the 22 admissible count vectors (within 0.24%), split and still falling 0.5–16.5% below it at the 10 inadmissible ones ($m \leq 60$); (H*) on the whole space at the shifted prior ($\kappa = 4.7 \cdot 10^{-4}$, $E_0 = 9.1 \cdot 10^{-10}$), the a-priori basins holding no particle in another cell below $m \approx 1.7 \cdot 10^5$ (matching) and $2.3 \cdot 10^6$ (shifted); the margin +0.0023 with 1/30 of the particles moved into another cell, negative from 1/15 | `exp_ot_basin.py` → `results/chaos/ot_basin.md`; `tests/test_ot_basin.py` (the identities for every assignment with the counts and their exact defects at other counts, the PL chain and its sharp constant, the decay from a particle in another cell below the a-posteriori bound with a positive margin, the admissible nearest-atom spurious minimum) |
| D‴.23 | the linear theory at the exact representation ($h_j = 3.06 \cdot 10^{-4}$–$0.195$; $\lambda_1^- = 2.68 \cdot 10^{-5}$–$3.06 \cdot 10^{-4}$ for $\eta = \sigma_0/10 \ldots \infty$; (v)'s radius at most $2.0 \cdot 10^{-5}$–$2.2 \cdot 10^{-4}$, 94–$6 \cdot 10^4$ times below the $K_0$-atom runs' starts, $2.2 \cdot 10^4$–$3.6 \cdot 10^5$ below the warm clouds'); the fixed prior's bias within 4.8% of the first-order formula in norm (0.85 and 0.73 of it in modes 1 and 3, the error first order in the shift); the $K_0$-atom loop: every mode above $10^{-6}$ decays at $\lambda_j^-$ (log-slopes 0.981–1.003 of it), fit error $9.78 \cdot 10^{-3} \to 1.8 \cdot 10^{-5}$ ($\eta = \sigma_0$) and $6.7 \cdot 10^{-6}$ ($\eta = \infty$) by $s = 6000$; separated starts (five draws): departures from the $K_0$-atom run second order in the separations, consistent with the same expected size at $m = 15$ and 510; $m = 15$–$510$ from the warm clouds: ungated the prior is dragged by the transit ($\|\bar{y}\|$ 0.032–0.71 at $s = 500$) and one $m = 15$ system at $\eta = \infty$ is carried onto a plateau of the unregularised loss (Hessian eigenvalues $-1.8 \cdot 10^{-5}$, $7.2 \cdot 10^{-5}$; still there at $s = 23000$, crossed by the gradient flow, below $10^{-12}$ after $7.97 \cdot 10^4$ more slow units); the transit gate leaves a residue in the weak modes ($\|\bar{y}\|$ $4.9 \cdot 10^{-3}$–$3.7 \cdot 10^{-2}$ against $1.3 \cdot 10^{-3}$, $\eta = \sigma_0$); the settled gate ($t_{\text{open}} = 1211$–$1304$) opens replicated and follows the $K_0$-atom run for every $m$ (per-seed ratios 0.995–1.000 at $\eta = \sigma_0$, 0.968–1.001 at $\eta = \infty$), fit errors $1.6 \cdot 10^{-4}$ and $3.4$–$3.5 \cdot 10^{-5}$ against $9.78 \cdot 10^{-3}$; the limits of (i): a non-replicated saddle held by the leak, and unbounded adaptive solutions, beyond the saturation and from an exact six-kernel fit (the width coordinate running off toward a Karush–Kuhn–Tucker point of the box-constrained loss) | `exp_adaptive_prior.py` → `results/chaos/adaptive_prior.md`; `tests/test_adaptive_prior.py` (the gradient structure, stationarity, the linearisation's rates and its self-adjointness in the gradient-flow metric, the first-order bias, the two gates, the limits of (i)) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.24 | the floor as the bias: $F$ equal to the bias form to the finite-difference error ($1.1 \cdot 10^{-6}$–$1.2 \cdot 10^{-8}$); the law (B) $1.36 \cdot 10^{-2}$, $6.17 \cdot 10^{-2}$, 0.229 per unit of $1/m - 1/M$ at $\sigma_0/\sigma_{\text{crit}} = 0.05$, 0.15, 0.5 (0.50–0.84 of the bound, which is $\sigma_0/4$ times the adaptive prior's coupled-distance law 46.3), modes 2–6 carrying 98–99% of it and 97% unweighted, the weight moving the shares toward $h \approx \sigma_0$; the parameter split to $1.4 \cdot 10^{-7}$, the adaptive prior's coupled-distance law 1.4–5.3 times the fixed prior's; 5 of the 12 sampled $m$-systems are fixed-prior saddles (class 1, $w_1 \geq 0.4$) and split (floors 0.81–1.01 of the replicated point's), the exact bias at the sampled weights reproduces the floor at the other seven (ratio 1.0000), the delta method misses it by 0.86–1.72; the nearest-atom leak over 6000 slow units: the fixed prior's floor stays, the adaptive prior's decreases without a floor (at most $2.0 \cdot 10^{-6}$ at $\eta = \sigma_0$ and $5.4 \cdot 10^{-8}$ at $\eta = \infty$ over the representable samples at $s = 6000$); $D$ 0.94–1.00 of the replicated stationary points' under both priors, the adaptive prior's 1.72–4.81 times the fixed prior's; $W^2$ the cross-class transport (0.67–0.99 of it), the two priors' $W^2$ within 1.00–1.32; the boundary sample $(8, 4, 3)/15$ decays like $1/s$ ($6000 \cdot \|\hat{f}_m - \hat{f}_M\|^2 = 0.88$ and 0.42 against 0.8 and 0.4); the design table at the misplaced prior (the fit error and the floor of the four designs; the OT designs' $D$ the largest, 6.0–1.0, the transit's mismatch; the ranking at $s = 1250$ and 2000); the real loop: the fixed prior at the averaged floor (ratios 0.996–1.001; the $m = 15$ run on a saddle orbit), the adaptive prior along the averaged decay (0.987–0.992), the exact representation invariant (five-period rms $\leq 9.0 \cdot 10^{-12}$); the instance's probabilities (outside $W$ 0.361, 0.074, 0.0021; Hoeffding 1.87, 0.60, 0.036; the fixed prior's saddles 0.369, 0.260, 0.109) and the adaptive prior's expected limiting floor ($\geq 7.3 \cdot 10^{-3}$, $2.2 \cdot 10^{-4}$, $6.6 \cdot 10^{-7}$: 1.8, 0.24, 0.005 times the fixed prior's law); Monte Carlo (600 nested samples): the exact floor 2.78, 1.53, 1.11 times the law; in two dimensions (four bumps, $p = 5$, $\sigma_0 = 0.05$, six seeds): every sample D‴.5-admissible and the fixed prior's floor the exact bias at the sampled weights (ratio 1.0000 in all 18 systems), the adaptive prior's at most 0.019 ($\eta = \sigma_0$) and 0.006 ($\eta = \infty$, the atoms reset every slow unit) of it at the 7 representable systems and still falling, the real loop ($\eta = \infty$) along the averaged prediction (0.979–1.009) and invariant from the exact representation ($\leq 4.7 \cdot 10^{-10}$) | `exp_adaptive_floor.py` → `results/chaos/adaptive_floor.md`; `tests/test_adaptive_floor.py` (the nearest-atom adaptive flow as a gradient flow, the free-mass energy as the minimum of the OT energy over the masses, also against masses moved by $\pm 1/m$, the floor law as the bias derivative with the weight's band-pass against the unweighted shares, the exact representation of a D‴.5-admissible sample stationary under the adaptive prior and the margins $\sigma_0$ at a fixed-prior saddle, the parameter split, the energy across a cell crossing and an empty cell) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.26 | the three-bump instance, $w = (6, 5, 4)/15$ (representable; a fixed-prior saddle), the real loop with the adaptive prior in the state, $\eta = \sigma_0$ and $\infty$, $k_e = 25$ and 50, the monodromy by central differences (step $10^{-5}$, DOP853 at rtol $10^{-13}$): return residuals $\le 1.7 \cdot 10^{-15}$; at $m = 15$ and 30 the 36 and 81 separation exponents at $\sigma_0/k_e$ to $\le 1.7 \cdot 10^{-6}$ (relative), the other 18 ($\eta = \sigma_0$) or 9 ($\eta = \infty$) equal to the weighted $K_0$-atom loop's to $1.4 \cdot 10^{-5}$–$6.4 \cdot 10^{-4}$; the collapsed variational equation (its generator the real loop's linearisation to $5.8 \cdot 10^{-11}$, its exponents the differences' to $4.1 \cdot 10^{-5}$) at $k_e = 12.5$–400 for $x_m = 2\sin t$ and $2\sin t + 0.8\sin 2t$ ($\|\bar{A}_1\| = 0$ and $5.6 \cdot 10^{-3}$, skew): the first-order deviations falling by $\to 4$ per doubling (the slowest mode 13.6% and 3.84% below at $k_e = 25$, 50; every mode below on the instance), the second-order residuals by $\to 16$ ($4.9 \cdot 10^{-7}$ at $k_e = 400$), $c_1/\lambda_1 = -100.6$ (the filter's part $-102.1 = -0.989\omega_1^2$, $\omega_1 = 10.16$), measured $-100.5$ at $k_e = 400$; the gain scaled by 30: $c_8/\lambda_8 = +4.17$, $c_9/\lambda_9 = +12.65$, those exponents above their prediction by $c_j/(\lambda_j k_e^2)$ to three digits; at $\eta = 10\sigma_0$ $\bar{A}_0$ self-adjoint in $\mathrm{diag}(I, (\sigma_0/\eta)I)$ to $8.7 \cdot 10^{-18}$ ($6.4 \cdot 10^{-2}$ in $I$); the error channel by Liouville's formula within $O(k_e^{-3})$ of $k_e - \mathrm{tr}\, H_L(w)/k_e$ | `exp_adaptive_floquet.py` (Part C: `--only-C`) → `results/chaos/adaptive_floquet.md`; `tests/test_adaptive_floquet.py` (at $w = q$, $\eta = \infty$, $m = 6$: the return of one period, the nine separation exponents, the $K_0$-atom loop's spectrum within the averaging correction; at $(2, 1, 1)/4$, $\eta = \sigma_0$, $m = 4$, the prior in the state: the real loop's Jacobian splitting into the separations and the collapsed variational equation, the monodromy's 3 separation and 18 collective exponents, the second-order prediction; on an asymmetric reference: $\bar{A}_1$ skew, $\ell_j \bar{A}_1 r_j = 0$, the residual $O(k_e^{-4})$, $c_1$ measured, Liouville's formula; at $\eta = 10\sigma_0$ $\bar{A}_0$ self-adjoint in the metric and not in $I$, $\bar{A}_1$ skew in it) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.26a | Corollary D‴.26's instance on $x_m = 2\sin t + 0.8\sin 2t$, $\eta = 0$ ($K = \hat{J}$, $\Lambda = \sigma_0 I$), $\sigma_0$ and $\infty$, $k_e = 12.5$–$1600$, a 512-point Fourier grid: the contraction for $P$ in 3–9 iterations (499 at $\eta = 0$, $k_e = 12.5$: stalled at rounding), invariance residual $\leq 3.8 \cdot 10^{-15}$, $\lvert P - P_2\rvert/\varepsilon^3 \to 3.34 \cdot 10^3$, $\lvert P - P_{(3)}\rvert/\varepsilon^4 \to 5.8 \cdot 10^4$; the slow block's rates the full variational equation's to $\leq 1.0 \cdot 10^{-8}$; the rates at $\pm\varepsilon$ equal to $\leq 5.9 \cdot 10^{-12}$ of the largest, $\leq 1.1 \cdot 10^{-9}$ per mode (first-order deviation $9.8 \cdot 10^{-2}$–$8.9 \cdot 10^{-6}$ of the largest, per mode up to 29.6%: the slowest mode −29.6%, −12.5%, −4.0%, −1.1% at $k_e = 12.5$–$100$, $\eta = \infty$), with a mismatch $1.1 \cdot 10^{-2}$–$7.8 \cdot 10^{-5}$ (per mode $6.1 \cdot 10^{-2}$–$3.5 \cdot 10^{-4}$); the Magnus parameter 0.48–0.004; the noise: decoupling residual $\leq 3 \cdot 10^{-16}$, $\lVert b - b_0\rVert/\lVert b_0\rVert$ 0.63–0.0074, $\lvert b - b_0 - \varepsilon b_1\rvert/\varepsilon^2 \to 3.34 \cdot 10^3$, $\mathrm{Cov}(v) \cdot 2\varepsilon^3 - 1$ from $-9.6 \cdot 10^{-2}$ to $-2.4 \cdot 10^{-5}$, $\lVert \mathbb{E}[vY'^{\top}] + b^{\top}\rVert/\lVert b\rVert$ 0.15–$1.2 \cdot 10^{-3}$, $\lVert \mathrm{Cov}(Y') - \bar{\Sigma}\rVert/\lVert\bar{\Sigma}\rVert$ 0.35–$9.4 \cdot 10^{-5}$ ($\eta = \infty$; $O(\varepsilon^2)$ to $k_e = 200$–$400$, $O(\varepsilon)$ beyond), the weakest mode's deviation $-34\% \to -6.6 \cdot 10^{-5}$ the sum of the noise's attenuation ($-56\% \to -1.1 \cdot 10^{-4}$) and the rate's correction ($+49\% \to +4.6 \cdot 10^{-5}$), the jitter with its next term at $O(\varepsilon^2)$ (0.30 → 0.001 at $k_e = 100$–$1600$), the floor at $t = 0$ at 0.763–1.0000 of the prediction, its period average at 0.766–0.99997 of D‴.30(ii)'s formula and pointwise in $[0.9952, 1.0091]$ at $k_e = 1600$, the jitter's predicted range | `exp_adaptive_floquet.py` (`--add-D`) → `results/chaos/adaptive_floquet.md`; `tests/test_adaptive_floquet.py` (Chang's $P$ and the exact slow multipliers, the $\varepsilon^3$ truncation on the instance; on a synthetic block with an $O(1)$, non-symmetric $\Lambda$, $P - P_2$ at $\varepsilon^3$ and $P - P_{(3)}$ at $\varepsilon^4$, a flipped $\Lambda$ and a dropped cubic term caught; the rates even in $\varepsilon$ at $\eta = \sigma_0$ and $\infty$, a mismatch breaking it; the slow noise decoupled, $b_0$ against $\dot{K}$ by the chain rule and the plant's own slope by differences, $b_1$ at $\varepsilon^2$, the covariances' orders) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.27 | the box $\mathcal{Z}$ (amplitudes and centres to 0.9 of $a_{\max}$, $c_{\max}$; the width coordinate in $[-6, 6]$): beyond the saturation (Part G(b)'s target) the projected prior converges at $\eta = \sigma_0$ (the projected midpoint rule, positions projected, $\mathrm{d}t = 0.25$ and 0.125, to $s = 10^5$; end states within $3.4 \cdot 10^{-12}$, the trajectory at $s = 300$ within $6.3 \cdot 10^{-7}$, $1.9 \cdot 10^{-7}$, $3.8 \cdot 10^{-8}$ of a reference for $\mathrm{d}t = 0.25$, 0.125, 0.0625) and $\eta = \infty$ (LSODA) to the same bounded stationary point (drift $\leq 3.5 \cdot 10^{-15}$), the prior's amplitude atom on the face, fit error 0.1415; the particles' largest distance from $\mathcal{Z}$ 0.92 and 0.89 against $B/\sigma_0 \leq 3686$; the amplitude bound 0.5–0.99 of $a_{\max}$: fit errors 0.1949–0.1101, between the fixed prior's 0.1750 and the unprojected 0.1050 for 0.7–0.99, above the fixed prior's for 0.5–0.65, the exact atoms outside the boxes up to 0.7; a Lloyd-consistent, non-optimal plan at $\eta = \infty$ along which the reduced energy rises from $1.251 \cdot 10^{-3}$ to $1.574 \cdot 10^{-3}$ while the selected energy falls (largest increment $-3.4 \cdot 10^{-6}$); on Part G(a)'s four-bump target the width coordinate stopped at the face $\kappa = 6$ from $s = 10^9$ (fit error within $9.3 \cdot 10^{-7}$ of the face minimiser's); the misplaced-prior run identical with and without $\mathcal{Z}$, the box never approached (margins $\geq 16.9\%$ of its width) | `exp_adaptive_prior.py` Part H → `results/chaos/adaptive_prior.md`; `tests/test_adaptive_prior.py` (the projected prior bounded and convergent at $\eta = \infty$, while the unprojected coordinate passes 11 by $s = 10^4$; at $\eta = \sigma_0$ and $\mathrm{d}t = 0.125$ the prior in its box with its amplitude atom on the face, the particles bounded and $\mathcal{J}$ non-increasing; inactive near the exact representation) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.28 | the three-bump instance plus $\varepsilon \cdot \exp(-(x-0.8)^2/(2 \cdot 0.3^2))$, $\varepsilon = 0$–$0.5$: the approximant's fit error $3.6 \cdot 10^{-3}$–$2.5 \cdot 10^{-2}$ ($\varepsilon = 0.05$–$0.5$), the gauge check (the gradient at the gauge transform $\leq 1.6 \cdot 10^{-15}$; Newton continued in $w$ from $\vartheta^b(q)$ at it to $\leq 1.7 \cdot 10^{-11}$, $\hat{f}$ to $\leq 5.5 \cdot 10^{-13}$; plain Newton there in 3–5 of 6 cases), the adaptive prior's sensitivity $1.4 \cdot 10^{-11}$–$2.7 \cdot 10^{-10}$ of the fixed prior's, $P(w^m \in W_g) = 0.52$–$0.64$, 0.87–0.88, 0.997 at $m = 15$, 60, 240 (0.26, 0.71, 0.98 at $\varepsilon = 0.5$), the branch at the saturation near $\varepsilon = 0.68$, the fixed prior's within $2 \cdot 10^{-9}$ of $\sigma_0 \hat{J}(\tilde{A} + \sigma_0 I)^{-1} \tilde{g}$ and $7.0 \cdot 10^{-4}$–$5.6 \cdot 10^{-3}$ of the Gauss–Newton form, the margins $\sigma_0 - (q_k/w_k)\lambda_k^+$ exact to $10^{-14}$ over all count vectors of $m = 15$, 60 (none negative: $\lambda^+ \leq 0.04\sigma_0$ at $\varepsilon \leq 0.3$, $\leq 0.10\sigma_0$ at $\varepsilon = 0.5$), the dynamics' separation eigenvalues equal to them to $\leq 5.8 \cdot 10^{-12}$; the dynamics ($m = 15$, 60, 240 against 510, four seeds, $\varepsilon = 0.1$ and 0.3, 6000 slow units): the fixed prior's floor within 8% of its value at $s = 600$ ($3.1 \cdot 10^{-4}$–$5.9 \cdot 10^{-3}$ at $m = 15$, $\varepsilon = 0.1$) and its fit error above the approximant's, the adaptive priors' floor falling ($m = 60$, $\varepsilon = 0.1$: to $1.0 \cdot 10^{-9}$–$1.4 \cdot 10^{-8}$ at $\eta = \sigma_0$, $3.2 \cdot 10^{-11}$–$3.8 \cdot 10^{-9}$ at $\eta = \infty$) and the fit error to the approximant's (median excess $1.2 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$, $1.6 \cdot 10^{-5}$ at $m = 15$, 60, 240 for the samples 0.02 inside $W_g$); at the sample outside $W_g$ the box minimum on the branch 1.107 and 1.586 times the approximant's fit error, a permuted gauge transform existing | `exp_gauge_floor.py` (`--only-A` recomputes Part A, the branch's end and the outside samples) → `results/chaos/gauge_floor.md`; `tests/test_gauge_floor.py` (the gauge's critical points and function, the continuation from $\vartheta^b(q)$, the two sensitivities, the residual curvature's structure, the margins and the dynamics' separation eigenvalues, the adaptive prior's convergence to the gauge transform at $\eta = \infty$ and $\sigma_0$) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.29 | the servo $\ddot{x} = -\tau_{\mathrm{cog}}(x) + u$, $\tau_{\mathrm{cog}} = 0.4\sin(3x + 0.3) + 0.1\sin 7x$, $x_m = \sin t$, $\lambda = 5$, $K_0 = 4$ kernels on the position, $\sigma_0 = 0.01$: $\vartheta^b(q)$ (the local minimum BFGS reaches from a fixed start) fit error $1.46 \cdot 10^{-2}$, $\tilde{A}$ spanning $2.9 \cdot 10^{-4}\sigma_0$–$30\sigma_0$ (four modes below $0.1\sigma_0$), $\lambda_k^+ \le 0.0098\sigma_0$, $W_g = \{w > (0.061, 0.085, 0.119, 0.008)\}$, the floor law $1.29 \cdot 10^{-2}$; 120 random starts: 1 at $\vartheta^b(q)$, 33 below it, all saturated, the best $6.39 \cdot 10^{-3}$, with its saturated amplitudes held at $\lvert\alpha\rvert = 6$–$14$ $7.14 \cdot 10^{-3}$–$6.40 \cdot 10^{-3}$; the misplaced prior's (the engineer's model, fit error 0.200) replicated point a saddle (margin $-3.97 \cdot 10^{-3}$, fit error 0.0545); the averaged loop ($m = 16$, 64, 128 against 256, three seeds, 2000 slow units) from clouds around $\vartheta^b(q)$: fixed nearest atom floor $1.85 \cdot 10^{-3}$, $2.24 \cdot 10^{-4}$, $4.49 \cdot 10^{-5}$ and fit error 0.174–0.038, adaptive ($\eta = \infty$) $4.85 \cdot 10^{-5}$, $2.88 \cdot 10^{-7}$, $6.36 \cdot 10^{-8}$ and 0.0298–0.0147, OT 0.01462 with no floor, $\sigma$-modification 0.328, no leak 0.0180, 0.0147, 0.0146 (floors $7.7 \cdot 10^{-6}$–$6.6 \cdot 10^{-8}$); misplaced: the fixed designs split and end at 0.048–0.10, the adaptive prior unaffected at $\eta = \infty$, 0.035–0.019 at $\eta = \sigma_0$; from the engineer's model: fixed 0.102, 0.060, 0.048, OT 0.048, adaptive 0.027–0.022 ($\eta = \infty$) and 0.032–0.028 ($\eta = \sigma_0$), no leak 0.0152–0.0147, the $K_0$-atom flow at $\eta = \infty$ 0.0297, 0.0213, 0.0169, 0.0155 at $s = 600$, 2000, $10^4$, $10^5$; phase-plane kernels: the circle 5 exact zeros and 8 below $10^{-6}\sigma_0$, ellipses $\times 1.1$ and $\times 2$ 8 and 6, a two-harmonic orbit 3, centred kernels 16 exact of 20; the real servo ($k_s = 25$, 1000 time units, records and targets every 0.02): position-error rms $1.7$–$2.1 \cdot 10^{-5}$ (adaptive, matching OT, no leak), $7.8 \cdot 10^{-5}$ (misplaced OT), $9.1 \cdot 10^{-5}$–$1.8 \cdot 10^{-4}$ (fixed), $5.4$–$5.5 \cdot 10^{-4}$ ($\sigma$-modification), at the exactly filtered prediction to 1.0001–1.0029 at $t \ge 500$ (the slaved 0.931–0.996) | `exp_servo_cogging.py` (`--only-A` recomputes Parts A and D, `--add-B` adds the missing Part B jobs, `--only-C [designs]` reruns Part C) → `results/servo/servo_cogging.md`; `tests/test_sliding.py` (the phase-plane loop as the gradient flow of $\bar{L}$, the zero Gauss–Newton eigenvalue per kernel on a circular orbit in general position, the servo following the averaged loop with the position error at both predictions) |

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*Table 47, continued*

| statement | what is checked | where |
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| D‴.30 | the three-bump instance at the lift of the exact representation of $q = 1/3$, $k_e = 25$: $h_j = 3.06 \cdot 10^{-4}$–0.195, $\operatorname{tr} Q = 8.784 = 8.105$ $(\langle \dot{\hat{J}}^\top \dot{\hat{J}} \rangle) + 0.679$ $(\langle a^2 \hat{J}^\top \hat{J} \rangle)$, $\operatorname{tr} H^2 = 0.0657$, $\langle \lvert \hat{J} \rvert^4 \rangle = 0.299$; the parameters' floor per unit $N_0$ 0.15967, 0.16271, 0.16897, 0.17567 at $\eta = 0$, $\sigma_0$, $10\sigma_0$, $\infty$, the weakest mode's variance 0.0865–2.09 ($\times$24.1); the second-order force (OU, $B = 100$) and its averaged stationary biases 491 and 2397 per $(\mathbb{E}\nu^2)^2$, equal to the floor at std 0.0026 and 0.0012, the full second-order mean's 2397.68, the exact periodic mean's 504 and 2653; the bias's local exponents in $k_e$ 1.93–1.99 ($B \geq 20k_e$), 1.69, 1.47 ($B = 8k_e$, $4k_e$); the separations' drift at the lift $\lambda_{\max}(S_k) = 0.0148$, 0.0101, 0.298 per $\mathbb{E}\nu^2$, the threshold std 0.031 there; along the second-order transient the mean error's rms 0.121 $\to$ 0.602 and $\lambda_{\max}(S_3)$ 0.307 $\to$ 0.349 over 300 time units; the exact periodic covariance of the linearised loop ($k_e = 25, \ldots,$ 800; $\eta = 0$, $\sigma_0$, $\infty$) at the prediction to 0.9762–1.00001, the control (noise input $-k_e\nu$) numerically even in $1/k_e$ (successive ratios $\to$ 4.00), the weakest mode's variance $\times$28.0 at $k_e = 25$, the exact stationary mean 10.7%–0.17% above the averaged (adaptive, $k_e = 25$–200); the real loop (RK4 $\mathrm{d}t = 10^{-3}$, OU $B = 100$, 12 common noise paths of 300 time units, std 0.003 and 0.01, $\eta = 0$ and $\infty$): $z$-scores $-0.99$ to 1.75 against the transient covariance plus the mean part, the mean part 42% and 89% at $t = 300$, the priors 6.0–6.4% apart (predicted 5.3–6.1%); the lift $m = 15$ on a common noise path identical to rounding ($5.6 \cdot 10^{-16}$ in float64); the separations of the lift's third class on eight antithetic pairs at $-2.50 \cdot 10^{-4}$, $+2.74 \cdot 10^{-6}$, $+4.53 \cdot 10^{-4}$ per time unit at std 0.01, 0.03, 0.05 ($\lambda_{\text{eff}}$ 0.3271, 0.3175, 0.2943; the lift's drift 0.2981, the transient's 0.3264, 0.3270, 0.3281), the fit's small-noise limit $0.3291 \pm 0.0033$, the rate vanishing at std 0.0299 (0.0308 by the lift's drift, 0.0294 by the transient's) | `exp_measurement_noise.py` (`--only-A`, `--only-B`, `--only-D` recompute a part) $\to$ `results/chaos/measurement_noise.md`; `tests/test_measurement_noise.py` (the closed form against the Lyapunov solution, the trace identity at $\eta = \infty$ and the monotonicity in $\eta$; the exact periodic floors at $k_e = 100$, $\eta = 0$, $\sigma_0$, $\infty$; a lift $m = 6$ staying replicated under a noise path and following the $K_0$-atom loop; a separation decaying at std 0.01 and growing at 0.05 (a sign test); the separations' drift against an independent quadrature; along the second-order transient the mean parameters at the averaged force and the mean error carrying the developing deviation's tracking error; the exact periodic mean against the filtered Gram, the fixed prior's branch and its leak in the bias; the mean function's curvature term; the OU noise's variance and correlation, the law seeing the measurement; the responses $(H + \sigma_0 I)^{-1}$ and $H^{-1}$ to a constant force, the quasi-static force against the full second-order mean's noise part) |

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*Table 47, continued*

| statement | what is checked | where |
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| D‴.31 | the instance of D‴.30 with an OU noise: the adaptive prior's bias per $(\mathbb{E}\nu^2)^2$ for MR, DR, DR* at $k_e = 25$, 50, 100, 200 and $B = 100$, 1000, from the noise's own moments and with the parameters' fluctuation (MR 2397–$1.66 \cdot 10^5$, its parameters' part $\leq 0.03\%$; DR and DR* $\leq 1.18 < \frac{1}{4}\|\partial_x^2 f\|_\rho^2 = 1.31$, their parameters' part $-8.7\%$ to $+3.3\%$ except $+56\%$ for DR at $k_e = B = 100$, its own response $3.9 \cdot 10^{-4}$, $1.1 \cdot 10^{-4}$, $2.9 \cdot 10^{-5}$, $7.4 \cdot 10^{-6}$ at $k_e = 25$–200 ($\propto k_e^{-2}$ on the one-dimensional reference); DR's noise part at its bound $\|g\|_\rho^2$ to 0.01% at $k_e = 25$; $k_e^2\|P\mathbb{E}[e]\|_\rho^2$ 2394 for MR, 0 for DR and DR* by construction, $\|P\mathbb{E}[e]\|_\rho^2$ 3.83 → 0.66; the bias reaching the floor at the std 0.0012, 0.085, 0.27); the prior family at $\eta = \sigma_0$, $\kappa/\eta = 0$, 0.1, 1, 10 and fixed (the weakest mode's response 3269–136, its bias factor 0–0.959, $RH + B = I$ to $1.1 \cdot 10^{-14}$; MR's bias 2397–491, DR's 0.471–0.211) and five priors with non-commuting leaks, $R$ and $B$ from their dynamic blocks (the definitions with $S_0$ by elimination to $1.9 \cdot 10^{-15}$, $RH + B = I$ to $1.4 \cdot 10^{-14}$, $HR + B - I$ 0.47–0.79); the real loop (RK4 d$t = 10^{-3}$, $B = 100$, std 0.01, $\eta = \infty$, 12 runs of 300 time units, common noise paths): $\mathbb{E}\|\hat{f} - f\|^2$ at $t = 300$ $2.43 \cdot 10^{-6}$ (MR), $2.80 \cdot 10^{-7}$ (DR), $2.75 \cdot 10^{-7}$ (DR*), $z$-scores $-2.04$ to 0.82; the MR − DR run mean at $\hat{J}(\bar{y}_{\mathrm{MR}} - \bar{y}_{\mathrm{DR}})$ to 0.5–5.3%, 0.43–1.83 sampling errors | `exp_eiv_remedy.py` (`--only-AB` recomputes Parts A and B, `--only-B` Part B) → `results/chaos/eiv_remedy.md`; `tests/test_eiv_remedy.py` (the washout identity along a real-loop solution and across a plan switch of the averaged loop; the block's transfer function, $RH + B = I$ along the prior family and, from their dynamic blocks, for non-commuting leaks; the response bound under random leaks, the smallest equivalent perturbation and the projection; the lift exact under DR and DR*, DR's run within the second order of MR's on a common path, the regressor's point; DR's and DR*'s noise forces least-squares with their pointwise bounds, MR's $F_1$ three orders of magnitude above; the quasi-static forces against the full second-order mean) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.32 | the three-bump instance ($\sigma_0 = 0.15\sigma_{\text{crit}}$), the averaged loop: (A) the lift of $\vartheta^e(w)$ at the 15 count vectors of $m = 15$: the largest drift $2.8 \cdot 10^{-17}$ (compact, $R = 1$) and $6.2 \cdot 10^{-10}$ (Gaussian), the spectrum (iii)'s to $2.8 \cdot 10^{-11}$ (the finite differences' error) and $4.6 \cdot 10^{-9}$ (the Gaussian kernel's cross-class terms) (the separations $1.41 \cdot 10^{-3}$–$3.77 \cdot 10^{-3}$, the slowest collective mode $2.2 \cdot 10^{-4}$–$4.2 \cdot 10^{-4}$); the Gaussian kernel's bias $2.1 \cdot 10^{-8}$, $1.3 \cdot 10^{-3}$, 0.068 (whitened) at $R = 1$, 1.5, 2, its first order to $6 \cdot 10^{-8}$–1.0% (56% off at $R = 3$); with the fourth bump the margins $\sigma_0 w_k - (q_k/w_k)\lambda_k^+$ the Jacobian's to $2 \cdot 10^{-11}$ at three count vectors, the class with the margin $-1.22 \cdot 10^{-4}$ at $\sigma_0/8$ splitting ($3 \to 4$ classes, the fit error $0.017 \to 0.0148$); twins merging under the Gaussian kernel, persisting under the compact one; the dead class joining a live one at $R = 2$ and 1.5 ($s \approx 2.8 \cdot 10^4$ against (v)'s $1.8 \cdot 10^4$), not moving at $R = 1$ (time scale $1.4 \cdot 10^7$); (I) the twins' contraction 1.2949 against (v)'s 1.2943; far twins (5.28 and 8.19 apart) not moving in $4 \cdot 10^4$ slow units; opposite-sign twins with a third class within reach not merging in $4 \cdot 10^5$ (the fit error $3.2 \cdot 10^{-4} \to 4.5 \cdot 10^{-3}$); (B) warm clouds, $m = 15$, 60, 240 against $M = 510$: $\|\hat{f}_m - \hat{f}_M\|^2$ at $s = 6000$ $1.9 \cdot 10^{-10}$–$5.4 \cdot 10^{-8}$ ($m = 15$, the representable samples), $2.4 \cdot 10^{-10}$–$7.0 \cdot 10^{-10}$, $5.1 \cdot 10^{-14}$–$2.2 \cdot 10^{-9}$, as under the adaptive prior, against the fixed prior's floor $3.0 \cdot 10^{-4}$–$2.4 \cdot 10^{-3}$, $1.1 \cdot 10^{-4}$–$9.9 \cdot 10^{-4}$, $4.9 \cdot 10^{-6}$–$8.2 \cdot 10^{-5}$; seed 3's sample $(8, 4, 3)$ on the boundary of $W$ at $7.0 \cdot 10^{-5}$–$7.1 \cdot 10^{-5}$ under every leak but the fixed prior; (C) broad starts: the annealed Gaussian kernel at three classes in 9 of 9 runs, the fit error $1.8 \cdot 10^{-7}$–$2.3 \cdot 10^{-6}$ (the adaptive prior told $K_0 = 3$: $4.2 \cdot 10^{-9}$–$2.1 \cdot 10^{-6}$, $K_0 = 2$: 0.13–0.19, $K_0 = 4$: $5.4 \cdot 10^{-4}$–$5.6 \cdot 10^{-3}$; the $\sigma$-modification 0.24; no leak 40–48 classes, the compact kernel without the broad stage 17–22); (J) at a fixed Gaussian scale 4–6 classes ($R = 1$, $4.8 \cdot 10^{-4}$–$1.6 \cdot 10^{-3}$) or three biased ($R = 2$, $7.8 \cdot 10^{-4}$–$1.4 \cdot 10^{-3}$), the schedules $1.5 \to 1$, $2 \to 0.5$, $3 \to 1$ at three classes ($2.4 \cdot 10^{-8}$–$5.9 \cdot 10^{-6}$); (D) the real loop, $k_e = 25$: the lift's return $7.8 \cdot 10^{-16}$, its 45 slow exponents (36 separations at $\sigma_0 w_k/k_e$) to $5.5 \cdot 10^{-6}$ (median $1.5 \cdot 10^{-7}$), the rms tracking error at $t \approx 2900$ $5.8 \cdot 10^{-4}$ and $2.4 \cdot 10^{-4}$ ($m = 15$, 60) against the fixed prior's $1.3 \cdot 10^{-3}$ and $3.9 \cdot 10^{-4}$; (E) the dither $\tau = 10^{-5}$: the within-class spread over the second half 0.068 (compact) and 0.063 (Gaussian) against $\sqrt{(1 - 1/n_k)\tau/(\sigma_0 w_k)} = 0.0635$, 0.036 under the fixed prior (0.0366), 0.44–0.56 without a leak; (F) the cogging servo: 7–8 classes, the fit error 0.14%–0.35% (the four-kernel designs 1.46% and 0.64%, the best equal-weight six- and eight-kernel fits $9.4 \cdot 10^{-6}$ and $3.3 \cdot 10^{-7}$); (G) $\sigma$ halved from $\sigma_0$ to $\sigma_0/64$ (the fourth bump, $m = 30$, three seeds): no split down to $\sigma_0/8$ (the predicted threshold $0.121\sigma_0$), the middle class splitting at $\sigma_0/16$ in every seed; no class with a positive margin split, all 8 with | `exp_cohesion_leak.py` (`--only A,...,L` recomputes parts, `--merge` merges parts computed elsewhere) → `results/chaos/cohesion_leak.md`; `tests/test_cohesion_leak.py` (the drift $-G\nabla J_m^{\text{coh}}$ and $J_m^{\text{coh}}$ decreasing; the forces internal and the mean-shift identity; the exact representation stationary with the predicted spectrum at two count vectors; the splitting margin at the best one-atom fit; the twins' contraction against (v)'s rate to 1% under the Gaussian kernel and their persistence beyond the compact range; the kernels' range and scale; the clusters of known configurations; the real loop's lift periodic with the separations' exact contraction) |

*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.33 | the instance of D‴.30 ($k_e = 25$, $B = 100$, OU noise): the separations' drift at the lift per class — MR $[-3.7 \cdot 10^{-1}, +1.5 \cdot 10^{-2}]$; $[-2.6 \cdot 10^{-2}, +1.0 \cdot 10^{-2}]$; $[-3.5 \cdot 10^{-2}, +3.0 \cdot 10^{-1}]$, DR $[-1.1 \cdot 10^{-2}, -5.0 \cdot 10^{-4}]$; $[-1.5 \cdot 10^{-2}, -1.5 \cdot 10^{-4}]$; $[-7.2 \cdot 10^{-3}, -3.4 \cdot 10^{-4}]$, DR* $[+1.6 \cdot 10^{-4}, +3.8 \cdot 10^{-3}]$; $[+5.1 \cdot 10^{-5}, +5.0 \cdot 10^{-3}]$; $[+1.1 \cdot 10^{-4}, +2.4 \cdot 10^{-3}]$ —, the threshold std 0.031, —, 0.238 (adaptive prior) and 0.018, —, 0.138 (cohesion leak); against $k_e$ at $B = 4k_e$ (25–800) MR's $\lambda_{\max}$ 0.2982 → 0.3045, $k_e$ times DR's −0.0038 and DR*'s 0.1245–0.1242, at $B = 100$ DR's threshold std 0.185, 0.138, 0.121 ($k_e = 200$, 400, 800), DR*'s 0.238 → 0.113, MR's 0.031 → 0.014, $\|S_k\| \le 0.37$ of (ii)'s bound, the floor 0.069; six targets at the lift (DR's centre entries negative in all, the whole drift indefinite for same-sign, wide, edge and close bumps, up to $+5.6 \cdot 10^{-4}$); the stationary point from the exact periodic mean — rms $\mathbb{E}[e]$ $6.70 \cdot 10^{-4}$ (DR), $6.44 \cdot 10^{-4}$ (DR*), the parameters' part $6.32 \cdot 10^{-4}$, $6.27 \cdot 10^{-4}$, the drift indefinite ($[-4.35, +3.68] \cdot 10^{-5}$ in class 1), under the fixed prior DR's negative definite and DR*'s positive definite (the threshold std 0.435); DR's transient over $5.2 \cdot 10^5$ time units (the largest eigenvalue first positive at $t = 4103$, 2947, 2714, at most $+1.05 \cdot 10^{-4}$), the cross term −8% (DR) and −17% (DR*) of the class-2 window drift over $[0, 50)$, DR*'s threshold std there 0.277; the real loop (four antithetic pairs per family, 300 time units): MR $+4.69 \cdot 10^{-4}$, $+1.25 \cdot 10^{-3}$ (std 0.05, 0.1; all runs growing), DR along $u$ $-2.9651 \cdot 10^{-4}$, $-3.3650 \cdot 10^{-4}$ (the noise part 0.98, 0.97 of the transient's prediction), along DR's top eigenvectors faster than $\sigma_0/k_e$ by 0.36%, 0.14%, 0.26% (std 0.05), DR* along its top eigenvector slower by 3.2% and 12.6% (0.97, 0.96 of the prediction) | `exp_eiv_remedy.py` (`--only-Dpred` recomputes Part D's predictions, `--only-D2` also the runs along the laws' own top eigenvectors) → `results/chaos/eiv_remedy.md`; `tests/test_eiv_remedy.py` (DR's drift MR's mean-error term alone at the lift, negative definite in every class, MR's 0.30 in class 3; DR*'s $O(1/k_e)$ and against an independent quadrature to $10^{-4}$; at the stationary point the exact periodic mean's two parts, the parameters' part dominant, the drift indefinite; the period maps against `second_order_transient` to $10^{-7}$ and DR's drift turning indefinite) |

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*Table 47, continued*

| statement | what is checked | where |
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| D‴.34 | the instances of D‴.32: (A) fixed positions (the five bumps' atoms, the masses (0.15, 0.30, 0.20, 0.25, 0.10)): the fit error at $s = 4 \cdot 10^4$ 0.388 ($\gamma = 0$; the least error at those masses 0.387), $2.9 \cdot 10^{-4}$ ($\gamma = 10^{-3}$), $3.0 \cdot 10^{-14}$ ($\gamma = 10^{-2}$); the amplitudes scaled by 1.25 ($\sum\lvert A_k\rvert/a_{\max} = 1.094$): 0.071 at both rates (the $\ell^1$ ball's least error 0.0684), every $\lvert a\rvert/a_{\max} \geq 0.992$; (B) the five bumps from broad starts, $m = 60$: $\gamma = 0$ 6–9 classes, 0.045–0.118; $\gamma = 10^{-3}$ 5–7, 0.0076–0.0139; $\gamma = 10^{-2}$ five in 3 of 3, the centres the bumps' to 0.01, $6.0 \cdot 10^{-4}$–$1.4 \cdot 10^{-3}$ (falling by 2–4% per 1000 slow units; seed 1 continued: $2.80 \cdot 10^{-4}$ after $10^5$ more under the Gaussian kernel, $2.85 \cdot 10^{-4}$ after $4 \cdot 10^4$ under the compact one); (D) the instance, $\gamma = 10^{-2}$: three classes, $2.4 \cdot 10^{-7}$–$9.7 \cdot 10^{-7}$ (without, seeds 0–2, $1.9 \cdot 10^{-7}$–$9.0 \cdot 10^{-7}$); (C) $\sigma_0/64$ from the lift of $\vartheta^b(q)$: 8–10 classes with and without the replicator (0.0055–0.0075 against 0.0059–0.0078); (F) three heavy bumps: three classes in 3 of 3, $7.8 \cdot 10^{-8}$–$1.6 \cdot 10^{-7}$; two close bumps: 1–3 classes, 0.068–0.13; (G) the class-level stationary points at $R = 1$: B's seed 1 under the Gaussian kernel $3.85 \cdot 10^{-4}$ (masses held) and $2.63 \cdot 10^{-4}$ ($u_k = \lambda$), under the compact one exact ($2.4 \cdot 10^{-16}$); three equal bumps $9.39 \cdot 10^{-5}$ and $9.17 \cdot 10^{-5}$, the masses moving toward (0.567, 0.217, 0.217), exact under the compact kernel; (E) the real loop at the instance's atoms ($k_e = 25$, $\gamma = 2$): $V$ 0.0418 $\to 2.1 \cdot 10^{-7}$ monotone, $\dot V + k_e e^2 \leq 1.1 \cdot 10^{-16} \max k_e e^2$, $\sum\beta - m \leq 2.4 \cdot 10^{-14}$; the averaged linearisation at the lift of $\vartheta^e(w)$, $(6, 5, 4)$: 15 eigenvalues below $10^{-9}$, the others in $[-0.1834, -3.08 \cdot 10^{-4}]$, at $\gamma = 0$ D‴.32's spectrum to $2.1 \cdot 10^{-11}$ | `exp_mass_transfer.py` (`--only A,...,G`) → `results/chaos/mass_transfer.md`; `tests/test_mass_transfer.py` (the gradient-flow identity by central differences of $J$, $\gamma = 0$ Proposition D‴.32's drift; the weights' drift against an independent quadrature of the fitness; the real loop's relative-entropy $\dot V = -k_e e^2$ pointwise from the vector field and $\sum\beta$ conserved; at fixed positions $\sum\beta = m$ kept, the exact fit reached with the replicator and not without it, and above the $\ell^1$ ball's least error when $\sum\lvert A_k\rvert/a_{\max} > 1$; at the lift the zero modes the predicted vectors — the within-class redistributions and one gauge per class —, a separation at $-\sigma_0 w_k$ with $\gamma > 0$, the rest D‴.32's spectrum) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.35 | Proposition D‴.32's resolution instance (the lift of $\vartheta^b(q)$, $(10, 10, 10)$, compact $R = 1$): (A) $\lambda_k^+ = 1.00 \cdot 10^{-4}$, $2.86 \cdot 10^{-4}$, $1.08 \cdot 10^{-4}$, $c_3 = 1.42 \cdot 10^{-5}$, $-1.257 \cdot 10^{-3}$, $1.20 \cdot 10^{-4}$, the middle class first at $0.1213\sigma_0$; (B) the split states $q = 1$–$4$ by Newton within 1.3%–15% ($\mu = 2 \cdot 10^{-6}$) and 0.3%–3.0% ($5 \cdot 10^{-7}$) of the prediction, the eigenvalues on the separations along $v$ $+1.2511\mu$ ($\times 8$) and $-1.023\mu$ for $q = 1$ at $5 \cdot 10^{-7}$ against $+1.25\mu$ and $-\mu$, 8, 7, 6, 5 unstable directions; (C) six seeds at $\mu = 5 \cdot 10^{-5}$ and at $\sigma_0/16$: the first excursion's particle the quadratic equation's in 12 of 12 (linear 6, flipped $c_3$ 2), its time to 1.1% (linear 22.4%), on the side $\operatorname{sgn}(c_3)v$ in 10 of 12 (the other two linear growth below $2\mu/\lvert c_3\rvert$); beyond the range at $\sigma_0/16$ a single particle in 3 of 6 and a bound pair in 3 of 6 (the top two within 1.2%–11%), the main piece's margin positive at the end; (D) Part G's schedule, seeds 0–2: 8–10 classes (the cohesion leak), 6–7 (replicator $\gamma = 0.1$, the first escapee's mass 0.105–0.106); (E) the replicator at $\gamma = 1$: 5, 6, 8 classes, the first escapee's mass 0.134–0.136, the fit error 0.0029–0.0030; (F) the mass-free leak continued to $\sigma_0/256$: 4 classes at $\sigma_0/64$ in 3 of 3 (the fit error 0.0149), then 9, 10, 9 at $\sigma_0/128$ and 9, 8, 9 at $\sigma_0/256$, the fit error 0.0028 — the cohesion leak's end state | `exp_split_bifurcation.py` (`--only A,...,F`) → `results/chaos/split_bifurcation.md`; `tests/test_split_bifurcation.py` (the full drift along $v$ against the reduced quadratic equation with $c_3$, the remainder cubic; the single-particle split state a saddle with $n - 2$ directions at $+\mu n/(n-2)$ and the radial one at $-\mu$; the predicted particle leaving first on the side $\operatorname{sgn}(c_3)v$, at $\mu = 10^{-4}$ with the other classes' margins positive; the reduced crossing against the linear closed form and an independent integration; the mass-free leak's blocks) |
| D‴.36 | Proposition D‴.32's three-bump instance, one particle per class at $\vartheta^e(1/3)$, $\tau = 10^{-4}$: (A) the predicted $D_{kk} = 2.85 \cdot 10^{-8}$, $1.63 \cdot 10^{-8}$, $1.81 \cdot 10^{-8}$ at $\gamma = 10^{-2}$ ($4.85 \cdot 10^{-10}$ to $4.23 \cdot 10^{-7}$ for the first class over $\gamma = 10^{-3}$–$10$); (B) 64 seeds, $5 \cdot 10^4$ slow units: the variance's slopes $(2.69 \pm 0.58, 2.21 \pm 0.47, 1.62 \pm 0.38) \cdot 10^{-8}$, the ratios 0.95, 1.36, 0.90; the means $0.3212 \pm 0.0046$, $0.3445 \pm 0.0041$, $0.3342 \pm 0.0038$ (the drift term); (C) 10 seeds, $2 \cdot 10^5$: the spread 0.057, 0.053, 0.052 ($\sqrt{Ds}$ 0.065, 0.049, 0.052), $\alpha_1$ up to 16.1 ($\lvert a\rvert/a_{\max} = 0.9994$, against 0.80 at $\gamma = 0$), the fit error the dither's (0.0253 against 0.0269); (D) the leak at $\varepsilon = 10^{-4}$, $10^{-3}$: the spreads 0.0152, 0.0103, 0.0117 and 0.0044, 0.0032, 0.0034 against the predicted 0.0139, 0.0099, 0.0105 and 0.0044, 0.0031, 0.0033, the means within 1.5 standard errors of 1/3 (4 seeds; the 16-seed test in D‴.37 (G)) | `exp_gauge_dither.py` → `results/chaos/gauge_dither.md`; `tests/test_gauge_dither.py` (the entropic leak a Fisher–Rao gradient; the two-class diffusion's closed form; the leak keeping the exact representation stationary and making the gauge stable; the leak's spread against the two-class Ornstein–Uhlenbeck closed form; $\gamma = 0$ leaving the weights at 1; $D$ against the projector of the code's own drift Jacobian; `run_dithered`'s noise $2\tau G\,\mathrm{d}t$) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.37 | Proposition D‴.36's instance, $\tau = 10^{-4}$: (A) the identities against the integrated limit map to $2.2 \cdot 10^{-5}$ and $1.1 \cdot 10^{-4}$; (B) at $W = 1/3$ the amplitudes' noise drains each class at 5.34, 3.47, $4.07 \cdot 10^{-7}$, the centres' and widths' at most $2.2 \cdot 10^{-10}$; (C) the drift over $\gamma\tau$ at $\gamma = 10^{-5}$ against $-\frac{1}{2} W_k(\partial_k \ell - \sum_l W_l \partial_l \ell)$ to 0.3% at four points; (D) the layer's zero at 0.66, 0.61, 0.59, 0.57, 0.57 $\delta^*$ for $\gamma = 10^{-2}$–$10^{-4}$, inside the drift $\tau G_\alpha \partial_\alpha^2 W_1$ to 0.6%; (E) at $\gamma = 10^{-2}$ a repelling node, three saddles, three attracting nodes $(0.284, 0.205, 0.511)$, $(0.284, 0.484, 0.232)$, $(0.562, 0.206, 0.232)$; (F) the 64-seed mean on the mean path within its errors; (G) 16 seeds: the leak's shift within 1–2 errors of $\kappa^{-1} b$ at $\varepsilon = 3\tau$, $10\tau$, 1.8 times it at $\varepsilon = \tau$, the second component above it at every $\varepsilon$; (H) 12 seeds over $10^6$: near-saturation fractions 0.50, 0.25, 0.33, the largest $\lvert\alpha_1\rvert$ by quarters 6.6, 10.1, 10.9, 11.3, 11–12% of the last quarter beyond the saturation | `exp_gauge_drift.py` → `results/chaos/gauge_drift.md`; `src/core/gauge_drift.py`; `tests/test_gauge_drift.py` (closed-form toy limit maps; an integrated limit map in a non-orthogonal metric; the instance's diffusion against D‴.36's projection; the slow limit against the log-determinant law; the log-determinant's independence of the split within a class) |

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*Table 47, continued*

| statement | what is checked | where |
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| D‴.38 | Proposition D‴.36's instance, $\tau = 10^{-4}$: (A) the corrected drift over $\gamma\tau$ at three points for $\gamma = 2 \cdot 10^{-2}$–$10^{-4}$ (at $W = 1/3$: $+0.063$, $-0.034$, $-0.029$ at $10^{-2}$, $-0.0007$, $+0.0006$, $+0.0001$ at $10^{-3}$, against the uncorrected $-0.206$, $+0.158$, $+0.048$ and $-0.350$, $+0.236$, $+0.115$; reversed from $\gamma = 2 \cdot 10^{-3}$ on); (B) $W^\sharp = (0.37652, 0.29876, 0.32472)$, D-optimal, the condition number 649.9 there against 648.6 at its minimiser along $M$; $\kappa_F = 2$'s zero within $1.8 \cdot 10^{-4}$, $4.4 \cdot 10^{-5}$, $5.4 \cdot 10^{-6}$ of it at $\gamma = 10^{-2}$, $10^{-3}$, $10^{-4}$, attracting at the rates 5.7–6.8$\gamma\tau$, the linearised stationary spread 0.040, 0.014, 0.0046; $\kappa_F = 1$'s zero attracting at $10^{-2}$ $(0.379, 0.297, 0.324)$, repelling at $10^{-3}$ and $10^{-4}$; (C) the exponents $-0.99$, $-0.00$, $+0.98$ in Katzenberger's limit; (H) across the floor's cap $\delta_{\text{cap}} = 0.00126$ the exponent at $\kappa_F = 2$ from $+0.99$ to $-0.22$ (unfloored $+0.99$ to $+1.15$); (D, E) 12 seeds $\times$ $10^6$ slow units at $\gamma = 10^{-2}$: the seeds' mean largest $\lvert\alpha_1\rvert$ in the last quarter 11.3 without the correction, 4.5 ($\kappa_F = 1$) and 4.1 ($\kappa_F = 2$) with it in either form, no record beyond a saturation in the last quarter with it, the second half's mean at $\kappa_F = 2$ within 1.07 standard errors of $W^\sharp$; (F) the leak at $\gamma = 3 \cdot 10^{-3}$, $\varepsilon = 10^{-4}$: the escapes in the same two of 16 seeds in every form, the masses 0.31–0.35 (masses form) against 0.21–0.46 (plug-in), the shift reversed ($+0.00145$, $-0.00083$, $-0.00061$ against the first order $+0.00111$, $-0.00073$, $-0.00037$ and the second $+0.00103$, $-0.00069$, $-0.00034$); (G) the leak at $\gamma = 10^{-2}$, $\varepsilon = 3 \cdot 10^{-4}$ (16 seeds, $\kappa_F = 0$ bit for bit Proposition D‴.37's Part G): the shift $(-0.00129, +0.00112, +0.00017)$, $(+0.00001, +0.00026, -0.00027)$, $(+0.00120, -0.00054, -0.00066)$ at $\kappa_F = 0$, 1, 2 against the second order $(-0.00134, +0.00089, +0.00045)$, $(-0.00006, +0.00004, +0.00001)$, $(+0.00112, -0.00075, -0.00038)$, within 1.03 s.e.; the paired differences within 0.93 s.e. of the second order; D‴.37's Part G at $\varepsilon = 10^{-4}$, $10^{-3}$ within 1.7 s.e. of it | `exp_gauge_fixman.py` → `results/chaos/gauge_fixman.md` (`--reanalyse` recomputes D–G from the stored records); `src/core/mass_transfer.py` (`fixman`, `fixman_tau`, `fixman_A`, `fixman_TA`, `fixman_floor`), `src/core/gauge_drift.py` (`F1`), `src/core/runcache.py` (the key with the code's version); `tests/test_gauge_fixman.py` (the correction as $(\tau/2)\partial_\beta \ell$ along $M$; zero without dither, $\sum \beta$ kept; the slow-limit cancellation; $\kappa_F = 2$'s zero at $W^\sharp$; the density exponents; the floor's cap and its exponent; the general form through the implicit function reducing to the closed form; the masses form equal to the plug-in form on $M$ and blind to an amplitude's excursion at fixed masses; the tracked contributions fixed on $M$ and relaxing at their time constant; `kappa()` as the leak's projected stiffness; the long runs' and the paired analysis' statistics on synthetic records), `tests/test_runcache.py` |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.39 | Proposition D‴.36's instance, $\tau = 10^{-4}$, the box $L = (5, 3, 4)$, $\sigma_B = 1$: (A) the plateau's level $7.72\tau$, $7.76\tau$ at $W = 1/3$ and $7.64\tau$, $7.71\tau$ at $W^\sharp$ (one amplitude at $\pm 20$), $7.69\tau$ and $7.67\tau$ on the escape boundary $\max\lvert\alpha_k\rvert = 6$, $7.66\tau$ and $7.63\tau$ at the box's edge; (B) the path at $W = 1/3$ monotone ($0 \to 7.71\tau$, the second width $0.40 \to 0.75$), at $W^\sharp$ a saddle at $8.21\tau$ toward the lowest plateau; (C) the escapes at $\tau = 1.5,\ 2,\ 3 \cdot 10^{-4}$ (10, 19, 24 of 24 within $4 \cdot 10^5$; the rates, the slope $6.31 \cdot 10^{-4}$ against $\Delta J = 7.69 \cdot 10^{-4}$, weighted by the escape counts $(0.81 \pm 0.14)\Delta J$); (D) the dead zone at $\tau = 2 \cdot 10^{-4}$ (the plateau's occupation by quarters with and without the box); (E) the full loop with the box at $\kappa_F = 0,\ 2$ (the masses against $P_B$'s vertices and $W^\sharp$; the records against the unboxed runs); (F) the two channels in the stored runs | `exp_gauge_plateau.py` → `results/chaos/gauge_plateau.md`; `src/core/mass_transfer.py` (`box`, `box_sigma`, `box_excess`); `tests/test_gauge_plateau.py` (the dead zone inactive inside the box and minus its $G$-gradient outside; $J$ bounded on the unconstrained coordinates and $J + V$ growing like $(\sigma_B/2)\,\mathrm{dist}^2$; the box's wall inside the saturation, $W^\sharp$ representable in it, the polytope's vertices; a dithered run inside the box bit for bit the unboxed one) |
| D‴.41 | Proposition D‴.36's instance, no dither, the measured error's noise of intensity $v$ in both laws: (A) at three points of $M$ and $\gamma = 10^{-2}$, $10^{-5}$ the Itô drift (Katzenberger, $\Sigma = v\langle UU^\top\rangle_\rho$) and the Wong–Zakai term; (A2) the noise's structure over the nodes at $\gamma = 10^{-1} \ldots 10^{-5}$ and the coloured-noise identity; (B) the targets ($W_\ell$, the $\gamma = 10^{-2}$ zero, the update in $a$); (C) the full loop under eleven schemes against the linear-response prediction averaged along the path, (C2) at the step 8; (D) $f = uv$'s exact rates to $\rho = 0.999$; (E) a diagonal network coordinate by coordinate; (F) a tanh network; (G) the selection's zero set against the correlation time (stability, basins), the slow limit against the dither, smooth against Markov noise; (C) 0.996–1.008 at $h = 0.25$, 0.979–1.049 at $h = 8$ ($\pm$ up to 0.05), (D) $\leq 0.3\%$, (E) 0.9988–0.9998, (F) 0.990–1.060 to $\rho = 0.9$; (G) three stable zeros from $\tau_c \approx 100$ | `exp_gauge_label.py` → `results/chaos/gauge_label.md`; `src/core/label_noise.py` (`fields`, `selection`, `psi`, `w_label`, `ar1_factor`, `corr_sums` (AR(1), block and Gaussian step correlations), `lr_factor`, `filtered_covariance`, `lr_potential`, `flat_fields`, `field_derivatives`, `wong_zakai`), `src/core/mass_transfer.py` (`drift(...,xi)`, `run_dithered`'s `label_v`, `label_scheme` (`'ito'`, `'ar1'`, `'heun'`, the last with AR(1) draws too), `label_rho`), `src/core/gauge_drift.py` (`limit_drift`'s Schur method, `spectral_projection`); `tests/test_gauge_label.py` (the noise entering along the fields; the fields orthogonal to the gauge in the flow's metric; the Wong–Zakai term cancelling the Itô drift; $W_\ell$ interior and minimising $\Psi_\ell$; the amplitude coordinates' target; the schemes' reductions; the AR(1) draws and Heun steps with correlated noise and with dither against hand-coded steps; the weighted and the covariance Wong–Zakai terms; the linear-response factor against the exact stationary moments, for AR(1) and block noise, and its continuum limits; the filtered covariance against the pairwise factors; Gaussian correlation sums; the finite-step slow limit) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.42 | Proposition D‴.36's instance with 2 and 3 particles per class: (A) $\log\det\nabla^2_\Theta J$ minus its one-per-class value, $-46.131219$ ($m=6$) and $-92.262438$ ($m=9$) at three class-mass points and three splits each; (B) Katzenberger's limit at $\gamma=10^{-5}$, $10^{-2}$ against $m=3$ (the ratios 0.5000 and 0.3333 in the slow limit, the diffusion's at every $\gamma$), $n=3$'s degenerate equal split included; (C) the correction per class at $\kappa_F=1$; (D) measurement noise at $m=6$ per component, the class Itô drift against $m=3$'s, the fractions' against the full trace; (E) the full loop; (F) anisotropic dither, $\gamma=10^{-3}\ldots 10^{-6}$; (A) to $4\cdot10^{-7}$; (B) 0.5000, 0.3333; (C) the class drift cancelled to $10^{-4}$; (D) 1.0000 at the equal split, 2.2–2.6% (up to 9.6% per component) at unequal ones; (E) the class masses at $m=3$'s law at half the speed, the fractions from an unequal split moving away from the equal one through the saturation; (F) 1.0000–1.0075 at the equal split, 1.4–3.5% off at unequal ones for the amplitudes-only mask | `exp_gauge_classes.py` → `results/chaos/gauge_classes.md`; `src/core/mass_transfer.py` (`fixman_classes`), `src/core/gauge_drift.py` (`limit_drift`'s Schur method at degenerate eigenvalues); `tests/test_gauge_classes.py` (the determinant blind to the split; the class masses at the law of $m$ and the fractions without diffusion; the correction per class and its tracked contributions per class; off $M$ the masses form against the plug-in one, each class its own $\tau_k$, the floor near a class's saturation; the degenerate equal split at $n=3$ and the two methods' agreement) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.43 | Kernel dictionaries and Gaussian designs, $w = u \odot u - v \odot v$ fitted with label noise: (A) the potential's identities (Cauchy–Binet, the gradient, concavity, the support count); (B) label-noise descent against the reduced flow on eight $10 \times 40$ instances where basis pursuit fails (16 paths, fresh and AR(1) draws at $\sigma = 0.1$, the AR(1) draws also at $\sigma = 0.3$, 0.5; the end interpolants); (C) recovery against sparsity, $10 \times 40$ and $20 \times 80$, the white flow run to convergence, $R$ at the slow flow's failures against the truth's; (D) Gaussian kernel dictionaries (two widths), recovery, and label-noise descent on three instances where basis pursuit fails, the reduced flow run to convergence; (E) other colours at a finite step, a smooth noise also at ten times the slow horizon; (F) the start's scale, the flows run to convergence; (G) sparse Bayesian learning (Tipping's EM), reweighted $\ell^1$ ($\varepsilon = 0.01$, 0.1, 1, from either start) and the log-det's majorisation–minimisation; (H) the dual certificates at the truth and at basis pursuit's solutions, on kernel dictionaries (widths 0.03–0.08) to $\tau_c = 10^5$ with the slow flow's return from $P = 10^{-3}$; (I) a fit floor ($y = \Phi w^* + \varepsilon e$); (J) Euler's step 0.5 and 0.125 against Heun's at the same physical noise; 40 instances per row (20 at $20 \times 80$) | `exp_colour_selection.py` → `results/chaos/colour_selection.md`; `src/core/colour_selection.py` (`gram`, `filter_of`, `weights`, `potential`, `det_subsets`, `project`, `rhs`, `reduced_flow`, `flow_to_convergence`, `uv_of`, `gd_interpolant`, `flow_interpolant`, `end_interpolants`, `basis_pursuit`, `label_noise_gd`, `label_noise_heun`, `sbl_em`, `rbf_dictionary`); `tests/test_colour_selection.py` (the subset expansion and the gradient, also with a finite step's filter; the white and slow potentials (finely sampled AR(1) draws tending to the slow one), $R$ linear (white) and concave (slow) along segments; the support count; the projection, singular cases included; the reduced flow on $M$, $P$'s rate, finite weights at a sparse fit, an overshooting trial stage rejected, white → basis pursuit, slow → the sparse truth; the flow as $R$'s gradient flow in its metric (a doubled or halved $w$-drift fails); the exact interpolant against small-step descent, the end interpolants (at infinity flagged, converging within the alive atoms' span); label-noise descent against hand-coded AR(1) steps; the dictionary; the $(u, v)$ round trip; the experiment's success threshold and dual certificate against hand computations; sparse Bayesian learning's EM on a 2-sparse truth; the white flow run to convergence; Heun's noiseless step on $M$) |

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*Table 47, continued*

| statement | what is checked | where |
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| D‴.45 | $5 \times 12$ Gaussian designs (2- and 3-sparse truths), $m = 12$, $G = 1$, $\gamma = 0.1$, $a_{\max} = 1$: (A) $\eta$'s closed form, the conjugate, the metric, $Q$ along the loop under a smooth node-correlated noise, the conserved law's shift integrated, $Q$'s drift along D‴.44's reduced flow; (B) the projection with its residual (failures counted) against the minimum-norm and basis-pursuit fits across the starting masses, the loop's end, its time at $s = 10^{-3}$, unequal masses' weighted limits, the branches' distributions (D‴.47); (C) Itô, Stratonovich and the mirror update against each other as $h \to 0$, the Euler step's exact mean change of $Q$, the selection from the lazy start where basis pursuit is and is not the truth, the Euler against the mirror update under sampled-data noise; (D) the real loop with two particles on one atom under no noise, an adaptation-only noise and a sensor noise; (E) the conserved law; (F) the rich limit's leading order from the certificate at $N$-atom vertices (unequal masses $\varepsilon r_i$, $\varepsilon = 10^{-5} \ldots 10^{-11}$), the closed form off the support, the small slacks; results: (A) $\eta$ to $1.5 \cdot 10^{-14}$, $Q$ to $1.4 \cdot 10^{-11}$ noiseless and $3.6 \cdot 10^{-10}$ under the noise, the shift to $1.5$–$4.4 \cdot 10^{-7}$, the drift to $9.1 \cdot 10^{-16}$; (B) the loop's end the projection to $4.7 \cdot 10^{-11}$ (no failure), its $\|w\|_1$ 1.37 times basis pursuit's at $s = 10$ and 1.0004 at $10^{-4}$ ($k = 2$), at $s = 10^{-3}$ within $10^{-8}$ from $t \approx 8.5 \cdot 10^6$, weighted basis pursuit to $1.7 \cdot 10^{-9}$ at $s = 10^{-7}$, the weighted minimum norm to $2.4$–$2.5 \cdot 10^{-9}$ at $s = 10^4$, at $s = 10^{-6}$ a truth atom on a wrong-signed branch in 13 and 18 of 20 instances; (C) the Itô ratios 1.0106, 1.0037, 1.0011 at $h = 0.1$, 0.03, 0.01, Heun's mean change $1.4 \cdot 10^{-5}$, $4.1 \cdot 10^{-6}$, $1.4 \cdot 10^{-6}$, the mirror update's $2.2 \cdot 10^{-16}$, the Euler step's closed form to 0.029–0.036% at $h = 0.005$, basis pursuit reached where it is the truth (6 instances), at $h = 0.5$ the projection moved by 0.030–0.043 (the mirror update: $9.0 \cdot 10^{-12}$); (D) $V$ monotone from 0.175 to 0.022, $\dot{V} = -k_e e^2$ to $3.0 \cdot 10^{-9}$, $Q$ to $7.6 \cdot 10^{-15}$, under noise $V$ not monotone and $Q$ kept to $6.8 \cdot 10^{-12}$ and $4.9 \cdot 10^{-13}$; (E) exact fits $1.6$–$4.1 \cdot 10^{-3}$ from the free law's projections; (F) the leading order within $1.8 \cdot 10^{-13}$ and $2.6 \cdot 10^{-11}$ at $\varepsilon = 10^{-7}$, the distance's factors 1068 and 984 from $10^{-7}$ to $10^{-9}$ ($\varepsilon^{3/2}$: 1000), $\mu_{\min} = 2.7 \cdot 10^{-4}$ still 0.17 away at $10^{-7}$ | `exp_mirror_free.py` → `results/chaos/mirror_free.md`; `src/core/mirror_free.py` (`FreeLaw`: `Q`, `beta`, `w`, `relevance`, `branch_point`, `eta`, `alpha_of_eta`, `psi`, `psi_star`, `bregman_div`, `alpha_of_w`, `rhs`, `bregman_projection`, `dQ_exact`, `dQ_euler`, `mirror_step`, `real_rhs`, `real_lyapunov`; `dual_newton`, `weighted_bp`); `tests/test_mirror_free.py` ($\eta$ against quadrature in every case and against a 60-digit reference; the inverse's round trips and range errors; the conjugate and the metric; $Q$ under a smooth node-correlated noise and the conserved law's shift; the noiseless limit as the projection; the rich projection as weighted basis pursuit; the Euler step's change of $Q$; $\dot{V} = -k_e e^2$ in the real loop, seeing the mass law; the mirror update on curves and branches) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.46 | (A) the Lyapunov identity with the leak, a disturbance and a measurement noise at random states; the averaged equilibrium against an $O(1)$ ridge (large masses and the saturation's wall) and the LASSO (small masses) on random $H$ at $k_e = 2$; (B) the real loop on a persistently exciting orbit under an approximation error and a biased disturbance against the filtered averaged limit, the comparison bound asserted, and a drift without excitation that the leak bounds; (C) the real loop's period means from the filtered equilibrium against it and the idealised one; (D) the regularisation paths on an orbit's Gram, the irrepresentable condition and the equilibrium's support over every 3-sparse pattern, the biased $\sigma \to 0$ limit, the rich start's time; results: (A) the identity to $9.5 \cdot 10^{-11}$ ($\sigma = 0$ and 0.3), the ridge ($\rho = 0.5$, moving $w$ by 31–55%) to $1.3 \cdot 10^{-8}$ and $1.3 \cdot 10^{-4}$ at $\beta^0 = 10^4$, $10^2$ and 0.15 off at $\beta^0 = 3$ ($w_\sqcap = 0.25$), the LASSO ($\sigma\lambda_1 = 0.02$) to 0.015, $1.5 \cdot 10^{-3}$, $1.5 \cdot 10^{-4}$ at $\beta^0 = 10^{-4}$, $10^{-6}$, $10^{-8}$ with its support in every draw; (B) without the leak the largest coefficient 0.309, 0.330, 0.337, 0.341 by quarters and $V$ 0.40 $\to$ 0.45, with $\sigma = 0.01$, 0.1 the coefficients at 0.081, 0.040 and $V$ at 0.085, 0.13 ($C/\lambda = 1.74$, 0.30), the late means within $2.5 \cdot 10^{-3}$ and $6.5 \cdot 10^{-4}$ of $w_\sigma^L$, the rms error 0.024 $\to$ 0.031, 0.037, without excitation 0.70, 2.31, 7.37, 23.3 at $t = 1$, $10^2$, $10^4$, $10^6$ against 0.46 at $10^6$ with the leak; (C) the filtered Gram 0.89% from the idealised one, the last period means within $1.6 \cdot 10^{-4}$ and $1.7 \cdot 10^{-4}$ of $w_\sigma^L$, 4.4 and 2.6 times closer than to $w_\sigma$ (the leak's bias 0.038, 0.12); (D) the rich start's equilibrium the LASSO to $3.7 \cdot 10^{-6}$–$6.5 \cdot 10^{-4}$ with exactly the truth's three atoms for $\mu \leq 10^{-2}$, the lazy start's the ridge to $3.9 \cdot 10^{-6}$–$2.5 \cdot 10^{-4}$, dense, the support exactly the truth's for 415 of 1760 patterns, the prediction for 416, agreement on 1745, the biased limit linear in $\sigma$ ($3.7 \cdot 10^{-8}$ at $\sigma = 10^{-8}$), the rich start's time $3.2 \cdot 10^5$ against $2 \cdot 10^6$ (quantity 1.22, five atoms) | `exp_mirror_leak.py` → `results/chaos/mirror_leak.md`; `src/core/mirror_free.py` (`real_rhs_leak`, `leak_equilibrium`, `filtered_gram`, `lasso`, `irrepresentable`); `tests/test_mirror_free.py` (the identity with the leak, a disturbance and a noise; $Q$ kept by the leak; the ridge, LASSO and biased limits at $k_e \neq 1$ with the solver's residual; the comparison bound under a worst-case disturbance; the filtered Gram's expansion) |

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| statement | what is checked | where |
|---|---|---|
| D‴.47 | (A) the slope $\mathrm{d}\eta/\,\mathrm{d}\log\lvert w\rvert$, the temperature at branch points, $B(\bar{w}\to 0, w)=\beta/(\gamma m)$, the fraction of small starts on branches, a $\pm$ pair against the hyperbolic entropy, the critical curve's power law; (B) the flow against the Bregman projection onto the sign-constrained fits on random designs (positive, random and mixed signs), the extinction rates; (C) the rich limits (branches to the sign-constrained weighted basis pursuit, $\alpha=0$ to basis pursuit, a mixed start) against the leading order from the certificates, down to $\varepsilon=10^{-128}$ (a decimal-arithmetic Newton step where the float curvatures underflow); (D) pairs per atom, to $u=10^{-256}$; (E) positive Gaussian dictionaries with few samples: the nonnegative certificate's margin, basis pursuit, the loop at six starts to $t=10^{9}$; (F) the real loop on positive branches; results: (A) the slope to $2.3\cdot 10^{-9}$, the temperature to ten digits, $B(\bar{w}, w)\to\beta/(\gamma m)$ ($5\cdot 10^{-10}$ at $\bar{\delta}=10^{-12}$), the fraction on branches 0.52, 0.95, 0.995, 0.9996 at $\beta(0)=10^{-3}$, $10^{-5}$, $10^{-7}$, $10^{-9}$, the pair to $2\cdot 10^{-5}$ for $\Delta\le 10$, $\eta\cdot w^{1/3}\to -3.218$; (B) the projection to $5.0\cdot 10^{-10}$, inexact fits on 5 of 8 positive and 2 of 8 random starts, the extinct masses below $10^{-103}$, the rates to 0.5–10%, with duplicated atoms the projection to $6.4\cdot 10^{-10}$, $1.6\cdot 10^{-4}$–0.20 from the equal split; (C) from branches 0.23 … $1.1\cdot 10^{-9}$ at $\varepsilon=10^{-4}$ … $10^{-64}$, the leading order to $3.4\cdot 10^{-14}$ ($\mu_{\min}=0.104$), from $\alpha(0)=0$ 0.29 … $4.3\cdot 10^{-11}$ at $\varepsilon=10^{-2}$ … $10^{-10}$, the mixed start's free particles $\approx\varepsilon^{2}$, or 2.27 where the branch atoms cannot fit, the projection failing there at $10^{-12}$ (residual 1.95), the decimal fallback in 37–55 steps; (D) 0.29 at $u=10^{-4}$ to $1.9\cdot 10^{-6}$ at $10^{-256}$ (slacks 0.016, 0.020), never both members of an atom active; (E) margins 0.015, 0.003, 0.027: the target from $\beta(0)=10^{-6}$ to 10 within $6.0\cdot 10^{-5}$ at $t=10^{9}$, 1.0–$1.3\cdot 10^{-3}$ away, on its atoms within $2.3\cdot 10^{-5}$, from $\alpha(0)=0$ a seven-atom fit 1 away (rich) or all 21 atoms (lazy), the quantiles' supports reached where the margin exceeds $10^{-3}$, 0.37–0.83 away below $10^{-4}$; (F) $\dot{V}=-k_e e^{2}$ to $1.7\cdot 10^{-8}$, $V$ 3.64, 2.67, 1.72, 0.87 by quarters, the unused atoms' largest mass 0.0055 → 0.78, 0.080 from $w^{*}$ at the end | `exp_mirror_branch.py` → `results/chaos/mirror_branch.md`; `src/core/mirror_free.py` (`ParticleSet`, `sign_ls_set`, `projection`, `dual_newton`'s decimal fallback, `certificate_slacks`, `rich_leading_order`, `weighted_bp`, `nonneg_certificate`, `logsinh`, `FreeLaw._eta_branch_log`); `tests/test_mirror_free.py` (the branch identities, the temperature, $B(0, w)=\beta/(\gamma m)$, the pair, the limit theorem with extinction, the fallback at $\varepsilon=10^{-32}$ against the leading order) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| D‴.48 | (A) from $\alpha = 0$: the adaptive inverse scale space path against the loop in $t/\Lambda(\varepsilon)$, $\varepsilon = 10^{-2} \dots 10^{-12}$, three designs ($\sup\lvert u - u_{\mathrm{ISS}}\rvert$, the residual, the entry times, the end); (B) on branches and $\pm$ pairs: the one- and two-sided paths against the loop in $t/\log(1/\varepsilon)$, $\varepsilon = 10^{-4} \dots 10^{-64}$; (C) moving centres: the sliding path against the loop ($\varepsilon = 10^{-3} \dots 10^{-8}$), the Wasserstein–Fisher–Rao metric's condensation against its limit, fixed centres, sixteen random targets, close and wider bumps, a lost near-tie's creep; (D) moving widths: bumps of widths 0.2 and 0.08 from kernels of width 0.12 ($\varepsilon = 10^{-5}$, $10^{-6}$) against fixed widths; results: (A) the paths' ends weighted basis pursuit to $5 \cdot 10^{-16}$, $u$'s largest distance to the path 0.54, 0.61, 1.15 at $\varepsilon = 10^{-2}$ and $6.7 \cdot 10^{-6}$, $8.8 \cdot 10^{-6}$, $2.0 \cdot 10^{-5}$ at $10^{-12}$ (slopes 0.500 between $10^{-6}$ and $10^{-8}$), the residual's difference $\propto \varepsilon^{3/2}$ away from the events, the end to $1.1 \cdot 10^{-16}$–$6.1 \cdot 10^{-15}$ at $10^{-12}$; (B) 15.7, 8.0, 4.0, 2.0, 1.0 at $\varepsilon = 10^{-4} \dots 10^{-64}$, the pairs 21, 7.9, 4.0, 2.0, 0.99; (C) the sliding path exact with three kernels, the loop 8, 4, then exactly three kernels from $\varepsilon = 10^{-5}$ on, the others displaced by at most 0.033, $2.6 \cdot 10^{-3}$, $2.3 \cdot 10^{-4}$, $2.3 \cdot 10^{-5}$, the Wasserstein–Fisher–Rao limit to $1.7 \cdot 10^{-5}$, $3.4 \cdot 10^{-7}$, $2 \cdot 10^{-7}$ and 15, 12, 8, 3, 3 kernels, the centres held on the grid 5 kernels, 0.019 from the target, the random targets on 11, 13, 16 of 16 at $\varepsilon = 10^{-5}$, $10^{-6}$, $10^{-8}$ (Wasserstein–Fisher–Rao: none, 10 and all sixteen at $\varepsilon = 10^{-6}$, $10^{-8}$, $10^{-10}$), the tie (0.4, 0.4), the near tie by two kernels at $10^{-6}$, close bumps' second entry earlier than on the path (16.2 against 33.5 at $10^{-6}$), wider bumps' 21 entries from $\tau = 9.1$ to $1.2 \cdot 10^{10}$, the creep's speed the profile's to 0.990–0.996 and 0.9999–1.0000, kernel 15 entering in place of kernel 12 at $\varepsilon = 10^{-8}$; (D) exactly three kernels at $\varepsilon = 10^{-6}$, with fixed widths six kernels (rms residual 0.024) or the amplitudes 0.71, −0.48, 0.59 (0.094) | `exp_mirror_transit.py` → `results/chaos/mirror_transit.md`; `src/core/rich_transit.py` (`Bumps`, `moving_rhs`, `critical_rhs`, `sliding_iss`, `wfr_inactive_rhs`, `transit_scale`), `src/core/mirror_free.py` (`iss_path`, `iss_eval`); `tests/test_rich_transit.py`, `tests/test_mirror_free.py` (the path and the transit at $\varepsilon = 10^{-6}$) |
| F(ii) along (L) | particle path length ($\geq$ Wasserstein length) $\leq \lambda_G^{1/2} L_1 \int \lVert e \rVert$ at every sample | `T::test_theorem_F_wasserstein_length_bounded_by_integrated_error` |
| F(ii′) along ($\mathrm{L}_\sigma$) | $W(\mu_{t_1}, \mu_{t_0}) \leq \int (\lambda_G^{1/2} L_1 \lVert e \rVert + \sigma_m W(\mu_r, \mu^0))$ on a grid of pairs; a tight instance with $e \equiv 0$ (ratio $\geq 0.99$), where the motion is $W \approx 1.05$ | `T::test_theorem_F_leak_clause_along_ot_sigma_mod`, `T::test_theorem_F_leak_clause_is_tight_when_the_leak_pays_for_all_motion` |
| E linearisation | Jacobian of the implemented loop equals $A(t)$; $QA + A^\top Q = \mathrm{diag}(-k_e I, 0)$ | `T::test_theorem_E_linearisation_and_lyapunov_identity` |
| E convergence | the matched error decays and the labelled error does not (qualitative); sliding-window excitation and fitted rates | `T::test_theorem_E_matched_error_converges_labelled_does_not`, `exp_pe_convergence.py` |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| Lemma E.0 | $p_{\mathrm{lo}}\|w\|^2 \le V \le p_{\mathrm{hi}}\|w\|^2$ and $\dot V \le -\gamma\|w\|^2$ at random $(t, w)$ for four regressors, closed-form and optimised $\varepsilon$; the envelope along trajectories; explicit rate $\le$ exact Floquet rate | `Ex::test_lemma_E0_pointwise_inequalities_hold_on_random_states`, `Ex::test_lemma_E0_envelope_and_floquet_ordering` |
| Lemma E.0(iii), Remark E.0a | the $\omega^{-2}$ law of the explicit rate and of the exact rate; the saturation of exact/explicit in the slow-adaptation limit; the tables | `Ex::test_remark_E1_omega_scaling_of_the_explicit_rate`, `Ex::test_slow_adaptation_limit_ratio_is_bounded`, `exp_explicit_rate.py` → `results/explicit_rate/table.md` |
| E, explicit constants | on the implemented loop, from a start inside the explicit ball, $\|w(t)\| \le (p_{\mathrm{hi}}/p_{\mathrm{lo}})^{1/2} e^{-\lambda t}\|w(0)\|$ at every sample ($L_J$ estimated from samples) | `Ex::test_theorem_E_explicit_nonlinear_envelope_on_the_implemented_loop` |
| Corollary E.0′.2 | the explicit envelope of Theorem E with Lemma E.0′'s constants on the implemented single-kernel loop at $k_e = 40$: 1.5 e-folds inside the envelope from a start in the explicit ball ($\mathrm{d}t = 0.005$, below the discretisation floor); $\lambda' > 10 \times$ Theorem E's rate from Lemma E.0 and $< 4 \times$ the exact linear rate; the constants table | `Ex::test_corollary_E0prime_2_envelope_on_the_implemented_loop`, `exp_explicit_rate.py` (table 5) |
| Corollary E.0′.3 | the explicit bound on the stationary parameter spread of the dithered (LTV) dominates the exact periodic covariance (sinusoid: ratios $2.7 \to 1.3$ ($\delta = 0.2$) and $2.5 \to 1.2$ ($\delta = 0.1$) at $k = 20 \to 200$, falling toward $1/(1-\delta)$; the Theorem E ideal: 15–18, the price of $q = 15$ directions at the worst excitation); the exact value equals the averaged $k \cdot \mathrm{tr}\, X$ within $2/k^2$ | `Ex::test_corollary_E0prime_3_dither_covariance_bound_against_the_periodic_lyapunov_solution`, `exp_explicit_rate.py` → `results/explicit_rate/table.md` (table 6) |
| Lemma E.0″ | on $\Psi = (\sin t, \cos\sqrt{2}\,t)$ (not periodic) with window $4\pi$: the exponential-average identities on the grid ($10^{-4}$), the conditions at $k = 20$, 50, 200 with $\delta_2 \le 5\%$, the certificate on 400 times, the guaranteed rate below the exact decay rate (top Lyapunov exponent) within a factor 2.6 and improving with $k$; the experiment: exact/guaranteed $2.5 \to 1.3$ on the quasi-periodic regressor and $5.0 \to 3.7$ on a slowed Lorenz-driven regressor ($k = 100$–$400$), the certificate holding at every admissible row | `Ex::test_lemma_E0dprime_quasi_periodic_regressor_certificate_and_rate`, `exp_pe_general.py` → `results/explicit_rate/general_pe.md` |
| Lemma E.0′ | the certificate $\sup_{t,x} \dot V/V \le -\gamma$ (a generalised eigenvalue problem on a time grid) on $\Psi = (\sin t, \cos t)$ at $k = 20$, 50, 200 and on the slow-adaptation ideal at $g = 0.01$, $k = 400$; $N_2 = 0$ numerically; the guaranteed rate below the exact Floquet rate and within a factor 2 (sinusoid) and 1.5 (ideal) of it, above Lemma E.0's by more than $10^4$ on the ideal; the table of $k$ and $\delta$ | `Ex::test_lemma_E0prime_sinusoidal_regressor_certificate_and_sharpness`, `Ex::test_lemma_E0prime_pays_for_ill_conditioning_once_on_the_theorem_E_ideal`, `exp_explicit_rate.py` → `results/explicit_rate/table.md` (table 4) |
| G(a), two scales | direct check (cosine $\ge 0.998$ in the $G^{-1}$ metric, `violating`; the weight block fails in `satisfying`, as explained in Section 19); $\bar L$ at period-mean parameters non-increasing | `exp_variable_projection.py` → `table_violating.md`, `table_satisfying.md` |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| G(b), (b′), (H2c) | VarPro holds when (H2b) and $S \ll 1$ hold; it fails when $S = O(1)$ although (H2b) holds; the induced-metric flow tracks the closed loop at every $S$ (started on the slow manifold; gains $\div 4$ isolates the averaging error) | `exp_variable_projection.py` → `table_satisfying.md`; `exp_slow_manifold_slope.py` → `table_slope.md` |
| H(i), algebraic identity | with $e := r/k_e$ and $x := x_m$, the averaged drift equals $-G\nabla V$ (code gradients against finite differences, $10^{-7}$); does not test averaging | `Lg::test_averaged_drift_is_the_mean_field_langevin_drift` |
| Remark H.0 | the frozen-$\mu$ averaging error decays like $1/k_e$, $n = 1$ and $n = 2$ | `Lg::test_averaging_error_of_the_formal_reduction_decreases_like_one_over_ke`, `Av::test_averaged_drift_error_decays_like_one_over_ke_for_an_orbit_enclosing_area` |
| H, need for entropy | finite-$m$ real loop: a noiseless symmetric start stays bit-identical (and moves); dither separates it | `Lg::test_noiseless_symmetric_initialisation_never_separates_but_dither_does` |
| H(ii), oscillation bound | $\lvert V_L[\mu](\vartheta)\rvert \le K_\Phi(F_\rho + K_\Phi)$ for random and adversarial $\mu$, $\vartheta$ | `Lg::test_first_variation_bound_used_in_holley_stroock` |
| H(ii) constant | formula transcription only | `Lg::test_lsi_lower_bound_formula` |
| H, global behaviour | collapsed against spread starts, with and without dither, two $(\sigma, \tau)$ settings | `exp_langevin_global.py` |
| H, speed–bias trade-off | escape time and loss floor against $\tau$; $\log(1/\tau)$ and Arrhenius fits, seed-averaged and per seed (neither identifies the mechanism) | `exp_langevin_tradeoff.py` |
| H, reference values | $\bar{L}(\mu^*_\tau)$ by a concave dual on a grid; the stationary state of the exactly averaged loop; their offset in the scaling of H.6 | `exp_gibbs_reference.py` → `results/langevin_global/gibbs_reference.md` |
| H.7 | expansion, Stokes area identity, parity under reversal, non-reciprocity of the exact second variation | `Av::test_frozen_measure_expansion_converges_at_rate_one_over_ke`, `Av::test_circulation_kernel_antisymmetric_vanishes_in_1d_and_equals_the_enclosed_area_integral`, `Av::test_shift_correction_is_even_and_lag_correction_is_odd_under_orbit_reversal`, `Av::test_exactly_averaged_dynamics_is_non_reciprocal_for_n1_and_n2` |
| H.7a, oscillation term | value against an independent one-period near-identity flow (within 3% at $k_e = 80$); odd under orbit reversal ($10^{-6}$) | `Av::test_oscillation_correction_matches_a_one_period_flow`, `Av::test_oscillation_correction_is_odd_under_orbit_reversal` |
| I(i), (ii) | $a^\top B(\xi, \xi')a'$ equals the antisymmetric part of $S_{\text{even}} + K$ ($10^{-10}$); $A(\vartheta, \vartheta_z) = 0$ and the cyclic sum over $(\vartheta_0, \vartheta_1, \vartheta_z)$ equals $A(\vartheta_0, \vartheta_1)$ ($10^{-14}$) | `NR::test_bilinear_form_B_reproduces_the_antisymmetric_second_variation`, `NR::test_zero_feature_collapses_the_cyclic_criterion_to_a_pairwise_one` |
| I(iii) | $n = 1$: $A \equiv 0$ at perfect fit ($< 10^{-13}$); $A \neq 0$ with a residual, equal to the closed form; the positivity witness | `NR::test_n1_reciprocal_iff_residual_vanishes_on_the_orbit`, `NR::test_positivity_witness_n1` |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| I(iv) | the closed form of $\kappa$ ($10^{-10}$); the second component of $F_s((0.5,0))$ is positive on the ellipse; $\kappa \equiv 0$ on a zero-current orbit; with $f \equiv \hat{f}_\mu$ on $\mathbb{R}^2$ ($Dr_\mu \equiv 0$) $B = 2\kappa I$, $A \neq 0$ on the ellipse and $A \equiv 0$ on the segment; with $f = \hat{f}_\mu + he_1$, $h$ zero on the ellipse, the curl term survives and equals $-\int \varphi\varphi' \partial_2 h \, \mathrm{d}\rho \cdot (e_1 e_2^\top - e_2 e_1^\top)$ | `NR::test_circulation_kernel_closed_form_and_vanishing_for_a_zero_current_orbit`, `NR::test_perfect_fit_n2_leaves_only_the_circulation_kernel`, `NR::test_perfect_fit_on_the_orbit_alone_keeps_the_curl_term` |
| J, averaged objective | $\nabla J_m$ against finite differences ($10^{-7}$); the critical point is a non-degenerate minimum with two distinct kernels | `Pe::test_gradient_of_J_m_matches_finite_differences_and_critical_point_is_nondegenerate` |
| J(a), (b), (d) | Newton shooting finds the periodic orbit (residual $< 10^{-10}$); $\|\bar{\Theta}_p - \Theta^*\|$ and $\max\|e\|$ halve from $k_e = 20$ to 40; the vector error of the first-order prediction of the offset halves as well (it is 0.55 at $k_e = 40$ in the fast-sweep test instance, and 0.041 at $k_e = 40$ in instance A of the experiment); the parameter Floquet exponents match $\frac{1}{k_e}\lambda(G_m^{1/2}\nabla^2 J_m G_m^{1/2})$ with a maximal relative error below 2% at $k_e = 40$ that halves with $k_e$; both instances at $k_e = 10$–$160$ in the experiment; the error-channel exponent recovered from Liouville's trace identity is $k_e - L$ with $\|L\| < 0.2$ | `Pe::test_theorem_J_periodic_orbit_offset_and_floquet_exponents`, `exp_periodic_orbit.py` → `results/periodic_orbit/table.md` |
| K(i), (ii) | the Hessian of $J_m$ at a collapsed configuration equals $I_m \otimes (\sigma_0 G^{-1} - R_2) + \frac{1}{m}(\mathbf{1}\mathbf{1}^\top) \otimes M_2$ (finite differences, $10^{-5}$ relative); $\sigma_{\mathrm{crit}}$ solves $\sigma = \lambda_{\max}(G^{1/2} R_2(\bar{\vartheta}(\sigma)) G^{1/2})$ ($10^{-8}$) and the separation block is singular there; the separation Floquet exponents of the collapsed orbit come in $(m-1)$-fold multiplets | `Pe::test_proposition_K_hessian_splitting_and_critical_leak`, `exp_orientation_stability.py` |
| K(iii), (iv) | the orientation difference of the softest separation exponent at $\sigma_0 = 0.5$: slope $-2.18$ against $k_e$, within 14% → 3% of the prediction at $\bar{\Theta}_2$ for $k_e = 10 \to 80$, while the prediction at $\Theta^*$ converges to a value 30% off; at $\sigma_{\mathrm{crit}}$ both orientations are unstable, predictions within 14% at $k_e = 40$; at the designed $\sigma_0$ the signs differ for $k_e = 10$ and 20 | `exp_orientation_stability.py` → `results/orientation_stability/table.md` |
| J′ averaged data, solver | the single-kernel $J$, $\nabla J$, $b_{\mathrm{fr}}$, $b_{\mathrm{osc}}$ against the general H.7/H.7a code ($2 \cdot 10^{-5}$); the weighted Poisson solver against the Ornstein–Uhlenbeck closed forms ($g$, the mean shift $(G\lambda)^{-1}b$, the Green identity, zero current for a gradient drift); the stationary Fokker–Planck solver of the first-order SDE: current ratio → 1 linearly in $\varepsilon$, $\|m - \pi(1+\varepsilon g)\|_1 \propto \varepsilon^2$ | `tests/test_dither.py` (3 tests) |

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*Table 47, continued*

| statement | what is checked | where |
|---|---|---|
| J′(b), (c), (d) | real loop, $2 \cdot 10^4$ replicas, $k_e = 10$–$80$, both orientations: the stationary means differ from the Gibbs values by $k_e^{-0.87}$ ($\alpha$), $k_e^{-0.83}$ ($\gamma$), $k_e^{-1.09}$ ($\alpha\gamma$); the first-order prediction removes 93–97% of the discrepancy at $k_e = 10$ and 70–85% at $k_e = 20$–$40$, the residual at the Monte Carlo floor at $k_e = 80$ (second order not resolved); the orientation-odd shift matches its prediction to 7–8% at $k_e = 10$ and within the 10–30% resolution at $k_e = 20$–$80$; $k_e^2 \mathbb{E}e^2/(2\pi(\bar{L})) - 1 = 0.5$–$0.7/k_e$ | `tests/test_dither.py::test_real_loop_means_are_closer_to_the_first_order_prediction_than_to_gibbs` ($k_e = 20$), `exp_dither_stationary.py` → `results/dither/table.md` |
| J′(e), J′.2 | the circulation of the real loop is below the Monte Carlo resolution at first order ($k_e \geq 20$) and dominated by higher orders at $k_e = 10$; the current and density expansion of the first-order averaged SDE checked on the grid (ratios $1.47 \to 1.03$, $\varepsilon^2$ remainder 0.31); the antisymmetric part of $D(G^{-1}b)$ against the formula of J′.2(iii) to six digits (referee round 6) | `exp_dither_stationary.py` → `results/dither/table.md`; `review/REVIEW_round6_Jprime.md` |
| Corollary J′.3 | the Gibbs measures $\pi_m$ (MALA, 4000 replicas, $m = 1$–$32$) converge to the mean-field minimiser $\mu_\tau^*$ (fixed point to $10^{-11}$) with a $1/m$ law: slopes $-1.01$ ($\mathbb{E}[\gamma_1]$) and $-0.99$ ($\mathbb{E}[\bar{L}(\mu_\Theta)]$), rescaled deviation $0.058 \pm 0.002$ at every $m$), $\mathbb{E}[\alpha_1^2]$ from 0.54 to 0.92 against 0.935 with rescaled deviation $-0.41 \pm 0.01$; the $m = 1$ row agrees with the quadrature of $\pi_1$ | `exp_gibbs_m.py` → `results/dither/gibbs_m.md`; `tests/test_dither.py::test_corollary_J3_mean_field_minimiser_and_gibbs_sampler_consistency` |
| H.7 + H.7a in the real closed loop | the orientation dependence of the learned kernels (same $\rho$) is predicted to within 3% in size and 11% as a vector, cosine 0.994 at $k_e = 40$; the measured difference scales like $k_e^{-1.01}$ | `exp_orbit_orientation.py` → `results/orbit_orientation/table_sigma2_slow16.md` |

Not checked in the repository:

- B.5 along (L), and B.6 and B.7;
- the Filippov form of C(i), the inequality of C(v), and Proposition C.5's $D^+V_{\mathrm{OT}} = \sigma_m(\delta - \delta^2)$;
- D(b), D's constant, and the one-sided constant of Remark D.1;
- the sharpness of the explicit rates of E beyond the tables of Remark E.0a and Lemma E.0′; Lemma E.0′ below its threshold $k_0$ (where it makes no claim) and for non-periodic PE; Corollary E.2 and Remark E.3;
- Theorem I(v) beyond the two instances (the analytic dichotomy is a proof only);
- Theorem J(c) (the basin; a single basin run is reported in `review/REVIEW_round5_J.md`);
- Proposition K(iii)–(iv) beyond the instance of `exp_orientation_stability.py`, and the second-order normal form it rests on;
- Theorem C′(iv) (its hypotheses are not met on any tested instance) and the convergence of the pure leak to a critical point beyond the tested instances; Theorem D′'s constant, its per-particle bound and its $k$-particle convergence (only the rate of (b) is checked);
- H.8, which is a proof only;
- Conjecture H.6.

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